# TELEGANA STATE
# &
# ANDHRA PRADESH
## PUBLIC SERVICE COMMISSION

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# CIVIL ENGINEERING
## [English Medium]
## Previous Years Objective Solved Papers


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# SYLLABUS

## Telangana State Public Service Commission, Hyderabad
## Scheme and Syllabus For Recruitment To The Post of Lecturers in Government Polytechnics in Technical Education

| Written Examination (objective Type | | No. of Questions | Duration (Minutes) | Marks |
|---|---|---|---|---|
| **Paper–I** | **General Studies and General Abilities** | **150** | **150** | **150** |
| **Paper–I** | **Concerned subject** | **150** | **150** | **300** |
| **TOTAL MARKS** | | | | **450** |

1. Current affairs– Regional, National and International.
2. International Relations and Events.
3. General Science, India's Achievements in Science and Technology.
4. Environmental issues, disaster Management-Prevention and Mitigation Strategies.
5. Economic and Social Development of India and Telangana.
6. Physical, Social and Economic geography of India.
7. Physical Social and Economic geography and Demography of Telangana.
8. Socio- economic, Political and Cultural History of Modern India with special emphasis on Indian National Movement.
9. Socio- economic, Political and Cultural History of Telangana with special emphasis on Telangana Statehood movement and formation of Telangana state.
10. Indian Constitution, India Political System Governance and Public Policy.
11. Social Exclusion, Rights issues such as Gender, Caste, Tribe, Disability etc. and inclusive policies.
12. Society, Culture, Heritage, Arts and Literature of Telangana.
13. Policies of Telangana State.
14. Logical reasoning, Analytical Ability and Data interpretation.
15. Basic English. ($10^{th}$ class Standard)

### PAPER–II CONCERNED SUBJECT
### CIVIL ENGINEERING (DEGREE LEVEL)

**1. Building Materials And Construction:**

Bricks–Types of Bricks, Indian standard classification, properties; Stones- Types of stones, classification, properties, dressing and polishing of stones, Methods of Quarrying, Cement- Different grades and types of cement, properties and IS specifications , Aggregates- coarse and fine aggregate, properties and IS specifications, Cement Mortar- Proportions of cement mortar for various applications, Concrete - Constituents of concrete, different grades of concrete, mix proportioning using IS code, Properties of fresh and hardened Concrete, Admixtures- Types of Admixtures.

**2. Strength of Materials And Theory of Structures:**

Strength of Materials: simple stresses and strains, elastic constants and relationship between then, Compound bars, Temperature stresses, Shear forces and bending moment diagrams for beams Principal stresses and Mohr's circle of stress, Theory of bending and bending stresses, Shear stress distribution, Theory of torsion Springs, Deflections of beams, Thin and thick cylinders,, Analysis of trusses, Betti-Maxwell theorem, shear centre and unsymmetrical bending.

Theory of Structure: Direct and bending stresses; Columns and struts, Strain energy method, Moving loads and influence lines Arches and suspension bridges, Static and kinematic indeterminacy Moment distribution, Slope deflection and Kani's methods applied to continuous beams and portal frames, matrix methods of analysis.

**3. Cement Concrete and Steel structures:**

**Concrete Structures:** Materials, permissible stresses and IS Specification Working stress methods, Limit State Method- Stress Blocks parameters, design of Beams, Slabs, Columns and Footing Design for shear and Torsion, Design of Retaining Walls, Water tanks, and T-Beam Slab bridges, Yield line theory.

**Steel Structures:** Properties of steel sections, permissible stresses, IS Specifications Riveted and welded joints and connections, Design of simple and compound Beams and columns, Column bases, Roof trusses, Plate and Gantry Girders, Plate Girder Lattice Girder Railway bridges, and Bearings. Plastic analysis.

**Pre-Stressed Concrete:** Basic Concepts, material for pre-stressing losses in Pre-stress, classification of pre- stressing system Analysis of PSC sections.

**4. Fluid Mechanics and Hydraulics:**

Fluid Properties, Measurement of Pressure Manometers; Fluid Kinematics classification of Fluids Stream function and Velocity potential, significance and used of Flownets, Fluid dynamics continuity equation Bernoulli's equations and impulse momentum equation, Laminar and Turbulent flow through pipes- significance of Reynolds number Hagen- Poiseuille's equation, Darcy-Weisbach equation, Friction factor, Water hammer concepts, Compressible flow Bernoulli's equation for isothermal and Adiabatic conditions, Mach Number Mach cone, stagnation properties, Steady uniform flow through open channels, Gradually varied flows significance of Froude number, classification and computation of Flow profiles, Hydraulic jump, Surges, Boundary layer-laminar and Turbulent Boundary layer Boundary layer thickness, rough and smooth Boundaries, Boundary layer separation Dimensional analysis and similarity laws, Hydraulic Turbines-classification, Velocity triangles, principles and design of reaction and impulse turbines, Centrifugal pumps specific speed, work done and efficiency, characteristic curves.

**5. Hydrology and Water Resources Engineering:**

Hydrological cycle; Rainfall - types and measurement, network design, lifiltration-$\phi$- index, Runoff- process factors and determination of runoff, dependable yield, Floods-flood hydrograph, computation of flood peak using rational formula, unit hydrograph method and Gumbel's extreme value methods, Groundwater- types of aquifer and properties, Darcy's law, specific yield, steady radial flow to wells in confined and unconfined aquifers, irrigation-types and advantages, soil water plant relationship, consumptive use, duty, delta, base period, crops and their water requirements, Single and multipurpose projects Dams- classification, forces and design of Gravity dam and Earth dam, Spillways types, energy dissipation stilling basin Appurtenances Canals alignment kennedy's and Lacey's theories, lining of canals; Weirs- components, design of vertical drop and sloping glacis weir, Seepage forces- Bligh's Theory.

Khosla's theory; Canal falls- types and design principles, cross drainage works classification and design principles of aqueducts Hydropower principles- classification and components of Hydroelectric power plants.

6. **Environmental Engineering:**

Water supply- objectives, rate of demand, population forecasts, Analysis of water classification design of coagulation, sedimentation filtration, disinfection and softening processes Methods of layout of distribution pipes-Hardy cross method, Waste water engineering- systems of sewerage, hydraulic formulae and design of sewers, BOD COD, self purification of natural streams, methods of sewage disposal, Treatment of sewage principles and design of grit chamber, sedimentation tanks trickling filters, activated sludge process, sludge digestion tanks, septic tanks Municipal solid waste characteristics, collection and transportation of solid wasters Air Pollution types and sources of pollutants, air quality standards, Noise pollution Impacts and permissible limits, measurement and control of noise pollution.

**7. Transportation Engineering:**

Highway Classification as per IRC Highway alignment, Engineering Surveys Geometric Design, Cross sectional elements of road Grades grade compensation, Traffic. Surveys speed volumes, origin and destination intersection at grade and grade separated, channelization Rotary intersection signal design-web star method, traffic signs, pavement marking Parking studies, accidental studies, pavement types Factors considered for pavement design, flexible and rigid pavements design concepts.

Railway Engineering Permanent way, rails, sleepers, ballast, Creep conning of wheel rail fixtures and fastenings, super elevation cant deficiency curves, turnout Points and crossings.

Airport Engineering Selection of site of Airport, runway orientation and design, wind rose diagram basic run way length correction to basic runway length.

**8. Soil Mechanics and Foundation Engineering:**

Soil Mechanics Physical properties of soils, Classification and identification, Permeability, Capillarity, Seepage Compaction, Consolidation shear Strength Mohr's circle, Earth pressure, Slope stability.

Foundation Engineering Site investigations, stress distribution in soils Bearing capacity, Settlement analysis, Types of Foundation, Pile foundations, Foundations on expansive soils, swelling and its preventions Coffer dams, Caissons, Dewatering Bracing for excavations, Newmark charts, machine foundations.

Engineering Geology Mineralogy structure Geology Groundwater Exploration methods, Engineering Geology applications for Tunnels, Dams and Reservoirs, Geological hazards and preventive measures.

**9. Estimation, Costing and Construction Management:**

Abstract estimate: Detailed estimate- centerline, long & short wall method, various items of Civil Engineering works as per Indian Standard, General Specifications Earth Work Brick/Stone Masonry in Cement Mortar RCC, Plastering in Cement Mortar, Floor finishes, white wash colour wash standard schedule of rate, lead and lift, preparation of lead statement, computation of earth work- mid-ordinate, mean sectional area, Trapezoidal method, Prismoidal Rule, Approximate estimate Plinth area and cubic rate estimate.

**10. Construction Management:**

Types of construction projects, Tendering and construction contracts, project planning and network analysis - PERT and CPM.

**11. Surveying:**

Principle and classification of surveying, chain surveying, Compass surveying Levelling and contouring, Theodolite surveying curves, Introduction and Fundamental concepts of electronic measuring instruments -EDM, Total station, components of GPS and basics of GIS.

# Syllabus

# Telangana State Public Service Commission, Hyderabad

## Scheme and Syllabus For Assistant Engineer, Municipal Assistant Engineer, Technical Officer and Junior Technical officer.

| Written examination (Objective type) | No. of Questions | Duration (minutes) | Maximum Marks |
|---|---|---|---|
| **Paper-I: General Studies and General abilities** | **150** | **150** | **150** |
| **Paper-II Civil Engineering (Diploma Level) OR Mechanical Engineering (Diploma Level) OR Electrical and Electronic Engineering (Diploma Level)** | **150** | **150** | **150** |
| | | **Total** | **300** |

| NAME OF THE PAPERS | LANGUAGE OF EXAMINATION |
|---|---|
| **Paper-I : General Studies and General Abilities** | **Bilingual i.e., English and Telugu** |
| **Paper-II Concerned Subject (diploma Level)** | **English** |

**PAPER-I: GENERAL STUDIES AND GENERAL ABILITIES–**

1. Current Affairs– Regional, National and International, 2. International Relations and Events.
3. General science, India's achievements in Science and Technology
4. Environmental issues and Disaster Management
5. Economy of India and Telangana, 6. Geography of India with a focus on Telangana
7. India Constitution and Polity with a focus on local self Government
8. Society, culture, Heritage, Arts and literature of Telangana
9. Policies of Telangana State 10. History of Modern India with a focus on India national Movement
11. History of Telangana with special emphasis on Movement for Telangana statehood
12. Logical Reasoning Analytical Ability and Data Interpretation
13. Basis English ($8^{th}$ Class Standard)

**PAPER-II: CIVIL ENGINEERING (DIPLOMA LEVEL)**

**1. Surveying :**

Fundamental concepts Classification of Surveys chain surveying compass surveying , Levelling and Contouring Theodolite Surveying Tacheometry Curves, Introduction and fundamental concepts of electronic measuring instrucments EDM, Total station GIS & GPS.

**2. Construction Materials & Practice :**

Properties and uses of construction materials stones, bricks, Tiles, Sand Cement, Timber, Plastics, Glass, Asbestos, Paints, Distempers, Enamels and Varnishes, Preparation of Cement mortar for various works.

Classification of Buildings as per NBC Site investigation for foundation as per NBC Trial Pit and auger boring classification of foundation, construction of spread footing and well foundation stone and brick masonry types and principles of construction doors and Windows types, fittings and fastenings, types and function of Lintels, Sunshades and Roofs flooring construction and types of material Types of Stairs Scaffolding types of plastering Pointing Painting and White/ colour wash.

**3. Engineering Mechanics and Strength of Materials :**

Forces- types of Forces, Parallelogram, Triangle and Polygon Law of forces, Lami's theorem Centre of Gravity and Moment of Inertial simple stresses and strains, Hooke's law stress strain diagram working strength elastic constants, Poisson's ratio Relationship between elastic constants, compound rods , temperature stresses, strain energy proof resilience, impact loading shear force and bending moment diagrams for simply supported, over hanging and cantilever beams, relation between intensity of loading shear force and bending moment Theory of simple bending modulus of section moment of resistance, distribution of shear stress in rectangular circular and I-section Deflection is cantilever and simply supported beams subjected to simple loading columns and struts- Euler's and Rankine's formulae, slenderness ratio simple built up columns Analysis of dams and retaining walls, simple plane and pin- jointed trusses, Stresses by method of joints and method of sections.

**4. Hydraulics :**

Properties of fluids, fluid pressure and its measurement Types of flows, energies in fluid motion, Bernoulli's theorem and its application- venture metre, pitot tube Orifice and mouthpiece, Notches and weirs, Flow through pipes, hydraulic gradient line and total energy line, laminar and turbulent flow in pipes, Reynolds number measurement of velocity open channels water turbines classification centrifugal and reciprocating pumps Layout of hydroelectric power plant.

**5. Quantity Surveying :**

Abstract estimate, detailed estimate centre line and long & short wall method, various items of Civil Engineering works as per India standards, General specification earth work brick/ stone masonry in cement mortar , RCC plastering in cement mortar Floor finishes with ceramic tiles and marbles white washing colour washing Standard schedule of rates, lead and lift preparation of lead statement computation of earth work Mid-ordinate mean sectional area Trapezoidal method, Prismoidal Rule approximate estimate plinth area and cubic rate estimate.

**6. Design of Structures (RCC and Steel) :**

RCC structures Design philosophies principles and concepts of working stress method and limit state method, loads and permissible stresses, IS specification, analysis and design rectangular beam slab T-beam column, footing and stair case. Steel structures. Properties of steel sections, loads and permissible stresses, IS specification, Analysis and design welded joints, beam column column base tension member design of roof truss.

**7. Irrigation Engineering :**

Definitions, duty, delta, base period, rainfall and its measurement factor affecting runoff, methods of computing maximum flood discharge, classification of head works component parts of a weir and barrage, factors influencing selection of site reservoirs and dams, classification of canals, canal lining cross drainage works soil erosion water logging soil water plant relationship necessity of irrigation advantages and disadvantages, irrigation methods.

**8. Environmental Engineering :**

Basics of ecosystem water supply scheme, Sources of water conveyance of water pipes joints and laying testing of water drinking water standards Treatment of water Distribution of water supply connection to a building. Quantity of sewage, surface drains, design of sewers running half full, limiting velocities, Laying of sewers sewer appurtenances collection of sewage samples, characteristics of domestic and industrial sewage-BOD-COD Sewage treatment septic tank & soak pit, sewage disposal dilution and sewage farming House drainage arrangements in building solid waste collection and disposal Air pollution sources, effects and controlling methods.

**9. Transportation Engineering :**

Alignment of roads- of roads- plain and hilly terrain surveys cross section of road structure width of pavement camber gradient super elevation transition curves horizontal and vertical alignment Pavement marking traffic signs traffic islands. Types of soil classification of soil Textural IS classification physical properties plasticity cohesion consolidation compaction permeability compressibility soil moisture content specific gravity density Bearing capacity of soil.

# Trend Analysis

## Telangana/Andhra Pradesh

| SL. No. | Paper name | Year | No. of question |
|---|---|---|---|
| 1. | TSPSC AEE | 22.01.2023 | 150 |
| 2. | APPSC (Env. Engg.) _Shift 1 | 14.05.2022 | 150 |
| 3. | APPSC Asst. Director Paper II | 22.10.2022 | 150 |
| 4. | APPSC AE shift 2 | 15.05.2022 | 150 |
| 5. | TSPSC HMWSSB Manager | 2020 | 150 |
| 6. | APPSC Poly Lect. | 2020 | 150 |
| 7. | APPSC Deputy Inspector of Survey | 27.09.2020 | 150 |
| 8. | TSPSC AEE | 2018 | 150 |
| 9. | TSPSCPCB (Asst. Env. Engg) | 2017 | 150 |
| 10. | TSPSC AEE (Civil/Mechanical) common | 2017 | 150 |
| 11. | TSPSC AEE | 2017 | 150 |
| 12. | APPSC-AEE-Mains | 2016 | 150 |
| 13. | APPSC AEE | 2016 | 150 |
| 14. | APPSC AE subordinate | 2016 | 150 |
| 15. | APPSC AE (Environmental & Civil) | 2016 | 150 |
| 16. | APPSC AE (Mains) Paper-III | 2016 | 150 |
| 17. | TSPSC Deputy Engineer | 2015 | 150 |
| 18. | TSPSC AE | 2015 | 150 |
| 19. | TSPSC AEE | 20.09.2015 | 150 |
| 20. | APPSC AEE( OBSERVERS ) | 2013 | 150 |
| 21. | APPSC AE | 2013 | 150 |
| 22. | APPSC AEE, Paper-1 | 2012 | 150 |
| 23. | APPSC AEE Paper-2 | 2012 | 150 |
| 24. | APPSC Public health & Municipal Engg. | 2012 | 150 |
| 25. | TSPSC AEE Paper-1 | 2011 | 150 |
| 26. | TSPSC AEE Paper-2 | 2011 | 150 |
| 27. | APPSC AEE Paper- 1 | 2004 | 150 |
| 28. | APPSC AEE Paper- 2 | 2004 | 150 |
| 29. | APPSC AEE | 2003 | 150 |
| 30. | APPSC AEE | 1993 | 100 |
| 31. | APPSC AEE | 1993 | 100 |
| 32. | APPSC AEE | 1987 | 100 |
| 33. | APPSC AEE | 1984 | 100 |
| | | **TOTAL** | **= 4600** |

# Telangana State Public Service Commission
# Assistant Engineer Exam. 2023
## Solved Paper

**Date - 22-01-2023**

**1. Which of the following statements cannot be used to describe free flow speed ($u_f$) a traffic stream?**

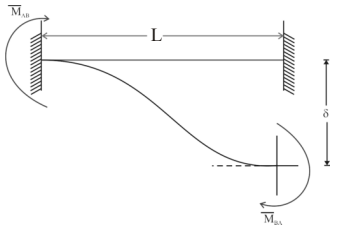
(a) $u_f$ is the speed when flow is negligible
(b) $u_f$ is the speed at which flow is maximum and density is optimum
(c) $u_f$ is affected by geometry of the rod
(d) $u_f$ is the speed when density is negligible

**Ans. (b) :** <u>**Free flow speed-**</u>
Free flow speed (FFS) is the speed of traffic as density approaches zero. Practically, it is the speed at which drivers feel comfortable traveling under the physical, environmental, and traffic control conditions exiting on an uncongested section of free way or multilane highway.

**2. What is the far end moment for a beam element of length L due to a settlement of δ at the near end? The beam bas the modulus of elasticity and moment of inertia as E and I, respectively.**

(a) $\frac{4EI\delta}{L^2}$ clockwise
(b) $\frac{6EI\delta}{L^2}$ anti – clockwise
(c) $\frac{6EI\delta}{L^2}$ anti – clockwise
(d) $\frac{4EI\delta}{L^2}$ anti – clockwise

**Ans. (c) :** Far end moment for a beam element of length L due to a settlement of δat the near end is –




$\overline{M}_{BA} = \frac{6EI\delta}{L^2}$ (clockwise)

**3. For sand bed, the Darcy's law would be valid if the Reynold's number defined by R= $\frac{\rho du}{\mu}$ is less than**

(a) 0.2 (b) 1000
(c) 10 (d) 1

**Ans. (d): Darcy's law** – in one dimensional flow, discharge through fully saturated soil is given by

$q = kiA$ or $v = ki$

$i = \frac{\Delta h}{\ell}$

Where,

A=cross sectional area of the soil corresponding to flow (q)
q = discharge
i = hydraulic gradient
Δh = loss of head in length 'L'
k = coefficient of permeability
v = Discharge velocity

Note - Darcy's law is valid only for laminar flow ( $R_e < 1$).

**4. Which method of Quarrying is suitable for costly, soft and stratified rocks?**

(a) Excavating (b) Heating
(c) Blasting (d) Wedging

**Ans. (d) : Wedging** –

- this method of quarrying is suitable for costly. Soft and stratified rocks such as sandstone, lime stone, late rite, marble and slate.
- About 10-15 cm deep holes, at around 10 cm spacing, are made vertically in the rock
- steel pins and wedges or plugs (conical wedges) and feathers are inserted in them.
- these plugs are then struck simultaneously with sledge hammer.

**5. A beam is prestressed by a straight tendon carrying a prestressing load P at an eccentricity e towards the bottom direction of the beam. Which match is the correct one for the beam?**

(a) Direct stress contribution i. minimum to resultant stress at the top fibre due to prestressing force
(b) Stress contribution to ii. subtractive resultant stress at the live load
(c) Maximum stress due iii. Additive to eccentricity iii. Additive
(d) Resulant stress at iv. Bottom the bottom fibre

(a) A- iii, B-i, C-iv, D-ii
(b) A- iv, B-ii, C-i, D-iii
(c) A- iii, B-ii, C-iv, D-i
(d) A- ii, B-i, C-iv, D-iii

**Ans. (c) :**

**6. If the Froude number of hydraulic jump is 5.5 it can be classified as**

(a) an oscillating jump (b) a steady jump
(c) a strong jump (d) a weak jump

**Ans. (b) :** Type of jump according to Froude number-

| Jump | $F_r$ (Number) |
|---|---|
| Undular Jump | $1 < F_r < 1.7$ |
| Weak Jump | $1.7 < F_r < 2.5$ |
| Oscillating Jump | $2.5 < F_r < 4.5$ |
| Steady Jump | $4.5 < F_r < 9$ |
| Strong Jump | $F_r > 9$ |

**7. The estimated value of the property at the end of its life period, without being dismantled is called as**

(a) Salvage value (b) Book value
(c) Scrap value (d) Market value

**Ans. (a) : <u>Salvage value</u>**

- It is value at the end of utility period without being this dismantle.
- It is the value of asset at the end of utility period. It implies that asset has further utility but due to some reason it is selling.

**8. Which of the following statements is/are correct for a plane stress isotropic material problem?**

(a) Three stress components can exist at a point.
(b) Six stress components can exist at a point.
(c) Three principal stresses can exist at a point.
(d) Two principal stresses can exist at a point.

(a) A and C only (b) A and D only
(c) D only (d) B and D only

**Ans. (b) : <u>Isotropic material</u>** –

- for an isotropic material, all planes are planes of material symmetry and are isotropic.
- there are 2 independent elastic constants associated with an isotropic material and 12 non zero terms in the stiffness matrix.

$\Rightarrow$ three stress components can exist at a point
$\Rightarrow$ two principal stresses can exist at a point

**9. A beam is designed using M 20 grade concrete and Fe 415 is used for the tension reinforcement. If the diameter of the main steel is 12 mm, what is the minimum value of the development length (Ld) in mm to be provided in the support section?**

(a) 300 (b) 470
(c) 564 (d) 260

**Ans. (c) :** Given,
M20
Fe415
$\phi$ = 12mm
$L_d$ = ?

$$L_d = \frac{0.87 f_y \phi}{4\tau_{bd}} \qquad \{\tau_{bd} = 1.2 N/mm^2$$

$$= \frac{0.87 \times 415 \times 12}{4 \times 1.6 \times 1.2}$$

$$= 564.14 \simeq 564mm$$

**10. The field capacity of a certain soil is 20% and the specific gravity is 1.6. Before applying irrigation water, a wet sample of soil was taken and its mass was found as 150 gm after over drying its mass was 136g. Determine the depth of water that must be applied to irrigate the soil to a depth o 0.9 m.**

(a) 90 mm (b) 180 mm
(c) 140 mm (d) 210 mm

**Ans. (c) :** Given d = 0.9m

**Given**

F.C = $\omega$ %

GS = 1.6 $= \frac{\gamma_d}{\gamma_\omega}$

$$\text{Water Contents} = \frac{150-136}{136} \times 100$$

$$= \frac{1400}{136} \approx 10\%$$

Solve- $\boxed{d_\omega = \frac{\gamma_d}{\gamma_\omega} \times d[\text{fc- omc}]}$

$= 1.6 \times 0.9 \times [.20 - .10]$

$\boxed{d_\omega = 0.144\,m} \simeq 144mm.$

**11. Le-Chateliers Apparatus is used to measure the unsoundness of cement caused due to the presence of**

(a) Unburnt lime (CaO)
(b) Unburnt lime (CaO) an unburnt magnesia (MgO) both
(c) Calcium sulphate ($CaSO_4$)
(d) Unburnt magnesia (MgO)

**Ans. (a) :** • Le-Chatelier Apparatus is used to measure the unsoundness of cement caused due to the presence of un burnt lime (CaO).

- the soundness of cement is determined either by 'Le chatelier's method' or by means of a 'Autoctave test.

No satisfactory test is available for deduction of soundness due to excess of calcium sulphate. But its content can be easily determined by chemical analysis.

- Le Chatelier's Method
- Auto clave Test.

**12. The specific gravity of particles of a sand is 2.64 and their porosity is 45% in loose state. the critical hydraulic gradient will be**

(a) 0.66 (b) 0.947
(c) 0.901 (d) 0.747

**Ans. (c) : Given**

$G = 2.64, \eta = 0.45, i_c = ?$

$$e = \frac{\eta}{1+\eta} = \frac{0.45}{1-0.45} = 0.818$$

$$i_c = \frac{G-1}{1+e} = \frac{2.64-1}{1+0.818} = 0.902$$

**13. Consider the following sets of derivatives for a prismatic beam.**

(a) Shear force (i) $\frac{d^2y}{dx^2}$
(b) Loading intensity (ii) $\frac{d^3y}{dx^3}$
(c) Bending moment (iii) $\frac{d^4y}{dx^4}$
(d) Slope (iv) $\frac{dy}{dx}$

Which combination is the correct combination?

(a) A-iv, B-i,C-ii,D-iii (b) A-ii, B-iii,C-i,D-iv
(c) A-ii, B-i,C-iv,D-iii (d) A-ii, B-i,C-iii,D-iv

**Ans. (b) :** slope - $\frac{dy}{dx}$

Bending moment - $\frac{d^2y}{dx^2}$

Shear force - $\frac{d^3y}{dx^2}$

Loading intensity - $\frac{d^4y}{dx^4}$

**14. The actual velocity of water flowing in a medium is observed as 0.30 m/s. The porosity of the medium is 0.40. The Darcy's velocity is**

(a) 0.12 m/s (b) 1.2 m/s
(c) 0.133 m/s (d) 0.075 m/s

**Ans. (a) :** Given,

$V_s = 0.30$m/sec

$\eta = 0.40$

$$v_s = \frac{v}{\eta}$$

$v = \eta V_s$

$= 0.4 \times 0.3 = 0.12$m/sec

**15. A steady, incompressible flow is given by $u = 2x^2 + y^2$ and $v = -4xy$. What is the convective acceleration along x- direction at point (1,2)?**

(a) -24 units (b) -8 units
(c) -24 units (d) -8 units

**Ans. (b) :** Given, point (1, 2)

$u = 2x^2 + y^2$

$= 2 \times 1 + 2^2$

$= 6$ units

$v = - 4xy$

$= - 4 \times 1 \times 2$

$$\frac{du}{dx} = 4x = 4 \times 1 = 4\text{units}$$

$$\frac{du}{dy} = 2y = 2 \times 2 = 4\text{ units}$$

$$ax = u\frac{du}{dx} + v\frac{du}{dy}$$

$= 6 \times 4 + (- 8 \times 4)$

$= 24 - 32$

$= - 8$ Units

**16. Match the name of stone in List 1 with the used of that stone in List2:**

| List -I | List- II |
|---|---|
| (a) Granite | (i) Ornamental work |
| (b) Limestone | (ii) Bridges |
| (c) Marble | (iii) Production of cement |
| (d) Slate | (iv) Flooring |

(a) A-iii, B-ii, C-i, D-iv (b) A-ii, B-iv, C-i, D-iii
(c) A-iii, B-iv, C-i, D-ii (d) A-ii, B-iii, C-i, D-iv

**Ans. (d) :** Type of rock and used,

| | | |
|---|---|---|
| Granite | - | Brides |
| Limestone | - | Production of cement |
| Marble | - | Ornamental |
| Slate | - | Flooring |
| Compacted sand stone | - | Fire proofing |

**17. The maximum number of vehicles observed in any five-minute period during the peak hour is 150. If the total flow in the peak hour is 1200 vehicles, the five-minute peak hour factor (round off to 2 decimal places) is**

(a) 0.67 (b) 0.45
(c) 0.52 (d) 0.78

**Ans. (a) : Given,**

maximum number of vehicle in 5 minute = 150 vehicles

Flow in the peak hour = 1200 vehicles

5 minute - peak hour factor

$$= \frac{\text{Peak hour flow}}{12(\text{5minute traffic})}$$

$$= \frac{1200}{12 \times 150}$$

$= 0.666 \simeq 0.67$

**18. In a model experiment with weir, if the dimension of the model weir are reduced by a factor k, the flow rate through the model weir is the following fraction of the flow rate through the prototype**

(a) $k^{5/2}$ (b) $k^{3/2}$

(c) 1 (d) $k^2$

**Ans. (a) :** Discharge (Q) = AV

$$= [L \times B]\left[\sqrt{2gH}\right]$$

Hence, flow ratio = $L_r^2\sqrt{L_r} = L_r^{5/2}$

Here $L_r = k$

Hence, discharge $\propto k^{5/2}$

**19. Which of the following statements is/are correct for Maxwell's law of deflection?**

(a) The work done by a set of external forces $F_m$ acting through displacement $\delta_{mm}$ produced by another set of forces $F_n$ is equal to the work done by the set of forces $F_n$ acting through displacements $\delta_{nm}$ produced by forces $F_m$. m and n are different set of points of a structure.

(b) The defection at any point $\delta_{nm}$ due to the force $F_m$ is equal to the deflection at the other point $\delta_{mm}$ due to the force $F_n$. m and n are two different points of a structure.

(c) The slope at any point $\theta_{nm}$ due to the force $F_m$ is equal to the slope at the other point $\theta_{mm}$ due to the force $F_n$. m and n are two different points of a structure.

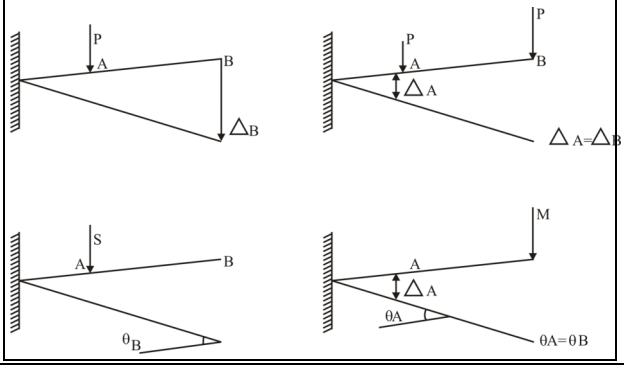
(d) The moment at any point $M_{nm}$ due to the force $F_m$ is equal to the moment at the other point $M_{mn}$ due to the force $F_n$. m and n are two different points of a structure.

(a) A and D only (b) C only

(c) A and B only (d) B only

**Ans. (d) : Maxwell's Reciprocal Theorem –**

In any beam, frame or truss, the defection at any point due to load p at any point A is equal to deflection at any point A due to load P at any point B.




**20. The type of surveying in which the curvature of the earth is considered is called**

(a) Preliminary surveying

(b) Plane surveying

(c) Geodetic surveying

(d) Topographical surveying

**Ans. (c) : Geodetic Survey-**

- It is done for fixing widely space control points.
- Geodetic surveying is done by Department of National Survey of India.
- Large area's earth's surface involved curvature are of earth is considered.

**21. The initial depth of hydraulic jump in a rectangular channel is 0.2 m and the sequent depth ratio in 10. The length of the jump is about**

(a) 4 m (b) 20 m

(c) 12 m (d) 6 m

**Ans. (c) :** Given,

$y_1 = 0.2M$

$\frac{y_2}{y_1} = 10$

$y_2 = 10 \times 0.2$

$= 2m$

Length of jump,

$L_j = 6.9 (y_2 - y_1)$

$= 6.9 (2-0.2)$

$= 12.42 \simeq 12m$

**22. Which of the following pairs are NOT correctly matched?**

(a) PERT- Probabilistic approach

(b) CPM- Float concept

(c) PERT- Three- time estimate

(d) PERT- Activity-oriented network model

**Ans. (d) :** • **PERT**- Stands for Programme evolution and review technique.

- it is based on probabilistic approach.
- three time estimates are made
- Time is directly proportional to time hence minimum cost will be achieved corresponding to minimum completion time.
- Suitable for new type of projects
- Each activity follows distribution.

**CPM -** Stands for critical path method .

this is based on deterministic approach in which only one time estimate is made for activity completion.

- A network diagram in CPM is activity is oriented.
- It is based on deterministic approach.
- there is only single time estimate for each activity.
- the probability of completion of activity in this estimated duration is 100 %

**23. In a Lacey's regime channel, the discharge is 100 $m^3/s$ and the silt factor is 1 : 1. The regime longitudinal slope is about**

(a) 1/5 100 (b) 1/4 100

(c) 1/7 100 (d) 1/6 140

**Ans. (d) : Given**

$Q = 100\ m^3/sec$

$f - 1.1$

$$S = \frac{(f)^{5/3}}{3340Q^{1/6}}$$

$$= \frac{(1.1)^{5/3}}{3340 \times (100)^{1/6}}$$

$$= \frac{1}{6138.92} \simeq \frac{1}{6140}$$

**24. Calculate the area in square meters enclosed between the survey line, the irregular boundary line, and the first and last offsets by Simpson's rule. The perpendicular offsets were taken at 20 m intervals from a survey line to an irregular boundary line: 2.35,6.50, 5.40, 6.65, 7.85, 6.2, 2.35, 6.5, 5.65**

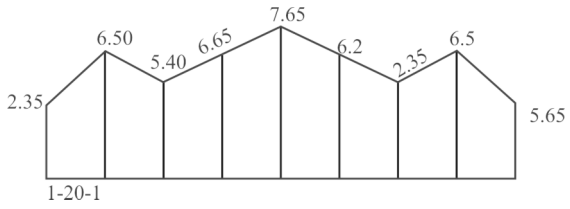
(a) 800 (b) 950

(c) 900 (d) 850

**Ans. (b) :** Using Simpson's one-third rule,

Given,

d = 20M

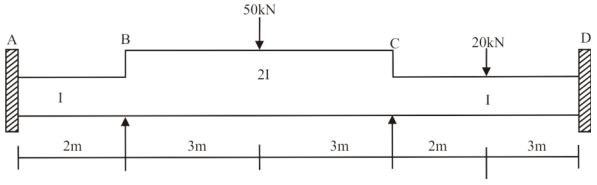



$$A = \frac{d}{3}\left[(0_1 + 0_n) + 4(0_2 + 0_4 + \text{---} + 0n\text{-}1) + 2(0_3 + 0_5 + \text{---} 0_{n\text{-}2})\right]$$

$$= \frac{20}{3}[(2.35 + 5.65) + 4(6.50 + 6.65 + 6.2 + 6.50) + 2(5.40 + 7.85 + 2.35)]$$

$$= 950.66 \simeq 95\ Om^2$$

**25. What is the distribution factor for the CB element at point C in the beam given below?**




(a) 0.50 (b) 0.62

(c) 0.40 (d) 0.57

**Ans. (d) :**

$$DF = \frac{\text{Stiffness of a member}}{\text{Sum of stiffness of all members at that joint}}$$

$$S_{CB} = \frac{4E(2I)}{6}$$

$$= \frac{8EI}{6} = \frac{4EI}{3}$$

$$S_{CD} = \frac{4EI}{4}$$

$$= EI$$

$$D.F = \frac{S_{CB}}{S_{CB} + S_{CD}}$$

$$= \frac{4EI/3}{\frac{4EI}{3} + EI}$$

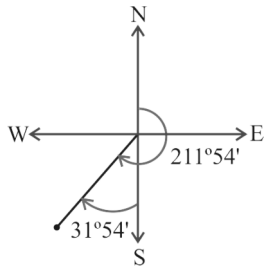
$$= 0.5714 \simeq 0.57$$

**26. Convert the circle bearing of 211°54' to quadrantal bearing.**

(a) S 31° 54' W (b) N 31° 54' W

(c) N 31° 54' E (d) S 31° 54' E

**Ans. (a) :**




**W.**C.B. = $211^0 54'$

quadrantal bearing,

= S 31° 54' W

**27. The 3-hr UH of a basin can be approximated as a triangle with a base period of 20 hour and peak coordinate of 300 m3/s. The area of the basin is**

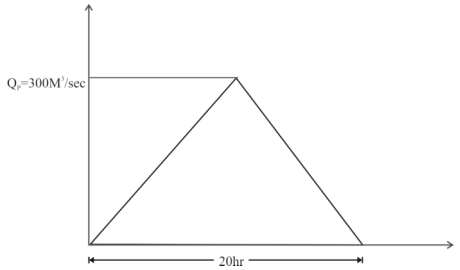
(a) 540 $km^2$ (b) 6000 $km^2$

(c) 300 $km^2$ (d) 1080 $km^2$

**Ans. (d) :** Given,

$Q_P = 300m^3$ / sec

base period = 20hr



$$A.\left(\frac{m}{100}\right)=\frac{1}{2}\times 20\,hr\times 300\frac{m^3}{Sec}$$

$$A=\frac{1}{2}\times 20\times 3600\times 300\times 100\ m^2$$

$$=1080 km^2$$

**28. Consider the following statement with respect to cements:**

(a) Vinsol resin or vegetable fats and oils and fatty acids are ground with ordinary cement to air entraining cement.

(b) Extra rapid hardening cement is prepared by adding 2% of calcium chloride.

Identify the correct statement.

(a) Statement B is true and A is false

(b) Both A and B are true

(c) Both A and B are false

(d) Statement B is false and A is true

**Ans. (b) :** • **Air Entraining cement –**

• Ordinary Portland Cement plus Vinsol resin or vegetable fats of oils or fatty acids.

• Small amount (0.1%) by weight of an air entrain agent.

**Extra Rapid hardening Cement :**

• Rapid hardening cement plus 2% $CaCl_2$ (also called calcium chloride cement)

• Especially used in cold weather but also give excessive shrinkage.

**29. On a section of a highway the speed denstity-relationship is linear and is given by $v=100-\frac{5}{6}k$; where v is in km/h and k is in veh/km. The capacity (in veh/h) of this section of the highway would be**

(a) Free flow speed is 100 kmph and jam density is 120 veh/km

(b) Maximum flow rate is 3000 veh/h and critical density is 60 veh/km

(c) Maximum flow rate is 6000 veh/h and critical density is 120 kmph

(d) Free flow speed is 120 kmph and jam density is 100 veh/km

(a) A and C only (b) A and B only

(c) D and C only (d) B and D only

**Ans. (b)** : Given

$$v=100-\frac{5}{6}k$$

Capacity, q =vk = ?

Jam density, kj = ?

$$q=\left(100-\frac{5}{6}k\right)k$$

$$q=100k-\frac{5}{6}k^2 \text{ ---------- (1)}$$

$$\frac{dq}{dk}=100-\frac{5}{6}\times 2k=100-\frac{5}{3}k$$

$$\frac{dq}{dk}=0$$

$$100-\frac{5}{3}k=0$$

$\boxed{k=60}$ veh/km

From equations (1),

$$q=100\times 60-\frac{5}{6}\times 60\times 60$$

$$=3000 \text{ veh / hr}$$

$$k_j=2k=2\times 60=120 \text{ veh / km}$$

$$v_f=2v=2\times\left(100-\frac{5}{6}\times 60\right)$$

$$=100 \text{ kmph}$$

**30. Which of the following statements is/are correct while analyzing an indeterminate structure?**

(a) The force method requires to satisfy equilibrium condition.

(b) The force method requires to satisfy compatibility condition.

(c) The displacement method requires to satisfy force-displacement for a structure.

(d) The displacement method requires to satisfy compatibility condition.

(a) B and C only (b) D only

(c) B only (d) A and C only

**Ans. (b) :**

| Force Method | Displacement method |
|---|---|
| •Also called compatibility method of consistent deformation, flexibility method. | •Also called equilibrium method, stiffness method |
| • Forces (BM, SF) and reactions are consider as unknown. | • Displacement ($\Delta\theta$) are consider as unknown |
| • Compatibility equation are used to find redundant | • Equilibrium equations are use to find redundant. |

**31. A fluid of jet of cross section Area A and velocity V strikes a flat plate moving with a velocity u. The fluid mass striking it per second is**

(a) ρAV (b) ρAu

(c) ρA(u–V) (d) ρA(V–u)

**Ans. (d):** A fluid of jet striking a flat vertical plate moving with a uniform velocity away from the jet.

V = absolute velocity of the jet

A = cross –sectional area of the jet, and

U = velocity of the flat plate held normal to the jet.

the relative velocity with the jet strikes the plate is (v-u).

Mass of water striking the plate per second = ρA (v-u)

Force exerted by the jet on the plate in the direction of jet

$F_x = \rho A (v - u)^2$

**32. In long and short wall method of estimation, the length of long wall is the centre to centre distance between the walls and**

(a) breadth of the wall on each side

(b) half the breadth of the wall on one side

(c) breadth of wall only on one side

(d) half breadth of wall on each side

**Ans. (d) :** In long and short wall method of estimation , the length of long wall is the centre to centre distance between the walls and half breadth of wall on each side.

**33. At a rated capacity of 44 cumecs, centrifugal pump develops 36m of head when operating at 1450rpm. The specific speed of the pump is**

(a) 65.44 (b) 100.44

(c) 1654.44 (d) 654.44

**Ans. (d) :** Given,

$Q = 44 m^3/sec$

$H = 36M$

$N = 1450rpm$

$$N_S = \frac{N\sqrt{Q}}{H^{3/4}} = \frac{1450 \times \sqrt{44}}{36^{3/4}} = 654.44$$

**34. Calculate the coefficient of permeability in m/sec, if the coefficient of consolidation and volume changes are 4.8 $mm^2$/min and** $1.02 \times 10^{-3} m^2/kN$.

(a) $4.8 \times 10^{-8}$ (b) $4.8 \times 10^{-10}$

(c) $0.8 \times 10^{-10}$ (d) $8.0 \times 10^{-9}$

**Ans. (c) :** Given,

$m_v = 1.02 \times 10^{-3} m^2/kN$

$$C_v = 4.8 \frac{mm^2}{min} = \frac{4.8 \times 10^{-6}}{60} m^2/sec$$

$k = ?$

$$C_v = \frac{k}{m_v \gamma_w}$$

$$k = C_v m_v . \gamma_\omega$$

$$= \frac{4.8 \times 10^{-6}}{60} \times 1.02 \times 10^{-3} \times 9.81$$

$= 8.0 \times 10^{-10} m/sec$

**35. The resistance of stones against the grinding action under traffic is measured by**

(a) Abrasion test

(b) Crushing strength test

(c) Hardness test

(d) Attrition test

**Ans. (d) :** The resistance of stones against the grinding action under traffic is measured by attrition test.

- In the Deval attrition test, particles of known weight are subjected to wear in an iron cylinder roted 10,000 times at certain speed.
- The proportion of material crushed finer than 1.7mm size is expressed as a percentage is taken as the attrition value of the aggregate.
- this test has been covered by IS 2386 (Part IV) 1963.

**36. A circular dug well in fine sand region has a yield of *15l*/s under a depression head of 4 m. Assuming specific capacity per unit area as 0.8/h, the bottom diameter of the dug well is**

(a) 2.65 m (b) 3.5 m

(c) 1.7 m (d) 4.65m

**Ans. (d) :** Given,

$Q = 15 \ell/_{se} = 15 \times 10^{-3} m^3/sec$

$H = 4m$

$C = 0.8/hr$

Area , $$A = \frac{\pi}{4} d^2 = \frac{Q}{H.C}$$

$$d^2 = \frac{4 \times 15 \times 10^{-3} \times 3600}{\pi \times 4 \times 0.8}$$

$d = 4.635 \simeq 4.64m$

**37. A steel pipe of length 3 m has outer and inner diameters of 150 mm and 125 mm, respectively. The pipe was held fixed at both ends and the pipe temperature was 20ºC when fixed. During the use, cold fluid moves through the pipe. As a result, the uniform temperature of the pipe becomes -15ºC. What will be the force on the fixed joint neglecting gravitational forces?**

**The coefficient of linear expansion and the modulus of elasticity for the pipe are** $10 \times 10^{-6}/°C$ **and 200 GPa respectively.**

(a) $120.31\pi kN$ (b) $130.31\pi kN$

(c) $125.31\pi kN$ (d) $110.31\pi kN$

**Ans. (a) :** Given, $D_o = 150 m$

$D_i = 125 m$

$\Delta T = T_1 - T_2 = 20^0C - (-15^0C)$

$\alpha = 10 \times 10^{-6} / {}^0C$

$E = 200 GP_a$

F = A EαΔT

$= \frac{\pi}{4}(150^2 - 125^2) \times 200 \times 10 \times 10^{-6} / {}^{0}_{C} \times 10^6 \times 35^0 C$

= 120.31 πkN

**38. A hydraulic jump occurs when there is a break in grade from a**

(a) Mild slope to steep slope
(b) Mild slope to milder slope
(c) Steep slope to steeper slope
(d) Steep slope to mild slope

**Ans. (d) :**

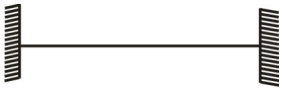
- A hydraulic jump occurs when there is a break in grade from a steep slope to mild slope.
- Hydraulic jump is a localized phenomenon that occurs when supercritical flow (Froude number, $F_r$ >1) Meets the subcritical flow ($F_r < 1$).

**39. Match the kinematic indeterminacy of the following structures neglecting axial strain.**

| | |
|---|---|
| (a) One span beam fixed at both ends | i.5 |
| (b) Two span continuous | ii.24 |
| (c) Five span continuous beam with fixed at one end and rollers at other supports | iii.0 |
| (d) One bay one storey space frame fixed at all supports | iv.3 |

(a) A–iv, B–iii, C–i, D–ii
(b) A–iv, B–i, C–iii, D–ii
(c) A–iii, B–i, C–iv, D–ii
(d) A–iii, B–iv, C–i, D–ii

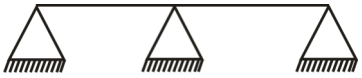
**Ans. (d) :** • One span beam fixed at both eds



$D_k = 3J\text{-}r_e$
$= 3 \times 2 - 6$
$= 0$

$\begin{cases} j = 2 \\ r_e = 6 \end{cases}$

• Two span continuous beam hinged at all supports



$D_k = 3_j - r_e =$

$= 3 \times 3 - 6 = 3$ $\quad \begin{cases} j - 3 \\ r_e = 6 \end{cases}$

• Five span continuous beam with fixed at one end and rollers at other supports

$D_k = 3_j - r_e$
$= 3 \times 6 - 8$
$= 10$ $\quad \begin{cases} j = 6 \\ r_e = 8 \end{cases}$

• one bay one storey space frame fixed at all Supports

$D_k = 6_j - r_e =$
$= 6 \times 4 - 0$
$= 24$ $\quad \begin{cases} j = 4 \\ r_e = 0 \end{cases}$

**40. Cube rate estimate is____as compared to the plinth area estimate.**

(a) equally accurate (b) Similar
(c) more accurate (d) less accurate

**Ans. (c) : Cube Rate estimate**

- Cube Rate estimate is more accurate as compared to the plinth area estimate.
- It is prepared on the basis of the cubical contents (LBH) of the building.

**Plinth Area Estimate –**

- Square meter Method this is prepared on the basis of plith area (BL) of building.

**41. A compacted fill was 8m thick has its void ratio as 1.2. The fill was loaded on surface by constructing an embankment covering a large area of fill. Some days after the embankment was constructed measurements of fill indicated an average void ratio of 0.8. Estimate compression of fill.**

(a) 2.5 (b) 2.45
(c) 1.35 (d) 1.45

**Ans. (d) :** Given ,

H = 8m

$e_0 = 1.2$ , $\Delta e = \frac{0.8}{2} = 0.4$

ΔH = ?

$\frac{\Delta H}{H} = \frac{\Delta e}{1 + e_0}$

$\frac{\Delta H}{8} = \frac{0.4}{(1 + 1.2)}$

$\Delta H = \frac{8 \times 0.4}{2.2}$

= 1.45m

**42. A prismatic circular bar is situated in the x-y-z coordinate system with the length L measured in the x direction. We measure the shear stress at a radius ρ from the center and the shear modulus of elasticity is G. The polar moment of inertia of the bar is J. What is the strain energy for the member when subjected to torsion T?**

(a) $\int_L \frac{T^2}{2GJ}\left[\iint \rho^2 dy\, dz\right]dx$

(b) $\iiint \frac{T^2}{2GJ} dx dy dz$

(c) $\iiint \frac{T^2}{2GJ^2} dx dy dz$

(d) $\int_L \frac{T^2}{2GJ^2}\left[\iint \rho^2 dy\, dz\right]dx$

**Ans. (d) :** strain energy, U

$$U = \int_L \frac{T^2}{2GJ}\left[\iint \rho^2 dy\, dz\right]dx$$

Where

T = Torque

G = Shear modulus

J = Polar moment of inertia

**43. A vehicle has a wheel base of 6.0 m. What is the off-tracking in metres while negotiating a curve path with a mean radius of 60 m?**

(a) 0.50 (b) 0.30

(c) 0.60 (d) 0.15

**Ans. (b) :** Given

$\ell = 6.0m$

R = 60m

Off tracking $\frac{\ell^2}{2R}$

$\frac{6^2}{2\times 60}$

= 0.3 m

**44. Arrange the steps of production of cement in chronological order.**

(a) blending the materials in the correct proportions

(b) grinding the clinker and gypsum

(c) crushing and grinding the raw materials

(d) burning the prepared mix in a kiln

(a) A,D,B,C (b) A,C,D,B

(c) A,B,C,D (d) C,A,D,B

**Ans. (d)** : there are four stages in the manufacture of Portland Cement –

**1.** crushing and grinding the raw materials

**2.** blending the materials in the correct proportions

**3.** burning the prepared mix in a kiln

**4.** grinding the burned product, known as ''clinker'' together with some 3% of gypsum (to control the time of set of the cement.)

**45. The silt factor f as used by Lacey's in his regime theory was adjusted to be equal to 1 for silt in an alluvium region. What is the median size of the alluvium soil?**

(a) 0.25 mm (b) 2.8 mm

(c) 3.2 mm (d) 0.32 mm

**Ans. (d) :** Given

f = 1

$f = 1.76\sqrt{d_{mm}}$

$d_{mm} = \left(\frac{1}{1.76}\right)^2$

= 0.32mm

**46. In channel routing using the Muskingum method, the value of the routing coefficients $C_0$ and $C_1$ are estimated as –0.3 and 0.6 respectively. The value of the $C_2$ is**

(a) 0.3 (b) –0.5

(c) 0.4 (d) 0.7

**Ans. (d) :** Muskingum equation

$C_0 + C_1 + C_{2-} = 1$

$-0.3 + 0.6 + C_2 = 1$

$C_2 = 1-0.3$

= 0.7

**47. Admixtures used to produce high strength concrete and reduce the water-cement ratio of concrete are called**

(a) Air entraining admixtures

(b) Super plasticizers

(c) Accelerators

(d) Retarders

**Ans. (b) :**

**<u>Super plasticizers (highrange water reducer)</u>**

• Super plasticizers produce extreme workability.

they achieve reduction in water content without loss of workability. Their use generally leads to an overall reduction in the cost. Super plasticizers are hydrodynamic lubricants which impart high workability by reducing friction between the grains thereby reducing the amount of water to be added.

• Super plasticizers are improved version of plasticizers.

- Use of Super plasticizers permits the reduction of water to the extent up to 30 percent without reducing workability in contrast to the possible reduction up to 15% in case of plasticizers.
- Some of the example of Super plasticizers are sulphonated metamine formaldehyde condensates.

**48. A road is being designed for a speed of 100 km/hr on a horizontal curve with a supper elevation of 7%. If the coefficient of side friction is 0.15, the minimum radius of the curve (in m) required for safe vehicular movement is**

(a) 144.8 (b) 357.9
(c) 178.9 (d) 202.1

**Ans. (b) :** Given,

$V = 100$ km/hr

$e_{max} = 7\% = 0.07$

$f_{max} = 0.15$

$$(e_{max} + f_{max}) = \frac{V^2}{127} R_{min}$$

$$R_{min} = \frac{V^2}{127(e_{max} + f_{max})}$$

$$= \frac{100^2}{127(0.07 + 0.15)}$$

$= 357.90$M

**49. Analysis of data on maximum one day rainfall depth at Hyderabad showed that a depth of 300 mm had a return period of 80 years. The probability of a one-day rainfall depth equal to or greater than 300 mm at Hyderabad occurring at least once is 80 years is**

(a) 0.314 (b) 0.0125
(c) 0.125 (d) 0.341

**Ans. (b) :** return period refers to the inverse of the probability of exceedance of an event.

$T = P^{-1}$

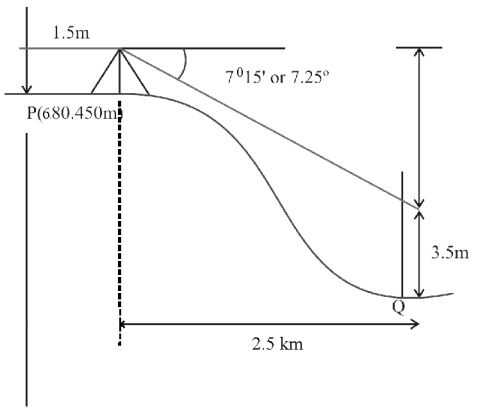
$$T = \frac{1}{P}$$

$$P = \frac{1}{T} = \frac{1}{80} = 0.0125$$

**50. A theodolite is placed at P and a 5 m long vertical staff is held at Q. The depression angle made at reading 3.5 m marking on staff is 7º15'. The horizontal distance between P and Q is 2500 m. The height of instrument at P is 1.5 m and reduced level of point P is 980.450 m. Using curvature correction and refraction correction determine the RL of point Q (in m).**

(a) 660.830 (b) 975.450
(c) 660.410 (d) 976.950

**Ans. (a) :**




V = tan ($7.25^0$) 2500 = 318.040m.

Apply correction i.e. Curvature

CC = - 0.06735 $[2.5]^2$ = 0.4209m

'Corrected distance (v) = 318.040-0.4209 = 317.61m

RL of pt Q = (980.450+1.5m) – 317.61m-3.5

= 660.84m

**51. Cavitation in fluid flow occurs when**

(a) the total energy suddenly increase
(b) the velocity head reduces to zero
(c) the pressure of flow decreases to a value close to its vapour pressure
(d) For total energy suddenly decreases

**Ans. (c) : Cavitation**

Cavitation is caused by,

- Vapourisation of the liquid and/ or release of dissolved air at low pressure. When the velocities and devations are high pressure can reach low values, according to Bernoulli's equation. this occurs when the pressure is lowered to the vapour pressure at the temperature considered.
- Movement of the vapour into a high-pressure region. the bubbles are carried along by the flow.

**52. For an isotropic material, which of the following statements is/ are correct?**

(a) The properties are same at all the points throughout the material, but different properties can be there in three direction at a points.
(b) The properties at a point are same in three different directions.
(c) The property in two directions is same, but it is different in one direction at a point.
(d) The properties are same at all the points throughout the material, and same properties observed in all three directions at a point.

(a) A and B only (b) A and D only
(c) C and D only (d) B and D only

**Ans. (d): Isotropic material –**

- An isotropic material is one for which the elastic properties are the same in all directions.
- the elastic behavior of an isotropic body is characterised by just two elastic constants-

usually the young's modulus E and the Poisson's ratio μ. the other relevant elastic.

Constant, namely, the shear modulus G and the bulk modulus k, can be expressed in terms of E and μ.

**53. In steady flow through a pipe the density, velocity and area of a section are 3 kg/m$^3$, 8m/s and 0.5m$^2$ respectively. The velocity at another section having area of 1m$^2$ and density of 4 kg/m$^3$ will be**

(a) 2 m/s (b) 1 m/s

(c) 4 m/s (d) 3 m/s

**Ans. (d) : Given**

$\rho_1 = 3kg/m^3$ - $A_2 = 1m^2$

$V_1 = 8m/sec$ - $\rho_2 = 1m^2$

$A_1 = 0.5m^2$ - $V_2 = ?$

$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$

$$V_2 = \frac{\rho_1 . A_1 . V_1}{\rho_2 A_2}$$

$$= \frac{3 \times 0.5 \times 8}{1 \times 4}$$

$= 3m/sec$

**54. In which type of contract, measurement of executed items are NOT required to keep?**

(a) Percentage Rate Contract

(b) Lump-Sum Contract

(c) Cost Plus Contract

(d) Item-Rate Contract

**Ans. (b) :** the type of contract, measurement of exeuted items are

- Percentage rate contract
- Cost plus percentage contract
- Item - Rate contract

**Percentage rate contract-**

⇒ In this form of contract, the department draws up' item rate tender' i.e. bill of quantities with-rate, amount, and total amount.

⇒ the contractors are required to offer to carry out the work as per with the rates shown in the specific price schedule or some percentage above or below the rates indicated in the schedule of work attached with tender.

**55. The coefficient of average rolling friction of a road is $f_r$ and its grade is + G%. If the grade of this road is doubled, what will be the percentage change in the braking distance (for the design vehicle to come to a stop) measured along the horizontal (assumer all other parameters are kept unchanged)?**

(a) $\frac{2f_r}{f_r + 0.01G} \times 100$ (b) $\frac{0.02G}{f_r + 0.01G} \times 100$

(c) $\frac{f_r}{f_r + 0.02G} \times 100$ (d) $\frac{0.01G}{f_r + 0.02G} \times 100$

**Ans. (d) :** the coefficient of average rolling friction of a road is $f_r$ and its grade is + G%. When grade of this road is doubled, then

Braking distance = $\frac{V_2}{2g(f_r + G)}$

When grade is doubled,

then braking distance= $\frac{V^2}{2g(f_r + 2G)}$

$$\% \text{ change} = \frac{\frac{V^2}{2g(f_r + 0.01G)} - \frac{V^2}{2g(f_r + 0.02G)}}{\frac{V^2}{2g(f_r + 0.01G)}} \times 100$$

$$= \frac{0.01G}{f_r + 0.02G} \times 100$$

**56. The total float can be expressed as**

(a) latest start time – earliest start time

(b) latest finish time– earliest finish time

**Identify the correct statement**.

(a) Statement B is true an A is false

(b) Both A and B are true

(c) Both A and B are false

(d) Statement B is false and A is true

**Ans. (b) : Total Float-**

It is the difference between maximum time available & actual time required for the completion of the artivity.

**Total Float** = latest start time - earliest start time

= latest finish time - earliest finish time

**57. A 1.25 m layer of soil having the porosity 0.35 and specific gravity 2.65 is subjected to an upward seepage head of 1.85 m. What depth of coarse sand would be required above the existing soil to provide a factor of safety of 2 against piping ? Assume that coarse sand has the same porosity and specific gravity as the soil and there is negligible head loss in sand.**

(a) 1.25 (b) 4.4

(c) 2.2 (d) 2.7

**Ans. (c) :** Given :

Upward seepage head = 1.85m

Factor of safety = 2

(1) Critical hydraulic gradient

$i_c = (G-1)(1-n)$

$= (2.65-1)(1-0.35)$

$i_c = 1.0725$

(ii) Factor of safety against piping

$$FOS = \frac{i_c}{i_{exit}}$$

$$2 = \frac{i_c}{i_{exit}} \Rightarrow \boxed{i_{exit} = \frac{1.0725}{2}} ---(i)$$

$$\Rightarrow \boxed{i_{exit} = \frac{H}{L}} W \rightarrow \text{See page head}$$

From $e_q^{n.}$ (i)

$$\frac{H}{L} = \frac{1.0725}{2}$$

$$L = \frac{2 \times 1.85}{1.0725} m$$

$$L = 3.4498 \text{ m}$$

(iii) Additional thickness of coarse sand layer

$$= (3.4498\text{-}1.25) \text{ m}$$

$$= 2.1998 \text{m}$$

$$\approx 2.2 \text{ m}$$

**58. Consider the following statements related to a horizontal venturimeter.**

(a) The velocity of flow in the main pipe is greater and the pressure is lesser than that at the throat section.

(b) The velocity of flow in the main pipe is lower and the pressure is larger than that at the throat section.

(c) The pressure difference between the main pipe and throat section is positive.

(d) The pressure difference between the main pipe an throat section is negative.

Which of these statements are NOT correct?

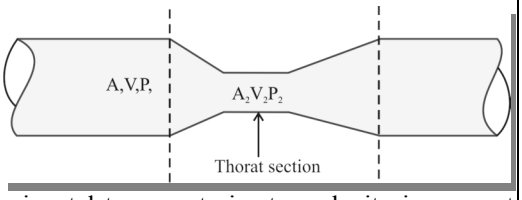
(a) A and D only (b) C and D only

(c) B and D only (d) A and B only

**Ans. (c) : Venturimeter -**

• To find discharge from a large diameter pipe

• It has much smaller head loss and having good accuracy.




• In a horizontal type venturimeter velocity is more at the throat section because area is less and pressure is also less .

**59. A water tank supported by a ring foundation having outer diameter of 8 m and thickness of 1m carries a uniform load intensity of 200 kN/m$^2$. The vertical stress (in kN/m$^2$) caused by the water tank at a depth of 4m below the center of foundation is**

(a) 15.8 kN/m$^2$ (b) 51.4 kN/m$^2$

(c) 10 k N/m$^2$ (d) 31.7 kN/m$^2$

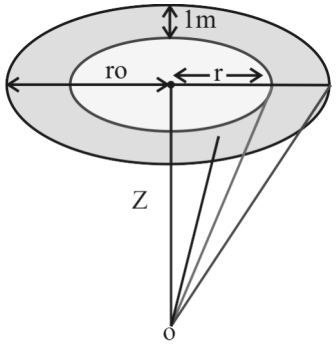
**Ans. (d) :** Given

Outer dia of tank = 8 m

Thickness of tank = 1m

Inner dia = (8m – 2m)

= 6m




(1) Vertical stress due to u.d. $\ell$ .

$\sigma_z = q(1 - \text{Cos}^3\theta)$ Where ;

$\text{Cos}\theta = \frac{Z}{\sqrt{r^2 + z^2}}$ $z \rightarrow$ depth

$r \rightarrow$ radius

$q \rightarrow$ load intensity

$$\Rightarrow \quad r_0 = \frac{8}{2} = 4m$$

$$r_i = \frac{6}{2} = 3m$$

$$\cos\theta_i = \frac{4}{\sqrt{3^2 + 4^2}} = 0.8; \quad \cos\theta_0 = \frac{4}{\sqrt{4^2 + 4^2}} = 0.707$$

Net Vertical stress (using super position).

$$\sigma_z = q(1 - \cos^3\theta_0) - q(1 - \cos^3\theta_i)$$

$$= q(1 - \cos^3\theta_0 - 1 + \cos^3\theta_i)$$

$$= 200\ [(0.8)^3\text{-}[(0.707)^3]$$

$$= 200 \times 0.1586$$

$$\boxed{\sigma_Z = 31.7 kN/m^2}$$

**60. Consider the following statements:**

(a) Efflorescence is the appearance of white powder on the surface of bricks.

(b) Efflorescence is caused by the presence of chalk in the brick earth.

(c) Moderate efflorescence– covering upto 50% area of the brick.

(d) Moderate efflorescence– covering less than 10% area of brick.

Which of the above statements is/are correct?

(a) A and B (b) B and C

(c) B and D (d) A and C

**Ans.(d):** Presence of efflorescence is classified as below.

**(1) Nil-**when the deposits of effloresces imperceptible.

**(2) Slight-**When the deposits of efflorescence does not cover more than 10% of exposed area of brick.

**(3) Heavy-**When the deposit of efflorescence cover more than 50% but do not powder or flake away the brick surface

**(4) Moderate** - When the deposits of efflorescence cover more than 10% but less than 50% of exposed area of the brick.

**61. The depth of scour according to Lacey's formula in a river represents the depth below the**

(a) maximum flood level in the river
(b) maximum depth of scour in the channel
(c) 1.2 m above the flood level
(d) bed level of the river

**Ans. (b) :** The depth of scour according to Lacey's formula in a river represents the depth below the maximum depth of scour in the channel.

$$\text{Lacey's regime} = 1.35\left(\frac{q^2}{f}\right)^{1/3}$$

(for any river width)

Lacey's regime source depth

$$= 0.48\left(\frac{Q}{f}\right)^{1/3}$$ (for alluvial regime width)

Where

q = discharge per unit width
f = Silt factor
Q = discharge

**62. A pitot tube is an instrument for measureing**

(a) Pressure of flow (b) Total energy
(c) Velocity of flow (d) Discharge of fluid

**Ans. (c) : Pitot tube -**

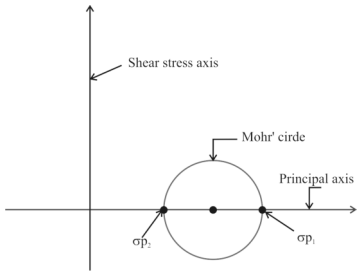
- A pitot tube is an instrument for measuring fluid flow velocity, water speed of a boat and also to measure ships speed relative to be water. they are used on both surface ship & submarine.

**63. Match the following for a body under plane stress condition.**

| | List -I | | List- II |
|---|---|---|---|
| (a) | Distance of the center of the Mohr's circle from origin of coordinates | (i) | Difference between normal stresses divided by 2 |
| (b) | The diameter of the Mohr's circle | (ii) | Shear stress |
| (c) | Distance of maximum normal stress from the centre of Mohr's circle | (iii) | Average normal stress |
| (d) | Vertical direction of the Mohr's circle | (iv) | Difference between major principal stresses |

(a) A- iii, B-iv, C-ii, D-i
(b) A- iv, B-i, C-iii, D-ii
(c) A- iii, B-iv, C-i, D-ii
(d) A- iv, B-iii, C-i, D-ii

**Ans. (b) :**




Co-ordinates of centre of Mohr's circle = $\sigma_2 + \frac{\sigma_1 - \sigma_2}{2} = \frac{\sigma_1 + \sigma_2}{2}$ = Average normal stress

Dia of Mohr's circle = $\sigma_1 - \sigma_2$ = Difference between major principal stresses.

Distance of maximum normal stress from centre of Mohr's circle = Radius of Mohr's circle

$$= \frac{\sigma_1 - \sigma_2}{2}$$

- Shear stress is shown on vertical direction where as normal stress is shown on horizontal direction.

**64. If the formation level of a highway has a uniform gradient for a particular length and the ground is also having a longitudinal slope, the earthwork cannot be calculated by**

(a) Mid- section formula
(b) Central line method
(c) Prismoidal formula
(d) Trapezoidal formula

**Ans. (b) :** If the formation level of a highway has a uniform gradient for a particular length and the ground is also having a longitudinal slope, the earthwork cannot be calculated by

(1) Mid - section formula
(2) Prismoidal formula

$$V = \frac{L}{3}\left[(A_1 + A_n) + 4(A_2 + A_4 + --A_{n-1}) + 2(A_3 + A_5 + --A_{n-2})\right]$$

(3) Trapezoidal formula

$$V = \left[\frac{A_1 + A_n}{2} + A_2 + A_3 + ---A_{n-1}\right]L$$

Where,

$A_1$, $A_2$, $A_3$----$A_n$ are the end consecutive areas and L is the distance between them.

**65. If duty is 1500 hec/ cumec and base period of 150 days for an irrigated crop, the delta of the crop will be**

(a) 86.4 cm (b) 0.864 mm
(c) 86.4 mm (d) 864 cm

**Ans. (a):** Given

D = 1500 ha/cumec

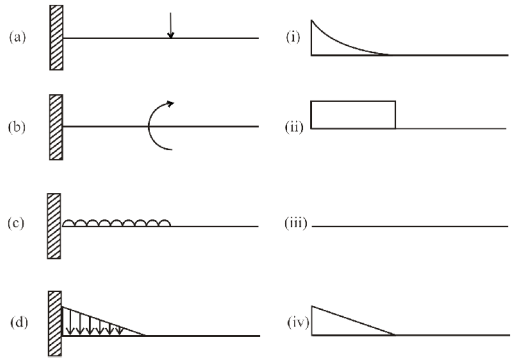
B = 150 days

$\Delta = ?$

$$\Delta = \frac{8.64B}{D}$$

$$= \frac{8.64 \times 150}{1500} = 86.4\,cm$$

**66. Match the shear force diagrams for the beams with different loadings.**




(a) A–i, B–ii, C-iv, D-iii

(b) A–i, B–iii, C-iv, D-ii

(c) A–ii, B–iii, C-iv, D-i

(d) A–ii, B–iii, C-iv, D-iv

**Ans. (c) :**

- For pure bending condition there is no shear force.
- For UVL, SF variation is parabolic and BM variation is cubic parabola.
- For UDL, SF variation is linear and BM variation is parabola.
- For point loads SF is constant where as BM variation is linear.

Hence option (c) is correct.

**67. The momentum correction factor is used to correct for**

(a) change in pressure

(b) non- uniform distribution of velocities at inlet and outlet sections

(c) change in total energy

(d) change in mass rate of flow

**Ans. (b) :** • The momentum correction factor is used to correct for non-uniform distribution of velocities at inlet and outlet sections.

**Momentum correction factor (β),**

$$\beta = \frac{\text{Actual linear momentum /sec}}{\text{Linar momentum/sec calculated from Average velocity}}$$

$$= \frac{\int_A V^2 dA}{A V_{avg}^2} \quad , \quad V_{avg} \int_A \frac{VdA}{A}$$

**68. Consider the following:**

| | | | |
|---|---|---|---|
| (a) | Honeycombing | (i) | Rise of water to surface of freshly placed concrete |
| (b) | Bleeding | (ii) | Separation of concrete constituent materials due to gravity |
| (c) | Thermal shrinkage | (iii) | Hollow spaces and cavities left in concrete mass |
| (d) | Segregation | (iv) | Volume reduction of concrete due to temperature variation |

**Choose the option in which given pairs are correctly matched.**

(a) A-i, B-iv, C-ii, D-iii

(b) A-ii, B-i, C-iv, D-iii

(c) A-iii, B-i, C-iv, D-ii

(d) A-iii, B-iv, C-i, D-ii

**Ans. (c) : Segregation–** It is the phenomenon of separation of concrete ingredients due to lack of cohesion is called as segregation.

**Honey combing–** It is the phenomenon in which hollow spaces and cavities are formed within concrete mass due to improper compaction.

**Bleeding–** It is phenomenon of rise of water to surface of freshly placed concrete.

**Thermal shrinkage–** Occurs due to temperature variation.

**69. In a 1 : 50 model of stilling basing filled with water, energy dissipation in model is 0.5 KW. The Corresponding energy dissipation in prototype for same fluid will be**

(a) 1.25 MW (b) 441.94 MW

(c) 321.5 MW (d) 62.5 MW

**Ans. (b) :** Given ;

Linear scale ratio $L_r = 50$

Energy dissipation in Model = 0.5kw

(i) Energy dissipation in model (P model) and proto type

$$\Rightarrow \frac{P_{prototype}}{P_{model}} = (L_r)^{3.5} \text{ putting givens}$$

$$P_{prototype} = 0.5 \text{ kw} \times (50)^{3.5}$$

$$= 441941.73 \text{ kw}$$

$$P_{prototype} = 441.94 \text{ Mw}$$

**70. The bending moment, modulus of elasticity and the moment of inertia of a beam are given by M, E and I, respectively. According to elastic beam theory, what is the radius of curvature?**

(a) $\frac{M}{EI}$ (b) WI

(c) $\frac{1}{EI}$ (d) $\frac{EI}{M}$

**Ans. (d):** According to elastic beam theory the radius of curvature

$$\frac{M}{I}=\frac{E}{R}=\frac{\sigma}{y}$$

$\rightarrow \boxed{R=\frac{EI}{M}} \rightarrow$ Radius of curvature

$\rightarrow \boxed{\frac{1}{R}=\frac{M}{EI}} \rightarrow$ Curvature of beam

**71. A vertical cut is made in saturated clay with $c = 30kN/m^2, \phi = 0^o$ and $\gamma = 20kN/m^3$.**

**What is the theoretical depth to which clay can be excavated without side collapse?**

(a) 1.4 m (b) 6 m

(c) 2.8 m (d) 12 m

**Ans. (b) :** Given

$C = 30KN/m^2$

$\phi = 0^0$

$\gamma = 20KN/m^3$

$$K_a = \frac{1-\sin\phi}{1+\sin\phi}=\frac{1-0}{1+0}=1$$

$$H_c = \frac{4C}{\gamma\sqrt{k_a}}=\frac{4\times 30}{20\times 1}$$

$= 6m.$

**72. Compressive strength of concrete mainly depends upon**

(a) Grading of aggregate

(b) Quantity of aggregate

(c) Water- cement ratio

(d) Quantity of cement

**Ans. (c) :** Compressive strength of concrete mainly depends upon water - Cement ratio

Compressive strength of concrete $\propto \frac{1}{w/c}$ ratio

**73. Which of the following pairs is/are incorrect?**

(a) Castigliano's theorem – Deflection

(b) Flexibility method – Consistent displacement method

(c) Stiffness method – Degree of static indeterminacy

(d) Betti's law – Reciprocal deflection

(a) C only (b) A and B only

(c) D and B only (d) C and D only

**Ans. (a):** Castigliano's theorem (deflection) –

(1) Castigliano's first theorem is applicable to linearly or non linearly elastic structures in which the temperature is constant and the supports are unyielding

(2) Castigliano's 2nd theorem is applicable to linearly elastic structure.

(3) **Flexibility method** - the matrix method is one of the two methods for analyzing the indeterminate structure, depending on the unknown is taken as forces, it is known as the force method (Flexibility method), and if the, unknown is taken as displacement method (stiffness method).

**74. The frequency distribution of duration of an individual activity takes the shape of _____as per the PERT analysis.**

(a) Normal distribution

(b) Beta distribution

(c) Log-Normal distribution

(d) Gamma distribution

**Ans. (b) :** The frequency distribution of duration of an individual activity takes the shape of Beta distribution as per the PERT analysis PERT stands for programme evaluation and review technique.

- Suitable for new type of projects
- Each activity follow β distribution
- it is based on probabilistic approach

**75. dy/dx is negative in the following GVF profiles**

(a) M1, S2, A2 (b) A3, A2, M2

(c) M2, A2, H2 (d) M2, A2, M2

**Ans. (c) :** $$\frac{dy}{dx}=\frac{So\left(1-\left(\frac{y_0}{y}\right)^{10/3}\right)}{\left(1-\left(\frac{y_c}{y}\right)^{3}\right)}$$

Where, NDL $\rightarrow$ Normal depth line

CDL $\rightarrow$ Critical depth line

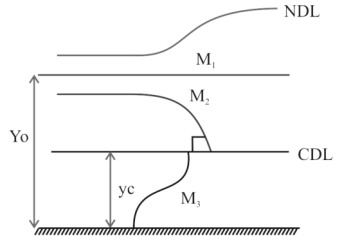
$S_o \rightarrow$ bed slope

$y_o \rightarrow$ Normal depth of flow

$y_c \rightarrow$ Critical depth of flow

$\frac{dy}{dx} \rightarrow$ slope (falling/rising)

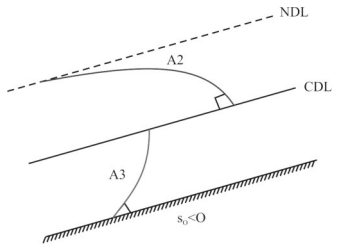
for the $M_2$ Profile



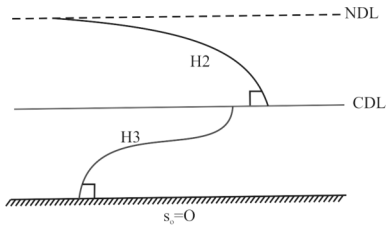

$y_c < y < y_o$

$$\frac{y_c}{y}<1, \frac{y_o}{y}>1, \frac{dy}{dx}=-(\text{falling})$$

$\Rightarrow$ for the $A_2$ Profile




$\Rightarrow$ for the $H_2$ Profile




**76. Consider the following statements from IS 800: 2007**

(a) The maximum value of the effective slenderness ratio of compression flange of a beam against lateral torsional buckling is 300.

(b) The partial safety factor for the combination of dead load, live load, crane load and wind load under the limit state of serviceability is 1.0.

(c) The partial safety factor for the resistance of members to buckling is 1.25

(d) The design strength for bolted connections due to block shear failure requires only net areas.

Which of the above- mentioned statements is/are correct?

(a) A and C only (b) B and C only

(c) A and B only (d) B and D only

**Ans. (c) :**

- The maximum value of the effective slenderness ratio of compression flange of a beam against lateral torsional buckling is 300.
- The partial safety factor for the combination of dead load, live load, crane load and wind load under the limit state of serviceability is 1.0.
- The partial safety factor for the resistance of members to buckling is 1.10
- The design strength for bolted connections due to block shear failure requires gross area, net area ultimate and yield strength of materials.

**$A_S$ per IS 800: 2007**

Maximum values of Effective slenderness Ratios

| Member | $\lambda$ |
|---|---|
| (i) A member carrying compressive load resulting from dead loads and imposed loads | 180 |
| (ii) Compression flange of a beam against lateral torsional buckling | 300 |
| (iii) A tension member in which a reversal of direct stress occurs due to loads other than wind or seismic forces | 180 |
| (iv) Members always under tension (other than pre-tensioned members) | 400 |

**Partial Safety factor for materials, $r_m$**

| Definition | Partial Safety factor |
|---|---|
| 1. Resistance, governed by yielding | 1.10 |
| 2. Resistance, of member to buckling | 1.10 |
| 3. Resistance, governed by ultimate stress | 1.25 |

**77. The average spacing between vehicles in a traffic stream is 20 m, then the density (in veh/km) of the stream is**

(a) 120 (b) 80

(c) 50 (d) 20

**Ans. (c) :** Given

S = 20m

K = ?

$$K = \frac{1000}{s}$$

$$K = \frac{1000}{20} = 50 \text{ veh/km}$$

**78. Convective Precipitation is caused when**

(a) rising of moist air due to surface heating

(b) When air mass pass over the mountain region and gets cooled

(c) When the cooler air mass forms wedge and lifts the warm air mass

(d) there is a frontal mechanism

**Ans. (a) : Convective Precipitation -**

- Convective Precipitation is caused by natural rising of warmer lighter air in colder, denser surroundings.

- the difference in temperature may result from unequal heating at the surface, unequal cooling at the top of the air layer, or mechanical lifting when air is forced to pass over a denser coldar air masses.
- Convective Precipitation is spotty and its intensity may vary from light showers to cloud bursts.

**79. The correct arrangement of different voids of concrete according to their size is.**

(a) entrapped voids > entrained voids> capillary voids> gel voids
(b) entrapped voids> entrained voids > gel voids> capillary voids
(c) capillary voids > entrapped voids > entrained voids > gel voids
(d) entrained voids > entrapped voids > capillary voids> gel voids

**Ans. (b) :** The correct arrangement of different voids of concrete according to their size is-
entrapped voids> entrained voids > gel voids> capillary voids

**Entrapped Air** - non uniform air distribution, due to insufficient compaction

**Entrained Air**-uniform air distribution, due to intentional creation with Air enteraining admixtures.

**80. A two-dimensional water jet strikes a fixed two-dimensional plate and splits into two streams such the ratio of discharge is 3. The angle of water jet striking normal to a plates is**

(a) 45º (b) 30º
(c) 60º (d) 75º

**Ans. (b) :** Given

$$\frac{Q_1}{Q_2} = 3$$

$$\frac{Q_1}{Q_2} = \frac{1+\cos\theta}{1-\cos\theta}$$

$$3 = \frac{1+\cos\theta}{1-\cos\theta}$$

$$3 - 3\cos\theta = 1 + \cos\theta$$

$$4\cos\theta = 2$$

$$\theta = \cos\text{-}1\left(\frac{1}{2}\right)$$

$$\boxed{\theta = 60^0}$$

→ Jet strikes at $(90^0\text{-}60^0) = 30^0$to the normal to the plate.

**81. The St. Venant Equations for unsteady open channel flow are**

(a) continuity and momentum equations
(b) energy an continuity equations
(c) momentum and energy equation
(d) momentum in two different forms

**Ans. (d): St. Venant assumption**

- The flow is one dimensional
- The streamline covatul is very small and the vertical fluids acceleration are negligible as a result, the present distribution are hydrostatic.
- The flow resistance and turbulent losses are the same as for a steady uniform flow for the same depth & velocity, regardness of trends of the depth.
- The Bed slope is small enough to satisfy following assumption

$$\cos\theta \simeq 1 \quad \& \quad \sin\theta \simeq \tan\theta$$

- The water density is constant

The flow can be described at any point and any time by two variables e.g V and d (or) Q and d, where V is the flow velocity, d is the water depth & Q is water discharges.

Hence the unsteady flow properties must be described by two equation; the consecration of mass & conservation of momentum.

(i-e continuity equation)

**82. What is the correct increasing order of the following materials with respect to their Poisson's ratio?**

(a) Magnesium alloys
(b) Cast iron
(c) Aluminum alloys
(d) Nickel

(a) D, B, A, C (b) A, C, B, D
(c) B, D, C, A (d) C, A, B, D

**Ans. (a) :**

| Materials | Poisson's ratio |
|---|---|
| Magnesium alloys | 0.35 |
| Cast iron | 0.21-0.26 |
| Aluminum alloys | 0.33 |
| Nickel | 0.31 |

**83. Premature stiffening of cement pastes due to conversion of hemihydrate ($CaSO_4$. $H_2O$) or anhydrite ($CaSO_4$) to gypsum can be eliminated by**

(a) Reducing $C_3A$ content of cement
(b) Adding accelerating admixture to concrete
(c) Continuous mixing or by reworking
(d) Reducing gypsum content of cement

**Ans. (c) :** The Premature stiffening of cement pastes due to conversion of hemihy drate or anhydrite to gypsum can be eliminated by continuous mixing or by reworking. Premature stiffening is caused by secondary gypsum and syngenite formation and it is also known as flash setting of cement.

**84. While aligning the hill road with a rolling gradient of 7 percent, a horizontal curve of radius 100 m is encountered. What is the compensated gradient in percentage as per IRC?**

(a) 5.5 (b) 6.0

(c) 5.7 (d) 6.25

**Ans. (d) :** Rolling gradient = 70%

R =100m.

Compensated gradient = gradient-grade compensation

$$\text{Grade Compensation} = \text{minmum}\left\{\frac{30+R}{R},\ \frac{75}{100}\right\}$$

$$= \text{minmum}\left\{\frac{30+100}{100},\ \frac{75}{100}\right\}$$

$$= \text{minmum}\left\{1.3\%,\ 0.75\%\right\}$$

$$\boxed{\text{grade compensation} = 0.75}$$

Compensated gradients = 7 - 0.75

$$= \boxed{6.25\%}$$

**85. A syphon aqueduct is provided when**

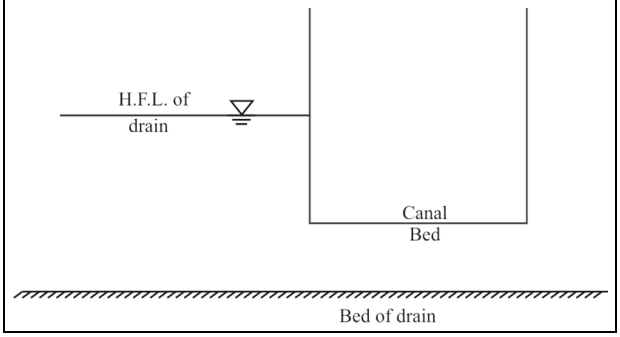
(a) the canal bed level is well below the flood level of

(b) the canal bed level is above the flood level of the drain

(c) The drain bed level is well below FSL of the canal

(d) The canal bed level lies between the bed level and HFL of the drain

**Ans. (d) : Syphon Aqueduct-** In this Canal bed level lies between the bed surface and HFL of the drain.




**86. For a plane strain problem, the number of non-zero stress components will be**

(a) 2 (b) 1

(c) 4 (d) 3

**Ans. (c) • Plane strain problem** - In long bodies which geometry and corresponding loading do not change significantly in the longitudinal direction are reffered to as plane strain problem.

In this problem non-zero strain components act in one plane only. The No of non zero stress components are four ie. $\sigma_x$, $\sigma_y$, $\tau_{yx}$, and $\tau_{xy}$,

**87. A surveyor measured the distance between two points on a plan drawn to a scale of 1 cm = 100 m and the result was 333 m. Later, however, he discovered that he used a scale of 1 cm = 25 m. Find the true scale distance between the points.**

(a) 666 (b) 999

(c) 1665 (d) 1332

**Ans. (d) :** RF of wrong scale = $\frac{1\text{cm}}{25000}$

RF of true scale = $\frac{1\text{cm}}{10000}$

Measured length = 333 m

Corrected length = ?

$$\text{Corrected length} = \frac{\text{R.F. of wrong scale}}{\text{R.F. of correct scale}} \times \text{Measured length} \times$$

$$= \frac{\frac{1}{2500}}{\frac{1}{10000}} \times 333$$

$$\boxed{\text{Corrected length} = 1332\text{m}}$$

**88. For a long pipe carrying liquid from one reservoir to another, at the exit section of the pipe, the energy grade line will**

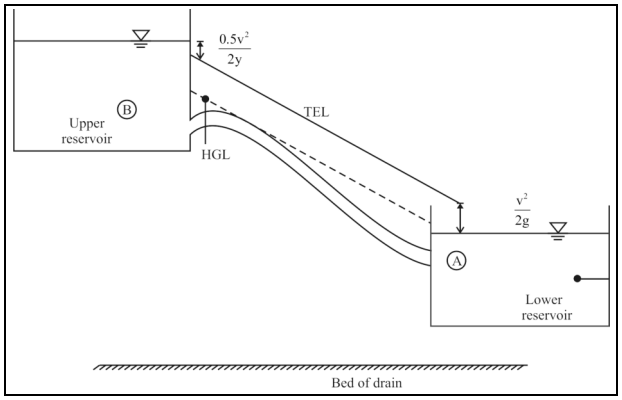
(a) meet the liquid surface

(b) lie at a distance ($v^2/2g$) below the liquid surface

(c) lie at a distance ($v^2/4g$) below the liquid surface

(d) lie at a distance ($v^2/2g$) above the liquid surface

**Ans. (d) : Energy grade line** - It is also known as total energy line which is a plate of total head. For a long pipe carrying liquid from a reservoir to another reservoir the energy grade line will be below $\left(\frac{0.5V^2}{2g}\right)$ to the liquid surface at the upper reservoir and above $\left(\frac{V^2}{2g}\right)$ to the liquid surface of the lower reservoir.




**89. If the plate load test show settlement of plate as 15 mm, for a 400 mm wide plate then how much more settlement a footing of width 1.2 m will have for a clayey soil compared to sand soil?**

(a) 74.4 mm (b) 12.4 mm
(c) 15.6 mm (d) 45.6 mm

**Ans. (c) :** • Settlement of foundation :-

(i) On Sandy soil

$$\frac{S_f}{S_P} = \left[\frac{B_f(B_P+0.3)}{B_P(B_f+0.3)}\right]^2$$

Where :
$S_f \rightarrow$ Settlement of foundation (mm).
$S_{p\rightarrow}$ Settlement of Plate (mm).
$B_{f\rightarrow}$ Width of foundation (m)
$B_{p\rightarrow}$ Width of plate (m).

Given ; Bf = 1.2m
Bp = 400mm = 0.4m
SP = 15mm

Putting givens

$$S_f = 15\times\left[\frac{1.2(0.4+0.3)}{0.4(1.2+0.3)}\right]^2$$

$S_f$ = 29.4 mm

(ii) On clay soil

$$\frac{S_f}{S_P} = \frac{B_f}{B_P}$$

$$S_f = S_P\times = \frac{B_f}{B_P} \Rightarrow 15\times\frac{1.2}{0.4}$$

$S_f$ =45 mm

(iii) difference is settlement of foundation clay soil to sandy soil.

$\Delta = S_{fclay} - S_{fsand}$
= 45mm-29.4

$$\boxed{\Delta = 15.6 \text{ mm}}$$

**90. According to IS 10262:2019, the relationship between target mean compressive strength ($f_t$) and characteristic compressive strength ($f_{ck}$) of concrete is**

(a) $f_t = f_{ck} + 1.05\times$ standard deviation
(b) $f_t = f_{ck} + 1.25\times$ standard deviation
(c) $f_t = f_{ck} + 0.65\times$ standard deviation
(d) $f_t = f_{ck} + 1.65\times$ standard deviation

**Ans. (d) :** As per is 10262 : 2019

Relationship between target mean compressive strength (ft) and characteristic compressive strength ($f_{ck}$).

$$\boxed{f_t = f_{ck} + 1.65\sigma}$$

Where ; $\sigma \rightarrow$ standard deviation (N/mm$^2$)

**91. Consider the following statements from IS 456 : 2000 code**

(a) The standard deviation for M 15 concrete is 4.0 N/mm2 in case of concrete mix.
(b) A minimum 2 samples should be tested for 15 m$^3$ of concrete.
(c) Basic value of span to effective depth ratio for a cantilever beam up to 10 m is 7.
(d) Design bond stress for M 35 concrete in limit state method for plain bars in tension is 1.5 N/mm$^2$.

Which of the above mentioned statements is/are incorrect?

(a) A and B only (b) B and D only
(c) A and D only (d) B and C only

**Ans. (*) :**

• The standard deviation for M15 concrete is 3.5 N/mm$^2$ in case of concrete mix.

• Minimum 2 samples should be lested for 15m$^3$ of concrete

• Span to effective depth ration for a cantilever beam up to 10m is 7.

• Design bond stress for M35 concrete in limit state method for plain bars in tension is 1.7. This value is increased by 25% for compression and 60% in case of deformed bars.

Hence, B, C & D are correct.

**92. A linear relationship is observed between speed and density on a certain section of a highway. The free flow speed is observed to be 100 km per hour and the jam density is estimated as 120 vehicles per km length. Based on the above relationship, the maximum flow expected on this section and the speed at the maximum flow will be respectively**

(a) 6000 vehicles per hour and 100 km per hour
(b) 6000 vehicles per hour and 50 km per hour
(c) 3000 vehicles per hour and 100 km per hour
(d) 3000 vehicles per hour and 50 km per

**Ans. (c):** Given Vf = 100 kmph.

Kj = 120 vph.

$\boxed{q_{max} = ?}$

Solve. $q_{max} = \frac{V_f K_j}{4}$

$q_{max} = \frac{100 \times 120}{4}$

$\boxed{q_{max} = 3000 \text{ veh/hr}}$

Speed at maximum flow.

$V_s = \frac{V_f}{2}$

$V_s = \frac{100}{2} = 50 km/hr$

**93. For a beam element, the bending moment, modulus of elasticity and the moment of inertia are given by M, E and I, respectively. What is the flexural rigidity of the beam element?**

(a) $\frac{M}{EI}$ (b) $\frac{EI}{M}$

(c) $\frac{1}{EI}$ (d) EI

**Ans. (d) :** If the bending moment, modulus of elasticity and the moment of inertial are given by M, E, I and σ respectively.

Then bending Equation is given as .

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

The multiple of EI is termed as the flexural rigidity because if it is more then the deflection of section will become less.

(i) Flexural rigidity→ E.I.

(ii) Axial rigidity→ EA

(iii)Torsional rigidity → GJ.

Where; A → Cross section area.

G → Share modulus.

J → Polar moment of inertia.

**94. The separation of the boundary layer takes place when the pressure gradient is**

(a) Negative

(b) Constant

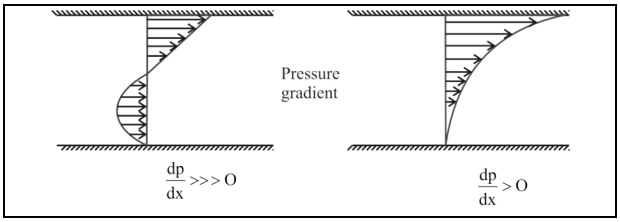
(c) Zero

(d) Positive

**Ans. (d) : Flow separation :-**

**Flow separation -**

When the pressure gradient is positive and the velocity gradient become negative than separation of the boundary layer takes place.




**95. The manometer shown in the given figure, connects two pipes carrying oil and water then**

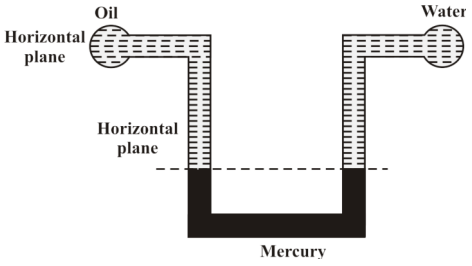
(a) If $S_1 < S_2$ the pressure in the pipes are equal

(b) Insufficient data to compare

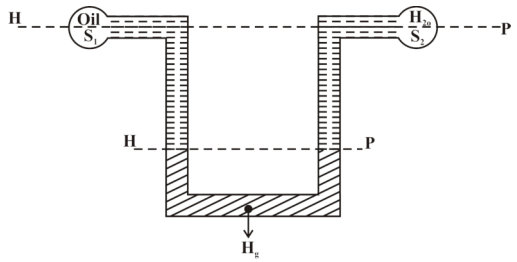
(c) If $S_1 < S_2$ the pressure in the water pipe is higher

(d) If $S_1 < S_2$ the pressure in oil pipe is higher

Where $S_1$ and $S_2$ are Specific Gravity of oil and water respectively.




**Ans. (d) :** →




if $S_1 < S_2$

Then : $\rho_1 < \rho_2 \Rightarrow \rho$ oil $< \rho_{water}$

So ; $P_1 < P_2$

$\boxed{Poil > Pwater}$

**96. A tube well commands 20 hectares of land under rice. When the well is discharging at a constant rate, the water level in the well drops to 120.2 m. The level of highest land is 129 m. The duty of rise is 800 ha/cumec on the field. For the pump efficiency of 60%, input power of the pump is**

(a) 4.32 Kw (b) 3.59 Kw

(c) 4.56 Kw (d) 4.82 Kw

**Ans. (b) :** Given = Area = 20 Ha

Height = 8.8m

Duty = 800 ha/cumec

$Q = \frac{\text{Area}}{\text{duty}} = \frac{20}{800}$

Solved-

$$\eta = \frac{\text{Output power}}{\text{Input power}}$$

$$\text{Input power} = \frac{\gamma QH}{\eta} = \frac{981 \times 20 \times 10 \times 8.8}{800 \times 0.6}$$

$$= 3597 \text{ watt}$$

Input power = 3.59 kw

**97. Arrange the steps of preparation of brick earth in chronological order.**

(a) Unsoiling (b) Cleaning
(c) Digging (d) Blending
(e) Weathering (f) Tempering
(a) A, B, C, D, E, F (b) A, C, B, D, E, F
(c) A, F, B, E, D, C (d) A, C, B, E, D, F

**Ans. (d) :** Flow chart of brick manufacturing–
Unsoiling → Digging → Cleaning → Weathering → Blending → Tempering → Moulding and burning

**98. Consider the following statements:**

(a) Running speed is higher than journey speed
(b) Journey speed includes delays
(c) Journey speed is the average speed maintained by a vehicle over a given course while vehicle is in motion
(d) Running speed is known as the overall travel speed.

Which one of the above statements is/ are correct?

(a) C and D only (b) A and D only
(c) A and B only (d) B and C only

**Ans. (c) :**

**99. The survey carried out to delineate natural features such as hills, rivers, forests and man made features, such as towns, villages, buildings, roads, transmission lines and canals is classified as**

(a) Land survey (b) Topographic survey
(c) Geological survey (d) Engineering survey

**Ans. (b) :• Topographic survey :**
The survey carried out to delineate natural features such as hills, rivers forests and man made features, such as towns, villages, roads canals and buildings etc. is known as land survey.

**• Land Survey :**
The Survey which is used to decide the area of property or boundary is known as land survey, cadastral surveyor property survey.

**• Engineering Survey :**
The survey carried out to collect requisite data for designing, planning, estimating and execution of engineering projects is called as engineering survey.

**• Geological Survey :**
The survey which is the systematic investigation of the geology beneath the ground for sur face in order to prepare a geological map model etc is known as geological survey.

**100. Write the correct decreasing order with respect to the number of stress components for the following case.**

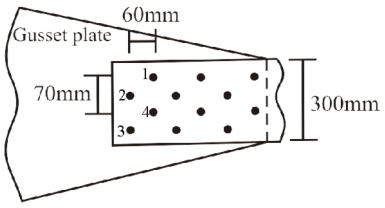
(a) An axially loaded member
(b) Plane strain problem
(c) Plane stress problem
(d) A single point of a three-dimensional structure

(a) D, B, C, A (b) C, D, B, A
(c) B, D, C, A (d) D, C, B, A

**Ans. (a) :**

| Condition | Stress components |
|---|---|
| (i) Axially loaded | 1 |
| (ii) Plane stress problem | 3 |
| (iii) Plane strain problem | 4 |
| (iv) Single point of 3D structure | 9 |

**101. A 300 mm plate with a thickness of 6 mm of grade Fe 410 is used as a tension member, as shown in the figure. the plate is connected to a 12 mm thick gusset plate using 20 mm diameter bolts of grade 4.6. The distance between the bolts are the same as the ones mentioned in the figure. What is the effective net area for the tension member?**




(a) 15.6.00 $mm^2$ (b) 1473.24 $mm^2$
(c) 1481.14 $mm^2$ (d) 1503.43 $mm^2$

**Ans. (c) :**

$$A_{net_{min}} = \left[ (B - n'd_o) + \frac{Ps^2}{4g} \right] \times t$$

Given,
p = 60 mm
g = 70 mm

$(A_{net})_{2-3} = (300 - 2 \times 21.5) \times 6 = 1542 \text{ mm}^2$

$(A_{net})_{1-2-3} = (300 - 3 \times 21.5 + \frac{60^2}{4 \times 70}) \times 6 = 1490.14 \text{ mm}^2$

$(A_{net})_{1-2-4-3} = (300 - 4 \times 21.5 + \frac{3 \times 60^2}{4 \times 70}) \times 6 = 1515.43 \text{ mm}^2$

Hence option (c) is most appropriate answer.

**102. Match the following:**

| | |
|---|---|
| (a) Tacheometry | (i) Angular Surveying |
| (b) Contouring | (ii) Horizontal and Vertical angles |
| (c) Levelling | (iii) Bearing |
| (d) Theodolite | (iv) Equal elevation |
| (e) Compass | (v) Elevation |

(a) A-i, B-iv, C-v, D-ii, E-iii
(b) A-ii, B-v, C-iv, D-iii, E-i
(c) A-iii, B-i, C-ii, D-v, E-iv
(d) A-ii, B-iii, C-i, D-iv, E-v

**Ans. (a) :**

| Parameter | Measuring instrument / Method |
|---|---|
| Equal elevation | Contouring |
| Elevation | Levelling |
| Horizontal and vertical angles | Theodolite |
| Bearing | Compass |
| Angular measurement in sloping terrain | Tacheometry |

**103. Match the following:**

| | |
|---|---|
| (a) Cautionary sign | (i) Downward equilateral triangle |
| (b) Indication sign | (ii) Octagonal |
| (c) Yield sign | (iii) Upward equilateral triangle |
| (d) Prohibitory sign | (iv) Rectangular |
| (e) Stop sign | (v) Circular |

(a) A-i, B-ii, C-iii, D-iv, E-v
(b) A-v, B-iv, C-ii, D-i, E-iii
(c) A-ii, B-i, C-v, D-iii, E-iv
(d) A-iii, B-iv, C-i, D-v, E-ii

**Ans. (d) :**

| Traffic signs | Shape |
|---|---|
| Cautionary sign | Upward equilateral triangle |
| Indicating sign | Rectangular |
| Yield sign | Downward equilateral Δ |
| Prohibitory sign | Circular |
| Stop sign | Octagonal |

**104. A dry sand specimen failed in a traxial test when the major and minor principal stresses were respectively 900kN/m$^2$ and 250kN/m$^2$. If same specimen is tested in direct shear test under a normal stress of 100 kN/m$^2$ shear stress will be equal to**

(a) 68.5 kN/m$^2$ (b) 575 kN/m$^2$
(c) 550 kN/m$^2$ (d) 650 kN/m$^2$

**Ans. (a) :** Given,
$\sigma_1 = 900$ kPa
$\sigma_3 = 250$ kPa

$$\sigma_1 = \sigma_3 \tan^2\left(45 + \frac{\phi}{2}\right)$$

$$900 = 250\tan^2\left(45 + \frac{\phi}{2}\right)$$

On solving $\phi = 34.42^0$
Now,

$$\tau = C + \sigma\tan\phi$$

$=100 \tan 34, 42^0$ (C = 0 for sand)
$= 68.51$ kPa

**105. Corrosion of reinforcement in concrete can be caused by the ingress of**

(a) Sulphates and chlorides
(b) Carbon dioxide and alkalis
(c) Alkalis and sulphates
(d) Chlorides and carbon dioxide

**Ans. (a) :**

**106. Calculate the critical depth corresponding to a discharge of 8 m$^3$/s in a rectangular channel of width 4m. Take g= 10m/s$^2$**

(a) $(0.5)^{1/3}$m
(b) $(0.4)^{1/3}$m
(c) $(0.2)^{1/3}$m
(d) $(0.8)^{1/3}$m

**Ans. (b) :** Given

$= 8m^3/sec$ $\boxed{q = \frac{8}{4} = 2m^2/\sec}$

B = 4m
g = 10m/sec2.
Sovled

$\boxed{y_c = \left(\frac{q^2}{g}\right)^{1/3}}$ for nectanglelar channel

$$\boxed{y_c = \left(\frac{2\times 2}{10}\right)^{1/3} = (0.4)^{1/3.}\text{m}}$$

**107. A precast concrete pile of size 50 cm× 50 cm is to be driven into clay strata whose unconfined compressive strength is 220 kN/m2 and adhesion factor = 0.6. The length of the pile required to carry safe working load o 450kN with factor of safety of 2.5 is**

(a) 6.647 m (b) 10.23 m
(c) 2 m (d) 3.3 m

**Ans. (a):** Square pile = B ×B = 50cm × 50cm

$$\text{Cu} = \frac{220kN/m^2}{2} = 110kpa$$

Q safe = 450

FOS = 2.5

$N_C = 9$

$\alpha = 0.6$

$$\text{Q safe} = \frac{Qu}{fos}$$

$$= \frac{CNc \times A_b + \alpha\bar{c}As}{fos}$$

$450 \times 2.5 = 110 \times 9 \times (0.5)^2 + 0.6 \times 110 \times 4 \times 0.5 \times L$

$\boxed{L = 6.647\,\text{m}}$

**108. What is the anchorage length of a diameter Fe 415 bars in M30 concrete column in compression? Assume $\sigma_s = 0.75f_y$**

(a) 41. 50 d (b) 51.87 d

(c) 32.42 d (d) 25.94 d

**Ans. (d) :** For M 30 $\tau_{bd}$ = 1.5 MPa

For fe 415 deformed bar is compression

$\tau_{bd} = 1.5 \times 1.6 \times 1.25 = 3$ MPa

Now,

$$\tau_{bd} \times (\pi \times \phi) \times L_d = \sigma_s \left(\frac{\pi}{4} \times \phi^2\right)$$

$$L_d = \frac{0.75 \times 415 \times \phi}{4 \times 3} = 25.935\phi$$

**109. The weight of water to be added in kN per m3 to make the soil saturated (with specific gravity =2.7) having moisture content+ 20% and void ratio = 0.72 is**

(a) 1.025 (b) 0.51

(c) 9.81 (d) 10.025

**Ans. (a) :** Given,

G = 2.7, W = 20% , e = 0.72

Water content required for full saturation.

Se = wG

$$w = \frac{1 \times 0.72}{2.7} = 0.267$$

Additional water content required = 26.67–20 = 6.67%

Dry density ($\gamma_d$) = $\frac{G\gamma w}{1+e} = \frac{2.7 \times 9.81}{1+0.72}$ = 15.40 kPa

Quantity of water to be added = $\frac{6.67}{100} \times 15.40 = 1.027$ kN

**110. A point load of 500 KN applied on the surface of thick layer of saturated clay. Using Boussinesq's elastic analysis the estimated vertical stress at a depth of 3 m and a radial distance of 2 m from the point of application of load is**

(a) 6.7 kPa (b) 2.3 kPa

(c) 5.5 kPa (d) 10.58 kPa

**Ans. (d) :** Q = 500 KN

z = 3m

r = 2m

**Solve** as P or Boussinesq's thorn

$$\sigma_z = \frac{3}{2\pi} - \frac{Q}{z^2}\left[\frac{1}{1+(r/z)^2}\right]^{5/2}$$

$$= \frac{3}{2\pi} \times \frac{500}{3 \times 3}\left[\frac{1}{1+\left(\frac{2}{3}\right)^2}\right]^{5/2}$$

$\boxed{\sigma_z = 10.58\,\text{k}P\text{a}}$

**111. Consider the following statements:**

(a) The pH value of water for making concrete should be more than 8.5 to produce corrosion - resistant concrete.

(b) The presence of sugar causes retardation in the setting of concrete.

(c) Entertained air voids reduce resistance against freeze- thaw deterioration.

(d) Chlorides should be less than 0.6 kg/m$^3$in the water used for construction of reinforced concrete.

Which of the above statements is/are correct?

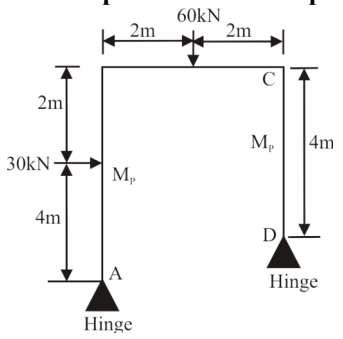
(a) A and B (b) B and C

(c) B and D (d) A and C

**Ans. (c) :**

- pH range of water for concrete (6 - 8.5)
- The presence of sugar causes retardation in the setting of concrete.
- Entertained air voids increase resistance against freezing and thawing.
- Chlorides should be less than 0.6kg/m$^3$ in the water used for construction of Rcc.

**112. What is the plastic moment for the frame as shown in the figure? All members of the frame have the same plastic moment capacity $M_P$.**




(a) 53.33 kNm (b) 36.92 kNm

(c) 48 kNm (d) 63.33 kNm

**Ans. (a) :** $D_s = R - 3$

$\boxed{= 4 - 3 = 1}$

(N) No of possible plastic Hinge is = 4,

Plastic Hinge required to form mechanism = $D_s$+1

= 1 + 1

= 2

No of Indiant Pend Mechanism = N-$D_s$

= u-1

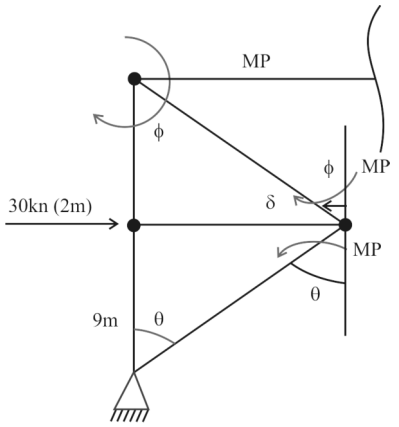
$\boxed{= 3}$

i.e 1) Beam mechanism in AB.

2) Beam mechanism in BC

3) Combined Mechaniosm

1) **Beam Mechanism in AB**




$$\begin{cases} \delta = \delta \\ 4\theta = 2\alpha \\ \alpha = 2\theta \end{cases}$$

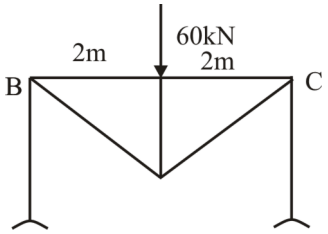
External W.D. = Internal W.D (WD=work done)

$30 \times \delta = M_P\theta + 2Mp\alpha.$

$30 \times 4\theta = M_P\theta + 2Mp2\theta.$

$\boxed{M_P = 30KN - M}$

Beam Mechanism in BC Span




$Wc = \frac{8Mp}{\ell.}$

$M_P = \frac{Wc.\ell}{8}$

$= \frac{60 \times 4}{82}$

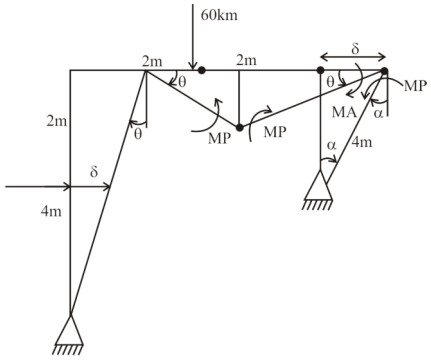
$\boxed{M_P = 30KN.M}$

**3) Combined Mechanism**

Figure

$$\begin{cases} \delta = \delta \\ \cancel{6}\theta = \cancel{4}\alpha \\ 3 \quad 2 \\ 3\theta = 2\alpha \\ \frac{3\theta}{2} = \alpha \end{cases}$$

$\boxed{1.50 = \alpha}$




External work done = Internel W.D.

$30 \times \delta + 60 \times s_1 = 3MP \times \theta + M_P\alpha$

$30 \times 40 + 60 \times 2\theta = 3MP \times \theta + M_P \times 1.5\theta$

$\boxed{Mp = 53.33kN - M}$

$\boxed{\text{Maxm. is } 53.33\,\text{kN.M}}$

**113. The extra widening required for a three-lane national highway at a horizontal curve of 900 m radius, considering a wheel base of 6 m and a design speed of 100 kmph is**

(a) 0.41 (b) 0.61

(c) 0.51 (d) 0.31

**Ans. (a) :** Given R = 900m

V = 100 kmph.

$\ell$ = 6m n = 3

Extra widening = $M_E$ + Mp

$$\boxed{W_E = \frac{n\ell^2}{2R} + \frac{V}{9.5\sqrt{R.}}}$$

$$= \frac{3 \times 6^2}{2 \times 900} + \frac{100}{9.5\sqrt{900}}$$

$= 0.06 + 0.35$

$\boxed{W_E = 0.41}$

**Note–** however no extra widening is required for curves having R > 300 m.

**114. A24 hr rainfall of 50 cm magnitude at a station located in hilly terrains has a return period of 50 years. The probability of having 3 succeeding years in which rainfall is greater than or equal to 50 cm is**

(a) $8 \times 10^{-4}$ (b) $8 \times 10^{-6}$

(c) $6 \times 10^{-6}$ (d) $4 \times 10^{-6}$

**Ans. (b) : Given**

T = 50 year

$P = \frac{1}{50}$

**Solution**

It is case of conditional Probability, when the results are independent.

Probability = (P). (P). (P)

$\boxed{\text{Probability} = P^3}$

$= \left(\frac{1}{50}\right)^3$

$= 8\times10^{-6}$

**115. For a three- dimensional problem, the stress components altogether is a tensor of order**

(a) 0 (b) 3

(c) 2 (d) 1

**Ans. (c) :** For three-dimensional problem the stress components altogether is a second order tensor. Those tensors which required magnitude, direction and point of application for their representation are known as 2nd order tensor.

**116. The observed bearings of a traverse are given below:**

| Line | Bearing | Line | Bearing |
|---|---|---|---|
| PQ | 45°15' | QP | 225°15' |
| QR | 107°15' | RQ | 286°15' |
| RS | 202°30' | SR | 21°30' |
| ST | 320°45' | TS | 141°45' |

**The stations (s) most likely to be affected by the local attraction is/are**

(a) only R (b) Q and R

(c) R and S (d) Only S

**Ans. (c) :** $PQ \rightarrow 225^0 15' - 45^0 15'$

$= 180^0$

P and Q are free from local attraction.

Q $\boxed{R}$ $\rightarrow 286^0 15' - 107^0 15'$

$= 179^0$

Q is free from local attraction so station R is affected by the local attraction.

RS $\rightarrow$ $202^0 30' - 21^0 30^0$

$= 181^0$

R is affected and S also can be affected .

ST $\rightarrow$ $320^0 45' - 141^0 45'$

$= 179^0$

Station S most likely to be affected

So the station R and S are most likely to be affected by the local attraction

**117. Find out the length of transition curve for a broad-gauge curve of 3 degrees having a cant of 12 cm. The maximum permissible speed on the curve us 100 km/hr and the allowable cant deficiency is 85 mm.**

(a) 86.40

(b) 89.40

(c) 87.60

(d) 62.04

**Ans. (c) :** length of transition curve

**1st Method**

$L = \max \begin{cases} 7.2\ e \text{ meter} \\ 0.073e.\ V_{max} \\ 0.073D,\ V_{max} \end{cases}$

**2st Method**

(i) Railway board formula $L = 4.4\sqrt{R}$

(ii) Rate of change of cant. $L = 3.6e$

(iii) Change of radial acularation $L = \frac{3.28V^3}{R}$

**Where ;**

$e \rightarrow$ cant in cm)

$D \rightarrow$ cant deficiency (cm)

$V_{max} \rightarrow$ max$^m$ Permissible speed (kmph)

**Given :**

Super elevation/cant = 12cm

Cant deficiency, D = 85mm=85.cm

max$^m$. Permissible speed $V_{max}$ = 100kmph.

1st Method

$L = \max \begin{cases} 7.2\times12 = 86.4\text{m} \\ 0.073\times12\times100 = 87.6\text{m} \\ 0.073\times8.5\times100 = 62.05\text{m} \end{cases}$

So L should be 87.6m

**118. M 20 grade of concrete is approximately**

(a) 1 : 1.5 : 3 (b) 1 : 4 : 8

(c) 1 : 1.5 : 2 (d) 1 : 2 : 4

**Ans. (a) :** M 5 $\rightarrow$ 1 : 5 : 10

M 7.5 $\rightarrow$ 1 : 4 : 8

M 10 $\rightarrow$ 1 : 3 : 6

M 15 $\rightarrow$ 1 : 2 : 4

M 20 $\rightarrow$ 1 : 1.5 : 3

M 25 $\rightarrow$ 1 : 1 : 2

**119. If in a gradually varied flow dy/dx is positive, then de/dx is equal to**

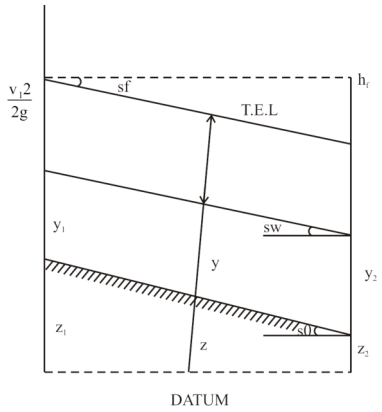
(a) $S_o - S_f$

(b) $S_o\left(1 - F^2\right)$

(c) $S_o - S_f - dy/dx$

(d) $S_f - S_o$

**Ans. (a):**




$$T.H = H \Rightarrow z + y + \frac{v^2}{zg.}$$

$$H = z + E$$

$$\frac{dH}{dx} = \frac{dz}{dx} + \frac{dE}{dx}$$

$$-S_f = -S_o + \frac{dE}{dx}$$

$\boxed{\frac{dE}{dx} = S_o - S_f}$ Energy Equation for gradually varied flow

**120. A retaining wall, 8m high with a smooth vertical back, retains a clay backfill with C'=15kN/m$^2$, $\phi' = 15^0$ and $\gamma = 18 kN/m^3$.**

**Assume that the tension cracks may develop to the full theoretical depth, the active thrust on the wall is**

(a) 23kN/m$^2$ (b) 20kN/m$^2$
(c) 46kN/m$^2$ (d) 61.67kN/m$^2$

**Ans. (d) :** Given = H = 8m

$C^1 = 15$ kPa

$\gamma = 18$kN/m

$\phi = 15^0$

$$k_a = \frac{1-\sin 15^0}{1+\sin 15^0} = 0.588$$

$$P_a = k_a(\sigma_v) - 2C\sqrt{k_a}$$

$$= 0.588 \times y \times H - 2C' \sqrt{0.588}$$

$$= 0.588 \times 18 \times 8 - 2 \times 18 \sqrt{0.588}$$

$\boxed{Pa = 61.78\,KN/m^2} \simeq 61.67\ KN/m^2$

**121. Consider the following:**

| | | | |
|---|---|---|---|
| (a) | Earnest money deposit | (i) | Settlement of a dispute by the decision of third person |
| (b) | Security deposit | (ii) | Offer in writing to execute some specified works |
| (c) | Arbitration | (iii) | Deposited by the contractor to safeguard the interests of the owner in the event of improper performance of the contract |
| (d) | Tender | (iv) | Guarantee in the form of cash on the part of the contractor to keep open the offer for consideration |

**Choose the option in which given pairs are correctly matched.**

(a) A-iv, B- iii, C-i, D-ii
(b) A-ii, B- i, C-iv, D-iii
(c) A-i, B- ii, C-iv, D-iii
(d) A-iii, B- iv, C-i, D-ii

**Ans. (a) : Arbitration–** It is the process in which dispute arised between two parties is settled by a third party trustable to both parties. .

**Earnest money–** It is the guarantee of contractor to owner in the form of cash/cheque.

- EM also shows the willingness of contractor to execute the project.

**Security deposit–** It is the money given by contractor to safeguard the interest of owner if any malpractices occur during execution of project.

**Tender–** It is a document inviting the interested particle in writing to execute projects.

**122. The diurnal variation is**

(a) More at magnetic poles and less at equator
(b) Does not changes from year to year
(c) More in winter and less in summer
(d) More in day time and less in night time

(a) A and D (b) B and D
(c) A and B (d) B and C

**Ans. (a) : Diurnal variation :**
It is generally more in summer season. Due to rotation of earth . about its own axis. Is more to pole, in day & summer time.

**123. An instantaneous unit hydrograph is a hydrograph of**

(a) unit duration and indefinitely small rainfall excess
(b) instant runoff from a unit area of catchment
(c) infinitely small duration and of unit rainfall excess
(d) instant duration and infinite rainfall excess

**Ans. (c) : • Instantaneous unit hydrograph -**
The instantaneous unit hydrograph is defined as a unit hydrograph produced by an effective rainfall of unit depth in infinitely small duration.

**124. Order the following structure with respect to their increasing static indeterminacy.**

(a) A two- span continuous beam hinged at one end and rollers at the other supports.

(b) A two span continuous beam fixed at one end and rollers at the other supports.

(c) A fixed- fixed beam with two internal hinges.

(d) A one bay one store fixed-fixed frame.

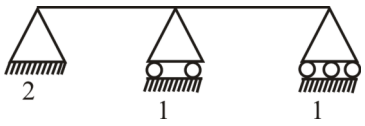
(a) C, A, D, B

(b) C, A, B, D

(c) C, B, D, A

(d) C, D, B, A

**Ans. (b) :**




**Note –** We assume general loeling i.e. only Horizontal Reactions Horizontal losing**.**

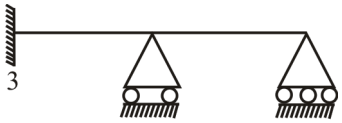
No of Reaction = 2 + 1 + 1 = 4

Equilibrium Eqn. = 3

$D_s$ = r - s = 4 - 3

$\boxed{D_S = 1}$

(b)

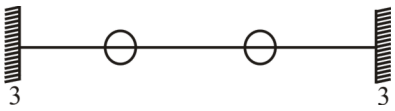



r = 5

s = 3

$\boxed{D_s = r - s = 5 - 3 = 2}$

(c)

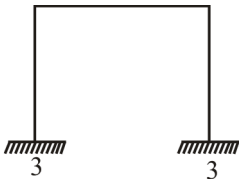



r = 6

s = 3 + 2 (for two in tonal Hinge)

$\boxed{D_s = r.s = 6 - 5 = 1}$

(d)



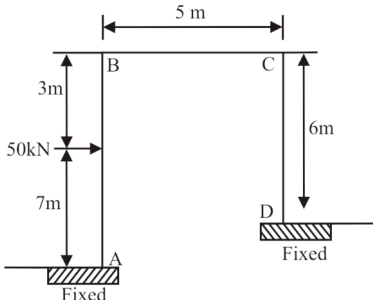

$D_s$ = r = s

= 6 - 3

$\boxed{D_s = 3}$

So option (b) $\boxed{C, A, B, D}$

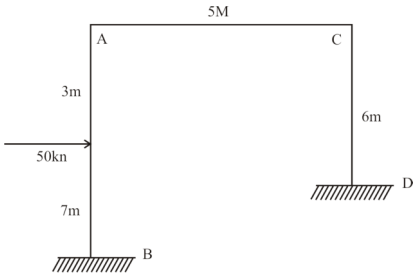
**125. What is the kinematic indeterminacy of the frame shown in the figure?**




(a) 2 (b) 4

(c) 1 (d) 3

**Ans. (d) :**




for fixed support $D_K$ = 0 {at A, D}

$D_K = 3$

for rigid Joint +3 {at B &C}

$\boxed{D_K = 6}$

$D_K$ (without Axial Deformation), deduct the No of Member in frame.

$D_K = 6 - 3$ [AB, BC & CD]

$\boxed{D_K = 3}$

**126. Isogonic line is line of**

(a) Positive declination

(b) Negative declination

(c) Same declination

(d) Zero declination

**Ans. (c) : Isogonic line-** The imaginary line joining points of equal declination.

**Agonic line-** The imaginary line joining zero declination

**127. What will be depth of d/s cutoff wall if maximum static head, length of floor and safe exit gradient is 2.4 m, 21.45 m and 1/5 respectively?**

(a) 2.56 m (b) 0.64 m

(c) 1.28 m (d) 0.9 m

**Ans. (c) :** H = 2.4

$G_E = \frac{1}{5}$

Length floor = 21.45

**Solved**

It is done by trial and Error Methods

$G_E = \frac{H}{d} \times \frac{1}{\pi\sqrt{\lambda}}$

$$\lambda = \frac{1+\sqrt{1+\alpha^2}}{2}$$

$$\alpha = \frac{b}{d.}$$

But putting d = 1.28 , $G_E$ = ?

$$\alpha = \frac{21.45}{1.28} = 16.75, \lambda = 1+\sqrt{\frac{1+(16.75)^2}{2}}$$

$$\boxed{\lambda = 8.39}$$

$$G_E = \frac{2.4}{1.28} \times \frac{1}{\Lambda\sqrt{8.39}}$$

$\boxed{G_E = \frac{1}{5}}$ So (B) Option is correct.

**128. For a pure shear problem with shear stress $\tau_{xy}$, what would be the maximum principal stress?**

(a) $\frac{\tau_{xy}}{2}$ (b) $\tau_{xy}$

(c) $\sqrt{\tau_{xy}}$ (d) $2\tau_{xy}$

**Ans. (b) :** In Pure shear case –

$$\sigma_{1/2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$\sigma_{1/2} = \pm\tau_{xy} \quad (\because \sigma_x = \sigma_y = 0)$

$\sigma_1 = +\tau_{xy}$

$\sigma_2 = -\tau_{xy}$

**129. A confined aquifer of thickness 5 m gives a steady discharge of 20 litre/sec through a well of 0.3 m radius. The height of water in the well dropped from 10 m to 8 m after pumping. What is the coefficient of permeability when the radius of influence is taken as 300 m?**

(a) 0.0001*(In 1000/π)

(b) 0.0011*(In 1000/π)

(c) 0.011*(in 1000/π)

(d) 0.101*(in 1000/π)

**Ans. (b) :** Given, Q = 20 × 10.3 m3/sec.

D = 5m

H-h = 2m

R = 300

$r_\omega$ = 0.3

Solv

$$Q = \frac{2\times\pi\times K\times 5\times 2}{\log e\left(\frac{300}{0.3}\right)}$$

$$20\times10^{-3} = \frac{2\times\pi\times K\times 5\times 2}{\log e\left(\frac{300}{0.3}\right)}$$

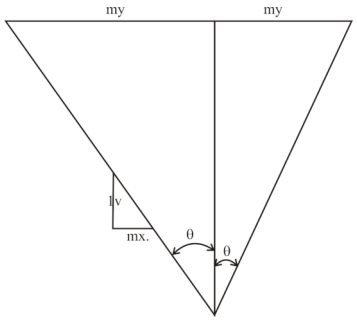
$$\boxed{k = \frac{0.001\log e(1000)}{\pi}}$$

**130. A triangular section is hydraulically efficient when the vertex angle is**

(a) 90º (b) 75º

(c) 30º (d) 120º

**Ans. (a) :**




θ is with vertical

$\boxed{\text{vertex} = 2\theta}$

For hydraulically efficient triangular section.

$\boxed{\theta = 45^0,\ m = \tan\theta = \tan 45^0 = 1}$

i.e vertex angle = 2θ = 2 ×$45^0$ = $90^0$

**131. Consider the following statements.**

(a) Cumulative errors are more important than compensating errors

(b) All cumulative errors are equally important

(c) The more times a line is measured, the more likely the accidental errors to disappear from mean

(d) Variation in temperature results in compensating error

Which of the above statements is/are correct?

(a) A and B (b) B and D

(c) A and C (d) B and C

**Ans. (b) :** In judging of the relative importance of error the following additional points are to be keep in mind.

(1) Cumulative errors are more important than compensating errors but, not all cumulative error as equally importance.

(2) One cumulative error some time balances another cumulative error

(3) In short line a compensating error may occur only once or twice and thus tail to be compensating.

**132. The length of a line measured with a 25 m chain was found to be 300m. Calculate the true length of the line if the chain was 15 cm too long.**

(a) 300.2 (b) 298.2

(c) 251.5 (d) 301.8

**Ans. (d) :** true chain length = 25m

Wrong chain length = 25 + 0.15= 25.15 m

Wrong length of line = 300M

True length of line = ?

$\Rightarrow$ True × True = Wrong × wrong

$25 \times$ True length of line $= 25.15 \times 300$

$$\text{True length line} = \frac{25.15 \times 300}{25}$$

$$\boxed{= 301.8\text{m}}$$

**133. Match the type of work in List 1 with the use requirements from fresh concrete in List 2 :**

| List I | List II |
|---|---|
| (a) Road | (i) high workability and semi fluid consistency |
| (b) Mass concrete | (ii) very low workability and very dry consistency |
| (c) Heavy reinforced RCC section | (iii) low workability and dry consistency |
| (d) RCC congested reinforcement | (iv) medium workability and plastic consistency |

(a) A-iii, B-ii, C-i, D-iv (b) A-ii, B-iv, C-i, D-iii
(c) A-iii, B-iv, C-i, D-ii (d) A-ii, B-iii, C-iv, D-i

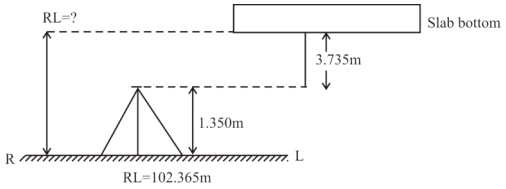
**Ans. (d) :**

| | | |
|---|---|---|
| 1 | Road | Very low workability and very dry consistency. |
| 2 | Mass concrete | Low workability and dry consistency. |
| 3 | Heavy reinforced RCC section | Medium work ability and plastic consistency |
| 4 | RCC Congested reinforcements | High workability and semi fluid consistency |

**134. A level instrument at a height of 1.350 m has been placed at a station having a Reduced Level (RL) of 102.365 m. The instrument reads– 3.735 m on a leveling staff held at the bottom of a bridge deck. The RL (in m) of the bottom of the bridge deck is**

(a) 104.75 (b) 116.72
(c) 107.45 (d) 99.98

**Ans. (c) :**




RL of instrument = (102.365+1.350) m
= 103.715m

RL of bottom of Slab = RL of instrument + 3.735m
= 103.715m + 3.735m

$$= \boxed{107.45\text{m}}$$

**135. While designing the abutments of a bridge, the lateral earth pressure to be considered is**

(a) Active earth pressure
(b) At the rest pressure
(c) At the bottom of the bridge
(d) Passive earth pressure

**Ans. (b) :**

- While the designing the abutments of a bridge the lateral earth pressure to be considered is at the rest pressure.
- Basement slabs are also designed for earth pressure at rest and retaining walls are designed for active earth pressure.

**136. For a catchment, the s- curve due to a rainfall intensity 1 cm/h is given by $Q=3-(2+t)\,t^{-2}$ where t is in hour, Q in m3/s. At t=1 hour, ordinate of instantaneous unit hydrograph will be**

(a) 5 $m^3/s$ (b) 50 $m^3/s$
(c) 3 $m^3/s$ (d) 500 $m^3/s$

**Ans. (a) :** Oridinate of IUH $= \frac{I}{D}\frac{dQS}{dt}$

$$= 1.\frac{d}{dt}\left[-\frac{-2}{t^2}-\frac{1}{t}\right]$$

$$= \frac{-2(-2)}{t^3}-\left(\frac{-1}{t^2}\right)$$

$$= \frac{4}{t^3}+\frac{-1}{t^2}$$

Put t = 1

$$= \frac{4}{1}+\frac{1}{1}$$

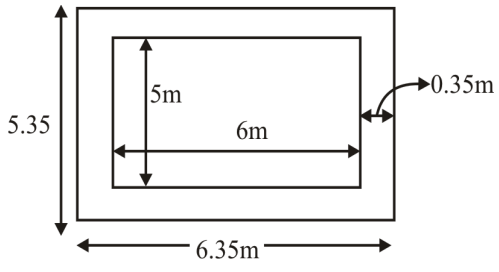
$= 4 + 1$

$$\boxed{= 5\,m^3/sec}$$

**137. The inner dimensions of a room are 6m × 5m with 35 cm wall thickness. The total central line length of the room will be**

(a) 21.2 m (b) 25.0 m
(c) 36.0m (d) 23.4 m

**Ans. (d) :**




Total centre to centre length = 2×6.35+2×5.35

$$= \boxed{23.4\text{m}}$$

**138. A rectangular channel 2 m wide carries a discharge of 10 $m^3$/s at a depth of 2m. Calculate the specific energy of the channel.**

(a) 1.8 m (b) 4.6 m

(c) 2.3 m (d) 3.0 m

**Ans. (c) :** Given B = 2m

Q = 10m3/sec

g = 2m

**Solution**

$$E = y + \frac{V^2}{2g}$$

$$V = \frac{Q}{k} = \frac{10}{2 \times 2} = 2.5\,\text{m/sec}$$

$$E = 2 + \frac{2.5^2}{2 \times 10}$$

$$\boxed{E = 2.31\,\text{m}}$$

**139. The design speed on a road is 72 kmph. Assuming the driver reaction time of 2.0 seconds and coefficient of friction of pavement surface as 0.30, the required stopping distance (in meters) for a two -lane two-way traffic is**

(a) 40.0 (b) 43.30

(c) 106.67 (d) 96.6

**Ans. (c) :** V = 72 kmph

$T_R$ = 2sec

$\mu$ = 0.30

SSD = ?

Solu.

$$SSD = 0.278\, VT_R + \frac{V^2}{254(f \pm S)}$$

$$= 0.278 \times 72 \times 2 + \frac{72 \times 72}{254(0.30)}$$

$$\boxed{SSD = 108.06\,\text{M}} \simeq 106.67$$

**140. The centre of buoyancy of a submerged body**

(a) Coincides with the center of gravity of body

(b) It is always above the centroid of the displaced volume of liquid

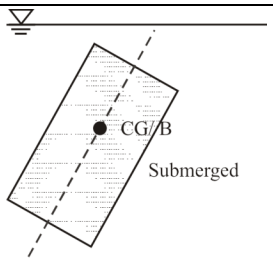
(c) It is always below the center of gravity of the body

(d) Coincides with the centroid of displaced volume of fluid

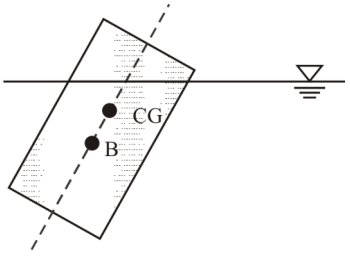
**Ans. (d) :** Solved

**Centre of buoyancy –**

(i) The Centre of buoyancy is the point at which buoyant force is acting. It is the centroid of the displaced volume of fluid.




(ii) In partially submerged condition center of buoyancy lies below the C.G of the body.




**141. Match the following errors in chaining.**

| | | | |
|---|---|---|---|
| (a) | Errors in reading the tape | (i) | Cumulative + or – |
| (b) | Errors due to temperature | (ii) | Cumulative + |
| (c) | Incorrect counting of tape lengths | (iii) | Mistake |
| (d) | Bad ranging | (iv) | Compensating ± |
| (e) | Variation in pull | (v) | Blunder |

(a) A-v, B-ii, C-iii, D-iv, E-i

(b) A-iv, B-iii, C-i, D-v, E-ii

(c) A-iii, B-i, C-v, D-ii, E-iv

(d) A-i, B-iv, C-ii, D-iii, E-v

**Ans. (c) :**

| Error | Nature |
|---|---|
| Reading the tap | Mistake |
| Due to temperature | Cumulative (+ve) or (- ve) |
| Incorrect counting of tap length | Blunder |
| Bad Ranging | Cumulative (+ ve) |
| Variation in pull | compensating (±) |

**142. For any material, the modulus of elasticity at a point can be given by $E_x$,$E_y$,$E_z$. Which of the following properties is true for a homogeneous material?**

(a) $E_x = E_y = E_z$ at a point and $E_x \neq E_y \neq E_z$ at the other points throughout the material

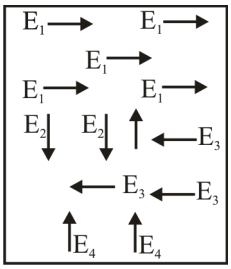
(b) $E_x = E_y \neq E_z$ at a point and $E_x = E_y = E_z$ the other points throughout the material

(c) $E_x \neq E_y \neq E_z$ at a point and $E_x = E_y = E_z$ at the other points throughout the material

(d) $E_x \neq E_y \neq E_z$ at a point and the same property exists at all the points throughout the material

**Ans. (d) : Homogenous material–** In homogeneous material property is same at every points in same direction. If direction changes then property will also change.

So $E_x \neq E_y \neq E_z$

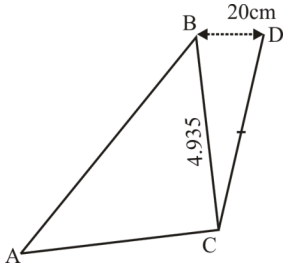



**143. A surveyor was reading a level staff as 4.935 m. Later he found that the staff was 20 cm off the vertical through the bottom. Find the error of reading.**

(a) 4.735 (b) 4.931

(c) 4..939 (d) 5.135

**Ans. (*) :** The observed value = 4.935m




In Δ BDC

$\Rightarrow BC^2 = CD^2 + BD^2$

vertical $\Rightarrow CD^2 = 4.935^{2-} (0.2)$

$CD^2 = 24.314$

$\boxed{CD = 4.939\,m}$

Hence, The correct staff reading 4.939m.

Error of reading $= 4.939 - 4.935 = 4 \times 10^{-3}$ m

**144. The saturated unit weight of sand in the bed of a pond 20 m deep is 20 kN/m$^3$. Unit weight of water is 10 kN/m$^3$. The effective stress at 4 m below bed level of pond is**

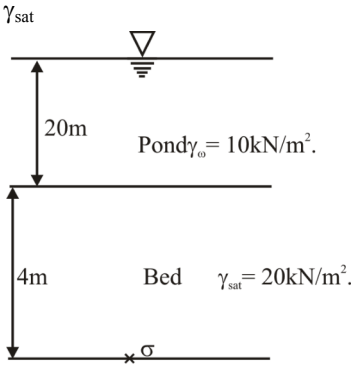
(a) 20 kN/m$^2$

(b) 80 kN/m$^2$

(c) 60 kN/m$^2$

(d) 40 kN/m$^2$

**Ans. (d) :** Given,




$\sigma^1 = \gamma^1 \times 4m$

$= (\gamma_{sat} - \gamma_\omega) \times 4$

$= (20 - 10) \times 4$

$\boxed{\sigma^1 = 40kN/m^2}$

**145. Consider the following statement with respect to cements:**

**(A) Low heat cement is produced by reducing the amount of tricalcium aluminate and increasing the amount of dicalcium silicate.**

**(B) Rapid hardening cement is produced by decreasing the amount of tricalcium silicate.**

(a) Statement B is true and A is false

(b) Both A and B are true

(c) Both A and B are false

(d) Statement B is false and A is true

**Ans. (d) : Low heat cement –**

- It is produced by reducing the amount of $C_3A$. $C_3S$ and increasing the amount of $C_2S$
- Initial setting time 1hr and expansion ≤ 10mm
- Widely used in mass concreting.

**Rapid hardening cement –**

- It is produced by increasing $C_3S$ and reducing $C_2S$ and fined grandring as compare to OPC. Initial and final setting time is same as OPC.

**146. An oven dry soil sample of volume 250 cc weight 430 g. If the specific gravity of solids is 2.70. What will be water content which willfully saturate the sample?**

(a) 29.8% (b) 73.1%

(c) 19.6% (d) 26.9%

**Ans. (c) :** V = 250CC

W = 430g

$G_S$= 2.00

w % = ? S = 1

Solv : Se = w.G.

$\boxed{e = w.G.}$

$$\boxed{\rho_d = \frac{430}{250} = 1.72\, gm/cc}$$

$$\gamma_d = \frac{G\rho_w}{1+e}$$

$$1+e = \frac{G\rho_w}{\rho_d}$$

$$\boxed{e = \frac{G\rho_w}{\rho_d}}$$

$$\boxed{\frac{G\rho_w}{\rho_d} - 1 = wG}$$

$$\frac{2.7\times 1}{1.72} - 1 = w \times 2.7$$

$$w = \frac{0.5697}{2.7}$$

$$\boxed{w = 21.10\% \simeq 19.6\%}$$

**147. In a particular season, the catchment was found to have a phi- index of 0.4 cm/h. If a rainfall of 2.4 cm occurs in that season at a uniform rate in a 8 hour storm, the resulting direct runoff depth on the catchment in cm will be**

(a) 0.8 cm (b) 2.8 cm
(c) 2 cm (d) 0

**Ans. (d) :** Given
$\phi = 0.4$ cm
P = 2.4 cm
t = 8hr

**Runoff** P- $\phi\times$ t
= 2.4 - 0.4×8
= 2.4 – 3.2
= -8.
= Runoff can't be Negatives So
Answer is zero

**148. Arrange the following in chronological order.**

(a) Integration and regrouping
(b) Modified Guarantee system
(c) Old Guarantee system
(d) State construction and ownership
(e) Nationalization

(a) A, C, B, D, E (b) D, E, A, B, C
(c) C, D, B, E, A (d) E, D, B, A, C

**Ans. (a) :** For project order are
1. Integration & Regrouping
2. Old Guarantee system
3. Modified Guarantee system
4. State construction and ownership
5. Nationalization

**149. The hydraulic radius of an efficient triangular section channel is**

(a) 2√2y (b) y
(c) y/2 (d) y/(2√2y)
Where, y is the depth of the water.

**Ans. (d):** hydraulic radius (R) = $\frac{\text{Area}}{\text{Peramiter}}$

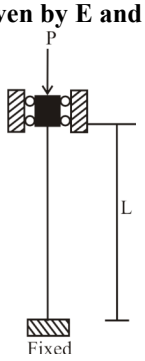
For efficient triangular section
$\theta = 45^0$, m ≃ $\tan\theta = \tan 45^0 = 1$

$$R = \frac{m\, y^2}{2y\sqrt{1+m^2}}$$

$$= \frac{y/2}{2y/\sqrt{2}}$$

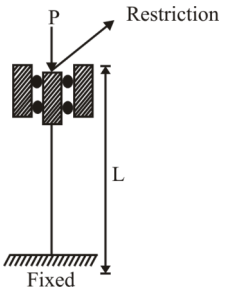
$$\boxed{R = \frac{y}{2\sqrt{2}}}$$

**150. What is the critical buckling load for the column shown in the figure? the modulus of elasticity and the moment of inertia of the column are given by E and I, respectively.**




(a) $\frac{\pi^2 EI}{L^2}$ (b) $\frac{4\pi^2 EI}{L^2}$
(c) $\frac{2.046\pi^2 EI}{L^2}$ (d) $\frac{\pi^2 EI}{4L^2}$

**Ans. (b) :**




In this case both end of column is fixed because of fixed and slider support with restriction in the direction of loading

So : effective length of column = $\frac{L}{2}$.

The buckling load

$$P_e = \frac{\pi^2 EI}{\left(\frac{L}{2}\right)^2}$$

$$\boxed{P_e = \frac{4\pi^2 EI}{L^2}}$$

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer (Environmental) Exam, 2022
## Solved Paper Date - 14.05.2022 (Shift-I)

**1. As per IS 456: 2000, the static modulus of elasticity of M 25 grade concrete is _____**

(a) 20000 N/mm$^2$ (b) 25000 N/mm$^2$
(c) 200 N/ mm$^2$ (d) 250 N/ mm$^2$

**Ans. (b) :** • Modulus of elasticity of concrete is recommended by IS 456 : 2000 –

$E_C = 5000 \sqrt{f_{ck}}$

Where, $f_{ck}$ is the characteristic strength of concrete.

$E_C = 5000 \sqrt{25}$

$E_C = 25000$ N/mm$^2$

**2. As per IS 456 : 2000, the partial factor of safety to be considered for steel reinforcement to assess the strength of a steel reinforced cement concrete structure, according to limit state method of design is _____**

(a) 1.5 (b) 1.15
(c) 1.65 (d) 2

**Ans. (b) :** As per IS 456 : 2000, partial factor of safety to be considered for steel reinforcement, according to limit state method (LSM) of design is 1.15.

According to LSM

Partial factor of safety for steel = 1.15

Partial factor of safety for concrete = 1.5

• According to WSM

Factor of safety for steel = 1.8

Factor of safety for concrete = 3

**3. Which of the following statements are INCORRECT?**

**Statement 1:** **Microsilica is a chemical admixture used in making concrete :**

**Statement 2:** **Microsilica particles are spherical in shape.**

**Statement 3:** **Mean particle size of micro silica ranges between 10 to 90 microns.**

**Statement 4:** **Specific surface are of micro silica is in the range of 20000 m$^2$/kg**

(a) Statements 1 and 2 are incorrect
(b) Statements 1 and 3 are incorrect
(c) Statements 2 and 3 are incorrect
(d) Statements 3 and 4 are incorrect j

**Ans. (b) :** Micro silica is also known as silica fume. It is an amorphous polymorph of silicon dioxide, silica. It is an ultrafine powder collected as a by - product of the silicon and ferrosilicon alloy production and consist of spheroid diameter with an average particle diameter of 150 mm. The main field of application is a pozzolanic material for high performance concrete.

**Properties–**

• The bulk density of silica varies from 130 to 600 kg/m$^3$.

• The specific gravity of silica fume is in range of 2.2 to 2.3

• Its specific surface area range from 13,000 to 30,000 m$^2$/kg

**4. As per IS 10262 : 2019, if angular coarse aggregate are replaced by sub-angular coarse aggregates in normal concrete, the quantity of mixing water per unit volume of concrete can be _____ to produce the same workability as that of angular coarse aggregates.**

(a) reduced by 10 kg (b) reduced by 20 kg
(c) increased by 10 kg (d) increased by 20 kg

**Ans. (a) :** As per clause 4, 2 of IS : 10262-2013, water can be reduced by 10 kg if sub-angular aggregates are used in place of angular aggregate.

| Aggregate used | Water reduction (kg) |
|---|---|
| Sub-angular | 10 kg |
| Gravel with crushed aggregate | 20 kg |
| Rounded gravel | 25 kg |

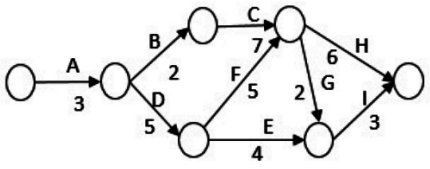
**5. As per national building code of India : 2016 (volume 1 - part 4). Which of the following civil engineering facility is classified as Group H buildings?**

(a) Warehouse (b) Railways
(c) Oil refineries (d) Highways

**Ans. (a) :** As per national building code of India : 2016 classification of building are–

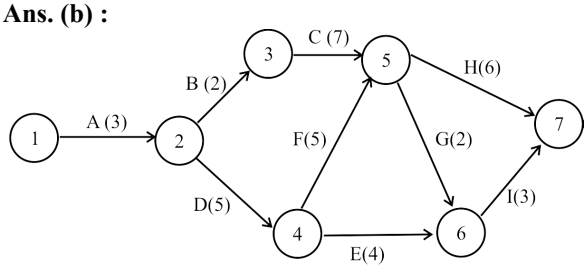
| Group | Type of bolding |
|---|---|
| A | Residential |
| B | Educational |
| C | Institutional |
| D | Assembly |
| E | Business |
| F | Merchant (will include both retail and wholesale stores) |
| G | Industrial include low, moderate and high fire hazards |
| H | Storage |
| I | Hazardous |

**6. The critical path for the given network diagram is _____**



(a) A-B-C-H (b) A-D-F-H
(c) A-D-F-G-I (d) A-D-E-I

**Ans. (b) :**




Time elapsed in path ABCH = 3+2+7+6 = 18 days
Path ADFH = 3+5+5+6 = 19 days
Path ADFGI = 3+5+5+2+3 = 18 days
Path ADEL = 3 + 5 + 4 + 3 = 15 days
Since path ADFH is timewise longest path hence it is critical path.

**7. As per IS 456-2000, what is the value of slump for degree of workability 'High'? Consider the placing condition as Tremic concrete.**
(a) 25-50 mm (b) 50-100 mm
(c) 10-150 mm (d) 100-150 mm

**Ans. (d) :**

| Placing condition | Degree of workability | Slump (mm) |
|---|---|---|
| Shallow section; pavement using power | Very low | 0-25 |
| Lightly reinforced section in slabs, beams, wall; canal lining | low | 25-75 |
| Heavy reinforced section in slab, beam, wall, column, slab form work | medium | 75-100 |
| Trench fill | high | 100-150 |
| Tremic concrete | very high | > 150 |

**8. D-cracking on concrete slabs is caused due to _____.**
(a) thermal expansion
(b) surface flow
(c) freezing and thawing
(d) crystallisation of salts in pores

**Ans. (c) :** Durability cracking (D-cracking) is caused by the concrete's inability to withstand environmental factors such as freeze thaw cycles.
- It usually appears as a pattern of cracks running parallel to a joint as linear crack.
- This type of cracking may eventually lead to disintegration of the concrete within 0.3 to 0.6 m of the joint or crack.

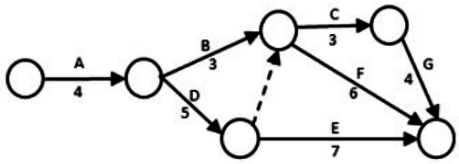
**9. Pre-planning phase of construction project does NOT involve _____**
(a) formulation of general frame work
(b) cost-benefit analysis
(c) investment alternatives
(d) placing and laying reinforcement

**Ans. (d) : Preplanning of construction projects-**
Pre planning of any construction project includes–
- To Defining the objective and scope of project as well as activities.
- Preparation of schedule.
- Cost estimation and comparison of alternatives.
- Defining the schedule of audit and inspection during construction.

**10. The function of management that involves proper communication, leadership and motivation to accomplish the set objective is __**
(a) coordination (b) controlling
(c) directing (d) organising

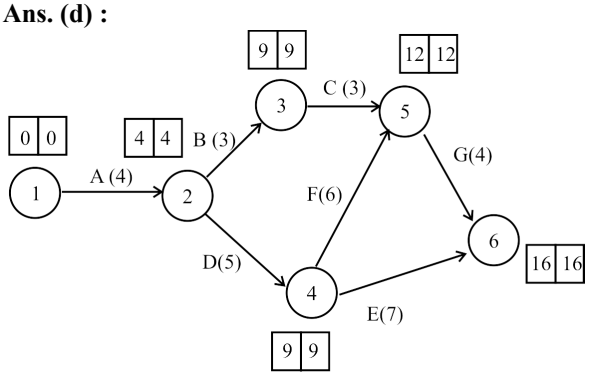
**Ans. (c) : Project co-ordination–**
- It is day to day management of activities within the department.
- Purpose of co-ordination is to streamline the workflow of activities so that it can be completed efficiently, economically and on time.
- It in valves communication between teams involved, leadership and control over teams.

**11. From the given network diagram, the total float for activity C and activity E is _____**




(a) 1 and 0, respectively
(b) 0 and 1, respectively
(c) 1 and 1, respectively
(d) 0 and 0, respectively

**Ans. (d) :**




Total float for C = LST–EST = 9–9 = 0
Total float for E = LST–EST = 9–9 = 0

**12. Which of the following property of water is determined by EDTA (ethylene-dia-minetetra acetate) titration?**
(a) hardness in water
(b) turbidity in water
(c) dissolved oxygen in water
(d) residual chlorine in water

**Ans. (a) :** EDTA (Ethylene Diamine Tetra Acetate) is used to determine the hardness of water. In this EBT (Eriochrome black T) indicate is used.

| Water quality parameter | Instrument used |
|---|---|
| Taste & odour | Osmoscope |
| Colour | Tintometer |
| Turbidity | • Turbidity rod<br>• Jackson turbidity meter<br>• Baylis turbidity meter<br>• Nephlometer |
| pH | Aquascope potentiometer |
| Oxygen | Winkler's method |

**13. The cleaning of slow sand filter is done by :**

(a) scrapping (b) back washing
(c) air cleaning (d) mechanical staining

**Ans. (a) :** The cleaning of slow sand filters is done by scrapping off top layers of sand and admitting water.

Slow, sand, filter–

Effective size of sand particle is taken to be 0.2 to 0.3 mm.

Coefficient of uniformity $\left(C_u = \frac{D_{60}}{D_{10}}\right)$ for filter medium should be 5.

- Design life of filter = 10 to 15 year.
- Frequency of cleaning is 1-3 months.

**14. When of the following comparison is correct with respect to rate of filtration in water purification? Where SSF = slow and filter, RSF = rapid sand filter, DMF = dual media filter and MMF = mixed media filter :**

(a) MMF $\geq$ RSF > DMF > SSF
(b) MMF $\geq$ DMF > RSF > SSF
(c) DMF < MMF $\geq$ RSF < SSF
(d) DMF $\geq$ MMF < RSF > SSF

**Ans. (b) : Slow sand filter–**

- Efficiency of bacteria removal = 98-99%
- Rate of filtration = 100–200 $\ell/m^2/hr$

**Rapid sand filters–**

- Depth of tank 7.5 to 3.5 m
- Area of tank 10 to 100 $m^2$ per unit.
- Rate of filtration = 3000 to 6000 $\ell/m^2/hr$
- Efficiency of filter = 90%

**Dual media filters–**

- It consist of fine sand layer tapped with a bed of attractive coal each generally 30 cm deep.
- Rate of filtration = 17000 to 15000 $\ell/m^2/hr$

**Mixed media filters–**

- It is a 3-layer filter of three property graded material; coal, sand and garnet in the proposition of 60, 30 and 10% respectively.
- Filtration rate – 15,000 $\ell/m^2/hr$

**15. In a water treatment process by sedimentation the flocculent particles follows _____**

(a) Type = I settling (b) Type = II settling
(c) Type = III settling (d) Type = IV settling

**Ans. (b) :** In a water treatment process by sedimentation the flocculent particles follow types II setting.

**Types of setting–**

**Discrete setting–** It assumes that particles settle as individual entities and their concentration in suspension.

**Flocculent setting–** It assumes that particles in liquid coalesce and flocculation occurs during setting which increase the mass and setting rate of particles.

**Hindered or zone setting–** It assumes that the concentration of particles in liquid is high enough to create sufficient inert particle force to hinder the setting of neighbouring particle.

**Compression setting–** It assumes that settled particle are further compressed due to setting particles at the top layer.

**16. Which of the following statements about the tube settler is incorrect?**

(a) It is the combination of multiple channels at an angle of $60^0$.
(b) Settling depth is less than conventional one
(c) Settling time is less than conventional one
(d) Effective settling area is decreased by providing multiple tubular channels sloped at an angle of $60^0$ and adjacent in each other

**Ans. (d) : Tube settlers–** Tube settlers use 25-50 mm diameter tubes having large wetted perimeter than wetted area so that flow is laminar.

- As per Hazen is theory the efficiency of sedimentation tank depends on surface area of basin.
- Tube settlers increase the surface area of basin and reduce the detention time to few minutes in comparison to 1 to 4 hours in case of convention at settling basins.
- Tube settlers are of two types.
  - $\rightarrow$ Horizontal tube settler.
  - $\rightarrow$ Steeply inclined settler ($\theta = 60^0$)
- Tube settlers reduces the settling depth.

**17. The horizontal tunnels constructed at shallow depths along the banks of a river to intercept ground water tables are called _____**

(a) canals (b) infiltration galleries
(c) springs (d) lakes

**Ans. (b) :** The horizontal tunnels constructed at shallow depths along the banks of a river to intercept ground water table are called infiltration galleries.

**Canal–** It is an artificial channel constructed on the ground to carry water to the field either from a reservation, tank or river

**Lake–** Lake is a large water body of stagnant water, within an area.

**Spring–** It is a place where water naturally flows out of the ground.

**18. Copper sulphate is most commonly used (added) to control_____ in lakes.**

(a) chlorine demand of water
(b) growth of algae
(c) dispersion of silt
(d) suspended particle

**Ans. (b):** Copper sulphate is most commonly used to control growth of algae in lakes.

- Copper sulphate is applied in a single dose usually less than 3 mg/$\ell$.
- It is applied by dragging sacks of crystals through water with the help of boats.
- It is also effective in killing many type of aquatic weeds.

**19. As per IS 10500 : 2012, the maximum permissible limit for fluoride content present in drinking water shall not exceed _____ in the absence of alternate source.**

(a) 0.015 ppm (b) 1.5 ppm
(c) 15 ppm (d) 150 ppm

**Ans. (b) :** As per IS 10500 : 2012, the maximum permissible limit for fluoride content present in drinking water shall not exceed 1.5 ppm in the absence of alternate source.

| Characteristics | Requirement (Acceptable limit) | permissible limit in the absence of alternate sources |
|---|---|---|
| Total dissolve solid, mg/l | 500 | 2000 |
| Turbidity, NTU | 1 | 5 |
| Colour, Hazen unit | 5 | 15 |
| pH | 6.5-8.5 | Not relaxation |
| Chloride | 250 | 1000 |
| Total hardness as $CaCO_3$ mg/$\ell$ | 200 | 600 |

**20. B-coli or E-coli are harmless organisms, but their presence in water indicates the :**

(a) presence of pathogenic bacteria
(b) absence of pathogenic bacteria
(c) presence of non-pathogenic bacteria
(d) absence of non-pathogenic bacteria

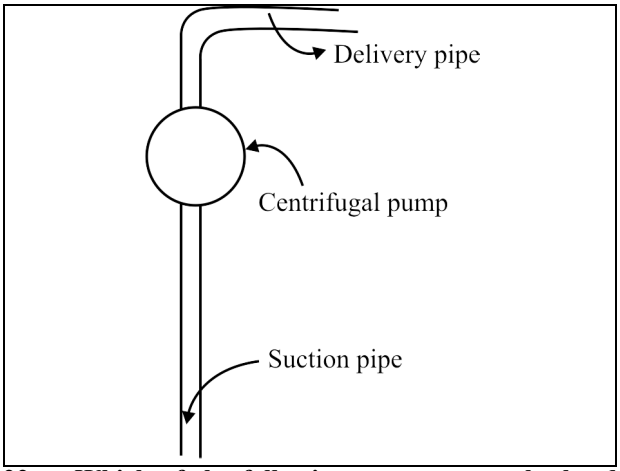
**Ans. (a) :** B-coli or E-coli are harmless organisms, but their presence in water indicates the presence of pathogenic bacteria.

- E-coli test is conducted to detect and count the presence of coliform.
- Bacteria coliform (B-coli) or escherichia coli (E-coli) are important harmless aerobic micro-organism.
- These bacteria live longer in water than pathogenic bacteria.
- Bacterial examination of water for E-coli is performed because they are used as indicator organisms for probable presence of pathogens.

**21. When centrifugal pump is used to lift water the direction of flowing water is changed by_____**

(a) $45^0$ (b) $60^0$
(c) $90^0$ (d) $180^0$

**Ans. (c) :** In centrifugal pump, direction of water is changed by $90^0$.




**22. Which of the following can measure the head low?**

(a) Weir (b) Notch
(c) Scrubber (d) Piezometer

**Ans. (d) :** Pizometer can measure the head loss.

**23. A surge tank is a storage reservoir at the_____ end of a close aqueduct to absorb a sudden_____ of pressure.**

(a) downstream, rise
(b) upstream, rise
(c) upstream, fall
(d) downstream, fall

**Ans. (a) :** A surge tank is a storage reservoir a the downstream end of a close aqueduct to absorb a sudden rise of pressure.

- Surge tank mitigates pressure variations due to rapid changes in the velocity of the water and protects the pipeline against water hammers.

**24. The valves which are used to remove the sediment in the pipelines are called :**

(a) butterfly valves (b) scour valves
(c) air valves (d) check valves

**Ans. (b) :** The valves which are used to remove the sediment in the pipelines are called scour valve. They are either located at the dead end or at the lowest point in the mains.

**Check valves–** It is also known as reflux valve or non-return valves. It is automatic device that allows water to flow in one direction only and avoids the back flow of water.

**25. To control the water hammer which of the following valves will be used?**

(a) Air valves (b) Pilot valves
(c) Scour valves (d) Blow off valves

**Ans. (a) :** Air relief valve is used to control the water hammer.

**Air relief valve–** The air relief valves are used to admit air to the pipe line in order to prevent the formation of vacuum in case of rapid flow negative pressure. This vacuum or negative pressure (known as water hammer) may hurt the pipe line at some lower point.

**26. In a water distribution system circular system, the main pipe line will be _____**

(a) in the centre of the city
(b) at any one side of city
(c) on the periphery of the city
(d) anywhere in the city

**Ans. (c) :** In a water distribution system circular system's the main pipe line will be on the periphery of the city.

- This system is very suitable for towns and cities having well-planned roads.
- It is economical and the pressure at the ends are reasonably equal.

**27. The rugosity coefficient value used in the design of sewer lines depends on _____**

(a) Longitudinal gradient
(b) hydraulic radius of pipe
(c) material used in sewer pipe
(d) cross sectional area of sewer pipe

**Ans. (c) :** The rugosity coefficient value used in the design of sewer lines depends on material used in sewer pipe.

- The manning's roughness coefficient is used in the manning's equation to calculate flow in open channel.

Manning's formula

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Where,
R = Hydraulic radius
S = Side slope
n = Rugosity coefficient

**28. Which of the following formula is used to find Chezy's constant?**

(a) Ryve's equation (b) Hazen's equation
(c) Bazin's equation (d) Continuity equation

**Ans. (c) :** Bazin's equation is used to find Chezy's constant.

Bazin formula, $C = \frac{157.6}{1.81 + \frac{k}{\sqrt{m}}}$

where,
k = Bazin's constant
m = hydraulic mean depth

Some other formula to chezy's constant are–

- Gangvillet-kutter formula
- Manning's formula

**29. According to general standards for discharge of environmental pollutants part - A : effluence, the maximum chemical oxygen demand for effluence from inland surface water shall be ____**

(a) 250 mg/*l* (b) 300 mg/*l*
(c) 350 mg/*l* (d) 500 mg/*l*

**Ans. (a) :** According to general standards for discharge of environmental pollutant part-A : effluents, the maximum chemical oxygen demand for effluent from inland surface water shall be 250 mg/*l*.

**General standards for discharge of environmental pollutants part : A : Effluents**

| Parameters | Standard | | | |
|---|---|---|---|---|
| | Inland surface water | Public sewers | Land for irrigation | Marine coastal area |
| Suspended solids mg/ℓ, max | 100 | 600 | 200 | - |
| pH value | 5.5 to 9 | 5.5 to 9 | 5.5 to 9 | 5.5 to 9 |
| Total residual chlorine mg/ ℓ | 1 | - | - | 1 |
| Biochemical oxygen demand [3 day at $27^0$C] mg/ℓ , max | 30 | 350 | 100 | 100 |
| Free ammonia | 5 | - | - | 5 |
| Total kjeldanl nitrogen | 100 | - | - | 100 |

**30. Sludge thickness will reduce the volume of sludge by _____ conditioning.**

(a) physical (b) chemical
(c) biological (d) physico-chemical

**Ans. (a) :** Sludge thickens will reduce the volume of sludge by physical conditioning.

- Sludge thickening is a process in which the concentration of solids in increased and the total sludge volume is decreased, but the sludge still behaves like a liquid instead of a solid.

**31. Fresh sludge has moisture content of 99%, and after thickening, its moisture content is reduced to 96%. The reduction in the volume of sludge is:**

(a) 197% (b) 5%
(c) 75% (d) 3%

**Ans. (c) :** The volume of the digested sludge at given moisture content is given by the formula as :

$$\frac{v_2}{v_1} = \left(\frac{100 - P_1}{100 - P_2}\right) \qquad \text{Given, } P_1 = 99\%, P_2 = 96\%$$

$$\frac{V_2}{V_1} = \frac{100 - 99}{100 - 96} = \frac{1}{4} = 0.25$$

$$\% \text{ decrease in volume} = \left(1 - \frac{v_2}{v_1}\right) \times 100$$

$$= (1 - 0.25) \times 100 = 75\%$$

**32. The two main gases liberated from an anaerobic sludge digestion tank would include:**

(a) $NH_3$ and $CH_4$ (b) $CO_2$ and $CH_4$
(c) $CH_4$ and Ne (d) $NH_3$ and $CO_2$

**Ans. (b):** The two main gases liberated from an anaerobic sludge digestion tank would include $CO_2$ and $CH_4$.

- Anaerobic digestion is the degradation of organic material by microbial activity in the absence of air transforming activity in the absence of air transforming it into biomass and biogas, a mixture of methane ($CH_4$), carbon dioxide ($CO_2$), and some truss gases.

Methane ($CH_4$) – 50 to 75%
Carbon dioxide ($CO_2$) – 25 to 50%
Hydrogen ($H_2$) – 5 to 10%

**33. On what media does the sludge drying bed work?**
(a) Sand
(b) polypropylene membrane
(c) Geotextile cloth
(d) plastic

**Ans. (a) :** Sludge drying bed work on the sand.

- Drying beds have been used since the beginning of the twentieth century. Drying of the sludge can be divided into two different sludges, namely drainage and evaporation.
- Drying is the process which can remove high amount of water from sludge.
- While thickening and dewatering can remove 7% and 35% of the total amount of water, respectively.

**34. When carbon monoxide reacts with hemoglobin, it forms_____**
(a) carboxy-haemoglobin
(b) carbonmin- haemoglobin
(c) amino acid
(d) carbon di oxide and hydrogen

**Ans. (a) :** When carbon monoxide reacts with hemoglobin, it forms carboxy-haemoglobin.

- Carboxy-haemoglobin is a stable complex of carbon monoxide and hemoglobin that forms in red blood cells upon contact with carbon monoxide.

**35. Collection efficiency of an electrostatic precipitator ranges between _____ for particles size between 0.1 μm to 0.2 μm.**
(a) 99.8% to 100% (b) 96% to 97%
(c) 86% to 8% (d) 83% to 85%

**Ans. (b) :** Electrostatic precipitators are used in thermal power plants, pulp and paper industry.

- It removes particles of size > 1 μ
- Its efficiency is 95-99%

**36. The collection efficiency of an electrostatic precipitator under ideal conditions is generally estimated by using_____**
(a) Vander Waals equation
(b) Dobson equation
(c) Deutsch-Anderson equation
(d) Schmutzdecke equation

**Ans. (c) :** The collection efficiency of an electrostatic precipitator under ideal conditions is generally estimated by using Deutsch-Anderson equation.

- As per Deutsch-Anderson equation, to determine the collection efficiency of the precipitates under ideal conditions.

$$\eta = 1 - e^{-w(A/Q)}$$

Where,
$\eta$ = collection efficiency of precipitator
w = migration velocity in cm/s
A = The effective collecting plate area of the precipitation in $m^2$.
Q = Gas flow through the precipitation in $m^3/s$.

**37. A venturi scrubber used to control air pollution is an example of _____**
(a) dry scrubbers (b) wet scrubbers
(c) gravity scrubbers (d) pressure scrubbers

**Ans. (b) :** A venturi scrubber used to control air pollution is an example of wet scrubbers.

- A venturi scrubber is a high-energy, high-efficiency wet scrubber for removing particulate matter from gas.
- A venturi includes a converging inlet section where the gas is accelerated to throat velocity, a throat, and a diverging outlet section where velocity pressure is recovered. .
- The venturi cross-section may be circular or rectangular.

**38. Which of the following wastes is not generated from domestic processes?**
(a) Coconut shell (b) Food waste
(c) Garden trimming (d) scrap lumber

**Ans. (d):**Scrap lumber is not generated from domestic process.

- Domestic waste is one of the most important components of municipal waste.

Domestic wastes include food waste, paper etc.

**39. When a retaining wall moves towards the backfill, the pressure exerted on the wall is termed as _____**
(a) passive earth pressure
(b) swelling pressure
(c) pore pressure
(d) active earth pressure

**Ans. (a) :** When a retaining wall moves towards the backfill, the pressure exerted on the wall is termed as passive earth pressure.

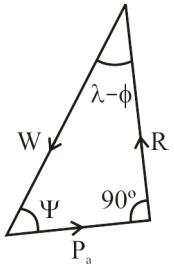
**Active earth pressure–** The lateral pressure exerted by the soil when the retaining wall tends to move away from the back fill due to excessive pressure of the retained soil is called active earth pressure.

| **Active Earth Pressure** | **Passive Earth Pressure** |
|---|---|
| Length of failure block $H\cot\left(45^0+\frac{\phi}{2}\right)$ | Length of failure block $H\cot\left(45^0-\frac{\phi}{2}\right)$ |
| $k_a=\frac{1-\sin\phi}{1+\sin\phi};$ $k_a=\tan^2\left(45-\frac{\phi}{2}\right)$ | $k_p=\frac{1+\sin\phi}{1-\sin\phi};$ $k_p=\tan^2\left(45+\frac{\phi}{2}\right)$ |
| $k_a$ = coefficient of active earth pressure | $k_p$ = coefficient of passive earth pressure |

**40. Which of the following expression is correct according to Coulomb's graphical method to find the angle (Ψ) between earth pressure line and the slope line? Where, δ is wall friction angle, and θ is inclination of back of wall to the vertical.**

(a) $\Psi = 90^0 + \theta + \delta$ (b) $\Psi = 90^0 + \theta - \delta$
(c) $\Psi = 90^0 - \theta + \delta$ (d) $\Psi = 90^0 - \theta - \delta$

**Ans. (d) :** According to Coulomb's graphical method to find the angle (Ψ) between earth pressure line and the slope line? Where, δ is wall friction angle, and θ is inclination of back of wall to the vertical.




$\Psi = 90^0 - \theta - \delta$

**41. The load carrying capacity of an individual friction pile is 200 kN. What is the total load carrying of a group of nine such piles with a group efficiency factor of 0.8?**

(a) 1800 kN (b) 1640 kN
(c) 1440 kN (d) 900 kN

**Ans. (c) :** Given, Number of piles = 9
Group efficiency factor = 0.8
Individual pile capacity = 200 kN
**Load carrying capacity of pile group-**
Group efficiency factor × Individual pile capacity × Number of piles
= 0.8 × 200 × 9 = 1440 kN

**42. In the static cone penetration method, for calculation of settlement in cohesionless soils, _**

(a) the sand layer is divided into small layers such that each small layer has approximately constant value of the cone resistance
(b) the sand layer is divided into small layers such that each small layer has different value of the cone resistance
(c) the sand layer considered to be a single layer so that there is zero value of the cone resistance
(d) the sand layer is divided into small layers of equal size such that each small layer undergoes same amount of settlement

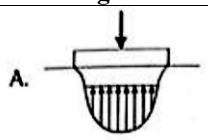
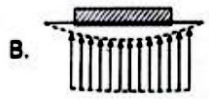
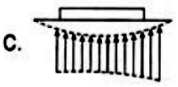
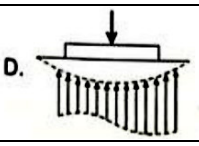
**Ans. (a) : Static cone penetration test–**
- useful on very soft and loose soil where transportation of heavy equipments required for S.P.T and DCPT test may not be possible.
- useful for determination of bearing capacity at different depths below foundation level.
- skin friction values required to be used for determining the length of piles.
- useful to determine water content in soil.

**43. The upward pressure developed by the soil on the bottom face of a footing is termed as ____**

(a) surface tension
(b) confining pressure
(c) contact pressure
(d) deviatoric pressure

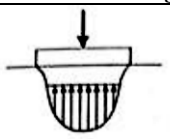
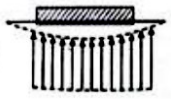
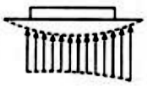
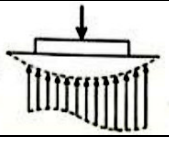
**Ans. (c) :** The upward pressure developed by the soil on the bottom face of a footing is termed as contact pressure.

**44. Match the columns**

| Contact pressure distribution diagrams | Description of footings |
|---|---|
| A.  | **1. Rigid footing on cohesive soil** |
| B.  | **2. Flexible footing on cohesive soil** |
| C.  | **3. Rigid footing on cohesion-less soil at ground level** |
| D.  | **4. Flexible footing on cohesionless** |

(a) A-3, B-1, C-4, D-2
(b) A-4, B-3, C-2, D-1
(c) A-3, B-2, C-4, D-1
(d) A-4, B-1, C-3, D-2

**Ans. (c) :**

| Contact pressure distribution diagrams | Description of footings |
|---|---|
| A.  | Rigid footing on cohesion-less soil at ground level |
| B.  | Flexible footing on cohesive soil |
| C.  | Flexible footing on cohesionless |
| D.  | Rigid footing on cohesive soil |

**45. In cohesive frictional soils, to obtain safe load by Static Pile load formula, a factor of safety of ___ in compression and _____ in uplift are recommended respectively.**

(a) 1.5 ; 3 (b) 3 ; 4.5
(c) 2.5 ; 3.0 (d) 3 ; 6

**Ans. (c):** As per IS : 2911-1-2-2010
FOS in compression = 2.5
FOS in uplift capacity = 3.0

**46. Match the following:**

| | |
|---|---|
| **A. Topographical survey** | **1. To survey for the features such as roads, railways, canals, towns, villages etc.** |
| **B. Engineering survey** | **2. For laying out plots and construction streets, water supply systems and sewers** |
| **C. Cadastral survey** | **3. To determine quantities and for collection of data for roads, railways, reservoirs, sewerage, water supply schemes etc,** |
| **D. City survey** | **4. To locate the boundaries of fields, houses, etc.** |

(a) A-1; B-3; C-4; D-2
(b) A-1; B-2; C-3; D-4
(c) A-4; B-3; C-2; D-1
(d) A-1; B-4; C-3; D-2

**Ans. (a) : Based on place of survey–**
**Topographical survey–** This is a survey conducted to obtain data to make a map and to locate the natural and artificial feature of the earth, e.g. river, woods, hills etc.
**Cadastral survey–** This is extensive urban and rural surveys made to plot the details such as boundaries of fields, houses and property lines.
**City survey–** A extensive survey of the area in and around a city for fixing reference monuments, locating and improving property lines, and determining the configuration and features of the land, is referred to as city survey.

**47. Which of the following is NOT an advantage of chain surveying?**
(a) It does not require costly equipment.
(b) It is suitable for small areas with a fair degree of accuracy
(c) Computation and plotting are simple
(d) It is suitable for large and hilly areas.

**Ans. (d) :** Chain surveying is not suitable for large and hilly area it is an disadvantage of chain survey.

**48. Which of the following is an advantage of chain surveying?**
(a) It is suitable for large areas because chaining for large areas is simple.
(b) It is suitable for undulated areas.
(c) It is suitable for small areas with a fair degree of accuracy.
(d) Results are always accurate

**Ans. (c) :** Advantage of chain surveying is that it is suitable for small areas with a fair degree of accuracy.

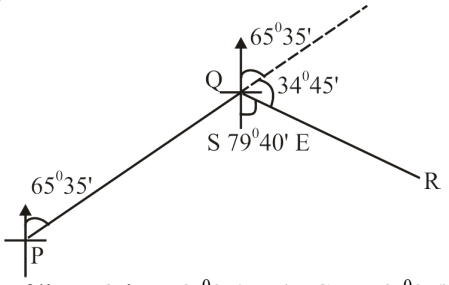
**Advantage of chain surveying–**
- It does not require any costly equipment and tools.
- Computations and plotting are also simple.

**49. The principle of compass surveying is ____**
(a) direct and indirect ranging
(b) triangulation
(c) traversing
(d) chaining

**Ans. (c) :** Principle of compass surveying is traversing.
- Principle of chain surveying is triangular.
- principle of plane table surveying is parallelism.

**50. If the bearing of line PQ is N $65^0$35'00"E and the deflection angle (clockwise) between the lines PQ and QR is $34^0$45'00", then the bearing of line QR is**
(a) N$30^0$ 50'E (b) N$34^0$ 45'E
(c) S$79^0$ 40'E (d) N$10^0$ 20'E

**Ans. (c) :**




Bearing of line PQ is N $65^0$35' E (WCB = $65^0$35')
Deflection between PQ and QR $34^0$45'
So, bearing of line QR in WCB system
QR = $65^0$35' + $34^0$45' = $100^0$20'
Bearing of line QR in RB system
= 180 – $100^0$20' = S $79^0$40' E

**51. Identify the incorrect statement about magnetic dip.**
(a) The horizontal angle made by the magnitude needle with vertical is known as dip of the needle.
(b) Angle of dip varies from $0^0$ at the equator to $90^0$ at the magnetic poles
(c) The value of dip varies from place to place
(d) Dip angle increase as we go from the equal to the poles

**Ans. (a) : Magnetic Dip–** It is the inclination of magnetic needle of compass with horizontal.
- It is 0 at equator and $90^0$ at poles.
- It is not constant.

**52. The variation in which magnetic meridian swings like a pendulum is known as _____**
(a) irregular variation (b) annual variation
(c) secular variation (d) diurnal variation

**Ans. (c) :** The variation in which magnetic meridian swings like a pendulum is known as secular variation.
It swings in one direction for about 100-150 years.
**Irregular variation–** It is caused due to magnetic disturbance or storms are listed under irregular variation.

**53. Select the option that is correct regarding the following two statements.**

**Statement 1 : Local attraction at any station is detected by observing the fore bearing and the back bearing of the line.**

**Statement 2 : If the difference between FB and BB is $180^0$, both the end stations are considered to be free from local attraction.**

(a) Both statement I and statement 2 are individually true and statement 2 is the correct explanation of statement I.

(b) Both statement 1 and statement 2 are individually true, but statement 2 is not the correct explanation of statements 1.

(c) Statement 1 is true, but statement 2 is false

(d) Statement 1 is false, but statement 2 is true

**Ans. (a) :** Local attraction by the magnetic materials is the disturbing influence on the magnetic needle of the compass. The amount of deviation of the magnetic needle is the measure of local attraction.

**Detection of local attraction–** The local attraction at any station is defined by observing the fore and back bearings of the line.

- If the difference between them is $180^0$, both the end stations are considered to be free from local attraction.

**54. Given are the bearings of the lines of an open traverse. Correct the bearings by finding local attraction at the affected stations.**

| | |
|---|---|
| **PQ: $36^0$00'** | **QP : $216^0$45'** |
| **QR: $98^0$15'** | **RQ : $276^0$00'** |
| **RS : $201^0$45'** | **SR : $23^0$15'** |
| **SP: $322^0$45'** | **PS : $142^0$45'** |

(a) QP : $215^0$00' : QR : $87^0$30' : RQ : $276^0$30' : RS : $201^0$15'

(b) QP : $216^0$10' : QR : $93^0$50' : RQ : $287^0$10' : RS : $208^0$45'

(c) QP : $210^0$30' : QR : $76^0$50' : RQ : $217^0$20' : RS : $210^0$15'

(d) QP : $216^0$00' : QR : $97^0$30' : RQ : $377^0$30' : RS : $203^0$15'

**Ans. (d) :**

| Line | F.B. | B.B. | Difference | Corrected F.B. | Corrected B.B. | Remark |
|---|---|---|---|---|---|---|
| PQ | $36^0$ | $216^0$45' | $180^0$45' | $36^0$ | $216^0$ | LA station 'Q' |
| QR | $98^0$15' | $276^0$ | $178^0$15' | $97^0$30' | $277^0$30' | LA at station 'R' |
| RS | $201^0$45' | $23^0$15' | $178^0$30' | $203^0$15' | $23^0$15' | |
| SP | $322^0$45' | $142^0$45' | $180^0$ | $322^0$45' | $142^0$45' | |

**55. The tangential method of tacheometry uses :**

(a) the readings against the top and bottom cross-hairs only

(b) the readings against the middle cross-hair only

(c) a constant intercept on the staff

(d) the readings against all the three cross-hairs

**Ans. (b) : Tangential method of tacheometry –**

This method is used when the staff is hold too far from the instrument station and it is difficult to read the staff.

- Here the staff is fitted with two target space as a fixed vertical distance (s) copy (say 2 m or 3m)
- By sighting the two targets, the vertical angle $Q_1$ and $Q_2$ are measured.

**56. The line joining points of equal depths of evapotranspiration is _____**

(a) isohyets (b) isopluvial

(c) isopleth (d) isothermal

**Ans. (c) :** The line joining points of equal depths of evatranspiration is isopleth.

- Isohytes line joining equal depth of rainfall.
- Isothermal line joining equal point of temperature.
- Isohaline line joining equal point of salinity.

**57. The daily flows of a river have 1050 data points, and the number of days the flow being equal to or greater than its class value is 205. Find the percentage probability of the flow magnitude being equaled or exceeded.**

(a) 0.15 (b) 0.195

(c) 0.255 (d) 0.295

**Ans. (b) :** Given,

Number of data points = 1050

Number data points exceeding design flood (class)= 205

Hence probability of flood (exceed) = $\frac{205}{1050} = 0.195$

**58. Which of the following is the Aridity index (AI)?**

**PET = Potential evapotranspiration**

**AET = Actual evapotranspiration**

(a) $\frac{AET - PET}{AET} \times 100$ (b) $\frac{PET - AET}{AET} \times 100$

(c) $\frac{AET - PET}{PET} \times 100$ (d) $\frac{PET - AET}{PET} \times 100$

**Ans. (d) : Aridity index–** It is a numerical indication which is used for measuring the degree of dryness of climate of a place.

- It is opposite to humidity index.

$$\text{Aridity index (A.I.)} = \frac{PET - AET}{PET} \times 100$$

Where,

PET = Potential evapotranspiration

AET = Actual evapotranspiration

**59. The daily discharged of the stream is given in the table. Find the percentage probability of the flow for the discharge 18 $m^3/s$.**

| **Class interval** | **5** | **6** | **10** | **2** | **3** |
|---|---|---|---|---|---|
| **Discharge $m^3/s$** | **15** | **25** | **32** | **18** | **12** |

(a) 30.33% (b) 22.33%

(c) 29.49% (d) 33.33%

**Ans. (d):** Given,
N = 5
M = 2

$$P_P = \frac{M}{N+1} \times 100\%$$

$$= \frac{2}{5+1} \times 100\% = 33.33\ \%$$

**60. The precipitation falling directly over the streams is called :**
(a) channel precipitation (b) through fall
(c) interception (d) run-off

**Ans. (a) :** The precipitation falling directly over the stream is called channel precipitation.
**Run-off →** It is the quantity of water that flows through the subsurface of the earth to any specific area or reservation.

**61. In the chart of weighing bucket rain gauge, the pen had reversed its path of travel at 4 hours of the next day after recording 100 mm of rainfall and the rainfall recorded in the next day at next two hours is 50 mm, then the cumulative rainfall recorded by gauge at say 6 hours of the next day is**
(a) 150 mm (b) 50 mm
(c) 200 mm (d) 100 mm

**Ans. (a) :** $t_1$ = 4 hours
$t_2$ = 2 hours
$p_1$ = 100 mm
$p_1$ = 50 mm
Cumulative rainfall = $p_1 + p_2$ = 100+ 50 = 150 mm

**62. Rate of transpiration depends upon the :**
(a) base period of the plant
(b) harvesting period of the plant
(c) atmospheric humidity and wind
(d) sowing period of the plant

**Ans. (c) :** Rate of transpiration in plant is depend upon humidity, light, temperature, wind etc.

**63. Match the characteristics of catchment :**

| | | | |
|---|---|---|---|
| **1.** | **Stream density** | **A** | **Elevation difference between the catchment outlet and the highest point on the basin perimeter** |
| **2.** | **Drainage density** | **B** | **Ratio of number of streams of all orders to the area of basin** |
| **3.** | **Relief** | **C** | **Ratio of total length of streams of all orders within the basin to the area of basin** |
| | | **D** | **Reflecting the pattern of branches that unit to form the trunk stream** |

(a) 1-A, 2-B, 3-C (b) 1-A, 2-D, 3-C
(c) 1-B, 2-D, 3-A (d) 1-B, 2-C, 3-A

**Ans. (d) : Stream density–** It is the ratio of number of streams of all orders to the area of basin.
**Drainage density–** It is the measure of the length of stream channel per unit area of drainage basin.
**Relief–** It is the elevation difference between the catchment outlet and the highest point on the basin perimeter.

**64. Two channels of order 2 and order 3 joined together produces an order of which of the following streams?**
(a) 3rd order (b) 2nd order
(c) 1st order (d) 5th order

**Ans. (a) : Stream order–** It is used to describe the hierarchy of streams from the top to the bottom of a catchment.
- When two streams with different order join, the resulting stream has the same order as the highest order of the two joining streams e.g. 2nd order + 3rd order stream join, the resulting stream is 3rd order.
- When two stream with same order join, the resulting stream has, the next highest order than the joining stream.

**65. What is the average value of duty for crop rice in India?**
(a) 775 hectares/cumec
(b) 1000 hectares/cumec
(c) 1500 hectares/cumec
(d) 1275 hectares/cumec

**Ans. (a) : Duty–** It is defined as the number of hectare of land that should be irrigated for full growth of crops by supplying cumec of water continuously to the base period of that crop.
**Average value of Duty is India–**

| Crop | Duty (hectare/cumec) |
|---|---|
| Rice | 775 |
| Sugarcane | 730 |
| Rabi | 1800 |
| Kharif | 1500 |

**66. A canal commands an irrigated area of 500 hectares. The intensity of irrigation of paddy in this area is 50%. The transplantation of paddy requires 30 days and total depth of water required by paddy is 50 cm on the field during transplantation. The available water during transplantation on the field is 15 cm. Calculate the duty of water, on the field for the given details.**
(a) 730.5 hectares/cumec
(b) 750 hectares/cumec
(c) 740.5 hectares/cumec
(d) 745.5 hectares/cumec

**Ans. (c) :** Relationship between duty, base period and delta is $\Delta = \frac{8.64B}{D}$
Where, $\Delta$ = delta in (m)
B = Base period in days
D = Duty in (hectaure/cumec)

Given,

$\Delta = 50 - 15 = 35 \text{ cm} = 0.35 \text{ m}$

B = 30 days

$D = \frac{8.64 \times B}{\Delta} \Rightarrow D = \frac{8.64 \times 30}{0.35}$

= 740.57 hectare/cumec

**67. Which of the following soil group has excellent workability as construction material?**

(a) SC (b) GM

(c) SM (d) GW

**Ans. (d) :** In the following well graded gravel (GW) has high workability.

- Aggregate size, well graded, texture, water-cement ratio is directly proportioned to workability.

**68. A soil having specific gravity 'G' equal to 2.66 has a theoretical maximum dry density of 1.865 g/cm$^3$ which corresponds to zero air voids at the OMC. Calculate the corresponding dry unit weight, if the density of water (ρw) = 1 g/cm$^3$.**

(a) 18.29 kN/m$^3$ (b) 21.56 kN/m$^3$

(c) 22.78 kN/m$^3$ (d) 24.79 kN/m$^3$

**Ans. (b) :** Given,

Specific gravity, G = 2.66

$\gamma_{d\max} = 1.865 \text{ g/cm}^3$

$\gamma_b = ?$

$\gamma_w = 1 \text{ g/cm}^3$

$\gamma_d = \frac{G\gamma_w}{1+e}$

$1.865 = \frac{2.66 \times 1}{1+e}$

e= 0.4262

Now put the value of void ratio (e) formula

$S \times e = w \times G$

$\therefore S = 1$

$w = \frac{e}{G} = \frac{0.4262}{2.66}$

Water content (w) = 0.1602

Now,

$\frac{\gamma_b}{\gamma_d} = (1+w)$

$\gamma_b = \gamma_d (1+w)$

$\gamma_b = 1.865(1+0.1602)$

$\gamma_b = 2.16 \text{ g/cm}^3$

$= \frac{2.16 \times 9.81 \times 10^{-3}}{(10^{-2})^3} = 21.23 \text{ kN/m}^3$

**69. A granular soil is subjected to major and minor principal stress of 8 MN/m$^2$ and 2 MN/m$^2$, respectively. What will be the radius of Mohr's circle?**

(a) 6 MN/m$^2$ (b) 3 MN/m$^2$

(c) 4 MN/m$^2$ (d) 10 MN/m$^2$

**Ans. (b) :** If $\sigma_x$ and $\sigma_y$ are two principal stress, then radius of mohr's circle = $\frac{\sigma_x - \sigma_y}{2}$

Radius = $\frac{8-2}{2} \text{ MN/m}^2 = 3 \text{ MN/m}^2$

**70. Determine the unconfined compressive strength of normally consolidated deposit of clay, if the effective stress at a depth of 15 m is 180 kN/m$^2$, plasticity index of the clay is 75% and the undrained cohesion value of clay equal to 69.75 kN/m$^2$**

(a) 34.87 kN/m$^2$ (b) 139.5 kN/m$^2$

(c) 69.75 kN/m$^2$ (d) 208.35 kN/m$^2$

**Ans. (b) :** Given,

$\bar{\sigma} = 180 \text{ kPa}$

$I_p = 75\%$

$C_u = 69.75 \text{ kPa}$

$\because q_u = 2C_u = 2 \times 69.75 = 139.75 \text{ kPa}$

**71. A dry sand specimen was tested in a triaxial machine with the cell pressure of 50 kPa. If the deviator stresses of failure was 100 kPa, calculate the major principal stress**

(a) 100 kPa (b) 150 kPa

(c) 50 kPa (d) 500 kPa

**Ans. (b) :** Given,

($\sigma_3$) cell pressure = 50 kPa

($\sigma_d$) deviator stress = 100 kPa

Major principal stress ($\sigma_1$)

= cell pressure ($\sigma_3$) + Deviation

= 50 + 100

$\sigma_1$ = 150 kPa

**72. During a triaxial test, the cell pressure is increased from 0.1 N/mm$^2$ to 0.25 N/mm$^2$ and the pore water pressure from 0.05 N/mm$^2$ to 0.15 N/mm$^2$. Calculate the Skempton's pore pressure parameter B.**

(a) $\frac{1}{3}$ (b) $-\frac{1}{3}$

(c) $-\frac{2}{3}$ (d) $\frac{2}{3}$

**Ans. (d) :** Skempton's pore pressure parameter B

$= \frac{\text{Change in pore pressure}(\Delta\bar{U})}{\text{Change in effective stress}(\Delta\bar{\sigma})}$

$\Delta\bar{U} = 0.15 - 0.05 = 0.10$

$\Delta\bar{\sigma} = 0.25 - 0.10 = 0.15$

$B = \frac{0.10}{0.15} = \frac{2}{3}$

**73. Which of the following expression is correct to calculate permissible stresses used in working stress method of design?**

(a) Permissible stress = Yield stress = Factor of safety

(b) Permissible Stress = Yield Stress / Factor of Safety

(c) Yield Stress = Permissible Stress / Factor of Safety
(d) Permissible Stress = Yield Stress - Factor of Safety

**Ans. (b) :** Permissibility stress = $\frac{\text{Yield stress}}{\text{Factor of safety}}$

Permissible stresses are defined as characteristics strength divided by factor at safety.

**74. Which of the following does not belong to the limit stated of strength?**
(a) Fracture due to fatigue
(b) Brittle fracture
(c) Rupture of the structures
(d) Vibration in structure

**Ans. (d) :** Vibration in structure does not belong to the limit state of strength.
• It belongs to serviceability limit states.

**75. As per IS 800 : 2007, the thickness of the lacing bar for single lacing should not be less than ____ of its effective length.**
(a) $\frac{1}{30}$ (b) $\frac{1}{40}$
(c) $\frac{1}{50}$ (d) $\frac{1}{60}$

**Ans. (b) :** As per IS 800 : 2007, the thickness of the during bar far single lacing should not be less than $\frac{1}{40}$ of its effective length.
• Thickens $\geq \frac{\ell}{60}$ for double lacing riveted or welded at intersection.
• The angle of inclination of the lacing with the longitudinal axis of the column should be between $40^0$ to $70^0$.
• Slenderness ratio of lacing bar should be less than 145.

**76. Which of the following is NOT a type of bolted joint?**
(a) Lap joint
(b) Single cover butt joint
(c) Double cover lap joint
(d) Double cover butt joint

**Ans. (c) :** Double cover lap joint is not a type of bolted joint.
**Type of bolted joint–**
(i) Lap joint
(ii) Butt joint
(iii) Single-cover butt joint
(iv) Double-cover butt joint

**77. In lightly stressed structures, where stiffness, rather than strength, controls the design and fatigue or brittle fracture is not a problem, the _____ welds are entirely adequate and generally more economical.**
(a) Fillet (b) butt
(c) plug (d) slot

**Ans. (a) : Fillet welds–** In lightly stressed structure when stiffness rather than strength control design and fatigue or brittle factor is not a problem, filled weld are adequate of economical.
• Fillet weld are provided when two members are to be joined are in different planes.
**Groove welds–** These are provided when the members to be joined are lined up (in same plane)
• Slot and plug weld are used to supplements the filled welds when required length of fillet weld cannot be provided.

**78. The reduction in the weld strength as per long joint is done when the length of the welded joint of splice or end connections in a compression or tension element exceeds_____ times the throat size of the weld.**
(a) 16 (b) 150
(c) 100 (d) 120

**Ans. (b) : Reduction factor for long joint (B/w)**
When the length of the welded joint, of a splice on end-connection in a compression or tension element is greater than 150 throat size, the design capacity of the weld shall be reduced by the factor.
Between $1.2-\frac{0.2\ell_j}{150\ t_t} \leq 1.0$
$\ell_j$ = Length of the joint in the direction of the force transfer.
$t_t$ = Throat size of the weld.

**79. Which of the following causes fatigue in a structure?**
(a) Stress due to self-weight
(b) Residual stress
(c) Large number of loading cycles
(d) Wide range of stress variation
(a) (a) and (b)
(b) (c) and (d)
(c) (b), (c) and (d)
(d) (a), (b) and (d)

**Ans. (b) :** Fatigue is the initiation and propagation of cracks in a material due to cyclic loading.
Residual stress, large number of loading cycles, wide range of stress variation.

**80. A metal specimen having a diameter of 20 mm is tested in universal testing machine. It was found that the strain in longitudinal, lateral direction was $4.5 \times 10^{-4}$ and $1.5 \times 10^{-4}$ respectively. What will be its fraction of Poisson's?**
(a) 0.31 (b) 0.30
(c) 0.25 (d) 0.5

**Ans. (a) :** Poisson's ratio = $\frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$$= \frac{1.5\times10^{-4}}{4.5\times10^{-4}} = 0.33$$

**81. Identify the correct statement with respect to the materials used in steel reinforced cement concrete structures.**

(a) Cement concrete has poor compressive strength when compare to its tensile strength
(b) Steel is weak in compression
(c) Steel has a good tensile strength
(d) Cement concrete has good tensile strength when compared to steel

**Ans. (c) :** Steel has a good tensile strength.
Cement concrete has good compressive strength when compare to its tensile strength.

**82. Let us consider two bars made of materials 'X' and 'Y'. The area of cross-section and length is the same for both materials. The free elongation produced by material 'X' and 'Y' individually, when subjected to a rise in temperature 'T' is 35 mm and 45 mm. If both the bars are combined to form a compound bar and are subjected to a rise in temperature $T_1$, the stresses setup within the material 'X' and 'Y' are :**

(a) tensile, compressive
(b) tensile, tensile
(c) compressive, compressive
(d) compressive, tensile

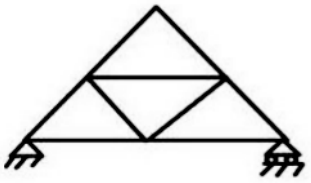
**Ans. (a) :** Material × Elongation is 35 mm
Material × Elongation is 45 mm
In order to have common expansion, tensile stress will be developed in material (x) and compression stress in material (y).

**83. At a location in a simply supported beam, the variation in the bending moment is given by $50x - 10x^2$ kN - m. The shear force at x = 2 is ____ kN.**

(a) 20 (b) 30
(c) 10 (d) 40

**Ans. (c) :** Shear force is the first derivation of bending moment.
Bending moment (M) = $50x - 10x^2$ kN-m
Shear force (V) = $\frac{dM}{dx} = 50 - 20x$
at x = 2 m
V = 50 – 20×2
= 10 kN

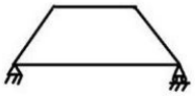
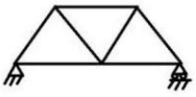
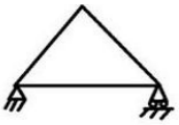
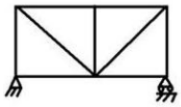
**84. Calculate the degree of external static indeterminacy for the plane truss shown in figure.**



(a) 0 (b) -1
(c) 1 (d) 3

**Ans. (a) :** $D_{se} = (2+1)–3 = 0$

**85. Identify the deficient truss from the following figures.**

(a) 
(b) 
(c) 
(d) 

**Ans. (a) :**For deficient truss: $m < 2j - 3$
$4 < 2 \times 4 - 3$
$4 < 5$

**86. The horizontal thrust in a cable structures is/will___**

(a) always zero
(b) decreases when there is increase in temperature
(c) decreases when there is decreases in temperature
(d) independent of temperature

**Ans. (b) :** Dip of cables increases due to rise in temperature as per the following equation.

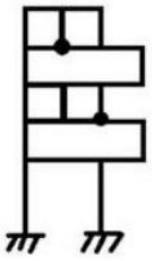
$$dh = \frac{3}{16} \propto t\left(\frac{\ell^2}{h}\right)$$

Since horizontal thrust in cable = $\frac{w\ell^2}{8h}$

Since $H \propto \frac{1}{h}$ & $h \propto \Delta T$

Hence horizontal thrust (H) reduces due to increases in temperature.

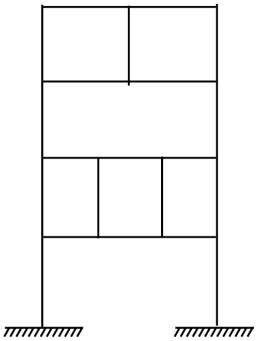
**87. Calculate the degree of external static indeterminacy for the given framed structure.**



(a) 2 (b) -1
(c) 0 (d) 1

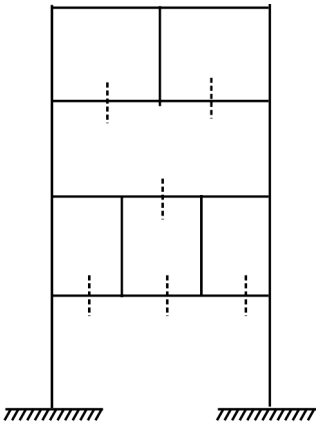
**Ans.(*):** $D_{se}$ = Unknown support reaction – Equilibrium equation
$D_{se} = (3+3) –3 = 3$
Hence none of the above options in correct.

**88. Calculate the internal static indeterminacy for the given plane frame shown in figure**



(a) 18 (b) 3
(c) 0 (d) 1

**Ans. (a) :**



Number of cuts required to make open tree structure =6
Hence $D_{si} = 3 \times 6 = 18$

**89. Which of the following statement is incorrect with respect to design consideration for taxiway layout?**

(a) As far as possible, the intersection of the taxiway and runway should be avoided.
(b) Exit taxiways are always placed at right angles to the runways.
(c) The route for the taxiway selected should provide the shorter practicable distance from the apron in the runway end
(d) Taxiways should be arranged so that aircraft taxing towards the apron do not interfere with the aircraft taxing for take-off

**Ans. (b) :** Taxiway is the road which runway to apron.
• It may connect to runway at right angle as well as on any other angle.
• Crossing (intersection) of runway and taxiway must be avoided as it will cause hindrance in take off and landing operations.

**90. The main function of taxiways is to provide access from the runways to the terminal area and service hangers.**
**Which of the following statements related to the length of a taxiway is/are true or false?**
**1. Taxiway length depends upon the distance between the apron and entry end or exit end of the runway.**
**2. To save fuel consumption, the length of the taxiway should be as short as practicable**

(a) 1 is true and 2 is false
(b) Both the statements are true
(c) 1 is false but 2 is true
(d) Both the statements are false

**Ans. (b) :** Taxiway is the road which runway to apron.
• It may connect to runway at right angle as well as on any other angle.
• Crossing (intersection) of runway and taxiway must be avoided as it will cause hindrance in take off and landing operations.
• Taxiway length shall be kept as short as possible for quick boarding de-boarding is as well as fuel consumption savings.

**91. The function of exit taxiways is to minimise the runway occupancy by the landing aircraft. The location of exit taxiways does not depend on_____**

(a) Touchdown speed
(b) Mix of aircrafts
(c) Size of apron
(d) Rate of deceleration of aircraft

**Ans. (c) : Aprons-**
■ On all air ports, airplane operates require areas on which they may position the airplanes during ground operates other than, those involving landing, taxiing and takeoff.
■ Such areas are generally referred to aprons and each defined according to its use as one of the following. Terminal, cargo parking, service, hangar, or holding apron (Referred to internationally as "holding bays")
■ The types and sized of aprons required vary for each airport depending on the amount of activity and size of airplane to be accommodated.
■ To complete the design of the airport, the need for these area as must be considered along with other important factors.

**92. The aircraft are berthed on the apron before they are loaded and unloaded. Hence, the loading apron is also known as the _____**

(a) landing apron
(b) parking apron
(c) service apron
(d) hanger

**Ans. (b) : Apron** are the placed where aircrafts are parked for boarding and de-boarding. These are also known as parking apron or loading apron.

**93. The relatively small aprons placed at a convenient location in the airport for the temporary storage of the aircraft is called___**

(a) the loading apron
(b) the holding apron
(c) the holding bay
(d) the parking apron

**Ans. (c):** The relatively small aprons placed at a convenient location in the airport for the temporary storage of the aircraft is called the holding bay.
**Holding apron** is the portion provided near the end of runway for engine run up and for hitching vehicle towing.
• They also hold the aircraft waiting for takeoff.

**94. In the case of flexible pavements, ineffective read surface drainage system is mainly due to:**
(a) reduced thickness of wearing coat
(b) poor quality of subgrade soil
(c) poor compaction of pavement layers
(d) inadequate cross slope of pavement surface as well as longitudinal slope of the roadside drains

**Ans. (d) :** Due to ineffective camber and longitudinal slope, road drainage is affected and mud pumping occur.

**95. Holes of 30 to 50 mm diameter are drilled through the CC pavement slab at a spacing of 1.5 to 3m. The sand-cement slurry is grouted under high pressure through these holes until the gaps under the slabs are filled up and cracked slab starts lifting. This process is called_____**
(a) mud pumping (b) mud jacking
(c) sealing (d) Raveling

**Ans. (b) : Mud jacking : Repair for sinking slab-**
■ Mud jacking or slab jacking is a simple process of pumping a special concrete mixtures underneath a slab to bring it back up to level.
■ A series of 11/2" holes are drilled down through the slab and the concrete is pumped into the holes under pressure. As the area beneath the slab fills with concrete, the slab rises.
■ The holes in the slab are then filled with a non shrinking grout to hide the repair.
■ Sinking slabs are often the result of poorly compacted soils or sub base underneath the slab and are sometimes caused by deeper, unavoidable soil erosion. Mud jacking is an economical alternative to replacing sunken slabs and often runs about half the cost of replacement.
eg- Foundation footings, Pool decks, Basement and garage slaps, Interior concrete floors.

**96. The turning zone is the area of the airport other than the approach area and it is intended for turning operations of the aircraft in case of emergencies like the failure of engine or trouble in the smooth working of aircraft experienced at the start of the take-off.**
**Which of the following statements related to the turning zone of the aircraft is/are true or false?**
**1. Any object located within a distance of 4.5 km from the airport reference point is considered as an obstruction, if its height exceeds 51 m above the ground or the established airport elevation, whichever is more.**
**2. An aircraft operates at a higher height in the turning zone, so it is necessary to assertion the fact that the area of the turning zone should be free from any obstruction.**
(a) 1 is true but 2 is false
(b) Both the statements are true
(c) 1 is false but 2 is true
(d) Both the statements are false

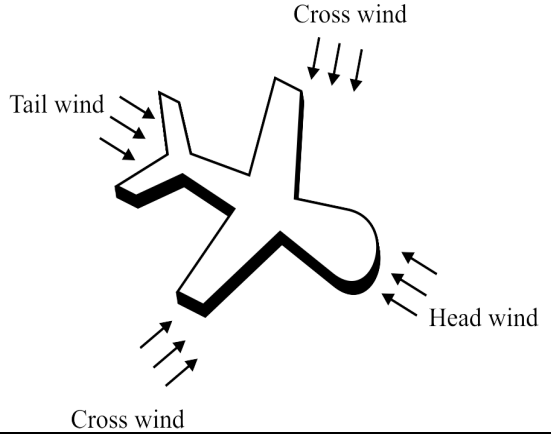
**Ans. (a) :** Any object located within a distance of 4.5 km from the airport reference point is considered as an obstruction, if its height exceeds 51 m above the ground or the established airport elevation, whichever is more.
An aircraft operates at a lower height in the turning zone, so it is necessary to assertion the fact that the area of the turning zone should be free from any obstruction

**97. The runway is usually oriented in the direction of the prevailing winds. The orientation of a runway is selected after a careful analysis of wind velocity and direction and duration over the past years.**
**Match the wind directions used in runway orientation with the definition.**

| Wind direction | Definition |
|---|---|
| **A. Headwind** | **1. It acts in the perpendicular direction of aircraft** |
| **B. Tailwind** | **2. It acts in the opposite direction of the head of the aircraft** |
| **C. Crosswind** | **3. It acts in the direction of the landing operation** |

(a) A-1, B-2, C-3
(b) A-2, B-1, C-3
(c) A-3, B-2, C-1
(d) A-2, B-3, C-1

**Ans. (d) : Headwind–** is the wind blowing towards the aircraft, it increase the lift and airplane hence pitot choose to take off and load in the headwind.
**Tail wind–** Tailwind is the wind blowing from behind the aircraft. It reduce the lift hence pitots avoid to take off in taxiwind.
**Cross wind–** It acts in the perpendicular direction of aircraft



**98. The ideal orientation of the runway decided from the study of the wind rose diagram may have to be altered or changed because of the following factors.**

**1. The orientation of the runway demands excessive grading and earthwork.**

**2. The location of the runway does not create excessive noise nuisance to the surrounding residential areas and public places of importance.**

(a) Both the statements are true
(b) 1 is false but 2 is true
(c) 1 is true but 2 is false
(d) Both the statements are false

**Ans. (c) :** Since airports are generally located away from the habitate and residential areas hence noise nuisance is not an issue to residential areas.

• Runway orientation depends on wind rose diagram as well cost of runway in that particular orientation.

**99. Which the following is an incorrect assumption in calculating the basic runway length?**

(a) The airport is at sea level
(b) No wind is blowing on the runway.
(c) The runway has a slope in the longitudinal direction.
(d) The en-route temperature is standard.

**Ans. (c) : Basic assumption is runway length–**

• No wind is blowing on runway.
• Aircrafts is loaded with full capacity.
• Airport is at see level.
• Runway is levelled and it has zero gradient.
• Temperature at the airport is standard.

**100. Given the following aircraft class, mix and average gate occupancy time, the capacity of 10 gates that serve the three classes of aircraft is :**

| Aircraft class | Mix(%) | Average occupancy time (min) |
|---|---|---|
| 1 | 15 | 20 |
| 2 | 30 | 40 |
| 3 | 55 | 60 |

(a) 18 aircrafts/hour
(b) 22 aircrafts/hour
(c) 12.5 aircrafts/hour
(d) 5.5 aircrafts/hour

**Ans. (c) :** Assume that each gate's available for all aircraft.

The gate capacity of a single gate is given by–

$c = 1 /$ weighted service time

$c = 1 / ( 0.15 \times 20 + 0.3 \times 40 + 0.55 \times 60)$

$= 0.02$ aircraft/min/gate

If G = total number of gates,

$C = G \times c$

$= 10 \times 0.02 = 0.2$ aircraft/min or 12 aircraft/hr

**101. As per Environment (Protection) rules, 1986, following the general standards for discharge of environments part-A : effluents, in the case of disposal to inland surface water, the temperature of disposing effluent water shall NOT exceed _____ above the receiving water temperature.**

(a) $5^0$C (b) $10^0$C
(c) $15^0$C (d) $20^0$C

**Ans. (a) :** Standard temperature for disposal of effluents is $5^0$C.

**102. Read the given statements related to secondary treatment of wastewater and identity whether they are correct or incorrect.**

**Statements :**

**(A) Secondary treatment of wastewater can be achieved by chemical unit processes such as chemical oxidation, coagulation-flocculation and sedimentation and chemical precipitation.**

**(B) Secondary treatment of wastewater cannot be achieved by employing biological processes (aerobic) where bacteria is used as catalyst for removal of pollutant.**

(a) Statement A is correct but B is incorrect
(b) Statement b is correct but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (c) :**

■ Secondary treatment of wastewater can be achieved by chemical unit processes such as chemical oxidation, coagulation-flocculation and sedimentation and chemical precipitation.

■ Secondary treatment of wastewater cannot be achieved by employing biological processes (aerobic) where bacteria is used as catalyst for removal of pollutant.

**103. The term 'sludge volume index' (SVI) used for the determination of sludge settleability is defined as :**

(a) volume occupied in ml by one gram of solids in the mixed liquor after settling for 30 min
(b) volume occupied in ml by one gram of solids in the mixed liquor after settling for 150 min
(c) percentage decrease in the volume of sludge after secondary treatment of waste water
(d) mass of sludge occupied in 1 ml of mixed liquor after settling for 150 min

**Ans. (a) : Sludge volume index (SVI) -** is the volume occupied in ml by 1 gm of solids in mixed liquor after settling for 30 minutes.

**104. Read the given statements related to biological treatment of waste water and identify whether they are correct or incorrect.**

**Statement :**

**(A) The objective of the biological treatment of wastewater is NOT to remove nutrients and phosphorous from the wastewater.**

**(B) The objective of the biological treatment of wastewater is to remove matter from the wastewater, which is present in present in soluble and colloidal form.**

(a) Statement A is correct but B is incorrect
(b) Statement B is correct but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) : Objective of Biological treatments-**
The objective of the biological treatment of wastewater is to remove matter from the wastewater, which is present in present in soluble and colloidal form.
(1) To transform dissolved and particulate biodegradable constituents into acceptable end products.
(2) To capture and incorporate suspended and nonsettleable colloidal solids into a biological floc or biofilm.
(3) To transform or remove nutrients. (Such as nitrogen and phosphorous)
(4) To remove specific trace organic constituents and compounds.

**105. Identify whether the given statements related to noise exposure forecast (NEF) are correct or incorrect.**

**Statements :**

**(A) NEF develops a noise contour diagram.**
**(B) NEF is associated with road traffic noise.**

(a) Statement A is correct but B is incorrect.
(b) Statement B is correct but A is incorrect
(c) Both statements are correct
(d) Both statements

**Ans. (a) :** The noise exposure forecast (NEF) is a methodology for predicting a single numerical rating for cumulative noise exposure of areas surrounding airports.
■ The contours produced by NEF can be superimposed on land use maps as a tool for the description, interpretation and prediction of noise exposure.

**106. Identify whether the given statements related to noise barrier are correct or incorrect.**

**Statements :**

**(A) series of trees between the source and receiver of a noise will decrease the sound level of the noise at the receiver point.**
**(B) For high-frequency sound, when diffraction occurs due to a noise barrier, the diffraction angle will be very large.**

(a) Statement A is correct but B is incorrect
(b) Statement B is correct but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (a) :** Noise barriers are being used with more frequency to solve problems of noise pollution from freeways in urban areas.
■ Noise barriers can have a substantial effect on the visual environment of highway and its surrounding environment. They are long continuous structures, often more than fifteen feet high made of various materials.
■ They can be significantly change the view from the road by blocking view of the road side and creating a monolithic uniformly of walls instead of changing urban scenery.

**107. Which of the following combination of noise exposure' has the maximum noise dose?**

(a) An average noise level of 90 dBA during a 4-hour period
(b) An average noise level of 90 dBA during an 8 hour period
(c) An average noise level of 80 dBA during a 6 hour period
(d) An average noise level of 85 dBA during an 8-hour period

**Ans. (b) :** An average noise level of 90 DB for 8 hours period is considered as maximum noise dose.

**108. Which of the following is the auditory effect caused by noise pollution?**

(a) Permanent hearing loss
(b) Loss in working efficiency
(c) Reduced depth and quality of sleep
(d) Interference with speech communication

**Ans. (a) :** Permanent hearing loss is the auditing effect caused by noise pollution.
Permanent hearing loss is also known as permanent threshold shift (PTS),
**Some other auditory health effects–**
**Acoustic trauma–** Sudden hearing damage caused by short burst of extremely loud noise such as gunshot.
**Tinnitus–** Ringing or buzzing in the car.
**Temporary hearing loss–** Also known as temporary threshold shift, which occur immediately after exposure to a high level of noise.

**110. Identify whether the given statements related to noise-induced hearing loss (NIHL) are correct or incorrect.**

**Statements :**

**(A) NIHL can be caused by a one-time exposure to an intense 'impulse' sound, such as an explosion.**
**(B) NIHL can be caused by continuous exposure to loud sounds over an extended period of time, such as noise generate in a woodworking shop.**

(a) Statement A is correct but B is incorrect.
(b) Statement B is correct but A is incorrect.
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (c) :** Noise-Induced Hearing Loss (NIHL)–
■ NIHL can be caused by a one-time exposure to an intense 'impulse' sound, such as an explosion.
■ NIHL can be caused by continuous exposure to loud sounds over an extended period of time, such as noise generate in a woodworking shop.

**111. Which of the following is the major gas produced from landfill and can be used as a renewable energy?**

(a) carbon monoxide
(b) methane
(c) oxygen
(d) hydrogen disulphide

**Ans. (b) :** $CH_4$ and $CO_2$ make upto 90-98% of the land fill gases, remaining gases consists of $N_2$, $O_2$ & $NH_3$ etc.

**112. As per CPHEEO-2016, the minimum sitting distance of land for sanitary landfill sites shall be _____ away from the centre line of a highway or railway line and water supply wells.**
(a) 100 m (b) 500 m
(c) 1000 m (d) 1500 m

**Ans. (b) :** As per CPHEEO-2016, the minimum sitting distance of land for sanitary landfill sites shall be 500 m away from the centre line of a highway or railway line and water supply wells.

**113. Which of the following is not an advantage of extended producer responsible (EPR) policy?**
(a) Creation of incentives for environmentally friendly product designs.
(b) Provision of a monetary incentive to the consumer to return the product or package
(c) Increase in natural resource demand for packaging and product containers
(d) Creation of infrastructure for collection and recycling of material.

**Ans. (c) :** Extended producer responsibility (EPR) is a system that impose a certain quota for the recyclable waste from products or packaging materials on the manufacture of the products.
**Advantage of EPR-**
■ Creation of incentives for environmentally friendly product designs.
■ Provision of a monetary incentive to the consumer to return the product or package.
■ Creation of infrastructure for collection and recycling of material.
**Disadvantage of EPR-**
■ Increase in natural resource demand for packaging and product containers

**114. The abbreviation 'MoEF&CC' is used to indicate a ministry dealing with Environmental impact assessment (EIA) studies in India. What does the abbreviation stand for?**
(a) Ministry of Environment, Forest and climate change
(b) Ministry of ecology, forests and climate correction
(c) Ministry of environment, foundry, charcoal and carbon
(d) Ministry of ecology, foundry and civil construction

**Ans. (a) :** The abbreviation 'MoEf&CC' stand for ministry of environment, forest and climate change.

**115. Identify the incorrect statement with respect to public participation in Environmental Impact Assessment (EIA) studies.**
(a) Public participation in EIA reduces communication gap between the experts and locals.
(b) Public participation will be able to provide advice on management, consideration local conditions.
(c) Opposition to management will be reduced if people are informed
(d) Public participation in EIA should be avoided as much as possible to reduce nuisance creation.

**Ans. (d) :** Environment impact assessment (EIA) is a process of evaluating the likely environmental impacts of a proposed project or development, taking into account inter related, socio-economic, cultural and human health impacts, both beneficial and adverse.
EIA cyclic comprises as four stages:
• Screening
• Scoping
• Public hearing
• Appraised

**116. Four ecosystems have been given, out of which three are alike in some manner and one, is different. Select the one that is different with respect to the nature of the ecosystem.**
(a) River (b) Forest
(c) Aquarium (d) Village

**Ans. (c) :** Aquarium is alike among four because is artificial or man-made ecosystem and other three are natural ecosystem.
**Natural ecosystem-** Are self sustaining ecosystem and contain natural factors and organism, e.g. river, forest, glass land, desert, village.
**Artificial ecosystem–** Dam, aquarium, garden etc.

**117. Select the incorrect statement with respect to permanent hardness of water.**
(a) It can be eliminated by boiling.
(b) It forms insoluble scum with soap.
(c) If affects the cleaning action of soap
(d) If contains dissolved salts of calcium and magnesium.

**Ans. (a) :** Boiling is used to remove temporary hardness of water.
Method of removing permanent - Hardness.
• Lime- soda process
• Demineralization process
• Zeolite or base exchange or cation-exchange.

**118. Which of the following audits is a type of management audit that comes under environment audit categories?**
(a) Compliance audit
(b) Policy audit
(c) Acquisition audit
(d) Health and safety audit

**Ans. (a) :** Compliance audit comes under environment audit categories.
• An environment audit provides an assessment of the environment performance of a business or organization.

**119. Which of the following heterotrophs is classified as a secondary consumer?**

(a) Rodents (b) Cows
(c) Goats (d) Sparrows

**Ans. (d) :** Among the following sparrows is a secondary consumer because all other feed on autotrophs and they are primary consumers because, all other feed on autotrophs and they are primary consumes.

$$\underset{\text{Autotraph}}{\text{Mulberry}} \rightarrow \underset{\text{Primary}}{\text{Caterpillar}} \rightarrow \underset{\text{Seconary}}{\text{Sparrow}} \rightarrow \underset{\text{Testiary}}{\text{kite}}$$

**120. Which of the following is an approach to design for sustainability?**

(a) Selection of resource with low environmental impact
(b) Design for safety and reliability
(c) Design for manufacturing
(d) Design for supply chain

**Ans. (a) :** Design for sustainability is a holistic design approach focuses on resource efficiency and the used the environmentally friendly material or selection or resource with low environmental impact.

**121. Which of the following ISO provisions describes the principles and framework life cycle assessment (LCA)?**

(a) ISO 14040 : 2006
(b) ISO 23362 : 2006
(c) ISO 15649 : 2006
(d) ISO 17985 : 2006

**Ans. (a) :** ISO 14040 : 2006 describes the principles and framework life cycle assessment (LCA).

**122. According to E Barbier, the primary objective of sustainable development is :**

(a) to reduce the absolute poverty of the world's poor by providing lasting and secure livelihoods that minimize resource depletion, environmental degradation cultural disruption and social instability
(b) to meet the needs of the present without compromising the ability of future generations to meet their own needs
(c) equitable use of resources for meeting the needs of the present and future generations without causing damage to the environment
(d) to conserve and nurture the biological diversity, gene pool and other resources for long-term food security

**Ans. (b) :** Primary objective of sustainable development is to reduce the absolute poverty of the world's poor by providing lasting and secure livelihoods that minimize resource depletion, environmental degradation cultural disruption and social instability

**123. Select the incorrect statement with respect to carbon offset, which is used as a financial instrument in sustainable development.**

(a) Carbon offset aims at reduction in greenhouse gas emissions.
(b) Use of renewable source of energy is one of the ways to generate carbon offset
(c) One carbon offsets equal to reduction of 1 metric ton of $CO_2$ or its equivalent in other greenhouse gases
(d) One metric ton of $CO_2$ reduction is equivalent to a reduction of around 450 kg carbon

**Ans. (d) :** One metric ton of $CO_2$ reduction is equivalent to reduction of around 450 kg carbon is an incorrect statement.
One metric ton of $CO_2$ reduction is equivalent to a reduction of around 272 kg carbon.

$CO_2 = 12 + 16 \times 2 = 44$ gm

12 gm of carbon in 44 gm of $CO_2$
In 1 kg of $CO_2$ = 0.272 kg of carbon

**124. Which of the following organizations was the first to work on life cycle assessment?**

(a) The coca-cola company
(b) pepsico
(c) 7UP company
(d) The peelle company

**Ans. (a) :** The coca-cola company was the first to work on life cycle assessment.

**125. Which of the following is NOT a sustainable development goal targeted to be achieved by 2030, according to the General Assembly of the United Nations?**

(a) No poverty
(b) Clean water and sanitation
(c) Quality education
(d) Life on Mars

**Ans. (d) :** Sustainable development Goals (SDGs) also known, as global goals, were adopted by the united nation in 2015 as a universal call to action to end poverty, protect the plane and ensure that by 2030 all people enjoy peace and prosperity.
Goal targeted to achieve are –

- No poverty
- Zero hunger
- Quality Education
- Gender equality
- Clean water and sanitation
- Good health and well-being
- Industry, innovation and instra-strucrture
- Reduced inequalities

**126. Who among the following introduced the concept of boundary layer and derived the equations for the boundary layer flow by correct reduction of Navier-Stokes equations?**

(a) Ludwing Prandtl (b) Dickens
(c) Ryve (d) Von Mises

**Ans. (a) :** Ludwig Prandtl introduced the concept of boundary layer and derived the equation for boundary layer flow by correct reduction of Navier-stokes equation.

**127. Identify whether the given statements related to application of the boundary layer theory are correct or incorrect.**

**Statements :**

**(A) This boundary - layer theory is valid beyond the point of separation.**

**(B) Mathematically, the application of the boundary - layer theory converts the character of governing Navier-Stroke equation from elliptic to parabolic.**

(a) Statements A is correct but B is incorrect
(b) Statements B is correct but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) :** Bandary layer equation are valid up to the point of separation. At the point of separation, the flow get detached iron the solid surface due to excessive adverse pressure gradient.

- Mathematically, the application of the boundary layer theory convert the character of governing Navier-stoke equation from elliptic to parabolic.

**128. Identify whether the given statements related to point of separation and the boundary layer theory in fluids are correct or incorrect.**

**Statements :**

**(A) The boundary layer theory does not speak about the location of the point of separation.**

**(B) At the point separation, the boundary layer thickness becomes quite large for the thin layer approximation to be valid.**

(a) Statement A is correct but B is incorrect
(b) Statement B is correct but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) :** Boundary layer theory speak about the location of the point of separation.

- The separation point S is determined from the condition $\left(\frac{dv}{dy}\right)_{y=0} = 0$
- At point of separation boundary layer thickness becomes quite large for the thin layer approximation to the valid.

**129. Which of the following is the correct expression, in case of open channel flow, to measure energy at a particular point in the channel?**

(a) $E = y + \frac{v^2}{2g}$, where E = specific energy, y = Pressure head $\frac{v^2}{2g}$ = velocity head

(b) $E = 2y + \frac{v^2}{g}$, where E = specific energy, y = Pressure head $\frac{v^2}{2g}$ = velocity head

(c) $E = y^2 + \frac{v^2}{2g}$, where E = specific energy, y = Pressure head $\frac{v^2}{2g}$ = velocity head

(d) $E = y + \frac{v^2}{g}$, where E = specific energy, y = Pressure head $\frac{v^2}{2g}$ = velocity head

**Ans. (a) :** Specific energy, $E = y + \frac{v^2}{2g}$

y = potential energy of the flow

$\frac{v^2}{2g}$ = Kinetic energy of the flow

**130. Calculate the specific energy at a point in a rectangular open channel whose width is 2 m, depth 1 m and discharge 2 $m^3$/sec. Take acceleration due to gravity as 10 m/$sec^2$.**

(a) 1.05 m (b) 1.20 m
(c) 1.50 m (d) 1.95 m

**Ans. (a) :** Given,
Width of rectangular channel, b = 2 m
Depth of water, y = 1 m
Discharge, Q = 2 $m^3$/s
Specific energy

$$E = y + \frac{V^2}{2g}$$

Where,

$$V = \frac{Q}{\text{Area}} = \frac{2}{b \times y} = \frac{2}{2 \times 1} = 1 \text{ m/s}$$

$$E = 1 + \frac{(1)^2}{2 \times 10}$$

E = 1 + 0.05
E = 1.05 m

**131. A lines channel of trapezoidal section carries a discharge of 6 $m^3$/sec, at a depth of 1 m, with bottom width 2 m and side slope of 1.5 horizontal to 1 vertical. Consider uniform flow and calculate the velocity of flow.**

(a) 1.312 m/sec (b) 1.714 m/sec
(c) 2.135 m/sec (d) 2.426 m/sec

**Ans. (b) :** Discharge = Area × Velocity
Q = A ×V
Area of trapezoidal section = (b+nd) × d
Where,
Width, b = 2m
depth, d = 1 m
Side slope 1.5 horizontal to vertical, n = 1.5
Discharge, Q = 6 $m^3$/s
A = (b + nd) × d
= (2+1.5 × 1) 1
= 3.5 $m^2$
Q = A × V
6 = 3.5 ×V
v = 1.714 m/sec

**132. Identify whether the given statements related to basic assumption in gradually varied flow analysis are correct or incorrect.**

**Statements:**

**(A) The head loss in a reach may be computed using an equation applicable to uniform flow, having the same velocity and hydraulic mean radius of the section.**

**(B) Channel bottom slope is large.**

(a) Statement A is correct but B is incorrect
(b) Statement B is correct but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (a) :** Assumption in gradually varied flow (GVF)
- Channel bed slope is small.
- Head loss is as for uniform flow
- A channel is prismatic
- Pressure distribution over the channel section is hydrostatic.

**133. Which of the following is an example of steady rapidly varied flow with broken flow profile?**

(a) The backwaters produced by a dam
(b) Hydraulic jump
(c) The flow of water in a pipe of constant diameter at constant velocity
(d) The flow of water through pipe of constant diameter connected to a pump pumping at a constant rate which in then switched off

**Ans. (b) :** Hydraulic jump is an example of steady rapidly varies flow.
**Hydraulic jump–** The rise of water level, which take place due to the transformation of the unstable shooting flow (super-critical) to the stable streaming flow (sub-critical flow).

**134. Identify whether the given statements related to applications of a hydraulic jump are correct or incorrect.**

**Statements :**

**(A) It is used to lower the water level on the d/s side of a metering flume and thus maintain low water level in the channel for irrigation etc.**

**(B) It is used to dissipate energy in the water flowing over the spillway and other hydraulic structures and thus prevent scouring downstream.**

(a) Statement A is correct but B is incorrect
(b) Statement B is correct but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) :** Application of Hydraulic jump–
- It is used to raised water level on downstream side of metering flume and thus maintain high water level needed for irrigation purpose.
- Energy dissipation – Sluice gate, spillway
- To mix chemical
- To avoid erosion in stilling basin
- Aeration in water treatment plant
- Reduce uplift pressure.

**135. In India, the south western monsoon called 'Kharif' spans mostly from_____**

(a) July to October
(b) April to June
(c) October to March
(d) January to March

**Ans. (a) :** In India, the south western monsoon called 'Kharif' spans mostly from July to October.
Rabi is sown in India during month of October and November and harvested in Feb or March.

**136. Which of the following crops is classified as Rabi crop in India?**

(a) Linseed (b) Sorghum
(c) Rice (d) Maize

**Ans. (a) :** Linseed is classified as Rabi crop in India.

| Rabi crops | Kharif crop |
|---|---|
| It is also known as winter crops. | Also known as monsoon crops. |
| Rabi is sown in India during month of October and November and harvested in Feb or March | This crop is soon in June or July and harvested around October. |
| E.g. wheat, Barley, Oats, Pulses, Mustard, linseed etc. | e.g. Rice, maize, sorghum, Bajra, soybean, cotton etc. |

**137. Calculate the delta for a crop of its duty is 1000 hectares/cumec on the field. Consider the base period of crop as 100 days.**

(a) 74.8 cm (b) 86.4 cm
(c) 100 cm (d) 106.56 cm

**Ans. (b) :** $\text{Delta}(\Delta) = \frac{8.64B}{D}$

Where,
Base period (B) = 100 day
Duty (D) = 1000 hectare/cumec

$$\Delta = \frac{8.64 \times 100}{1000}$$

$$\Delta = \frac{8.64 \times 100}{1000}$$

$$= 86.4 \text{ cm}$$

**138. Identify whether the given statements related to diversion head works constructed at the head of the canal are correct or incorrect.**

**Statements :**

**(A) If the major part or entire ponding of water is achieved by a raised crest and smaller part or nil part of it is achieved by the shutters, then this barrier is called a barrage.**

**(B) If the major part of entire ponding of water is achieved by a raised crest and a smaller part or nil part of it is achieved by the shutters, then this barrier is called a weir.**

(a) Statement A is correct but B is incorrect.
(b) Statement B is correct but A is incorrect.
(c) Both statements are correct.
(d) Both statements are incorrect.

**Ans. (b) :** The work which are constructed at the head of canal in order to divert the river water towards the canal so as to ensure regulated silt free continuous supply of water, with a certain minimum head into the canal is termed as Diversion Headwork e.g. weir, barrage.

| Weir | Barrage |
|---|---|
| Major ponding is achieved from raised crest. | Major ponding is achieved from gate/shutter |
| It can't be used as means of transport | It can be used as means of transfer. |
| Cheap but lack control | Casting but effective control |
| Afflux cost during flood is very high | Afflux cost during flood is very less. |

**139. Identify whether the given statements related to diversion head works are correct or incorrect.**

**Statements :**

**(A) Fish ladder arrangement is provided in the weir or dam for interrupting/preventing the movement of fishes either to the upstream or downstream.**

**(B) Silt ejector is used for removal of silt which has entered the canal.**

(a) Statements A is correct but B is incorrect
(b) Statements B is correct but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) : Fish ladders–** It is provided in the weir or dam for movement of fish either to the upstream or downstream.

- Fish ladder dissipates the energy of water by provision of baffle wall in it.

**Silt ejector–** These are device which extract silt from the canal water after silted water has travelled a certain distance in to the off-take canal.

**140. Identify the incorrect statements with respect to vertical curves used in highways.**

(a) The parabolic curve produces excellent riding qualities and is preferred in highways.
(b) The circular arc produces good riding quantities and is preferred in highways.
(c) Provision of parabolic curves is not recommended in highways.
(d) Abrupt change in the rate of grade is not recommended in highways.

**Ans. (c) :** Vertical curves are introduced at the intersection of two gradient either as summit curves as sag curve.

- About change in rate of grade is not recommended in highway.
- Square parabola is generally preferred due to excellent riding quality, simplicity of calculation and uniform rate of change of gradient.

**141. In setting out of vertical curve, the difference in elevation between a point on the tangent and the corresponding point on the curve varies as :**

(a) the square of its horizontal distance from the point of tangency
(b) 2 times the horizontal distance from the point of tangency
(c) 0.5 times the horizontal distance from the point of tangency
(d) the square root of its horizontal distance from the point of tangency

**Ans. (a) :** In a vertical curve the offsets from a tangent to the curve always vary as the square of the distance along the tangent from the point of tangency.

**142. A simple circular curve of radius(r) 150 m is to be set on the ground with a central angle $60^0$. Calculate its length of long chord.**

(a) 100 m (b) 125 m
(c) 150 m (d) 175 m

**Ans. (c) :**

Given,

Radius (R) = 150 m

Central angle ($\Delta$) = $60^0$

Length of chord = $2 \times R \sin\frac{\Delta}{2}$

$= 2 \times 150 \times \sin\frac{60^0}{2}$

$= 2 \times 150 \times \frac{1}{2}$

= 150 m

**143. Identify the incorrect statement with respect to electromagnetic, radiation used in remote sensing.**

(a) If produces a time-varying magnetic field and vice versa.
(b) Once electromagnetic radiation is generated, it remains self-propagating
(c) It is capable of travelling across space.
(d) Human eyes are sensitive to all parts of the electromagnetic system.

**Ans. (d) :** Electromagnetic radiation (EMR) consist of waves of the electromagnetic field, which propogate through space and carry electromagnetic radiant energy.

- Once generated its remain self propagating.
- It consist of magnetic and electric field.
- It produce a time varying magnetic field and vice-versa.

**144. Identify the incorrect statements with respect to airborne remote sensing system.**

(a) Very high spatial resolution images (20 cm or less) can be obtained through airborne remote sensing.
(b) Aircraft remote sensing is better than satellite remote sensing to map a large area at a less cost.

(c) In airborne remote sensing, downward sideward looking sensors mounted on aircrafts are used to obtain images of the earth's surface.
(d) The major disadvantage of airborne remote sensing is the high cost per unit area of ground coverage.

**Ans. (b) :** In airborne remote sensing, downward or sideward looking sensor are mounted on an aircraft to obtain images of the earth's surface.
- Analog aerial photography, videography and digital photography are commonly used in airburne remote sensing.

**Advantage–**
- It is capable of offering very high spatial resolution image (20 cm or less).

**Disadvantage–**
- Low coverage area and high cost per unit area of ground coverage.
- It is not cost-effective to map a large area.

**145. Which of the following parameters is not covered in the design of structures by limit state of serviceability?**
(a) Deformation and deflection
(b) Vibration in the structure
(c) Corrosion and durability
(d) Equilibrium of the structures

**Ans. (d) : Limit state–** State at which one of the condition pertaining to the structure has reached a limit value.
**Limit state at serviceability–** Deflection, vibration, corrosion, fire resistance.

**146. As per IS : 456-2000, while doing the design for the limit state of collapse in flexure, the maximum strain in concrete at the outermost compression fibre is taken as____ in bending.**
(a) 0.35 (b) 0.035
(c) 0.00035 (d) 0.0035

**Ans. (d) :** As per clause no 38.1 of IS 456 : 2000, the maximum strain in concrete at the outermost compression fibre is taken as 0.0035 in bending.
- As per clause no; 39.1 of IS 456 : 2000, the maximum compressive strain in concrete in axial compression is taken as 0.002.

**147. What is the failure pattern of a column with low slenderness ratio (i.e., relatively short and stocky column) subjected to an axial load which can cause failure?**
(a) Fall under ultimate loads, with the material reaching its ultimate strength
(b) Fail under relatively low compression loads and susceptible to buckling failure
(c) Fail due to development of considerable secondary moments
(d) Fails due to development of both bending and shear stresses.

**Ans. (a) :** Column with low slenderness ratio, fails as a result of failure of material i.e., this structure fails when concrete or steel reaches its ultimate strength.
- Column with high slenderness ratio, the structure fails mainly due to buckling along with large lateral deflection.

**148. As per IS 456-2000, the minimum cross-sectional area of longitudinal reinforcement to be provided in a RCC column of size 200 mm × 600 mm is _____**
(a) 750 $mm^2$ (b) 960 $mm^2$
(c) 1060 $mm^2$ (d) 1250 $mm^2$

**Ans. (b) :** Minimum area of steel = 0.8% of gross area
$$= \frac{0.8}{100} \times 200 \times 600$$
$$= 960 \text{ mm}^2$$

**149. As per IS 456-2000, in the case of RCC beams what is the maximum depth of the web, beyond which side face reinforcement shall be provided along the two faces of beam?**
(a) 450 mm (b) 550 mm
(c) 650 mm (d) 750 mm

**Ans. (d) :** As per IS 456 : 2000
- If the depth of web exceeds 750 mm then side face reinforcement equal to 0.1% of cross sectional area and is equally distributed on both faces of the beam.

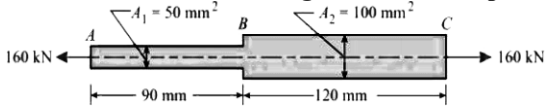
**150. According to IS : 800-2007, the maximum effective slenderness ratio for tension members in which members in which reversal of direct stresses occurs due to loads other than wind or seismic force is _____**
(a) 100 (b) 140
(c) 180 (d) 220

**Ans. (c) :** As per IS : 800-2007
Maximum value of effective slenderness ratio

| Member | Maximum effective slenderness ratio |
|---|---|
| Member carrying compressive load resulting from dead load (DL) and imposed loads (IL) | 180 |
| A member in which reversal of direct stress other than wind or seismic forces | 180 |
| A member subjected to compression force resulting only from combination with wind/earthquake section. | 250 |
| Compression flange of a beam | 300 |
| Tie in a roof truss or reversed of stress due to wind/earthquake force | 350 |
| Member always under tension | 400 |

# Andhra Pradesh Public Service Commission
# Assistant Director Exam, 2022
## Solved Paper Date - 22.10.2022 (Shift-II)

**1. An automobile component made of metal with a modulus of elasticity of 200 GPa is subjected to a tensile load of magnitude 160 kN as shown in the figure.**
**Determine the total elongation of the compound**




(a) 1.2 mm (b) 0.8 mm
(c) 2.4 mm (d) 1.8 mm

**Ans. (c) :** Given,
$A_1 = 50\ mm^2$, $A_2 = 100\ mm^2$
$L_1 = 90$ mm, $L_2 = 120$ mm
$P_1 = P_2 = 160$ kN; $E_1 = E_2 = 200$ GPa
$= 200 \times 10^3$ MPa (N/mm²)
Total elongation ($\delta$) = $\delta_1 + \delta_2$

$$\text{Total elongation} = \frac{P_1L_1}{A_1E} + \frac{P_2L_2}{A_2E_2}$$

$$\text{Total elongation} = \frac{160\times10^3\times90}{50\times200\times10^3} + \frac{160\times10^3\times120}{100\times200\times10^3}$$

Total elongation = 1.44 + 0.96 = 2.4 mm.

**2. An alloy wire of 2 mm² cross-sectional area and 15 N weight hangs freely under its own weight. Find the maximum length of the wire if its extension is not to exceed 0.6 mm. Take modulus of elasticity (E) for the wire material as 150 GPa**

(a) 10 m (b) 24 m
(c) 15 m (d) 30 m

**Ans. (b) :** Given, A = 2 mm², $\delta L$ = 0.6 mm
Self weight w = 15 N, E = 150 GPa = $150 \times 10^3$, MPa= L?
**Elongation due to self weight–**

$$\delta_L = \frac{wL}{2AE}$$

$$0.6 = \frac{15\times L}{2\times2\times150\times10^3}$$

L = 24000 mm = 24 m

**3. A point in a strained material is subjected to two mutually perpendicular tensile stresses of 200 MPa and 100 MPa. Determine the intensities of shear stress on a plane inclined at 30° with the axis of minor tensile**

(a) 25.00 MPa (b) 43.30 MPa
(c) 32.42 MPa (d) 50.00 MPa

**Ans. (b) :** Given,
$\sigma_1$ = 200 MPa (Tensile)
$\sigma_2$ = 100 MPa (Tensile)
$\theta = 30^0$ (With the axis of minor tensile)
On the inclined plane
Shear stress or tangential stress for tensile stress –

$$\sigma_t = \frac{\sigma_1 - \sigma_2}{2}\sin 2\theta$$

$$\sigma_t = \frac{200-100}{2}\sin 60$$

$$= \frac{100}{2}\times\frac{\sqrt{3}}{2} = \frac{100\sqrt{3}}{4}$$

$\sigma_t = 43.30$ MPa

**4. A plane element in a body is subjected to a tensile stress of 80 MPa accompanied by a shear stress of 25 MPa. Find the normal stress on a plane inclined at an angle of $15^0$ with the tensile stress**

(a) 2.56 MPa (b) –1.89 MPa
(c) –7.14 MPa (d) 10.82 MPa

**Ans. (c) :** Given,
$\sigma_x$ = 80 MPa, $\tau_{xy}$ = 25 MPa, $\theta = 15^0$
(with the tensile stress).

$$\sigma_n = \frac{\sigma_x}{2} - \frac{\sigma_x}{2}\cos 2\theta - \tau_{xy}\sin 2\theta$$

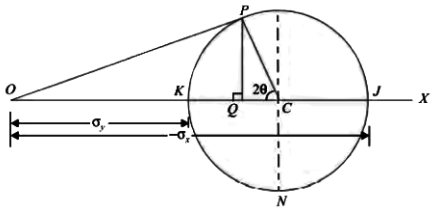
$$\sigma_n = \frac{80}{2} - \frac{80}{2}\cos 30^0 - 25\sin 30^0$$

$$\sigma_n = 40 - 40\times\frac{\sqrt{3}}{2} - 25\times\frac{1}{2}$$

$$\sigma_n = 40 - 20\times\sqrt{3} - 12.5$$

$\sigma_n = -7.14$ MPa

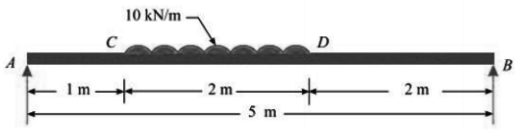
**5. Mohr's circle for stresses on an oblique section of a body subjected to direct stresses in two mutually perpendicular directions is shown in the figure. The line 'QP' perpendicular to the line 'OX' gives ____ according to the chosen scale.**




(a) normal stress
(b) maximum shear stress
(c) resultant stress
(d) shear stress

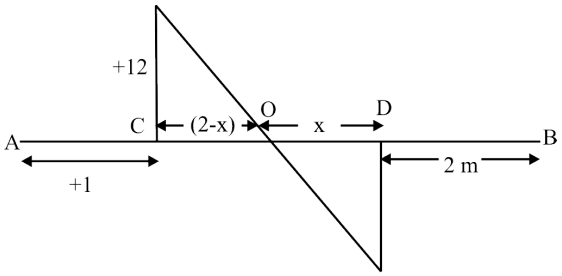
**Ans. (d) :** Mohr's circle for stress on an oblique section of a body subjected to direct stresses in two mutually perpendicular direction is shown in the figure. The line 'QP' perpendicular to the line 'OX' gives shear stress according to the chosen scale.

**6. A simply supported beam of span 5 m is loaded with a uniformly distributed load of intensity 10 kN/m over a length of 2 m shown in the given figure. By considering the shear fore at 'C' as +12 kN and that at 'D' as -8 kN, determine the position (distance measured from A) at which shear force is zero.**




(a) 2.5 m (b) 2.2 m
(c) 2.35 m (d) 2.0 m

**Ans. (b) :** Given,
Shear force at C = +12 kN
Shear force at D = –8 kN
Determine the position from A at which shear force is zero = ?




Shear force diagram of UDL

By similarity of triangle

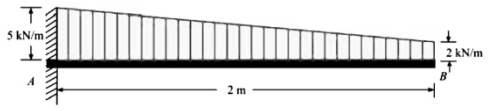
$$\frac{12}{8}=\frac{2-x}{x}$$

$12x = 16–8x \rightarrow 20x = 16 \rightarrow x = 0.8$ m

- O is the point at which shear force is zero.
- Distance of a from A is–

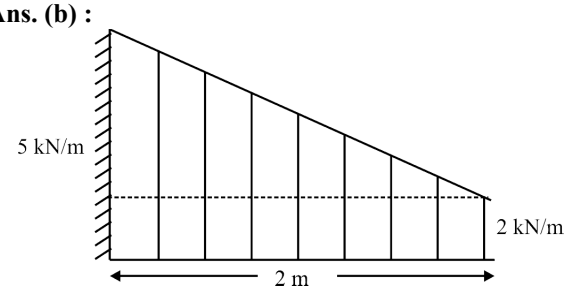
$1+(2–x)$
$= 1 + (2–0.8)$
$= 2.2$

**7. A cantilever beam of 2 m span is subjected to a gradually varying load from 2 kN/m to 5 kN/m as shown in the figure. Calculate the magnitude of the moment at 'A'.**




(a) 3.5 kN/m (b) 6 kN/m
(c) 7 kN/m (d) 4.5 kN/m

**Ans. (b) :**




Let $V_s$ divided whole load in to the two parts –
(i) A UDL of 2kN/m at whole span figure
(ii) A UDL of 3 kN/m at whole span
Load due to UDL = $2 \times 2 = 4$ kN

Load due to UVL = $3\times\frac{2}{2}=3$ kN

Moment at A

$$M_A = 4\times1+3\times\frac{2}{3} \qquad \left[w\times L+w\times\frac{L}{3}\right]$$

$M_A = 4 + 2 = 6$ kN/m

**8. Consider the given statements with respect to shear force and bending moment diagrams for beams and identify whether the statements are correct.**

**Statements :**

**(A) If there is a point load at a section on the beam, then the shear fore and bending moment suddenly change (i.e., the variation line is vertical).**

**(B) If there is a uniformly distributed load between two points along the length of a beam, then the shape of bending moments diagram for that segment is parabolic in nature.**

(a) Statement A is correct and B to incorrect
(b) Statement B is correct and A to incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) :** If there is a uniformly distributed load between two points along the length of a beam, then the shape of bending moment diagram for that section is parabolic in nature.
For UVL $\rightarrow$ Bending moment diagram is cubic parabola.
For point load $\rightarrow$ Bending moment diagram is linear (inclined).

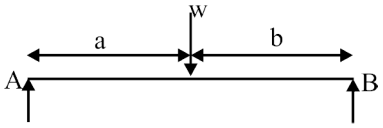
**9. The maximum deflection in a simply supported beam of span 'I' subjected to a uniformly distributed load with intensity 'w' kN/m is equal to ____. Where, 'EI' is flexural rigidity of beam and is constant throughout its length.**

(a) $\frac{5W\ell^4}{384EI}$ (b) $\frac{W\ell^3}{48EI}$
(c) $\frac{W\ell^4}{24EI}$ (d) $\frac{W\ell^3}{36EI}$

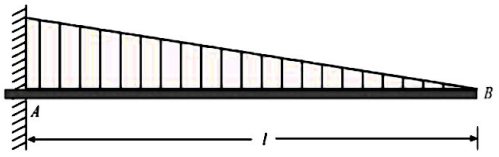
**Ans. (a) :** Maximum deflection of simply supported beam subjected to UDL $\frac{5W\ell^4}{384EI}$.

- Maximum defection of simply supported beam subjected to concentrated load at mid point of span $=\frac{w\ell^3}{48EI}$

• Maximum deflection of simply supported beam subjected to concentrated load at any point = $\frac{wb\left(a^2+2ab\right)^{3/2}}{9\sqrt{3}\times EI(a+b)}$




**10. Calculate the slope at the free end of a cantilever beam of length $\ell$ = 5 m subjected to a uniformly varying load with intensity varying from 0 kN/m (at B) to 20 kN/m in (at A) shown in the figure. Consider that the flexible rigidity (EI) of the beam is constant throughout its length.**




(a) $\frac{43.33}{EI}$ rad (b) $\frac{85.52}{EI}$ rad

(c) $\frac{104.16}{EI}$ rad (d) $\frac{154.89}{EI}$ rad

**Ans. (c) :** Given, $\ell$ = 5 m
w =20 kN/m
**Solution–** Slope for cantilever beam carrying UVL from free end to fixed end is;

$$\theta = \frac{w\ell^3}{24EI}$$

Where,
w = load intensity
$\ell$ = Span of beam
E = young modulus of elasticity
I =Moment of inertia

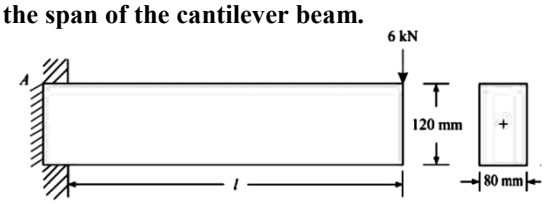
• $\theta = \frac{20\times(5)^3}{24EI} = \frac{104.16}{EI}$ radian

**11. Consider the given statements with respect to the conjugate beam method and select the correct answer.**
**Statement A : Deflection at any section of an actual beam is equal to shear force at the corresponding section.**
**Statement B : Loading system at any section of an actual beam is equal to $\frac{M}{EI}$ diagram at the corresponding section.**

(a) Statement A is correct, but B is incorrect
(b) Statement B is correct, but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) :** Conjugate beam is an imaginary beam of length equal to that of the original beam but for which the load diagram is the $\frac{M}{EI}$ diagram. That's means. The load at any point on the conjugate beam is equal to BM at that point divided by EI.
• The slope at any section of real beam is equal to the shear force at the corresponding section of the conjugate beam.
• The deflection at any section of real beam is equal to the bending moment at the corresponding section of the conjugate beam.

**12. A cantilever beam with a rectangular cross-section having section modulus 192 × $10^3$ $mm^3$ is shown in the figure. If the cantilever is subjected to a point load of 6 kN at the free and the bending stress is not exceed 50 MPa, find the span of the cantilever beam.**




(a) 1.2 m (b) 1.6 m
(c) 0.8 m (d) 2.4 m

**Ans. (b) :** Given,
Bending stress($\sigma$) = 50 MPa
Depth of neutral axis, $y = \frac{120}{2} = 60$ mm, F = 5kN
b = 80 mm, d = 120 mm
By flexural formula–

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

$$\frac{50}{60} = \frac{M}{\frac{80\times(120)^3}{12}} \Rightarrow M = 9.6\times10^6 \text{ N.mm}$$

Moment due to point load = Bending moment
$6 \times 10^3 \times \ell = 9.6 \times 10^6$
$\ell = 1.6 \times 10^3$ mm or 1.6 m

**13. A circular beam of 100 mm diameter is subjected to a shear force of 25 kN. Calculate the value of maximum shear stress**

(a) $\frac{10}{\pi}$MPa (b) $\frac{13.3}{\pi}$MPa

(c) $\frac{15}{\pi}$MPa (d) $\frac{18.5}{\pi}$MPa

**Ans. (b) :** Given, D = 100 mm
Force = 25 kN
Solution
Average shear stress $\tau_{avg} = \frac{\text{Shear force}}{\text{area}}$

$$\tau_{avg} = \frac{25\times10^3}{\frac{\pi}{4}\times(100)^3} = \frac{10}{\pi}\text{MPa}$$

We know that,

$$\tau_{max} = \frac{4}{3}\tau_{avg}$$

$$\tau_{max} = \frac{4}{3} \times \frac{10}{\pi} \text{MPa} = \frac{13.3}{\pi} \text{MPa}$$

**14. A circular shaft of 20 mm diameter is required to transmit torque from one shaft to another. Find the safe torque, which the shaft can transmit, if the shear stress is not to exceed 30 MPa.**

(a) 12000 π N-mm (b) 18000 π N-mm
(c) 15000 π N-mm (d) 20000 π N-mm

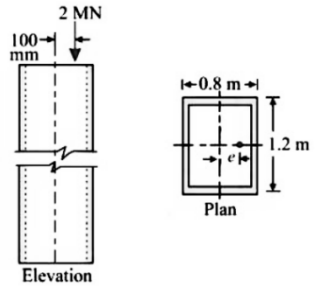
**Ans. (c) :** Given,
diameter of circular shaft = 20 mm
τ = 30 MPa
Using the relation $\frac{T}{J} = \frac{\tau}{R}$
Where,
T = Torque applied to the shaft
J = Polar moment of inertia = $\frac{\pi}{32} \times D^4$
τ = Shear stress in material
R = Radius of shaft = $\frac{D}{2} = \frac{20}{2} = 10$ mm

$$\frac{T}{J} = \frac{\tau}{R}$$

$$\frac{T}{\frac{\pi}{32} \times (20)^4} = \frac{30}{10}$$

T = 15000 π N-mm.

**15. A hollow rectangular masonry pier, 1.2m × 0.8 m and 150 mm thick, carries a vertical load of 2 MN in the vertical plane bisecting 1.2 m side at an eccentricity of 100 mm from the geometric axis of the section shown in the given figure. Calculate the maximum stress intensity in the section if the section if the section modulus of the given section is $90.5 \times 10^6$ mm$^3$.**




(a) 1.71 MPa (b) 6.13 MPa
(c) 3.42 MPa (d) 8.56 MPa

**Ans. (b) :** Given,
B = 0.8, m = 800 mm
D = 1.2 m = 1200 mm
t = 150 mm
Inner width b = B – 2t
= 800 – 2 × 150 = 500 mm
Inner depth (d) = D – 2t
= 1200 – 2 × 150
= 900 mm
Area of wall A = B × D – b × d
= 800 × 1200 – 500 × 900 = 510000 mm$^2$
Given,
eccentric load P = $2 \times 10^6$ N
e = 100 mm

$$\frac{I}{y} = 90.510^6 \text{ mm}^3$$

$$M = P.e. = 2 \times 10^6 \times 100$$

$= 2 \times 10^8$ N-mm
Bending stress

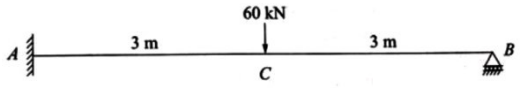
$$\sigma_b = \frac{M}{I} \times y = \frac{2 \times 10^8}{90.5 \times 10^6} = 2.2099 \text{ N/mm}^2$$

Direct stress

$$\sigma_d = \frac{P}{A} = \frac{2 \times 10^6}{510000} = 3.921 \text{ N/mm}^2$$

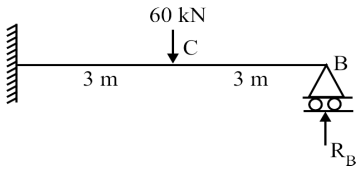
Maximum stress = $\sigma_d + \sigma_b = 3.921 + 2.209$
= 6.130 N/mm$^2$

**16. A propped cantilever beam is subjected to a point load of magnitude 60 kN at its midspan shown in the given figure.**
**Calculate the bending moment at C.**




(a) 42.50 kN-m (b) 56.25 kN-m
(c) 34.75 kN-m (d) 67.50 kN-m

**Ans. (b) :**

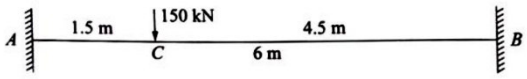



Given,
Length of beam = 6 m
Point load = 60 kN
Propped reaction $R_B = \frac{5}{16} W$

$$= \frac{5}{16} \times 60 = 18.75 \text{ kN}$$

BM at centre C = $R_B \times 3 = 18.75 \times 3 = 56.25$ kN

**17. A fixed beam of span 6 m is subjected to a point load of magnitude 150 kN as shown in the figure. identify the correct compatibility condition by considering the change of slope between the end supports is zero (which gives the area of $\frac{M}{EI}$ diagram zero). Where $R_B$ is vertical reaction at B and $M_B$ is the moment at B.**




(a) 72 $R_B$ – 18 $M_B$ = 928.13

(b) $26\ R_B - 12\ M_B = 244.25$
(c) $18\ R_B - 6\ M_B = 168.75$
(d) $36\ R_B - 12\ M_B = 244.25$

**Ans. (c) :**

$$M_A = \frac{wab^2}{\ell^2} = \frac{150 \times 1.5 \times 4.5^2}{6^2} = 126.5625 \text{ kN-m}$$

$$M_B = \frac{wba^2}{\ell^2} = \frac{150 \times 4.5 \times 1.5^2}{6^2} = 42.1875 \text{ kN-m}$$

$$\varepsilon M_A = 0$$

$$150 \times 1.5 - 126.56 + 42.1875 = R_B \times 6$$

$$R_B = 23.4374 \text{ kN}$$

$$18\ R_B - 6\ M_B = 168.75$$

$$18 \times 23.4379 - 6 \times 42.1875 = 168.715$$

**18. According to Castigliano's theorem, the displacement at the point of application of the load, in its own line of action, will be obtained by evaluating the ____ of the structure with respect to the applied load.**

(a) first partial derivative of the total internal strain energy
(b) initial lack of fit
(c) second partial derivative of deflection
(d) first partial derivative of deflection

**Ans. (a) :** According to Castigliano's theorem, the displacement at the point of application of the load, in its own line of action, will be obtained by evaluating the first partial derivative of the total internal strain energy of the structure with respect to the applied load.

**Castigliono's first theorem–** It is a type of force method and it states that the first partial derivative of the total internal energy in a structure, with respect to any particular deflection component at a point is equal to the force applied at that point and in the direction corresponding to that deflection component.

$$\Delta = \frac{\partial u}{\partial f}$$

$$U = \sum_{i=1}^{n} \frac{p_i^2 dx}{2AE}$$

**19. Consider the given statements with respect to the slope-deflection method and identify whether the statements are correct.**

**Statements -**

**(A) All internal reaction components are determined from the force displacement equations, by knowing the independent displacement components.**

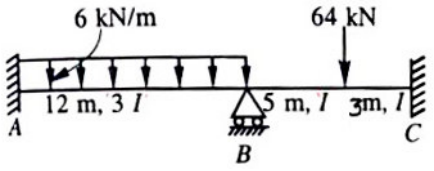
**(B) Internal reaction components are expressed as functions of the independent displacement components.**

(a) Statement A is correct and B is incorrect
(b) Statement B is correct and A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (c) :** With respect to the slope-deflection method–

- All internal reaction components are determined from the force displacement equations, by knowing the independent displacement components.
- Internal reaction components are expressed as functions of the independent displacement components.

**20. A condition beam, as shown in the given figure, is to be analyzed by using the moment distribution method. What is the distribution factor to be applied at 'B' for the member 'BA'?**




(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{4}{3}$ (d) $\frac{5}{3}$

**Ans. (b) :** According to figure-

$$DF = \frac{k_i}{\sum k_i}$$

$$= \frac{\frac{4E(3I)}{12}}{\frac{4E(3I)}{12} + \frac{4EI}{8}}$$

$$= \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{8}}$$

$$DF = \frac{2}{3}$$

**21. Consider the given statements with respect to Kani's method used for the analysis of multi-storied frames and identify whether the statements are correct.**

**Statements:**

**(A) In Kani's method, the unbalanced joint moments are distributed.**

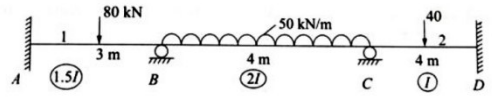
**(B) kani's method has been developed on the basis of the slope deflection method with a different from of moment distribution.**

(a) Statement A is correct and B is incorrect
(b) Statement B is correct and A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) :** With respect to Kani's method used for the analysis of multi-storied frames the kani's method has been developed on the basis of the slope deflection method with a different from of moment distribution.

- In Kani's method, the balanced joint moments are distributed.

**22. Find the rotation factor for member 'BA' which is required to analyze a continuous beam, as shown in the figure, by using Kani's method**




(a) –0.25 (b) –0.50
(c) 0.50 (d) –0.75

**Ans. (a) :** Distribution factor by Kani method

$$DF = -0.5\left[\frac{ki}{\sum ki}\right]$$

$$= 0.5\left[\frac{\frac{4E(1.5)I}{3}}{\frac{4EI(1.5)}{3}+\frac{4E(2EI)}{4}}\right]$$

$$= 0.5\left[\frac{2EI}{2EI+2EI}\right]$$

$$= 0.5\left[\frac{1}{2}\right] = -0.25$$

**23. Consider the given statements with respect to two and three hinged arches and identify whether the statements are correct.**
**Statements :**

**(A) The horizontal thrust is very much affected in two-hinged arches due to change in temperature when compared to three-hinged arches.**

**(B) From the deflection point of view, a three-hinged arch for a given span is much stiffer then that for two-hinged arches.**

(a) Statement A is correct and B is incorrect
(b) Statement B is correct and A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

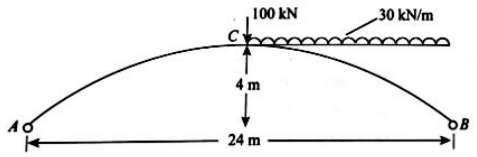
**Ans. (a) :** With respect to two and three hinged arches the horizontal thrust is very much affected in two-hinged arches due to change in temperature when compared to three-hinged arches.

**24. A two hinged parabolic arch will span 24 m is subjected to a uniformly distributed load of intensity 0.5 kN/m. Calculate the bending moment at its mid span (crown).**

(a) 0 kN/m (b) 36 kN/m
(c) 6 kN/m (d) 12 kN/m

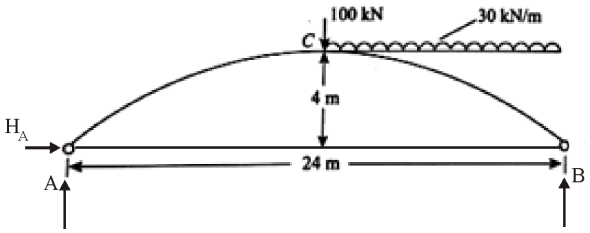
**Ans. (a) :** Bending moment at it's mid span = 0
• If two - hinged parabolic is subjected to uniformly distributed load of intensity q per unit length.
The bending moment at anywhere in the arch is zero.

**25. A two hinged parabolic arch of span 24 m with a central rise of 4 m is subjected to a point load and a uniformly distributed load as shown in the figure. Calculate the ratio of the vertical support reaction at 'B' that of 'A'.**




(a) 1.825 (b) 2.285
(c) 2.625 (d) 3.420

**Ans. (b) :**




$H_A + H_B = 0$
$R_A + R_B = 30 \times 12 + 100$
$R_A + R_B = 360 + 100 = 460$
$R_B \times 24 = 100 \times 12 + 30 \times 12 \times 18$
$R_B = 50 + 15 \times 18$
$= 270 + 50$
$R_B = 320,\ R_A = 140$
$\frac{R_B}{R_A} = \frac{320}{140} = 2.285$

**26. Consider the given statements with respect to the buckling of columns and select the correct answer.**
**Statement A : Local buckling in steel columns can be prevented by providing suitable width-to-thickness ratios to the compression elements.**
**Statement B : Flexural torsional buckling failure in steel columns can occur in those sections with one axis of symmetry and also sections with both axis of symmetry.**

(a) Statement A is correct, But B is incorrect
(b) Statement B is correct, But A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (a) :** Statement A is correct, but B is incorrect.
**Local bulling–** It occurs by buckling of one ore more individual plate elements like flange or web with no overall deflection is called local buckling. Plate elements of cross section may buckle locally due to compressive stresses.
• It can be avoided before the limit state is achieved by limiting the width to thickness ratio of each element of a cross-section subjected to compressive due to axial force, moment or shear.
**Flexural - torsional buckling–** This type of buckling only occurs in compression members that have unsymmetrical cross-section with one axis of symmetry. Flexural-torsional buckling is the simultaneous bending and twisting of a member. This mostly occurs in channels, structural tees double-angle shapes, and equal leg single angles.

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**27. For bolted gusset base, the thickness of the gusset plate should NOT be less than____**

(a) 16 mm (b) 18 mm
(c) 20 mm (d) 22 mm

**Ans. (a) :** For bolted gusset base, the thickness of gusset plate should not be less than 16 mm in thickness and gusset angles should be of unequal legs in which the vertical leg should accommodate two rows of bolts and one row of bolts in horizontal leg.

**28. As per IS 456 : 2000, the minimum area of flexural reinforcement in RCC beam of width 200 mm and effective depth 425 mm shall not be less than ____ Use M 20 grade concrete and Fe 500 steel.**

(a) 144.5 $mm^2$ (b) 102.5 $mm^2$
(c) 202 $mm^2$ (d) 240 $mm^2$

**Ans. (d) :** Given,
In R.C.C. beam
b = 200 mm, d=425 mm, grade of concrete = M20, steel $F_c$ 500 .
According to IS 456 : 2000–

Minimum tension reinforcement $\frac{A_{st}}{bd} = \frac{0.85}{f_y}$

Where,
$A_{st}$ = area of steel in tension/flexure
b = width of beam
d = effective depth of beam
$f_y$ = grade of steel

$$\frac{A_{st}}{200 \times 425} = \frac{0.85}{500} \Rightarrow A_{st} = 144.5 \text{ mm}^2$$

**29. Find the effective width of flange in a RCC T-beam, if the distance between points of zero moments in the beam is 3.6 m, breadth of web 200 mm and thickness of flange 150 mm.**

(a) 1.3 m (b) 1.7 m
(c) 2.1 m (d) 2.5 m

**Ans. (b) :** Given, In RCC T-beam,
$\ell_o$= 3.6 m, $b_w$ = 200 mm, $D_f$ = 150 mm
To find– Effective width of flange $b_f$ = ?
According to IS 456 : 2000–
Clause 23.1.2. Effective width of flange–

(a) For T-beam, $b_f = \frac{\ell_0}{6} + b_w + 6D_f$

Where, $b_f$ = Effective width of flange, $\ell_o$ = Distance between points of zero moments, $b_w$ = breadth of web, $D_f$ = thickness of flange.

(b) $b_f = \frac{3.6}{6} + 0.2 + 6 \times 0.15$

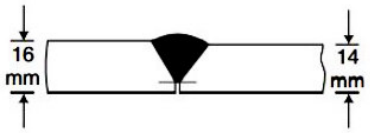
$b_f$ = 1.7 m

**30. The maximum compressive strain in concrete at limit state of collapse in compression to be considered for the design of axially loaded short column is _____**

(a) 0.002 (b) 0.0035
(c) 0.02 (d) 0.035

**Ans. (a) :** According to IS 456 : 2000
Clause 38.1- (a) The maximum compressive strain in concrete in axial compression is taken as 0.002.

**31. Calculate the strength of the weld if two plates of 16 mm and 14 mm thickness are to be joined by a single-V groove weld as shown in the figure. Due to some reasons the effective length of the weld that could be provided was 175 mm only. Take partial factor of safety 1.25 and grade of steel Fe 410.**




**Options :**

(a) 306.25 kN (b) 224.62 kN
(c) 272.89 kN (d) 378.98 kN

**Ans. (a) :** $t_t = \frac{5}{8} \times 14 = 8.75$

Given, Effective length of weld = 175 mm
Effective area of weld effective = 18.75 × 175
= 1531.25 $mm^2$

Strength of weld = $t \times \ell_{eff} \times \frac{f_y}{\gamma}$

$$= 8.75 \times 175 \times \frac{250}{1.25 \times 1000} = 306.25 \text{ kN}$$

**32. Which of the following is NOT the factor used to calculate the effective net area of tension members?**

(a) Ductility factor (b) Geometric factor
(c) Shear lag factor (d) Imperfection factor

**Ans. (d) :** Imperfection factor is not the factor used to calculate the effective net area of tension members.

- A concrete almost always weakness the member affecting its efficiency. The net area determined for plates and flats and the other section is, therefore, modified to account for factors such as ductility of the material, method of fabrication, the geometry factor, and the shear lag effect that influence the effectiveness of the member.
- All these factors reduce the effectiveness of the member, but the shear lag is the most important. The modification net area is called effective net area and may be defined by the following equation–

Anet =$K_1$ $K_2$ $K_3$ $K_4$ $A_n$
Where,
$K_1$ = ductility factor
$K_2$ = hole forming factor
$K_3$ = geometry factor
$K_4$=shear lag factor and $A_n$ is the net area of the section.

**33. Consider the given statements will respect to influence lines and select the correct answer.**

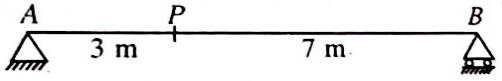
**Statement A: An influence line represents the variation of deflection at a specific point in a member as a unit concentrated force moves over the member.**

**Statements B : An influence line represents the variation of either the reaction, shear force and bending moment at a specific point in a member as a unit concentrated force moves over the member.**

(a) Statement A is correct, but B is incorrect
(b) Statement B is correct, but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

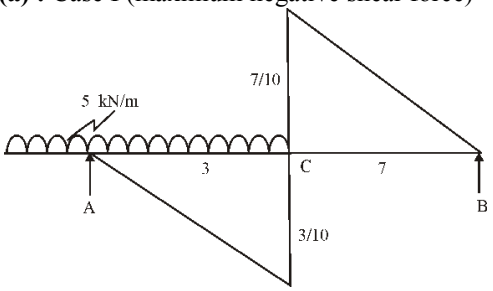
**Ans. (c):**

**34. A simply supported beam of span 10 m is shown in the given figure. Calculate the maximum positive and negative shear, respectively, at P, when a uniformly distributed rolling load of intensity 5 kN/m spread over a length of 6 m traverse the span from left to right (A to B).**




(a) 12 kN, –2.25 kN (b) 10 kN, –4.25 kN
(c) 8 kN, –6.25 kN (d) 9 kN, –5.25 kN

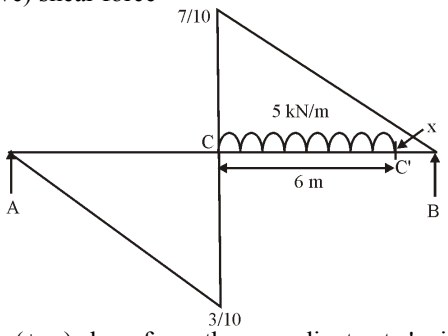
**Ans. (a) :** Case I (maximum negative shear force)




max(-ve) shear fore $\Rightarrow$ Area involved to A–C $\times$ intensity of load

$$-\frac{\frac{3}{10}+0}{2}\times 3\times 5 = -2.25 \text{ kN}$$

Case II
Max (+ve) shear force




For max (+ve) shear force the co-ordinate at c' will be

$$\frac{7/10}{7}=\frac{x}{1},\ x=\frac{1}{10}$$

So the max (+ve) shear force = Area enclosed cc' $\times$ intensity of load

$$=\frac{7/10+1/10}{2}\times 6\times 5$$
$$= 12 \text{ kN}$$

**35. A post-tensioned concrete beam, 100 mm wide and 300 mm deep, is pre-stressed by three cables, each with a cross-sectional area of 50 $mm^2$ and with initial stress of 1200 $N/m^2$. All three cables are straight and located 100 mm from the soffit of the beam. If the modular ratio is 6, calculate the loss of pre-stress in the three cables due to elastic deformation of concrete under simultaneous tensioning and anchoring of all the three cables.**

(a) 6.8 $N/mm^2$ (b) 16.2 $N/mm^2$
(c) 0 $N/mm^2$ (d) 22.56 $N/mm^2$

**Ans. (c) :** In post tensional number no elastic shortening loss if cable are stretched and anchored simultaneously.

**36. Consider the given statements with respect to basic assumptions made in strain compatibility method of estimating flexural strength of pre-tensioned concrete members. identify whether the statements are correct.**

**Statements :**

**(A) The distribution of stress in concrete in linear.**
**(B) The resistance of concrete in tension is neglected.**

(a) Statement A is correct and B is incorrect.
(b) Statement B is correct and A is incorrect.
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (c) : Both statement are correct–** The distribution of concrete strain is linear (Plane section normal to axis remains plane even after bending) the basic theory is applicable to all structural concrete sections whether reinforced or pre-stressed and some assumption are made. .

- The resistance of concrete in tension in neglected in strain compatibility method at the end of the section, the maximum compressive strain in concrete at failure reaches a particular values are the assumptions made in strain compatibility method.

**37. Which of the following laboratory tests is NOT used to determine the shearing resistance of soil?**

(a) Unconfined compression test
(b) Vane shear test
(c) Direct shear test
(d) Recuperation test

**Ans. (d) :** Recuperation test is not used to determine the shearing resistance of soil.
To test the shear strength of soil following types of test is performed.
(i) Direct shear test
(ii) Tri axial shear test
(iii) Un confined compression test
(iv) Vane shear test

**38. An excavation is made with a vertical face in clay soil, which has mobilized unit cohesion ($C_u$) = 50 kN/m$^2$. Unit weight of soil ($\gamma$) = 18 kN/m$^3$. The maximum depth of excavation shall be_____ so that it is stable. (consider the Taylor's stability number as 0.26).**

(a) 10.64 (b) 12.62
(c) 15.67 (d) 18.69

**Ans. (a) :** $C_u = 50$ kN/m$^2$, $\gamma = 18$ kN/m$^3$
Sn = 0.0261, $\mu_c$ = ?

Taylor is stability number, $S_n = \frac{C_u}{\gamma H_c}$

Where, $C_u$ = cohesion, $\gamma$ = unit weight of soil $H_C$ = safe height.

Column, $0.261 = \frac{50}{18 \times H_c}$

$H_c$ = 10.64 m

**Note–** Taylers stability number is based on the friction circle analysis.
It is valid for homogenous, finite slapes and no seepage conditions.

**39. A masonry dam has pervious sand as the foundation. Calculate the maximum permissible upward gradient (rounded up to 2 decimals after point), if the factor of safety 4 is required against boiling of sand. (Take porosity (n) = 45% and specific gravity (G) of sand = 2.65.**

(a) 0.23 (b) 0.35
(c) 0.55 (d) 0.65

**Ans. (a) :** Given, G = 2.65, n = 45%, FOS = 4
Solution
FOS against boiling, FOS =

$$\frac{\text{critical hydraulic gradient}(i_{cr})}{\text{Permissible hydraulic gradient}(i)}$$

$$i_{cr} = \frac{G-1}{1+e}$$

Where, $i_{cr}$ = critical hydraulic gradient
i = existing gradient
G = specific gravity, e = void ratio

n = 45% = 0.45

$$e = \frac{n}{1-n} = \frac{0.45}{1-0.45}$$

e = 0.82

$$i_{cr} = \frac{G-1}{1+e} = \frac{2.65-1}{1+0.82}$$

$i_{cr} = 0.91$

FOS against boiling, FOS =

$$\frac{\text{critical hydraulic gradient}(i_{cr})}{\text{Permissible hydraulic gradient}(i)}$$

$$4 = \frac{0.91}{i}$$

$$i = \frac{0.91}{4} = 0.2275$$

$i \approx 0.23$

**40. The shearing strength when compacted and saturated for soil group GW (Well graded gravel) is _____.**

(a) excellent (b) good
(c) fair (d) poor

**Ans. (a) :** The shearing strength when compacted and saturated for soil group GW (well graded gravel) is excellent.

**41. The activity value of mineral Na-montmorillonite ranges between _____**

(a) 0.2–0.6 (b) 0.6–1.1
(c) 1.5–3 (d) 4–7

**Ans. (d) :** The activity value of mineral Na-Montmorillonite range between 4-7.
Activity, that is represents the compressibility of volume change in soil with a change in water content.

- Activity defined as the ratio of plasticity index ($I_P$) of soil to the percentage of particles in it having a size less than 2 $\mu$ (Clay size).

$$A_f = \frac{I_P}{\%C}$$

Where, $A_f$ = Activity of clay
$I_p$ = Plasticity of index
%C = % of clay size particle

| Mineral | Activity |
|---|---|
| Na – Montmorillonite | 4–7 |
| Ca – Montmorillonite | 1.5 |
| Na – illite | 0.5 – 1.3 |
| Na – kaolinite | 0.3 – 0.5 |

**42. Sensitivity values for 'sensitive clays' lies in the range of _____**

(a) 4-8 (b) 2-1
(c) 9-12 (d) >13

**Ans. (a) : Sensitivity of clay–** The ratio of unconfined compressive strength of the natural or undisturbed soil to the unconfined compressive strength of soil in a remoulded state, without change in the water content, is known as sensitivity.

$$S_f = \frac{(q_u)_{undisturbed}}{(q_u)_{remoulded}}$$

$q_u$ = 2 Cu where Cu = cohesion of pure clay
Sensitivity term is used only for cohesive soil.

| Sensitivity | %water content | Consistency |
|---|---|---|
| 1-4 | Normal | Gravel, coarse sand |
| 4-8 | Sensitive | sand |
| 8-15 | Extra sensitive | Flocculent structure soil |
| >15 | Quick | Fine clay |

**43. The plasticity Index for low plastic clays is :**

(a) 8–11 (b) > 15
(c) < 7 (d) 11 – 15

**Ans. (c):** The plasticity index for low plastic clay is < 7.
**Plasticity index ($I_P$) :** It is the range of moisture content over which a soil exhibits plasticity.

$$I_P = W_L - W_P$$

Where, $W_L$ = water content at liquid limit
$W_P$ = water content at plastic limit
If $w_p \geq wL$ , $I_P$ is zero.
Value of $I_P$ can not be (-ve).

| Ip | Type of clay |
|---|---|
| 0 | No plastic |
| 0-7 | Low plastic clay |
| 7-17 | Medium plastic clay |
| > 17 | High plastic clay |

**44. The general shear failure of soil is characterized by**
(a) a well-defined failure pattern
(b) a poorly defined shear plane
(c) slip surface not visible
(d) the load-settlement curve is continuous

**Ans. (a) :** The general shear failure of soil is characterized by a well-defined failure pattern.
There are three modes of shear failure of soil–
(i) General shear failure
(ii) Local shear failure
(iii) Punching shear failure
Depending upon the compressibility of soil & depth of fasting with respect to it's breadth (D/B ratio).
When the ultimate bearing capacity of the soil is reached, It may fail in any one of the failure above mentioned.
**(i) General shear failure–** In this made a slight downward movement of the footing develops fully plastic zones and a sudden failure takes with a considerable bulging of the ground surface adjacent to the footing.
- Characterized by well defined failure pattern, consisting of a wedge and slip surface and bulging (leaving) of soil surface adjacent to the footing.
- It occurs in case of dense sand or stiff cohesive soil supporting the footing.
- Failure load is well defined in this case.
- The load settlement diagram is similar to stress-strain for dense sand or over. Consolidated clay as shown.
- The ultimate load in well defined on this cause as shown typically in figure given below.

**45. The coefficient of permeability is :**
(a) proportional to the effective particle size
(b) proportional to the square of an effective particle size
(c) proportional to the cube of an effective particle size
(d) proportional to the square root of an effective particle size

**Ans. (b) :** Coefficient of permeability (k) is proportional to the square of an effective particle size.

$$K \propto D_{10}^2$$

It is also directly proportion to void ratio, $\gamma_w$(unit weight of water) and intentionally proportional to $\mu$ (Dynamic viscosity constant).

**46. What is the standard temperature range and duration to be maintained in an oven drying method for determination of water content of soil sample?**
(a) 24 hours at a temperature of 105-110$^0$C
(b) 10 hours at a temperature of 75-80$^0$C
(c) 12 hours at a temperature of 85-90$^0$C
(d) 5 hours at a temperature of 95-100$^0$C

**Ans. (a) :** Oven drying method –
Simplest and most accurate method.
Soil sample is dried in a controlled temperature (IOS-110$^0$C) for 24 hrs.
Water content is calculated as–

$$w = \frac{w_2 - w_3}{w_3 - w_1} \times 100$$

Where, $w_1$= weight of contains
$w_2$ = weight of contains + moist sample
$w_3$ = weight of contains + dried sample
Weight of water = $w_2 - w_3$
weight of solids = $w_3 - w_1$

**47. The method used to determine in situ unit weight of a natural soil deposit is_____**
(a) oven dry method
(b) core-cutter method
(c) hydrometer analysis
(d) sand-bath method

**Ans. (b) : Core-cutter method–** Used in cohesive soil
- Can't be used in case of hard and gravelly soil.
- This method consists of driving a core-cutter (volume = $10^3$cc) in to the soil and removing it, the cutter filled with soil is weighed. Volume of cutter is also known from its dimensions and in-situ unit-weight is obtained by dividing soil weight by volume of cutter.

$$\gamma = \frac{w}{V}$$

**48. The group symbol 'SM' in Unified Soil classification chart refers to :**
(a) Clayey sand, poorly graded sand-clay mixtures
(b) Silty sand, poorly graded sand-silt mixtures
(c) well graded sand
(d) well graded gravel

**Ans. (b) :** The group symbol 'SM' in unified soil classified chart refers to silty sand, poorly graded sand-silt mixture.

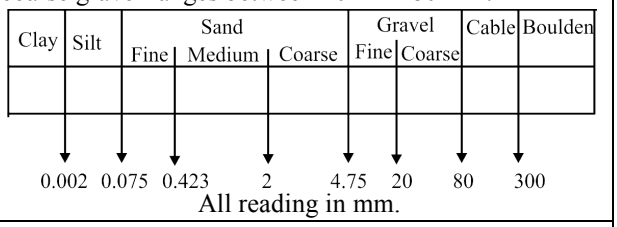
| Soil type | Profile | Subgroup | Suffix |
|---|---|---|---|
| Gravel | G | well graded | W |
| Sand | S | poorly graded | P |
| Silt | M | silty | M |
| Clay | C | clayey | C |

**49. The permeability of soil decreases sharply with:**
(a) increase in water content on dry side of optimum
(b) decrease in water content on dry side of optimum
(c) increases in water content on wet side of optimum
(d) decrease in water content on wet side of optimum

**Ans. (a):** The permeability of soil decreases shapely with increase in water content on dry side of optimum.
Factor affecting permeability–
- Permeability of soil directly proportional to the particle size ($D_{10}$), unit weight of water ($\gamma_w$), void ratio (e), temperature.
- Permeability of soil intestinally proportional to the viscosity of water ($\mu$).

**50. The grain size distribution of soil fraction 'Coarse gravel' ranges between_____**
(a) 20 mm – 80 mm
(b) 0.425 mm – 2.00 mm
(c) 2.00 mm – 4.75 mm
(d) 3.67 mm – 6.3 mm

**Ans. (a) :** The grain size distribution of soil fraction coarse gravel ranges between 20 mm- 80 mm.



All reading in mm.

**51. As per IS soil classification, which of the following symbols is used for poorly graded sands?**
(a) GP (b) SP
(c) MH (d) SC

**Ans. (b) :** SP – poorly graded sand.
GP – poorly graded gravel
SC – Clayey sand

**52. According to Terzaghi, the shape factor with cohesion used to calculate the safe bearing capacity of soil for square footing if _____**
(a) 1.1 (b) 0.9
(c) 1.3 (d) 1.2

**Ans. (c) :** According to Terzaghi, the shape factor with cohesion used to calculate the safe bearing capacity of soil for square footing if 1.3.
For square footing,
$q_u = 1.3\ CN_C + q(q-1)+0.4b\gamma N_\gamma$

**53. The coefficient of earth pressure at rest for dense sand may lie in the range of _____**
(a) 0.40–0.45 (b) 1.0–1.5
(c) 0.8– 1.2 (d) 1.5–1.8

**Ans. (a) :**

| **Soil** | **$k_0$** |
|---|---|
| Dense sand | 0.40 to 0.45 |
| Loose sand | 0.45 to 0.50 |
| NCC | 0.50 to 0.60 |
| OCC | 1.0 to 4.0 |

**54. If the void ratio of a soil sample is 0.5, then the porosity is _____**
(a) 0.25 (b) 0.33
(c) 0.5 (d) 1

**Ans. (d) :** Given, e = 0.5, n = ?
We know that, Porosity (n) = $\frac{e}{1+e}$
Where, e = void ratio
$$n = \frac{0.5}{1+0.5} = 0.33$$

**55. Which of the following is NOT an application of Reynold's number?**
(a) Airplanes flying in air
(b) Flow of water in canals
(c) Ground improvement technique
(d) Flow of water on spillway

**Ans. (c) : Application of Reynold's number–** The reynolds number is used to study fluid as they flow, It determines whether a fluid is laminar and turbulent. It is used when modeling the movement of organisms. Swimming through water. It is based on various factors like velocity, pressure, heat temperature.
$$Re = \frac{\text{Inertia force}}{\text{Viscous force}} = \frac{\rho V d}{\mu}$$

| **Flow condition** | **Pipe flow** | **Open channel flow** |
|---|---|---|
| Laminar flow | $R_e \le 2000$ | $R_e \le 500$ |
| Transition flow | $2000 < R_e < 7000$ | $500 < R_e < 1000$ |
| Turbulent flow | $R_e > 7000$ | $R_e > 1000$ |

**56. The pressure represented by a column of 9 cm of water is ____. (Consider that specific weight of water = 9790 N/m$^3$)**
(a) 9790 N/m$^2$ (b) 881.1 N/m$^2$
(c) 979.1 N/m$^2$ (d) 1000 N/m$^2$

**Ans. (b) :** Given– (Coloumn height of water) h = 9 cm
w = 9790, P = ?
We know that
$w = \rho g$
Where, w = specific weight
$\rho$ = density
g = gravity
$9790 = \rho \times 9.81$
$\rho = 997.96$ kg/m$^3$
Pressure,
We know that,
$P = \rho g h$
$P = 997.98 \times 9.81 \times 0.09$
$P = 881.1$ N/m$^2$

**57. A 2.0 m wide Rectangular channel carries water for a depth of 1.0 m. The bed slope of the channel is 0.0036. The average shear stress on the boundary is _____ (consider that specific weight of water = 9790 N/m$^3$)**
(a) 22 Pa (b) 20.53 Pa
(c) 17.62 Pa (d) 15.26 Pa

**Ans. (c) :** Given,
Width of rectangular channel = 2m
Depth of rectangular channel = 1m

Bed slope =0.0036

$$R = \frac{A}{P} = \frac{B.y}{B+2y}$$

$$= \frac{2\times1}{2+2\times1} = \frac{2}{4} = 0.5 \text{ m}$$

$$\tau_0 = \gamma_R S_0$$

$$= 9810 \times 0.50 \times 0.0036$$

$\tau_0 = 17.62$ Pa

**58. A stationary hydraulic jump occurs in a rectangular channel with the initial and sequent depth being equal to 0.10 m and 1.10 m respectively. The estimated energy loss is ____**

(a) 2.27 m (b) 2 m
(c) 1.5 m (d) 1.1

**Ans. (a) :** Given,

$y_1 = 0.10$ m, $y_2 = 1.10$ m

Solution,

Energy loss in a hydraulic jump is given as–

$$E_L = \frac{\left(y^2 - y^1\right)^3}{4.y_1 y_2}$$

$$E_L = \frac{(1.10-0.10)^3}{4\times0.10\times1.10} = 2.27 \text{ m}$$

**59. The separation of boundary layer takes place with the pressure gradient is ____ and the velocity gradient is _____ respectively.**

(a) positive negative
(b) maximum, zero
(c) zero, maximum
(d) negative, positive

**Ans. (a) :** The separation of boundary layer takes place with the pressure gradient is positive and the velocity gradient is negative, respectively.

- A favourable pressure gradient is one in which the pressure decrease in the flow direction i.e.–

$$\frac{dp}{dx} < 0$$

- An adverse pressure gradient is our in which pressure increase in the flow direction.

i.e. $\frac{dp}{dx} > 0$

**60. The dimensional of specific weight is _____**

(a) $ML^{-2}T^{-2}$ (b) $ML^{-1}T^{-2}$
(c) $ML^{-1}T^{-1}$ (d) $ML^{-2}T^{-1}$

**Ans. (a) :** The dimension of specific weight is $ML^{-2}T^{-2}$.

$$\text{Specific weight} = \frac{\text{weight}}{\text{volume}} = \frac{\text{kg}\times\frac{\text{m}}{\text{s}^2}}{\frac{\text{m}^3}{1}} = \frac{\text{kg}}{\text{m}^2\text{s}^2}$$

Dimension of Specific weight = $[ML^{-2}T^{-2}]$

**61. The linear momentum equation is based on :**

(a) Newton's third law
(b) Newton's second law
(c) Bernoulli's equation
(d) continuity equation

**Ans. (b) :** Newton's second low is based on linear momentum equation.

- The momentum equation is a statement of Newton's second low and relates the sum of the forces acting on an element of fluid to its acceleration or rate of change of momentum.
- Bernoulli's equation is based on conservation of energy.
- Continuity equation is based on conservation of mass.

**62. The difference between the total head line and the hydraulic grade line in an open channel flow represents_____**

(a) the velocity head (b) the pressure head
(c) the elevation head (d) the total energy head

**Ans. (a) :** The difference between the total head line and the hydraulic grade line in an open channel flow represents the velocity head.

$$HGL = \frac{P}{\rho g} + z$$

$$TEL = \frac{P}{\rho g} + \frac{v^2}{2g} + z$$

$$TEL - HGL = \frac{v^2}{2g}$$

**63. Bourdon gauge measures___**

(a) gauge pressure
(b) atmospheric pressure
(c) Strain in solids
(d) absolute pressure

**Ans. (a) :** Bourdon gauge measures gauge pressure.

- Bourdon gauges are purely mechanical devices utilizing the mechanical deformation of a flattened but bent tube that winds or unwinds depending on the pressure deference between the inside and out side. The motion is against a spring torque such that a needle attached to the shaft indicates directly the pressure difference.

**64. Flow in an open channel is classified as turbulent if the Reynold's number of flow is__**

(a) less than 1200 (b) less than 600
(c) more than 1000 (d) more than 2000

**Ans. (d) :** For open channel flow reynold's number–

| **Type of flow** | **Reynolds number** |
|---|---|
| Laminar | less than 500 |
| Transition | 500-2000 |
| Turbulent | Greater than 2000 |

**65. Consider the given statements with respect to triangular notch. identify whether the statements are correct.**

**Statements :**

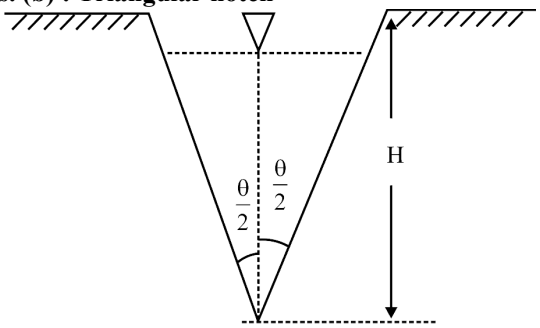
**(A) A triangular notch gives less accurate results than a rectangular notch for measuring low discharge.**

**(B) In case of triangular notch with known values of the angle of notch and co-efficient of**

**discharge, the height of water over notch is only data required for the computation of discharge.**

(a) Statement A is correct and B is incorrect.
(b) Statement B is correct and A is incorrect.
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (b) : Triangular notch–**




A V-notch weir is also called the triangular notch or weir. The discharge over a triangular weir or notch is given by the.

$$Q = \frac{8}{15} C_d \tan\frac{\theta}{2} \sqrt{2g}\ H^{5/2}$$

Where,
a = discharge of fluid, $C_d$ = Coefficient of discharge
θ = notch angle and H = height of water above the notch.
For measuring the low discharge a triangular notch gives more accurate results than a rectangular notch.

**66. Which of the following minor losses in pipe flow is taken as $\frac{v^2}{2g}$ ? (where V is velocity of liquid in pipe and g is acceleration due to gravity.**

(a) Loss of head at the exit of the pipe
(b) Loss of head at the entrance of a pipe with a sharp connect entrance
(c) Loss of head due to friction
(d) Loss of head due to obstruction in a pipe.

**Ans. (a) :** Loss of head at the exit of the pipe =

$$h_\ell = \frac{V^2}{2g}$$

Loss of head at entrance to pipe, $h_e = \frac{0.5V^2}{2g}$

Loss due to bends–

$$h_e = \frac{kv^2}{2g}$$

$k \approx 1.2$ for $90^0$ bend

$k \approx 0.4$ for $45^0$ bend

**67. When is Bernoulli's equation applicable between any two points?**

(a) In any type of rotational flow of a fluid
(b) In any type of irrotational flow of a fluid
(c) In steady rotational flow of an incopressible fluid
(d) In steady irrotational flow of an incompressible fluid

**Ans. (d) :** Bernoulli's equation is applicable for following type of condition–

- Fluid is ideal
- Flow is steady
- Flow is continuous
- Imcompressible
- Flow is alon-viscous & irrotational
- Bernoulli's equation in applicable along a stream line.

**68. As per IS 1130 : 1969, marble slabs shall be supplied with the thickness ranges between___.**

(a) 5 to 15 mm (b) 15 to 90 mm
(c) 20 to 150 mm (d) 75 to 210 mm

**Ans. (c) :** Marble is a regular and over used material for numerous buildings, monuments and sculptures. It is used for conventional & non conventional uses of colourful marble at the global level.
According to IS 1130 : 16969-
Size - Standard slab length = 70-250 cm
Width = 30 cm – 100 cm
Thickness = 20 – 150 mm

- Moisture absorption shall be 0.4% max according to IS 1124-1974 testing method.
- Hardness shall be 3 min with moh's scale testing method.
- Specific gravity shall be 2.5 with the IS 1122-1974 testing method.

**69. The specific gravity of structural granite when tested, according to IS 1121 (part I) : 1974 shall not be less than_____**

(a) 2.6 (b) 2.9
(c) 3.2 (d) 3.6

**Ans. (a) :** The specific gravity of structural granite tested, according to IS 1121 (Part 1) : 1974 shall not be less than 2.6.

- The compression strength when tested according to IS : 1121 (Part I) : 1974 shall not be less than 1000 kg/cm$^2$. The water absorption when tested according to IS : 1228-1974 shall not be more 0.50%.

**70. Consider the given statements with respect to seasoning of timber and select the correct answer.**
**Statement A : Seasoning of timber increase the strength, durability, and workability of timber.**
**Statements B : Seasoning of timber makes it difficult to paint to the surface of timber.**

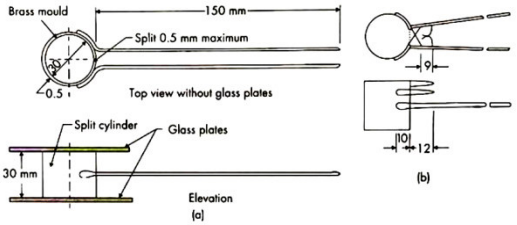
(a) Statement A is correct, but B is incorrect
(b) Statement B is correct, but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (a) : Seasoning of timber–** Seasoning allows timber to burn readily, if used as fuel.

- Seasoning decreases the weight of timber and thereby to lower the cost of transport and handling.

• Seasoning makes timber safe from the attack of fungi in set.
• Seasoning reduces the tendency of timber to crack, shrink and warp.
• Seasoning makes timber fit for receiving treatment of points, preservatives, varnishes.
• Seasoning imparts hardness, stiffness, strength and better electrical resistance to timber.

**71. Identify the given apparatus used for testing of cement.**




(a) vicat apparatus used to determine consistency test of cement
(b) Blaine's air permeability apparatus used to determine the fineness of cement
(c) Le - Chatelier's apparatus used to determine the soundness test of cement
(d) Vee-Bee test apparatus used to determine the permeability test of cement.

**Ans. (c) :** Le-Chatelier's apparatus used in determine the soundness test of cement.
• Expansion of Handle shows soundness, for OPC it is 10 mm. For PPC, RHC, IRST-40, its value remain.
• Le-chatelier's method is used to test unsoundness due to lime only.
• Autoclave method is used to test unsoundness due to lime and magnesia both.

**72. Which of the following properties of bitumen is determined using the ring and ball apparatus?**
(a) Viscosity of bitumen
(b) Softening point of bitumen
(c) Penetration resistance of bitumen
(d) Specific gravity of bitumen

**Ans. (b) : Softening print test of bitumen–** It is the temperature at which attains a particular degree of consistency. It is conducted by ring and ball apparatus.
• The temperature rise during the test is at the uniform rate of $5 \pm 0.5^0$C per minute.
• Viscosity test of bitumen measures the resistance to flow. Viscosity value is the time taken by50 ml bitumen to flow trough orifice of 10 mm at $23^0$C.
• Penetration test of bitumen is used to determine the grade of bitumen and also determines the hardness and softness of bitumen penetration value is determined in 1/10th of mm of std needle having weight 100 gm at $25^0$C.

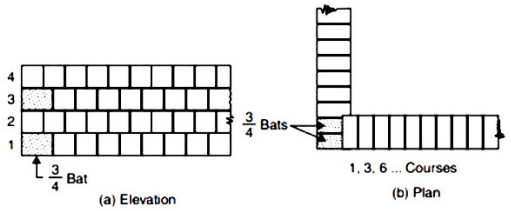
**73. Which of the following components of oil point makes the paint film opaque, possesses binding properties and reduces shrinkage cracks in the film on drying?**
(a) Adultant (b) Base
(c) Vehicle (d) Pigment

**Ans. (b) : Bass–** It is a metallic oxide which is the principal constituent of a paint. It make the paint film opaque and process binding properties which reduce the shrinkage crack in the film on drying.
E.g.– While lead, red led, zinc oxide, Iron oxide etc.
• Vehicle is a liquid that binds the pigment to the surface and protect pigment from decay l.g. linseed oil, dehydrated castor oil, etc.
• Pigments are the colour producing compound.

**74. A solution of resin or resinous substance dissolved in alcohol, turpentine or spirit is called_____**
(a) resin paint (b) varnish
(c) distemper (d) aluminium paint

**Ans. (b) :** A solution of resin or resinous substances dissolved in alcohol, turpentine or spirit is called varnish.
**Distemper –** They are made by mixing chalk, essential colour and water. They have no effect on the weather.
**Aluminum paint–** They are made by mixing fine aluminum powder with varnish. It is shiny and corrosion resistant.

**75. identify the type of brick bond used for brick masonry as shown in the figure.**




(a) stretcher bond
(b) header bond
(c) English bond
(d) single Flemish bond

**Ans. (b) : Header bond–** All the bricks are laid as header or full brick which is laid with it's width parallel to face the wall.
• The elevation of the wall laid in header bonds.
• Used for caused surface in work such as well lining or well foundation etc.

**76. According to SP 20 (S & T) 1991 'handbook on Masonry Design and construction, a horizontal force of____ percent of vertical loads need not be considered for elements of constructing that provide lateral stability to the structure as a whole.**
(a) 2.5 (b) 10
(c) 15 (d) 25

**Ans. (a) :** According to SP 20 (S & T) 1991 'handbook on Masonry Design and construction, a horizontal force of 2.5 percent of vertical loads need not be considered for elements of constructing that provide lateral stability to the structure as a whole.

**77. Which of the following type of concretes is manufactured under the trade name 'Wirand Concrete?**
(a) Fibre reinforced concrete
(b) Ready mix concrete

(c) Geopolymer concrete
(d) Light weight concrete

**Ans. (a) :** Fibre reinforced concrete is manufactured under the trade name 'Wirand concrete'

- Wirand is maccaferri steel fibric reinforcement for concrete. It acts directly on the concrete matrix. Improving it's mechanical properties ductility, structural strength, toughness and preventing cracking. Steel fibres enables the redistribution of tensile forces, increasing the flexibility of the concrete.

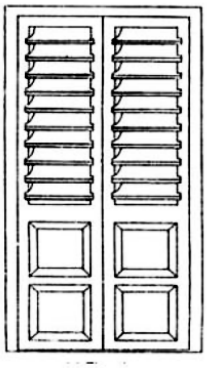
**78. Which of the following material consists of closely spaced wire meshes that are impregnated with rich cement mortar?**

(a) Ferrocement
(b) Shotcrete
(c) Self compacting concrete
(d) Geopolymer concrete

**Ans. (a) :** Ferrocement is a relatively new material consisting of wire meshes and cement (mortar. It consists of closely spaced wire meshes which are impregnated with rich cement mortar mix.

- Shotcrete is defined as mortar conweyed through a hose and pneumatically projected at a high velocity on to a surface.
- Geopolymer concrete is concrete based on an inorganic binder polymarized from AI-Si rich materials of geological or industrial origin, such as fly-ash, Geopolymer is used as the binder, instead of cement paste to produce concrete.

**79. Identify the type of door shown in the figure.**

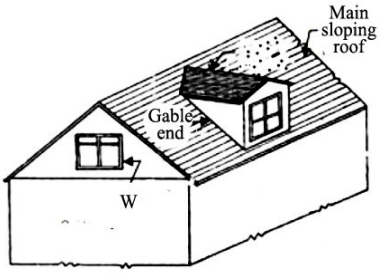


(a) Flush door
(b) Louvered door
(c) Battened, ledged and braced door
(d) Battened and ledged door

**Ans. (b):** Louvered door- Shutters are provided with louvers.

- Louverse are arranged in such a way that horizontal vision is obstructed.
- It allows face passage of air.

**80. Identify the type of window that is labeled 'W', as shown in the figure.**




(a) Dormer window (b) Gable window
(c) Bay window (d) Skylight windows

**Ans. (b) : Gable window–** Gable windows are provided for sloped roof buildings. These windows are provided at the gable end of slapped roof so, they are called gable window. They also improve the appearance of building.

**Dormer window–** Dormer windows are provided for sloped roofs these are projected from the sloping surface.

- They provide ventilation as well as lighting to the room. They also enhance aesthetic sense of room.

**81. Consider the given statements the given statements with respect to pointing and select the correct answer.**

**Statement A : Pointing consists of raking the joints to a depth of 10 mm to 20 mm and filling it with richer mortar mixes.**

**Statement B : Pointing is ideally suited for stone masonry because stones have attractive colours and good resistance to penetration by water.**

(a) Statement A is correct, but B is incorrect
(b) Statement B is correct, but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (c) :** Pointing consists of raking the joints to a depth of 10 mm to 20 mm and filling it with richer mortar mixes.

- Pointing is ideally suited for stone masonry because stones have attractive colours and good resistance to penetration by water.

**Note :** Finishing of mortar parts of either stone masonry or brick masonry is called pointing.

**82. Consider the given statements with escape to sources of information for writing specification and select the correct answer.**

**Statement A : The contract drawings prepared in the initial stage of construction is a main source of information for writing specification.**

**Statement B : For specifications of proprietary commodities, a reference can be made to the catalogues supplied by the manufacturer of that item.**

(a) Statement A is correct, but B is is incorrect
(b) Statement B is correct, but A is incorrect

(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (c) :** with escape to sources of information for writing specification–

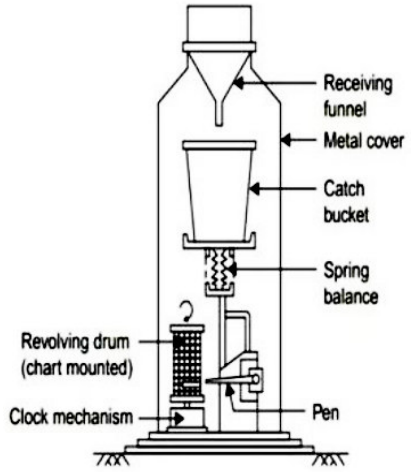
- The contract drawings prepared in the initial stage of construction is a main source of information for writing specification.
- For specifications of proprietary commodities, a reference can be made to the catalogues supplied by the manufacturer of that item.

**83. Consider the given statements with respect to PERT and CPM methods and select the correct answer.**
**Statement A: CPM was developed for planning, scheduling and control of civil works.**
**Statement B : The event-oriented networks using these time estimates for activity durations having uncertainties can be termed as a PERT network.**
(a) Statement A is correct but B is is incorrect
(b) Statement B is correct, but A is incorrect
(c) Both statements are correct
(d) Both statements are incorrect

**Ans. (c) :** With respect to PERT and CPM methods–

- CPM was developed for planning, scheduling and control of civil works.
- The event-oriented networks using these time estimates for activity durations having uncertainties can be termed as a PERT network.

**84. Identify the type of rain gauge shown in the given figure.**




(a) Symon's rain gauge
(b) Weighing bucket rain gauge
(c) Tipping bucket rain gauge
(d) Float type of rain gauge

**Ans. (b): Weighing bucket rain gauge–** This rain gauge weights the rain, which falls into bucket set on a platform of a spring or level balance. The increasing weight of bucket and it's counts are recorded on the chat held by a clock driven drum. The record shows the accumulation of precipitation with time in the shops of a mass curve of precipitation.

- The gage must be revised about one a weak when the clock is reward and the chart is replaced. For high rainfall, the recording mechanism reverse the direction of record immediately or reaching the upper edge of the recording chart.

**85. Which of the following scientists gave the runoff calculate formula 'R = 0.85P – 30.5' for the Ghat regions of western India?**
**Where R = runoff in cm, P = Rainfall in cm**
(a) Parker (b) Rohwer
(c) Ingis and desouza (d) Lacey

**Ans. (c) :** Inglis and derouza gave the runoff calculate formula, between annual runoff R in cm and annual rainfall P and cm as follows-

- **For Ghat region of western India-**

Highlands
$R = 0.85\,P - 30.5$

- For deccan plateau–

Plain curves –

$$R = \frac{1}{254}P(P - 17.8)$$

**86. In surface source of water, 'firm yield' is also known as_____**
**Where, yield is the amount of water that can be supplied from a reservoir in a specififed interval of time.**
(a) safe yield (b) secondary yield
(c) tertiary yield (d) average yield

**Ans. (a) :** Firm yield is also known as safe yield.
It is the maximum quantity of water that can be supplied from the reservoir with fall guarantee during the worst dry period.
**Secondary yield–** The quantity of water available more than safe yield is known as secondary yield.

- This yield is available during a period of high inflows.
- This secondary yield of the reservoir can be used either to generate extra hydroelectric power or for irrigation of extra lands.

**87. Which of the following types of sewer are used when sewers are carried over piers while crossing low lying areas**
(a) Brick sewers
(b) Cast iron sewers
(c) Asbestos cement sewers
(d) Lead sewers

**Ans. (b) : Cast iron sewers–** These pipes are stronger and capable to with stand greater tensile, compression, as well as bending stresses. However, these are costly. Cast iron pipes are used for outfall sewers, rising mains of pumping stations, and inverted siphons, where pipes are running under pressure. These are also suitable for sewers under heavy traffic load, such as such as sewer below railways and highways. They are used for carried over piers in case of low lying areas.

- These sewers are 100% lake proof sewers but shows less resistant to corrosions hence, generally lined from inside with cement concrete, coal far paint, epoxy etc.

These are joined together by bell and spigot joint. IS : 1536-1989 and vertically cost pipes, respectively.

**88. Activated sludge process is an :**

(a) Anaacrobic attached growth process
(b) anacrobic suspended growth process
(c) aerobic attached growth process
(d) aerobic suspended growth process

**Ans. (d) :** Followings are the classification of secondary treatment units–

| Method | Contact mechanism | Decomposition |
|---|---|---|
| Trickling filler | Attached growth | Aerobic |
| Rotating biological Contactor | Attached growth | Aerobic |
| Activated sludge process | Suspended growth | Aerobic |
| Oxidation pond | Suspended growth | Aerobic |
| Septic tank | Suspended growth | Anaerobic |
| Imhoff tank | Suspended growth | Anaerobic |

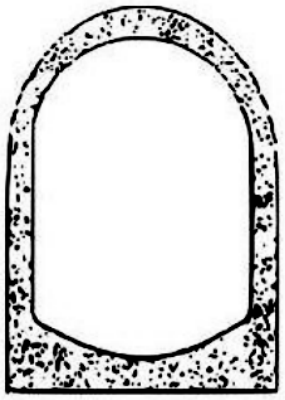
**89. Which of these chemical characteristic is detected by Winkler's method?**

(a) Chemical oxygen demand is sewage
(b) Dissolved oxygen content of sewage
(c) Hydrogen sulphide gas content of sewage
(d) Total organic carbon content of sewage

**Ans. (b) :** Dissolved oxygen content of sewage is the chemical characteristic detected by Winkler's method.

- The dissolved oxygen concentration of sea water is defined as the number of micromoles of dioxgen gas ($O_2$) per kilogram of seawater (m mol.kg$^{-1}$)

**90. Identify the type of sewer shown in the given figure.**



(a) New egg-shaped sewer
(b) Horseshoe shaped sewer
(c) U-shaped section
(d) Semi-circular shaped sewer

**Ans. (d) : Semic circular shaped sewer–** This section gives a wider base at bottom and hence it becomes suitable for constructing large sewers with less available headroom.

It is out dated.

**U shaped sewer–** The section may have true shape of letter 'U' or a small trench of U-shape can be set up in the large section.

- That trench is known as cunette.
- Such type of sewer used for combined sewer having maximum flow of storm water.
- It is used for larger sewers and especially in open cuts.
- The invert is form in the form of semi-circular arch.

**91. What is the design period for the treatment units of a sewerage scheme?**

(a) 5-10 years (b) 10-15 years
(c) 15-20 years (d) 20-25 years

**Ans. (c) :** As per manual on sewerage and sewage treatment systems - 2013; the design period for the treatment units of sewerage scheme 15-20 years.

- Design period for the design of a water supply project generally takes as 20-30 years.
- For distribution systems, pump house and pipe connections to several treatment units = 30 years.
- Water treatment units– 15 years
- Storage reservoirs/dams = 30 years

**92. As per BIS code, the acceptable indoor noise level is libraries is :**

(a) 30 dB (b) 45 dB
(c) 65 dB (d) 75 dB

**Ans. (b) :** The acceptable noise levels for indoor levels in different types of residential areas is given in below table-

| Location | Noise level dB |
|---|---|
| Radio and TV studios | 25-30 |
| Music room | 30-35 |
| Hospitals, classrooms, auditoria | 35-40 |
| Apartments, hotels, homes, conference, rooms, small offices | 35-40 |
| Counterooms, private offices libraries | 40-45 |
| Large pubic offices, banks stores etc | 45-50 |
| Restaurants | 50-55 |

**93. The stadia readings with horizontal sight at a vertical staff held 50 m away from the tacheometer were 1.385 and 2.380. The focal length of the object glass was 25 cm. The distance between the object glass and the trunion axis of a tacheometer was 15 cm. Calculate the stadia interval.**

(a) 4 mm (b) 5 mm
(c) 6 mm (d) 7 mm

**Ans. (b):** $\Delta = KS + C$

$$\Delta = \left(\frac{f}{i}\right)S + (f + d) \quad \text{..... (i)}$$

Given,
D = 50 m
S = 2.380-1.385 = 0.995
f = 25 cm = 0.25 m
d = 15 cm = 0.15 m
Put all the value in equation number (i)

$$50 = \left(\frac{0.25 \times 0.995}{i}\right) + (0.25 + 0.15)$$

i = 0.005 m
i = 5 mm

**94. Select the suitable option regarding the function of transition curve.**

**A : To minimise super elevation and the discomfort of passengers, these type of curve are generally provided on both sides of circular curves.**

**B : To introduce the designed super elevation gradually at the starting point of the curve.**

(a) Only A is true
(b) Only A is true
(c) Both A and B are false
(d) Both A and B are true

**Ans. (d) :** Regarding the function of transition curve. –
- To minimise super elevation and the discomfort of passengers, these type of curve are generally provided on both sides of circular curves.
- To introduce the designed super elevation gradually at the starting point of the curve.

Basic equation of super-elevation

$$e + f = \frac{v^2}{gR}$$

**95. In an old map, a line AB was drawn to a magnetic bearing of $5^0 30'$, the magnetic declination at the time being $1^0$ east. To what magnetic bearing should be line be set now, if the present magnetic declination is $8^0 30'$ East?**

(a) $358^0$ (b) $298^0$
(c) $287^0$ (d) $200^0$

**Ans. (a) :** Given,
Initial magnetic bearing = $5^0 30'$
Initial magnetic declination = $1^0$ East
Present magnetic declination = $8^0 30'$ East
True bearing of line = $5^0 30' + 1^0 = 6^0 30'$
Now,
T.B. = MB ± declination [For east (+) ve for west (-v) ve]
(a) $6^0 30'$ = M.B. + $8^0 30'$
(b) MB = $-2^0 = 360^0 - 2^0 = 358^0$

**96. What is the type of control flooding, where the agricultural land is divided into small strips by a series of field channels, which are connected to the supply channel?**

(a) Free flooding (b) Basin method
(c) Check flooding (d) Border flooding

**Ans. (d): Border flooding–** This is the type of control flooding, where the agricultural land is divided in to small strips by a series of field channels, which are connected to the supply channel.
- The land area confined in each strip is of order 10 to 20 meters wide and 100 to 400 miters long.

Water is allowed to flow from the supply ditch in to each strip. Hence the relationship to obtain the maximum area irrigated with supply ditch is given by-

$$t = 2.303 \frac{y}{f} \log_{10}\left(\frac{Q}{Q - fA}\right)$$

**97. Calculate the evaporation (in mm) from a lake, if the pan evaporation is 50 mm and the pan coefficient is 0.60**

(a) 25 mm (b) 30 mm
(c) 35 mm (d) 40 mm

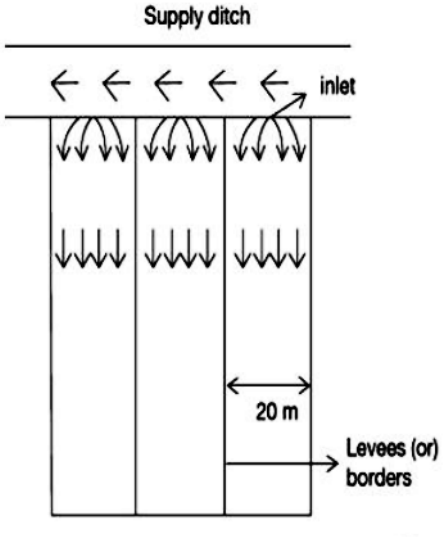
**Ans. (b) :** Given,
Pan evaporation = 50 mm
Pan coefficient = 0.60
Lake evaporation = 0.60 × 50 = 30 mm

**98. An area in which the crop is NOT sown in the a particular season, and the area is kept under no cultivation, is known as __**

(a) gross command area
(b) cultural cultivable area
(c) culturable unculivated area
(d) net command area

**Ans. (c) :** An area is which the crop is not sown in a particular season, and the area is kept under cultivation is known as culturable uncultivated area.
- Gross commanded area, it is the total area, bounded with the irrigation boundaries of the projects which can be economically irrigated without considering the limitation of the quantity of available water.

**99. Identify the method of surface irrigation shown in the given figure.**




(a) Border strip method
(b) Basin flooding
(c) Check flooding
(d) Zigzag method

**Ans. (a):** In it the land is divided into a no. strips (Each strip is of the order of 10 to 20 m in width & 100 to 400 m in length separated by low levees called borders.

- Water is allowed to flow the supply ditch into each strip. The water flows slowly towards the lower end & it infiltrates into the soil as it advance.

**100. A canal is designed for a maximum discharge of 60 cumes, but the average discharge is 40 cumec, what is the capacity factor?**

(a) 0.8 (b) 1.3
(c) 0.67 (d) 1.23

**Ans. (c) :** Given, Average discharge = 40 $m^3/s$
Maximum discharge = 60 $m^3/s$

$$\text{Capacity factor} = \frac{\text{Average discharge}}{\text{Maximum dishcarge}}$$

$$= \frac{40}{60} = 0.67$$

**101. Which method of irrigation is used to irrigate orchards?**

(a) Check method
(b) Basin method
(c) Furrow method
(d) Border strip method

**Ans. (b) :** Basin method is used to irrigate orchord.
**Basin method–** It is special type of check flood.
Basin may be regular or irregular shape.
**Furrow irrigation–** Furrow or narrow field ditches excavated between the row of crops and irrigation water is filled in then.

- Furrow may be vary from 8-30 cm in depth and may 400 m long.

**102. The ratio of the amount of water applied to land, to the amount of water supplied from the reservoir, is known as :**

(a) water conveyance efficiency
(b) water application efficiency
(c) water use efficiency
(d) consumptive use efficiency

**Ans. (a) :** The ratio of the amount of water applied to land, to the amount of water supplied from the reservoir, is known as water conveyance efficiency.
**Irrigation efficiency–** It is the ratio of output water to input water.
Higher the loss, lesser is the efficiency

$$\left(\eta \propto \frac{1}{\text{loss of } H_2O}\right)$$

Water applicable efficiency $(\eta_a)$

$$= \frac{\text{Water stored in root zone (ws)}}{\text{Water delivered to the field (wf)}} \times 100$$

**103. What is the discharge capacity required at the outlet to irrigate 2200 hectares of sugarcane having a kor depth of 14 cm and a kor period of 20 days?**

(a) 1.56 $m^3/sec$ (b) 1.63 $m^3/sec$
(c) 1.78 $m^3/sec$ (d) 1.91 $m^3/sec$

**Ans. (c) :** Given,
Area = 2200 hectare
Kor depth ($\Delta$) = 14 cm = 0.14 m
Kor period (B) = 20 days
We know that,

$$\Delta = \frac{8.64\ B}{D}$$

$$D = \frac{8.64 \times 20}{0.14}$$

D = 1234.28 hectare/cumec

$$\text{Duty} = \frac{\text{Area}}{\text{Discharge}}$$

$$\text{Discharge} = \frac{2200}{1234.28}$$

Discharge = 1.782 $m^3/s$

**104. Which of the following types of canal is a classified type of canal, based on the discharge through it?**

(a) irrigation canal (b) Navigation canal
(c) Power canal (d) Main canal

**Ans. (d) :** Main canal is a type of canal, based on discharge.
Classification of canal based on different factors

| Based on the nature of supply source | Based on function | Based on discharge | Based on alignment |
|---|---|---|---|
| Permanent canal | Irrigation canal | Main canal | Ridge canal |
| Inundation canal | Power canal | Branch canal | Contour canal |
| | Feeder canal | Major distributary | Side-slope canal |
| | Carrier canal | Minor distributary | |
| | Navigation canal | Field canal | |

**105. Which of the following statements is true in case of perennial irrigation?**

(a) The irrigation water is available only to the rainy season
(b) No hydraulic structure is required
(c) The canal water contains practically no silt and hence chemical manure is essential
(d) Initial cost and maintenance cost is loss for this system

**Ans. (c) : Prennial irrigation–** In this system of irrigation constant and continuous water supply is assured to the crops in accordance with the requirement of the crop throughout the crop system.

**106. The ratio of the quantity of water stored into the root zone of the crops, to the quantity of water delivered to the field, is known as**

(a) water application efficiency
(b) water conveyance efficiency

(c) water use efficiency
(d) Water surface efficiency

**Ans. (a) :** The ratio of the quantity of water stored into the root zone of the crops, to the quantity of water delivered to the field, is known as water application efficiency.

- For loss due to surface runoff and deep percolation.

$$n_a = \frac{w_f}{w_f} \times 100$$

**107. The amount of irrigation water that is required to meet the evapo-transpiration needs of the crop during its full growth is known as :**
(a) effective rainfall
(b) consumptive irrigation requirement
(c) net irrigation requirement
(d) field irrigation requirement

**Ans. (b) :** Consumptive irrigation requirement (CIR) : It is defined as the amount of irrigation water that is required to meet the evapo-transpiration needs of the crop during its full growth.
CIR = Consumptive use– Effective rainfall.

**108. Which of the following is a rigid dam?**
(a) Earthen dam (b) Rockfill dam
(c) Tailing dam (d) Masonry dam

**Ans. (d) :** Based on material of construction dams are classified in to two category.
**Rigid dam–** It is constricted with a rigid material such as a stone, masonry, concrete, steel or timber.

**109. The volume of water that is stored below the minimum pool level of a reservoir, which cannot be used for normal operating conditions, is called_____**
(a) dead storage (b) live storage
(c) valley storage (d) surcharge storage

**Ans. (a) : Dead storage–** The volume of water that is stored below the minimum pool level of a reservoir, which can not be used for normal operating condition.
**Live storage–** The storage of water in a reservoir upto top level of sluice gate.

**110. If the circular main scale of a theodolite is divided into degrees and each degree is divided into $\frac{1}{4}$ parts, then the main scale division is divided into _____ vernier scale divisions to have a least count of 0.005 degrees.**
(a) 50 (b) 500
(c) 5000 (d) 5

**Ans. (a) :** Given,
Least count = $0.005^0$

$$S = \frac{1^0}{4} = \frac{60 \times 60}{4} = 900$$

$$0.005 \times 3600 = \frac{900}{n} \Rightarrow n = 50$$

**111. A 20 m chain was found to be 10 cm too long after chaining distance of 1500 m, calculate the true distance.**
(a) 1503.75 m (b) 1500.75 m
(c) 1503.55 m (d) 1503.25 m

**Ans. (a) :** Given –
True length of chain = 20 m
Incorrect length of chain = 20 + 0.10 = 20.10 m
Incorrect measured length = 1500 m
True/correct measured length = ?
Solution,
Correct/Time length of line =
Incorrect length/measured length

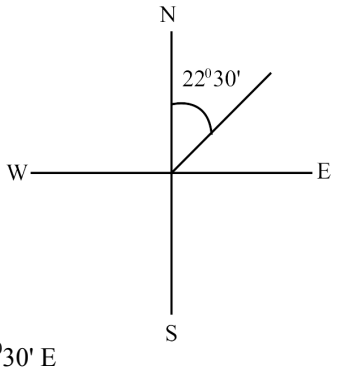
$$\times \frac{\text{Incorrect length of chain}}{\text{True length of ....}}$$

$$\text{Correct length} = 1500 \times \frac{20.10}{20}$$

Correct length = 1507.5 m

**112. What will be conversion of whole circle bearing $22^0$30 minutes into quadrantal bearing?**
(a) North $22^0$30 minutes east
(b) North $9^0$48 minutes east
(c) South $31^0$36 minutes west
(d) North $32^0$36 minutes west

**Ans. (a) :** Given,
Bearing in WCB = $22^0 90^0$
Solution




R.B. = N $22^0$30' E

**113. Which of the following statements is true in case of prismatic compass when compared to surveyor's compass?**
(a) The magnetic needle is of edge bar type
(b) The magnetic needle acts as an index
(c) The graduations engraved arc inverted
(d) The readings are taken by direct observation through the top of the glass.

**Ans. (c) :**

| Prismatic compass | Surveyor compass |
|---|---|
| Graduated card ring is attached with needle. The ring does not rotate along with the line of sight. | The graduated card is attached to the box and not the needle. The card rotates along with the line of sight. |
| Graduation are engraved in vertical | Graduation are engraved erect |
| Reading is taken with the help of prism provided at dry silt | Reading is taken by directly seeing through top of glass |
| Tripod may or may not be provided | The instrument can not be used without tripod. |

**114. Which of the following is a temporary adjustment surveyor' compass?**
(a) Centering
(b) Adjustment of levels
(c) Adjustment of needle
(d) Adjustment of the sight vane

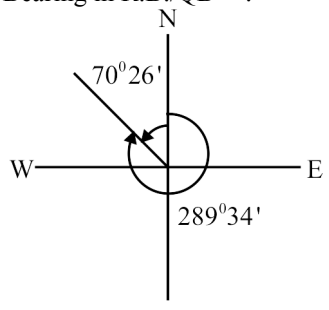
**Ans. (a) :** The following temporary adjustments are done after fixing the surveyor's compass on the tripod.
**(i) Centring** - Centring is the process of setting up the instrument exactly over the station peg.
• The centering is done by moving the legs of the tripod.
**(ii) Levelling–** Levelling is required so that the graduated ring is horizontal and it swings truly on the pivot.
**(iii) Focusing–** Focusing on the target has to be done utilizing the dry vane and object vane.

**115. The line joining the points having zero magnetic declination is known as _____**
(a) annual variation (b) diurnal variation
(c) isogonic line (d) agonic line

**Ans. (d) :** The line joining the points having zone magnetic declination is known as agonic line.
The line joining same magnetic declination is known as isogonic line.

**116. The whole circle bearing of a line is found to be $289^0 34^0$, calculate its bearing in quadrantal system.**
(a) N $70^0$ 26' E (b) N $70^0$ 26' W
(c) S $70^0$ 26' E (d) S $70^0$ 26' W

**Ans. (b) :** Given,
Bearing in W.C.B. = $289^0$34'
Bearing in R.B./QB = ?




R.B. = $360^0$–$289^0$34'
R.B. = N $70^0$ 26' W

**117. A back sight takes taken on a benchmark (Point A) is found to be 1.535 and the fore sight at the next point (Point B) is 2.650 m. If the reduced level at point A is 936.00 m. Calculate the reduced level at point B.**
(a) 933.350 m (b) 934.885 m
(c) 936.540 m (d) 940.185 m

**Ans. (b) :** R.L. at point A = 936.00 m (Bench mark)
B.S. of point A = 1.535 m
F.S. of point B = 2.650 m
R.L. of point = ?
Height of instrument (H.I.) = Benchmark (BM) + Back sight (B.S.)
H.I. = 936.00 + 1.535
H.I.= 937.535 m
Reduced level of forward station (B) = μI – Force sight (FS)
RL at point B = 937.535-2.650
RL at pont B = 934.885 m

**118. Which is the second most important cereal crop grown in the north and north-western parts of India?**
(a) Rice
(b) Jawar
(c) Wheat
(d) Maize

**Ans. (c) : Wheat-** It's Rabi crop which is mostly consumed by the whole world; and Punjab and Uttar Pradesh are one of the largest produces of wheat and is the staple diet of the north Indians.
• Wheat is the most important create crop grown in the north-western parts of India.
• Rice in the second most important crop in the north & north-western part of India.

**119. Which of following is NOT a chemical characteristics of raw water?**
(a) Turbidity
(b) pH
(c) Alkalinity
(d) Acidity

**Ans. (a) :** Turbidity, colour, taste and odour are the physical characteristic of water.
• Alkanity, pH, Hardness, dissolved solids, chloride content, nitrogen content fluoride content and oxygen are the chemical characteristics of water.
• MPN test, membrane filter technique and collier m index test on the biological test to check colliform; which is biological parameter of water.

**120. Which of the following is an example of mobile source of air pollutant?**
(a) channel vessels
(b) industrial processing
(c) power plants
(d) dust storm

**Ans. (a) :** Mobile sources are primarily on read motorized which inducing cars, trucks, buses and motorcycles.
• They also include non-read equipment such as construction equipment, recreational vehicles, small-powered equipment, marine vessels/channel vessels and locomotive.

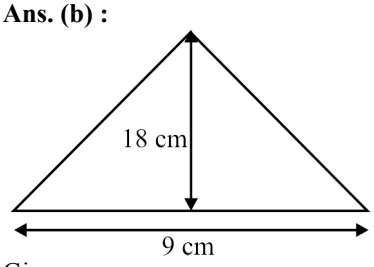
**121. Electronics precipitators are used as pollution control devices for the separation of_____ from gas stream.**
(a) $SO_2$ (b) $NO_x$
(c) Hydrocarbons (d) Particulate matter

**Ans. (d) :** Electronics / electrostatic precipitators are used as pollution control devices for the separation of particulate matter from gas stream using the force of an induced electrostatic charge.

• ESP (Electrostatic precipitator) can be operated at high temperature and pressures and it's power requirement is low.
• An ESP is a type of filter that users static electricity to remove soot and ash from exhaust fumes before they exit the smoke stacks.

**122. The difference in elevation between the edge of the pavement and the crown point at mid-width of the pavement is 18 cm, by considering width of pavement as 9 m, determine the value of camber provided.**
(a) 1 in 20 (b) 1 in 25
(c) 1 in 30 (d) 1 in 35

**Ans. (b) :**




Given,
Y = 18 cm
Width of pavement, w = 9 m

$$\text{Camber} = \frac{2y}{w} = \frac{2\times 18}{9\times 100}\text{cm}$$

$$\text{Camber} = \frac{1}{25}$$

**123. An intersection that usually consists of structures that provide for traffic to cross at different levels without interruption is known as**
(a) grade separated intersection
(b) at grade intersection
(c) compound intersection
(d) simple intersection

**Ans. (a) :** An intersection that usually consists of structures that provide for traffic to cross at different levels without interruption is known as grade separated intersection. It is also known as interchange.
• Grade separated intersection cause less hazard and delay than grade intersections.

**124. Which of the following is NOT a classification type of urban road?**
(a) Major district road
(b) Arterial road
(c) Collector street
(d) Local street

**Ans. (a) :** Major district road is not a classification type of urban road.

| Rural road | Urban road |
|---|---|
| National highway | Express-way |
| State highway | Arterial road |
| MDR / major district road | sub-arterial road |
| ODR (other district road) | Local street |
| Village road | |

**125. Which of the following parts of a highway is provided near public conveyances with guide maps to enable drivers to stop clear of the carriageway?**
(a) Lay bay (b) Bus bay
(c) Cycle track (d) Frontage road

**Ans. (a) : Lay bay–** Lay bay are provided near public conveniences with guide maps to unable drivers to stop clear off the carriage way.
• Lay byes should normally be of 3 m width and at least 30 m length with 15 m tapers on both sides.
**Bus bay–** They are provided by recessing the kerb to conflict with the moving traffic. Bus bays should be located at least 75 m away from the intersection.
**Cycle tracks–** It is recommended along busy connecting roads where the intensity and speed of material traffic make it unsafe to allow cyclists on the carriageway.

**126. _____ is a viscous liquid obtained when natural organic materials (wood/coal) are destructively distilled in the absence of air.**
(a) bitumen (b) asphalt
(c) tar (d) cutback

**Ans. (c) :** Tar is a viscous liquid obtained when natural organic materials (word/coals) are destructively distilled in the absence of air.
Tar is soluble in Toluene ($C_6H_5CH_3$), less resistance to water, more susceptible to temperature change, more free carbon content specific gravity of tar (1.10-1.25).

**127. Which of the following is NOT an economic study which is done during planning surveys of highway planning?**
(a) Per capita income
(b) Trend in population growth
(c) Standard of living of different population groups and trend in the changes
(d) Industrial and agricultural products and their listing in classified groups, area wise.

**Ans. (c) :** Major economical study under the highway development programme–
• Per capita income
• Trend in population growth
• Industrial and agricultural products and their listing in classified groups, area wise.

**128. The extra widening of pavement at horizontal curves on two lane roads, is NOT required if the radius of curve is greater than**
(a) 200 m (b) 250 m
(c) 300 m (d) 150 m

**Ans. (c) :** If radius is greater than 300 m, IRC has suggested not to provide any extra widening.
• If $300 \geq R \geq 50$; IRC has suggested to provide extra widening on both inner or outer edge.
• If $R < 500$; IRC has suggested to provide extra widening only at inner edge.
• For a single lane road IRC has not considered any psychological widening for overtaking operation.

Extra widening–

$$w_e = w_{mechanical} + w_{psychological}$$

$$w_e = \frac{n\ell^2}{2R} + \frac{v}{9.5\sqrt{R}}$$

Where,
n = no. of lanes, R = radius (m)
v = velocity (km/h)

**129. While aligning a hill road with ruling gradient of 6%, a horizontal curve of radius 68 m is encountered. Find the compensated gradient at the curve.**

(a) 4% (b) 4.6%
(c) 5% (d) 5.2%

**Ans. (*) :** Given,
Ruling gradient, G = 6 % (which is > 4%) hense grade compensation is required).
• Radius of horizontal curve, R = 68 m
Grade compensation is given by,

$$G.C. = \frac{30+R}{R} \not> \frac{75}{R}$$

$$G.C. = \frac{20+68}{68} \not> \frac{75}{68}$$

$$G.C. = 1.44 \not> 1.10$$

$$G.C. = 1.10$$

Grade compensated gradient is given by,
Compensated gradient = gradient – G.C.
= 6–1.10 = 4.89%

**130. Improved enforcement is considered as a remedy for which of the following situation in order to prevent accidents?**

(a) Roadside obstacles hazard by present?
(b) Excessive speed
(c) Wet road skidding
(d) Splash obscuring visibility poor delineation

**Ans. (b) :** Improved enforcement is considered as a remedy for excessive speed situation in order to prevent accidents.

**131. As per IRC, what is the minimum carriagway width of a highway of 2 lanes, without raised kerbs?**

(a) 4 m (b) 5.5 m
(c) 7 m (d) 9 m

**Ans. (c) :**

| No. of lanes | width (minimum) |
|---|---|
| Single lane road | 3.75 m |
| Two lane road with out kerb | 7 m |
| Two-lane road with kerb | 7.5 m |
| Intermediate - lane | 5.5 m |

**132. Which of the following traffic survey schemes is more relevant when deciding location of major routes of a city, and is also used for the design of the Mass Rapid Transit System?**

(a) Speed survey
(b) Traffic capacity survey
(c) Origin and destination survey
(d) Traffic volume survey

**Ans. (c) :** Origin and destination survey is more relevant when deciding location of major routes of a city, and is also used for the design of the mass rapid transit system.
• Origin and destination study are not important in planning new highway facilities or for improving existing road system.

**133. On a T-intersection with two-way traffic, the total number of conflict points are____**

(a) 8 (b) 18
(c) 20 (d) 24

**Ans. (b)** On a T-intersection with two-way traffic, the total number of conflict points are 18.
• For two way traffic on a right angled road intersection, the conflict points are 24 (crossing = 4, weaving = 12, merging=8).

**134. The horizontal distance between two points on two consecutive contours is known as**

(a) horizontal equivalent
(b) contour interval
(c) contour line
(d) contour gradient

**Ans. (a) : Horizontal equivalent–** The horizontal distance between two points on two consecutive contours is known as horizontal equivalent.
**Contour elevation–** It is the line joining points of equal elevation on the surface of the earth.
**Contour internal–** The difference in elevation between successive contour lines on a given map is fixed. This vertical distance between any two contour lines on a map is called the contour interval.

**135. Which of the following chains used for linear measurement has length of 66 ft and 100 numbers of links?**

(a) Revenue chain (b) Engineers chain
(c) Surveyor chain (d) Metric chain

**Ans. (c) : Types of chain–**
• Gunter's chain/Surveyor's chain = 66 ft (100 links)
• Engineer's chain = 100 ft (100 links)
• Revenue chain = 33 ft (16 links)
• Metric chain = 30 m (150 links)
20 m (100 link)

**136. What is the action by the assistant if the code of signal for ranging by the surveyor is 'Left arm extended?**

(a) Continue to move to the right
(b) Continue to move to the left
(c) Move slowly to the left
(d) Move slowly to the right

**Ans. (b) :** The codes of signals which are used to direct the assistant while ranging a line.

| Action | Meaning |
|---|---|
| Rapid sweeps with the right hand | Move considerably to the right |
| Rapid sweeps with the left hand | Move considerably to the left. |

| | |
|---|---|
| Slow sweeps with the right hand | Move slowly to the right |
| Slow sweeps with the left hand | Move slowly to the left. |
| Right arm extended | Continue to move to the right |
| Left arm extended | continue to move the the left |
| right arm up and moved to the right | plump the rod to the right |
| left arm up and moved to the left | plump the road to the left |
| Both arms above the head and then brought down | correct |
| Both arms extended forward and depressed briskly | Fix |

**137. Which of the following survey instruments is made fully with stout timber?**

(a) Cross staff (b) Offset rods
(c) Pegs (d) Plumb bob

**Ans. (c) : Pegs–** Wooden pegs are used to mark the position of the stations or terminal points of survey lines. They are made of stout timber. These are available about 2.5 to 3 m cm square and 15 cm length. They are driven in the ground with the help of a wooden hammer and kept about 4 cm projection above the ground.

**138. Which of the following instruments is used for measuring offsets, which consists of 2 laths each of 1 m in length and riveted together?**

(a) Butt rod (b) Pegs
(c) Whites (d) Plumb bob

**Ans. (a) :** Butt rod is used for measuring offsets, which consists of 2 laths each of 1 m in length and riveted together.

**Plumb bob–** Plumb bob is used for transferring the points to the ground incase of chaining along a slope. Plumb bob is also used for centering in plane table surveying, compass surveying and theodolite surveying. It is also used to check if the inserted ranging rod is vertical or not.

**139. Left arm up and moved to the left sign by the surveyor indicates which of the following?**

(a) Move considerably to the left
(b) Fix the rod
(c) Continue to move to the left
(d) Plumb the rod to the left

**Ans. (d) :** The codes of signals which are used to direct the assistant while ranging a line.

| Action | Meaning |
|---|---|
| Rapid sweeps with the right hand | Move considerably to the right |
| Rapid sweeps with the left hand | Move considerably to the left. |
| Slow sweeps with the right hand | Move slowly to the right |
| Slow sweeps with the left hand | Move slowly to the left. |
| Right arm extended | Continue to move to the right |
| Left arm extended | continue to move the the left |
| right arm up and moved to the right | plump the rod to the right |
| left arm up and moved to the left | plump the road to the left |
| Both arms above the head and then brought down | correct |
| Both arms extended forward and depressed briskly | Fix |

**140. Which of the following is NOT an assumption made in unit hydrograph theory?**

(a) Effective rainfall is uniformly distributed within its duration
(b) Effective rainfall is uniformly distributed throughout the whole area of the drainage basin
(c) Ordinates of direct runoff of common base time are inversely proportional to the total amount of direct runoff represented by each hydrograph
(d) Time duration of hydrograph of direct runoff due to an effective rainfall of unit duration is constant.

**Ans. (c) : Assumption made in unit hydrograph theory–**

- Effective rainfall is uniformly distributed within its duration.
- Effective rainfall is uniformly distributed throughout the whole area of the drainage basin.
- Time duration of hydrograph of direct runoff due to an effective rainfall of unit duration is constant.

**141. The firm yield of a reservoir is also called____ yield.**

(a) unsafe (b) secondary
(c) average (d) safe

**Ans. (d) : Safe yield–** It is also known as firm yield.

- It is the maximum quantity of water that can be supplied from the reservoir with full guarantee during the worst dry period.

**142. Which type of sewer is highly resistant to sulphide corrosion?**

(a) Brick sewer
(b) Vitrified clay sewer
(c) Asbestos cement sewer
(d) RCC sewer

**Ans. (b) :** There are manufactured from days and shapes of special qualities, which are pulverized then mixed thoroughly with water.

- These pipes are highly resistant to sulphide corrosion and therefore, preferred for carrying polluted sewage and industrial wastes.

**143. What is the other name of a jack well?**

(a) Dry intake structure
(b) Submerged intake structure
(c) Wet intake structure
(d) Canal intake structure

**Ans. (c) :** Well intake structure is the other name of a jack well.

- It is a type of intake tower in which the water level is practically the same as the level of the source of supply.

**Dry intake structure–** In case of dry intake there is no water in the water tower.

- Water enters through entry port directly into conveying pipes. It is simply used for operation of values.

**144. The flowing through velocity in an Imhoff tank should not exceed_____.**

(a) 0.3 m/min (b) 1.5 m/min
(c) 2 m/min (d) 0.9 m/min

**Ans. (a) :** The flowing trough velocity in an imhoff tank should not exceed 0.3 m/min.

- Its design depth should be between 3 to 3.5 m.

**145. Earth work details are given in which phase of scopes of highway engineering?**

(a) Development planning and locations
(b) Highway design, geometrics and structures
(c) Traffic performance and its control
(d) Materials, construction and maintenance

**Ans. (d) : Scope of the highway engineering–**
Road are generally constructed on small embankment, slightly above the general ground level, in order to avoid the difficult drainage and maintenance problems.

- Such roads are therefore termed as highway and science and technology dealing with roads are generally called highway engineering.

**Highway engineering deals with various phase like–**
Development, planning alignment, highway material, traffic control, highway geometric design and location, pavement design, construction and maintenance, economic consideration, financial and administration.

**146. The Indian Road Congress recommends the use of ____ as ideal transition curve in the horizontal alignment of highways.**

(a) Cubic parabola (b) Parabola
(c) Lemniscate (d) Spiral curve

**Ans. (d) :** The Indian Road Congress recommends the use of spiral or clothoid as ideal transition curve, because–

- Rate of change of centrifugal accelerate is constant.
- Simple calculation and easy to implement.

**147. The recommended camber for cement concrete roads in areas with light rainfall is _____**

(a) 1.7% (b) 2.7%
(c) 3.7% (d) 4.7%

**Ans. (a) :** Recommended camber–

| Types of pavement | Rainfall | |
|---|---|---|
| | Heavy | Light |
| Cement concrete or height type bitumen | 2% | 1.7% |
| Thin bituminous | 2.5% | 2% |
| W.B.M. or gravel water bond macadam | 3% | 2.5% |
| Earth road | 4% | 3% |

**148. Which of the following is an example of active remote sensing instrument?**

(a) Hyperspectrical radiometer
(b) Spectrometer
(c) Radiometer
(d) RADAR

**Ans. (d) : Active remote sensing–** Active remote sensing instruments operate with their own source of emission or light. Active sensors used internal stimuli to collect data about earth.

- Radar is an example of active remote sensing instrument.
- Spectrometer, spectro radiometer, radiometer, sounder, accelerometer are the example of passive remote sensing.

**149. The imaginary point on the celestial sphere directly beneath a given position or observer, and diametrically opposite the zenith, is knwon**

(a) nadir point (b) zenith
(c) pole (d) celestial point

**Ans. (a) :** The imaginary point on the celestial sphere directly beneath a given position or observer, and diametrically opposite the zenith, is known as nadir point.

**Zenith–** It is the point above observer that is directly opposite the nadir on the celestial sphere.

Pole– Pole is the imaginary point at which the axis of rotation of the celestial sphere intersects.

**Celestial point–** Celestial points are the imaginary points on the celestial sphere.

**150. If the height of an object above the datum is 30 m, the flying height above the datum is 2000 m, and the radial distance of the image of the object top is 20 m, what will be the relief displacement?**

(a) 0.3 m (b) 0.5 m
(c) 0.6 m (d) 0.7 m

**Ans. (a) :**

$$d = \frac{r \times h}{H}$$

Given,

$h = 30$ m, $H = 2000$, $r = 20$ m

$$d = \frac{20 \times 30}{2000} = 0.3$$

# Andhra Pradesh Public Service Commission
# Assistant Engineer (Common, Civil/Mech.) Exam, 2022
## Solved Paper Date - 15.05.2022 (Shift-II)

**1. Of the following concepts of classical mechanics, which is not an independent one?**

(a) Space (b) Time
(c) Mass (d) Force

**Ans. (d) :** Classical mechanics is the mathematical study of the motion of everyday objects and the forces that affect them.

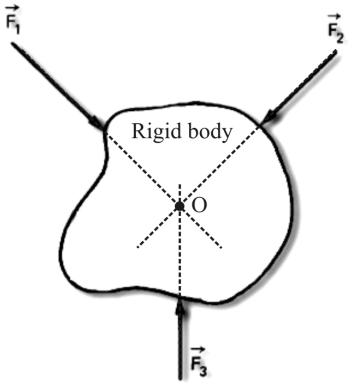
• Force is not an independent one as it depend upon external action.

**2. If the line of action of all the forces in the system lies on the same plane, then it is called a_____**

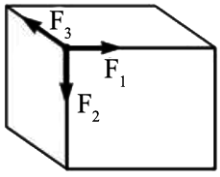
(a) coplanar force system
(b) non-coplanar force system
(c) concurrent force system
(d) parallel force system

**Ans. (a) :** If the line of action of all the forces in the system lies on the same plane, then it is called a coplanar force system.

**Concurrent forces–**If lines of action of forces in a system of force meet at a point this is called concurrent forces

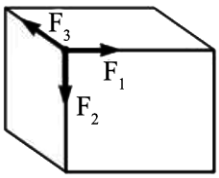



**3. The system of force represented in the following figure is the_____**



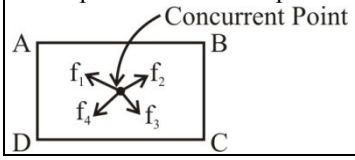

(a) coplanar concurrent force system
(b) non-coplanar concurrent force system
(c) coplanar non-concurrent force system.
(d) non-coplanar non-concurrent force system

**Ans. (b) :**




non-coplanar concurrent force system

A number of forces lying in one plane and intersecting at one point are called coplanar concurrent force.




**4. A planet, having mass and radius half of that of earth, will have a value of acceleration due to gravity_____ that on earth (assume that constant of gravitation is the same for both the plastic).**

(a) half of (b) the same as
(c) double (d) four times

**Ans. (c) :** $g = \frac{GM_e}{R_e^2} = 9.81\ m/s^2$

$g' = \frac{GM_p}{R_p^2} = \frac{G(M_e/2)}{(R_e/2)^2} = 2g = 19.62\ m/s^2$

So, its double.

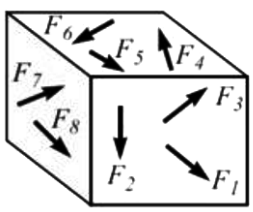
**5. The resultant of any two_____ forces may be found by the parallelogram Law.**

(a) collinear concurrent
(b) non-collinear concurrent
(c) collinear non-concurrent
(d) non-collinear non-concurrent

**Ans. (b) :** The resultant of any two non-collinear concurrent forces may be found by the parallelogram Law.

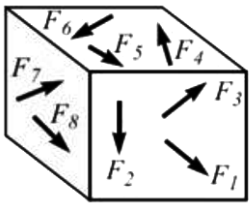
From the principle of parallelogram of forces, it follows that two forces applied at one point can always be replaced their resultant which is equivalent to them.

**6. The system of forces represented in the following figure is the_____**



(a) coplanar unlike parallel force system
(b) non-coplanar unlike parallel force system
(c) coplanar non-concurrent and non-parallel force system
(d) non-coplanar non-concurrent and non-parallel force system

**Ans. (d) :**




non-coplanar non-concurrent and non-parallel force system.

**7. A block rests on a horizontal frictional surface. Of the conditions given in the options, in which case can the equation $F = \mu_S N$ be applied (where F is the frictional force, $\mu_S$ is the coefficient of static friction and N is the normal reaction?**

(a) The forces applied to the block do not tend to move it along the surface of contact
(b) The applied forces tend to move the block along the surface of contact, but are not large enough to set it in motion.
(c) The applied forces are such that the block is just about to slide.
(d) The block is sliding under the section of the applied forces.

**Ans. (c) :** $F = \mu_S N$, where F = friction force
The applied forces are such that the block is just about to slide.
• Static friction is what keeps the block from moving without being pushed and it must be over come with a sufficient opposing force before the block will move.
• Kinetic friction is the force that resists the relative movement of the surfaces once they are in motion.

**8. Based on the two statements given below, choose the correct answer.**
**Statement A : A point where the whole weight of the body is assumed to act is called the centre of gravity.**
**Statement B : For a non-generous plate, the coordinates for the centre of gravity and the centroid of the area are the same.**

(a) Both statements A and B are correct
(b) Both statements A and B are incorrect
(c) Statement A is correct, but statement B is incorrect
(d) Statement A is incorrect, but statement B is correct

**Ans. (c) :** A point where the whole weight of the body is assumed to act is called the centre of gravity.
• For a non-generous plate, the coordinates for the centre of gravity and the centroid of the area are different.

**9. A shaft runs at 80 rpm and drives another shaft at 150 rpm through belt drive. If the diameter of the driving pully is 600 mm, the diameter of the driven pulley is (neglect the belt thickness).**

(a) 320 mm (b) 520 mm
(c) 850 mm (d) 1125 mm

**Ans. (a) :** Velocity ratio ,

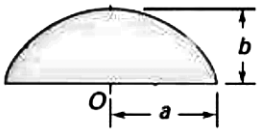
$$\frac{N_2}{N_1} = \left(\frac{d_1 + t}{d_2 + t}\right)\left(1 - \frac{S}{100}\right)$$

Slip (S) = 0

$$\frac{150}{80} = \frac{600 + 5}{d_2 + 5},$$

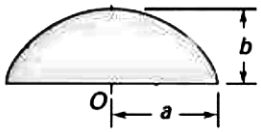
$d_2 = 320$ mm

**10. The coordinates of the centre of gravity (x,y) of the semi-elliptical area shown in the given figure, with respect to the origin O, are :**



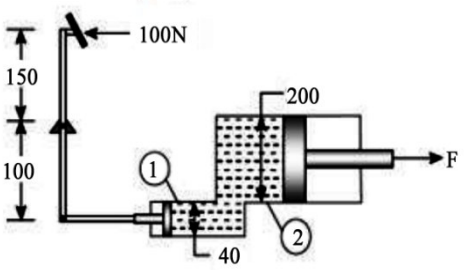

(a) $\left(0, \frac{4a}{3\pi}\right)$ (b) $\left(0, \frac{4b}{3\pi}\right)$
(c) $\left(\frac{4a}{3\pi}, \frac{4b}{3\pi}\right)$ (d) $\left(\frac{4b}{3\pi}, \frac{4a}{3\pi}\right)$

**Ans. (b) :**




Centre of gravity = $\left(0, \frac{4b}{3\pi}\right)$

**11. A schematic diagram of a hydraulic braking system is shown in the following figure. If the applied a force of 100 N, the force F available at the brakes is (take specific gravity of the fluid to be 0.8 and units of length as mm).**

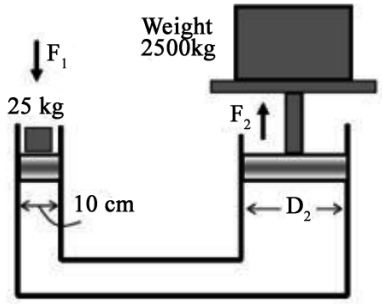



(a) 1.67 kN (b) 2.5 kN
(c) 2.75 kN (d) 3.75 kN

**Ans. (b) :** $\frac{F_1}{A_1} = \frac{F}{A_2}$
Where, $F_1 = 100$ N,

Diameter, $D_1 = 40$ mm
F = ?
$D_2 = 200$ mm
So, F = 2.5 kN

**12. A hydraulic lift is used to lift a weight, as shown in the following figure. The diameter of the piston ($D_2$) on which the weight is to be placed is ;**




(a) 0.1 m (b) 1 m
(c) 1 mm (d) 1 cm

**Ans. (b) :** $\frac{F_2}{F_1} = \frac{A_2}{A_1}$

$$\frac{2500}{25} = \frac{\frac{\pi}{4}(D_2)^2}{\frac{\pi}{4}\times(10)^2} \Rightarrow D_2 = 1 \text{ m}$$

**13. The efficiency, η, of the screw jack is given by (where α is the helix angle and ϕ is the angle of friction).**

(a) $\eta = \frac{\tan\alpha}{\tan(\alpha+\phi)}$

(b) $\eta = \frac{\tan\alpha}{\tan(\alpha-\phi)}$

(c) $\eta = \frac{\tan(\alpha+\phi)}{\tan\alpha}$

(d) $\eta = \frac{\tan(\alpha-\phi)}{\tan\alpha}$

**Ans. (a) :** The efficiency, η, of the screw jack is given by $\eta = \frac{\tan\alpha}{\tan(\alpha+\phi)}$

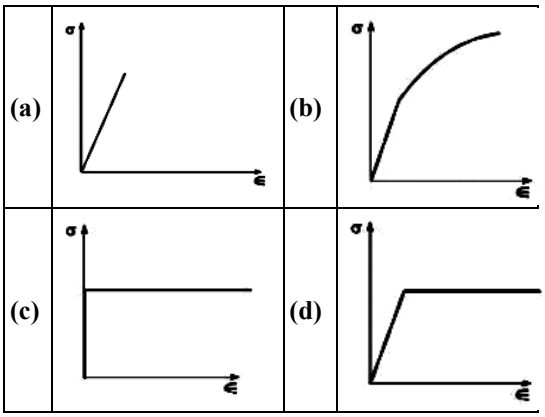
Where,
α is the helix angle and ϕ is the angle of friction

**14. When one object presses against another, the stresses developed at the contact surface is referred to be**

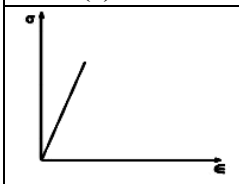
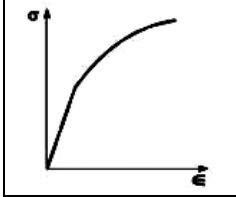
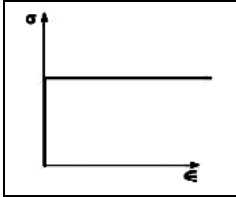
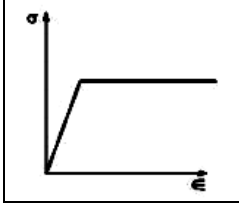
(a) bearing stress (b) tensile stress
(c) normal stress (d) shear stress

**Ans. (a) :** When one object presses against another, the stresses developed at the contact surface is referred to be bearing stress.

**15. The stress (σ) – strain (ε) diagram for a Linear-Elastic material is :**




**Ans. (a) :**

| | |
|---|---|
|  | Linear elastic material |
|  | Linear elastic plastic |
|  | Perfectly plastic |
|  | Elastic perfectly plastic |

**16. The relationship between the engineering strain (e) and the true strain (ε), for materials with no changes in volume during deformation, is (where in represents natural log).**

(a) $\varepsilon = \text{In}(e+1)$ (b) $\varepsilon = \text{In}(e-1)$
(c) $e = \text{In}(\varepsilon+1)$ (d) $e = \text{In}(\varepsilon-1)$

**Ans. (a) : Engineering strain–** It is the static of change in dimension to the original dimension.

$$\varepsilon_e = \frac{\Delta L}{L_0}$$

**True strain–** It is the ratio of instantaneous elongation to instantaneous dimension.

$$d\varepsilon_t = \frac{dL}{L}$$

$$\int d\varepsilon_t = \int_{L_0}^{L_f} \frac{dL}{L}$$

$$\varepsilon_t = \left[\log_e L\right]_{L_0}^{L_t}$$

$$\varepsilon_t = \log_e\left[\frac{L_f}{L_0}\right]$$

$$= \log_e\left[\frac{L_0 + \Delta L}{L_0}\right] = \log_e\left[1 + \varepsilon_e\right]$$

**17. The relationship between the engineering stress (s) and the true stress (σ), for materials, with no changes in volume during deformation, is (where e is the engineering strain and ε is the true strain).**

(a) $\sigma = s(e+1)$ (b) $\sigma = s(e-1)$

(c) $s = \sigma(\varepsilon+1)$ (d) $s = \sigma(\varepsilon+1)$

**Ans. (a) :** Engineering stress-strain is lower than true stress-strain because we consider initial area only throughout the process while in true stress-strain curve we consider the deformed area & which is deformed drastically, therefore stress coil increases.

Engineering stress $\left(\sigma_E\right) = \frac{\text{Applied force}}{\text{Initial area}}$

& true stress $\left(\sigma_T\right) = \frac{\text{Applied force}}{\text{Deforming area}}$

• Also, the curved traced by true stress will be above the engineering stress and moves right with increases in load & corresponding strain.

$$\sigma = s(e+1)$$

Where,
s= engineering stress
σ = true stress

**18. The relationship between Young's modulus of rigidity (E), bulk modulus of elasticity (K) and modulus of rigidity (G) is :**

(a) $E = \frac{9GK}{(3K+G)}$ (b) $E = \frac{(3K+G)}{9GK}$

(c) $E = \frac{3GK}{(3K+2G)}$ (d) $E = \frac{(3K+2G)}{3GK}$

**Ans. (a) :** • Modulus of elasticity (E) $= \frac{\text{Stress}}{\text{Strain}}$

• Bulk Modulus (K) $= \frac{\text{Direct stress}}{\text{Volumetric strain}}$

• Modulus of rigidity (N) $= \frac{\text{Shear stress}}{\text{Shear strain}}$

**By relation between E, K and G.**

$$\left[E = \frac{9GK}{G+3K}\right]$$

**19. The relationship between poisson's ratio (γ), bulk modulus of elasticity (k) and modulus of rigidity (G) is :**

(a) $\gamma = \frac{2(G-3K)}{(3K+2G)}$ (b) $\gamma = \frac{(3K-2G)}{2(G+3K)}$

(c) $\gamma = \frac{2(G+3K)}{(3K-2G)}$ (d) $\gamma = \frac{(3K+2G)}{2(G-3K)}$

**Ans. (b) :** $\gamma = \frac{(3K-2G)}{2(G+3K)}$

Where,
γ = Poisson's ratio
k = Bulk modulus of elasticity
G = Modulus of rigidity

**20. A rod is 2 m long at a temperature of $10^0$C. The temperature of it is raised to $80^0$C. If the expansion due to temperature rise is prevented, the stress developed in the rod is (take young's modulus of rigidity, E=1.0 ×$10^5$ MN/$m^2$ and coefficient of thermal expansion, α = 0.000012 per $^0$C)**

(a) 64 N/$mm^2$ (b) 64 N/$mm^2$

(c) 84 N/$mm^2$ (d) 84 kN/$mm^2$

**Ans. (c) :** Given,

$\alpha = 0.000012$, $\Delta T = 70^0$C, $E = 1\times10^5$ MN/$m^2$

$\sigma = \alpha\Delta TE$

$= 0.000012 \times (80-10) \times 1 \times 10^5$

$= 84$ N/$mm^2$

**21. The cylindrical portion of the rivet is called the ____ and the lower portion is known as the ____**

(a) shank; tail (b) head; tail

(c) body; base (d) head; base

**Ans. (a) :** The cylindrical portion of the rivet is called the shank and the lower portion is known as the tail.

**22. A riveted joint formed between two plates, when they are brought face to face such that an overlap exists, is called a _____**

(a) lap joint (b) butt joint

(c) tee joint (d) edge joint

**Ans. (a) :** A riveted joint formed between two plates, when they are brought face to face such that an overlap exists, is called a lap joint. Edge of each piece is welded to the surface of the other.

**But joint–** The two members to be connected are place end to end. Additional plates provided on either one or both sides, called cover plates are placed and are connected to the main plates.

**23. Which riveted joint has the highest efficiency, which is expressed in percentile of the commercial boiler joint?**

(a) Double riveted lap joint
(b) Triple riveted lap joint
(c) Double riveted butt joint
(d) Triple riveted butt joint

**Ans. (d):**

| Joints | | % |
|---|---|---|
| Lap | Single riveted | 50-60 |
| | Double riveted | 60-72 |
| | Triple riveted | 72-80 |
| Butt (double strap) | Single riveted | 55-60 |
| | Double riveted | 76-84 |
| | Triple riveted | 80-88 |

**24. The distance between the _____ of the rivet hole and the -_____ of the plate is known as the marginal pitch.**

(a) centre, nearest edge (b) centre, farthest edge
(c) edge; centre (d) edge ; nearest edge

**Ans. (a) :** The distance between the centre of the rivet hole and the nearest edge of the plate is known as the marginal pitch.

**25. The basic symbol used for the single V-butt weld joint is :**

| | | | |
|---|---|---|---|
| (a) | | (b) | |
| (c) | | (d) | |

**Ans. (c) :**

| | |
|---|---|
| | Single U |
| | Single V butt weld |
| | Fillet |
| | Square |

**26. A plate, 100 mm wide and 10 mm thick, is to be welded plate by means of double parallel fillets. The plates are subjected to a static load of 80 kN. The length of weld (neglecting allowance), if the permissible shear stress in the weld does not exceed 55 MPa, is :**

(a) 103 cm (b) 103 mm
(c) 115.5 mm (d) 115.5 cm

**Ans. (b) :** Given, P = 80 kN
s = 10 mm
For double parallel fillet weld $\ell_e = 2\ell$
Permissible shear stress $\tau_{max}$ = 55 MPa

$$\tau_{max} = \frac{P}{0.707\ s\ell_e}$$

$$55 = \frac{80\times10^3}{0.707\times10\times\ell_e}$$

$$\ell = 103\,\text{mm}(\text{weld})$$

**27. Two plates, 200 mm wide and 10 mm thick are to be welded by means of transverse welds at the ends. If the plates are subjected to a load of 70 kN, the size of the weld (neglecting allowance), assuming the allowable tensile stress of 70 MPa, is :**

(a) 141.42 mm (b) 141.42 cm
(c) 153.92 mm (d) 153.92 cm

**Ans. (a) :** $\sigma_t = \frac{P}{0.707s\ell}$

$$70 = \frac{70\times10^3}{0.707\times s\times100}$$

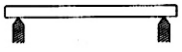
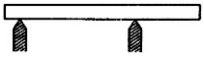
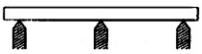
S = 141.42 mm

**28. Uniformly varying load on a beam is also known as :**

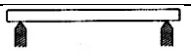
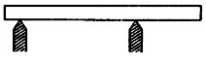
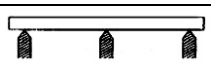
(a) concentrated load (b) sinusoidal load
(c) triangular load (d) unequal load

**Ans. (c) :** Uniformly varying load on a beam is also known as triangular load.
It is a type of load which consist of zero intensity at one end and linearly various to the other end.

**29. Which of the following is the overhanging beam?**

(a) 
(b) 
(c)
(d) 

**Ans. (b) :** j

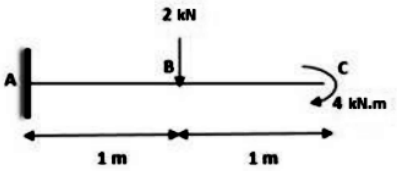
| | |
|---|---|
|  | **Simply supported beam** |
|  | **Over hanged beam** |
| | **Encastre beam** |
|  | **Continuous beam** |

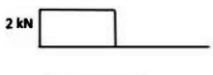
**30. In a loaded beam, the point where the bending moment is zero is known as :**

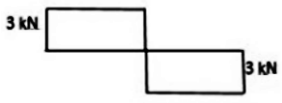
(a) the contraflexure point
(b) the zero point
(c) the bottom point
(d) the Unwin point

**Ans. (a) :** In a loaded beam, the point where the bending moment is zero is known as the contraflexure point.

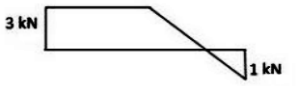
**31. A cantilever beam AB carries loading as shown in the following figure. Which of the following is the SFD for the beam?**




(a) 


(b) 

(c)




(d)

**Ans. (a) :** The reaction at support,

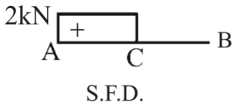
$R_A = 2kN$ (upwards)

$(S.F)_A = +2kN$

$(S.F)_C = +2-2=0$

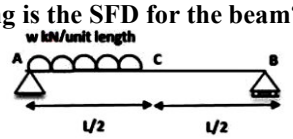
& $(S.F)_C = +2$ (in left)

$(S.F)_B = 0$

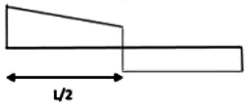



S.F.D.

**32. A simply supported beam has span L as shown in the following figure. Point C is the centre of the beam. It is the subjected to UDL, w/unit length, in the portion A to C. Which of the following is the SFD for the beam?**

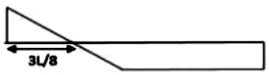



(a)



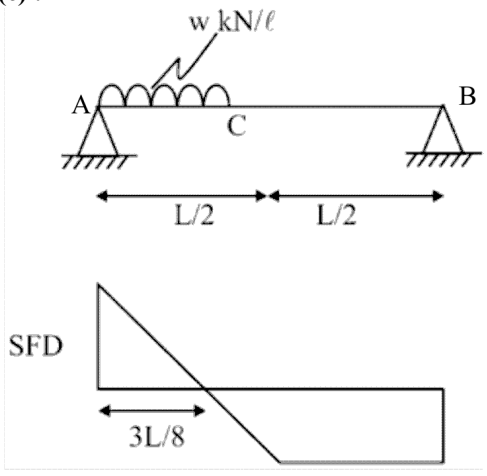

(b)

(c)




(d)

**Ans. (c) :**




taking moment about point 'B', $\sum M_B = 0$

$$R_A \times L - w \times \frac{L}{2}\left(\frac{L}{4}+\frac{L}{2}\right) = 0$$

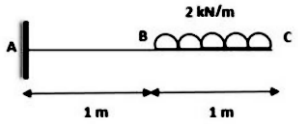
$$R_A = \frac{3}{8}wL$$

Let shear force is zero at a distance x from left hand support 'A' then :

$$\frac{3}{8}wL - w.x = 0$$

$$x = \frac{3L}{8}$$

**33. A cantilever beam is loaded as shown in the following figure. What is the shape of the bending moment diagram for portion AB?**



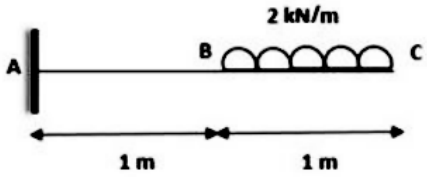

(a) Parabolic

(b) Linearly varying with maximum bending moment at B

(c) Constant bending moment from A to B

(d) Linearly varying with maximum bending moment at A

**Ans. (d) :**




$R_A + R_B = 2kN$

$\sum M_A = 0$

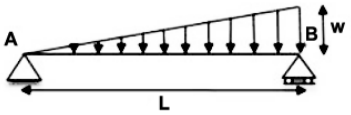
$R_c \times 2 = 2 \times 1.5$

$R_c = 1.5kN$

$R_A = 0.5$ kN

the bending moment diagram for portion AB linearly varying with maximum bending moment at A.

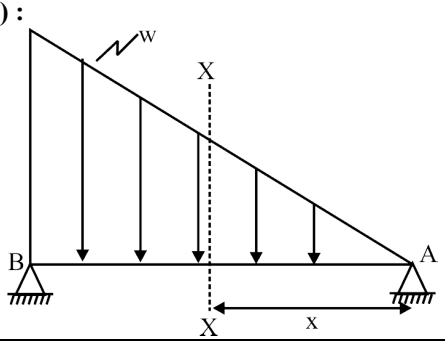
**34. For the simply supported beam shown in the following figure, at what distance from the support A is the shear force zero?**




(a) L/4 (b) L/3

(c) L/2 (d) $L/\sqrt{3}$

**Ans. (d) :**



$$\frac{dM}{dx}=0$$

$$=\frac{d}{dx}\left(\frac{wx^3}{6L}-\frac{wLx}{6}\right)$$

$$=\frac{-3wx^2}{6L}+\frac{wL}{6}=0$$

$$x=\frac{L}{\sqrt{3}}$$

**35. Which of the following statements is not true about pure bending in beams?**

(a) Inner edge of cross-section of the beam undergoes compression.

(b) Outer edge of cross-section of the beam undergoes tension

(c) Internal stresses are present along the length of the beam

(d) Shear stress is present in the beam

**Ans. (d) : Pure bending in beams–**

- Inner edge of cross-section of the beam undergoes compression.
- Outer edge of cross-section of the beam undergoes tension.
- Internal stresses are present along the length of the beam.
- Shear stress is not present in the beam.

**36. Which of the following is NOT an assumption used in the theory of simple bending?**

(a) The material is homogenous and isotropic

(b) Transverse planes remain plane after bending

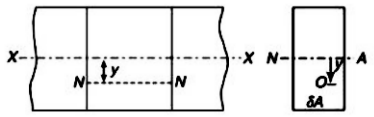
(c) Initially the beam is straight

(d) The radius of curvature is small compared with the dimensions of the cross-section

**Ans. (d) : Assumption for simple bending–**

- The material is homogenous and isotropic
- Transverse planes remain plane after bending
- Initially the beam is straight
- The radius of curvature is large compared with the dimensions of the cross-section.

**37. The longitudinal stress ($\sigma$) for the section of beam (N-N) shown in the following figure, subjected to simple bending having moment M, is given by the formula (where E = Young's modulus of rigidity, I = second moment of area)**




(a) $\sigma=\frac{M}{I}y$ (b) $\sigma=\frac{I}{M}y$

(c) $\sigma=\frac{y}{MI}$ (d) $\sigma=\frac{MI}{y}$

**Ans. (a):** The longitudinal stress ($\sigma$) for the section of beam (N-N) shown in the following figure, subjected to simple bending having moment M, is given by the formula $\sigma=\frac{M}{I}y$

Where E = Young's modulus of rigidity, I = second moment of area.

**38. Section modulus (Z) of a beam in simple bending is (where M = bending moment, = second moment of area, $\sigma$ = longitudinal stress).**

(a) $Z=\frac{M}{\sigma}$ (b) $Z=\frac{\sigma}{M}$

(c) $Z=\frac{M}{I}$ (d) $Z=\frac{I}{M}$

**Ans. (a) :** Bending equation,

$$\frac{M}{I}=\frac{\sigma}{y}=\frac{E}{R}$$

$$Z=\frac{M}{\sigma}$$

**39. Section modulus (Z) of a beam in simple bending is (where M = bending moment, = second moment of area, $\sigma$ = longitudinal stress).**

(a) $M=\sigma Z$ (b) $\sigma=ZM$

(c) $M=IZ$ (d) $I=ZM$

**Ans. (a) :** Bending equation,

$$\frac{M}{I}=\frac{\sigma}{y}=\frac{E}{R}$$

$$M=\sigma Z$$

**40. A 120-mm-wide and 10mm-thick steel plate is bent into circular are of 8-m radius. The maximum value of stress produced is (take Young's Modulus of rigidity, E = 200 GPa) :**

(a) 12.5 MPa (b) 25 MPa

(c) 125 MPa (d) 250 MPa

**Ans. (c) :** Given,

Width, B = 120 mm, Thickness, D =10 mm

R = 8 m.

E = 200 GPa

$I = BD^3 / 12$

$= 120 \times 10^3 / 12\ mm^4$

$$\frac{\sigma}{y}=\frac{E}{R}$$

$$y=\frac{D}{2}$$

$$\sigma=125\ MPa$$

**41. The variation of bending moment due a point load on a simply supported beam is governed by the _____**

(a) cubic law (b) parabolic law

(c) linear law (d) quartic law

**Ans. (c) :** The variation of bending moment due a point load on a simply supported beam is governed by the linear law

When there is no external load is present the variation in bending moment is linear.

Here, $R_A = R_B = \frac{w}{2}$

B.M. at section x from support A

$B.M. = R_A \times x$ (linear variation)

$B.M. = \frac{wx}{2}$

At $x = \frac{\ell}{2}$, $B.M._{(max)} = \frac{w\ell}{4}$

**42. The variation of bending moment between two sections of a beam is equal to :**

(a) the area under the shear force diagram
(b) the difference of shear force between the two sections
(c) the area under the bending moment diagram
(d) the difference of bending moment between the two sections

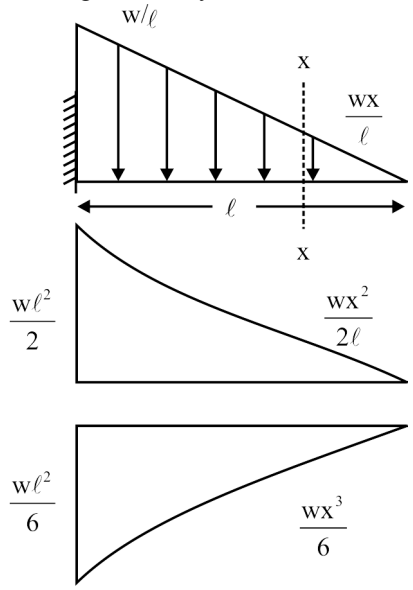
**Ans. (a) :** The variation of bending moment between two sections of a beam is equal to the area under the shear force diagram.

• The change in bending moment between two points on the beam equals the area under the shear force diagram between the same two points.

**43. Variation in bending moment is a cantilever beam, carrying a load, the intensity of which varies uniformly from zero at the free end to w per unit run at the fixed end, is governed by the**

(a) Cubic law (b) Parabolic law
(c) Linear law (d) Quartic law

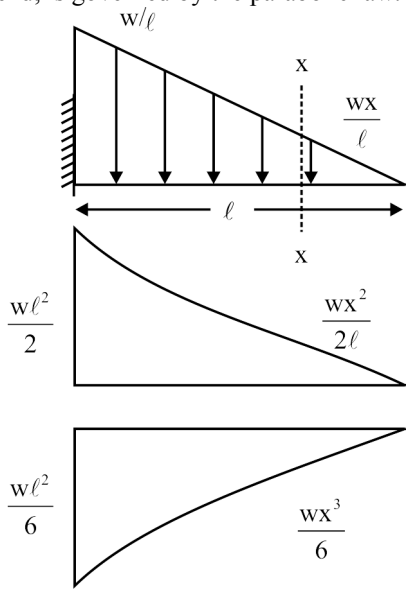
**Ans. (a) :** Variation in bending moment is a cantilever beam, carrying a load, the intensity of which varies uniformly from zero at the free end to w per unit run at the fixed end, is governed by the cubic law.




**44. Variation in shear force in a cantilever beam, carrying a load, the intensity of which varies uniformly from zero at the free end to w per unit run at the fixed end, is governed by the**

(a) Cubic law (b) Parabolic law
(c) Linear law (d) Quartic law

**Ans. (b) :** Variation in shear force in a cantilever beam, carrying a load, the intensity of which varies uniformly from zero at the free end to w per unit run at the fixed end, is governed by the parabolic law.

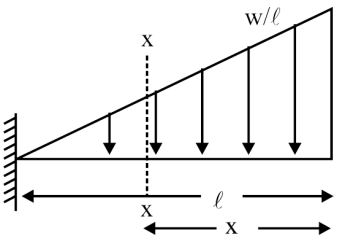



**45. Variation in shear force in a cantilever beam (length - ℓ), carrying a load, the intensity of which varies uniformly from zero at the fixed end to w per unit run at the free end, is (x is distance from free end) :**

(a) $w\left(x^2 - \frac{x^2}{2\ell}\right)$ (b) $w\left(x - \frac{x^2}{2\ell}\right)$

(c) $w\left(x^2 - \frac{x}{2\ell}\right)$ (d) $w\left(x^2 - \frac{x^3}{2\ell}\right)$

**Ans. (b) :**




Total load = $\frac{1}{2} \times w\ell$

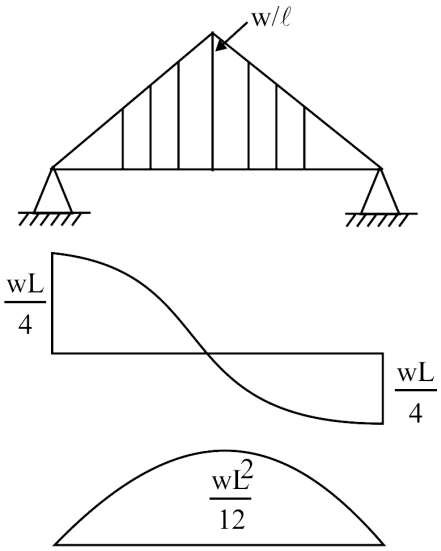
$S_x = wx - \frac{1}{2}x\left(\frac{wx}{\ell}\right)$

**46. A simply supported beam (length=ℓ) has a distributed load of intensity varying with zero at each end to w per unit run at midspan. The variation in shear force will be**

(a) $w\left(\frac{\ell}{8}-\frac{2x^2}{\ell}\right)$ (b) $w\left(\frac{\ell}{8}-\frac{4x^2}{\ell}\right)$

(c) $w\left(\frac{\ell}{4}-\frac{x^2}{\ell}\right)$ (d) $w\left(\frac{\ell}{4}-\frac{2x^2}{\ell}\right)$

**Ans. (c) :**




Total load $= \frac{1}{2}\times L\times w = \frac{w\ell}{2}$

$S_x = \frac{wL}{4}-\frac{1}{2}x\left(\frac{w}{\ell/2}\times x\right)$

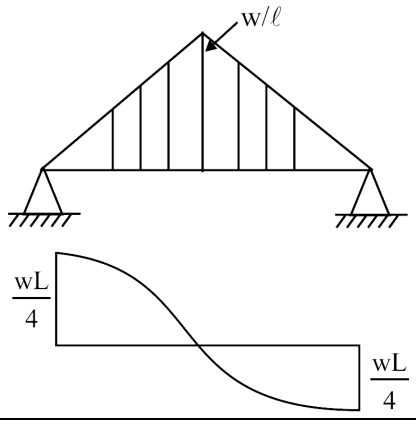
$= \frac{w\ell}{4}-\frac{wx^2}{\ell}$

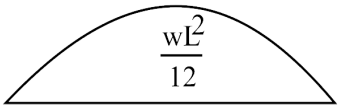
**47. A simply supported beam (length = ℓ), has a distributed load of intensity varying with zero at each end to w per unit run at midspan. The variation in bending moment will be (x is distance from left end).**

(a) $w\left(\frac{\ell}{4}-\frac{x^2}{3\ell}\right)x$ (b) $w\left(\frac{3\ell}{4}-\frac{x^2}{\ell}\right)x$

(c) $w\left(\frac{3\ell}{4}-\frac{x^2}{3\ell}\right)x$ (d) $w\left(\frac{\ell}{4}-\frac{x^3}{3\ell}\right)x$

**Ans. (a) :**




Variation in BM $= w\left(\frac{\ell}{4}-\frac{x^2}{3\ell}\right)x$

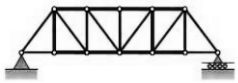
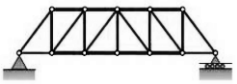
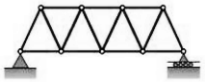
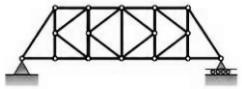
**48. In an analysis of plane frame structure, members are assumed to by joined together with.**

(a) riveted joints (b) welded joints

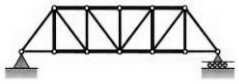
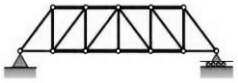
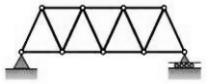
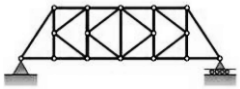
(c) rigid joints (d) pin joints

**Ans. (d) :** In an analysis of plane frame structure, members are assumed to by joined together with pin joints.

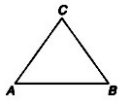
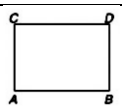
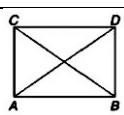
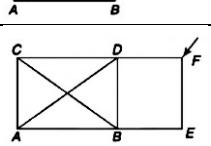
**49. Which of the is the Pratt type truss?**

| | |
|---|---|
| **(a)** |  |
| **(b)** |  |
| **(c)** |  |
| **(d)** |  |

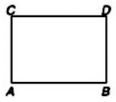
**Ans. (a) :**

| | |
|---|---|
|  | Pratt |
|  | Howe |
|  | Warren |
|  | K-truss |

**50. Which of the following is a deficiency frame?**

| | |
|---|---|
| **(a)** |   |
| **(b)** |   |
| **(c)** |   |
| **(d)** |   |

**Ans. (b) :** deficiency frame

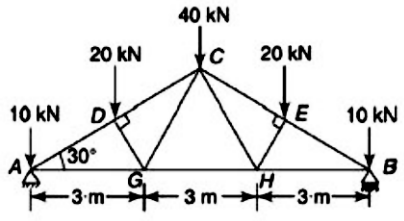



**51. The method of joints is preferred in the analysis of plane frames if forces are required to be determined in :**

(a) one member only (b) two member only
(c) a few members (d) in all the members

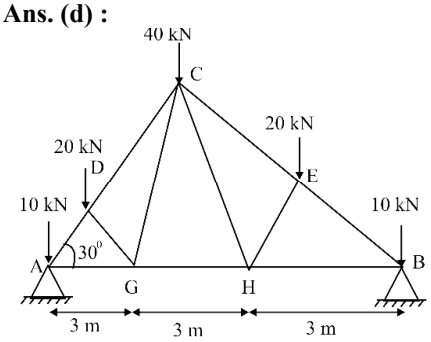
**Ans. (d) :** The method of joints is preferred in the analysis of plane frames if forces are required to be determined in all the members.

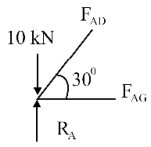
**52. The force in member AD of the truss shown in the following figure is (take D and E as the mid-points of AC and BC.**




(a) 120 kN (b) -120 kN
(c) 80 kN (d) -80 kN

**Ans. (d) :**




$\sum M_A = 0$

$10\times 9 + 20\times 7.5 + 40\times 4.5 + 20\times 1.5 = 9R_B$

$R_B = 50$ kN

$R_A + R_B = 100$ kN

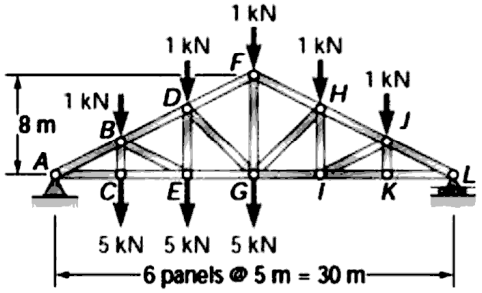
$R_A = 50$ kN

$\sum V = 0$

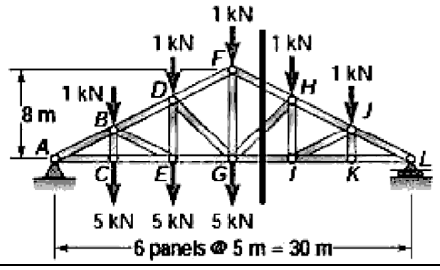
$R_A + F_{AD}\sin 30^0 = 10$

$F_{AD} = -80$ kN

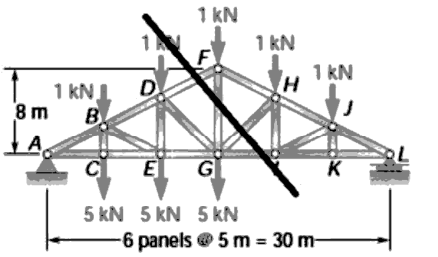
**53. To determine the force in members FH, GH and GI of the truss shown in the following figure, using the method of sections, the correct way to draw the section will be :**



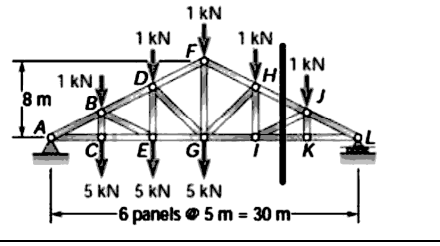

**(a)**



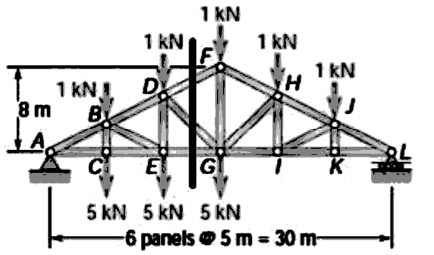

**(b)**



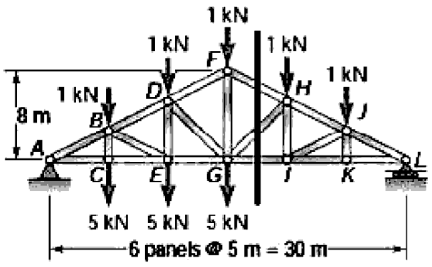

**(c)**




**(d)**




**Ans. (a) :** Using the method of section, the correct way to draw the section will be–




**54. The variation of shear stress in a circular shaft, subjected to torsion, with radial distance is :**

(a) linear (b) parabolic
(c) hyperbolic (d) quartic

**Ans. (a) :** The variation of shear stress in a circular shaft, subjected to torsion, with radial distance is linear.

**55. The relation governing the torsional torque in circular shafts is (where T=Torque, $\tau$=shear stress, r = radial distance, $\ell$ = length of shaft, J = polar moment of inertia, G = modulus of rigidity, $\theta$ = angle of twist.**

(a) $\frac{T}{\tau}=\frac{\tau}{\ell}=\frac{G\theta}{J}$ (b) $\frac{T}{J}=\frac{\tau}{r}=\frac{G\theta}{\ell}$

(c) $\frac{T}{J}=\frac{\tau}{\ell}=\frac{G\theta}{r}$ (d) $\frac{T}{\ell}=\frac{\tau}{r}=\frac{G\theta}{J}$

**Ans. (b) :** $\frac{T}{J}=\frac{\tau}{r}=\frac{G\theta}{\ell}$

Where,
T=Torque, τ=shear stress, r = radial distance,
$\ell$ = length of shaft, J = polar moment of inertia,
G = modulus of rigidity, θ = angle of twist

**56. The angle of twist of a circular shaft is given by (where T = Torque, $\ell$ = length of shaft, J = polar moment of inertia, G = modulus of rigidity).**

(a) $\frac{GJ}{T\ell}$ (b) $\frac{T\ell}{GJ}$

(c) $\frac{TJ}{G\ell}$ (d) $\frac{TG}{J\ell}$

**Ans. (b) :** $\frac{T}{J}=\frac{\tau}{r}=\frac{G\theta}{\ell}$

Where,
T=Torque, τ=shear stress, r = radial distance,
$\ell$ = length of shaft, J = polar moment of inertia,
G = modulus of rigidity, θ = angle of twist

$\theta=\frac{T\ell}{GJ}$

**57. The maximum shear stress of a solid shaft is given by (where T = Torque, d = shaft diameter)**

(a) $\frac{16T}{\pi d}$ (b) $\frac{16T}{\pi d^2}$

(c) $\frac{16T}{\pi d^3}$ (d) $\frac{16T}{\pi d^4}$

**Ans. (c) :** $\tau=\frac{16T}{\pi d^3}$

Where,
T = Torque, d = shaft diameter,

Bending stress, $\sigma=\frac{32\,M}{\pi d^3}$

**58. The maximum shear stress in a hollow shaft subjected to a torsional moment_____**

(a) is at the middle of thickness
(b) is at the inner surface of the shaft
(c) is at the at the outer surface of the shaft
(d) can be anywhere on the shaft

**Ans. (c) :** The maximum shear stress in a hollow shaft subjected to a torsional moment is at the at the outer surface of the shaft.

**59. The ratio of the maximum stress of a hollow shaft to that of a solid shaft, subjected to torsion, if both are of the same material and of the same outer diameter, is (k is the ratio of the inner diameter to the outer diameter of the hollow shaft),**

(a) $\frac{1}{1-k^4}$ (b) $1-K^4$

(c) $\frac{1}{1-k^3}$ (d) $(1-(k)^3)$

**Ans. (a) :** The ratio of the maximum stress of a hollow shaft to that of a solid shaft, subjected to torsion, if both are of the same material and of the same outer diameter, is $\frac{1}{1-k^4}$

Where,
k is the ratio of the inner diameter to the outer diameter of the hollow shaft

**60. The torsional stiffness is given by ($\ell$ = length of shaft, J= polar moment of inertia, G = modulus of rigidity)**

(a) $\frac{JG}{\ell}$ (b) $\frac{J}{G\ell}$

(c) $\frac{G}{J\ell}$ (d) $\frac{\ell}{JG}$

**Ans. (a) :**

$\frac{T}{J}=\frac{\tau}{r}=\frac{G\theta}{\ell}$

$\frac{T}{\theta}=\frac{JG}{\ell}$

Where,
T=Torque, τ=shear stress, r = radial distance,
$\ell$ = length of shaft, J = polar moment of inertia,
G = modulus of rigidity, θ = angle of twist

**61. A cylinder-shell may be termed as thin if the ratio of thickness of the wall to the diameter of the shell is less than :**

(a) 1 : 5 (b) 1 : 10
(c) 1 : 15 (d) 1 : 20

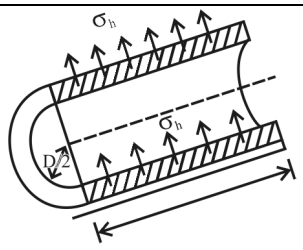
**Ans. (c) :** A cylinder-shell may be termed as thin if the ratio of thickness of the wall to the diameter of the shell is less than $\frac{1}{15}$

Thick shell $\frac{t}{D}>\frac{1}{15}$

**62. The circumferential stresses in thin cylinders are also known as**

(a) hoop stresses (b) radial stresses
(c) tangential stresses (d) longitudinal stresses

**Ans. (a) :** The circumferential stresses in thin cylinders are also known as hoop stresses.
Consider a thin cylindrical vessel subjected to an internal fluid pressure. The circumferential stress will be set up in the material of the cylinder, if the bursting of the cylinder takes place.




P = Internal pressure of fluid
d = Internal diameter of the cylinder
t = Thickness of the wall of the cylinder
$\sigma_h$ = Circumferential or hoop stress in material

- The bursting will takes place if the force due to fluid pressure is more than the resisting force due to circumferential stress set up in the material.
- **Force due to fluid pressure** = P × Area on which P is acting

$= P \times (d \times L)$ ..... (i)

- **Force due to circumferential stress** = $\sigma_1$ × Area on which $\sigma_1$ is acting

$= \sigma_1 \times (L \times t + L \times t)$

$= \sigma_1 \times 2Lt = 2\sigma_1 \times L \times t$ .....( ii)

**Equation (i) and (ii) we get–**

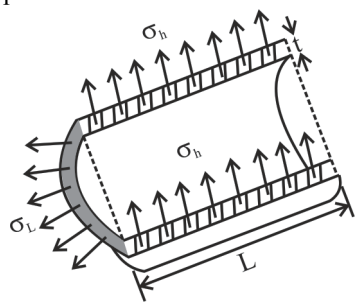
$P \times d \times L = 2\sigma_1 \times L \times t$

$$\boxed{\sigma_h = \frac{Pd}{2t}}$$

**63. Longitudinal stresses in the body of a thin cylinder (with uniform thickness) having spherial end are _____ hoop stresses.**

(a) half of (b) the same as
(c) double of (d) triple of

**Ans. (a) :** Let the diameter of cylindrical shell be 'd' and pressure is 'p' and thickness is 't' then–




($\sigma_L$) Longitudinal stress in shell $= \frac{pd}{4t}$

($\sigma_h$) Hoop stress in shell $= \frac{pd}{2t}$

$$\frac{\text{Longitudinal stress}}{\text{Hoop stress}} = \frac{\frac{pd}{4t}}{\frac{pd}{2t}} = \frac{pd}{4t} \times \frac{2t}{pd} = \frac{2}{4} = 0.5$$

**64. Hoop strain in the body of a thin cylinder (with uniform thickness 't') having spherical ends is (d=internal diameter, p = internal pressure, E = Young's modulus, v = poisson's ratio**

(a) $\frac{pd}{2tE}(2-v)$ (b) $\frac{pd}{4tE}(2-v)$
(c) $\frac{pd}{4tE}(4-v)$ (d) $\frac{pd}{2tE}(4-v)$

**Ans. (b) :** Hoop strain in the body of a thin cylinder (with uniform thickness 't') having spherical ends is

$\frac{pd}{4tE}(2-v)$

d=internal diameter, p = internal pressure,
E = Young's modulus, v = poisson's ratio

**65. The ratio of hoop stress to hoop strain in the hemispherical portion of a thin cylinder (with uniform thickness 't') is (E = Young's modulus, v = Poisson's ratio).**

(a) $\frac{E}{(1-v)}$ (b) $\frac{E}{(2-v)}$
(c) $\frac{2E}{(1-v)}$ (d) $\frac{2E}{(2-v)}$

**Ans. (a) :** The ratio of hoop stress to hoop strain in the hemispherical portion of a thin cylinder (with uniform thickness 't') is $\frac{E}{(1-v)}$.

Where, E = Young's modulus, v = Poisson's ratio

**66. Neglecting the effect of internal pressure, the volumetric strain of a thin cylinder is ($\sigma$ = hoop stress, E = Young's modulus, v = Poisson's ratio)**

(a) $\frac{\sigma}{E}(2.5-2v)$ (b) $\frac{\sigma}{E}(2.5-v)$
(c) $\frac{\sigma}{E}(1.5-2v)$ (d) $\frac{\sigma}{E}(1.5-v)$

**Ans. (a) :**
Neglecting the effect of internal pressure, the volumetric strain of a thin cylinder is $\frac{\sigma}{E}(2.5-2v)$

Where, $\sigma$ = hoop stress, E = Young's modulus,
v = Poisson's ratio

**67. If the load on a column is increased to a value that on its removal the deflection remains and the column doesn't return to its original position, the load is known as :**

(a) critical load (b) ultimate load
(c) yield load (d) breaking load

**Ans. (a) :** If the load on a column is increased to a value that on its removal the deflection remains and the column doesn't return to its original position, the load is known as critical load.
The axial load which is sufficient to keep the column in a slight deflected shape is called crippling, buckling or critical load.

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$

**68. The ratio of the equivalent length of a column fixed at one end with the other free to that of a column with both ends hinged is (consider the length of the two columns to be the same).**

(a) 1 : 1 (b) 2 : 1
(c) 3 : 1 (d) 1 : 2

**Ans. (b) :** The ratio of the equivalent length of a column fixed at one end with the other free to that of a column with both ends hinged is 2 : 1.

Effective length of column with different end condition.

| End condition | Both end hinged | One end fixed other free | Both end fixed | One end fixed and other hinged |
|---|---|---|---|---|
| Effective length ($l_{eff}$) | L | 2L | L/2 | $L/\sqrt{2}$ |

**69. Euler's Crippling Load for a column with both ends fixed is _____ that for a column fixed at one end with the other end hinged.**

(a) the same as (b) half of
(c) double of (d) triple of

**Ans. (c) :** Euler's Crippling Load for a column with both ends fixed is double of that for a column fixed at one end with the other end hinged.

| End condition | Both end hinged | One end fixed other free | Both end fixed | One end fixed and other hinged |
|---|---|---|---|---|
| Effective length ($l_{eff}$) | L | 2L | L/2 | $L/\sqrt{2}$ |

**70. Perry's approximate formula is (where $\sigma_{max}$ = maximum compressive stress, $\sigma_0 = \frac{P}{A}, \sigma_e = \frac{P_e}{A}$ e=eccentricity, $y_c$ = distance of the extreme compressive fibre from the neutral axis, k = radius of gyration, p = Rankine's crippling load, $P_e$ = Euler's crippling load, A = area.**

(a) $\left(\frac{\sigma_{max}}{\sigma_0}-1\right)\left(1-\frac{\sigma_0}{\sigma_e}\right)=\frac{(1.2e)y_c}{k^2}$

(b) $\left(1-\frac{\sigma_{max}}{\sigma_0}\right)\left(1-\frac{\sigma_0}{\sigma_e}\right)=\frac{(1.2e)y_c}{k^2}$

(c) $\left(\frac{\sigma_{max}}{\sigma_0}-1\right)\left(\frac{\sigma_0}{\sigma_e}-1\right)=\frac{(1.2e)y_c}{k^2}$

(d) $\left(\frac{\sigma_{max}}{\sigma_0}+1\right)\left(1+\frac{\sigma_0}{\sigma_e}\right)=\frac{(1.2e)y_c}{k^2}$

**Ans. (a) :** Maximum compressive stress

$$\left(\frac{\sigma_{max}}{\sigma_0}-1\right)\left(1-\frac{\sigma_0}{\sigma_e}\right)=\frac{(1.2e)y_c}{k^2}$$

Where,
e=eccentricity, $y_c$ = distance of the extreme compressive fibre from the neutral axis,
k = radius of gyration,
p = Rankine's crippling load,
$P_e$ = Euler's crippling load, A = area

**71. In a columns, what is the relationship between the slenderness ratio and the critical stress?**

(a) Slenderness ratio increase with the increase in critical stresses
(b) Slenderness ratio decrease with the increase in critical stress
(c) Slenderness ratio remains constant with increase in critical stress
(d) Slenderness ratio first increase and after a maximum value, starts to decrease, with the increase in critical stress

**Ans. (b) :** Critical stress = $\frac{\text{Critical load}}{\text{Area}}$

Critical load as per Euler's formula

$$P=\frac{\pi^2 EI}{L^2}$$

$$I = Ar^2$$

Both sides division with A

$$\frac{P}{A}=\frac{\pi^2 E\, Ar^2}{A.(l)^2}$$

Hence $\frac{P}{A}$ = Critical stress

So, Critical stress = $\frac{\pi^2 E}{\left(\frac{l}{r}\right)^2}$

Slenderness ratio decrease with the increase in critical stress.

**72. Secant formula is applicable for :**

(a) short columns under axial loading
(b) long columns under axial loading
(c) short columns under eccentric loading
(d) long columns under eccentric loading

**Ans. (d) :** ■ Secant formula was adopted in the earlier IS 800 prior to 1984

■ IS 800 : 1984 recommended the use of Marchant Rankine Formula.

■ IS 800 : 2007 recommends different column curbs viz a, b, c, & d in non-dimensional form based on Perry Robertson approach based on cross-section approach.

It is applicable for long column under eccentric loading

**73. On application of shear stress, the fluid will :**

(a) start to flow
(b) not flow
(c) flow or not depending on the value of the shear stress
(d) flow or not depending on other factors apart from the shear stress

**Ans. (a) :** On application of shear stress, the fluid will start to flow

■Shear stress is primarily caused by friction between fluid particles due to fluid viscosity.

■When a shear stress is applied to a fluid at rest, the fluid will not remain at rest, but will move because of the shear stress.

**74. _____ is the ratio of dynamic viscosity to kinematic viscosity**

(a) Density (b) Virtual viscosity

(c) Specific weight (d) Specific volume

**Ans. (a) :** Density is the ratio of dynamic viscosity to kinematic viscosity.

The dynamic viscosity of a fluid is the measure of its resistance to flow when an external force is applied.

**75. The dimensions of specific weight are same as that of :**

(a) pressure (b) force/volume

(c) work/volume (d) density

**Ans. (b) :** The dimensions of specific weight are same as that of force/volume

The specific weight is also known as unit weight is the weight per unit volume of a material.

**76. Which of the following statements is correct about the density of water?**

(a) Density of water increase with temperature

(b) Density of water decreases with temperature

(c) Density of water is maximum at $4^0$C

(d) Density of water is maximum at $0^0$C

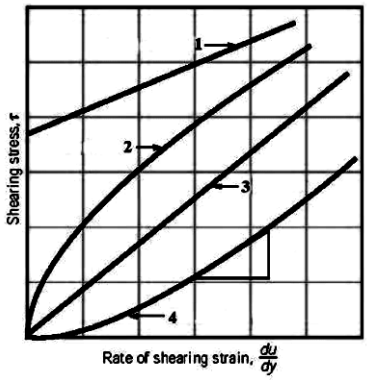
**Ans. (c) :**

Density of water is maximum at $4^0$C

The maximum density of water occurs at $4^0$C because at this temperature two opposing affects are in balance .

When the temperature decrease, the water becomes frozen at $0^0$C and its density is reduced and its volume increasers, due to which the ice floats in the water.

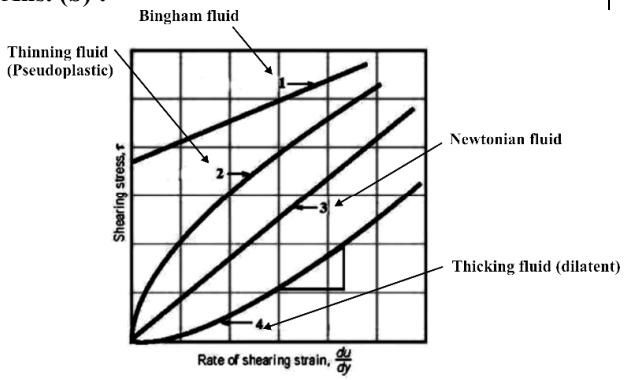
**77. In the graph of shearing stress vs rate of shearing strain, which of the following lines represents shear**




(a) 1 (b) 2

(c) 3 (d) 4

**Ans. (b) :**




**78. If the specific gravity of a fluid is 1.26, its specific weight will be (take density of water = 1000 kg/m$^3$ and acceleration due to gravity = 10 m/s$^2$)**

(a) 12.6 N/m$^3$

(b) 12.6 kN/m$^3$

(c) 1.26 N/m$^3$

(d) 1.26 kN/m$^3$

**Ans. (b) :** Specific gravity, $G = \frac{\rho_{liq}}{\rho_{water}}$

$\rho_{liq} = 1.26 \times 1000 = 1260\,kg/m^3$

Specific weight = $\rho_{liq} g$

$= 12.6\ kN/m^3$

**79. For liquids or gases at rest, the pressure gradient in the vertical direction at any point in a fluid depends only on the _____ of the fluid at that point.**

(a) viscosity

(b) specific weight

(c) temperature

(d) thermal conductivity

**Ans. (b) :** For liquids or gases at rest, the pressure gradient in the vertical direction at any point in a fluid depends only on the specific weight of the fluid at that point.

Pressure in a fluid at rest is independent of the shape or cross section of the container

**80. Which of the following statements is NOT correct about gauge pressure?**

(a) It can be more than absolute pressure.

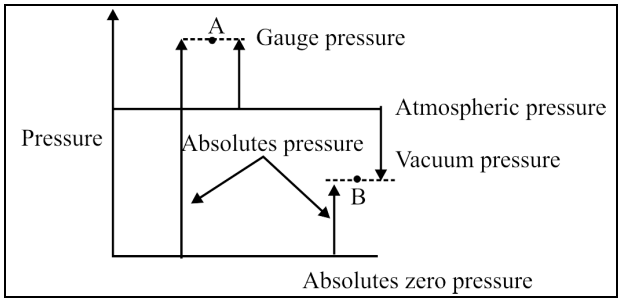
(b) For a pressure value more than atmospheric pressure, gauge pressure will be positive

(c) It can have negative values

(d) For a pressure value less than atmospheric pressure, gauge pressure will be negative.

**Ans. (a) :** statement about gauge pressure-

- For a pressure value more than atmospheric pressure, gauge pressure will be positive
- It can have negative values
- For a pressure value less than atmospheric pressure, gauge pressure will be negative.




**81. The variation of atmospheric pressure with altitude is given by (where $P_a$ and $T_a$ are the pressure and temperature at sea level, β is the lapse rate, is the acceleration due to gravity and R is gas constant).**

(a) $P = P_a\left(1-\frac{\beta z}{T_2}\right)^{\frac{g}{R\beta}}$

(b) $P = P_a\left(1-\frac{\beta z}{T_2}\right)^{\frac{R\beta}{g}}$

(c) $P = P_a\left(\frac{\beta z}{T_2}-1\right)^{\frac{g}{R\beta}}$

(d) $P = P_a\left(\frac{\beta z}{T_2}-1\right)^{\frac{R\beta}{g}}$

**Ans. (a) :** The variation of atmospheric pressure with altitude is given by

$$P = P_a\left(1-\frac{\beta z}{T_2}\right)^{\frac{g}{R\beta}}$$

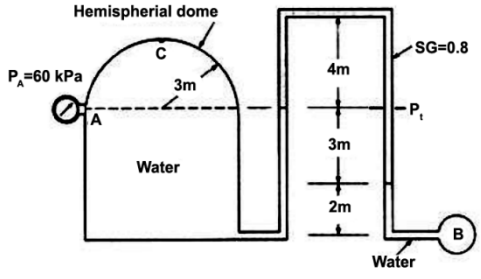
where $P_a$ and $T_a$ are the pressure and temperature at sea level, β is the lapse rate, is the acceleration due to gravity and R is gas constant

**82. Any two points at _____ elevation in a continuous mass of the same static fluid will be at ____ pressure(s).**

(a) the same; the same (b) the same; difficult
(c) different; the same (d) the same; absolute

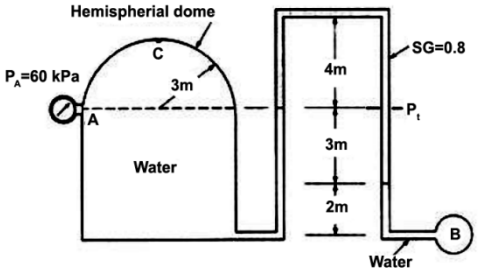
**Ans. (a) :** Any two points at same elevation ina continous mass of the same static fluid will be at same pressure thus we can jump across U-tube filled with the same fluid.

**83. With reference to the given figure, if p represents pressure, which of the following is correct?**



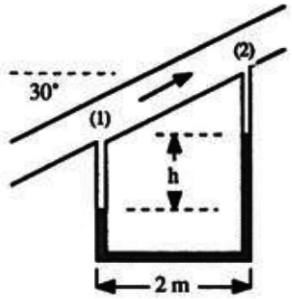

(a) $P_A = P_t$ (b) $P_A = P_t - P_B$
(c) $P_A < P_t$ (d) $P_A > P_t$

**Ans. (a) :**




Pressure,
$P_A = P_t$
In a liquid at same level the pressure will be same at all points

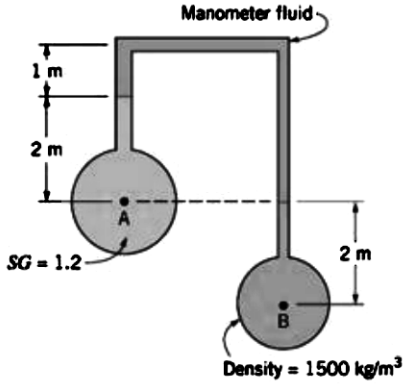
**84. Water flows upward in a pipe slanted at $30^0$, as shown in the given figure. If the mercury manometer reads h = 12 cm, the pressure difference between points (1) and (2) in the pipe is (take specific weight of water as 9790 N/m$^3$ and of mercury as 133100 N/m$^3$)**




(a) 2.4 kPa (b) 16.0 kPa
(c) 26.1 kPa (d) 34.4 kPa

**Ans. (c) :** The vertical distance between points (1) and (2) equals (2 m). tan $30^0$ or 1.155 m.
$P_1$+ 9790 h – 133100h – 9790 (1.155 m) = $P_2$
$P_1$–$P_2$ = (133100–9790) (0.12) + 11300= 26100 Pascal
= 26.1 KPa

**85. For the stationary fluid shown in the following figure, if the pressure at point B is 20 kPa greater than that at point A. the specific weight of the manometer fluid is (take specific weight of water as 9810 N/m$^3$ and acceleration due to gravity as 9.81 m/s$^2$.**



(a) 7.1 kN/m$^3$ (b) 7.1 N/m$^3$
(c) 16.5 kN/m$^3$ (d) 16.5 N/m$^3$

**Ans. (a) :** Given,

$P_B = P_A + 20$

$P_B - P_A = 20$

$P_{\text{left limb}} = P_A - (1.2 \times 9.81 \times 2)$

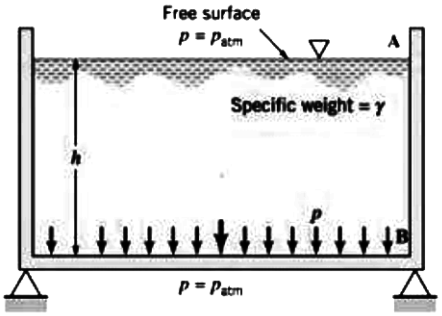
$P_{\text{right limb}} = P_B - (1.5 \times 9.81 \times 2) - (s \times 9.81 \times 2)$

$= P_B - 29.43 - 19.02\ s\ m$

By solving we get

$S = 7.1\ kN/m^3$

**86. For the tank shown in the following figure, the resultant force due to the pressure on side wall AB will be :**




(a) located at the mid-point of the wall
(b) located at the upper half of the wall
(c) located at the lower half of the wall
(d) located at the base of the wall (at point B)

**Ans. (c) :** , The resultant force due to the pressure on side wall AB will be located at the lower half of the wall

$P_A = P_{atm}$

**87. Based on the two statements given below, choose the correct answer.**

**Based on the two statements given below, choose the correct answer.**

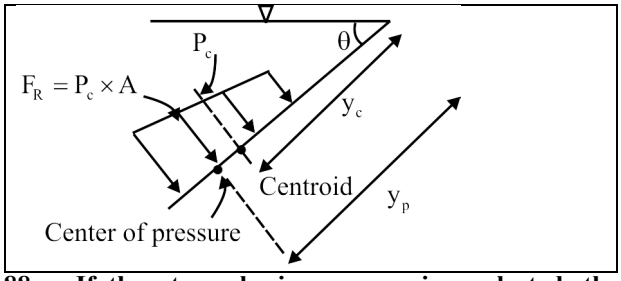
**Statement A : The magnitude of the resultant fluid force is equal to the pressure acting at the centroid of the area multiplied by the total area.**

**Statement B : The resultant fluid force, acting on a fully submerged inclined plane surface, does not pass through the centroid of the area.**

(a) Both statements A and B are correct
(b) Both statements A and B are incorrect
(c) Statement A is correct but statement B is incorrect
(d) Statement A is incorrect but statement B is correct

**Ans. (a) :** The magnitude of the resultant fluid force is equal to the pressure acting at the centroid of the area multiplied by the total area.

■The resultant fluid force, acting on a fully submerged inclined plane surface, does not pass through the centroid of the area.




**88. If the atmospheric pressure in neglected, the coordinates for the centre of pressure ($x_{CP}$, $y_{CP}$) for a fully submerged inclined plane surface is (where $I_{xx}$ is the area moment of inertia of the plate area about its centroidal x axis, $I_{xy}$ is the product of inertia of the plane surface, θ is the angle of inclination of the plane surface, $h_{CG}$ is the depth straight down from the surface to the plate centroid, A is the area of plane surface :**

(a) $X_{cp} = \dfrac{I_{xy}\sin\theta}{h_{co}A}, y_{cp} = \dfrac{I_{xx}\sin\theta}{h_{co}A}$

(b) $X_{cp} = \dfrac{I_{xy}\sin\theta}{h_{co}A}, y_{cp} = \dfrac{I_{xy}\sin\theta}{h_{co}A}$

(c) $X_{cp} = \dfrac{I_{xy}\sin\theta}{h_{co}A}, y_{cp} = \dfrac{I_{xx}\cos\theta}{h_{co}A}$

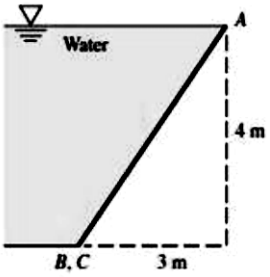
(d) $X_{cp} = \dfrac{I_{xx}\cos\theta}{h_{co}A}, y_{cp} = \dfrac{I_{xy}\sin\theta}{h_{co}A}$

**Ans. (a) :** If the atmospheric pressure in neglected, the coordinates for the centre of pressure ($x_{CP}$, $y_{CP}$) for a fully submerged inclined plane surface is

$$X_{cp} = \frac{I_{xy}\sin\theta}{h_{co}A}, y_{cp} = \frac{I_{xx}\sin\theta}{h_{co}A}$$

where $I_{xx}$ is the area moment of inertia of the plate area about its centroidal x axis, $I_{xy}$ is the product of inertia of the plane surface, θ is the angle of inclination of the plane surface, $h_{CG}$ is the depth straight down from the surface to the plate centroid, A is the area of plane surface

**89. Panel ABC in the slanted side of a water tank (shown in the following figure) is an isosceles triangle with vertex at A and base BC = 2m. The water force on the panel is (take specific weight of water as 9790 N/m$^3$**




(a) 131.10 N (b) 131.0 N
(c) 32.6 N (d) 32.6 N

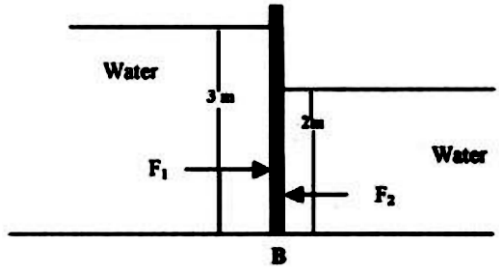
**Ans. (b):** The centroid of ABC is $\frac{2}{3}$ of the depth down or 8.3 m from the surface.

The panel area is $\left(\frac{1}{2}\right)\times 2\times 5 = 5\ \text{m}^2$

The water force is

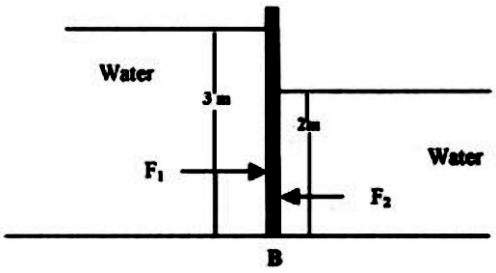
$$F_{ABC} = \gamma h_{CG} A_{panel} = 9790\times 2.67\times 5$$
$$= 131\ \text{N}$$

**90. A vertical lock gate is 4 m wide and separates water levels of 2 m and 3m, respectively, as shown in the following figure. The moment about the bottom required to keep the gate stationary is (take specific weight of water as 9790 N/m$^3$.**




(a) 104 kN.m (b) 114 kN.m
(c) 124 kN.m (d) 134 kN.m

**Ans. (c) :**




$$F_1 = \gamma h_{CG_1} A_1 = (9790)(1.5)(3)(4)$$
$$= 176220\ \text{N}$$
$$F_2 = \gamma h_{CG_2} A_2 = (9790)(1)(2)(4)$$
$$= 78320\ \text{N}$$
$$y_{cp_1} = \frac{\left[-\left(\frac{1}{12}\right)(4)(3)^3 \sin 90^0\right]}{1.5\times 4\times 3}$$
$$= -0.5\ \text{m}$$

So, $F_1$ acts at $1.5 - 0.5 = 1$ m above B

$$y_{cp_2} = \frac{\left[-\left(\frac{1}{12}\right)(4)(2)^3 \sin 90^0\right]}{1\times 4\times 2}$$
$$= -0.33\ \text{m}$$

So, $F_2$ acts at $1 - 0.33 = 0.67$ m above B

Taking moment about point B

$$\sum M_B = (176220\text{N})(1\text{m}) - (78320\text{N})(0.667\text{m})$$
$$= 124\ \text{kNm}$$

**91. Flow at varying rates through a long straight pipe of uniform cross-section is a___**

(a) steady and uniform flow
(b) steady and non-uniform flow
(c) unsteady and uniform flow
(d) unsteady and non-uniform flow

**Ans. (c) :** Unsteady and uniform flow – long straight pipe of uniform cross section.
Steady and non-uniform flow – tapering pipe
Unsteady and non-uniform flow– wave travelling along a channel.

**92. A____ is a set of fluid particles that form a line at a given instant.**

(a) streamline (b) pathline
(c) streak line (d) timeline

**Ans. (d) :** A timeline is a set of fluid particles that form a line at a given instant.
A streak line is a curved line formed by a string of fluid particles which have passes through a certain point.

**93. Which of the following statements is NOT correct about flow streamline?**

(a) Fluid particles accelerate normal to streamlines.
(b) Fluid particles accelerance along streamlines.
(c) The component of weight along a streamline does not depend on the streamline angle
(d) The lines that are tangent to the velocity vectors throughout the flow field are called streamlines.

**Ans. (c) :** Related to flow stream –
- Fluid particles accelerate normal to streamlines.
- Fluid particles accelerance along streamlines.
- The component of weight along a streamline depend on the streamline angle
- The lines that are tangent to the velocity vectors throughout the flow field are called streamlines.

**94. The Bernoulli equation can be obtained by integrating F = ma along a_____**

(a) streamline (b) pathline
(c) streakline (d) timeline

**Ans. (a) :** The Bernoulli equation can be obtained by integrating F = ma along a stream line.

**95. Which of the following assumption for the Bernoulli equation $\frac{P}{\rho}+\frac{v^2}{2}+gz =$ constant, is NOT correct?**

(a) Flow is steady
(b) Flow is incompressible
(c) Flow is viscous
(d) Flow is along single streamline

**Ans. (c) :** Assumption for the Bernoulli equation $\frac{P}{\rho}+\frac{v^2}{2}+gz =$ constantm is-
- Flow is steady
- Flow is incompressible
- Flow is inviscous
- Flow is along single streamline

**96. The pitot formula, used for velocity measurement using pitot tube, is (where V is the velocity, $p_0$ is the stagnation pressure, $p_s$ is the static pressure and ρ is the density**

(a) $V \approx \left[2\frac{(p_0 - p_s)}{\rho}\right]^{1/2}$ (b) $V \approx \left[\frac{(p_0 - p_5)}{\rho}\right]^{1/2}$

(c) $V \approx \left[2\frac{\sqrt{(p_0 - p_s)}}{\rho}\right]^{1/2}$ (d) $V \approx \left[\frac{\sqrt{(p_0 - p_s)}}{\rho}\right]^{1/2}$

**Ans.(a):** Static pressure + Dynamic pressure = Total pressure

$$P_s + \rho\frac{V^2}{2} = P_0$$

$$V \approx \left[2\frac{(p_0 - p_s)}{\rho}\right]^{1/2}$$

**97. Which of the following is NOT a constriction meter?**

(a) Thin-plate orifice (b) Flow nozzle
(c) Venturi tube (d) Rotameter

**Ans. (d)** Constriction meter such as thin-plate orifice, flow nozzle and venturi tube.

**98. Which of the following statements is NOT true about nozzle meter discharge coefficient, $C_d$ (Re is Reynolds number and β is ratio of nozzle diameter to pipe diameter)?**

(a) $C_d$ generally increases with increase in Re at constant β.
(b) $C_d$ generally increases with increases to β at constant $R_e$.
(c) $C_d$ for nozzle meter is more than that for an orifice meter at the same β and Re.
(d) Formation of Vena contracts does not take place.

**Ans. (a) :** $C_d$ generally increases with increases to β at constant $R_e$.
- $C_d$ for nozzle meter is more than that for an orifice meter at the same β and Re.
- Formation of Vena contracts does not take place.

**99. If the actual velocity in the contracted section of a jet of liquid flowing from a 50-mm-diameter orifice is 8.91 m/s under a head of 5 m, the value of the coefficient of velocity will be (take acceleration due to gravity as 10 m/s$^2$.**

(a) 0.891 (b) 0.861
(c) 0.901 (d) 0.821

**Ans. (a) :** $C_v = \frac{\text{Actual velocity}}{\text{Theoretical velocity}}$

$$C_v = \frac{8.91}{\sqrt{2\times10\times5}} = 0.891$$

**100. The flow velocity (V) from an orifice, attached to a tank of sufficiently large cross-sectional area, is (where $a_c$ is the area of vena contraction, $a_1$ is the lank area, h is the height of the fluid in the tank g is the acceleration due to gravity.**

(a) $V = \sqrt{2gh\frac{a_c}{a_1}}$ (b) $V = \sqrt{\frac{2gh}{\left(\frac{a_1}{a_c}\right)^2 - 1}}$

(c) $V = \sqrt{\frac{2gh}{1-\left(\frac{a_c}{a_1}\right)^2}}$ (d) $V = \sqrt{2gh}$

**Ans. (d) :** The flow velocity (V) from an orifice, attached to a tank of sufficiently large cross-sectional area, is $V = \sqrt{2gh}$ where $a_c$ is the area of vena contraction, $a_1$ is the lank area, h is the height of the fluid in the tank g is the acceleration due to gravity.

**101. The actual discharge (Q) through an orifice, attached to a tank, is given by (where V is the flow velocity from the orifice $a_c$ is the area of contraction, a is the orifice area, h is the height of the fluid in the tank, $C_c$ is the coefficient of contraction, Cv is the coefficient of velocity and g is the acceleration due to gravity).**

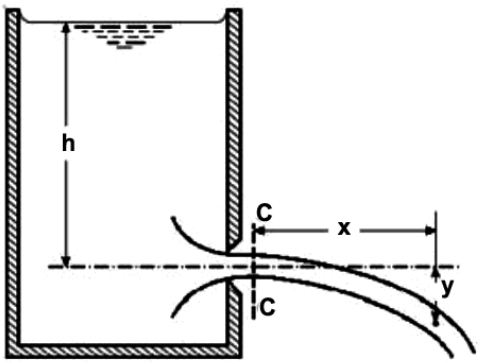
(a) $Q = a \times V$
(b) $Q = (C_c \times a) \times (C_v\sqrt{2gh})$
(c) $Q = (C_c \times a_c) \times (C_v\sqrt{2gh})$
(d) $Q = a\sqrt{2gh}$

**Ans. (b) :** The actual discharge (Q) through an orifice, attached to a tank, is given by

$$Q = C_c \times a \times C_v\sqrt{2gh}$$

Where V is the flow velocity from the orifice $a_c$ is the area of contraction, a is the orifice area, h is the height of the fluid in the tank, $C_c$ is the coefficient of contraction, Cv is the coefficient of velocity and g is the acceleration due to gravity.

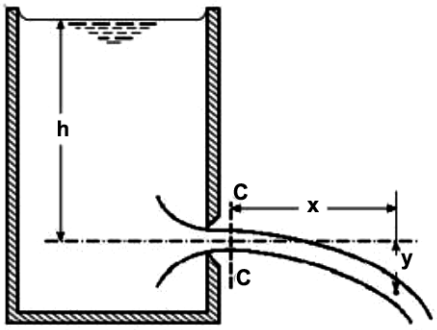
**102. In reference to the following figure, the coefficient of velocity is given by**




(a) $\sqrt{\frac{x^2}{4hy}}$ (b) $\sqrt{\frac{y}{4hx^2}}$

(c) $\sqrt{\frac{x}{4hy^3}}$ (d) $\sqrt{\frac{y^2}{4hx}}$

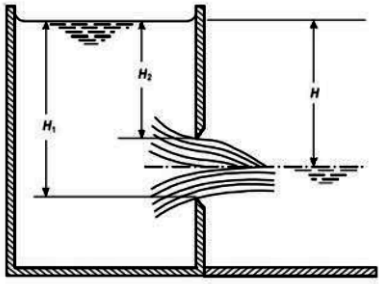
**Ans. (a) :**




$$C_v = \sqrt{\frac{x^2}{4hy}}$$

$$C_d = \frac{Q}{a\sqrt{2gh}}$$

**103. In reference to the following figure, the total discharge, Q, through the partially submerged orifice is given by (where $C_{d1}$ and $C_{d2}$ represents the coefficients of discharge for the free portion and submerged portion of the orifice, respectively and the breadth of the orifice is b)**



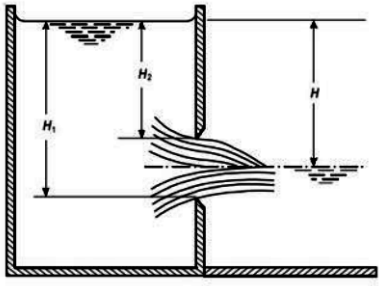

(a) $Q = \frac{2}{3}C_{d1}b\sqrt{2g}\left\{H^{3/2} - H_2^{3/2}\right\} + C_{d2}b\left(H_1 - H\right)\sqrt{2gH}$

(b) $Q = \frac{2}{3}C_{d1}b\sqrt{2g}\left\{H_1 - H\right\} + C_{d2}b\left(H^{3/2} - H_2^{3/2}\right)\sqrt{2gH}$

(c) $Q = \frac{2}{3}C_{d1}b\sqrt{2gH}\left\{H^{3/2} - H_2^{3/2}\right\} + C_{d2}b\left(H_1 - H\right)\sqrt{2gH}$

(d) $Q = C_{d1}b\sqrt{2g}\left\{H^{3/2} - H_2^{3/2}\right\} + \frac{2}{3}C_{d2}b\left(H_1 - H\right)\sqrt{2gH}$

**Ans. (a) :**




Total discharge,

$$Q = \frac{2}{3}C_{d1}b\sqrt{2g}\left\{H^{3/2} - H_2^{3/2}\right\} + C_{d2}b\left(H_1 - H\right)\sqrt{2gH}$$

Where $C_{d1}$ and $C_{d2}$ represents the coefficients of discharge for the free portion and submerged portion of the orifice, respectively and the breadth of the orifice is b.

**104. Water discharge at the rate of 98 litres per second through a vertical sharp-edged orifice of area 0.01 $m^2$ placed under a constant head to 10 m. A point on the jet measured from the vens contracta of the jet has coordinates 3.85 m horizontal and 0.4 m vertical. The value of coefficient of contraction is (take acceleration due to gravity as 10 $m/s^2$.**

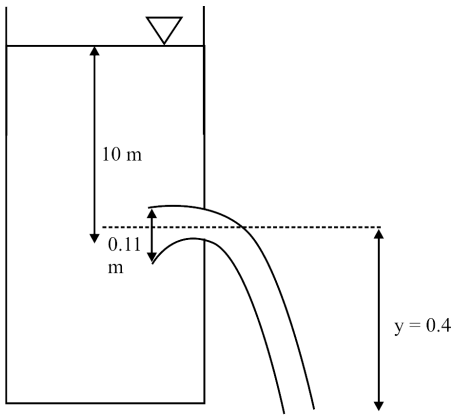
(a) 0.03 (b) 0.73
(c) 0.78 (d) 0.96

**Ans. (b) :** Given,

$Q_a$ = 0.0982 ℓ/sec

x = 3.85 meter

y = 0.4 meter




$$Q_a = C_d a\sqrt{2gH}$$

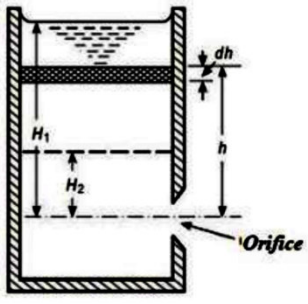
$$C_d = \frac{Q_a}{a\sqrt{2gH}}$$

$$C_d = \frac{0.0982}{0.01\sqrt{2\times 9.81\times 10}} = 0.70$$

$$C_v = \sqrt{\frac{x^2}{4yH}} = \sqrt{\frac{(3.85)^2}{4\times 0.4\times 10}} = 0.96$$

$$C_c = \frac{C_d}{C_v} \Rightarrow C_c = 0.72$$

**105. In reference to the following figure, the time to completely empty the vertical cylinderical tank is (where A is the cross-sectional area of the tank, a is the area of orifice and $C_d$ is the coefficient of discharge.**



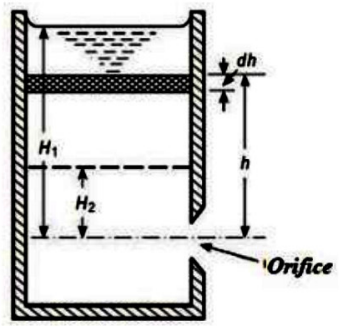
(a) $\dfrac{2A\left(H_1^{1/2}\right)}{C_d a\sqrt{2g}}$

(b) $\dfrac{2a\left(H_1^{1/2}\right)}{C_d A\sqrt{2g}}$

(c) $\dfrac{2A\left(h^{1/2}\right)}{C_d a\sqrt{2g}}$

(d) $\dfrac{2a\left(H_1\right)}{C_d A\sqrt{2gh}}$

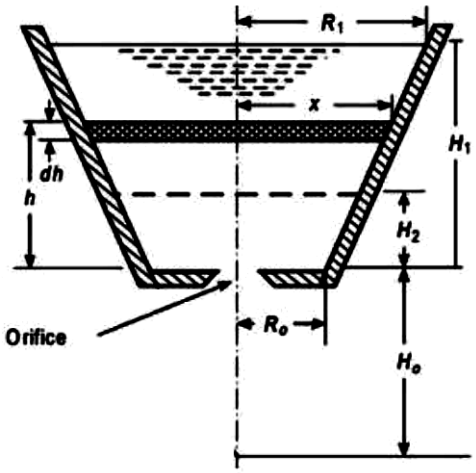
**Ans. (a) :**




the time to completely empty the vertical cylinderical tank is $\dfrac{2A\left(H_1^{1/2}\right)}{C_d a\sqrt{2g}}$

Where, A is the cross-sectional area of the tank, a is the area of orifice and $C_d$ is the coefficient of discharge

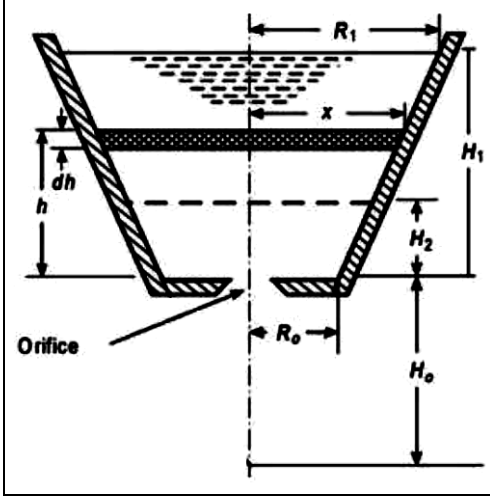
**106. In reference to the conical tank shown in the given figure, which of the following expression for $H_0$ is correct?**




(a) $H_0 = \dfrac{R_0 H_1}{\left(R_1 - R_0\right)}$ (b) $H_0 = \dfrac{R_1 H_2}{\left(R_1 - R_0\right)}$

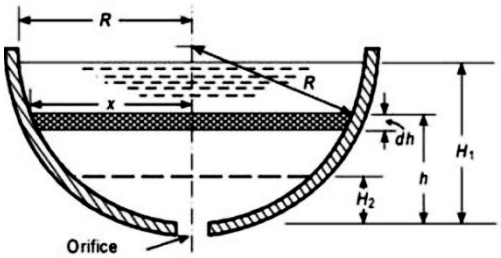
(c) $H_0 = \dfrac{\left(R_1 - R_0\right)}{R_0 H_1}$ (d) $H_0 = \dfrac{\left(R_1 - R_0\right)}{R_1 H_2}$

**Ans. (a) :**




$$H_0 = \frac{R_0 H_1}{\left(R_1 - R_0\right)}$$

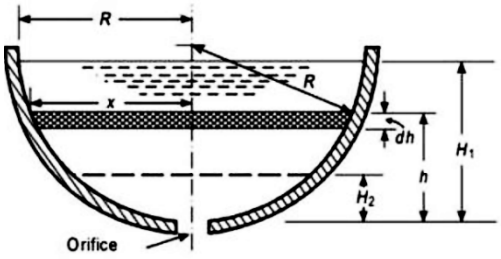
**107. In reference to the following figure, if the hemispherical tank was full at the beginning and it is completely emptied, the time taken is given by (where g is the acceleration due to gravity, a is the area of orifice and $C_d$ is the coefficient of discharge)**




(a) $\dfrac{14\pi R^{3/2}}{15C_d 2\sqrt{2g}}$ (b) $\dfrac{15\pi R^{1/2}}{14C_d a\sqrt{2g}}$

(c) $\dfrac{14C_d a\sqrt{2g}}{15\pi R^{3/2}}$ (d) $\dfrac{15C_d a\sqrt{2g}}{14\pi R^{3/2}}$

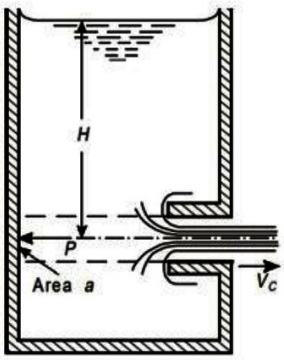
**Ans. (a) :**




The time taken is given by $\dfrac{14\pi R^{3/2}}{15C_d 2\sqrt{2g}}$

Where g is the acceleration due to gravity, a is the area of orifice and $C_d$ is the coefficient of discharge.

**108. In a Borda's mouthpiece as shown in the following figure, 40 mm diameter discharge under a constant head of 1.5 m. if the coefficient of a velocity for the entrance section of the mouthpiece is 0.95, the coefficient of countraction, when the mouthpiece is running free, is**




(a) 0.554 (b) 0.39
(c) 0.63 (d) 0.68

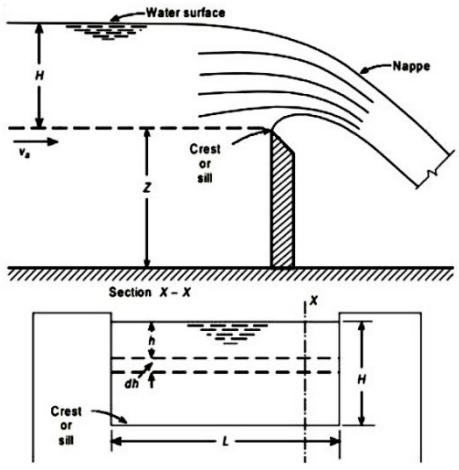
**Ans. (a) :**

$$C_c = \frac{\text{Area of the jet at vena-contracta}}{\text{Area of the orifice}}$$

$$C_d = C_c \times C_v$$

So, we get coefficient of cntraction is equal 0.554

**109. The discharge Q passing over the rectangular notch shown in the following figure is given by (where $C_d$ is the coefficient of discharge $h_a$ is the approach velocity based head and g is the acceleration due to gravity.**




(a) $Q = \frac{2}{3} C_d \sqrt{2gL}\left[(H + h_a)^{3/2} - h_a^{3/2}\right]$

(b) $Q = C_d \sqrt{2gL}\left[(H + h_a)^{3/2} - h_a^{3/2}\right]$

(c) $Q = \frac{3}{2} C_d \sqrt{2gL}\left[(H + h_a)^{2/3} - h_a^{3/3}\right]$

(d) $Q = \frac{2}{3} C_d \sqrt{2gL}\left[(H + h_a)^{2/3} - h_a^{2/3}\right]$

**Ans. (a):** The discharge Q passing over the rectangular notch

$$Q = \int_0^H C_d b dh \sqrt{2gh}$$

$$= C_d b \sqrt{2g}\left[\frac{h^{3/2}}{3/2}\right]_0^H$$

$$= \frac{2}{3} C_d b \sqrt{2g}\left[h^{3/2}\right]_0^H$$

$$= \frac{2}{3} C_d b \sqrt{2g}\left[h^{3/2}\right]_0^H$$

$$= \frac{2}{3} C_d b \sqrt{2g} H^{3/2}$$

$$Q = \frac{2}{3} C_d \sqrt{2gL}\left[(H + h_a)^{3/2} - h_a^{3/2}\right]$$

where $C_d$ is the coefficient of discharge $h_a$ is the approach velocity based head and g is the acceleration due to gravity

**110. Francis formula for the discharge over a rectangular weir neglecting the approach velocity is (here H and L are the height and width of the weir, respectively and n is the number of end contraction for the weir.**

(a) $Q = 1.84\ (L – 0.1(nH)H^{3/2}$
(b) $Q = 1.84\ (L – 0.1(nH)H^{2/3}$
(c) $Q = 18.4\ (L – 0.1(nH)H^{3/2}$
(d) $Q = 1.84\ (L – 0.1(nH)H^{4/3}$

**Ans. (a) :** Discharge over a rectangular weir

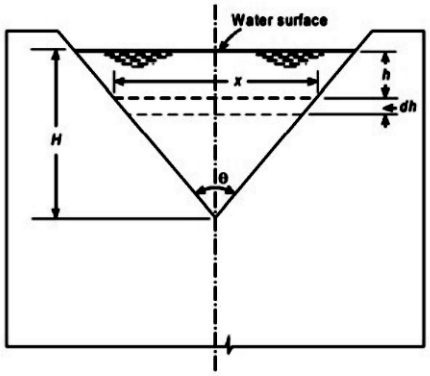
$$Q = \frac{2}{3} C_d \left(L - 0.1nH\right)\sqrt{2g} H^{3/2}$$

$C_d = 0.623$, $g = 9.81\ m/s^2$

$Q = 1.84\ (1 – 0.1nH)^{3/2}$

where H and L are the height and width of the weir, respectively and n is the number of end contraction for the weir

**111. For flow over the triangular notch shown in the following figure, constant for the notch is given by (where $C_d$ is the coefficient of discharge, and g is the acceleration due to gravity.**




(a) $\frac{51}{8} C_d \sqrt{2g} \sin\frac{\theta}{2}$ (b) $\frac{8}{15} C_d \sqrt{2g} \sin\frac{\theta}{2}$

(c) $\frac{15}{8} C_d \sqrt{2g} \sin\frac{\theta}{2}$ (d) $\frac{8}{15} C_d \sqrt{2g} \tan\frac{\theta}{2}$

**Ans. (d):** Standard results for triangular notch

$$Q = \int_0^H C_d 2(H-h)\tan\frac{\theta}{2}dh\sqrt{2gh}$$

$$= 2C_d\sqrt{2g}\tan\frac{\theta}{2}\int_0^H (H-h)\sqrt{h}dh$$

$$= 2C_d\sqrt{2g}\tan\frac{\theta}{2}\int_0^H \left[\left(Hh^{1/2}-h^{3/2}\right)\right]dh$$

$$Q = 2C_d\sqrt{2g}\tan\frac{\theta}{2}\left[\frac{Hh^{3/2}}{3/2}-\frac{h^{5/2}}{5/2}\right]_0^H$$

$$= \frac{8}{15}C_d\sqrt{2g}\tan\frac{\theta}{2}H^{5/2}$$

Where $C_d$ is the coefficient of discharge, and g is the acceleration due to gravity.

**112. Which of the following statements is NOT true about a triangular weir?**

(a) The nappe emerging from a triangular weir or notch has the same shape for nearly all different heads.

(b) For measuring low discharge a triangular weir or notch is more useful as compared to a rectangular weir.

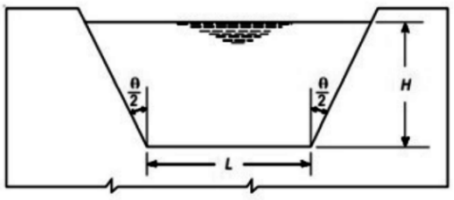
(c) In most of the cases of flow over a triangular weir or notch, the velocity of approach may be neglected without introducing an appreciable error

(d) Ventilation of a triangular weir is must

**Ans. (d) :** For triangular weir–

- The nappe emerging from a triangular weir or notch has the same shape for nearly all different heads.
- For measuring low discharge a triangular weir or notch is more useful as compared to a rectangular weir.
- In most of the cases of flow over a triangular weir or notch, the velocity of approach may be neglected without introducing an appreciable error.

**113. In a typical Cipolletti weir shown in the figure, the value of $\frac{\theta}{2}$ is :**




(a) $14^0$ (b) $22^0$

(c) $30^0$ (d) $38^0$

**Ans. (a) :** Discharge $Q = \frac{2}{3}C_d\sqrt{2g}H^{3/2}$,

Total discharge, Q =

$$\frac{2}{3}C_d\sqrt{2g}(L-0.2H)^{3/2}+\frac{8}{15}C_d\sqrt{2g}\tan\frac{\theta}{2}H^{5/2}$$

Equating both

$$L = L-0.2H+\frac{4}{5}\tan\frac{\theta}{2}\times H$$

$$\frac{4}{5}\tan\frac{\theta}{2}\times H = 0.2H$$

$$\tan\frac{\theta}{2} = 0.2\times\frac{5}{4} = \frac{1}{4}$$

$$\frac{\theta}{2} = 14^0$$

**114. The darcy friction factor (f) for laminar flow in circular pipes is given by ($Re_d$ is pipe diameter-based Reynolds number).**

(a) $f = \frac{64}{Re_d}$ (b) $f = \frac{Re_d}{64}$

(c) $f = 0.316Re_d^{-1/4}$ (d) $f = \left(1.8\log\left(\frac{Re_d}{6.9}\right)\right)^{-2}$

**Ans. (a) :** darcy friction factor (f) for laminar flow in circular pipes is given by $f = \frac{64}{Re_d}$.

($Re_d$ is pipe diameter-based Reynolds number)

**115. Darcy-Weisbach equation used for computing the loss of head due to friction ($h_f$) is pipes is given by (where V is the velocity of flow, L and D are the length and diameter of the pipe, respectively, f is the friction factor and g is the acceleration due to gravity?**

(a) $h_f = \frac{fLV^2}{2gD}$ (b) $h_f = \frac{2gD}{fLV^2}$

(c) $h_f = \frac{fLV}{\sqrt{2gD}}$ (d) $h_f = \frac{\sqrt{2gD}}{fLV}$

**Ans. (a) :** Darcy-Weisbach equation used for computing the loss of head due to friction ($h_f$) is pipes is given by

$$h_f = \frac{fLV^2}{2gD}$$

**116. Which of the following losses occurring in pipe flow does NOT belong to the category of minor loss?**

(a) Loss is sudden expansion

(b) Loss in bends

(c) Loss in flow through valves, open or partially closed

(d) Frictional loss in pipes

**Ans. (d) : Minor loss–** • Loss is sudden expansion

- Loss in bends
- Loss in flow through valves, open or partially closed

**Major loss–**

Frictional loss in pipes

**117. In pipe flows the minor loss coefficient, K, and the Darcy friction factor, f, are related by the equation (where $L_{eq}$ is the pipe equivalent length and d is the pipe diameter)**

(a) $L_{eq} = \frac{Kd}{f}$ (b) $L_{eq} = \left(\frac{K}{f}\right)^{1/2}d$

(c) $L_{eq} = \left(\frac{K}{f}\right)^{1/2}d$ (d) $L_{eq} = \left(\frac{K}{f}\right)^{2}d$

**Ans. (a):** In pipe flows the minor loss coefficient, K, and the Darcy friction factor, f, are related by the equation $L_{eq} = \frac{Kd}{f}$

**118. What is the correct relationship of loss coefficients (K) for $90^0$ bend radius (R) to pipe diameter (d) ratio?**

(a) K increases with increase in R/d
(b) K decrease with increase in R/d
(c) K first decrease till a certain value and then increases with increase in R/d
(d) K first increase till a certain value and then decreases with increase in R/d

**Ans. (c) :** K first decrease till a certain value and then increases with increase in R/d of loss coefficient for $90^0$ bend radius to pipe diameter ratio.

**119. The loss coefficient $K_{SE}$, of flow entering from a pipe of smaller diameter (d) to a pipe of larger diameter (D), known as sudden expansion, is**

(a) $K_{SE} = \left(1 - \frac{d^2}{D^2}\right)^{1/2}$ (b) $K_{SE} = \left(1 - \frac{d^2}{D^2}\right)$

(c) $K_{SE} = \left(1 - \frac{d}{D}\right)^2$ (d) $K_{SE} = \left(1 - \frac{d^2}{D^2}\right)^2$

**Ans. (d) :** The loss coefficient $K_{SE}$, of flow entering from a pipe of smaller diameter (d) to a pipe of larger diameter (D), known as sudden expansion, is $K_{SE} = \left(1 - \frac{d^2}{D^2}\right)^2$.

**120. The empirical formula of loss coefficient K, for a flow entering from a finite reservoir of diameter D to a pipe of diameter d is (valid upto d/D = 0.76**

(a) $K = 0.42\left(1 - \frac{d^2}{D^2}\right)^2$ (b) $K = 0.42\left(1 - \frac{d}{D}\right)^2$

(c) $K = 0.42\left(1 - \frac{d^2}{D^2}\right)$ (d) $K = 0.42\left(1 - \frac{d^2}{D^2}\right)^{1/2}$

**Ans. (c) :** The empirical formula of loss coefficient K, for a flow entering from a finite reservoir of diameter D to a pipe of diameter d is $K = 0.42\left(1 - \frac{d^2}{D^2}\right)$.

**121. For gradual conical expansion from a pipe of diameter $d_1$ to a pipe of diameter $d_2$, the loss coefficient K is (where $C_p$ is the pressure-recovery coefficient).**

(a) $D = C_p - \frac{d_1^4}{d_2^4}$ (b) $K = 1 - \frac{d_1}{d_2} - C_p$

(c) $K = 1 - \frac{d_1^2}{d_2^2} - C_p$ (d) $K = 1 - \frac{d_1^4}{d_2^4} - C_p$

**Ans. (d):** For gradual conical expansion from a pipe of diameter $d_1$ to a pipe of diameter $d_2$, the loss coefficient K is $K = 1 - \frac{d_1^4}{d_2^4} - C_p$
Where $C_p$ is the pressure-recovery coefficient.

**122. The pipe-head loss is equal to the change in the..**

(a) height of the hydraulic grade line
(b) pressure head only
(c) gravity head only
(d) velocity head only

**Ans. (a) :** The pipe-head loss is equal to the change in the height of the hydraulic grade line.

**123. In frictionless flow, with no work or heat transfer, the energy grade line**

(a) linearly increase
(b) linearly decrease
(c) has constant height
(d) first increases then decreases

**Ans. (c) :** In frictionless flow, with no work or heat transfer, the energy grade line has constant height .

**124. The height to which a liquid would rise in a piezometer tube attached to the flow in the same as the _____ .**

(a) height of the hydraulic grade line
(b) height of the energy grade line
(c) gravity head only
(d) velocity head only

**Ans. (a) :** The height to which a liquid would rise in a piezometer tube attached to the flow in the same as the height of the hydraulic grade line .

**125. In an open-channel flow, the hydraulic grade line is :**

(a) below the free surface of the water
(b) identical to the free surface of the water
(c) above the free surface of the water
(d) can be above or below the free surface of the water

**Ans. (b) :** In an open-channel flow, the hydraulic grade line is identical to the free surface of the water.
The HGL is the height to which to which liquid would rise in a piezometric tube attached to the flow.

**126. The relationship between hydraulic grade line (HGL) and the energy grade line (EGL) is**

(a) HGL = EGL – velocity head
(b) HGL = EGL – potential head
(c) HGL = EGL – presser head
(d) HGL = EGL – (pressure head + potential head)

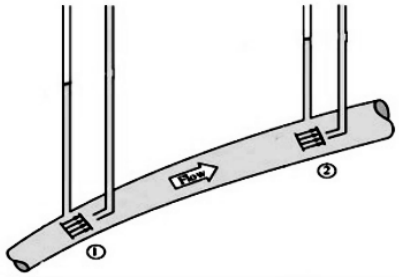
**Ans. (a) :** The relationship between hydraulic grade line (HGL) and the energy grade line (EGL) is HGL = EGL – velocity head.

**127. The head given by a pitot stagnation- velocity tube corresponds to the _____.**

(a) hydraulic grade line (b) energy grade line
(c) gravity head line (d) velocity head only

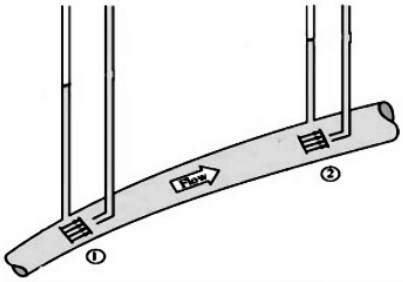
**Ans. (b) :** The head given by a pitot stagnation- velocity tube corresponds to the energy grade line.

**128. The hydraulic grade line (HGL) and the energy grade the line (EGL) for a frictionaless flow in the duct shown in the following figure are ____ respectively.**




(a) of gradually increases height and straight
(b) of gradually decreases height and straight
(c) of gradually increases height and gradually increasing higher
(d) straight and straight

**Ans. (a) :**




The hydraulic grade line (HGL) and the energy grade the line (EGL) for a frictionaless flow in the duct of gradually increases height and straight.
The energy gradient line shows the height of the total Bernoulli equation
In frictionless flow with no work or heat transfer the EGL has constant height.

**129. The power available at the outlet of a pipe is (where Q is the discharge through the pipe, V is the velocity of flow, L and D are the length and the diameter of the pipe, respectively, w is the specific weight, f is the friction factor and g is the acceleration due to gravity).**

(a) $w\left(\frac{\pi D^2}{4}\times V\right)\left(H-\frac{fLV^2}{2gD}\right)$

(b) $w\left(\frac{\pi D^2}{2}\times V\right)\left(H-\frac{fLV^2}{2gD}\right)$

(c) $w\left(\frac{\pi D^3}{4L}\times V\right)\left(H-\frac{fLV^2}{2gD}\right)$

(d) $w\left(\frac{\pi D^3}{2L}\times V\right)\left(H-\frac{fLV^2}{2gD}\right)$

**Ans. (a) :** The power available at the outlet of a pipe is

$w\times A\times V\left(H-h_f\right)$

$w\left(\frac{\pi D^2}{4}\times V\right)\left(H-\frac{fLV^2}{2gD}\right)$

Where Q is the discharge through the pipe, V is the velocity of flow, L and D are the length and the diameter of the pipe, respectively, w is the specific weight, f is the friction factor and g is the acceleration due to gravity

**130. The water from a reservoir at a high attitude is conveyed by a pipeline. The efficiency of power transmission in this case is given by (where Q is the volume flow rate, R is the hydraulic resistance of the pipeline and H is the potential head of water in the reservoir).**

(a) $\eta_p=1-\frac{RQ^2}{H}$ (b) $\eta_p=1-\frac{H}{RQ^2}$

(c) $\eta_p=1-\frac{RQ^2}{H^2}$ (d) $\eta_p=1-\frac{H^2}{RQ^2}$

**Ans. (a) :** The water from a reservoir at a high attitude is conveyed by a pipeline. The efficiency of power transmission in this case is given by $\eta_p=1-\frac{RQ^2}{H}$
Where Q is the volume flow rate, R is the hydraulic resistance of the pipeline and H is the potential head of water in the reservoir.

**131. The efficiency of power transmission through a pipe, at the condition of maximum power delivered, is (where H is the total head supplied at the entrance to the pipe and hf is the loss of head due to friction)**

(a) 1/2 (b) 1/4
(c) 1/3 (d) 2/3

**Ans. (d) :** The efficiency of power transmission through a pipe, at the condition of maximum power delivered, is 2/3.
Where H is the total head supplied at the entrance to the pipe and hf is the loss of head due to friction.

**132. Corresponding to the maximum power transmitted through a pipeline, the efficiency of power transmission is :**

(a) 50.0% (b) 56.7%
(c) 66.7% (d) 76.7%

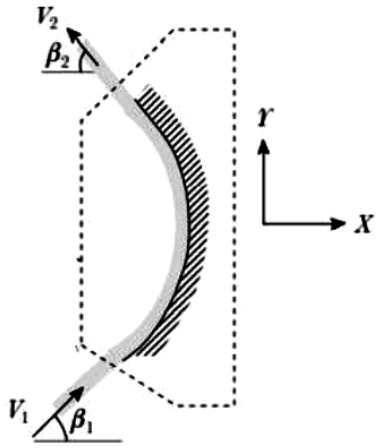
**Ans. (c) :** Corresponding to the maximum power transmitted through a pipeline, the efficiency of power transmission is 66.7% .

**133. The relation between force F and jet velocity V, for a high-velocity jet impinging on a stationary flat vertical plate (neglecting friction), is**

(a) $F\propto V^{1/2}$ (b) $F\propto V^{3/2}$
(c) $F\propto V^2$ (d) $F\propto V^3$

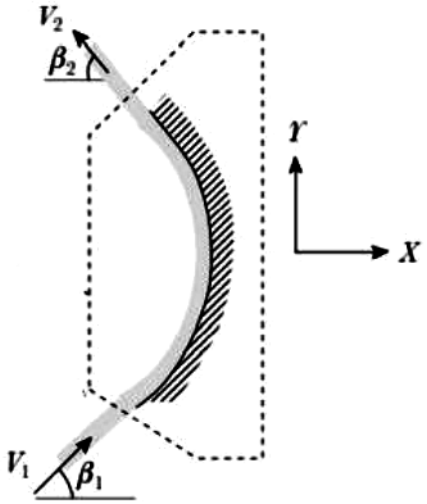
**Ans. (c) :** The relation between force F and jet velocity V, for a high-velocity jet impinging on a stationary flat vertical plate (neglecting friction), is $F\propto V^2$

**134. Which of the following statements is true about force F due to a high-velocity jet impingement on a stationary curved plate (neglecting friction) as shown in the given figure?**




(a) Force in X-direction is dependent on $V_1\cos\beta_1$
(b) Force in X-direction is dependent on $V_2\cos\beta_2$
(c) Force in Y-direction is dependent on $V_1\cos\beta_1$
(d) Force in Y-direction is dependent on $V_2\cos\beta_2$

**Ans. (a) :**




Force in X-direction is dependent on $V_1\cos\beta_1$
Force in Y-direction is dependent on $V_2\sin\beta_2$

**135. The force, F, generated when a jet strikes at the middle of one of the flat plate mounted on a wheel is (where V is the jet velocity and u is the tangential velocity of the wheel at the middle of the plate, D is the diameter of the wheel at the middle of the plate and A is the area of the plate).**

(a) $F = \rho AV(V-u)$ (b) $F = \rho AV^2$

(c) $F = \rho AVu$ (d) $F = \rho Au^2$

**Ans. (a) :** The force, F, generated when a jet strikes at the middle of one of the flat plate mounted on a wheel is

$$F = \rho AV(V-u)$$

Where V is the jet velocity and u is the tangential velocity of the wheel at the middle of the plate, D is the diameter of the wheel at the middle of the plate and A is the area of the plate

**136. A number of flat plates are mounted on a wheel and the jet strikes at the middle of the plate. The efficiency of this wheel is (where V is the jet velocity and u is the tangential velocity of the wheel at the middle of the plate).**

(a) $\dfrac{u^2}{2V(V-u)}$ (b) $\dfrac{2V(V-u)}{u^2}$

(c) $\dfrac{V^2}{2u(V-u)}$ (d) $\dfrac{2u(V-u)}{V^2}$

**Ans. (d) :** Power of jet = $F_x.u$

$$= \rho AV(V-u)u$$

Efficiency of system

$$\eta = \frac{\text{output of jet}}{\text{input of jet}}$$

$$= \frac{\text{work done by jet}}{\text{kinetic energy}}$$

$$= \frac{\rho AV(V-u)u}{\frac{1}{2}\rho AV^2} \times 100$$

$$\eta = \frac{2(V-u)u}{V^2} \times 100$$

For maximum efficiency

$$\frac{d\eta}{du} = 0$$

$$\frac{2}{V^2}(V-2u) = 0$$

$$u = \frac{V}{2}$$

$$\eta_{max} = \frac{2\left\{V.\frac{V}{2} - \left(\frac{V}{2}\right)^2\right\} \times 100}{V^2}$$

$$\eta_{max} = 50\%$$

**137. A number of flat plates are mounted on a wheel and the jet strikes at the middle of the plate. The efficiency of this wheel is maximum when (where V is the jet velocity and u is the tangential velocity of the wheel at the middle of the plate)**

(a) $u = \dfrac{V}{\sqrt{2}}$ (b) $u = \dfrac{V}{2}$

(c) $u = \dfrac{\sqrt{2}V}{3}$ (d) $u^2 = 2V(V-u)$

**Ans. (b) :** Power of jet = $F_x.u$

$$= \rho AV(V-u)u$$

Efficiency of system

$$\eta = \frac{\text{output of jet}}{\text{input of jet}}$$

$$= \frac{\text{work done by jet}}{\text{kinetic energy}}$$

$$= \frac{\rho AV(V-u)u}{\frac{1}{2}\rho AV^2} \times 100$$

$$\eta = \frac{2(V-u)u}{V^2} \times 100$$

For maximum efficiency

$$\frac{d\eta}{du} = 0$$

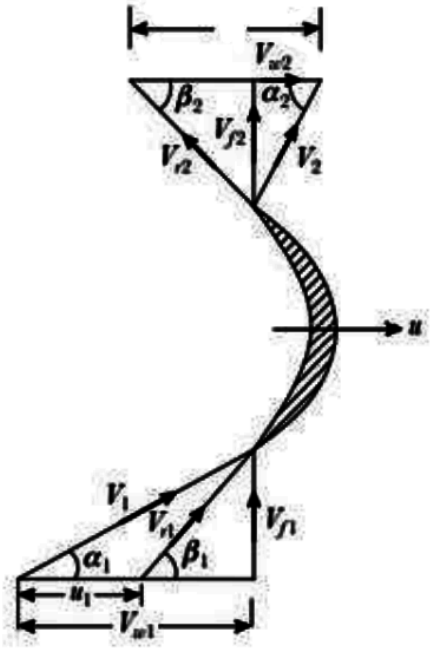
$$\frac{2}{V^2}(V-2u) = 0$$

$$u = \frac{V}{2}$$

$$\eta_{max} = \frac{2\left\{V.\frac{V}{2} - \left(\frac{V}{2}\right)^2\right\} \times 100}{V^2}$$

$$\eta_{max} = 50\%$$

**138. The jet at a velocity $V_1$ at an angle $\alpha_1$ strikes a curved blade moving with a velocity u, as shown in the following figure. The Euler's head is given by :**



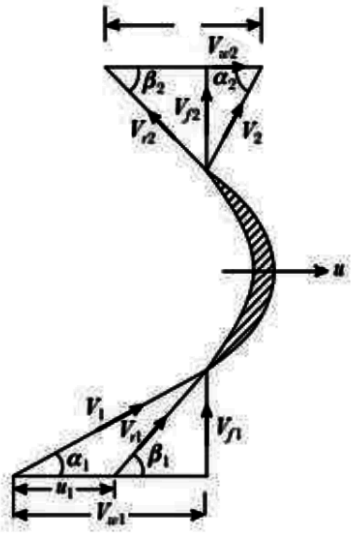

(a) $\frac{V_{w1}u_1 - V_{w2}u_2}{g}$

(b) $\frac{V_{w1}u_2 - V_{w2}u_1}{g}$

(c) $\frac{V_{f1}u_1 - V_{f2}u_2}{g}$

(d) $\frac{V_{f1}u_2 - V_{f2}u_1}{g}$

**Ans. (a) :**




Euler's head= $\frac{V_{w1}u_1 - V_{w2}u_2}{g}$

**139. Based on the two statements given below, choose the correct answer.**
**Statement A : Reaction turbines are low-head, high flow devices.**
**Statement B : In a reaction turbines, flow enters at the larger-diameter section and discharge through the eye.**

(a) Both statements A and B are correct
(b) Statement A is correct, but statement B is incorrect
(c) Statement A is incorrect, but statement B is correct
(d) Both statement A and B are incorrect

**Ans. (a) :** Reaction turbines are low-head, high flow devices.
In a reaction turbines, flow enters at the larger-diameter section and discharge through the eye.
Pelton turbine– high head, low discharge

**140. Head coefficient $C_H$ of a turbine is (where H = head, D = impeller diameter, g = acceleration due to gravity, n = shaft speed).**

(a) $\frac{gH}{n^2D^2}$ (b) $\frac{n^2D^2}{gH}$ (c) $\frac{gH^2}{n^3D^3}$ (d) $\frac{n^3D^3}{gH^2}$

**Ans. (a) :** Head coefficient $C_H$ of a turbine is $\frac{gH}{n^2D^2}$ where,
H = head, D = impeller diameter, g = acceleration due to gravity, n = shaft speed

**141. Power coefficient Cp of a turbine is (where bhp = available power, D = impeller diameter, ρ = fluid density, n = shaft speed).**

(a) $\frac{\rho n^3D^3}{bhp}$ (b) $\frac{bhp}{\rho n^5D^3}$

(c) $\frac{\rho n^3D^5}{bhp}$ (d) $\frac{bhp}{\rho n^3D^5}$

**Ans. (d) :** Power coefficient Cp of a turbine is $\frac{bhp}{\rho n^3 D^5}$
where bhp = available power, D = impeller diameter, ρ = fluid density, n = shaft speed

**142. The power specific speed $N_{sp}$ for a turbine is (where bhp = available power, H = head, ρ = fluid density, g = acceleration due to gravity, n= shaft speed) :**

(a) $\frac{n(bhp)^{1/2}}{\rho^{1/2}(gH)^{3/4}}$ (b) $\frac{\rho^{1/2}(gH)^{5/4}}{n(bhp)^{7/4}}$

(c) $\frac{n(bhp)^{3/2}}{\rho^{3/2}(gH)^{7/4}}$ (d) $\frac{\rho^{3/2}(gH)^{7/4}}{n(bhp)^{3/2}}$

**Ans. (a) :** As with pump, the use of a specific speed parameter can help provide this information.
The power specific speed $N_{sp}$ for a turbine is $\frac{n(bhp)^{1/2}}{\rho^{1/2}(gH)^{3/4}}$
where
bhp = available power, H = head, ρ = fluid density, g = acceleration due to gravity, n= shaft speed

**143. The peripheral-velocity factor for an impulse turbine is given by $\phi = \frac{u}{(2gH)^{1/2}}$ . The maximum efficiency for an impulse turbine occurs at an approximate value of ϕ equal to ____ (where, u = bucket linear velocity, H = head, g = acceleration due to gravity).**

(a) 0.27 (b) 0.37
(c) 0.47 (d) 0.57

**Ans. (c) :** The peripheral-velocity factor for an impulse turbine is given by $\phi = \frac{u}{(2gH)^{1/2}}$ . The maximum efficiency for an impulse turbine occurs at an approximate value of ϕ equal to 0.47
Where, u = bucket linear velocity,
H = head,
g = acceleration due to gravity).

**144. For a centrifugal pump, the ratio of the power available at the impeller to the power available at the shaft of the pump is known as :**

(a) overall efficiency (b) volumetric efficiency
(c) hydraulic efficiency (d) mechanical efficiency

**Ans. (d) :** Mechanical efficiency is the ratio of the power available at the impeller to the power available at the shaft of the pump .
Overall efficiency, $\eta_o = \eta_m \times \eta_h \times \eta_v$
$= \frac{S.P}{W.P}$

**145. For a centrifugal pump, the ratio of the actual flow rate to the theoretical flow rate is known as :**

(a) overall efficiency
(b) volumetric efficiency
(c) hydraulic efficiency
(d) mechanical efficiency

**Ans. (b) :** For a centrifugal pump, the ratio of the actual flow rate to the theoretical flow rate is known as volumetric efficiency

**146. For a centrifugal pump, for the blade exit angle less than $90^0$, the pump head_____**

(a) decrease with increases discharge
(b) remains constant with increasing discharge
(c) increase with increasing discharge
(d) first increase and then decreases with increases discharge

**Ans. (a) :** For a centrifugal pump, for the blade exit angle less than $90^0$, the pump head decrease with increases discharge

**147. The efficiency of a typical centrifugal pump is maximum at :**

(a) zero discharge
(b) 40% of maximum discharge
(c) 60% of maximum discharge
(d) maximum discharge

**Ans. (c) :** The efficiency of a typical centrifugal pump is maximum at 60% of maximum discharge.

**148. Capacity coefficient $C_Q$ of a pump is-**

(a) $\frac{Q}{n^2D^3}$ (b) $\frac{nD^3}{Q}$ (c) $\frac{Q}{nD^3}$ (d) $\frac{n^2D^3}{Q}$

**Ans. (c) :** Capacity coefficient $C_Q$ of a pump is $\frac{Q}{nD^3}$
Were Q=discharge, D=impeller diameter, n = shaft speed

**149. The head required at the centrifugal pump inlet to keep the liquid from cogitating or boiling is known as :**

(a) minimum head (b) threshold head
(c) ultimate head
(d) net positive-suction head

**Ans. (d) :** The head required at the centrifugal pump inlet to keep the liquid from cogitating or boiling is known as net positive-suction head .
The margin of pressure over vapor pressure at the pump suction nozzle is NPSH

**150. For a fluid flowing through a centrifugal pump having density ρ, discharge Q, circumferential speed $u_1$, tip speed $u_2$, and absolute circumferential velocity components of the flow $V_{t1}$ and $V_{t2}$, the power delivered to the fluid is given by**

(a) $\rho Q(u_1V_{t1} - u_2V_{t2})$ (b) $\rho Q(u_2V_{t2} - u_1V_{t1})$
(c) $\rho Q(u_1V_{t2} - u_2V_{t1})$ (d) $\rho Q(u_2V_{t1} - u_1V_{t2})$

**Ans. (b) :** For a fluid flowing through a centrifugal pump having density ρ, discharge Q, circumferential speed $u_1$, tip speed $u_2$, and absolute circumferential velocity components of the flow $V_{t1}$ and $V_{t2}$, the power delivered to the fluid is given by $\rho Q(u_2V_{t2} - u_1V_{t1})$ .

# Telangana State Public Service Commission
# HMWSSB Manager Exam, 2020
## Solved Paper

**1. Identify the expression useful for the calculation of circumferential stress for a given thin cylinder.**

(a) $\frac{pd}{8t}$ (b) $\frac{pd}{4t}$

(c) $\frac{pd}{2t}$ (d) $\frac{pd}{6t}$

**where p = internal pressure, d= internal diameter of the cylinder, t = thickness of the wall of the cylinder**

**Ans. (c) :** Circumferential stress or Hoop stress $= \frac{pd}{2t}$

Longitudinal stress $= \frac{pd}{4t}$

p = internal pressure
d = internal diameter
t = thickness of the wall of the cylinder

**2. Identify the equations for the calculations of radial pressure ($P_x$) and hoop stress ($\sigma_x$) at any radius 'x' in case of a thick cylinder are**

(a) $P_x = \frac{b}{x^2} - a \quad \sigma_x = \frac{b}{x^2} + a$

(b) $P_x = \frac{b}{x^2} + a \quad \sigma_x = \frac{b}{x^2} - a$

(c) $P_x = \frac{b}{x^2} - 2a \quad \sigma_x = \frac{b}{x^2} + 2a$

(d) $P_x = \frac{b}{x^2} + 2a \quad \sigma_x = \frac{b}{x^2} - 2a$

**Ans. (a) :** In case of thick cylinder

radial pressure (Px) = $\frac{b}{x^2} - a$

hoop stress ($\sigma_x$) = $\frac{b}{x^2} + a$

**3. The Crippling load for a column by Euler's formula for different end conditions is given. Identify the correct pair from the following pairs.**

(a) When both the ends are hinged $= \frac{\pi^2 EI}{\ell^2}$

(b) When boh the ends are fixed $= \frac{\pi^2 EI}{4\ell^2}$

(c) When one end is fixed and the other is free $= \frac{4\pi^2 EI}{\ell^2}$

(d) When one end is fixed and the other is hinged $= \frac{\pi^2 EI}{2\ell^2}$

**Ans. (a) :** Euler's crippling load $P_{cr} = \frac{\pi^2 EI}{(L_{eff})^2}$

| S. No. | Description | Effective | Euler's crippling load |
|---|---|---|---|
| 1. | One end fixed and other end hinged | $\frac{l}{\sqrt{2}}$ | $\frac{2\pi^2 EI}{l^2}$ |
| 2. | Both ends fixed | $\frac{l}{2}$ | $\frac{4\pi^2 EI}{l^2}$ |
| 3. | Both ends hinged | $l$ | $\frac{\pi^2 EI}{l^2}$ |
| 4. | One end fixed and other end free | $2l$ | $\frac{\pi^2 EI}{4l^2}$ |

**4. Read the following with respect to the analysis of frames:**

1. **If the number of members in a frame is less than (2j – 3) then the frame is known as deficient frame.**
2. **If the number of members in a frame is more than (2j – 3) then the frame is known as redundant frame.**
3. **If the number of members in a frame is equal to (2j – 3) then the frame is known as perfect frame.**

**Identify the correct statement from the following:**

(a) A, B and C (b) A and B only
(c) A and C only (d) B and C only

**Ans. (a) :** if m = 2j–3 → perfect frame
if m > 2j–3 → Redundant frame
if m < 2j–3 → Deficient frame

**5. A mild steel bar of 706 mm$^2$ area and length 2.4 m is subjected to a tensile load of 90 kN. Find the strain energy stored in the bar, if the load is applied gradually.**

(a) 68838.5 Nm (b) 6883.85 Nm
(c) 68.838 Nm (d) 688.3 J

**Ans. (b) :** A = 706 mm$^2$
L = 2.4 m = 2400 mm
P = 90 × 10$^3$ N

$U_{gradually} = \frac{P^2 L}{2AE}$

$= \frac{(90\times10^3)^2 \times 2400}{2\times2\times10^3\times706} = 6883852.691$ Nmm

=6883.85 N-m

**6. Read the following statements related to continuous beams:**

1. **Clapeyron's theorem of three moments method is used for the analysis of continuous beams.**
2. **The moment distribution method is an interative technique and is very useful in analyzing continuous beams.**
3. **Continuous beams have to be designed for much less bending moments than those in the case of simply supported beams.**
4. **Maximum bending moment is generally over the supports.**

**Identify the correct statements**

(a) 1 only (b) 1 and 2 only
(c) 1, 2 and 3 only (d) 1, 2, 3 and 4 only

**Ans. (d) :**
- Clapeyron's theory provides a simple method to analyse continuous beams.
- Continuous beam are statically indeterminate structure and simples supported beam are statically.

determinate structure. Indeterminate structure has less B.M. compared to determinate structure.

**7. Muller Breslau's principle for obtaining influence lines is applied to**

1. **Statically determinate beams and frames**
2. **Statically indeterminate structures**
3. **Statically indeterminate structures, the material of which is elastic and follows Hook's law.**

**Identify the correct statement**

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

**Ans. (c) :** Muller Breslau's principle used to draw qualitative ILD for any structure either statically determinate and indeterminate structures, the material of which is elastic and follows Hook's law.

**8. The ratio of the effective length of a column to the least radius of gyration of the column is known as**

(a) Crippling stress (b) Polar modulus
(c) Slenderness ratio (d) Section modulus

**Ans. (c) :** Slenderness ratio = $\dfrac{\text{Effective length}}{\text{Least radius of gyration}}$

**9. Assuming that the concrete is uncracked, compute the bending stress in the extreme fibers of the beam having a size of 600 mm × 300 mm for a bending moment of 60 kNm.**

(a) 0.33 MPa (b) 333 MPa
(c) 33.3 MPa (d) 3.33 MPa

**Ans. (d) :** When the section is uncracked then the whole section is considered in the analysis.

$\dfrac{M}{I}=\dfrac{f}{y}=\dfrac{E}{R}\rightarrow$ (Simply bending equation)

Bending stress, $f=\dfrac{M}{I}.y$

$$f=\frac{M}{Z}=\frac{60\times10^6}{\frac{1}{6}\times300\times600^2}=3.33\ \text{MPa}$$

**10. The maximum strain in the tension reinforcement in the section at failure shall not be less than**

(a) equal to 0.0035 (b) $\dfrac{f_y}{1.15E_s}+0.002$
(c) $\dfrac{1.15f_y}{E_s}+0.002$ (d) $\dfrac{f_y}{1.15E_s}+0.0002$

**Ans. (b) :** As per IS : 456-2000, the maximum strain in the tension reinforcement shall not be less than 0.002 + $\dfrac{fy}{1.15E_S}$.

**11. A compression member may be considered as short when both slenderness ratios $\dfrac{\ell_{ex}}{D}$ and $\dfrac{\ell_{ey}}{b}$ are less than**

(a) 10 (b) 6
(c) 12 (d) 14

**where $l_{ex}$ = effective length in respect of the major axis, $l_{ey}$ = effective length in respect of the minor axis, D = depth in respect of the major axis, b = width of the member.**

**Ans. (c) :** As per IS : 456-2000, if it is said to be a short column then its slenderness ratio $\dfrac{L_{ex}}{D}$ and $\dfrac{L_{ey}}{b}$ shall be less than or equal to 12.

**12. Match List-I with List-II and select correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| A. Fineness | 1. Indicates the degree of density or stiffness of cement |
| B. Consistency | 2. It is the stage when the paste completely loses its property |
| C. Final setting | 3. It refers to the ability of the cement paste to retain its volume after setting |
| D. Soundness | 4. The overall particle size distribution of cement |

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 4 | 3 | 1 | 2 |
| (b) | 4 | 1 | 2 | 3 |
| (c) | 4 | 1 | 3 | 2 |
| (d) | 4 | 3 | 2 | 1 |

**Ans. (b) :** Fineness indicates the over all particle size distribution of cement particles consistency indicates the degree of density or stiffness of cement paste, final setting is the stage when the cement paste completely loses its plasticity and soundness is the ability of the cement paste to retain its volume after setting.

**13. In case of RCC footings, the minimum cover should be**

(a) 60 mm (b) 25 mm
(c) 40 mm (d) 50 mm

**Ans. (d) :** As per IS : 456-2000, the minimum cover provided for the footing is 50 mm.

**14. Side face reinforcement shall be provided on both sides if the depth of the beam exceeds in mm**

(a) 750 (b) 500
(c) 650 (d) 400

**Ans. (a) :** Side face reinforcement shall be provided on both sides if the depth of the beam exceeds in 750 mm.

**15. Which Indian code book is used for the design of water tanks (storage of liquids)**

(a) IS 3370 (b) IS 1786
(c) IS 456 (d) IS 13920

**Ans. (a) :**

| | |
|---|---|
| IS : 456 | Code of practice for plain and reinforced concrete. |
| IS : 13920 | Ductile design and detailing of reinforced concrete structures subjected to seismic force code of practice. |
| IS : 3370 : | Concrete structure for storage of liquids code of practice |
| IS : 1786 | High strength deformed steel bars and wires for concrete reinforcement. |

**16. Equivalent shear shall be calculated for shear and tension by using the following formula:**

(a) $V_e = V_u + T_u/b$
(b) $V_e = 1.6\ V_u + T_u/b$
(c) $V_e = V_u + 1.6\ T_u/b$
(d) $V_e = 1.6\ T_u/b$

**where $V_e$ = equivalent shear, $V_u$ = shear, $T_u$ = torsional moment, b = breadth of beam.**

**Ans. (c) :** Equivalent shear force can be estimated by the following formula $V_e = V_u + \frac{1.6\ Tu}{b}$

**17. The minimum thickness of braced walls as per IS 456 is**

(a) 250 mm (b) 100 mm
(c) 120 mm (d) 80 mm

**Ans. (b) :** As per IS : 456-2000, the minimum thickness of wall is 100 mm.

**18. Lap splices shall not be used for bars longer than**

(a) 38 mm (b) 40 mm
(c) 46 mm (d) 36 mm

**Ans. (d) :** As per IS : 456-2000, splicing should not be done if bar diameter in more than 36 mm.

**19. Development length of bars can be calculated using the following formula:**

(a) $\frac{\phi\sigma_s}{\tau_{bd}}$ (b) $\frac{\phi\sigma_s}{4\tau_{bd}}$
(c) $\frac{\sigma_s}{4\tau_{bd}}$ (d) $\frac{\phi\sigma_s}{2\tau_{bd}}$

**where $\phi$ = nominal diameter of the bar, $\sigma_s$ = stress in the bar at the section considered at design load, $\tau_{bd}$ = design bond stress.**

**Ans. (b) :** The development length can be estimated by the following formula.

$$L_d = \frac{\phi\sigma_{st}}{4\tau_{bd}}$$

When, $\phi$ = bar diameter
$\sigma_{st}$ = design stress in steel.
$\tau_{bd}$ = Average bond stress.

**20. According to IS 456, the thickness of RCC footing on soils at its edges is kept not less than**

(a) 100 mm (b) 150 mm
(c) 175 mm (d) 300 mm

**Ans. (b) :** As per IS 456-2000, the minimum edge thickness of footing when it is resting on soil is 150 mm.

**21. Which code book gives guidelines for concrete mix design proportioning**

(a) IS 10262:2009 (b) IS 4326:1993
(c) IS 13920:1993 (d) IS 1893:2002

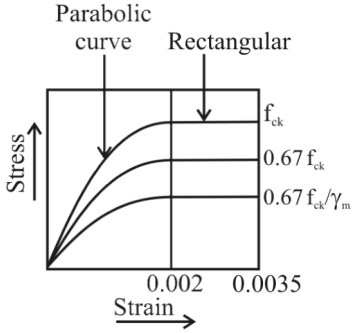
**Ans. (a) :**

| | |
|---|---|
| IS : 13920 – 1993 | – Ductility detailing |
| IS : 1893 – 2002 | – Design of earthquake resistance structures |
| IS : 4326 – 1993 | – Earthquake resistance design |
| IS : 10262 | – Mix design |

**22. The shape of the stress-strain curve of concrete upto a strain 0.002 is**

(a) exponential (b) linear
(c) parabolic (d) hyperbolic

**Ans. (c) :** The stress-strain curve for concrete is parabola upto 0.002 strain and constant till failure.




**23. Which method is used for calculating the loss of pre-stress due to creep of concrete?**

(a) Creep coefficient method
(b) Coefficient method
(c) Pre-stress coefficient method
(d) Magnitude coefficient method

**Ans. (a) :** Loss of prestress due to creep of concrete can be determined from the following.
(1) Ultimate creep strain method.
(2) Creep coefficient method.

**24. Concrete in which pre-stress is imparted to concrete through bond between the tendons and surrounding concrete is known as**

(a) bonded pre-stressed concrete
(b) full pre-stressing
(c) limited pre-stressing
(d) pre-tensioning

**Ans. (d) :** In pretensioning method the stresses are transferred through bond.

**25. What is the unit mass of steel?**

(a) 9850 kg/m$^3$ (b) 8850 kg/m$^3$
(c) 7850 kg/m$^3$ (d) 6850 kg/m$^3$

**Ans. (c) :** The unit mass of steel is 7850 kg/m$^3$ .
Modulus of elasticity (E) = $2 \times 10^5$ N/mm$^2$
Poisson's ratio = 0.27 to 0.30
Coefficient of thermal expansion = $12 \times 10^{-6}/^0$C
Shear rigidity = $0.769 \times 10^5$ N/mm$^2$

**26. The minimum number of longitudinal steel bars in helically reinforced RCC columns must be**

(a) 6 (b) 8
(c) 4 (d) 10

**Ans. (a) :** Helically reinforcement is provided in circular columns only.
As per IS : 456-2000, minimum number of longitudinal bars in circular column are 6.
As per IS : 456-2000, minimum number of longitudinal bars in rectangular column are 4.

**27. In bolted/riveted construction, the minimum width of lacing bars shall be ..... times the diameter of the end bolt/rivet.**

(a) one (b) three
(c) two (d) four

**Ans. (b) :** Minimum width of lacing bar $\not< 3 \times$ diameter of bolt/rivet.

**28. Column bases are mainly subjected to and designed for**

(a) bearing and tension
(b) compression and tension
(c) bearing and compression
(d) bending and compression

**Ans. (c) :** Steel columns shall be provided with column bases capable of distributing the compressive forces in the compressed steel parts of the column over a bearing area such that the bearing pressure on the foundation does not exceed the design strength of the point.

**29. The dimensions of viscosity and surface tension are**

(a) $FLT^{-2}$ and $FL^{-1}$ (b) $FL^{-1}T^{-1}$ and $FL^{-1}$
(c) $FLT^{-1}$ and $FL^{-2}$ (d) $FLT^{-1}$ and $FL^{-1}$

**Ans. (*) :** Viscosity is taken as dynamic viscosity ($\mu$) is

units : $\frac{N-s}{m^2} = Pascal - sec = kg/m - sec$

F = force (Newton)
L = length (meter)
T = time (second)

$\mu$ directions = $\frac{F.T}{L^2} = FL^{-2}T$

(Number method with options)

Surface tension ($\sigma$) :

'$\sigma$' Units force / length = $\frac{Newton}{meter}$

'$\sigma$' dimensions = $\frac{F}{L} = FL^{-1}$ Matching (1) and (2)

• No option is matching for dynamic viscosity.

**30. Match List-I with List-II and select correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| A. A member carrying compressive loads resulting from dead loads from imposed loads | I. 400 |
| B. Compression flange of a beam against lateral torsional buckling | II. 350 |
| C. Member always under tension | III. 300 |
| D. A member normally acting as a tie in a roof truss | IV. 180 |

| | A | B | C | D |
|---|---|---|---|---|
| (a) | IV | II | I | III |
| (b) | IV | III | I | II |
| (c) | IV | III | II | I |
| (d) | IV | II | III | I |

**Ans. (b) :**

| List - 1 Member | List - 2 Maximum effective slenderness ratio |
|---|---|
| (A) A member carrying compressive loads resulting from dead loads and imposed loads. | 180 |
| (B) Compression flange of a beam against lateral buckling torsional. | 300 |
| (C) Member always under tension. | 400 |
| (D) A member normally as a tie in a roof truss. | 350 |

**31. The pressure at any point in a fluid at rest has the same magnitude in all directions. This law is known as**

(a) Newton's law (b) Darcy's law
(c) Pascal's law (d) Stoke's law

**Ans. (c) :** The pressure at any point in a fluid at rest has the same magnitude in all direction. This law is known as pascal's law.

**32. The value of kinetic energy correction factor and momentum correction factor for laminar flow in pipes is**

(a) 2.0 and 1.4 (b) 2.0 and 1.5
(c) 2.2 and 1.5 (d) 2.0 and 1.333

**Ans. (b) :** For laminar flow (pipe) kinetic energy factor $(\propto) = 2$

Momentum correction factor ($\beta$) = 1.333

**33. In case of a turbulent flow, which of the following statement is correct:**

(a) proportional to the velocity of flow
(b) independent of the pressure
(c) independent of the nature of the surface in contact
(d) proportional to the square root of the velocity

**Ans. (*) :** Question is not specific
Assumption bases like
• Head loss $\propto$ (velocity)$^2$.

**34. In a pressure penstock 4500 m long water is flowing at 4.0 m/sec. If the velocity of the pressure wave travelling in the pipe due to sudden complete closure of a valve at the downstream end is given as 1500 m/sec. Find the maximum pressure rise and the period of oscillation**

(a) 60 MN/m$^2$ and 6 seconds
(b) 60 N/m$^2$ and 6 seconds
(c) 600 N/m$^2$ and 6 seconds
(d) 6 MN/m$^2$ and 6 seconds

**Ans. (a) :** L = 4500 km
V = 4 m/s
C = 1500 m/s
Sudden complete closure of a value, pressure rise $(\Delta P)$
Water hammer = $\rho CV$
$= 1000 \times 4 \times 1500$
$= 6 \times 10^6$ N/m$^2$ = 60 MN/m$^2$
Period of oscillation (T) = $\frac{2L}{C}$
$= \frac{2 \times 4500}{1500} = 6\text{sec}$

**35. At a certain point in castor oil the shear stress is 0.316 N/m$^2$ and the velocity gradient is 0.316 sec$^{-1}$. Identify the correct dynamic viscosity from the following:**

(a) 0.1 Ns/m$^2$ (b) 1.0 Ns/m$^2$
(c) 10 Ns/m$^2$ (d) 0.001 Ns/m$^2$

**Ans. (b) :** $\tau = 0.316$ N/m$^2$
$\frac{dv}{dy} = 0.316$ sec$^{-1}$
Newton's law of viscosity $(\tau) = \mu \left(\frac{dV}{dy}\right)$
$0.316 = \mu \times 0.316$
$\mu = 1 \frac{N-s}{m^2}$

**36. According to the studies of US Bureau of Reclamation, depending upon the valve of Froude Number of the incoming flow, there are various types of hydraulic jump which may occur on a horizontal flow. Consider the following statements.**

1. **For Froude Number 1.0 to 1.7, the jump is called as undular jump.**
2. **For Froude Number 1.7 to 2.5. the jump is called as weak jump.**
3. **For Froude Number 2.5 to 4.5, the jump is called as oscillating jump.**
4. **For Froude Number 4.5 to 7.0, the jump is called as steady jump.**
5. **For Froude Number 9.0 and more, the jump is called as strong jump.**

**Choose the option in which all the statements are correct:**

(a) 1, 2, 3, 4 and 5 (b) 1, 2, 3 and 4 only
(c) 1, 2 and 4 only (d) 1, 2, 3 and 5 only

**Ans. (d) :** Classification of hydraulic jump based on froud number of incoming flow.

| Range of froud number | Type of Jump |
|---|---|
| 1 < Fr < 1.7 | Undular jump |
| 1.7 < Fr < 2.5 | Weak jump |
| 2.5 < Fr < 4.5 | Oscillating jump |
| 4.5 < Fr < 9 | Steady jump |
| Fr > 9 | Strong or choppy jump |

**37. Match List-I with List-II and select correct answer using the codes given below the lists:**

| List-I (Dimensionless) number | List-II (Definition) |
|---|---|
| A. Reynold's number | 1. Ratio of inertia force to the viscuous force |
| B. Froude number | 2. Ratio of inertia force to the surface tension force |
| C. Weber number | 3. Ratio of inertia force to the elastic force |
| D. Mach number | 4. Ratio of inertia force to the pressure force |
| E. Euler's number | 5. Ratio of inertia force to the gravitational force |

**Identify the correct match:**

| | A | B | C | D | E |
|---|---|---|---|---|---|
| (a) | 1 | 5 | 4 | 2 | 3 |
| (b) | 1 | 5 | 2 | 3 | 4 |
| (c) | 1 | 5 | 3 | 2 | 4 |
| (d) | 1 | 5 | 2 | 4 | 3 |

**Ans. (b) :** j

| List-I (Dimensionless) number | List-II (Definition) |
|---|---|
| A. Reynold's number | Ratio of inertia force to the viscuous force |
| B. Froude number | Ratio of inertia force to the gravitational force |
| C. Weber number | Ratio of inertia force to the surface tension force |
| D. Mach number | Ratio of inertia force to the elastic force |
| E. Euler's number | Ratio of inertia force to the pressure force |

**38. Von Karman momentum integral equation is applicable to**

(a) laminar boundary layer only
(b) turbulent boundary layer only
(c) transition boundary layer only
(d) both laminar and turbulent boundary layers

**Ans. (c,d) :** $\frac{\tau_0}{\rho U_\infty^2} = \frac{dQ}{dx}$

Equation is known as Von-Karman momentum integral equation for boundary layer flow, this applicable
(i) Laminar boundary layer
(ii) Transaction boundary layer
(iii) Turbulent boundary layer

**39. The concept of boundary layer was first introduced by**

(a) Hagen-Poiseuille (b) L. Prandtl
(c) J. Boussinesq (d) Nikurade

**Ans. (b) :** boundary layer was first introduced by L. Prandtl ( Ludwing Prandtl).

**40. Determine the number of the impellers required for multistage pump to lift 4200 liters/minute against a total head of 186 m, at a speed of 750 rpm. The specific speed is not to exceed 700. The manometric head available per stage is 18.6 m.**

(a) 10 (b) 100
(c) 1 (d) 7

**Ans. (a) :** Number of impellers required to rise liquid

$$= \frac{\text{Total head}}{\text{Manometric head}} = \frac{186\,\text{m}}{18.6\,\text{m/impeller}}$$

= 10 impeller.

**41. Water at 15ºC flows at 4 m/sec in a 150 mm pipe. At what velocity must oil at 30ºC flow in a 75 mm pipe for the two flows to be dynmically similar? Take kinematic viscosity for water at 15ºC as $2.0 \times 10^{-6}$ m²/sec and that for oil at 30ºC as $3.0 \times 10^{-6}$ m²/sec**

(a) 120 m/sec (b) 1200 m/sec
(c) 12 m/sec (d) 1.2 m/sec

**Ans. (c) :** For dynamic similarity,

$(R_e)_1 = (R_e)_2$

$$\left(\frac{\rho v D}{\mu}\right)_1 = \left(\frac{\rho v D}{\mu}\right)_2$$

$$\frac{v_1 D_1}{\nu_1} = \frac{v_2 D_2}{\nu_2}$$

$$\frac{4 \times 0.15}{2 \times 10^{-6}} = \frac{v_2 \times 0.075}{3 \times 10^{-6}}$$

$v_2 = 12$ m/s

**42. In the boundary layer, the flow separation depends on**

**1. The curvature of the surface**
**2. The Reynolds number of flow**
**3. The roughness of the surface**

**Consider the above statements and identify the correct option from the following:**

(a) 1 only (b) 1 and 2 only
(c) 1, 2 and 3 (d) 1 and 3 only

**Ans. (a) :** • Flow separation depends on the curvature of the surface.
• Separation can take place in both laminar and turbulent boundary layer.
• Roughness of surface reduces flow separation.

**43. The correct sequence of the centrifugal pump components through which the fluid flow is**

(a) Suction pipe, delivery pipe, impeller and foot valve and strainer
(b) Suction pipe, foot valve and strainer, impeller and delivery pipe
(c) Foot valve and strainer, impeller, suction pipe and delivery pipe
(d) Strainer and foot valve, suction pipe, impeller and delivery pipe

**Ans. (d) :** Correct sequence of the centrifugal pump : strainer and foot valve, suction pipe, impeller and delivery pipe.

**44. A Pelton wheel has to be designed for the following data. Power to be developed is 6000 kW, net head available is 300 m, speed is 550 rpm, discharge is 3 m³/sec, ratio of jet diameter to wheel diameter is 1/10. Find the overall efficiency of the turbine. Take specific weight of water as 10000 N/m³.**

(a) 66.67% (b) 60%
(c) 56.67% (d) 50%

**Ans. (a) :** $P_0 =$ 6000 kw
H = 300 m
$Q = 3m^3$ / sec

$\frac{d}{D} = \frac{1}{10}$, x = 10 kN/m3

$$\eta_o = \frac{\text{output power}}{\text{water power}} = \frac{6000\ \text{kW}}{\text{r.QH}}$$

$$\eta_o = \frac{6000}{10 \times 3 \times 300}$$

$$= \frac{6000}{9000} = 0.667 = 66.7\%$$

**45. The equation of a velocity distribution over a plate is given by $u = 2y - y^2 + 20$. Asuming $\mu$ = 8.00 poise, the shear stress at the boundary and shear stress at a point 15 cm from the boundary in N/m².**

(a) 1.6 at the boundary and 1.36 at 15 cm from the boundary
(b) 26 at the boundary and 20.6 at 15 cm from the boundary
(c) 36 at the boundary and 24.6 at 15 cm from the boundary
(d) 18 at the boundary and 14.6 at 15 cm from the boundary

**Ans. (a):** Newton's law of viscosity.

$$\tau_y = \mu\left(\frac{du}{dy}\right)_y$$

$$\tau_y = (8\times 0.1)\frac{d}{dy}\left(2y - y^2 + 20\right)$$

$\tau_y = (0.8)(2-2y)$

$y = 0.15$ m

$\tau_{0.15} = (0.8)(2 - 2\times 0.15)$

$\tau_{0.15} = 1.36$ N/m$^2$

At boundary i.e. $y = 0$

$\tau_{y=0.8} = 0.8(2 - 2\times 0)$

$= 0.8 \times 2$

$\tau_{y=0} = 1.6$ N/m$^2$

**46. Consider the following statements:**

1. **Pelton wheel is a tangential flow impulse turbine.**
2. **Francise turbine is an axial flow reaction turbine.**
3. **Kaplan turbine is a radial flow reaction turbine.**
4. **Draft tube is used in a reaction turbine to convert residual kinetic energy into pressure energy.**

**Which of the above statements are not correct?**

(a) 1 and 4 only (b) 2 and 3 only
(c) 2 and 4 only (d) 1 and 3 only

**Ans. (b) :** • Francis turbine is radial flow reaction turbine.
• Kaplan turbine is axial flow reaction turbine.

**47. A control volume refers to**

(a) a fixed region in space
(b) a closed system
(c) a reversible process only
(d) a specified mass

**Ans. (a) :** Control volume refers to a fixed region in space.

**48. Navier - Strokes equation is useful in the analysis of**

(a) non viscous flows
(b) turbulent flows
(c) viscous flows
(d) both non-viscous and turbulent flows

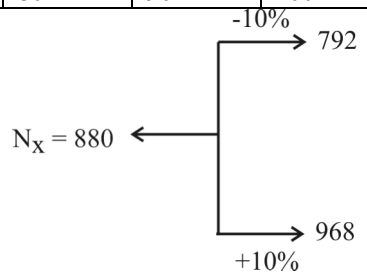
**Ans. (c) :** Navier-Stokes equation is used for viscous flows analysis.

**49. Raingauge station 'X' did not function for a part of a month during which a storm occurred. The storm produced rainfalls of 80, 90 and 100 mm at three surrounding stations A, B and C respectively. The normal annual rainfalls at the stations X. A, B and C respectively 880, 800, 880 and 1000 mm. Estimate the missing storm rainfall at station X.**

(a) 98.67 mm (b) 88.67 mm
(c) 86.67 mm (d) 86.87 mm

**Ans. (b) :** j

| Station | A | B | C | X |
|---|---|---|---|---|
| N(mm) | 800 | 880 | 1000 | 880 |
| P(mm) | 80 | 90 | 100 | $P_x$ = ? |




$N_C$ = 1000 which is not within 10% of $N_x$ use normal ratio method.

$$\frac{P_x}{N_x} = \frac{1}{n}\left[\frac{P_1}{N_1} + \frac{P_2}{N_2} + \frac{P_3}{N_3}\right]$$

$$\frac{P_x}{880} = \frac{1}{3}\left[\frac{80}{800} + \frac{90}{880} + \frac{100}{1000}\right]$$

$P_x = 88.67$ mm

**50. Identify standard non-recording raingauge among the following raingauges used for measuring rainfall which was prescribed by IMD.**

(a) Tipping bucket type
(b) Weighing bucket type
(c) Symons raingauge
(d) Floating type with siphon arrangement

**Ans. (c) : Non - recording type raingauge :**
(1) Symons raingauge
**Recording type raingauge –**
(1) Tipping bucket type
(2) Weighing bucket type
(3) Floating type with siphon arrangement

**51. The observed annual runoff from a basin of area 1000 km$^2$ is 600 cm$^3$ and the corresponding annual rainfall over the basin during the same year is 1500 mm. What is the runoff coefficient?**

(a) 0.50 (b) 0.60
(c) 0.45 (d) 0.40

**Ans. (d) :** Area = 1000 km$^2$
Run off volume = 600 cm$^3$
P = 1500 mm

$$\text{Run off depth (R)} = \frac{\text{Run off volume}}{\text{Area}}$$

$$\text{Runoff depth (R)} = \frac{600\times(100)^3}{1000\times(1000)^2} = 0.6 \text{ m} = 600 \text{ mm}$$

$$\text{Run off coefficient} = \frac{R}{P} = \frac{600}{1500} = 0.4$$

**52. Laminar sublayer exits in**

(a) laminar boundary layer region
(b) transition boundary layer region
(c) turbulent boundary layer region
(d) between laminar boundary layer region and transition boundary layer region

**Ans. (c):** Laminar sublayer exits in turbulent boundary layer region.

**53. Isohytes are defined as**
(a) line joining points of equal rainfall intensity
(b) line joining points of equal storm duration
(c) line joining points of equal relative humidity
(d) line joining points of equal amount rainfall

**Ans. (d) : Isohyets :** A line on a map connecting points having the same amount of rainfall [equal rainfall depth] in a given duration.

**54. The basic principles of unit hydrograph are based on**
(a) non linear time variance and linearity
(b) non-linearity and time invariance
(c) linearity and time variance
(d) linearity and time invariance

**Ans. (d) :** Basic principles of unit hydrograph theory :
(1) Linear response
(2) Time invariance

**55. The whole circle bearing of the lines of a closed traverse are given below:**

| Line | AB | BC | CD | DA |
|---|---|---|---|---|
| Bearing | 78º40'20" | 152º31'40" | 251º18'40" | 3º44'15" |

**Find the included angles of the traverse:**
(a) 105º03'55", 106º08'40", 61º13'10" and 67º34'25"
(b) 105º03'55", 106º08'40", 81º13'00" and 67º34'25"
(c) 78º40'20", 152º31'40", 251º18'40" and 3º44'15"
(d) 105º03'55", 106º08'40", 91º13'10" and 67º34'25"

**Ans. (b) :** Including angles –
$\angle A = 90^0 - 78^0 40'20" + 90^0 + 3^0 44'15"$
$\angle A = 183^0 44'15" - 18^0 40'20"$
$\angle A = 105^0 3'55"$

$\angle B = 78^0 40'20" + (180^0 - 152^0 31'40"$
$\angle B = 258^0 41'20" - 152^0 31'40"$
$\angle B = 106^0 8'40"$

$\angle C = (152^0 31'40" - 90^0 + 270^0 - 251^0 18'40"$
$\angle C = 62^0 31'40" + (270^0 - 251^0 18'40")$
$\angle C = 81^0 13'00"$

$\angle D = (251^0 18'40" - 180^0) - 3^0 44'15"$
$\angle D = (71^0 18'40" - (3^0 44'15")$
$\angle D = 67^0 34'25"$

**56. Which of the following formations doesn't contain any groundwater?**
(a) Aquifuge (b) Aquiclude
(c) Aquifer (d) Aquitard

**Ans. (a) :** A rock neither containing water not transmitting water.
Eg : solid granite.

**57. Darcy's law for groundwater movement states that the velocity is proportional to**
(a) the logarithm of hydraulic gradient
(b) the hydraulic gradient
(c) the reciprocal of the hydraulic gradient
(d) the square of the hydraulic gradient

**Ans. (b) :** $v \propto i$
Velocity of flow is proportional to the hydraulic gradient.

**58. Flow duration curve is the graph drawn between**
(a) cumulative volume of flow and time
(b) the discharge in the stream and time
(c) the accumulated discharge and time
(d) the discharge and the percent of time such discharge is equaled or exceeded

**Ans. (d) :** Flow duration curve of stream is a plot of discharge against the percent of time the flow was equaled or exceeded. This curve is also known as discharge - frequency curve.
**Flow duration curve –**
- The flow duration curve is highly useful in the planning and design of water resources project.
- The flow duration curve also finds use in the design of drainage system and in flood control studies.
- Flow duration curve can be used to estimate what minimum level of flow can be maintained through a reservoir so that minimum demand at all times can be met.

**59. Water is released at the rate of 12 cumec at the head of a canal. If duty at the field is 1250 hectares/cumec and loss of water ink transmit is 0.25%. Find the area of the land that can be irrigated in hectares.**
(a) $1.125 \times 10^8$ (b) $1.125 \times 10^6$
(c) $1.125 \times 10^4$ (d) $1.125 \times 10^2$

**Ans. (c) :**
$Q_{HC} = 12$ cumecs
$D_f = 1250$ ha/cumecs
Losses = 0.25 = 25%
n = 75% (efficiency)
QE = 75% 12 = 9 cumec

$$Q = \frac{A}{D}$$

$A = Q \times D = 9 \times 1250 = 11250$ ha
$A = 1.1250 \times 10^4$ ha

**60. From which river the water has been lifting to Konda Pochamma reservoir?**
(a) Kaleswaram (b) Krishna
(c) Godavari (d) Musi

**Ans. (c) : Godavari–** The godavari is the largest peninsular river system in India.
It is also called the Dakshin Ganga.
It rises in the Nasik district of Maharashtra, and discharge its water into the bay of Bengal.

**61. Outlets which maintain a constant discharge irrespective of the fluctuations in the water supply channel and water course is known as**
(a) Rigid module
(b) Non-modular outlet
(c) Semi modular outlet
(d) Kennedy's gauge outlet

**Ans. (a) :** In the rigid module, Discharge (Q) is constant irrespective of fluctuations on upstream and downstream.

**62. The rate of seepage (q) through an earth dam obtained from a flow net is given by the following question**

(a) $q = k\,h\,(N_f \times N_d)$ (b) $q = h\left(\frac{N_f}{N_d}\right)$

(c) $q = kh\left(\frac{N_f}{N_d}\right)$ (d) $q = k\left(\frac{N_f}{N_d}\right)$

**Where k = coefficient of permeability of soil, h = total head causing the flow. $N_f$ = total number of low channels, $N_d$ = total number of potential drops.**

**Ans. (c) :** The rate of seepage (q) through an earth dam obtained from a flow net is given by–

$$q = kh\left(\frac{N_f}{N_d}\right)$$

Where,
k = coefficient of permeability of soil
h = total head causing the flow
$N_f$ = total number of flow channel
$N_d$ = total number of potential drops
• Applicable only for isotropic soils.

**63. Lacey's general regime flow equation is given by the following equation.**

(a) $V = \frac{1}{6}R^{3/4}S^{1/2}$ (b) $V = 1260\,R^2S$

(c) $V = 35.5\left(\frac{R}{S}\right)^{\frac{1}{2}}\sqrt{RS}$ (d) $V = 10.8\,R^{2/3}\,S^{1/2}$

**where S = bed slope, R = hydraulic mean radius in m.**

**Ans. (d) :** $V = 10.8\,R^{2/3}\,S^{1/2}$
The Lacey's equations are–

1. $R = 2.5\frac{V^2}{f}$

2. $V = \left(\frac{Qf^2}{140}\right)^{\frac{1}{6}}$

f = friction factor
Q = discharge

**64. Identify from the following which Indian code book is used for determination of evaporation from reservoirs**
(a) 6939-1992 (b) 6955-1973
(c) 6966-1989 (d) 6926-1973

**Ans. (a) :** Indian Code Book 6939-1992 is used for determination of evaporation from reservoirs.

**65. A gate consists of two leaves of timber or steel hinged and sealed to the spillway crest is called as .....**
(a) bear-trap gate (b) drum gate
(c) tainter gate (d) rolling gate

**Ans. (c) :** A gate consists of two leaves of timber or steel hinged and sealed to the spillway crest is called as tainter gate or radial gate.

**66. Consider the following items.**
**1. Intakes 2. Water treatment plant**
**3. Service reservoirs 4. Control valves**
**5. Hydrants**
**Choose the correct answer related to components of public water supply system:**
(a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 1, 2, 3 and 4 only (d) 1, 2, 3, 4 and 5 only

**Ans. (d) :** Components of public water supply system are–
1. Intake
2. Water treatment plant
3. Service reservoirs
4. Control valves
5. Hydrants.

**67. The total hardness value obtained from the complete analysis of a water sample is found to be 117 mg/*l*. The analysis further showed that the concentrations of all the three principal cations causing hardness are numerically same. Value of the carbonate hardness is 57 mg/*l*. Calculate the value of non-carbonate hardness is**
(a) 50 mg/*l* (b) 60 mg/*l*
(c) 6 mg/*l* (d) 117 mg/*l*

**Ans. (b) :** Total hardness = 117 mg/*l*
Carbonate hardness = 57 mg/*l*
Non-carbonate hardness = ?
TH = CH + NCH
117 = 57 + NCH
NCH = 60 mg/*l*

**68. Units of turbidity is**
(a) gram/*l*
(b) unit less
(c) Nephelometric turbidity units
(d) threshold number

**Ans. (c) :** Units of turbidity is Nephelometric turbidity units.

**69. Identify water distribution system among the following:**
(a) Combined gravity and pumping system
(b) Tree system
(c) Reticulation system
(d) Radial system

**Ans. (a) : Methods of distribution–** The method of distribution depends upon the topography of the area. The following methods or systems may be adopted for distribution.

1. Gravity system.
2. Combined gravity and pumping system.
3. Pumping system.

**There are four type layout system –**

- Dead end system
- Grid Iron system
- Circular system
- Radial system

**70. During certain period the load on the hydel power plant varies from 15000 to 35000 kW. Calculate the load factor.**

(a) 0.620 (b) 0.7143
(c) 0.510 (d) 0.810

**Ans. (b) :**

$$\text{Load factor} = \frac{\text{Average load}}{\text{Peak load}}$$

$$= \frac{\frac{15000+35000}{2}}{35000}$$

$$= \frac{25}{35} = \frac{5}{7} = 0.7143$$

**71. Which valve is placed in a pumping main so that if pump fails or stops, what is prevented from flowing back to the pump and thus pumping equipment is saved from possible damage?**

(a) sluice valve
(b) pressure relief valve
(c) reflux valve
(d) air relief valve

**Ans. (c) :** To prevent back flow reflux valve is provided.

**Check valve or reflux valve or non-returning valve:** The posses some automatic device which allows the water to flow in one direction only.

**Sluice valve/gate valve, sheet off valves:** These valves are provided to regulate the flow of water through the pipe and are essential to divide the main line into several sections.

**Air relief valve:** Air relief valve are also known as air valve the air relief valve is provided at the summit to release the air pressure.

**72. Assertion (A) :** Slow sand filters are more efficient in removal of bacteria than rapid sand filters.

**Reason (R) :** The sand used in slow and filters is finer than that in rapid sand filters.

Select correct answer based on the coding system given below:

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

**Ans. (a) :** Slow sand filter media consists sand grains of very fine size could able to trap even microorganisms i.e. bacteria and higher forms of life.

**73. The horizontal flow velocity in a sedimentation tank passing a discharge 'Q' and having length 'L', width 'B' and depth 'H' is given by**

(a) $\frac{Q}{BH}$ (b) $\frac{Q}{BL}$
(c) $\frac{Q}{B \times I}$ (d) $\frac{Q}{L \times H}$

**Ans. (a) :** $V_H = \frac{Q}{\text{c/s area}} = \frac{Q}{B \times H}$

Velocity of flow $(V_f) = \frac{Q}{BH}$

Over flow rate $V_0 = \frac{\text{Discharge}}{\text{Surface area}} = \frac{Q}{BL}$

Detention time 'T' = $\frac{LBH}{Q}$

**74. Particulate matter (fly ash) carried in effluent gasses from the furnaces burning fossils fuels are better removed by**

(a) wet scrubber
(b) cyclone
(c) cotton bag house filter
(d) electrostatic precipitator

**Ans. (d) :** Fly ash from thermal power stations effectively removed by electrostatic precipitator.
Mostly used in thermal power plant, mining, metallurgical, industries, pulp and paper industries, iron & steel plant etc.

**75. Two primary air pollutants are**

(a) sulphur oxide and ozone
(b) nitrogen oxide and peroxyacetylintarte
(c) sulphur dioxide and hydrocarbons
(d) ozone and peroxyacety nitrate

**Ans. (c) :** Among the given two primary air pollutions are sulphur oxide and hydrocarbons.

**76. The design technique adopted in design of large water supply networks as an aid to supply and separate the smaller loops is**

(a) electric network analyzer method
(b) equivalent pipe method
(c) circle method
(d) Hardy Cross method

**Ans. (d) :** Complex pipe networks consisting the closed loops solved by hardy cross method.

**77. Read the following statements related to road alignment:**

1. **Improper alignment would result in increase in construction cost.**
2. **Improper alignment would result in increase in maintenance cost.**
3. **Improper alignment would result in decrease in vehicle operation cost.**
4. **Improper alignment would result in increase in accident rate.**

**Identify the wrong statement:**

(a) 1 only (b) 2 only
(c) 3 only (d) 4 only

**Ans. (c) :** Good alignment reduces vehicle operation, maintenance and construction costs.

**78. The required camber of a pavement depends on**

1. **the type of pavement surface**
2. **the amount of rainfall**
3. **longitudinal slope of the road**
4. **traffic volume**

**Identify the correct statement:**

(a) 1 and 2 only (b) 1 and 3 only
(c) 1 and 4 only (d) 1, 2, 3 and 4

**Ans. (a) :** Camber depends on
(i) Type of road surface.
(ii) Rainfall.

**79. According to Indian Roads Congress, the maximum width of road vehicles is**

(a) 2.65 m (b) 2.44 m
(c) 2.00 m (d) 3.8 m

**Ans. (b) :** According to Indian Roads Congress, the maximum width of road vehicles is 2.44 m.
Maximum width of the pavement is for single lane 3.75 m.

**80. In which year Motor Vehicle Act was brought into effect by Government of India to regulate the road traffic in the form of traffic laws, ordinances and regulations.**

(a) 1959 (b) 1929
(c) 1949 (d) 1939

**Ans. (d) :** In year 1939, meter vehicle act was brought into effect by government of India to regulate the road traffic in the form of traffic laws, ordinances and regulations.

**81. The second twenty year road development plan 1961-81 is also called as .....**

(a) Lucknow Road Plan
(b) Delhi Road Plan
(c) Bombay Road Plan
(d) Nagpur Road Plan

**Ans. (c) :** 1943-63 Nagpur road plan (This plan was completed in 1961).
1961-81 Bombay road plan
1981-2001 Lucknow road plan.

**82. What type of noise can be abated by providing lining on walls and ceiling with sound absorbing materials?**

(a) Direct air borne noise (b) Source noise
(c) Reflection noise (d) Structural noise

**Ans. (c) :** Reflection noise is controlled by (or) about by providing lining on walls and ceiling with sound observing materials.

**83. What is the maximum contact shear stress (F) at the contact point between the wheel and the rail is given by the empirical formula.**

(a) $4.13\left(\frac{P}{R}\right)^{\frac{3}{2}}$ (b) $2.13\left(\frac{P}{R}\right)^{\frac{1}{2}}$
(c) $4.13\left(\frac{P}{R}\right)^{\frac{1}{2}}$ (d) $2.13\left(\frac{P}{R}\right)^{\frac{3}{2}}$

**Ans. (a) :** Maximum contact shear stress (F) at the contact point between the wheel and the rail is given by,

$F = 4.13\left(\frac{P}{R}\right)^{\frac{3}{2}}$

R = radius of fully worn out wheel
P = wheel load

**84. Calculate the rate of super-elevation for a horizontal highway curve of diameter 100 m and speed 50 kmph.**

(a) 0.25 (b) 0.022
(c) 0.22 (d) 0.0022

**Ans. (b) :** Super elevation

$R = \frac{D}{2} = \frac{1000}{2} = 500\,m$

V = 50 kmph

$e = \frac{V^2}{225R} = \frac{50^2}{225(500)} = 0.022$

**85. Indian railways is the world leader in the manufacture of which type of sleepers?**

(a) Cast iron (b) Wooden
(c) Concrete (d) Steel

**Ans. (c) :** Indian railways is the world leader in producing prestress concrete sleepers. Appropriate answer is concrete sleepers.

**86. Identify the correct formula for the calculation of safe speed on transitioned curves for broad gauge and meter gauge in km/h using the following equation.**

(a) $V = 4.4\sqrt{D-70}$ (b) $V = 2.4\sqrt{D-70}$
(c) $V = 2.4\sqrt{R-70}$ (d) $V = 4.4\sqrt{R-70}$

**Ans. (d) :** Safe speed on transition curve is
For BG and MG = $V = 4.4\sqrt{R-70}$

**87. Wear on rails is more prominent at some locations. Study the following points related to this**

1. **On sharp curves**
2. **On steep gradients**
3. **On approaches to railway stations**
4. **In tunnels and coastal areas.**

**Identify the correct statement from the following:**

(a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 1, 2, 3 and 4 only (d) 1, 3 and 4 only

**Ans. (c) :** Were on rails is more prominent on
1. Curves due to super elevation
2. Steep gradients due to impact
3. On approaches to railway sections (due to braking effect)
4. In tunnels (pollution) and in costal areas (corrosion) effect.

**88. Calculate the safe stopping distance for design speed of 36 kmph for two way traffic on a two lane road. Assume coefficient of friction as 0.25 and reaction time of driver as 2.5 seconds. Take g = 10 $m^2$/sec.**

(a) 20 m (b) 25 m
(c) 45 m (d) 40 m

**Ans. (c) :** V = 36 kmph = 10 m/s
Two lane (T/W)
f = 0.25, t = 2.5 sec, g = 10 $m/s^2$

$$SSD = V.t + \frac{V^2}{2gf}$$

$$SSD = 10 \times 2.5 + \frac{10 \times 10}{2 \times 10 \times 0.25} = 25 + 20 = 45\,m$$

**89. Match List-I with List-II and select correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| **Term** | **Definition** |
| **A. Porosity** | **1. Ratio of the volume of air voids to the total volume of voids** |
| **B. Air content** | **2. Ratio of the volume of voids to the volume of solids** |
| **C. Void ratio** | **3. Ratio of the volume of water in the voids to the volume of voids** |
| **D. Degree of saturation** | **4. Ratio of volume of voids to the total volume of the soil mass** |

**Identify the correct match:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 4 | 2 | 3 | 1 |
| (b) | 4 | 1 | 2 | 3 |
| (c) | 4 | 1 | 3 | 2 |
| (d) | 4 | 2 | 1 | 3 |

**Ans. (b) :**

| List-I | List-II |
|---|---|
| Porosity | $\frac{V_v}{V}$ |
| Air content | $\frac{V_a}{V_v}$ |
| Void ratio | $\frac{V_v}{V_s}$ |
| Degree of saturation | $\frac{V_w}{V_v}$ |

**90. If $k_x$ and $k_z$ are the permeabilities in the X and Z directions respectively in a two-dimensional flow situation, the effective permeability $k_e$ is given by**

(a) $\sqrt{k_x k_z}$ (b) $\frac{k_x}{k_z}$
(c) $\frac{k_z}{k_x}$ (d) $k_x + k_z$

**Ans. (a) :** If $k_x$ and $k_z$ are the permeabilities in the X and Z directions respectively in a two-dimensional flow situation, the effective permeability $k_e$ is given by

$$k_e = \sqrt{k_x k_z}$$

**Essential formula** – Basic equation for a two-dimensional laminar flow of water through soil.

Rate of seepage 'q' = $k.H\left(\frac{N_f}{N_d}\right)$

Critical hydraulic gradient. $(i_c) = \frac{G-1}{1+e}$

**91. The critical hydraulic gradient for a soil with a grain specific gravity 'G' and void ratio 'e' is given by the following equation.**

(a) $\frac{G+e}{1+e}$ (b) $\frac{G+1}{1+e}$
(c) $\frac{G-1}{1+e}$ (d) $\frac{G+e}{1-e}$

**Ans. (c) :** Critical hydraulic gradient, $i_c = \frac{G-1}{1+e}$
Where, G = specific gravity of soil
e = void ratio

**92. One cubic meter of wet soil weights 19.80 kN. If the specific gravity of soil particles is 2.70 and water content is 1%. Find the dry unit weight of the soil.**

(a) 18 $kN/m^3$ (b) 18 $N/m^3$
(c) 1.8 $kN/m^3$ (d) 1.8 $N/m^3$

**Ans. () :** Given,
Wet soil weights $\gamma_t$ = 19.80 kN
G = 2.7
Water content 'w' = 1%= 0.01
Dry unit weight '$\gamma_d$' = ?

$$\gamma_d = \frac{\gamma_t}{1+w} = \frac{19.80}{1.01} = 19.60\ kN/m^3$$

**93. Read the following statements related to compaction of soil**

1. **compaction increases the dry density of soil**
2. **compaction increases its shear strength**
3. **compaction increases decrease the tendency for settlement of the soil**
4. **compaction brings high permeability of the soil**

**Identify the correct statement from the following:**

(a) 1, 2 and 4 only (b) 1, 3 and 4 only
(c) 1 and 4 only (d) 1, 2 and 3 only

**Ans. (d) :** Compaction reduces permeability of soil.
**Compaction of soil–**

- Compaction increases the dry density of soil.
- Compaction increases its shear strength
- Compaction increases decreases the tendency for settlement of the soil.
- Compaction reduces permeability of soil.

**94. A sand fill compacted to a bulk density of 20 kN/m$^3$ is to be placed on a compressible saturated marsh deposit 3.5 m thick. The height of the sand fill is to be 3 m. If the height of the sand fill is to be 3 m. If the volume compressibility of the deposit is $7 \times 10^{-4}$ m$^2$/kN. Estimate the final settlement of the soil.**

(a) 147 mm (b) 1470 mm
(c) 14.70 mm (d) 1.47 mm

**Ans. (a) :**

$m_v = 7 \times 10^{-4} m^2 / kN$

$\Delta\sigma = q = r \times 3$

$\Delta\sigma = 20 \times 3 = 60 kPa$

$\Delta H = m_v \times H.\Delta\sigma$

$= 7 \times 10^{-4} \times 3.5 \times 60$

$= 0.147 = 147$ mm.

**95. The stresses at failure on the failure plane in cohesionless soil mass were shear stress = 3 kN/m$^3$, normal stress = 4 kN/m$^2$. Determine the resultant stress on the failure plane.**

(a) 3 kN/m$^2$ (b) 5 kN/m$^2$
(c) 4 kN/m$^2$ (d) 7 kN/m$^2$

**Ans. (b) :** $\tau_f = 3$ kN/m$^2$

$\sigma = 4$ kN/m$^2$

$\sigma_R = \sqrt{\sigma^2 + \tau_f^2}$

$= \sqrt{4^2 + 3^2} = 5 kPa$

**96. Strap footing is also called**

(a) pump handle foundation
(b) raft foundation
(c) combined foundation
(d) pile foundation

**Ans. (c) :** Strap footing is also called combined foundation.

**97. The time factor for a particular average degree of consolidation**

(a) depends upon the drainage path
(b) depends upon the coefficient of consolidation
(c) is independent of the distribution of initial excess hydrostatic pressure
(d) depends upon the distribution of initial excess hydrostatic pressure

**Ans. (c) :** Initially, the excess pore pressure distribution is uniform and degree of consolidation is zero.

$$\text{Time factor} = T_v = \frac{C_v t}{d^2}$$

- Depends upon–

(i) Drainage path
(ii) Coefficient of consolidation
(iii) Time (t)

**98. Rebhann's graphical method is based on**

(a) Rankine's theory
(b) Westergaard's theory
(c) Culamnn's theory
(d) Coulomb's theory

**Ans. (d) :** Rebhann's graphical method is based on Coulomb's theory.

**99. The geophone is**

(a) useful for drawing a bore well lag
(b) to measure the electrical resistivity of soil
(c) an electromechanical device that detects vibrations and converts them into measurable electric signals
(d) to measure the magnitude of the velocity

**Ans. (c) :** The geophone is an electromechanical device that detects vibrations and converts them into measurable electric signals.

**100. In a semi-infinite homogeneous isotropic material, the electrical resistivity is measured using the following equation.**

(a) $2\pi D\frac{E}{I}$ (b) $\pi D\frac{E}{I}$
(c) $2\pi D\frac{I}{E}$ (d) $\pi D\frac{I}{E}$

**Where D = distance between electrodes in m, E = potential drop between the inner electrodes (volts) and I = current flowing between the outer electrodes (A).**

**Ans. (a) :**

Electricity resistivity = $2\pi D\frac{E}{I}$

D = distance between electrodes in m.
E = potential drop between the inner electrodes (volts)
I = current flowing between the outer electrodes (A)

**101. Who is the father of modern soil mechanics?**

(a) Rankine (b) Cassagrande
(c) Newmark (d) Terzaghi

**Ans. (d) :** The father of modern soil mechanics is Terzaghi.

**102. Working out the exact quantities of various items of works is known as**

(a) estimating (b) mensuration
(c) quantity surveying (d) valuation

**Ans. (c) :** Quantity surveying is the process of finding out exact quantities of different items of work, so answer is quantity surveying.
After finding out exact quantities we multiply them with unit rates (found out from rate analysis or SSR) to get detailed estimate.

**103. The quantity of wood for the shutters of doors and windows is measured in**

(a) m (b) m$^2$
(c) m$^3$ (d) lumpsum

**Ans. (b) :** Units of measurement for shutters of doors and windows is done in m$^2$.

| Item | Units of measurement |
|---|---|
| • Earth work in excavation | cum |
| • Lime concrete in foundation | cum |
| • Door and windows frames or chowkhats rafters beams | cum |

**104. Newmark influence chart is applicable to the following soil mass**

1. **semi-infinite** 2. **homogeneous**
3. **isotropic** 4. **elastic**

**Identify the correct statement:**

(a) 1, 2 and 3 only (b) 1 and 2 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

**Ans. (d) :** Newmark influence chart is applicable to the following soil mass–
1. Semi-infinite 2. Homogeneous
3. Isotropic 4. Elastic

**105. Estimate the cost of brickwork for a wall 4 m long, 3 m high and 30 cm thick. The rate of brick work is ₹ 320/m$^3$.**

(a) ₹ 1,15,200 (b) ₹ 11,520
(c) ₹ 115.2 (d) ₹ 1,152

**Ans. (d) :**

**Fig.**

Volume = $(3 \times 4 \times 0.3)$ m$^3$
$= 3.6$ m$^3$
Estimate = Quantity × Rate
$= 3.6 \times 320 = 1152$ Rs.

**106. P, Q, R and S are four points at the corners of an excavation. The heights of the excavation at the four points are 3.21 m, 1.47 m, 2.15 m and 3.51 m. PS and QR are parallel sides of lenghts 3 m and 7.5 m respectively. The vertical distance between PS and QR is 6 m. Find the volume of the excavation.**

(a) 8142 m$^3$ (b) 8.142 m$^3$
(c) 81.42 m$^3$ (d) 814.2 m$^3$

**Ans. (c) :**

**Fig.**

$$V = A \times \left( \frac{h_1 + h_2 + h_3 + h_4}{4} \right)$$

$$V = (7.5 + 3)\frac{6}{2}\left( \frac{3.21 + 2.15 + 3.51 + 1.47}{4} \right)$$

$V = 81.4275$ m$^3$

**107. To determine the whole volume formed by given sections over a given length, the Prismoidal formula can be applied**

(a) irrespective of the number of sectional areas being odd or even
(b) only when the number of sectional areas are odd
(c) only when the number of sectional areas are even
(d) only for the three sections at a time

**Ans. (b) :** The prismoidal formula is applicable when there is an odd number of section.

**108. Read the following statements:**

1. **Lead is the average horizontal distance between site of earthwork and the area of disposal.**
2. **Lift is the average vertical distance between level of excavation to the place of spreading or heaping.**
3. **Lift is the average horizontal distance between site of earthwork and the area of disposal.**
4. **Lead is the average vertical distance between level of excavation to the place of spreading or heaping.**

**Identify the correct statements:**

(a) 1 and 2 only (b) 3 and 4 only
(c) 1 and 3 only (d) 1 and 4 only

**Ans. (a) :** Lead is the average horizontal distance and lift is the average vertical distance.
This terminology is commonly used in movement of earth (soil), meaning transport of soil from borrow pit to embankment and also while compacting the embankment in different layers.

**109. Identify the size of standard modular bricks in cms**

(a) 19 × 9 × 9 (b) 19 × 19 × 9
(c) 12 × 6 × 6 (d) 12 × 12 × 6

**Ans. (a) :** Standard size = $(19 \times 9 \times 9)$ cm$^3$
Nominal size = $(20 \times 10 \times 10)$ cm$^3$

**110. The joints of the brick work shall be raked out to a depth of 20 mm and the surface of the wall washed and cleaned and kept wet for two days. This process is known as**

(a) jointing (b) pointing
(c) soiling (d) plastering

**Ans. (b) :** The preparing the brick surface for pointing, first all the joints in masonry are racked down to a depth of 20 mm while the still soft. The joints and surface are cleaned and then thoroughly welted.

**111. The following are related to units of measurements for various items and materials. Identify the WRONG pair:**

(a) Thin partition wall – m$^2$
(b) Sand filling – m$^3$
(c) Rock excavation – m$^3$
(d) Damp proof course – m$^3$

**Ans. (d) :** Damp proof course has a fixed thickness and it is measure in m$^2$.

**112. A scale of 1 cm = 5 km is represented as a representative fraction is**

(a) 1 : 5000000 (b) 1 : 500
(c) 1 : 5000 (d) 1 : 500000

**Ans. (d) :** 1 cm = 5 km
1 cm = 5 × 1000 m
= 5 × 1000 × 100 cm
1 cm = 500000 cm
1 : 500000.

**113. A 30 m chain was tested before a survey and found to have a length of 30.3 m. If the length of a line measure with this chain was 290 m, find the true length of the line.**

(a) 290.0 m (b) 287.70 m
(c) 292.90 m (d) 290.30 m

**Ans. (c) :** $\ell$ = true length of the line
$\ell'$ = true length of the chain
L = wrong length of the line
L' = wrong length of the chain

$$\ell = \ell'\left(\frac{L'}{L}\right)$$

$$= 290\left(\frac{30.3}{30}\right) = 292.90\,m$$

**114. A prism square is used to**
(a) check the alignment of survey lines
(b) set a line at right angles to a survey line
(c) set an enlarged view of station marks
(d) marking of survey stations

**Ans. (b) :** Instrument used to set out right angles in chain survey.
1. Cross - staff
2. Prism - square
3. Optical - square

**115. Identify wrong statement from the following:**
(a) Prismatic compass measures whole circle bearings
(b) Surveyors compass measures whole circle bearing
(c) In Prismatic compass, readings are taken from the South end
(d) In Surveyors compass, readings are taken from North end.

**Ans. (b) :** Surveyors compass–
1. measures reduced bearings
2. uses edge bar type needle
3. Tripod is essential.

**116. The formwork which is used for vertical structural member such as columns, RCC walls etc known as**
(a) Staging (b) Centering
(c) Formwork (d) Shuttering

**Ans. (d) :** The formwork (shuttering) serves as mould for concrete structure components.

**117. The daily variation of the position of the magnetic needle as it points to the North is called as**
(a) Irregular variation (b) Secular variation
(c) Diurnal variation (d) Annual variation

**Ans. (c) :** Diurnal variation - 24 hours
Annual variation - 365 days
Secular variation - 250 - 300 years
Irregular variation - Random.

**118. Match List-I with List-II and select correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| **Terms** | **Definitions** |
| **A. Telescope axis** | **1. It is the line joining the intersection of the cross hairs to the optical center of the object glass and its continuation.** |
| **B. Line of Collimation** | **2. It is the operation of changing from free left to face right and vice versa.** |
| **C. Face left observation** | **3. It is the line joining the optical center of the object glass to the center of the eye piece** |
| **D. Changing face** | **4. Setting the theodolite exactly over the station mark** |
| **E. Centering** | **5. It is the reading taken when the instrument is in the inverted or face night position** |

**Identify the correct match:**

| | A | B | C | D | E |
|---|---|---|---|---|---|
| (a) | 3 | 1 | 4 | 5 | 2 |
| (b) | 3 | 1 | 4 | 2 | 5 |
| (c) | 3 | 1 | 5 | 2 | 4 |
| (d) | 3 | 1 | 5 | 4 | 2 |

**Ans. (*) : Telescope Axis**–Line joining the optical centers of eye-piece of objective.
**Line of collimation**–Line joining intersection of cross hair to optical center of objective and its continuation.
**Face left**–Vertical circle towards left face of the observer.
**Changing face**–Changing from face left to face right of vice-versa.
**Centering**–Settling instrument above station on ground.

**119. Match List-I with List-II and select correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| **Quadratic bearing** | **Whole circle bearing** |
| **A. N 60º25' E** | **1. 94º30'** |
| **B. S 85º30' E** | **2. 277º15'** |
| **C. S 64º30' W** | **3. 244º30'** |
| **D. N 82º45' W** | **4. 60º25'** |

**Identify the correct match:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 4 | 1 | 2 | 3 |
| (b) | 4 | 2 | 1 | 3 |
| (c) | 4 | 1 | 3 | 2 |
| (d) | 4 | 2 | 3 | 1 |

**Ans. (c) :**

**Conversion of Q.B. Into W.C.B.**

| Q.B. | Rule | W.C.B. between |
|---|---|---|
| N 60º25' E | WCB = QB | $= 60^0 25'$ |
| S 85º30' E | WCB = $180^0$–QB | $= 180^0 - 85^0 35'$<br>$= 94^0 30'$ |
| S 64º30' W | WCB = $180^0$+QB | $= 180^0 + 64^0 30'$<br>$= 244^0 30'$ |
| N 82º45' W | WCB = $360^0$–QB | $= 360^0 - 82^0 45'$<br>$= 277^0 15'$ |

**120. A reverse curve is one**
(a) having circular and half cubic parabola as a compound curve
(b) with a simple circular curve and a transition curve

(c) where the simple circular curve is set from the second tangent point in the reverse direction
(d) having two simple circular curves with centers in opposite directions

**Ans. (d) :** A reverse curve is a combination of a circular curve with centers at opposite sides of the curve.

**121. Electromagnetic distance measurement measuring instruments are uses**
(a) radio waves
(b) radiation frequencies like gamma rays
(c) radiation frequencies like x rays
(d) radiation frequencies from visible light to microwaves

**Ans. (d) :** EDM–
1. Electro-optic instruments uses modulated light wave.
2. Microwave emitting instrument.
3. Infra-red emitting instrument.

**122. A total station can measure**
(a) distances electrically only
(b) horizontal and vertical angles and distances
(c) horizontal angles only
(d) vertical angles only

**Ans. (b) :** Basic instrument of total station–
1. Horizontal angle
2. Vertical angle
3. Distance.

**123. The number of satellites used in global positioning system operated by the US department of defence is**
(a) 18 (b) 24
(c) 30 (d) 7

**Ans. (b) :** GPS requires 24 active satellite in 6 fixed orbit.

**124. Determine the radius of a curve if it is designated as $3^0$ curve as 30 m arc. $\pi = 3.14$**
(a) 563.25 m (b) 573.25 m
(c) 583.25 m (d) 593.25 m

**Ans. (b) :** D = 3º (30 m)

$$R = \frac{1719.75}{D^o} = \frac{1719.75}{3^o} = 573.25 \text{ m}$$

**125. What is efflorescence?**
(a) Swelling of brick due to presence of carbonaceous matter and gas
(b) Deformation of brick due to exposure to rain
(c) Formation of white patches on the brick surface due to insoluble salts in the brick clay
(d) Impurities in the brick clay which show after burning

**Ans. (c) :** Efflorescence is the formation of white patches on the brick surface due to insoluble salts in the brickway.

**126. What is the example for the stratified rock from the following?**
(a) Slate (b) Granite
(c) Basalt (d) Laterite

**Ans. (a) :** Straitified rocks are consisting of different layers in its structure and these layers are separated by planes of stratification. These planes are also called cleavage planes or bedding planes. These rocks can easily split up along these bedding planes.
Majority of slates are formed by the dynamic metamorphism of shales. Their characteristic slaty change may or may not be parallel to the bedding planes of the original shales.

**127. Some of the minerals are listed below**
**1. Talc 2. Gypsum**
**3. Flourite 4. Quartz**
**5. Diamon 6. Calcite**
**Identify correct sequence hardness of minerals from soft to hard**
(a) 1, 2, 6, 3, 4 and 5 (b) 1, 2, 6, 4, 3 and 5
(c) 1, 6, 2, 3, 4 and 5 (d) 1, 2, 3, 6, 4 and 5

**Ans. (a) :** The standard set of the reference minerals used to determine the hardness of any unknown mineral is called Moh's scale of hardness. The actual minerals of the set and their hardness are as follows–

| | |
|---|---|
| Talc = 1 | Feldspar = 6 |
| Gypsum = 2 | Quartz = 7 |
| Calcite = 3 | Topaz = 8 |
| Fluorite = 4 | Corundum = 9 |
| Apalite = 5 | Diamond = 10 |

**128. Identify the wrong pair based on chemical characteristics of rock**
(a) Argillaceous - principal constituent is clay
(b) Siliceous - principal component is silica
(c) Calcareous - principal component is lime
(d) Calcareous - principal component is Clay

**Ans. (a) :** Chemically calcareous rocks are mainly composed by calcium carbonate.

**129. Consider the following statements of preparation of brick earth**
**1. Digging 2. Weathering**
**3. Tempering 4. Moulding**
**5. Blending 6. Un-soiling**
**Identify the correct sequence of these operations for preparation of brick earth**
(a) 6, 1, 2, 5, 3, 4 (b) 6, 1, 2, 5, 4, 3
(c) 6, 2, 1, 5, 3, 4 (d) 6, 1, 5, 2, 3, 4

**Ans. (a) :** Preparation of brick earth as follows–
1-Unsoiling,
2-Digging,
3-Weathering,
4-Blending,
5-Tempering,
6-Moulding.

**130. The Ordinary Portland Cement (OPC) has been classified into the following types of cement**
(a) OPC 33 only
(b) OPC 33 and OPC 43 only
(c) OPC 33, OPC 43 and OPC 53 only
(d) OPC 33, OPC 43, OPC 53 and OPC 63 only

**Ans. (c):** Ordinary Portland Cement is classified into three grades i.e., 33, 44 and 53 grades.

**131. Identify correct pair from the following:**

| Chemical component of OPC | Function |
|---|---|
| (a) CaO | Controls strength and soundness |
| (b) $Al_2O_3$ | Gives colour |
| (c) $Fe_2O_3$ | Responsible for quick setting |
| (d) $SiO_2$ | Imparts colour |

**Ans. (a) :** Lime (CaO) controls the strength and soundness aspects of cement.

**132. The percentage bulking for fine aggregate is calculated using the following formula.**

(a) $\frac{\text{Bulked volume} - \text{Dry volume}}{\text{Dry volume}} \times 100$

(b) $\frac{\text{Bulked volume}}{\text{Dry volume}} \times 100$

(c) $\frac{\text{Bulked volume} - \text{Dry volume}}{\text{Bulked volume}} \times 100$

(d) $\frac{\text{Dry volume}}{\text{Bulked volume}} \times 100$

**Ans. (a) :**

$$\% \text{ Bulking} = \frac{\text{Bulked volume} - \text{Dry volume}}{\text{Dry volume}} \times 100$$

**133. The difference in 7 days compressive strength of cubes/cylinders prepared with impure and pure waters should not differ by more than**

(a) 12% (b) 10%
(c) 8% (d) 15%

**Ans. (b) :** The difference in 7 days compressive strength of cubes/cylinders prepared with impure and pure waters should not differ by more than 10%.

**134. The upper limit of suspended particles in water for the preparation of concrete is**

(a) 2500 ppm (b) 500 ppm
(c) 2000 ppm (d) 200 ppm

**Ans. (c) :** The upper limit of suspended particles in water for the preparation of concrete is 2000 ppm.

**135. The operations involved in obtaining minerals are called as**

(a) Wedging (b) Quarrying
(c) Mining (d) Excavation

**Ans. (c) :** Mining is the extraction of valuable minerals of other geological materials from the earth. Usually from an ore body, dode, vein, seam, ray or placer deposit.

**136. Read the following statements related to concrete:**

1. **A rounded spherical or cubical shaped aggregate when computed contains less voids and higher strength than an irregular aggregate of the same nominal size.**
2. **High strength concrete gives lower strength as compared to lean concrete if larger size aggregate is used.**
3. **The larger aggregates have lower total surface area and require lower water to cement ratio resulting in higher strength.**
4. **Large size aggregates give heterogeneous concrete causing non-uniform distribution of load when stressed.**

**Identify the correct statement:**

(a) 1, 2 and 3 only (b) 1 and 2 only
(c) 1, 3 and 4 only (d) 1, 2 and 4

**Ans. (c) :**
- A rounded spherical or cubical shaped aggregate when computed contains less voids and higher strength than an irregular aggregate of the same nominal size.
- The larger aggregates have lower total surface area and require lower water to cement ratio resulting in higher strength.
- Large size aggregates give heterogeneous concrete causing non-uniform distribution of load when stressed.

**137. What is the characteristic compressive strength of 150 mm cube at 28 days for M40 grade**

(a) 40 $N/mm^2$ (b) 40 $N/m^2$
(c) 40 $N/cm^2$ (d) 40 $kN/m^2$

**Ans. (a) :** The characteristic compressive strength of 150 mm cube at 28 days for M40 grade is 40 $N/mm^2$.

**138. Match List-I with List-II and select correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| **Admixtures** | **Chemicals** |
| **A. Water reducing admixture** | **1. Sulphonated melanin** |
| **B. Air entering agent** | **2. Calcium chloride** |
| **C. Super plasticizer** | **3. Lingosulphonate** |
| **D. Accelerator** | **4. Neutralized Vinsol resin** |

**Identify the correct match:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 2 | 4 | 1 | 3 |
| (b) | 1 | 3 | 4 | 2 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 3 | 4 | 2 | 1 |

**Ans. (c) :**

| Admixtures | Chemicals |
|---|---|
| Water reducing admixture | Lingo-sulphonate |
| Air entering agent | Neutralized Vinsol resin |
| Super plasticizer | Sulphonated melanin |
| Accelerator | Calcium chloride |

**139. The maturity of concrete is defined as**

(a) the summation of product of time and temperature
(b) the summation of time and temperature
(c) the product of time and temperature
(d) the product of time and curing time

**Ans. (a) :** Maturity of concrete is the summation of product of curing period (time) and curing temperature.

**140. The modulus of elasticity (E) and modulus of rigidity (C) are related by**

(a) $C = \frac{mE}{3(m-2)}$ (b) $C = \frac{mE}{2(m+1)}$

(c) $C = \frac{3(m-2)}{mE}$ (d) $C = \frac{2(m+1)}{mE}$

**where $\frac{1}{m}$ = Poisson's ratio**

**Ans. (d) :** $E = 2C(1+\mu)$

$$E = 2C\left(1+\frac{1}{m}\right)$$

$$E = 2C\left(\frac{m+1}{m}\right)$$

$$C = \frac{mE}{2(m+1)}$$

**141. For a given material, Young's modulus is 1.5 × $10^5$ N/mm$^2$ and Poisson's ratio is 1/4. Calculate the bulk modulus.**

(a) 1× $10^5$ N/cm$^2$ (b) 0.5× $10^5$ N/mm$^2$

(c) 1× $10^5$ N/mm$^2$ (d) 0.8× $10^5$ N/mm$^2$

**Ans. (c) :**

$$E = 1.5\times10^5 \frac{N}{mm^2} \qquad \left(\mu = \frac{1}{\mu}\right)$$

$E = 3k(1-2\mu)$

$$1.5\times10^5 = 3k\left(1-2\times\frac{1}{4}\right)$$

$$1.5\times10^5 = 3k(0.5)$$

$$k = \frac{1.5\times10^5}{3\times0.5} = 1\times10^5 N/mm^2$$

**142. Identify false statement from the following:**

(a) The total strain energy stored in a body is commonly known as resilience

(b) The maximum strain energy stored in a body is known as proof resilience

(c) Modulus of resilience is defined as the proof resilience of a matter per unit volume

(d) Modulus of resilience is calculated using $\frac{\sigma^2}{4E}$ equation, where $\sigma$ = stress induced in the body

**Ans. (d) :**

Modulus of resilience $= \frac{\sigma^2}{2E}$

$\sigma$ = stress induced in the body

modulus of resilience is defined as the proof resilience of a matter per unit volume.

**143. A rod is 2 m long at a temperature of 10ºC and temperature is raised to 80ºC. If expansion is prevented, find the stress induced in the material of the rod. Take E = 1.0 × $10^5$ MN/m$^2$ and $\propto$ = 0.000012 per degree centrigrade.**

(a) 84 MN/m$^2$ (b) 84 N/mm$^2$

(c) 96 N/mm$^2$ (d) 84 MN/mm$^2$

**Ans. (b):**

L = 2m, $t_i$ = 10ºC, $t_f$ = 80ºC,

$\Delta T = 80 - 10 = 70^\circ C$

$E = 1\times10^5 MN/m^2$

$E = 1 \times 10^5$ N/mm$^2$

$\alpha = 0.000012^\circ/C$

$\sigma = E \propto \Delta T$

$$\sigma = 1\times10^5 \times 0.000012 \times 70 = 84\frac{N}{mm^2}$$

**144. Identify the correct pair from the following?**

(a) Poisson's ratio: Ratio of lateral strain to the longitudinal strain.

(b) Pascal's law : The relationship between stress and strain for a unidirectional stress

(c) Factor of safety: Ratio between ultimate stress to shear stress

(d) Lateral strain : Ratio of increase in the length to the length of the body

**Ans. (a) :**

$$\text{Poisson's ratio} = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$$

**145. Read the following statement and identify the wrong statement from the following. These are related to theory of simple bending with assumptions.**

(a) The material of the beam is homogeneous and isotropic

(b) The value of Young's modulus of elasticity is not equal in tension and compression

(c) Each layer of the beam is free to expand or contract

(d) The transverse sections which were plane before bending remain plane after bending also

**Ans. (b) : Theory as simple bending–** The filament / fibers of the material are subjected to neither compression nor tension.

- Pure bending means there is no difference in bending moment across the section of beam.
- The line of intersection of the neutral axis layer with transverse section is called neutral axis.
- Hence the theory of bending (pure) state that the amount by which a layer in a beam subjected to pure bending. Increase or decrease in length depend upon the position of the layer w.r.t. neutral axis.
- The value of Young's modulus of elasticity is equal in tension and compression.

**146. Match List-I with List-II and select correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| **Cross section** | **Section Modulus** |
| **A. Rectangular** | **1.** $\frac{1}{6D}(BD^3 - bd^3)$ |
| **B. Circular** | **2.** $\frac{\pi}{32D}(D^4 - d^4)$ |

| | |
|---|---|
| **C. Hollow Rectangular** | **3.** $(BD^2/6)$ |
| **D. Hollow circular** | **4.** $(\pi D^3/32)$ |

**Where 'D' is outer diameter and 'd' is the inner diameter.**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 3 | 1 | 2 | 4 |
| (b) | 3 | 4 | 2 | 1 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 3 | 1 | 4 | 2 |

**Ans. (c) :**

| Cross section | Section Modulus |
|---|---|
| Rectangular | $(BD^2/6)$ |
| Circular | $(\pi D^3/32)$ |
| Hollow Rectangular | $\frac{1}{6D}(BD^3 - bd^3)$ |
| Hollow circular | $\frac{\pi}{32D}(D^4 - d^4)$ |

**147. A cantilever beam of length 2 m carries a uniformly distributed load of 2 kN/m over the whole length and a point load of 3 kN at the free end. Calculate the maximum shear force and bending moment.**

(a) 7 kN and 10 kN-m (b) 7 kN and 10 N-m
(c) 7 N and 10 N-m (d) 7 N and 10 kN-m

**Ans. (a) :** $\sum f_y = 0$

$-3 - (\alpha \times \alpha) + R_A = 0$

$R_A = 3 + 4 = 7$ kN

$SF_A = R_A = 7$ kN

$\sum M_A = 0$

$3 \times 2 + (2 \times 2) \times \frac{2}{2} - M_A = 0$

$M_A = 6+4 = 10$ kN-m

$BM_A = M_A = 10$ kN-m

**148. Identify the correct pair from the following:**

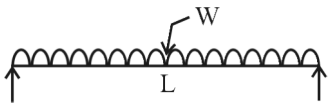
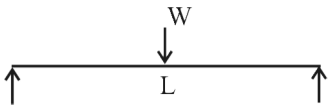
(a) Slope at the supports of a simply supported beam (length = L) carrying a point load (W) at the center $= \frac{WL^2}{64EI}$

(b) Deflection at the center of a simply supported beam (length = L) carrying a point load (W) at the center $= \frac{WL^3}{48EI}$

(c) Maximum slope of a simply supported beam carrying a UDL of W/unit length over the entire span $= \frac{WL^3}{34EI}$

(d) Maximum deflection of a simply supported beam carrying a UDL of W/unit length over the entire span $= \frac{5WL^4}{284EI}$

**Ans. (b) :**

| | |
|---|---|
| $\theta_{max} = \frac{wL^3}{24\ EI}$ |   |
| $y_{max} = \frac{5wL^4}{384\ EI}$ | |
| $\theta_{max} = \frac{wL^2}{16\ EI}$ |   |
| $y_{max} = \frac{wL^3}{48\ EI}$ | |

**149. A leaf spring carries a central load of 3000 N. The leaf spring is to be made of 10 steel plates each 5 cm wide and 6 mm thick. Length of the spring is 600 mm and E = 2 × $10^5$ N/$mm^2$. Determine the deflection at the center of the spring.**

(a) 12.25 mm (b) 15.25 mm
(c) 11.25 mm (d) 1.25 mm

**Ans. (c) :** w = 3000 N, n = 10, b = 5 cm = 50 mm
t = 6 mm, L = 600 mm, E = 2 × $10^5$ N/$mm^2$

$(\sigma_b)_{limiting} = 150 \frac{N}{mm^2}$

$\Delta_{centre} = \frac{3}{8} \frac{WL^3}{Enbt^3}$

$= \frac{3}{8} \times \frac{3000 \times (600)^3}{2 \times 10^5 \times 10 \times 50 \times 6^3}$

$= 11.25$ mm

**150. Read the following:**

1. **Polar modulus is defined as the ratio of the polar moment of inertia to the radius of the shaft.**
2. **Polar modulus is also called as section modulus.**
3. **Polar modulus of a solid circular shaft is calculated using $\frac{\pi}{16}D^3$ formula.**
4. **Polar modulus of a hollow circular shaft is calculated using $\frac{\pi}{16D}(D^4 - d^4)$.**

**Identify the correct statement from the following:**

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1 and 4 only

**Ans. (b & d) :** Polar section modulus $(Z_p) = \frac{J}{r_{max}}$

$(Z_p)_{solid} = \frac{J}{r_{max}} = \frac{\frac{\pi}{32}D^4}{D/2}$

$(Z_p)_{hollow} = \frac{\frac{\pi}{32}(D^4 - d^4)}{D/2} = \frac{\pi}{16D}(D^4 - d^4)$

Statement A, C, D are correct.

# Andhra Pradesh Public Service Commission
# Polytechnic Lecturer Exam, 2020
## Solved Paper

**1. What was/is the standard length (in ft.) of Gunter's chain and in how many parts was/is it sub-divided ?**

(a) 66,100 (b) 60,100
(c) 66,50 (d) 60,50

**Ans. (a) :** The various types of survey chain
1. Revenue chain – 33ft, 16 links
2. Gunter's chain – 66 ft, 100 links
3. Engineer's chain–100 ft, 100 links
4. Metric chains :
(a) 20m–100Links
(b) 30m– 150 Links

**2. The lines passing through points of zero declination are said to be :**

(a) Isogonic lines
(b) Agonic lines
(c) Isometric lines
(d) Isochronous lines

**Ans. (b) : Agonic lines:** The Lines passing through points of zero declination are said to be Agonic lines.
**Isogonic line :** The Passing through the points on earth surface at which the declination is same at a given time.
**Isoclinic lines :** These are the lines which joint points on earth with same value of dip.
**Aclinic line :** These are special case of isoclinic lines Which points of zero dip.

**3. Sanjeev wants to do levelling measurements for a river tributary, wherein he is finding it impossible to balance backsight and foresight distances. He only has access to a simple level. Which of the following methods of levelling would you recommended ?**

(a) Benchmark levelling
(b) Reverse levelling
(c) Reciprocal levelling
(d) Cross-section levelling

**Ans. (c) : Type of levelling–**
1. Barometric levelling
2. Trigonometric levelling
3. Spirit levelling/Direct levelling
4. Differential levelling
5. Reciprocal levelling
6. Profile and cross section levelling/ Longitudinal sectioning.

**Reciprocal Levelling :** It is a method of levelling used for the determination of the difference of elevations of the two points which are situated quite apart and it is not possible to set up the instrument midway between these points.

**Example:** Opposite banks of river as opposite sides of valley.
With the help of reciprocal levelling we can eliminate error due to.
(a) The curvature error
(b) Refraction error
(c) collimation error

**Cross levelling:** It is a type of differential levelling done to determine difference of elevations of ground surface along the lines perpendicular to the centre line.

**4. In which case the vertical angle measured using Theodolite will NOT pass through the station mark and the measured horizontal angle will be in error?**

(a) If the theodolite is not correctly centered over the ground station mark at which the horizontal angles are to be measured.
(b) If the ambient temperature is above $45^0$C
(c) If the pegs are of differential lengths
(d) If ambient temperature is $20^0$C

**Ans. (a) :** If the theodolite is not correctly centered over the ground station mark at which the horizontal angles are to be measured then there will be error in the vertical angle measured by theodolite.

**5. A point temporarily used to transfer an elevation is called :**

(a) Benchmark
(b) Turning point
(c) Temporary benchmark
(d) Foresight

**Ans. (b) :**
**Turning point:** A point temporarily used to transfer an elevation is called turning point..
**Benchmark:** Benchmark is a selectively permanent point of senesce whose elevation with respect to some assume datum is known
**Temporary Benchmark:** These benchmarks are temporarily established whenever required generally these are the points at which is closed your a day and next day work is started.
**Fore sight reading :–** Fore sight is the reading taken on a last point of survey on change point i.e., after your sight reading either the instrument is shifted as work is closed.

**6. One nautical mile equals :**

(a) 1.732 km (b) 1.781 km
(c) 1.852 km (d) 1.897 km

**Ans. (c):** A nautical mile is 1852 meters or 1.852 km. In English measurement system, a nautical mile is 1.1508 miles or 6076 feet.

**7. If *l* is the length of the line, then according to Bowditch's method, the linear and angular errors are proportional to-**

(a) $l$ & $1/(l)$ (b) $l^{0.5}$ & $1/(l^{0.5})$

(c) $l^{0.67}$ & $1/(l^{0.67})$ (d) $l^{2}$ & $1/(l^{2})$

**Ans. (b) :** Bowditch's rule is used to balance a traverse when the linear and angular measurements are equally precise.

For angular measurement $\propto \frac{1}{\sqrt{l}}$

$\propto \frac{1}{l^{0.5}}$

For linear measurement $\propto \sqrt{l}$

$\propto l^{0.5}$

**8. The angle subtended at the center of the earth north by the arc of meridian intercepted between the place and the equator is called _____ of a place.**

(a) Latitude
(b) Longitude
(c) Co-longitude
(d) Azimuth

**Ans. (a) :** The angle subtended at the centre of the earth north by the place and the equator is called latitude of a place.
Latitude of a line is its orthographic projection on N-S meridian.

**9. Which of the following methods of contour interpolation is very crude and used for very small-scale works?**

(a) Graphical method
(b) Arithmetical calculation
(c) Conveniently taking tracking cloth and dividing it into desired intervals
(d) Estimation

**Ans. (d) :** Estimation method of contour interpolation is very crude and used for very small scale works.
**Interpolation of Contour Estimation–** The point on the required contour are located by eye judgment or estimating between whose elevation are already known.
**Use**- small scale map.

**10. The sensitivity of a level tube is expressed in terms of :**

(a) the ground level where the levelling instrument is placed
(b) angle in seconds subtended with respect to the meridian
(c) the perpendicularity of horizontal plate with the instrument
(d) angle in seconds subtended at the centre by the arc of one division length of the level tube

**Ans. (d) :** The sensitivity of a level tube is expressed in terms of angle in seconds subtended at the centre by the arc of one division length of the level tube.

**11. Which of the following is a WRONGLY stated application of triangulation survey ?**

(a) Assisting in the determination of mean sea level
(b) Determining accurate locations for setting out of civil engineering works
(c) Establishing accurate control for photogrammetric surveys for large areas
(d) Establishing accurate control for plane and geodetic surveys covering large areas

**Ans. (a) :** Application of triangulation survey :
1. Determining accurate locations for setting out of civil engineering works.
2. Establishing accurate control for photogram metric surveys for large areas.
3. Establishing accurate control for plane and geodetic surveys covering large area.

- Triangulation is a surveying method that measures the angles in a triangle formed by three survey control points.

**12. If the focal length of camera is 20 cm and the map distance is 10 cm representing a ground distance of 200 m, then what is the height of the camera above the selected data ?**

(a) 380 m (b) 395 m
(c) 400 m (d) 420 m

**Ans. (c) :** 400m

$$\frac{y}{Y} = \frac{f}{H}$$

$$\frac{10cm}{200m} = \frac{20cm}{H}$$

$$H = 400m$$

**13. Which of the following instruments is used either to enlarge or to reduce a plan already drawn ?**

(a) Eidograph
(b) Box sextant
(c) Burel hand level
(d) Mining dial

**Ans. (a) :** Ediograph is used either to enlarge or to reduce a plan already drawn.
Box sextant is small pocket instrument used for horizontal angle measurement.

**14. Which of the following is NOT a usually used method for orienting a plain table ?**

(a) Orientation with a trough compass
(b) Orientation by three point problem
(c) Orientation by five point problem
(d) Orientation by back sighting

**Ans. (c): Three point problem–** In this condition, three points and their position in the field are known the plane table is placed at as opposition from where all three points are visible. So by sighting those three points we can locate the point where equipment is located.
This can be achieved by many methods as follow–
(i) Tracing paper method
(ii) Lehman's method
(iii) Analytical methods
(iv) Graphical method.

**15. Which of the following is NOT an indirect method of contouring ?**
(a) Radial line method
(b) Cross-section method
(c) Hand level method
(d) Grid method

**Ans. (c) : Hand level method–** Hand level method is not a method of indirect contouring. A hand level is essentially a sprit level inside of telescope. The device is generally used by grading contractors and land surveyors to compare multiple points against a reference point to get approximations of level or distance.
Hand level save time compared to setting up a tripod and levelling instrument.

**16. The length of a line, originally 100 mm long on a map plotted to a scale of 1/1000, was found to be 95 mm due to shrinkage of the map. The map prepared using a tape of length 20 m was later found to be actually 21 m. If a certain area on the map, measured using a planimeter, is 100 mm$^2$, determine the correct area on the ground (in m$^2$) :**
(a) 244.3 (b) 183.2
(c) 152.7 (d) 122.2

**Ans. (d) :** 122.2

$$\text{S.F. (Shrunk factor)} = \frac{\text{Shrunk length}}{\text{Original length}}$$

$$\text{S.F.} = \frac{95}{100} = 0.95$$

Shrunk scale = S.F. × original scale

$$= 0.95 \times \frac{1}{1000} = \frac{1}{1052.63}$$

1 mm = 1.05263m

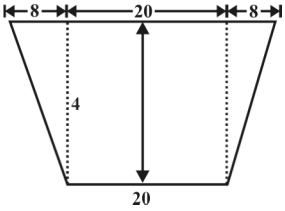
$100\text{mm}^2 = 11080.33\text{m}^2$

$$\text{So, corrected Area} = \left(\frac{21}{20}\right)^2 \times 11080.33$$

$$= 122.16\text{m}^2$$

**17. Calculate the earthwork (cum) required to fill a trapezoidal cross-section that has breadth of formation as 20 m, center height as 4 m and side slopes as 1 vertical to 2 horizontal. The length of this section is 2 km :**
(a) 168000 (b) 224000
(c) 246400 (d) 268800

**Ans. (b) :** Length of the trapezoidal c/s = 2 km
Breath of trapezoidal c/s = 20m
Height of trapezoidal c/s = 4m
Side slop = 1V : 2H




Area of Trapezoid

$$= \frac{1}{2}(36+20)\times 4 = 112\text{m}^2$$

Earth work to required to fill a trapezoidal cross section
= 112×2000
= 224000m$^3$

**18. Which of the following is NOT a typical principal used in electromagnetic distance measurement (EDM) equipment used for surveying ?**
(a) Light waves
(b) Infrared waves
(c) Supersonic waves
(d) Micro waves

**Ans. (c) :** Supersonic wave is not a typical principal used in electromagnetic distance measurement (EDM) equipment.
These are these types of EDM instrument based an wavelength–
1. light waves
2. Infrared waves
3. Micro waves

**19. The drag that permeating water exerts in the upward direction, which in-turn tends to oppose the force of gravity and when water flows upward under the critical hydraulic gradient, it completely neutralize the force on account of weight of particles, and thus leaves the particles suspended in water. Soil in such a state does not behave like soil but like a very viscous liquid popularly known as :**
(a) Quick sand
(b) Bulking
(c) Effervescence
(d) Swelling

**Ans. (a) :** Quick sand is a hydraulic condition in which due to upward flow, soil such as sand looses all its shearing strength and it cannot support any lead. The soil is said to have become quick as alive and boiling will occur, and the gradient at which this condition occurs is known as critical hydraulic gradient quick

sand condition generally occurs in cause silt or fine sand, not in coarse sand or gravel, because these soils are highly pervious. Hence as per Darcy's law, a critical hydraulic gradient and such flow surely materialize in practice.

**20. An engineer wanting to find suitable sand for embankment filling observes that a particular type of sand has a relative density of 0%. What can he conclude from this ?**

(a) Sand is in its loosest state
(b) Sand is in its densest state
(c) Sand is in its intermediate state of compaction
(d) This sand cannot be further compacted

**Ans. (a) :** Sand is in its loosest state.
when, relative density ($I_D$) = 100%, Then soil is in densest condition.
when, relative density ($I_D$) = 0% i.e. soil is in loosest condition.

| **State of soil** | **$I_D$ (%)** |
|---|---|
| Very loose | < 15 |
| Loose | 15 to 35 |
| Medium dense | 35 to 60 |
| Dense | 65 to 85 |
| Very dense | 85 to 100 |

**21. What would best describe the degree of saturation of the soil in which water exists in inter-connected lenses around particle contacts ? Air exists in interconnected channels and both, air and water phrases are continuous.**

(a) Very high saturation
(b) High saturation
(c) Medium saturations
(d) Very low saturation

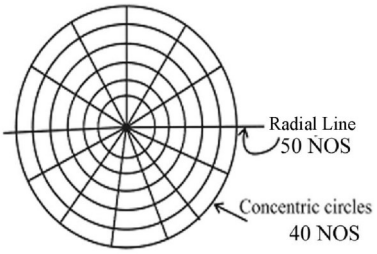
**Ans. (c) :**

| **S.N.** | **Degree of saturation** | **Description of state** |
|---|---|---|
| 1. | Very high | Air in the form of occluded bubble in the pore water. No continuity in the air phase. |
| 2. | high | Transition - H |
| 3. | medium | Water exists in inter-connected lenses around particles contacts. Air exists in interconnected channel. Both air and water phase continuous. |
| 4. | low | Transition - L |
| 5. | very low | No continuity in water phase. |

**22. Who proposed influence chart for vertical pressure distribution in soil ?**

(a) Newman
(b) Newmark
(c) Skempton
(d) Darcy

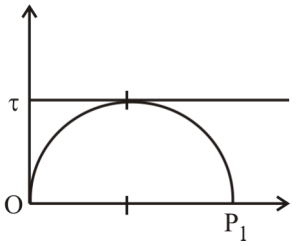
**Ans. (b) :** Newmark proposed influence chart for vertical pressure distribution in soil. It is a simple, graphical procedure for computing the vertical stress at any depth in a soil loaded uniformly at the surface in an irregular shape.




**23. What can be said for soil in which the failure envelope for the total stress Mohr's circles becomes a horizontal line ?**

(a) Angle of friction is zero
(b) Angle of friction is greater than 15
(c) Angle of friction is greater than 30
(d) Angle of friction is greater than 45

**Ans. (a) :** When angle of friction is zero, then the failure envelope for the total stress mohr's circle becomes a horizontal line for soil.




**24. If the weight of a hammer is 10 kN and the height of drop is 2 m, then what is the theoretical input energy ?**

(a) 5 kN-m (b) 10 kN-m
(c) 15 kN-m (d) 20 kN-m

**Ans. (d) :**
Theoretical input Energy = weight of hammer × height of drop
= 10 kN × 2m = 20kN-m

**25. Darcy's law of water flow through soils is valid for :**

(a) only sands under laminar flow
(b) only clays under laminar flow
(c) all soils under laminar flow
(d) all soils under turbulent flow

**Ans. (c) :** Darcy's law of flow through soil assumes that the flow is Laminar.
In one dimensional flow discharge through fully saturated soil is given by Darcy :

$q = kiA$
or $V = k\,i$
$q$ = discharge
A = cross sectional area of soil corresponding to flow 'q'

k = coefficient of permeability

i = hydraulic gradient $= \frac{\text{loss of head}}{\text{length}}$

V = average velocity or discharge velocity

**26. A Geotechnical engineer tests a soil and find that its liquidity index is 1.2. Which of the following states is the soil in ?**

(a) At Liquid limit
(b) At plastic limit
(c) In liquid state
(d) In oven dry sate

**Ans. (c) :** In liquid state

- When, liquidity index ($I_L$) > 1; soil is in the liquid state.
- Liquidity index ($I_L$) = $0 \le I_L \le 1$; soil is in the plastic state.
- Liquidity index ($I_L$) = $I_L \le 0$; soil is in the semi-solid or solid state.

**27. In order to take a call for commencing construction, the project manager requests saturation tests to be conducted on a clay. If the volume of voids is found to be 55 volume units and the volume of water in the voids is found to be 49.5 volume units, what is the current saturation level of the soil ?**

(a) 111% (b) 90%
(c) 10% (d) 900%

**Ans. (b) Given,**

$$\text{Degree of saturation (S)} = \frac{\text{Volume of water } (V_W)}{\text{Volume of voids } (V_V)} \times 100$$

$$= \frac{49.5 \text{ Volume unit}}{55 \text{ Volume unit}} \times 100$$

$$= 0.9 \times 100\%$$

$$= 90\%$$

**28. Which of the following is a WRONGLY stated design criteria for machine foundations ?**

(a) The settlements should be within prescribed limits.
(b) The center of gravity of the machine should preferably lie in the same vertical line as the center of gravity of the foundation system.
(c) There must be resonance.
(d) The dynamic amplitudes of the machine-foundation-soil system must be within the prescribed limits under service conditions.

**Ans. (c) :** Resonance should be avoided in all circumstances.

- High amplitude vibration are generated in resonance which is a prime concern for safety of building.

**29. A lab technician estimated the voids ratio of a soil as 3. What will be the porosity of this soil ?**

(a) 1.25 (b) 1.00
(c) 0.75 (d) 0.33

**Ans. (c) :** Given,

Void ratio (e) = 3

$$\text{Porosity (n)} = \frac{e}{1+e}$$

$$= \frac{3}{1+3} = \frac{3}{4} = 0.75$$

**30. The Skempton's pore water pressure parameter for a normally consolidated clay ranges between :**

(a) –0.5 to 0 (b) 0 to 1.0
(c) 0.5 to 1.0 (d) 1.0 to 1.5

**Ans. (c)** The Skempton's pore water pressure parameter for –

| | |
|---|---|
| • Normally consolidated clays | 0.5-1.0 |
| • Over consolidated clays | 0.5-0 |
| • Sandy Clays | 0.5-0.7 |

**31. Which of the following is NOT an advantage of well foundation ?**

(a) It can resist scouring better
(b) Possible to ensure its resting on suitable bearing stratum of uniform nature
(c) It can withstand large lateral loads.
(d) Sinking of the well causes vibrations that are good for adjacent structures.

**Ans. (d) : Advantage of well foundation** :

– It can resist scouring better

– Possible to ensure its resting on suitable bearing stratum of uniform nature.

– It can withstand large lateral loads.

**32. According to the Indian standard, the maximum settlement allowed for isolated foundation of a reinforced concrete structure constructed on plastic clay is limited to :**

(a) 25 mm (b) 50 mm
(c) 75 mm (d) 100 mm

**Ans. (c) :** According to the indian standard, the maximum settlement allowed for isolated foundation of a reinforced concrete structure constructed on plastic clay is limited to 75mm.

**33. Which of the following is NOT an advantage of driven concrete precast piles ?**

(a) Quality of concrete is never ensured
(b) Suitable for aggressive environment
(c) High displacement piles
(d) Suitable for underwater installations

**Ans. (a): Advantages of driven precast concrete piles**–

- Better quality control is possible as casting is done in yard.
- More suitable in aggressive environment such as sea water spray, sulphate water etc.
- Generally used in onshore areas due to ease in driving.

**34. Which of the following compactors will typically provide maximum average output in clean sand and gravel for an average thickness in the range of 300-500 mm?**

(a) Static steel drum roller
(b) Vibratory smooth drum roller
(c) Sheepfoot roller
(d) Pneumatic tyre roller

**Ans. (b) :** Vibratory smooth drum roller will yield maximum average output in clean sand and grand for an average thickness in the range of 300-500 mm.
Sheep foot rollers are ideally suitable for compaction of cohesive soils. (compacts by combination of tamping and kneading action)
Pneumatic tyre rollers are affective for both cohesive as well as cohesion less soils.

**35. If a soil compactor runs at an efficiency factor of 0.8, has a drum width of 2 m, operates at 10 km/h speed and compacts a soil layer of 300 mm in 6 passes, what will be its output (cum/h)?**

(a) 800 (b) 750
(c) 680 (d) 520

**Ans. (a) :** Efficiency factor = 0.8
drum width = 2 m
speed = 10 Km/h

$$\text{soil layer in 1 passes} = \frac{300}{6} = 50mm$$

output = width × speed × soil layer × efficiency factor
$2 \times 10 \times 10^3 \times 0.05 \times 0.8 = 800\ m^3/h$

**36. A gravity retaining wall has an overturning moment of 200 kNm/m about the left hand bottom edge. If the resisting moment is 500, what is the factor of safety for overturning ? Is it safe or unsafe for overturning only ?**

(a) 2.5, unsafe (b) 2.5, safe
(c) 0.4, unsafe (d) 1.5 safe

**Ans. (b) :** Given
overturning moment = 200 kN-m
resisting moment = 500 kN-m

$$\text{factor of safety for overturning} = \frac{\text{resisting moment}}{\text{overturning moment}}$$

$$\text{F.O.S} = \frac{500}{200} = 2.5$$

and, resisting moment 500kN-m is
grater than the over turning moment 200 kN-m
hence the gravity retaining wall is safe.

**37. In which of the following samplings is care taken to ensure that at least the composition of the soil as it exists in nature remains unaltered?**

(a) Undisturbed samples
(b) Disturbed representative samples
(c) Disturbed non-representative samples
(d) Intact samples

**Ans. (b):** Disturbed representative sampling is taken to ensure that at least the composition of the soil as it exists in nature remains unaltered.
• In this soil structure gets modified but compressive is unaltered.

**38. Which of the following is NOT a soil classification system employed in pavements ?**

(a) Burmister's descriptive system
(b) Civil Aeronautics administration system
(c) Casagrande soil classification
(d) US department of agriculture system

**Ans. (d) :** US department of Agriculture system is not a soil classification system employed in pavement.

**39. As per Indian standard, the diameter (mm) and height (mm) of a CBR mould should be;**

(a) 150, 150 (b) 150, 175
(c) 175, 150 (d) 175, 175

**Ans. (b) :** As per Indian standard, the diameter (mm) and height (mm) of CBR mould shoed be 150, and 175 respectively.
• Cylindrical mould with inside dia 150 mm and height 175 mm provided with a detachable extension collar 50 mm height and a detachable perforated base plate 10 mm thick.
• Spacer disc 148 mm in dia and 47.7 mm in height alongwith handle.

**40. The zero mile stone in India is located at :**

(a) Patna (b) Chhindwara
(c) Seoni (d) Nagpur

**Ans. (d):** The zero mile stone in India is located at Nagpur.

**41. According to the Indian standard, which of the following Bitumen is suitable for 7-day average maximum air temperature of 46°C :**

(a) VG10 (b) VG20
(c) VG30 (d) VG40

**Ans. (d) :**

| **Grade** | **Suitable for 7 day average maximum air temperature (°C)** |
|---|---|
| VG10 | < 30 |
| VG20 | 30–38 |
| VG30 | 38–45 |
| VG-40 | > 45 |

**42. Which distress is caused by the consolidation of one or more layer of flexible pavement ?**

(a) Ravelling
(b) Alligator formation
(c) Rut
(d) Shrinkage cracking

**Ans. (c):** Rutting is longitudinal deformation or depression of the pavement surface along the wheel paths of heavy vehicle. Longitudinal ruts are formed due to the repeated application of heavy loads along the same wheel path resulting in cumulative non-recoverable or permanent deformation of the pavement layer including.

(i) Subgrade
(ii) One or more pavement layers.
**The various cause of rutting may be seemed up as –**
(i) Inadequate stability of subgrads or sub-base or base course or surface course or a few of these pavement layer.
(ii) inadequate compaction of subgrades or any of the pavement layer.
(iii) Channelized movement of heavy wheel loads causing significant vertical strain an the subgrades.
(iv) improper specification or design of bituminous mix.
(v) inadequate thickness of the flexible pavement or weak pavement structure.

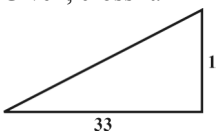
**43. Which of the following is NOT a common type of culvert ?**
(a) Slab (b) Pipe
(c) Hyperbolic (d) Arch

**Ans. (c) : Types of culverts–**
- Box culvert
- Pipe culvert
- Slab culvert
- Arch culvert

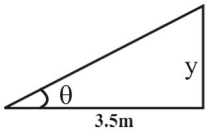
Hyperbolic type is not a culvert.

**44. If a 7 m wide Bituminous road is to be constructed with a cross fall of 1 in 33, then what will be the rise of crown with regard to the edges ?**
(a) 0.11 (b) 0.21
(c) 0.03 (d) 4.71

**Ans. (a) :** 0.11
Given, cross fall = 1 in 33




Crown is at centre
so it is 7/2 = 3.5 m from the edge




$$\tan\theta = \frac{y}{3.5}$$

$$y = \frac{3.5}{33} = 0.106 \approx 0.11$$

**45. If overtaking sight distance is 250 m, then what is the desirable length (km) of the overtaking zone ?**
(a) 0.75 (b) 1
(c) 1.25 (d) 1.5

**Ans. (c) :** Overtaking sight distance (OSD) = 250m
Length of the Overtaking zone = 5 × OSD
= 5 × 250m
= 1250 m
= 1.250 km

**46. If R is the radius of the circular curve (in m), then the maximum grade compensation recommended is:**
(a) 50/R (b) 75/R
(c) 100/R (d) 125/R

**Ans. (b) :** The Maximum grade compensation recommended is 75/R

$$\text{Grade compensation} = \min\left\{\frac{30+R}{R}, \frac{75}{R}\right\}$$

**47. What is the tentative capacity (Capacity PCU per day, both directions) of a single lane rural road with 3.75 m wide carriageway and normal earthen shoulders ?**
(a) 500 (b) 1000
(c) 2500 (d) 5000

**Ans : (b)** Capacity of different types of roads in rural areas.

| Types of road | Capacity PCU per day (both direction) |
|---|---|
| Single lane with 3.75 m wide carriageway and normal earthen shoulders | 1000 |
| Single lane road with 3.75 m wide carriageway and 1.0m wide hard shoulders. | 2500 |
| Road with intermediates lanes of width 5.5 m and normal earthen shoulder | 5000 |
| Two lane roads with 7.0 m wide carriageway and earthen shoulders | 10000 |
| Four lanes divided highway (depending on traffic, access control etc). | 20000 to 30000 |

**48. If the pressure on the plunger at 2.5 mm penetration of soil is 14 units, while the corresponding pressure at 2.5 mm for standard crushed stone is 70 units, then the CBR value (%) at 2.5mm will be:**
(a) 50 (b) 20
(c) 70 (d) 80

**Ans. (b) :**

$$\text{CBR, value} = \frac{\text{load on soil sample}}{\text{standard}} \times 100$$

or

$$\text{CBR, value} = \frac{\text{Pressuse on soil sample}}{\text{standard pressure}} \times 100$$

$$= \frac{14}{70} \times 100 = 20$$

**49. For a broad gauge (BG) in India, what is the maximum height of rolling stock ?**

(a) 2.745 m (b) 3.25 m
(c) 3.45 m (d) 4.14 m

**Ans. (d) :** According to Indian railway, the maximum height and maximum width of rolling stock for BG is 4140 mm and 3250 mm respectively and for meter gauge, the value are 3455 mm and 2740 mm.

**50. In case of mountain alignment of railway track, in order to remain within the ruling gradient the length of the railway line is increased artificially by the development process. Of the following, which is NOT one such development technique?**

(a) Zigzag line method
(b) Switch back method
(c) Complete loop method
(d) Winkler's method

**Ans. (d) :** The levels in mountain alignment vary considerably, so in order to remain within the ruling gradient, the length of railway line is increased artificially by the following development process–
(i) Zig-zag line method
(ii) Switch back method
(iii) Spiral or complete loop method.
• Winkler's method is not development technique

**51. In case of cuttings in good soils, road bed should be given a cross slope of at least 1 in ____ from the centre towards the drains on either side.**

(a) 5 (b) 10
(c) 15 (d) 30

**Ans. (d) :** In case of cuttings in good soils, road bed should be given a cross slope of at least 1 in 30 from the centre towards the drains on either side.

**52. If a vertical summit curve is formed at the intersection of two gradients +2.5% and –3%, what is the deviation angle (N)?**

(a) 0.5 (b) 1.5
(c) 4.5 (d) 5.5

**Ans. (d) :** $n_1 = +2.5\%$
$n_2 = -3\%$
N = deviation angle $= |n_1 - n_2|$
$= |(2.5\%) - (-3\%)| = 5.5\%$

**53. An automotive company designs a truck with a wheelbase of 8 m. What is the off tracking (m) while negotiating a curved path with a mean radius of 32 m ?**

(a) 0.125 (b) 0.25
(c) 0.625 (d) 1

**Ans. (d) :** Given,
Length of wheel base ($l$) = 8 m
Radius of curved path (R) = 32m
New
$$\text{Off tracking} = \frac{l^2}{2R} = \frac{8^2}{2\times 32} = 1 \text{ m}$$

**54. A traffic flow stream operates at a speed of 40 km/h in a traffic lane. If the average center to center spacing of the vehicles in 15 m, when the vehicles follow one behind the other as a queue, then the theoretical capacity of the single lane (number of vehicles per hour) is :**

(a) 178 (b) 25
(c) 2667 (d) 2780

**Ans. (c) :** Given,
Speed = V = 40 km/hr
Average center to center spacing i.e. S = 15m
$$\text{density, } k = \frac{1000}{S} = \frac{1000}{15} = \frac{200}{3}$$
Now, Traffic flow = q = K.V
$$= \frac{200}{3} \times 40 = 2667$$

**55. If the average spacing between vehicles under stopped condition is 5 m and the free mean speed on a roadway is 75 km/h, then what are the values of jam density (vehicles/km/lane) and maximum flow (vehicles/h/lane) respectively ?**

(a) 200, 3750 (b) 200, 3000
(c) 250, 3000 (d) 250, 3750

**Ans. (a) :** Average spacing, S = 5m
Free mean speed, $V_f$ = 75km/hr
$$\text{Jam Density} = k_J = \frac{1000}{5} = 200 \text{ veh/km/lane}$$
$$\text{Maximum flow} = q_{max} = \frac{k_J V_f}{4}$$
$$= \frac{75 \times 200}{4} = 3750 \text{Vehicles/h/lane}$$

**56. If the thickness of sub-base for a specific group index is 15 cm and the combined thickness of surface, base and sub-base course is 450 mm, what is the thickness (mm) of the base and surfacing ?**

(a) 30 (b) 300
(c) 400 (d) 40

**Ans. (b) :** Total thickness– i.e., surface + base + sub base = 450 mm
Given thickness of sub base = 15 cm = 150 mm
So, thickness of base + surface = 450-150 = 300 mm.

**57. A substance's thermodynamic propensity to escape form one environmental compartment to another is termed as :**

(a) Escape velocity (b) Fugacity
(c) Dispersion coefficient (d) Dissolving ratio

**Ans. (b) :** A substance's thermo dynamic propensity to escape from one environmental compartment to another is termed as fugacity.

- Fugacity has units of pressure (Pascal) and can be regarded physically as the partial pressure exerted by one environment phase or compartment on another.

**58. In the language of air pollution, which of the following is NOT a technique for controlling odour ?**

(a) Dilution
(b) Removal
(c) Conversion
(d) Amplification

**Ans. (d) :** Amplification is not a technique for controlling odour.

**59. The ratio of the volume of water that can be drained off by gravity to the total volume of soil after attaining the point of saturation is called :**

(a) Specific yield
(b) Specific capacity
(c) Yield
(d) Drawdown

**Ans. (a) :** The ratio of the volume of water that can be drained off by gravity to the total volume of soil is known as specific yield.
Specific capacity of well is its yield per unit drawdown.

**60. According to Indian standard, the acceptable pH value of drinking water is :**

(a) < 6 (b) 6.5 to 7
(c) 6.5 to 8.5 (d) > 8

**Ans. (c) :** According to Indian standard, the acceptable pH value of drinking water is 6.5 to 8.5

**61. During water purification, which of the following processes reduces the net electrical repulsive forces at particle surfaces by adding specific chemicals ?**

(a) Coagulation
(b) Agglomeration
(c) Flocculation
(d) Oxidation

**Ans. (a) :** During water purification, coagulation process reduces the net electrical repulsive forces at particle surfaces by adding specific chemicals.

**62. The biological oxygen demand is based on the assumption that the amount of organic material remaining at any time (t) is governed by :**

(a) first order function
(b) second order function
(c) third order function
(d) fourth order function

**Ans. (a) :** The biological oxygen demand, is based on the assumption that the amount of organic material remaining at any time (t) is governed by first order function.

**63. The number density of the gas divided by number density of all gases in dry air is termed as :**

(a) Domination ratio
(b) Plank's constant
(c) Metcalf index
(d) Mixing ratio

**Ans. (d) :** Mixing ratio $\Rightarrow$
The number density of the gas divided by number density of all gases in dry air is termed as mixing ratio.

**64. Which of the following is NOT true ?**

(a) Ground water plume flows in turbulent way
(b) Air mass plume flows in turbulent way
(c) Ground water plume is incompressible
(d) Air mass plume is compressible

**Ans. (a) :** Ground water plume flows in tubular way is not true.

**65. In a small town with a population of 10,000, if the per capita water demand is 100 1/d, then what is the yearly water consumption (in million litres) ?**

(a) 91.3 (b) 182.5
(c) 243.3 (d) 365.0

**Ans. (d) :**
Population of town = 10000
per capita water demand = 100 *l*/d
Yearly water consumption
$= 1000000 \times 365$
$= 365 \times 10^6 l = 365$ million liters

**66. Which of the following intake structures is INCORRECTLY paired?**

(a) Rectangular sump well – river intake
(b) Submersible rectangular chamber – lake intake
(c) Intake well on dam – reservoir intake
(d) Intake well at the center of canal – canal intake

**Ans. (d) :** Intake well at canal bank– canal intake, rectangular sump well-River intake, Submersible rectangular chamber–Lake intake, Intake well on dam–reservoir intake.

**67. Which of the following is NOT an advantage of chloramine-ammonia treatment of water ?**

(a) It is less effective than chlorine alone
(b) It prevents bad taste and odour
(c) There is no danger of overdose
(d) Quantity of chlorine required is reduced especially if organic matter is present in large amounts

**Ans. (a):** Advantages of chloramines– Ammonia treatment of water :
1. It prevents bad taste and odour
2. These is no danger of overdose
3. Quantity of chlorine required is reduced especially is organic matter is present in large amounts.

**68. Which of the following treatments is especially applicable to the removal of tastes resulting from dissolved gases ?**
(a) Activated carbon
(b) Lime
(c) Chloramine
(d) Fluorine

**Ans. (a) :** Activated carbon treatment is especially applicable to the removal of tastes resulting from dissolved gases.

**69. Which of the following reactions leads to the formation of volatile fatty acids ?**
(a) Iron reduction
(b) Aerobic oxidation
(c) Acid fermentation
(d) Anammox

**Ans. (c) :** Acid fermentation section leads to the formation of volatile fatty acids.

**70. For which of the following minimum air quality index should people with lung disease (asthma) avoid long exertion ?**
(a) 0 to 50 (b) 51 to 100
(c) 101 to 150 (d) 151 to 200

**Ans. (c) :** 101 to 150 minimum air quality index should people with lung disease (asthma) avoid long exertion.

**71. In a wastewater treatment process, the biomass produced is 12,000 units, while the substrate consumed in 54,000 units. Estimate the biomass synthesis yield.**
(a) 0.22 (b) 0.35
(c) 0.45 (d) 0.78

**Ans. (a) :** Given,
biomass produced = 12000 unit
substrate consumed = 54000 unit
biomass synthesis yield = 12000/54000
= 0.22

**72. The wastewater treatment engineer of a small town wants to dewater raw sludge. Which amongst the following options available to him should he practically NOT use ?**
(a) Filtering
(b) Drying using flat bed
(c) Drying using a lagoon
(d) Centrifugal action

**Ans. (c) :** Drying using a lagoon not used.

**73. If the peak flow rate is 183 cum/d and long-term average flow rate is 100 cum/d, then what is the sustained peaking factor ?**
(a) 0.45 (b) 0.83
(c) 1.83 (d) 2.83

**Ans. (c) :** The peak flow rate = 183 cum/d
long term average flow = 100 cum/day
$$\text{Sustained peaking factor} = \frac{183}{100} = 1.83$$

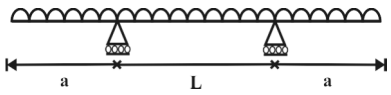
**74. If the detention time of 20 min is used for pre-aeration basin with grit removal for an average flow of 1000 lpm, then the net volume of basin (cum) required will be :**
(a) 6.67 (b) 10.00
(c) 20.00 (d) 24.42

**Ans. (c) :**
$$\text{Discharge (Q)} = 1000 \text{ lpm}$$
$$= 1000 \times 10^{-3} \text{ m}^3/\text{min}$$
$$= 1 \text{ m}^3/\text{min}$$
$$t = 20 \text{ min}$$
$$\text{Discharge} = \frac{\text{volume}}{\text{time}}$$
$$\text{Volume} = Q \times t = 1\text{m}^3/\text{min} \times 20\,\text{min}$$
$$\text{Volume} = 20\text{m}^3$$

**75. If specific capacity of a well is 3.5 cum/h/unit head and the cross-sectional area of the bottom of the well is 7 $m^2$, then the specific yield (cum/h/sq.m) and yield (cum/h) of the well at a constant depression head of 3 m will be respectively :**
(a) 0.03 and 3.33 (b) 0.05 and 10.5
(c) 0.5 and 1.05 (d) 0.5 and 10.5

**Ans. (d) :**
$$\text{Specific capacity} = \frac{\text{discharge (yield)(Q)}}{\text{drawadown}} \quad \frac{3.5}{7} = 0.5$$
Yield of the wall = specific capacity × depression head
= 3.5 × 3 = 10.5 cum/hr

**76. A simply supported beam with equal overhang on both sides carry Uniformly Distributed Load on entire length. What would be the percentage overhang on each side so that maximum bending moment is minimum ?**
(a) 17.7% (b) 20.7%
(c) 25.2% (d) 33.3%

**Ans. (b) :**


Now for maximum, positive bending moment = negative bending moment.

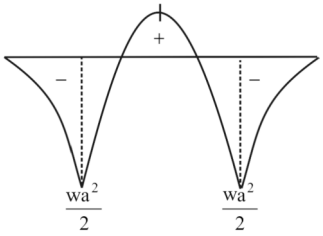
Maximum negative bending moment will be in overhang portion and its magnitude will be $\frac{wa^2}{2}$

Maximum positive bending moment will be on the center of mid span and its magnitude will be $\frac{w}{8}\left(\ell^2 - 4a^2\right)$

Now $\frac{w}{8}\left(\ell^2 - 4a^2\right) = \frac{wa^2}{2}$

$$\frac{\ell^2}{8} = a^2$$

$$\boxed{a = \frac{\ell}{2\sqrt{2}}}$$




$$\% \text{ overhang} = \frac{a}{L + 2a} \times 100$$

$$= \frac{\frac{L}{2\sqrt{2}}}{L + \frac{L}{\sqrt{2}}} \times 100 = 20.71\%$$

**77. Which of the following theory gives best result for ductile materials ?**

(a) Rankine's Theory
(b) St. Vanant's Theory
(c) Mises – Henky Theory
(d) Gusset of Tresca's Theory

**Ans. (c) : Maximum shear strain energy theory or maximum Distortion energy theory ( von mises-henky theory)**–This assumes that failure occur when shear strain energy (distortion energy) in the complex system is equal to that at the yield point in tensile compression test. This is also most economical theory.
– Mises-Henky theory does not apply to brittle material for which elastic limit stress in tension and compression is quite different. Failure of most ductile material is ground by the distortion energy criteria or von mises theory.
**Maximum principal stress theory (Rankine's theory)–I**t is well suited to brittle materials.
**Maximum shear stress theory (Guest-Tresca)**. This assumes that maximum shear stress in the complex stress system becomes equal to that at the yield point in simple tensile test. This theory holds good for ductile materials.

**Note–** Maximum shear stress theory is most conservative theory.

**78. Number of independent elastic constants for homogeneous and isotropic material is :**

(a) 6 (b) 2
(c) 9 (d) 21

**Ans. (b) :** The number of independent elastic constants for a homogeneous and isotropic material in elastic constant is 2.

| **Material** | **No. of elastic constant (Independent)** |
|---|---|
| Isotropic | 2 (E and μ) |
| Orthotropic | 9 |
| Anisotropic or Non-Isotropic | 21 |

**79. The Influence Line Diagram for bending moment for determinate structure is :**

(a) Curved
(b) Linear
(c) Combination of curved and linear
(d) Irregular

**Ans : (b)** The variation of influence line for stress function in a statically determinate structure is linear.

ILD for shear stress

**80. A circular bar of diameter 50 mm is subjected to pull of 50 KN. Take poission's ratio = 0.33 and Young's Modulus of Elasticity = $7.07\times10^4$ N/mm$^2$. Value of shear modulus is :**

(a) $2.65\times10^4$ N/mm$^2$ (b) $3.6\times10^4$ N/mm$^2$
(c) $6.93\times10^4$ N/mm$^2$ (d) $9.36\times10^4$ N/mm$^2$

**Ans. (a) :** Given

$$E = 7.07\times10^4 \text{ N/mm}^2$$

$$\mu = 0.33$$

$$E = 2G(1+\mu)$$

$$G = \frac{E}{2(1+\mu)} = \frac{7.07\times10^4}{2(1+0.33)} = 2.65\times10^4 \text{ N/mm}^2$$

**81. If an area has any axis of symmetry then product of inertia will be :**

(a) Zero
(b) Positive
(c) Negative
(d) Either positive or negative

**Ans. (a) :** If an area has any axis of symmetry then product of inertia will be zero.

**82. Slope deflection method is used when :**

(a) $D_k = D_s$ (b) $D_k = 0, D_s = 0$

(c) $D_k < D_s$ (d) $D_s < D_k$

**Ans. (c) :** Slope deflection method is displacement method. It is generally used

When $D_K < D_S$

Example of displacement method are

– moment distribution method

– stiffness matrix method

– Kani's method

**83. For triangular section maximum shear stress occurs at :**

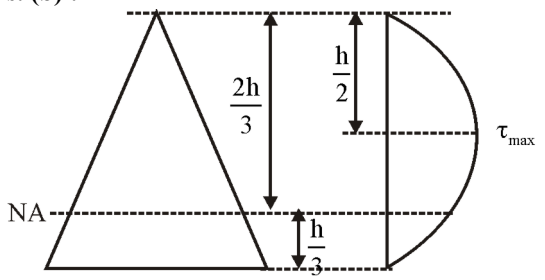
(a) Neutral axis

(b) At h/2 from apex

(c) At h/3 from apex

(d) At h/3 from bottom

**Ans. (b) :**




For triangular section,

$$\tau_{max} = \frac{3}{2}\tau_{avg}$$

For triangular section $\tau_{max}$ is also equal to $\frac{3}{2}\tau_{avg}$ but $\tau_{max}$ occurs at $\frac{h}{2}$, not at neutral axis like rectangular section.

**84. Radial Shear ($S_x$) for three hinge arch is given by :**

(a) $R_A.x - H_y$

(b) $R_A \cos\theta - H \sin\theta$

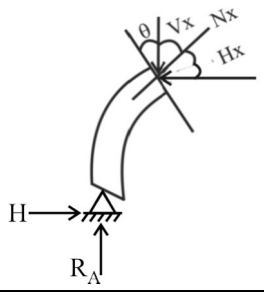
(c) $R_A \sin\theta + H \cos\theta$

(d) $R_A \sin\theta - H \cos\theta$

**Ans. (b)** Radial shear $(S_x) = R_A \cos\theta - H\sin\theta$

and Normal shear $(N_x) = R_A \sin\theta + H\cos\theta$

$R_A$ and H are important for the design of the cross-section of the arch.




**85. Degree of freedom available for space – rigid frame is:**

(a) 2 (b) 3

(c) 4 (d) 6

**Ans. (d): Degree of freedom (DOF) –** The Number of independent parameters (Measurements) that are needed to uniquely finds its position in space and at instant of time.

– Each joint of space rigid jointed frame has 6 degree of freedom.

– Each joint of plane rigid jointed frame has 3 degree of freedom.

– Each joint of plane pin-jointed frame has 2 degree of freedom.

– Each joint of space pin-jointed frame has 3 degree of freedom.

**86. In a plane strain problem in xy plane, the shear strain = $15\times10^{-6}$ and the normal strain in x and y direction is zero. For this strain, what is the diameter of the Mohr's circle of strain ?**

(a) $5\times10^{-6}$ (b) $7.5\times10^{-6}$

(c) $15\times10^{-6}$ (d) $30\times10^{-6}$

**Ans. (c) :**

Radius of Mohr's circle $= \frac{\phi_{max}}{2}$

$$= \frac{15\times10^{-6}}{2}$$

$$= 7.5\times10^{-6}$$

So, diameter of Mohr's circle strain $= 15\times10^{-6}$

**87. At 20°C, Water has a vapour pressure of :**

(a) 10.3 KPa (b) 1 KPa

(c) 2.34 KPa (d) 760 KPa

**Ans. (c) :**

At $20^0$C water has a vapour pressure by 17.5 torr and also 1 torr = 133.33 Pa

So, vapour pressure = 17.5×133.33=2.33KPa

**88. In a geometrically similar model of a spill way the discharge per meter length is 0.4 $m^3$/s. If the scale of model is 1 : 42. Find the discharge per meter run of prototype.**

(a) 10.87 $m^3$/sec (b) 108.087 $cm^3$/sec

(c) 108.87 $m^3$/sec (d) 1.087 $m^3$/sec

**Ans. (c) :**

$$\frac{Q_{model}}{Q_{prototype}} = (L_r)^{5/2}$$

$$\frac{Q_{model}}{Q_{prototype}} = \left(\frac{1}{42}\right)^{5/2} \qquad \frac{0.4 m^3/sec}{Q_P} \qquad \left(\frac{1}{42}\right)^{5/2}$$

$$Q_{prototype} = 4572.81 m^3/sec$$

For per meter length $= \frac{4572.81}{42} = 108.87 m^3/sec$

**89. What will be the value of convective acceleration and local acceleration for a flow water in uniform straight pipe of constant diameter with unsteady flow condition?**

(a) Convective acceleration = 0,
Local acceleration = 0

(b) Convective acceleration = 0,
Local acceleration $\neq 0$

(c) Convective acceleration $\neq 0$,
Local acceleration $\neq 0$

(d) Convective acceleration $\neq 0$,
Local acceleration = 0

**Ans. (b) :** For unsteady and uniform flow convective acceleration = 0,

Local acceleration $\neq 0$

$$a_x = \underbrace{\frac{\partial u}{\partial t}}_{\text{Local or temporal acceleration}} + \underbrace{u.\frac{\partial u}{\partial x} + v.\frac{\partial u}{\partial y} + w.\frac{\partial u}{\partial z}}_{\text{convective or advective acceleration}}$$

similarly

$$a_y = \frac{\partial v}{\partial t} + u.\frac{\partial v}{\partial x} + v.\frac{\partial v}{\partial y} + w.\frac{\partial v}{\partial z}$$

$$a_z = \frac{\partial w}{\partial t} + u.\frac{\partial w}{\partial x} + v.\frac{\partial w}{\partial y} + w.\frac{\partial w}{\partial z}$$

| Type of flow | Convective acceleration | Temporal (local) Acceleration |
|---|---|---|
| Steady and uniform | 0 | 0 |
| Steady of non-uniform | Exists | 0 |
| Unsteady and uniform | 0 | Exists |
| Unsteady and non-uniform | Exists | Exists |

**90. A pump is used to transfer water from sump to an over head reservoir has following H-Q. characteristic $H = 23 + 11Q + 110Q^2$. The system curve is given by $H = 25 + 90Q^2$. What is the shut-off head of the pump?**

(a) 25 (b) 23
(c) 0.229 (d) 19.74

**Ans. (b) :** Shut-off head of the pump is where the discharge become zero in centrifugal pump.
The actual shut off head of a centrifugal pump is much less than the theoretical part, So

$H_{actual} = 23 + 11Q + 110Q^2$

$H_{th} = 25 + 90Q^2$ (For shut off head Q = 0 )

So the $H_{actual} = 23 + 11 \times 0 + 110 \times 0^2 = 23$ unit

$H_{th} = 25 + 90 \times 0^2 = 25$ unit

Since the $H_{actual} < H_{th}$
Hence 23 is the right answer.

**91. The water is flowing in a pipe of cross sectional area 19.625 $m^2$ and perimeter 15.7 m. The hydraulic mean diameter is :**

(a) 4 m (b) 5 m
(c) 6 m (d) 7 m

**Ans. (b) :**

$$\text{Hydraulic mean diameter} = \frac{4\text{Area}}{\text{Wetted perimeter}}$$

$$= \frac{4 \times 19.625 m^2}{15.7 m} = 5m$$

**92. What will be the compressibility of water if an increase of pressure by 20 $MN/M^2$ caused 1% reduction in the volume ?**

(a) 2.5 $cm^2/MN$ (b) 5 $cm^2/MN$
(c) 7.5 $cm^2/MN$ (d) 10 $cm^2/MN$

**Ans. (b) :** 5 $cm^2/MN$

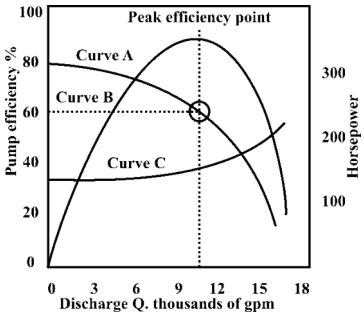
$$\text{(k) Bulk modulus} = \frac{\Delta p}{\text{volumetric strain}} = \frac{\Delta p}{\frac{\Delta V}{V}}$$

$$(k) = \frac{20 MN/m^2}{\frac{0.01V}{V}} = 2000 MN/m^2$$

$$\text{Compressibility} = \frac{1}{k} = \frac{10^4}{200} cm^2/MN$$

$$= 5 cm^2/MN$$

**93. Given figure shows the performance curve for a centrifugal pump at various flow. Curve A is for :**




(a) Head
(b) Efficiency
(c) Power input
(d) Power output

**Ans. (a) :** From given figure–
(i) Curve A denotes "Head"
(ii) Curve B denotes "Efficiency"
(iii) Curve C denotes "Power"

**94. A rectangular deep channel has 5.0m width and 3.0m water depth. The velocity at different depth is given below.**

| Depth (m) | 0 | 0.6 | 1.2 | 1.6 | 2.4 |
|---|---|---|---|---|---|
| Velocity (m/s) | 0 | 4 | 4.2 | 3.5 | 3 |

(a) 4 m/s (b) 4.2 m/s
(c) 3.5 m/s (d) 3 m/s

**Ans. (c) :**
we know,
In deep channel

$$\overline{V} = \frac{V_{0.2y} + V_{.8y}}{2}$$

where y = depth of water

$Y_{.2y} = V_{.2\times3} = V_{0.6} = 4\text{m/s}$

$Y_{0.8y} = V_{.8\times3} = V_{2.4} = 3\text{m/s}$

$$\overline{V} = \frac{4+3}{2} = 3.5\text{m/s}$$

**95. An Effluent stream is defined as:**
(a) A stream which contributes some flow to ground water discharge
(b) A stream which always carry some flow
(c) A stream which receives some flow from the ground water discharge
(d) A stream which does not have any base flow contribution

**Ans. (c) :** An Effluent stream is defined as a stream which receive some flow from the ground water discharge.
**Perennial stream :** one which always carries some flow and ground water contributes throughout the year.
**Intermittent stream :** one which has limited contribution from the ground water during dry season the water table falls below the stream bed and dries up.
**Ephemeral stream :** one which does not have any base flow contribution. The stream dries up soon after the end of steam flow.

**96. The peak of a flood hydrograph due to one hour storm in a catchment is 170m$^3$/sec. The total depth of the runoff generated due to this storm is 50mm. Assume a constant of base flow of 20m$^3$/sec. The peak discharge of 1 hour unit hydrograph for this catchment is:**
(a) 34 m$^3$/sec (b) 3 m$^3$/sec
(c) 30 m$^3$/sec (d) 3.4 m$^3$/sec

**Ans. (c) :** 30 m$^3$/sec
Peak discharge of DRH = 170 – 20
= 150 m$^3$/sec
Direct run off = 50mm = 5cm
The peak value of 1 hr unit hydrograph

$$= \frac{\text{Peak Discharge}\left(\frac{\text{m}^3}{\text{sec}}\right)}{\text{Runoff depth}\,(\text{cm})}$$

$$= \frac{150}{5} = 30\text{m}^3/\text{sec}$$

**97. When the left support of a simply supported beam rotates, the fixed end moment at left support is:**
(a) 3 EI $\theta_L$/L (b) 4 EI $\theta_L$/L
(c) 2 EI $\theta_L$/L (d) 3 EI $\theta_L$/L

**Ans. (a) :** Allowing left end or support rotation by an amount $\theta_A$, the fixed end moment will be (at right support)–

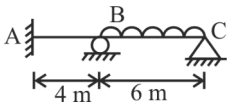
$$\text{FEM} = \frac{3EI\theta_L}{L}$$

**98. A statically indeterminate structure :**
(a) can be analysed using equations of compatibility only
(b) can be analysed using equations of statics and compatibility equation
(c) can be analysed using equations of statics only
(d) cannot be analysed at all

**Ans. (b) :** A statically indeterminate structure can be analyzed using equations of static equilibrium and compatibility equation.

**99. A two span continuous beam ABC with end A fixed and end C hinged is having AB = 4 m and BC = 6 m, $I_{AB} : I_{BC} = 1 : 2$. It is subjected to udl (uniformly distributed load) of 10 kN/m over the entire right span. The moment at the end C is :**
(a) 25 kNm hogging
(b) 11.25 kNm hogging
(c) 11.25 kNm sagging
(d) 0

**Ans. (d) :**

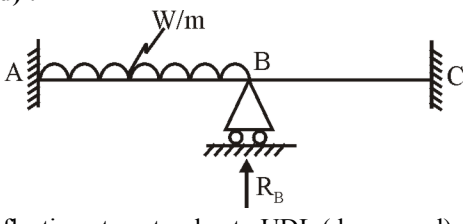



Given point C is hinged, as we know that B.M about a hinge is zero

$BM_C = 0$

**100. A fixed beam having length L is propped at midway. It carries udl (uniformly distributed load) w/m extending from one fixing to centre prop. Calculate the prop reaction.**
(a) WL/2 (b) WL/8
(c) 3WL/4 (d) WL/4

**Ans. (d) :**




Net deflection at centre due to UDL (downward)

$$\Delta_1 = \frac{1}{2}\left[\frac{1}{384} \times \frac{WL^4}{EI}\right]$$

Net deflection to propped reaction–

$$\Delta_2 = \frac{R_B L^3}{192EI}$$

From capability equation –

$$\Delta_1 = \Delta_2$$

$$\frac{1}{2}\left[\frac{1}{384}\times\frac{WL^4}{EI}\right] = \frac{R_B L^3}{192EI}$$

$$R_B = \frac{WL}{4}$$

**101. The strain energy of a pin-jointed structure can be expressed as :**

(a) $\Sigma F^2L/2AE$ (b) $\Sigma F^2L/AE$

(c) $\Sigma L^2F/2AE$ (d) $\Sigma FL^2/AE$

**Ans. (a) :** The strain energy of a pin-jointed structure can be expressed as $\sum F^2L/2AE$

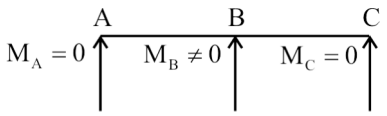
• Pin jointed frame will take only axial load. Hence strain energy due to axial load will be counted only.

**102. The Clapeyron's method formulates :**

(a) loading conditions
(b) equilibrium conditions
(c) moment conditions
(d) compatibility conditions

**Ans. (d) :** Clapeyron's theorem is also called three moment theorem.

• When there are three supports in continuous beam there the Clapeyron's theorem is used to get the moments.




• Here the equation can't solved by only equilibrium condition but the compatibility conditions are also applied to get moment and the reaction.

**103. The proof load of a leaf spring is :**

(a) deflection per unit load
(b) load required to flatten the spring
(c) load required to cause 2% permanent deflection
(d) load per unit deflection

**Ans. (b) :** The proof load of a leaf spring is load required to flatten the spring.

The load at which the plates become straight is called proof load.

**104. A water main of 180 cm diameter contains water at a pressure head of 300 m. The thickness of the metal shell required for water main, given that max permissible stressing metal is 600 kg/cm$^2$, will be :**

(a) 4.5 cm (b) 3.5 cm

(c) 2.5 cm (d) 1.5 cm

**Ans. (a) :**

Given,

$$\sigma_h = 600\ kg/cm^2$$

$$d = 180cm$$

$$P = 300\times10^2\ g/cm^2$$

$$P = 30kg/cm^2$$

$$\sigma_h = \frac{Pd}{2t}$$

$$t = \frac{30\times180}{600\times2}$$

$$t = 4.5cm$$

**105. The perry-Rebertson's formula considers :**

(a) crushing mode
(b) initial curvature
(c) initial eccentricity and curvature
(d) buckling mode

**Ans. (c) : Perry-Rebertson's assumption–** Any imperfection in the strut or column through faulty workmanship or material or eccentricity of loading can be allowed for giving the strut an initial curvature. For ease of calculation, this is assumed to be a cosine curve.

**106. An allowable axial load 3 m long column with hinged ends is 30 kN. Another column of same material, same cross-section and same length – but one end fixed and other end hinged – suffers buckling. What is the buckling load for the column ?**

(a) 60 kN (b) 30 kN

(c) 300 kN (d) 600 kN

**Ans. (a) :**

case-1

When both ends hinged

$$P_E = \frac{\pi^2 EI}{L^2} = 30kN$$

case-2

When one end fixed and other end hinged

$$P_E = 2\ \frac{\pi^2 EI}{L^2} = 2\times30kN$$

$$P_E = 60kN$$

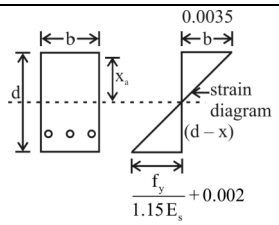
**107. The maximum strain in tension reinforcement at the time of failure should not be less than :**

(a) $0.002 + (f_y/E_s)$ (b) $f_y/1.15E_s$

(c) $0.002 + (f_y/1.15E_s)$ (d) 0.0035

**Ans. (c):** The maximum strain in the tension reinforcement in the section at failure shall not be less than–

$$\varepsilon_{su} = \left(\frac{f_y}{1.15E_s}\right) + 0.002$$




or $\varepsilon_{su}=\left(\frac{0.87f_y}{E_s}\right)+0.002$

$\varepsilon_{su}$ = Strain in steel at time of failure or limiting strain in steel.

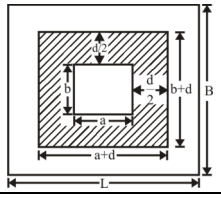
$f_y$ = Characteristic strength of steel

$E_s$ = Modulus of elasticity (steel) = $2 \times 10^5$ N/mm$^2$

**108. In case of a footing, the critical section for punching shear is considered at :**

(a) the face of the column
(b) 'd' distance from the face of the column
(c) 'd/2' distance from the face of the column
(d) '2d' distance from the face of the column

**Ans. (c) :** Punching shear or two way action of the footing, with potential diagonal cracking along the surface of truncated cone or pyramid around the concentrated load. The distance of critical section from the periphery of the column shall be taken at d/2.




**109. During earthquakes, the corner and edge columns may be subject to :**

(a) uniaxial bending
(b) biaxial bending
(c) combined biaxial bending and tension
(d) combined biaxial bending and torsion

**Ans. (d) :** During earthquakes, the corner and edge columns may be subject to combined biaxial bending and torsion.

**110. The minimum grade of concrete for pre-tensioned pre-stressing concrete member is :**

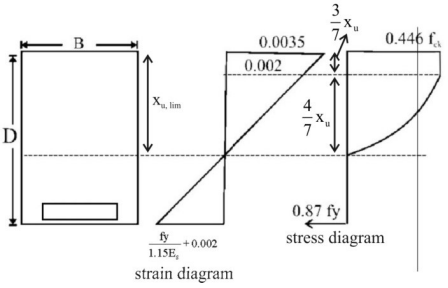
(a) M30
(b) M40
(c) M60
(d) M70

**Ans : (b)** As per IS 1343-1980, The minimum grade of concrete for pre-tensioned pre-stressing concrete member is M40

**111. The limiting compressive strain of concrete in case of bending is :**

(a) 0.002
(b) 0.0025
(c) 0.003
(d) 0.0035

**Ans : (d)**




So, By IS: 456-2000

Maximum strain in concrete = 0.0035 and that of in steel is $0.002+\frac{f_y}{1.15E_s}$

**112. An axially loaded column is of 450×450 mm size. The effective length of the column is 5 m. What is the minimum eccentricity of the axial load for the column ?**

(a) 25 mm
(b) 16 mm
(c) 20 mm
(d) 30 mm

**Ans. (a) :** $e_{min}$ (Minimum eccentricity)

$$= \max.\ \frac{l}{500}+\frac{'B'\text{or}'D'}{30}$$
$$20\text{mm}$$

$$= \max.\ \frac{5000}{500}+\frac{450}{30}=25\text{mm}$$
$$20\text{mm}$$

$e_{min}$ = 25mm

**113. The deflection of beams and one-way slabs can be controlled by using the appropriate :**

(a) modular ratio
(b) water/cement ratio
(c) aspect ratio
(d) span/depth ratio

**Ans. (d) :** The deflection of beams and one-way slabs can be controlled by using the appropriate span/depth ratio.

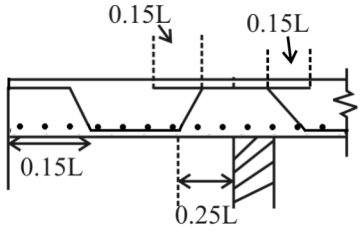
- Effective span to overall depth ratio to control deflection criteria is as follows–

| | |
|---|---|
| Cantilever beam | 7 |
| simply supported beam | 20 |
| continuous beam | 26 |
| one way slab | 30 |
| two way slab | 35 |
| continuous slab | 40 |

**114. In case of a two-way slab, for positive moment reinforcement, 100% reinforcement must be provided up to a distance of _____ from the simply-supported edge.**

(a) 0.10 L (b) 0.15 L
(c) 0.20 L (d) 0.25 L

**Ans. (b):** In case of a two-way slab for positive moment reinforcement, 100% reinforcement must be provided upto a distance of 0.15L from the simply-supported edge.




**115. The end or side covers for a steel bar in RCC work may be taken as : _____**

(a) 1.2 cm to 2 cm
(b) 2.5 cm to 5 cm
(c) 4 cm to 5 cm
(d) 3 cm to 7 cm

**Ans. (c) :** In RCC work the end or side covers for steel bar may be 4 cm to 5 cm $\left(1\frac{1}{2}" \text{ to } 2"\right)$ and the bottom and top covers may be 1.2 cm to 2 cm for slab and 2.5 cm to 5 cm for beam.

**116. A compression member has a centre-to-centre length of 5 m. It is fixed at one end & hinged at other end. The recommended effective length of the column is :**

(a) 4.0 m (b) 2.8 m
(c) 3.2 m (d) 2.6 m

**Ans. (a) :** A compression member has a centre-to-centre length of 5m. It fixed at one end & hinged at other end. The recommended effective length of the column is 0.8L

Then, effective length = 0.8L
= 0.8×5
= 4m

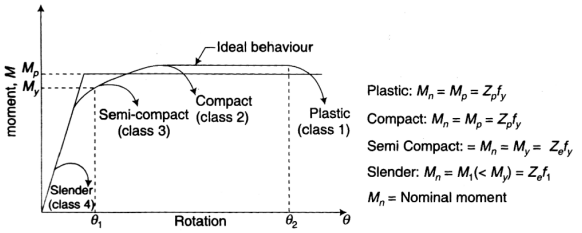
**117. A section has the capacity to develop plastic hinge, but does not have the capacity to form collapse mechanism. Such a section is called :**

(a) plastic section
(b) compact section
(c) Semi-compact section
(d) slender section

**Ans. (b) : Compact section–** A cross-section, which can develop plastic moment, but has inadequate plastic rotation capacity needed for formation of a plastic collapse mechanism of the member or structure.

**Semi-compact section–** Cross-section, which can attain the yield moment, but not the plastic moment before failure by plate buckling.

**Plastic section–** Cross-section which can develop a plastic hinge and sustain plastic moment over sufficient plastic rotation required for formation of plastic failure mechanism of the members of structure.




*Moment–rotation behaviour of four classes of cross-sections*

**118. For an electrically-operated cranes up to 50 ton capacity, the maximum permissible deflection is given by :**

(a) Span/500 (b) Span/750
(c) Span/1000 (d) Span/1250

**Ans. (b) :** For an electrically-operated cranes upto 50 ton capacity, the maximum permissible deflection is given by Span/750.

**Deflection limit on gantry girder-**

| category | maximum deflection |
|---|---|
| Manually operated crane | Span/500 |
| Electrically operated crane upto 500kN capacity | Span/750 |
| Electrically operated crane more than 500 kN capacity | Span/1000 |

**119. If the slope of the truss is 15º, the live load on truss is taken as :**

(a) 0.5 (b) 0.55
(c) 0.6 (d) 0.65

**Ans : (d)** According to IS: 875-1975
Angle of inclination = $15^0$
Imposed load = 0.75 kN/m$^2$
By using formula,

$$\text{Live load} = \text{imposed load} - .02\left(\text{angle of inclination} - 10^o\right)$$

$= 0.75 - 0.02\ (15-10)$
Live load = $0.75 - 0.02 \times 5 = 0.65$kN/m$^2$

Note– IS 675 specifies the following live loads to be assumed in the analysis of an industrial building.

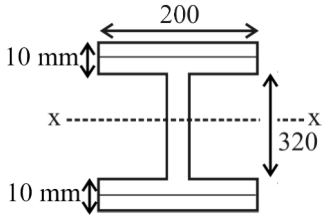
| Roof slope | Access | Live load |
|---|---|---|
| $\le 10^0$ | Provided | 1.5 kN/m$^2$ for plain area |
| | Not provided | 0.75 kN/m$^2$ for plain area |
| $\ge 10^0$ | | For roof membrane sheets or purlins 0.75 kN/m$^2$ less 0.02 kN/m$^2$ for every degree increased in slope over $10^0$ $\not<$ 400 N/m$^2$ |

**120. An ISMB 300 has section modulus value of 600 × $10^3$ $mm^3$, Plates of dimensions 200×10 mm are welded over an each flange of width 320 mm. What is modulus of the plated section ?**

(a) $462\times10^3$ $mm^3$ (b) $550\times10^3$ $mm^3$

(c) $710\times10^3$ $mm^3$ (d) $1220\times10^3$ $mm^3$

**Ans. (d) :**




Section modulus due to plates about

$x\text{-}x = \frac{I}{y}$

$I = 2\times\left(\frac{200\times10^3}{12}+200\times10\times165^2\right)$

$= 108933333.3$

$y = 170$

$z_{plates} = \frac{I}{y} = \frac{108933333.3}{170}$

$= 640.782\times10^3\ mm^3$

section modulus of built-up section = $Z_{plates} + Z_I$

$= (640.78+600)\ 10^3\ mm^3$

$= 1240.78\times10^3 mm^3$

**121. A 12mm thick plate is connected to two 6 mm plates on either side and subjected to tension then maximum value of pitch 'P' is :**

(a) 72mm (b) 200mm

(c) 120mm (d) 96mm

**Ans. (d) :** Given,

Thickness of plate, t = 6mm

And, for tension member

$max^m$ pitch $\not>$ 16t

or {whichever is less

200mm

16t = 16 × 6 = 96 mm

So, $max^m$ pitch = min. {96mm, 200mm

= 96 mm

**122. A circular shaft of diameter 150mm is welded to a rigid plate by an external all-round fillet weld of size 10mm. If a torque of 10kNm is applied to the shaft, then find maximum shear stress developed in the fillet weld.**

(a) $80.8N/mm^2$ (b) $40.0N/mm^2$

(c) $56.5N/mm^2$ (d) $65.4N/mm^2$

**Ans. (b) :** Given,

Round fillet weld of size = 10mm

Torque (T) = 10kNm

D = 150mm $\Rightarrow$ r = 75 mm

Polar moment of inertia, $I_P = Ar^2$

$= (KS)\ (\pi D)\ r^2$

$= (0.707\times10)\times\pi\times150\ (75)^2$

$= 18740581.93\ mm^4$

Now, From $eq^n$ of Torsion

$\frac{T}{I_P} = \frac{\tau}{r}$

$\tau = \frac{Tr}{I_P}$

$= \frac{(10\times10^6)\times75}{18740581.93} = 40\ N/mm^2$

**123. The retaining wall is considered to be safe against sliding if which of the following conditions is satisfied ?**

(a) $(0.9\times F_{Resisting\ force}/F_{Sliding\ force}) \geq 1.40$

(b) $(0.9\times F_{Resisting\ force}/F_{Sliding\ force}) \geq 1.70$

(c) $(0.9\times F_{Resisting\ force}/F_{Sliding\ force}) \leq 1.40$

(d) $(F_{Resisting\ force}/F_{Sliding\ force}) > 2$

**Ans. (a) :** The retaining wall is considered to be safe against sliding if

$(0.9\times F_{Resisting\ force}/F_{Sliding\ force}) \geq 1.40$ is satisfied.

**124. The clear spans of a two-way slab are 4.6m×6.20 m, and it is subjected to the factored load of 19kN/$m^2$ (including dead loads also). The shear force along the longer edge will be–**

(a) 15.5kN (b) 25.7 kN

(c) 29.13 kN (d) 35.06 kN

**Ans. (d) :** Load = 19 kN/$m^2$

The shear force along the longer span

$\frac{\ell_y}{\ell_x} = r \Rightarrow \frac{6.20}{4.6} = 1.347$

For longer edge–

$V_y \Rightarrow \frac{r}{r+2} w\times\ell_x$

$V_y \Rightarrow \frac{1.347}{1.347+2}\times19\times4.6 = 35.17\ kN$

**125. In case of a continuous beam, the shear force coefficients at the support next to the end support and inner side of mid span (Where $w_d$ = Dead load and imposed load(fixed), $W_L$ = live load) :**

(a) $(0.40w_d + 0.4w_L)L_C$

(b) $(0.60w_d + 0.60w_L)L_C$

(c) $(0.50w_d + 0.60w_L)L_C$

(d) $(0.55w_d + 0.60w_L)L_C$

**Ans. (d) :** In case of a continuous beam, the shear force coefficients at the support next to the end support and inner side of mid span is $(0.55w_d + 0.60w_L)L_C$

**126. Which of the following statements is incorrect ?**

(a) The compressive strength of concrete increases with increase in water cement ratio of the mix.

(b) The modulus of elasticity of concrete increases with increase in the compressive strength of concrete.

(c) The shear strength of concrete increases with increase in the compressive strength of concrete.

(d) Creep and the shrinkage of concrete depends upon the water-cement ratio of the concrete mix.

**Ans. (a) : Correct statements are–**

- The compressive strength of concrete decreases with increase in water cement ratio of the mix.
- The modulus of elasticity of concrete increases with increase in the compressive strength of concrete.
- The shear strength of concrete increases with increase in the compressive strength of concrete.
- Creep and the shrinkage of concrete depends upon the water-cement ratio of the concrete mix.

**127. The atmospheric pressure is given as 680 mm of Hg at a mountain location. Convert this to kPa and metre of water forms respectively.**

(a) 90.7 kPa, 92.5 m of $H_2O$

(b) 907 kPa, 9.25 m of $H_2O$

(c) 90.7 kPa, 9.25 m of $H_2O$

(d) 907 kPa, 92.5 m of $H_2O$

**Ans. (c) :** $90.7\text{kPa}, 9.25\text{m of } H_2O$

$P = \rho_{Hg} \times g \times h$

$P = 13.6 \times 1000 \times 9.81 \times 0.680$

$P = 90.72 \text{ kPa}$

$$\text{Pressure in meter of water} = \frac{90.72 \times 10^3}{P_w g}$$

$$= \frac{90.72 \times 10^3}{10^3 \times 9.81}$$

$$\text{Pressure in meter of water} = 9.248\text{m of } H_2O$$

**128. The muschel curves belong to the category of :**

(a) main characteristic curve of the turbine

(b) operating characteristic curves of the turbine

(c) constant efficiency curves of the turbine

(d) operating characteristic of a pump

**Ans. (c)** Muchel curve / constant efficiency curve / iso efficiency curve.

The curve obtained from the speed vs. efficiency and speed vs. discharge curves from different gate openings. For a given efficiency from $N_u$ vs. $Q_u$ curves, corresponding to values of speeds there are two values of discharge.

Characteristic curves of turbine are characteristic of particulars turbines, which help in studying the performance of the turbine under various conditions.

The characteristic curves are obtained, are following–

- Constant head curves or main characteristic curves.
- Constant speed curves or operating characteristic curves.
- Constant efficiency curves or Muschel curves.

**129. A double acting duplex reciprocating pump has:**

(a) 4 pistons

(b) 4 cylinders

(c) 2 pistons

(d) 1 pistons

**Ans. (c) :** Double acting pump - Water is connected with two side of piston

Single acting pump - Water is connected with one side of piston

**130. In a pelton turbine, the degree of reaction is :**

(a) One

(b) Zero

(c) In the range 0 to 0.5

(d) In the range 0.5 to 1

**Ans. (b) :** For Pelton Turbine $\Rightarrow$ DOR = 0

For Francis turbine $\Rightarrow 0 < \text{DOR} < 1$

For propeller and kaplan turbine $\Rightarrow \frac{1}{2} < \text{DOR} < 1$

**131. A new design of a valve is to be tested in liquid benzene flows though the valve, which of the following parameters is most important.**

(a) Froude No

(b) Reynolds No

(c) Mach No

(d) Euler No

**Ans. (b) :** A new design of a valve is to be tested if liquid benzene through the valve, Reynold no. is most important parameter.

**Reynolds Number :** Indicates type of flow in pipe or channel.

**Froude Number :** Sleeping interface exists between fluids of different densities (Gravity force are predominant)

**Mach No.** High speed flow of a gas (effect of compressibility is significant)

**Weber number :** Formation of liquid droplets (surface tension is predominant).

**132. A velocity field in a particular flow is given by $V = 20y^2i - 20\,xy\,j$ m/s. The average velocity at points (1, -1, 2).**

(a) 0, 0, 30 rad/s
(b) 0, 30 rad/s, 0
(c) 30 rad/s, 0, 0
(d) 0, 0, 0

**Ans. (a) :** $\vec{v} = \frac{1}{2}\text{curl}\,v = ui + vj + w\vec{k}$

$$\vec{v} = \frac{1}{2}\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 20y^2 & -20xy & 0 \end{vmatrix}$$

$$\vec{u} = \frac{1}{2}\begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -20xy & 0 \end{vmatrix} = 0, \quad w = \frac{1}{2}\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ 20y^2 & -20xy \end{vmatrix}$$

$$\vec{v} = \frac{1}{2}\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ 20y^2 & 0 \end{vmatrix} = 0$$

$$w = -\frac{1}{2}(20y + 40y)$$

$$w_{(1,-1,2)} = \frac{-1}{2}\left(20 \times (-1) + 40(-1)\right)$$

$$w = 30 \text{ rad/sec}$$

So, $u = 0, v = 0, w = 30$ rad/sec

**133. To model a river, the horizontal scale is 1 : 500 and vertical scale is 1 : 50. Determine model bed slope if prototype bed slope is 0.0003.**
(a) 0.0003 (b) 0.003
(c) 0.03 (d) 0.00003

**Ans. (b) :** 0.003

$$\frac{H_m}{H_P} = \frac{1}{500} \qquad H_m = x$$
$$H_P = 500x$$

also $\frac{V_m}{V_P} = \frac{1}{50} \qquad V_m = Y$
$V_P = 50y$

for Prototype $\frac{V_P}{H_P}$ 0.0003

$$\frac{50y}{500x} = 0.0003$$

$$\frac{y}{x} = 0.003$$

**134. A submarine is moving horizontally in seawater having density of 1026 kg/m$^3$, with its axis 20 m below the surface of the water. A pilot tube is placed in front of submarine along its axis and is connected to differential mercury manometer showing a deflection of 20 cm. Determine the velocity of submarine.**
(a) 0.69 m/s (b) 69.3 m/s
(c) 6.93 m/s (d) 0.069 m/s

**Ans. (c) :** Reading on Manometer(y)= 20 cm = 0.2m of mercury.
Specific Gravity of Mercury= 13.6
Specific Gravity of Sea Water = 1.026

$$\text{Head(h)} = y\left(\frac{S_{hg}}{S_{\text{sea water}}} - 1\right)$$

$$= 0.2\left(\frac{13.6}{1.026} - 1\right) = 2.45\text{m}.$$

Velocity of submarine–

$$V = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 2.45}$$
$$= 6.93 \text{ m/sec}.$$

**135. A plunger is fitted to a vertical pipe filled with water. The lower end of the pipe is submerged in a sump. The plunger is drawn up with an acceleration of 5m/s$^2$. Find maximum height above the water level of the sump at which the plunger will work without cavitations. Assume atmospheric pressure = 10.0 m(abs). Limiting pressure from cavitations consideration = 2m(abs). Take acceleration of gravity as 10m/s$^2$.**
(a) 0.533m (b) 53.3 m
(c) 5.33m (d) 0.053 m

**Ans. (c) :** Acceleration $a_p$= 5.0 m/s$^2$

Acceleration head $H_a = \frac{L_p}{g} a_p$

$$H_a = \frac{L_p}{g} \times 5 = \frac{L_p}{2}$$

Here suction head $H_s = L_p$, $H_{pmin}$ = 2.0 (abs)
and $H_{atm}$ = 10.0 m (abs)

Hence, $2.0 = 10.0 - L_p - \frac{L_p}{2}$

$$L_p = \frac{8 \times 2}{3} = 5.35\text{m}$$

**136. The drag force on a streamlined shape is due primarily to the :**
(a) wake
(b) component of pressure force acting in flow direction
(c) shear stress
(d) separate region near the trailing edge

**Ans. (c) :** Total drag on streamlined body will be due to friction (shear) only. Through the boundary layer will start at the leading edge, will become turbulent to laminar, yet it does not separate up to rear most part of body in case of streamline body. Thus behind streamlined body, wake formation zone will very small and consequently pressure drag will be very small.

**137. The percentage of total quantity of fresh water available in the world of either in the liquid state or in the frozen state is about :**

(a) 2.5% (b) 10%
(c) 15% (d) 30%

**Ans. (a) :** The percentage of total quantity of fresh water available in the world of either in the liquid state or in the frozen state is about 2.5%.

**138. Evapotranspiration can be found out by using:**

(a) phytometer
(b) hygrometer
(c) anemometer
(d) lysimeter

**Ans. (d) :** A lysimeter is used to measure evapotranspiration.

- Evapotranspiration– Evaporation + Transpiration.
- Measurement of evapotranspiration is done in the ways either by using lysimeter or by the use of field plots.
- Lysimeter studies are time consuming and expensive.
- wind velocity is measured by Anemometer,
- Humidity is measured by Hygrometer.

**139. A reservoir is called linear reservoir when the storage is a function of _____**

(a) only inflow
(b) only outflow
(c) both inflow and outflow
(d) neither inflow nor outflow

**Ans. (b) :** A reservoir is called linear reservoir when the storage is a function of only outflow.

**140. The type of the geological formations that are highly porous but impermeable are called :**

(a) aquifer (b) aquiclude
(c) aquitard (d) aquifuge

**APPSC Poly. Tech. Lect. 2020**

**Ans. (b) :**

**Aquifer:** An Aquifer is a saturated formation of earth material which not only stores water but yield it in sufficient quantity relatively eerily once to its high permeability. Deposits of sand and cruel form good aquifers.

**Aquitard:** it is a formation through which only seepage is possible and thus the yield is insignificant compared to an aquifer. A sandy clay unit is an example of Aquitard.

**Aquiclude:** formation like clay which is highly porous but not permeable due to very small size of pores.

**Aquifuge:** it is a geological formation which is neither porous for permeable massive compact rock without any fractures is an Aquifuge:

**141. Manning's formula, which is used for calculating mean velocity of open channel flow, is given by :**

(a) $V = (1/n)R^{3/4}S^{1/2}$
(b) $V = (1/n)R^{2/3}S^{3/2}$
(c) $V = (1/n)R^{2/3}S^{1/2}$
(d) $V = (1/n)R^{2/3}S^{1/2}$

**Ans. (c) :**

$$V = \frac{1}{n} R^{2/3}S^{1/2}$$

V = Average velocity of flow

$\eta$ = Manning's constant or snugness coefficient

$$R = \frac{A}{P} = \frac{\text{cross - sectional Area}}{\text{Wetted perimeter}}$$

S = However, since in uniform flow, bed slope energy line slope and water surface slope are equal

So, $So = S_f = S_e$

**142. Which of the following stage measurement instruments employs the stilling well ?**

(a) Bubble gauge
(b) Staff gauge
(c) Wire gauge
(d) Float gauge recorder

**Ans. (d) : Float gauge recorder-** It is the most common type of automatic stage recorder in use. In this a float operating in a stilling well is balanced by means of a counter weight over the pulley of a recorder.

**143. The storage coefficient of a well is defined as the :**

(a) discharge per unit draw down of a well
(b) measure of water that can be removed by pumping
(c) limit of withdrawal from well without depletion of the aquifer
(d) draw down per unit discharge of the well

**Ans. (a) :** The storage coefficient (also known as storativity) represents the volume of water released by a column of confined aquifer of unit cross-section area under a unit decrease in piezometric head.

**144. A storm of 6 hour duration gives a direct runoff of 6 cm and has an average intensity of 3 cm/hr. Calculate the $\phi$ index.**

(a) 1.5 cm/hr (b) 2 cm/hr
(c) 2.5 cm/hr (d) 3 cm/hr

**Ans. (b) :**

Intensity or rainfall = 3 cm/hr

P = total rainfall = $3 \times 6 = 18$cm

Q = direct runoff = 6cm

t = duration of rainfall = 6 hour

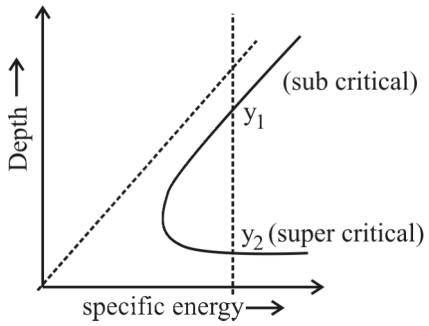
$$\phi \text{index} = \frac{P - Q}{t} = \frac{18 - 6}{6} = \frac{12}{6}$$

$= 2\text{cm/hr}$

**145. The term 'alternate depth' is used in open channel flow to denote the paired flow depths _____.**

(a) having the same kinetic energy for a given discharge
(b) having the same specific force for a given discharge
(c) having the same Froude number
(d) having the same specific energy for a given discharge

**Ans. (d) :** 'Alternate depth' is used in open channel flow to denote the paired flow depths having the same specific energy for a given discharge.




**146. For a hydraulically efficient trapezoidal channel with a depth of flow y, the wetted perimeter is equal to :**

(a) $4y/\sqrt{3}$ (b) $\sqrt{3}y^2$
(c) $2\sqrt{3}y$ (d) $y/2\sqrt{3}$

**Ans. (c) :** For a hydraulically efficient trapezoidal channel with a depth of flow y, the wetted perimeter is equal to $2\sqrt{3}y$.

for most economical trapezoidal section following conditions should be satisfied.

1. T/2 = length of end of side slope.
2. R = y/2
3. A circle of radius y should be inscribed in trapezoidal channel section.
4. $\theta = 60^0$
5. Above condition implies that most economical transformation section should be half of rectangular section.

**147. In a triangular channel, the critical depth is given by :**

(a) $(2Q^2/gm^2)^{1/5}$
(b) $(gQ^2/2m^2)^{1/5}$
(c) $(2Q^2/gm^2)^{1/7}$
(d) $(gQ^2/gm^2)^{1/7}$

**Ans. (a) :** $(2Q^2/gm^2)^{1/5}$

critical depth = $Y_C = (2Q^2/gm^2)^{1/5}$

**148. Which of the following conditions represents the gradually varied steady flow ?**

(a) Flood flow in a river
(b) Overland flow due to rain fall
(c) Backwater curve due to any obstruction such as weir
(d) Hydraulic jump

**Ans. (c) :**
**Gradually varied flow:** if depth of flow changes gradually our a long distance along the length of channel such that curvature of free surface is mild, then flow is called as gradually varied flow (GVF).
**Example:**
i. Flow at upstream of sluice gate.
ii. The back water curve produced by an obstruction to flow such as weir.
–in GVF the loss of energy is mainly due to boundary friction
–In GVF, the pressure distribution in vertical direction is taken as hydrostatic.

**149. An unconfined aquifer with c/s area of $3 \times 10^4$ $m^2$, having a hydraulic gradient of 1 in 100, discharges ground water into stream at the rate of 1200 $m^3$/day. The corresponding hydraulic conductivity of this aquifer will be :**

(a) 4 m/day (b) 4.5 m/day
(c) 5 m/day (d) 3.5 m/day

**Ans. (a) :** Given, $A = 3 \times 10^4 m^2$

$$i = \frac{H_L}{L} = \frac{1}{100}$$

$$Q = 1200 m^3/day$$

$$= \frac{1200}{24 \times 3600} \frac{m^3}{sec} = \frac{1}{72} m^3/sec$$

Now, $Q = KiA$

$$\frac{1}{72} = k \ \frac{1}{100} \ (3 \times 10^4)$$

$$K = \frac{100}{72 \times 3 \times 10^4}$$

$$= 4.6296 \times 10^{-5} m/sec$$

$$= 4 m/day$$

**150. A bridge has an expected life of 25 years, and is designed for a flood magnitude of return period 50 years. What is the risk associated with this hydrologic design ?**

(a) $(0.98)^{25}$ (b) $(0.98)^{50}$
(c) $(0.5)^{0.25}$ (d) $1-(0.98)^{25}$

**Ans. (d) :** The probability of the event occurring at least once in n successive year.

$$P_1 = 1 - (1-P)^n$$

$$P = \frac{1}{T} = \frac{1}{50} = 0.02, n = 25$$

$$P_1 = 1 - (1 - 0.02)^{25} = 1 - (0.98)^{25}$$

# Andhra Pradesh Public Service Commission
# Deputy Inspector of Survey Exam, 2020
## Solved Paper
**Date - 27.09.2020**

**1. Mild steel is _____ material.**
(a) plastic (b) brittle
(c) ductile (d) alloy

**Ans. (c) : Ductile material–** If post elastic strain is greater than 5%, it is called ductile material.
Ex= lead, mild steel, copper.

**2. The elongation of a conical bar due to its self-weight is:**
**(where, $\ell$ = length of the conical bar, E= modulus of elasticity, $\gamma$ = unit weight)**
(a) $\frac{\gamma\ell^2}{6E}$ (b) $\frac{\gamma\ell^2}{2E}$
(c) $\frac{\gamma\ell^2}{3E}$ (d) $\frac{\gamma\ell^2}{E}$

**Ans. (a) :** Elongation of a conical bar due to its self weight –

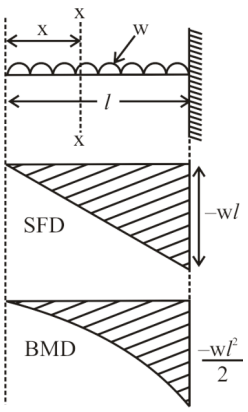
$$\Delta = \frac{\gamma\ell^2}{6E}$$

Where, $\ell$ = length of conical bar,
E = modulus of elasticity
$\gamma$ = unit weight
Elongation of prismatic bar due to its self weight –

$$\Delta = \frac{\gamma\ell^2}{2E}$$

**3. The shear force (SF) variation of a cantilever beam of length (L) carrying a uniformly distributed load (UDL), is :**
(a) linear from free end
(b) non-linear from free end
(c) Parabolic
(d) uniform throughout

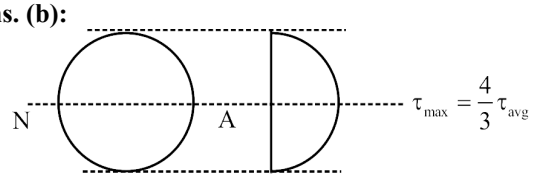
**Ans. (a):**




$SF_{xx} = -w.x$
$SF_{x=0} = 0$, $S.F_{x=L} = w.l$

**4. For circular cross section, shear stress is maximum at :**
(a) the peripher (b) the neutral axis
(c) $\frac{1}{2}$ radius from the centre
(d) $\frac{1}{3}$ radius from the centre

**Ans. (b):**




• For circular cross section, shear stress is maximum at the neutral axis.

**5. Using differential equation of elastic curve M = $EI\left(\frac{d^2y}{dx^2}\right)$, the slope and deflection at may section can be found on :**
**(where, M - moment, EI-flexural rigidity)**
(a) The integration only
(b) double integration only
(c) first and second integration, respectively
(d) by successive differention

**Ans. (c) :**

| Deflection equation | EI.y |
|---|---|
| Slope equation | $EI\left(\frac{dy}{dx}\right)$ |
| Moment equation | $EI\left(\frac{d^2y}{dx}\right)$ |
| Shear equation | $EI\left(\frac{d^2y}{dx^3}\right)$ |
| Load equation | $EI\left(\frac{d^4y}{dx^4}\right)$ |

**6. A solid shaft is experiencing applied torque T and its diameter is d. For the torque NOT exceeding proportional limit, the maximum sharing stress is :**
(a) $\frac{16T}{\pi d^3}$ (b) $\frac{8T}{\pi d^3}$
(c) $\frac{4T}{\pi d^3}$ (d) $\frac{2T}{\pi d^3}$

**Ans. (a) :** Maximum shearing stress = $\tau_{max} = \frac{16T}{\pi d^3}$

Principal stress '$\sigma$' = $\frac{32M}{\pi d^3}$

**7. A long column fails by :**

(a) crushing (b) buckling

(c) shearing (d) tension

**Ans. (b) :**

| Types of column | Fails in |
|---|---|
| Short column | Crushing |
| Intermediate column | Combined, crushing and buckling |
| Long column | Buckling |

**8. If two forces A and B are acting at an angle θ and their resultant R makes an angle α with the force A, then :**

(a) $\text{Tan}\,\alpha = \dfrac{A\cos\theta}{(B + A\sin\theta)}$

(b) $\text{Tan}\,\alpha = \dfrac{A\sin\theta}{(A + B\cos\theta)}$

(c) $\text{Tan}\,\alpha = \dfrac{B\sin\theta}{(A + B\cos\theta)}$

(d) $\text{Tan}\,\alpha = \dfrac{B\sin\theta}{(B + A\sin\theta)}$

**Ans. (c) : Parallelogram law of forces–**

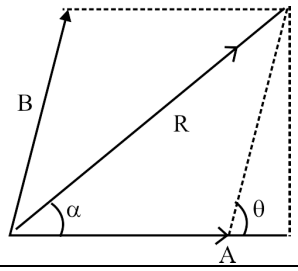
figure

**Resultant force–**

$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Direction of resultant

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

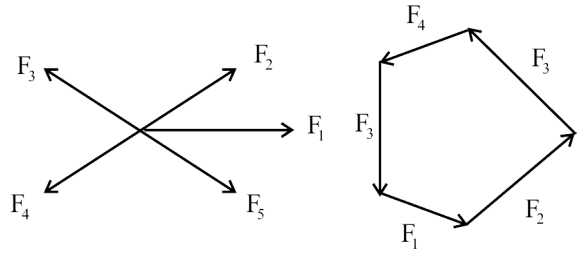



**9. When a system of forces in a plane from a closed polygon, the system is called.**

(a) equilibrium (b) non-equilibrium

(c) stable (d) unstable

**Ans. (a) : Polygonal low–** If more than three forces are in equilibrium, then, they form a closed polygon when represented in a tip to tail arrangement.

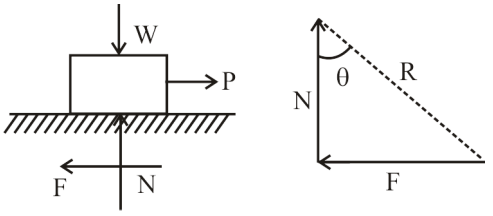



**10. If the force of friction is zero and the normal reaction is NOT zero, then the angle of friction is:**

(a) $0^0$ (b) $45^0$

(c) $60^0$ (d) $90^0$

**Ans. (a) :** The angle of friction b/w any two surfaces in contact is defined as the angle which the resultant of the force of limiting friction F and normal reaction N makes with the direction of normal reaction N.




$$\mu = \tan\theta = \frac{F}{N}$$

■ If the force of friction is zero then the resultant is along the direction of normal.

Hence, if the angle of friction is zero, the body will experience zero friction.

**11. In the general in first order pulley system, the velocity ratio is given by (where n is the number of available pulleys) :**

(a) $n^2$ (b) $2^n$ (c) 2n (d) 2n – 1

**Ans. (b) :** In general, in the first oder pulley system, velocity ratio (VR) is given by $2^n$,

Where, n is the number of movable pulleys present in the system.

**12. The strain along longitudinal direction is 0.25. if the poisson's ratio is 0.2, then the lateral strain is :**

(a) 0.05 (b) 0.005 (c) 0.5 (d) 5

**Ans. (a) :** Poisson's ratio 'μ' $= \dfrac{\text{lateral strain}}{\text{longitudinal strain}}$

$$0.2 = \frac{\text{lateral strain}}{0.25}$$

Lateral strain = 0.05

**13. The stress due to suddenly applied load is _____ times that of a gradually applied load.**

(a) 2 (b) 0.2

(c) 20 (d) 0.02

**Ans. (a) :** The stress due to suddenly applied load is 2 times that of a gradually applied lead.

**14. The most efficient connection in transmitting the forces of steel member is :**

(a) riveted connection (b) bolted connection

(c) welded connection (d) pin connection

**Ans. (c) :** The most efficient connection in transmitting the forces of steel member is welded connection.

- Welded connections show limited ductility and undesirable mode of failure.
- A welded structure are comparatively lighter.

**15. A cylindrical pipe of diameter 1.5 m and thickness 1.5 cm is subjected to an internal fluid pressure of 1.5 N/mm$^2$. The hoop stress developed in the pipe is :**

(a) 75 N/mm$^2$ (b) 60 N/mm$^2$
(c) 90 N/mm$^2$ (d) 1.5 N/mm$^2$

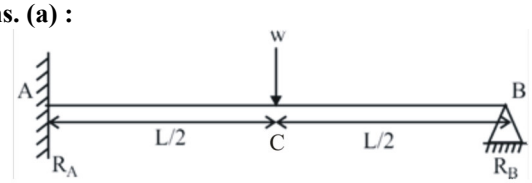
**Ans. (a) :** Given,
Diameter 'D' = 1.5 m = 1500 mm
Thickness 't' = 1.5 cm = 15 mm
Pressure 'P' = 1.5 N/mm$^2$

$$\text{Hoop stress} = \frac{PD}{2t}$$
$$= \frac{1.5\times1500}{2\times15}$$
$$= 75 \text{ N/mm}^2$$

**16. If a propped cantilever beam carries a point load at the centre, then the number of points of contra flexure will be**

(a) 1 (b) 2
(c) 3 (d) 4

**Ans. (a) :**

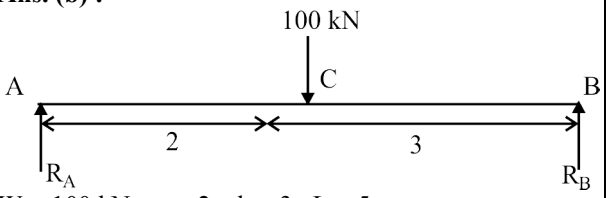



• Two span continuous beam with UDL over both spans will have two points of contraflexure.
• Fixed beam with UDl will have two points of confraflexure.
A propped cantilever beam with UDL will have one point of centraflexure.

**17. If a simply supported beam of span 5 m carries a point load of 100 kN at 2 m from the left-hand side, then the maximum bending moment on it is :**

(a) 62.5 kNm (b) 120 kNm
(c) 125 kNm (d) 312.5 kNm

**Ans. (b) :**




W = 100 kN, a = 2, b = 3, L = 5 m

$$\text{Maximum bending moment} = \frac{W\times a\times b}{L}$$
$$= \frac{100\times2\times3}{5}$$
$$= 120 \text{ kNm}$$

**18. The polar moment of inertia of a circular cross-sectional area of diameter d is :**

(a) $\frac{\pi d^4}{32}$ (b) $\frac{\pi d^4}{64}$
(c) $\frac{\pi d^4}{4}$ (d) $\frac{\pi d^4}{128}$

**Ans. (a) :** Polar moment of inertia $I_P = \frac{\pi d^4}{32}$

• Polar section modulus/torsional strength $(Z_P) = \frac{\pi d^3}{16}$

**19. The principal planes at a point are inclined at:**

(a) $45^0$
(b) $60^0$
(c) $90^0$
(d) $180^0$

**Ans. (c) :** The principal plane at a point are inclined at $90^0$.
**Principal plane–** A plane that is acted upon by a normal stress only is known as a principal plane;

**20. If the hoop strain and cylinderical strain in case of a thin cylinderical shell are $e_1$, and $e_2$, then the volumetric strain is :**
**(where, $e_h$ = circumferential strain, $e_1$ = longitudinal strain)**

(a) $e_h + e_1$ (b) $e_h + 2e_1$
(c) $2(e_h+e_1)$ (d) $2e_h+e_1$

**Ans. (d) :** Given hoop strain = $e_1$
Cylinderical strain = $e_2$
Circumferential strain = $e_n$
Volumetric strain = $(2e_n + e_1)$

**21. Euler's formula CANNOT be used in case of :**

(a) short columns
(b) long columns
(c) small (k/1) values
(d) large (1/k) values

**Ans. (a) :** Euler's formula cannot be used in case of short columns–
It is application for long column.
Euler's formula

$$P_e = \frac{\pi^2 E I_{min}}{\ell_e^2}$$

Where,
$P_e$ = Buckling load
$I_{min}$ = Minimum moment of inertia
$\ell_e$ = effective length

**22. The maximum eccentricity of the vertical load on a short column of external diameter 'D' and the internal diameter 'd' for no tension to occur is :**

(a) $\frac{8D}{(D^2+d^2)}$ (b) $\frac{(D^2+d^2)}{8D}$
(c) $\frac{(D^2+d^2)}{D}$ (d) $\frac{(D^2-d^2)}{8D}$

**Ans. (b) :** The beam is subjected to a compressive load P acting with an eccentricity 'e'
So, there will be two types of stresses i.e. direct stress and bending stress.

Thus for no stress at extreme left fiber, Direct compressive stress + Tensile bending stress= 0

$$= -\frac{P}{A} + \frac{My}{I} = 0$$

$$= \frac{P}{\frac{\pi}{4}\left(D^2 - d^2\right)} + \frac{(P.e)\times\frac{D}{2}}{\frac{\pi}{64}\left(D^4 - d^4\right)} = 0$$

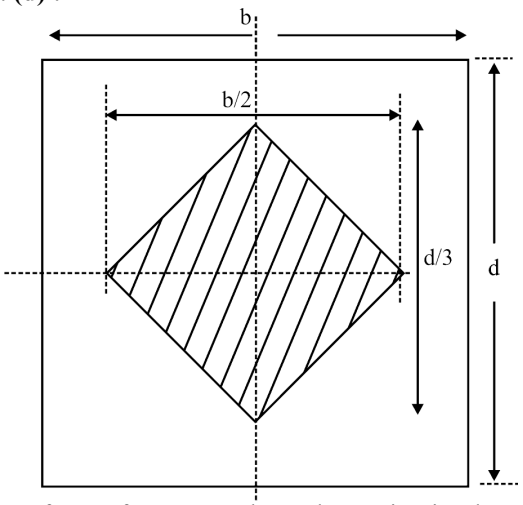
$$= \frac{1}{\left(D^2 - d^2\right)} = \frac{16e\times\frac{D}{2}}{\left(D^4 - d^4\right)}$$

Maxi. eccentricity 'e' $= \frac{D^2 + d^2}{8D}$

**23. The shape of the kern area of a rectangular section is a:**

(a) circle (b) square
(c) rectangle (d) rhombus

**Ans. (d) :**



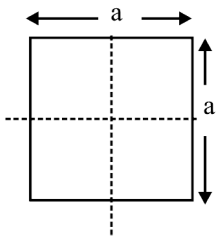

Shape of Kern for rectangular and I-section is Rhombus and for square section shape is square.

**24. The ratio of the moment of inertia of a square plate to that of a circular plate of the same depth is :**

(a) 1.563 (b) 1.571
(c) 1.698 (d) 3.141

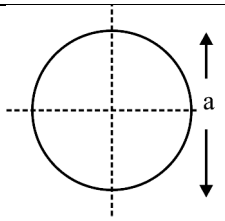
**Ans. (c) :** Moment of inertia of square plate–

$$I_s = \frac{a^4}{12}$$




Moment of inertia of a circular plate

$$I_c = \frac{\pi a^4}{64}$$




Ratio $= \frac{I_s}{I_c} = \frac{\frac{a^4}{12}}{\frac{\pi a^4}{64}} \Rightarrow \frac{64}{12\times\pi} = 1.698$

**25. If the hoop stress in a thin cylinder is 24 N/mm$^2$, then the longitudinal stress is equal to:**

(a) 12 N/mm$^2$ (b) 6 N/mm$^2$
(c) 24 N/mm$^2$ (d) 36 N/mm$^2$

**Ans. (a) :** Given,

Hoop stress '$\sigma_n$' $= \frac{Pd}{2t}$

$\sigma_n = 24$ N/mm$^2$

Longitudinal stress '$\sigma_L$'$= \frac{Pd}{4t} = \frac{1}{2}\left(\frac{Pd}{2t}\right)$

$$= \frac{1}{2}\times 24$$

$= 12$ N/mm$^2$

**26. The difference between the bending moment value at any two sections will be equal to :**

(a) the area of shear force diagram between those two sections
(b) the difference in slope of shear force diagram at the same section
(c) the moment of area of loading diagram, between those two sections
(d) the area of deflection diagram between those two sections

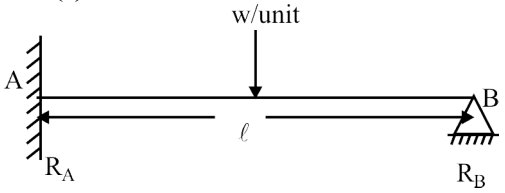
**Ans. (a) :** Change of bending moment between two section is equal to the area under the shear force diagram between those two section.

• Change of shear force between two section is equal to the area under the load intensity diagram between those two section.

**27. A cantilever of length *l* carries a UDL of w per unit over the whole length. It the free end be supported over a rigid prop. the reaction of the prop will be :**

(a) 2wl/8 (b) 5wl/8
(c) 3wl/8 (d) 7wl/8

**Ans. (c) :**




Reaction of prop 'B' $= \frac{3w\ell}{8}$

**28. Strain energy (U) caused by bending is given by the relations :**

(a) $U = \int\left(\frac{M}{2EI}\right)dx$ (b) $U = \int\left(\frac{M^2}{2EI}\right)dx$

(c) $U = \int\left(\frac{M^2}{EI}\right)dx$ (d) $U = \int\left(\frac{M^2}{3EI}\right)dx$

**Ans. (b) :** The strain energy is the energy stored in a body due to its elastic deformation.

- Strain energy due to bending is–

$$U = \int\left(\frac{M^2}{2EI}\right)dx$$

- Strain energy due to torsion is–

$$U = \int\frac{T^2L}{2GJ}dx$$

**29. The slenderness ratio of a free stable column of effective length 3 m and 30 mm diameter, is:**

(a) 100.6 (b) 200.8

(c) 692.8 (d) 400

**Ans. (d) :** Given,

Effective length '$\ell_e$' = 3m = 3000 mm

diameter of column 'd' = 30 mm

Least radius of gyration = $r_{min} = \sqrt{\frac{I_{min}}{A}}$

$$r_{min} = \sqrt{\frac{\frac{\pi d^4}{64}}{A}}$$

$$r_{min} = \sqrt{\frac{\pi \times 30^4 \times 4}{64 \times \pi \times 30^2}} = 7.5 \text{ mm}$$

Slenderness ratio '$\lambda$' = $\frac{\ell_e}{r_{min}} = \frac{3000}{7.5} = 400$

**30. The reciprocal of specific volume is known as**

(a) specific gravity (b) specific weight

(c) mass density (d) viscosity

**Ans. (c) :** The reciprocal of the mass density is known as specific volume.

**31. The pressure at any point in a fluid is defined as__**

(a) force per unit width

(b) force per unit volume

(c) force per unit length

(d) force per unit area

**Ans. (d) :** Pressure = $\frac{\text{Force}}{\text{Area}}$

Unit = $\frac{N}{m^2}$

**32. Gauge pressure ($P_{gauge}$) is equal to : (where, $P_{abs}$ = absolute pressure, $P_{vae}$ = vacuum pressure, $P_{atm}$= atmospheric pressure)**

(a) $P_{gague} = P_{abs} - P_{vae}$ (b) $P_{gague} = P_{abs} + P_{vae}$

(c) $P_{gague} = P_{abs} + P_{atm}$ (d) $P_{gague} = P_{abs} - P_{atm}$

**Ans.(d): Gauge pressure ($\rho$gauge) = $\rho_{abs} - \rho_{atm}$**

- Gauge pressure can be positive, negative or zero.
- Atmospheric pressure varies with altitude, temperature and local conditions.

**33. To measure the difference in pressures between two points in a pipe, which of the following flowing instruments is used?**

(a) Current meter

(b) Differential manometer

(c) U-tube manometer

(d) Micrometer

**Ans. (b) :** Differential manometers are the devices used for measuring the difference of pressure between two points in a pipe or in two different pipes.

**34. The total surface on an immersed surface inclined at an angle $\theta$ with the liquid surface, is: (where W = specific weight of the liquid (Newton/cubic meter), A = wetted plane surface area, X = depth of the centroid below the free surface)**

(a) W.A. (b) W.X

(c) W.A.X (d) W.A.X/sin$\theta$

**Ans. (c) :** Inclined plane surface submerged in liquid–

- The plane of the surface makes an angle $\theta$ with the free surface of the liquid.

Total pressure force 'F' = $\rho$gAx (W=$\rho$g)

= WAX

Centre of pressure $h = \frac{IG\sin^2\theta}{A\bar{x}} + x$

**35. The piezometric head is equal to :**

(a) pressure head + datum head

(b) pressure head + velocity head

(c) pressure head + datum velocity

(d) pressure head – datum velocity

**Ans. (a) :**

Piezometric head = Pressure head + datum head

Pressure head = $\frac{\rho}{w}$ (w = $\rho$g)

Datum head = z

**36. An orifice fitted with some kind of extension is known as :**

(a) notch (b) weir

(c) nozzle (d) mouth piece

**Ans. (d) : Mouthpiece–** A mouthpiece in an attachment in the form of a small tube or pipe fixed to the orifice (the length of pipe extension is usually 2 to 3 times the orifice diameter) and is used to increase the amount of discharge.

**37. The loss of head due to sudden contraction is : (where, V = velocity, G = acceleration due to gravity)**

(a) $\frac{V^2}{4g}$ (b) $\frac{V^2}{2g}$

(c) $\frac{V^2}{3g}$ (d) $\frac{V^2}{g}$

**Ans. (a):** Loss due to sudden contraction–

$$h_f = \frac{V_2}{4g}$$

Losses at exit of pipe $h_f = \frac{V^2}{2g}$

Losses at entrance, $t = pipe = h_f = \frac{0.5\ V^2}{2g}$

**38. Chezy's formula is given by :**
**(Where, V = velocity, i = hydraulic gradient, m = hydraulic mean depth, c = chezy's constant)**

(a) $V = i\sqrt{mc}$ (b) $V = c\sqrt{mi}$

(c) $V = m\sqrt{ci}$ (d) $V = m^3\sqrt{ci}$

**Ans. (b) :** Chezy formula–

$$V = c\sqrt{mi}$$

Where,
C = Chezys constant
m = Hydraulic mean depth
i = Hydraulic slope

- Relation between chezys constant and friction factor

$$C = \sqrt{\frac{2g}{f}}$$

**39. To correct poise into SI units the poise should be divided by :**

(a) 0.1 (b) 0.01

(c) 10 (d) 100

**Ans. (c) :** Poise and stokes are the units of viscosity and kinematic viscosity respectively.
To convert the unit of viscosity from poise to MKS units, poise should be divided by 98.1 and to convert poise into SI units the poise should be divided by 10.

**40. The maximum efficiency of power transmission**

(a) 90% (b) 83.3%

(c) 65.8% (d) 66.67%

**Ans. (d) :** Maximum efficiency of transmission of power–

$$\eta = \frac{H - h_f}{H}$$

For maximum power transmission through pipe the condition of give by equation–

$$\eta_f = \frac{H}{3}$$

$$\eta_{max} = \frac{H - H/3}{H} = 1 - \frac{1}{3} = \frac{2}{3}$$

$= 66.67\%$

**41. For laminar flow through a circular tube, the average velocity at a section is (where $V_{max}$=maximum velocity)**

(a) $\frac{1}{3}V_{max}$ (b) $\frac{1}{2}V_{max}$

(c) $\frac{2}{3}V_{max}$ (d) $\frac{4}{3}V_{max}$

**Ans. (b):** For laminar flow through a circular tube (pipe), the maximum velocity is equal to twice of the average velocity.

$$V_{max} = 2 \times V_{avg}$$

$$V_{avg} = \frac{1}{2}V_{max}$$

**42. A turbine is a device which converts :**

(a) hydraulic energy into mechanical energy
(b) mechanical energy into hydraulic energy
(c) kinetic energy into mechanical energy
(d) electric energy into mechanical energy

**Ans. (a) :** A turbine is a device or machine that converts hydraulic energy to mechanical energy, finally which is converted to electric energy.

- The hydraulic machines which convert the mechanical energy into hydraulic energy are called pump.
- The mechanical energy so obtained its supplied to the generator coupled to the runner, which then generate electrical energy.

**43. If 'D' is the diameter of the Pelton wheel, 'd' is the diameter of the jet, then the number of buckets on the periphery of the Pelton wheel is equal to :**

(a) $\frac{D}{2d}$ (b) $\frac{D}{2d} + 10$

(c) $\frac{D}{2d} + 15$ (d) $\frac{D}{2d} + 20$

**Ans. (c) : Pelton wheel turbine–**
Number of buckets on a runner–

$$Z = 15 + \frac{D}{2d}$$

**Jet Ratio–** It is defined as the ratio of the pitch diameter (D) of the pelton wheel to the diameter of the jet (d). It is denoted by (m')

$$m = \frac{D}{d}\left(= 12 \text{ for most cases}\right)$$

**44. Cavitation is caused by :**

(a) low distance (b) high pressure

(c) low velocity (d) low pressure

**Ans. (d) : Cavitations–** It is the formation of gas bubbles of a flowing liquid in a region where the pressure of the liquid falls below its vapour pressure that is low pressure.

**45. If mass and volume of fluid are 1000 kg and 100 $m^2$, respectively, then the density of the fluid is:**

(a) 10 $kg/m^3$ (b) 150 $kg/m^3$

(c) 120 $kg/m^3$ (d) 100 $kg/m^3$

**Ans. (a) :** Given,
mass 'm' = 1000 kg
volume 'V' = 100 $m^2$

Density '$\rho$' $= \frac{m}{V} = \frac{1000}{100}$

'$\rho$' = 10 $kg/m^3$

**46. The head loss is more in a:**
(a) nozzle meter
(b) venturimeter
(c) inclined venturimeter
(d) orifice meter

**Ans. (d) : Orifice meter–** An orifice meter is used to measure the discharge in a pipe.
• It is suitable in location where venturimeter cannot be filled due to space restrictions.

$$Q_{act} = \frac{Cd\ a_1 a_0 \sqrt{2gh}}{\sqrt{a_1^2 - a_0^2}}$$

**47. The coefficient of discharge $C_d$, velocity $C_v$ and contraction $C_e$, are related as :**
(a) $C_d = C_e \div C_v$ (b) $C_d = C_e + C_v$
(c) $C_d = C_e - C_v$ (d) $C_d = C_e \times C_v$

**Ans. (d) : Co-efficient of contraction ($C_e$)-**
It is defined as the ratio of the area of the jet at vena-contracta to the area of the orifice $C_e = \frac{A_c}{A}$

**Coefficient of velocity ($C_v$)**

$$C_v = \frac{\text{Actual velocity of a jet of liquid at vena conracta}}{\text{Theoretical velocity of jet}}$$

**Coefficient of discharge ($C_d$)**
It is defined as the ratio of the actual discharge from an orifice to the theoretical discharge from the orifice.

$$C_d = \frac{Q_{act}}{Q_{th}}$$
$$C_d = C_e \times C_v$$

**48. Power transmitted through a pipe is maximum when the loss of head due to friction is _____ of the total head supplied.**
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{1}{5}$

**Ans. (b) :** Power transmitted through a pipeline is maximum when the head lost due to friction in the pipe is equal to one-third of the total supply head.

i.e., $H_f = \frac{H}{3}$

where, H = Total supply head,
$H_f$= Head loss due to friction.
Power transmitted at outlet of the pipe–

$$P = \gamma Q(H - H_f)$$

$$P = \gamma \times \frac{\pi d^2 v}{4}(H - H_f)$$

$$P = \gamma\left(\frac{\pi}{4} d^2 \times V\right)\left(H - \frac{f\ell V^2}{2gd}\right)$$

Condition for maximum transmission of power

$$\frac{dp}{dv} = \gamma \times \frac{\pi}{4} d^2 \left(H - \frac{3f\ell V^2}{2gd}\right)$$

$$\frac{dP}{dv} = 0$$

$$\frac{dP}{dv} = H - 3 \times \frac{f\ell v^2}{2gd} = 0$$
$$H = 3H_f$$
$$H_f = \frac{H}{3}$$

**49. A change of angular momentum in a fluid flow along a curved path produced a :**
(a) change in momentum
(b) change in energy
(c) change in pressure
(d) torque

**Ans. (d) :** A change of angular momentum in a fluid flow along a curved path produced a torque.

**50. The ratio between power developed by the runner, and net power supplied by the water at the entrance of a turbine, is known as**
(a) hydraulic efficiency
(b) volumetric efficiency
(c) overall efficiency
(d) mechanical efficiency

**Ans. (a) : Hydraulic efficiency ($\eta_h$) =** The hydraulic efficiency of the turbine is the ratio of the power developed by the runner to the net power supplied by the water at the entrance to the turbine.

$$\text{Mechanical efficiency } (\eta_m) = \frac{\text{Power available at the turbine shaft}}{\text{Power developed by the runner}}$$

**51. Francis turbines work on the principle of**
(a) tangential flow (b) impulse
(c) axial flow (d) reaction

**Ans. (d) : Reaction turbines–** As stated earlier the principal distinguishing features of a reaction turbine are that total head of water is converted into velocity head before it reaches the runner, and that the water completely fills all the passage in the runner.
• The two reaction type of turbines which are predominantly used these days are francis turbine and kaplan turbine.

**52. The function of governor in a turbine is to**
(a) give maximum overall efficiency
(b) give maximum hydraulic efficiency
(c) control the speed of the runner
(d) control the flow through the runner

**Ans. (c) : The working of the governor–**
• Control the speed of the runner
• When the load on the generator drops, the speed of turbine runner increases. Now since the actuator or pendulum is driven by the turbine main shaft, due to increase in the speed the balls move outward, resulting in an upward movement to the sleeve.

**53. A centrifugal pump is the reverse of a/an :**
(a) axial flow reaction turbine
(b) outward radial flow reaction turbine
(c) radial inward flow reaction turbine
(d) axial flow impulse turbine

**Ans. (c):** The action of a centrifugal pump is just the reverse of a radially inward flow reaction turbine.

- Similarly the axial flow pumps are reverse of propeller or kaplan turbines and the mixed flow pumps are the reverse of mixed flow type turbines such as Francis turbine.

**54. If net head is H and net suction head is $H_1$ for a centrifugal pump, then its cavitation parameter $\sigma$ is given**

(a) $\frac{H_1}{H}$ (b) $\frac{H}{H_1}$

(c) $H \times H_1$ (d) $H^2 \times H_1$

**Ans. (a) : Cavitation in centrifugal pumps =** Cavitation begins to appear in centrifugal pumps when the pressure at the suction falls below the vapour pressure of the liquid.

Cavitation parameter 'd' = $\frac{H_1}{H}$

**55. The depth of weld in case of a butt weld is :**

(a) equal to the thickness of the plate
(b) 0.5 times the thickness of the plate
(c) 1.5 times the thickness of the plate
(d) 2 times the thickness of the plate

**Ans. (a) :** The depth of weld in case of a butt weld is equal to the thickness of the plate.

**56. The ratio of mean velocity to maximum velocity for turbulent flow is about :**

(a) 0.8 (b) 0.08
(c) 0.008 (d) 8.0

**Ans. (a) :** The ratio of mean velocity to maximum velocity for turbulent flow is 0.8.

Common mean velocity distribution for turbulent flow.

$$\frac{u-\bar{u}}{v^*} = 5.75\log_{10}\frac{y}{R} + 3.75$$

Where,

V* – Shear velocity

u = Flow velocity

$\bar{u}$ = Avg. velocity

$v^* = \bar{u} \times \sqrt{\frac{f}{8}}$

f = Friction factor

**57. The difference in elevation between the total energy line and the hydraulic gradient line of pipe flow at a point, is equal to :**

(a) piezometric head (b) velocity head
(c) pressure head (d) datum head

**Ans. (b) :** In an uniform diameter pipe, HGL always be parallel to TEL and vertical gap b/w these two lines at any section of pipe is the velocity head.

HGL = TEL – Velocity head

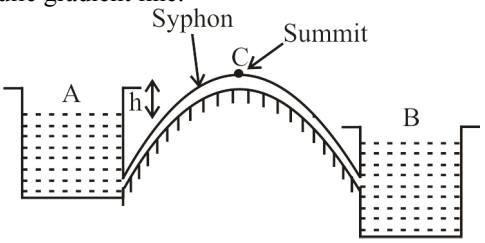
Velocity head = TEL – HGL

- The difference b/w the total head lines and hydraulic gradient line is always constant for particular pipe flow and it is equal to the velocity head.

**58. A syphon pipe is :**

(a) any bent pipe
(b) the part of pipe above the hydraulic gradient line
(c) the part of pipe below the hydraulic gradient line
(d) useful for pumping water to high levels

**Ans. (b) :** A syphon pipe is the part of pipe above the hydraulic gradient line.




- A syphon is a long bend pipe that is used to transfer liquid from the reservoir at a higher elevation to another reservoir at a lower elevation when these reservoirs are separated by high level ground.

**59. If the specific weight of an oil is 20 kN/m$^2$, and the specific weight of a standard fluid is 10 kN/m$^3$, then the specific gravity of the oil is :**

(a) 2 (b) 2.3
(c) 2.4 (d) 3

**Ans. (a) :** Specific weight of an oil = 20 kN/m$^2$

Specific weight of a standard fluid = 10 kN/m$^3$

$$\text{Specific gravity} = \frac{\text{Weight of a substance}}{\text{Weight of an equal volume of water}}$$

$$= \frac{20}{10}$$

Specific gravity = 2

**60. 10 m of water column is equal to :**

(a) 100 kN/m$^2$ (b) 10 kN/m$^2$
(c) 1 kN/m$^2$ (d) 0.1 kN/m$^2$

**Ans. (a) :** Given,

h = 10m, $\rho_w$ = 1000 kg/m$^3$, g = 10 m/s$^2$

$P = \rho_w \times g \times h$

Where,

$\rho_w$ = is density of water and h is the height of water column.

P = 1000×10×10

P = 100000 N/m$^2$

P = 100 kN/m$^2$

**61. The depth of the centre of pressure of a vertical rectangular plate of depth 3 m and width 4 m when the water surface coincides with the top of the gate, is :**

(a) 1 m (b) 2 m
(c) 1.5 m (d) 3 m

**Ans. (b) :** For rectangular plate,

$$h^* = \frac{I_G \sin^2\theta}{A\bar{h}} + \bar{h} = \frac{\frac{bh^3}{12}}{bh \times \frac{h}{2}} + \frac{h}{2} = \frac{2h}{3}$$

Given, b = 4m and depth = 3m

Depth of the center of pressure on a vertical rectangular gate.

$$h^* = \frac{2h}{3} = \frac{2\times 3}{3} = 2m$$

**62. The pressure head at a point is 20 m, velocity at that point is 0.5 m/s, and the point is 5 m above the datum. The total head at the point is :**

(a) 25 m (b) 25.5 m
(c) 20.5 m (d) 19.5 m

**Ans. (b) :** Total head = Pressure head + Velocity head + Datum head

$$\text{Velocity head} = \frac{V^2}{2g} = \frac{(0.5)^2}{2\times 9.81} = 0.01m$$

Total head = 20m + 0.01 + 5 = 25.01 m
Note:- Total head is taken as greater than 25m
So, The option (b) is considered.

**63. The velocity of flow at vena contraction when the head is 1 m is :**

(a) 2.21 m/s (b) 1.41 m/s
(c) 1 m/s (d) 4.43 m/s

**Ans. (d) :** The velocity of flow at vena contraction

$$= \sqrt{2gh}$$

Head (h) = 1m
The velocity of flow at vena contraction

$$= \sqrt{2\times 9.81\times 1}$$
$$= 4.43 \text{ m/sec}$$

**64. It the head of water supply to a nozzle is 15 m, then the loss of head due to friction at maximum power transmission is :**

(a) 3 m (b) 1.5 m
(c) 5 m (d) 50 m

**Ans. (c) :** Head of water supply at a nozzle (H) = 15 m
The loss of head due to friction at maximum power transmission = H/3

$$= 15/3$$
$$= 5 \text{ m}$$

**65. Water is flowing in a pipe at a velocity of 10 m/sec, the velocity head of water is _____**

(a) 0.05 m (b) 0.5 m
(c) 5 m (d) 50 m

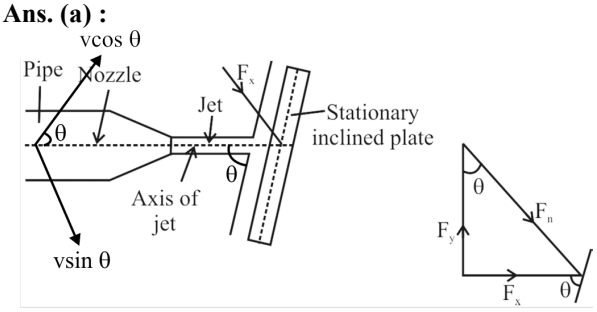
**Ans. (c) :** From given,
Velocity (V) = 10 m/s

$$\text{Velocity head (h)} = \frac{V^2}{2g}$$
$$= \frac{10\times 10}{2\times 10}$$
$$= 5 \text{ m}$$

**66. The force exerted by a jet of water on stationary inclined plate in the direction of jet is given by :**
**(where, A = area of jet, V = velocity in m/s**

(a) $Fx = \rho AV^2\sin^2\theta$
(b) $Fx = \rho AV^2(1+\cos\theta)$
(c) $Fx = \rho AV^2$
(d) $Fx = \rho AV^2(1-\cos\theta)$

**Ans. (a) :**




We know that,
$F = \rho\times a\times v^2$
Let up apply the impulse momentum eq$^n$ in the direction normal to the plate.

$$F_n = -\left[\rho Q(0) - \rho Q v\sin\theta\right]$$

$$\rho Q v\sin\theta = \rho A v^2\sin\theta$$

$F_x = F_n\sin\theta = \rho v^2\sin\theta\times\sin\theta = \rho Av^2\sin^2\theta$
$F_y = F_n\cos\theta = \rho v^2\sin\theta\times\cos\theta = \rho v^2\sin\theta.\cos\theta$

**67. A water jet of 0.015 m2 in area has a velocity of 1.5 m/s. If this jet impinges normally on a plate which is moving at a velocity of 5 m/s in the direction of the jet, then the force of the plate due to this impact is :**

(a) 1000 N (b) 150 N
(c) 1500 N (d) 15000 N

**Ans. (c) :** Given,
Area of jet (A) = 0.015 m$^2$
Absolute velocity of jet (V) = 15 m/sec
Velocity of plate (u) = 5 m/sec
Density of water ($\rho$) = 1000 kg/m$^3$
Force on the plate due to impact of jet ($F_x$)
$= \rho\times A\times(V-u)^2 = 1000\times 0.015\times(15-5)^2$
$\therefore F_x = 1500$ N

**68. The number of buckets required for a pelton wheel having jet ratio of 24 is about :**

(a) 7 (b) 15
(c) 21 (d) 27

**Ans. (d) :** Number of buckets on a runner,

$$Z = 15 + \frac{D}{2d}$$

Given jet ratio = 24

$$\text{i.e. } \frac{D}{d} = 12$$

$$Z = 15 + \frac{1}{2}\times 24$$

Z = 15 + 12 = 27
$\therefore$ Number of buckets on a runner = 27

**69. A turbine is operated under a head of 25 m. It runs with a speed of 240 RPM when the discharge is 12 m$^2$/sec. If the overall efficiency is 80%, then what is the power generated?**

(a) 1920 kW (b) 3000 kW
(c) 2354 kW (d) 960 kW

**Ans. (c):** Given, Head (H) = 25m
Run speed (N) = 240 RPM
Discharge (Q) = 12 $m^3$/sec
Overall efficiency ($\eta_o$) = 80%
Power available at the turbine shaft-
$P = wQH \times \eta_o$
$= 9810 \times 12 \times 25 \times 0.80$
$= 2354.4 \times 10^3$ w or 2354.4 kw

**70. If Net Positive Suction Head (NPSH) is 5 m and Net Head (H) is 20 m, then the cavitation parameter ($\sigma$) for a centrifugal pump is :**
(a) 0.25 (b) 0.025
(c) 25 (d) 250

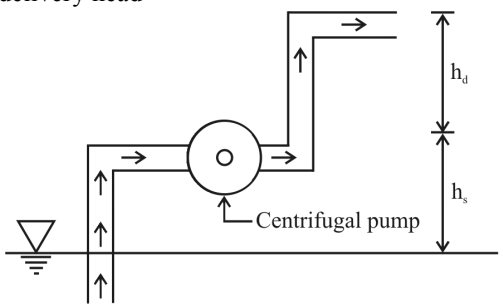
**Ans. (a) :** Net positive suction head (NPSH) = 5 m
Net head (H) = 20 m

$$\text{Cavitation parameter (6)} = \frac{\text{Net positive suction head}}{\text{Net head}}$$

$$= \frac{5}{20} = 0.25$$

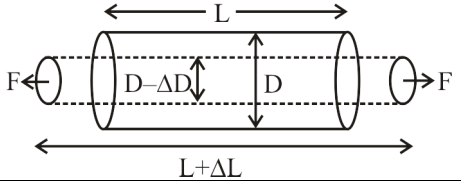
**71. Static head of a pump is the :**
(a) sum of suction height and delivery lift
(b) total length of a pipe line
(c) sum of friction losses and length of pipe
(d) sum of depth of water in the pump and water tank

**Ans. (a) : Static head (Hs)**
$H_s = h_s + h_d$
$h_s$ = suction head
$h_d$ = delivery head




**72. The ratio of decrease in length to the original length of a body is known as :**
(a) tensile strain
(b) compressive strain
(c) volumetric strain
(d) shear strain

**Ans. (b) :** $\text{Strain} = \frac{\text{Change in dimension}}{\text{Original dimension}}$




• The force making a change in length is compressive in nature the corresponding strain is called a compressive strain.
• When we take the strain in longitudinal direction i.e. (change in length/original length) then it is called the longitudinal strain.

**73. Modulus of elasticity is defined as :**
(a) the ratio of compressive stress to tensile strain
(b) the ratio of tensile stress to compressive strain
(c) the ratio of compressive stress to tensile stress
(d) the ratio of compressive stress to compressive strain

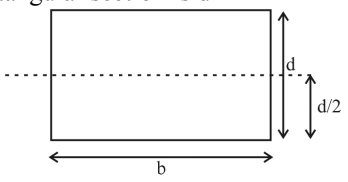
**Ans. (d) :** According to Hooke's law,
$\sigma \propto \in$
$\sigma = E \in$
So, $\boxed{E = \frac{\sigma}{\in}}$
So, modulus of elasticity (E) is defined as the ratio of compressive stress ($\sigma$) to compressive strain ($\in$).

**74. The mathematical expression for Modulus of resilience is :**
(a) Modulus of resilience = $\frac{\text{proof resilience}}{\text{volume of the body}}$
(b) Modulus of resilience = proof resilience × volume of the body
(c) Modulus of resilience = Proof resilience - volume of the body
(d) Modulus of resilience = Proof resilience + volume of the body

**Ans. (d) :** The mathematical expression for Modulus of resilience is $\frac{\text{proof resilience}}{\text{volume of the body}}$

**75. In rectangular section, let 'b' be the width and 'd' be the depth. Then the moment of inertia 'I' of the rectangular section about a line passive through the base is :**
(a) $I = \frac{bd^2}{6}$ (b) $I = \frac{bd^3}{12}$
(c) $I = \frac{bd^3}{3}$ (d) $I = \frac{bd^2}{9}$

**Ans. (c) :** Let breadth of rectangular section is b and depth of rectangular section is d




$I_{CG} = \frac{bd^3}{12}$
Moment of inertia of a rectangular sections about base
$= \frac{bd^3}{12} + bd\left(\frac{d}{2}\right)^2$
$= \frac{bd^3}{12} + \frac{bd^2}{4} = \frac{bd^3}{3}$

**76. Radius of gyration is defined as _____**

(a) the ratio of the moment of inertia and area
(b) the ratio area and moment of inertia
(c) the square root of the area by moment of inertia
(d) the square root of the moment of inertia by area

**Ans. (a) :** The radius of gyration of an area with respect to a particular axis is the square root of the moment of inertia divided by the area.
Radius of gyration describes the way in which the total cross-sectional area is distributed around its centroidal axis.

$I = Ar^2$

r =radius of gyration

$$r = \sqrt{\frac{I}{A}}$$

**77. A cantilever of length "L" is fixed at "A" and carrying a gradually varying load from zero at the free end "B" to "w" per unit length at the fixed end. Calculate the shear force by taking a section 'X' at a distance 'x' from the free end B.**

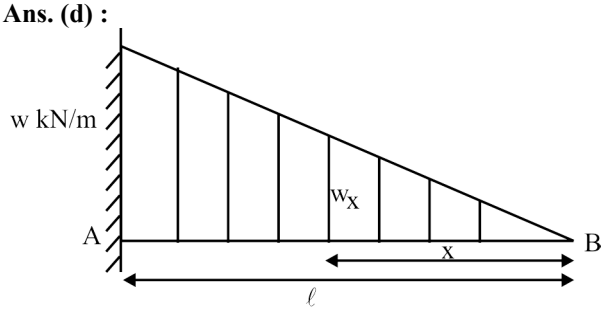
(a) Shear force = $\frac{wx}{L}$

(b) Shear force = $\frac{wx^2}{L}$

(c) Shear force = $\frac{wx}{2L}$

(d) Shear force = $\frac{wx^2}{2L}$

**Ans. (d) :**




$$\frac{w_x}{x} = \frac{w}{\ell}$$

$$w_x = \left(\frac{w}{\ell}\right)x$$

(SF) at distance x from B

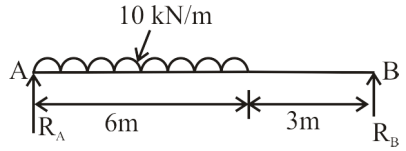
$$= \frac{1}{2}(x) \times \frac{wx}{\ell}$$

$$= \frac{1}{2}\frac{wx^2}{\ell}$$

**78. A simply supported beam AB is of length 9 m and carrying a UDL of 10 kN/m for a distance of 6 m from the left end. Calculate the reaction at support A.**

(a) 10 kN (b) 25 kN
(c) 30 kN (d) 40 kN

**Ans. (d) :**




$R_A + R_B = 60$ kN.....(1)

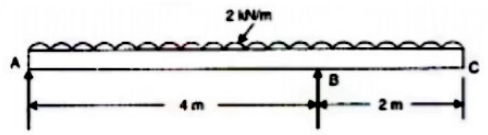
$\Sigma M_A = 0$

$9R_B = 60 \times 3$

$R_B = 20$ kN

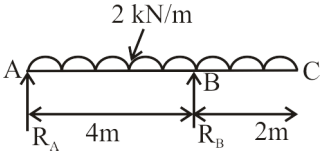
$R_A = 40$ kN

**79. An over-hanging beam is carrying a UDL of 2 kN/ m over the entire length as shown in the figure. Calculate the reaction at A.**




(a) 2 kN (b) 3 kN
(c) 5 kN (d) 10 kN

**Ans. (b) :**




$R_A + R_B = 12$ kN.....(1)

$\Sigma M_A = 0$

$4 \times R_B = 12 \times 3$

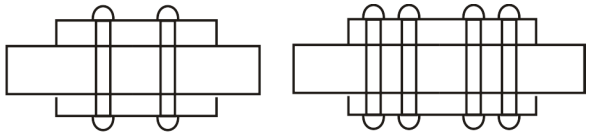
$R_B = 9$ kN

$R_A = 3$ kN

**80. In case of _____ the edges of the plates to be jointed together overlap each other.**

(a) single riveted butt joint
(b) double riveted butt joint
(c) triple riveted butt joint
(d) lap joint

**Ans. (d) :** In case of lap joint the edges of the plates to be jointed together overlap each other.



Single riveted butt joint Double riveted butt joint

• In the single riveted joint the plates to be joined are overlapped at the ends and a single row of rivets placed in the middle of the lap.

**81. The number of rivets (n) in a small joint are obtained from the relation :**

(a) $n = \frac{\text{Higher rivet value for shearing and bearing}}{\text{Pull to be transmitted across the joint}}$

(b) $n = \frac{\text{Pull to be transmitted across the joint}}{\text{Hihger rivet value for shearing and bearing}}$

(c) $n = \frac{\text{Least rivet value for shearing and bearing}}{\text{Pull to be transmitted across the joint}}$

(d) $n = \frac{\text{Pull to be transmitted across the joint}}{\text{Least rivet value for shearing and bearing}}$

**Ans. (d) :** In a small joint–
Number of rivets,

$$n = \frac{\text{Pull to be transmitted across the joint}}{\text{Least rivet value for shearing and bearing}}$$

$$\text{Number of rivets} = \frac{\text{Force}}{\text{Rivet value}}$$

**82. What is the disadvantage of welded connections?**

(a) Welded structure are comparatively lighter
(b) Welding takes less time
(c) Welding requires skilled labour
(d) A welded structure has a better finish and appearance

**Ans. (c) : Advantage of welded connection–**
- Welded structure are comparatively lighter
- Welding takes less time
- A welded structure has a better finish and appearance

**Disadvantage of welded connection–**
- Welded connection does not allow any form of expansion. Contraction in the connection could make it weak.
- Internal and external distortion can happen while the areas of connection are exposed to uneven heating during the process of welding.
- Welding requires skilled labour.

**83. If 't' is the size of weld, then what is the throat thickness of a fillet weld?**

(a) Throat thickness = 0.5 t
(b) Throat thickness = 0.707 t
(c) Throat thickness = 1.35 t
(d) Throat thickness = 1.7 t

**Ans. (b) :** j Thickness of throat (t) = K.S
where,
t = effective throat thickness
k = constant
S = Size of weld

$$\text{Throat thickness of fillet weld} = \frac{S}{\sqrt{2}} = 0.7075$$

| Angle between fusion face | 60° to 90° | 91° to100° | 101° to 106° | 107° to 113° | 114° to 120° |
|---|---|---|---|---|---|
| K | 0.7 | 0.65 | 0.60 | 0.55 | 0.5 |

**84. The planes which have no shear are known as :**

(a) shear planes (b) principal planes
(c) normal planes (d) inclined planes

**Ans. (b) : Principal planes:**
- These planes on which normal stresses acted alone (i.e. no shear stress) are called principal planes.

**Principal stresses-**
- Normal stresses acting on mutually perpendicular principal planes on which shear stress is zero are called principal stress.

**85. Which of the following is the correct formula to determine the power transmitted by the shafts (p)?**
**(where, T=mean torque transmitted and ω = angular speed of shaft)**

(a) $P = T + \omega$ (b) $P = T - \omega$
(c) $P = T \times \omega$ (d) $P = T \div \omega$

**Ans. (c) :** Power transmitted by the shaft(P)
= Mean torque transmitted × Angular speed of shaft

$$P = T \times w$$

Power transmitted by shaft subjected to torque (τ) is–

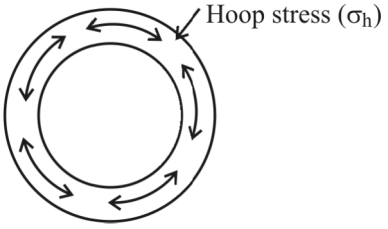
$$P = \frac{2\pi NT}{60} \text{ Watt}$$

Where, N = Number of revolutions per minute
T = Applied torque (N-m)

**86. Let 'p' be the internal fluid pressure, 'd' be the internal diameter of the thin cylinder and 't' be thickness of the wall of the cylinder. Then the circumferential stress ($\sigma_c$) in the thin cylinder is given by :**

(a) $\sigma_c = \frac{pd}{t}$ (b) $\sigma_c = \frac{pd}{2t}$
(c) $\sigma_c = \frac{pd}{3t}$ (d) $\sigma_c = \frac{pd}{4t}$

**Ans. (b) :** Hoop stress is also called circumferential stress as it is directed along the tangent to the circumference and is tensile in nature.




$$\text{Hoop stress, } \sigma_h = \frac{pd}{2t}$$

**87. Which of the following is the correct expression for factor of safety?**

(a) Factor of safety = Ultimate tensile stress + Permissible stress
(b) Factor of safety = Ultimate tensile stress – Permissible stress
(c) Factor of safety = Ultimate tensile stress × Permissible stress
(d) Factor of safety = $\frac{\text{Ultimate tensile stress}}{\text{Permissible stress}}$

**Ans. (d) :** The ratio between ultimate stress and Permissible or working stress is called factor of safety.

$$\text{Factor of safety} = \frac{\text{Ultimate tensile stress}}{\text{Permissible stress}}$$

**88. Determine the change in length of a steel bar, which is 4 m long, 30 mm wide and 20 mm thick, and is subjected to an axial pull of 30 kN in the direction of its length. Take E = 2 × $10^5$ N/$mm^2$ and Poisson's ratio = 0.3.**

(a) 1 mm (b) 3 mm
(c) 4.5 mm (d) 6 mm

**Ans. (a) :** Change in length of steel bar $(\delta\ell) = \frac{PL}{AE}$

Given,

L = 4×1000 mm
b = 30 mm
d = 20 mm
P = 30 kN = 30×$10^3$N
E = 2×$10^5$ N/$mm^2$

$$\delta\ell = \frac{30\times10^3\times4\times10^3}{30\times20\times2\times10^5} = 1\text{mm}$$

$\therefore \quad \delta\ell = 1\text{mm}$

**89. A steel rod is 2 m long and 50 mm diameter. An axial pull of 100 kN is suddenly applied to the rod. Calculate the instantaneous elongation produced in the rod. (Take E = 200 GN/$m^2$)**

(a) 1.018 mm (b) 2.867 mm
(c) 3.078 mm (d) 5.450 mm

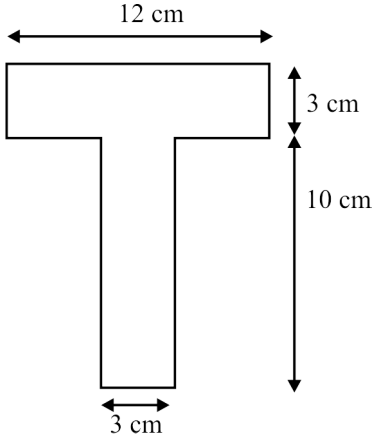
**Ans. (a) :** Length ($l$) = 2m = 2000mm
Diameter (d) = 50 mm
Pull load (P) = 100 kN = 100×$10^3$N

Elongation $(\delta l) = \frac{PL}{AE}$

$$(\delta l) = \frac{100\times10^3\times2000}{\frac{\pi}{4}(50)^2\times2\times10^5} = 0.509\,\text{m}$$

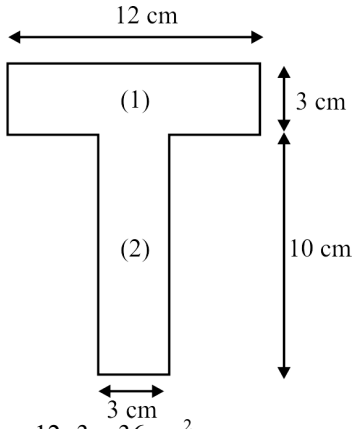
Instantaneous elongation = 2$\delta l$ = 2×0.509 = 1.018 mm

**90. Find the centre of gravity from the bottom of the T-section shown in the figure.**




(a) 5.56 cm (b) 6.78 cm
(c) 8.54 cm (d) 10.77 cm

**Ans. (c):**




$A_1$ = 12×3 = 36 $cm^2$
$y_1$ = 10 + 1.5= 11.5 cm
$A_2$ = 10×3 = 30 $cm^2$
$y_2$ = 5 cm

Taking x- axis as the reference axis

$$\bar{y} = \frac{A_1y_1 + A_2y_2}{A_1 + A_2}$$

$$= \frac{36\times11.5 + 30\times5}{30+36} = 8.54\,\text{cm}$$

**91. The failures due to tearing of the plate between the rivet hold and the edge may be avoided if the margin or distance between the centre of the rivet and nearest edge of the plate is _____ times the diameter or the rivet.**

(a) 1 (b) 1.5
(c) 2 (d) 3.5

**Ans. (b) :** The failures due to tearing of the plate between the rivet hold and the edge may be avoided if the margin or distance between the centre of the rivet and nearest edge of the plate is 1.5 times the diameter or the rivet.

**92. The tensile stresses at a point across two mutually perpendicular planes are 120 N/$mm^2$ and 60 N/$mm^2$, Determine the normal stress on a plane inclined at $30^0$ to the axis of the minor stress.**

(a) 99 N/$mm^2$ (b) 105 N/$mm^2$
(c) 203 N/$mm^2$ (d) 300 N/$mm^2$

**Ans. (b) :** $P_1$= 120 N/$mm^2$
$P_2$ = 60 N/$mm^2$
θ = 30

$$\text{Normal stress} = \frac{P_1+P_2}{2} + \left(\frac{P_1-P_2}{2}\right)\cos2\theta$$

$$= \frac{120+60}{2} + \left(\frac{120-60}{2}\right)\cos(2\times30)$$

= 90 + 30 × cos60
= 105 N/$mm^2$

**93. When a member is subjected to a direct stress (σ) in one plane, then the normal stress on an oblique plane (which is inclined at an angle θ with the normal cross-section) is given by :**

(a) Normal stress = $\sigma \cos^2\theta$

(b) Normal stress = $\frac{\sigma}{2} \sin^2\theta$

(c) Normal stress = $\sigma \cos\theta$

(d) Normal stress = $\sigma \sin^2\theta$

**Ans. (a) :** Member subjected to direct stress in one plane–

When a member is subjected to direct stress ($\sigma$) in one plane, the normal & shear stress on an oblique plane at an angle $\phi$ to the direction 1er of the direction of force.

Normal stress ($\sigma_n$) = $\sigma\cos^2\phi$

Tangential or shear stress $= \frac{\sigma}{2}\sin 2\phi$

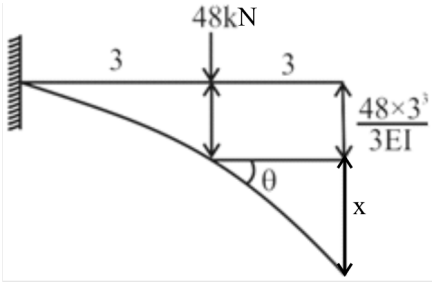
Where $\sigma = \frac{P}{A}$

**94. A cantilever of length 6 m carries a point load of 48 kN at its centre. The cantilever is propped rigidity at the free end.**
**Determine the reaction at the rigid prop.**

(a) 10 kN (b) 15 kN

(c) 20 kN (d) 35 kN

**Ans. (b) :**



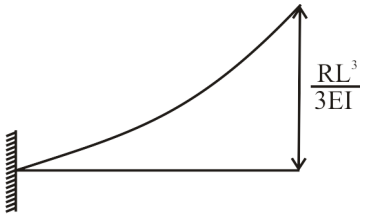

$\theta = \frac{x}{3}$

$x = 3\theta$

$$\frac{48\times3^3}{3EI} + \frac{48\times3\times3}{2EI}\times3 = \frac{RL^3}{3EI}$$

$$\frac{48\times3^3}{3EI} + \frac{48\times3\times3}{2EI}\times3 = \frac{R\times6^3}{3EI}$$

R = 15 kN




**95. Let 'd' be the dimater of circular section, then the modulus 'Z' for a circular section is :**

(a) $Z = \frac{\pi d^2}{64}$ (b) $Z = \frac{\pi d^3}{32}$

(c) $Z = \frac{\pi d^2}{6}$ (d) $Z = \frac{\pi d^3}{4}$

**Ans. (b) : Section modulus (Z)** – It is the ratio of moment of inertia (I) of the beam cross-section about the neutral axis of the distance ($y_{max}$) of extreme fiber from the neutral axis.

- In case of a circular section,

Section modulus (Z) = $\frac{I}{y_{max}}$

$I = \frac{\pi}{64}d^4$

y = distance from the centroid to top edge i.e. y = d/2

$$Z_{\text{circular section}} = \frac{I}{y_{max}} = \frac{\frac{\pi d^4}{64}}{\frac{d}{2}}$$

$$Z_{\text{circular section}} = \frac{\pi d^3}{32}$$

**96. A cylinderical vessel, whose ends are closed by means of rigid flange plates, is made of steel plates 3 mm thick. The length and the internal diameter of the vessel is 50 cm and 25 cm, respectively. Determine the longitudinal and hoop stresses in the cylinderical shell due to an internal fluid pressure of 3 N/mm$^2$**

(a) Longitudinal stress= 53.2 N/mm$^2$ and hoo stress = 100 N/mm$^2$

(b) Longitudinal stress = 60 N/mm$^2$ and Hoop stress = 110 N/mm$^2$

(c) Longitudinal stress = 62.5 N/mm$^2$ and Hoop stress = 125 N/mm$^2$

(d) Longitudinal stress = 70.9 N/mm$^2$ and Hoop stress = 132 N/mm$^2$

**Ans. (c) :** $\sigma_h = \frac{Pd}{2t}$

$$= \frac{3\times250}{2\times3}$$

$$= 125 \text{ N/mm}^2$$

$$\sigma_l = \frac{Pd}{4t} = \frac{3\times250}{4\times3}$$

$$= 62.5 \text{ N/mm}^2$$

**97. A solid shaft of 150 mm diameter is used to transmit torque. Find the maximum torque transmitted by the shaft if the maximum shear stress induced to the shaft is 45 N/mm$^2$.**

(a) 29820.58 N-m (b) 4600.77 N-m

(c) 5670.67 N-m (d) 8445.23 N-m

**Ans. (a) :** $\tau_{max} = \frac{16T}{\pi d^3}$

$$T = \frac{\tau_{max}\pi d^3}{16}$$

$$= \frac{45\times\pi\times150^3}{16}$$

$$= 29820.586 \text{ N-m}$$

**98. Which of the following statements if Incorrect regarding the assumptions made in Euler's Columns theory?**

(a) The column is initially perfectly straight and the load is applied axially.
(b) The cross-section of the column is uniform throughout its length
(c) The self-weight of the column is negligible
(d) The direct stress is very large as compared to the bending stress.

**Ans. (d) :** The following assumptions are made in Euler's column theory.

- The column is initially straight and load is applied axially.
- The cross-section of the column is uniform throughout its length.
- The column of material is perfectly elastic homogeneous and isotropic and obeys Hooke's law.
- The length of the column is very large as compared to its lateral dimension.
- The direct stress is very small as compared to the bonding stress.

**99. In columns, when both ends are fixed, then the relation between effective ($L_e$) and actual length ($\ell$) is given by**

(a) $L_e = \ell$ (b) $L_e = 2\ell$
(c) $L_e = \frac{\ell}{2}$ (d) $L_e = \frac{\ell}{\sqrt{2}}$

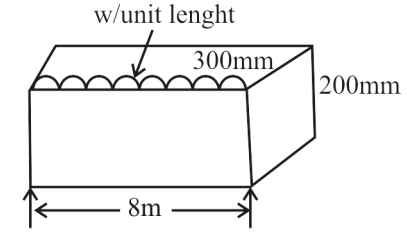
**Ans. (c) :**

| **Support conditions** | **Effective length (Left)** |
|---|---|
| Both ends hinged/pinned | $L_e = L$ |
| One end hinged other end fixed | $L_e = L/\sqrt{2}$ |
| Both ends fixed | $L_e = L/2$ |
| One end fixed and the other end free | $L_e = 2L$ |

**100. A rectangular beam 200 mm deep and 300 mm wide is simply supported over span of 8 m. What UDL per meter the beam may carry, if the bending stress is NOT to exceed 120 N/mm$^2$?**

(a) 10 kN/m (b) 15 kN/m
(c) 25 kN/m (d) 30 kN/m

**Ans. (d) :**




$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y}$$

$$M = \frac{wl^2}{8}, \quad \sigma = 120\,N/mm^2$$

$$\frac{M}{I} = \frac{\sigma}{y}$$

$$\frac{wl^2}{8.\frac{bd^3}{12}} = \frac{120}{d/2}$$

$$w = \frac{8\times 300\times 200^2\times 120}{8\times 8\times 6}$$

w = 30 kN/m

**101. Find the minimum diameter of a steel wire, which is used to raise a load of 4000 N, if the stress in the rod is NOT to exceed 95 MN/m$^2$.**

(a) 5.4 mm (b) 6.8 mm
(c) 7.32 mm (d) 9 mm

**Ans. (c) :** Given,

F = 4000N, $\sigma$ = 95 MN/m$^2$ = 95N/mm$^2$

$$\sigma = \frac{F}{A}$$

$$95 = \frac{4000}{A}$$

$$A = 42.10\ mm^2$$

Let the diameter is d

$$A = \frac{\pi}{4}(d)^2 \Rightarrow 42.10 = \pi\times\frac{d^2}{4}$$

$\therefore$ d = 7.32 mm

**102. Find Young's modulus of a brass rod of diamter 25 mm and of length 250mm which is subjected to a tensile load of 50 kN when the extension of the rod is equal to 0.3 mm.**

(a) 56.78 GN/m$^2$ (b) 68.66 GN/m$^2$
(c) 74.78 GN/m$^2$ (d) 84.88 GN/m$^2$

**Ans. (d) :** $\text{Stress} = \frac{\text{Force}}{\text{Area}} = \frac{50000N}{\frac{\pi}{4}\times(25)^2}$

Stress = 101.91 N/m$^2$

$$\text{Strain} = \frac{\text{Extension in rod}}{\text{Original length of rod}}$$

$$= \frac{0.3}{250} = 0.0012$$

$$\text{Modulus of elasticity} = \frac{\text{Stress}}{\text{Stain}}$$

$$= \frac{101.91}{0.0012} = 84.88\,GN/m^2$$

**103. A rod is 2 m long at a temperature of $10^0$C. Find the expansion of the rod when the temperature is raised to $80^0$C. Take E = 1.0 $\times 10^5$ MN/m$^2$ and $\alpha$ = 0.00012 per degree centrigrade.**

(a) 0.168 cm (b) 0.175 cm
(c) 0.196 cm (d) 1.23 cm

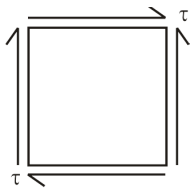
**Ans. (a) :** Given data-
• A rod of length L = 2m at a temp $T_1$ = 10ºC is heated to a final temp $t_2$ = 80ºC.
• Young's modulus of elasticity = $1\times10^5$ MN/m$^2$
$\alpha$ = 0.00012 per degree centimeter
Expansion of the rod = $\Delta L = L \times \alpha(T_2 - T_1)$
$= 2\times0.00012\times(80-10)$
= 0.168 cm

**104. Determine the bulk modulus of a material for which Young's modulus is $1.2 \times 10^5$ N/mm$^2$ and modulus of rigidity is $4.8 \times 10^4$ N/mm$^2$ and**
(a) $4 \times 10^4$ N/mm$^2$ (b) $5.6 \times 10^4$ N/mm$^2$
(c) $7.3 \times 10^4$ N/mm$^2$ (d) $8 \times 10^4$ N/mm$^2$

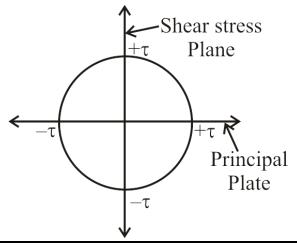
**Ans. (d) :** Given,
$E = 1.2 \times 10^5$ N/mm$^2$
$G = 4.8 \times 10^4$ N/mm$^2$
Relation between Young modulus and modulus of rigidity, $E = 2G(1+\mu)$
$1.2 \times 10^5 = 2 \times 4.8 \times 10^4 (1+\mu)$
$1+\mu = 1.25$
$\mu = 0.25$
Relation between bulk modulus and modulus of elasticity $k = \dfrac{E}{3(1-2\mu)} = \dfrac{1.2\times10^5}{3(1-2\times0.25)}$
$k = 8 \times 10^4$ N/mm$^2$

**105. When an element is subjected to simple shear stresses, then :**
(a) the planes of maximum normal stresses are parallel to each other
(b) the planes of maximum normal stresses are inclined at 60 degrees to the plane of pure shear
(c) one of the maximum normal stress is tensile while the other maximum normal stress is compressive
(d) the maximum normal stresses are of different magnitude and are not equal to the shear stress on the plane of pure shear

**Ans. (c) :**




$$\sigma_{P_1}/\sigma_{P_2} = \frac{\sigma_x+\sigma_y}{2} \pm \frac{1}{2}\sqrt{(\sigma_y-\sigma_x)^2+4\tau^2}$$
$\sigma_x = 0$, $\sigma_y = 0$
$$= \frac{0+0}{2} \pm \frac{1}{2}\sqrt{(0-0)^2+4\tau^2} = \pm\frac{1}{2}\times2\tau = \pm\tau$$




**106. Which of the following is the expression for modulus of resilience (where $\sigma$ = stress induced in the body, E= Young's modulus?**
(a) Modulus of resilience = $\dfrac{\sigma^2}{E}$
(b) Modulus of resilience = $\dfrac{\sigma^2}{2E}$
(c) Modulus of resilience = $\dfrac{\sigma^2}{2E}.V$
(d) Modulus of resilience = $\dfrac{\sigma^2}{E}.V$

**Ans. (b) :** Modulus of resilience = $\dfrac{\sigma^2}{2E}$
• It is defined as the proof resilience per unit volume.
Resilience is basically the energy stored by material within the elastic limit.
$$\text{Resilience} = \left(\frac{\sigma^2}{2E}\times V\right)$$
$$\frac{\text{Resilience}}{\text{Volume}} = \frac{\sigma^2}{2E}$$
Modulus of resilience = $\dfrac{\sigma^2}{2E}$

***107. A tensile load of 60 kN is gradually applied to circular bar of 4 cm diameter and 5 m length. If the value of $E = 2 \times 10^5$/mm$^2$, then determine the strain energy absorbed by the rod.**
(a) 20.67 N-m (b) 38.81 N-m
(c) 42.66 N-m (d) 50.90 N-m

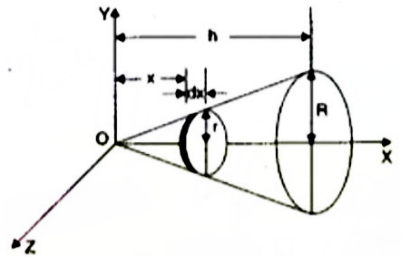
**Ans. (b) :** Given, D = 4 cm = 40 mm
L = 5m
$E = 2\times10^5$ N/mm$^2$
P = 60 kN
$$\text{Strain energy stored} = \frac{1}{2}P.\delta$$
$$= \frac{1}{2}\frac{P^2L}{AE}$$
$$= \frac{1}{2}\times\frac{(60\times1000)^2\times5000}{\frac{\pi}{4}(40)^2\times2\times10^5}$$
$$= 35.809 \approx 38.81 \text{ N-m}$$

**108. From the given two statements, which is/are valid in case of a centroid of simple plane figures.**
**Statement 1 : The centre of gravity of a triangle lies at the point where the three medians of the triangle meet.**
**Statement 2 : The centre of gravity of a rectangular or of a parallelogram is at the point, where its diagonals meet each other.**
(a) Statement 1 is true and statement 2 is false
(b) Statement 1 is false and statement 2 is true
(c) Both statements are true
(d) Both statements are false

**Ans. (c):** The centre of gravity of a triangle lies at the point where the three medians of the triangle meet.

- The centre of gravity of a rectangular or of a parallelogram is at the point, where its diagonals meet each other.

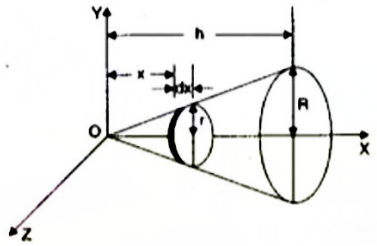
**109. A right circular cone of radius 'R' at the base and of height 'h' is placed as shown in the figure. Find the location of the centroid of the volume of the cone.**




(a) $\frac{h}{2}$ (b) $\frac{h}{4}$

(c) $\frac{3h}{2}$ (d) $\frac{3h}{4}$

**Ans. (d) :**




Given,
Radius of cone = R
Height of cone = h
Radius of the plate = r
Thickness of the plate = dx
Volume of the thin plate

$$dv = \pi r^2 \times dx$$

from similar $\Delta s$, , we get

$$\frac{R}{r} = \frac{h}{x}$$

$$r = \frac{Rx}{h}$$

$$dv = \pi\left(\frac{Rx}{h}\right)^2 dx$$

$$\overline{x} = \int \frac{x dv}{dv} = \frac{\int x\pi\left(\frac{Rx}{h}\right)^2}{\int \pi\left(\frac{Rx}{h}\right)^2 dx}$$

$$\frac{\frac{\pi R^2}{h^2}\int_0^h x^3 dx}{\frac{\pi R^2}{h^2}\int_0^h x^2 dx}$$

$$\left[\frac{x^4/4}{x^3/3}\right]_0^h = \frac{3h}{4}$$

**110. In double riveted lap joints, the diameter of rivet is 2 cm and pitch of rivets is 6 cm. If the tensile stress = 120 N/mm$^2$, shear stress = 90 N/mm$^2$ and crushing stress = 160 N/mm$^2$, then find the efficiency. The thickness of the plates is 1.2 cm.**

(a) 40.99% (b) 52.67%

(c) 65.45% (d) 79.08%

**Ans. (c) :** Shearing resistance of rivets per pitch length
$P_a = 2 \times n \times \pi/4 \times d^2 \times \tau_s$
For double riveted joints, n = 2

$$P_a = 2 \times 2 \times \frac{\pi}{4} \times (20)^2 \times 90$$

= 113.09 kN
Crushing strength per pitch length
$P_c = n \times d \times t \times \sigma_c = 2 \times 20 \times 12 \times 160$
= 768 kN
Tearing resistance per pitch length
$P_t = (p-d) \times t \times \sigma_t$
= (60–20)×12×120
= 576 kN
Strength of solid plate = $P \times t \times \sigma_t$
= 60×12×120
= 864 kN

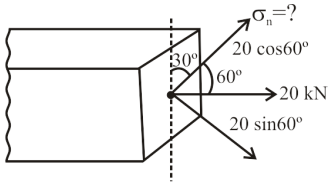
$$\eta = \frac{864}{576} = 66\%$$

**111. A rectangular bar of cross sectional area 10000 mm$^2$ is subjected to an axial load of 20 kN. Determine the normal stress on a section which is inclined at an angle of 30$^0$ with normal cross section of the bar.**

(a) 0.5 N/mm$^2$ (b) 1 N/mm$^2$

(c) 1.5 N/mm$^2$ (d) 3 N/mm$^2$

**Ans. (a) :** Given, A = 10000 mm$^2$
P = 20 kN
θ = 30º




$$\sigma_n = \frac{P\cos 60^\circ}{A}$$

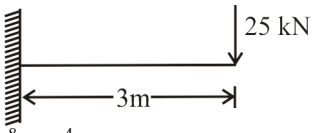
$$= \frac{20 \times 1000 \times \cos 60^\circ}{10000}$$

= 1 N/mm$^2$

**112. A cantilever of length 3 m is carrying a point load of 25 kN at the free end. If the moment of inertia of the beam = 10$^4$ mm$^4$ and value of E = 2.1 × 10$^5$ N/mm$^2$, then find the slope and deflection at the free end respectively.**

(a) 0.00234 rad and 6 mm
(b) 0.00453 rad and 8.64 mm
(c) 0.00535 rad and 10.71 mm
(d) 0.00747 rad and 6.89 mm

**Ans. (c):**




$I = 10^8\ mm^4$

$E = 2.1\times10^5\ N/mm^2$

$\theta = ?$

$\delta = ?$

$$\theta = \frac{PL^2}{2EI} = \frac{25\times1000\times(3\times1000)^2}{2\times2.4\times10^5\times10^8}$$

$= 0.00535$ rad

$$\delta = \frac{PL^3}{3EI} = \frac{25\times1000\times(3\times1000)^3}{3\times2.1\times10^5\times10^8}$$

$= 10.714$ mm

**113. Which of the following statements is incorrect regarding the assumptions made in the theory of simple bending?**

(a) The material of the beam is homogenous and isotropic

(b) The value of Young's modulus of elasticity is the same in tension and compression

(c) The transverse sections which were plane before bending, remain plane after bending also

(d) The radius of curvature is small compared with the dimensions of the cross-section

**Ans. (d) :** The following assumptions are made in the theory of simple bending–

- The material of the beam is perfectly homogenous isotropic.
- The value of E (young's modulus of elasticity) is the same in tension and compression.
- The transverse section which were plane before bending remains plane after bending.
- The radius of curvature is large compared with the dimensions of the cross-section.
- The beam is in equilibrium i.e. there is no resultant pull or push in the beam sections.
- The beam material is stressed within its elastic limit and thus obeys Hook's law.

**114. A steel plate of width 120 mm and thickness 20 mm is bear into a circular are of radius 10 m. Determine the maximum stress induced. Take $E = 2\times105\ N/mm^2$.**

(a) $100\ N/mm^2$ (b) $200\ N/mm^2$

(c) $350\ N/mm^2$ (d) $500\ N/mm^2$

**Ans. (b) :** Given,

$E = 200\ GPa = 200000\ MPa$

$B = 120\ mm$ & $D = 20\ mm$

$y = D/2 = 10\ mm$

$R = 10m = 10000\ mm$

$$\sigma = \frac{E}{R}.y = \frac{200000\times10}{10000} = 200\,MPa$$

**115. Which of the following statements is incorrect regarding the assumptions made in the derivation of shear stress produced in a circular shaft subjected to torsion?**

(a) The material of the shaft is uniform throughout

(b) The twist along the shaft is non-uniform

(c) Cross-sections of the shaft, which are plane before twist, remain plane after twist

(d) All radii which are straight before twist, remain straight after twist.

**Ans. (b) :** Assumptions made in the derivation of shear stress produced in a circular shaft subjected to torsion–

- The material of the shaft is uniform throughout
- The twist along the shaft is uniform
- Cross-sections of the shaft, which are plane before twist, remain plane after twist
- All radii which are straight before twist, remain straight after twist.

**116. When one end is fixed and the other is free, then what is the crippling load (P) for a column by Euler's formula? (where, 1 = actual length of the column, E=Young's modulus of the material of the column. I= least moment of inertia of the column?**

(a) $P = \frac{\pi^2 EI}{\ell^2}$ (b) $P = \frac{\pi^2 EI}{4\ell^2}$

(c) $P = \frac{4\pi^2 EI}{\ell^2}$ (d) $P = \frac{2\pi^2 EI}{\ell^2}$

**Ans. (b) :** Given condition-

One end fixed and other end free.

$$P_{cr} = \frac{\pi^2 EI}{(L_e)^2}$$

$Leff = 2L$

$$P_{cr} = \frac{\pi^2 EI}{(2L)^2} = \frac{\pi^2 EI}{4L^2}$$

**117. The MKS unit of dynamic viscosity is :**

(a) $\frac{dyne-sec}{cm^2}$ (b) $\frac{kgf-sec}{m^2}$

(c) poise (d) $Ns/m^2$

**Ans. (b) :** Viscosity can be defined as the measure of a fluid's resistance to deformation at a given flow–

$$\text{Shear stress} = \mu.\frac{du}{dy}$$

Where,

$\mu$ = viscosity of fluid

$\frac{du}{dy}$ = rate of shear deformation

$$\text{Kinematic viscosity} = \frac{\text{dynamic viscosity}}{\text{density of fluid}}$$

Dynamic viscosity ($\mu$)

SI unit = $NS/m^2$ or Pa-S

CGS unit = Poise

MKS unit = $\frac{\text{kgf - sec}}{\text{m}^2}$

1 poise = 0.1 Pa-S

Kinematic viscosity (ν)

MKS unit = SI unit = $m^2/s$

CGS unit = Stokes or $cm^2/s$

1 Stoke = $10^{-4}$ $m^2/s$

**118. The devices which are used for measuring the pressure by balancing the fluid column by the spring or dead weight are called :**

(a) simple manometers
(b) differential manomteres
(c) mechanical gauges
(d) piezometers

**Ans. (c) :** Mechanical gauges are defined as the device used for measuring the pressure by balancing the fluid column by the spring or dead weight.

The commonly used mechanical pressure gauges are–

- Diaphragm pressure gauge.
- Bourdon tube pressure gauge
- Dead-weight pressure gauge
- Bellows pressure gauge

**119. A device used for measuring the velocity of flow at any point in a pipe or channel is called a/an:**

(a) pitot tube (b) orifice meter
(c) venutrimeter (d) piezometer

**Ans. (a) :** Pitot tube is a device used for calculating the velocity of flow at any point in a pipe or a channel.

- The pitot tube is used to measure velocity at a point.
- Pitot tube is based on the principle that if the velocity of flow at a point becomes zero, the pressure is increased due to the conversion of the kinetic energy into pressure energy.

**120. From the following options, select the INCORRECT statement regarding the advantages of triangular notch over rectangular notch.**

(a) The expression for discharge for a right angled V-notch is very simple
(b) For measuring low discharge, a triangular notch gives more accurate results than a rectangular notch.
(c) In case of triangular notch, H (depth) and angle of notch are required for the computation of discharge.
(d) Ventilation of a triangular notch is necessary

**Ans. (d) : Advantages of triangular notch over rectangular notch –**

- The expression for discharge for a right angled V-notch is very simple
- For measuring low discharge, a triangular notch gives more accurate results than a rectangular notch.
- In case of triangular notch, H (depth) and angle of notch are required for the computation of discharge.
- Ventilation of a triangular notch is not necessary.

**121. A_____ is a device used for measuring the rate of flow of a liquid through a small channel or a tank.**

(a) weir
(b) notch
(c) nappe
(d) crest

**Ans. (b) :** A notch is a device used for measuring the rate of flow of a liquid through a small channel of the tank.

- Weirs can be used to measure flow rates in open channels and rivers.
- Venturimeter is a device to measure the rate of flow of fluid flowing through a pipe.

**122. Let 'V' be the velocity at outlet of pipe and 'g' be the acceleration due to gravity, then the loss of head at the exit of pipe is :**

(a) $h_o = \frac{V^2}{g}$ (b) $h_o = \frac{V^2}{2g}$

(c) $h_o = 0.5\frac{V}{g}$ (d) $h_o = 0.5\frac{V}{2g}$

**Ans. (b) :** Head loss at exit

$$h_{Le} = \frac{V^2}{2g}$$

Head loss due to expansion

$$h_{Le} = \frac{(V_1 - V_2)^2}{2g}$$

Head loss at entry

$$h_{Le} = \frac{0.5V_1^2}{2g}$$

Where,

$V_1$ = Velocity at inlet

g = acceleration due to gravity

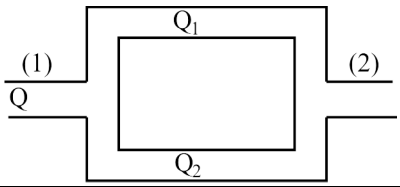
$V_2$ = Velocity at outlet

**123. When three or more reservoirs are connected by means of pipes, having one or more junction, then the system is called a :**

(a) series pipe system
(b) parallel pipe system
(c) branched pipe system
(d) connected pipe system

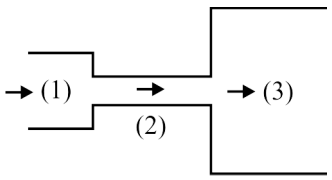
**Ans. (c) :** When three or more reservoirs are connected by means of pipes, having are or more junction, then the system is called a branched pipe system.

**Parallel pipe–** The main pipelines divides into two or more parallel pipes which again join together downstream.



**Series pipes–**
Pipes in series or compound pipes are defined as the pipes of different length and different diameters connected end to end.

Pipes in series




$Q_1 = Q_2 = Q_3$

**124. The ratio of inertia force of a flowing fluid and the viscous force of the fluid is known as :**
(a) Euler's number
(b) Froude number
(c) Reynolds number
(d) Weber number

**Ans. (c) :**

$$\text{Renolds number} = \frac{\text{Inertia force}}{\text{Viscous force}}$$

$$\text{Eulers number} = \sqrt{\frac{\text{Inertia force}}{\text{Pressure force}}}$$

$$\text{Mach number} = \sqrt{\frac{\text{Inertia force}}{\text{Elasticity force}}}$$

$$\text{Froude number} = \sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}}$$

$$\text{Weber number} = \sqrt{\frac{\text{Inertia force}}{\text{Surface tension force}}}$$

**125. Calculate the specific weight of one litre of petrol of specific gravity 0.7.**
(a) 7857 N/m$^2$ (b) 7000 N/m$^2$
(c) 6867 N/m$^2$ (d) 6000 N/m$^2$

**Ans. (c) :** Specific gravity = 0.7
Density of water ($\rho_w$) = 1000 kg/m$^3$
Specific weight = $\rho_w \times g \times$ specific gravity
= 1000 × 9.81 × 0.7
Specific weight = 6867 N/m$^2$

**126. The temperation lapse rate is denoted by L and hence L = $L = \frac{-g}{R}\left(\frac{K-1}{K}\right)$, where 'g', 'K' and 'R' are constant. In the expression if K > 1, then the lapse rate is :**
(a) zero which means temperature is constant
(b) positive which means temperature increase
(c) negative which means temperature increases
(d) zero and no change in the temperature

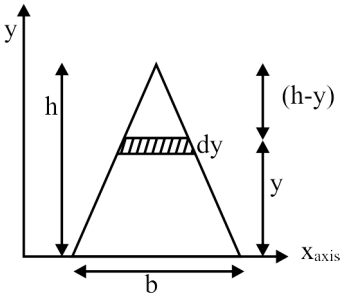
**Ans. (c) :** Temperature lapse rate

$$L = \frac{-g}{R}\left(\frac{k-1}{k}\right)$$

Where, g, k & R are constant.
If k > I → Lapse rate is negative which means temperature is increases
K < 1 ⇒ Lapse rate is positive which means temperature is decreases.
k = 0 → Lapse rate is zero which means temperature is constant.

**127. For triangle, the moment of inertia about the base ($I_O$) is (where, b= width of triangle, h = height of triangle)**
(a) $\frac{bh^3}{3}$ (b) $\frac{bh^3}{12}$
(c) $\frac{bh^3}{36}$ (d) $\frac{bh^3}{64}$

**Ans. (b) :**




$dA = xdy$

From similar Δ

$$\frac{x}{b} = \frac{h-y}{h}$$

$$x = \frac{b}{h}(h-y)$$

$$I_x = \int y^2 dA = \int_0^h y^2 x dy$$

$$I_x = \int_0^h y^2 \frac{b}{h}(h-y)dy$$

$$I_x = \frac{b}{h}\int_0^h \left(y^2 h - y^3\right)dy$$

$$I_x = \frac{b}{h}\left|\frac{y^3 h}{3} - \frac{y^4}{4}\right|_0^h$$

$$= \frac{b}{h}\left(\frac{h^3}{3} - \frac{h^4}{4}\right)$$

$$= \frac{bh^4}{h}\left(\frac{1}{3} - \frac{1}{4}\right) = bh^3\left(\frac{1-3}{12}\right)$$

$$I_x = \frac{bh^3}{12}$$

**128. A flow in which the velocity at any given time does NOT change with respect to space is known as :**
(a) laminar flow
(b) turbulent flow
(c) uniform flow
(d) non-uniform flow

**Ans. (c): Uniform flow–** If the velocity vector at all the points in the flow is the same at any instant of time then the flow is uniform flow.

**Steady flow–** If the fluid and flow characteristics at a point may change with time, then the flow in unsteady.

- Laminar flow occurs when the fluid flows in infinites parallel layer with no disturbance between them.

**Uniform flow–**.If for a given length of the channel, the velocity of flow, depth of flow, slope of the channel and cross-section remains constant, the flow is said to be uniform flow.

**129. Water is flowing through a pipe of 5 cm diameter under a pressure of 29.43 N/cm$^2$ (gauge) and with mean velocity of 2 m/s. Find the total head or total energy per unit weight of the water at a cross section, which is 5 m above the datum line.**

(a) 20.05 m
(b) 25.78 m
(c) 35.20 m
(d) 40.80 m

**Ans. (c) :** $H = \frac{P}{\rho g} + Z + \frac{V^2}{2g}$

$= \frac{29.43\times10^4}{9.81\times1000} + 5 + \frac{2^2}{2\times9.81}$

$\frac{P}{\rho g} = \frac{29.43\times10^4 N}{9.81\times1000 m^3}$

= 35.203 m

**130. Which of the following statements is/are true regarding an orifice meter?**

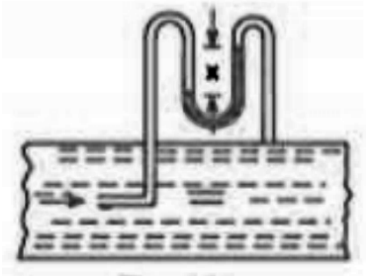
(A) Orifice meter is a device used for measuring the rate of flow of fluid through a pipe.
(B) Orifice meter works on the same principle as that of venturimeter
(C) In this, the orifice diameter is kept generally 0.5 times the diameter of the pipe, through it may vary from 0.4 to 0.8 times the pipe diameter.

(a) Only statement (A)
(b) Both statement (A) & (B)
(c) Both statement (B) & (C)
(d) All statement (A) (B) & (C)

**Ans. (d) : Orifice meter–**

- Orifice meter is a device used for measuring the rate of flow of fluid through a pipe.
- Orifice meter works on the same principle as that of venturimeter
- In this, the orifice diameter is kept generally 0.5 times the diameter of the pipe, through it may vary from 0.4 to 0.8 times the pipe diameter.

**131. Identify the following arrangement which is used to determine the velocity at any point in a pipe by pitot-tube.**



(a) Pitot-tube along with a vertical piezometer tube
(b) Pitot-tube connected with piezometer tube
(c) Pitot-tbe and vertical piezometer tbe connected with a differential U-tube manometer
(d) Pitot-static tube, which consists of two circular concentric tubes one side, the other with some annular space in between and outlet of these two tubes, are connected to the differential manometer.

**Ans. (c) :** Pitot-tbe and vertical piezometer tbe connected with a differential U-tube manometer .

**132. A pitot-static tube is used to measure the velocity of water in a pipe. The stagnation pressure head is 6 m and static pressure head is 5 m. Calculate the velocity of flow assuming the co-efficient of tube = 0.98.**

(a) 2.45 m/s (b) 3.56 m/s
(c) 4.34 m/s (d) 5.55 m/s

**Ans. (c) :** Given,
$h_s$= 6m
$h_t$ = 5m
$h = h_s - h_t = 6–5 = 1m$
The velocity of flow $(V) = Cd\sqrt{2\times9.81\times1}$
Where, h = pressure head difference.
$\therefore$ The velocity of flow (V) $= 0.98\sqrt{2\times9.81\times1}$
$\therefore$ V = 0.98×4.42
= 4.34 m/s

**133. For a triangular weir, the error in discharge due to the error in the measurement of head is given by (where, Q = discharge, H=head over the notch) :**

(a) $\frac{dQ}{Q} = \frac{1}{2}\frac{dH}{H}$ (b) $\frac{dQ}{Q} = \frac{3}{2}\frac{dH}{H}$
(c) $\frac{dQ}{Q} = \frac{5}{2}\frac{dH}{H}$ (d) $\frac{dQ}{Q} = \frac{7}{2}\frac{dH}{H}$

**Ans. (c) :**

For rectangular weir the error in discharge due to the error in the measurment of head.

$\frac{dQ}{Q} = \frac{3}{2}\frac{dH}{H}$

- For a triangular weir

$\frac{dQ}{Q} = \frac{5}{2}\frac{dH}{H}$

**134. Hydraulic gradient line is defined as the line which gives :**

(a) the sum of pressure head, datum head and kinetic head of flowing fluid in pipe with respect to some reference line

(b) the difference of pressure head, datum head and kinetic head of a flowing fluid in a pipe with respect to some reference line

(c) the sum of pressure head and datum head of flowing fluid in a pipe with respect to some reference line

(d) the difference of pressure head and datum head of a flowing fluid in a pipe with respect to some reference line

**Ans. (c) : Total Energy Line (T.E.L)**

• It is defined as the line which gives the sum of pressure head ($P/\rho g$), Kinetic head ($V^2/2g$) and datum (Z)

$$\text{T.E.L.} = \frac{P}{\rho g} + \frac{V^2}{2g} + Z$$

**Hydraulic Gradient Line (H.G.L.)**

• It is defined as the line which gives the sum of pressure head ($P/\rho/g$) and datum (Z)

$$\text{H.G.L} = \frac{P}{\rho g} + Z$$

**135. When a main pipe divides into 2 branches and again joins together downstream to form a single pipe, then the branch pipes are said to be connected in parallel. The rate of flow in the main pipe is equal to**

(a) the difference of the rate of flow through the branch pipes

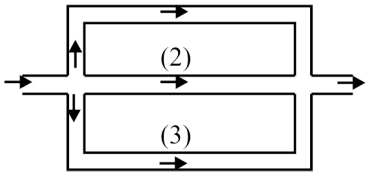
(b) the sum of the rate of flow through the branch pipes

(c) the product of the rate of flow through the branch pipes

(d) the rate of flow through one branch pipe by the rate of flow through the second branch pipe

**Ans. (b) :** If the main pipe divides into two or more branches and again join together downstream to form a single pipe, then the branched pipes are said to be connected in parallel pipe.

(1) Pipes in parallel




$Q_{total} = Q_1 + Q_2 + Q_3$

$h_{L_1} = h_{L_2} = h_{L_3}$

• The rate of flow in the main pipe is equal to the sum of the rate of flow through branch pipes.

• The loss of head for each branch pipe is the same.

**136. A nozzle is fitted at the end of a pipe of length 300 m and of diameter 100 mm. For the maximum transmission of power through the nozzle, find the diameter of the nozzle. (Take co-efficient of friction f= 0.009)**

(a) 24.78 mm (b) 26.08 mm

(c) 34.77 mm (d) 45.9 mm

**Ans. (b) :** Given, D = 10 cm = 0.1 m

$l$ = 300m

f = 0.009

$$d^4 = \frac{D^5}{8fl} = \frac{(0.1)^5}{8 \times 0.009 \times 300}$$

d = 0.02608 m

d = 26.08 mm

**137. Which of the following is the correct expression for retarding force in gradual closure of valve of water hammer in pipes?**

(a) Retarding force = Mass ÷ Retardation

(b) Retarding force = Mass × Retardation

(c) Retarding force = Mass + Retardation

(d) Retarding force = Mass - retardation

**Ans. (b) :** A retarding force causes the acceleration of an object to be negative.

F = m.a

Where, F is the resultant force, the force acts against the direction of the object's current velocity is the retarding force.

∴ Retarding force = Mass × Retardation

**138. Water is flowing with a velocity of 1.5 m/s in a pipe of length 2500 m and of diameter 500 mm. At the end of the pipe, a valve is provided. Find the rise in the pressure if the valve is closed in 25 seconds. Take the value of C (velocity of pressure wave) = 1460 m/s.**

(a) 12.5 $N/cm^2$ (b) 13 $N/cm^2$

(c) 13.9 $N/cm^2$ (d) 15 $N/cm^2$

**Ans. (d) :** Given,

V = 1.5 m/s, L = 2000 m, D = 500 mm = 0.5 m

C = 1460 m/s and t = 25 sec.

Reference time

$$T = \frac{2L}{L}$$

$$T = \frac{2 \times 2500}{1460} = 3.425 \text{sec}$$

t' = 25 sec

t' > T → It corresponds to the gradual closure of the value.

For gradual closure

$$p = \frac{\rho V L}{t'}$$

$$p = \frac{1000 \times 1.5 \times 2500}{25}$$

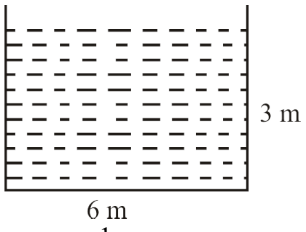
$p = 15 \times 10^4$ $N/m^2$

p = 15 $N/cm^2$

**139. Find the velocity of flow of water through a rectangular channel, 6 m wide and 3 m deep, when it is running full. The channel is having bed slope as 1 in 2000. (Take Chezy's constant C = 55.)**

(a) 1.5 m/s (b) 2 m/s
(c) 4.5 m/s (d) 6.7 m/s

**Ans. (a) :** Rectangular channel
Wide b = 6 m
Depth d = 3m




Channel bed slope $= \frac{1}{2000}$
Chezy's constant C = 55
Now, we know that

$$V = C\sqrt{mi}$$

Where, i = Slope
m = Hydraulic radios

$$m = \frac{A}{P} = \frac{6\times3}{6+3+3} = \frac{18}{12} = \frac{3}{2}$$

Put all value in $V = C\sqrt{mi}$

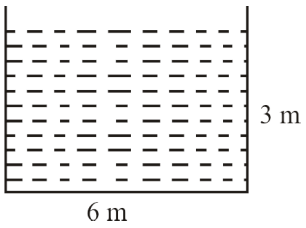
$$V = 55\sqrt{\frac{3}{2}\times\frac{1}{2000}}$$

V = 55 × 0.0273
V = 1.5 m/s

**140. Find the critical velocity of the water flowing through a rectangular channel of width 5 m, when the discharge is 15 $m^3/s$.**

(a) 2.55 m/s (b) 3.08 m/s
(c) 5.76 m/s (d) 6.8 m/s

**Ans. (b) :** V = ?




$$i = \frac{1}{2000}, C = 55$$

$$V = C\sqrt{mi} \qquad m = \frac{6\times3}{12} = \frac{18}{12} = \frac{3}{2}$$

$$V = 55\sqrt{\frac{3}{2}\times\frac{1}{2000}}$$

V = 55 × 0.0273
V = 1.5 m/s

$$y_C = \left(\frac{3^2}{9.81}\right)^{1/3}$$

$y_C$ = 3.08 m/s

**141. Find the kinematic viscosity of an oil having density 981 $kg/m^3$. The shear stress at a point in the oil is 0.2452 $N/m^2$ and velocity gradient at that point is 0.2 per second.**

(a) 15 stokes (b) 14.2 stokes
(c) 13 stokes (d) 12.5 stokes

**Ans. (d) :** Given data-

$$V = ?$$
$$g = 981\ kg/m^3$$
$$\tau = 0.2452\ N/m^2$$
$$\frac{du}{dy} = 0.2/\sec$$

$\because$ Shear stress is given by-

$$\tau = u.\frac{du}{dy}$$
$$0.2452 = u \times 0.2$$
$$u = \frac{0.2452}{0.2} = 1.226\ N^5/m^2$$

$\because$ Density is given by

$$\rho = \frac{u}{V}$$
$$\therefore V = \frac{u}{\rho} = \frac{1.226}{951}$$

$\therefore$ V = $0.125\times10^{-2}$ $m^2/s$
$\therefore$ V = 12.5 Stokes

**142. Calculate the gauge pressure and absolute pressure at a point 3 m below the free surface of a liquid having a density of $1.53 \times 10^3$ $kg/m^3$ if the atmospheric pressure is equivalent to 750 mm of mercury. The specific gravity of mercury is 13.6 and density of water is 1000 $kg/m^3$.**

(a) 25870 $N/m^2$ and 125080 $N/m^2$, respectively
(b) 30070 $N/m^2$ and 120000 $N/m^2$, respectively
(c) 38902 $N/m^2$ and 135080 $N/m^2$, respectively
(d) 45028 $N/m^2$ and 145090 $N/m^2$, respectively

**Ans. (d) :** Given,
h = 3m, $\rho_{liq} = 1.53\times10^3$ $kg/m^3$
$h_{atm}$ [750 mm of mercury (Hg)] = 0.75 m
Specific gravity ($S_{Hg}$) = 13.6
Density ($\rho$) = 13.6×1000 $kg/m^3$
$P_{atm} = \rho.g.h_{atm} = 13.6\times1000\times9.81\times0.75$
$P_{atm}$ = 100,062 $N/m^2$
$P_{gauge} = (\rho.g.g)_{liquid} = 1.53\times10^3\times9.81\times3$
$P_{gauge}$ = 45027.9 $N/m^2 \approx$ 45028 $N/m^2$
$P_{abs} = P_{atm} + P_{gauge}$
= 100062 + 45027.9
$P_{abs}$ = 145090 $N/m^2$

**143. In U-tube differential manometers, consider the two points A and B are at different levels and also contains liquids of different specific gravity. These points are connected to the U-tube differential manometer. Let the pressure at A and B be PA and PB, h = difference of mercury level in the U-tube, y = distance of the**

**centre of B from the mercury level in the right limb, x= distance of the centre of A from the mercury level in the right limb, $\rho_1$ = density of liquid at A, $\rho_2$ = density of liquid at B, $\rho_2$ = density of heavy liquid or mercury. What is the difference of pressure at A and B?**

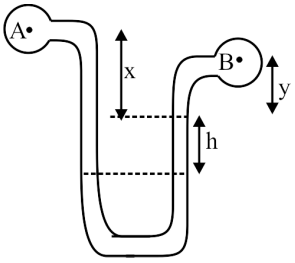
(a) $P_A - P_B = g.h\left(\rho_g - \rho_l\right)$

(b) $P_A - P_B = g.h\left(\rho_g - \rho_l\right) - \rho_l gx$

(c) $P_A - P_B = g.h\left(\rho_g - \rho_l\right)\rho_2 gy - \rho_l gx$

(d) $P_A - P_B = g.h\left(\rho_g - \rho_l\right) + \rho_2 gy - \rho_l gx$

**Ans. (d) :**




$$P_A + \rho_l gx + \rho_l gh = P_B + \rho_2 gy + \left(\rho_g\right)gh$$

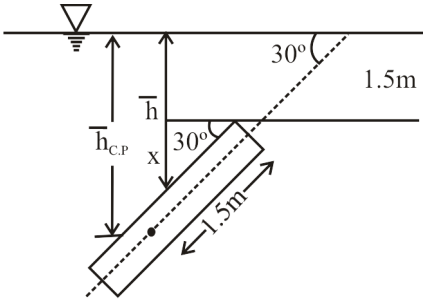
$$P_A - P_B = gh\left(\rho_g - \rho_l\right) + \rho_2 gy - \rho_l gx$$

**144. A rectangular plane surface, 2 m wide and 3 m deep, lies in water in such a way that its plane makes an angle of $30^0$ with the free surface of water. Determine the position of the centre of pressure when the upper edge is 1.5 m below the free water surface.**

(a) 1.56 m (b) 2.33 m

(c) 3.98 m (d) 5.80 m

**Ans. (b) :**




$$\sin 30^o = \frac{x}{1.5}$$

$$\frac{1}{2} \times 1.5 = x$$

$$x = 0.75m$$

$$\bar{h} = \bar{x} + 1.5 = 0.75 + 1.5$$

$$= 2.25m$$

$$\bar{h}_{C.P} = \bar{h} + \frac{I_{GG}\sin^2\theta}{A\bar{h}}$$

$$= \bar{h} + \frac{bd^3\sin^2\theta}{12 \times bd.\bar{h}}$$

$$= \bar{h} + \frac{d^2\sin^2\theta}{12\bar{h}}$$

$$= 2.25 + \frac{(3)^2\sin^2(30^o)}{12 \times 2.25}$$

$$= 2.33m$$

**145. A venturi meter having inlet diameter of 30 cm, throat diameter of 15 cm is inserted in a vertical pipe carrying water, flowing in the upward direction. A differential mercury manometer connected to the inlet and throat gives a reading of 20 cm. Find the discharge. Take co-efficient of discharge= 0.98.**

(a) 125.75 litre/second

(b) 137.89 litre/second

(c) 156.99 litre/second

(d) 170.67 litre/second

**Ans. (a) :** In the venturimeter,

Rate of flow

$$Q = \frac{C_d.A_1.A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}$$

Here, $h = 20\left(\frac{\rho_m}{\rho} - 1\right)$

Here, $h = 20(13.6–1) = 252$ cm

$$\therefore Q = \frac{0.98 \times \pi/4 \times (30)^2 \times \pi/4 \times (15)^2}{\sqrt{\left[\pi/4 \times 30^2\right]^2 - \left[\pi/4 \times 15^2\right]^2}} \times \sqrt{2 \times 9.81 \times 252}$$

$$\therefore Q = \frac{0.98 \times (30)^2 \times \pi/4 \times (15)^2}{\sqrt{(30)^4 - (15)^4}} \times \sqrt{2 \times 981 \times 252}$$

$\therefore Q = 125.75$ liter/sec.

**146. Determine the height of a rectangular weir of length 6 m to be built across a rectangular channel. The maximum depth of water on the upstream side of the weir is 1.8 m and discharge is 2000 litres/second. Take co-efficient of discharge = 0.6 and neglect end contractions.**

(a) 0.328 m (b) 1.8 m

(c) 1.472 m (d) 2.54 m

**Ans. (c) :** Given,

Length of weir (L) = 6m

Depth of weir ($H_1$) = 1.8 m

Discharge (Q) = 2000 lit/sec = 2 $m^3$/sec

Coefficient of discharge ($C_d$) = 0.6

Let its height of water above the crest of weir

Then,

The discharge is given by the eq$^n$-

$$Q = \frac{2}{3}C_d.L\sqrt{2g}.H^{3/2}$$

$$Q = \frac{2}{3} \times 0.6 \times 6 \times \sqrt{2 \times 9.81} \times H^{3/2}$$

$$2 = \frac{2}{3} \times 0.6 \times \sqrt{2 \times 9.81} \times H^{3/2}$$

$$H^{3/2} = \left(\frac{2}{10.623}\right)^{2/3} \quad \therefore H = 0.328\text{m}$$

Height of weir ($H_2$) = $H_1 - H$
= 1.8 – 0.328
= 1.472 m

$\therefore$ Height of the rectangular weir is 1.472 m

**147. An oil of specific gravity 0.7 is flowing through a pipe of diameter 300 mm at the rate of 500 litres/s. Find the head lost due to friction and if the flow for a length of 1000 m. Take viscosity = 0.29 stokes.**

(a) 163.18 m (b) 174.78 m
(c) 180.99 m (d) 200 m

**Ans. (a) :** Given,

Kinematic viscosity, $\nu = 0.29 \times 10^{-4}$ m$^2$/s

$\ell$ = 1000 m, dia (d) = 0.3m, Q = 0.5 m$^3$/s

$$V = \frac{Q}{A}$$

$$= \frac{0.5 \times 4}{\pi \times 0.3^2} = 7.073\,\text{m/s}$$

Reynold's no,

$$R_e = \frac{Vd}{\nu}$$

$$= \frac{7.073 \times 0.3}{0.29 \times 10^{-4}} = 7.31 \times 10^4$$

$R_e$ lies between 4000 to 10000

$$F = \frac{0.079}{R_e^{1/4}} = 0.0048$$

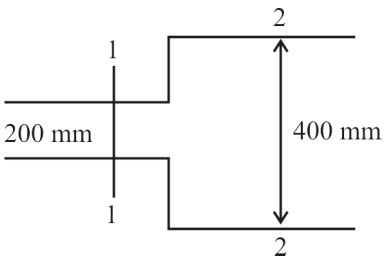
$$h_F = \frac{4F\ell V^2}{2gd}$$

$$= \frac{4 \times 0.0048 \times 1000 \times 7.073^2}{2 \times 9.81 \times 0.3} = 163.18\text{ m}$$

**148. Find the loss of head when a pipe of diameter 200 mm is suddenly enlarged to a diameter of 400 mm. The rate of flow of water through the pipe is 250 litres/s.**

(a) 1.234 m of water (b) 1.816 m of water
(c) 2.456 m of water (d) 2.987 m of water

**Ans. (b) :**




$h_L$ = ?

$$Q = 250\text{ Litres/sec} = \frac{250}{1000}\text{m}^3/\text{sec} = 0.25\text{ m}^3/\text{sec}$$

$$A_1 = \frac{\pi}{4}(0.2)^2 = 0.03141\text{m}^2$$

$$A_2 = \frac{\pi}{4}(0.4)^2 = 0.12566\,\text{m}^2$$

$Q = A_1V_1$

$$V_1 = \frac{0.25}{\frac{\pi}{4}(0.2)^2} = 7.9577\,\text{m/sec}$$

$$h_L = \frac{V_1^2}{2g}\left[1 - \frac{A_1}{A_2}\right]^2$$

$$h_L = \frac{7.957}{2 \times 9.81}\left[1 - \frac{0.03141}{0.12566}\right]^2 = 1.815\text{m}$$

**149. Which of the following statements is/are true regarding use of a syphon?**

(A) To carry water from one reservoir to another reservoir separated by a hill or ridge
(B) To take out the liquid from a tank which is not having outlet
(C) To empty a channel provided with outlet sluice

(a) Only statement A
(b) Both statement A and B
(c) Both statements B and C
(d) All statements A, B and C

**Ans. (b) : Syphon is used in following cases-**

(a) To take out water from one reservoir to another reservoir separated by a hill or ridge.

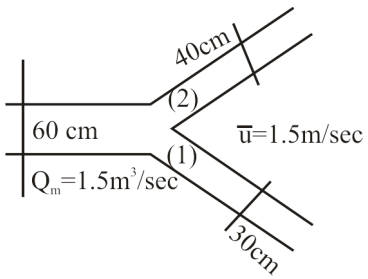
(b) To drain out water from a channel without any outlet.

(c) To take out the water from a tank which does not have any outlet.

**150. A pipe line 60 cm in diameter bifurcates at a Y-junction into two branches of 40 cm and 30 cm in diameter each. If the rate of flow in the main pipe is 1.5 m/s and mean velocity of flow in the 30 cm diameter pipe is 7.5 m/s, determine the rate of flow in the 40 cm diameter pipe.**

(a) 0.97 m$^3$/s (b) 1.23 m$^3$/s
(c) 1.78 m$^3$/s (d) 3.4 m$^3$/s

**Ans. (a) :**




$Q_m = Q_1 + Q_2$
$= A \times \bar{u} + Q_2$

$$1.5\text{m}^3/\text{sec} = \frac{\pi}{4}(0.3)^2 \times 7.5\text{m/sec} + Q_2$$

$Q_2 = 0.9698 \sqcup 0.97$ m$^3$/sec

# Telangana State Public Service Commission
# Assistant Executive Engineer Exam, 2018
## Solved Paper

**1. Identify the incorrect statement**
**In a Lacey regime, channel carrying a discharge Q in an alluvial of silt factor f**
(a) the area of flow is function of Q only
(b) the wetted perimeter is a function of Q only
(c) the side slopes are independent of Q and f
(d) the longitudinal slope is a function of Q and f

**Ans. (a) :** According to Lacey's the design formula to build a canal as follows –
- Silt factor (f) = $1.76\sqrt{d_{mm}}$
- Velocity of flow (V) = $\left[\frac{Qf^2}{140}\right]$
- Hydraulic mean depth (R) = $\frac{5V^2}{2f}$
- Wetted perimeter (P) = $4.75\sqrt{Q}$
- Bed slope (S) = $\frac{f^{5/3}}{3340\times Q^{1/6}}$
- From the above equations and continuity equation, we can say that, discharge (Q) depends on area (A) and velocity (V).

**2. Match the followings :**
**I. Ratio of long chord to tangent length of a simple circular curve of radius R and deflection angle Δ**
**II. ratio of long chord length to the simple circular curve of radius R and deflection angle Δ**
**A. sin (Δ/2)**
**B. cos (Δ/2)**
**C. 2 cos (Δ/2)**
**D. $360^0$ sin (Δ/2) / Δ.π**
**E. $360^0$ cos (Δ/2) / Δ.π**
(a) I-A, II-D (b) I-C, II-E
(c) I-D, II-C (d) I-C, II-D

**Ans. (d) :**

For the given curve–
- Tangent length (T) = $R\tan\left(\frac{\Delta}{2}\right)$
- Long chord (L) = $2R\sin\left(\frac{\Delta}{2}\right)$
- Length of curve ($\ell$) = $\left(\frac{2\pi R}{360}\right)\Delta$

I- Ratio of long chord to tangent length

$$=\frac{2R\sin\frac{\Delta}{2}}{R\tan\frac{\Delta}{2}}=2\cos\left(\frac{\Delta}{2}\right)$$

II- Ratio of long chord to length of curve

$$=\frac{2R\sin\frac{\Delta}{2}}{\frac{2\pi R}{360}\Delta}=360\sin\left(\frac{\Delta}{2}\right)/\Delta.\pi$$

**3. Ratio of bearing capacity of double under reamed pile to that of single reamed pile is nearly, for sandy soils.**
(a) 1.5 (b) 2.0
(c) 1.7 (d) 1.2

**Ans. (a) :** Ratio of bearing capacity of double under-reamed pile to that of single under-reamed pile is 1.5, for sandy soils.

**4. Shear centre is a point through which the load should act in a member such that**
(a) shear and bending acts but no twisting
(b) only shear acts and no bending and twisting
(c) bending and twisting act
(d) only twisting acts no bending

**Ans. (a) :** In case of unsymmetrical loading if load does not pass through shear centre the section will twist however when if load passes through shear centre section will have only bending & no twisting.
- If section has only one axis of symmetry then shear centre would lie on the axis of symmetry.

**5. An overflow spillway is 10 m long, between two square abutments and has two piers of 0.25 m width on its crest, when the head over the weir is 0.6 cm. the effective lengths of spillway for calculation of the discharge by the weir formula is**
(a) 9.40 m (b) 9.14 m
(c) 9.50 m (d) 9.26 m

**Ans. (c) :** Effective length of spillways

$$L_e = L - 2h_e\left[k_P \times N + k_a\right]$$

L = 10 m
$h_e$ = 0.6 m
N = 2 Pier
$K_a$ = 0.2 (square abutment)
$k_p$ = 0.1 (For Pier)

$L_e = 10 - 2\times 0.6\left[0.1\times 2 + 0.2\right]$
= 10 – 1.2 [0.2 + 0.2]
= 10 – 1.2 × 4
= 10–0.48
= 9.52
= 9.5 m

**6. The tellurometer which is used for EDM uses**
(a) modulated infer-red waves
(b) micro waves
(c) modulated light waves
(d) light waves

**Ans. (b): Microwave instrument-** These instruments come under the category of long range instruments, where in the carrier frequencies of the range of 3 to 30 GHz (1 GHz = $10^9$) enable distance measurements upto 100 km range. Tellurometer come under this category.

**7. temperature stresses in a bar are**

(a) the stress resulted in it when its temperature is maintained constant

(b) the stresses in it when the strains due to temperature changes are prevented

(c) the stresses resulted in it when the temperature is increased

(d) the tresses resulted in it when the temperature is decreased

**Ans. (b) :** Temperature stresses in a bar are the stresses in it when the strains due to temperature changes are prevented.

$$\sigma_{th.stress} = E\,\alpha\Delta T$$

Here,

E = Modulus of elasticity

$\alpha$ = Co.eff of thermal expansion

$\Delta T$ = Temperature change

**8. What is the value of gradient of the line Rs, if the horizontal distance between the two points R and S is 200.0 m and the reduce level of the point R = 103.0 m and the reduced level of the point S = 99.0 m**

(a) 1 in 25 (b) 1 in 10

(c) 1 in 50 (d) 1 in 15

**Ans. (c) :** Given,

Distance between point R and S (L) = 200 m

R.L. of point 'R' = 103 m

R.L. of point 'S' = 99 m

The difference in the elevation of R and S (h)

= 103–99 = 4 m

$$\text{Gradient} = \tan\theta = \frac{h}{L} = \frac{4}{200} = \frac{1}{25}$$

**9. The dimensions storage coefficient 'S' are**

(a) $L^3$ (b) $L^2/T$

(c) $L^3/T$ (d) dimensionless

**Ans. (d) :** Storativity or the storage coefficient is the volume of water released from storage per unit decline in hydraulic head in the aquifer, per unit area of the aquifer.

- Storativity is a dimensionless quantity and is always greater than 0.

$$S = \frac{d\gamma_w}{dh}\frac{1}{A} = S_S b + S_y$$

Where,

$V_w$ = Volume of water released from storage ($L^2$)

h = Hydraulic head (L)

$S_S$ = The specific storage

$S_y$ = The specific yields

b = The thickness of aquifer

A = Area ($L^2$)

**10. Choose the option in which the items of list I are correctly matches with the list II**

**List I**

**A. secondary road system**

**B. economic studies**

**C. engineering studies**

**D. road use studies**

**List-II**

**1. demographic study**

**2. traffic study**

**3. state highways**

**4. topographic studies**

(a) A-3, B-1, C-2, D-4

(b) A-3, B-1, C-4, D-2

(c) A-1, B-3, C-2, D-4

(d) A-1, B-3, C-4, D-2

**Ans. (b) : Secondary road system–** It is consist of state highway and major district roads.

- Economic studies = Demographic steady
- Engineering studies = Topographic studies.
- Road are studies= Traffic study

**11. For the same solid content, if the quantity of sludge with moisture content of 98% is x, then the quantity of sludge with moisture content of 96% will be**

(a) 2x (b) x/2 .

(c) x (d) x/4

**Ans. (b) :** $\frac{V_2}{V_1} = \frac{100 - P_1}{100 - P_2}$

Where,

$V_1$ and $V_2$ are the volume of the sludge of the moisture content of $P_1$ and $P_2$

Given,

$V_1$= X, $P_1$ = 98%, $P_2$ = 86%

then,

$$\frac{V_2}{x} = \frac{100 - 98}{100 - 96} = \frac{2}{4}$$

$$V_2 = \frac{X}{2}$$

**12. Based on the governing equation for the theory of pure bending which of the following is valid**

(a) bending moment = the product of bending stress and the section modulus

(b) bending strength = product of modulus of elasticity and curvature

(c) moment resistance = the product of flexural rigidity and the curvature

(d) bending moment = flexural rigidity per unit radius

**Ans. (a) :** Bending equation

$$\frac{f}{y} = \frac{M}{I} = \frac{E}{R}$$

Where,

f = bending stress

y = distance from the neutral axis

M= bending moment

I = moment of inertia
E = Modulus of elasticity
R= Radius of curvature

$$\frac{f}{y}=\frac{M}{I}\Rightarrow M=\frac{f\times I}{y}\left(\because \frac{I}{y}=\text{section modulus}\right)$$

$$M = f\times z \qquad \left(\therefore \frac{I}{y}=z\right)$$

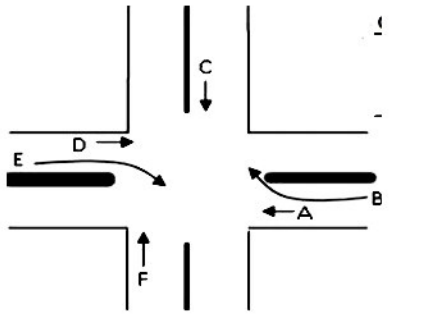
Bending moment = The product of bending stress and the section modulus
Moment o resistance = The product of flexural rigidity and the curvature

$$\frac{M}{I}=\frac{E}{R}\Rightarrow M=\frac{EI}{R}$$

Bending moment = Flexural rigidity per unit radius.

**13. A 3 phase traffic signal at a 4-legged intersection is designed for flow as show below.**

| Choice | Phase-I | Phase-II | Phase-III |
|---|---|---|---|
| | A, B | D, E | C, F |
| | A, D | C, F | E, B |
| | A. D | C. E | B. F |
| | A, B | E, F | C, D |




**There are 6 groups of flows identified by the alphabets, A through F, among these A, C, D and F are through flows, and B and E are right turnings. Identify which phasing scheme is not recommended**

(a) Z (b) W
(c) Y (d) X

**Ans. (a) :** j

**14. At a section of a shaft, a bending moment of 8 kN-m and a twisting moment of 6 kN-m act together. The equivalent twisting moment in kN-m is given by**

(a) 14 (b) 2
(c) 10 (d) 48

**Ans. (c) :** Equivalent twisting moment–
It is the twisting moment that alone produces maximum shear stress equal to the maximum shear stress produce due to combined bending and torsion.

$$T_{ev}=\sqrt{M^2+T^2}$$

M = 8 kN-m, T=6 kN- m

$$T_{ev}=\sqrt{(8)^2+(6)^2}$$

$$T_{eq}=\sqrt{100}$$

$T_{eq}$ = 10 kN-m

**15. An I-beam section is of 100 mm width flange and 20 mm thick web. Under a given shear force, the increase in the shear stress at the junction of flange and web would be**

(a) 80 times (b) 120 times
(c) 2000 times (d) 5 times

**Ans. (d) :** Shear stress at the junction of flange and web-

$$\tau=\frac{FAy}{Ib}$$

F = Force applied
A = Area of the flange
I = moment of inertia of I-section
y = distance of centroid of the flange from N.A.

- Let shear stress at flange at the junction of flange and web = $\tau_1$

Width of flange, $b_1$ = 100 mm
Width of flange, $b_2$ = 20 mm

- Let shear stress at the web at the junction of flange and web = $\tau_2$

then,

$$\tau_1=\frac{FAY}{Ib_1} \quad \text{..... (i)}$$

$$\tau_2=\frac{FAY}{Ib_2} \quad \text{..... (ii)}$$

$$\therefore \frac{\tau_1}{\tau_2}=\frac{b_2}{b_1}\Rightarrow \frac{\tau_1}{\tau_2}=\frac{20}{100}=\frac{1}{5}$$

$$\tau_2=5\times\tau_1$$

**16. According to the Indian standards institute, the standard size of modular bricks is**

(a) 23 cm × 10 cm × 7.5 cm
(b) 20 cm × 10 cm × 10 cm
(c) 10 cm × 10 cm × 9 cm
(d) 25 cm × 15 cm × 7.5 cm

**Ans. (b) :** Standard size bricks
19 cm × 9 cm × 9 cm
Nominal size bricks
20 cm × 10 cm × 10 cm

**17. Trap efficiency of a reservoir is a function of the ratio of**

(a) dead storage capacity to live storage capacity
(b) dead storage capacity to annual total sediment inflow volume
(c) reservoir capacity to that of annual sediment inflow
(d) reservoir capacity to the total annual inflow volume of water

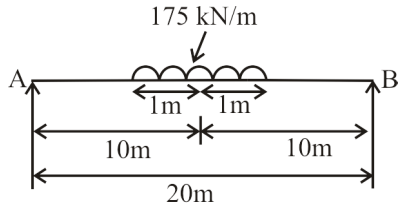
**Ans. (d) :** Trap efficiency of a reservoir is a function of the raio of reservoir capacity to the total annual inflow volume of water.

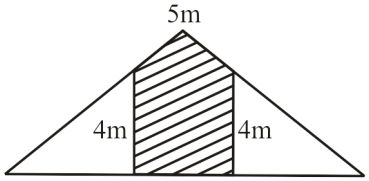
- Trap efficiency of a storage reservoir is defined as the ratio of the total sediment deposited in a given period to the total sediment in flow in that period.

**18. A track load of length 4 m and magnitude 700 kN is crossing a bridge girder of 20 m span. What would be the maximum BM (in kN-m) at the center of the girder**

(a) 3000 (b) 5600
(c) 3150 (d) 4500

**Ans. (c) :**




ILD for $MB_C$

$$w = \frac{700}{4} = 175\,kN/m$$

Max bending moment at C = Intensity × Area of shaded ILD

$$= 175\left(\frac{4+5}{2}\right)\times 2 + 175\left(\frac{4+5}{2}\right)\times 2$$

= 3150 kN-m

**19. Quartzite is a**

(a) metamorphic rock (b) argillaceous rock
(c) siliceous rock (d) calcareous rock

**Ans. (a) : Metamorphic rocks-**
Metamorphic rocks are formed when rock changes over a period of time due to a lot of physical changes like pressure heat and different chemical activity.

**20. For a simply supported beam of given material and cross section, if the udl throughout is replaced by a central point load equal to the total load due to udl, the central deflection of the beam would**

(a) would increases by 1.6 times
(b) would be double
(c) would decreases by 1.6 times
(d) remains same

**Ans. (a) :** Deflection due to point load–

$$\delta_1 = \frac{W\ell^3}{48EI} \qquad (\text{If } W = w\times\ell),\ w \rightarrow \text{total load}$$

Deflection due to u.d.l

$$\delta_1 = \frac{5W\ell^3}{384EI}$$

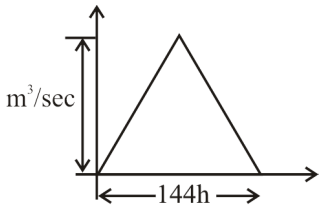
$$\frac{\delta_2}{\delta_1} = \frac{W\ell^3}{48EI} \div \frac{5W\ell^3}{384EI}$$

$$\delta_2 = 1.6\delta_1$$

**21. A 596 $km^2$ catchment has a 12 h unit hydrograph which can be approximated as a triangle. It its time base is 144h, its peak ordinate in $m^3$/sec is**

(a) 50 (b) 13
(c) 2 (d) 23

**Ans. (d) :**




Given,

$A = 596\ km^2$. Base time = 144 hour

$$\text{Total volume} = \frac{1}{2}\times 144\times 3600\times Q_{peak}$$

Total volume

= Effective rainfall × Area of the catchment

$$\frac{1}{2}\times 144\times 3600\times Q_{peak} = \left(1\times 10^{-2}\right)\times\left(596\times 10^{6}\right)$$

$$Q_{peak} = 22.99$$

$$Q_{peak} \approx 23\ m^3/sec$$

**22. The diameter of the disc for used for compulsory control sign which is having blue background colour is**

(a) 900 mm (b) 470 mm
(c) 600 mm (d) 1200 mm

**Ans. (c) :** Mandatory signs-
• Violation of mandatory signs can lead to penalties.
**Example=** stop, give way, limits prohibited no parking and compulsory sign.
The diameter of a mandatory sign disc must be 60 cm or 600 mm.

**23. One end of a two dimensional water tank has the shape of a quadrant of a circle of radius 2m. when the tank is full, the vertical component of the force per unit length on the curved surface will be**

(a) 19.62 π kN (b) 32.24 π kN
(c) 9.81 π kN (d) 29.43 π kN

**Ans. (c) :** $F_V = \rho g(V)$

$$= 9.81\times\left(\frac{\pi}{16}\times(4)^2\times 1\right)$$

$$= 9.81\pi kN$$

**24. The Indian railways use what sizes of crossing, which facilities to cross the vehicles at higher speed?**

(a) 1 in 12 and 1 in 16
(b) 1 in 16 and 1 in 20
(c) 1 in $8\frac{1}{2}$ and 1 in 12
(d) 1 in 12 and 1 in 20

**Ans. (b): Crossing–** A crossing is a device introduced at the point where two gauge faces cross each other to permit flanges of a railway vehicle to pass from one track to another.

- For the higher speed, we used 1 in 16 and 1 in 20.
- Most commonly used crossing on broad gauge track is 1 in 8.5 and 1 in 12.

**25. The shear stress in a circular pipe with a laminar flow in it**

(a) varies parabolically across cross section
(b) varies inversely as the distance from mid plane
(c) remains constant over cross section
(d) varies directly as the distance from the mid plane

**Ans. (d) : Laminar flow in horizontal pipes–** A fully developed laminar pipe flow is merely a balance between pressure and viscous forces, for the steady fully developed laminar flow of a fluid through a horizontal circular pipe of radius, R the shear stress distribution is

$$\tau = \frac{-\partial p}{\partial x}.\frac{r}{2} \Rightarrow \tau \times r$$

Shear stress is proportional to the radial distance from centre of the pipe.

**26. What is the value of mid-ordinate, if the radius of a simple circular curve-560 m and the deflection angle = $120^0$.**

(a) 186.66 m (b) 280 m
(c) 50 m (d) 140 m

**Ans. (b) :** $\text{Mid}-\text{ordinate}(m) = R\left(1-\cos\frac{\Delta}{2}\right)$

Given, R = 560 m , $\Delta = 120^0$

$M = 560\left(1-\frac{\cos 120}{2}\right)$

M = 280 m

**27. The chlorine demand of a water sample was found to be 12 ppm. The amount of bleaching powder containing 30% available chlorine to be added to treat one litre of such a water sample is**

(a) 1.33 mg (b) 0.67 mg
(c) 0.34 mg (d) 0.06 mg

**Ans. (b) :** Chlorine demand = 0.2 mg/$\ell$

Quantity of water treated = 1 $\ell$

Amount of chlorine needed = 0.2 mg

Amount bleaching powder = $\frac{0.20}{0.30} = (0.67\ \text{mg})$

**28. Standard EDTA (ethylene diamine tetra acetic acid) solution is used to determine the**

(a) hardness of water
(b) residual chlorine in water
(c) dissolved oxygen in water
(d) turbidity of water

**Ans. (a):** Standard EDTA (ethylene diamine tetra acetic acid) solution is used to determine the hardness of water.

- **Hardness of water–** The hardness of water is due to the presence of soluble bicarbonates chlorides and sulphates of calcium and magnesium.

**29. The settling velocity of a particle in a sedimentation tank depends on**

(a) surface area of tank
(b) both depth and surface area of tank
(c) thickness of tank
(d) depth of tank

**Ans. (a) :** Settling velocity–

$$V_s = \frac{(\gamma_s - \gamma_w)d^2}{18\mu}$$

Applicable for Re < 1 , (stoke's law1

$D \le 0.2\text{mm}$

Time of horizontal flow = $\frac{\text{Length of tank}}{\text{Velocity of flow}} = \frac{L}{V_f}$

Velocity of flow = $\frac{Q}{B \times H}$

Detention time ($t_D$) = $\frac{L \times B \times H}{Q}$

$= \frac{\text{Volume of tank}}{\text{Discharge}}$

Surface over flow tank ($V_s$) = $\frac{Q}{B \times L}$

So, settling velocity depends on the surface area of tank.

**30. A gravity dams has two alternate designs having the same base width one with drainage gallery at 1/10 base width from the upstream end and the other without the gallery. If there is no tail water, the reduction in the total uplift force in the former case as a percentage of the latter**

(a) 13.3% (b) 43.3%
(c) 66.7% (d) 56.7%

**Ans. (*) :** Note: For this question, discrepancy is found in question/answer. So, This question is ignored for all candidates.

**31. Which of the following statements is wrong?**

(a) in Flemish bond, the alternate headers of each course are centrally supported over the stretchers in the course below
(b) in English bond, vertical joints in the header courses come over each other and vertical joints in the stretcher course are also in the same line
(c) in Flemish bond, every alternate course starts with a header at the corner
(d) in English bond, the heading course should start with a queen closer

**Ans. (d): English bond and flemish bond–**

- English bond and flemish bond are the two most common brick masonry patterns used in wall construction.
- A brick construction pattern with alternate courses of brick laid as stretchers and headers forms an English bond.
- A Flemish bond is a brick construction pattern that consists of alternate stretchers and headers for each course.

**32. In hydrometer analysis for a soil mass**

(a) meniscus correction is subtractive and dispersing agent correction additive
(b) meniscus correction is additive and dispersing agent correction is subtractive
(c) both meniscus correction and dispersing agent correction are subtractive
(d) both meniscus corrosion and dispersing agent correction are additive

**Ans. (b) :** A hydrometer is a device used to measure the specific gravity of liquids.

The following three corrections are necessary for-

**(a) Meniscus correction–** Since the hydrometer reading increase downward on the stem, the meniscus correction ($C_m$) is always positive.

**(b) Temperature correction–** If the temperature at the time of the test is more than that of calibration of the hydrometer, the observed reading will be less and the correction ($C_t$) would be positive and vice-versa.

**(c) Deflocculating agent correction–** The addition of the deflocculating agent increases the density of the suspension and thus a correction ($C_d$) is applied which is always negative.

**33. For analysis of pavements and determination of modulus of sub grade reaction by plate load test, the most representative value of modulus reaction can be obtained at a pressure of ..... kg/cm$^2$.**

(a) 0.70 (b) 0.20
(c) 2.50 (d) 1.25

**Ans. (*) :** Note: For this question, discrepancy is found in question/answer. So, This question is ignored for all candidates

**34. Match list-I with list-II and select correct answer using the codes given below**

**List-I (shape tunnel)**
**A. horse shoe section**
**B. circular**
**C. egg shaped**
**D. segmental roof**
**List-II (attribute preference)**
1. Gives self cleaning velocity even in dry weather
2. suitable for soft rocks
3. best suited for non-cohesive soils
4. suitable for soft materials
5. suitable for slab bays

(a) A-2, B-3, C-4, D-5
(b) A-5, B-4, C-1, D-3
(c) A-2, B-3, C-1, D-5
(d) A-3, B-2, C-4, D-1

**Ans. (b) : Shape of tunnel–** The tunnel shape is governed by the structural efficiency for sustaining external loads as also the hydraulic efficiency.

- The tunnel cross-section may be circular, rectangular or any combination of these forms.

**Circular section–** It is the most suitable and stable shape from structural consideration where the tunnel is subjected to high inwards or outwards radial pressure.
Suitable for soft material.

**Horse-shoe section–** It is a compromise between circilar and D-shaped section.
It is structurally strong to withstand external rock and water pressure.
It is suitable in a moderately good rock as the tunnel has to resist internal pressure also.

**Egg-shaped–** Provides least cross-section area at the bottom

- Used for carrying sewerage.
- Effective in resisting external internal pressure.

**Segmental roof–** Best suited for non-cohesive soils.

**35. Stokes law valid only if the size of particle is**

(a) between 0.2 mm and 1.0 mm
(b) less than 0.0002 mm
(c) between 0.0002 mm and 0.2 mm
(d) greater than 0.2 mm

**Ans. (c) :** Stoke's law valid for

- Particle size– Between 0.0002 mm to 0.2 mm

**36. The Hardy Cross method of hydraulic analysis of pipe networks, besides analysis of pipe networks, besides satisfying the continuity and energy principles must also satisfy the condition that**

(a) flow in to any junction equals the outflow from it
(b) flow in each pipe has been loss according to Darcy's Weisbach or any other pipe head loss equation
(c) algebraic sum of the head losses around any closed loop is zero
(d) momentum principle is followed

**Ans. (c) :** Hardy cross-method

The hardy cross-method is an interactive method for determining the flow in pipe network system where the inputs and outputs are known but the flow inside the network is unknown.

- The Hardy cross analysis is based on the principle that–

→ At each junction, the total in flow must be equal to he total outflow.

→ The algebraic sum of the head losses around any closed loop is zero.

**37. Consider the following statements**

1. undisturbed samples may be obtained with the help of augers
2. auger drilling is most effective in clayey soils
3. hollow stem augers are sometimes used to drill holes in silty sand

**Which of these statements are correct?**

(a) 3 and 2 only (b) 3 and 1 only
(c) 3, 2 and 1 only (d) 2 and 1 only

**Ans. (a) : Auger drilling–** It is a simple and cost-effective boring technique which can be used for almost all types of soil apart from gravelly soil and rocks.

- Auger boring is used to collect disturbed soil specimen. It collects the soil sample from a maximum practical depth of nearly 35 m based on the available time and equipment type.
- Soil samples collect from auger flight are disturbed making it difficult to determine the precise depth of sample.
- Difficult drilling in extremely dry, fine materials.
- Hollow-stem auger drilling is difficult in saturated soils and soils containing very coarse gravles, cobbles or boulders.

**38. Maximum number of vehicles can be parked with**

(a) 75 degree angle parking
(b) parallel parking
(c) perpendicular parking
(d) 45 degree angle parking

**Ans. (c) :** A maximum number of vehicles can be parked at $90^0$ parking angle.

- The best angle in angle parking, considering the road obstruction and the convenience of driver is $45^0$.

**39. What is the length of each link in a revenue chain**

(a) 0.6 feet (b) 20 cm
(c) 1 feet (d) 2.0625 feet

**Ans. (d) :** Revenue chain–
This chain is mainly used during cadastral survey (The survey in which determination of land boundaries and area takes place is called cadastral or land survey).

- It is 33 feet long and divided into 16 links.
- Each links is 2.0625 feet long.
- The links of chain is made up of steel and handle is made of brass.

**40. As a designer, what would be your estimate (in $N/m^2$) of the flexural tensile strength of a concrete, whose characteristics cube compressive strength is 25 $N/mm^2$**

(a) 3.0 (b) 2.5 (c) 3.5 (d) 4.0

**Ans. (c) : Modulus of rupture–**

- It is measure of the tensile strength of concrete beams or slabs.
- Flexural strength of concrete ($f_{cr}$)

$f_{cr} = 0.7\sqrt{f_{ck}}$

$f_{ck} \rightarrow$ characteristic cube compressive strength of concrete ($N/mm^2$)

$f_{cr} = 0.7 \times \sqrt{25} = 3.5\ N/mm^2$

**41. The assumption not made in the derivation of bernoullis equation**

(a) steady flow (b) 2 dimensional flow
(c) uniform flow (d) inviscid flow

**Ans. (b, c) :** The following are the assumption made in the derivation of Bernoulli's equation.

- The fluid is ideal, i.e viscosity is zero.
- The flow is steady
- The flow is incompressible
- The flow is irrotational
- There is no assumption that the flow should 2-dimensional or uniform.

**42. If the total hardness and alkalinity of a sample of water are 300 mg/*l* and 100 mg/*l* ($CaCO_3$ scale) respectively. Then its carbonate and non carbonate hardness in terms of mg/*l* will be respectively**

(a) 100 and 400 (b) 100 and 200
(c) 400 and zero (d) 400 and 300

**Ans. (b) :** Given,
Total hardness = 300 mg/$\ell$
Alkalinity = 100 mg/$\ell$
If total hardness > Alkalinity
Carbonate hardness = alkalinity = 100 mg/$\ell$
Non-carbonate hardness = Total hardness – Alkalinity
= 300 – 100
= 200 mg/$\ell$

**43. In a field two pumps are provided each with capable of delivering Q amounts of discharge under a head H. there is a requirement of delivery of 2Q amounts of discharge. How do you arrange the pumps to get required discharge.**

(a) arrangements is not possible
(b) half series remaining parallel
(c) series arrangement
(d) parallel arrangement

**Ans. (d) : When pump in series–**

- Discharge will remain the same.
- Total head = sum of heads of all the pumps connected in series.

**When pump in parallel**

- Heads against pumps are working remain the same.
- Discharge will be the sum of the discharge in an individual pump.
- Resulting discharge for two pumps = 2Q and total head = H for parallel arrangement.

**44. without which of the following basic ingredients, concrete can still be called a concrete?**

(a) water (b) cement
(c) coarse aggregate (d) fine aggregate

**Ans. (d) :** Concrete is made up of three basic components, water, aggregate (rock sand or gravel) and portland cement).

• Mortar is madeup of three basic components- water, binding material (cement) and fine aggregate.
• Thus, without fine aggregate concrete can still be called a concrete.

**45. Indian roads congress recommends, softening point of ordinary road marking material hot applied thermoplastic compound shall be.. degree centigrade**
(a) 90 + 10.1 (b) 100.0 + 10.0
(c) 102.5 + 9.5 (d) 120 + 10.5

**Ans. (c) :** Properties of thermoplastic material, when tested in accordance with ASTM $D_{36}$/BS – 3262– (Part-I) shall be as below-
(a) Luminance
**(b) White –** Daylight luminance at 45 degrees-65 percent min as per AASTOM249
**(c) Drying time–** When applied at a temperature specified by the manufacture and to the required thickness, the material shall set to bear traffic in not more than 15 min.
**(d) Skid resistance–** Not less than 45 as per BS 6044.
**(e) Cracking resistance at low temperature**
The material shall show no cracks an application to concrete blocks.
**(f) Softening point–** $102.5 \pm 9.5^0$C as per ASTMD36
**(g) flow resistance–** Not more than 25 percent as per AASHTO M249
**(h) Yellowness index–** Not more than 0.12 as per AASHTO M 249.

**46. The standard temperature gradient considered by the international civil aviation organization, the temperature decrease linearly be .... per every 1000 feet increase in altitude**
(a) $0.5^0$C (b) $2^0$C
(c) $1^0$C (d) $1.5^0$C

**Ans. (b) :** The international civil aviation organization (ICAO) created a standard temperature for aviation in order to establish a baseline from which manufactures could develop performance data usable for pilots around the globe.
• The standard temperature decreases $2^0$C or 3.5 $^0$F for every 1000 feet gained and this is reasonably accurate upto 36000 feet MSL.

**47. For an elementary gravity dam, profile is made up of concrete of relative density 2.5 the limiting height of low dam without considering the uplift was found to be Hv and with consideration of full uplift was Hs, then Hs/Hv will have a value of**
(a) 1.8 (b) 1.0
(c) 1.4 (d) 0.7

**Ans. (c) :** Given,
Relative density (G) = 2.5
Considering uplift condition ($H_S$)

$$H_s = \frac{f}{\gamma_w (G - C + 1)}$$

f = maximum allowable stress of dam material.
$\gamma_w$= density of rotating fluid
G = relative density
C = constant
C = 1 (for recommended by V.S.B.R.)
• Without considering uplift condition $(H_v)$

$$H_v = \frac{f}{(G+1)\gamma_w} \qquad C = 0$$

$$\frac{H_s}{H_v} = \frac{\frac{f}{\gamma_w (G - C + I)}}{\frac{f}{\gamma_w (G+1)}}$$

$$\frac{H_s}{H_v} = \frac{G+1}{G-C+1} = \frac{2.5+1}{2.5-1+1} = \frac{3.5}{2.5}$$

$$\frac{H_s}{H_v} = 1.40$$

**48. In order to avoid the danger of stress corrosion in pre-stress corrosion in pre-stresses concrete works, which of the following is not permitted to be used**
(a) sodium carbonate (b) fluoro silicate
(c) calcium chloride (d) triethanolamine

**Ans. (c) :** Whenever there is chloride in concrete there is on increased risk of corrosion of embedded steel.
• The higher the chloride content or if subsequently exposed to worm moist condition. The greater the risk of corrosion.

**49. In a siphon aqueduct the worst condition of uplift on the floor occurs when**
(a) the canal and drainage are full
(b) the canal is empty, the drainage at H.F.L. and the water table as at HFL of the stream
(c) the canal is full and drainage is empty
(d) the canal is full and the drainage is empty and the water table is at the stream bed

**Ans. (b) : Drainage is full–** H.F.L. of drain is above canal bed, so maximum upward water, pressure underneath the canal floor.
**Canal is empty–** Minimum downward, so counteracting pressure is minimum.
**Natural water table at drainage bed level–** Ensuring availability of maximum net water head,

**50. In R.C.C. cantilever retaining wall, if the check for overturning is not satisfied, which of the following would serve the better in meeting the check?**
(a) increasing the heel slab span
(b) decreasing the toe slab span
(c) providing a sloping backfill
(d) providing a shear key

**Ans. (a) : Retaining wall–** A retaining wall is a structure designed and constructed to resist the lateral pressure of soil when there is desired change in ground elevation that excess the angle of repose of the soil.

• Retaining walls are used for supporting soil laterally so that it can be retained at different levels on the two sides.
• Cantilever type retaining wall has the following components– Item
Toe slab
Heel slab
• Resisting moment against overturning is generated due to the–
• Self-weight of the wall
• Back fill over the heel slab
• Resisting moment against over turning > Over turning moment
then retaining wall is safe against over turning.
• If the heel slab length increases, the resisting moment also increases as more backfill over the heel slab tends to increases the resisting of wall against overturning thus wall becomes safe against overturning.

**51. The slenderness ratio for masonry wall should not more than**
(a) 20 (b) 30
(c) 40 (d) 25

**Ans. (b) :** Slenderness ratio of masonry wall is defined as the effective height divided by the effective thickness or its effective length divided by the effective thickness, which ever is less.
• For load-bearing wall, the maximum permissible slenderness ratio ($\lambda$) is–
$\lambda = 27$
• For non-load bearing wall, the maximum permissible slenderness ratio ($\lambda$) is 30.
Hence, the slenderness ratio for masonry walls should not be more than 30.

**52. The term alternate depths in open channel is associated with**
(a) at the beginning and end of a hydraulic jump
(b) at the beginning and end of gradually varies flow profile
(c) having the same specific energy for a given discharge
(d) having the same kinetic energy for a given discharge

**Ans. (c) :**
**Alternate depth –** The two possible depth ($y_1$, $y_2$) that have the same specific energy for a given discharge are known as alternate depths.

**53. Which of the following building stones absorbs water more?**
(a) quartzite (b) slate
(c) shale (d) granite

**Ans. (c) :** The water absorption for different stones is as follows–

| Types of stone | Water absorption (% not greater than) |
|---|---|
| Sand stone | 10 |
| Shale | 10 |
| Quartzite | 3 |
| Slate | 1 |
| Granite | 1 |
| Lime stone | 10 |

**Characteristic of good building stone–**
Compressive strength > 1000 kg/cm$^2$
High durability and coefficient of hardness > 14
Specific gravity > 2.7
Water absorption < 5%
Toughness index > 13

**54. In the design of storm sewers, "time of concentration- "is relevant to determine the**
(a) time of travel
(b) area served by the sewer
(c) rainfall intensity
(d) velocity in the sewer

**Ans. (c) : Rainfall intensity–** Rainfall intensity is related to rainfall duration and design storm recurrence interval.
• Rainfall intensity at a duration equal to the time of concentration ($T_c$) is used to calculate the peak flow in the rational method.

**55. In Reynolds law of similitude the discharge ratio Qr is.**
(a) $L_r^3\rho_r$ (b) $L_r^3\rho_r/\mu_r$
(c) $L_r^{5/2}$ (d) $L_r\mu_r/\rho_r$

**Ans. (d) :** Reynold number is the ratio of inertia forces to viscous forces.

$$R_e = \frac{\text{Inertia force}}{\text{Viscous force}} = \frac{\rho vL}{\mu}$$

Reynold's model law–

$$R_e = \frac{\rho VL}{\mu}$$

$$V_r = \frac{\mu_r}{\rho_r L_r}$$

$$Q_r = A_r.V_r$$

$$= L_r^2.\frac{\mu_r}{\rho_r.L_r} = \frac{L_r\mu_r}{\rho_r}$$

**56. The shear strength of a soil**
(a) is directly proportional to the angle of internal friction of the soil
(b) decrease with increase in normal stress
(c) decreases with decreases in normal stress
(d) in inversely proportional to the angle of internal friction of the soil

**Ans. (a, c) : Shear strength of soil–** The strength of the soil resists the shear due to applied load called shear strength.
• The tangent of the angle of shearing resistance.
$S = C+\sigma\tan\phi$
$S \propto \tan\phi$
It is directly proportional to the angle of internal friction of the soil, also normal stress.

**57. Which of the following properties of aggregate that are influenced by porosity, permeability and water absorption of aggregates, is considered to be less important in general in the Indian context?**

(a) resistance to abrasion
(b) bound between aggregate and cement paste
(c) resistance to aggressive chemical agencies
(d) freezing and thawing

**Ans. (d) :** Freezing and thawing properties of aggregate that are influenced by porosity, permeability and water absorption of aggregates, is considered to be less important in general in the Indian context.

**58. Disinfection efficiency is**

(a) highest at pH value of 7
(b) reduced at higher pH value of water
(c) unaffected by pH value of water
(d) increased at higher pH value of water

**Ans. (b) :** Disinfection efficiency is reduced at higher pH value at water.

**59. The discharge capacity required at the outlet to irrigate 2600 ha of sugar cane having a kor depth of 17 cm and a kor period of 30 days is**

(a) 14.7% (b) 1.71%
(c) 0.18% (d) 2.3%

**Ans. (b) :** Discharge capacity required at the outlet to irrigation $(Q) = \frac{A \times \Delta}{B \times 86400}$

Where,

$A \rightarrow$ Area = 2600 ha

$\Delta \rightarrow$ Kore depth = 17 cm = 0.17 m

$B \rightarrow$ Kore period = 30 days

$$Q = \frac{2600 \times 10^4 \times 0.17}{30 \times 86400}$$

$$Q = 1.71 \quad m^3/sec$$

**60. Stopping sight distance is the minimum distance available on highway which is the**

(a) distance of sufficient length along centre line of the roadway to stop the vehicle without collision
(b) longest distance of sufficient length to stop the vehicle without collision
(c) shortest sight distance of sufficient length across the curve to stop the vehicle without collision
(d) shortest sight distance of sufficient to stop the vehicle without collision

**Ans. (a) : Stopping sight distance (SSD)–** The minimum sight distance available in a highway at any spot should be of sufficient length to stop a vehicle traveling as design speed, safety without collision with any other obstruction.

**61. A leaf spring is an example of**

(a) A beam of uniform strength
(b) A bar of uniform strength
(c) A closely coiled spring
(d) A beam of uniform geometry

**Ans. (a) :** Leaf springs– Leaf spring (some time also known as laminated spring) are made up of beams of uniform strength, and are normally used in automobiles to absorb the shock and vibration produced by road undulations. thereby providing comfort to passengers.

**62. Poissons ratio and bulk modulus of a material are defined in which of the following stress states, respectively**

(a) uniaxial and uniaxial
(b) uniaxial and triaxial
(c) tiraxial and triaxial
(d) triaxial and uniaxial

**Ans. (b) : Poissons ratio (μ) –** Poisson ratio is the ratio of transverse contraction strain to longitudinal extensive strain in the direction of the stretching force.

- Compressive deformation is considered negative.
- Tensile deformation is considered positive.

$$\mu = \frac{-\text{Lateral strain}}{\text{Longitudinal strain}}$$

**Bulk modulus of elasticity (k)–**

- It is describes volumetric elasticity or the tendency of an object to deform in all directions when uniformly loaded in all directions.
- It is defined as volumetric stress over volumetric strain and is the inverse of compressibility.

$$k = \frac{-p}{\Delta V / V} \qquad \text{Compressibility } (\beta) = \frac{1}{k}$$

The bulk modulus is an extension of young's modulus to three dimension.

So, Poisson's ratio and bulk modulus of material is defined in uniaxial and triaxial stress states.

**63. Property of fluid that describes its internal resistance is known as**

(a) internal energy (b) viscosity
(c) resistance (d) friction

**Ans. (b) : Viscosity–** Viscosity is a measure of a fluid's resistance to flow. It determine the strain rate that is generated by a given applied shear stress.

**64. A hydraulic turbine has a discharge of $3m^3/s$ operating under a head of 16 m with a speed of 500 rpm. If it is to be operated under 36 m then the rotational speed will**

(a) 447 rpm (b) 600 rpm
(c) 750 rpm (d) 400 rpm

**Ans. (c) :** Given,

$N_1 = 500$ rpm, $H_1 = 16$ m, $H_2 = 36$ m

$N_2 = ?$

$$N_u = \frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$$

$$N_2 = N_1 \frac{\sqrt{H_2}}{\sqrt{H_1}} = 500 \times \frac{\sqrt{36}}{\sqrt{16}}$$

$$N_2 = 500 \times \frac{6}{4}$$

$N_2 = 750$ rpm

**65. Rolled steel beams are designed by Indian standards in terms of**

(a) depth of the section and weight per meter
(b) width of the flange and weight per meter
(c) weight per meter and width of the flange
(d) weight per meter and depth of the section

**Ans. (a) :** The rolled steel beam are designated by the series to which become sections belong (abbreviated reference symbols), followed by depth in mm of the section and weight in kN per meter length of the beam.
Ex - MB 225@0.312 kN/m

**66. The gross bearing capacity of a footing is 460 kN/m$^2$. If the footing is 1.5 wide and is a depth of 1.0 m in a clayey soil which has a unit weight of 20 kN/m$^3$, then the net bearing capacity is**

(a) 440 kN/m$^2$ (b) 430 kN/m$^2$
(c) 23 kN/m$^2$ (d) 480 kN/m$^2$

**Ans. (a) :** Given,
Gross bearing capacity of footing ($q_u$) = 460 kN/m$^2$
Depth of foundation ($D_f$) = 1m
Unit weight ($\gamma$) = 20 kN/m$^3$
Net bearing capacity = ?
Net bearing capcity ($Q_{nu}$)= $q_u$ – Over burden pressure(q)

$q = \gamma D_f$

$Q_{nu} = q_u - \gamma D_f$

$= 460 - 20 \times 1$

$Q_{nu} = 440$ kN/m$^2$

**67. Specific gravity of most of the building stone lies between**

(a) 3.0 to 2.5 (b) 2.5 to 2.0
(c) 2.0 to 1.5 (d) 3.5 to 3.0

**Ans. (a) :** Specific gravity of most of the building stones lies between 2.5 to 3.
• Mostly specific gravity should be taken at least 2.7.

**68. Two sources generate noise levels of 90 dB and 94 dB respectively. The cumulative effective of these two noise levels on the human ear is**

(a) 184 dB (b) 94 dB
(c) 95.5 dB (d) 95 dB

**Ans. (c) :** $L_{eq} = 10 \log \sum_{i=1}^{i=n} (10)^{\frac{L_i}{10}} \times t_i$

$n \rightarrow$ Total number of sound sample
$L_i \rightarrow$ The noise sample of any $i^{th}$ sample
$L_1 \rightarrow 90$ dB
$L_2 \rightarrow 94$ dB
$t_i \rightarrow$ Time duration of $i^{th}$ sample

$$l_{eq} = 10\log_{10}\left[\left(10^{L_1/10} \times \frac{t_1}{T}\right) + \left(10^{L_2/10} \times \frac{t_2}{T}\right)\right]$$

$$= 10\log_{10}\left[\left(10^{90/10}\right) + \left(10^{94/10}\right)\right]$$

$L_{eq} = 95.5$ dB

**69. The following data are available for a cross drainage work project**

| Item | Canal | Drainage |
|---|---|---|
| FSL/HFL | 105 | 104 |
| Bed level | 100 | 102 |
| Discharge | 80 m$^3$/s | 12 m$^3$/s |

**The most appropriate cross drainage works for this situation**

(a) aqueduct (b) syphon
(c) siphon aqueduct (d) super passage

**Ans. (b) :** Bed level of canal = 100 m
Bed level of drainage = 102 m
HFL of drainage = 106 m
HFL of canal = 105 m
HFL of the canal (105 m) > Bed level of drainage (102 m)
So,
For the given data system is recommended
**Canal syphon–** A canal syphon (or simply syhpon) is structure in which the canal is taken below the drainage and the canal water flows under symphonic action.
• A canal syphon is constructed when the F.S.L. of the canal is above the drainage bed level.

**70. The kernel of square and circular sections will be respectively**

(a) circle and rhombus (b) rhombus and circle
(c) circle and square (d) square and circle

**Ans. (d) : Kern–** The kern of a section is the region in which a compressive point load may be applied without producing any tensile stress on the cross-section.
• The kernal of the square section is also a square, while the shape of kernal for the rectangle is rhombus.
• The kern of the circular section is a circle with a diameter equal to d/4.

**71. Shingle is**

(a) water bound pebbles (b) air weathered rock
(c) decomposed Laterite (d) crushed granite

**Ans. (a) :** Shingle is a rectangular tile (wooden or asphalt) used on a roof.
• Shingle is water bound pebble, it is used as a roofing material, surfacing of the boundary walls etc.

**72. The limit of crushing value of aggregative as specific (in percentage) by IS : 383-1970 for aggregate used for runways and such other wearing surface is**

(a) 45 (b) 25
(c) 20 (d) 30

**Ans. (d) : Aggregate crushing value-** The aggregate crushing value, when determined in accordance with IS : 2386 (part IV)-1963 shall not exceed 45 percent for aggregate used for concrete other than for wearing surface and 30 percent for concrete for wearing surfaces, such as runways, roads and pavements.

**73. The observed runoff due to a 12h storm of uniform intensity of 0.5 cm/h was a triangular direct runoff hydrograph of base 64 h and maximum ordinate 30 m$^3$/s. If the catchment area is 120 km$^2$, the $\phi$ index for the basin, cm/h is**

(a) 1.20 (b) 0.26
(c) 0.52 (d) 0.88

**Ans. (b):** Given,
Intensity of rainfall (i) = 0.5 cm/hr
Duration of rainfall ($t_c$) = 12 hours
Q = 30 $m^3$/sec
A = 120 $km^2$, T = 64 hrs.
From the hydrograph, we can get the effective rainfall (R)
$6400 \times 3600 \times 0.5 \times 30 = 120 \times 10^6 \times R$

$$R = \frac{6400 \times 3600 \times 0.5 \times 30}{120 \times 10^6}$$

R = 0.0288 m = 2.88 cm
Total rainfall (P) = Intensity of rainfall × duration of rainfall
= 0.5 × 12 = 6 cm

$$\phi \text{ - index} = \frac{P - R}{t_e}$$

$$= \frac{6 - 2.88}{12}$$

$\phi$-index = 0.26 cm/hrs.

**74. 76 in jack arch floor the rise is**
(a) 1/6 th of the span
(b) 1/8 th of the span
(c) 1/12 th of the span
(d) 1/10 th of the span

**Ans. (c) : Jack arch floors–** Jack arch is an arch of either brick or concrete, supported on lower flange of mild steel joist (R.S.J.).
The joists are spaced 1 to 1.5 m centre to centre, and are supported at their ends either, on walls or on longitudinal girders.
The rise of the arch is kept equal to $\frac{1}{12}$th of the span.
• The minimum depth of concrete at crown is kept equal to 15 cm.

**75. At a horizontal joint of width 70 m in a gravity dam, the factor of safety against sliding is 1.05. if the sum of all the horizontal forces above the joint is 70 MN and the average shear strength of joint is 1.40 MPa, the friction factor of the joint**
(a) 3.24 (b) 2.30
(c) 2.45 (d) 1.85

**Ans. (c) :** Given,
FOS against sliding = 1.05 = F
Sum of horizontal forces ΣH = 70 MN

$$F = \frac{\mu \Sigma V}{\Sigma H}$$

$$1.05 = \frac{\mu \Sigma V}{\Sigma H}$$

$\mu \Sigma V = 73.5$ MN

$$\text{Shear friction factor (S.F.F.)} = \frac{\mu \Sigma V + bq}{\Sigma H}$$

$\Sigma V \rightarrow$ net vertical forces
$\mu \rightarrow$ coefficient of friction
$q \rightarrow$ average cohession or shear strength of joint
$b \rightarrow$ width of joint

$$SFF = \frac{73.5 + 70 \times 1.40}{70}$$

SFF = 2.45

**76. The following types of GVF profiles do not exit**
(a) $C_2H_1A_1$ (b) $C_2H_2A_1$
(c) $C_2H_1A_2$ (d) $C_1H_1A_1$

**Ans. (*) :** GVF profile
Mild $\rightarrow M_1, M_2, M_3$
Steep $\rightarrow S_1, S_2, S_3$
Critical $\rightarrow C_1, C_3$
Horizontal slope $\rightarrow H_2, H_3$
Adverse slope $\rightarrow A_2, A_3$

**77. The type of pointing in which the mortar is first pressed into the rakes joint and then finished off flush with the face of the bricks or stones is called**
(a) V-grooved pointing
(b) Flush pointing
(c) tuck pointing
(d) stuck pointing

**Ans. (b) : Flush pointing–** This type of pointing is formed by pressing mortar in the rocket joint and by finishing off flus with edge of masonry units.
• The edges are neatly trimmed with trowel and straight edge.
• It does not give good appearance.
• It is more durable because of resisting the provision at space for dust, water etc.

**78. Which type of pre stressing is used in pre stressed concrete pavements?**
(a) longitudinal (b) contractions
(c) traverse (d) expansion

**Ans. (a) :**

**79. In a triangular notch there is an error of 4% in observing the head, the error in the computation of discharge**
(a) 10% (b) 4%
(c) 6% (d) 2.5%

**Ans. (a) :** For triangular notch or weir–

$$Q = \frac{8}{15} C_d . \tan\frac{\theta}{2} \sqrt{2g} \times H^{5/2}$$

$$Q = KH^{5/2} \quad \text{..... (i)} \left( \text{Where, } K = \frac{8}{15} C_d, \tan\frac{\theta}{2}, \sqrt{2}g \right)$$

Differentiating equation w.r.t. 'H' we get

$$dQ = K\frac{5}{2} H^{3/2} \times dH \quad \text{..... (ii)}$$

Dividing equation (ii) by (i)

$$\frac{dQ}{Q} = \frac{K\frac{5}{2} H^{3/2} \times dH}{KH^{3/2}}$$

$$\frac{dQ}{Q} = \frac{5}{2}\frac{dH}{H}$$

$$\text{Given} = \frac{dH}{H} = 4\%$$

$$\frac{dQ}{Q} = 2.5 \times \frac{dH}{H}$$

$= 2.5 \times 4 = 10\%$

Percentage error in computing discharge = 10%

**80. Water is pumped through a pipeline to a height of 10 m at the rate of 0.1 $m^3/s$, friction and other minor losses are 5 m, pumping power required in kW is**

(a) 13.3 (b) 14.7
(c) 20 (d) 9.8

**Ans. (b) :** Given,
Discharge through pump (Q) = 0.1 $m^3$/sec
Head (H) = 10 m
Friction head ($h_f$)= 5 m
g = 9.81 m/$sec^2$
Power required of pump (P) = $\rho \times Q \times g \times (H + h_f)$
P = 1000 × 9.81 × 0.1 × (10+5) = 1475 WaH
P = 14.715 kW
P = 14.7 kW

**81. a formula for the peak flood discharge from a catchment area A is Q = 450 $A^{2/3}$ where Q is in $Ft^3/s$ and A in in square miles. If Q is to be in $m^3/s$ and A is to be square km. the formula world be Q= $KA^{2/3}$ where K is**

(a) 62.1 (b) 137.1
(c) 25.1 (d) 248.4

**Ans. (a) :** Given,
$Q = 450\ A^{2/3}$
Where, Q → Discharge in $Ft^3$/sec
A → Catchment area in square miles
1 Ft = 0.3048 meter
1 mile = 1.609 km

$$1\frac{Ft^3}{sec} = (0.3048)^3 = 0.0283\ m^3/sec$$

1 square mile = $(1.609)^2 = 2.588\ km^2$
For discharge in $m^3$/sec and area in square km the formula would be–

$$Q = \left[\frac{(1.6092)^{2/3}}{(0.3048)^3}\right] A^{2/3}$$

$Q = 66.48\ A^{2/3}$

**82. A soil has a bulk density of 22 $kN/m^3$ and water content of 10% the dry density of the soil is**

(a) 22.0 $kN/m^3$ (b) 23.2 $kN/m^3$
(c) 18.6 $kN/m^3$ (d) 20.0 $kN/m^3$

**Ans. (d) :** Given,
Bulk density ($\gamma_b$) = 22 $kN/m^3$
Water content (w) = 10%

$$\text{Dry unit weight } (\gamma_d) = \frac{\gamma_b}{1+w}$$

$$\gamma_d = \left(\frac{22}{1+0.10}\right) = 20\,kN/m^3$$

**83. In designing hydraulic structure in alluvial rivers, the equation that is used to calculate the normal depth of scour R for a discharge intensity of q $m^3/s/m$ is**

(a) $R = 1.35\ (q^2/f)^{1/3}$
(b) $R = 1.20\ (q^2/f)^{2/3}$
(c) $R = \sqrt{g}$
(d) $R = 1.35\ (q/f)^{2/3}$

**Ans. (a) :** $R = 1.35\left(\frac{q^2}{f}\right)^{1/3}$

Here, R = Normal depth of scour
q = Discharge per unit width
f = Silt factor

**84. At a joint where three members are meeting, the possible combination of rotation contribution factors is**

(a) -0.05, -0.35, -0.1 (b) -0.2, -0.3, -0.5
(c) -0.1, 0.4, -0.2 (d) 0.2, 0.3, 0.5

**Ans. (a) :** Kani method –
- Kani method is also known as rotation contribution method.
- Kani method rotation factor for joints

$$R_{ij} = -0.5\frac{K_{ij}}{\sum K_{ij}}$$

Where,

$$K_{ij} = \frac{I_{ij}}{L_{ij}} \rightarrow \frac{\text{Moment of inertia}}{\text{Length the member}}$$

And the sum of the rotation factor at any joints – 0.5
- The above criteria are filled only by the rotation factor = 0.05, –0.35, –0.1

**85. The best position for laying a brick, in a masonry construction work is**

(a) with the frog down
(b) with the frog up or down but not to the sides
(c) with the frog to the sides
(d) with the frog up

**Ans. (d) : Frog–** It is indicates the trade name of the brick.
- In brick work, the bricks are laid with frog uppermost. It thus affords a key for mortar when the next brick is placed over it.
- Size of frog = 100 mm × 40 mm × 10 mm

Depth of frog = 10 mm to 20 mm
- The purpose of providing frog is to form a key for holding the mortar.
- The brick are laid with frog on top.

**86. Aggregate comprising particles falling essentially within a narrow limit of size fractions are called**

(a) gap graded aggregates
(b) all-in-all aggregates
(c) uniform aggregates
(d) single size aggregates

**Ans. (d) : Single size aggregates–** Aggregate comprising particles falling essentially within a narrow limit of size fractions are called single size aggregate.

**87. While designing a hill road with a ruling gradient of 1 in 20 on sharp horizontal curve radius of 60 m is encountered, the compensated gradient as per Indian road congress standard is**

(a) 5.0% (b) 1.2%
(c) 3.5% (d) 1.5%

**Ans. (*) :** Given,
Rulling gradient = 1 in 20 = 5%
Horizontal curve radius = 60 m
Grade compensation –

$$\text{Grade compensation} = \frac{30+R}{R}\% = \frac{30+60}{60} = 1.5\%$$

$$\text{But maximum compensation} = \frac{75}{R}\% = \frac{75}{60} = 1.25\%$$

Compensation gradient = Ruling gradient – Grade compensation
= 5% – 1.25%
= 3.75%

**Note-** At horizontal curves, the percentage of compensation in gradient may be provided using the formula (30+R)/R, subjected to a maximum of (75/R). The compensated gradients may not however be flatter than 4 percent.

**88. Bending moment in a abeam is not a function of**

(a) position of the load
(b) type of beam
(c) cross section of the beam
(d) span of the beam

**Ans. (c) :** Bending moment in a beam is not a function of cross-section of the beam.

**89. In a converging steady flow, there is**

(a) no acceleration
(b) only for convective acceleration
(c) no temporal acceleration
(d) convective and temporal acceleration

**Ans. (b) :** When u, v and w are velocity components along x, y, and z direction
Acceleration along the x-axis, y-axis, z-axis

$$a_x = u\frac{du}{dx} + v\frac{du}{dy} + w\frac{du}{dz} + \frac{du}{dt}$$

$$a_y = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}$$

$$a_z = u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}$$

There are two components of the acceleration

$u\frac{du}{dx} + v\frac{du}{\partial y} + w\frac{du}{\partial z}$ this group is called convective acceleration.

$\frac{du}{dt}$ this group is called temporal acceleration.

**90. Which of the following unit work in anaerobic conditions?**

(a) active sludge treatment
(b) sedimentation tank
(c) sludge digestion tank
(d) trickling filter

**Ans. (c) : Sludge digestion tank–** The decomposition of complex organic matters in sludge digestion.
• The sludge can be digested by a sludge digestion tank or an imhoff tank.

**91. St. venant equation represent**

(a) continuity equation in two different forms
(b) continuity and momentum equation
(c) momentum and energy equation
(d) momentum equation in two different forms

**Ans. (b) :** St. Venant equation represent continuity and momentum equation.
Continuity equation–

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$$

Momentum equation–

$$\frac{1}{A}\frac{\partial Q}{\partial t} + \frac{1}{A}\frac{\partial}{\partial x}\left(\frac{Q^2}{A}\right) + g\frac{\partial y}{\partial x} - g\left(S_0 - S_f\right) = 0$$

**92. If an unconfined aquifer released 5 $m^3$ of water of a water table drop of 2 m over a horizontal area of 10 $m^2$, the specific yield is**

(a) 25% (b) 50%
(c) 75% (d) 10%

**Ans. (a) :** Specific yield

$$= \frac{\text{Volume of water drained by gravity}}{\text{Total volume}} \times 100$$

$$= \frac{5}{2\times 10} \times 100$$

$$= \frac{1}{4} \times 100$$

$$= 25\%$$

**93. The maximum velocity occurs in an egg shaped sewer when the ratio of depth of flow to vertical diameter is**

(a) 0.95 (b) 0.81
(c) 0.33 (d) 0.50

**Ans. (b) :** The maximum velocity occurs in an egg shaped sewer when the ratio of depth of flow to vertical diameter is 0.81.

**94. Arrange the order of permanent adjustments of a theodolite**

**i. plate level test**
**ii. cross-hair ring test**
**iii. bubble tube adjustment test**
**iv. spire test**
**v. collimation in azimuth test**
**vi. vertical are test**

(a) I-IV-II-III-VI
(b) III-II-I-IV-VI
(c) II-I-V-IV-VI-III
(d) I-II-V-IV-III-VI

**Ans. (d):** Permanent adjustment in case of a transit theodolite are–

**Adjustment of the horizontal plate levels–** The axis of the horizontal plate levels must be perpendicular tothe vertical axis.

**Cross-hair ring test–** This test is done to make the vertical cross-hair lie in $\alpha$ plane perpendicular tothe horizontal axis.

**Collimation adjustment–** The line of collimation should coincide with the axis of the telescope and axis of the objective slide and should be at right angles to the horizontal axis.

**Horizontal axis adjustment–** The horizontal axis must be perpendicular to the vertical axis.

**Adjustment of the telescope level or altitude level–**

- The axis of the telescope level or altitude level must parallel to the line of collimation.
- **Vertical circular index adjustment–** The vertical circle vernier must read zero when the line of collimation is horizontal.

**95. With reference to degree of precision in measuring offsets in chain surveying what is the value of the offsets should be measured to the nearest, if the proposed scale of plotting is 1 : 200 and a point on the paper having thickness of 0.25 mm?**

(a) 20 cm (b) 15 cm
(c) 5 cm (d) 10 cm

**Ans. (c) :** Given,

Offset $\le 0.025$

Offset $\le 0.025 \times 5$

Offset $\le 0.025 \times 200$

Offset $\le$ 5 cm

- Therefore, the maximum offset that can be measured is 5 cm.

**96. A clayeye soil has field capacity of 35% and permanent wilting point of 20% if the specific weight of soil is 12.75 kN/m$^3$, the available holding capacity in 0.80 m depth of soil, constituting the root zone depth of a crop, is**

(a) 17.5 cm (b) 15.6 cm
(c) 20.8 cm (d) 36.4 cm

**Ans. (b) :** Given,

Field capacity ($F_c$) = 35%

Permanent wilting point (pwp) = 20%

Specific weight of soil = 12.75 kN/m$^3$

Depth of soil (d) = 0.80 m

$\gamma_w$ = 9.81 kM/m

The depth of water stored ($d_w$) in the root zone is

$$d_w = \frac{\gamma_d}{\gamma_w} \times d \times (F_c - PWP)$$

$$d_w = \frac{12.75}{9.81} \times 0.8 \times (0.35 - 0.20)$$

$d_w$ = 0.1559 m

$d_w$ = 15.50 cm $\approx$ 15.6 cm

**97. From the contour map of a pond the following contour areas were plan metered**

| **Contours (m)** | **1107** | **1113** | **1119** |
|---|---|---|---|
| **Area (m$^2$)** | **90** | **88** | **70** |

**By prismoidal rule, the volume of the water in pond is.... m$^3$.**

(a) 1008 (b) 1024
(c) 336 (d) 2880

**Ans. (b) : Prismoidal rule–**

Given,

$A_1$ = 90, D = 6

$A_2$ = 88

$A_3$ = 70

The volume of water in the pond (V)

$$V = \frac{D}{3}\left[1^{st} \text{ area} + \text{last area} + 4\,\text{Even area} + 2 \text{ odd area}\right]$$

$$V = \frac{6}{3}\left[(90 + 70) + 4 \times (88)\right]$$

V = 1024 m$^3$

**98. The plinth area of a building Not includes**

(a) area of cantilevered porch
(b) are of the walls at the floor level
(c) area of stair cover
(d) lift and wall including landing

**Ans. (a) : Plinth area includes –**

- Wall area
- Area of lift (upto 2m$^2$)
- Internal shaft area (upto 2 m$^2$)
- Area of barsati and area of mum by at trace level.
- Area of porch (other than cantilever)
- Area of lofts (shelves or almirah)

**Plinth area does not includes–**

- Lift area (above 2 m$^2$)
- Internal shaft area (above 2 m$^2$)
- Area of cantilever balcony, sunshade device, courtyard, area of loft are of porch not included by support.
- Open parking area
- Open play ground.

**99. If the pitch is 60 mm and rivet value is 40 kN, the number of rivets required for a riveted connection carrying an eccentric load of 150 kN at a distance of 300 mm from the centre line is**

(a) 10 (b) 12
(c) 6 (d) 9

**Ans. (a) :** Given,

Pitch ($\phi$) = 60 mm = 0.06 m

Rivet value ($R_v$) = 40 kN

Eccentric load (P) = 150 kN

e = 300 mm = 0.30 m

M = P $\times$ e

The number of eccentric connection is

$$n = \sqrt{\frac{6M}{pR_v}}$$

$$n = \sqrt{\frac{6\times150\times0.30}{0.060\times40}}$$

$n = 10.6 \approx 10$

$n = 10$

**100. In a subcritical flow in a channel $\Delta Z$ m is the maximum height of a smooth hump that can be installed to cause critical flow over the hump. If a hump of height $\Delta Z > \Delta Z$ m is installed then**

(a) the flow over the hump will be super critical
(b) the flow over the hump will be critical and the lowering of the upstream water
(c) the flow over the hump will be sub critical
(d) the flow over the hump will be critical and the up stream water surface will rise

**Ans. (d) :** In a subcritical flow in a channel $\Delta Z$ m is the maximum height of a smooth hump that can be installed to cause critical flow over the hump. If a hump of height $\Delta Z > \Delta Z$ m is installed then the flow over the hump will be critical and the up stream water surface will rise.

**101. A catchment of area 100 hectares has a runoff coefficient of 0.5. A storm of duration larger than the time of concentration of the catchment and of intensity 7.2 cm/hr causes a peak discharge in $m^3/s$ of**

(a) 360 (b) 50
(c) 100 (d) 10

**Ans. (d) :** Given,
Coefficient of run off (k) = 0.5
Intensity ($P_I$) = 7.2 cm/hr
Catchment area (A) = 100 hectare

$$\text{Peak discharge } (Q_P) = \frac{1}{36}K\times P_I\times A$$

$$Q_P = \frac{1}{36}\times0.5\times7.2\times100$$

$Q_P = 10\ m^3/sec$

**102. GPS uses 24 satellites**

(a) 24 orbits (b) 8 orbits
(c) 12 orbits (d) 6 orbits

**Ans. (d) :** The satellites in the GPS constellation are arranged into six equally-spaced orbital planes surrounding the earth.

**103. What is the allowable limit of variation in overall length of a 30 m metric chain, when measured at 8 kg pull and checked against a steel tape standardized at $20^0$C?**

(a) ± 7 mm (b) ± 8 mm
(c) ± 5 mm (d) ± 6 mm

**Ans. (b) : Chain surveying–**

- The measurements in chain surveying are linear.
- Measurement for this survey are taken on the field and other supplementary works like plotting and calculation are carried out in the office.
- Chain, tape, ranging rod, arrows and cross staff, offset rods, plumb bob are the tools required for chain surveying.

The maximum tolerance (at $20^0$C with 8 kg pull) allowable for the metric chain are as follows–

05 m chain $\rightarrow \pm 3$ mm
10 m chain $\rightarrow \pm 3$ mm
20 m chain $\rightarrow \pm 5$ mm
30 m chain $\rightarrow \pm 8$ mm

**104. If r is the normal depth of scour by Lacey equation, the depths below HFL up to which the upstream and downstream sheet piles in a weir are usually preferred to protect against the effect of scour are respectively**

(a) 2.0 R and 1.5 R (b) 1.0 R and 1.25 R
(c) 2.5 R and 1.25 R (d) 1.5 R and 2.0 R

**Ans. (c) :** The value of scour depth R, that is the normal depth of scour below HFL may be determined according to Lacey's scour relations.
Location maximum scour depth for upstream curved head of guide bund is 2.5 R and for straight reach of guide bund to nose at downstream guide bund is 1.5 R

**105. A canal was designed to supply the irrigation needs of 1200 ha of growing rice of 140 days base period and having a delta of 134 cm. If the canal waters are used to irrigate wheat of base period 120 days and having delta of 52 cm, the area that can be irrigated is**

(a) 399 ha (b) 2651 ha
(c) 543 ha (d) 3608 ha

**Ans. (b) :** Delta $(\Delta) = \frac{8.64B}{D}$

$$D_{rice} = \frac{8.64\times140}{1.34} = 902.6 \text{ ha/cum}$$

Discharge from canal (Q)

$$Q = \frac{\text{Area}}{\text{Duty}} = \frac{1200}{902.6} = 1.3294 \text{ cumec}$$

Now duty for wheat $\{\text{Delta}(\Delta) = 52\text{cm} = 0.52\text{m}\}$

$$D_{wheat} = \frac{8.64\times B}{\Delta} = \frac{8.64\times120}{0.52} = 1993.846 \text{ ha/cumec}$$

Now,

$$Q = \frac{\text{Area}}{\text{Duty}} \Rightarrow \text{Area} = Q\times\text{Duty}$$

$A = 1993.846 \times 1.3294$

$A = 2650.62$ heet □ 2651 ha

**106. Activated carbon is used for**

(a) removing hardness
(b) removing corosiveness
(c) removing odours
(d) disinfection

**Ans. (c) : Activated carbon is used–** This is a specially treated carbon, which possesses the property of absorbing and attracting impurities.

• Due to absorption property is removes taste, colour and odour.
• It removes organic matter present in water.
• It removes taste, odours and colour caused by the presence of iron magnese phenols, excess chlorine, hydrogen sulphide etc.
• It averades is not harmful.
• The activated carbon is mostly used in the powdered form and may be added to water either before or after the coagulation but before filtration.

**107. U tube manometers are employed to measure pressure of which type of fluids**
(a) liquids (b) liquids and gases
(c) solids (d) gases

**Ans. (b) : U-tube manometer :** A U-tube manometer consists of two tubes joined at one end to form a U-shaped tube.
• The two tubes are said to constitute the limbs of the U-tube manometer.
• A U-tube manometer can be employed to measure the pressure in any fluid liquid or gases.

**108. In an irrigated plot, the net irrigation requirement of a crop is found to be 14.9 cm, the application efficiency is 80% and the water conveyance efficiency is 70% the gross irrigation requirement is**
(a) 26.6 cm (b) 8.3 cm
(c) 17.0 cm (d) 21.3 cm

**Ans. (a) :** Given,
Net irrigation requirement (NIR) = 14.9 cm
Application efficiency ($n_a$) = 80% = 0.80
Conveyance efficiency ($n_c$) 70% = 0.70
Gross irrigation requirement (GIR) = $\frac{NIR}{n_a \times n_c}$
$GIR = \frac{14.9}{0.8 \times 0.70}$
GIR = 26.60 cm

**109. One litre of sewage, when allowed to settle for 30 minutes gives a sludge volume of 27 $cm^3$, of the dry weight of this sludge is 3 grams, the sludge volume index will be**
(a) 30 (b) 81
(c) 9 (d) 24

**Ans. (c) :** Given,
Volume of settled sludge = 27 $cm^3$
= 27 m*l*
Dry weight of the sludge = 3 gm
Sludge volume Index (SVI) =
$\frac{\text{Volume of settled sludge}}{\text{Dry weight of sludge}}$
$SVI = \frac{27}{3} = 9 \text{ m}l/\text{gram}$

**110. If E and G represents the moduli of elasticity and shear of a material respectively, which of the following is not a possible relation**
(a) E = 3G (b) E = 2G
(c) E < G (d) E = G

**Ans. (c,d) :** Young modulus (E), bulk modulus (K) and rigidity modulus (G) at an elastic solid are together called elastic constants.
Young modulus (E) can be expressed using K, and is (Poisson's ratio)
$E = 3K(1 - 2\mu)$
E can also be expressed using G (rigidity) and μ
$E = 2G(1+\mu)$
For μ = 0 For μ = 0.5
E = 2G E = 3G
E value lies between 2G & 3G
So option (a) & (b) will be possible and option (c) & (d) not possible.

**111. Acccording to the Indian roads congress practice (IRC : 37-2012), the standard axle load prescribed for tridem-axles with dual wheels on either side configuration is**
(a) 148 kN (b) 240 kN
(c) 224 kN (d) 80 kN

**Ans. (c) :** According to the Indian roads congress practice (IRC 37-2012)

| Types of axle | Standard axle load in kN |
|---|---|
| Single axle with single wheel on either side | 65 kN |
| Single axle with dual wheel on either side | 80 kN |
| Tandem axle with dual wheels on either side | 148 kN |
| Tridem axles with dual wheels on either side | 224 kN |

**112. The critical depth equation for a triangular channel is**
(a) $v/v\sqrt{gy}$ (b) $(2Q^2/gZ^2)^{1/5}$
(c) $(q^2/g)^{1/3}$ (d) $(3Q^3/gZ^2)^{1/3}$

**Ans. (b) :** Critical depth for triangular channel–
$y_c = \left[\frac{2Q^2}{g\tan^2\theta}\right]^{1/5}$
$y_c = \left[\frac{2Q^2}{gz^2}\right]^{1/5}$
Where,
$y_c$ = critical depth
Q = discharge
$z = \tan\theta$

**113. The population of a town in three consecutive years is 5000, 7000 and 8400 respectively. The population of the town in the fourth consecutive year according to geometrical increase method is**
(a) 10100 (b) 9500
(c) 9800 (d) 10920

**Ans. (d):** Population in I[st] year is $P_1$= 5000
Population in II[nd] year is $P_2$ = 7000
Increment (I) = 7000–5000 = 2000

Geometrical rate growth in $G_1 = \frac{2000}{5000} = 0.4$

Population in 3[rd] year $P_3$= 8400
Increment = (8400–7000) = 1400

Geometrical rate growth is $G_2 = \frac{1400}{7000} = 0.20$

Geometrical mean ($G_m$)

$$G_m = \sqrt{G_1 \times G_2} = \sqrt{0.40 \times 0.20}$$

$G_m = 0.282 = 28.28\%$

Population in fourth (4th) year $P_4$

$P_4 = P_3 (1+ G_m)$
$= 8400 (1 + 0.2828 )$
$= 10775.52 = 10776$

So, nearest option (d) 10920 is correct.

**114. For a laminar flow the reduction of radius to half will increase the pressure gradient by a factor**
(a) 4 (b) 8
(c) 2 (d) 16

**Ans. (d) :** Laminar flow–

Flow rate $Q = \frac{\pi R^4 \Delta P}{8nL}$

Q = flow rate
Δ P = pressure difference
R = Pipe radius
n = viscosity
L = Pipe length
Pressure gradient

$$\Delta P = \frac{Q \times 8nL}{\pi R^4}$$

Keeping all other parameter constant
We have

$$\frac{\Delta P}{L} \propto \frac{1}{R^4}$$

Radius reduction half $R' = \frac{R}{2}$

$$\left(\frac{\Delta P}{L}\right)' = 16\left(\frac{\Delta P}{L}\right)$$

When the reduction of radius to half will increase the pressure gradient by a factor of 16.

**115. Which of the following is a good example of a composite bar?**
(a) a steel stanchion
(b) a stone pillar
(c) a reinforced concrete column
(d) a wooden post

**Ans. (c) : Composite bars –**
A composite bar made of two bars of different materials rigidity fixed together so that both bars strain together under external load.

Since strain in the two bars are same, the stresses in the two bars depend on their young's modulus of elasticity.
Example → Reinforced concrete column
Flitched beam

**116. The terminal velocity of a small sphere falling under the action of gravity in a very viscous fluid values**
(a) inversely as the viscosity of fluid
(b) directly as the viscosity of fluid
(c) directly as the diameter of sphere
(d) inversely as the diameter of sphere

**Ans. (a) :** $V_s = \frac{(G-1)\rho g d^2}{18\mu}$

$$V_s \propto \frac{1}{\mu}$$

Here,
$V_s$ = Settling velocity/Terminal velocity
d = Diameter of sphere particles
μ = Viscosity of fluid
G = S.p.g. of sphere particle

**117. what is the capillary rise of water above the water table, if the water table is at a depth of 10 m in a silty soil, void ratio (e) of the soil is 0.50 and the effective diameter $D_{10}$ of the soil mass is 0.05 mm, assume the emprict constant (C) value = 0.5 sq.cm**
(a) 0.50 cm (b) 200 cm
(c) 50 cm (d) 196 cm

**Ans. (b) :** Given,
Void ratio (e) = 0.50
Effective diameter of soils ($D_{10}$) = 0.05 mm
= 0.005 cm
Constant (c) = 0.5 sq.cm.
The height of capillary in soils is ($h_c$)

$$h_c = \frac{C}{e \times D_{10}} = \frac{0.5}{0.50 \times 0.005}$$

$h_c$ = 200 cm

**118. Torque per unit twist of a shaft is known as**
(a) stiffness of the shaft
(b) strength of the shaft
(c) deflection of the shaft
(d) curvature of the shaft

**Ans. (a) :** Torque per unit twist of a shaft is known as stiffeners of the shaft.
Torsional equation

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

**119. In the case of normal concrete, approximately how much quantity of water is required to hydrate 1 gram of cement?**
(a) 0.235 (b) 0.253
(c) 0.532 (d) 0.325

**Ans.(a):** On an average 23% of water by weight of cement is required for chemical reaction with portland cement compounds, about 15% water by weight of cement is required to fill the gel-pore.

- Thus 38% water by weight is required for chemical reaction and gel pores.
- If more than 38% of water is used, the excess will cause undesirable cavities.

**120. What is the airport referenced temperature, if the monthly mean of average daily temperature for the hottest month of a study year = $24^0$C and the monthly mean of the maximum daily temperature for the same month of the same year = $30^0$C.**

(a) $26^0$C (b) $24^0$C
(c) $30^0$C (d) $27^0$C

**Ans. (a) :** Reference temperature = $t_a + \frac{t_m - t_a}{3}$

Where,

$t_a \rightarrow$ Monthly mean of average daily temperature for the hottest of a study year = $24^0$C

$t_m \rightarrow$ Monthly mean of the maximum daily temperature for the same month of the some year = $30^0$C

Reference temperature = $24 + \frac{30-24}{3}$

$= 24 + 2 = 26^0$C

**121. At a work site 25 mm diameter bars of grade-2 mild steel where used under inevitable situation. The permissible stress in tension (in MPa) should be taken as**

(a) 126 (b) 130
(c) 140 (d) 117

**Ans. (d):** The permissible stress in steel as per IS 456-2000

| Types of stress in steel reinforcement | Permissible stress | | |
|---|---|---|---|
| | Mild steel grade-I | Mild steel grade-II | High yield deformed bars (Fe 415) |
| UP to 20 mm | 140 | 90% of 140 = 126 | 230 |
| Over 20 mm | 130 | 90% of 130= 117 | 230 |
| Compression in lumn bars | 130 | 90% of 130= 117 | 190 |
| Compression in beams/slabs where compressive resistance of concrete is not taken. | | | |
| UP to 20 mm | 140 | 20% of 140 = 126 | 190 |
| Over 20 mm | 130 | 90% of 130 = 117 | 190 |

**122. The hydrostatic law**

(a) holds unaltered to for a liquid in a vessel subjected to constant rotation
(b) holds unaltered for a liquid in a vessel subjected to constant acceleration
(c) doesn't hold for the atmosphere
(d) holds for a liquid in equilibrium only

**Ans. (a) : Hydrostatic law –** According to the hydrostatic law, at any point inside a static fluid, the vertical rate of increase of pressure must be equal to the local specific weight of the fluid.

- The hydrostatic law holds unaltered to for a liquid in a vessel subjected to constant rotation.

**123. The predominant fluid property associated with cavitation phenomenon**

(a) surface tension (b) vaour pressure
(c) mass density
(d) bulk modulus of elasticity

**Ans. (b) :** Cavitation is a phenomenon in which static pressure of liquid reduces to below the liquid's vapour pressure, leading to the formation of small vapour filled cavities in the liquid.

**124. At the foot of a spillway, the jump rating curve is above the tail water rating curve at low discharges and below it is at higher discharges. The suitable type of energy dissipater for this situation is**

(a) a horizontal apron
(b) a roller bucket
(c) ski jump bucket
(d) a secondary dam combined with a sloping apron

**Ans. (d) : Spillway–** A spillway is a structure constructed at a dam site, for effectively disposing of the surplus water from upstream to down stream.

- The frequency of flow over a spillway depends upon

$\rightarrow$ Runoff characteristics of the drainage area

$\rightarrow$ Reservoir storage

- At the foot of a spillway, the jump rating curve is below the tail water curve at low flows and above it at high flows. The suitable types of energy dissipater for this situation is a sloping apron partly above and partly below the river bed.

**125. Two irrigation channels are A and B are designed to using Lacey's theory. If the median diameter of the silt particles are the same for both the channel and if the discharge in channel A is 50% greater than at B, then between these two channels**

(a) channel A has smaller hydraulic radius
(b) channel B has smaller longitudinal slope
(c) channel B has larger wetted perimeter
(d) channel A has smaller longitudinal slope

**Ans. (d) :** We know that,

$S = \frac{f^{5/3}}{3340 Q^{1/6}}$ $\qquad V = \left(\frac{Qf^2}{140}\right)^{1/6}$

$f = 1.76\sqrt{d}$

Given,

$Q_A = 1.5 Q_B$

$Q_A > Q_B$

$S \propto \frac{1}{Q}$

So channel A has smaller longitudinal slope.

**126. In GIS, interpolation is made possible by a principle called**

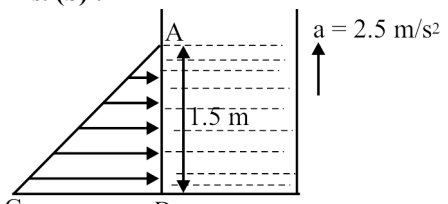
(a) Spatial autocorrelation
(b) Thermatic autocorrelation
(c) Spatial auto-correction
(d) thematic auto-correction

**Ans. (a) : Spatial auto correlation–** In GIS, interpolation is made possible by a principle called spatial auto-correlation.

**127. An open tank contains water to a depth of 1.5 m. the tank is put in an elevator which accelerates at 2.5 m/sec$^2$. The pressure force per meter length on one side of tank when the acceleration is upward is**

(a) 9.8 kN (b) 13.82 kN
(c) 55.2 kN (d) 15.85 kN

**Ans. (b) :**




$\rho gh\left(1+\frac{a}{g}\right)$

Gauge pressure at any point subject constant according in upward direction.

$$= \rho gh\left(1+\frac{a}{g}\right)$$

$$= 1000\times 9.81\times 1.5\left(1+\frac{2.5}{9.81}\right)$$

$P = 18465\ N/m^2$

$$\text{Pressure force} = \frac{1}{2}\times CB\times BA\times \text{width of tank}$$

$$= \frac{1}{2}\times 18465\times 1.5\times 1$$

$= 13848.75$ N

$F = 13.84$ kN

Note

If downward according act

$$P = \rho gh\left(1-\frac{a}{g}\right)$$

**128. Which of the following statements is not true in case of IS code method of mix design?**

(a) Water cement ratio is selected based on the computed target mean strength
(b) Limitation are kept on maximum water cement ratio and minimum cement content
(c) Vee bee time is used for workability
(d) Type and maximum size of aggregate are to be specified

**Ans. (c) :** IS code method of concrete mix design procedure–

- Target mean strength of concrete
- Selection of water cement ratio
- Selection of water and sand content.
- Determination of cement content.
- Determination of coarse and fine aggregate content.
- Determination of total volume of all the materials.

Factors depend on mix design-

- Grade designation
- Type of cement
- Maximum nominal size of aggregate
- Minimum water cement ratio
- Workability
- Minimum cement content

So, option (c) is correct.

Workability is generally checked by slump test during concrete mix design.

**129. Leachate is coloured liquid that comes out of**

(a) aerated lagoons
(b) septic tanks
(c) compost plants
(d) sanitary landfill

**Ans. (d)** : During rainy season, when excess water seeping through the area, may come out of the dump, as a coloured liquid, called Leachate. This is highly poisonous and polluted.

**130. When the Mohr's circle drawn for a certain stress state on a plane in a body, is tangential to the X-axis, which of the following may be true?**

(a) both the bi-axial normal stresses are of equal magnitude but of same nature
(b) shear stress on that plane is zero
(c) both the bi-axial normal stresses are of equal magnitude but of opposite nature
(d) one of the bi-axial normal stresses is zero

**Ans. (d) :** Case I : biaxial same nature -

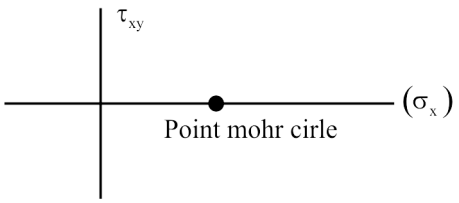
$$\sigma_x = \sigma,\ \sigma_y = \sigma,\ \tau_{xy} = 0$$

$$C = \left[\left(\frac{\sigma_x+\sigma_y}{2}\right), 0\right]$$

$$= (\sigma, 0)$$

$$R = \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}$$

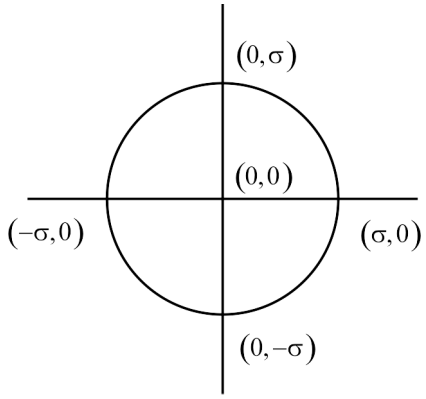
$$= \sqrt{\left(\frac{\sigma-\sigma}{2}\right)^2+0} \quad = 0$$




Case II : Biaxial opposite nature

$$C = \left[\left(\frac{\sigma_x + \sigma_y}{2}\right), 0\right] = (0, 0)$$

$$R = \sqrt{\left(\frac{\sigma - (-\sigma)}{2}\right)^2 + 0} \quad = \sigma$$

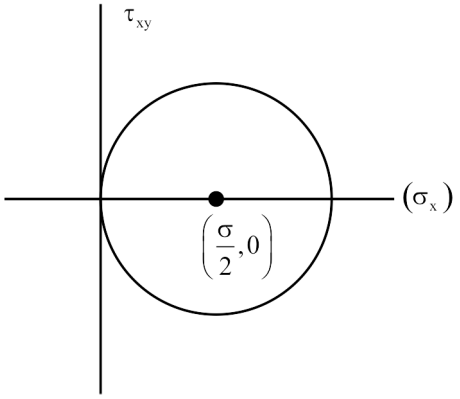



Case III

$$\sigma_x = \sigma,\ \sigma_y = 0,\ \tau_{xy} = 0$$

$$C = \left[\left(\frac{\sigma + 0}{2}\right), 0\right] = \left[\frac{\sigma}{2}, 0\right]$$

$$R = \sqrt{\left(\frac{\sigma - 0}{2}\right)^2 + 0} \quad = \left(\frac{\sigma}{2}\right)$$




**131. Generally the detention period of grit chamber is kept as**

(a) 2-4 hrs (b) 12 hrs
(c) 5 minute (d) 1 minute

**Ans. (d) :**

Detention time for different units –

| Units | Detention time |
|---|---|
| Grit chamber | 40-60 sec |
| Detritus | 3-4 min |
| Oxidation pond | 20-30 days |
| Imhoff tank | 1-2 hrs |
| Septic tank | 12-36 hrs |
| Aerated lagoon | 1-2 month |

**132. In a field test of a formation having a porosity of 25%, the hydraulic gradient was formed to be 0.04, and the velocity of a tracer added to the ground water was 6 m/h, the permeability of the aquifer is**

(a) 0.04 cm/s (b) 1 cm/s
(c) 3 cm/s (d) 0.01 cm/s

**Ans. (b) :** Given,
Porosity (n) = 25% = 0.25
Hydraulic gradient (i) = 0.04
Seepage velocity ($V_s$) = 6 m/h

$$V_s = \frac{600}{3600} = \frac{1}{6}\text{cm/sec}$$

Discharge velocity ($V_d$) = ki (k → coefficient of permeability)
Relation between discharge velocity and seepage velocity.

$$V_d = nV_s$$

$$ki = nV_s$$

$$k \times 0.04 = 0.25 \times \frac{1}{6}$$

Permeability of the aquifer (k) = $\frac{0.25}{0.04} \times \frac{1}{6}$

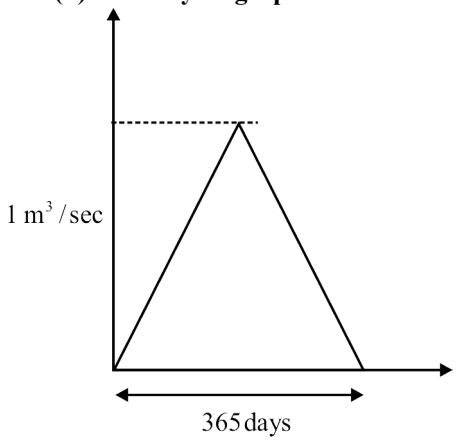
$$k = 1.042 = 1\text{ cm/sec}$$

k = 1 cm/sec

**133. A mean annual runoff 1 $m^3$/s from a catchment of area 10 $km^2$ represents an effective rainfall of**

(a) 10.0 cm (b) 315.4 cm
(c) 100 cm (d) 31.54 cm

**Ans. (b) : Unit hydrograph method –**




Volume of catchment area = volume of unit hydrograph
10 $km^2$ × effective rain fall depth (d)

$$= \frac{1\text{ m}^3}{\text{sec}} \times 365\text{ days}$$

$$d = \frac{1\text{ m}^3}{\text{sec}} \times \frac{365 \times 24 \times 60 \times 60\text{sec}}{10 \times 10^6\text{ m}^2}$$

d = 3.1530 m
d = 315.36 cm = 315.4 cm

**134. In a sudden contraction the velocity head changes from 0.5 m to 1.25 m. the coefficient of contraction is 0.66, the head loss in this contraction is**

(a) 0.648 (b) 0.133 m
(c) 0.6444 m (d) 0.332 m

**Ans. (d) :** Given,

Velocity head after contraction point $\frac{V_2^2}{2g} = 1.25$ m

Coefficient of contraction ($C_c$) = 0.66

Head loss due to sudden contraction ($H_L$)

$$H_L = \frac{(V_c - V_2)^2}{2g} = \frac{V_2^2}{2g}\left(\frac{1}{C_c} - 1\right)^2$$

$$H_L = 1.25 \times \left(\frac{1}{0.66} - 1\right)^2$$

$H_L = 0.331726$ m

$H_L = 0.332$ m

**135. A wheel of 114 kN axle load is coming on to the cantilever portion of a T-beam bridge with it's centre of gravity located at 0.6 m from the end longitudinal girder. In what width (in meters) this wheel load is effective**

(a) 1.31 (b) 1.13
(c) 2.33 (d) 3.11

**Ans. (b) :**

**136. Which one of the following solid waste disposal methods is ecologically most acceptable?**

(a) sanitary landfill (b) composting
(c) pyrolysis (d) incineration

**Ans. (b) : Composting–** Composting is a biological method for the decomposition of organic wastage.

- This is an environmentally, and ecologically acceptable method.

**137. In a BOD test, 1.0 ml of raw sewage was diluted to 100 ml and the dissolved oxygen concentration of diluted sample at the beginning was 6 ppm and it was 4 ppm at the end of 5 days incubation at $20^0$C. The BOD of raw sewage will be**

(a) 400 ppm (b) 200 ppm
(c) 300 ppm (d) 100 ppm

**Ans. (b) :** Given,

Initial dissolved oxygen ($D_i$) = 6 ppm

Final dissolved oxygen ($D_f$) = 4 ppm

The BOD of the raw sewage at the end of 5 days incubation period.

$$BOD = (D_i - D_f) \times D_F$$

$D_F \rightarrow$ Dilution factor

$$D_F = \frac{\text{Volume of diluted sample}}{\text{Volume of raw sewage}}$$

$$D_F = \frac{100}{1}$$

$$BOD = (6-4) \times \frac{100}{1}$$

BOD = 200 ppm

**138. The primary pollutant caused by incomplete combustion of organic matter is**

(a) sulphur dioxide (b) carbon monoxide
(c) ozone (d) nitrogen oxides

**Ans. (b) : Carbon monoxide–** Carbon monoxides is a colourless, odourless and tasteless toxic gas produced by the incomplete combustion of carbonaceous fuels such as wood, petrol, charcoal, natural gas and kerosene.

**139. In an old map, a line RS was drawn to a magnetic bearing of $5^0$30' and the magnetic declination at that time being $1^0$East. To what magnetic bearing should the line be set now, if the present magnetic declination is $8^0$30' East?**

(a) $13^0$ (b) $354^0$
(c) $360^0$ (d) $358^0$

**Ans. (d) :** Given,

Magnetic bearing (M.B.) = $5^0 30^0$

Magnetic declination ($D_E$) = $1^0$ East

TB = $M.B + D_E$

TB = M.B. + Magnetic declination (Eest)

= $5^0$30' + $1^0$

If present magnetic declination (D) = $8^0$30' East

Then find magnetic bearing (M.B.) = ?

T.B. = M.B. + Magnetic declination

$6^0$30' = M.B. + $8^0$30'

M.B. = $6^0 30^0 - 8^0$30'

M.B. = $2^0$

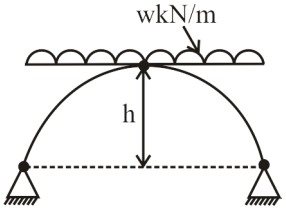
M.B. = $360^0 - 2^0$

M.B. = $358^0$

**140. A three hinged parabolic arch of span L and rise h carries a UDL of w kN/m over the entire span. At any section**

(a) bending moment alone is zero
(b) shear force alone is zero
(c) both bending moment and shear force are zero
(d) both bending moment and shear force are not zero

**Ans. (c) :**




$$-RA \times \frac{L}{2} + H \times h + \frac{w\ell^2}{8} = 0$$

$$-\frac{w\ell^2}{4} + H \times h + \frac{w\ell^2}{8} = 0$$

$H = \frac{w\ell^2}{8H}$

Now taking moment at a section, L = x

$$M_x = \frac{wx^2}{2} + \frac{w\ell^2}{8n} \times y - \frac{w\ell}{2}x$$

Putting $x = \frac{L}{4}$

So by parabolic equation

$$y = \frac{4hx(\ell - x)}{\ell^2} = \frac{4h \times \frac{\ell}{4}\left(\ell - \frac{\ell}{4}\right)}{\ell^4}$$

$$y = \frac{3h}{4}$$

How putting value of x and y in equation.

$$M_x = \frac{wx^2}{2} + \frac{w\ell^2}{8n}\frac{4h(\ell x - x^2)}{\ell^2} - \frac{w\ell x}{2}$$

$$M_x = \frac{wx^2}{2} + \frac{w\ell x}{2} - \frac{wx^2}{2} - \frac{w\ell x}{2}$$

$$M_x = \frac{wx^2}{2} + \frac{w\ell x}{x} - \frac{wx^2}{2} - \frac{w\ell x}{2}$$

$$\frac{\partial M_x}{dx} = wx + \frac{w\ell}{2} - wx - \frac{w\ell}{2}$$

$V_x = 0$

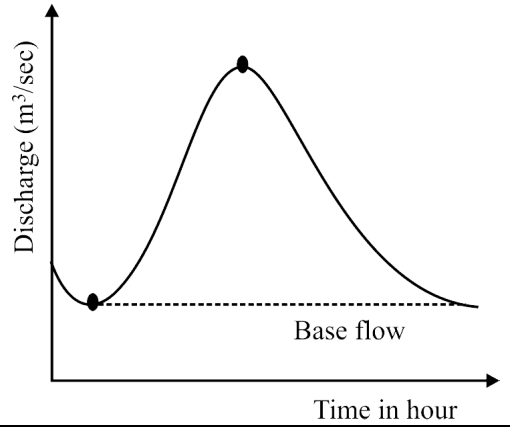
**141. A hydrograph is a plot of**

(a) stream discharge against time
(b) rainfall intensity against time
(c) cumulative rainfall against time
(d) cumulative runoff against time

**Ans. (a) : Hydrograph–** A hydrograph is a continuous plot of instantaneous discharge V/S time.

It results from combination of physiographic and meteorological conditions in a watershed and represents the integrated effects of climate, hydrologic losses. surface runoff, interflow and ground water flow.

- It is also known as storm hydrograph or flood hydrograph.




**142. The double mass curve technique adopted to**

(a) estimate the missing data
(b) obtain intensities of rainfall at various duration
(c) to obtains the amount of storage needed to maintain a demand pattern
(d) to check the consistency of data

**Ans. (d) :** The double mass curve technique adopted to check the consistency of data.

- Mass curve technique adopted to reservoir capacity or total amount of rainfall.

**143. What would be the possible number of total conflicts points with a rotary 3-legged intersection having 2-lane 2-way operation**

(a) 12 (b) 6
(c) 9 (d) 3

**Ans. (b) :** 6 possible number of total conflicts points with a rotary 3-legged intersection having 2-lane 2-way operation.

**144. The process of making the background rough, before plastering is**

(a) hacking (b) doubbing
(c) blistering (d) peeling

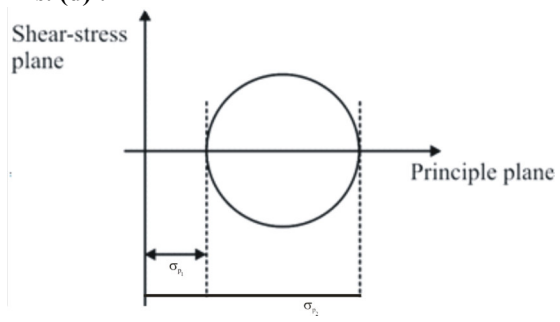
**Ans. (a) : Hacking–** The process of making the background rough, before plastering is hacking.

- Hacking is done on the finished wall before application of plaster of Paris or tiles.
- It is the process of roughening the wall surface so that plaster can rest on it otherwise it would get shear off due to plane surface of the wall.
- Generally it is adopted for renovation purpose (applying pop) earlier mason prepare it with cement plaster and later on if owner wants to renovates wall of their house with tiles or pop they use this process called hacking.

**145. If $P_1$ and $P_2$ are the bi-axial normal stresses and q is shear stress on a plane in a strained body then what is condition that both the principal stresses will be of the same sign?**

(a) $(p_1p_2)$ is equal to $q_2$
(b) $(p_1p_2)$ is equal to $q_2$
(c) $(p_1p_2)$ is equal to $q_2$
(d) $(p_1p_2)$ is equal to $q_2$

**Ans. (d) :**




Major principal stress $(\sigma_{p_1})$

$$\left(\sigma_{p_1}\right)=\frac{P_1+P_2}{2}+\frac{1}{2}\sqrt{\left(P_2-P_1\right)^2+4q^2}$$

Minor principal stress $\left(\sigma_{p_2}\right)$

$$\sigma_{p_2}=\frac{P_1+P_2}{2}-\frac{1}{2}\sqrt{\left(P_2-P_1\right)^2+4q^2}$$

To have $\sigma_{p_1}$ and $\sigma_{p_2}$ are the same sign,

$$\sigma_{p_1}.\sigma_{p_2}\geq 0$$

$$\left(\frac{P_1+P_2}{2}+\frac{1}{2}\sqrt{\left(P_2-P_1\right)^2+4q^2}\right)\left(\frac{P_1+P_2}{2}-\frac{1}{2}\sqrt{\left(P_2-P_1\right)^2+4q^2}\right)\geq 0$$

$$\left(\frac{P_1+P_2}{2}\right)^2-\left(\frac{1}{2}\sqrt{\left(P_2-P_1\right)^2+4q^2}\right)^2\geq 0$$

$$P_1P_2\geq q^2$$

So, $P_1$, $P_2$ is greater than or equal to $q^2$.

**146. Which of the following statements is not true in case of portland slag cement**

(a) better sulphate resistance

(b) slag should not be in glassy form

(c) lower heat of hydration

(d) slower early strength

**Ans. (b) : Portland slag cement–**

- Mixture of portland cement granulated blast furnace slag and gypsum.
- It requires more time for curing and shuttering.
- High sulphate ristance and it is used in mass concreting.
- Slag should be in glassy form, generally the glass content of the slag, should be in excess of 90% to show satisfactory properties.

**147. Hearting is the portion is a**

(a) wall between facing and backing

(b) brick cut in such a manner that is one long face remains uncut

(c) brick cut across the width

(d) wall of exposed to weather

**Ans. (a) :** Hearting is the portion is a wall between facing and backing.

**148. A weir across an alluvial river has a horizontal floor length 54 m and retains 6 m of water under full pond conditions. If the downstream sheet pile is driven to a depth of 6 m below the average bed level, the exit gradient for this case is**

(a) $1/(\pi\sqrt{5})$ approx

(b) $1/(\pi\sqrt{2})$ approx

(c) $1/(\pi\sqrt{3})$ approx

(d) $1/(\pi\sqrt{6})$ approx

**Ans. (a) :** Given,

Horizontal floor length, (b) = 54 m

Depth of sheet pile, (d) = 6 m

Head causing the flow (H) = 6 m

The exit gradient is

$$G_e=\frac{H}{d}\times\frac{1}{\pi\sqrt{\lambda}}$$

$$\lambda=\frac{1+\sqrt{1+\left(\frac{b}{d}\right)^2}}{2}=\frac{1+\sqrt{1+9^2}}{2}$$

$$\lambda=\frac{10.055}{2}=5.02769$$

$\lambda\approx 5$ approx

$$G_e=\frac{6}{6}+\frac{1}{\pi\sqrt{5}}$$

$$G_e=\frac{1}{\pi\sqrt{5}}\text{ approx}$$

**149. The layout of distribution of which water flows towards the outer periphery is**

(a) Dead end system

(b) Grid iron system

(c) Ring system

(d) Radial system

**Ans. (d) : Radial system–** The supply pipes are laid rapidly ending towards the periphery.

- The area is divided into different zones.
- The water pumped into the distribution reservoir kept in the middle of each zone.

**150. Lower food to Micro-organism (F/M) ratio in a conventional activated treatment plant will mean**

(a) Lower BOD removal

(b) Higher BOD removal

(c) No effect on BOD removal

(d) Medium BOD removal

**Ans. (b) :** Food (F) to micro-organism (m) ratio =

- It is also called organic loading.

$$\frac{F}{M}=\frac{\text{kg of BOD applied per day}}{\text{kg of mass in aeration tank}}$$

- Lower the F/M ratio, higher the BOD removal in plant.

# Telanagana State Public Service Commission
# Assistant Environmental Engineer Exam, 2017
## Solved Paper
**Date - 17.05.2017**

**1. Under what legal enactment are Indian Forests protected?**
(a) Forest Conservation Act
(b) Environmental Protection Act
(c) Wildlife Protection Act
(d) Reserved Forests Act

**Ans. (a) : Forest Conservation Act 1980**
■ Forest conservation act came into existence on 25th Oct 1980.
■ It extended to pan India except Jammu & Kashmir at that time however it extends to pan India now after revocation of Article 370.
■ It ensure conservation of forest & forest resources & prevents the loss of forest biodiversity.
**Environmental Protection Act-** This act came into force from 19th November, 1986. It is widely considered as a response to Bhopal gas leak.
**Wildlife Protection Act-** This act came into force in 1972. It provides for protection wild animals, birds & plants etc.

**2. What would be the preferred food of Dugong dugong, the sea cow?**
(a) Small fish (b) Coral reefs
(c) Sea weed (d) Sea grass

**Ans. (d) :** Dugong - Dugong (Sea Cow)
■ It is a marine mammal commonly known as seller's sea cow.
■ It largely depends on sea grass for its food, hence forth these are found near to coastal areas.

**3. Which were the top timber species that were harvested by the British?**
(a) Teak, sal and deodar
(b) Teak, bamboo and sal
(c) Bamboo, sal and rosewood
(d) Bamboo and rosewood

**Ans. (a) :** An FAO report claims that in colonial times it was believed that forest it a national resource which should be utilized for the interest of government. Forest areas become a source of revenue for Britishers. Teak was extensively exploited for shipbuilding while cal, pine & deodar were harvested for making railway sleepers.

**4. Which list protects animals in India?**
(a) Red list (b) Scheduled list
(c) Protection list (d) Endangered list

**Ans. (b) :** The wild life protection at (WPA) of 1972 is divided into six schedule list that provide varying degree of protection.
• Poaching, smuggling and illegal trade in animal listed in schedule 1-4 are all prohibited.
• The WPA 1972 govern wild life conservation and protection in India.
• India is the first country in the world to include measure for environment preservation and conservation in its constitution.

**5. From the list below, identify the most appropriate set of keystone species–**
(a) Ficus, Termite Mound
(b) Ficus, Gaur
(c) Peacock, Cobra
(d) Cobra, Neem

**Ans. (a) :** A keystone species is an organism that helps to define an entire ecosystem, without keystone species the ecosystem will be dramatically different or cease to exist altogether.
■ Ficus (Fig trees) are keystone species as key provide food & shelter to many birds, reptiles, frogs, bats insects etc.
■ Termites build massive natural structures that provides habitats for plants & animals.
■ Termites feed on dead plant material which helps to clean up habitats & cycle nutrients through ecosystem.

**6. What is MRR technique?**
(a) Mark, Release, Recapture
(b) Mark, Rescue, Restore
(c) Mark, Revive, Rescue
(d) Mark, Recapture, Release

**Ans. (a): MRR Technique-** It is a technique to measure size of population of small mammals in an area.
■ It stands for mark, release & recapture.

**7. What does the Ramsar Convention address?**
(a) Protection of water bodies
(b) Protection of rivers
(c) Protection of wetlands
(d) Protection o seas

**Ans. (c) :** RAMSAR convention address for conservation of wetlands signed between countries at small Iranian town named Ramsar in 1971.

**8. Who drafts or develops the Red List of Species**
(a) Government of India
(b) The United National Environment Programme
(c) The International Union for the Conservation of Nature
(d) Biodiversities

**Ans. (c) : Red list of species-** International union for conservation of natures red list of threatened /Endangered species was established in 1964.
■ It has evolved as most comprehensive information source on global conservation status of animals, fungi & plant species.

**9. How many international conventions govern the protection of nature?**

(a) 7 (b) 8
(c) 6 (d) 4

**Ans. (a) :** There are total 7 international conventions governing the protection of nature till date-

| Name of convention | Year of establishment |
|---|---|
| Ramsar convention | 1971 |
| Stockholm convention | 2001 |
| Boon convention | 1979 |
| Vienna convention | 1985 |
| Basel convention | 1989 |
| Minamata convention | 2013 |
| Rotter dam convention | 1998 |

**10. Which of the following is NOT considered a planetary boundary**

(a) Ozone depletion in the upper atmosphere
(b) Global biodiversity loss
(c) Worsening air pollution in cities
(d) Increasing levels of nitrogen in terrestrial ecosystems

**Ans. (c) :** Planetary boundary concept is aimed to define the environmental limits within which humanity can safely survive.

■ This concept was introduced in 2009 & it has proved beneficial in global sustainability policy development.

■ There are nine boundaries as listed below-

(i) Climate change
(ii) Ocean acidification
(iii) Stratospheric ozone depletion
(iv) Biochemical flows in nitrogen cycle
(v) Global freshwater use
(vi) Land system change
(vii) Erosion of biosphere integrity
(viii) Chemical pollution
(ix) Atmospheric aerosol loading

**11. Which of the following is NOT a major risk associated with global warming:**

(a) Increase in the incidence of malaria
(b) Increase in water shortage
(c) Increase in the incidence of nitrogen loading in streams and rivers
(d) Increased morbidity from heat waves, droughts and floods

**Ans. (c) : Effect of global warming-**

(i) More hotter & several weather
(ii) Higher death rates
(iii) Dirtier air
(iv) Wild life extinction
(v) Increasing acidity in ocean
(vi) Rising sea levels

**12. The interaction of economic and environment considerations gives rise to ..... development**

(a) Bearable (b) Equitable
(c) Sustainable (d) Viable

**Ans. (d) : Viable development-** For a project to be viable its economic benefits should be equal or more than its economic cost.

■ A viable development is tested on following parameter-

(i) Economic feasibility
(ii) Legal feasibility
(iii) Technical feasibility
(iv) Environmental & social sustain ability

**13. One of the major negative outcomes of construction of embankments on both sides of Kosi river in Bihar is:**

(a) Degradation of river bed
(b) Aggradations of river bed
(c) Reduced availability of water in the downstream reaches
(d) Political problems between the states of Bihar and West Bengal

**Ans. (b) :** Kosi river is an effluent river in Bihar in which ground water supplies water to river as well as due to erosion of banks silting (aggradations) occurs on river bed.

■ Due to this aggradations of river bed, water level in the river topples the banks & results in flood in surrounding areas.

**14. Construction of a large number of dams in Columbia river basin has affected the livelihood of Native Americans in that area because**

(a) There is a significant decrease in the population of anadromous fish
(b) Many of them became sick because of malaria
(c) The lands where they were growing crops were inundated
(d) They were relocated to areas which were inhabitable

**Ans. (a) :** There are more then 60 dams on Columbia river basin in USA & Canada.

■ There dams are the backbone of pacific north west economy, providing power, irrigation to farmlands. However these benefits have come at an exceptionally high environmental cost.

■ Salmon & steel head other fishes pop have declined from 16 million to lesser than 1 million.

**15. Salmon population has come down drastically in Columbia river basin due to construction of large number of dams. This is because**

(a) People had easy access to the river and this resulted in over
(b) Salmon could not jump over the dams!
(c) Increased depth of water in reservoirs exerted tremendous pressure and salmon could not withstand this
(d) Construction of dams resulted in the increase in GHGs, and this killed salmon

**Ans. (b) :** Copy salmon of 14

■ Dams block passage of salmon & steelhead between spawning & rearing habited & the pacific ocean.

■ Wherever fish ladder (fish passage) is not provided the habitat is permanently blocked.

■ As per some reports more than 40% of the habitat is permanently blocked by dams.

**16. There are many proponents for "busting of dams" because they have caused an ecological havoc. However, from an eco-hydrological point of view, one needs to be cautions about dam removal because**

(a) The new eco system created may not match with what on had before construction of dam
(b) Removal of dams may decrease the availability of water in the basin
(c) Neither reason (a) nor reason (b)
(d) Both reasons (a) and (b)

**Ans. (a) :** Those dams which are not economically & ecologically viable are removed in order to restore the rivers. However dam can't restore the river completely to historic condition.
■ Fishes return to their historic habitats. Downstream reaches can be restored & reconnect oceans to head waters.

**17. Pick the wrong statement from the set below, with regard to DDT**

(a) It is an endocrine disrupting compound
(b) Causes breast cancer
(c) Causes egg shell thinking in birds
(d) It does not remain in the environmental for a long time

**Ans. (d) : Negative impacts of DDT exposure-**
■ Exposure of DDT in people occurs from eating foods including meet, fish & dairy products.
■ DDT can convert into DDE & both persist into body and environment.
■ DDT & DDE exposure in regnant women can cause breast cancer as it is present in breast milk.
■ Laboratory animal studies have shown that DDT exposure can effect the liver & reproduction & also can cause carcinogens.

**18. Which of the following emissions will increase if we increase the use of bio-diesel blends**

(a) Total unburned hydrocarbons
(b) Carbon monoxide
(c) Particulate matter
(d) Oxides of nitrogen

**Ans. (d) :** $NO_x$ emissions with heat biodiesel ($B_{100}$) typically increases by about 10-30% relative to conventional diesel fuel.
■ $NO_x$ emissions with $B_{20}$ biodiesel blends typically increase by 1-7%.

**19. Opponents of increased usage of bio-diesel in India are most likely to cite the following reason in their arguments**

(a) There is no real benefit in terms of GHG reduction
(b) There is a likelihood of increase in food insecurity
(c) Fuel insecurity will increase
(d) There is a likelihood of increase in air pollution

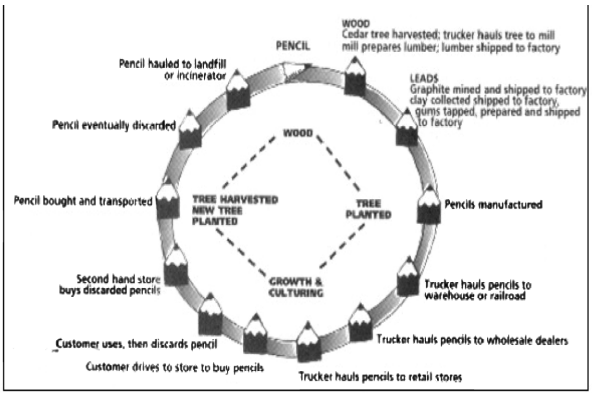
**Ans. (b) :** Opponents of increased wage of bio diesel in India are most likely to cite the reasons in their arguments are–
• There is likelihood of increase to food insecurity
• Monocutture of crops year after year.
• Industrial pollution
• High cost of producing.

**20. In a given process, which of following is ideally LEAST of the relevant factors from a life cycle analysis perspective**

(a) Toxic exposure to workers
(b) Long term environmental impacts
(c) Proximity to raw materials
(d) Cost of pollution control

**Ans. (d) :** Like cycle analysis is a method of quantifying the environmental impacts associated with given factor from a life cycle analysis perspective is cost of pollution control.

**21. Consider the lifecycle analysis (LCA) of a simple manual pencil shown in the figure below.**




**The following are statements that are made with reference to the above figure :**

**(i) Pencil manufactures has no long-term ecological impacts.**
**(ii) Raw materials are all locally available.**
**(iii) Transportation costs are externalized and do not belong in this LCA.**
**(iv) Customer has a responsibility in managing waste from pencil usage.**
**(v) Waste management during pencil manufacture is required.**

**Which of the statement above explicitly false based on the LCA shown?**

(a) i, ii, iii
(b) i, ii, iv
(c) i, ii, v
(d) i, ii, iii, iv, v

**Ans. (a) : i, ii, iii**

**22. Which component of the environmental management system does the term 'green chemistry' primarily apply to?**

(a) Remediation (b) Process Design
(c) Safety (d) Risk Assessment

**Ans. (b):** Green chemistry also called sustainable chemistry is an area of chemistry & chemical engineering focused on the design of products & processes that minimize or eliminate the use of hazardous substances.
Principles of green chemistry are-
(i) Design of process to maximize the output
(ii) Promoting the use of renewable materials
(iii) Design of energy efficient processes
(iv) Use of safe & environmental friendly substance
(v) Avoiding the production of water

**23. Which of the following is not likely to be a part of the risk assessment process of the use of hazardous materials (pollutants) in the environment?**
(a) Cost of material
(b) Exposure of material to a receptor
(c) Atmospheric transport of material
(d) Biodegradability of the material

**Ans. (a) : Risk Assessment of pollutants-**
Risk Assessment of pollutants include
(i) Effect of pollutant on receptor
(ii) Level of risk
(iii) Its concentration in the atmosphere
(iv) Biodegradability of pollutant
(v) Identification of hazards

**24. Which of the following is an example of responsible environmental management?**
(a) Dumping wastewater is sea.
(b) Disposal of wastewater in local lakes.
(c) Burning of solid waste in local communities.
(d) Decentralized community waste management facilities.

**Ans. (d) :** Dumping waste water directly into seawater without any treatment, disposal to waster water into local lakes and burning to solid waste in local residential areas are the example of poor environmental management.

**25. Which of the following can be considered as an "End of Pipeline" treatment process?**
(a) Green Chemistry approach to reduce pollution
(b) Catalytic converters in automobiles to reduce emissions
(c) Soil excavation to remove historical contamination
(d) Usage of personal respiratory protection masks

**Ans. (b) : End of pipe technology-** There are the methods/processes to remove already formed contaminants from a stream of air, water, water or in any other similar surrounding.
■ As these technology are implemented at the last stage hence these are named as end of pipe technology.
Hence (b) of the above options is correct.

**26. Which of the following is NOT TRUE with reference to environmental liability for a historically contaminated site?**
(a) The regulatory agency always takes responsibility for site remediation.
(b) People living in the affected area bear the primary liability.
(c) Current owner may take the liability costs
(d) The original polluter may take responsibility

**Ans. (b) :** Environmental liabilities for a historically contaminated site should not be bear by the people living in the affected area. There should be a regular agency that always takes responsibilities for the site remediation beside it two original polluter may likes responsibilities and current owner may also takes the liabilities cots.

**27. Which of the following environmental option is potentially the least expensive and time consuming?**
(a) Dredging and relocation to another site
(b) Dumping and containing hazardous waste in landfill
(c) Incineration of biomedical waste
(d) Monitored natural recovery

**Ans. (d) :** Natural environment recovery which is also known as natures healing is the phenomenon of cleaning the pollutants & other disrupting substance by nature itself.
Eg- Restoration of aquatic life after oil spills. Restoration of coral reefs after tsunami, wetland & grass land restoration etc.
■ Natural recovery is time taking process and it may also take years depending upon the level of disturbance, but it is least expansion.

**28. In which world region do we currently witness the highest growth rates in urban population?**
(a) Latin America (b) Australasia
(c) Asia (d) Europe

**Ans. (c) :** Currently growth rate of urban population is highest in Asia because of rapid industrialization & development activities.

**29. Changes in the level of urbanization describe .....**
(a) The proportion of the population living in urban centers.
(b) The number of urban centers in a country.
(c) A factor of reclassification from rural to urban areas.
(d) The net balance of rural-urban migration.

**Ans. (a) :** Urbanization refers to the population shift from rural area to the urban areas.
■ Level of urbanization shone the fraction of population living in urban centers.

**30. Two aspects of the rapid growth in the world's urban population over the last 50-100 years are .....**
(a) significantly higher morality rate and the unprecedented size of the largest cities
(b) the increase in the number of large cities and the disproportionately high fertility rates in cities

(c) the youth bulge and environmental pollution
(d) the increase in the number of large cities and the historically unprecedented size of the largest cities

**Ans. (d) :** Two aspects of rapid growth in world's urban population over the last 50-100 years are the increase in the number of large cities and the historically unprecedented size of the largest city.

**31. What is meant by the "urban environment"?**
(a) the natural elements (air, water, land, climate, flora, fauna) as part of the city
(b) the built environment (physical environment constructed/modified for human habitation & activity)
(c) the social environment as determined by urban lifestyles
(d) the physical environment in urban areas: mix of natural elements and the environment

**Ans. (d) : Urban Environment-** Urban environment/ urban area is the region surrounding a city.
■ Most people of urban environment have non agricultural jobs, urban areas are highly developed hence density situated human structures such as houses, commercial buildings, roads, bridges etc.

**32. The city of Freilburg in Germany became known as the "environmental capital" as far back as 1992. Which one of the following policies was part of the new agenda?**
(a) restoration and preservation of water bodies to secure clean drinking water
(b) afforestation in and around the city to reduce the effects of acid rain
(c) implementation of a smog and ozone warning system
(d) construction of the first municipal photovoltaic power plant in Germany

**Ans. (c) :** Freilburg is considered as single best city for sustainable urban development.
■ Its planning was started in easily 1970 & it has tacked its energy needs, climate change, transport & land use its salient planning feature are.
(i) Car less transportation system
(ii) Promotion of public transport, walking & cycling culture.
(iii) Compact urban design to leave $2/3^{rd}$ of area for green use.
■ Strict air quality standard, smog + Ozone warning system

**33. The Sustainable Development Goals (SDGs) are considered the Post 2015 Global Agenda aimed at resolving the social, economic and environmental problems troubling the world. What is the role assigned to cities and urban development in this new agenda?**
(a) cities and urban development is viewed as a key policy area of cross-cutting nature that is relevant in all the 17 goals
(b) urban development is considered in specific targets of some goals like good jobs and economic growth, good health, clean water and sanitation, responsible consumption, clean energy, and poverty alleviation
(c) sustainable cities and communities are considered as very important and thus are a goals by itself
(d) cities and urban development are not worth to be considered as there are too many other problems that need to be addressed first and relate indirectly to improve cities

**Ans. (c) :** Sustainable development goals (SDGs) nine adopted by the united nations in 2015 as a universal call to action to end poverty, protect the planet and ensure lace and prosperity by 2030. The role assign to cities and urban development is sustainable cities and communities are considered as very important and thus are a goal by itself.

**34. Identify the three factors that are seen as crucial in determining the generation of $CO_2$ emissions in cities.**
(a) lifestyle, consumption patterns and urban form
(b) energy sources, spatial form and proportion of poor population
(c) spatial form, application of technology and lifestyles
(d) lifestyles, spatial form and energy sources

**Ans. (d) :** The following factors are given that seen as critical in determining the generation of $CO_2$ emission in cities-life style i.e. motorized transport that use huge quantities of forsil fuel, spatial form, energy sources like thermal pavour plant and industrialization.

**35. "Resilience" is one of the central concepts and approaches to prepare cities and their population for climate-related disasters. What does "resilience" mean applied to climate-related disasters in an urban area?**
(a) it refers to the capacity of a city's infrastructure and services to withstand disasters as well as how communities and institutions are expected to cope with such an event
(b) it refers to the adaptive capacity of the urban authorities in disaster preparedness and management
(c) it refers to the reduction of disaster risks and the vulnerability of infrastructures and communities
(d) it is a two-pronged long-term strategy focusing on mitigation of the urban carbon footprint and improvements in urban governance

**Ans. (a) :** Resilience to climate related disasters applied in an urban area tries to identify the capacity of city's urban infrastructures and service to withstand disasters on one hand and how communities and institutions are expected to deal with such an event on the other.

**36. Turbidity in raw water**

(a) makes the disinfection process ineffective
(b) results in corrosion
(c) results in odor problems
(d) results in alkalinity

**Ans. (a) :** Turbidity in raw water is caused due to suspended impurities present in it.

■ These suspended impurities absorb the disinfectants on its surface hence less or nil amount of disinfect is available for killing of pathogens hence disinfection process becomes ineffective.

**37. Hypochlorite is used in water treatment for**

(a) Disinfection (b) a coagulant
(c) an auxiliary chemical (d) a coagulant aid

**Ans. (a) :** Disinfectants used in raw water treatment are-
(i) Ozone
(ii) Chlorine
(iii) Hypochlorite
(iv) Reverse osmosis
(v) UV Rays

**38. Mottling teeth is associated with the presence of**

(a) Chlorides in water (b) Calcium in water
(c) Magnesium in water (d) Fluorides in water

**Ans. (d) : As per 15:10500-2012 -** Upto 1 mg/l of fluoride is needed In water to prevent dental cavities.

■ **If fluoride is upto 1.5-2 mg/l -** Causes discoloration/ mottling of teeth.

■ **If > 5 mg/l -** Causes bone flourosis

**39. Infectious hepatitis is**

(a) a bacterial disease
(b) a protozoal disease
(c) a viral disease
(d) due to blood infection

**Ans. (c) :** Hepatitis is often caused by a virus however it can also because by heavy alcohol dose, toxins, some medications & certain medical conditions.

**40. Reverse osmosis process can remove only**

(a) Turbidity
(b) Turbidity, dissolved solids and bacteria
(c) bacteria
(d) colour

**Ans. (b) :** Reverse Osmosis is capable of removing upto 99% of the dissolved salts, particles, colloids & bacteria & pathogens from feed water.

**41. Attributes of a good water supply system should include**

(a) quality and quantity, affordability, reliability
(b) affordability and reliability only
(c) reliability only
(d) quantity and affordability only

**Ans. (a) : Properties of good water supply system-**

■ It should full fill the current & future demands of community.

■ Water quality should conform to is : 10500- 2012.

■ It should be cheap hence every body can afford it.

■ It should be safe against future recontamination.

**42. In a constructed wetland the oxygen is supplied by**

(a) Surface aerators (b) natural aeration
(c) Compressed aerators (d) oxidizing agents

**Ans. (b) :** Constructed wet land is an economical efficient and environment friendly technology for the treatment of waste water.

• The aquatic, plant, and micro-organism in constructed wetand play a key role in the treatment of waste water. Plant add oxygen during the process of photo-synthesis or by direct transport from the atmosphere.

**43. Manganese is a pollutant commonly found**

(a) Ground water (b) Surface water
(c) Sea water (d) Lake water

**Ans. (a) :** Manganese is found naturally in ground water, but its level can be increased by human activities like steel production & mining.

■ Manganese can turn the water, brown or rusty, causes staining of faucets sinks etc.

**44. A census town in India is:**

(a) A continues urban spread comprising one or more towns and their adjoining out growth(s)
(b) Areas around a cone city or town, such well recognized places, like, Railway colony, university campus, port area, etc., lying outside the limit of town
(c) All villages with a minimum population of 5,000 persons in the preceding Census, at least 75% of male main working population engaged in non-agricultural activities and a population density of at least 400 persons per sq km.
(d) All places with a municipality, corporation, cantonment board, notified town area committee, etc

**Ans. (c) :** Census towns are defined as places that satisfy the following criteria.

• Minimum population of 5000.

• At least 75% of male working population engaged in non agricultural pursuite.

• Density of population at least 400/km$^2$

**45. Severe demand supply gap in water is expected in the following river basin in India in future**

(a) Godavari (b) Subernarekha
(c) Brahmaputra (d) Ganga

**Ans. (d) :** Severe demand supply gap is expected in the Ganga river basin demand of water in Ganga basin is increasing substantially due to increasing population, growing urbanization and rapid industrialization, combined with the need for raising agricultural production and the supply of water is inadequate compare do its demand. Hence demand-supply gap is expected in future.

**46. Following is not a principle of Integrated Water Management**

(a) Provide equilitable access to water for all stake holders.
(b) Manage in an integrated way, recognizing that water has multiple uses

(c) Water should not be treated as an economic good
(d) Manage at the lowest appropriate level

**Ans. (c) : Principles of integrated water management system-** Integrated water resource management system is based on three principles
(i) Social equity
(ii) Economic efficiency
(iii) Environmental sustainability
(iv) Water is treated as an economic entity

**47. When an area gets urbanized, following component in the water balance gets reduced**
(a) Evapo-transpiration
(b) Run-off
(c) Imported potable water
(d) Wastewater generation

**Ans. (a) :** When an area is urbanized, buildings, houses & roads are increased which lead to cutting of trees & reduction of green land.
■ Due to reduction is green land use evapo transpiration is reduced.

**48. In Water Sensitive Urban Design (WSUD), storm water is**
(a) directly let into nearby streams/water bodies through efficient drainage system
(b) mixed with domestic waste water to reduce the concentrations by dilution and then let into water bodies
(c) storm water is treated and majority of it is used for recharging aquifers
(d) storm water is treated and is then let into streams/water bodies

**Ans. (a) :** In water sensitive urban design storm water is directly let into nearly streams/water bodies through/efficient drainage system. WSUD seeks to minimize this impact by integrating development with site's natural features and promoting the integration of stormwater, water supply and sewage management.

**49. In stakeholders meetings, city officials are typically concerned with**
(a) job creation
(b) costs & ease of maintenance
(c) provision of basic services
(d) awareness

**Ans. (b) :** In stakeholder meeting, city officials are typically concerned with costs and ease of maintenance, social change, safety and security etc.

**50. Dimensions of kinematic viscosity**
(a) $L^2T^{-1}$ (b) $L^2T$
(c) $ML^2T^{-1}$ (d) $ML^{-1}T^{-2}$

**Ans. (a) :** Kinematic viscosity= $\frac{\text{Dynamic Viscosity}(\mu)}{\text{Mass density}(\rho)}$

SI unit→ $\frac{m^2}{s}$

Dimension →$[M^o L^2 T^{-1}]$

CGS unit → $cm^2/s$ or stoke

1 stoke = $10^{-4}$ $cm^2/s$

**51. A flow field in which the divergence of velocity vector is zero represents always**
(a) an irrotational flow
(b) an incompressible flow
(c) an unsteady compressible flow
(d) a steady and compressible flow

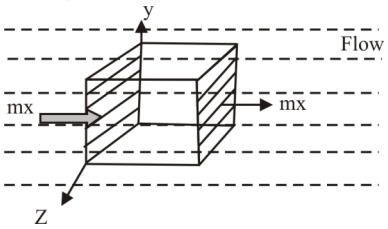
**Ans. (b) :** divergence $\boxed{\vec{V}\cdot\vec{v}=o}$ or $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=o$

Proof:- In flow field the divergence of velocity vector is zero by CONSERUATION of MASS:-
Continuity eh:- In 3-D flows-

$$\vec{V}=u\hat{i}+v\hat{j}+w\hat{k}$$

U, V, W → Velocity components is x y & z direction.

$$\Rightarrow \quad \vec{V}=f^n(x,y,z,t)$$




Conservation of mass-

$$\dot{m}_{in}-\dot{m}_{out}=\dot{m}_{st} \text{ .....eq}^n \quad (A)$$

↓ (strange)

In x direction-

$$(\dot{m}_{in}-\dot{m}_{out})_x=\dot{m}_x-\left(\dot{m}_x+\frac{\partial \dot{m}_x}{\partial_x}\right)\partial_x$$

$$=-\frac{\partial \dot{m}_x}{\partial_x}.d_x$$

$$=\frac{\partial}{\partial_x}\left[(dy.dz)u.\rho\right].d_x$$

$$(\dot{m}_{in}-\dot{m}_{out})x=\frac{-\partial}{\partial_x}(\rho u).d\forall \text{ .....(1)}$$

Similarly,

$$(\dot{m}_{in}-\dot{m}_{out})y=\frac{-\partial}{\partial_y}(\rho v).d\forall \text{ .....(2)}$$

$$(\dot{m}_{in}-\dot{m}_{out})z=\frac{-\partial}{\partial_z}(\rho w).d\forall \text{ .....(3)}$$

& $\boxed{\dot{m}_{st}=\frac{\partial \rho}{\partial t}.d\forall}$

Putting the value in eq$^n$ (A), we get

$$\dot{m}_{in}-\dot{m}_{out}=\dot{m}_{st}$$

$$-\left[\frac{\partial}{\partial_x}(\rho_u)+\frac{\partial}{\partial y}(\rho v)+\frac{\partial}{\partial z}(\rho w)\right].d\forall=\frac{\partial \rho}{\partial t}.d\forall$$

$$\frac{\partial}{\partial x}(\rho u)+\frac{\partial}{\partial y}(\rho v)+\frac{\partial}{\partial z}(\rho.w)+\frac{\partial \rho}{\partial t}=0$$

"Generd 3D continuity equation"
Article-

$$u.\frac{\partial\rho}{\partial x}+\frac{\rho.\partial u}{\partial x}+\frac{v.\partial\rho}{\partial y}+\frac{\rho.\partial v}{\partial y}+\frac{w.\partial\rho}{\partial z}+\frac{\rho.\partial w}{\partial z}+\frac{\partial\rho}{\partial t}=0$$

$$\text{So, } \frac{u.\partial\rho}{\partial x}+\frac{v.\partial\rho}{\partial y}+\frac{w.\partial\rho}{\partial z}+\frac{\partial\rho}{\partial t}+\rho\left(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}\right)=0$$

$$\frac{\Delta\rho}{\Delta t}+\rho(\vec{\nabla}.\vec{V})=0$$

By Assumption-steady and incompressible flow
$\rho$ = constant

$$\boxed{\frac{\partial u}{\partial x}+\frac{\partial w}{\partial z}+\frac{\partial v}{\partial y}=0} \quad \text{or} \quad \boxed{\vec{\nabla}.V=0}$$

**52. Mach number represents a ratio of**
(a) Inertia to elastic force
(b) Inertia to gravity force
(c) Inertia to viscous force
(d) Inertia to pressure force

**Ans. (a) :** Mach number (M)- Mach number is defined as the square root of the ratio of the inertia force of a flowing fluid to the elastic force.
Mathematically,

$$M=\sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}}=\sqrt{\frac{Fi}{Fe}}$$

Where,

$Fi = \rho AV^2$

Fe = Elastic force = Elastic stress × Area
$= K \times A = K \times L^2$

$$\therefore \quad M=\sqrt{\frac{\rho A.V^2}{K\times L^2}}=\sqrt{\frac{\rho\times L^2\times V^2}{K\times L^2}}=\sqrt{\frac{V^2}{K/\rho}}=\frac{V}{\sqrt{K/\rho}}$$

But $\sqrt{\frac{K}{\rho}}=C$ = Velocity of sound in the fluid

$$\therefore \quad \boxed{M=\frac{V}{C}}$$

**53. The x and y components of velocity in a two-dimensional incompressible flow are given by**
**u = 3x + 2y; v = 2x–3y**
**The vorticity in the flow field is given by**
(a) 1 (b) 1/2
(c) 0 (d) –1

**Ans. (d) :** Given,
u = 3x + 2y; v = 2x–3y
Formula vorticity = 2 × ω
Where , ω = Angular velocity

$$\text{So, } \omega_z=\frac{1}{2}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)$$

$$=\frac{1}{2}\left[\frac{\partial(2x-3y)}{\partial x}-\frac{\partial}{\partial y}(3x+2y)\right]$$

$$=\frac{1}{2}(2-2)$$

$$=0$$

Vorticity = 2ω
$= 2\times0 = 0$
Note- So flow is irrotational

**54. The laminar boundary layer thickness on flat plate varies as (in terms of distance x from the leading edge)**
(a) $\frac{1}{x}$ (b) $\frac{1}{\sqrt{x}}$
(c) $\sqrt{x}$ (d) $x^2$

**Ans. (c) :** Laminar boundary layer thickness on flat plate

$$\Rightarrow \delta=\frac{5x}{\sqrt{R_{ex}}}$$

This formula used in the absence of velocity profit

Using $R_{ex}=\frac{\rho V_x}{\mu}$

$$\text{So, } \quad \delta=\frac{5x}{\sqrt{\frac{\rho V_x}{\mu}}}$$

i.e. $\boxed{\delta\propto\sqrt{x}}$

Note- for turbulent region

$$\delta\propto x^{4/5}$$

**55. In a pipeline, the flow turns turbulent when the Reynolds number crosses a critical value which lies between**
(a) 500 – 800 (b) 1000 – 1200
(c) 1500 – 1800 (d) 2000 – 2700

**Ans. (d) :** Limiting value of Reynold's number for pipe flow.

| **Flow condition** | **Pipe flow** |
|---|---|
| Laminar flow | Re ≤ 2000 |
| Transitional flow | 2000 < Re < 4000 |
| Turbulent flow | Re > 4000 |

**56. In a steady flow through a straight nozzle, the center line velocity changes from 1 m/s to 10 m/s in 0.3 m length. Change in the magnitude o convective acceleration is**
(a) 200 m/s$^2$ (b) 235 m/s$^2$
(c) 270 m/s$^2$ (d) 305 m/s$^2$

**Ans. (c) :** $u_1$ = 1 m/s
$u_2$ = 10 m/s
dx = 0.3 m
Find- change in magnitude of convective acceleration.

$$a_x=\frac{u.\partial u}{\partial x}$$

Change in magnitude is given as

$$a_x=u_2\times\frac{dv}{dx}-u_1.\frac{du}{dx}$$

$$=10\times\frac{(10-1)}{0.3}-1\times\frac{(10-1)}{0.3}$$

$$=270 \text{ m/s}^2$$

**57. The Reynolds number of a flow through a pipe is 32. The Darcy-Weisbach friction factor, f is equal to**

(a) 1.0 (b) 2.0
(c) 4.0 (d) 8.0

**Ans. (b) :** Given- Reynold's number ($R_e$) = 32
Darcy weinbacs friction factor f = ??
Formula- for laminal flow

$$f = \frac{64}{R_e} \quad \{\text{Since value of } R_e \text{ is below } 200\}$$

$$f = \frac{64}{32}$$

$$f = 2$$

**58. The specific speed of a pump is defined as (Q = flow rate in $m^3/s$; N = speed of the pump in rpm; H = head developed by the pump in meters)**

(a) $\frac{N\sqrt{Q}}{H^{3/4}}$ (b) $\frac{NQ}{H^{3/4}}$
(c) $\frac{\sqrt{N}Q}{H^{3/4}}$ (d) $\frac{NQ^{3/4}}{\sqrt{H}}$

**Ans. (a) :** Specific speed of a centrifugal pump ($N_s$)-
It is defined as the speed of a geometrically similar pump which would deliver one cubic meter of liquid per second against a head of one meter.
It is denoted by $N_s$

$$\boxed{N_s = N\frac{\sqrt{Q}}{H_m^{3/4}}}$$

Expression for specific speed for a pump-
The discharge Q, for a centrifugal pump is given by the relation-

$$Q = \text{Area} \times \text{Velocity of flow}$$
$$= \pi D \times BV_f$$
$$Q \propto BV_f \text{.....(i)}$$

D = Diameter of the impeller of the pump
B = Width of the impeller
We know that B & D
$\therefore$ From eq$^n$ (i), we have $Q \propto D^2V_f$.....(ii)
We also know that tangential velocity is given by

$$u = \frac{\pi DN}{60} \propto DN \text{ .....(iii)}$$

Now the tangential velocity (u) and velocity of flow ($V_f$) are related to the manometric head ($H_m$) as

$$u \propto V_f \propto \sqrt{H_m} \text{ .....(iv)}$$

Substituting the value of u in eq$^n$ (iii), we get

$$\sqrt{H_m} \propto DN \text{ or } D \propto \frac{\sqrt{H_m}}{N}$$

Substituting the values of D in eq$^n$ (ii)

$$Q \propto \frac{H_m}{N^2} \times \sqrt{H_m} \quad [\because \text{from eq}^n \text{ (iv) } V_f \propto \sqrt{H_m}]$$
$$\propto \frac{H_m}{N^2} \times \sqrt{H_m}$$
$$\propto \frac{H_m^{3/2}}{N^2}$$
$$\therefore \quad Q = K.\frac{H_m^{3/2}}{N^2} \text{ .....(v)}$$

Where K is constant of proportionality.
If $H_m = 1$ and $Q = 1m^3/s$, N becomes = $N_s$
If $H_m = 1$ and $Q = 1m^3/s$, N becomes = $N_s$
Substituting the value in eq$^n$ (V),

$$\text{We get } 1 = \frac{K \times 1^{3/2}}{N_s} = \frac{K}{N_s^2}$$

$\therefore K = N_s^2$
Substituting these value of K is eq$^n$ (V) get

$$Q = N_s^2 \frac{H_m^{3/2}}{N^2} \text{ or } N_s^2 = \frac{N^2Q}{H_m^{3/2}}$$

$$\boxed{\therefore N_s = \frac{N\sqrt{Q}}{H_m^{3/4}}}$$

**59. Draft tube is a conduit**

(a) in a pumping scheme and connects the sump and the pump impeller
(b) in a pumping scheme and connects the air chamber with the delivery line
(c) in a hydro-electric scheme and connects the runner exit to the tail race
(d) in a hydro-electric scheme and connects the penstock and the surge tank

**Ans. (c) : DRAFT TUBE-** The draft tube is a pipe of gradually increasing area which connects the outlet of the runner to the tail race. This pipe of gradually increasing area is called a draft tube.

It is used for discharging water from exit of the turbine to the tail race level. One end of the draft tube is connected to the outlet of the runner while other end is submerged below the level of water in the tail race.

**It has following purpose-**

(i) It permits a negative head to be established at the outlet of the runner and thereby increase the net heat on the turbine. The turbine may be placed grove the tail race without any loss of net head and here turbine may be inspected properly.

(ii) It converts a large proportion of the kinetic energy $\left(\frac{V_2^2}{2g}\right)$ rejected at the outlet of the turbine into useful pressure energy. Without the draft tube, the Kinetic energy rejected at the outlet of the turbine will go waste to the tail race.

Hence, by using draft tube, the net head on the turbine increases. The turbine develops more power and also the efficiency of the turbine increase.

**60. The Froude number, F in an open channel is defined as (V = velocity; P = wetted perimeter; D = hydraulic depth; R = hydraulic radius; g = acceleration due to gravity)**

(a) $\frac{V}{\sqrt{gP}}$ (b) $\frac{V}{\sqrt{gR}}$

(c) $\frac{V}{\sqrt{gD}}$ (d) $\frac{V}{\sqrt{gP^2/R}}$

**Ans. (c) : Froude's number ($F_e$)-** The Froude's number is defined as the square root of the ratio of inertia force of a flowing fluid to the gravity force.

Mathematically,

$$F_e = \sqrt{\frac{F_i}{F_g}} \quad .....(i)$$

$F_i = \rho AV^2$

$F_g$ = Force due to gravity

= Mass × Acceleration due to gravity

$= \rho \times \text{volume} \times g = \rho \times L^3 \times g$

$= \rho \times L^2 \times L \times g = \rho \times A \times L \times g$

From eq$^n$ (i)

$$F_e = \sqrt{\frac{F_i}{F_g}} = \sqrt{\frac{\rho AV^2}{\rho ALg}}$$

$$= \sqrt{\frac{V^2}{Lg}} = \frac{V}{\sqrt{Lg}} \; \square \; \frac{V}{\sqrt{gD}}$$

**61. The flow in a rectangular channel is subcritical. If the width is contracted at a certain section, the water surface**

(a) at a downstream section will drop

(b) at a down stream section will rise

(c) at an upstream stream section will drop

(d) remains same at the down stream section

**Ans. (a) :** The flow in a rectangular channel is sub-critical. If width is contracted at a curtain section, the water surface at the downstream section will drop.

**62. In a uniform open channel flow in wide rectangular channel, the flow depth is 1 m and the bed slope is 0.001. The average bed shear stress is**

(a) 9.9810 N/m$^2$

(b) 9.810 N/m$^2$

(c) 98.10 N/m$^2$

(d) 981.0 N/m$^2$

**Ans. (b) :** Given,

Flow depth = 1m

$$\text{bed slope} = \frac{1}{1000}$$

Wide rectangular channel. So hydraulic mean depth = d

Average sheer stress = ?

Formula- $\tau_0 = r_w \times R \times S$

$$= 9.81 \times 1 \times \frac{1}{1000} = 9.81 \times 10^{-3} \text{ kN/m}^2$$

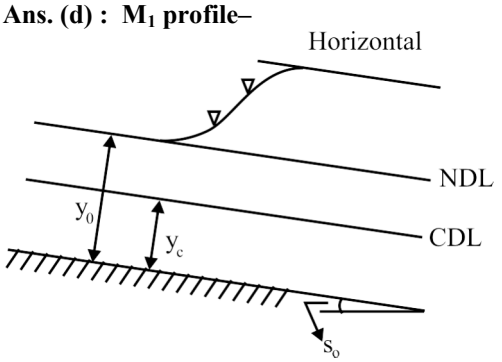
$= 9.81$ N/m$^2$

**63. The flow profile on the upstream side of sluice gate on a mild channel is classified as**

(a) H2 (b) M3

(c) M2 (d) M1

**Ans. (d) : $M_1$ profile–**




$y > y_0 > y_c$

- It is the most common of all GVF profile.
- It is found in sub critical flow condition.
- It is observed when obstruction to the flow is carried out such as weir, dam, control structure.

**$M_2$ profile–** $M_2$ profile occurs at a sudden drop in the bed of channel at transition and it the canal outlet into the pool.

**$M_3$ profile–** When a super critical stream enter a mild slope channel $M_3$ type of flow profile occur.

**64. The sequent depth ratio in a hydraulic jump formed in a horizontal rectangular channel is 2.0. The Froude number of supercritical stream is**

(a) 1/2 (b) $\sqrt{2}$

(c) $\sqrt{3}$ (d) 2

**Ans. (c) :** Given sequent depth ratio in a hydraulic jump

$\frac{y_1}{y_2} = 2.0$

Froude No. $F_r$ = ?

Formula $\frac{y_2}{y_1} = \frac{1}{2}\left[-1+\sqrt{1+8F_{r1}^2}\right] \rightarrow$ sub critical

$$\frac{y_1}{y_2} = \frac{1}{2}\left[-1+\sqrt{1+8F_{r2}^2}\right] \rightarrow \text{super critical}$$

In question super critical condition is given.

So, $\frac{y_1}{y_2} = \frac{1}{2}\left[-1+\sqrt{1+8F_r^2}\right]$

$$2.0 = \frac{1}{2}\left[-1+\sqrt{1+8F_r^2}\right]$$

By solving we get,

$$\boxed{F_r = \sqrt{3}}$$

**65. COD > $BOD_5$ for domestic wastewater because**

(a) Nitrification of $NH_3$

(b) Presence of non-biodegradable organic matter

(c) Presence of chemically degradable organic matter

(d) Presence of colour

**Ans. (b) :** $BOD_5 \rightarrow$ Biodegradable organic matter

COD $\rightarrow$ Non- Biodegradable and Biodegradable organic matter

So $\boxed{COD > BOD_5}$

**66. The Stoke's Law of settling is applicable for**

(a) Flocculent suspenson
(b) Discrete particles
(c) Zone settling
(d) Compression settling

**Ans. (b) : Type–I (Discrete settling)**

- It occur when there is no changes in shape, size and mass of particle during settling.
- Settling velocity of discrete particle can be computed using stokes law.
- Settling in PST (raw water plant) or in grit chamber (WWP) is type-I settling.

**Type-II (Flocculent settling)**

Found in clarifflloculator (RWP)

**Type-III (zone/hindered settling)**

Found in SST (Secondary sedimentation tank) After ASP (activated sludge process).

**Type-IV (Compression settling)**

Found at bottom of deep sludge mass as in bottom SST flowing trickling filter.

**67. The water distribution system is designed for**

(a) Annual average water demand
(b) Peak daily demand
(c) Peak hourly demand
(d) Peak weekly demand

**Ans. (c) :** Design period of various components of water supply project.

| Units | Design (Parameters) Discharge | Design period |
|---|---|---|
| Water treatment unit | Maximum daily demand | 15 years |
| Main supply pipes (water mains) | Maximum daily demand | 30 years |
| Well and tube wells | Maximum daily demand | 30-50 years |
| Demand reservoir | Average annual demand | 50 years |
| Distribution system | Maximum hourly demand coincident draft | 30 years |

**68. The ultimate BOD of a wastewater having $BOD_5$ of 200 mg/L and rate constant of 0.23 d is**

(a) 192 mg/L (b) 292 mg/L
(c) 392 mg/L (d) 92 mg/L

**Ans. (b) :** Use eq$^n$

$y_t = L[1-(10)^{-kot}]$

Hole $K = 0.23/day$ (given)

$\therefore$ $Ko = 0.434k = 0.434 \times 0.23 = 0.1$

Using t = 5 days, we have

$y_s$ = BOD of 5 days = 200 mg/l = $L[1-10^{-0.1\times5}]$

$$= L\left[1-\frac{1}{10^{0.5}}\right]$$

$$200 = L\left[1-\frac{1}{3.16}\right] = 0.684L$$

$$L = \frac{200}{0.684} = 292.3976\,mg/L$$

$\approx$ 292 mg/L

**69. The volume of an extended aeration plant with a biomass concentration of 4 kg $m^3$, to remove 400 kg/d of BOD at a BOD removal rate of 0.1 d is**

(a) 100 $m^3$
(b) 1000 $m^3$
(c) 10,000 $m^3$
(d) 100,000 $m^3$

**Ans. (b) :** Volume of an extended aeration plant = ?

Biomass concentration = 4 $kg/m^3$

BOD remove = 400 kg/d

BOD removal rate = 0.1d

Empirically or by units rule

$$\text{Volume } (m^3) = \frac{400\,kg/d}{4gk/m^3} \times \frac{0.1\times1}{d}$$

$$= 1000\ m^3$$

**70. The two major components of the atmosphere are :**

**1. Nitrogen and Helium**
**2. Oxygen and Ozone**
**3. Nitrogen and carbon dioxide**
**4. Nitrogen and Oxygen**

**Ans. (d) :**

| Constituent | | Percent by volume |
|---|---|---|
| Nitrogen ($N_2$) | → | 78.084 |
| Oxygen ($O_2$) | → | 20.946 |
| Argon (Ar) | → | 0.934 |
| Carbon dioxide ($CO_2$) | → | 0.036 |
| Neon (Ne) | → | 0.00182 |
| Helium (He) | → | 0.000524 |
| Methane ($CH_4$) | → | 0.00015 |
| Krypton (Kr) | → | 0.00014 |
| Hydrogen ($H_2$) | → | 0.00005 |

So Nitrogen and oxygen are major components of the atmosphere.

**71. The primary pollutant formed due to incomplete combustion of carbon in fuel is**

(a) Sulphur trioxide
(b) Peroxygacetyl nitrate
(c) Ozone
(d) CO

**Ans. (d) :** The primary pollutant formed due to incomplete combustion of carbon in fuel is- Carbon monoxide (CO)

**72. The pollutant responsible for silvering of lower surface of leaves is**

(a) $SO_2$ (b) $NO_2$
(c) PAN (d) $O_3$

**Ans. (c) :** PAN → Peroxy Acetyl Nitrate.

It is powerful respiratory and eye irritants present in photo chemical smog.

PAN formula → $CH_3$ C(O) $OONO_2$

It is also responsible for silvering of lower surface of leaves.

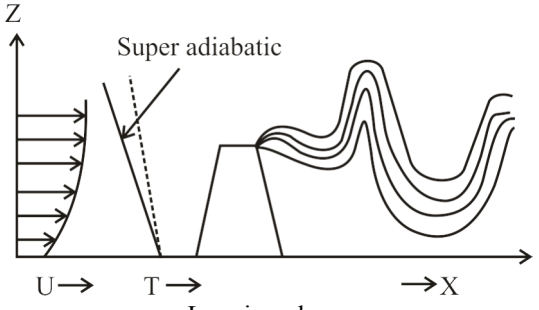
**73. The plume shape during neutral or slightly stable atmospheric conditions is**

(a) Fanning (b) Coning

(c) Looping (d) Lofting

**Ans. (b) : Plume Behavior-**

**(1) Looping plume-** It has wavy character and occurs in reaper adiabatic environment which produces highly unstable environment because of rapid mixing.

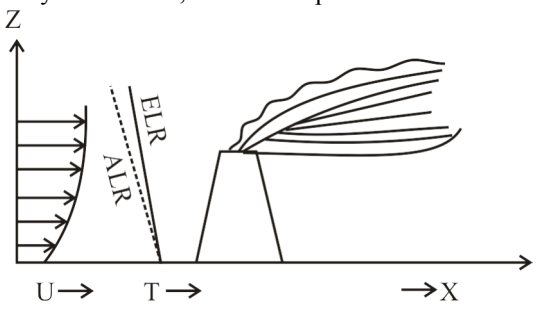



Looping plume

**(2) Coning plume-** It occurs under sub-adiabatic condition

■ The plume dispersion is known as coning, because the plume makes a cone like shape about the plume line.

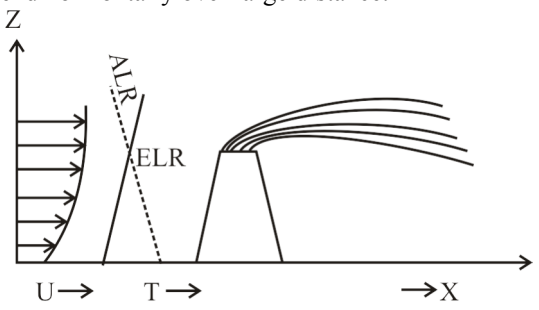
■ It occur on cloudy day or night with strong wind velocity > 32 km/hr, when the lapse rate is near neutral.




Coning Plume

**(3) Fanning plume-** under extreme condition of inversion, banning plume occurs.

■ These coil be no vertical mixing and the plume will extend horizontally over large distance.

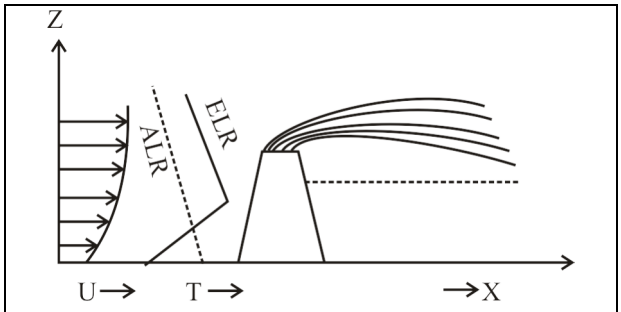



Fanning Plume

**(4) Lofting plume-** When there is strong super adiabatic copse rate above a surface in version.

■ It is most favorable plume.

■ It prevent downward motion by inversion, but upward mixing will be cutie turbulent and rapid.




Lofting Plume

**74. In urban air pollution the most poisonous gas is**

(a) CO (b) $SO_2$

(c) HC (d) $NO_2$

**Ans. (a) :** In urban air pollution the most poisonous gas is– Carbon Monoxide (CO)

When the concentration of carboxy- hemoglobin reaches about 3-4% in blood, the ability of hemoglobin to bind with oxygen is largely reduced. Thus the reason for the poisonous nature of CO is its ability to folem a STABLE COMPLEX with hemoglobin.

**75. The united nations conference on environmental held at Stockholm in the year**

(a) 1952 (b) 1962

(c) 1972 (d) 1982

**Ans. (c) :** The UNITED NATIONS CONFERENCE on environment hold at Stockholm in 1972.

It is first world conference to make the environment a major issue.

The Stockholm declaration, which contained 26 principles, placed environmental issues at the forefront of international concerns and marked the start of a dialogue between industrialized and developing counties on the link between economic growth, the pollution of the air, water and oceans and well being of the people around the world.

**76. Indian national ambient air quality standard for 8 hour average carbon monoxide pertaining to sensitive area is**

(a) 5 mg $m^3$ (b) 2 mg $m^3$

(c) 1 mg $m^3$ (d) 10 mg $m^3$

**Ans. (b) :**

| **Pollutant** | **Time weighted average** | **Indus trip Residential rural and other areas** | **Sensitive area** |
|---|---|---|---|
| Carbon Monoxide (CO) μg/$m^3$ | 24 hours<br>8 hours<br>1 hour | 1<br>2<br>4 | 1<br>2<br>4 |

According to table answer should be 2μg/$m^3$ for 8 hours for Carbon Monoxide.

**77. Indian started adopting European emission and fuel regulations for four-wheatear vehicles form the year**

(a) 1988 (b) 2000

(c) 2001 (d) 2005

**Ans. (b) :** India started adopting European emission and fuel regulations for four- wheeler vehicles form the year-2000

**78. The environmental (Protection) Act is passed in the year**
(a) 1986 (b) 1990
(c) 1974 (d) 1981

**Ans. (a) :** The various regulations are–
(1) The forest Act, 1927
(2) The wildlife (protection) Act, 1972
(3) The water (Prevention and control of pollution) Act, 1974
(4) The water (Prevention and control of pollution) Cass Act, 1977
(5) The forest conservation Act, 1980
(6) The air (Prevention and control of pollution) Act, 1981
(7) The environment (protection) Act, 1986
(8) The public liability insurance Act, 1991

**79. The various unit operations exist in a sewage treatment plant as follows : (i) Screening (ii) girt chamber (iii) Secondary sedimentation (iv) aeration (v) primary sedimentation.**
(a) i, ii, iii, iv, v (b) i, ii, v, iv, iii
(c) iv, i, ii, iii, v (d) i, ii, iii, v, iv

**Ans. (b) :** The various unit operations are as follows-
(i) Screening
(ii) Grit chambers
(iii) Primary sedimentation
(iv) Aeration
(v) Secondary sedimentation

**80. Color of fresh sewage is**
(a) green (b) brown
(c) grey (d) white

**Ans. (c) :** Color of fresh sewage- Grey
Color of septic sewage- Dark brown

**81. If the sewage contains greases and fatty oils, they are removed in**
(a) grit chambers
(b) detritus tanks
(c) sedimentation tanks
(d) skimming tanks

**Ans. (d) : Skimming Tanks-** They are employed for removing oil and grease from the sewage.
■ They are placed before the sedimentation tank.
■ Skimming tanks are provided in particular industrial towns or at colder hill stations like Simla, Mussoorie, Nainital etc. where greasy materials may congaed early.
■ In not countries line India, it is difficult to use because they donot Coagulate and congeal easily.

**82. Activated sludge is the**
(a) aerated sludge in the aeration unit
(b) sludge settled in the humus tank
(c) sludge in the secondary tank after aeration and rich in microbial mass
(d) sludge in the secondary tank after aeration and rich in nutrients

**Ans. (c) :** Activated sludge process- Aerobic suspended culture sludge in the secondary tank after aeration and rich in microbial mass.

**83. Secondary treatment of sewage removes**
(a) suspend ended organic matter
(b) dissolved and colloidal organic matter
(c) harmful bacteria
(d) fats and grease

**Ans. (b) :**

| Operation | Purpose |
|---|---|
| Primary sedimentation tank | Suspended matter |
| Secondary treatment of sewage | Dissolved and Colloidal organic matter |
| Chlorination | Harmful micro organizers |
| Skimming tank | Fats and grease |
| Grit chamber | Inorganic matter |

**84. This is not part of advanced wastewater treatment processes**
(a) electro dialysis
(b) ion exchange
(c) carbon adsorption
(d) screening

**Ans. (d) :** Screening- Primary treatment operation
Electrolysis, Ion encharge, Carbon absorption } Advanced work water treatment process

**85. The minimum DO prescribed for a river stream to avoid fish kill is**
(a) 2 ppm (b) 4 ppm
(c) 6 ppm (d) 8 ppm

**Ans. (b) :** The minimum DO for a river stream to avoid fish kill is- 4 ppm

**86. High COD to BOD ratio of an organic pollutant represents**
(a) high biodegradability of the pollutant
(b) low biodegradability of the pollutant
(c) presence of free oxygen for aerobic decomposition
(d) presence of toxic material in the pollutant

**Ans. (b) :** High COD to BOD ratio of an organic pollutant represents- low biodegradability of the pollutant

**87. Dried sludge is a good**
(a) fuel (b) soil builder
(c) manure (d) filter for low lying area

**Ans. (c) :** Dried sludge is a food as manure.
Dried sludge effectively treated can be used as organic fertilizers for growth of plants of better efficiency results in productivity.

**88. Which one of the following substances stores more heat energy than all other atmospheric constituents combined?**
(a) Carbon dioxide (b) Ozone
(c) Water vapor (d) Nitrogen

**Ans. (c) : (a) $CO_2$** → Most important gas is causing global worming. Basically, it is acidic in nature.
**(b) Ozone** → They are two type of ozone
(1) Good ozone (2) Bad ozone
Stratospheric ozone
■ **Good Ozone-** It is 90% of the atmospheric ozone. Generally found in stratosphere.
■ **Bad ozone-** It is about 10% of the ozone.
Found in troposphere which generated due to green house gases emission and effects our atmosphere by tiring the ozone layer due to harmful U.V. radiations.
**(c) Water vapour**- It stores moves heat energy than all other atmospheric constituent.
It is most abundant in atmosphere. If water vapour is absent from our atmosphere then there will always cold no heat will be there so water vapour's play key role in our atmosphere.
**(d) Nitrogen ($N_2$)-** Most abundant gas in our atmosphere. It is about 78% in our whole atmosphere.

**89. Vertical mixing due to buoyancy is increased when atmospheric conditions are :**
(a) Unstable (b) Neutral
(c) Stable (d) Extremely stable

**Ans. (a) :** Vertical mixing due to buoyancy is in creased when atmospheric conditions are- UNSTABLE

**90. Main photochemical oxidant is**
(a) Ozone
(b) Hydrogen peroxide
(c) Nitrogen dioxide
(d) alkyl nitrate

**Ans. (a) :** Common or well known photochemical oxidants include Ozone ($O_3$), Hydrogen peroxide ($H_2O_2$) and peroxyacetyl nitrate (PAN)

**91. Green house effect is the result of**
(a) Over crop production
(b) Large amount of fuel consumption
(c) Volcanic eruption
(d) Green cover

**Ans. (b) : Activity** **Gases**
(a) Volcanic eruption → $SO_2$
(b) Large amount of fuel consumption → Green house gases
(c) Green cover → Afforestation process
(d) Marshy load → $CH_4$

**92. Photochemical smog is produced by**
(a) Complex chemical reaction between hydrocarbon and oxides of nitrogen
(b) Complex chemical reaction between carbon monoxide and hydrocarbon
(c) Complex chemical reaction between carbon monoxide and nitrogen
(d) Complex chemical reaction between SOx and NOx

**Ans. (a) : Photochemical smog-** Photochemical smog is a brownish gray haze caused by the action of solar ultraviolet radiation on atmosphere polluted with hydrocorbons and oxides of nitrogen.
It is also known as 'LOS ANGELES SMOG'.
The components of photochemical smog were established during 1950's
**Reaction of occurs-**
(1) $NO + VOC \rightarrow NO_2$
(2) $NO_2 + UV \rightarrow NO + O$
(3) $O + O_2 \rightarrow O_3$
(4) $NO_2 + VOC \rightarrow PAN$
Net results-
$NO + VOC + O_2 + UV \rightarrow O_3$, PAN and other oxidants.

**93. Carbon monoxide emissions are high when**
(a) Vehicle is cruising
(b) Vehicle is accelerating
(c) Vehicle is decelerating
(d) Vehicle is idling

**Ans. (d) : Vehicle is idling-** Commission are high.
Maximum permissible carbon monoxide emission while idling.
= 3% for cars
= 4.5% for 2 & 3 vehicle

**94. Method used for measurement of nitrogen dioxide concentration is**
(a) Chemi-luminescence method
(b) Non-dispersive Infrared analyzer
(c) Flame ionization detection method
(d) Obscuration method

**Ans. (a) : $NO_2$ Concentration-**
(i) Chemi-Luminescence method.
(ii) Modified Jaws & Hochheisel
**$SO_2$ Concentration-**
(i) Improved west and grade
(ii) Ultraviolet fluorescence
**Co Concentration-**
(i) Non dispersive Infrared (NDIR) spectroscopy

**95. If the CO content in the air breathed is 100 ppm, the saturation value of COHb in the blood is :**
(a) 20% (b) 15%
(c) 10% (d) 5%

**Ans. (b) :**

| Carbon monoxide exposure concentration (PPM) | Steady-state blood COHb (percent) |
|---|---|
| 0.1 | 0.25 |
| 0.5 | 0.32 |
| 10 | 1.8 |
| 20 | 3.2 |
| 80 | 1.1 |
| 100 | 14 |
| 200 | 24 |
| 1000 | 61 |

**96. The effect of earth's rotation on wind speed and direction is called as**

(a) Coriolis effect
(b) Gravitataional effect
(c) Centrifugal effect
(d) La Nina

**Ans. (a) : Coriolis effect–** The earth's rotation creates an apparent force (Coriolis force) that defects moving air to the right of its initial direction in the Northern hemisphere and to the left of its initial direction in the southern hemisphere.

- The magnitude of the deflection or coriolis effect, varies significantly with latitude.
- The effect is proportional to wind speed, that is deflection increases as wind strengthens.

**97. Ringlemann's scale is used to measure**

(a) Grade of exhaust emission
(b) Grade of stack emission
(c) Grade of smoke density
(d) Turbidity

**Ans. (c) :** Ringlemann's scale is used to measure density of smoke.
It was developed by a French professor of agricultural engineering maximum ringelman of La station of Essais demachines in Paris, whom first specified the scale in 1888.

**98. The photochemical reaction results in formation of**

(a) Only CO (b) Both $O_3$ and PAN
(c) Both PAN and CO (d) Both $O_3$ and CO

**Ans. (b) :** The secondary air pollutant can be more dangerous than original basis primary air pollutant.
Ex PAN (Peroxy acetyl nitrate)

$$NO_x + HC + H_2O \xrightarrow{\text{sunligh}} O_3 + \text{Formaldehydes} + PAN$$

**99. Commonly used controlling unit for removal of particulates of size > 50 micron is**

(a) Gravity settling chamber
(b) Cyclone
(c) Fabric filter
(d) Electrostatic precipitator

**Ans. (a) : Gravitational settling chamber–**
The emitted pollutant, when made to pass through a settling chamber, drop some of their large size particles in the chamber (under stokes law)

- It remove particle size greater than 50 μm.
- It efficiency < 50%
- It require large space.

**Cyclone collector–** It remove particle in range of 5-25 μm.
**Electrostatic precipitator** – It remove particle of size greater than 1 μm
**Fabric filler–** It removes the particle size <1 μm

**100. In unit operations, removal of pollutant is taking place due to**

(a) Physical forces
(b) Chemical process
(c) Biological process
(d) Combination of chemical and biological process

**Ans. (a) : Unit operation–** The means of treatment in which the application of physical forces predominates are known as unit operation, major treatment methods fallings under this category are screening, mixing, sedimentation etc.
**Unit process–** The type of treatment in which the removal of contaminants is brought about by the addition of chemicals are use of biological mass or microbial activities are known as unit process.

**101. Theoretical oxygen demand of one mole of glucose is**

(a) 180 g (b) 192 g
(c) 32 g (d) 1.06 g

**Ans. (b) :** $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$
Molecular weight of $O_2$ = 32 gm
1 mole of Glucose required = 6 mole of $O_2$
$= 6 \times 32 = 192$ gm

**102. Clariflocculator is mainly used for the removed of**

(a) Colloidal particles
(b) Easily settleable particles
(c) Pathogens
(d) Organic Matter

**Ans. (a) :** Flloculation is the process in which neutralize particle and sticky are brought in contact with each others so as to form a bigger size particle (floc) which can get easily settled in the following sedimentation tank.

- If all process coagulation, flocculation and sedimentation are carried out in single tank then it is called clarifflocculator.

**103. For the treatment of wastewater containing large quantity of colloidal organic matter, the following system is most suitable**

(a) Conventional activated sludge process
(b) Extended aeration process
(c) Contact stabilization process
(d) Trickling filter

**Ans. (c) :** For the treatment of waste water containing large quantities of collidal organic matter, the contract stabilization process is most suitable. in this method a modification of the conventional activated sludge process that uses two aeration tanks for separation of sludge and further process.

**104. If the sludge volume index is 100, then the concentration of biomass in the returned sludge is**

(a) 10,000 mg/L (b) 2000 mg/L
(c) 1000 mg/L (d) 100 mg/L

**Ans. (a) :** Concentration of biomass $= \frac{10^6}{SVI} = \frac{10^6}{100}$
$= 10000$ mg/L

**105. There is no emission standard for**
(a) Primary pollutants
(b) Secondary pollutants
(c) Volatile organic compounds
(d) Particulate matter

**Ans. (b) :** Secondary pollutants are pollutants which form in the atmosphere. These pollutants are not emitted directly from a source. Ex-ozone, particulates matter, PANs. There is no emission standards for secondary pollutants.

**106. For the removal of gaseous pollutants, one can use**
(a) Setting chambers
(b) Bag filters
(c) Electrostatic precipitators
(d) Absorption towers

**Ans. (d) :** Removal of gaseous pollutants → Absorption towers.

**107. Landfill leachate is**
(a) Solid waste generated in the landfill
(b) Gaseous waste generated in the land
(c) Liquid waste oozing out of the landfill
(d) Solid waste deposited in the landfill

**Ans. (c) : Landfill leachate-** In most landfills, the liquid portion of the leachate is composed of the liquid produced from the down position of the wastes and liquid that has entered the landfill from external sources, such as surface drainage rainfall, groundwater and water from underground springs. Under normal conditions, leachate is found at the bottom of landfills from there, it moves through the underlying strata detesting on the characteristics of the surrounding material.
The use of clay lines or synthetic lines like goo-textiles has been the most favorable method for reducing and eliminating the percolation of teachate

**108. The two most important primary pollutants that contribute to the formation of acid rain air**
(a) $CO_2$ and $SO_2$ (b) $NO_x$ and $SO_2$
(c) NO and $NO_2$ (d) $NO_x$ and $CO_2$

**Ans. (b) : Acid rain–** Acid rain results when gaseous emission of sulphur oxides ($SO_X$) and nitrogen oxide ($NO_x$) interact with water vapour and sunlight and are chemically converted to strong acidic compounds ($H_2SO_4$ & $HNO_3$), along with other organic and inorganic chemical.

**109. The presence of nitrates in excess of permissible limit causes**
(a) Skeletal fluorosis
(b) Methemoglobinemia
(c) Lauxative action
(d) Respiratory problem

**Ans. (b) :** The presence of nitrates in excess of permissible limit i.e. 45 mg/L causes → Blue baby syndrome OR Methemoglobinemia.

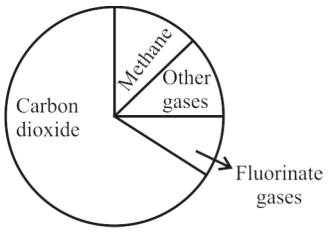
Presence of fluoride is excess of permissible limit
If < 1 ppm → Dental cavities
If > 1.5 ppm → Fluorsis and discoloration of teeth
If > 5 ppm → Bone Fluorsis

**110. Solar radiation that reaches the surface of the Earth comprises of**
(a) X-rays and Visible rays
(b) Y-rays, X-rays and UV-rays
(c) X-rays, UV-rays, visible rays and IR rays
(d) UV rays, visible rays and IR rays

**Ans. (d) :** Solar radiation that reaches the surface of earth comprises of→ UV rays
IR rays (49.40%)
Visible rays (42.3%)

**111. Major green House Gases due to human activity are**
(a) $CH_4$ and $H_2O$
(b) $N_2O$, and $H_2O$
(c) $N_2O$ and $CH_4$
(d) $N_2O$, $CH_4$ and $H_2O$

**Ans. (d) :** Major green house gases from people's activities→




**112. Oxidation states of sulphur based air pollutants responsible for Acid Rain are**
(a) +2 and +4 (b) -2, +2 and +4
(c) -2, +4 and +6 (d) +2, +4 and +6

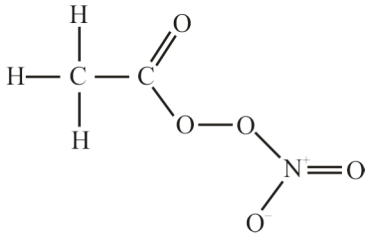
**Ans. (c) :** Oxidation state of sulphur is sulphur based air pollutants responsible for Acid Rain are → –2 +4 +6

**113. The sequence of different radicals produced in the photochemical smog formation reactions are**
(a) $RCH_2^* \rightarrow RCH_2O_2^* \rightarrow HO_2^* \rightarrow HO^*$
(b) $OH^* \rightarrow HO_2^* \rightarrow RCH_2O^* \rightarrow RCH_2O_2^* \rightarrow RCH_2^*$
(c) $RCH_2O_2^* \rightarrow RCH_2O_2^* \rightarrow HO_2^* \rightarrow HO^*$
(d) $RCH_2^* \rightarrow RCH_2O_2^* \rightarrow HO_2^* \rightarrow HO^*$

**Ans. (a) :** Sequence (or order) of different radicals produced in the photochemical smog formation reactions are-
$RCH_2 \rightarrow RCH_2O_2 \rightarrow RCH_2O \rightarrow HO_2 \rightarrow HO$

**114. Different types of chemical bonds present in a molecule of proxy acetyl nitrate (PAN) the dangerous carcinogenic pollutants are**
(a) 7 single + 2 double + 1 coordinate bonds
(b) 7 single + 1 double + 2 coordinate bonds
(c) 7 single + 3 double bonds
(d) 5 single + 1 double bonds and a nitrate ion

**Ans. (a) :** PAN structure → $C_2H_3O_5N$



7 single bond + 2 double bond + 1 co-ordinate bond

**115. The precise estimation of total organic loading of water is given by the following method**
(a) BOD
(b) COD
(c) Wet oxidation with peroxodisulphate
(d) Gas chromatography

**Ans. (c) :** Wet oxidation with peroxodisulphate → Precise measurement of total organic leading.
BOD → Biodegradable organic matter measurement.
COD → Non Biodegradable + Biodegradable organic matter measurement

**116. The naturally occurring chelating agent that holds the metal ions in natural water and soil is**
(a) Sodium tripholyphosphate
(b) Fluvic acid
(c) EDTA
(d) Sodium nitrilotriacetate

**Ans. (b) :** The naturally occurring chelating agent that holds the metal ions in natural water and soil is → Fluvic acid

**117. The following gas can be estimated by fluorescence technique after is with ozone**
(a) $NO_2$ (b) $CO_2$
(c) CO (d) NO

**Ans. (d) :**

| Pollutant | | Method of measurement |
|---|---|---|
| NO | → | Estimated by fluorescence technique after reaction with Ozone |
| $NO_2$ | → | Jacob & Hochheiser method chemi-luminescence method. |
| CO | → | Non dispersive infrared (NDIR) spectroscopy |
| $SO_2$ | → | Wast and Greke method ultraviolet fluorescence |

**118. The exhaust gas of an automobile was found to contain 3% by volume of CO at a temperature of $80^0$C. Express the concentration in μg $m^3$.**
(a) 19.39 (b) 29.09
(c) 38.60 (d) 14.48

**Ans. (b) :**

**119. In the determination of elements by atomic absorption spectroscopy, the line source is made intermittent by**
(a) golly cell (b) nebulizer
(c) Chopper (d) thermo couple

**Ans. (c) :** In the determination of the elements by atomic obsorption spectroscopy, the line source is made intermittent by the chopper. The chopper system contains two disk chopper and two fermi chopper and the measurement can be switched between modes.

**120. SADAR is useful for the study of**
(a) lower atmosphere
(b) middle atmosphere
(c) upper atmosphere
(d) ionosphere

**Ans. (a) :** SADAR is useful for the study of → lower atmosphere

**121. Solar constant refers to**
(a) the amount of solar radiation incident on a plane perpendicular to be incident beam
(b) the amount of total radiation emitted by sun
(c) average radiation emitted from sun
(d) average radiated received at any angle by a receiving surface

**Ans. (a) : Solar constant–** Solar constant is amount of solar radiation incident normally per second on 1 $m^2$ area of a perfectly black surface at mean distance of earth from Sun.

$$\text{Solar constant} = \frac{\text{Solar energy required}}{\text{area} \times \text{time}}$$

**122. Aerosol is**
(a) dispersion of solid and liquid particles suspended in gas
(b) dispersion of air molecules in atmosphere
(c) dispersion of the fine solid particles in a liquid
(d) dispersion of colloids in liquids

**Ans. (a) : Aerosol**- dispersion of solid and liquid particles suspended in gas.
Ex- Fog, mist, dust, forest exudates and
Geyser steam- Natural Aerosol
Air pollutants and smoke- Anthropogenic aerosol

**123. Abnormal, surface temperature on Venus is explain by**
(a) Normal green house effect
(b) Runoff green house effect
(c) Stefan Bolzmann law
(d) Wiens law

**Ans. (b) :** Abnormal, surface temperature on Venus is explain by runoff green house effect.

**124. An effective pH for a good coagulation is**
(a) 5 to 5.5 (b) 9 to 9.5
(c) 7.5 to 8 (d) 10 to 10.5

**Ans. (c) :** An effective pH value for good coagulation- 6.5 to 8.5
But is question paper TSPSC answer by- 7.5 to 8 consider right

**125. MPN is a measure of**
(a) Turbidity in water
(b) Fecal coliforms in water
(c) Nitrate in water
(d) Phosphates in water

**Ans. (b) :** MPN→ Most probable number (By multiple tube fermentation).

■ Most probable number is a number, which represents the bacterial density, which is most likely to be present in water.

■ Different dilution of sample of water is mixed with lactose broth and incubated in test tubes for 48 hours at 37ºC.

■ Presence of acid or carbon dioxide gas in the test tube indicates the presence of coli form bacteria.

■ The coli form actually is gram-negative, non-spore forming, rod shaped bacteria that ferment lactose and the end products are acid and $CO_2$.

■ Then referring to standard table MPN of E-coli per 100 ml water is found out.

■ When tables are not available, the Thomas equation can be used to estimate the MPN-

$$\text{MPN/10 ml} = \frac{\text{Number of positive tube} \times 100}{\sqrt{\left(\begin{matrix}\text{ml of sample in}\\ \text{negative tube}\end{matrix}\right) \times \left(\begin{matrix}\text{ml of sample in}\\ \text{all tubes}\end{matrix}\right)}}$$

The applying the Thomas equation, the court of positive tube should begin with the highest dilution in which at least one negative results has occurred.

**126. The influent turbidity to mechanical filters should not exceed**

(a) 1 mg/l (b) 10 mg/l
(c) 100 mg/l (d) 50 mg/l

**Ans. (b) :** The influent turbidity to mechanical filters should not exceed 10 mg/l.

**127. Which one of the following is not a coagulant**

(a) Copper sulphate (b) Alum
(c) Ferric chloride (d) Ferrous sulphate

**Ans. (a) :** The common coagulants added in water-
(a) Alum
(b) Copperas
(c) Chlorinated copperas
(d) Sodium Aluminate
(e) Lime

**128. Prechlorination in water is done to**

(a) To kill bacteria and virus
(b) to reduce inorganic matter
(c) to kill algae and protect filters
(d) to increase alkalinity

**Ans. (c) : Types of Chlorination-**
(i) Plain chlorination
(ii) Pre- chlorination
(iii) Post chlorination
(iv) Double chlorination
(v) Break point chlorination
(vi) Super chlorination
(vii) Dechlorination

**(i) Plain chlorination-** Used to indicate that only the chlorine treatment and no other treatment has been given to the raw water.

■ This helps in removing bacteria, organic matter and colour from the saw water.

■ It is used to clean water i.e. turbidity between 20-30 mg/L

■ Dose- 0.5 mg/L

**(ii) Pre- chlorination-** It is the process of applying chlorine to the water before filtration or rather before sedimentation- coagulation.

■ Dose- 0.1 to 0.5 mg/l when comes to filters
Normal dose- 5 to 10 mg/l

■ Prechlorination is always followed by post chlorination to ensure find safety.

**(iii) Post chlorination-**

■ Applying chlorine at the end when all the treatment is completed is called post chlorination.

■ The dose should be such that 0.1 to 0.2 mg/l should be left after a contact period of 10 min.

**(iv) Double chlorination-** It means water is chlorinated twice.

■ Pre chlorination and post chlorination are generally used in double chlorination.

**(v) Break point chlorination-** Break point chlorination is a term which gives in an idea of the extent of chlorine added to water.

In fact, it represents that must dose of chlorination, beyond which any further additional chlorine will appear as free residual chlorine.

In general practice, chlorine is added beyond break point to ensure a residual of 0.2-0.3 mg/l as free chlorine. This residual takes care of future recontamination.

**(vi) Super-chlorination-** When excess chlorine (5 to 15 mg/l) is added in cooler during epidemic such that it gives a residual of 1-2 mg/l beyond break point is called super chlorination.

**(vii) Dechlorination-** When chlorine residue is high encase chlorine will be removed by dechlorinating agent various dechlorinating agent-
(1) $Na_2S_2O_3$ (Sodium thio-sulphate)- cheapest in all.
(2) Activated carbon
(3) Sulphur dioxide ($SO_2$)
(4) Sodium bisulphate ($N_aHSO_3$)
(5) Ammonia ($NH_4OH$)

**129. Chlorides in waste water is**

(a) Toxic
(b) Biodegradable
(c) Non biodegradable
(d) Corrosive

**Ans. (c) :** Chlorides in waste water is → Non biodegradable

**130. Entamoeba histolitica is a**

(a) Bacteria
(b) Cyst of protoza
(c) Virus
(d) Fungus

**Ans. (b) :** Entamoeba histolitica → Cyst of protoza

**131. Activated sludge process is an**

(a) attached growth system
(b) anaerobic system
(c) suspended growth system
(d) chemical treatment system

**Ans. (c) :** Suspended growth process- Activated sludge process aerated lagoon oxidation pond Aerobic Digester Anaerobic Digester.
Attached growth process- Trickling filter rotating biological contractors Bio towers

**132. Bacillus is a**

(a) Virus
(b) Bacteria
(c) Protozoa
(d) Medicinal drug

**Ans. (b) :** Bacillus - Bacteria

**133. Clay liners in land filling are provided**

(a) To strength the base
(b) To construct bunds
(c) As a mix with solid wastes
(d) To prevent leachate

**Ans. (d) :** Clay liners in land filling are provided- to prevent leachate.
Ex. of clay lines or synthetic liners like geo-textiles (clay, bentonite, zeolite and kaoline)

**134. Langley's is a measure of**

(a) Solar radiation
(b) Rainfall intensity
(c) Wind speed
(d) Earth quake intensity

**Ans. (a) :** The langley is a unit of heat transmission, especially used to express the rate of solar radiation received by earth.

**135. The size dependent collection efficiency of particles is the following**

(a) Grade efficiency
(b) Overall collection efficiency
(c) Murphree efficiency
(d) Structure efficiency

**Ans. (a) : Grade efficiency–**
Classification is the art of separating solid particles in a mixture of solids and fluid into fraction according to particle size or density by method other than screening. Most methods of separation are not 100% effective. There is usually a range of particle size that are separated with various degree of efficiency. The variation in the efficiency is referred to as the grade efficiency curve.

**136. Cyclons separators use the following force to separate the particulates**

(a) Gravity
(b) Centrifugal
(c) Elector-static
(d) Electromagnetic

**Ans. (b) :** Centrifugal collectors in duding cyclone collectors in duding cyclone collector or separator and dynamic-
Minimum particle size that- 5-25 μm
Can be removed in μm
Efficiency- 50-90%
A cyclone depurator is a specially designed closed chamber, in which the velocity of the inlet gas is transformed into spinning vortex and the particles from the gas are thrown out under the centrifugal force on the walls of chamber, slide down the hopper and thus removed.
It's efficiency depends on the generated centrifugal force, which in turns depends on mass of the particles (Mp), inlet gas velocity (V) and radius of cyclone (R) and is given by the equation-
Fc = Centrifugal force generated

$$= Mp.\frac{V^2}{R}$$

**137. In wet scrubbers, the particles separation takes place on**

(a) Fibers
(b) Droplets of liquid
(c) Filter medium
(d) Surface

**Ans. (b) : Wet scrubber-** In these devices, the flue gas is made to push up against a down failing water (liquid) current. The particulate matter mixes up with water droplets and thus, falls down and gets removed.

**138. In the sufficiency equation for electrostatic precipitators 'u' represents the following**

(a) Setting velocity of particle
(b) Migration velocity of particle
(c) Relative velocity of gas and particle
(d) Size of the particle

**Ans. (b) : Electrostatic precipitators-** In Electrostatic precipitators, the flue gas is made to pass through a highly ionised zone, where the particles get electrically charged and are separated out (migrated out) from the gas, with the help of electrostatic forces in the powerful electric field.

**139. As per hazardous waste (Management and handling) Rules 1989, the hazardous waste has been put into the following number of categories**

(a) 16 (b) 14
(c) 18 (d) 10

**Ans. (c) :** Hazardous waste (management & handling) rule 1989.
- It is divided in to 18 categories.
- Type of waste in category one is cyanide waste .
- Type of waste in category 18 is discarded container and container liners of hazardous and toxic chemicals and waste.

**140. The following is a chemical method of hazardous waste treatment**

(a) Neutralization
(b) Distillation
(c) Filtration
(d) Absorption

**Ans. (a) :** Hazardous waste can be treated by chemical, thermal, biological and physical method.

• Chemical method included ion exchange, precipitation oxidation and reduction and neutralization.

**141. Cyanide removal by adsorption is done on**

(a) Silica gel
(b) Activated alumina
(c) Activated carbon
(d) Fly ash

**Ans. (c) :** Cyanide removal by adsorption is done on activated carbon.

**142. In ecosystem consumers are also known as**

(a) Heterotrophs
(b) Phagotrophs
(c) Photosynthetic autocross
(d) Prdators

**Ans. (a) :** In ecosystem a consumer is a heterotroph and a produces is an autotraph.

Produces $\rightarrow$ Priamry consumer $\rightarrow$ Secondary consumer $\rightarrow$ Tertiary consumer $\rightarrow$ decomposer.

**143. An ecosystem is a layered structure with the following number of layers**

(a) Single layered
(b) Double layered
(c) Multi layered
(d) Five

**Ans. (b) :** An ecosystem is a layered structure with the following number of layers double layered.

**144. In general the colour of pigment comes from the Wavelength of light**

(a) Absorbed
(b) Reflected
(c) Deflected
(d) Refracted

**Ans. (b) :** In general the colour of pigment comes from the Wavelength of light reflected.

**145. In photosynthesis, the conversion of unusable sunlight energy into usable chemical energy is associated with the action of green pigment called**

(a) Chlorobenzene
(b) Chloroamine
(c) Chlorophyll
(d) Corotenoid

**Ans. (c) :** In photosynthesis, the conversion of unusable sunlight energy into usable chemical energy is associated with the action of green pigment called Chlorophyll.

**146. Ecosystem productivity is measured in terms of**

(a) Keal/$m^2$.year
(b) Lumens/$m^2$.year
(c) Kg/$m^2$.year
(d) $m^3/m^2$.year

**Ans. (a) :** Ecosystem productivity is measured in the term of kcal/$m^2$.year. It refers to the rate of generation of biomass is an ecosystem over a period by plant during photosynthesis.

**147. Biogeochemical cycles mean cycling of the following through a system**

(a) Energy
(b) Biological organisms
(c) Matter
(d) Chemicals

**Ans. (c) :** A biochemical cycle is the pathway by which a chemical substance cycle the biotic and the abiotic compartments of earth. It more generally known as cycle of matter.

**148. When fresh water from rivers meets the ocean water, it is called**

(a) Bay
(b) Straight
(c) Delta
(d) Estuary

**Ans. (d) :** When a fresh water from rivers meets the ocean water is called estuary. Estuaries are places where rivers meet the sea and may be defined as areas where salt water is measurably diluted with fresh water.

**149. Generally, adsorbent is a solid can be used many times by a process of**

(a) Dissolution
(b) Oxidation
(c) Reduction
(d) Regeneration

**Ans. (d) :** Generally, adsorbent is a solid can be used many times by a process of regeneration.

**150. In the dilution technique, lapse rate is defined as**

(a) Change of moisture content with height
(b) Change of air pressure with height
(c) Change of air temperature with height
(d) Change of flow direction

**Ans. (c) :** In the dilution technique, lapse rate is defined as Change of air temperature with height

# Telangana State Public Service Commission
# Assistant Executive Engineer (Common, Civil/Mech.) Exam, 2017
# Solved Paper

**1. Reaction turbine requires**

(a) High head and low discharge
(b) High head and high discharge
(c) Low head and low discharge
(d) Low head and high discharge

**Ans. (d) :** Francis and kaplan turbines are reaction turbines francis turbine works on medium head and medium discharge and Kaplan turbine works on low head and high discharges.

**2. In mixed flow turbines, the water enters the blades _____ and comes out _____.**

(a) radially, axially
(b) radially, radially
(c) axially, radially
(d) axially, axially

**Ans. (a) :** In mixed flow turbines like modern francis turbines water enters radially and comes out axially.

**3. Which of the following statement is correct?**

(a) In an impulse turbine, the water impinges on the buckets with pressure energy
(b) In a reaction turbine, the water glides over the moving vanes with kinetic energy
(c) In an impulse turbine, the pressure of the flowing water remains unchanged and is equal to atmospheric pressure
(d) In a reaction turbine, the pressure of the flowing water increase after gliding over the vanes

**Ans. (c) :** In impulse turbine pressure of water remains constant as it flows our the blade. As the where is exposed to atmosphere, the pressure is equal to atmospheric pressure.
In reaction turbine the pressure continuously decreases as water flows guide vanes as well as runners blades.

**4. In an Inward flow reaction turbine**

(a) the water flows parallel to the axis of the wheel
(b) the water enters at the centre of the wheel and then flows towards the outer periphery of the wheel
(c) the water enters the wheel at the outer periphery and then flows towards the centre of the wheel
(d) the flow of water is partly radial and partly axial

**Ans. (c) :** In inward flow reaction turbine like conventional francis turbine, water enters radially inward at outer periphery and then flows towards the centre.

**5. The flow ratio of Francis turbine is defines as the ratio of the**

(a) velocity of flow at inlet to the theoretical jet velocity
(b) theoretical velocity of jet to the velocity of flow at inlet
(c) velocity of runner at inlet to the velocity of flow at inlet
(d) velocity of runner at outlet to the velocity of flow at inlet

**Ans. (a) :** $k_f = \frac{V_{f_1}}{\sqrt{2gH}} = \frac{\text{Velocity of flow at Inlet}}{\text{Theoretical jet velocity}}$

$k_f = [0.6 \text{ to } 0.9]$

**6. The unit speed of the turbine runner is**

(a) $\frac{N}{\sqrt{H}}$ (b) $\frac{N}{H}$

(c) $N/H^{3/2}$ (d) $\frac{N}{H^2}$

**Ans. (a) : Unit speed –** It is defined as the speed of a turbine working under a unit head (i.e. unit a head of 1m). It is denoted by $N_u$.

let, N = speed of turbine under a head (H)

H = Head under which a turbine is working

u = Tangential velocity

$u \propto \sqrt{H}$

and $u = \frac{\pi DN}{60}$ [For given turbine D = constant.]

$u \propto N$

$N \propto \sqrt{H}$

$N = k_1\sqrt{H}$ ..... (i)

It head on the turbine become unity, the speed become unit speed.

$H = 1, \quad N = N_u$

$N_u = k_1\sqrt{1}$

$k_1 = N_u$

From equation (i)

$N = N_u\sqrt{H}$

$N_u = \frac{N}{\sqrt{H}}$

**Note-** unit discharge & unit power

$Q_u = \frac{Q}{\sqrt{H}}, P_u = \frac{P}{H^{3/2}}$

**7. The specific speed of a turbine is given by the equation.**

(a) $\frac{N\sqrt{P}}{H^{3/2}}$ (b) $\frac{N\sqrt{P}}{H^{2}}$

(c) $\frac{N\sqrt{P}}{H^{5/4}}$ (d) $\frac{N\sqrt{P}}{H^{3}}$

**Ans. (c) :** The specific speed of turbine is given by

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

N = speed in rpm
P = power output in kW
H = operating head in m
The specific speed of the turbine depends only on the turbine geometry and is used select most suitable turbine at given site.

**8. A Pelton wheel develops 1750 kW under a head of 100 metres while running at 200 rpm and discharging 2500 litres of water per second. The unit power of the wheel is**

(a) 0.25 kW (b) 0.75 kW
(c) 1.75 kW (d) 3.75 kW

**Ans. (c) :** $P_u = \frac{P}{H^{3/2}} = \frac{1750\,kW}{(100)^{3/2}} = 1.75\,kW$

**9. The speed of an imaginary turbine, identical with the given turbine, which will develop a unit power under a unit head, a known as**

(a) normal speed (b) unit speed
(c) specific speed (d) pitot speed

**Ans. (c) :** $N_s = \frac{N\sqrt{P}}{H^{5/4}}$

When = P = 1 kW, H = 1 m then N = $N_S$

**10. Which of the following turbine is preferred for 0 to 25 m head of water?**

(a) Pelton wheel
(b) Kaplan turbine
(c) Francis turbine
(d) Modern Francis turbine

**Ans. (b) :** For very low heads Kaplan turbine is most suitable and high discharge.

| Turbine | Specific speed | Head |
|---|---|---|
| Pelton single jet | 10-35 | 250-1000 m |
| Pelton multijet | 35-60 | 250-1000 m |
| Francis | 60-300 | 60-150 |
| kaplan | >300 | <30 m |

**11. The cavitations in centrifugal pump can be reduced by**

(a) reducing discharge
(b) reducing suction head
(c) increasing the discharge
(d) increasing the flow velocity

**Ans. (b) : Cavitations in centrifugal pump** – Cavitation begins to appears in centrifugal pumps when the pressure at the suction falls below the vapour pressure of the liquid.
The intensity of cavitation increases with decrease in valve of NPSH. The cavitation in a pump can be noted by a sudden drop in efficiency, head and power requirement.

**12. The specific speed of a turbine is the speed of an imaginary turbine, identical with the given turbine, which**

(a) delivers unit discharge under unit head
(b) delivers unit discharge under unit speed
(c) develops unit power under unit head
(d) develops unit power under unit speed

**Ans. (c) :** $N_s = \frac{N\sqrt{P}}{H^{5/4}}$

When P = 1 kW and H = 1 m
then N = $N_S$

**13. In the casing of a centrifugal pump, the kinetic energy of the water is converted into _____ energy before the water leaves the casing.**

(a) Pressure (b) Magnetic
(c) Electronic (d) Dynamic

**Ans. (a) :** Pump casing has gradually increasing area it performs similar to deffuser. As area increases, velocity decreases and pressure increases. Thus the pump casing converts kinetic energy of water partially into pressure energy.

**14. In a centrifugal pump with usual notations wit radial entry of liquid, the manometric efficiency ($\eta_H$) is given by**

(a) $\frac{u_2 v_{w_2}}{gH}$ (b) $\frac{gH}{u_2 v_{w_2}}$

(c) $\frac{H}{g u_2 v_{w_2}}$ (d) $\frac{2gH}{u_2 v_{w_2}}$

**Ans. (b) :** Hydraulic (or) manometric efficiency ($\eta_{meno}$) : it is ratio of manometric head to head imparted to the fluid to the, mark done by the impeller per unit weight to fluid.

$$\eta_{mano} = \frac{H}{H_e} = \frac{H}{\frac{u_2 v_{w_2}}{g}} = \frac{gH}{u_2 V_{w_2}}$$

**15. Mechanical efficiency of a centrifugal pump is the ratio of**

(a) energy available at the impeller of the energy supplied to the pump by the prime mover
(b) actual workdone by the pump to the energy supplied to the pump by the prime mover
(c) energy supplied to the pump to the energy available at the impeller
(d) manometric head to the energy supplied by the the impeller per kN of water

**Ans. (a) : Mechanical efficiency–** The ratio of the power delivered by the impeller to the liquid to the power input to the pump shaft is known as mechanical efficiency.

$$\eta_m = \frac{\text{Power delivered by the impeller to the liquid}}{\text{Power input to the pump shaft}}$$

**16. Overall efficiency of a centrifugal pump is the ratio of**

(a) energy available at the impeller to the energy supplied to the pump by the prime mover

(b) actual work done by the pump to the energy supplied to the pump by the prime mover.

(c) energy supplied to the pump to the energy available at the impeller

(d) manometric head to the energy supplied by the impeller per kN of water

**Ans. (b) :** Overall efficiency of a centrifugal pump is the ratio of actual work done by the pump to the energy supplied to the pump by the prime mover

$$\eta_0 = \frac{\text{Actual energy added by pump to water}}{\text{Work done by prime mover}}$$

**17. Discharge (Q) of a centrifugal pump is given by**

(a) $Q = \pi.D.V_f$ (b) $Q = \pi.b.V_f$

(c) $Q = \pi.D.b.V_f$ (d) $Q = D.b.V_f$

Where D = Diameter of impeller at inlet

b = Width of impeller at inlet, and

$V_f$ = Velocity of flow at inlet.

**Ans. (c) :** Discharge (Q) = $\pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$

D = diameter of impeller at inlet

b = width of impeller at inlet, and

$V_f$ = velocity of flow at inlet.

**18. Multi-stage centrifugal pumps are used to**

(a) given high discharge

(b) produce high heads

(c) pump viscous fluids

(d) high velocity

**Ans. (b) : Multistage pump–** A multistage centrifugal pump is one which has two or more identical impeller mounted on the same shaft or on different shaft important function of multistage pump are–

(i) To produce greater head

(ii) To discharge a large quantity of liquid.

**19. The specific speed of pump has following dimensions :**

(a) $L^{3/4}T^{3/2}$ (b) $M^0L^0T^0$

(c) $M^{-1/2}L^{1/2}T^{1/4}$ (d) $L^{3/4}T^{1/2}$

**Ans. (*) :** For pumps,

$$N_S = \frac{N\sqrt{Q}}{H^{3/4}} = \frac{T^{-1}\sqrt{L^3T^{-1}}}{L^{3/4}} = \frac{T^{-1}L^{3/2}T^{-1/2}}{L^{3/4}}$$

$= M^oL^{3/4}\,T^{-3/2}$

**Note:** (–) Minus sign is missing in the dimension of T in option.

**20. A centrifugal pump will start delivering liquid only when the pressure rise in the impeller is equal to the**

(a) kinetic head (b) velocity head

(c) manometric head (d) static head

**Ans. (c) :** A centrifugal pump will start delivering liquid only when the pressure rise in the impeller is equal to the manometric head.

• The manometric head is defined as the head against which a centrifugal pump has to work.

**21. Head developed by a centrifugal pump is**

(a) Proportional to diameter of impeller

(b) Proportional to speed of impeller

(c) Proportional to diameter and speed of impeller

(d) Proportional to diameter of delivery pipe

**Ans. (c) :** Head developed by the centrifugal pump is proportional to impeller diameter and impeller speed.

$$\frac{\sqrt{H_m}}{DN} = \text{constant}$$

**22. Discharge of a centrifugal pump is**

(a) directly proportional to diameter of its impeller

(b) inversely proportional to diameter of its impeller

(c) directly proportional to (diameter)$^2$ of its impeller

(d) inversely proportional to (diameter)$^2$ of its impeller

**Ans. (a) :** $Q = A_f V_f = \pi D_1 B_1 V_{f_1} = \pi D_2 B_2 V_{f_2}$

i.e. $Q \propto D$

$D_1$ = Diameter of impeller at inlet

$B_1$ = width of impeller at inlet

$V_{f_1}$ = velocity of flow at inlet

**23. Power required to drive a centrifugal pump is directly proportional to _____ of its impeller.**

(a) diameter

(b) square of diameter

(c) cube of diameter

(d) fourth power of diameter

**Ans. (b) :** $P \propto \rho g Q h$

$\propto (D^2\sqrt{H})H$

$\propto D^2 H^{3/2}$

$\therefore$ $P \propto D^2$ if head is constant

**24. The specific speed ($N_S$) of a centrifugal pump is given by**

(a) $\frac{N\sqrt{Q}}{H^2}$ (b) $\frac{N\sqrt{Q}}{H^{3/4}}$

(c) $\frac{N\sqrt{Q}}{H^{1/2}}$ (d) $\frac{N\sqrt{Q}}{H^{5/4}}$

**Ans. (b) :** for pump

$$N_S = \frac{N\sqrt{Q}}{H^{3/4}}$$

Where,

Q = discharge ($m^3$/sec)

N = Actual speed of pump in rpm

H = delivery head total manomentric in (meters)

**25. The specific speed from 160 to 500 rpm of a centrifugal pump indicates that the pump is**

(a) slow speed with radial flow at outlet

(b) medium speed with radial flow at outlet

(c) high speed with radial flow at outlet

(d) high speed with axial flow at outlet

**Ans. (d) :** The ranges of specific speeds for different types of pumps are tabulated below-

| Types of pump | Specific speed |
|---|---|
| Slow speed radial flow | 10-30 |
| Medium speed radial flow | 30-50 |
| High speed radial flow | 50-80 |
| Mixed flow | 80-160 |
| Axial flow | 160-500 |

**26. Which of the following pump is suitable for small discharge and high heads?**
(a) centrifugal pump
(b) axial flow pump
(c) mixed flow pump
(d) Reciprocating pump

**Ans. (d) :** For small discharge at high pressure heads, Reciprocating pump is preferred.
Where as centrifugal pump is preferred for large discharge and less heads.

**27. The centrifugal pump preferred for a specific speed between 80 to 160 rpm is**
(a) slow speed with radial flow at outlet
(b) medium speed with radial flow at outlet
(c) high speed with radial flow at outlet
(d) high speed with mixed flow at outlet

**Ans. (d) :** The ranges of specific speeds for different types of pumps are tabulated below-

| Types of pump | Specific speed |
|---|---|
| Slow speed radial flow | 10-30 |
| Medium speed radial flow | 30-50 |
| High speed radial flow | 50-80 |
| Mixed flow | 80-160 |
| Axial flow | 160-500 |

**28. In a laminar flow through a circular pipe of diameter 20 cm, the maximum velocity is found to be 1 m/s. The velocity at a radial distance 5 cm from the axis of the pipe will be**
(a) 0.25 m/s (b) 0.50 m/s
(c) 0.75 m/s (d) 1.0 m/s

**Ans. (c) :** $V = V_{max}\left(1-\frac{r^2}{R^2}\right) = 1\left(1-\frac{5^2}{10^2}\right) = 0.75\ m/s$

**29. In a simple tensile test, Hooke's law is valid up to the**
(a) elastic limit
(b) limit of proportionality
(c) ultimate stress
(d) breaking point

**Ans. (b) : Hooke's law–** According to this law for homogeneous, isotropic and linear elastic material stress is directly proportional to strain upto limit of proportionality $\sigma \propto \varepsilon$
$\sigma = E\varepsilon$

**30. The percentage elongation of a material as obtained from static tension test, depends on**
(a) diameter of the test specimen
(b) gauge length of the specimen
(c) nature of end grips of the testing machines
(d) geometry of the test specimen

**Ans. (b) :** The percentage elongation of a material as obtained from static tension test, depends on gauge length of the specimen. % elongation $= \frac{\Delta\ell}{\ell}\times 100$

**31. If a block of material of length 25 cm, breadth 10 cm and height 5cm undergoes volumetric strain of $\frac{1}{5000}$ then change in volume will be**
(a) 0.50 $cm^3$ (b) 0.25 $cm^3$
(c) 0.20 $cm^3$ (d) 0.75 $cm^3$

**Ans. (b) :** Volumetric strain = $e_v = \frac{\Delta V}{V}$
Initial volume V = (25 × 10 × 5) $cm^3$
$e_v = \frac{1}{5000}(\text{given})$
$\Delta V = e_v \times V$
$= \frac{1}{5000}\times(25\times10\times5)$
$= 0.25\ cm^3$

**32. A steel rod of diameter 1 cm and 1 m long is heated from 200 ºC to 1200 ºC. Its $\alpha = 12\times10^{-6}$/K and E = 200 GN/$m^2$. If the rod is free to expand, thermal stress developed on it is**
(a) $12\times10^3$ N/$m^2$ (b) 240 kN/$m^2$
(c) zero (d) infinite

**Ans. (c) :** If a body is free and is subjected to change in temp [either increasing or decreasing] the body undergoes deformation either expansion or contraction.
Hence, strain = $\frac{DL}{L} = \frac{L\alpha\Delta T}{L}$
Strain = $\alpha\Delta T$
and the stress generalized is zero. [since body is free]
• If however the body is restrained either fully or partially and temperature is changes the stress will occur termed as thermal stress and the strain would depends upon extent of restrainment.

**33. A steel rod 10 mm in diameter and 1 m long is heated from 20 ºC to 120 ºC,E = 200 GPa and $\alpha = 12\times10^{-6}$ per ºC. If the rod is not free to expand, the thermal stress developed is**
(a) 120 MPa (tensile) (b) 240 MPa (tensile)
(c) 120 MPa (comp. (d) 240 MPa (comp.)

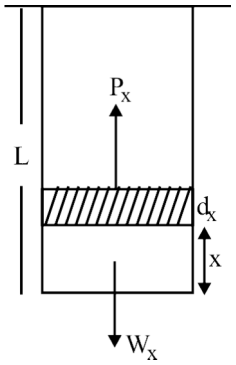
**Ans. (d) :** Thermal strain = e = $\propto\Delta T$
Thermal strain = $\sigma$ = E.e = E $\propto$ $\Delta$T
= (200 × $10^9$) × (12 × $10^{-6}$) × (120 – 20)
= 200 × $10^9$ × 12 × $10^{-6}$ × 100
= 240 MPa (compressive)
Since body tries to expand, but it's expansion is restricted.
$\therefore$ compressive stresses will be induced.

**34. A heavy uniform rod of length 'L' and material density δ is hung vertically with its top end rigidly fixed. How is the total elongation of the bar under its own weight expressed?**

(a) $\frac{2\delta L^2 g}{E}$ (b) $\frac{\delta L^2 g}{E}$

(c) $\frac{\delta L^2 g}{\sqrt{2E}}$ (d) $\frac{\delta L^2 g}{2E}$

**Ans. (d) :**




Deformation due to self weight–

$$\delta_{strip} = \frac{P_{(x)}dx}{AE}$$

$$P_{(x)} = W_x = V_x\gamma$$

$\gamma$ = unit weight of material.

$$P_{(x)} = A \times x \times \gamma$$

$$\delta_{strip} = \frac{A.x.\gamma d_x}{AE}$$

$$= \left(xdx\frac{\gamma}{E}\right)$$

$$\Delta = \int_0^L \delta_{strip} = \int_0^L \frac{\gamma}{E}xdx$$

$$= \frac{\gamma}{E}\left[\frac{x^2}{2}\right]_0^L$$

$$\Delta = \frac{\rho g L^2}{2E}$$

**35. A round bar length $\ell$, elastic modulus E and Poisson's ratio is μ, is subjected to an axial pull 'P'. What would be the change in volume of the bar?**

(a) $\frac{P\ell}{(1-2\mu)E}$ (b) $\frac{P\ell(1-2\mu)}{E}$

(c) $\frac{P\ell\mu}{E}$ (d) $\frac{P\ell\mu}{E}$

**Ans. (b) :** Volumetric strain for cylinder

$$e_v = e_\ell + 2e_d$$

$$e_\ell = \frac{P/A}{E}$$

$$e_d = -\mu\frac{P/A}{E}, \quad e_v = \frac{P}{AE} + \left(-\frac{\mu P}{AE} - \frac{\mu P}{AE}\right)$$

$$\therefore e_v = \frac{P}{AE}(1-2\mu)$$

$$e_v = \frac{\Delta V}{V}$$

$$\Rightarrow \Delta V = (A \times \ell)\frac{P}{AE}(1-2\mu)$$

$$\Delta V = \frac{P\ell}{E}(1-2\mu)$$

**36. Which one of the following is rupture stress?**

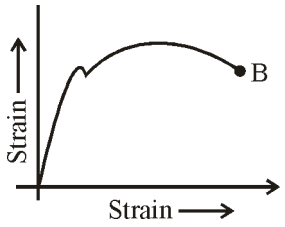
(a) Breaking stress

(b) Maximum load/original cross-sectional area (A)

(c) Load breaking point /A

(d) Load at breaking point/neck area

**Ans. (a) :**




B = rupture stress or Breaking stress.

**37. A 2m long rod of diameter 2 mm is subjected to an axial pull of 12 kN. The rod is extended by 0.5 cm. What is the approximate value of the modulus of elasticity of the material of the rod?**

(a) 12 GPa (b) 180 GPa

(c) 125 GPa (d) 200 GPa

**Ans. (*) :** Length of rod '$\ell$' = 2 m

Diameter 'd' = 2 mm = $2 \times 10^{-3}$ m

Axial pull 'p' = 12 kN

Changed in length '$\Delta\ell$' = 0.5 cm = $0.5 \times 10^{-2}$ M

$$\Delta\ell = \frac{P\ell}{AE}$$

$$E = \frac{P\ell}{A\Delta\ell} = \frac{\left(12\times10^3\right)\times 2}{\frac{\pi}{4}\left(2\times10^{-3}\right)^2 \times \left(0.5\times10^{-2}\right)}$$

$= 1520 \times 10^9$ N/m$^2$

= 1520 Gpa

**38. Which one of the following information cannot be obtained from the static tensile test of a mild steel specimen?**

(a) Modulus of elasticity

(b) Qualitative determination of toughness

(c) Ductility

(d) Weldability

**Ans. (d) :** Weldability cannot be obtained from static tensile test.

**39. If a material has numerically the same value for its modulus of rigidity and bulk modulus, then what is Poisson's ratio?**

(a) 0.25 (b) 0.2
(c) 0.15 (d) 0.125

**Ans. (d) :** $E = 3k(1-2\mu)$ & $E = 2G(1+\mu)$
Given,
Modulus of rigidity = Bulk modulus
$G = K$,
Also
$2G(1+\mu) = 3K(1-2\mu)$
$2 + 2\mu = 3 - 6\mu$
$8\mu = 1$
$\mu = \frac{1}{8} = 0.125$

**40. What property of material enables it to be drawn into wires with the application of tensile force?**

(a) Plasticity (b) Elasticity
(c) Ductility (d) Malleability

**Ans. (c) :** Ductility is the property of material, which shows its ability to be drawn into wires.

**41. The phenomenon of decreased resistance of a material due to reversal of stress is called**

(a) resilience (b) elasticity
(c) creep (d) fatigue

**Ans. (d) : Fatigue–** Deterioration of the material under repeated cycle of stress or strain resulting in progressive cracking that finally produce fracture of material is termed as fatigue.
• On microscopic level typically fatigue initiate the the formation of the crack at the point of high stress which gradually increases due to repeated cycle of load and finally lead to the sudden failure.

**42. As soon as the external forces causing deformation in a perfectly elastic body are withdrawn, the elastic deformation disappears**

(a) Only partially
(b) completely over a prolonged period of time
(c) completely and instantaneously
(d) completely after an initial period of rest

**Ans. (c) :** If a body regains its original configuration immediately and completely after removal of deformation force it is called as perfectly elastic body.

**43. What is the maximum possible value of Poisson's ratio for a non-dilatants material?**

(a) 0.67 (b) 0.50
(c) 0.33 (d) 0.25

**Ans. (c) :** For non dilatants material, maximum value of poisson's ratio is 0.33.

**44. In a pure tensile member, the normal stress on a plane at right angles to the direction of load is 100 N/mm$^2$. What is the normal stress at a plane whose normal is inclined at 60$^o$ to the direction of the load?**

(a) 075N/mm$^2$ (b) 100 Nmm$^2$
(c) 125 N/mm$^2$ (d) 150 N/mm$^2$

**Ans. (*) :** All option are incorrect
$\sigma_\theta = \frac{\sigma}{2} + \frac{\sigma}{2}\cos 2\theta$
$\sigma_\theta = \frac{100}{2} + \frac{100}{2}\cos(2\times 60)$
$\sigma_\theta = 25$ N/mm$^2$

**45. What is the normal stress on a plane inclined at 45$^0$ to the axis of a square rod of side a subjected to an axial tensile force of**

(a) T/a$^2$ (b) T/2a$^2$
(c) T/4a$^2$ (d) T/8a$^2$

**Ans. (b) :**
$\sigma_\theta = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta$
$\sigma_x = \frac{T}{a^2}$
$\sigma_y = 0,\ \tau_{xy} = 0$
$\therefore\ \sigma_\theta = \frac{\sigma_x}{2} + \frac{\sigma_x}{2}\cos 2\theta$
$\sigma_{45^o} = \frac{T}{2a^2} + \frac{T}{2a^2}\cos 90^o$
$\sigma_{45^o} = \frac{T}{2a^2}$

**46. Modulus of elasticity and Poisson's ratio of a material are 2.1 × 10$^5$ N/mm$^2$ and 0.25 respectively. What is the value of modulus of rigidity of the same material in 10$^5$ N/mm$^2$?**

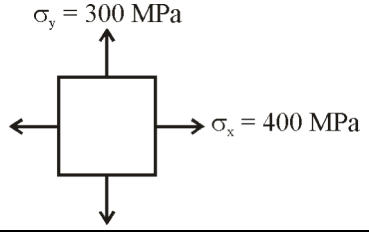
(a) 0.84 (b) 0.70
(c) 1.40 (d) 0.50

**Ans. (a) :** $E = 2G(1+\mu)$
$2.1 \times 10^5 = 2G(1 + 0.25)$
$G = \frac{2.1\times 10^5}{2\times 1.25} = 0.84\times 10^5\ \text{N/mm}^2$
$G = 0.84 \times 10^5$ N/mm$^2$

**47. Principal stresses at a point in a stressed solid are 400 MPa and 300 MPa respectively. The normal stress on plane inclined at ± 45$^o$ to the principal planes will be**

(a) 200 MPa and 5 MPa
(b) 350 MPa and both planes
(c) 100 MPa and 600 MPa
(d) 1500 Mpa and 500 MPa

**Ans. (b) :**



$$\sigma_\theta = \frac{(\sigma_x + \sigma_y)}{2} + \frac{\sigma_x + \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$\tau_{xy} = 0$

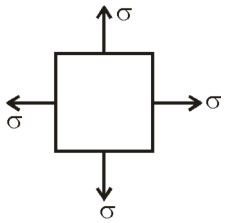
$$\sigma_\theta(\pm 45^\circ) = \frac{400+300}{2} + \frac{400+300}{2}\cos 90^\circ$$

= 350 MPa on both planes.

**48. When the two principal stresses are equal and like, the resultant stress on any plane is**

(a) equal to principal stress
(b) zero
(c) one half of the principal stress
(d) one third of the principal stress

**Ans. (a) :** When two principal stresses are equal and like




Resultant stress on any plane

$\sigma_R = \sqrt{\sigma_\theta^2 + \tau_\theta^2}$

$\sigma_\theta = \sigma$ (irrespective of any plane)

Shear stress, $\tau_\theta = \frac{\sigma_1 - \sigma_2}{2} = \frac{\sigma - \sigma}{2} = 0$

$\tau_\theta = 0$

$\sigma_R = \sigma$ (Magnitude of principal stress)

**49. On a plane, resultant stress is inclined at an angle of 45º to the plane, if the normal stress is 100 N/mm$^2$, shear stress on plane will be**

(a) 71.5 N/mm$^2$ (b) 100 N/mm$^2$
(c) 86.6 N/mm$^2$ (d) 120.8 N/mm$^2$

**Ans. (b) :** Given : Angle of obliquity $\phi = 45^\circ$

$\sigma_\theta = 100$ N/mm$^2$

$\tan\phi = \frac{\tau_\theta}{\sigma_\theta}$

$\tan 45^\circ = \frac{\tau_\theta}{100}$

$\tau_\theta = 100$ N/mm$^s$

**50. The maximum shear stress occurs on**

(a) principal planes
(b) plane at 45º to the principal planes
(c) planes at 90º to the principal planes
(d) planes independent of the inclination to the principal planes

**Ans. (b) :** Maximum shear stress occurs on planes at 45º to the principal planes and each separated by 90º.

**51. The ratio of the maximum shear stress to the difference of the two principal stresses is**

(a) 1/2 (b) 1/3
(c) 1/4 (d) 1.6

**Ans. (a) :** Maximum shear stress on any plane

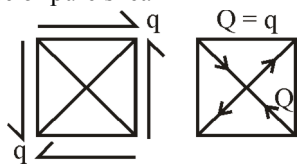
In plane $\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$

$$\frac{\tau_{max}}{\sigma_1 - \sigma_2} = \frac{\sigma_1 - \sigma_2}{2(\sigma_1 - \sigma_2)} = 1 : 2$$

**52. A specimen is subjected to pure shear, the shear stress being q. Tensile and compressive stresses of intensity, occur on planes inclined at 45º to the shear stress. What is the value of the ratio Q/q ?**

(a) 2 (b) 1.5
(c) 1.25 (d) 1

**Ans. (d) :** Case of pure shear

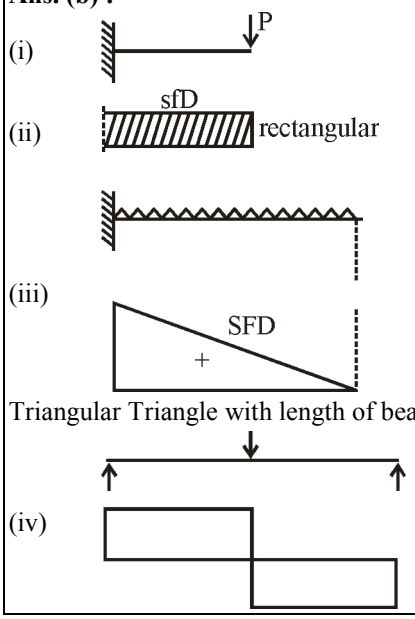



$\therefore \frac{Q}{q} = 1$

**53. If the SF diagram for a beam is a triangle with length of the beam as its base, the beam is**

(a) A cantilever with a concentrated load at its free end
(b) A cantilever with UDL over its whole span
(c) Simply supported with a concentrated with a concentrated load at its mid-point
(d) Simply supported with a UDL over its whole span

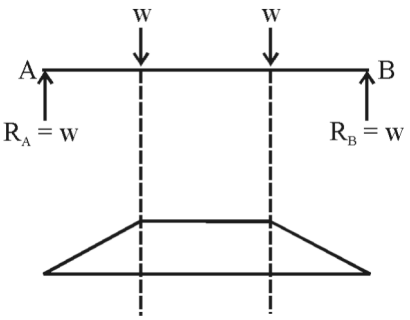
**Ans. (b) :**




**54. The bending moment diagram for a simply supported beam is a rectangle over a larger portion of the span except near the supports. What type of load does the beam supports. What type of load does the beam carry?**

(a) A uniformly distributed symmetrical load over a larger portion of the span except near the supports
(b) A concentrated load at mid span
(c) Two identical concentrated loads equidistant from the supports and close to mid-point of the beam.
(d) Two identical concentrated loads equidistant from the mid span and close to supports

**Ans. (d) :**




Two identical concentrated loads equidistant from the mid span and close to supports.

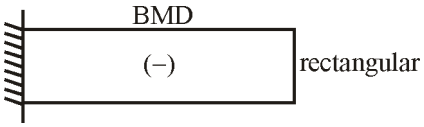
**55. Which one of the following is the correct statement?**
**If for a beam, dM/dx = 0 for its whole length, the beam is a cantilever**
(a) free from any load
(b) subjected to a concentrated load at its free end
(c) subjected to an end moment
(d) subjected to a UDL over its whole span

**Ans. (c) :** if for a beam $\frac{dM}{dx} = 0$

$\Rightarrow$ M = const (say k)

Then cantilever is subjected to an end moment with no point loads along its length.

$M_x = M$




**56. Which one of the following is the correct answer?**
**The point of contraflexure in a beam is a point on its length where**
(a) the shear force is zero
(b) the bending moment is maximum
(c) the bending moment changes its algebraic sign and is zero at the point
(d) the shear force changes its algebraic sign

**Ans. (c) :** Point of contraflexure in a beam is a point where bending moment changes its sign and is zero at that point.

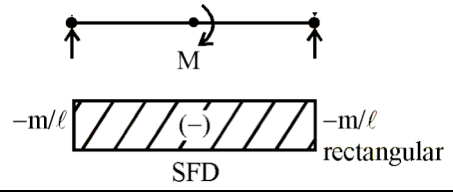
**57. For a cantilever $\frac{dM}{dx}$ = a constant for its whole length. What is the shape of the SF diagram for the beam?**
(a) Rectangle (b) Triangle
(c) A parabola (d) A hyperbola

**Ans. (a) :** $\frac{dM}{dx}$ = constant = k = shear force

M = kx + C,
$M \propto x$
BMD is a straight line
SFD is a constant line
$\therefore$ Shape of SFD is rectangular.

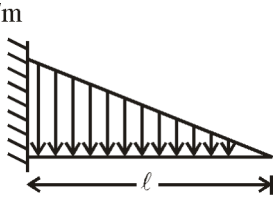
**58. SF diagram for a simply supported beam is a rectangle with its longer side equal to beam length. What type of load is acting on the beam?**
(a) Concentrated load at it mid span
(b) UDL over its whole span
(c) Concentrated load along with a couple at a point on beam length
(d) Couple at a point on the beam length

**Ans. (d) :**




**59. A cantilever beam of span/carries a uniformly varying load of zero intensity at the free end and w per metre length at the fixed end. What does the integration of the ordinate of the load diagram between at the limits of free and fixed ends of the beam give?**
(a) Bending moment at the fixed end
(b) Shear force at the fixed end
(c) Bending moment at the free end
(d) Shear force at the free end

**Ans. (b) :**




Shear force between any two points an the loaded beam is equal to total area of the loading diagram between these two points.

$(SF)_{\text{fixed end}} = \frac{1}{2} \times \ell \times w$

**60. Due to some point load anywhere on a fixed beam, the maximum free bending moment is M. The sum of fixed end moment is**

(a) M (b) 1.5 M
(c) 2.0 M (d) 3.0 M

**Ans. (a) :** Due to some point load any where on a fixed beam, the maximum free bending moment is M. The sum of fixed end moment is M.

**61. A cantilever beam of rectangular cross-section is 1 m deep and 0.6 m thick. If the beam were to be 0.6 m deep and 1 m thick, then the beam would**

(a) be weakend 0.5 time
(b) be weakend 0.6 time
(c) be strengthed 0.6 time
(d) have the same strength on the original beam because the cross-sectional area remains the same

**Ans. (b) :** Flexural strength of beam depends upon it's section modulus.

$$Z_I = \frac{\frac{1}{12}\times 0.6\times 1^3}{0.6/2} = 0.1\,m^3$$

$$Z_{II} = \frac{\frac{1}{12}\times 1\times 0.6^3}{1/2} = \frac{(0.6)^2}{6} = 0.6\times 0.1$$

$= 0.06\ m^3$

$$Z_{11} < Z_I = \frac{0.06}{0.1} = 0.6$$

Beam would be weakened by 0.6 times.

**62. A straight bimetallic strip of copper and steel is heated. It is free at ends. The strip will**

(a) Expand and remain straight
(b) Not expand but bend
(c) Expand and bend also
(d) First only

**Ans. (c) :** if heated and free to expand

since $\propto_c = 18 \times 10^{-6}$ /°c

$\propto_s = 11 \times 10^{-6}$ /°c

Copper expands more than steel and hence the binetamic strip will bend with outer fiber as copper.

**63. A pipe of external diameter 3 cm and internal diameter 2 cm and length 4m supported at its ends. If carries a point load of 65 N at its center-section, modulus of pipe will be**

(a) $\frac{65\pi}{64}cm^3$ (b) $\frac{65\pi}{32}cm^3$
(c) $\frac{65\pi}{96}cm^3$ (d) $\frac{65\pi}{124}cm^3$

**Ans. (c) :** External diameter = 3 cm = $D_0$
Internal diameter = 2 cm = $D_i$
Modulus of e/s of pipe

$$Z = \frac{I}{y_{max}}$$

$$I = \frac{\pi}{64}\left(D_0^4 - D_i^4\right)$$

$y_{max} = D^o/2$
(distance of extreme fiber from neutral axis)

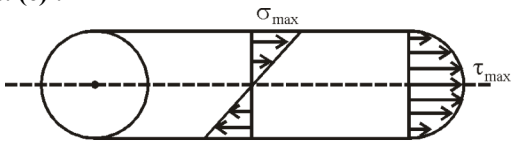
$$I = \frac{\frac{\pi}{64}\left[3^4 - 2^4\right]}{3/2}$$

$$z = \frac{65\pi}{96}$$

**64. In a loaded beam under bending**

(a) both the maximum normal and maximum shear stresses occur at the skin fibers.
(b) both the maximum normal and maximum shear stresses occur at the neutral axis
(c) the maximum normal stress occurs at the outer fibers
(d) the maximum normal stress occurs at the neutral axis while the maximum shear stress occurs at the fibers

**Ans. (c) :**



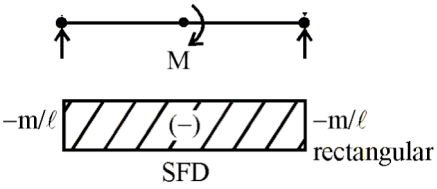

Bending stress distribution — Shear stress distribution

Normal stress occurs maximum at extreme fiber while maximum shear stress occurs at neutral axis.

**65. For the shear force to be uniform throughout the span of a simply supported beam, it should carry which one of the following loadings?**

(a) A concentrated load at mid span
(b) UDL over the entire span
(c) A couple anywhere within its span
(d) Two concentrated loads equal in magnitude and placed at equal distance from each support

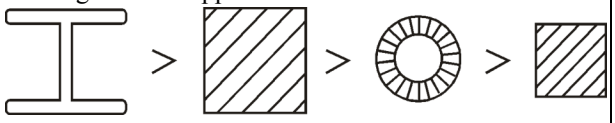
**Ans. (c) :**




**66. Which one of the following is the preferable cross-section of a beam for bending loads?**

(a) Circular (b) Annular circular
(c) Rectangular (d) I-section

**Ans. (d) :** For a given cross-sectional area material and bending moment applied.



'Z' value decrease → $M_R$ decreases →
Bending stress $(\sigma_b)_{max}$ increases
I section is preferred for bending applications.

**67. Three beams of 100 mm × 1200 mm section are made from RCC, aluminium and timber respectively. These three beams are loaded identically with similar supports, Maximum bending stress will occur in which beam?**

(a) Timber beam
(b) Aluminium beam
(c) RCC beam
(d) The three beams will have same maximum bending stress

**Ans. (d) :** Bending stress does not depend upon material.

**68. A beam has rectangular section 100 mm × 200 mm. If it is subjected to a maximum BM of 4 × $10^7$ Nmm, then the maximum bending stress developed would be**

(a) 30 N/mm$^2$ (b) 60 N/mm$^2$
(c) 90 N/mm$^2$ (d) 120 N/mm$^2$

**Ans. (b) :** Rectangular section = 100 mm × 200 mm

$$I = \frac{1}{12}\times 100\times 200^3$$

$$y_{max} = \frac{200}{2} = 100$$

$$Z = \frac{I}{y_{max}} = \frac{\frac{1}{2}\times 100\times 200^3}{100} = \frac{2}{3}\times 10^6 \text{ mm}^3$$

$M_R = \sigma_{max} \times Z$

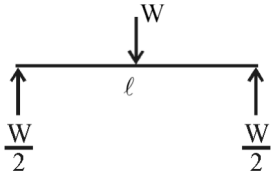
$$\sigma_{max} = \frac{M_R}{z} = \frac{4\times 10^7}{\frac{2}{3}\times 10^6} \text{ N–mm}$$

$\sigma_{max}$ = 60 N/mm$^2$

**69. A timber beam is simply supported at the ends and carries a concentrated load at mid span. The maximum longitudinal stress 'f' is 12 N/mm$^2$ and the maximum shear stress 'q' is 1.2 N/mm$^2$. The ratio of span to depth would be**

(a) 10 (b) 6
(c) 5 (d) 4

**Ans. (c) :**




Maximum shearing stress $q = \frac{3}{2}q_{avg}$

(for rectangular such)

$$q = \frac{3}{2}\left(\frac{W}{2}\right)\times\frac{1}{bd}$$

$$q = \frac{3W}{4bd} \qquad \text{———(1)}$$

Maximum bending stress $f = \frac{M}{Z}$

$$f = \frac{\frac{W}{4}\ell}{bd^2/6} = \frac{3W\ell}{2bd^2} \qquad \text{———(2)}$$

$$\frac{f}{q} = \frac{2\ell}{d}$$

$$\frac{\ell}{d} = \frac{1}{2}\times\frac{12}{1.2} = 5$$

**70. The moment of inertia of a given rectangular area is minimum about**

(a) its longer centroidal axis
(b) its polar axis
(c) its axis along the diagonal
(d) its shorter centroidal axis

**Ans. (d) :** $I_{xx} = \frac{1}{12}bd^3$

$$I_{yy} = \frac{1}{12}db^3$$

since b < d

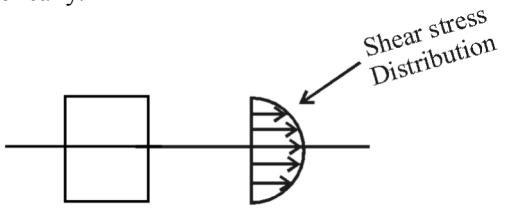
$\therefore I_{xx} > I_{yy}$

Hence MOI of a given rectangular section is minimum about shorter centroidal axis.

**71. For a beam rectangular section under bending the shear stress across the depth varies**

(a) Linearly (b) Exponentially
(c) Hyperbolically (d) Parabolically

**Ans. (d) :** for a beam of rectangular section under bending the shear stress across the depth varies parabolically.




Variation is paraboic

**72. In the cross-section of rectangular beam, what is the ratio of the average shear stress to the maximum shear stress?**

(a) 3/2 (b) 2/3
(c) 4/3 (d) 3/4

**Ans. (b) :** For a rectangular section

($q_{max}$) Maximum shear stress

$$= \frac{3}{2}\times \text{average shear stress}\left(q_{avg}\right)$$

$$q_{avg} = \frac{2}{3}q_{max}$$

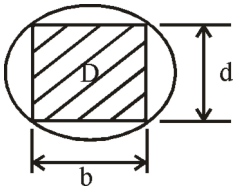
$$\frac{q_{avg}}{q_{max}} = \frac{2}{3}$$

**73. A beam of rectangular cross-section is to be cut from a circular beam of diameter D. What is the ratio of the depth of the beam to its width for maximum moment of resistance?**

(a) $\sqrt{3}$ (b) $\sqrt{2}$

(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{3}{\sqrt{2}}$

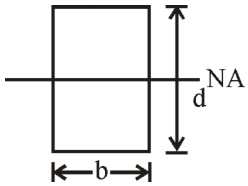
**Ans. (b) :**




$b^2 + d^2 = D^2$

$d^2 = D^2 - b^2$

for the strongest beam in bending its 'z' should be minimum.




$$\frac{dz}{db} = 0$$

$$z = \frac{bd^2}{6}$$

$$z = \frac{b\left(D^2 - b^2\right)}{6}$$

$$\frac{D^2 - 3b^2}{6} = 0 \Rightarrow b = \frac{D}{\sqrt{3}}$$

$b^2 + d^2 = D^2$

$$d^2 = D^2 - \frac{D^2}{3} \Rightarrow d = \sqrt{\frac{2}{3}D^2}$$

$d = b.\sqrt{2}$

$$\frac{\text{depth}}{\text{width}} = \sqrt{2}$$

**74. In a beam of solid circular cross-section, what is the ratio of maximum shear stress to the average shear stress?**

(a) 3/4 (b) 4/3

(c) 3/2 (d) 2/3

**Ans. (b) :** For circular cross-section

$$q_{max} = \frac{4}{3}q_{avg}$$

$$\frac{q_{max}}{q_{avg}} = \frac{4}{3}$$

**75. In case of a beam of I-section subjected to transverse shear force 'F', the maximum shear stress occurs at the**

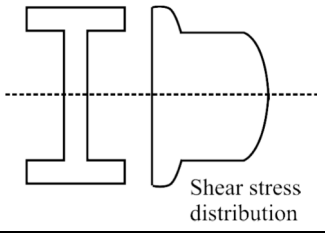
(a) top edge of the flange

(b) centre of the web

(c) junction of the flange and web

(d) bottom edge of the flange

**Ans. (b) :** Maximum shear stress occurs at centre of the web.




**76. A beam of uniform strength refers which the following?**

(a) A beam in which extreme fibre stresses are same at all cross-section along the length of the beam

(b) A beam in which the moment of inertia about the axis of bending is a constant at all cross-section of the beam

(c) A beam in which the distribution of bending stress across the depth of cross-sections of the beam

(d) A beam in which the bending stress is uniform at the maximum bending moment cross-section

**Ans. (a) :** Beams of uniform strength are those beams whose cross-section varies along length of beam, such that maximum bending stress at each cross-section reaches permissible stress and is constant through out the length of beam.

**77. A mild steel plate is subjected to a moment M each at its ends such that it bends into an arc of a circle of radium 10 m. The plate has width 60 mm and thickness 10 mm. $E = 2 \times 10^5$ N/mm$^2$. What is the maximum bending stress produced in the plate?**

(a) 100 MPa (b) 200 MPa

(c) 300 MPa (d) 400 MPa

**Ans. (a) :** Thickness of plats = 10 mm

Radius of curvature = 10 m

$E = 2 \times 10^5$ N/mm$^2$

Flexure formula $\frac{E}{R} = \frac{f}{y}$

$$f = \frac{E \times y}{R}$$

$$f = \frac{2 \times 10^5 \times \frac{10}{2}}{10 \times 1000} \text{ N/mm}^2$$

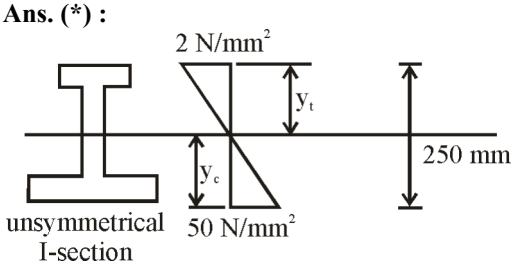
$f = 100$ N/mm$^2$ = 100 MPa

**78. A mild steel structural beam has cross-section which is an unsymmetrical I-section. The overall depth of the beam is 250 mm. The flange stresses at the top and bottom are 2 N/mm$^2$ and 50 N/mm$^2$ respectively. What is the depth of the neutral axis from the top of a beam?**

(a) 50 mm (b) 100 mm

(c) 150 mm (d) 200 mm

**Ans. (*) :**




$$\frac{\sigma_t}{\sigma_b} = \frac{y_t}{y_c}$$

$$\frac{y_t}{y_c} = \frac{2}{50} = \frac{1}{25}$$

$y_c = 25\ y_t$

Also,

$y_t + y_c = 250$ mm

$y_t + 25\ y_t = 250$ mm

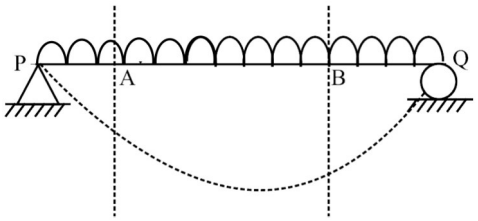
$$y_t = \frac{250}{26} = 9.61\text{mm}$$

**79. The ratio of area under the bending moment diagram to the flexural rigidity between two points along a beam gives the change in**

(a) deflection (b) slope

(c) shear force (d) bending moment

**Ans. (b)**




**Mohr's first theorem–**

$$\frac{d^2y}{dx^2} = \frac{M}{EI}$$

as $\frac{dy}{dx} = \theta$

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{M}{EI}$$

$$\frac{d\theta}{dx} = \frac{M}{EI}$$

$$\int_A^B d\theta = \int_A^B \frac{M}{EI} dx$$

$\theta_B - \theta_A$ = area under $\frac{M}{EI}$ diagram between A & B.

**Mohr's 2nd theorem–**

Deflection of any point A on the elastic curve w.r.t. tangent drawn at another point 'B' is equal to the moment of area under $\frac{M}{EI}$ diagram between point A & B about A.

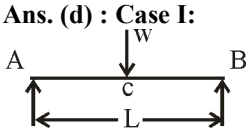
$$\left[t_{AB} = \bar{x}\int_A^B \frac{M}{EI} dx\right]$$

**80. In a simply supported beam AB of span L, the mid-point is C. In case–I, the beam is loaded by a concentrated load W. In case–2, the beam is subjected to a UDL of intensity W such wL = W. The ratio of central deflection in case–1 to that in case–2 is**

(a) $\frac{5}{3}$ (b) $\frac{3}{5}$

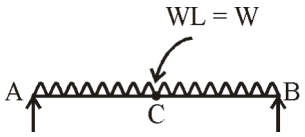
(c) $\frac{5}{8}$ (d) $\frac{8}{5}$

**Ans. (d) : Case I:**




deflection at c $\Rightarrow \delta_1 = \frac{WL^3}{48EI}$

**Case II:**




Deflection at C $\Rightarrow \delta_2 = \frac{5}{384EI} W\ell^4$

Now,

$$\frac{\delta_1}{\delta_2} = \frac{WL^3}{48EI} \times \frac{384EI}{5WL^4}$$

$$\frac{\delta_1}{\delta_2} = \frac{384}{48 \times 5} \times \frac{WL^3}{5WL^4}$$

$$\frac{\delta_1}{\delta_2} = 8:5$$

**81. A simply supported beam AB of span L is subjected to a concentrated load W at the centre C of the span. According to Mohr's moment area method, which of the following gives the deflection under the load?**

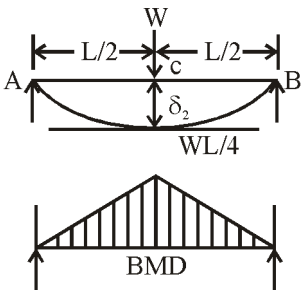
(a) Moment of the area of M/EI diagram between A and C taken about C

(b) Moment of the area of M/EI diagram between A and B taken about B

(c) Moment of the area of M/EI diagram between A and B taken about A

(d) Moment of the area of M/EI diagram between A and C taken about A

**Ans. (d) :**




Moment of the area of M/EI diagram between A and C taken about A.

**82. Two steel shafts, one solid of diameter D and other hollow of outside diameter D and inside diameter D/2 are twisted to the same angle of twist per unit length. Ratio of maximum shear stress in solid shaft to that in the hollow shaft is**

(a) $\frac{4}{9}$ (b) $\frac{8}{7}$

(c) $\frac{16}{15}$ (d) 1

**Ans. (d) :** Torsion formula

$$\frac{\tau}{r_{max}} = \frac{G\theta}{\ell}$$

for solid shaft $r_{max} = D/2$

for hollow shaft $r_{max} = D/2$

Since both are made of steel

$\therefore$ G remains same

Given angle of twist per unit length is ame

$\therefore \frac{\theta}{\ell}$ for both shafts is same

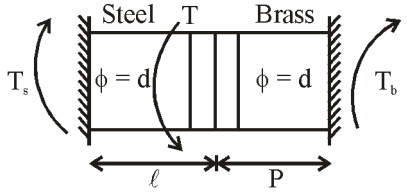
$$\frac{\delta_s}{\delta_h} = \frac{(r_{max})_s}{(r_{max})_h} = \frac{D/2}{D/2} = 1$$

**83. Steel shaft and brass shaft of same length and diameter are connected by a flange coupling. The assembly is rigidly held at its ends and is twisted by a torque through the coupling. Modulus of rigidity of steel is twice that of brass. If torque of the steel shaft is 500 Nm, then the value of the torque in brass shaft will be**

(a) 250 Nm (b) 354 Nm

(c) 500 Nm (d) 708 Nm

**Ans. (a) :**




$G_S = 2\ G_B$

$T_S = 500$ NM

$T_S + T_b = T$ ——— (1)

Also angle of twist at coupling is equal for both shafts

$$\frac{T_s \ell}{G_s j} = \frac{T_b \ell}{G_b J}$$

($\ell$, are equal for both shafts)

$$\frac{T_s}{T_b} = \frac{G_s}{G_b} = 2 \quad \text{——— (2)}$$

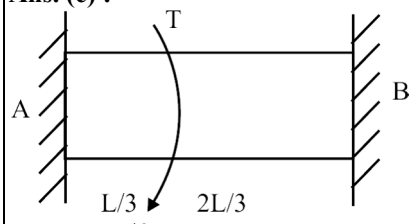
$$T_b = \frac{500}{2} = 250 \text{ NM}$$

**84. A hollow shaft of length L is fixed at its both ends. It is subjected to torque T at a distance of L/3 from one end. What is the reaction torque at the other end of the shaft?**

(a) $\frac{2T}{3}$ (b) $\frac{T}{2}$

(c) $\frac{T}{3}$ (d) $\frac{T}{4}$

**Ans. (c) :**



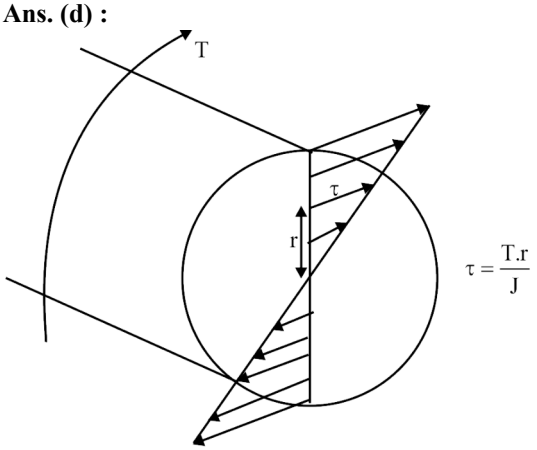

$$T_B = \frac{T \times L/3}{L} = \frac{T}{3}$$

**85. Which one of the following is true for torsional shear stress at the axis of a circular shaft?**

(a) Minimum (b) Maximum

(c) Negative (d) Zero

**Ans. (d) :**




So, Torsional shear stress at centre of shaft is zero.

**86. A circular shaft of length 'L' a uniform cross-sectional area 'A' and modulus of rigidity 'G; is subjected to a twisting moment that produces maximum shear stress ' $\tau$' in the shaft. Strain energy in the shaft is given by the expression AL/kG, where k is equal to**

(a) 2 (b) 4

(c) 8 (d) 16

**Ans. (b) :** Strain energy stored in a shaft due to torque

$$U = \left(\frac{\tau^2}{4G}\right) \text{vol.}$$

$$U = \frac{\tau^2}{4G} \times A \times L$$

$$U = \frac{AL}{kG}$$

Then k = 4

**87. When subjected to a torque, a circular shaft undergoes a twist of $1^0$ in a length of 1200 mm, and the maximum shear stress induced is 80 N/mm$^2$. The modulus of rigidity of the material of the shaft is $0.8 \times 10^5$ N/mm$^2$. What is the radius of the shaft?**

(a) $90/\pi$mm (b) $108/\pi$mm
(c) $180/\pi$mm (d) $216/\pi$mm

**Ans. (d) :** $\tau = 80$ N/mm$^2$

$$\frac{\theta}{\ell} = \frac{10}{1200} = \frac{1^0}{1200} \times \frac{\pi}{180} \quad \text{(in radian)}$$

$G = 0.8 \times 10^5$ N/mm$^2$

Torsion formula

$$\frac{\tau}{r_{max}} = \frac{G\theta}{\ell}$$

$$\frac{80}{r_{max}} = \frac{0.8\times10^5 \times 1^0 \times \pi}{1200\times180}$$

$$r_{max} = \frac{80\times1200\times180}{0.8\times10^5\times\pi} = \frac{216}{\pi}$$

**88. Power is transmitted through a shaft, rotating at 2.5 Hz (150 rpm). The mean torque on the shaft is $20 \times 10^3$ Nm. What magnitude of power in kW is transmitted by the shaft?**
(a) $50\pi$ (b) $120\pi$
(c) $100\pi$ (d) $150\pi$

**Ans. (c) :** N = 150 rpm

$T = 20 \times 10^3$ Nm

$$P = \frac{2\pi NT}{60\times1000}\text{kW}$$

$$= \frac{2\pi\times150\times20\times10^3}{60\times1000}$$

$$= \frac{100\pi\times10^3}{1000} = 100\pi \text{ kW}$$

**89. Which one of the following statements is correct?**
(a) Euler's formula holds good only for short columns
(b) A short column is one which has the ratio of its length to least radius of gyration greater than 100
(c) A column with both ends fixed has minimum equivalent length or effective length
(d) The equivalent length of column with one end fixed and other end hinged is half of its actual length

**Ans. (c) :** Euler's formula holds good for long columns only.
- Slenderness ratio less than 12 short column and greater than 12 long column.

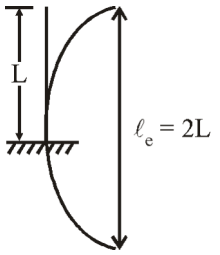
| end condition | One end fixed other free | Both end fixed | One end fixed other hinged | Both end hinged |
|---|---|---|---|---|
| effective length | $2\ell$ | $\ell/2$ | $\ell/\sqrt{2}$ | $\ell$ |

**90. What is the cause of failure of a short MS strut under an axial load?**
(a) Fracture stress (b) Shear stress
(c) Buckling (d) Yielding

**Ans. (b) :** The cause of failure of a short MS strut under an axial lead is shear stress.

**91. Which one of the following columns has effective length twice the value of actual length?**
(a) Hinged-Hinged column
(b) Fixed-Fixed column
(c) Fixed-Hinged column
(d) Fixed-Free column

**Ans. (d) :** fixed end column




**92. Critical Euler buckling load for a long column of diameter D was evaluated as P. If the diameter of the section is reduced to D/2, what is the load carrying capacity of the modified column ?**
(a) P/2 (b) P/4
(c) P/8 (d) P/16

**Ans. (d) :** $P_{cr} = \frac{\pi^2 EI}{\ell_e^2}$

$P_{cr} \propto I$

$P_{cr} \propto D^4$

$$\frac{P_{cr1}}{P_{cr2}} = \frac{D^4}{\left(\frac{D}{2}\right)^4}$$

$$P_{cr2} = \frac{P}{16}$$

**93. The axial load which just produces the condition of elastic instability in a column is**
(a) Rankine load (b) Euler load
(c) Yield load (d) Crushing load

**Ans. (b) :** The axial load which just produces condition of elastic instability in a column is called Euler load or Buckling load on crippling load.

**94. If the stress on the cross-section of a circular short column of diameter D is to be wholly compressive, the load should be applied within a concentric circle of diameter.**
(a) D/2 (b) D/8
(c) D/4 (d) D/6

**Ans. (c) :** For short columns for no tension condition
Direct stress + Bending stress $\le 0$ (compression)
To avoid any tensile stresses in the column load should be applied with is a concentric circle of diameter D/4.

$$I = \frac{\pi}{64}D^4$$

$y_{max} = D/2$

$z = \frac{\pi}{32}D^3$

$\frac{-P}{\frac{\pi}{4}D^2} + \frac{P \times e}{\frac{\pi}{32}D^3} \le 0$

$e \le D/8$

Hence kern of the section $= \frac{D}{8} + \frac{D}{8} = \frac{D}{4}$

**95. Column $C_1$, has both the ends hinged while the column $C_2$ has one end hinged and other end fixed. What is the ratio of the critical load for $C_1$ to that of $C_2$ according to the Euler's formula?**

(a) 2 (b) 1/2

(c) 4 (d) 1/4

**Ans. (b) :**

$P_{cr} \propto \frac{1}{\ell_e^2}$

$\frac{(P_{cr})_{\text{hinged-hinged}}}{(P_{cr})_{\text{hinge-fixed}}} = \frac{\left(\frac{L}{\sqrt{2}}\right)^2}{(L)^2} = 1/2$

**96. In an axially loaded compressive member with a circular cross-section of radius, r, what is the radius of the core section which is proof against tensile stress?**

(a) r/2 (b) r/3

(c) r/4 (d) r/6

**Ans. (c) :** Core section which is proof against tensile stresses in case of circular reaction is of diameter $\frac{D}{4}$

$\therefore$ Core section radius will be $\frac{D}{8}$

if radius of circular section is 'r' then radius of core section will be

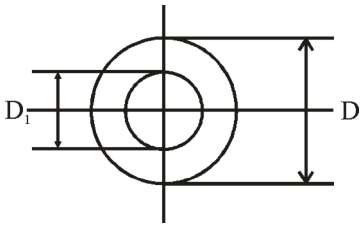
$s'r' = \frac{2r}{8} = \frac{r}{4}$

**97. A short column of external diameter D and internal diameter $D_1$ is subjected to an eccentric load P at an eccentricity 'e' thereby causing tensile stress at the extreme fibre. What is the magnitude of eccentricity?**

(a) $\left(D_1 + D_1^2\right)/8\pi D$ (b) $(D^2 + D_1^2)/8D$

(c) $(D^2 - D_1^2)/8D$ (d) $(D^2 - D_1^2)/8D$

**Ans. (b) :**




for short columns eccentric load P will cause no tension.

if $\frac{-P}{A} + \frac{Pe}{Z} \le 0$

$A = \frac{\pi}{4}\left[D^2 - D_1^2\right]$

$I = \frac{\pi}{64}\left[D^4 - D_1^4\right]$

$y_{max} = \frac{D}{2}$

$\therefore \frac{-P}{\frac{\pi}{4}\left[D^2 - D_1^2\right]} + \frac{Pe}{\frac{\frac{\pi}{64}\left(D^4 - D_1^4\right)}{D/2}} \le 0$

$e = \frac{D^2 + D_1^2}{8D}$

**98. A column of length 4 m, an area of cross-section 2000 $mm^2$. moments of inertia, $I_{xx}$ = 720 cm4, $I_{yy}$ = 80 $cm^4$, is subjected to a buckling load. Both the ends of the column are fixed. What is the slenderness ratio of the column?**

(a) 200 (b) 120

(c) 100 (d) 80

**Ans. (c) :** Length of column = L = 4 m area of cross – section = A = 2000 $mm^2$

$I_{xx} = 720\ cm^4$

$I_{yy} = 720\ cm^4$

$\ell_e$ fixed-fixed $= \frac{L}{2} = 2m$

Slenderness ratio = $\lambda = \frac{\ell_e}{k_{min}}$

$K_{min} = \sqrt{\frac{I_{min}}{A}}$

$= \sqrt{\frac{80\ cm^4}{2000 \times 10^{-2}\ cm^2}}$

$= \sqrt{\frac{100}{25}} = 2\,cm$

$\therefore \lambda = \frac{\ell_e}{k_{min}} = \frac{200\,cm}{2\,cm} = 100$

**99. The diagram showing the variation of axial load along the span is called**

(a) shear force diagram

(b) bending moment diagram

(c) thrust diagram

(d) influence line diagram

**Ans. (c) :** Diagram showing variation of axial load along the span is called thrust diagram.

**100. The variation of the bending moment due to the moving load on a fixed ended beam occurs**

(a) linear (b) parabolic

(c) cubic (d) constant

**Ans. (b) :** The variation of the Bending moment due to a moving load on a fixed ended beam occurs parabolic.

**101. The relationship between the radius of curvature R, bending moment M and flexural rigidity EI is given by**

(a) $R = \frac{M}{EI}$ (b) $M = \frac{EI}{R}$

(c) $EI = \frac{R}{M}$ (d) $E = \frac{MI}{R}$

**Ans. (b) :** Bending equaiton, $\frac{M}{I} = \frac{E}{R}$

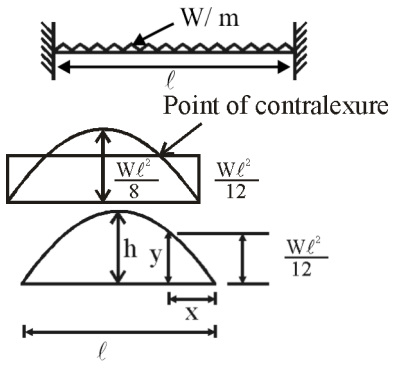
$\Rightarrow M = \frac{EI}{R}$

**102. A prismatic beam of length L is fixed at both ends carries a uniformly distributed load. The distance of points of contra flexure from either end is**

(a) 0.207 L (b) 0.211 L

(c) 0.277 L (d) 0.25 L

**Ans. (b) :**




$y = \frac{W\ell^2}{12}$

$y = \frac{4h}{\ell^2} x(\ell - x)$

$\frac{W\ell^2}{12} = \frac{4}{\ell^2}\left(\frac{W\ell^2}{8}\right) x(\ell - x)$

$\frac{\ell^2}{6} = x(\ell - x)$

$x^2 - \ell x + \frac{\ell^2}{6} = 0$

$x = 0.211\ \ell$

**103. If a Mohr circle is drawn for a fluid element inside a fluid body at rest, it would be**

(a) a circle not touching the origin

(b) a circle touching the origin

(c) a point on the normal stress axis

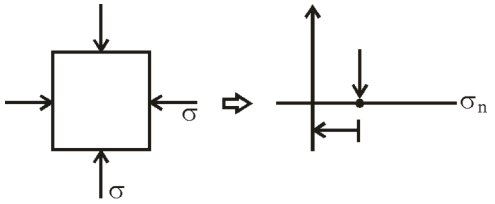
(d) a point on the shear stress axis

**Ans. (c) :** Fluid element inside a fluid body at rest experience hydrostatic pressure.

Radius of mohr circle, $R = \frac{\sigma_1 - \sigma_2}{2} = \tau_{max}$

$\tau_{max} = \frac{\sigma - \sigma}{2} = 0$

It means mohr circle reduce at a point lies on the stress axis.




**104. Local atmospheric pressure is measured by**

(a) Hydrometer (b) Barometer

(c) Hygrometer (d) Altmeter

**Ans. (b) :** local atmospheric pressure is measured by 'Barometer'.

- The atmospheric air exerts a normal pressure upon all surface with which it is in contact and it is known as atmospheric pressure. The atmospheric pressure is also known as barometric pressure.
- The atmospheric pressure at sea level (above absolute zero) is called standard atmospheric pressure.

**105. A barometer at a given location**

(a) shows the local atmospheric pressure which is invariant with time

(b) always shows the local atmospheric pressure which may change with time

(c) shows the standard atmospheric pressure, if it is of aneroid type

(d) shows the local temperature if it is of mercury column type

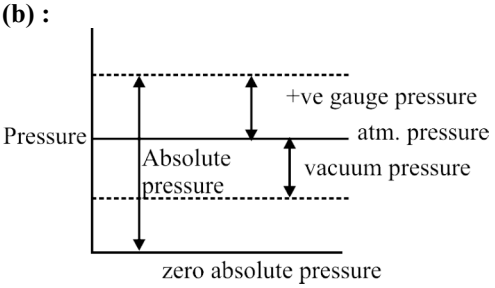
**Ans. (b) :** Barometer stress local Atmospheric pressure Barometer always shows the local atmospheric pressure which may change with time.

**106. The standard atmospheric pressure is 760 mm of mercury. At a certain location the barometer reads 710 mm of mercury. At this place an absolute pressure of 360 mm of mercury corresponds to a gauge pressure, in mm of mercury**

(a) 400 mm of vacuum (b) 350 mm of vacuum

(c) 360 mm of vacuum (d) 710 mm

**Ans. (b) :**




Vacuum pressure = atm. pressure – absolute pressure

= 710 – 360 =350 mm.

**107. An inclined manometer contains a liquid of relative density 0.8 and has an inclination of 30° to the horizontal. For a certain pressure the column length was 10 cm. If there is an uncertainty of 1° in the measurement of the angle of inclination, the calculated pressure would have an uncertainty of**

(a) 1% (b) 0.28%
(c) 1.75% (d) 3.33%

**Ans. (d) :** $P = \rho gL \sin\theta$

$P \propto \rho gL \sin\theta$ ————(1)

$dP = k.\cos\theta\, d\theta$ ————(2)

(2) ÷ (1) for small Angles

$$\frac{dP}{P} = \frac{d\theta}{\theta} = \frac{1^\circ}{30^\circ} \times 100\% = 3.33\%$$

**108. For a submerged plane in a liquid, the resultant hydrostatic force F on one side of the plane is related to area A, censorial depth h, depth of the centre of pressure $h_{cp}$ and depth of bottom edge $h_b$ as**

**F = (where γ is specific weight)**

(a) $\gamma A h_{cp}$ (b) $\gamma A h$

(c) $\gamma A_b$ (d) $Ah/\gamma$

**Ans. (b) :** Hydrostatic force F on one side of plane area submerged = $\rho ghA = rhA$

**109. For an inclined plane submerged in a liquid, the centre of fluid pressure on one side of the plane will be**

(a) above the top edge of the area
(b) vertically below the centre of gravity
(c) below the centre of gravity
(d) in the same horizontal plane as the centre of gravity

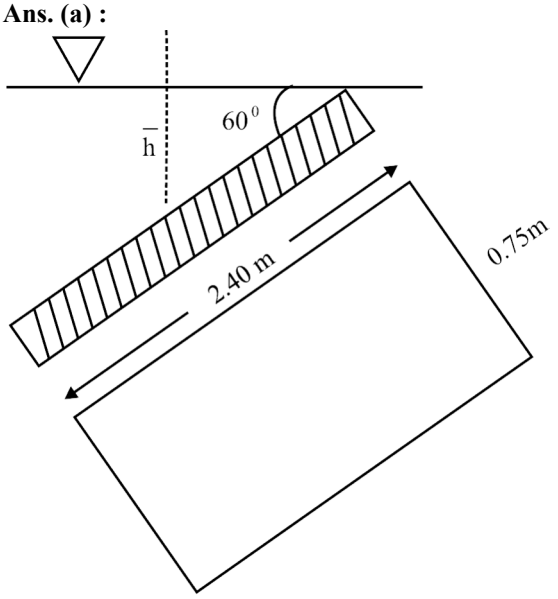
**Ans. (c) :** Centre of pressure $(h^*) = \bar{h} + \frac{I_G \sin^2\theta}{\bar{h}.A}$

h* is always below centroid of submerged surface.

**110. A rectangular plate 0.75 m × 2.4 m is immersed in a liquid of relative density 0.85 with its 0.75 m side horizontal and just at the water surface. If the plane of the plate makes an angle of 60º with the horizontal, the pressure force on one side of the plate, in kN, is**

(a) 15.6 (b) 7.8
(c) 24.0 (d) 18.0

**Ans. (a) :**




$$\bar{h} = \frac{2.4}{2}\sin 60$$

Total pressure force, $F = \rho.g.\bar{h}.A$

$$= (0.85 \times 1000)(9.81)\left(\frac{2.4 \times \sin 60}{2}\right)(0.75 \times 2.4)$$

= 15598.1 N = 15.6 kN

**111. A rectangular plate 30 cm × 50 cm is immersed vertically, in water with its longer side vertical. The total force on one side of the plate is estimated as 17.6 kN. If the plate is turned in the vertical plane at its centre of gravity by 90º and if all other factors remain the same, the total force on one side of the plane would now be**

(a) 8.8 kN (b) 15.6 kN
(c) 17.6 kN (d) 19.6 kN

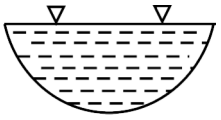
**Ans. (c) :** $\bar{h}$ is the distance from free surface to centroid of the submerged surface. When $\bar{h}$ is changed, hydrostatic free also not changed by tilting plane 90º, about centroidal line, $\bar{h}$ is remains same.

**112. A hollow hemispherical object of diameter D was immersed in water with its plane surface coinciding with the free surface where γ is the specific weight. The vertical component of force on the curved surface is given by $F_V$ =**

(a) $\frac{3}{8}\gamma\pi D^3$ (b) $\frac{1}{12}\gamma\pi D^3$

(c) $\frac{1}{24}\gamma\pi D^3$ (d) zero

**Ans. (b) :** Weight of water component occupied in Hollow semispherical



$$V = \frac{2}{3}\pi r^3$$

$$= \frac{2}{3}\pi r^3 \gamma$$

$$= \frac{2}{3}\pi \frac{D^3}{8}\gamma$$

$$= \frac{1}{12}\gamma\pi D^3$$

**113. A 2.0 m diameter penstock pipe carries water under a pressure head of 100 m. If the wall thickness is 7.5 mm, the tensile stress in the pipe wall, in MPa, is**

(a) 65.3 (b) 130.5
(c) 231.0 (d) 1305.0

**Ans. (b) :** D = 2 m, H = 100 m, t = 7.5 mm = 0.0075 m

$P = \rho gh$ (N/m²)

$$\sigma_t = \frac{PD}{2t} = \frac{(1000 \times 9.81 \times 100) \times 2}{2 \times 0.0075}$$

$= 130.8 \times 10^6$ N/m²

= 130.8 MPa

**114. A liquid in a circular container is given a rigid body rotation about the axis of the cylinder. The piezometric line in a cross section is**

(a) a horizontal line (b) a circular arc
(c) a parabola (d) a vertical line

**Ans. (b) :** Constant pressure surface in forced vortex is a paraboloid of revolution. A circle drawn on the surface such that its centre is an the axis of paraboloid has a constant elevation. As the pressure is already same on the parabolic surface the pizometric head is constant along the circle.

**115. An open cylindrical tank with its axis vertical is 0.89 m high and is 0.8 m in diameter. It is filled with an oil of density 800 kg/m$^3$ and is rotated at 120 rpm about the axis of the cylinder. The gauge pressure at the centre of the bottom of the tank is**

(a) 0.49 m of oil (b) zero
(c) 1.29 m of oil (d) 6.27 mPa

**Ans. (b) :** $Z_{max} = \frac{\omega^2 R^2}{2g} = \left(\frac{2\pi \times 120}{60}\right)^2 \times \frac{0.4^2}{2 \times 9.81}$

i.e. $Z_{max} = 1.29$ m $> 0.89$ m

$\therefore$ surface at the centre of base exposed to atmosphere.

$\therefore \rho_{gauge} = 0$

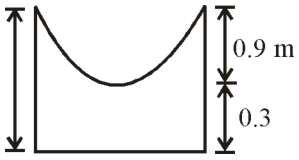
**116. An open circular cylinder 1.2 m high is filled with a liquid to its top. The liquid is given a rigid body rotation about the axis of the cylinder and the pressure at the centre of the bottom is found to be 0.3 m of liquid. The ratio of the volume of liquid spilled out of the cylinder to the original volume is**

(a) 3/8 (b) 3/4
(c) 1/2 (d) 1/4

**Ans. (a) :** $V_{spilled} = V_{paraboloid}$

$\frac{1}{2}\pi R^2 \times 0.9$

$V_{original} = \pi R^2 \times 1.2$




$$\frac{V_{spilled}}{V_{original}} = \frac{\frac{1}{2} \times \pi \times R^2 \times 0.9}{\pi R^2 \times 1.2} = \frac{9}{24} = \frac{3}{8}$$

**117. A 20 cm diameter open cylindrical container contains kerosene (relative density = 0.80) to a height of 20 cm. It is rotation about a vertical axis of the cylinder. If the bottom of the cylinder at the axis is just exposed, the speed of rotation is**

(a) 94.6 rpm (b) 267.5 rpm
(c) 133.8 rpm (d) 535.0 rpm

**Ans. (b) :** $Z_{max} = \frac{\omega^2 R^2}{2g}$

Given, $y/2 = 0.2$m

$\therefore$ y = 0.4 m

$0.4 = \frac{\omega^2 \times 0.1^2}{2 \times 9.81}$

$\omega = 28.01$ rad/sec

$N = \frac{\omega \times 60^o}{2\pi}$

$= 267.5$ rpm

**118. The velocity in a cross section is non-uniform. In one—dimensional method of analysis, the kinetic energy of fluid per unit weight is**

(a) $V^2/2g$ (b) $\alpha V^2/2g$
(c) $\beta V^2/2g$ (d) $p/y$

**Ans. (b) :** If velocity at a cross-section not uniform the actual kinetic energy is different from kinetic energy calculated based on average velocity.

Hence kinetic correction factor (usually denoted by $\alpha$) must be used to get actual kinetic energy.

i.e. actual K.E. $= \alpha \frac{V^2}{2g}$

**119. Indicate the incorrect statement:**
**For Bernoulli equation as $p/\gamma + V^2/2g + Z$ = constant to be applicable (where $\gamma$ is specific weight)**

(a) the flow must be steady
(b) the fluid must be an ideal gas
(c) the flow must be irrotational
(d) the fluid must be incompressible

**Ans. (b) :** The fluid must be ideal (frictionless) but not necessarily ideal gas. Hence second statement is incorrect.

- Assumption of Bernoulli equation –

(i) The liquid is ideal and incompressible.
(ii) Flow is steady and continuous
(iii) Flow is along stream line

**120. In a flow of a real fluid with no addition of energy**

(a) the energy line will be horizontal or sloping upward in the direction of the flow
(b) the energy line can never be horizontal or sloping upward in the direction of the flow
(c) the piezometric line can never be horizontal or sloping downward in the direction of the flow
(d) the centre line of the pipe can never be above the energy line

**Ans. (b) :** Energy line can never be horizontal or sloping upward because energy always decreases in the direction of flow due to friction. The piezometric line can increase in the direction of flow if velocity in diffuser velocity decreases, pressure increases hence piezometric line can rise in the direction of flow.

Pipe centerline can be above energy line for example in siphon. In such a case pressure for the pipe portion above energy line will be below atmospheric.

**121. The difference between the total head line and the hydraulic grade line represents**

(a) the velocity head
(b) the piezometric head
(c) the pressure head
(d) the elevation head

**Ans. (a) :** The total energy line represents variation of total energy $\left(\frac{P}{\rho g}+\frac{V^2}{2g}+z\right)$ and hydraulic grade line represents variation of piezometric head $\left(\frac{P}{\rho g}+z\right)$ in the direction of flow. Hence the difference between the two line is kinetic energy head $\left(\frac{V^2}{2g}\right)$.

**122. The kinetic energy correction factor $\alpha$ is defined as $\alpha$ =**

(a) $\frac{1}{AV^3}\int v^3 dA$ (b) $\frac{1}{A}\int v^3 dA$
(c) $\frac{1}{AV^2}\int v dA$ (d) $\frac{1}{AV^2}\int v^3 dA$

**Ans. (a) :** $\alpha = \frac{\text{actual K.E.}}{\text{K.E. based on avg. velocity}}$

$= \frac{\int v^3 dA}{AV^3}$

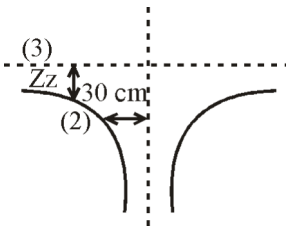
**123. At a distance of 10 cm from the axis of a whirlpool in a ideal liquid, the velocity is 5 m/s. At a radius of 30 cm the depression of the free surface below the surface of the liquid at a very large distance is**

(a) 7.98 cm (b) 3.33 cm
(c) 14.16 cm (d) 21.37 cm

**Ans. (c) :** Assuming the voetex as s free vortex

i.e. $V = \frac{k}{r}$

$\frac{V_2}{V_1} = \frac{r_1}{r_2}$




$\frac{V_2}{5} = \frac{10}{30}$

$V_2 = \frac{5}{3}$ m/s

As the free vortex is an rotation vortex flow the Bernoulli's equation can be applied between any two points. Considering undisturbed free surface as a reference level application of Bernoulli's equation between the given point and far away point from the centre gives.

$\frac{P_2}{\rho g}+\frac{v_2^2}{2g}+z_2 = \frac{P_3}{\rho g}+\frac{v_3^2}{2g}+z_3$

$\frac{P_{atm}}{\rho g}+\frac{v_2^2}{2g}+z_2 = \frac{P_{atm}}{\rho g}+0+z_3$

$\because Z_3 - Z_2 = \frac{v_2^2}{2g} = \frac{\left(\frac{5}{3}\right)^2}{2\times 9.81}$

$= 0.1416$ m $= 14.16$ cm

**124. A 20 cm diameter orifice discharging from a tank issues out a jet of 15.75 cm diameter at vena contracta. The coefficient of contraction is**

(a) 0.520 (b) 0.620
(c) 0.790 (d) 0.887

**Ans. (b) :** Coefficient contraction ($C_C$)

$= \frac{\text{Area Vena Contracta}}{\text{Area of orifice}}$

$= \frac{\frac{\pi}{4}\times 15.75^2}{\frac{\pi}{4}\times 20^2}$

$= 0.62$

**125. An orifice is discharging under a head of 1.25 m of water. A pitot tube kept at its centre line at the vena contracta indicates a head of 1.20 m of water. The coefficient of velocity of the orifice is**

(a) 0.990 (b) 0.980
(c) 0.965 (d) 0.960

**Ans. (b) :** $V = C_v\sqrt{2gH}$

Where, H = head at orifice

$V = \sqrt{2gh}$

Where h = head measured by pitot tube

$C_v = \sqrt{\frac{h}{H}} = \sqrt{\frac{1.2}{1.25}} = 0.98$

**126. Identify the incorrect statement:**
**A flow nozzle**

(a) has a contraction coefficient of unity
(b) is less costly than a venturimeter
(c) is more efficient than an orifice meter
(d) has overall losses much smaller than in a venturimeter

**Ans. (d) :** Losses in flow nozzle (nozzle meter) are more as compared to venturimeter but less as compared orifice meter.

**127. A venturimeter has a $C_d$ = 0.95. For a differential head of 2.8 m across the inlet and the throat, the loss of head between the inlet and throat is**

(a) 0.273 m (b) 0.140 m
(c) 0.302 m (d) 0.95 m

**Ans. (a):** $c_d = \sqrt{\frac{H - H_L}{H}}$

$0.95 = \sqrt{\frac{2.8 - H_L}{2.8}}$

$H_L = 0.273$ m

**128. If any venturimeter carrying a liquid at sufficiently high discharges, cavitation conditions can occur at the throat. This occurs due to the pressure at the throat reaching a critical value controlled by**

(a) the viscosity of the fluid
(b) eddy viscosity of the flow
(c) volumetric bulk modulus of the fluid
(d) surface tension of the fluid

**Ans. (d) :** In cavitations happens when pressure inside flow falls below vapour pressure. The vapour pressure for particular fluid depends only on temperature of the fluid and surface tension of the fluid.

**129. The stagnation pressure in front of an object in a fluid flow is equal to**

(a) static pressure
(b) dynamic pressure
(c) sum of the static and dynamic
(d) Piezometric head

**Ans. (c) :** The pressure on the front of an object rises as the object moves through a fluid. This pressure is called the stagnation pressure, and is equal to the pressure at the stagnation point.

- The different between the stagnation pressure and the static pressure is called dynamic pressure.
- Stagnation pressure = Dynamic pressure + Static pressure.

**130. The pitot-static tube measures**

(a) the dynamic pressure
(b) the static pressure
(c) the total head
(d) the difference in static and dynamic pressure

**Ans. (a) :** The pitot static tube measures difference between stagnation pressure and static pressure. This difference is equal to dynamic pressure.

**131. A pitot-static tube indicates a differential head of 0.75 m of water between its two openings when inserted in a stream of water. If the coefficient of the tube is 0.99, the velocity in m/s, at the location of the tube is**

(a) 4.43 (b) 0.78
(c) 3.84 (d) 3.80

**Ans. (d) :** $v = C_V\sqrt{2gh}$

$= 0.99\sqrt{2 \times 9.81 \times 0.75}$

$= 3.8$ m/s

**132. In a triangular notch there is an error of 4% in observing the head. The error in the computed discharge is**

(a) 4% (b) 10%
(c) 6% (d) 2.5%

**Ans. (b) :** for Triangular notch

$$Q = \frac{8}{15} C_d\sqrt{2g} \tan\frac{\theta}{2} H^{5/2}$$

Assumption only H to be the variable

$$dQ = \frac{8}{15} C_d\sqrt{2g} \tan\frac{\theta}{2} \times \frac{5}{2} H^{5/2} dH$$

$$\frac{dQ}{Q} = \frac{dH}{H} \times \frac{5}{2}$$

$= 0.04 \times 2.5 = 0.1$

i.e. $\frac{dQ}{Q} \times 100 = 10\%$

**133. In a 90º triangular notch, for a given head, the error in the estimated discharge due to a 2% error in the measurement of the vertex angle is**

(a) $\pi\%$ (b) 5.0%
(c) 3.0% (d) $\pi/2\%$

**Ans. (a) :** For triangular notch

$$Q = \frac{8}{15} C_d\sqrt{2g} \tan\frac{\theta}{2} H^{5/2}$$

Assumption only $\theta$ to be the variable-

$$dQ = \frac{8}{15} C_d\sqrt{2g} \sec^2\frac{\theta}{2} \times \frac{d\theta}{2} \times H^{5/2}$$

$$\text{i.e. } \frac{dQ}{Q} = \frac{\text{Sec}^2\frac{\theta}{2}}{\tan\frac{\theta}{2}} \times \frac{d\theta}{2}$$

$$= \frac{1}{2}\left(\frac{\text{Sec}^2\frac{\theta}{2}}{\tan\frac{\theta}{2}}\right) \times \frac{d\theta}{\theta} \times \theta$$

taking $\theta = \frac{\pi}{2}$

$$\frac{dQ}{Q} = \frac{1}{2} \times \frac{\sec^2\left(\frac{\pi}{4}\right)}{\tan\left(\frac{\pi}{4}\right)} \times 0.02 \times \frac{\pi}{2} = 0.01\pi$$

$$\frac{dQ}{Q} \times 100 = \pi\%$$

**134. The discharge over a 90º V-notch is written as $Q = 1.37\ H^{5/2}$ where Q is in $m^3$/sec and H is in metres. The $C_d$ of this notch is**

(a) 0.611 (b) 0.580
(c) 0.464 (d) 0.710

**Ans. (b) :** for Triangular notch

$$Q = \frac{8}{15} C_d\sqrt{2g} \tan\frac{\theta}{2} H^{5/2}$$

But, $Q = 1.37\, H^{5/2}$

$$\therefore \quad \frac{8}{15} C_d\sqrt{2g} \tan\frac{\theta}{2} = 1.37$$

$$\text{i.e. } \frac{8}{15} \times C_d \times \sqrt{2 \times 9.81} \times \tan\frac{\pi}{4} = 1.37$$

$C_d = 0.58$

**135. A Cipolletti weir has a side slope of**

(a) 1 vertical : 4 horizontal
(b) 1 vertical : 2 horizontal
(c) 1 horizontal : 4 vertical
(d) 1 horizontal : 2 vertical

**Ans. (c) :** Cipolletti weir is a trapezoidal weir with side slope of 1 : 4 (horizontal to vertical)

**136. A submerged weir is one in which the water level on the downstream of the weir is**

(a) just at the crest level
(b) below the crest level
(c) is above the crest level
(d) at the same elevation as the upstream water surface

**Ans. (c) :** When down stream water level is above crest level of the weir then the weir is called submerged weir.

**137. The specific heat ratio k is given by the following expression :**

(a) $1 + (C_P/R)$ (b) $\frac{1}{1-(R/C_v)}$

(c) $1 + (R/C_v)$ (d) $1 + (C_v/R)$

**Ans. (c):** $C_p - C_v = R$

$$\frac{C_p}{C_v} - 1 = \frac{R}{C_v}$$

$$K - 1 = \frac{R}{C_v}$$

$$K = 1 + \frac{R}{C_v}$$

**138. An isentropic process is**

(a) adiabatic and irreversible
(b) adiabatic and frictionaless
(c) reversible and isothermal
(d) any adiabatic process

**Ans. (b) :** Isentropic thermodynamic process is considered to be adiabatic, reversible and frictionless process.

**139. The speed of sound in air varies as**

(a) $\sqrt{T}$ (b) $\sqrt{p}$

(c) $1/\sqrt{p}$ (d) p

**Ans. (a) :** $C = \sqrt{KRT}$

$$C \propto \sqrt{T}$$

**140. In an atmosphere the speed of sound is 300 m/s. If a plane travels at 1620 km/h in this atmosphere, the Mach angle is**

(a) 30.5º (b) 56.3º
(c) 10.7º (d) 41.8º

**Ans. (d) :**

$$\sin\alpha = \frac{1}{m} = \frac{1}{\frac{v}{c}} = \frac{c}{v}$$

$$\alpha = \sin^{-1}\left(\frac{c}{v}\right)$$

$$= \sin^{-1}\left(\frac{300}{1620 \times \frac{5}{18}}\right) = \sin^{-1}\left(\frac{300}{450}\right)$$

$$= \sin^{-1}(0.667) = 41.8^0$$

**141. In a standard atmosphere the temperature is 15 ºC at sea level and –56ºC at an altitude of 20 km. A supersonic plane has the same speed at the sea level as well as at an altitude of 20 km. If its Mach number is 1.5 at sea level, its Mach number at an altitude of 20 km in standard atmosphere is**

(a) 2.80 (b) 1.30
(c) 1.99 (d) 1.73

**Ans. (d) :** $T_1 = 15°C = 15 + 273 = 288k$

$T_2 = -56°C = -56 + 273 = 217k$

$M_1 = 1.5$

$M_2 = ?$

$$M_1 = \frac{V}{C_1} = \frac{V}{\sqrt{KRT_1}}$$

$$M_2 = \frac{V}{C_2} = \frac{V}{\sqrt{KRT_2}}$$

$$\frac{M_2}{M_1} = \sqrt{\frac{T_1}{T_2}}$$

$$M_2 = 1.5\sqrt{\frac{288}{217}} = 1.73$$

**142. For air (k = 1.4) the critical pressure ratio $P_1/P_0$ for isentropic flow is**

(a) 0.833 (b) 0.728
(c) 0.628 (d) 0.528

**Ans. (d) :** Critical pressure in nozzle flow

$$\frac{P_1}{P_o} = \left(\frac{2}{k+1}\right)^{k/k-1}$$

Given,

$k = 1.4$

$$\frac{P_1}{P_o} = \left(\frac{2}{1.4+1}\right)^{\frac{1.4}{1.4-1}} = 0.528$$

**143. In a normal shock in a gas**

(a) the upstream flow is supersonic
(b) the upstream flow is subsonic
(c) the downstream flow is sonic
(d) the downstream flow is supersonic

**Ans. (a) :** Normal shock wave is characterized by upstream super sonic flow and down stream is subsonic flow.

**144. In isentropic flow between two points**

(a) stagnation pressure and stagnation temperature may vary

(b) the stagnation pressure decreases in the direction of the flow

(c) the stagnation temperature and stagnation pressure decrease with increase in the velocity

(d) the stagnation temperature and stagnation pressure remain constant

**Ans. (a) :** Between two points of isentropic flow, stagnation pressure and stagnation temperature varied.

**145. In a normal shock occurring in air (k = 1.4), if the upstream Mach number is 3.52, the Mach number after the shock is**

(a) 0.61

(b) 0.45

(c) 0.13

(d) 0.28

**Ans. (b) :** Mach number relations is normal shock wave pattern.

Given,

$m_1 = 3.52, \qquad k = 1.4$

$$m_2 = \sqrt{\frac{2+(k-1)m_1^2}{2k\ m_1^2-(k-1)}}$$

$$m_2 = \sqrt{\frac{2+(1.4-1)\ (3.52)^2}{2\times1.4\times3.5\times2^2-(1.4-1)}}$$

$$m_2 = \sqrt{\frac{6.9562}{34.293}} = \sqrt{0.203} = 0.45$$

**146. In a normal shock wave in a gas with k = 1.4, one of Mach numbers is 0.5. The other Mach number is**

(a) 2.65 upstream of the shock

(b) 0.06 downstream of the shock

(c) 0.02 upstream of the shock

(d) 3.75 upstream of the shock

**Ans. (a) :**

$$m_1 = \sqrt{\frac{2+(k-1)m_2^2}{2k\ m_2^2-(k-1)}}$$

Given,

$m_2 = 0.5$

(After normal shock wave, subsonic flow)

$$m_1 = \sqrt{\frac{2+(1.4-1)\ (0.5)^2}{2\times1.4\times0.5^2-(1.4-1)}}$$

$$m_1 = \sqrt{\frac{2.1}{0.3}} = \sqrt{7}$$

$m_1 = 2.65$ (upstream)

**147. An impulse turbine is used for**

(a) low head of water

(b) high head of water

(c) medium head of water

(d) high discharge

**Ans. (b) :** Pelton turbine is an impulse turbine and it is suitable for high head and low discharge.

**148. The hydraulic efficiency of an impulse turbine is the**

(a) ratio of the actual power produced by the turbine to the energy actually supplied by the turbine

(b) ratio of the actual work available at the turbine to the energy imparted the wheel

(c) ratio of the work done on the wheel to the energy of the jet

(d) ratio of power developed at the buckets to energy produced by the shaft

**Ans. (c) :** For impulse turbine (i.e. pelton turbine)

$$\eta_h = \frac{\text{Power developed by wheel}}{\text{K.E. supplied by nozzle}}$$

**Note-** Actual work is different from the power developed.

Actual work = power developed – Mechanical losses.

**149. The maximum hydraulic efficiency of an impulse turbine is**

(a) $(1+\cos\phi)/2$

(b) $(1-\cos\phi)/2$

(c) $(1+\sin\phi)/3$

(d) $(1-\sin\phi)/2$

**Where $\phi$ = Angle of blade tip at outlet**

**Ans. (a) :** The condition for max hydraulic efficiency for the pelton turbine is $\frac{u}{v}=\frac{1}{2}$ and the corresponding max. efficiency is-

$$\eta_{max} = \frac{(1+\cos\phi)}{2}$$

**150. The function of guide vanes in a reaction turbine is to**

(a) allow the water to enter the runner without shock

(b) allow the water to flow over them, without forming eddies

(c) allow the required quantity of water to enter the turbine

(d) allow the water to flow to the draft tube

**Ans. (a) :** The Primary function of guide vanes is to direct water towards the runner in such a way that the flow enters into the runner without shock.

**Note-** By changing the flow area between the guide vanes the discharge entering into the turbine can also be controlled.

Hence both (1) and (3) are correct but (1) is most appropriate were as it is the main function.

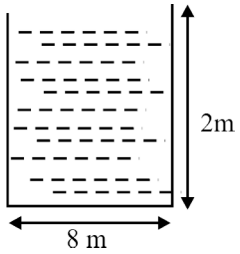
# Telangana State Public Service Commission
## Assistant Executive Engineer Exam., 2017
## Solved Paper

**1. A surge tank is provided in a hydropower scheme to :**
(a) Provide additional storage close to the penstock.
(b) Take care of change of slope alignment or size of the water conductor system.
(c) Reduce the pressure under transient conditions.
(d) Provide convenient overspills.

**Ans : (c)** A surge tank is provided in a hydropower schemes to reduce the pressure under transient conditions or to reduce water hammer in penstock.

**2. A regime channel has a full supply depth of 2.0 m and a bed width of 8.0 m. The full supply discharge in this channel is :**
(a) 6.9 $m^3/s$ (b) 18.0 $m^3/s$
(c) 2.8 $m^3/s$ (d) 1.8 $m^3/s$

**Ans : (a)**




$P = 8 + 2 \times 2 = 12$

$P = 4.75\sqrt{Q}$

$12 = 4.75\sqrt{Q}$

$Q = 6.38\ m^3/s$

$Q \approx 6.4\ m^3/s$

**3. In an irrigation plot the net irrigation requirement of a crop is found to be 14.9 cm, the application efficiency is 80% and the water conveyance efficiency is 70%. The gross irrigation requirement is :**
(a) 8.3 cm (b) 26.6 cm
(c) 17.0 cm (d) 21.3 cm

**Ans : (b)** Irrigation efficiency 80% and conveyance efficiency 70%.
The gross irrigation requirement is –

$$X \times \frac{80}{100} \times \frac{70}{100} = 14.9 \text{ cm}$$

$$X \times 0.80 \times 0.70 = 14.9 \text{ cm}$$

$$X = \frac{14.9}{0.56} = 26.6 \text{ cm}$$

**4. If the depletion of oxygen is found to be 2 mg/*l* after incubating 3 ml of sewage diluted to 300 ml at 20°C for 5 days, then the $(BOD)_5$ of the sewage would be:**
(a) 200 mg/*l* (b) 300 mg/*l*
(c) 500 mg/*l* (d) 600 mg/*l*

**Ans : (a)** $(BOD)_5 = (DOi - D.Of) \times D.F.$

$D.Di - D.Of = 2\ mg/\ell$

$$\text{Dilution factor 'D.F.'} = \frac{300}{3} = 100$$

$$(BOD)_5 = 2 \times 100 = 200\ mg/\ell$$

**5. The natural process under which the flowing river water gests cleaned is known as:**
(a) Oxidation (b) Photosynthesis
(c) Reduction (d) Self-purification

**Ans. (d) :** The self cleaning capacity of a river refers to the process by which pollutants discharged into a river gets cleaned by natural processes factors like the flow of the river and the presence of bacteria which decomposes pollutants into harmless substances contributes to the self cleaning capacity of a river.

**6. If a sewer carrying a discharge of 3 cumec outfalls into a river having a discharge of 10 cumec and DO equal to 9.1 mg/*l*, the resultant DO of the mix will be equal to**
(a) 5 mg/*l* (b) 6 mg/*l*
(c) 7 mg/*l* (d) 8 mg/*l*

**Ans. (c) :** Resultant DO after mix will be given by–

$$DO = \frac{Q_1DO_1 + Q_2DO_2}{Q_1 + Q_2}$$

Discharge of river, $Q_1 = 10$ cumec
Discharge of sewer, $Q_2 = 3$ cumec
Dissolved oxygen of river, $DO_1 = 9.1$ mg/L
Dissolved oxygen of sewer '$DO_2$' = 0 mg/L

$$DO = \frac{10 \times 9.1 + 3 \times 0}{10 + 3} = 7\ mg/L$$

**7. The phenomenon by virtue of which soil pores gets clogged with sewage matter is called**
(a) Sewage farming (b) Sewage bulking
(c) Sewage sickness (d) Sewage irrigation

**Ans. (c) :** When sewage is applied continuously on a piece of land the soil pores or voids may get filled up and clogged with sewage matter retained in then known as sewage sickness.

**8. Surface loading for rectangular sedimentation tank of length L, width B and height H is given by -**

(a) $\frac{Q}{BH}$ (b) $\frac{Q}{BL}$

(c) $\frac{BH}{Q}$ (d) $\frac{BL}{Q}$

**Ans. (b) :** Surface loading = $\frac{\text{discharge}}{\text{surface area}} = \frac{Q}{B.L}$

Where, B×L = Surface area

• Detention time = $= \frac{\text{Volume of tank}}{\text{Discharge}} = \frac{LBH}{Q}$

**9. For a grit chamber if the recommended velocity is 0.2 m/s and detention time is 1 minute, the length of the grit chamber is**

(a) 8 m (b) 10 m

(c) 12 m (d) 15 m

**Ans. (c) :**
Length of grit chamber 'L' = detention time × velocity
detention time = 1 minute.
Velocity 'v' = 0.2 m/sec = 12 m/minute.
L = 1 × 12
L = 12 m

**10. Lower $\frac{F}{M}$ ratio in a conventional activated sludge treatment plant will mean**

(a) Lower BOD removal
(b) Higher BOD removal
(c) No effect on BOD removal
(d) Sometimes is lower and sometimes higher removal

**Ans. (b) :** Food to micro-organism ratio $\left(\frac{F}{M}\right)$

• It is defined as ratio of daily BOD load applied to the aerator system in gm to total microbial mass in the system in gm.

• $\frac{F}{M}$ ratio for an activated sludge plant is the main factors controlling BOD removal. Lower the $\frac{F}{M}$ valve, the higher will be the BOD removal in the plant. The $\frac{F}{M}$ ratio can be varied by varying the MLSS concentration in the aeration tank.

**11. Leachate is a colored liquid that comes out of:**

(a) Septic tanks (b) Sanitary land fills

(c) Aerated lagoons (d) Compost plants

**Ans. (b) :** Leachate is liquid that comes out of compost in environmental context it is a liquid material which drains from stockpiled or landfills and contains elevated concentration of unwanted material obtained from the material it has entered through.

**12. During temperature inversion in atmosphere air pollutants tends to**

(a) Accumulate above inversion layer
(b) Accumulate below inversion layer
(c) Disperse laterally
(d) Disperse vertically

**Ans. (b) : Negative lapse Rate–** When the temperature of the environment (i.e ambient air) increase with altitude then lapse rate becomes inverted or negative from normal slate.

• Negative lapse rate occur under conditions usually referred to as inversion (a shall in which the warmer air lies over the colder air below).

i.e. air pollutant accumulated below the inversion.

**13. If P represents the pressure of sound ware and $P_{ref}$ represent the reference pressure then Sound Pressure Level (SPL) is equal to**

(a) $20 \log_{10}\left(\frac{P}{P_{ref}}\right)$ (b) $\frac{1}{20} \log_{10}\left(\frac{P}{P_{ref}}\right)$

(c) $20 \log_{10}\left(\frac{P_{ref}}{P}\right)$ (d) $\frac{1}{20}\log_{10}\left(\frac{P_{ref}}{P}\right)$j

**Ans. (a) :**

Sound Pressure Level (SPL) = $20 \log_{10}\left(\frac{P}{P_{ref}}\right)$

where,
P = pressure of sound ware and
$P_{ref}$ = reference pressure

**14. The true specific gravity (G) of a given soil sample was determined in the laboratory at 10ºC and 45ºC. Answer which of the following is true?**

(a) $G_{10ºC} > G_{45ºC}$ (b) $G_{10ºC} < G_{45ºC}$

(c) $G_{10ºC} = G_{45ºC}$ (d) $G_{10ºC} = 2G_{45ºC}$

**Ans. (b) : True specific gravity–** It is also called absolute, apparent bulk or mass specific gravity.

The soil solids have permeable and impermeable voids inside them, the permeable voids being capable of getting filled with water.

If all the internal voids of soil particles are excluded for determining the true volume of solids, the specific gravity is called true specific gravity, $G_m = \frac{\gamma}{\gamma_w}$

The true specific gravity (G) of a given soil sample was determined in the laboratory at 10ºC and 45ºC, $G_{10ºC} < G_{45ºC}$ is true for above condition.

**15. In a sieve analysis, 70% of the soil mass is retained on ISS 2.00 mm and 60% is finer than ISS 4.00 mm. Determine effective size of the soil mass if its coefficient of curvature Cc = 2.00.**

(a) 0.50 mm (b) 1.00 mm

(c) 1.50 mm (d) 2.00 mm

**Ans. (a) :** Given,
$D_{10}$ = ?, $D_{30}$ = 2mm, $D_{60}$ = 4mm
Coefficient of curvature Cc = 2 mm

$$C_C = \frac{(D_{30})^2}{D_{60} \times D_{10}} \Rightarrow 2 = \frac{2^2}{4 \times D_{10}} \Rightarrow D_{10} = 0.50\,\text{mm}$$

**16. At a site, the ground water table is coinciding with the ground level. If saturated unit weight of soil is 20 kN/cum, determine the effective stress at 3 m below the ground level.**

(a) 30 kPa (b) 60 kPa
(c) 45 kPa (d) 100 kPa

**Ans. (a) :** Effective stress = σ–u

Tata stress $(\sigma) = \gamma_{sat} \times z$

$= 20 \times 3 = 60$ kPa

Pore water stress $(u) = \gamma_w \times z$

$= 10 \times 3 = 30$ kPa

$\bar{\sigma} = 60 - 30$

$\bar{\sigma} = 30$ kPa

**17. The time required for a consolidating medium with single drainage to undergo 90% of its primary consolidation settlement was estimated as 36 years. All other conditions remaining same, estimate the time required if the medium has double drainage.**

(a) 3.6 years (b) 36 years
(c) 72 years (d) 9 years

**Ans. (d) :** Time factor $T_v = \frac{C_v t}{z^2}$, $t \propto z^2$

$$\frac{t_{Single\,drainage}}{t_{Double\,drainage}} = \frac{z^2}{(z/2)^2}$$

$$\frac{t_s}{t_d} = 4$$

$$t_d = \frac{t_s}{4} = \frac{36}{4} = 9 \text{ days}$$

**18. The shear strength of a Clayey soil (ϕ = 0) is found to be 60 kPa when tested in and unconfined compressive strength test. Then, the major and minor principal stresses at failure respectively are**

(a) 120 kPa, 60 kPa (b) 120 kPa, 0 kPa
(c) 60 kPa, 120 kPa (d) 120 kPa, 120 kPa

**Ans. (b) :** Given, shear strength = 60 kPa

Major principal stress $\sigma_1$ = 2× shear strength

= 2×60

= 120 kPa

Minor principal stress, $\sigma_3 = 0$

**19. The load carrying capacity of an end bearing pile is found to be 400 kN. All other parameters remaining same, what will be capacity if its diameter is halved.**

(a) 100 kN (b) 500 kN
(c) 1000 kN (d) 2000 kN

**Ans. (a) :** Ultimate bearing capacity (Q) = $N_c \times C \times$ pile tip area.

$$Q = N_c \times C \times \frac{\pi}{4} \times d^2$$

$$Q \propto d^2$$

$$\frac{Q_1}{d_1^2} = \frac{Q_2}{d_2^2}$$

$$\frac{400}{d^2} = \frac{Q_2}{(d/2)^2} \Rightarrow Q_2 = 100 \text{ kN}$$

**20. Which type of caissons are preferred in emergency conditions where time for sinking of caissons is not available?**

(a) Box caisssons
(b) Pneumatic caisssons
(c) Open caissons
(d) Box, Pneumatic and Open caissons

**Ans. (a) :** Box caissons are preferred in emergency conditions where time for sinking of caissons is not available.

**21. If a field compaction process, if Maximum Dry Density, Field Bulk Density and Field Moisture Content are 20 kN/m$^3$, 22 kN/m$^3$ and 10% respectively, the relative compaction is:**

(a) 90% (b) 95%
(c) 100% (d) 85%

**Ans. (c) :** $\gamma_{d\,(max)} = 20$ kN/m$^3$

$\gamma_t = 22$ kN/m$^3$

Field moisture content 'w' = 10%

$$\gamma_{d(Field)} = \frac{\gamma_t}{1+w} \Rightarrow \frac{22}{1+0.1} \Rightarrow \frac{22}{1.1}$$

$\gamma_{d(field)} = 20$ kN/m$^3$

$$\text{Relative compaction} = \frac{\gamma_{d(Field)}}{\gamma_{d(max)}} \times 100$$

$$= \frac{20}{20} \times 100$$

$=100\%$

**22. A 10 m thick clay layer with single drainage undergone 90% of primary consolidation settlement in 10 years. Considering the corresponding time factor as 0.848, the coefficient of consolidation is:**

(a) 1.0 m$^2$/year (b) 10 m$^2$/year
(c) 8.48 m$^2$/year (d) 84.8 m$^2$/year

**Ans. (c) :** Thick of clay layer = 10 m

Single drainage undergone 'd' = 10 m

t = 10 year

$T_v = 0.848$

$C_v = ?$

$$T_v = \frac{C_v t}{d^2}$$

$$0.848 = \frac{C_v \times 10}{(10)^2}$$

$$C_v = \frac{84.8}{10}$$

$$C_v = 8.48 \text{ m}^2/\text{year}$$

**23. What of the following statements is incorrect?**

(a) Even the slight movement of wall away from backfill develops full active earth pressure.

(b) Earth pressure at rest will be present when the wall does not move relative to the back fill.

(c) Full passive earth pressure develops only after considerable movement of wall towards the backfill.

(d) For a given wall and backfill conditions, the magnitude of active earth pressure will always be higher than passive earth pressure.

**Ans. (d) :** • Even the slight movement of wall away from backfill develops full active earth pressure.

• Earth pressure at rest will be present when the wall does not move relative to the back fill.

• Full passive earth pressure develops only after considerable movement of wall towards the backfill.

• For a given wall and backfill conditions, the magnitude of active earth pressure will always be less than passive earth pressure.

**24. In a direct shear test on a dry clean sandy soil specimen with c = 0 kPa, $\phi = 45^o$, determine the shear stress at failure when the normal stress was 100 kPa.**

(a) 45 kPa (b) 75 kPa

(c) 145 kPa (d) 100 kPa

**Ans. (d) :** Given, C = 0 kPa, $\phi = 45^o$

Shear stress, $\tau = C + \sigma \tan \phi$

$= 0+100 \tan 45$

$= 100 \times 1$

$= 100$ kPa

**25. In a standard penetration test, the blows are recorded as 1/20/30. If correction factor for overburden is 0.90, the standard penetration resistance, N-value corrected for overburden and dilatancy is**

(a) 57 (b) 32

(c) 37 (d) 30

**Ans. (d) :** Recorded value $(N_R) = 20 + 30 = 50$

Corrected value after overburden correction $(N_C)_{overburden} = 0.9 \times 50 = 45$

Now, $(N_c)_{dilatancy} = 15 + \frac{1}{2}(45 - 15)$

$= 15 + 15 = 30$

**26. A vertically upward seepage flow is taking place under a head of 3 m in a cohessionless medium of 2 m thick having G = 2.65 and e = 0.65. Determine factor of safety against quicksand.**

(a) 2 (b) 3

(c) 0.67 (d) 0.50

**Ans. (c) :** Factor of safety $= \frac{I_c}{I_i}$

$I_c = \frac{G-1}{1+e} = \frac{2.65-1}{1+0.65} = 1$

$I_i = \frac{h}{L} = \frac{3}{2} = 1.5$

Factor of safety $= \frac{1}{1.5} = \frac{10}{15} = \frac{2}{3} = 0.67$

$= 0.67$

**27. The co-efficient of permeability (k) of a soil mass was found to be $1 \times 10^{-3}$ cm/s for flow of water when the test was performed at 20ºC. When the test was repeated at 50ºC, what will be the "k" if viscosity and unit weight of the fluid are viscosity and unit weight of the fluid are decreased to 60% and 90% respectively?**

(a) $1 \times 10^{-3}$ cm/s (b) $1.50 \times 10^{-3}$ cm/s

(c) $0.50 \times 10^{-3}$ cm/s (d) $2 \times 10^{-3}$ cm/s

**Ans. (b) :** Given,

$k_1 = 1 \times 10^{-3}$ cm/s

Unit weight of fluid, $\gamma_{w_1}$

Viscosity, $\eta$

$k_2 = ?$

$\gamma_{w_2} = 0.9\ \gamma_{w_1}$

$\eta_2 = 0.6\ \eta_1$

$k = \frac{\gamma_w}{\eta}$

$\frac{k_1}{k_2} = \frac{\gamma_{w_1}}{\eta_1} \times \frac{\eta_2}{\gamma_{w_2}} = \frac{0.6}{0.9} = \frac{2}{3}$

$k_2 = 1 \times 10^{-3} \times \frac{3}{2} = 1.5 \times 10^{-3}$ cm/sec

**28. The flow net of an earthen dam consists of 7 flow lines and 12 potential drops. What is the discharge per meter length if k = 0.1 m/s, H = 10m?**

(a) 0.50 cum/sec (b) 1.0 cum/sec

(c) 0.25 cum/sec (d) 2 cum/sec

**Ans. (a) :** Given,

k = 0.1 m/sec

H = 10 m

$N_f = 7 - 1 = 6$ $\begin{cases} N_f = \text{No. of flow channel} \\ N_d = \text{No. of potential drop} \end{cases}$

$N_d = 12$

$Q = kH\frac{N_f}{N_d}$

$= 0.1 \times 10 \times \frac{6}{12} = 0.5$ cum/sec

**29. In a laboratory compaction test, it was required to use 1600 cc capacity mould in place of the standard 1000 cc mould. To ensure application of standard compaction effort, how many blows do you recommend in place of the standard 25 blows per layer?**

(a) 15 (b) 56

(c) 40 (d) 100

**Ans. (c) :** Compactive effort (E) $= \frac{Nnwh}{V}$

$$\left(\frac{Nnwh}{V}\right)_1 = \left(\frac{NnwH}{V}\right)_2$$

$$\frac{25 \times n \times w \times h}{1000} = \frac{N_2 \times n \times w \times h}{1600}$$

$$N_2 = \frac{25}{10} \times 16$$

$N_2 = 40$ blow

**30. In a tri-axial compression test, a specimen failed at a deviatoric stress of 180 kPa when the cell pressure was 100 kPa. Determine the major and minor principal stresses**

(a) 180, 100 (b) 80, 20
(c) 280, 100 (d) 100, 80

**Ans. (c) :** Major principal stress, $\sigma_1 = \sigma_c + \sigma_d$
$= 100 + 180 = 280$ kPa
Minor principal stress, $\sigma_3 = \sigma_c = 100$ kPa

**31. The increment in vertical stress ($\sigma_z$) directly 1 m below a point load was found to be 477.50 kPa. Determine the magnitude of the point load**

(a) 100 kN (b) 500 kN
(c) 1000 kN (d) 2000 kN

**Ans. (c) :** Vertical stress '$\sigma_2$' = 471.5 kPa

$$\sigma_2 = 0.4775 \times \frac{Q}{d^2}$$

$$477.5 = 0.4775 \times \frac{Q}{1^2}$$

$$Q = \frac{477.5}{0.4775} = 1000 \text{ kPa}$$

**32. The net ultimate bearing capacity of a 2m wide square footing laid on pure cohesive soil is estimated at 400 kPa. All other parameters remaining same what will be the net ultimate bearing capacity if size of the footing is doubled?**

(a) 100 kPa (b) 200 kPa
(c) 400 kPa (d) 50 kPa

**Ans. (c) :** Net ultimate bearing capacity of pure cohesive soil is given by-
$(q_{nu})_{clay} = CN_C$
• Net bearing capacity of clay independent on size of footing.
So, Answer C is correct.

**33. As per IS : 3955–1967, tilt of a well foundation should generally be limited to**

(a) 1 in 20 (b) 1 in 40
(c) 1 in 60 (d) 1 in 100

**Ans. (c) :** As per IS : 3955–1967, tilt of a well foundation should generally be limited to 1 in 60.
The shift in well foundation should not be more than 1% of depth of sunk.

**34. The ability of soil particles to undergo rehabilitation and regain a part of the lost strength when left undisturbed is known as:**

(a) Dilation (b) Thixotropy
(c) Plasticity (d) Swelling

**Ans. (b) :** Thixotropy = It is regain of loss in strength of soil with passage of time.
During pile driving to clayey soil, frictional resistance by soil increases due to property of thixotropy.

**35. Which of the following statements about Dynamic Formula is correct?**

(a) Applicable for the bored cast in situ piles only.
(b) Overestimate the capacity in sensitive clays.
(c) Underestimate the capacity wherein localized momentary liquefaction takes place while driving the pile.
(d) Set value is average of the first few blows.

**Ans. (c) :** The correct statements about Dynamic Formula is underestimate the capacity wherein localized momentary liquefaction takes place while driving the pile.

**36. The end product of weathering that has been transported downslope under the influence of gravity and is accumulated at the base of the hills and bare slopes is known as:**

(a) Eluvium (b) Deluvium
(c) Sediments (d) Peat

**Ans. (b) :** The end product of weathering that has been transported downslope under the influence of gravity and is accumulated at the base of the hills and bare slopes is known as deluvium.

**37. Which of the following is a deposit of glacial origin consisting of un-assorted mixture of boulders and clay particles?**

(a) Loess (b) Talus
(c) Till (d) Eskers

**Ans. (c) : Till–** It is an unstratified deposit formed by melting of a glacier. The deposit consists of particles of different sizes, ranging from boulders to clay.
• The soil is generally well-graded. It can be easily densified by compaction. Till is also known as boulder clay.

**38. Groups of small sized faults that overlap each other are named as:**

(a) Parallel faults (b) Peripheral faults
(c) Radial faults (d) Enechelon faults

**Ans. (d) : Enechelon faults –** Enechelon faults is group of small size faults that over lap each other are named enechelon faults. The term 'enechelon' refers to closely spaced, parallel or subparallel, over lapping or step like minor structural features in rock (faults, tension fractures), which lie oblique to the overall structural trend.

**Parallel fault–** The terms is used to express the occurrence of a number of fault that strike essentially in the same direction and have the same dip.

**Peripheral faults–** When in any given region the majority of faults show curved or arcuate pattern, the faulting is termed as peripheral and the group of circular faults as peripheral or arcuate faults.

**Radial faults–** Radial faults if comprise a group of fault that spring out or radiate from a common point or direction.

**39. Which of the following conditions is not favorable for construction of a Tunnel?**

(a) Tunnels parallel to the strike with beds dip in the tunnel.
(b) Tunnel parallel to the dip direction.
(c) Small sized tunnel in massive thick horizontal layers normal to the tunnel direction.
(d) Tunnels in Hard, Crystalline, Massive Rocks.

**Ans. (a) :** Tunnels parallel to the strike with bed dip in the tunnel is not favorable condition for tunneling.

**40. Which of the following is not ideal as foundation of Dams as it is prone to dissolution and results in formation of cavities?**

(a) Granite (b) Limestones
(c) Dolerites (d) Sandstones

**Ans. (b) :** Lime stone is affected by frost and atmosphere, it is suitable for flooring, paving and roofing and in the manufacturing of lime and cement.

- It is not suitable for foundation of dam.

**41. The super elevation (in meters) for mixed traffic conditions with a design speed of 100 kmph for a horizontal curve of 738 m, is**

(a) 0.23 (b) 0.06
(c) 0.10 (d) 0.45

**Ans. (b) :** $e = \frac{V^2}{225R}$

R = 738 m

V = 100 km/h

$e = \frac{(100)^2}{225 \times 738} = 0.06$

**42. Bitumen of grade 100/120 means**

(a) Its compressive strength is 100 to 120 kPa.
(b) Its tensile strength is 100 to 120 kPa.
(c) Its penetration value is 10 to 12 mm.
(d) Both compressive and tensile strength is 100 to 120 kPa.

**Ans. (c) :** Bitumen grade 100/120 indicates the penetration value of bitumen from 10 mm to 12 mm. This distance is measured in one-tenth of a millimeter.

$\text{Penetration} = \frac{1}{10} \times \text{Reading scale}$

$= \frac{1}{10} \times (100\text{to}120) = (10\text{to}12)\,\text{mm}$

**43. The transition curve adopted by Indian Railways with standard notations is of the form**

(a) $y = 6RLx$ (b) $y = x^3/6\,RL$
(c) $y = 64\,RL\,e^x$ (d) $y = 36\,RL\,x^2$

**Ans. (b) :** Indian Railways mostly uses the cubic parabola for transition curves. The equation of the cubic parabola is – $\boxed{y = \frac{x^3}{6Rl}}$

$l$ = distance of any point on transition curve from take off point.

x = horizontal coordinate on transition curve.

y = vertical coordinate on transition curve

R = Radius of circular curve.

**44. What is track capacity of a permanent way if 360 trains pass in a day?**

(a) 3.6 (b) 36
(c) 1.5 (d) 15

**Ans. (d) :** Track capacity $= \frac{360}{24} = 15$ trains/hr.

**45. The maximum design gradient for vertical profile of a road is:**

(a) Exceptional gradient (b) Ruling gradient
(c) Limiting gradient (d) Hydraulic gradient

**Ans. (b) : Ruling gradient:** It is the maximum gradient within which the designer attempts to design the vertical profile of a road. It is also known as design gradient.

**Limiting gradient–** The gradient steeper than the ruling gradient, which may be used for a limited road length, is called limiting gradient or maximum gradient.

**Minimum gradient–** The minimum desirable slope essential for effective drainage of rainwater from the purpose is 0.5% if the side drains are lined and 1% if the side drains are unlined.

**Exceptional gradient-** The gradient steeper than the limiting gradient which may be used in short length of road, only in an extraordinary situation is called exceptional gradient.

**46. Which of the following tests measures the toughness of road aggregates?**

(a) Shape test
(b) Crushing strength test
(c) Impact test
(d) Abrasion test

**Ans. (c) :** Toughness of road aggregates measured by impact test.

- It is defined as the ability of aggregate to resist impact loading.
- For pavement construction, AIV $\not>$ 30%
- For general construction, AIV $\not>$ 45%
- For bituminous macadam, AIV $\not>$ 40%

**47. The equilibrium cant for a railway curve of radius 344 m for an average speed of 60 kmph is:**

(a) 6.88 cm (b) 8.24 cm
(c) 9.36 cm (d) 5.43 cm

**Ans. (b):** As the gauge length is not given so we can choose any gauge. here for calculation simplicity we consider M.G.

so, cant= $\frac{GV^2}{127R}$

cant = $\frac{1\times 60^2}{127\times 344}$ =8.24cm

**48. The maximum speed on a 3º broad gauge curve according to Martin's formula is**

(a) 67.45 kmph (b) 88.97 kmph
(c) 97.88 kmph (d) 122.59 kmph

**Ans. (c) :** Radius R = $\frac{1720}{D}=\frac{1720}{3^0}=573.33$

For broad gauged transition curve martin's formula is –

$V = 4.4\sqrt{R-70}$

Where, V in kmph and R in meter.

$V = 4.4\sqrt{573.33-70} = 98.71$ kmph

Nearest option is (c).

**49. Enoscope is used to fine**

(a) Average speed (b) Time mean speed
(c) Space mean speed (d) Spot speed

**Ans. (d) :** One of the simplest methods of finding the spot speed is by using enoscope which is just a mirror box, supported on a tripod stand. The main advantage of this method is that it is simple and cheap and is easy to use. Other equipment used are photoelectric meter, radar speedometer etc.

Spot speed is used to ensure the speed regulation on roads.

**50. In cement concrete pavements tie bars are provided at:**

(a) Warping joints (b) Longitudinal joints
(c) Contraction joints (d) Expansion joints

**Ans. (b) :** The tie bars in concrete pavement are provided across the longitudinal joint.

- Tie bar are not load transfer devices but serve as a means to tie two slabs.
- Longitudinal joint– these are provided along the length of the pavement.

**51. In an electric railway line, the height of catenary contact above the rail top is:**

(a) 4.35 m (b) 6.78 m
(c) 5.70 m (d) 3.12 m

**Ans. (c) :** In OHE (Overhead electrification system), the supply of electricity is through an overhead system of suspended cables known as catenary. These wire is generally at about 5.5 m from the rail level. The minimum height is around 4.8 m (e.g. under bridge or over-passes etc.) In yards, sheds or line loading up to sheds etc, the catenary contact wire may be higher, as 5.8m.

**52. In a permanent way, the cement concrete sleepers of 250 mm width are placed at a spacing of 700 mm. Determine the depth of ballast**

(a) 15 cm (b) 17.5 cm
(c) 20 cm (d) 22.5 cm

**Ans. (d) :**

Depth of Ballast $\boxed{D=\frac{S-B}{2}}$

Where, S = Sleeper spacing
B = Width of sleeper

$=\frac{700-250}{2}=225$ mm = 22.5 cm

**53. In normal landing conditions, the aircraft should come to stop within ___ percentage of the landing distance.**

(a) 50 (b) 60
(c) 80 (d) 90

**Ans. (b) :** In normal landing conditions, the aircraft should come to stop within 60 % of the landing distance.

**54. The width of runway pavement for a runway of 1500 m length (Classified as Airport Type-B of ICAO) is**

(a) 45 m (b) 30 m
(c) 22.5 m (d) 18 m

**Ans. (a)**

| Airport type | Basic runway length (m) | | Runway width (m) |
|---|---|---|---|
| | Max | Min | |
| A | Over 2100 | 2100 | 45 |
| B | 2099 | 1500 | 45 |
| C | 1499 | 900 | 30 |
| D | 899 | 750 | 22.5 |
| E | 749 | 600 | 18 |

**55. Mud Jacking is a type of failure of rigid pavements due to:**

(a) Stagnation of water and hence softening of clayey subgrade.
(b) The rigid pavement transformed into the form of mud.
(c) Jack used to lift the mud beneath the rigid pavement.
(d) Mud injected beneath the rigid pavement.

**Ans. (a) :** Mud Jacking is a type of failure of rigid pavements due to infiltration of water through the joints, cracks or edge of the pavement forms soil slurry. Poor joint sealer allowing infiltration of water. Repeated wheel loading causing erosion of underlying material.

**56. Which of the following pavements layers serve the drainage function?**

(a) Wearing course
(b) Subgrade
(c) Granular Sub Base
(d) Wet Mix Maccadam

**Ans. (c) :** The granular sub-base (GSB) course has to serve as an effective drainage layer of the pavement and also has to sustain lower magnitude of compressive stresses than base course.

- The GSB cum drainage layer is laid above the subgrade covering the full width of the formation between the longitudinal drains.
- As per the specifications laid down by the ministry of road transport and highways on a national highway, GSB may be caused as sub-base and drainage layer.

**57. When a new topping (wearing surface) is proposed on an old wearing surface, which of the following measure is recommended to ensure bond between the old and new layers?**

(a) Heating the old surface
(b) Scarifying the old surface
(c) Cooling the old surface
(d) Providing a layer spearator

**Ans. (b) :** When a new topping (wearing surface) is proposed on an old wearing surface, Scarifying the old surface is recommended to ensure bond between the old and new layers.

Basically this process consists of four steps:

1. Softening of the asphalt pavement surface with heat;
2. Scarification and/or mechanical removal of the surface material;
3. Mixing of the material with recycling agent, asphalt binder, or new mix;
4. Lay down and paving of the recycled mix on the pavement surface.

**58. The magnetic declination at a place if the magnetic bearing of sun at noon is 352°40' is:**

(a) 352°40'E (b) 172°40'W
(c) 7°20'E (d) 7°20'W

**Ans. (c) :** At noon, the sun is exactly on the geographical meridian. Hence the true bearing of the sun at noon is zero or $180^o$ depending upon whether it is to the north of the place or to the south of the place.

$M.B._{noon} = 352^o40'$

$T.B. = M.B. + \delta$

$360^o = 352^o\ 40' + \delta$

$\delta = 7^o20'E$

**59. The operation of forming the clear image of the object in the plane of cross hairs is known as**

(a) Elimination of parallax
(b) Focussing
(c) Ranging
(d) Traversing

**Ans. (b) : Focusing–** Focusing is to be done in order to form image through objective lens at the plane of the diagram and to view the clear image of the object through eye-piece. This is being carried out by removing parallax by proper focusing of objective and eye-piece.

**60. Which of the following statement is incorrect?**

(a) Two contour lines having the same elevation cannot unite and continue as a single line.
(b) A single contour line cannot split into two lines.
(c) A series of straight, parallel and equally spaced contours represent a steep undulating surface.
(d) Contour lines of different elevations can unite to form one line only in the case of a vertical cliff.

**Ans. (c) : Characteristics of contours–** Contour lines of different elevation can unite to form one line only in the case of a vertical cliff.

- Two contour lines having the same elevation cannot unite and continue as one line. Similarly, a single contour cannot split into two lines.
- A series of straight, parallel and equally spaced contours represent a uniform surface.

**61. The vertical distance between the line of sight and the horizontal line (V) in an inclined sight using a Tacheometer with staff held vertical (in standard notations) is given by:**

(a) $V = ks \cos\theta.\sin\theta + C \sin\theta$
(b) $V = ks \tan 2\theta + C \sin\theta$
(c) $V = ks \cos\theta.\sin\theta - C \sin\theta$
(d) $V = ks \cot 2\theta + C \sin\theta$

**Ans. (a) :** Vertical distance,

$$V = ks \cos\theta.\sin\theta + C \sin\theta$$

Horizontal distance,

$$D = ks \cos^2\theta + C \cos\theta$$

**62. The operation "plunging" of a Theodolite is also known as:**

(a) Swinging (b) Transiting
(c) Centering (d) Changing face

**Ans. (b) : Transiting :** It also known as plunging or reversing. It is the process of turning the telescope about its horizontal axis through 180° in the vertical plane thus bringing it upside down and making it point exactly in opposite direction.

**63. A ring type of foundation is constructed with internal and external diameters of 8 m and 12 m respectively. If thickness of the slab is 10 cm and cost of RCC per cubic meter is ₹ 1000, the cost of RCC used in the slab is**

(a) ₹ 8628 (b) ₹ 6280
(c) ₹ 2680 (d) ₹ 1000

**Ans. (b) :** Internal diameters = 8 m

External diameter = 12 m

Total volume of RCC slab is 'V'

$$V = \frac{\pi}{4} \times \left(12^4 - 8^2\right) \times \frac{10}{100}$$

Total cost = V × 1000

$$= \frac{\pi}{4} \times (12+8) \times (12-8) \times 100$$

= 6580 Rs.

**64. A fund which is gradually accumulated and set aside to reconstruct the property after the expiry of its period of utility is known as**

(a) Annuity (b) Dividend
(c) Mutual fund (d) Sinking fund

**Ans. (d) : Sinking Fund**–A certain amount of gross rent is set aside annually as sinking fund to accumulate the total cost of construction when the life of the building is over. This annual sinking fund is also taken as outgoings

$$(S) = \frac{S_n \times R}{(1+R)^n - 1}$$

$S_n$ = Sinking fund to be accumulated in 'n' years
R = Rate of interest in decimal
S = Yearly instalment of sinking fund

**65. The system in which the owner acquires land, prepare drawings & estimation and the contractor execute the work, operate and collect fee from the users till the full cost is realized, is known as:**

(a) Rate Contract (b) BOT
(c) Lumpsum contract (d) Turnkey

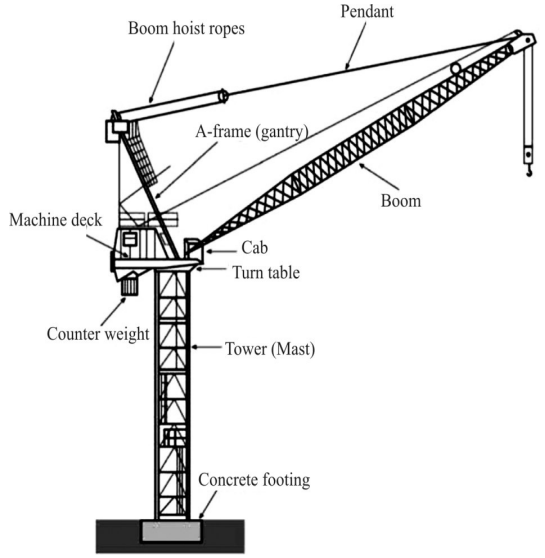
**Ans. (b) :** Build-operate – transfer (BOT) is a form of project delivery method; usually for large scale infrastructure project like road etc.

**66. The most commonly used cranes in high rise construction sites where concentration of work is in a limited area of the site**

(a) Tower cranes (b) Overhead cranes
(c) Crawler cranes (d) Goliat cranes

**Ans. (a) :** Tower cranes is the most commonly used cranes in high rise construction sites where concentration of work is in a limited area of the site.

- Tower cranes are a common fixture at any major construction site.
- Tower cranes are essential for moving heavy goods, materials or tools around a site.




**Tower Crane**

**67. In compaction specification if MDD, FDD stands for Maximum Dry Density and Field Dry Density, the term "Relative Compaction" is defined as**

(a) MDD + FDD (b) MDD – FDD
(c) MDD/FDD (d) FDD/MDD

**Ans. (d) :** Relative compaction $\frac{FDD}{MDD}$

**Relative compaction–** The objective of compaction in field is to achieve the maximum dry density that was obtained in the laboratory compaction test performed on the same soil.

**68. When additional works were found necessary during construction, which were not contemplated at the beginning, the estimate prepared for such items is known as:**

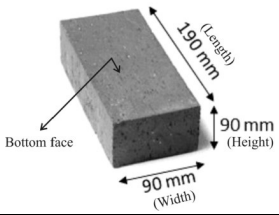
(a) Preliminary estimate
(b) Complementary estimate
(c) Supplementary estimate
(d) Revised estimate

**Ans. (c) : Supplementary estimate**–This is fresh detailed estimate in addition to the original sanctioned estimate prepared when additional works are deemed necessary during the progress of a work to supplement the original works.

**69. The standard modular size of common building bricks shall be:**

(a) 190 mm 90 mm × 90 mm
(b) 200 mm × 100 mm × 100 mm
(c) 200 mm × 90 mm × 90 mm
(d) 190 mm × 100 mm 100 mm

**Ans. (a) :** Actual dimension of a brick used in construction is = 19×9×9 cm




**70. For a beam of cross sectional size 100 mm 200 mm, the nominal maximum size of coarse aggregate should NOT be greater than**

(a) 10 mm (b) 20 mm
(c) 25 mm (d) 40 mm

**Ans. (c) :** Nominal size of aggregate is the smallest size of sieve through which entire amount of aggregate can pass through it. But is may retain 10% of total weight of aggregate. The nominal size of aggregate not exceed by 25 mm. in beam of cross-section 100 mm × 200 mm.

$$\text{Maximum size of aggregate} = \frac{\min\{b, d\}}{4}$$

$$= \frac{100}{4} = 25\text{mm}$$

$$\text{Maximum size of reinforcement} = \frac{\min\{b, d\}}{8}$$

**71. The Ph value of water used for mixing and curing of concrete shall NOT be less than**

(a) 5 (b) 6
(c) 7 (d) 8

**Ans. (b) :** The pH value of water shall not be less than 6. Mixing or curing of concrete with seawater is not recommended because of presence of harmful salts in the seawater.

**72. The minimum grade of concrete for reinforced concrete in sea water and exposing to coastal environment is:**

(a) M 20 (b) M 25
(c) M 30 (d) M 35

**Ans : (c)** Minimum grade of concrete for structure in sea water should be M30. Minimum grade of concrete depends upon atmospheric conditions.

| Sr. No. | Exposure | Min Grade of concrete | Min Cement Content ($kg/m^3$) | Max Free w/c ratio |
|---|---|---|---|---|
| 1 | Mild | M 20 | 300 | 0.55 |
| 2 | Moderate | M 25 | 300 | 0.50 |
| 3 | Severe | M 30 | 320 | 0.45 |
| 4 | very severe | M 35 | 340 | 0.45 |
| 5 | Extreme | M 40 | 360 | 0.40 |

**73. The degree of workability of concrete for heavily reinforced sections of slab should be**

(a) Very high (b) High
(c) Low (d) Medium

**Ans : (d)** The workability of concrete required for the heavily reinforced sections in slab, beam and column is medium.

**As per IS 456 : 2000**

| Reinforcement | Workability |
|---|---|
| Light reinforcement | Low |
| Heavily reinforcement | Medium |
| Trench fill or pilling | High |

**74. The short term static modulus of elasticity for concrete of grade M 25 is**

(a) 25 GPa (b) 2500 MPa
(c) 2.5 GPa (d) 250 MPa

**Ans. (a) :** The short term static modulus of elasticity

$= E_c = 5000\sqrt{f_{ck}}$

$E_c = 5000\sqrt{25} = 25000\ N/mm^2 = 25\ GPa$

**75. The target mean strength of concrete mix for characteristic compressive strength of 40 MPa is**

(a) 45.8 MPa (b) 48.3 MPa
(c) 46.6 MPa (d) 49.9 MPa

**Ans. (b) :** Target mean strength = $f_{cm} = f_{ck} + 1.65\times \sigma$ for M40 grade of concrete –

$\sigma = 5$ (by referring table 1 of IS 10262:2009

$f_{cm} = 40 + 1.65 \times 5$

$f_{cm} = 48.25$

**76. Nominal mix concrete can be used for concrete of grade**

(a) M 35 (b) M 30
(c) M 25 (d) M 20

**Ans. (d) :** Nominal mixes for grades of concrete such as M20 are generally used for small scale construction.

**77. For compressive strength determination the minimum number of cubes required in a sample are**

(a) 2 (b) 3
(c) 4 (d) 5

**Ans. (b) :** A sample consist of three cubes/specimens it is for the same concrete sample cast more than three cubes, so that any defective cube should not be tested or any doubt full test result due to testing may be rejected and not included in the average of three cubes.

**78. What will be the compressive strength of concrete if the flexural tensile strength of concrete is 3.5 MPa?**

(a) 25 MPa (b) 12.5 MPa
(c) 5 MPa (d) 2.5 MPa

**Ans. (a) :** Flexural tensile strength = $f_{cr} = 0.7\sqrt{f_{ck}}$

$3.5 = 0.7\sqrt{f_{ck}}$

$f_{ck} = \frac{(3.5)^2}{(0.7)^2} = 25\ MPa$

**79. If the elastic modulus for the material is two times its shear modulus, then its bulk modulus is**

(a) Equal to elastic modulus
(b) One-third of elastic modulus
(c) Three times of elastic modulus
(d) Indeterminate

**Ans. (b) :** $E = 2G(1+\mu)$

Given,

$E = 2G$

$2G = 2G(1+\mu)$

$\mu = 0$

$E = 3K(1-2\mu)$

$E = 3K$

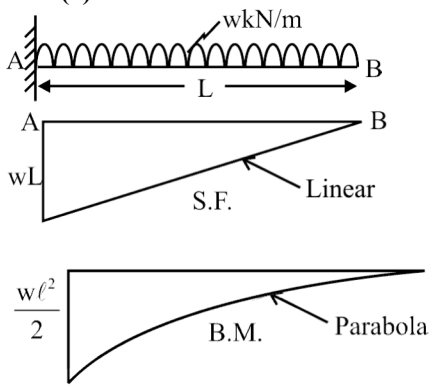
$K = \frac{1}{3}\times E$

Where, K = Bulk modulus
G = Shear modulus
E = Elastic modulus
$\mu$ = Poisson's ratio

**80. The shape of the shear force diagram for cantilever beam carrying a uniformly distributed load throughout its length is:**

(a) a straight line (b) a parabola
(c) a hyperbola (d) an ellipse

**Ans. (*):**




The shape of the shear fore diagram set cantilever beam carrying a uniformly distributed load throughout its length is triangle.

**81. A simply supported beam of rectangular cross-section 100 mm × 200 mm carrying a concentrated load at mid-span. If the maximum shear stress developed in the beam is 3 MPa, what is the magnitude of concentrated load?**

(a) 20 kN (b) 40 kN
(c) 80 kN (d) 160 kN

**Ans. (b) :**

$$\tau_{max} = \frac{3}{2}\tau_{avg}$$

$$3 = \frac{3}{2} \times \frac{P}{bd}$$

$$3 = \frac{3}{2} \times \frac{P}{200 \times 100}$$

$$P = 40\,KN$$

**82. The radius of Mohr's circle of stress is 35 MPa and minor principle stress is –20 MPa. The major principle stress is:**

(a) 15 MPa (b) 50 MPa
(c) 60 MPa (d) 95 MPa

**Ans. (b) :** $R = \tau_{max} = 30$ MPa, $\sigma_2 = -20$ MPa

$$R = \tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

$$35 = \frac{\sigma_1 - (-20)}{2}$$

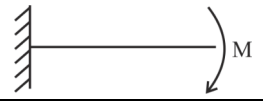
$$70 = \sigma_1 + 20$$

$$\sigma_1 = 50 \text{ MPa}$$

**83. A cantilever beam of span 'L' is subjected to a clockwise moment at free end. What is the ratio of deflection to rotation at the free end?**

(a) 2L (b) 3L/2
(c) L (d) L/2

**Ans. (d) :**




Deflection at free end $\delta = \frac{ML^2}{2EI}$

Rotation at free end $\theta = \frac{ML}{EI}$

$$\frac{\delta}{\theta} = \frac{\frac{ML^2}{2EI}}{\frac{ML}{EI}} = \frac{L}{2}$$

**84. A thin cylinder is subjected to an internal pressure of 40 MPa. If the longitudinal stress is 60 MPa, then the circumferential stress is**

(a) 120 MPa (b) 30 MPa
(c) 20 MPa (d) 10 MPa

**Ans. (a) :** $\sigma_L = \frac{Pd}{4t} = \frac{Pd}{2 \times 2t} = \frac{1}{2} \times \sigma_h$

$$60 = \frac{1}{2}\sigma_h \Rightarrow \sigma_h = 120 \text{ MPa}$$

**85. The inside diameter of a hollow shaft is one-third of its outside diameter. The ratio of its torque carrying capacity to that of a solid shaft of the same material and the same outside diameter is**

(a) 2/3 (b) 8/9
(c) 26/27 (d) 80/81

**Ans. (d) : Hollow shaft–**

$$D_i = \frac{D_o}{3}$$

$$\frac{T}{J} = \frac{\tau}{r}$$

$$T_1 = \frac{\tau \times \frac{\pi}{32}\left(D_o^4 - D_i^4\right)}{D_o/2} \quad \text{..... (i)}$$

**Solid shaft–**

$$T_2 = \frac{\tau \times \frac{\pi}{32} D_o^4}{D_o/2} \quad \text{..... (ii)}$$

$$\frac{T_1}{T_2} = \frac{D_o^4 - D_i^4}{D_o^4} \Rightarrow \frac{\left(D_o^4 - \frac{D_o^4}{81}\right)}{D_o^4} = 1 - \frac{1}{81} = \frac{80}{81}$$

**86. The buckling load for a column fixed at both ends is 50 kN. If the ends of same column are changed to hinged condition, then the buckling load will be:**

(a) 37.5 kN (b) 25 kN
(c) 12.5 kN (d) 100 kN

**Ans. (c) :** Given :

$(P_c)_1 = 50$ kN, $(P_c)_2 = ?$

$$(L_e)_1 = \frac{L}{2} \quad (L_e)_2 = L$$

$$\because \quad P_c \propto \frac{1}{L_e^2}$$

$$\therefore \quad \frac{(P_c)_2}{(P_c)_1} = \left(\frac{(L_e)_1}{(L_e)_2}\right)^2$$

$$\therefore \quad (P_c)_2 = (P_c)_1 \times \left(\frac{\frac{L}{2}}{L}\right)^2$$

$$\therefore \quad (P_c)_2 = 50 \times \frac{1}{4} = 12.5\,\text{kN}$$

**87. The unit of proof resilience are**

(a) $N/m^2$ (b) N-m

(c) $N/m^3$ (d) $N\text{-}m^2$

**Ans. (b) : Proof resilience–** Maximum energy absorbing capacity of a material in the elastic region is called proof resilience.

Unit of proof resilience are N-m

$$\text{Modulus of resilience} = \frac{\text{Proof resilience}}{\text{Volume of the body}}$$

**88. If the load does NOT passes through shear centre of the section of the beam, then there will be**

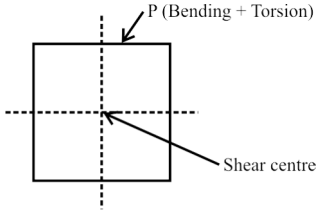
(a) No bending of the beam

(b) Only bending of the beam

(c) Only twisting of the beam

(d) Both bending and twisting of the beam

**Ans. (d) :** If the load does NOT passes through shear centre of the section of the beam, then there will be bending and twisting of the beam.




**89. A rod of length 1 m, area 300 sq.mm and self weight 6 kN is suspended vertically and is firmly fixed at the top. Modulus of elasticity is 200 GPa. What is the total elongation of the bar due to its self weight?**

(a) 1 mm (b) 0.1 mm

(c) 0.05 mm (d) 0.5 mm

**Ans. (c) :**

Total elongation of the bar due to its self weight

$E = 200\ \text{GPa} = 200 \times 10^3\ \text{MPa}$

$$= \frac{wL}{2AE} = \frac{6 \times 1000 \times 1000}{2 \times 300 \times 200 \times 10^3} = 0.05\ \text{mm}$$

**90. A short hollow cylindrical column has 200 mm external diameter and 100 mm internal diameter. What is the diameter of core for the column?**

(a) 62.5 mm (b) 25 mm

(c) 50 mm (d) 33.3 mm

**Ans. (a) :** Given :

$D = 200\ \text{mm}, d = 100\ \text{mm}$

$\because$ core diameter of hallow cylindrical column

$$= \frac{D^2 + d^2}{4D}$$

$$= \frac{(200)^2 + (100)^2}{4 \times 200} = 62.5\text{mm}$$

**91. The degree of redundancy of a fixed beam subjecting to vertical loads only is**

(a) One (b) Two

(c) Three (d) Four

**Ans. (b) :** For a general system of loading, for a fixed beam,–

- For end, the no unknowns are 3 which are horizontal reaction, vertical reaction and moment at the fixed end, So a total of 6 unknown–

We have 3 available equation of equilibrium which are $\sum F_x = 0$, $\sum F_y = 0$ and $\sum M_z = 0$,

Static Indeterminacy, E= Number of unknowns (R) – Number of equilibrium equations (r)

Static indeterminacy = 6 – 3 = 3

**But** For fixed beam loaded transversely or only vertical loading.

R = 4

r = 2

Static indeterminacy = 4 – 2 = 2

**92. The basic unknowns of slope deflection method are**

(a) Slopes of joints only

(b) Deflections of joints only

(c) Both slopes and deflections of joints

(d) Moments at joints

**Ans. (c) :** The basic unknowns of slope deflection method are both slopes and deflection of joints.

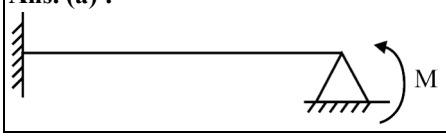
- The basis of the slope-deflection method lies in the slope-deflection equations, which relate-the end moment of each member in terms of end rotation and deflection other member.
- In nutshell, these equations represent the force displacement relationship which is one of three principles of structural analysis.

**93. Moment required to rotate prop end of a propped cantilever beam by 2 radians is 40 kNm. What will be the moment required to rotate the same end by 1 radian if length of the beam is doubled and the other end of the beam is changed to hinge?**

(a) 7.5 kNm (b) 20 kNm

(c) 10 kNm (d) 5 kNm

**Ans. (a) :**



$$M = \frac{4EI}{L}\theta$$

$$40 = \frac{4 \times 2}{L}EI$$

$$\frac{EI}{L} = 5$$

$$M = \frac{3EI}{L}\theta$$

$$M = \frac{3 \times 5}{2} \times 1$$

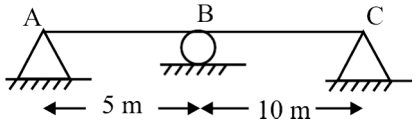
$= 7.5$ kN-m

**94. A two span simply supported continuous beam 'ABC' of constant flexural rigidity 'EI' is subjected to transverse loading. The span length of the members 'AB' and 'BC' are 5 m and 10 m respectively. What is the rotation factor at joint B for the member 'BA'?**

(a) 2/3 (b) 1/3

(c) –1/6 (d) –1/3

**Ans. (d) :**




$$\text{Rotation factor} = 0.5 \times \frac{k_i}{\sum k_i}$$

$$= -0.5 \times \frac{\frac{3}{4} \times \frac{I}{5}}{\frac{3}{4} \times \frac{I}{5} + \frac{3}{4} \times \frac{I}{10}}$$

$$= -0.5 \times \frac{3\frac{I}{20}}{9\frac{I}{40}} = -\frac{1}{3}$$

**95. A square beam of cross sectional size 100 mm and span 3 m is fixed at both the ends. Young's modulus of the beam material is 200 GPa. What is the area of analogous column for the beam?**

(a) $6 \times 10^{-4}$ m/kNm$^2$ (b) $18 \times 10^{-4}$ m/kNm$^2$

(c) $6 \times 10^{4}$ m/kNm$^2$ (d) $18 \times 10^{4}$ m/kNm$^2$

**Ans. (b) :** Given,

L = 3m, a = 100 mm E = 200 GPa

$$\text{Area of analogous column} = \frac{L}{EI}$$

$$= \frac{3}{200 \times 10^6 \times \frac{100^4}{12}}$$

$$= 18 \times 10^{-4} \text{ m/kNm}^2$$

**96. Which principle is used for finding the influence lines for determined and indeterminate structures?**

(a) Castigliano's theorem

(b) Principle of virtual work

(c) Muller-Breslau principle

(d) Theorem of least work

**Ans. (c) :** To get influence lines for statically indeterminate structure equations of static equilibrium must be augmented with deflection equations. Muller-Breslau's principles can be used to determine the influence line for indeterminate structure like beam of frame.

**97. A horizontal thrust 'H' is developed in a two-hinged semi-circular arch of radius 'R' carrying a concentration load 'W' at the crown. What will be the horizontal thrust if radius of the arch is doubled?**

(a) H (b) 2H

(c) H/2 (d) H/4

**Ans. (a) :** For two hinged semi-circular arches with load 'W' applied at any section.

$$\left[\text{Horizontal thrust} = H = \frac{W}{\pi}\right]$$

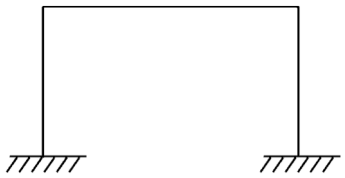
So, the horizontal thrust will remain same, not depend upon radius of the arch, so it is equal to H

**98. What is the size of stiffness matrix for a single-bay single-story rigid-jointed plane frame? Neglect axial deformations.**

(a) 6 (b) 5

(c) 4 (d) 3

**Ans. (d) :**



$D_s = 3$

Stiffeners matrix = Degree of static indeterminacy

Flexibility matrix

= Degree of kinematic indeterminacy.

**99. The anchorage value of 90º bend for the bar of 20 mm diameter is:**

(a) 80 mm (b) 160 mm

(c) 240 mm (d) 320 mm

**Ans. (b) :** Anchorage values of $90^0$ bend

$\phi = 20$ mm

Anchorage value = $8\phi$

$= 8 \times 20$

$= 160$ mm

**100. The minimum nominal concrete cover to all steel reinforcement for footings is**

(a) 20 mm (b) 30 mm

(c) 40 mm (d) 50 mm

**Ans. (d):**
Nominal cover for different member

| Member | Cover |
|---|---|
| Beam | 25 mm |
| Slab | 15mm |
| Column | 40 mm |
| Footing | 50 mm |

**101. What is the minimum area of tension reinforcement required for a concrete beam of width 200 mm, effective depth 500 mm and with Fe 250 grade steel?**
(a) 340 sq.mm
(b) 170 sq.mm
(c) 3400 sq.mm
(d) 1700 sq.mm

**Ans. (a) :** Given,
width of beam = 200 mm
effective depth = 500 mm

$$A_{st_{min}} = \frac{0.85bd}{f_y}$$

$$A_{st_{min}} = \frac{0.85 \times 200 \times 500}{250} = 340 \text{ mm}^2$$

**102. A reinforced concrete slab is 100 mm thick. The maximum diameter of reinforcement bar that can be used is:**
(a) 6 mm (b) 8 mm
(c) 12 mm (d) 14 mm

**Ans. (c) :** As per IS 456 : 2000
maximum dia of main bar in concrete slab

$$\phi_{main} = \frac{1}{8} \times \text{thickness}$$

$$\phi_{main} = \frac{1}{8} \times 100$$

$$\phi_{main} = 12.5 \approx 12\text{mm}$$

**103. The limiting value of depth of neutral axis for concrete beam of overall depth 300 mm and Fe 415 grade steel with effective cover 50 mm is:**
(a) 144 mm (b) 132 mm
(c) 115 mm (d) 120 mm

**Ans. (d) :** Limiting depth of neutral axis =

$$X_u, \lim = \frac{700}{0.87 f_y + 1100} \times d$$

$$X_u, \lim = \frac{700}{0.87 \times 415 + 1100} \times (300 - 50)$$

$X_u$, lim = 119.776 □ 120 mm

Or

We know that standard values of the ratio of limiting depth of neutral axis to the effective depth (k) of the beam for different steel sections as following–

| Grade of steel | Fe 500 | Fe 415 | Fe 250 |
|---|---|---|---|
| K value | 0.46 | 0.48 | 0.53 |

Effective cover = 50 mm
Over all depth D = 300 mm
Effective depth d = D – 50
300 – 50 = 250 mm
Limiting depth of neutral axis
$X_u = 0.48 \times d = 0.48 \times 250 = 120$ mm

**104. The unsupported length of column of size 150 mm × 300 mm is 3.0 m. What is the minimum eccentricity for which the column has to be designed?**
(a) 11 mm (b) 16 mm
(c) 20 mm (d) 25 mm

**Ans. (c) :** Minimum eccentricity of column–

$$e_{min} = \max. \text{ of } \begin{cases} \frac{\ell_e}{500} + \frac{LLD}{30} \\ \text{or} \\ 20\text{mm} \end{cases}$$

$$= \frac{3000}{500} + \frac{150}{30} = 6 + 5 = 11 \text{ mm less than 20 mm.}$$

So, $e_{min}$ = 20 mm.

**105. The structural steel cross-section which can develop plastic moment of resistance but have inadequate plastic hinge rotation capacity is classified as**
(a) Plastic (b) Compact
(c) Semi-compact (d) Slender

**Ans. (b) : Compact–** Cross-section which can develop plastic moment of resistance, but have inadequate plastic hinge rotation capacity for formation of plastic mechanism due to local buckling. • The width to thickness ratio of plate elements shall be less than that specified under class-2 (compact), but greater than that specified under class-1 (plastic) in table 2.1 of IS 800-207.

**106. The recommended limit of vertical deflection for gantry girder carrying EOT crane of capacity 50 tons is**
(a) Span/750 (b) Span/500
(c) Span/300 (d) Span/240

**Ans. (a) :** For an electrically-operated cranes upto 50 ton capacity, the maximum permissible deflection is given by Span/750.
**Deflection limit on gantry girder-**

| category | maximum deflection |
|---|---|
| Manually operated crane | Span/500 |
| Electrically operated crane upto 500kN capacity | Span/750 |
| Electrically operated crane more than 500 kN capacity | Span/1000 |

**107. A steel plate is 200 mm wide and 10 mm thick. A bolt of nominal diameter 16 mm is driven. The net effective sectional area of the plate is**

(a) 2000 sq.mm (b) 1840 sq.mm
(c) 1820 sq.mm (d) 1740 sq.mm

**Ans. (c) :** Given,
Width (B) = 200 mm
Thickness (t) = 10 mm
$A_{net} \Rightarrow (B–d)\ t$
d = dia of hole = d= 16+ 2 = 18 mm
$A_{net}$ = (B-d)t = (200–18) 10 = 1820 sq. mm

**108. The stiffener which prevents local crushing of the web of plate girder due to concentrated loading is:**
(a) Diagonal stiffener
(b) Intermediate transverse stiffener
(c) Load carrying stiffener
(d) Bearing stiffener

**Ans. (d) :** Bearing stiffeners are placed over the end bearing of welded plate girder and over intermediate bearing of continuous welded plate girders. They consist of two or more plates placed on both sides of the web.

**109. A compression member is subjected to an axial force of 2000 kN. The lacings shall be designed to resist a transverse shear of**
(a) 50 kN (b) 40 kN
(c) 30 kN (d) 20 kN

**Ans. (a) :** Lacing can be designed to resist a total transverse shear at any point in the member equal to 2.5% of axial force in member.
So, Axial force 2000 kN
Resist transverse shear force = 2.5% of axial force
$= 2.5\% \times 2000$
$= \frac{2.5}{100} \times 2000 = 50\ kN$

**110. A fillet weld of size 5 mm and total length 100 mm is used to connect two steel plates. What is the effective area of fillet weld?**
(a) 500 sq.mm (b) 315 sq.mm
(c) 350 sq.mm (d) 450 sq.mm

**Ans. (b) :** Given,
Weld size (s) = 5 mm
Total length (L) = 100 mm
effective length ($l_{eff}$) = L – 2s = 100 – 2 × 5 = 90 mm
effective area of filled weld = t × $l_{eff}$ = k.s × $l_{eff}$
$= 0.7 \times 5 \times 90$
$= 315\ mm^2$

**111. Which movements of the bridge deck will be permitted by an elastomeric bearing?**
(a) No movement is permitted
(b) Translation only
(c) Rotation only
(d) Both translation and rotation

**Ans. (d) : Elastomeric bearing for bridge structure.**
- It consist of elastomeric manufactured from synthetic or natural rubber and can take both translation and rotation movement through elastomer deformation.
- The ability of elastomer to carry large vertical loads is because of reinforcement provision that prevent lateral bulging of elastomer.

**112. The minimum grade of concrete to be used for pre-tensioned and post-tensioned prestressed concrete work respectively is**
(a) M 30, M 30 (b) M 40, M 40
(c) M 40, M 30 (d) M 30, M 40

**Ans. (c) :**

| **Minimum grade** | **Types of construction** |
|---|---|
| M-5, M-75 | Lean concrete base |
| M-15 | Plain cement concrete |
| M-20 | R.C.C. (General construction) |
| M-30 | R.C.C. in sea water and post tensioned |
| M-40 | pre-tensioned |

**113. The type of prestressed concrete member in which tensile stresses are allowed but no visible cracking is**
(a) Type 1 (b) Type 2
(c) Type 3 (d) Type-4

**Ans. (b) :**
- **Type-1**– Refers to a fully prestressed concrete and no tensile stresses.
- **Type-2**–To limit prestressing of concrete where tensile stresses do not exceed the cracking stresses of concrete.
- **Type-3**–The structure are those which allow tensile stresses to occur leading to limited crack widths and that too under extraordinary loading conditions which may occur early on the structure.

**114. Newton's law of viscosity for a fluid states that the shear stress is**
(a) Proportional to angular deformation
(b) Proportional to rate of angular deformation
(c) Inversely proportional to angular deformation
(d) Inversely proportional to rate of angular deformation

**Ans. (b) : Newton's law of viscosity –** It states that the shear stress ($\tau$) on a fluid element layer is directly proportional to the rate of shear strain or rate of angular deformation.

$$\tau = \mu \frac{du}{dy} = \mu \frac{d\theta}{dt}$$

$\frac{du}{dy}$ = velocity gradient

$\frac{d\theta}{dt} \Rightarrow$ Rate of shear strain or rate of angular deformation.

**115. A flow of a viscous fluid with $\mu = 1.0 \frac{N-S}{m^2}$ has a velocity distribution given by $u = 0.9y – y^2$. The shear stress at y = 0.45 m is**
(a) 0.90 N/m$^2$ (b) $\infty$
(c) Zero (d) – 0.90 N/m$^2$

**Ans. (c) :** Velocity distribution equation

$u=0.9y - y^2$

$\frac{du}{dy} = 0.9 - 2y, \qquad \mu = 1.0\frac{N-s}{m^2}$

$\left(\frac{du}{dy}\right) = 0.9 - 2\times 0.45 = 0$

$(\tau) = \mu\left(\frac{du}{dy}\right) = 1.0\times 0 = 0$

**116. When the barometer reads 740.0 mm of mercury, a pressure of 10 kPa suction at that location is equivalent to**

(a) 10.2 m of water (abs)
(b) 9.87 m of water (abs)
(c) 88.72 kPa (abs)
(d) 0.043 kPa (abs)

**Ans. (c) :** Atmospheric pressure at that location

$= 13.6 \times 1000 \times 9.81 \times 0.74$

$= 98.72$ kPa

Absolute pressure = $P_{atm} + P_{vacuum,\ suction}$

as given pressure is suction (i.e., vacuum)

So,

$P_{abs} = 98.72 - 10 = 88.72$ kPa

**117. The pressure represented by a column of 5 cm of oil of relative density 0.75 is**

(a) 7357.5 N/m$^2$ (b) 367.87 N/m$^2$
(c) 0.0225 N/m$^2$ (d) 3.75 N/m$^2$

**Ans. (b) :** $P = 0.75 \times 1000 \times 9.81 \times 0.05$

$= 367.875$ N/m$^2$ .

**118. If stream function $\psi = 2xy$, the magnitude of the velocity vector at (2, –2) is**

(a) $4\sqrt{2}$ (b) 4
(c) – 8 (d) $\sqrt{2}$

**Ans. (a) :** Given $\psi = 2xy$

$\frac{\partial\psi}{\partial x} = v \qquad v = 2y$

$\frac{\partial\psi}{\partial y} = -u \qquad u = -2x$

Velocity at (2,2)

$V = \sqrt{(2\times(-2))^2 + (-2\times 2)^2}$

$V = 4\sqrt{2}$

**119. A velocity potential exist**

(a) Whenever the real fluid flow exist
(b) When the flow is real and rotational
(c) When the flow satisfies the condition of irrotational motion
(d) When the flow satisfies the equations of continuity

**Ans. (c) :** If velocity potential function exists, the flow should be irrotational.

- If the velocity potential function satisfies the laplace equation.

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$$

- It is a case of steady incompressible irrotational flow.

**120. The total head in a flow is the sum of**

(a) Piezometric head and datum head
(b) Piezometric head and pressure head
(c) Piezometric head and velocity head
(d) Piezometric head, velocity head and datum head

**Ans. (c) :** Total head = Piezometric head + Velocity head $\left(\frac{V^2}{2g}\right)$

Piezometric head = Pressure head $\left(\frac{p}{\rho g}\right)$ + datum head (Z)

**121. A water jet has an area of 0.03 m$^2$ and impinges normally on a plate. If a force of 1 kN is produced as a result of this impact, the velocity of the jet in m/s is**

(a) 15 (b) 33.4
(c) 3.4 (d) 5.78

**Ans. (d) :** Area = 0.03 m$^2$

Force 'F' = 1 kN = 1000 N

$F = \rho AV^2$

$1000 = 1000 \times 0.03 \times V^2$

$V^2 = \sqrt{\frac{1000}{30}}$

V= 5.78 m/s

**122. If Froude law of similitude exists between a model and a prototype, then the force ratio ($F_r$) is**

(a) $L_r^3$ (b) $L_r\rho_r$
(c) $L_r^3\,\rho_r$ (d) $L_r^3\rho_r^{-1}$

**Ans. (c) :** Froude model law will be applicable in this case.

$$F_{r_m} = \frac{V_m}{\sqrt{gL_m}} = \frac{V_p}{\sqrt{gL_p}} = F_{r_p}$$

if $\quad L_r = \frac{L_m}{L_p}$

$$V_r = \frac{V_m}{V_p} = \sqrt{L_r}$$

$$\frac{(Force)_m}{(Force)_p} = \rho_r L_r^2 V_r^2$$

$\Rightarrow \quad \frac{(Force)_m}{(Force)_p} = \boxed{L_r^3\rho_r}$

**123. The pressure drop in an 8 cm diameter horizontal pipe is 75 kPa in a distance of 15 m. The shear stress at the pipe wall, in kPa is**

(a) 0.1 (b) 2.0

(c) 5.0 (d) 0.4

**Ans. (a) :** The wall shear stress (z) =

$$\tau = -\frac{\partial p}{\partial t} \times \frac{R}{2}$$

$$= -\left(\frac{75}{15}\right) \times \frac{4\times10^{-2}}{2}$$

$$= \frac{5}{50}$$

$\tau = 0.1$ kPa

**124. A laminar boundary layer has a velocity distribution given by $\frac{u}{U} = \frac{y}{\delta}$. The displacement thickness δ* for this boundary layer is**

(a) δ

(b) δ/2

(c) δ/4

(d) δ/6

**Ans. (b) :**

$\frac{u}{v} = \frac{y}{\delta}$ and

Displacement thickness = $\int_0^{\delta}\left(1-\frac{u}{v}\right)dy$

$$= \int_0^{\delta}\left(1-\frac{y}{\delta}\right)dy$$

$$= \left[y - \frac{y^2}{2\delta}\right]_0^{\delta}$$

$$= \left[\delta - \frac{\delta^2}{2\delta}\right]$$

$$= \frac{\delta}{2}$$

**125. Separation of boundary layer takes place when**

(a) $\left(\frac{\partial u}{\partial y}\right)_{y=0} > 0$

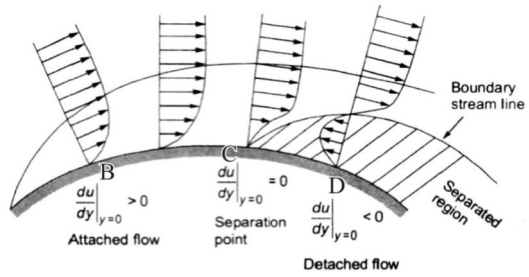
(b) $\left(\frac{\partial^2 u}{\partial^2 y}\right)_{y=0} > 0$

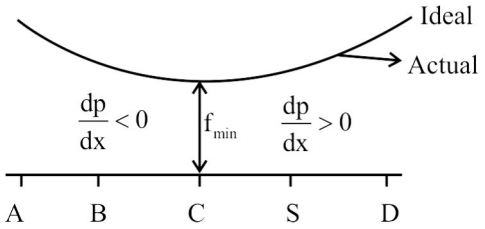
(c) $\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$

(d) $\left(\frac{\partial u}{\partial y}\right)_{y=\delta} > 0$

**Ans. (c) :** Due to this the boundary layer no more sticks to the boundary but is shifted away from the boundary, this phenomenon is called as "Boundary layer separation."

• An adverse pressure gradient is on which pressure increase in the flow direction $\left(i,e\frac{dp}{dx} > 0\right)$




At the verge of separation $\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$

$\therefore$ Shear stress $\tau = \mu\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$

**126. In a pipe flow the shear velocity $u_*$ is related to friction factor f and mean velocity V as $\frac{u_*}{V}$ =**

(a) $\sqrt{\frac{f}{8}}$ (b) $\sqrt{\frac{8}{f}}$

(c) $\sqrt{\frac{8g}{f}}$ (d) $\frac{1}{\sqrt{f}}$

**Ans. (a) :** Friction factor 'f' = $\frac{8\tau_0}{\rho V^2}$

Shear velocity , $u_* = \sqrt{\frac{\tau_0}{\rho}}$

$$f = \frac{8}{V^2}u_*^2$$

$$\frac{u_*^2}{V^2} = \frac{f}{8}$$

$$\frac{u_*}{V} = \sqrt{\frac{f}{8}}$$

**127. A triangular channel with side slopes of 1.5 horizontal : 1 vertical carries a discharge of 2.0 $m^3/s$. The critical depth for this flow is**

(a) 1.129 m (b) 0.816 m

(c) 0.742 m (d) 0.711 m

**Ans. (b) :** Given,

$m = \tan\theta = \frac{1.5}{1} = 1.5$

Discharge = 2 $m^3$/sec

Critical depth ($y_c$) = ?

$$y_c = \left[\frac{2Q^2}{g.\tan^2\theta}\right]^{1/5}$$

$$= \left[\frac{2\times 2^2}{9.81\times(1.5)^2}\right]^{1/5} = 0.816\text{m.}$$

**128. In a hydraulic jump occurring in a horizontal rectangular channel, the sequent depths are 0.25 m and 1.25 m. The energy loss in this jump is**

(a) 0.8 m (b) 1.0 m
(c) 1.25 m (d) 1.50 m

**Ans. (a) :** $y_1 = 0.25$m
$y_2 = 1.25$ m

Energy loss due to hydraulic jump $'E_v' = \frac{(y_2 - y_1)^3}{4y_1y_2}$

$$= \frac{(1.25-0.25)^3}{4\times 1.25\times 0.25}$$

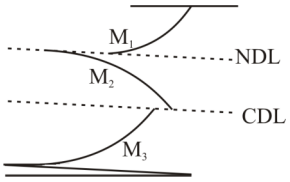
$$= \frac{1^3}{1.25} = \frac{1}{1.25}$$

$$= 0.8 \text{ m}$$

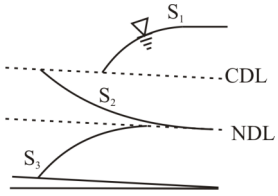
**129. The flow will be supercritical state in the following profiles:**

(a) $M_3$, $S_3$ and $M_1$ (b) $M_2$, $S_1$ and $M_3$
(c) $S_2$, $S_3$ and $M_3$ (d) $S_1$, $S_2$ and $S_3$

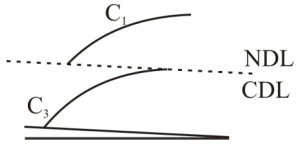
**Ans. (c) :** In super critical state of flow, the depth of flow is below the critical depth line.
flow profile for various sections–
(a) Mild slope




(b) steep slope




(c) critical slope

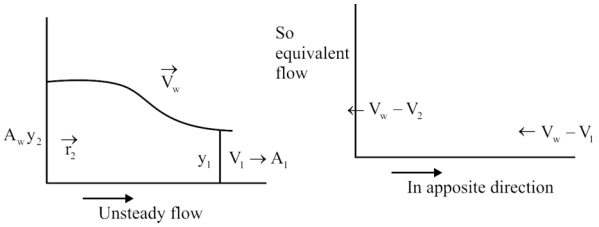



Therefore $M_3$, $S_2$, $S_3$, $C_3$ are flow profiles for super critical state of flow.

**130. A positive surge travels upstream in a canal with an absolute velocity $V_w$. With suffixes 1 and 2 referring to sections upstream and downstream of the surge respectively, the continuity equation is written as**

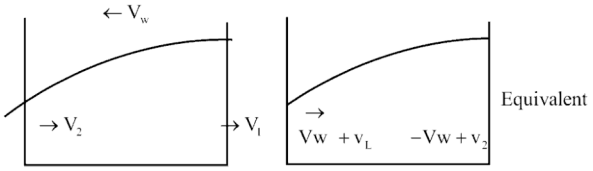
(a) $A_1V_1 = A_2V_2$
(b) $A_1(V_1 + V_w) = A_2(V_2 - V_w)$
(c) $A_2(V_w - V_2) = A_1(V_w + V_1)$
(d) $A_1(V_1 + V_w) = A_2(V_2 + V_w)$

**Ans. (d) : Positive surge moving downstream–** A positive surge moving with absolute velocity $V_w$ in downstream direction. Assuming a person standing at the bank of channel also moves with velocity $V_w$ in the same direction and if $V_w$ is more than $V_1$ and $V_2$ both, he will observe as the section 1 and 2 are moving with velocity $(V_w - V_1)$ and $(V_w - V_2)$ respectively towards upstream direction and the surge is stationary.




Appling continuity equation and momentum for the transformed steady state flow.
$(V_w - V_2)\, A_2 = (V_w - V_1)A_1$
$f_1 - f_2 = PQ\,(V_2 - V_1)$
So, $\gamma A_1\bar{y}_1 - \gamma A_2\bar{y}_2 = \rho(V_w - V_1)A_1(V_w - V_1)$

**Positive surge moving upstream–**




**From continuity equation–**
$(V_w + V_2)\, A_2 = (V_w + V_1)A_1$
$(f_1 - f_2) = PQ\,(V_2 - V_1)$

**131. An aircraft moves at 1580 Km/h in an atmosphere where the temperature is –60°C. If k = 1.4 and R = 287 J/(kg-K) the Mach number of the plane is:**

(a) 0.67 (b) 1.50
(c) 2.10 (d) 5.4

**Ans. (b) :**
V = 1580 km/h = 438.88 m/s
$T = -60^0C = 213$ k
Velocity of sound $C = \sqrt{kRT}$

$$= \sqrt{1.4\times 287\times 213} = 292.54$$

Now mach no. $= \frac{V}{C} = \frac{438.88}{292.54} = 1.50$

**132. The unit power $P_u$ of a turbine developing a power P under a head H is equal to**

(a) $\frac{P}{H^{5/2}}$ (b) $\frac{P}{\sqrt{H}}$

(c) $P.H^{3/2}$ (d) $\frac{P}{H^{3/2}}$

**Ans. (d) : For turbine –**

Unit power, $P_u = \frac{P}{H^{3/2}}$

Unit discharge, $Q_u = \frac{Q}{\sqrt{H}}$

Unit speed, $N_u = \frac{N}{\sqrt{H}}$

**133. If two pumps identical in all respects and each capable of delivering a discharge 'Q' against a head 'H' are connected in parallel, the resulting discharge is**

(a) 2Q against a head 2H
(b) 2Q against a head H
(c) Q against a head 2H
(d) Q against a head h

**Ans. (b) : Two pump in parallel–**

- Head against pumps are working remain the same.
- Discharge will be the sum of the discharge in an individual pump.
- Resulting discharge for two pumps = 2Q and total head = H for parallel arrangement.

**134. The science which deals with occurrence, distribution and circulation of water is called**

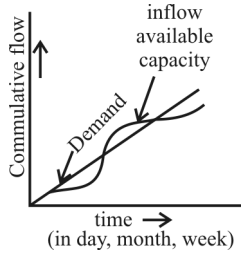
(a) Hydrography
(b) Hydrometry
(c) Hydrology
(d) Hydraulics

**Ans. (c) :** Hydrology is the science which deals with the occurrence, circulation and distribution of water on earth and its atmosphere the movement of water from one phase i.e liquid or gaseous to another phase is known as hydrological cycle.

**135. The rainfall mass curve shows the variation of**

(a) Rainfall intensity with time
(b) Rainfall intensity with cumulative rainfall
(c) Rainfall excess with time
(d) Cumulative rainfall with time

**Ans. (d) :** The rainfall mass curve shows the variation of cumulative rainfall with time.




| Type of curve | Curve between |
|---|---|
| Mass curve | Cumulative rainfall v/s time |
| Hyetograph | Rainfall intensity v/s time |
| Hydrographs | Discharge v/s time |

**136. The Theissen weights of 4 rain-gauges A, B, C and D covering a river basin are 0.15, 0.25, 0.30 and 0.30 respectively. If the average depth of rainfall for the basin is 5 cm and the rainfalls recorded at B, C and D are 5 cm, 4 cm and 5 cm respectively, what is the rainfall at A?**

(a) 5 cm (b) 6 cm
(c) 7 cm (d) 8 cm

**Ans. (c) :** Here,

$\frac{A_1}{A} = 0.15$ and $P_1$= ?

$\frac{A_2}{A} = 0.25$ and $P_2$= 5 cm

$\frac{A_3}{A} = 0.30$ and $P_3$= 4 cm

$\frac{A_4}{A} = 0.30$ and $P_4$ = 5 cm

The average depth of precipitation for the basin is given by

$$P_{avg} = P_1 \times \left(\frac{A_1}{A}\right) + P_2 \times \left(\frac{A_2}{A}\right) + P_3 \times \left(\frac{A_3}{A}\right) + P_4 \times \left(\frac{A_4}{A}\right)$$

$$5 = X \times 0.15 + 5 \times 0.25 + 4 \times 0.30 + 5 \times 0.30$$

$$\boxed{X = 7\text{cm}}$$

**137. A 6 hr storm with hourly intensities of 7, 18, 25, 12, 10 and 3 mm/hr produced a runoff of 33 mm. Then the ϕ-index is**

(a) 7 mm/hr
(b) 3 mm/hr
(c) 10 mm/hr
(d) 8 mm/hr

**Ans. (d) :**

Total rainfall = 7 + 18 + 25 + 12 + 3 + 10 = 75 mm

Effective rainfall = 33 mm

Infiltration = 75–33 = 42

$t_e$ = 6 hrs

$\phi$– index $= \frac{42}{6} = 7$ mm/hr

But 7 mm/hr ≥ Rainfall intensity of 7 mm/h and 3 mm per hour.

So $\phi$ index calculated by neglected 7 mm and 3mm

$$\phi \text{ index} = \frac{18+25+10+12-33}{4} = \frac{65-33}{4} = 8\,\text{mm/h}$$

**138. A 6 hr storm with a uniform intensity of 1.5 cm/hr produced a runoff depth of 72 mm. The average infiltration rate during this storm is**

(a) 3 mm/hr (b) 6 mm/hr
(c) 9 mm/hr (d) 12 mm/hr

**Ans. (a):** The average infiltration rate–

$$w = \frac{P - R - \ell_a}{t}$$

$P = 1.5 \times 10 \times 6 = 90$ mm

$R = 72$ mm

$t = 6$ hr

$\ell_a = 0$

$$= \frac{90 - 72 - 0}{6} = \frac{18}{6} = 3 \text{ mm/hr}$$

Where, P = Total storm precipitation depth

R = Total storm runoff depth

t = Time of rainfall excess

$\ell_a$= initial losses in terms of depth.

**139. The dimensions of transmissibility (T) of the aquifer are:**

(a) $LT^{-1}$ (b) $L^2T^{-1}$

(c) $L^2T^2$ (d) $L2T^{-2}$

**Ans. (b) :** The transmissibility (T) of the aquifer is the product of the coefficient of permeability and the thickness of the aquifer (T) = k.b.

- The dimensions of transmissibility (T) of the aquifer are $L^2T^{-1}$ .

**140. The elevation of water table in an confined aquifer at two locations separated by a distance of 100 m is 1028.2 m and 1027.0 m respectively. If the permeability of the aquifer is 12 m/day and porosity is 15 percent, what is the actual velocity of flow in the aquifer?**

(a) 0.66 m/day

(b) 0.45 m/day

(c) 0.96 m/day

(d) 0.85 m/day

**Ans. (c) :**

$$V = ki = 12 \text{ m/day} \times \frac{1028.2 - 1027}{100}$$

$$= \frac{12\text{m}}{\text{day}} \times \frac{102}{100}$$

$$V_{actual} = \frac{V \times 100}{n}$$

$$= \frac{100}{15} \times \frac{12 \times 1.2}{100}$$

$$= 0.96 \text{ m/day}$$

**141. The term unit in the unit hydrograph refers to:**

(a) Unit depth of runoff

(b) Unit duration of the storm

(c) Unit base period of the hydrograph

(d) Unit area of the basin

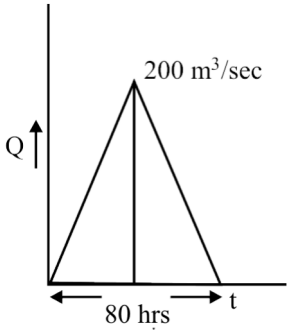
**Ans. (a) :** A unit hydrograph is defined as the hydrograph of direct runoff resulting from one unit depth (1 cm) of rainfall excess occurring uniformly over the basin and at a uniform rate for a specified duration.

**142. The direct runoff hydrograph of a basin can be approximately as a triangle with a base period of 80 hr and peak flow of 200 $m^3/s$ occurring at 16th hour. If the area of the basin is 1440 $km^2$, what is the depth of runoff indicated by the hydrograph?**

(a) 1 cm (b) 10 cm

(c) 2 cm (d) 20 cm

**Ans. (c) :**




Volume = Area × depth

$$= \frac{1}{2} \times 80 \times 200 \times 3600 = d \times 1440 \times 10^6$$

$d = 0.02$ m

$d = 2$ cm

**143. In Gumbel's distribution the relation between the reduced variate y and the return period Tr is given by**

(a) $e^{-e^{-y}} = 1 - \frac{1}{T_r}$ (b) $e^{-e^{-y}} = \frac{1}{T_r}$

(c) $e^{-e^{-y}} = 1 + \frac{1}{T_r}$ (d) $e^{-e^{-y}} = \frac{1}{T_r} - 1$

**Ans. (a) :** E.J. Gumbel's in 1941 was the first to consider that annual flood peaks are extreme values of floods in each of the annual series of recorded flood. Hence, floods should follow the extreme value distribution.

Return period = $T_r$year

Reduced variable = y

Naperian logarithm = e

$$P = -e^{-e^{-y}}$$

$$P = \frac{1}{T_r}$$

$$\frac{1}{T_r} = 1 - e^{-e^{-y}}$$

**144. Rational formula is used to arrive at the designated peak flow of Qs. If a storm intensity is doubled and the duration is increased three times, the resulting peak discharge is**

(a) P (b) 2 Qp

(c) 3 Qp (d) $\frac{Qp}{2}$

**Ans. (b):** Peak discharge = QP = $CI_iA$
Where, C = runoff coefficient
$I_i$ = Average rainfall intensity (in/hr)
A = Area
$Q_p \alpha I_i$ (If Ii is increased 2 time)
Then,
Peak discharge increased 2 time $2Q_p$.

**145. For the irrigation of a crop, the base period B in days and delta (Δ) in metres are related to duty D in ha/($m^3$/sec) at the field as**

(a) D = [0.864 B/Δ]
(b) D = [8.64 B/Δ]
(c) D = [0.864 Δ/B]
(d) Δ = [8.64 D/B]

**Ans. (b) :** Base period in number of days for which irrigation is required and Δ is depth of water required during period and D is the irrigating capacity of the water in hectare /cumec.

D → Duty in hectare per cumec

Δ → Total depth of water supplied

B → Base period in days

Total volume of water in D hectare of land = D × Δ hectare metre

$D \times \Delta \times 10^4\ m^3$ ..... (i)

If 1 cumec water is discharge for B days, then volume of water

$= 1 \times 24 \times 60 \times 60\,B$ cubic m..... (ii)

From equation (i) and (ii)

$$D \times \Delta \times 10^4 = 1 \times 24 \times 60 \times 60\,B$$

$$\Delta = \frac{1 \times 24 \times 60 \times 60 \times B}{10^4 \times D}$$

$$\boxed{\Delta = \frac{8.64B}{D}\text{ metre}} \text{ or } \frac{864B}{D}\text{ cm}$$

- 1 hectare = $10^4$ sq. meters
- 1 cumec - day = 8.64 hectare - meters

**146. The centre of pressure of wave pressure due to a wave of height $h_w$ acting on a gravity dam will be at a height above the maximum still water level of**

(a) $h_w/2$ (b) $3\,h_w/8$
(c) $h_w/3$ (d) $2h_w/3$

**Ans. (b) :** The centre of pressure of wave pressure due to a wave of height $h_w$ acting on a gravity dam will be at a height above the maximum still water level of $\frac{3}{8}h_w$.

**147. Provision of a filter and a rock toe in an earthen dam is done**

(a) to prevent the piping action in the dam section.
(b) to collect and drain out the seeping water.
(c) to reduce the seepage velocity.
(d) to have the resultant in the middle third of the base

**Ans. (b) :** Provision of a filter and a rock toe in an earthen dam is done to collect and drain out the seeping water.

**148. If the head of water over an Ogee spillway is less than the design head,**

(a) the pressure on the spillway crest will be negative
(b) the coefficient of discharge will be less than the design coefficient of discharge
(c) the coefficient of discharge will be larger than the design coefficient of discharge.
(d) the cavitation phenomenon will occur.

**Ans. (b) :** If the head of water over an Ogee spillway is less than the design head the coefficient of discharge will be less than the design coefficient of discharge.

**149. A stable channel has three degrees of freedom in a sense that the depth, width and slope can adjust to the given discharge and sediment condition. In this sense which of the following methods of canal design gives a true regime channel for silt laden flow in fine alluvium?**

(a) Kennedy's method
(b) Tractive force method
(c) Lacey's method
(d) Manning's formula and Kennedy's equation

**Ans. (c) :** True regime condition of Lacey's theory
- discharge is constant
- flow is uniform
- silt charge and silt grade is constant
- Soil is incoherent alluvium.

**150. The following data are available for a cross drainage project:**

| Item | Canal | Drainage |
|---|---|---|
| FSL/HFL | 105.00 m | 104.00 m |
| Bed level | 100.00 m | 102.00 m |
| Discharge | 80 $m^3$/s | 12 $m^3$/s |

**The most appropriate cross drainage works for this situation is:**

(a) Aqueduct
(b) Syphon aqueduct
(c) Syphon
(d) Super passage

**Ans. (c) :** The most appropriate cross drainage works for this situation is siphon.
$FSL_{canal} > FSL_{drain}$
$\text{Bed level}_{canal} < \text{Bed level}_{drain}$

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer (Mains.) Exam, 2016
## Solved Paper

**1. The resultant of two perpendicular forces each equal to P/2 will be equal to**

(a) $\frac{P}{\sqrt{2}}$ (b) $\sqrt{2}P$

(c) 2P (d) $\sqrt{2}P$

**Ans. (a) :** Given,

$\theta = 90^o$

$P = Q = \frac{P}{2}$

$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$

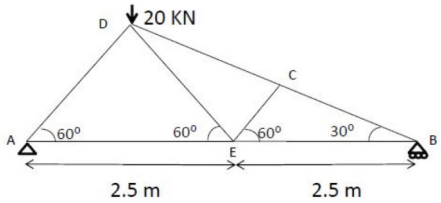
$= \sqrt{\left(\frac{P}{2}\right)^2 \times \left(\frac{P}{2}\right)^2 + \frac{2\times P}{2} \times \frac{P}{2} \times \cos 90^o}$

$= \sqrt{\frac{P^2}{4} + \frac{P^2}{4}}$

$= \sqrt{\frac{2P^2}{4}} = \sqrt{\frac{P^2}{2}}$

$= \frac{P}{\sqrt{2}}$

**2. The axial forces in the members CE and DE of the plane truss shown in the figure below are equal to**




(a) 10, 10 in KN (b) 20, 20 in KN

(c) 8.7, 8.7 in KN (d) 0, 0

**Ans. (d) :** In a three member force system of two members are parallel to each other then the force in a third member is zero.

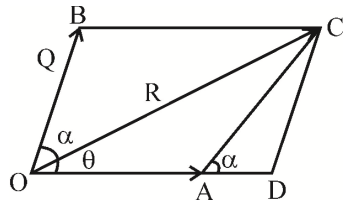
- DC and CB is collinear so force in member CE = 0
- AE and EB is collinear so force in member ED = 0

**3. If two concurrent forces P and Q having an angle α between them, then the direction of their resultant defined by angle θ with reference to the directions of P is given by tan θ which is equal to**

(a) $\frac{Q\sin\alpha}{P + Q\cos\alpha}$ (b) $\frac{P\sin\alpha}{Q + P\cos\alpha}$

(c) $\frac{P\sin\alpha}{Q + Q\cos\alpha}$ (d) $\frac{Q\sin\alpha}{P + P\cos\alpha}$

**Ans. (a) :** Parallelogram Law of forces– This Law is used for finding the resultant of two forces acting at a point.




$R\sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$

$\tan\theta = \frac{Q\sin\alpha}{P + Q\cos\alpha}$

**4. The maximum angle of inclination of the plane (with horizontal) at which a body remains in equilibrium under the action of friction only is termed as**

(a) Angle of wedge
(b) Angle of equilibrium
(c) Angle of repose
(d) All the given answers

**Ans. (c) : Angle of repose–** The maximum angle of inclination of the plane (with horizontal) at which a body remains in equilibrium under the action of friction only is termed as angle of repose.

**5. If the number of members (m) are more than that required by the equation m = 2j – 3 (j is number of joints) such frames are called**

(a) Redundant frame
(b) Weak frame
(c) Perfect frame
(d) Ideal frame

**Ans. (a) :** m = 2j – 3 (Perfect frame)
m > 2j – 3 (Imperfect redundant frame)
m < 2j – 3 (Deficient frame)

**6. A compound bar having two members X and Y of length L when subjected to tensile force P would have elongation equal to (in usual notations)**

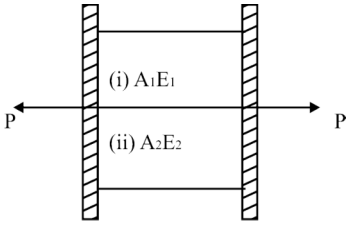
(a) $\frac{PL}{A_X E_X + A_Y E_Y}$

(b) $\frac{PL}{A_X E_X} + \frac{PL}{A_Y E_Y}$

(c) $\frac{PL}{A_X E_X - A_Y E_Y}$

(d) $\frac{PL}{A_X E_X} - \frac{PL}{A_Y E_Y}$

**Ans. (a):**




$P_1 + P_2 = P$ .....(i)

For composite section–

$$\Delta_1 = \Delta_2$$

$$\frac{P_1 L}{A_1 E_1} = \frac{P_2 L}{A_2 E_2}$$

$$P_2 = \frac{P_1 A_2 E_2}{A_1 E_1}$$

From equation (i)

$$P_1 + \frac{P_1 (A_2 E_2)}{A_1 E_1} = P$$

$$P_1 = \frac{P A_1 E_1}{A_1 E_1 + A_2 E_2}$$

$$P_2 = \frac{P A_2 E_2}{A_1 E_1 + A_2 E_2}$$

$$\Delta_1 = \Delta_2 = \Delta = \frac{P_1 L}{A_1 E_1} = \frac{P_2 L}{A_2 E_2}$$

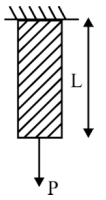
$$\Delta = \frac{PL}{A_1 E_1 + A_2 E_2}$$

**7. A vertical prismatic bar fixed at the top end and loaded with P at the bottom free end, is having a unit weight of w and its length is L. If σ is the working stress, the safe sectional area for the bar when P and self weight are considered is given by**

(a) $\frac{P}{\sigma + wL}$ (b) $\frac{P}{\sigma - wL}$

(c) $\frac{\sigma + wL}{P}$ (d) $\frac{\sigma - wL}{P}$

**Ans. (b) :**




Safe cross section area = $\frac{P}{\sigma - wL}$

**8. If temperature changes from $t_0$ to t in a simple bar with both end constrained and having modulus of elasticity E and thermal coefficient α, the thermal stress due to the temperature rise is**

(a) $\frac{E(t - t_0)}{\alpha}$ (b) $\alpha \frac{(t - t_0)}{E}$

(c) $E\alpha(t - t_0)$ (d) $\frac{E\alpha}{(t - t_0)}$

**Ans. (c) :** when a bar of length ($l$) is subjected to an increase in temp(t) then the increase in length when the bar is free to expand is given by-

$\delta\ell = \ell\alpha t$

α = coeff. of linear expansion

If the ends of the bar are fixed to rigid supports so that its expansion is prevented, then compressive thermal strain induced in the bar,

$$\varepsilon = \frac{\delta\ell}{\ell} = \ell\alpha(t - t_0)$$

$$= \alpha(t - t_0)$$

And thermal stress

$= \varepsilon E$

$= E\alpha(t - t_0)$

**9. A circular solid uniform metal of cross section area A and length L is hanging vertically from its upper end. If the bar is having weight of W and modulus of rigidity E, the total elongation of the bar due to its own weight will be**

(a) $\frac{2WL}{AE}$ (b) $\frac{WL}{AE}$

(c) $\frac{WL}{2AE}$ (d) $\frac{WL}{3AE}$

**Ans. (c) :** Deformation in the bar due to self weight is half of deformation due to applied load equal to self weight.

$$\delta = \frac{w\ell}{2AE}$$

(a) Prismatic bar–

$$\delta\ell = \frac{\gamma L^2}{2E}$$

(b) Conical bar

$$\delta\ell = \frac{\gamma L^2}{6E} = \frac{1}{3} \times \text{ Deflection of prismatic bar}$$

Here, γ = specific weight

L = Length of bar

E = Young's modulus

Deflection ($\delta l$) of tapered bar

$$\delta\ell = \frac{4WL}{\pi E D_1 D_2}$$

Here,

W = Load Applied

D = diameter

L=Length of bar

Deflection of rectangular tapering bar

$$\delta\ell = \frac{PL \log_c \left(\frac{B_2}{B_1}\right)}{E.t(B_1 - B_2)}$$

Here,

t = thickness

W = Load, applied

E = Young's modulus

**10. Volumetric strain of a sphere of diameter increasing from $d_1$ to $d_2$ is equal to**

(a) $\frac{d_2 - d_1}{d_1}$ (b) $3\frac{(d_2 - d_1)}{d_1}$

(c) $3\frac{d_2 + d_1}{2d_1}$ (d) $\frac{d_2 + d_1}{2d_1}$

**Ans. (b) :** Volumetric strain of a sphere of diameter increasing from $d_1$ to $d_2$ is equal to

$$= \frac{3(d_2 - d_1)}{d_1}$$

**11. If a circular rod having Poisson's ratio μ is subjected to an axial pull and if the strain is e, then the lateral strain is equal to**

(a) $-\mu e$ (b) $\mu e$

(c) $-e/\mu$ (d) $e/\mu$

**Ans. (a) :** $\mu = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$

$$\mu = \frac{-\varepsilon_y}{\varepsilon_x} = \frac{-\varepsilon_z}{\varepsilon_x}$$

Lateral strain

$$\varepsilon_y = \varepsilon_z = \frac{-\mu\sigma x}{E}$$

$$\varepsilon_y = \varepsilon_z = -\mu e$$

**12. The maximum tangential stress acting on a plane inclined to the direction of the axial stress (p) in a tensile bar is equal to**

(a) p (b) 2p

(c) p/2 (d) 3p/2

**Ans. (c) :** $\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$

$$= \frac{p - 0}{2}$$

$$\tau_{max} = \frac{p}{2}$$

**13. If an element in a body is in equilibrium under shearing stress only, such a state of stress is called**

(a) Pure rotation (b) Pure shear

(c) Pure bending

(d) None of the given answers

**Ans. (b) :** If an element in a body is in equilibrium under shearing stress only, such, state of stress is called pure shear.

**14. A steel bar is heated from 20ºC to 35ºC and it is free to expand. Then bar will have**

(a) No stress (b) Tensile stress

(c) Compressive stress (d) Shear stress

**Ans. (a) :** If a material expands or contract freely due to heating or cooling then no stress will develop in material.

**15. If a solid steel sphere of 10,000 $cm^3$ subjected to a uniform hydrostatic pressure of 700 × $10^4$ kg(f)/sq.m, then decrease in volume of the sphere for 1/K = 6×$10^{-11}$ (kg(f)/sq.m)$^{-1}$**

(a) 2.4 $cm^3$ (b) 8.4 $cm^3$

(c) 4.2 $cm^3$ (d) 1.2 $cm^3$

**Ans. (c) :** V = 1000 $cm^3$

P = 700 × $10^4$ kg(f)/$m^2$

$$\frac{1}{k} = 6\times10^{-11}\left(\text{kg(f)}/\text{m}^2\right)^{-1}$$

$$k = \frac{1}{6}\times10^{11}\left(\frac{\text{kgf}}{\text{m}^2}\right)$$

$$\varepsilon_v = \left(\frac{-P}{k}\right)$$

$$\frac{\Delta V}{V} = \left(\frac{7\times10^6}{1/6\times10^{11}}\right)$$

$$\Delta v = \frac{100\times1000}{10^4/6} = 4.2\,\text{cm}^3$$

**16. Shear force at a section of a horizontal beam is the**

(a) algebraic sum of the vertical forces to any one side of the section

(b) algebraic sum of the vertical forces on both sides of the section

(c) algebraic sum of the moments of all forces on any one side of the section

(d) algebraic sum of the moments of all forces on both sides of the section

**Ans. (a) :** Shear force at a section of a horizontal beam is the algebraic sum of the vertical force to any one side of the section.

**17. For UDL (uniformly distributed load) on a beam, the bending moment diagram (BMD) shows**

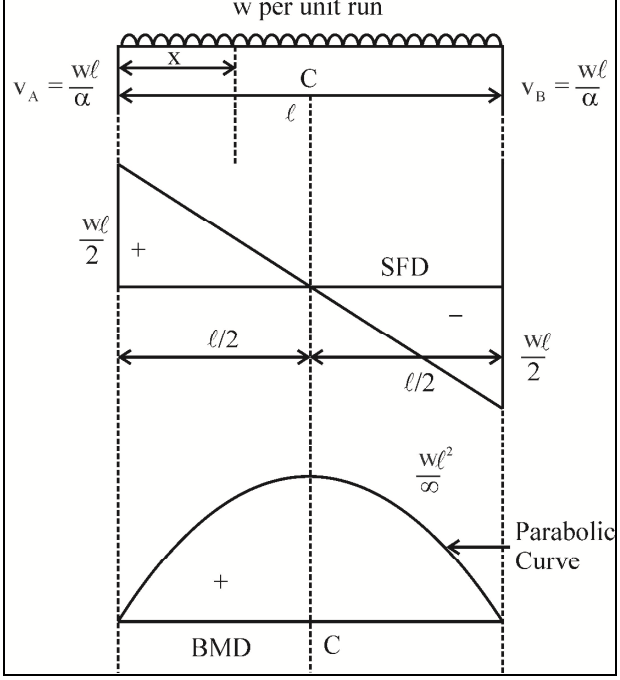
(a) Linear variation

(b) Parabolic (curve) variation

(c) Rectangle or constant value

(d) Trapezoidal variation

**Ans. (b) :** Simply supported beam carrying a uniformly distributed of 'w' per unit run over the whole span.



$$S.F_x = \frac{+w\ell}{2} - w_x$$

$$BM_x = \frac{+w\ell}{\alpha} - \frac{w_x^2}{2}$$

$$B.M_{max} = m_c = \frac{w\ell^2}{8}$$

**18. Bending moment is maximum at a section of a beam where**

(a) Shear force changes from –ve to +ve
(b) Shear force is zero
(c) Shear force changes from +ve to –ve
(d) All the given answers

**Ans. (d) :** Bending moment is maximum at a section of a beam where-
: Shear force changes from –ve to +ve.
: Shear force is zero.
: Shear force changes from +ve to –ve

**19. Which of the following is giving correct relation between load (w), shear force (F) and bending moment (M)**

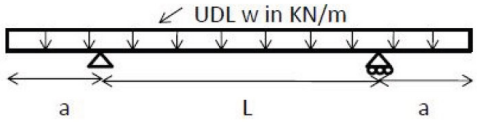
(a) $M = \frac{dF}{dx}$ (b) $F = \frac{dM}{dx}$

(c) $w = \frac{dM}{dx}$ (d) $M = \frac{dw}{dx}$

**Ans. (b) :** Correct relation between load(w) shear force (F) and bending moment (m).

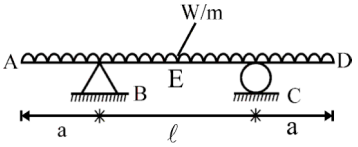
$$\boxed{F = \frac{dM}{dx}}$$

**20. A beam with equal overhangs is carrying UDL as shown in the following figure. The bending moment at the centre of the beam will be zero for the condition**




(a) a = L/3 (b) a = L/2
(c) a = 3L/2 (d) a = L

**Ans. (b) :**




Since beam is symmetrical

$R_B = R_C$

$R_B + R_C - w(l + 2a) = 0$

$$R_B = R_C = \frac{w(\ell + 2a)}{2}$$

BM at point (E)

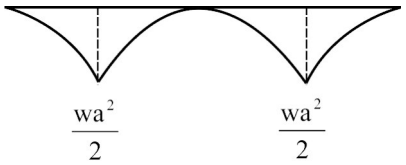
$$(BM)_E = \frac{w}{8}(\ell^2 - 4a^2)$$

If $\ell = 2a$, Bending moment at centre = 0

$$\frac{w}{8}(\ell^2 - 4a^2) = 0$$

$$\ell^2 = 4a^2$$

$$a = \frac{\ell}{2}$$

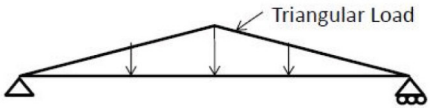



**Note –** For BM to be zero at centre of the beam, % of overhang required on both side.

$$\% \text{ of over hang} = \frac{a}{\ell + 2a} \times 100$$

$$= \frac{a}{4a} \times 100 = 25\%$$

**21. A simply supported beam is carrying a linearly varying load from zero at either ends to the maximum value at the mid span. Then the shape of shear force diagram (SFD) is**




(a) Rectangle
(b) Triangular
(c) Second Degree Parabola
(d) Cubic

**Ans. (c) :**

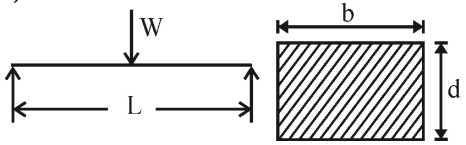
| Type of load | SF diagram | BM diagram |
|---|---|---|
| Point load | Straight line | Triangular |
| UDL | Triangular | 2nd degree parabola |
| UVL | 2nd degree parabola | 3rd degree parabola |

**22. A simply supported beam of span L and constant width b carries a point load W at mid span. The depth of the beam required at the mid span for maximum extreme fiber stress p**

(a) $d = \frac{3WL}{2bp}$ (b) $d = \sqrt{\frac{3WL}{2bp}}$

(c) $d^3 = \frac{3WL}{2bp}$ (d) $d = \frac{3WL}{2bp^2}$

**Ans. (b) :**




Constant width b

$$M = \frac{WL}{4}$$

$$\frac{M}{I} = \frac{p}{y}$$

$$\frac{WL}{4\times\frac{1}{2}bd^3}=\frac{p}{\frac{d}{2}}$$

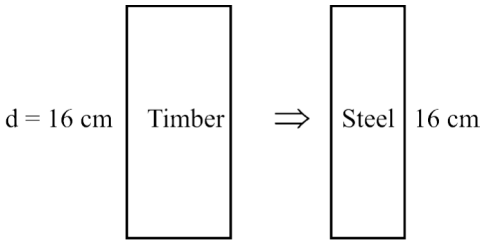
$$\frac{3WL}{bd^2}=2p$$

$$d=\sqrt{\frac{3WL}{2bp}}$$

**23. If a timber beam 8 cm wide and 16 cm deep is to be converted into an equivalent steel section of the same depth for analysis purpose, then the width of the equivalent section for a modular ratio of 20 will be**

(a) 160 cm (b) 2.5 cm
(c) 0.4 cm (d) 12 cm

**Ans. (c) :** Transformed section is drawn by converting one material into another equivalent material.




Width of equivalent section (steel) $=\frac{b}{m}$

$=\frac{8}{20}$

b' = 0.4 cm

**24. If the average shear stress in a rectangular section beam is 5 N/sq.mm, then maximum shear stress for the circular section of the equal area is**

(a) 7.50 N/sq.mm (b) 6.65 N/sq.mm
(c) 10 N/sq.mm (d) 2.50 N/sq.mm

**Ans. (b) :** For circular section

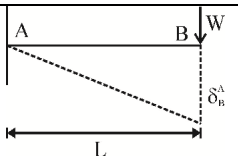
$$\tau_{max}=\frac{4}{3}\tau_{avg} \quad \text{(for same area)}$$

$$=\frac{4}{3}\times 5=6.65\,N/sq.mm$$

**25. A cantilever beam of length L is subjected to a point load W at free end. Then the slope at the free end is equal to**

(a) $\frac{WL^3}{3EI}$ (b) $\frac{WL^2}{24EI}$
(c) $\frac{WL^3}{60EI}$ (d) $\frac{WL^2}{2EI}$

**Ans. (d) :** A cantilever beam of length L is subjected to a point load W at free end-




Slope $\theta_B^A=\frac{WL^2}{2EI}$

Deflection $\left(\delta_B^A\right)=\frac{WL^3}{3EI}$

**26. Flexural rigidity in usual notations is**

(a) $\frac{M}{EI}$ (b) $\frac{EI}{R}$
(c) EI (d) MR

**Ans. (c) :** Bending equation $\frac{\sigma}{y}=\frac{M}{I}=\frac{E}{R}$

EI = flexural rigidity.

**27. The shape of the shear stress distribution diagram for a rectangular beam is**

(a) Parabola (b) Rectangle
(c) Hyperbola (d) Triangular

**Ans. (a) :** shear force for rectangular cross section-

$$=\frac{6F}{bd^3}\left[\left(\frac{d}{2}\right)^2-y^2\right]$$

This shows that the variation is parabolic.

**28. The ability of material to absorb energy when it is deformed elastically and release energy upon unloading is called**

(a) Ductility (b) Hardness
(c) Resilience (d) Toughness

**Ans. (c) :** The ability of material to absorb energy when it is deformed elastically and release energy upon unloading is toughness of a material decreased when it is heated. This property is desirable in parts subjected to shock and impact loads.

**29. The strain energy stored in a solid circular shaft, under pure torque, per unit its volume (take max. shear stress = q and modulus of rigidity = C) is expressed as**

(a) $q^2/C$ (b) $q/(4C)$
(c) $q^3/(4C)$ (d) $q^2/(4C)$

**Ans. (d) :** Strain energy stored in a solid circular shaft under pure torque per unit its volume

$$U=\frac{T^2L}{2CJ}=\frac{q^2}{4C}$$

■ Strain energy in a prismatic bar under pure axial loading.

$$U=\frac{T^2L}{2AE}=\frac{\sigma^2}{2E}(Vol.)$$

$$=\frac{1}{2}\sigma E(Vol.)$$

■ Strain energy in a tapered bar under purely axial load

$$U=\frac{1}{2}P.\delta, \text{ where } \delta=\frac{4PL}{\pi E d_1 d_2}$$

**30. The principle of super position is valid only for**

(a) Any elastic materials
(b) All metals
(c) Linear elastic materials
(d) Homogeneous materials

**Ans. (c) :** Linear elastic materials.

**31. The value of Poisson's ratio for which bulk modulus of a material will be equal to its Young's modulus**

(a) 0.33 (b) 0.15
(c) 0.45 (d) 0.25

**Ans. (a) :** $E = 3k(1 - 2\mu)$

Given as (k = E)

$E = k$

$E = 3E(1 - 2\mu), \quad 2\mu = 1 - \frac{1}{3}$

$\frac{1}{3} = (1 - 2\mu) \qquad \mu = \frac{1}{3} = 0.33$

**32. For Euler formula to be valid for mild steel struts which has yield stress of 3200 kg(f)/sq.cm, the slenderness ratio should not be less than**

(a) 80.48 (b) 40.28
(c) 160.96 (d) 53.65

**Ans. (a) :**

$\sigma = \frac{\pi^2 E}{\lambda^2}$,

$\sigma = \frac{3200 \times 9.81}{100}\ N/mm^2$

$E = 2.1 \times 10^5\ N/mm^2$

$\frac{3200 \times 9.81}{100} = \frac{\pi^2 \times 2.1 \times 10^5}{\lambda^2}$

$\lambda = 81.25 \approx 80.48$

**33. Rankine's formula proposed for crippling load ($P_r$) to cover all cases from short to very long column in terms of ultimate load for short column ($P_s$) and Euler's crippling load for long column ($P_c$) is**

(a) $P_r = P_s + P_c$ (b) $P_r = \frac{P_s + P_c}{2}$

(c) $\frac{1}{P_r} = \frac{1}{P_s} + \frac{1}{P_c}$ (d) $P_r = \frac{P_s - P_c}{P_s P_c}$

**Ans. (c) :** Rankin's formula

$\frac{1}{P_r} = \frac{1}{P_s} + \frac{1}{P_c}$

Rankin's load = $P_r$

Buckling load $P_e = \frac{\pi^2 E I_{min}}{L_e^2}$

$$P_r = \frac{A\sigma_c}{1 + a\left(\frac{L_e}{k}\right)^2}$$

Here, A = Area of column

$a = \frac{\sigma_c}{\pi^2 E}$ = Rankine's constant

- This formula is valid for both long and short column.
- Effect of crushing and buckling is considered in this formula.

**34. The torque that produces a twist of one radian in a shaft per unit length is called**

(a) Modulus of elasticity (b) Bulk modulus
(c) Torsion (d) Torsional rigidity

**Ans. (d) :** $\frac{T}{J} = \frac{G\theta}{\ell} = \frac{\tau}{r}$

$\theta = 1$ radian

$\ell =$ unit length

$T = GJ$ = Torsional rigidity

**35. Power transmitted by a shaft (kW) subjected to an average torque T (kN-m), rotating at N rpm is equal to**

(a) $\frac{2\pi NT}{3600}$ (b) $\frac{2\pi NT}{4500}$

(c) $\frac{2\pi NT}{60}$ (d) $\frac{\pi NT}{3060}$

**Ans. (c) :**

Power transmitted by shaft–

$P = T \times \omega$

$\omega = \frac{2\pi N}{60}$

$P = \frac{2\pi NT}{60}$ kW

T = applied torque

N = rotation per minute

w = speed of rotation

**36. A brass tube has enclosed a steel bar and they have equal cross-sectional area. The Young's modulus of elasticity is 200 GPa and 100 GPa for steel and brass respectively. Then the ratio of stress developed in the steel bar to that in the brass tube under compression is**

(a) 0.5 (b) 1
(c) 1.5 (d) 2

**Ans. (d) :** By using compatibility equation

$\left(\frac{\sigma_S}{E_S}\right) = \left(\frac{\sigma_B}{E_B}\right)$

$\frac{\sigma_S}{200} = \frac{\sigma_B}{100}$

$\left(\frac{\sigma_S}{E_B}\right) = 2$

**37. Condition for no tension in a solid column of dia D the eccentricity e must be less than or equal to**

(a) D/8 (b) D/6
(c) D/4 (d) D/2

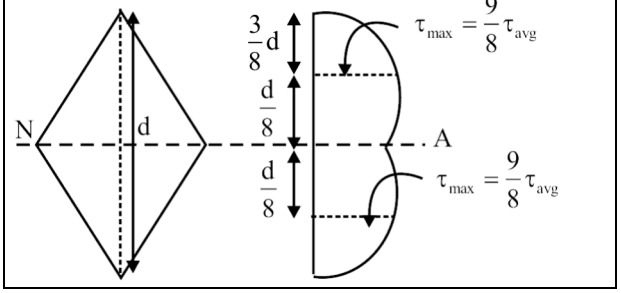
**Ans. (a) :**

| Section | Eccentricity | Dimension of kern | Shape of kern |
|---|---|---|---|
| Solid circular | $e = \frac{D}{8}$ | $\frac{D}{4}$ | circular |
| Rectangular | $e = \frac{b}{6}, \frac{d}{6}$ | $\frac{b}{3} \times \frac{d}{3}$ | Rhombus |
| Square | $e = \frac{d}{6}$ | $\frac{d}{3} \times \frac{d}{3}$ | Square |

**38. A horizontal beam of square section is placed with one diagonal (d) placed horizontally. The maximum shear stress occurs at**

(a) d/8 from neutral axis
(b) d/4 from neutral axis
(c) neutral axis
(d) 3d/8 from neutral axis

**Ans. (a) :** A horizontal beam of square section is placed with one diagonal (d) placed al axis.




**39. The mathematical expression for moment area theorem I for angle $\theta_{KL}$ between tangents drawn at K and L to elastic curve is**

(a) $\theta_{KL} = \int_K^L \frac{MI}{E} dx$ (b) $\theta_{KL} = \int_K^L \frac{M}{EI} dx$

(c) $\theta_{KL} = \frac{M}{EI}$ (d) $\theta_{KL} = \frac{E}{MI}$

**Ans. (b) :** The change in the slope from any pint K & L is equal to the area of $\frac{M}{EI}$ diagram between K & L.

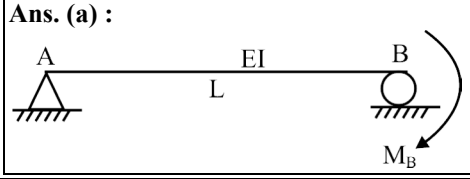
$$\theta_{KL} = \int_K^L \frac{M}{EI} dx$$

**40. A simply supported beam AB of L span and EI flexural rigidity is subjected to $M_B$ moment at B. Then the rotation at A is given by**

(a) $\theta_A = \frac{ML}{6EI}$ (b) $\theta_A = \frac{ML}{4EI}$

(c) $\theta_A = \frac{ML}{3EI}$ (d) $\theta_A = \frac{ML}{2EI}$

**Ans. (a) :**




$$M_A = \frac{ML}{6EI}$$
$$M_B = \frac{ML}{3EI}$$

**41. In a thick cylinder subjected to internal pressure, the hoop stress is ..... at the outer surface and is ..... at the inner surface**

(a) min, min (b) max, max
(c) max, min (d) min, max

**Ans. (d) :** In a thick cylinder subject to internal pressure the hoop stress in minimum at the outer surface and is maximum at the inner surface.

**42. An element is subjected to $p_x$ = 35 Mpa (tensile) and $p_y$ = 20 MPa (tensile) and shear stress q = 7.5 MPa. Then the direction of principal stresses is**

(a) 45º (b) 22.5º
(c) 30º (d) 15º

**Ans. (b) :** $\tan 2\theta p = \left(\frac{2\tau_{xy}}{\sigma_x - \sigma_y}\right)$

$$\tan 2\theta p = \frac{2 \times 7.5}{35 - 20}$$
$$\tan 2\theta p = \frac{15}{15} = 1$$
$$\tan 2\theta p = \tan 45$$
$$2\theta p = 45$$
$$\theta p = 22.5^0$$

**43. Shearing stress on principal planes will be**

(a) Equal to normal stresses
(b) Half the normal stresses
(c) Greater than normal stresses
(d) zero

**Ans. (d) :** Shearing stress on principal planes will be zero.
The planes over which principle stress act are term as principle plane.

**44. If normal stresses of same nature $p_x$ and $p_y$ and shear stress q are acting on two perpendicular planes and $q = (p_x p_y)^{1/2}$, then the major and minor principal stresses respectively are**

(a) $p_x + p_y$ and $p_x - p_y$
(b) $p_x$ and $p_x - p_y$
(c) $0.5(p_x + p_y)$ and $0.5(p_x - p_y)$
(d) $p_x + p_y$ and zero

**Ans. (d) :** $\sigma_x = p_x, \sigma_y = p_y, \tau_{xy} = \sqrt{p_x p_y}$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{1,2} = \frac{p_x + p_y}{2} \pm \sqrt{\left(\frac{p_x - p_y}{2}\right)^2 + \left(\sqrt{p_x p_y}\right)^2}$$

$$\sigma_{1,2}=\frac{p_x+p_y}{2}\pm\sqrt{\frac{p_x^2}{4}-\frac{2p_x\times p_y}{4}+\frac{p_y^2}{4}+p_xp_y}$$

$$\sigma_{1,2}=\frac{p_x+p_y}{2}\pm\sqrt{\frac{1}{4}\left(p_x^2+2p_xp_y+p_y^2\right)}$$

$$\sigma_{1,2}=\frac{p_x+p_y}{2}\pm\frac{p_x+p_y}{2},\sigma_1=p_x+p_y,\sigma_2=0$$

**45. The maximum shearing stress produced by shrinkage between two cylinders (mounted one inside the other) of E = $2\times10^6$ kg(f)/m$^2$ and shrinkage factor = 0.002, is equal to**

(a) 2000 kg(f)/m$^2$ (b) 2500 kg(f)/m$^2$
(c) 4000 kg(f)/m$^2$ (d) 1000 kg(f)/m$^2$

**Ans. (a) :** For maximum shearing stress

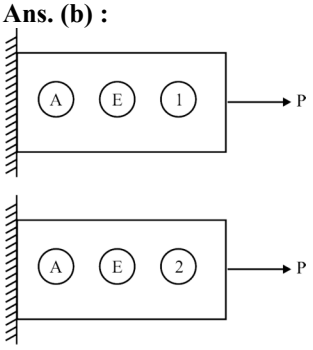
$$=\frac{2\times10^6\times0.002}{2}$$

$$=2000\text{kg(f)/m}^2$$

**46. Two bars of same area and length but of different materials are subjected to same tensile force. If the bars have their axial elongation in the ratio of 4:6, then the ratio of modulus of elasticity of the two materials would be**

(a) 4 : 6 (b) 6 : 4
(c) $2:\sqrt{6}$ (d) $\sqrt{6}:2$

**Ans. (b) :**




Elongation due to load P, $\delta=\frac{PL}{AE}$

$$\delta\ell_1=\left(\frac{PL}{AE}\right)_1$$

$$\delta\ell_2=\left(\frac{PL}{AE}\right)_2$$

$$\frac{\delta\ell_1}{\delta\ell_2}=\frac{\left(\frac{PL}{AE}\right)_1}{\left(\frac{PL}{AE}\right)_2}=\frac{4}{6}$$

$$\frac{E_2}{E_1}=4:6\Rightarrow\frac{E_1}{E_2}=6:4$$

**47. The bulk modulus of elasticity of a materials is twice its modulus of rigidity. The poisson's ratio of the material is**

(a) 1/7 (b) 2/7
(c) 3/7 (d) 4/7

**Ans. (b):** Relation between $\mu,K,G$ Given by

$$\mu=\frac{3k-2G}{6k+2G}$$

Given, k = 2G

$$\mu=\frac{3\times2G-2G}{6\times2G+2G}$$

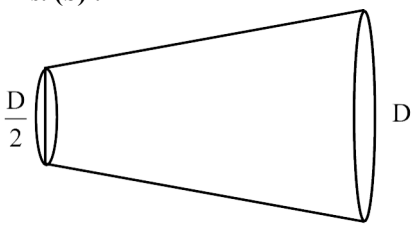
$$\mu=\frac{6G-2G}{12G+2G}$$

$$\mu=\frac{4G}{14G}=\frac{2}{7}$$

**48. The diameter of a tapering rod varies from 'D' to 'D/2' in length of 'L' m. If it is subjected to an axial tension of 'P' the change in length is**

(a) $4PL/(\pi ED^2)$
(b) $8PL/(\pi ED^2)$
(c) $2PL/(\pi ED^2)$
(d) None of the given answers

**Ans. (b) :**



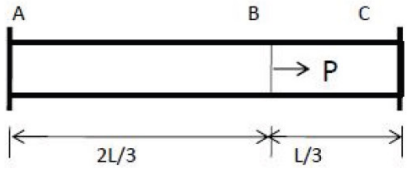

Elongation of tappered bar

$$\delta=\frac{4PL}{\pi ED_1D_2}$$

$$\delta=\frac{4PL}{\pi ED\frac{D}{2}}$$

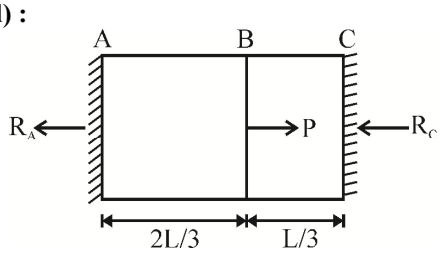
$$=\frac{8PL}{\pi ED^2}$$

**49. The ratio of loads shared by parts 'AB' and 'BC' of the bar shown below is**



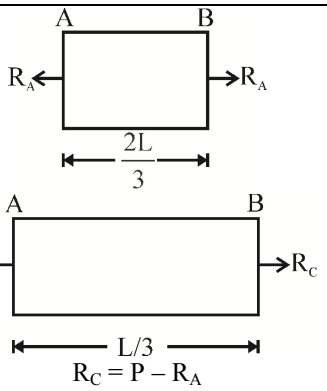

(a) 1 : 1 (b) 2 : 1
(c) 3 : 1 (d) 1 : 2

**Ans. (d) :**




Rigid section so deformation between A and C is zero.

$$\delta_{AB}+\delta_{BC}=0$$




$$\frac{R_A(2L/3)}{AE} - \frac{R_C(L/3)}{AE} = 0$$

$$\frac{R_A \times 2L}{3AE} = \frac{R_C \times L}{3AE} = \frac{R_A}{R_C} = \frac{1}{2}$$

**50. If a steel tyre is heated and struck on a rigid wheel, after cooling the tyre will be subjected to**

(a) Bending (b) Tension
(c) Hoop stress (d) Compression

**Ans. (c) :** The tyre is first heated due to which it expands. The heated tyre is then fitted on the wheel. When the tyre cools, it contracts and makes a tight fit on the wheel i.e. after cooling the tyre will be subjected to tension.

**51. Let the strain produced in length and diameter of cylindrical rod be 0.02 and 0.005 respectively. Then the volumetric strain is given by**

(a) 0.03 (b) 0.025
(c) 0.015 (d) 0.01

**Ans. (d) :**

let, $\in_v$ = Volumetric strain for cylindrical rod.

$\in_L$ = Longitudinal strain = 0.02

$\in_d$ = Diameter strain = –0.005

$$\in_v = \in_L + 2 \in_d$$
$$= 0.02 + \left[20 \times (-0.005)\right]$$
$$= 0.02 - 0.01$$
$$\in_v = 0.01$$

**52. A material with large deformation at failure is termed**

(a) Brittle (b) Elastic
(c) Ductile (d) Elasto-plastic

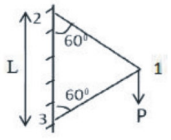
**Ans. (c) :** Ductility– It is property of a material enabling it to be drawn into wire with the application of a tensile force.

**53. Creep of a material is**

(a) Continued deformation with time under sustained loading
(b) Disappearance of deformation on removal of load
(c) Not being ductile
(d) To become brittle

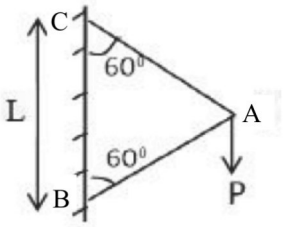
**Ans. (a): Creep–** When a part is subjected to a constant stress at high temp. for a long period of time, it will undergo a slow and permanent deformation called creep. This property is considered in designing internal combustion engines, boilers and turbines.

**54. A system of two prismatic bars of equal length (L) and equal cross-section (A) carries a vertical load P as shown in the figure given below. If their modulus of elasticity is E, the vertical displacement of the hinge 1 is**




(a) $\frac{PL}{AE}$ (b) $\frac{2PL}{AE}$
(c) $\frac{3PL}{AE}$ (d) $\frac{2.5PL}{AE}$

**Ans. (b) :**




Using Lami's theorem

$$\frac{F_{AC}}{\sin 60} = \frac{P}{\sin 60} = \frac{F_{AB}}{\sin 240}$$

$F_{AC} = P$

$F_{AB} = -P$

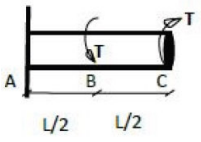
When unit vertical load is applied

$F_{AB} = -1$

$F_{AC} = 1$

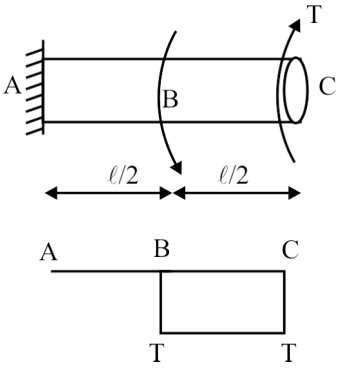
$$\Delta = \frac{\sum pk\ell}{AE} = \left[P(1) + (-p)(-1)\right]\frac{\ell}{AE} = \frac{2P\ell}{AE}$$

**55. A shaft of J polar moment of Inertia and C modulus of rigidity is fixed at one end and subjected to torque T at the free end and the same torque at mid twist between the free end and the midpoint is equal to**




(a) $\frac{T}{2CJ}$ (b) $\frac{T}{3CJ}$
(c) $\frac{T}{CJ}$ (d) $\frac{T}{4CJ}$

**Ans. (a):**




Since these μ no torque between B and A

$$\theta_C - \theta_B = \frac{TL}{GJ} = \frac{T \times \frac{1}{2}}{CJ}$$

$$\theta_C - \theta_B = \frac{TL}{2CJ}$$

**56. If a shaft is simultaneously subjected to a torque T and bending moment M, the maximum shear stress is**

(a) $\frac{16}{\pi D^3}(M+T)$

(b) $\frac{16}{\pi D^3}(M+\sqrt{M^2+T^2})$

(c) $\frac{16}{\pi D^3}(M-\sqrt{M^2+T^2})$

(d) $\frac{16}{\pi D^3}\sqrt{M^2+T^2}$

**Ans. (d) :** Combined effect of bending and torsion

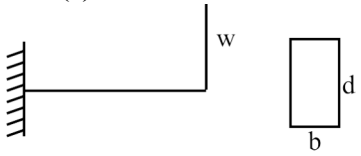
- Principal stress $= \frac{16}{\pi D^3}\left[M \pm \sqrt{M^2+T^2}\right]$
- Maximum shear stress $= \frac{16}{\pi D^3}\sqrt{M^2+T^2}$
- Equivalent Bending moment $= \frac{1}{2}\left[M+\sqrt{M^2+T^2}\right]$
- Equivalent torque $= \sqrt{M^2+T^2}$

**57. The ratio of energy stored in a rectangular cantilever beam loaded at the free end without producing permanent set to the energy stored in the same bar in simple tension is**

(a) 1/2 (b) 1/3

(c) 1/6 (d) 1/9

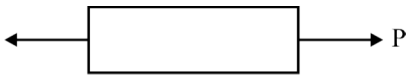
**Ans. (*) :** Case- I




Strain energy $U = \frac{1}{2}P\Delta$

$$U_1 = \frac{1}{2}P\left(\frac{P\ell^3}{3EI}\right) = \left(\frac{P^2\ell^3}{6EI}\right)$$

Case II




$$U_2 = \frac{1}{2}P\Delta = \frac{1}{2}P\left(\frac{P\ell}{AE}\right)$$

$$U_2 = \frac{P^2\ell}{2AE}$$

Now

$$\frac{U_1}{U_2} = \frac{\frac{P^2\ell^3}{6EI}}{\frac{P^2\ell}{2AE}} = \frac{\ell^2 \times 2A}{6I}$$

$$\frac{U_1}{U_2} = \frac{\ell^2 \times 2(bd)}{6 \times \frac{bd^3}{12}}$$

$$\frac{U_1}{U_2} = \left(\frac{4\ell^2}{d^2}\right)$$

**58. The theory which is in good agreement with experiments for ductile materials is**

(a) Maximum stress theory

(b) Maximum strain theory

(c) Maximum shear theory

(d) All the given answers

**Ans. (c) :**

| Theory | Given by | Remark |
|---|---|---|
| Maximum principle stress | Rankine | Suitable for brittel |
| maximum principle strain | St Venant | Suitable for brittel |
| Maximum shear stress | Guest & Treesca | Suitable for ductile |

**59. The factor of safety in bending for rectangular beams in terms of yield point stress ($\sigma_{yp}$) and working stress ($\sigma_W$) is**

(a) $1.5\frac{\sigma_{yp}}{\sigma_W}$ (b) $1.3\frac{\sigma_{yp}}{\sigma_W}$

(c) $1.33\frac{\sigma_{yp}}{\sigma_W}$ (d) $\frac{\sigma_{yp}}{\sigma_W}$

**Ans. (d) :** the factor of safety in benign for rectangular beams in terms of yield point stress ($\sigma_{yp}$) and working stress ($\sigma_W$) is

$$F.O.S. = \frac{\sigma_{yp}}{\sigma_W}$$

**60. The phenomenon of the decreased resistance of a material to repeated stresses is called**

(a) fatigue (b) creep

(c) freezing (d) melting

**Ans. (a): Fatigue–** When a material is subjected to repeated stress, it fails at stresses below the yield point stresses.

**61. A log of wood is of 3m diameter circular section. Then the width of the strongest rectangular section is bending that can be cut out from this log of wood is**

(a) $\sqrt{6}\,m$ (b) $\sqrt{3}\,m$

(c) $1/\sqrt{3}\,m$ (d) $\sqrt{2}/3m$

**Ans. (b) :**

$D^2 = b^2 + d^2$

$$z = \frac{bd^2}{6}$$

$$z = \frac{b\left(D^2 - b^2\right)}{6}$$

$$z = \frac{D^2 b - b^3}{6}$$

For strongest section $\frac{dz}{db} = 0$

$$\frac{D^2 - 3b^2}{6} = 0$$

$$3\,b^2 = D^2$$

$$b = \frac{D}{\sqrt{3}}$$

$$b = \frac{3}{\sqrt{3}} = \sqrt{3}m$$

**62. A column that fails essentially by direct crushing at ultimate load is called**

(a) Euler's column

(b) long column

(c) short column

(d) None of the given answers

**Ans. (c) :** A column that fails essentially by direct crushing at ultimate load is called short column

- Intermediate column fails in both buckling and crushing.
- Long column fails in buckling.

**63. If the hoop stress in a thin cylinder is 24 N/sq.mm, then its longitudinal stress is equal to**

(a) 36 N/sq.mm (b) 24 N/sq.mm

(c) 12 N/sq.mm (d) 6 N/sq.mm

**Ans. (c) :** Hoop stress $\left(\sigma_h\right) = \frac{Pd}{2t}$

Long. stress $\left(\sigma_\ell\right) = \frac{Pd}{4t}$

$$= \frac{Pd}{(2t)}\frac{1}{2}$$

$$= \frac{24}{2} = 12N/mm^2$$

**64. The Rankine's constant for a cast iron column with both ends hinged**

(a) 1/7500 (b) 1/1600

(c) 1/9000 (d) 0.00016

**Ans. (b):** The value of Rankin's constant are as follows

$\alpha$ = Rankine constant = $\left(\frac{\sigma_c}{\pi^2 E}\right)$

(a) For wrought iron $(a) = \frac{1}{9000}$

(b) For mild steel $(a) = \frac{1}{7500}$

(c) For cast iron $(a) = \frac{1}{1600}$

(d) For timber $(a) = \frac{1}{750}$

**65. A compression member effectively held in position and restrained in direction at one end but not held in poison or restrained in direction at the other end. If its actual length is L, then its effective length is equal to**

(a) 0.67 L (b) L

(c) 1.5 L (d) 2 L

**Ans. (d)**

| End condition | Effective length |
|---|---|
| Effective held in position and restrained against rotation at both ends | 0.65 L |
| Effective held in position at both ends restrained against rotation at one end | 0.80 L |
| Effective held in position at both end, but not restrained against rotation | 1.00 L |
| Effective held in position and restrained against rotation at one end, and at the other end restrained against rotation but not held in position. | 1.20 L |
| Effective held in position and restrained against rotation at one end, and at the other end partially restrained against rotation but not held in position | 1.5 L |
| Effective held position and restrained against rotation at one end but not held in position nor restrained against rotation at the other end | 2.0 L |

**66. The width b and depth d of a cantilever beam carrying a point load at its free end are changed into 0.5b and 2d respectively. Then its maximum deflection for the same load condition reduces by**

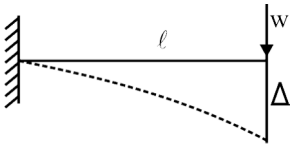
(a) one-fourth

(b) half

(c) three-fourth

(d) zero

**Ans. (a) :**



$$\Delta = \frac{w\ell^3}{3EI} = \frac{w\ell^3}{3E\left(\frac{bd^3}{12}\right)}$$

$$\Delta \propto \frac{1}{bd^3}$$

$$\frac{\Delta_1}{\Delta_2} = \frac{0.5\, b(2d)^3}{bd^3} = \frac{8 \times 0.5\, bd^3}{bd^3} = 4$$

$$\Delta_2 = \frac{\Delta_1}{4}$$

**67. A prismatic bar 1 m long and 4 sq. cm in cross sectional area is compressed by a force of 80 kN. If E = 200 kN/sq.mm, the total strain energy stored in the bar is equal to**

(a) 40 kN-mm (b) 0.05 kN-mm
(c) 400 kN-mm (d) 80 kN-mm

**Ans. (a) :** strain energy stored in prismatic bar

$$U = \frac{P^2L}{2AE}$$

$$= \frac{80\times10^3 \times 80\times10^3 \times 1\times1000}{2\times400\times200\times1000}$$

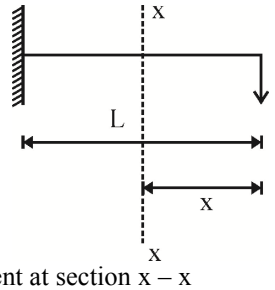
$$= 40000 \text{ N} - \text{mm}$$

$$= 40 \text{ kN-mm}$$

**68. A cantilever beam of length L is subjected to a point load P at free end. Then strain energy in usual notations is expressed as**

(a) $PL^3/(3EI)$ (b) $PL/(6EI)$
(c) $PL^2/(36EI)$ (d) $P^2L^3/(6EI)$

**Ans. (d) :**




Bending moment at section x – x

$M_{x-x} = P_x$

Strain energy = $\int_0^L \frac{M_{x-x}^2 dx}{2EI}$

$$= \int_0^L \frac{(Px)^2 dx}{2EI}$$

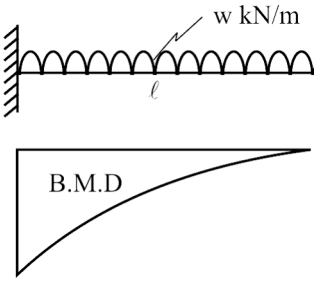
$$= \frac{P^2}{2EI}\left[\frac{L^3}{3}\right]$$

$$= \frac{P^2L^3}{6EI}$$

**69. The number of contra flexure points that occur in a cantilever beam subjected uniformly distributed load is**

(a) 3 (b) 2
(c) 1 (d) 0

**Ans. (d) :** The number of contra flexure points that occur in a cantilever beam subjected uniformly distributed load is zero.




Point of contra flexure =0

**70. Maximum value of shear stress in thin cylinders is equal to**

(a) hoop stress (b) 2(hoop stress)
(c) 0.5(hoop stress) (d) 0.25(hoop stress)

**Ans. (d) :** $\tau_{max} = \frac{\sigma_1 - \sigma_3}{2}$

$$\tau_{max} = \frac{\frac{pd}{2t} - \frac{pd}{4t}}{2} = \frac{pd}{8t}$$

$$\tau_{max} = \frac{1}{4}\left(\frac{pd}{2t}\right) = 0.25 \times \text{hoop stress}$$

**71. Euler's buckling load for both ends fixed condition is equal to (in usual notations)**

(a) $\pi^2 EI/l^2$ (b) $4\pi^2 EI/l^2$
(c) $2\pi^2 EI/l^4$ (d) $0.25\pi^2 EI/l^5$

**Ans. (b) :** Case –1

Both ends fixed

$$P = \frac{4\pi EI}{L^2}$$

$$Le = \frac{L}{2}$$

Case – 2 Both ends hinged

$$P = \frac{\pi^2 EI}{L^2}$$

$$Le = L$$

Case– 3 One end fixed and other end free

$$P = \frac{\pi^2 EI}{4L^2}$$

$$Le = 2L$$

Case– 4 One end fixed and other ends hinged

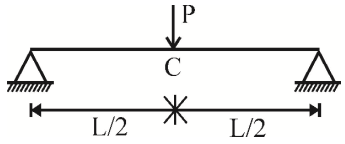
$$P = \frac{2\pi^2 EI}{L^2}$$

$$Le = \frac{L}{\sqrt{2}}$$

**72. A simply supported beam of span L carries a point load P at center, the slope at left end is**

(a) $PL^3/(16EI)$ (b) $PL^2/(48EI)$
(c) $PL^2/(16EI)$ (d) $PL^3/(48EI)$

**Ans. (c):**




$$\theta_{A,B} = \frac{PL^2}{16EI}$$

$$\delta_C = \frac{PL^3}{48EI}$$

**73. The elastic curve of a beam means**

(a) BMD (b) SFD

(c) Deflection curve (d) Stress-strain curve

**Ans. (c) :** The elastic curve of a beam means deflection.

**74. Plastic response of a material to compressive force is known as**

(a) Elasticity (b) Ductility

(c) Plasticity (d) Malleability

**Ans. (d) : Malleability–** It is a special case of ductility which permits materials to be rolled or hammered into thin sheets. A malleable material should be plastic but it is not essential to be so strong. The malleable materials commonly used in engineering practice (In order of diminishing malleability) are lead, soft steel, wrought iron, copper and aluminium.

**75. When equations of statics are not sufficient to determine all the reactive forces at the supports, such beams are called**

(a) statically indeterminate

(b) statically determinate

(c) Imperfect

(d) Defective

**Ans. (a) :** When equations of static's are not sufficient to determine all the reactive forces at the supports such beams are called statically indeterminate.

**76. In Newtonian fluids, for a given shear stress, the rate at which the fluid deforms is**

(a) Inversely proportional to the dynamic viscosity

(b) Directly proportional to the dynamic viscosity

(c) Independent of kinematic viscosity

(d) None of the given answers

**Ans. (a) :** For Newtonian fluid

$$\tau \alpha \frac{d\theta}{dt}$$

$\tau$ = Shear stress opposing the movement of fluid.

$$\tau = \mu \frac{d\theta}{dy}$$

For given shear stress ($\tau$) $\frac{d\theta}{dy} = \frac{\tau}{\mu}$

$$= \frac{d\theta}{dy} \propto \frac{1}{\mu}$$

$\mu$ = Absolute viscosity, or coefficient of viscosity or dynamic viscosity

**77. The pressure difference between inside and outside of a soap bubble of diameter d in terms of surface tension $\sigma$ is**

(a) $2\sigma/d$ (b) $8\sigma/d$

(c) $4\sigma/d$ (d) $\sigma/d$

**Ans. (b) :** Pressure inside a soap bubble $p = \frac{8\sigma}{d}$

Pressure inside a jet = $\frac{2\sigma}{d}$

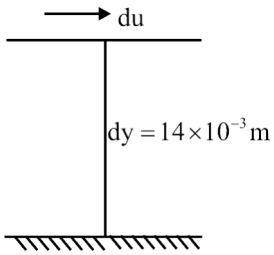
Pressure inside a droplet (P) = $\frac{4\sigma}{d}$

**78. Two horizontal plates are placed 14 mm apart and the space between them is fluid with an oil of viscosity of 14 poise. If the upper plate moves with 2.5 m/s velocity, the shear stress in the oil is**

(a) 2500 N/sq.m (b) 250 N/sq.m

(c) 25 N/sq.m (d) 2.5 N/sq.m

**Ans. (b) :**




$$\tau = 14 \text{ poise} = \frac{14}{10} \text{N sec/m}^2$$

$dy = 14 \times 10^{-3}$ m, dv = 2.5 m/sec

$$\tau = \mu \frac{du}{dy}$$

$$= \frac{14}{10} \times \frac{2.5 \times 1000}{14}$$

$$= 250 \text{N/m}^2$$

**79. With reference to the containers of different shapes having the same base area and filled with the same liquid for equal depths, the apparent contradiction in the hydrostatic force on the base of a liquid container and the weight of liquid in the container is known as**

(a) D' Alembert's paradox

(b) Hydrodynamic paradox

(c) Elevator paradox

(d) Hydrostatic paradox

**Ans. (b) : Hydrodynamic paradox–** According to the hydrostatic equation, p = wh; the intensity of pressure (p) depends only on the height of the column and not at all upon the size of the column. Thus, in all vessels of different shapes and sizes, the same intensity of pressure would be exerted on the bottom of each of these vessels. Since each of the vessels has the same area A at the bottom, the pressure force P = p × A on the base of each vessel would be same. This is independent of the fact that the weight of liquid in each vessel is different. This situation is referred to as **hydrostatic paradox.**

**80. The existence of velocity potential in fluid-flow indicates that**

(a) The flow must be irrotational
(b) The flow is rotational and satisfies the continuity equation
(c) The vorticity must not be zero
(d) The circulation around any closed curve must have a finite value

**Ans. (a) : Rotational flow and irrotational flow–**
rotational flow is that type of flow of which fluid particle while moving direction of flow rotate about their centre of mass otherwise irrotational flow.

**81. The scale ratio in model of spillway is 1:9. If the discharge in the prototype is 2430 cumecs, the discharge in the model is (in cumecs)**

(a) 270 (b) 90
(c) 30 (d) 10

**Ans. (d) :** Discharge ratio $Q_r = (Lr)^{5/2}$

$$\frac{Qm}{Qp} = \left(\frac{Lm}{Lp}\right)^{5/2}$$

$$Qm = Qp = \left(\frac{1}{9}\right)^{5/2}$$

$$= 2430 \times \frac{1}{243} = 10 \text{ m}^3/\text{sec}$$

**82. Hydraulic jump is expected when slope of a channel changes from**

(a) steep to steeper (b) steep to mild
(c) mild to critical (d) mild to steep

**Ans. (b) :** Hydraulic jump is expected when slope of a channel changes from steep to mild.

**83. If Thoma's cavitation factor is less than the critical factor, then the overall efficiency of a reaction turbine will have**

(a) abnormal increase
(b) no change
(c) sharp fall
(d) gradual increase

**Ans. (c) :** It Thoma's cavitation factor is less than the critical cavitation factor then the overall efficiency of a reaction turbine will have sharp fall.

**84. The angle of contact between pure water and clean glass is**

(a) 0º (b) 45º
(c) 90º (d) 130º

**Ans. (a) : Angle of contact–**
Water and glass = $0^0$
Water and mercury = $120^0$
kerosines and glass = $26^0$

**85. Parshall flume is used to measure discharge in**

(a) pipes (b) wells
(c) canals (d) spillways

**Ans. (c) : Metering flumes–**
• It is structure constructed in canal for measuring its discharge accurately.
• It is an artificially flumed section of the channel which can be utilized for calculation of discharge.
Ex - Venturiflume, standing wave flume, parshall flume etc.

**86. Fluid flow analysis is valid as long as the smallest length dimension of the problem is much larger than the distance between molecules. This concept is known as**

(a) Control volume (b) Ideal fluid
(c) Fluid continuum (d) Homogeneous fluid

**Ans. (c) :** Fluid flow analysis is valid as long as the smallest length dimension of the problem is much larger than the distance between molecules. This concept is known as fluid continuous.

**87. A body weighing 40 N is losing 5N weight when submerged in water. Then its specific gravity of the body is equal to**

(a) 8 (b) 6
(c) 4 (d) 2

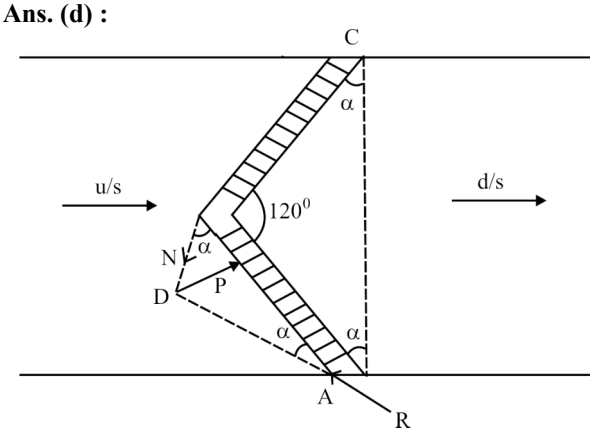
**Ans. (*) :** N = 40 N, Weight in water = 5 N
$w - F_B = 5$ N
$40 - F_B = 5$ N
$F_B = 35$ N $F_B$ = Bouyant force
$V \times \rho_w \times g = 35$ ..... (i)
Weight of body = 40 N
$V\rho.g = 40$ N ..... (ii)
Equation (ii) / (1)

$$\frac{\rho}{\rho_w} = \frac{40}{35} = 1.14$$

**88. For a closed single pair of lock gates having 120º between them when symmetrically placed, the resultant pressure force (P) acting on a gate and the reaction force (F) between the butting edges of the two gates are related as**

(a) F = P/2 (b) F = 2P
(c) $F = \sqrt{3}P/2$ (d) F = P

**Ans. (d) :**




$$120 + \alpha + \alpha = 120$$

$$2\alpha = 60 \Rightarrow \alpha = 30^0$$

Resolving the force at at D in direction parallel to gate along AB
$N\cos\alpha = R\cos\alpha$
N=R

Resolving normally to gate

$P = R\sin\alpha + N\sin\alpha$

$P = R\sin\alpha + R\sin\alpha$

$R = \dfrac{P}{2\sin\alpha}$

$R = \dfrac{P}{2\sin 30^0}$ $\quad R = F$ (given)

$F = \dfrac{P}{2\times\frac{1}{2}}$ $\quad F = P$

**89. A liquid jet of 30 m/s velocity is striking a single symmetrical moving curved vane at the centre. For maximum efficiency, the velocity of the vane should be**

(a) 30 m/s (b) 20 m/s
(c) 15 m/s (d) 10 m/s

**Ans. (d) :** Jet striking moving curved vane at centre.

$$\eta = \frac{2u(v-u)^2(1+\cos\theta)}{v^3}$$

v = velocity of jet
u = velocity of vane

For constant jet velocity $= \dfrac{d\eta}{du} = 0$

$\dfrac{2}{v^3}(1+\cos\theta)(v^2+3u^2-4uv) = 0$

$v^2+3u^2-4uv = 0$

$(u-v)(v-3u) = 0$

$v = 3u$

$u = \dfrac{v}{3}$

$v = \dfrac{30}{3} = 10\ \text{m/sec}$

**90. If the velocity distribution over parallel planes is identical in a flow region, then the flow is treated as**

(a) one dimensional (b) two dimensional
(c) three dimensional (d) Axi-Symmetric

**Ans. (b) :** It the velocity distribution over parallel planes is identical in a flow region, then the flow is treated as two dimensional.

**91. A merchant ship has metacentric height of 2.7m. After unloading it increased to 3.6 m. The ratio of periodic times before and after unloading the cargo ship if radius of gyration is not altered is**

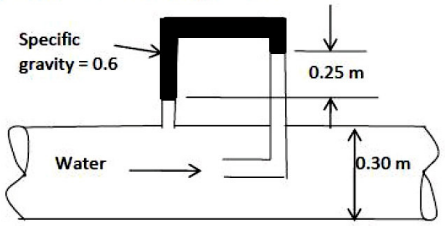
(a) 1 (b) 0.75
(c) $\sqrt{3/4}$ (d) $\sqrt{4/3}$

**Ans. (d) :** $T = 2\pi\sqrt{\dfrac{k^2}{g.GM}} \Rightarrow T \propto \dfrac{1}{\sqrt{GM}}$

$\dfrac{T_1}{T_2} = \sqrt{\dfrac{GM_2}{GM_1}} = \sqrt{\dfrac{3.6}{2.7}}$

$= \dfrac{6}{3\sqrt{3}} = \left(\dfrac{2}{\sqrt{3}}\right) = \sqrt{\dfrac{4}{3}}$

**92. The velocity of head of water flowing through the pipe of 0.3m dia for the conditions shown in the figure below is**




(a) 0.25 m (b) 0.15 m
(c) 0.44 m (d) 0.1 m

**Ans. (d) :** Velocity at point $V = \sqrt{2gh}$

x = 0.25 = Monometric reading

$h = x\left(1-\dfrac{\rho_{mano}}{\rho_w}\right)$

$= x\left(1-\dfrac{600}{1000}\right)$

$= 0.25(1-0.6) = 0.25\times 0.4 = 0.1$

$V = \sqrt{2\times 9.81\times 0.1}$

Velocity head $= \dfrac{V^2}{2g}$

$= \dfrac{2\times 9.81\times 0.1}{2\times 9.81} = 0.1$ m

**93. Performance characteristic curves indicate that unit discharge is independent of unit speed for**

(a) Pelton wheel
(b) Low specific speed Francis turbine
(c) High specific speed Francis turbine
(d) Kaplan turbine

**Ans. (a) :** Performance characteristic curves indicate that unit discharge is independent of unit speed for pelton wheel.

**94. Hot wire anemometer is used for measuring**

(a) velocity of gases
(b) velocity of liquids
(c) pressure of gases
(d) temperature of fluids

**Ans. (a) :** Hot wire anemometer is used for measuring velocity of gases.

- Venturimeter – Rate of flow (discharge)
- Flow hazzle – Rate of flow
- Orifice meter – Rate of flow
- Bend meter – Rate of flow
- Rotameter – Rate of flow
- Current meter – velocity in open channels

**95. A pipe boundary will behave as hydrodynamically smooth if the relative magnitude of average height of the surface protrusions and the thickness of the laminar sublayer is**

(a) $\le 0.25$ (b) $< 6$
(c) $\ge 6$ (d) $> 11.6$

**Ans. (a):** Through experiment Nikurandse found that the boundary behave as

Hydrodynamically smooth boundary $\frac{k}{\delta'} < 0.25$

Hydrodynamically transition boundary $0.25 < \frac{k}{\delta'} < 6$

Hydrodynamically rough boundary $\frac{k}{\delta'} > 6$

**96. If a sphere of 10 mm dia is falling in a fluid medium of kinematic viscosity of 10 stokes with terminal velocity of 12 mm/s, then coefficient of drag on the sphere will be**

(a) 12 (b) 24
(c) 100 (d) 200

**Ans. (d) :** Reynold number $= \frac{Vd}{\nu}$

$$= \frac{12\times10^{-3}\times10\times10^{-3}}{10\times10^{-4}}$$

Re = 0.12

$$Cd = \frac{24}{Re} = \frac{24}{0.12} = 200$$

**97. The boundary-layer separation occurs when**

(a) the pressure gradient is zero
(b) the pressure gradient is negative
(c) the pressure gradient is adverse
(d) velocity increases

**Ans. (c) :** When the pressure increases in the direction of flow $\left(\frac{dp}{dx} > 0\right)$, the pressure forces act opposite to the direction of flow and further increase the retarding effect of the viscous forces, subsequently the thickness of the boundary layer increases rapidly in the direction of flow. If these forces act over a long stretch, the boundary layer gets separated from the surface and moves into the main stream.

**98. Compressibility of fluid is insignificant if Mach number is**

(a) greater than 1 (b) equal to 1
(c) between 0.4 - 1 (d) less than 0.4

**Ans. (d) :** Compressibility of fluid is insignificant if mach number is less than 0.4.

**99. The pressure drop in a laminar flow through circular pipe is dependent on**

(a) velocity of flow (b) viscosity of the fluid
(c) diameter of the pipe (d) All the given answers

**Ans. (d) :** Pressure drop in laminar flow –

$$\left[\frac{P_1 - P_2}{\rho g} = \frac{32\mu vL}{\rho g d^2}\right]$$

From above relation, pressure drop depend on velocity of flow, viscosity of fluid, diameter of pipe and length of pipe.

**100. The deformation drag experienced by a sphere of d dia, moving at V velocity through a fluid of μ dynamic viscosity is given by**

(a) $3\pi$ Vd/μ (b) $3\pi\mu$Vd
(c) $3\pi\mu Vd^2$ (d) $3\pi\mu V^2 d$

**Ans. (b) :** The deformation drag experienced by a sphere of d dia, moving at V velocity through a fluid of μ dynamic viscosity is given by $3\pi\mu$Vd.

**101. Intensity of turbulence is**

(a) the root mean square value of velocity fluctuations
(b) The average kinetic energy of turbulence
(c) The mean time interval between the reversals in the sign of velocity fluctuations
(d) the square root of Reynolds number

**Ans. (a) :** The turbulence intensity is defined as the ratio of the root mean square of the velocity fluctuation (V') to the mean flow velocity ($V_{avg.}$).

$$I = \frac{V'}{V_{avg.}}$$

**102. The average coefficient of friction drag for turbulent boundary layer is expressed by**

(a) $0.664/\sqrt{(Re)}$ (b) $0.664/(Re)^{1/5}$
(c) $0.074/(Re)^{1/5}$ (d) $0.074/\sqrt{(Re)}$

**Ans. (c) :** For flat plate

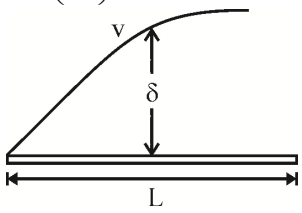
The average coefficient of friction drag for turbulent boundary larger.

$$(C_D) = \frac{F_D}{\frac{1}{2}\rho A v^2}$$

Where,

$F_D$ = Drag force
v = Free stream velocity of fluid
ρ = Density of fluid
A = L × b
L = Length of plate
b = Width of plate

$$F_D = 0.072\times\frac{\rho v^2}{2}\times\left(\frac{\mu}{\delta v}\right)^{1/5}\times b\times L^{4/5}$$




$$C_D = \frac{0.072\times\rho v^2\times\left(\frac{\mu}{\rho v}\right)^{1/5}\times b\times L^{4/5}}{\frac{\rho v^2}{\alpha}\times b\times L}$$

$$= 0.072\times\left(\frac{\mu}{\rho v}\right)^{1/5}$$

$$\because Re = \frac{\rho vL}{\mu}$$

$$C_D = \frac{0.072}{Re^{1/5}}$$

This is valid for $Re > 5\times10^5$ but less than $10^7$

**103. The drag coefficient at supersonic velocities depends on**

(a) Reynolds number only
(b) Mach number only
(c) Weber number only
(d) Euler number only

**Ans. (b) :** The drag coefficient at supersonic velocities depends on mach number only.
Mach Number represented by 'M'
M < 1 ; flow is subsonic
M = 1; flow is sonic
1< M < 5 ; Flow is supersonic
M > 5 ; flow is hypersonic

**104. The entrance length or length of establishment of turbulent flow in pipes is**

(a) Inversely proportional to pipe dia
(b) proportional to Reynolds number (Re)
(c) proportional to $Re^{0.25}$
(d) proportional to $Re^{0.6}$

**Ans. (c) :** From the ..... equation

$$h_f = \frac{fLv^2}{2gD} \quad .....(i)$$

Where,
L = Length of pipe
D = Dia of pipe
v = mean velocity of flow
f = friction factor
$h_f$ = Head loss due to friction.

for Turbulent flow

$$f = \frac{0.316}{(R_e)^{0.25}} \quad .....(ii)$$

from equation (i) and (ii)

$$h_f = \frac{0.316}{(R_e)^{0.25}} \times \frac{Lv^2}{2gD}$$

$$L \propto (R_e)$$

hence, entrance length directly proportional to $(R_e)^{0.25}$

**105. In a rectangular channel of 1m width the critical depth for a flow of 3.132 cumecs is approximately equal to**

(a) 2 m (b) 1 m
(c) 0.1 m (d) 3.14 m

**Ans. (b) :** Given that
Width of rectangular channel (b) = 1m
Total discharge (Q) = 3.312 .....
= 3.132 $m^3$/sec

critical depth (hc) $= \sqrt[3]{\frac{q^2}{g}}$ or $\left(\frac{q^2}{g}\right)^{1/3}$

Where, q = discharge per unit width

$$= \frac{Q}{b}$$

$$= \frac{3.132}{1} = 3.132$$

$$g = 9.81\ m/s^2$$

$$h_c = \left[\frac{(3.132)^2}{9.81}\right]^{1/3} = 0.998$$

$$h_c \approx 1.0m$$

**106. The control which will limit pump outlet pressure to a predetermined level and adjust pump outlet flow to the level needed to maintain the set pressure, is**

(a) pressure compensator control
(b) Load sense control
(c) Meter-In circuit
(d) Meter-Out circuit

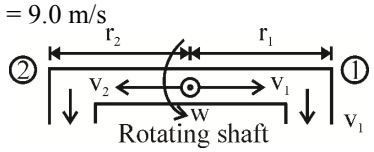
**Ans. (a) :** Pressure compensator control will limit pump outlet pressure to a predetermined level and object pump outlet flow to the level needed to mention the set pressure.

**107. An unsymmetrical sprinkler has frictionless and equal flow through each of the two nozzles. If the absolute velocities are 6.0 and 9.0 m/s, the ratio of the distances of the nozzles from the rotating shaft**

(a) 9 : 6 (b) $9^{1/2} : 6^{1/2}$
(c) $9^2 : 6^2$ (d) $9^3 : 6^3$

**Ans. (a) :** Given unsymmetrical sprinkler has frictionless-

$v_1 = 6$ m/s
$v_2 = 9.0$ m/s




Let,
$v_1$ = velocity of fluid at section 1
$r_1$ = distance of nozzle from rotating shorft.
$v_2$ = velocity of fluid at section 2,
$r_2$ = distance of the nozzle from the rotating shaft.
v = rw

$$w = \frac{v}{r}$$

$$w_1 = w_2$$

$$\frac{v_1}{r_1} = \frac{v_2}{r_2}$$

$$\frac{r_2}{r_1} = \frac{v_2}{v_1}$$

$$\frac{r_2}{r_1} = \frac{9}{6}$$

$$r_2 : r_1 = 9 : 6$$

**108. The pressure variation along the radial direction for vortex flow along a horizontal plane is given by**

(a) $\partial p / \partial r = \rho v^2 / r$ (b) $\partial p / \partial r = \rho v / r$
(c) $\partial p / \partial r = \rho v^2 / r^2$ (d) $\partial p / \partial r = \rho v^3 / r^2$

**Ans. (a) :** The pressure variation at any radius in horizontal plane is given by

$$\partial p / \partial r = \rho v^2 / r$$

**109. The speed of a geometrically similar turbine such that when it is developing 1 kW the total suction head is equal to 1 meter. The speed is known as**

(a) specific speed (b) unit speed
(c) suction specific speed (d) critical speed

**Ans. (c) : Suction specific speed–** The speed of a geometrically similar turbine such that when it is developing 1 kw the total suction head is equal to meter. The speed is known as suction is known as suction specific speed.

**110. Relation between shear and pressure gradients in a two dimensional laminar flow under steady and uniform conditions**

(a) $\partial\tau/\partial x = \partial p/\partial x$

(b) $\partial\tau/\partial y = \partial p/\partial x$

(c) $\partial^2\tau/\partial x^2 = \partial^2 p/\partial x^2$

(d) $\partial^2\tau/\partial x^2 = \partial^2 p/\partial y^2$

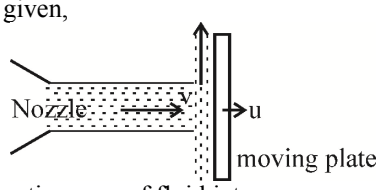
**Ans. (c) :** The pressure gradient in the direction of flow is equal to the shear gradient in the direction normal to the direction of flow.

$$\partial\tau/\partial y = \partial p/\partial x$$

**111. A fluid jet of cross sectional area of a and velocity of V strikes (normal to) moving plate arranged in series with velocity u and u < V, the actual fluid mass striking is equal to**

(a) ρa(V – u) (b) ρau
(c) ρaV (d) ρa(V + u)

**Ans. (a) :** given,




a = cross section area of fluid jet
v = velocity of fluid jet
u = velocity of moving plate

u < v

Hence,

Relative velocity of the fluid jet respect to moving plate. = (v – u)

therefore,

the actual fluid mass striking the plate

= ρ × Area of jet × velocity with which jet strikes the plate

= ρ × a × (v – u)

**112. In order to have a continuous flow through a siphon, no portion of the pipe should be higher than ..... above HGL.**

(a) 2.55 m (b) 0.775 m
(c) 7.75 m (d) 25.5 m

**Ans. (c) :** In the siphon, summit pressure should not be less than vapour pressure of liquid to avoid air lock formation at summit. Due to this a continuous flow through a siphon, no portion of the pipe should be higher than 7.75m above HGL (Hydraulic gradient line).

**113. The loss of energy in Orifice meter is more than in the case of Venturimeter because**

(a) Suddent obstruction
(b) Turbulence
(c) More friction
(d) pressure variation

**Ans. (a) :** Due to sudden obstruction (sudden reduced in cross-section area) In orifice meter, the loss of energy in orifice meter is more than in the case of venturimeter. Because in the ventruimeter cross area of pipe in which fluid is flowing gradually reduce.

**114. For a body completely submerged in a fluid, the centre of gravity (G) and centre of buoyancy (O) are known. The body is said to be in stable equilibrium, if**

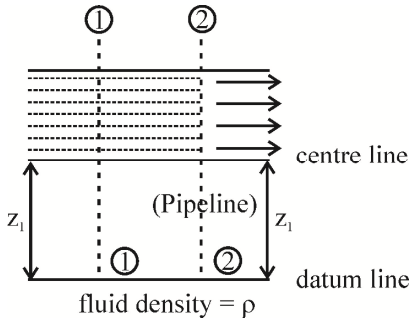
(a) O does not coincides with centre of mass of the displaced fluid
(b) G coincides with centre of mass of the displaced fluid
(c) O lies above G
(d) O lies below G

**Ans. (c) :** For a body completely submerged in a fluid. the centre of buoyancy (O) lies above the centre of gravity (G)
Example – balloon

**115. At two points 1 and 2 in a pipeline, the velocities are V and 2V respectively. Both the points are at the same elevation. The fluid density as ρ. The difference between pressure at 1 and 2 is**

(a) $\frac{1}{2}\rho V^2$ (b) $\frac{3}{2}\rho V^2$
(c) $2\rho V^2$ (d) $3\rho V^2$

**Ans. (b) :**




Given,
$v_1$ = velocity (v)
$v_2$ = velocity (2v)
Let,
$\rho_1$ = pressure at section 1
$\rho_2$ = pressure at section ?
Applying the Bernoulli' s equation –

$$\frac{p_1}{\rho g}+\frac{v_1^2}{2g}+z_1=\frac{p_2}{\rho g}+\frac{v_2^2}{2g}+z_2$$

$$\left[\because z_1 = z_2, \text{given}\right]$$

$$\Rightarrow \frac{p_1 - p_2}{\rho g} = \frac{v_2^2 - v_1^2}{2g}$$

$$\Rightarrow (p_1 - p_2) = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right)$$

$$= \frac{1}{2}\rho\left[(2v)^2 - (v)^2\right]$$

$$= \frac{1}{2}\rho\left(4v^2 - v^2\right)$$

$$(p_1 - p_2) = \frac{3}{2}\rho v^2$$

**116. The velocity field for a flow is given by $\mathbf{v = (5x + 6y + 7z)i + (6x + 5y + 9z)j + (3x + 2y + \lambda z)k}$ and density varies as $\rho = \rho_o e^{-2t}$. In order that the mass is conserved, the value of $\lambda$ should be**

(a) –12 (b) –10
(c) –8 (d) 10

**Ans. (b) :**
Continuity equation (since mass is conserved)

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \text{.....(i)}$$

$u = 5x + 6y + 7z$
$v = 6x + 5y + 9z$
$w = 3x + 2y + \lambda z$

$$\frac{\partial u}{\partial x} = 5, \frac{\partial v}{\partial y} = 5, \frac{\partial w}{\partial z} = \lambda$$

From equation (i)
$5+5+\lambda = 0$
$\lambda = -10$

**117. A 10 cm side cube weighing 5N is immersed in a liquid of relative density 0.8 contained in a rectangular tank of cross-sectional area 15cm × 15 cm. If the tank contained liquid to a height of 8cm before the immersion. The buoyant force on the cube is**

(a) 10 N (b) 5000 N
(c) 5 N (d) zero

**Ans. (c) :** Given: weight of the floating body = 5 N
According to principle of floatation weight of the body floating in a fluid is equal to boyant force
So bouyant force = 5 N.

**118. Match List I and List II and select the correct answer using the codes given below the lists**

| **List I description** | **List II property of fluid** |
|---|---|
| P. Property which explains the spherical shape of the drop of a liquid | 1. Viscosity |
| Q. Property which explains cavitation in fluid flow | 2. Surface tension |
| R. Property which explains rise of sap in free | 3. Compressibility |
| S. Property which explains flow of oil jet | 4. Vapour pressure |
| | 5. Capillarity |

| | **P** | **Q** | **R** | **S** |
|---|---|---|---|---|
| (a) | 1 | 2 | 4 | 5 |
| (b) | 2 | 4 | 5 | 1 |
| (c) | 3 | 4 | 5 | 1 |
| (d) | 1 | 2 | 3 | 4 |

**Ans. (b) :** (P) property which explains the spherical shape of = surface tension the drop of a liquid.
(Q) property which explains cavitation in fluid flow = vapour pressure.
(R) Property which explains rise of sap in free = capillarity.
(S) Property which explains flow of oil jet = viscosity.

**119. Uniform flow occurs**

(a) when the spatial rate of the change of velocity is zero
(b) when the temporal rate of the change of velocity is zero
(c) when the velocity changes steadily along the direction of the flow
(d) only when velocity vector at any point remains constant

**Ans. (a) :** Uniform flow– A flow is uniform, if the velocity and property of flow do not change from point to point in space at particular time i.e., when the spatial rate of change of velocity is zero.

**120. The x-component of velocity in a 2D incompressible flow is given by u = 1.5x. At the point (x,y) = (1,0), the y-component of velocity v = 0. The equation for the y-component of the velocity is**

(a) v = 0 (b) v = 1.5y
(c) v = –1.5y (d) v = –1.5x

**Ans. (c) :** x- component of velocity in a 2D incompressible flow

$u = 1.5\,x$ ..... (i)

point (x, y) = (1, 0)
y – component of velocity
the equation for the y – component of the velocity is = ?
Applying the continuity equation in 2D–

$$\frac{\partial y}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\Rightarrow \frac{\partial (1.5x)}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\Rightarrow 1.5 + \frac{\partial v}{\partial y} = 0$$

$$\Rightarrow \frac{\partial v}{\partial y} = 1.5 \Rightarrow \partial v = -1.5\partial y$$

Integrating both side

$v = -1.5\,y + c$ ..... (iii) where c = const.

at point (x, y) = (1, 0); v = 0
$0 = -1.5 \times 0 + c \Rightarrow c = 0$
Putting the value of c in equation (iii)
$v = -1.5y + c$
$v = -1.5y$

**121. For a 2D irrotational flow, the velocity potential is defined as $\varphi = \log_e(x^2 + y^2)$. Which of the following is possible stream function $\psi$, for this flow?**

(a) $\frac{1}{2}\tan^{-1}\left(\frac{y}{x}\right)$ (b) $\tan^{-1}\left(\frac{y}{x}\right)$

(c) $2\tan^{-1}\left(\frac{y}{x}\right)$ (d) $2\tan^{-1}\left(\frac{x}{y}\right)$

**Ans. (c) :** Given

velocity potential function

$\phi = \log_e (x^2 + y^2)$ ..... (i)

$\therefore$ velocity potential function is defined as a scalar function of space and time such that negative derivative w.r.to. any axis give component of velocity along that axis.

$$u = \frac{-\partial\phi}{\partial x}, v = \frac{-\partial\phi}{\partial y}$$

$$w = \frac{-\partial\phi}{\partial z} \text{.....(ii)}$$

from equation (i)

$$\phi = \log_e (x^2 + y^2)$$

$$\frac{\partial\phi}{\partial x} = \frac{2x}{x^2 + y^2}$$

$$\begin{cases} \because\ u = \dfrac{\partial\phi}{dx} \\ \text{or } \dfrac{\partial\phi}{\partial x} = -u \end{cases}$$

$$\Rightarrow -u = \frac{2x}{x^2 + y^2} \quad \text{.....(iii)}$$

Since, stream function

$$\frac{\partial\phi}{dy} = -u \quad \text{.....(iv)}$$

from equation (iii) and (iv)

$$\frac{\partial\psi}{\partial y} = \frac{2x}{x^2 + y^2}$$

$$\Rightarrow \partial\psi = \frac{2x}{x^2 + y^2}.\partial y$$

Integrating on both side–

$$\psi = 2x\int \frac{dy}{x^2 + y^2}$$

$$= 2x \times \frac{1}{x}\tan^{-1}\left(\frac{y}{x}\right)$$

$$\psi = 2\tan^{-1}\left(\frac{y}{x}\right)$$

**122. A pitot static tube having coefficient of velocity 0.98, measures velocities of water in a pipe. The stagnation pressure recorded in 3m and static pressure 2m. The actual velocity in pipe is**

(a) 4.429 m/s (b) 4.341 m/s

(c) 4.431 m/s (d) zero

**Ans. (b) :** Actual velocity of flow

$$V = C_v\sqrt{2g(\text{stagnation head} - \text{static head})}$$

$$V = C_v\sqrt{2g(3-2)}$$

$$V = 0.98\sqrt{2\times 9.81\times 1}$$

V = 4.341 m/sec

**123. The velocity in m/s at a point in a 2D flow is given as $v = 2yi + 3xj$. The equation of stream line passing through the point is**

(a) $3dx - 2dy = 0$ (b) $2x + 3y = 0$

(c) $3dx + 2dy = 0$ (d) $xy = 6$

**Ans. (a) :** the velocity in m/s at a point in a 2D flow.

Given,

$v = 2yi + 3xj$ .....(i)

Hence, equation of stream line in 2D flow is

$$\frac{\partial x}{u} = \frac{\partial y}{v}$$

from equation(i)

$u = 2y$ and $v = 3x$

therefore ,

$$\frac{dx}{2y} = \frac{dy}{3x}$$

$$\Rightarrow 3xdx = 2y.dy$$

$$\Rightarrow 3xdx - 2ydy = 0 \quad \text{.....(ii)}$$

Hence, equation of stream line passing through the point is

$3dx - 2dy = 0$ (iii)

According to given option, (i) is correct.

**124. Select the correct statement**

(a) Absolute pressure = Gage pressure – Atmospheric pressure

(b) Gage pressure = Absolute pressure – Atmospheric pressure

(c) Absolute pressure = Atmospheric pressure + vacuum pressure

(d) Gage pressure = Atmospheric pressure + vacuum pressure

**Ans. (b) :** we known that

Absolute pressure = Gauge pressure + Atmospheric pressure

Gauge pressure = Absolute pressure – Atmospheric pressure

**125. In a laminar flow of a liquid down in an inclined plane, the central line velocity is found to be 30 cm/s. The average velocity of the flow, in cm/is**

(a) 20 (b) 30

(c) 15 (d) 10

**Ans. (a) :** For Laminar flow in a pipe –

$$\frac{u_{max}}{u_{avg}} = \frac{3}{2}$$

given,

$u_{max} = 30$ cm/s

$$u_{avg.} = \frac{2}{3} \times u_{max}$$

$$= \frac{2}{3} \times u_{max}$$

$$= \frac{2}{3} \times 30$$

$$= u_{avg} = 20 cm/s$$

Where,

$u_{avg.}$ = Average velocity of flow

$u_{max}$ = centre line velocity

**126. Match List I and List II and select the correct answer using the codes given below the lists**

| List I | List II |
|---|---|
| P. Moment of momentum equation | 1. Equation to find energy loss in pipeline having laminar flow. |
| Q. Bernoulli's equation | 2. Equation of motion for 1-D steady flow |
| R. Euler's equation | 3. Equation based on momentum conservation principle |
| S. Hagen-poiseulle equation | 4. 3-D equation of motion based on momentum conservation |

| | P | Q | R | S |
|---|---|---|---|---|
| (a) | 2 | 3 | 4 | 1 |
| (b) | 3 | 2 | 1 | 4 |
| (c) | 2 | 3 | 1 | 4 |
| (d) | 3 | 2 | 4 | 1 |

**Ans. (d) :** (P) moment of momentum equation – Equation based on momentum conservation principle.
(Q) Bernoulli's equation– equation of motion for 1–D steady flow.
(R) Euler's equation – 3–D equation of motion based on momentum conservation.
(S) Hagen poiseulle equation – equation to find energy loss in pipe line having laminar flow.

**127. When a pressure of $2\times10^4$ kN/m$^2$ is applied to 50 litres of a liquid, its volume is decreased by 0.5 litre. The bulk modulus of liquid in N/m$^2$ is**

(a) $20 \times 10^9$ (b) $2 \times 10^9$
(c) $4 \times 10^9$ (d) $40 \times 10^9$

**Ans. (b) :** Bulk modulus (k) $= \frac{vdp}{dv}$

$dp = 2\times10^4 \text{ kN/m}^2 = 2\times10^7 \text{ N/m}^2$

$v = 50 \text{ litres} = 0.05 \text{m}^3$

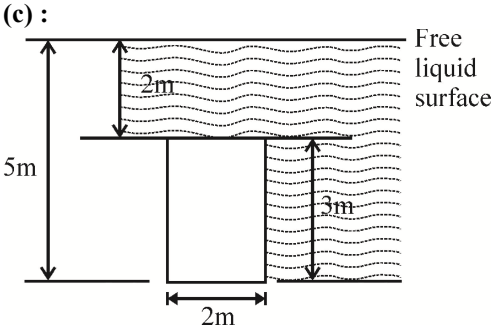
$dv = 0.5 \text{ Litre} = 0.5\times10^{-3} \text{m}^3$

$$k = \frac{0.05\times2\times10^7}{0.5\times10^{-3}} = 2\times10^9 \text{ N/m}^2$$

**128. A rectangular plate of 2 m wide is submerged in water vertically such that its top and bottom surfaces are 2m and 5m respectively below the free surface. The total pressure on plate is**

(a) 20601 N (b) 20601 kN
(c) 206.01 kN (d) 206.01 kN

**Ans. (c) :**




Total pressure force on plate (F) = $\rho g A \bar{x}$

$\rho = 1000 \text{ kg/m}^3$

$g = 9.81 \text{ m/s}^2$

$A = 2 \times 3 = 6\text{m}^2$

$$\bar{x} = 2 + \frac{3}{2} = 3.5\text{m}$$

$\therefore$ $F = 1000 \times 9.81 \times 6 \times 3.5$

$= 206.01 \times 10^3$ Newton

$= 206 .01$ kN

**129. An iceberg floats in sea water. If the specific gravity of iceberg and sea water are 0.9 and 1.03 respectively. The percentage of total volume of the iceberg below the sea water surface is**

(a) 87.38% (b) 8.738%
(c) 7.38% (d) 78.38%

**Ans. (a) :** given,

Specific gravity of iceberg ($S_{iceberg}$)

$S_{Ice} = 0.9$

$\therefore$ density of Iceberg ($\rho_{Ice}$) = 900 kg/m$^3$

Specific gravity of sea water (Sea water)

$= S_{sw} = 1.03$

$\therefore$ density of sea water ($\rho_{sw}$) = 1030 kg/m$^3$

Now, weight of Iceberg = $\rho_{Ice} \times g \times$ volume of Iceberg ($V_{Ice}$) $= 900 \times 9.81 \times V_{Ice}$ .....(i)

In equilibrium condition,

The weight of sea water displaced

= water of Iceberg

$= 900 \times 9.81 \times V_{Ice}$ ..... (ii)

Therefore

Volume of sea water displaced ($V_{sw}$) =

$$\frac{\text{Weight of sea water displaced}}{\text{weight density of sea water}}$$

$$= \frac{900\times9.81\times V_{Ice}}{1030\times9.81}$$

$$= \frac{90}{103} V_{Ice}$$

Now total percentage of Iceberg above the sea water

$$= \frac{V_{Ice} - V_{sw}}{V_{Ice}} \times 100$$

$$= \frac{V_{Ice} - \frac{90}{103}V_{Ice}}{V_{Ice}} \times 100$$

$= 12.62\%$

Now, total percentage of Iceberg below the sea water = $100 - 12.62 = 87.37\%$

$\approx 87.38\%$

**130. A control volume implies**

(a) an isolated system
(b) a closed system
(c) a specified mass in fluid flow
(d) a fixed region in space

**Ans. (d) :** control volume– A control volume implies a fixed region in space. At steady state, a control volume can be though of as an arbitrary volume in which the mass of the continuum remains constant.

**131. The reading of differential manometer of a venturimeter, placed at 45º to the horizontal is 11 cm. If the venturimeter is turned to horizontal position, the manometer reading will be**

(a) zero (b) $\frac{1}{\sqrt{2}}$ cm

(c) 11 cm (d) $11\sqrt{2}$ cm

**Ans. (c) :** In case of the venturimeter, the reading of deferential manometer will not be change when venturimeter turned from inclined position at $45^0$ to horizontal position that is reading will be same (11 cm).

**132. If a tank discharges water from an orifice under variable head h, the water will be lowered at a constant velocity, if the surface area of the tank varies as**

(a) $\sqrt{h}$ (b) $\frac{1}{\sqrt{h}}$

(c) $\frac{1}{h}$ (d) $h$

**Ans. (d) :** $\int dQ = \int A\, dh$

The surface area of the tank varies as h.

**133. An orifice discharges under a head of 1.25m of water. A pitot tube kept at its centre line at the vena contracta indicates a head of 1.20 m of water. The coefficient of velocity of the orifice is**

(a) 0.990 (b) 0.980
(c) 0.956 (d) 0.965

**Ans. (b) :** Given,

orifice discharge head (h) = 1.25 m

theoretical velocity $(V_{th}) = \sqrt{2gh}$

$= \sqrt{2 \times 9.81 \times 1.25}$

Now, actual velocity $(V_{ac})$ at vena contracta is–

$V_{ac} = \sqrt{2gh} \qquad \{\because h' = 1.20m$

$V_{ac} = \sqrt{2 \times 9.81 \times 1.20} m/s$

Hence, coefficient of velocity $(Cv) = \frac{V_{ac}}{V_{th}}$

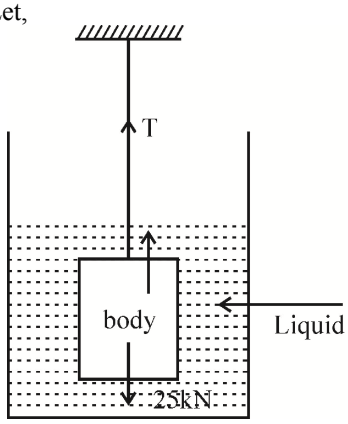
$= \frac{\sqrt{2 \times 9.81 \times 1.20}}{\sqrt{2 \times 9.81 \times 1.25}} = 0.979$

$C_v \approx 0.980$

**134. A solid is suspended by a thread and the body is submerged in a liquid. If V and $\gamma_b$ all the volume and specific weight of the body are 1 $m^3$ and 25 kN and the specific weight of the liquid is 10 kN, the tension in the thread is**

(a) 15 kN (b) 10 kN
(c) 5 kN (d) 25 kN

**Ans. (a) :** Let,




T = Tension in the thread

$F_B$ = Buoyant force = 10kN

= $S_p$ weight of liquid

$S_p$ weight of body = 25 kN

At equilibrium condition

$T + F_B = 25$ kN

$\Rightarrow T + 10$ kN $= 25$ kN

T = 15 kN

**135. Which of the following is used for the measurement of rate of flow in open channel such as flow of water in river?**

(a) Notches (b) weir
(c) orifices (d) Mouthpieces

**Ans. (b) :** "Weir" is used for the measurement of rate of flow in open channel such as flow of water in river.

**136. A stagnation point is a point in the fluid flow where**

(a) total energy is zero
(b) pressure is zero
(c) velocity of flow is zero
(d) total energy is maximum

**Ans. (c) :** At a stagnation point kinetic energy of the fluid converted into the pressure energy. Hence. at stagnation point velocity of flow is zero.

**137. Each term of Bernoulli's equation stated in the form $\frac{p}{\gamma} + \frac{v^2}{2g} + z =$ constant, has units of**

(a) N (b) Nm/kg
(c) Nm/s (d) Nm/N

**Ans. (d) :** Bernoulli's equation.

$$\frac{p}{\rho g}+\frac{v^2}{2g}+2=\text{constant}$$

Potential head (in meter)

Kinetic heat (in meter)

Pressure head (in meter)

Hence, Each term of Bernoullis equation is in energy per unit weight (Nm/N or meter (m)

**138. The specific speed ($N_s$) of pump is given by the expression**

(a) $N_s=\frac{N\sqrt{Q}}{H_m^{5/4}}$ (b) $N_s=\frac{N\sqrt{P}}{H_m^{3/4}}$

(c) $N_s=\frac{N\sqrt{Q}}{H_m^{3/4}}$ (d) $N_s=\frac{N\sqrt{P}}{H_m^{5/4}}$

**Ans. (c) :** In case of pump–

specific speed $(N_S)=\frac{N\sqrt{Q}}{(H)^{3/4}}$

Where,

N = Speed in RPM

Q = Discharge

H = Head

**139. Which of the following statement is correct?**

(a) centrifugal pump convert mechanical energy into hydraulic energy by sucking liquid into chamber

(b) reciprocating pumps convert mechanical energy into hydraulic energy by means of centrifugal energy

(c) centrifugal pump convert mechanical energy into hydraulic energy by means of centrifugal energy

(d) reciprocating pumps convert hydraulic energy into mechanical energy

**Ans. (c) :** Centrifugal pumps convert mechanical energy in to hydraulic energy by means of centrifugal force.

**140. The internal and external diameter of the impeller of a centrifugal pump are 200 mm and 400 mm respectively. The pump is running at 1200 r.p.m. The vane angles of the impeller at inlet and outlet are 20° and 30° respectively. The water enters the impeller radially and velocity of flow is constant. The tangential velocity at the inlet and outlet in m/s is**

(a) 125.6, 25.13 (b) 12.56, 25.13

(c) 12.56, 2.513 (d) 1.256, 25.13

**Ans. (b) :** Internal dia of impeller ($D_1$) = 200 mm = 0.2m

External Dia of impeller ($D_2$) = 400 mm = 0.4m

speed of impeller (N) = 1200 r.p.m.

$Q = 20^o$

$\phi = 30^o$

$u_1$ = tangential velocity at inlet

$$=\frac{\pi D_1 N}{60}=\frac{\pi\times 0.2\times 1200}{60}$$

$u_2$ = Tangential velocity at outlet

$$=\frac{\pi D_2 N}{60}=\frac{\pi\times 0.4\times 1200}{60}$$

$$=25.13\text{m/s}$$

**141. A centrifugal pump having outer diameter equal to two times the inner diameter and running at 1000 rpm. Works against a total head of 40 m. The velocity of flow through the impeller is constant and equal to 2.5 m/s. The vanes are set back at an angle 40° at outlet. If the outer diameter of the impeller is 500 mm and width at outlet is 50 mm, the discharge is.**

(a) 1.963 $m^3$/sec (b) 0.1963 $m^3$/min

(c) 0.1963 $m^3$/sec (d) 0.1963 $cm^3$/sec

**Ans. (c) :** Discharge through centrifugal pump

$$(Q)=\pi D_1 B_1\times v_{f_1}$$

The velocity of flow through the impeller is constant means.

$$v_{f_1}=v_{f_2}=2.5\text{m/s}$$

$$N=1000\text{r.p.m.}$$

$$D_2=500\text{mm}=0.5\text{m}$$

(outer dia of impeller)

Width at outlet ($B_2$) = 50 mm

= 0.05m

Therefore,

Discharge (Q) = $\pi D_2 B_2\times v_{f_2}$

$$=\pi\times 0.5\times 0.5\times 2.5=0.1963\text{m}^3/\text{sec}$$

**142. If two pumps, identical in all respects and each capable of delivering a discharge Q against a head H are connected in parallel, the resulting discharge is**

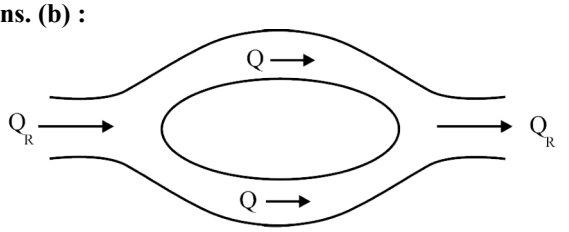
(a) Q against a head 2H

(b) 2Q against a head H

(c) 2Q against a head 2H

(d) 2Q against a head $H^{1/2}$

**Ans. (b) :**




Given,

Each pipe discharge = Q

In the parallel pipe connection discharge is cumulative.

Now,

Resultant discharge ($Q_R$) by two pumps = $Q_1 + Q_2$

$\because Q_1 = Q_2 = Q$

$\therefore Q_R = Q + Q$

$Q_R = 2Q$

Head remains same in parallel connection

$H_1 = H_2 = H$

Hence, resultant Discharge is 2 Q against a head H.

**143. In laminar flow through a pipe discharge varies**

(a) linearly as the viscosity
(b) as the square of radius
(c) inversely at the pressure drop
(d) inversely as the viscosity

**Ans. (d) :** discharge (Q) = $\frac{\pi}{8\mu}\left(\frac{-\partial p}{\partial x}\right).R^4$

$$Q \propto \frac{1}{\mu}$$

Where, R = radius of pipe

$\mu$ = viscosity

$\frac{\partial p}{\partial x}$ = Pressure gradient

therefore, In laminar flow through a Pipe discharge varies inversely proportional to the fluid viscosity.

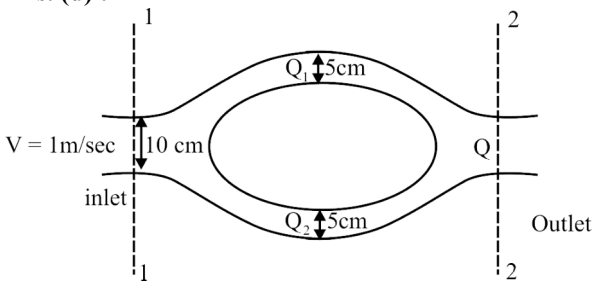
**144. A pipeline of diameter 10cm carries a discharge at a velocity of 1 m/s. It branches into two pipes each of diameter 5 cm. Consider the following statements**

1. **velocity in each branch is 2 m/s**
2. **discharges in the two branched pipes are equal**
3. **rate of flow before and after branching is same**
4. **velocity in 5cm pipe is half of that in 10 cm pipe**

**Select the correct answer using codes given below:**

(a) 1 and 2 (b) 1 and 4
(c) 2, 3 and 4 (d) 1, 2 and 3

**Ans. (d) :**




$$Q = Q_1 + Q_2$$

Discharge in main pipe = Q = AV

$$= \frac{\pi}{4} \times 0.1^2 \times 1 = 7.85 \times 10^{-3} m^3/sec$$

- Since dia of pipes is same so discharge in each pipe will be same $Q_1 = Q_2 = \frac{Q}{2}$
- Velocity in each pipe = $\frac{Q_1 \text{ or } Q_2}{\frac{\pi}{4}(0.05)^2}$

$$= \frac{\frac{\pi}{4}(0.1)^2 \times 1 \times \frac{1}{2}}{\frac{\pi}{4}(0.05)^2} = 2 \text{ m/sec}$$

- Discharge at inlet = discharge at outlet

**145. A penstock is 3000 m long. Pressure wave travels in it with a velocity of 1500 m/s. If the turbine gates are closed uniformly and completely in a period of 4.5 seconds, then it is called**

(a) rapid closure (b) slow closure
(c) sudden closure (d) uniform closure

**Ans. (b) :** Time taken by the pressure wave from valve to tank

$$t = \frac{2L}{C} = \frac{2 \times 3000}{1500} = 4 sec$$

Since time of closure = $4.5 sec > 4 sec$

= Slow closure of valve.

**146. A reaction turbine discharge 35m$^3$/s under a head of 9m and with an overall efficiency of 91%. The power developed in kW is**

(a) 286.65 (b) 37.49
(c) 3.822 (d) 28.665

**Ans. (*) :** overall efficiency $(\eta_0) = 91\% = 0.91$

Discharge (Q) = 35m$^3$/s

Head (H) = 9m

$$\eta_0 = \frac{\text{Shaft power in kw}}{\text{water power in kw}}$$

$$= \frac{p}{\frac{\rho \times g \times Q \times H}{100}}$$

$$p = \frac{\eta_0 \times \rho \times g \times Q \times H}{100}$$

$$= \frac{0.91 \times 1000 \times 9.81 \times 35 \times 9}{100}$$

$$= 2812.03 kw$$

No option is correct

**147. A draft tube is fitted to the exit of a reaction turbine, located at 1.82 m above tail race level. The efficiency of draft tube is 75% and velocity of flow at the entry to draft tube is 10 m/s. The pressure at the entrance section of draft tube is**

(a) –5 m (b) –3.82 m
(c) 10.2 m (d) 1.82 m

**Ans. (a) :** $H_S$ = vertical height of draft tube above the tail race,

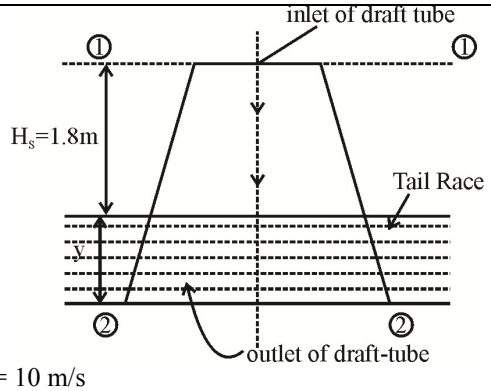
Efficiency of draft tube

$$= \frac{\text{Conversion of kinetic head into pressure head}}{\text{Kinetic head at the inlet of draft tube}}$$

$$\eta_d = \frac{\left(\frac{v_1^2}{2g} - \frac{v^2}{2g}\right)}{\left(\frac{v_1^2}{2g}\right)} = \frac{v_1^2 - v_2^2}{v_1^2}$$

Given, $\eta_d = 75\% = 0.75$

Velocity at entry of the draft tube




$(v_1) = 10$ m/s

$$0.75 = \frac{(10)^2 - v_2^2}{(10)^2}$$

$$75 = 10 - v_2^2$$

$$v_2^2 = 100 - 75$$

$$v_2^2 = 25,\ v_2 = 5 \text{ m/sec}$$

Applying Bernoulli's equation to inlet (section 1–1) and outlet (section 2-2) of the draft tube and taking section 2-2 as the datum line, we get,

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + (H_s + y) = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

$$\frac{p_2}{\rho g} = \text{Atmospheric head} + y$$

$$= \frac{p_a}{\rho g} + y$$

$$z_2 = 0$$

If $\frac{p_a}{\rho g}$ (i.e., atmospheric pressure head is taken zero)

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + (H_s + y) = \left(\frac{p_a}{\rho g} + y\right) + \frac{v_2^2}{2g} + 0$$

$$\frac{p_1}{\rho g} = \frac{v_2^2 - v_1^2}{2g} - H_s$$

$$= \frac{(5)^2 - (10)^2}{2 \times 9.81} - 1.82$$

$$= -3.823 - 1.82$$

$$= -5.643\text{m}$$

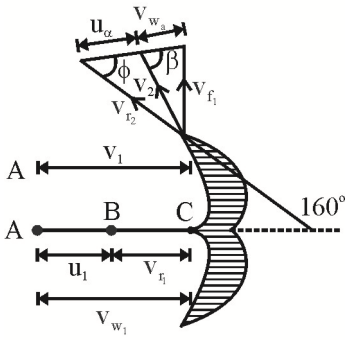
$$\frac{p_1}{\rho g} \approx -5\text{m}$$

Where, $\frac{p_1}{\rho g}$ = Pressure head at the entrance section of draft - tube

**148. A Pelton wheel has a mean bucket speed of 10 m/s with a jet of water flowing at the rate of 700 litre/sec under a head of 30m. The bucket deflects the jet through an angle of 160°. Theoretical velocity of jet is 24.26 m/s. Assuming the coefficient of velocity 0.98, the velocity of whirl at inlet in cm/sec is**

(a) 2377 (b) 23.77
(c) 237.7 (d) 23770

**Ans. (a) :**




Velocity triangles for pelton wheel –
theoretical velocity of jet ($v_{th}$) = 24.26 m/s
coefficient of velocity ($c_v$) = 0.98
$\therefore$ velocity of jet inlet ($v_j$) = $c_v \times v_{th}$
= 0.98 × 24.26 =23.77 m/s
Let,
$V_{w1}$ = velocity of whirl at inlet from the velocity triangle velocity of jet inlet ($v_1$) = velocity of whirl at inlet ($v_{w1}$)

$$v_1 = v_{w1}$$
$$= 23.77 \text{ m/s}$$
$$= 2377 \text{ cm/sec}$$

**149. Two jets strikes the buckets of a Pelton wheel, which is having shaft power as 15450 kW. The diameter of each jet is given as 200 mm. If the net head on the turbine is 400m, the power at the inlet of turbine is (Take $C_v$ = 1.0)**

(a) 21817.44 kW (b) 2181.1744 kW
(c) 21.81 kW (d) 21817.44 W

**Ans. (a) :** Shaft power SP = 15450 kW
Dia of jet = 200 mm
Net head ($H_n$)= 400 m
$C_V = 1$

Absolute velocity of nozzle, $V_1 = C_v\sqrt{2gH}$

$$= 1\sqrt{2 \times 9.81 \times 400}$$

$V_1$ = 88.58 m/sec

Discharge through jet–

$$Q = AV_1$$

$$Q = \frac{\pi}{4}(0.2)^2 \times 88.58$$

$Q = 2.78\ m^3/sec$

Power of inlet of turbine of both jet

$$= 2 \times \frac{1}{2}\rho Q V_1^2$$

$= 1000 \times 2.78 \times 88.50^2$
= 21843 kW

**150. The hydraulic grade line is**

(a) always above the energy grade line
(b) the velocity head below the energy grade line
(c) always above the closed conduit
(d) always sloping downward in the direction of flow

**Ans. (b) :** The hydraulic grade line is the velocity head below the energy grade line.

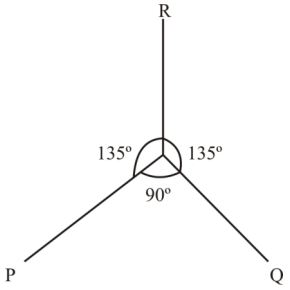
# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 2016
# Solved Paper

**1. If three forces P, Q and R keep a body in equilibrium, the angle between P & Q is 90° and R is acting vertically downward, then magnitude of forces R and Q respectively in terms of P will be**

(a) $P/\sqrt{2}, \sqrt{2}P$ (b) $\sqrt{2}P, P/\sqrt{2}$

(c) $P, \sqrt{2}P$ (d) $\sqrt{2}P, P$

**Ans. (d) :**




Using lanci's theorem

$$\frac{R}{\sin 90^\circ} = \frac{Q}{\sin 135^\circ} = \frac{P}{\sin 135^\circ}$$

$$R = \frac{P}{\sin 135^\circ} = \frac{P}{1/\sqrt{2}} = \sqrt{2}P$$

$Q = P \quad R = \sqrt{2}P$

**2. The algebraic sum of moments of two unlike parallel forces about any point in their plane will be**

(a) Zero

(b) Constant

(c) Always anticlockwise

(d) Always clockwise

**Ans. (b) :** The algebraic sum of moments of two unlike parallel forces about any point in their plane will be constant and is equal to the moment of the couple. .

**3. If the number of members (m) in a frame having joints (j) is more than that required by the equation m=2j-3, then such frames called**

(a) Ideal frame (b) Imperfect frame

(c) Perfect frame (d) Redundant frame

**Ans. (d) :** In frames

$m = 2j - 3$ (Perfect frame)

$m > 2j - 3$ then called Redundant frame.

$m < 2j - 3$ called deficient frame.

**4. A steel bar of 500mm length is under tensile stress of 100 N/mm$^2$. If the modulus of elasticity is $2\times10^{11}$ N/m$^2$, then the total elongation of the bar will be**

(a) 0.25 mm (b) 2.50 mm

(c) 5.00 mm (d) 0.50 mm

**Ans. (a) :** $\text{Strain} = \dfrac{\text{stress}}{\text{modulus of elasticity}}$

$$\frac{\Delta l}{l} = \frac{\sigma}{E}$$

$$\Delta l = \frac{\sigma}{E} \times l$$

$$= \frac{100}{2\times10^{11}\times10^{-6}} \times 500$$

Total elongation ($\Delta l$) = .25mm

**5. Hoop stress for a wooden wheel with steel flat tyre in terms of outside diameter of the wheel *D*, inside diameter of the tyre *d* and Young's modulus *E* is**

(a) E(D-d)/d (b) E(D/d)

(c) E(D/(D-d)) (d) E(d/D)

**Ans. (a) :** Hoop stress for wooden wheel,

$$= \frac{E(D-d)}{d}$$

E = modulus of elasticity

D = External diameter of wheel

d = Internal diameter of wheel

**6. A tensile force of P is applied on a compound bar having two members X and Y. Then the load shared by the member X is**

(a) $\dfrac{PA_XE_X}{A_XE_X + A_YE_Y}$ (b) $\dfrac{PA_YE_Y}{A_XE_X + A_YE_Y}$

(c) P/2 (d) $\dfrac{PE_X}{E_Y}$

**Ans. (a) :** Tensile force P is applied on a compound bar having two members X and Y then

load shared by $X = \dfrac{PA_xE_x}{A_xE_x + A_yE_y}$

load shared by $Y = \dfrac{PA_yE_y}{A_xE_x + A_yE_y}$

**7. Relation between the elastic constants E and C for poisson's ratio of 0.25 is**

(a) E=10C (b) E=5C

(c) E=2.5C (d) E=0.25C

**Ans. (c) :** Relation between E and C with $\mu$ is

$E = 2C(1+\mu)$

$E = 2C(1+.25)$

$E = 2.5C$

**8. The radius of Mohr's circle represents**

(a) Minimum normal stress

(b) Maximum normal stress

(c) Minimum shear stress

(d) Maximum shear stress

**Ans. (d) :** Radius mohr's circle represent the value of maximum shear stress.

- Normal stress on the plane of maximum shear stress is represented by coordinate of center of mohr circle.

**9. For an inclined plane in a rectangular block subjected to two mutually perpendicular normal stresses 1000 MPa and 400 MPa and shear stresses 400 MPa, the maximum normal stress will be**

(a) 1200 MPa (b) 700 MPa

(c) 600 MPa (d) 200 MPa

**Ans. (a) :** use of max. principal stress formula,

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{1000+400}{2} + \sqrt{\left(\frac{1000-400}{2}\right)^2 + 400^2}$$

$$= 700 + \sqrt{300^2 + 400^2}$$

$$= 700 + 500$$

$$= 1200 \text{ MPa}$$

**10. At a hinge in a beam**

(a) Shear force is zero

(b) Bending moment is maximum

(c) Bending moment is zero

(d) Bending moment changes sign

**Ans. (c) :** At a hinge in a beam bending moment is zero.

**11. The bending moment for a beam with equal overhangs and carrying equal point loads each of P at the free ends, the shear force for the beam portion between the supports is**

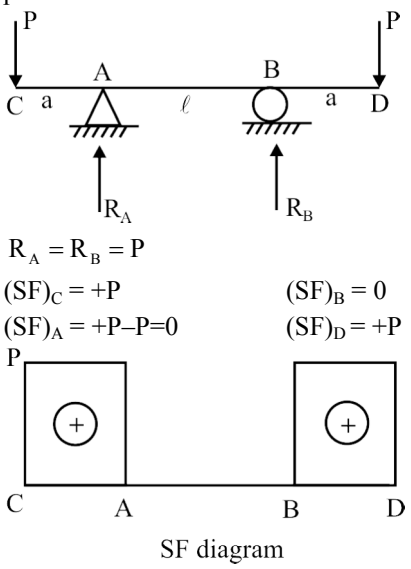
(a) Zero

(b) equal to P

(c) Linearly varying from –P to +P

(d) Linearly varying from 0 to P

**Ans. (a) :** If load applied at overhang portion of beam than shear force is zero between supports of beam portion.




$R_A = R_B = P$

$(SF)_C = +P$ $(SF)_B = 0$

$(SF)_A = +P-P=0$ $(SF)_D = +P$




SF diagram

**12. Assumption made in the theory of bending is**

(a) Radius of curvature is small

(b) Radius of curvature is large

(c) Transverse sections of the beam do not remain plane after bending

(d) Doesn't follow Hook's law in bending

**Ans. (b) :** Assumptions in theory of Bending,

1. The material of beam is homogeneous and isotropic.
2. The value of young's modulus of plasticity is same in tension and compression.
3. The transverse sections which were plane before bending, remain plane after bending also.
4. The radius of curvature is large as compared to dimensions of the cross section.
5. Each layer of the beam is free to expand or contract, independently of the layer, above or below it.

**13. In a beam of rectangular cross section, the ratio of the maximum shear stress to the average shear stress is**

(a) 2.50 (b) 2.00

(c) 1.50 (d) 1.33

**Ans. (c) :** In rectangular beam section

$$\frac{\tau_{max}}{\tau_{avg.}} = \frac{3}{2}$$

in circular section

$$\frac{\tau_{max}}{\tau_{avg.}} = \frac{4}{3}$$

**14. A simply supported beam of 1 m span is carrying a load of 4.8 kN at mid span. If Young's modulus of elasticity (E) is $2\times10^8$ kN/m$^2$ and moment of inertia (I) is 20 cm$^2$, then the maximum deflection will be**

(a) 5.00 mm (b) 2.50 mm
(c) 0.50 mm (d) 0.25 mm

**Ans. (b) :** When loading at center is SSB then deflection $\Delta = \frac{wl^3}{48EI}$

$$= \frac{4.8\times\left(10^3\right)^3}{48\times 2\times 10^8\times 10^{-6}\times 20\times 10^4}$$

$\Delta = 2.5\text{mm}$

**15. The differential equation of flexure is**

(a) $EI\frac{d^2y}{dx^2} = -M$ (b) $EI\frac{dy}{dx} = M^2$

(c) $EI\frac{d^2y}{dx^2} = M^2$ (d) $M\frac{d^2y}{dx^2} = -EI$

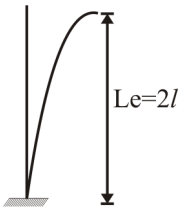
**Ans. (a) :** Differential equation of flexure is

$$EI\frac{d^2y}{dx^2} = -M$$

**16. The slenderness ratio of free standing column of length 4 m and 40mm×40mm section is**

(a) 115.2 (b) 230.4
(c) 693.24 (d) 346.8

**Ans. (c) :** Slenderness ratio




$$(\lambda) = \frac{kl}{r}$$

$$= \frac{2\times 4000}{\sqrt{\frac{I}{A}}}$$

$$= \frac{8000}{\sqrt{\frac{40^4}{12\times 40^2}}} = 693.24$$

**17. Euler's crippling load for a column of length l with both ends hinged and flexural rigidity EI is given by**

(a) $P=2\pi^2EI/\ell^2$ (b) $P=\pi^2EI/\ell^2$
(c) $P=4\pi^2EI/\ell^2$ (d) $P=\pi^2EI/(4\ell^2)$

**Ans. (b) :** When both ends hinged then $\ell_e = \ell$

Euler's crippling load $P = \frac{\pi^2EI}{\ell^2}$

- When both fixed

$\ell_e = \ell/2$

- When one end free and one fixed

$\ell_e = 2\ell$

- One end hinged and one fixed

$\ell_e = \frac{\ell}{\sqrt{2}}$

**18. For shaft in torsion**

(a) $\frac{q}{r} = \frac{T}{J}$ (b) $\frac{C\theta}{l} = \frac{T}{J}$

(c) $\frac{q}{r} = \frac{C\theta}{l}$ (d) All the above

**Ans. (d) :** Torsion equation

$$\frac{q_{max}}{R} = \frac{q}{r} = \frac{T}{J} = \frac{C\theta}{L}$$

q = shear stress
C = shear modulus
r = distance from center of shaft
T = torque
L = length of shaft
θ = angle of twist
J = polar moment of Inertia

**19. The ratio of maximum shear stress of a solid shaft of diameter D to that of a hollow shaft having external diameter D and internal diameter 0.5D is**

(a) 0.50 (b) 1.00
(c) 1.50 (d) 0.94

**Ans. (d) :** max shear stress for solid shaft

$$\tau_{max_1} = \frac{16T}{\pi D^3}$$

max. shear stress for hollow shaft

$$\tau_{max_2} = \frac{16T}{\pi D^3\left[1-\left(\frac{d}{D}\right)^4\right]}$$

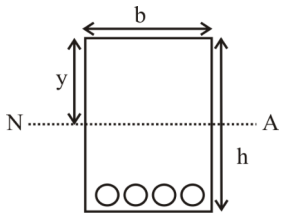
$$\text{Ratio} = \frac{\tau_{max_1}}{\tau_{max_2}} = \frac{16T/\pi D^3}{16T/\pi D^3\left[1-\left(\frac{.5D}{D}\right)^4\right]}$$

$= .94$

**20. Section modulus (Z) for a rectangular cross section is**

(a) $bh^3/12$ (b) $bh^3/6$
(c) $bh^2/12$ (d) $bh^2/6$

**Ans. (d) :** Section modulus $Z = \frac{I}{Y}$



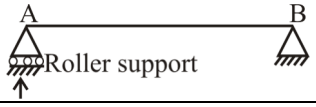

$$Z = \frac{\frac{bh^3}{12}}{\frac{h}{2}} = \frac{bh^2}{6}$$

**21. The reaction at a roller support of beam will be**

(a) Tangential to support (b) Normal to support

(c) Unknown in direction (d) Inclined to support

**Ans. (b) :** Reaction at Roller support of beam will be always Normal to support.




**22. For a beam subjected to point loads, the shear force between point loads**

(a) is constant (b) varies linearly

(c) has parabolic variation

(d) always remain zero

**Ans. (a) :** When a beam is subjected to points load shear force between point loads is constant.

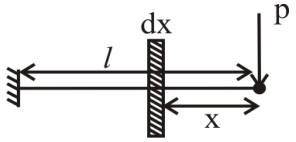
• Any beam segment where no loads are applied the shear force remains constant (Horizontal line).

**23. The Strain energy stored due to bending for a cantilever beam of span *l*, Modulus of elasticity E, moment of inertia I and a point load P at free end is**

(a) $P^2l^3/(6EI)$ (b) $P^2l^5/(40EI)$

(c) $P^2l^3/(96EI)$ (d) $P^2l^5/(240EI)$

**Ans. (a) :**




Moment Mx = P × x

$$\text{Strain energy} = \int Ux = \int_{\partial}^{l} \frac{(P \times x)^2}{2EI} dx$$

$$U = \frac{P^2l^3}{6EI}$$

**24. The method in which a single equation is formed for all loads on the beam and the equation constructed in such a way that the integral constants apply to all the sections of the beam is**

(a) Moment area method

(b) Conjugate beam method

(c) Macaulay's method

(d) 6method of super position

**Ans. (c) :** In Macaulay's method a single equation is formed for all loads on the beam and the equation constructed is such a way that the integral constants apply to all the sections of beam.

The starting point is the relation from Euler Bernoulli beam theory.

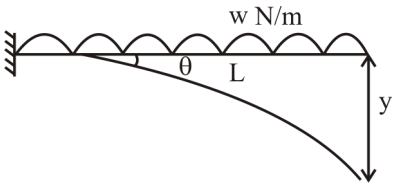
$$EI\frac{d^2y}{dx^2} = M$$

**25. A cantilever beam with UDL of w N/m for entire length of L will have maximum deflection equal to**

(a) $wL^4/(48EI)$ (b) $5L^4/(384EI)$

(c) $wL^4/(8EI)$ (d) $wL^4/(3EI)$

**Ans. (c) :**




Max deflection $y = \frac{wL^4}{8EI}$

Max slope $\theta = \frac{wL^3}{6EI}$

**26. What is poise?**

(a) Unit of viscosity (b) Dyne-sec/$cm^2$

(c) 0.1 times Ns/$m^2$ (d) All the above.

**Ans. (d) :** Poise is the unit of dynamic viscosity in the centimeter-gram-second system of units.

1 poise = 1 Dyne-sec/$cm^2$ = 0.1 N-s/$m^2$

**27. Surface tension has the units of**

(a) Nm (b) N/m

(c) N/$m^2$ (d) N/$m^3$

**Ans. (b) :** Surface tension (T) $= \frac{\text{force per unit}}{\text{length}}$

$$T = \frac{F}{L} = \frac{N}{m} = N/m$$

Unit of T = N/m

**28. The ratio of specific heat of a gas at constant pressure and the specific heat of the gas at constant volume**

(a) is always constant

(b) varies with temperature

(c) varies with pressure

(d) is not constant

**Ans. (a) :** Ratio of specific heat of gas at constant pressure of the constant volume is $K = \frac{C_p}{C_v}$ is always constant for a gas.

- Ratio of specific heat is dimensionless and the value is the same in SI and the imperial system of units.

**29. A perfect gas obeys**

(a) Boyle's law only
(b) Charles' law only
(c) both Boyle's law and Charles' law
(d) None of the above

**Ans. (c) :** The ideal gas or perfect gas is a combination of all the individual gas laws. The ideal gases perfectly obey the ideal gas laws (Boyle's law and Charles law).

**30. The thermal efficiency of an engine, which is supplied heat at the rate of 15000 Nm/s and gives an output of 4500 W is**

(a) 45% (b) 50%
(c) 30% (d) 33%

**Ans. (c) :** input = 15000 N/m/s or watt

output = 4500w

$$\text{efficiency } \eta = \frac{\text{output}}{\text{input}} \times 100$$

$$= \frac{4500}{15000} \times 100$$

$$= 30\%$$

**31. Micro-manometer is used to**

(a) determine low pressure difference
(b) determine higher pressure difference
(c) pressure measurement for gases only
(d) measure pressure in congested areas

**Ans. (a) : Micromanometer-** The micromanometer is another variation of liquid column manometer that is based on the principle of inclined tube manometer and is used for the measurement of extremely small differences of pressure.

**32. Prandtl's Pitot tube measures**

(a) Stagnation head only
(b) Pressure head only
(c) Velocity head and Stagnation head
(d) Pressure head and Stagnation head

**Ans. (d) :** Pitot static tube or Prandtl Pitot tube It measures both stagnation pressure and static pressure. It consist two holes in the tube. The inner hole is known as stagnation tube, which measures stagnation pressure. The outer one is known as static tube, which measures static pressure.

**33. Based on principle of conservation of energy, the equation derived for an ideal fluid is**

(a) Bernoulli's Equation
(b) Newton's Law of Viscosity
(c) Impulse – momentum equation
(d) Continuity equation

**Ans. (a) :** Bernoulli's Equation is based on conservation of energy and equation is for an ideal fluid.

$$h + \frac{V^2}{2g} + \frac{P}{\rho g} = \text{constant}$$

**34. Mouthpiece has increased net head when compared to small orifice due to**

(a) turbulence in the tube
(b) vena contracta occurred within the tube
(c) smaller length of the tube
(d) smooth boundary of the tube

**Ans. (b) :** Mouthpiece has increased net head because of Vena Contracta occurred within the tube as compared to small orifice.

- The discharge rate is increased due to a decrease in the pressure at the venacontracta with the mouthpiece , resulting in an increase in the effective head, causing the flow.

**35. If Velocity potential satisfies Laplace equation the flow will be**

(a) Irrotational flow (b) Free vortex flow
(c) Continuous flow (d) Forced vortex flow

**Ans. (c) :** Velocity potential function ($\phi$)

$$u = \frac{-\partial\phi}{\partial x}$$

$$v = \frac{-\partial\phi}{\partial y}$$

$$w = \frac{-\partial\phi}{\partial z}$$

**Laplace equation-**

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0$$

- Any function $\phi$ that satisfy the laplace equation will correspond to some case of fluid flow (continuous).
- If velocity potential function satisfy laplace equation it represent the possible steady, in compressible irrotational flow.

**36. The ratio of viscous forces to inertial forces is called**

(a) Mach number (b) Weber number
(c) Reynolds number (d) Froude number

**Ans. (c) :** $\text{Reynolds number} = \frac{\text{Inertia force}}{\text{Viscous force}}$

$$\text{Mach number} = \sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}}$$

$$\text{Weber Number} = \sqrt{\frac{\text{Inertia force}}{\text{Surface tension force}}}$$

$$\text{Froude Number} = \sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}}$$

$$\text{Euler Number} = \sqrt{\frac{\text{Inertia force}}{\text{Pressure force}}}$$

**37. If the position of meta centre remain lower than centre of gravity of the floating body, the body will remain in a state of**

(a) Stable equilibrium
(b) Unstable equilibrium
(c) Neutral equilibrium
(d) None of the above

**Ans. (b) :** M is above G - Stable equilibrium
M is coincide G - Neutral equilibrium
M is below G - Unstable equilibrium

**38. The error in discharge due to error in the measurement of head over a rectangular notch is given by**

(a) $\frac{dQ}{Q} = \frac{1}{2}\frac{dH}{H}$ (b) $\frac{dQ}{Q} = \frac{3}{2}\frac{dH}{H}$

(c) $\frac{dQ}{Q} = \frac{3}{4}\frac{dH}{H}$ (d) $\frac{dQ}{Q} = \frac{5}{2}\frac{dH}{H}$

**Ans. (b) :** In rectangular Notch $\frac{dQ}{Q} = \frac{3}{2}\frac{dH}{H}$

In triangular Notch $\frac{dQ}{Q} = \frac{5}{2}\frac{dH}{H}$

**39. Impulse momentum equation gives relationship between force (F), mass (m) and velocity (v) as**

(a) $F = v\frac{dm}{dt}$ (b) $F = \frac{1}{m}\frac{dv}{dt}$

(c) $F = \frac{d(mv)}{dt}$ (d) $F = v\frac{dv}{dt}$

**Ans. (c) :** The impulse experienced by the object equals the change in momentum of the object in equation

$$f = \frac{d(mv)}{dt}$$

In a collision, objects experience an impulse, the impulse causes and is equal to the change in momentum.

**40. In turbulent flow, the loss of head is approximately proportional to**

(a) Velocity (b) $(\text{Velocity})^{1/2}$
(c) $(\text{Velocity})^{3/4}$ (d) $(\text{Velocity})^2$

**Ans. (d) :** In case of turbulent flow through pipe it has been observed through experiment that the viscous friction effects associated with fluid are proportional to
(i) Length of the pipes
(ii) Wetted perimeter
(iii) $V^n$, where V is the average velocity of flow and n is an index varying from 1.5 to 2 [For commercial pipe = 2]

**41. A rigid pipe of length L, diameter D, wall thickness t and modulus of elasticity E is conveying water with a mean velocity of V. If mass density and bulk modulus of water are ρ and K respectively, then water hammer due to sudden closure in the pipe will cause increase in pressure equal to**

(a) $\rho LV/T$ (b) $V\sqrt{\rho K}$

(c) $V\sqrt{\frac{\rho}{\left(\frac{1}{K}+\frac{D}{Et}\right)}}$ (d) None of the above

**Ans. (b) :** Water hammer due to sudden closure of value $= V\sqrt{\rho K}$ or $P = \rho VC$

$C = \sqrt{\frac{k}{\rho}}$, C = velocity of pressure wave

- If gradual closure of valve $P = \frac{\rho LV}{t}$
- If sudden closure of elastic pipe

$$P = V\sqrt{\frac{\rho}{\left(\frac{1}{k}+\frac{D}{Et}\right)}}$$

**42. The discharge from a centrifugal pump is 100 lpm and the leakage is 11 lpm. Then the volumetric efficiency of the pump is**

(a) 89% (b) 90%
(c) 99% (d) 80%

**Ans. (a) :** Volumetric efficiency

$$\eta_v = \frac{\text{Actual discharge}}{\text{Theoretical discharge}}$$

$$\eta_v = \frac{100-11}{100} = 89\%$$

**43. Draft tube for a reaction turbine is arranged for**

(a) Safety purpose only
(b) Increasing velocity head
(c) Converting kinetic head into pressure head
(d) Diverting water only

**Ans. (c) :** Draft tube is a connecting pipe which is filled generally at the outlet or exit of turbines which and convert kinetic energy of water at outlet of turbine to static pressure.

**44. Which hydraulic turbine more efficient at part load operation?**

(a) Pelton wheel (b) Francis turbine
(c) Propeller turbine (d) Kaplan turbine

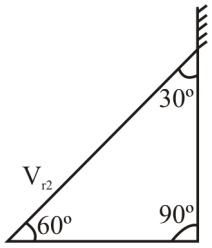
**Ans. (d) : Kaplan turbine–** A propeller turbine is quite suitable, when the load on the turbine remain constant at part load its efficiency is very low since the blade is fixed, the water entre with shock and eddies are formed which reduce the efficiency.

• This defect of the propeller turbine removed in kaplan turbine. In a kaplan turbine the runner blades are adjustable, So high efficiency maintained even at part load.

**45. In an inward flow reaction turbine, velocity of flow at the inlet is 2 m/s. if the guide vanes make an angle of 30º to the tangential velocity direction of the runner, the absolute velocity of the water leaving the guide vanes is**

(a) 1 m/s (b) 2 m/s
(c) 3 m/s (d) 4 m/s

**Ans. (d) :**




$V_{r1}$ = 2 m/s at inlet

$\phi = 60^\circ$

$V_{r1} = V_{r2}\cos\phi$

$$V_{r2} = \frac{V_{r1}}{\cos\phi}$$

$$= \frac{2}{\cos 60} = \frac{2}{1/2} = 4$$

$V_{r2}$ = 4 m/s

**46. In a laminar boundary layer over a flat plate, the growth of the boundary layer with distance x from leading ledge is given by δ/x is proportional to Reynolds number ($Re_x$) as**

(a) $Re_x^{1/2}$ (b) $Re_x^{-1/2}$
(c) $Re_x^{1/5}$ (d) $Re_x^{-1/5}$

**Ans. (b) :** $\frac{\delta}{x} = \frac{5}{\sqrt{Re}}$

The growth of boundary layer with distance x from leading edge is given by $\frac{\delta}{x}$ is proportional to Reynolds number ($Re_x$) as ($Re_x^{-1/2}$).

**47. The cavitation parameter σ is defined in terms of net positive suction head (NPSH) and net head H as**

(a) NPSH/√H (b) H/NPSH
(c) H/√NPSH (d) NPSH/H

**Ans. (d) :** Thoma's cavitation number is defined as

$$\sigma = \frac{NPSH}{H}$$

where H is the head across the turbo machine.

**48. The momentum correction factor is used to account for**

(a) Change in direction of flow
(b) Change in pressure
(c) Non uniform distribution of velocity
(d) Unsteady flow

**Ans. (c) :** Momentum correction factor is used to account for Non. uniform distribution of velocity.

**49. Centre of Pressure is**

(a) Always below the centroid of the submerged plane
(b) Always at the centroid of the submerged plane
(c) Always above the centroid of the submerged plane
(d) Anywhere with respect to the centroid of the submerged plane

**Ans. (a) :** Center of pressure is always below the centroid (center of Gravity) of the submerged plans.

$$\bar{h}_{cp} = \bar{h} + \frac{I_G \sin^2\theta}{A\bar{h}}$$

**50. What is a condition for a hydraulically efficient channel**

(a) Minimum flow rate.
(b) Maximum wetted perimeter.
(c) Constant velocity.
(d) Minimum wetted perimeter.

**Ans. (d) :** If we take area as constant, hydraulic radius is maximum if the wetted perimeter is minimum.

**51. The Indian standard brick size is**

(a) 19 cm × 9 cm × 9 cm
(b) 18 cm × 9 cm × 9 cm
(c) 20 cm × 10 cm × 10 cm
(d) 20 cm × 10 cm × 9 cm

**Ans. (a) :** Standard brick size = 19 cm × 9 cm × 9 cm

**52. The first class brick immersed in water for 24 hours should not absorb water (by weight) more than**

(a) 5% (b) 10%
(c) 15% (d) 20%

**Ans. (d) :** 1st class brick water content = 20%
2nd class brick water content = 22%
3rd class brick water content = 25%

**53. Seasoning of timber is done for**

(a) To reduce weight only
(b) To make it soft only
(c) To prevent shrinkage and warping
(d) To increase moisture content

**Ans. (c) :** Seasoning is the process of drying timber to remove the bound moisture contained in walls of the wood cells to produce seasoned timber. It is done for prevent shrinkage and warping.

**54. IS 12269 deals with**

(a) 53 grade Ordinary portland cement
(b) 43 grade Ordinary portland cement
(c) 33 grade Ordinary portland cement
(d) Sleeper cements

**Ans. (a) :** for opc 33 grade - IS 269
for opc 43 grade - IS 8112
for opc 53 grade - IS 12269

**55. Workability of concrete may be measured by**

(a) Slump test
(b) Minimum void method
(c) Maximum density method
(d) Fineness modulus method

**Ans. (a) :** Workability of concrete measured by,
(1). Slump test
(2) Compaction factor test
(3) Vee-bee consistometer test

**56. Ultrasonic pulse velocity measurements may be used to establish**

(a) Homogeneity of concrete
(b) Presence of cracks and voids
(c) Quality of concrete in relation to standard requirements
(d) All the above

**Ans. (d) :** Ultrasonic pulse velocity is a non-destructive test which is used to establish,
(1) Homogeneity of concrete
(2) Presence of crack and voids
(3) Quality of concrete in relation to standard requirements.

**57. In the analyses of plane frame, the stiffness factor at a joint is taken equal to the sum of stiffness factors the members connected to the joint by using**

(a) Principle of virtual work
(b) Principle of angular momentum
(c) Principle of super position
(d) Principle minimum work done

**Ans. (c) :** For analysis of plane frame the stiffness factor at a joint is taken equal to the sum of stiffness factors the member connected to the joint by using principle of super position.

**58. The principle of virtual work is applied to elastic system by considering virtual work done by**

(a) Internal forces only
(b) External forces only
(c) Internal as well as external forces
(d) None of the above

**Ans. (c) :** The principle of virtual work done states that in equilibrium the virtual work of the forces applied to a system is zero (external as well as internal forces).

**59. The deflection at any point of a perfect frame can be obtained by applying a unit load at the joint in**

(a) The direction in which the deflection is required
(b) Vertical direction
(c) Horizontal direction
(d) Inclined direction

**Ans. (a) :** For finding the deflection at any point of a perfect frame can be obtained by applying a unit load at the joint in the direction in which the deflection is required.

**60. In the procedure of Macaulay's method, for getting the deflection equation**

(a) The differential equation of flexure is integrated for once
(b) The differential equation of flexure is integrated for twice
(c) The differential equation of flexure is integrated for thrice
(d) None of the above

**Ans. (b) :** In Macaulay's method

$$\frac{d^2y}{dx^2} = \frac{M}{EI}$$

- $1^{st}$ Integration of equation of flexure is getting slope equation.
- $2^{nd}$ Integration of equation of flexure is getting deflection equation.

**61. The Castigliano's second theorem can be used compute deflections**

(a) In statistically determinate structures only
(b) At the point under the load only
(c) for beams and frames only
(d) For any type of structure

**Ans. (d) : Castigliano's second theorem:-** The first partial derivatives of the total energy in a structure with respect to the force applied at any point is equal to the deflection at the point of application of that force in the direction of its line of action. It is applicable for all types of structure.

**62. The width of analogous column in the method of analogy is**

(a) 2/(EI) (b) 1/(EI)
(c) 1/2(EI) (d) 1/4(EI)

**Ans. (b) :** Analogous column is a short column with cross section width L and breadth 1/EI.
Where,
L is the length of beam and E is young's modulus of beam and I is moment of Inertia of beam.

**63. In slope deflection equations, the deformations are considered to be caused by**

(a) Axial force (b) Shear force
(c) Bending moment (d) All the above

**Ans. (c):** By forming slope deflection equations and applying joint and shear equilibrium conditions, the rotation angles are calculated. Substituting them back into the slope deflection equations, member and moment are ready determined. Deformation of member is due to bending moment.

**64. As per IS-800, the minimum pitch of bolts in a row of bolts is recommended as the diameter of the bolt times**

(a) 2.0 (b) 2.5
(c) 3.0 (d) 4.0

**Ans. (b) :** Pitch p shall not less than 2.5d, where d is the nominal diameter of bolt.

- Pitch shall not be more than-

(A) 16t or 200mm whichever is less in case of tension member.

(B) 12t or 200mm which ever is less in case of compression member, where t is thickness of thinnest member.

**65. In a fillet weld, the weakest section is the**

(a) smaller side of the fillet
(b) throat of the fillet
(c) side perpendicular to force
(d) side parallel to force

**Ans. (b) :** Throat is the section of weld with minimum thickness, therefore, it will carry maximum stress and hence it is the weakest section.

**66. In rolled steel beams, major part of bending moment is resisted by**

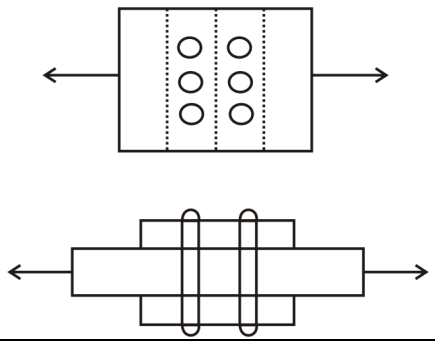
(a) Flanges
(b) Web
(c) Flanges and web
(d) Flanges, web and fillets

**Ans. (a) :** In beam element I or H shaped the web resists the shear forces while the flanges resisted most of the bending moment experienced by the beam.

**67. Load on connection is not eccentric for**

(a) Lap joint
(b) Single cover butt joint
(c) Double cover butt joint
(d) All the above

**Ans. (c) :** Load on double cover butt joint in not eccentric



**68. A steel column in a structure carries a load of 125 kN. It is built up of 2 ISMC 350 channels connected by lacing. The lacing carries a load of**

(a) 125 kN (b) 12.5 kN
(c) 3.125 kN (d) Zero

**Ans. (c) :** Lacing carries the 2.5% of loading that carries the column structure.

$$\text{Lacing carries the load} = \frac{125 \times 2.5}{100}$$
$$= 3.125 \text{ kn}$$

**69. The section in which concrete is not fully stressed to its maximum permissible value while stress in steel reaches its maximum value, is called**

(a) Under reinforced section
(b) Critical section
(c) Over reinforced section
(d) Balanced section

**Ans. (a) : Under reinforced beam-** Reinforced concrete beam sections in which the steel reaches yield strain at loads lower than the load at which the concrete reaches failure strain are called under reinforced sections.

**70. The centre to centre spacing of vertical stirrups, in a rectangular reinforced concrete beam is**

(a) increased towards the centre of the span of the beam.
(b) decreased towards the centre of the span of the beam.
(c) increased at the ends
(d) maintained nonuniformly.

**Ans. (a) :** In rectangular concrete beam center to center spacing of vertical stirrups increased towards the centre of the span of the beam because shear force is reducing towards center of span and maximum at ends.

**71. Maximum reinforcement in an RCC beam of dimension b x D shall not exceed**

(a) 0.06 bD (b) 0.04 bD
(c) 0.02 bD (d) 0.08 bD

**Ans. (b) :** • Maximum reinforced in an RCC beam is 4%.

• Minimum reinforced in an RCC beam is $\frac{0.85}{fy}bd$

**72. The diameter of longitudinal bars in a column should not be less than**

(a) 4 mm (b) 8 mm
(c) 12 mm (d) 16 mm

**Ans. (c) :** • Diameter of longitudinal bar in column should not be less than 12mm

• Diameter of vertical stirrups in column should not be less than 6 mm.

**73. The profile of the prestressing steel in prestressed concrete member follows**

(a) Axial force diagram
(b) Shear force diagram
(c) Bending moment diagram
(d) None of the above

**Ans. (c) :** Profile of prestressing steel in prestressed concrete member follows bending moment diagram.

**74. The technique for establishing and maintaining priorities among various jobs of a project, is known**

(a) Event flow scheduling technique
(b) short interval scheduling
(c) Critical ratio scheduling
(d) Slotting technique for scheduling

**Ans. (c) :** Critical ratio scheduling is the technique for establishing and maintaining priorities among various job of a project.

$$\text{Critical ratio} = \frac{\text{Remaining time}}{\text{Remaining process}}$$

**75. The performance of a specific task in CPM, is known**

(a) Dummy (b) Event
(c) Contract (d) Activity

**Ans. (d) :** Critical path activities are the project tasks that must start and finish on time to ensure that the project ends on schedule.

**76. If a is the optimistic time, b is the pessimistic time and m is most likely time of an activity, the expected time of the activity, is**

(a) (a+m+b)/6 (b) (a+2m+b)/6
(c) (a+4m+b)/6 (d) (a+5m+b)/6

**Ans. (c) :** Expected time $(t_e) = \frac{a + 4m + b}{6}$

a = optimistic time
m = most likely time
b = pessimistic time

**77. The field capacity of a soil is 25%, its permanent wilting point is 15% and specific dry unity weight is 1.5, if the depth of root zone of a crop, is 80 cm, the storage capacity of the soil, is**

(a) 8 cm (b) 10 cm
(c) 12 cm (d) 14 cm

**Ans. (c) :** Storage capacity of soil

$$d' = \left(\frac{\gamma_d}{\gamma_w}\right).d.(FC - PWP)$$

$= 1.5 \times 80(.25 - .15)$

$= 12$ cm

**78. One of the basic assumptions made in unit hydrograph theory is**

(a) Rainfall is uniform all over the catchment
(b) Base flow is considered
(c) Rainfall varies with time
(d) All the above

**Ans. (a) : Assumptions in unit hydrograph theory-**

(1) The effective rainfall is uniformly distributed over the entire drainage basin.

(2) The effective rainfall occurs uniformly within its specifies duration.

(3) The effective rainfalls of equal (unit) duration will produce hydrographs of direct runoff having same or constant time base.

**79. A process determining outflow pattern from reservoirs for any given pattern of inflow, storage and tail-water condition is known as**

(a) Hydrograph (b) Flood discharge
(c) Runoff estimate (d) Flood routing

**Ans. (d) : Flood Routing-** procedure to compute output hydrograph when input hydrograph and physical dimensions of the storage are known, used for flood forecasting, design of spillways, reservoirs and flood protection work etc.

**80. The recording type rain-guage gives**

(a) Mass curve of rainfall
(b) Hydrograph
(c) Total rainfall of 24 hours period
(d) Monthly rainfall

**Ans. (a) :** Recording type rain guage (weighing bucket type rain guage) instrument gives a plot of accumulated rainfall values against the elapsed time and the curve so formed is called mass curve.

**81. Lacey's regime width (W) for a channel having flood discharge of Q is given by**

(a) $W = 1.35(Q)^{1/2}$ (b) $W = 4.75(Q)^{1/2}$
(c) $W = 1.35(Q)^{1/3}$ (d) $W = 4.75(Q)^{1/3}$

**Ans. (b) :** Lacey's regime width $(W) = 4.75\sqrt{Q}$

Silt factor $f = 1.76\sqrt{D_{10}}$

Lacey's normal scour depth $= 1.35\left(\frac{q^2}{f}\right)^{1/3}$

**82. The solid roller bucket is arranged at the end of a spillway to**

(a) measure the discharge
(b) provide stability to the spillway
(c) trap silt from flowing water
(d) dissipate energy through hydraulic jump

**Ans. (d) :** Roller bucket type energy dissipater is an effective method of dissipating energy is by, stilling basin for forming a jump at low discharges and to combine with the basin a sloping apron for developing a satisfactory jump at high discharge.

**83. To avoid gravity dam failure by crushing for the dam material of allowable stress 300 t/sq.m and specific gravity 2.4, the limiting height is equal to**

(a) 88.23 m (b) 125.25 m
(c) 214.29 m (d) >214.29 m

**Ans. (a) :** Limiting height of gravity dam

$$H_{max} = \frac{\sigma}{\gamma_w (S_c + 1)}$$

$$= \frac{3000 kN/m^2}{10(2.4+1)}$$

= 88.23 m

**84. The recommended maximum water supply needs as per CPHEEO guidelines for metropolitan and mega cities where sewerage system is existing or contemplated in litres per capita per day (lpcd)**

(a) 45 (b) 70
(c) 136 (d) 150

**Ans. (d) :** According to CPHEEO guidelines, water demand for metropolitan and mega cities in lpcd is 150.

**85. Dental caries, a disease caused by drinking water due to**

(a) Excess fluorine
(b) Absence of fluorine
(c) Excess of nitrates
(d) Presence of lead

**Ans. (b) :** Dental caries- Permanently damaged area in teeth that develop into tiny holes. It is due to absence of flourine in drinking water.

**86. If the BOD of a diluted sample in 1:100 ratio is 11 ppm in the beginning and 8 PPM at the end of 5 days, then BOD of the sample is**

(a) 300 PPM (b) 950 PPM
(c) 137.5 PPM (d) 73 PPM

**Ans. (a) :** Initial DO = 11PPM
Final DO = 8PPM
Dilution factor = 100
BOD after 5 days = (initial DO – final DO) × D.F.
= (11 – 8)×100
= 300 PPM

**87. The peak factor suggested by CPHEEO for computing carrying capacity in the design of sewers for the contributory population of 20,000 is**

(a) 2.00 (b) 2.25
(c) 2.50 (d) 3.00

**Ans. (d) :**

| Contributory population | Peak factor |
|---|---|
| upto 20000 | 3.00 |
| 20001 to 50000 | 2.50 |
| 50001 to 750000 | 2.25 |
| above 750000 | 2.00 |

**88. The tolerance limit of pH for industrial effluents discharged into inland surface waters**

(a) Less than 5.5 (b) 5.5 to 9.0
(c) Above 9.0 (d) Must be 7 only

**Ans. (b) :** Tolerance pH limit for industrial effluent is 5.5 to 9.0
- BOD for 5 days = 30 – 100 PPM
- temperature $\not> 40^\circ C$

**89. Relationship involving voids ratio (e), degree of saturation (S), water content (w) and specific gravity of soil solids (G) is given by**

(a) wSe=G (b) we=SG
(c) wS=Ge (d) wG=Se

**Ans. (d) :** Se = wG
S = degree of saturation
e = void ratio
w = water content
G = specific gravity

**90. A clay soil sample has unconfined compression strengths in the undisturbed state and remoulded state are 200 kN/sq.m and 60 kN/sq.m respectively. Then its sensitivity is**

(a) 260 (b) 140
(c) 0.3 (d) 3.33

**Ans. (d) :** $\text{Sensitivity} = \frac{\text{UCS in undisturbed soil}}{\text{UCS in remoulded soil}}$

$$= \frac{200 kN/m^2}{60 kN/m^2}$$

= 3.33

**91. The core-cutter method for determining in-situ unit weight is suitable for**

(a) Soils containing gravel particles
(b) Stiff clays
(c) Soft cohesive soils
(d) Sandy soils

**Ans. (c) : Core cutter method** - soft cohesive soil.
**Sand Replacement method**- suitable for gravelly, sandy and dry soil.

**92. The average permeability for flow perpendicular to bedding planes when compared to the average permeability for the flow parallel to the bedding planes**

(a) Always less (b) Always greater
(c) Always equal (d) Lesser or greater

**Ans. (a) :** Flow parallel to bedding plane permeability is greater than flow perpendicular to bedding plane, because soil strata mostly deposited in horizontal layers.

Coefficient of permeability to bedding plane

$$k_H = \frac{k_1z_1 + k_2z_2 + k_3z_3 + ....}{z_1 + z_2 + z_3 + .....}$$

Coefficient of permeability perpendicular to bedding plane

$$Kv = \frac{z_1 + z_2 + z_3 + .....}{\frac{z_1}{k_1} + \frac{z_2}{k_2} + \frac{z_3}{k_3} + ....}$$

**93. The earth pressure theory that is used for the design of cantilever retaining wall is**

(a) Meyerhof's theory (b) Rankine's theory
(c) Terzagi's theory (d) Skempton's theory

**Ans. (b) :** Earth pressure theory used for design of cantilever retaining wall–

(1) Rankine theory

(2) Coulomb's theory

**94. California bearing ratio (CBR) is a**

(a) measure of soil strength
(b) procedure for designing flexible pavements
(c) method of soil identification
(d) measure to indicate the relative strengths of paving materials

**Ans. (a) :** California bearing ratio (CBR) is a penetration test used for evaluate the subgrade strength of roads and pavements and strength of soil.

**95. Correct length of a 100 m tape, weighing 24 N when pull of 200 N is applied at the ends is freely suspended is**

(a) 100.12 m (b) 100.06 m
(c) 99.88 m (d) 99.94 m

**Ans. (d) :** Sag correction $Se = \frac{L}{24}\frac{w^2}{p^2}$

$$= \frac{100}{24} \times \frac{24^2}{200^2}$$

= .06 (Negative)

Correct length = 100 – .06

= 99.94m

**96. If the radius of a circular is 100 m and deflection angle is 90°, then the length of backward tangent is**

(a) 141.4 m (b) 70.7 m
(c) 100 m (d) 50 m

**Ans. (c) :** Length of backward tangent in circular curve

$$(T) = R \tan\left(\frac{\Delta}{2}\right)$$

$$= 100 \tan\left(\frac{90^o}{2}\right)$$

= 100m

**97. Northing, easting and elevation are obtained in**

(a) Chain surveying
(b) Compass surveying
(c) Auto Level surveying
(d) Total station surveying

**Ans. (d) :** Total station is a surveying equipment combination of electromagnetic distance measuring instrument and electronic theodolite. The instrument can be used to measure horizontal and vertical angles as well as sloping, horizontal, elevation distance of object.

**98. The type of Bitumen for which the viscosity has been decreased by a volatile dilutant is known as**

(a) Super Bitumen (b) Cutback Bitumen
(c) Lighter Bitumen (d) Fast Bitumen

**Ans. (b) :** Cutback bitumen (liquid bitumen) is bitumen that is dissolved in a solvent. Typical solvents include Naptha, gasoline and kerosene, white spirit etc. these are volatile in nature, amount of solvents controls the viscosity of cutback bitumen.

**99. Benkelman beam deflection method is used for design of**

(a) Rigid overlay on rigid pavement
(b) Flexible overlay on flexible pavement
(c) Flexible overlay on rigid pavement
(d) Rigid overlay on flexible pavement

**Ans. (b) : Benkelman beam deflection-** Benkelman beam deflection technique is widely used all over the world for evaluating of structural capacity of existing flexible pavements and also for estimation and design of overlays for strengthening of any weak pavement.

Pavement deflection measurements are non destructive.

**100. The absolute minimum radius for a horizontal curve designed for a speed of 80 KMPH, given the permissible values of super elevation and coefficient of friction are 0.04 and 0.16**

(a) 458 m (b) 252 m
(c) 150 m (d) 105 m

**Ans. (b) :** $(e + \mu) = \frac{V^2}{127R}$

$$(.04 + .16) = \frac{80^2}{127 \times R}$$

R = 251.96

R ≅ 252 m

# Andhra Pradesh Public Service Commission
# Assistant Engineer (Subordinate) Exam, 2016
## Solved Paper

**1. Two forces A and B are acting at an angle θ and their resultant R makes on angle α with the force A, then**

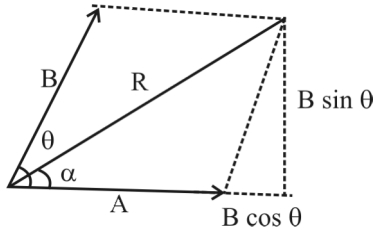
(a) $\tan\alpha = \dfrac{B\sin\theta}{A + B\cos\theta}$

(b) $\tan\alpha = \dfrac{A\sin\theta}{A + B\cos\theta}$

(c) $\tan\alpha = \dfrac{A\sin\theta}{B + A\sin\theta}$

(d) $\tan\alpha = \dfrac{B\sin\theta}{B + a\cos\theta}$

**Ans. (a) :**




According to law of parallelogram,

Magnitude of resultant force (R)

$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

From diagram,

$\therefore \tan\alpha = \dfrac{\text{Prependicular}}{\text{Base}}$

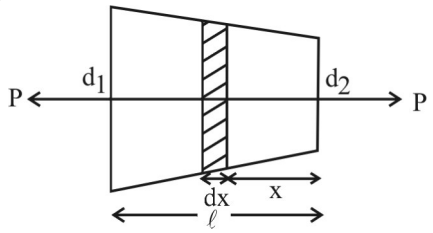
Direction of resultant, $\tan\alpha = \dfrac{B\sin\theta}{A + B\cos\theta}$

**2. A bar uniformly tapering from a diameter $d_1$ at one end to a diameter $d_2$ at the other end, is subjected to an axial tensile load P. The length of the bar is $\ell$ and the young's modulus is E. The extension of the bar is**

(a) $\dfrac{4P\ell}{\pi E d_1 d_2}$ (b) $\dfrac{P\ell E}{4\pi d_1 d_2}$

(c) $\dfrac{P\ell}{4\pi E d_1 d_2}$ (d) $\dfrac{4P\ell d_1 d_2}{\pi E}$

**Ans. (a) :**




Consider a small strip of thickness 'dx' at a distance 'x' from smaller dia side.

Let the dia of the bar be d' at distance x

$d' = d_1 - \left(\dfrac{d_1 - d_2}{\ell}\right)x$

Let, $\left(\dfrac{d_1 - d_2}{\ell}\right) = K$

$\therefore d' = d_1 - Kx$

$\therefore$ cross sectional area at distance x from the smaller end $= A' = \dfrac{\pi}{4}d^{12} = \dfrac{\pi}{4}(d_1 - kx)^2$

Intensity of stress on the section = σ'

$$= \frac{P}{A'} = \frac{4P}{\pi(d_1 - kx)^2}$$

$\therefore$ Strain $= e' = \dfrac{\sigma'}{E} = \dfrac{4P}{\pi E(d_1 - kx)^2}$

$\therefore$ Extension of the length dx $= e'dx$

$$= \frac{4P}{\pi E(d_1 - kx)^2}.dx$$

$\therefore$ Total extension of the bar $= \delta = \dfrac{4P}{\pi E}\int_0^{\ell}\dfrac{dx}{(d_1 - kx)^2}$

$$\delta = -\frac{4p}{\pi EK}\left[\frac{1}{d_1 - kx}\right]_0^{l}$$

$$\delta = \frac{4p}{\pi EK}\left[\frac{1}{d_1 - k\ell} - \frac{1}{d_1}\right]$$

put, $K = \dfrac{d_1 - d_2}{\ell}$

$$\delta = \frac{4p\ell}{\pi E(d_1 - d_2)}\left[\frac{1}{d_1 - d_1 + d_2} - \frac{1}{d_1}\right]$$

$$\delta = \frac{4p\ell}{\pi E(d_1 - d_2)}\left[\frac{1}{d_2} - \frac{1}{d_1}\right]$$

$$\delta = \frac{4p\ell}{\pi E(d_1 - d_2)}.\frac{d_1 - d_2}{d_1 d_2}$$

$$\delta = \frac{4p\ell}{\pi E d_1 d_2}$$

For the particular case when the rod is of uniform dia.

$d_1 = d_2 = d$

and for this case, $\delta = \dfrac{4p\ell}{\pi E d^2}$

**3. It is desired to shrink a thin steel tyre on to a rigid wheel of 3600 mm diameter. What is the internal diameter of the tyre, if after fitting the hoop stress in the tyre is 95 N/mm$^2$. Take E=2x 10$^5$ MPa**

(a) 3598.29 mm (b) 3592.46 mm
(c) 3614.27 mm (d) 3618.38 mm

**Ans. (a) :** D = 3600 mm, $\sigma_H$ = 95 N/m$^2$
E = 2 × 10$^5$ MPa
According to Hook's law

$$\sigma = E\varepsilon$$

$$\varepsilon = \frac{\sigma}{E} = \frac{95}{2\times10^5}$$

$$\varepsilon = 4.75\times10^{-4}$$

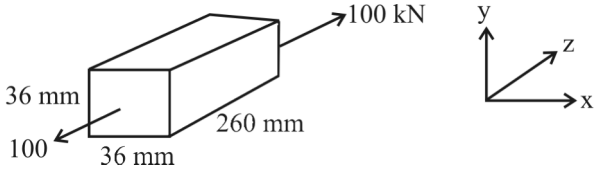
$$\frac{\delta D}{D} = \varepsilon = 4.75\times10^{-4}$$

$\delta D = 4.75 \times 10^{-4} \times 3600$
$\delta D = 1.71$ mm
D – d = 1.71 mm
d = D–1.71
d = 3600 – 1.71
d = 3598.29 mm

**4. A bar 36 mm × 36 mm × 260 mm long is subjected to a pull of 100 kN, in the direction of length. The decrease in each internal dimension is 0.004 mm. The lateral strain is**

(a) 1.11 × 10$^{-4}$ (b) 1.25 × 10$^{-4}$
(c) 1.32 × 10$^{-4}$ (d) 1.45 × 10$^{-4}$

**Ans. (a) :**




Decrease in lateral dimension = 0.004 m (Given)

$$\text{Lateral strain} = \frac{\text{change in length of internal dimension}}{\text{Original length of lateral dimension}}$$

$$\varepsilon_y = \frac{0.004}{36} = \boxed{1.11\times10^{-4}}$$

**5. Kinematic viscosity of air at 20$^0$C is given to be 1.6 × 10$^{-5}$ m$^2$/s. Its kinematic viscosity at 70$^0$C will be approximately.**

(a) 2.2 × 10$^{-5}$ m$^2$/s (b) 1.6 × 10$^{-5}$ m$^2$/s
(c) 1.2 × 10$^{-5}$ m$^2$/s (d) 1.0 × 10$^{-5}$ m$^2$/s

**Ans. (a) :**
Given – $\nu_{20^0C} = 1.6 \times 10^{-5}$ m$^2$/s
$\nu_{70^0C}$ = ?
Initial temperature of air ($T_1$) = 20$^0$C = 293 K
Intial kinematic viscosity of air ($\nu_1$) = 1.6 × 10$^{-5}$ m$^2$/s
Final temperature of air ($T_2$) = 70$^0$C = 343 K

We know that,

$$\nu = \frac{\mu}{\rho} = \frac{\text{dynamic viscosity}}{\text{density}}$$

The dynamic viscosity ($\mu$) of gases increase with increase in temperature

i.e. $\mu \times \sqrt{T}$ & $\upsilon \propto T^{3/2}$

$$\frac{\nu}{T^{3/2}} = \text{constant}$$

$$\frac{\nu_1}{(T_1)^{3/2}} = \frac{\nu_2}{(T_2)^{3/2}}$$

$$\frac{1.6\times10^{-5}}{(293)^{3/2}} = \frac{\nu_2}{(343)^{3/2}}$$

$$\nu_2 = 2.026\times10^{-5}\ \text{m}^2/\text{s}$$

Density of gases decreased with increase in temperature at constant pressure.

**6. The shear developed in lubricating oil, of viscosity 9.81 poise, filled between two parallel plates 1 cm apart and moving with relative velocity of 2 m/s is :**

(a) 196.2 N/m$^2$ (b) 20 N/m$^2$
(c) 40 N/m$^2$ (d) 29.62 N/m$^2$

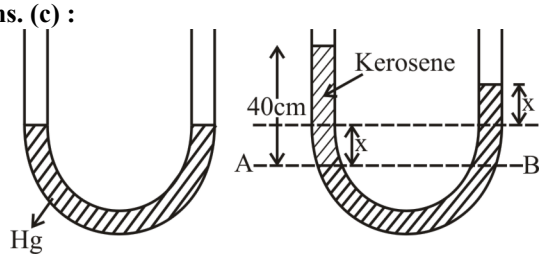
**Ans. (a) : Given :** Viscosity ($\mu$) = 0.981 Ns/m$^2$
Distance between plates = dy = 1 cm =1 × 10$^{-2}$ m
Relative velocity of plates = du = 2 m/s
According to Newton's law of viscosity

$$\tau = \mu\frac{du}{dy} = 0.981\times\frac{2}{1\times10^{-2}} = 196.2\ \text{N/m}^2$$

**7. An open U tube is containing mercury, Kerosene of specific gravity 0.8 is poured into one of its limbs so that the length of column of kerosene is about 40 cm. The level of mercury column in that limb is lowered approximately by how much?**

(a) 2.35 cm (b) 3.6 cm
(c) 1.2 cm (d) 0.6 cm

**Ans. (c) :**




At : AB
0.8 × 40 = 13.6 × 2x

$$x = \frac{32}{13.6\times2}$$

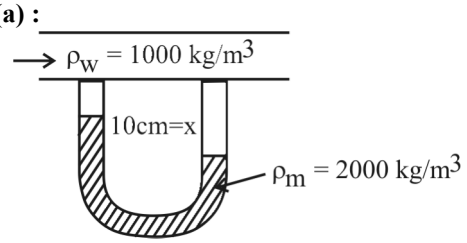
$$x = \frac{16}{13.6}$$

$$\boxed{x = 1.17 \approx 1.2\,\text{cm}}$$

**8. Pressure drop of water, having a density of 1000 kg/m$^3$, flowing through a pipe between two points is measured by using a vertical U--tube manometer. Manometer uses a liquid with density 2000 kg/m$^3$. The difference in height of Manometric liquid in the two limbs of the manometer is observed to be 10 cm. The pressure drop between the two points is :**

(a) 981 N/m$^2$ (b) 98.1 N/m$^2$
(c) 1962 N/m$^2$ (d) 19620 N/m$^2$

**Ans. (a) :**




Density of water ($\rho_\omega$) = 1000 kg/m$^3$
Density of manometric liquid ($\rho_n$) = 2000 kg/m$^3$
Difference in height of manometric liquid (x) = 10 cm
pressure drop between two points = $P_1 - P_2$

Difference in pressure head = $h = \frac{P_1 - P_2}{\rho g}$

When manometer contains heavier liquid, the difference in pressure head is given by

$$h = x\left[\frac{S_n}{S_o} - 1\right]$$

Where,
$S_n$ = special gravity of heavier liquid
$S_o$ = special gravity of liquid flowing through pipe
z = difference in height of heavier liquid in manometer

$S_n = \frac{2000}{1000} = 2$ ; So = $\frac{1000}{1000} = 1$; x = 10 cm = 0.1 m

$$h = 0.1\left[\frac{2}{1} - 1\right] = 0.1$$

Now, $\frac{P_1 - P_2}{\rho g} = h \Rightarrow P_1 - P_2 = 0.1 \times 1000 \times 9.81$

$P_1 - P_2 = 981$ N/m$^2$

Pressure drop between two points = 981 N/m$^2$

**9. Streamlines, path lines and streak lines are virtually identical for**

(a) Steady flow (b) Uniform flow
(c) Flow of ideal fluids (d) Non uniform flow

**Ans. (a) :**
**Stream line –** It is a curved imaginary line such that the tangent at any point on it will give the direction of velocity.
**Path line –** Actual path traveled by any individual fluid particle over some time period is called path line.
**Streak line –** It is the locus of particles that have passed sequentialy through a prescribed point in the flow.
**Note :** For steady flow stream line pathline and streakline are same or virtually identical.

**10. A specimen 160mm$^2$ in cross section stretches by 0.06 mm over a 50 mm gauge length under an axial load of 35 kN. The strain energy is**

(a) 1050 N-mm (b) 780 N-mm
(c) 1435 N-mm (d) 1298 N-mm

**Ans. (a) :** Given,
Area of specimen = A = 160 mm$^2$
Gauge length = L = 50mm
Change in gauge length = 0.05 mm = $\Delta$L
Axial load applied = P = 35 kN = 35 × 10$^3$ N

Now, strain energy (u) = $\frac{1}{2} \times P \times \Delta$

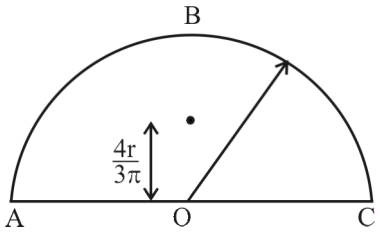
$\Delta = \delta l = 0.06$ mm

$u = \frac{1}{2} = 35 \times 10^3 \times 0.06 = 1050$ N – mm

**11. What is the moment of inertia of a semicircular lamina with the base horizontal about the centroidal axis parallel to x-axis, if the radius is R.**

(a) 0.10976 R$^4$ (b) 0.2084 R$^4$
(c) 0.23 R$^4$ (d) 0.3142 R$^4$

**Ans. (a) : Moment of inertia of a semicircular section–**




We know that,
MOI of semicircular section about base AC is equal to half of the MOI of circular section about AC.

$$I_{AC} = \frac{1}{2} \times \frac{\pi}{64} \times d^4 = 0.393 r^4$$

We also know that area of semicircular section

$$a = \frac{1}{2} \times \pi r^2 = \frac{\pi r^2}{2}$$

and distance between centre of gravity of the section and the base, AC = h = $\frac{4r}{3\pi}$

MOI of the section through its C.G. and parallel to x-x axis is,

$I_G = I_{AC} - ah^2$

$$= \left[\frac{\pi}{8} \times r^4\right] - \left[\frac{\pi r^2}{2} \times \left(\frac{4r}{3\pi}\right)^2\right]$$

$$= \left[\frac{\pi}{8} r^4\right] - \left[\frac{8}{9\pi} r^4\right]$$

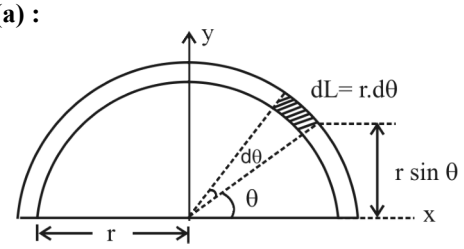
$$\boxed{I_G = 0.1097 R^4}$$

**Note :** The MOI about y-y axis will be the same as that about base AC i.e 0.393 r$^4$

**12. The centre of gravity of a semi circular arc of radius 'r' is on the central radius from the base diameter at a distance of**

(a) $2r/\pi$ (b) $3r/\pi$
(c) $4r/\pi$ (d) $r/\pi$

**Ans. (a) :**




$$\bar{y} = \int \frac{y.dL}{dL}$$

$$\bar{y} = \frac{\int_0^{\pi} (r.\sin\theta)(r.d\theta)}{\int_0^{\pi} r.d\theta}$$

$$\bar{y} = \frac{r^2 \int_0^{\pi} \sin\theta.d\theta}{r\int_0^{\pi} r.d\theta} = \frac{r[-\cos\theta]_0^{\pi}}{[\theta]_0^{\pi}}$$

$$\bar{y} = \frac{-r[\cos\theta]_0^{\pi}}{\pi - 0} = \frac{-r[\cos\pi - \cos\theta]}{\pi} = \frac{-r(-1-1)}{\pi}$$

$$\boxed{\bar{y} = \frac{2r}{\pi}}$$

Alternatively, we can solve it by Pappus-Guldinus equation

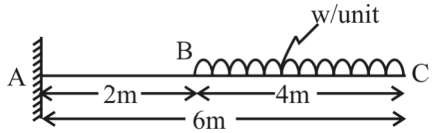
$$\bar{y} = \frac{\text{surface area of sphere}}{2\pi \times \text{length of arc}} = \frac{4\pi r^2}{2\pi \times \pi r} = \frac{2r}{\pi}$$

**Note :** Centroid of semicircular area = $\frac{4r}{3\pi}$

**13. A cantilever of length 6 m is carrying a uniformly distributed load of ω per unit run for a distance 4m from the free end. The value of maximum bending moment is**

(a) 16ω (b) 22ω
(c) 18.5ω (d) 21ω

**Ans. (a) :**

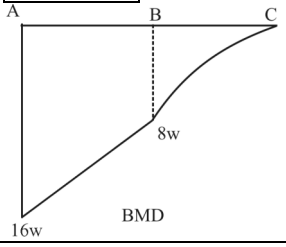



Total load = 4w and act at distance 2 m from free end for bending moment calculation.

$M_C = O$

$M_B = 4w \times 2 = 8\ w$

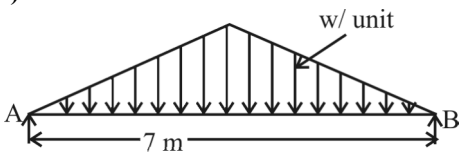
$M_A = 4w \times 4 = \boxed{16\ w = M_{max}}$




**14. A simply supported beam of length 7m is carrying a load whose intensity varies uniformly from zero at each end to ω per unit run at the mid span. The value of maximum shear force is**

(a) 1.75 ω (b) 2.38 ω
(c) 2.96 ω (d) 3.42 ω

**Ans. (a) :**




Total load = Area of loading diagram

$$= \frac{1}{2} \times 7 \times w = 3.5\,w$$

Reactions at A & B will be equal due to symmetry of load, so this total load (3.5 w) is divided equal on both the supported.

$$R_A = R_B = \frac{3.5w}{2} = 1.75\,w$$

for uniformly varying loading, the maximum shear force (S.F.) will be at supports.

So, $\boxed{\text{Max. S.F.} = 1.75w}$

**15. A two-dimensional flow field has velocities along the x and y directions given by $u = x^2t$ and $v = -2xyt$ respectively, where t is time. The equation of streamlines is :**

(a) $x^2y$ = constant
(b) $xy^2$ = constant
(c) xy = constant
(d) $x/y^2$ = constant

**Ans. (a) :** Given,

Velocity along x direction = $u = x^2t$

Velocity along y direction = $v = -2xyt$

Equation of stream line,

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

$$\frac{dx}{x^2t} = \frac{dy}{-2xyt} = \frac{dz}{w}$$

$$\int \frac{dx}{x} = -\frac{1}{2}\int \frac{dy}{y}$$

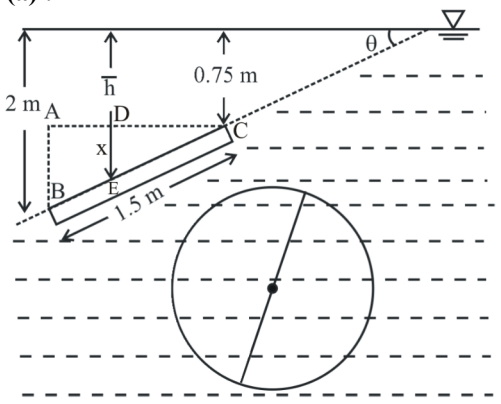
$$\ell n\,(x^2y) = 0$$

$$\boxed{x^2y = \text{constant}}$$

**16. A circular plate 1.5 m diameter is submerged in water with its greatest and least depths below the surface being 2 m 0.75 m respectively. What is the approximate total pressure on one face of the plane?**

(a) 24 kN (b) 12 kN
(c) 16 kN (d) 32 kN

**Ans. (a) :**




Total pressure on the face of plate,

$F = \rho g A \bar{h}$

$\rho = 1000$ kg/m$^3$, $g = 9.81$ m$^2$/s, $A = \frac{\pi}{4} \times 1.5^2$ m$^2$

Similar $\Delta ABC$ & CDE

$$\frac{AB}{BC} = \frac{DE}{EC}$$

$$\frac{(2-0.75)}{1.5} = \frac{x}{(1.5/2)}$$

$$\frac{1.25}{1.5} = \frac{x}{0.75}$$

$x = 0.625$ m

$\bar{h} = 0.75 + x = 0.75 + 0.625 = 1.375$ m

$$F = 1000 \times 9.81 \times \frac{\pi}{4} \times (1.5)^2 \times 1.375$$

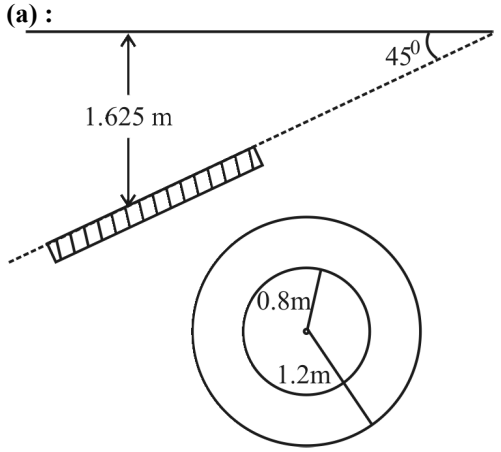
$F = 23836.59$, N= 23.84 kN ⊔ 24 kN

So, F = 24 kN

**17. A circular annular plate bounded by two concentric circles of diameter 1.2 m and 0.8 m is immersed in water with its plane making an angle of $45^0$ with the horizontal. The centre of the circle is 1.625 m below the free surface. What will be the total pressure force on the face of the plate?**

(a) 10.00 kN (b) 7.07 kN
(c) 14.141 kN (d) 18.00 kN

**Ans. (a) :**




Total pressure force (F) = $\rho g A \bar{h}$

projected area (A) = $\frac{\pi}{4}(1.2^2 - 0.8^2)$

$\bar{h} = 1.625$ m

$$F = 1000 \times 9.81 \times \frac{\pi}{4}\left(1.2^2 - 0.8^2\right) \times 1.625$$

F = 10016.18 N = 10.05 kN ⊔ 10 kN

So, total pressure force = F = 10 kN

**18. The position of centre of pressure on a plane surface immersed vertically in a static mass of liquid is given by (where A is the total area of the surface, $I_G$ is the moment of inertial of the area about an axis passing through the centroid of the area and parallel to the free liquid surface, and $\bar{x}$ is the vertical depth of centroid of surface below the free surface of liquid)**

(a) $\bar{X} + \frac{I_G}{A\bar{x}}$ (b) $\bar{X} - \frac{I_G}{A\bar{x}}$
(c) $\bar{X} + \frac{A\bar{x}}{I_G}$ (d) $\bar{X} - \frac{A\bar{x}}{I_G}$

**Ans. (a) : Total pressure and centre of pressure –**

| Inclined surface | Vertical surface | Horizontal surface |
|---|---|---|
| Total pressure (F) = $\rho g A \bar{h}$<br>Centre of pressure (h*) =<br>$h^* = \frac{I_G \sin^2\theta}{A\bar{h}} + \bar{h}$ | $F = \rho g A \bar{h}$<br>When $\theta = 90^0$<br>$\sin 90^0 = 1$<br>$h^* = \frac{I_G}{A\bar{h}} + \bar{h}$ | $F = \rho g A \bar{h}$<br>When $\theta = 0^0$<br>$\sin 0 = 0$<br>$h^* = \bar{h}$ |

**19. In a flow field, the streamlines and equipotential lines**

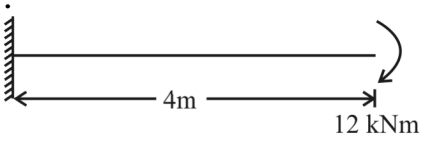
(a) Cut orthogonally except at the stagnation points
(b) Are parallel
(c) Are orthogonal everywhere in the flow field
(d) Cut at any angle

**Ans. (a) :** In a flow field, the stream lines and equipotential lines cut orthogonally at all points except at the stagnation point because at stagnation point the velocity of flow is zero, that means there will be no flow in that direction and hence no flow lines or stream lines.

**20. A cantilever of length 4m is subjected to clockwise couple of 12kN-m at free end. What is the shear force at fixed end?**

(a) zero (b) 12 kN
(c) 3 kN (d) 4 kN

**Ans. (a) :**

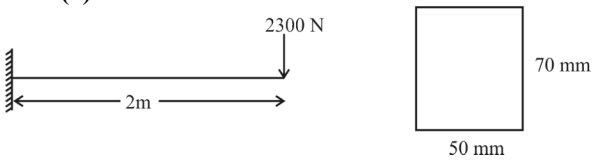



As S.F. is due to load only and here there is no load on the beam so there will not be any S.F. in beam.

**21. A cantilever of length of 2 m fails, when a load of 2300 N is applied at free end. What is the stress at failure, if the section of cantilever is 50 mm × 70 mm.**

(a) 112.65 MPa (b) 121.38 MPa
(c) 132.93 MPa (d) 141.64 MPa

**Ans. (a) :**




From bending equation,

Bending stress at failure = $\sigma_f = \frac{M_{max}}{Z}$

$M_{max} = 2300 \times 2 \times 10^3$

$Z = \frac{bd^2}{6} = \frac{50 \times 70^2}{6}$

$\sigma_f = \frac{2300 \times 2 \times 10^3 \times 6}{50 \times 70^2}$

$= 112.65\,N/mm^2$

$= 112.65$ MPa

**22. A beam of symmetrical section is subjected to bending moment which procedures a maximum stress of 110 N/mm$^2$. If the beam is 260 mm deep find radius to which the longitudinal axis of beam will be bent. Take E= 2 × 10$^5$ MPa**

(a) 236.36 m (b) 21.46 m
(c) 192.71 m (d) 28.9 m

**Ans. (a) :** Given,

$\sigma_{max} = 110 N/mm^2$

d = 260 mm,

$y = \frac{d}{2} = \frac{260}{2} = 130\,mm$

E = 2 × 10$^5$ MPa

From bending equation

$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$

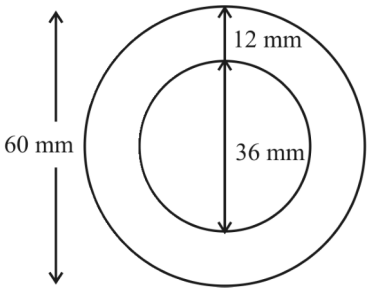
$R = \frac{E.y}{\sigma} = \frac{2 \times 10^5 \times 130}{110} = 236363.63$ mm

$\boxed{R = 236.36m}$

**23. A hollow cast iron pipe is of external diameter 60 mm and 12 mm thick. The section modulus is**

(a) 18448.13 mm$^3$ (b) 17017 mm$^3$
(c) 19362.24 mm$^3$ (d) 16496.75 mm$^3$

**Ans. (a) :**




External dia of pipe (D) = 60 mm

Thickness of pipe (t) = 12 mm

Internal dia. of pipe (d) = D – 2t = 60 – 2 × 12 = 36 mm

Section modulus of hollow circular section,

$Z = \frac{I}{y}$

$I = \frac{\pi}{64}\left(D^4 - d^4\right)$

$y = \frac{D}{2}$

$z = \frac{\pi}{32D}\left[D^4 - d^4\right]$

$Z = \frac{\pi}{32 \times 60}\left[60^4 - 36^4\right] = 18448.13\ mm^2$

taking $\pi = 3.14$

$Z = 18448.13\ mm^2$

**24. A steady, incompressible flow, is given by u = 2x$^2$ + y$^2$ and v = –4xy. What is the convective acceleration along x- direction at point (1, 2)?**

(a) $a_x = -8$ unit (b) $a_x = 6$ unit
(c) $a_x = 24$ unit (d) $a_x = -24$ unit

**Ans. (a) :** Given,

$u = 2x^2 + y^2$

$v = -4xy$

Convective acceleration along x - direction

$a_x = u\frac{du}{dx} + v\frac{du}{dy}$ ..... (i)

$\frac{du}{dx} = 4x$ , $\frac{du}{dy} = 2y$

at (1, 2) , at (1, 2)

$\left(\frac{du}{dx}\right)_{(1,2)} = 4$ , $\left(\frac{du}{dy}\right)_{(1,2)} = 4$

$u_{(1,2)} = 2 \times 1^2 + 2^2 = 6$

$v_{(1,2)} = -4 \times 1 \times 2 = -8$

Put the values in equation (i)

$a_x = 6 \times 4 + (-8) \times 4 = 24 - 32 = \boxed{-8\ units}$

**25. For irrotational and incompressible flow, the velocity potential and stream function are given by ϕ and ψ respectively. Which one of the following sets is correct?**

(a) $\nabla^2\phi = 0; \nabla^2\Psi = 0$ (b) $\nabla^2\phi \neq 0; \nabla^2\Psi = 0$
(c) $\nabla^2\phi = 0; \nabla^2\Psi \neq 0$ (d) $\nabla^2\phi \neq 0; \nabla^2\Psi \neq 0$

**Ans. (a):** Velocity potential function ($\phi$)

$\phi = f(x, y, z, t)$

$$\frac{-\partial\phi}{\partial x} = u, \frac{-\partial\phi}{\partial x} = v, \frac{-\partial\phi}{\partial z} = w$$

Continuity equation for steady incompressible flow–

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\frac{-\partial^2\phi}{\partial x^2} - \frac{\partial^2\phi}{\partial y^2} - \frac{\partial^2\phi}{\partial z^2} = 0$$

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0$$

$$\nabla^2\phi = 0$$

Stream function ($\psi$)

$$\frac{\partial\psi}{\partial x} = v, \qquad \frac{\partial\psi}{\partial y} = -u$$

For no rotation (Irrotational)

$$w_z = \frac{1}{2}\left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right] = 0$$

$$\frac{\partial v}{\partial x} = \frac{\partial u}{\partial x}$$

$$\frac{\partial^2\Psi}{\partial x^2} = -\frac{\partial^2\Psi}{\partial y^2}$$

$$\frac{\partial^2\Psi}{\partial x^2} + \frac{\partial^2\Psi}{\partial y^2} = 0$$

$$\nabla^2\psi = 0$$

**Note :** Velocity potential function exists only for irrotational flow

Stream function exists both for rotational and irrotational flow.

**26. A horizontal pipe of cross sectional area of 5 $cm^2$ is connected to venturimeter of throat are of 3 $cm^2$. The Manometer reading is equal to 5 cm of water. The discharge in $cm^3$/sec is nearly**

(a) 370 (b) 0.45

(c) 55 (d) 150

**Ans. (a) :** Discharge through venturimeter

$$Q = \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}}\sqrt{2gh}$$

$a_1 = 5\ cm^2$; $h = 5$ cm

$a_2 = 3\ cm^2$ $g = 981\ cm/sec^2$

$$Q = \frac{5 \times 3 \times \sqrt{2 \times 981 \times 5}}{\sqrt{5^2 - 3^2}} = 371.42\ cm^3/sec$$

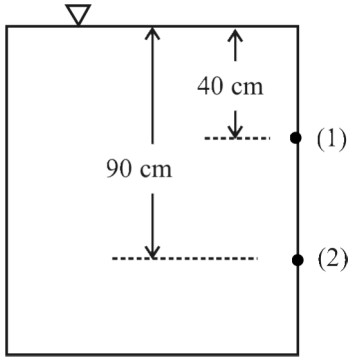
$Q = 370\ cm^3/sec$ (Approx)

**27. A tank containing water the has two orifices of the same size of depths of 40 cm and 90 cm below the free surface of water. The ratio of discharge through these orifices is :**

(a) 2 : 3 (b) 1 : 1

(c) 4 : 8 (d) 16 : 81

**Ans. (a) :**




$$\frac{Q_1}{Q_2} = \frac{A_1V_1}{A_2V_2}, \quad \boxed{\frac{Q_1}{Q_2} = \sqrt{\frac{h_1}{h_2}}}$$

$A_1 = A_2 = A$

$V_1 = \sqrt{2g \times 40}$

$V_2 = \sqrt{2g \times 90}$

$$\frac{Q_1}{Q_2} = \frac{\sqrt{40}}{\sqrt{90}} = \frac{2}{3} \text{ or } 2:3$$

**28. In flow through a pipe, the transition from Laminar to turbulent flow does not depend on**

(a) Length of the pipe

(b) Velocity of the flow

(c) Density of the fluid

(d) Diameter of the pipe

**Ans. (a) :** In flow through pipe the transition from laminar to turbulent flow depends upon.

(i) Velocity of the flow

(ii) Density of the fluid

(iii) Diameter of the pipe

(iv) Reynolds Number

$$R_e = \frac{\rho v D}{\mu}$$

if $R_e < 2000 \rightarrow$ Laminar flow

$R_e = 2000 - 4000 \rightarrow$ Transitional flow

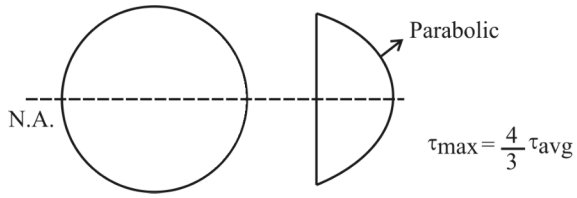
$R_e > 4000 \rightarrow$ Turbulent flow

**29. For a solid circular section, the shear stress distribution is according to a**

(a) parabolic law (b) constant

(c) cubic law (d) linear law

**Ans. (a) :** Shear stress distribution for solid circular section.

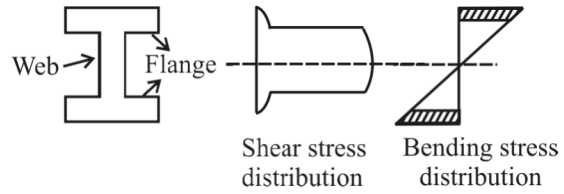



**30. In an I section, most of the shear force will be taken by**

(a) Web (b) top flange

(c) bottom flange (d) near to web

**Ans. (a):**



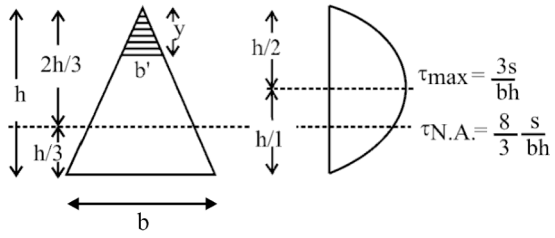

In a I - section as most of the shear is in web portion hence web restrain about 80-90% shear and as most of the bending stresses are at the outer fiber near flange portion hence flange restrain about 80-90% bending moment.

**31. A beam is triangular in section and is of base 'b' and height 'h'. If it is placed with its base horizontal, what is maximum shear stress if the shear force is S.**

(a) $\frac{8S}{3bh}$ (b) $\frac{3S}{4bh^3}$

(c) $\frac{4S}{5b^2h}$ (d) $\frac{8S}{5bh}$

**Ans. (a) :**




MOI about N.A. = $I = \frac{bh^3}{36}$

Let the shear stress intensity be at a depth 'y' from the top.

Width of beam at depth 'y' from the top = $b' = \frac{b}{h}y$

$$\tau = \frac{Sa\bar{y}}{Ib} = \frac{S\left(\frac{y}{2}\frac{by}{h}\right)\left(\frac{2}{3}h - \frac{2}{3}y\right)}{\left(\frac{bh^3}{36}\right)\left(\frac{by}{h}\right)}$$

$$\tau = \left(\frac{12S}{bh^3}\right)y(h-y)$$

For $\tau$ to be max, $\frac{d\tau}{dy} = \left(\frac{12S}{bh^3}\right)(h-2y) = 0 \quad \therefore y = \frac{h}{2}$

$$\tau_{max} = \left(\frac{12S}{bh^2}\right)\frac{2}{3}h \times \frac{h}{3}$$

$$\boxed{\tau_{max} = \frac{3S}{bh}}$$

To find the shear stress at N.A, put $y = \frac{2}{3}h$

$$\tau_{N.A.} = \frac{12S}{bh^3} \times \frac{2}{3}h \times \frac{h}{3}$$

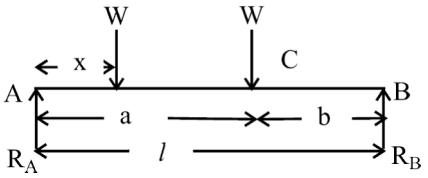
$$\boxed{\tau_{N.A.} = \frac{8S}{3bh}}$$

**32. A beam of length simply supported at the ends carries a point load W at a distance 'a' from the left end. Also *l* -a= b. The deflection under load is given by :**

(a) $\frac{Wa^2b^2}{3EIl}$ (b) $\frac{Wabl}{3EI}$

(c) $\frac{Wab}{6EIl}$ (d) $\frac{Wa^2bl}{3EI}$

**Ans. (a) :**




Reaction at A = $\frac{Wb}{l}$

Reaction at B = $\frac{Wa}{l}$

The strain energy stored by the Beam AB (U)

$U_{AB} = U_{AC} + U_{BC}$

$$= \int_0^a \left(\frac{Wb}{l}x\right)^2 \frac{dx}{2EI} + \int_0^b \left(\frac{Wb}{l}x\right)^2 \frac{dx}{2EI}$$

$$= \frac{W^2b^2a^3}{6EIl^2} + \frac{W^2a^2b^3}{6EIl^2}$$

$$= \frac{W^2a^2b^2}{6EIl^2} + \frac{W^2a^2b^3}{6EIl^2}$$

$$= \frac{W^2a^2b^2}{6EIl^2}(a+b)$$

$\because a + b = l$

$$U_{AB} = \frac{W^2a^2b^2}{6EIl}$$

Deflection under load 'W' is given by –

$$\delta = \frac{\partial u}{\partial W} = \frac{(2W)a^2b^2}{6EIl}$$

$$\boxed{\delta = \frac{Wa^2b^2}{3EIl}}$$

**33. In a lock-gate, the reaction between two gates is (Where F = Resultant pressure on the lock gate, and $\alpha$ = Inclination of the gate with normal to the side of the lock)**

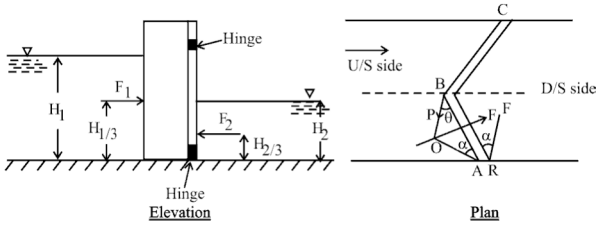
(a) $\frac{F}{2\sin\alpha}$

(b) $\frac{F}{\sin\alpha}$

(c) $\frac{2F}{\sin\alpha}$

(d) $\frac{2F}{\sin(\alpha/2)}$

**Ans. (a):** Lock gates are devices used for changing the water level in a canal or a river for navigation.




Let, AB and BC are two lock gates hinged at A & C.

F = Resultant force due to water on the gate AB or BC acting at right angle to the gate.

R = Reaction at the lower and upper hinge

P = Reaction at the common contact surface of the two gates and acting perpendicular to the contact surface.

Let, the force P and F meat at O. Then the reaction R must pass through O as the gate AB, is in the equilibrium under the action of three forces. Let θ is the inclination of the lock gate with the normal to the side of the lock.

In ∠ ABO, ∠BAB = ∠ABO = O

Resolving all forces along the gate AB and putting equal to zero, we get,

$R\cos\alpha - P\cos\alpha = 0$ or $R = P$

Resolving forces normal to the gate AB

$R \leq \sin\alpha + P\sin\alpha - F = 0$

$F = R\sin\alpha + P\sin\alpha$

$F = 2P\sin\alpha$

$$\boxed{P = \frac{F}{2\sin\alpha}}$$

**34. From a reservoir, water is drained through two pipes of 10 cm and 20 cm diameter respectively. If frictional head loss in both the pipes is same, then the ratio of discharge through the larger pipe to that through the smaller pipe will be**

(a) $\sqrt{2}$ (b) $2\sqrt{2}$

(c) 4 (d) $4\sqrt{2}$

**Ans. (d) :** $h_{f_1} = h_{f_2}$ (According to question)

Dia of largest pipe ($d_1$) = 20 cm

Dia of smaller pipe ($d_2$)= 10 cm

$Q_1$ = Discharge through larger pipe

$Q_2$ = Discharge through smaller pipe

$$\frac{fLV_1^2}{2gd_1} = \frac{fLV_2^2}{2gd_2^2}$$

$$\frac{fLQ_1^2}{12.1d_1^5} = \frac{fLQ_2^2}{12.1d_2^5}$$

$$\left(\frac{Q_1}{Q_2}\right)^2 = \left(\frac{d_1}{d_2}\right)^5$$

$$\left(\frac{Q_1}{Q_2}\right) = \left(\frac{20}{10}\right)^{5/2} = (2)^{5/2}$$

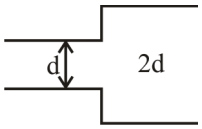
$$\frac{Q_1}{Q_2} = \sqrt{32} = 4\sqrt{2}$$

**35. Two pipelines of equal lengths are connected in series. The diameter of the second pipe is two times that of the first pipe. The ratio of frictional head losses between the first pipe and the second pipe is :**

(a) 1 : 32 (b) 1 : 16

(c) 1 : 8 (d) 1 : 4

**Ans. (a) :**




Dia of first pipe = d

Dia of second pipe = 2d

$$\frac{h_{f_1}}{h_{f_2}} = \frac{\frac{fLQ^2}{12.1d^5}}{\frac{fLQ^2}{12.1(2d)^5}}$$

$$h_f \propto \frac{1}{d^5}\quad \frac{h_{f_1}}{h_{f_2}} = \left(\frac{d_2}{d_1}\right)^5$$

$$\frac{h_{f_1}}{h_{f_2}} = \left(\frac{2d}{d}\right)^5 = \frac{2^5}{1} = 32:1$$

**36. A jet of water issues from a nozzle with a velocity of 20 m/s and it impinges normally on a flat plate moving away from it at 10 m/s. If the cross-sectional area of the jet is 0.02 $m^2$ and the density of water is taken as 1000 kg/m2, then the force developed on the plate will be**

(a) 2000 N (b) 10 N

(c) 100 N (d) 1000 N

**Ans. (a) :** $\rho = 1000$ kg/$m^3$

Velocity of jet (V) = 20 m/s

Velocity of plate (u) = 10 m/s

Cross-sectional area of jet = (A) = 0.02 $m^2$

Force exerted by the jet on a moving flat vertical plate is given by

$F_x = \rho A (V-u)^2$

$F_x = 1000 \times 0.02 (20 - 10)^2$

$\boxed{F_x = 2000\ N}$

**37. Water is pumped out of deep well under a total head of 98 m. Number of similar pumps with a design speed of 1000 rpm and specific speed of 900 with a rated capacity of 0.15 $m^3$/s are available. How many numbers of pumps will be needed to pump the water from the well?**

(a) 3 (b) 9

(c) 6 (d) 2

**Ans. (a) :** Given,

Total head = 98 m

Design speed = N = 1000 r.p.m.

Specific speed = $N_s$ = 900

Rate capacity = Q = 0.15 $m^3$/s

Let, $H_m$ = Head developed by each pump

$$Ns = \frac{N\sqrt{Q}}{H_m^{3/4}}$$

$$900 = \frac{1000 \times \sqrt{0.15}}{H_m^{3/4}}$$

$$H_m^{3/4} = \frac{10}{9} \times \sqrt{0.15} = 0.430$$

$$H_m = (0.43)^{4/3} = 0.325\,m$$

$$\text{No. pumps required} = \frac{\text{Total head}}{\text{Head developed by one pump}}$$

$$n = \frac{98}{0.325} = 301.53$$

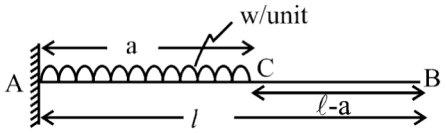
*** Mistake in question data.**

**38. A cantilever of length $\ell$ is carrying a uniformly distributed load of w per unit run of a distance's from fixed end. The slope at the free end is (where EI is the flexural rigidity)**

(a) $\frac{wa^3}{6EI}$ (b) $\frac{wa^4}{8EI}$

(c) $\frac{wa^4}{6EI}$ (d) $\frac{wa^3}{8EI}$

**Ans. (a) :**




Slope at C = slope at B

$$\theta_C = \theta_B = \frac{wa^3}{6EI}$$

Deflection at C,

$$y_C = \frac{wa^4}{8EI}$$

Deflection at B,

$$y_B = y_C + (\ell - a)\theta_C$$

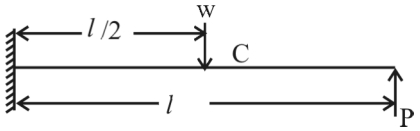
$$y_B = \frac{wa^4}{8EI} + \frac{wa^3}{6EI} \times (l - a)$$

**39. A point load 'W' is acting at mid span of cantilever of length '$l$'. If the free end is supported on a rigid prop, the reaction of the prop is**

(a) $\frac{5W}{13}$ (b) $\frac{5W}{11}$

(c) $\frac{5W}{16}$ (d) $\frac{7W}{9}$

**Ans. (c) :**




To find reaction due to prop,

Deflection at free end due to load 'W' at mid span
= Deflection at free end due to prop reaction

$$\frac{W(l/2)^3}{3EI} + \frac{W(l/2)^2}{2EI} \times \frac{l}{2} = \frac{Pl^3}{3EI}$$

$$\frac{Wl^3}{24EI} + \frac{Wl^3}{16EI} = \frac{Pl^3}{3EI}$$

$$\frac{2W + 3W}{48} = \frac{P}{3}$$

$$\boxed{P = \frac{5W}{16}}$$

**40. The Unwin's formula for the diameter of rivet 'd' considering thickness of the plate 't' is given by**

(a) $d = 6.05\sqrt{t}$ (b) $d = 7.125\sqrt{t}$

(c) $d = 5.36\sqrt{t}$ (d) $d = 6.46\sqrt{t}$

**Ans. (a) :** Unwin's formula for diameter of rivet 'd'

$$d = 6.05\sqrt{t}$$

Where, t = thickness of plate in mm.
and d is the nominal diameter of the rivet in mm.

**41. The stress along the contact surface of a rivet and the plate is**

(a) bearing stress
(b) shearing stress
(c) temperature stress
(d) hoop stress

**Ans. (a) : Bearing stress**– It is a contact pressure between two separate bodies. In this case it is the stress between the contact surface of a rivet and the plate.
**Shearing stress –** The stress which acts parallel to a plane on which a force has been applied. Shear stresses tends to promote siding along a plane.
**Temperature stress –** Stresses due to change in temperature.
**Hoop stress –** This stress is directed along the tangent to the circumference of the cylinder. This stress is tensile in nature. It is also called circumferential stress.

**42. To replace a pipe of diameter D by n parallel pipes of diameter d, the formula is**

(a) $d = \frac{D}{n^{2/5}}$ (b) $d = \frac{D}{n}$

(c) $d = \frac{D}{n^{1/2}}$ (d) $d = \frac{D}{n^{3/2}}$

**Ans. (a) :** We know that, in parallel connection head loss in each pipe remains same.

$$h_{f_1} = h_{f_2} = h_{f_3} = h_{f_n}$$

$$Q_n = Q_1 + Q_2 + Q_2 + .....$$

$$h_f = \frac{fLQ^2}{12.1D^5}$$

$$Q = \left(\frac{12.1D^5}{fL}\right)^{1/2} = n\left(\frac{12.1d^5}{fL}\right)^{1/2}$$

$$D^5 = n^2 d^5 \Rightarrow d = \frac{D}{n^{2/5}}$$

**43. Consider the following statements :**

(1) Pelton wheel is a tangential flow impulse turbine
(2) Francis turbine is an axial flow reaction turbine
(c) Kaplan turbine is a radial flow reaction turbine
Which of the above statements is /are correct?
(a) 1 alone (b) 1 and 3
(c) 2 alone (d) 3 alone

**Ans. (a) :** Pelton wheel is a tangential flow impulse turbine.
Francis turbine is an mixed flow reaction turbine.
Kaplan and propeller turbines are axial flow reaction turbine.
Hence statement (b) and (c) are incorrect.

**44. A reaction turbine discharge 30 $m^3/s$ of water under a head of 10 m with an overall efficiency of 92%. The power developed is :**
(a) 2707 kW (b) 2952 kW
(c) 2870 kW (d) 2652 kW

**Ans. (a) :** Given,
$Q = 30\ m^3/s$, $H = 10m$, $\eta_0 = 92\%$
Power developed = $\rho g Q H \eta_0$
$P = 1000 \times 9.81 \times 30 \times 10 \times 0.92$
$= 2707.56$ kW
$P = 2707$ kW

**45. An error of 1% in measuring the head of water over the crest of a triangular notch, produces an error in the discharge which is equal to**
(a) 2.5% (b) 1.25%
(c) 1.5% (d) 2.0%

**Ans. (a) :** Effect of discharge due to error's in head measurement.
If there is x% error in measurement of head, error is measurement of discharge is given by $\frac{dQ}{Q} \times 100$

$Q = KH^n$
$dQ = KnH^{n-1}\, dH$
$\frac{dQ}{Q} = n \times \frac{dH}{H}$
$\left(\frac{dQ}{Q} \times 100\right) = n\left(\frac{dH}{H} \times 100\right)$
$\frac{dQ}{Q} \times 100 = \%$ error in discharged
$\frac{dH}{H} \times 100 = \%$ error in head measurement

% error in discharge = n × % error in head measurement

$n = \frac{3}{2} \rightarrow$ for rectangular weir
$n = \frac{5}{2} \rightarrow$ for triangular weir
$n = 1 \rightarrow$ for proportial weir

Hence an error of 1% in measuring the head of water over the crest of a triangular notch, produces an error in the discharge equal to 2.5%

**46. In a centrifugal pump, the liquid enters the pump**
(a) At the centre
(b) At the bottom
(c) At the top
(d) From side

**Ans. (a) :** The centrifugal pump acts as a reverse of an inward radial flow reaction turbine. This means that the flow in centrifugal pumps is in the radial outward directions. The liquid enters the pump at the centre and leaves through casing.

**47. The maximum number of reaction components possible for a hinged support for all kinds of loading is**
(a) 2
(b) 0
(c) 3
(d) 1

**Ans. (a) :** Reaction components possible for hinged support = 2
Reaction components possible for fixed support = 3
Reaction components possible for roller support = 1

**48. The volumetric strain produced in a sphere is, how many times the strain of it's diameter**
(a) three
(b) two
(c) four
(d) six

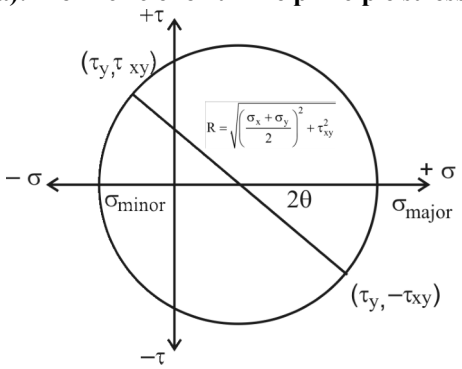
**Ans. (a) :**
Volumetric strain of sphere = 3 × diametric strain
Volumetric strain of cylinder = longitudinal strain + (2 × Diametric strain)

**49. If the number of members in a truss are more than that of a perfect truss. It is known as**
(a) Redundant truss
(b) Deficient truss
(c) Full truss
(d) General type truss

**Ans. (a) :** If the number of members in a truss (m) is equal to 2j-3 then it is called perfect truss.
**Stability and determinacy of truss**
**For 2-D truss**
$m = 2j - 3 \rightarrow$ statically determinate (perfect truss)
$m < 2j-3 \rightarrow$ statically determinate but unstable (deficient truss)
$m > 2j - 3 \rightarrow$ statically indeterminate (redundant truss)

**50. When the principal stresses are unlike, in Mohr's circle, the normal resultant stresses will be**
(a) Negative for points on the circle to the left of the vertical axis through the origin
(b) Negative for points on the circle to the right of the vertical axis through the origin
(c) Positive for points on the circle to the left of the vertical axis through the origin
(d) Either negative or positive depends on the shear stress and radius of the Mohr's circle.

**Ans. (a): Mohr circle for unlike principle stresses–**



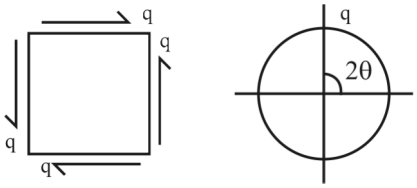

When the principal stresses are unlike in Mohr's circle, the normal and resultant stresses will be negative for points on the circle to the left of the vertical axis through the origin.

**51. Complementary shear stresses of intensity q are induced at a point on a body, the plane carrying the maximum shear stress are at the following angle with the principle planes**

(a) $45^0$ (b) $30^0$

(c) $60^0$ (d) $90^0$

**Ans. (a) :**




From mohr circle for max shear

$2\theta = 90^0$

$\theta = 45^0$

Alternatively by analytical method

$$\tau_{xy} = \frac{\sigma_1 - \sigma_2}{2} \times \sin 2\theta$$

For τxy to be max,

$\sin 2\theta = 1$ (Numerically)

$2\theta = 90^0$ or $270^0$

$\theta = 45^0$ or $135^0$

**52. Which of the following is/are related to measure the discharge by a triangular notch? (where θ is the central angle; H is the head; $C_d$ is coefficient of discharge; and g is acceleration due to gravity.)**

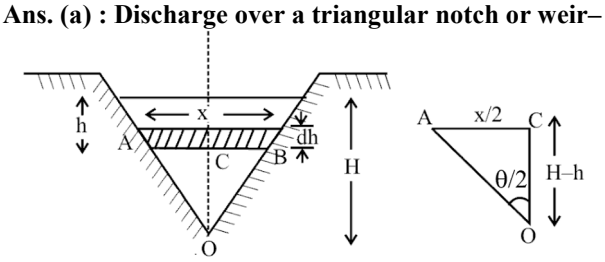
(a) $\frac{8}{15} C_d \sqrt{2g} \tan\frac{\theta}{2} H^{5/2}$

(b) $\frac{4}{15} C_d \sqrt{2g} \tan\frac{\theta}{2} H^{5/2}$

(c) $\frac{8}{15} C_d \sqrt{2g} \tan\frac{\theta}{2} H^{3/2}$

(d) $\frac{2}{3} C_d \sqrt{2g} \tan\frac{\theta}{2} H^{3/2}$

**Ans. (a) : Discharge over a triangular notch or weir–**




In Δ AOC

$$\tan\frac{\theta}{2} = \frac{x}{2(H-h)} \Rightarrow \frac{x}{2} = (H-h)\tan\frac{\theta}{2}$$

Area of strip = x.dh

Theoretical velocity of water through strip = $\sqrt{2gh}$

Discharge through strip,

dQ = $C_d$ × Area of strip × Velocity

$= C_d \times (x.dh)\sqrt{2gh}$

$= C_d \times 2\,(H\text{-}h) \tan\frac{\theta}{2}\, dh\, \sqrt{2gh}$

$$\therefore \text{ total discharge} = Q \int_0^H 2C_d (H-h)\tan\frac{\theta}{2} \times \sqrt{2gh} \times dh$$

$$Q = 2C_d \times \tan\frac{\theta}{2} \times \sqrt{2g} \int_0^H (H-h) h^{1/2}, dh$$

$$= 2C_d \times \tan\frac{\theta}{2} \times \sqrt{2g} \int_0^H \left(H \times h^{1/2} - h^{3/2}\right) dh$$

$$= 2C_d \times \tan\frac{\theta}{2} \times \sqrt{2g} \left[\frac{Hh^{3/2}}{3/2} - \frac{h^{5/2}}{5/2}\right]^H$$

$$= 2C_d \times \tan\frac{\theta}{2} \times \sqrt{2g} \left[\frac{2}{3}H^{5/2} - \frac{2}{5}H^{5/2}\right]^H$$

$$= 2C_d \times \tan\frac{\theta}{2} \times \sqrt{2g} \left[\frac{4}{15}H^{5/2}\right]$$

$$\boxed{Q = \frac{8}{15} C_d \times \tan\frac{\theta}{2} \times \sqrt{2g} \times H^{5/2}}$$

**Note :** Discharge through rectangular notch.

$$\boxed{Q = \frac{2}{3} C_d \times L \times \sqrt{2g} \times H^{3/2}}$$

**53. To avoid vaporization pipe lines are laid over the ridge so that these are above the hydraulic gradient line, not more than**

(a) 6.4 m (b) 2.4 m

(c) 5.0 m (d) 10.0 m

**Ans. (a) : Hydraulic gradient line (HGL)–**

It is defined as the line which gives the sum of presure head (P/ρg) and datum head (z)

$$HGL = \frac{P}{\rho g} + z$$

HGL line lies below TEL with a difference of velocity head ($V^2/2g$)

• In order to avoid vaporization in the pipelines, the pipelines over the ridge is laid in such a way that it is not more than 6.4 m above the hydraulic gradient.

**54. The temperature influence of dynamic viscosity on a fluid is as follows :**

(a) For gases viscosity increase with increase in temperature and for liquids it decreases with increase in temperature

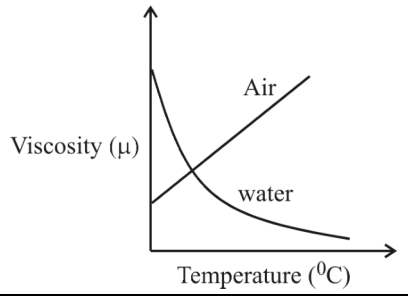
(b) For gases viscosity increases with increase in temperature and for liquids it increases with increase in temperature

(c) For gases viscosity decreases with increases in temperature and for liquids it increases with increase in temperature

(d) For gases viscosity decreases with increases in temperature and for liquids it decreases with increases in temperature.

**Ans. (a) : Effect of temp on viscosity of fluids –**
In case of liquid, cohesive forces between molecules cause viscosity. Hence at higher temp when molecule possesses more energy, they can oppose the large intermolecular cohesive force more strongly. As a result, energized liquid molecule can move more freely, thus viscosity of liquid decreases with increase in temperature.

- In case of gases, viscosity is caused by molecular collisions. At high temp, more molecular collision per unit time occurs. This results in greater resistance to flow.




**55. Which of the followings is the appropriate equation used to compute minimum speed for starting a centrifugal pump?**
**(Where $H_m$ is the manometric head, g is acceleration due to gravity, $D_1$ is the outside diameter of the impeller and $D_2$, is the inter diameter of the impeller.**

(a) $\frac{60\sqrt{2gH_m}}{\pi\sqrt{D_1^2 - D_2^2}}$ (b) $\frac{12\sqrt{gH_m}}{\pi\sqrt{D_1^2 - D_2^2}}$

(c) $\frac{60\sqrt{2H_m}}{\pi\sqrt{D_1^2 - D_2^2}}$ (d) $\frac{48\sqrt{2gH_m}}{\sqrt{D_1^2 - D_2^2}}$

**Ans. (a) : MM speed for starting a centrifugal pump**– When a centrifugal pump is started, it will start delivering liquid only if the pressure rise in the impeller is more than or equal to monometric head flow will commence only if, $\frac{u_2^2 - u_1^2}{2g} \geq H_m$

For minimum speed

$$\frac{u_2^2 - u_1^2}{2g} = H_m \quad \text{..... (i)}$$

also, $\eta_{man} = \frac{gH_m}{v_{w_2} u_2} \Rightarrow H_m = \eta_{man.} \times \frac{v_{w_2} u_2}{g}$ ..... (ii)

equating equation (i) and (ii)

$$\frac{u_2^2 - u_1^2}{2g} = \eta_{man.} \times \frac{v_{w_2} u_2}{g} \quad \text{..... (iii)}$$

$$u_2 = \frac{\pi D_2 N}{60} \text{ and } u_1 = \frac{\pi D_1 N}{60}$$

Substituting the values of $u_2$ and $u_1$ in equation (iii)

$$\left(\frac{\pi D_2 N}{60}\right)^2 - \left(\frac{\pi D_2 N}{60}\right)^2 = 2\eta_{man.} \times v_{w_2} u_2$$

$$\eta_{man} \times v_{w_2} \times u_2 = H_m \times g$$

$$\left(\frac{\pi N}{60}\right)^2 \left[D_2^2 - D_1^2\right] = 2gH_m$$

$$\boxed{N = \frac{60\sqrt{2gH_m}}{\pi\sqrt{D_2^2 - D_1^2}}}$$

**56. The specific speed of a turbine running with a speed of N developing a power of P under working head of H is given by**

(a) $\frac{N\sqrt{P}}{H^{5/4}}$ (b) $\frac{N\sqrt{P}}{H^{5/2}}$

(c) $\frac{N\sqrt{P}}{H^{2/4}}$ (d) $\frac{N\sqrt{P}}{H^{3/2}}$

**Ans. (a) : Specific speed of turbine –** It is defined as the speed of a turbine which is identical in shape, geometrical dimensions, blade angle, gate opening etc., with the actual turbine but of such a size that it will develop unit power when working under unit head.

$$\boxed{N_s = \frac{N\sqrt{P}}{H^{5/4}}}$$

**57. A kaplan turbine with a runner diameter of 9m is operating under a head of 10m. At optimum regime the turbine discharge 500 cumecs of water to develop the power. What is the unit discharge of the turbine?**

(a) 1.952 (b) 0.617

(c) 1.667 (d) 2.68

**Ans. (*) :** Discharge $Q = A_f \times V_f$

$$Q = \left(\frac{\pi}{4}\right)\left(d^2\right)\sqrt{2gh}$$

$$Q = \frac{\pi}{4} \times 9^2 \left(\sqrt{2gH}\right)$$

$$Q \propto \sqrt{H}$$

$$Q = K\sqrt{H} \quad \text{..... (i)}$$

for

$H = 1, Q = Q_u,$

$Q_u = K$ ..... (ii)

From (i) and (ii)

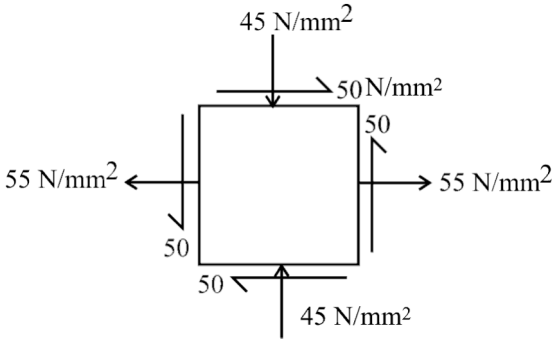
$$Q = Q_u\sqrt{H}$$

$$Q_u = \frac{500}{\sqrt{10}} = 158.11$$

**58. The normal stresses of 55 N/mm$^2$ tensile and 45 N/mm$^2$ compressive on two mutually perpendicular planes are acting at a point in a piece of elastic material. These planes also carry shear stresses of 50 N/mm$^2$. The minor principal stress will be**

(a) 65.71 N/mm$^2$ compressive
(b) 75.71 N/mm$^2$ tensile
(c) 65.71 N/mm$^2$ tensile
(d) 75.71 N/mm$^2$ compressive

**Ans. (a) :**




Sign conversion, Tensile = + ve
Comp. = –ve

$$\sigma_{major/minor} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau^2_{xy}}$$

Considering positive sign for major principle stress and negative sign for minor principle stress.

$\sigma_x = 55$ N/mm$^2$, $\sigma_y = -45$ N/mm$^2$, $\tau_{xy} = 50$ N/mm$^2$

$$\sigma_{minor} = \frac{55 + (-45)}{2} - \sqrt{\left[\frac{55 - (-45)}{2}\right]^2 + 50^2}$$

$\sigma_{minor} = 5 - 70.71 = -65.71$ N/mm$^2$

$\sigma_{minor} = 65.71$ N/mm$^2$ (compressive)

**59. In case of crystalline and brittle materials criterion of failure, that takes place under maximum principal stress, is put forwarded by**

(a) Theory of Rankine
(b) Theory due to Guest
(c) Theory of St. Venant
(d) Theory due to Beltrami and Haigh

**Ans. (a) :**
**(i)** Max. principle stress theory / Rankine's theory / Lames theory / Maximum stress theory.
- Applicable for brittle materials (Best for brittle materials)
- Not suitable for ductile materials
- Not suitable for pure shear case.

**(ii)** Maximum principal strain theory / Saint Venant theory.
- Satisfactory for brittle materials
- Not suitable for pure shear case and hydrostatic loading.

**(iii)** Maximum shear stress theory / Tresca theory / Guess theory.
- Applicable for ductile materials
- Most conservative theory
- Not suitable for hydrostatic loading.

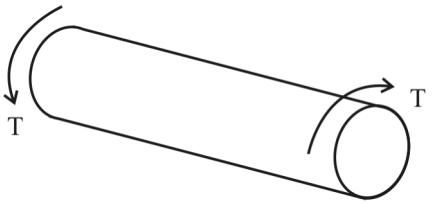
**(iv)** Maximum strain energy theory / Heigh theory.
- Applicable for ductile material
- Not suitable for brittle material and pure shear case.

**(v)** Maximum shear strain energy theory / Distortion energy theory / Hencky theory / Von-mises theory.
- Most appropriate theory for ductile materials
- Perfect for pure shear case.

**60. A shaft of circular section is said to be in pure torsion when it is subjected to equal and opposite end couples whose axes coincides with the**

(a) axis of the shaft
(b) normal line to the axis of the shaft
(c) any line passing through the section of the shaft
(d) parallel line to the axis of the shaft

**Ans. (a) :** A shaft of circular section is said to be in pure torsion when it is subjected to equal and opposite end couples whose axis coincides with the axis of the shaft.




**61. Polar modulus is the ratio of**

(a) Polar moment of inertia of the shaft to the maximum radius
(b) Moment of inertia of the shaft to the maximum radius
(c) Polar moment of inertia of the shaft to the permissible shear stress
(d) Moment of inertia of the shaft to the maximum shear stress

**Ans. (a) :** Polar modulus is defined as the ratio of polar moment of inertia of the shaft to the maximum radius of the shaft. It is also called as torsional section modulus.

$$Z_p = \frac{J}{R}$$

**62. Torsional stiffness is defined as**

(a) torque required to produced a twist of one radian over the length of the shaft
(b) torque required to produce a twist of one radian per limit length of the shaft
(c) torque required to produce a twist of one radian per unit cross sectional area of the shaft
(d) torque required to produce a twist of one radian net unit polar modulus of the shaft

**Ans. (a): Torsional stiffness –** It is defined as torque required to produce a twist of one radian over the length of the shaft.

$$\frac{T}{J}=\frac{G\theta}{L}\Rightarrow GJ\rightarrow \text{Torsional Rigidity}$$

$$\text{Torsional stiffness}=\frac{\text{Torsional Rigidity}}{\text{Length}}=\frac{GJ}{L}$$

$$K=\frac{GJ}{L}=\frac{T}{\theta}$$

**63. The function of governor in turbine is**

(a) keeping the speed of rotation of runner constant for varying loads
(b) keeping the load constant for varying speed of rotations
(c) keeping the speed of rotation of runner in proportion to the varying loads
(d) keeping the discharge through the runner constant for varying loads

**Ans. (a) :** A steam turbine governor is a component of turbine control system that regulates rotational speed in response to changing load conditions. The governor output - signal manipulates the position of steam inlet valve or nozzles which in turn regulates the steam flow to the turbine.

**64. The hydraulic efficiency of Francis turbine is calculated using which of the following equation**
**(Where $V_w$ is the velocity of whirl at inlet : u is the tangential velocity at the inlet. H is head and g is the acceleration due to gravity.)**

(a) $V_wu/gH$ (b) $V_wH/gu$
(c) $V_wu/2gH$ (d) $V_wu^2/gH$

**Ans. (a) :** Hydraulic efficiency of Francis turbine –

$$\eta_h=\frac{R.P.}{W.P.}=\frac{\frac{W}{1000g}\left[V_{w_1}u_1\pm V_{w_2}u_2\right]}{\frac{W\times H}{1000}}$$

$$\eta_h=\frac{V_{w_1}u_1\pm V_{w_2}u_2}{gH}$$

If discharge at outlet in radial direction then $V_{w_2}=0$

$$\boxed{\eta_h=\frac{V_{w_1}u_1}{gH}}$$

**65. Bernoulli's equation represent the**

(a) Energies at any point in the flow field and is obtained by integrating the Euler equations
(b) Forces at any point in the flow field and is obtained by integrating the momentum equation for viscous flows
(c) Momentum at any point in the flow field and is obtained by integrating the equation of continuity
(d) Moment of momentum and is obtained by integrating the energy equation

**Ans. (a) :** Euler's equation of motion–

$$\frac{dP}{\rho}+gdz+vdv=0$$

Bernoulli's equation is obtained by integrating the Euler's equation

$$\int\frac{dP}{\rho}+\int gdz+\int v\,dv=\text{constant}$$

If flow is incompressible ρ is constant

$$\frac{P}{\rho}+gz+\frac{v^2}{2}=\text{constant}$$

$$\boxed{\frac{P}{\rho g}+z+\frac{v^2}{2g}=\text{constant}}$$

Each term in Bernoulli's equation represents energy per unit weight

**66. In a centrifugal pump casing, the flow of water leaving the impeller is**

(a) free vortex motion (b) rectilinear flow
(c) radial flow (d) forced flow

**Ans. (a) :** Flow of water at the impeller of a centrifugal pump is forced vortex and the flow of water learning the impeller through casing is free vortex due to its shape.

**67. The relationship between various efficiencies of turbine is expressed as (where $\varepsilon_0$ is the overall efficiency. $\varepsilon_m$ is the mechanical efficiency, $\varepsilon_h$ is the hydraulic efficiency and $\varepsilon_v$ is the volumetric efficiency.)**

(a) $\varepsilon_0=\varepsilon_h\times\varepsilon_v\times\varepsilon_m$ (b) $\varepsilon_0=\varepsilon_h+\varepsilon_v+\varepsilon_m$
(c) $\varepsilon_h=\varepsilon_0-\varepsilon_0/(\varepsilon_v\,\varepsilon_m)$ (d) $\varepsilon_m=\varepsilon_h-\varepsilon_0/(\varepsilon_v\,\varepsilon_h)$

**Ans. (a) : Efficiencies of turbines –**

**(i) Hydraulic efficiency ($\eta_h$)**

$$\eta_h=\frac{\text{Power delivered to runner}}{\text{Power supplied at inlet}}=\frac{R.P.}{W.P.}$$

**(ii) Mechanical efficiency ($\eta_m$)**

$$=\frac{\text{Power at the shaft of the turbine}}{\text{Power delivered by water to the runner}}=\frac{S.P.}{R.P.}$$

**(iii) Volumetric efficiency ($\eta_v$)**

$$\eta_v=\frac{\text{Volume of water actually striking the runner}}{\text{Volume of water supplied to the turbine}}$$

**(iv) Overall efficiency ($\eta_0$)**

$$\eta_0=\frac{\text{Volume available of the shaft of the turbine}}{\text{Power supplied at the inlet of the turbine}}$$

$$\boxed{\eta_0=\eta_h\times\eta_m\times\eta_v}$$

**68. A column that fails due to direct stress is called**

(a) short column (b) long column
(c) Medium column (d) Slender column

**Ans. (a)** • Short column fails due to direct stress and failure is crushing failure.
• Long column fails due to bending stresses and failure is buckling failure.
• Intermediate medium column fails due to combination of both direct stresses and bending stress and mode of failure is crushing and buckling.

**69. The principal compression member in a crane is called as**

(a) boom (b) post
(c) stanchion (d) strut

**Ans. (a) : Boom –** It is a principal compression member in a crane.
**Post –** It is a compression member in small structure with no roof.
**Stanchion –** A structural steel member usually larger than a strut, whose main function is to withstand axial compressive force.
**Strut –** It is one inclined member subjected to the axial load and supported by hinged or pin jointed support at both the ends.

**70. The load at which the member just buckles is called**

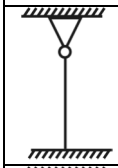
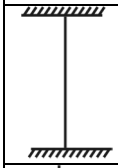
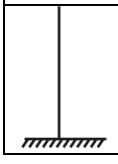
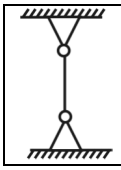
(a) Critical load (b) crushing load
(c) equilibrium lad (d) ultimate load

**Ans. (a) :** The critical load is the greatest load that will not cause lateral deflection (buckling). For loads greater than the critical load, the column will deflect laterally. The critical load puts the column in a state of unstable equilibrium. A load beyond the critical load causes the column to fail by buckling.
**Crushing load –** It is the intensity of load required to crush the material.
**Ultimate local –** Absolute maximum load that a structure can bear without failing.

**71. When both ends of the column are hinged the crippling load at which the column just buckled is**
**(Where E is Young's modulus of column, I is the Moment of inertia of column and L is effective length of column)**

(a) $\frac{\pi^2 EI}{L^2}$ (b) $\frac{\pi^2 EI}{2L^2}$
(c) $\frac{\pi EI}{L^2}$ (d) $\frac{2\pi^2 EI}{L^2}$

**Ans. (a) :**

| Description | Crippling load | Eff. length | Eff. length as per IS code |
|---|---|---|---|
|  | $\frac{2\pi^2 EI}{L^2}$ | leff = $\frac{l}{\sqrt{2}}$ | 0.8 $l$ |
|  | $\frac{4\pi^2 EI}{L^2}$ | leff = $\frac{l}{2}$ | 0.65 $l$ |
|  | $\frac{\pi^2 EI}{4L^2}$ | leff = 2$l$ | 2 $l$ |
|  | $\frac{\pi^2 EI}{L^2}$ | leff = $l$ | $l$ |

**72. The value of angle of repose is the same as the value of _____**

(a) Limiting angle of friction
(b) Coefficient of friction
(c) Static friction
(d) normal reaction between the contact surface

**Ans. (a) : Limiting friction –** The maximum value of static friction upto which body does not move and the angle which it makes with horizontal is called angle of limiting friction.
**Angle of repose –** It is the angle of the inclined plane with horizontal such that a body placed on it is just begins to slide.
**In limiting condition –**

$F = mg \sin\alpha$
$R = mg \cos\alpha$

Where, $\alpha$ = angle of repose

So, $\frac{F}{R} = \tan\alpha$

$\therefore \frac{F}{R} = \mu_s = \tan\theta = \tan\alpha \quad (\because \tan\theta = \mu_s)$

or $\boxed{\theta = \alpha}$

**73. For turbulent flows, the boundary may be classified as hydro-dynamically smooth boundary when k/δ is**

(a) < 0.25 (b) < 0.75
(c) < 1.5 (d) < 6

**Ans. (a) :** In turbulent flows in pipes, very close to the wall, effect of viscosity is maximum. Hence it is said that a laminar sublayer exist near the boundary.
If thickness of laminar sublayer is large and eddies are not able to penetrate till the boundary, boundary act as hydrodynamically smooth where if thickness of laminar sublayer is small and eddies penetrates upto the boundary then boundary is said to be hydrodynamically rough.
**As per Nikuradse experiment –**

$\frac{K}{\delta'} < 0.25 \Rightarrow$ hydrodynamically smooth boundary

$\frac{K}{\delta'} > 6 \Rightarrow$ rough boundary

$0.25 < \frac{K}{\delta'} < 6 \rightarrow$ transition

Where, δ' = thickness of laminar sublayer
K = Avg. size of roughness

**74. According to Rouse the distance required for the establishment of a fully developed turbulent flow is given as**
**(where x is the length of the pipe from the entrance of the pipe upto the section where the flow is established and d is the diameter of the pipe).**

(a) (x/d) = 50 (b) (x/d) = 20
(c) (x/d) = 30 (d) (x/d) = 40

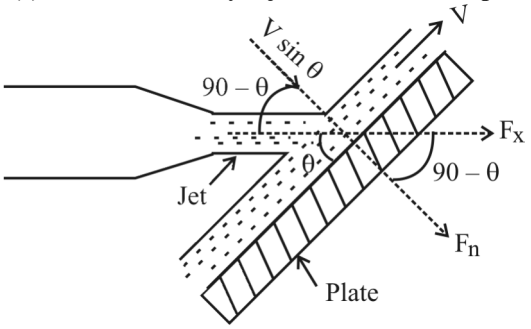
**Ans. (a) :** A fluid flow need some length to fully develop the velocity profile after passing through components like bends, valves, pumps, turbine etc, the length after obstruction until the flow velocity profile is fully developed is called establishment length or entrance length.

For laminar flow $\frac{x}{d} = 0.05\ Re$

For turbulent flow, $\frac{x}{d} = 50$

**75. The normal force of thrust exerted by a jet of water of cross sectional area 'a' and velocity 'V' on a flat plate inclined at an angle θ to the jet is given by**
**(where Q is the discharge)**
(a) $QV \sin^2\theta/g$ (b) $QV \cos\theta/g$
(c) $QV \sin\theta/g$ (d) $QV (1+\sin\theta)/g$

**Ans. (c):Force exerted by a jet on inclined flat plate–**




Force exerted by the jet on the plate in the direction normal to the plate

$F_n$ = Mass of jet per second × Initial velocity of jet before striking in the direction of n – Final velocity of jet after striking in the direction of n.

$= \rho aV [V \sin\theta - 0]$

$\boxed{F_n = \rho aV^2 \sin\theta}$

**Force per unit wt.**

$$\boxed{F_n = \frac{QV\sin\theta}{g}} \qquad \left(\because av = Q,\ \gamma = \rho g\right)$$

$F_x = F_n \sin\theta$

$F_y = F_n \cos\theta$

**76. For the turbulent flow in a pipe of radius R, what will be the value of 'y' at which the point of velocity is equal to the mean velocity?**
(a) 0.223 R (b) 0.45 R
(c) 0.523 R (d) 0.67 R

**Ans. (a) :** Point velocity = Average Velocity

$$5.75 \log_{10}\left(\frac{y}{R}\right) + 8.5 = 5.75 \log\left(\frac{R}{K}\right) + 4.75$$

$$5.75 \log_{10}\left(\frac{y}{R}\right) + 3.75 = 0$$

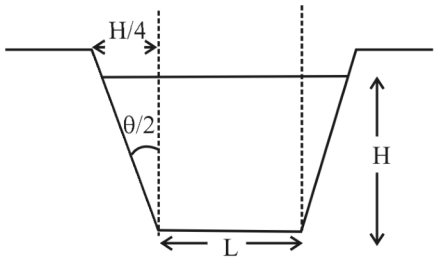
$$\log\left(\frac{y}{R}\right) = -0.6521$$

$$\frac{y}{R} = 0.2228$$

$\boxed{y = 0.223\ R}$

**77. A Cippoletti notch is a trapezoidal notch having inclination of slopping sides as**
(a) 1 horizontal to 4 vertical
(b) 1 horizontal to 2 vertical
(c) 1 horizontal to 3 vertical
(d) 1 horizontal to 1 vertical

**Ans. (a) :** A Cippoletti notch is a trapezoidal notch having inclination of sloping sides as 1 horizontal to 4 vertical.




$$\tan\frac{\theta}{2} = \frac{1}{4}$$

$$\boxed{\frac{\theta}{2} = 14^0 2'}$$

Discharge through Cipolletti weir–

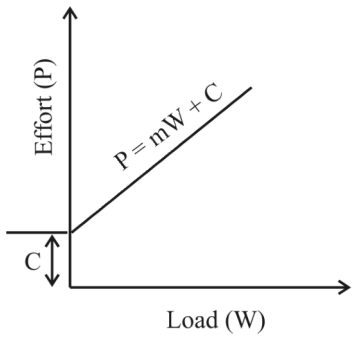
$$\boxed{Q = \frac{2}{3} C_d \times L \times \sqrt{2g}\ H^{3/2}}$$

**78. The law of machine can be expressed mathematically in the form of**
(a) linear (b) exponential
(c) parabolic (d) hyperbolic

**Ans. (a) : Law of machines –** Machines which are used to lift a load are governed by the **'Law of Machine'** which states that the effort to be applied on the machine (P) is related to the weight (w) which it can lift as –

$\boxed{P = mW + C}$ → [Equation of straight line]

Where,
m and C are positive constants which are characteristics of the machine




m = slope of line
C = Effort required to start the machine

**79. In lifting machines, as load increases the mechanical advantage**

(a) increases
(b) decreases
(c) remains same
(d) initially increases and then decreases

**Ans. (a) :** Lifting machines are chose machines that are used for lifting loads. The force (or effect) is applied is applied at one point of one machine and weight (or load) is lifted at the other point of the machine as the load increase the mechanical advantage increases.

**80. What is the velocity ratio of a lifting machine that is working at an efficiency of 60% in raising a load of 7700 N by an effort of 150N?**

(a) 85.55 (b) 51.33
(c) 61.72 (d) 55

**Ans. (a) :** $\text{Efficiency} = \dfrac{\text{Force ratio}}{\text{Velocity ratio}}$

$\text{Velocity Ratio} = \dfrac{\text{Force ratio}}{\text{Efficiency}}$

$VR = \dfrac{7700 \times 100}{150 \times 60}$

$VR = 85.55$

**81. The velocity ratio as per law of machine is equal to**
**(Where m = slope of line and is equal to coefficient of friction, W = load lifted and c= amount of effort required by the m/c to overcome the friction)**

(a) $m + \dfrac{c}{W}$ (b) $mW + c$

(c) $\dfrac{m}{W} + c$ (d) $mc + W$

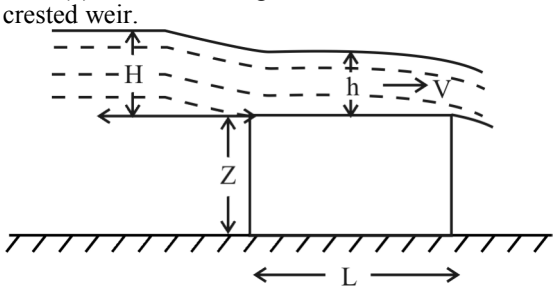
**Ans. (a) :** Velocity Ratio as per law of machine –

$$VR = m + \frac{C}{W}$$

**82. The maximum discharge over a broad crested weir is**
**(Where H is the height of water surface above the crest of the weir on the upstream side; L is the length of the crest of the weir, and $C_d$ is the coefficient of discharge)**

(a) $1.7\ C_d\ LH^{3/2}$ (b) $1.7\ C_d\ LH^{2/3}$
(c) $1.7\ LH^{3/2}$ (d) $1.98\ C_d\ LH^{3/2}$

**Ans. (a) :** A weir having a wide crest is known as broad crested weir.




If 2 L > H, the weir is called broad crested.
if 2L < H, the weir is called narrow crested.

**Discharge over a broad crested weir –**

H= Height of water above the crest
L = Length of crest
h = Head of water at the middle of weir.
v = velocity of flow over the weir.

Applying Bernoulli's equation to the still water surface on the upstream side and running water at the end of weir.

$$O + 0 + H = 0 + \frac{V^2}{2g} + h$$

$$\frac{V^2}{2g} = H - h$$

$$V = \sqrt{2g(H-h)}$$

$Q = C_d \times \text{Area of flow} \times \text{Velocity}$

$$= C_d \times L \times h \times \sqrt{2g(H-h)}$$

$$= C_d \times L \times \sqrt{2g\left(Hh^2 - h^3\right)}$$

The discharge will be maximum, if $(Hh^2 - h^2)$ is max.

$$\frac{d}{dh}\left(Hh^2 - h^3\right) = 0$$

$$2h \times H - 3h^3 = 0$$

$$2H = 3h$$

$$\boxed{h = \frac{2}{3}H}$$

$$Q_{max} = C_d \times L \times \sqrt{2g\left[H \times \left(\frac{2}{3}H\right)^2 - \left(\frac{2}{3}H\right)^2\right]}$$

$$= C_d \times L \times \sqrt{2g}\sqrt{\left[H \times \frac{4}{9}H^2 - \frac{8}{27}H^3\right]}$$

$$Q_{max} = C_d \times L \times \sqrt{2g}\sqrt{\frac{4}{27}H^3}$$

$$= C_d \times L \times \sqrt{2g} \times 0.3849 \times H^{3/2}$$

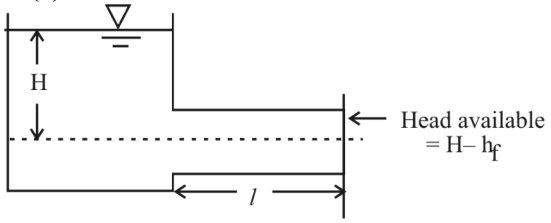
$$= 0.3849 \times \sqrt{2 \times 9.81} \times C_d \times L \times H^{3/2}$$

$$Q_{max} = 1.705 \times C_d \times L \times H^{3/2}$$

**83. The power available at the outlet of the pipe carrying water with a rate of flow of Q and head of H supplied at the entrance of the pipe is (γ is the specific weight : and $h_f$ is the loss of head due to friction)**

(a) $\gamma Q(H - h_f)$ (b) $\gamma QH$
(c) $\gamma Q(H + h_f)$ (d) $\gamma QH(H + h_f)$

**Ans. (a) :**

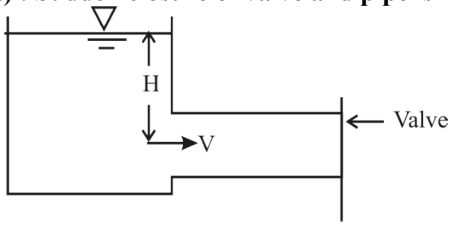


Power supplied at inlet of the pipe
= Weight of water per sec × Head at outlet
$\boxed{P_{inlet} = W \times H = \gamma QH}$
Power available at outlet of pipe
= Weight of water per sec × Head at outlet
$= W \times (H-h_f)$
$\boxed{P_{outlet} = \gamma Q(H-h_f)}$

**84. If the valve in water pipe is instantaneously closed and the pipe is considered rigid, the intensity of pressure rise due to water hammer is given by**
**(Where V is the velocity of flow; γ is the specific weight of water, t is the time taken close the value; K is the bulk modulus of water and g is the acceleration due to gravity)**

(a) $V\sqrt{\frac{\gamma K}{g}}$ (b) $Vt\sqrt{\frac{\gamma K}{g}}$
(c) $Vt\sqrt{\frac{K}{g}}$ (d) $V\gamma\sqrt{\frac{K}{g}}$

**Ans. (a) : Sudden closure of valve and pipe is rigid–**




When valve is closed suddenly, the K.E. of flowing water is converted into strain energy of water if the effect of friction is neglected and pipe wall is assumed perfectly rigid.

Loss of kinetic energy = $\frac{1}{2}\times$ mass of water in pipe $\times V^2$

$$= \frac{1}{2}\times\rho AL\times V^2$$

Gain of strain energy = $\frac{1}{2}\left(\frac{P^2}{K}\right)\times$ Volume

$$= \frac{1}{2}\frac{P^2}{K}\times AL$$

Equating loss of K.E. to gain of strain energy

$$\frac{1}{2}\rho AL\times V^2 = \frac{1}{2}\frac{P^2}{K}\times AL$$

$$P^2 = \frac{1}{2}\rho AL\times V^2\times\frac{2K}{AL}$$

$$P^2 = \rho KV^2$$

$$P = V\sqrt{K\rho}$$

$$p = v\sqrt{\frac{K\rho^2}{\rho}}$$

$$P = \rho V\times C \qquad \left(\because \sqrt{\frac{K}{\rho}} = C\right)$$

Where, C = Velocity of pressure wave

**85. The water pressure at the summit point of a siphon is**
(a) minimum
(b) equal to atmospheric pressure
(c) maximum
(d) zero

**Ans. (a) :** The highest point of syphon is called as summit and a syphon can be defined as a long bend pipe, used to carry water from a reservoir at a higher level to another reservoir at lower level two reservoirs are separated by a hill.
Pressure at summit is below atmospheric as it lies above the free water surface in the reservoir at higher level.
So, the pressure at summit point of a siphon is minimum.

**86. In practice the formation of air lock occurs at _____ water pressure at the summit of a siphon**
(a) 2.5 m of water absolute
(b) 7.8 m of water absolute
(c) 10.3 m of water
(d) 13.6 m of water

**Ans. (a) :** A pipe which raises above its hydraulic gradient line has negative pressure and is known as siphon. Theoretically this negative pressure may be reduced to – 10.3 m of water but in actual practice this pressure is only –7.6 m of water or 10.3 – 7.6 = 2.7 m of water absolute.
If pressure at summit becomes less than 2.7 m of water absolute, the dissolved air and other gases would come out from water and collect at the summit and flow of water will be obstructed.

**87. A lifting machine will be reversible, if the efficiency**
(a) > 50% (b) equal to 100%
(c) >75% (d) <50%

**Ans. (a) : Reversible machine –** If a machine is capable of doing some work in the reverse direction, after the effort is removed, then the machine is known as reversible machine. The condition for a machine to be reversible is its efficiency should be more than 50%.

**88. In general, in first order pulley system velocity ratio is given by (Where n is the number of movable pulleys.)**
(a) $2^n$ (b) 2n
(c) $n^2$ (d) 2n–1

**Ans. (a) :** Velocity ratio of first order pulley system = $2^n$.
Where, n = number of movable pulley.

**89. Velocity ratio in wheel and differential axle is given by**
**(Where D= diameter of effort A, $d_2$ = diameter of the axle B and $d_1$ = diameter of the axle C.)**

(a) $\frac{2D}{d_2-d_1}$ (b) $\frac{2D}{d_2+d_1}$
(c) $\frac{d_2-d_1}{2D}$ (d) $\frac{d_2+d_1}{2D}$

**Ans. (a) :** Differential wheel and Axle

Velocity ratio = $\frac{2D}{d_1 - d_2}$

Mechanical advantage = $\frac{W}{P}$

Efficiency = $\frac{M.A.}{V.R.}$

**90. In a first order system of pulleys, there are 3 movable pulleys. What is the mechanical advantage of the system if the efficiency of the system is 80%.**

(a) 6.4 (b) 4.6
(c) 6.0 (d) 8.0

**Ans. (a) :** V.R $= 2^n = 2^3$

$\eta = \frac{MA}{VR}$

$0.80 = \frac{MA}{2^3}$

MA = 0.80× 8 = 6.4

**91. If `the screw in a screw jack is double threaded then lead of the screw is**

(a) 2 × pitch (b) equal to pitch
(c) pitch / 2 (d) $(pitch)^2$

**Ans. (a) :** If the screw in a screw jack is double threaded them load of the screw is equal to 2 times of pitch.

**92. The point through which the resultant gravitational force acts for any orientation of the body is termed as**

(a) Centre of gravity
(b) Centroid
(c) Origin
(d) Centre pair of plane area

**Ans. (a) : Centre of gravity / Centre of mass –** It is the point where the whole mass of the body is concentrate. This is where the gravitational force (Weight) of the body acts for any orientation of the body.

**Centroid –** It is the geometrical centre of the object. It is the centre of gravity for objects of uniform density.

**93. In order of avoid Cavitation in centrifugal pumps, Thoma's equation for determination of cavitation factor is**

**(Which $N_\varepsilon$ is the specific speed of the pump; N is speed of the pump; and Q is the rate of flow)**

(a) $0.103\ (N_s/1000)^{4/3}$ (b) $0.103\ (QN_s/1000)^{4/3}$
(c) $0.103\ (QN_s/1000)^{3/4}$ (d) $0.103\ (N_s/1000)^{3/4}$

**Ans. (a) : Cavitation in centrifugal pumps –** In centrifugal pumps the Cavitation may occur at the inlet of the impeller of the pump or at the suction side of the pumps. where the pressure is considerably reduced. Hence if the pressure at the suction side of the pump drops below the vapour pressure of the liquid then the cavitation may occur. The cavitation in a pump can be noted by sudden drop in efficiency and head. In order to determine whether cavitation will occure in any portion of the suction side of the pump, the critical value of Thoma's cavitation factor ($\sigma$) is calculated.

**Thomos cavitation factor for centrifugal pumps –**

$$\sigma = \frac{(H_b) - H_s - h_{LS}}{H} = \frac{(H_{atm} - H_v) - H_s - H_{LS}}{H}$$

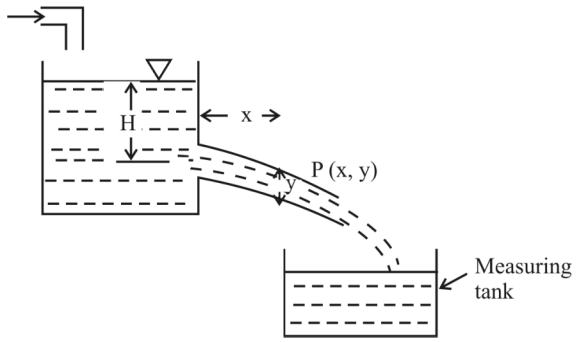
**If in terms of specific speed –**

$$\sigma = 0.103\left(\frac{N_s}{1000}\right)^{4/3}$$

**94. The coefficient of velocity for an orifice is given by (where x and y are the coordinates of a point in the jet measured from the vena - contraction and h is the orifice head)**

(a) $\sqrt{\frac{x^2}{4hy}}$ (b) $\sqrt{\frac{y^2}{4hx}}$
(c) $\sqrt{\frac{hx^2}{4y}}$ (d) $\sqrt{\frac{3x^2}{4hy}}$

**Ans. (a) : Determination of coefficient of velocity –**




Consider a liquid particle which is at vena contracta at any time and takes the position at P along the jet in time 't'.

Let, x = horizontal distance traveled by the particle in time 't'

y = vertical distance between P and orifice.

v = actual velocity of jet at vena contracta.

$x = v \times t$

$y = \frac{1}{2}gt^2$ .....(i)

$t = \frac{x}{v}$ ..... (ii)

Put the value of 't' in equation ..... (i)

$y = \frac{1}{2}g \times \left(\frac{x}{v}\right)^2$

$v^2 = \frac{gx^2}{2y}$

$v = \sqrt{\frac{gx^2}{2y}}$

But theoretical velocity, $V_{th} = \sqrt{2gH}$

Coefficient of velocity $(C_v) = \frac{v}{v_{th}}$

$$C_v = \sqrt{\frac{gx^2}{2y}} \times \frac{1}{\sqrt{2gH}} = \sqrt{\frac{x^2}{4yH}}$$

Coefficient of velocity $(C_v) = \frac{v}{v_{th}}$

$$C_v = \sqrt{\frac{gx^2}{2y}} \times \frac{1}{\sqrt{2gH}} = \sqrt{\frac{x^2}{4yH}}$$

$$C_v = \frac{x}{\sqrt{4yH}}$$

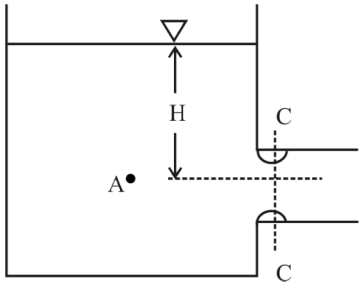
**95. For external mouthpiece absolute pressure head at vena-contracta is given as (where Ha is the atmospheric pressure head and H is the height of liquid surface above the centre of the mouthpiece)**

(a) $H_a - 0.89\ H$ (b) $H_a + 0.33\ H$

(c) $H_a + 0.95\ H$ (d) $H_a - H$

**Ans. (a) :** A mouthpiece is a short length of a pipe which is two or three times its diameter in length. If this pipe is fitted externally to the orifice, the mouthpiece is called external cylindrical mouthpiece and if it is fitted internally then mouthpiece is called internal mouthpiece or Borda's or Re-entrant mouthpiece.

**Pressure head at Vena - contracta –**




Applying Bernoulli's equation at A and C-C, we get

$$\frac{P_A}{\rho g} + \frac{V_A^2}{2g} + Z_A = \frac{P_C}{\rho g} + \frac{V_c^2}{2g} + Z_c$$

$$\frac{P_A}{\rho g} = H_a + H \qquad \begin{bmatrix} V_A = 0 \\ Z_A = Z_C \end{bmatrix}$$

$$H_a + H + 0 = \frac{P_C}{\rho g} + \frac{V_C^2}{2g} = H_c + \frac{V_C^2}{2g}$$

$$H_c = H_a + H - \frac{V_C^2}{2g}$$

$C_C$ For external mouthpiece = 0.62

Hence, $v_c = \frac{V_1}{0.62}$

$$H_c = H_a + H - \left(\frac{V_1}{0.62}\right)^2 \times \frac{1}{2g}$$

$$H_a = H + H - \left(\frac{V_1}{2g}\right)^2 \times \frac{1}{(0.62)^2}$$

$$H = 1.375\ \frac{V_1^2}{2g}$$

$$\frac{V_1^2}{2g} = \frac{H}{1.375} = 0.7272\ H$$

$$H_c = H_a + H - 0.7272H \times \frac{1}{(0.62)^2}$$

$= H_a + H - 1.89H$

$H_c = H_a - 0.89\ H$

**96. A 6m long rectangular weir is divided into 3 bays by two vertical posts each 0.3 m wide. What will be discharge through the weir for head of 0.45m?**

(a) 2.85 $m^3/s$ (b) 2.5 $m^3/s$

(c) 3.85 $m^3/s$ (d) 1.85 $m^3/s$

**Ans. (a)**

Discharge through rectangular weir with end contraction–

$$Q = \frac{2}{3} \times Cd \times [L - 0.1nH] \times \sqrt{2g} H^{3/2}$$

n = 6 = no. of end contractions, H = 0.45m

$C_d = 0.62$

Effective length, $L = 6 - 2 \times 0.3 = 5.4$ m

$$Q = \frac{2}{3} \times 0.62 [5.4 - 6 \times 0.1 \times 0.45] \times \sqrt{2g} \times (0.45)^{3/2}$$

$Q = 2.85\ m^3/s$

**97. In an orifice, the coefficient of resistance is defined as the ratio of**

(a) Loss of kinetic energy as the liquid flow through an orifice and the actual kinetic energy possessed by the flowing fluid.

(b) Theoretical kinetic

(c) Actual kinetic energy possessed by the flowing fluid and the loss of kinetic energy as the liquid flow through an orifice

(d) Theoretical kinetic energy as the liquid flow through an orifice and the loss of kinetic energy as the liquid flow through an orifice

**Ans. (a) : Coefficient of resistance ($C_r$) –** It is ratio of loss of head (loss of K.E.) in the orifice to the head of water (actual K.E.) available at the exit of the orifice.

$$C_r = \frac{\text{Loss of K.E.}}{\text{Actual K.E.}} = \frac{C_c^2 - C_d^2}{C_d^2}$$

$C_r$ is generally neglected, while solving problems.

**98. Maximum efficiency of the screw will occurs at an helix angle of the thread ($\alpha$) equal to (Where $\phi$ = helix angle.)**

(a) $\frac{\pi}{4} - \frac{\phi}{2}$ (b) $\frac{\pi}{2} - \frac{\phi}{2}$

(c) $\frac{\pi}{4} - \frac{\phi}{4}$ (d) $\frac{\pi}{2} - \frac{\phi}{4}$

**Ans. (a):** The efficiency of a screw jack is maximum when the helix angle is equal to $40^0$ to $45^0$. However a reasonable efficiency is achieved above $15^0$. Due to difficulties in forming the thread, helix angle greater than $30^0$ are rarely used.
For maximum efficiency.

$$\text{Helix angle} = \frac{\pi}{4} - \frac{\phi}{2}$$

**99. Eulers critical load is proportional to (Where L = length of the column k = constant, k = constant depends on end conditions and EI = flexural rigidity)**

(a) EI (b) $L^2$
(c) 1/EI (d) $k^2/EI$

**Ans. (a) :** Crippling load = $\frac{\pi^2 EI}{L^2}$

$$P \propto \frac{1}{L^2} \propto EI$$

**100. Doubling the length of a column brings down the load it can carry without buckling by a factor of**

(a) 4 (b) 2
(c) 1/2 (d) 1/4

**Ans. (d) :** Crippling load = $\frac{\pi^2 EI_{min}}{L_e'^2}$

$$\therefore P \propto \frac{1}{L_e'^2}$$

$$L_e' = 2L_e$$

If length of calcium is doubled than buckling is reduced by 1/4 times.

**101. In a thin cylindrical shell, the longitudinal stress is (Where d is the diameter of thin cylinder and t is the thickness of thin cylinder)**

(a) $\frac{pd}{4t}$ (b) $\frac{pd}{2t}$
(c) $\frac{pd}{t}$ (d) $\frac{pd}{6t}$

**Ans. (a) :** For thin cylinder,

Longitudinal / Axial Stress = $\sigma_l = \frac{pd}{4t}$ (tensile)

Circumferential / Hoop stress = $\sigma_h = \frac{pd}{2t}$ (tensile)

**102. The ratio of hoop stress to longitudinal stress**

(a) 2 (b) 1
(c) 2/3 (d) 3/2

**Ans. (a) :** $\frac{\text{Hoop stress}}{\text{Longitudinal stress}} = \frac{\frac{pd}{2t}}{\frac{pd}{4t}} = 2$

**103. The Darcy's friction factor in the pipe of uniform diameter in which a viscous flow is taking place is given in terms of Reynolds number ($R_e$)**

(a) $\frac{64}{R_e}$ (b) $\frac{1}{R_e}$ (c) $\frac{4}{R_e}$ (d) $\frac{16}{R_e}$

**Ans. (a) :** Loss of head due to friction in viscous flow

$$h_f = \frac{32\mu\bar{u}L}{\rho g D^2} = \frac{4fL\bar{u}^2}{2gD}$$

$$f = \frac{16\mu}{\bar{u}\rho D} = 16 \times \frac{\mu}{\rho v D}$$

$$\boxed{f = \frac{16}{Re}}$$ f = coefficient of friction

Friction factor (f ') = 4f

$$f' = \frac{64}{Re}$$

For turbulent flow –

$$f = \frac{0.079}{Re^{1/4}}$$

Friction factor = f'= $\frac{0.316}{Re^{1/4}}$

**104. The theoretical value of coefficient of contraction of sharp edged orifice is equal to**

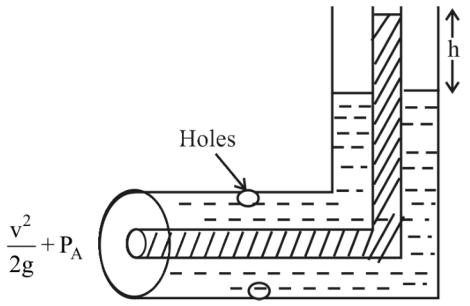
(a) $\pi/(\pi+2)$ (b) $(\pi+2)/\pi$
(c) $(\pi-2)/\pi$ (d) $\pi/(\pi-2)$

**Ans. (a) :** The theoretical value of coefficient of contraction for a sharp edge orifice as equal to 0.611 or $\frac{\pi}{\pi+2}$

**105. The coefficient of Prandtl Pitot tube is equal to**

(a) 1.0 (b) 0.80 (c) 0.87 (d) 0.95

**Ans. (a) :** Prandtl tube or pitot static tube is used to measure velocity at a point in fluid flow. The front portion of the tube is rounded to avoid separation of flow and on its shaft, holes are provided at certain distance where stream lines become parallel. The two tube are enclosed in the same tube.




$$h = \frac{v_A^2}{2g}$$

$$v_A = \sqrt{2gh}$$

Actual velocity, $V_{actual} = C_v\sqrt{2gh}$

Where $C_v$ is the coefficient of prandtl pitot tube.

$$\boxed{C_v = 0.99 \text{ to } 1.0}$$

**106. The energy loss in an**

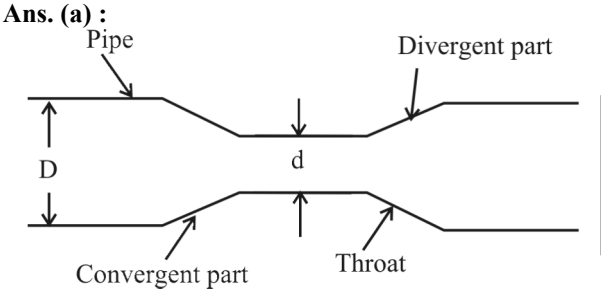
(a) orifice meter is more than that of venturimeter
(b) venturimeter is more than that of orifice meter
(c) orifice meter is same as that of venturimeter
(d) orifice meter and that of venturimeter is zero

**Ans. (a) :** The energy losses or head loss in orifice meter is more than venturimeter due to flow separation.
$C_d = 0.98 \rightarrow$ For venturimeter
$C_d = 0.60 \rightarrow$ For orifice meter

**107. The length of the convergent cone parallel to the axis of the venturimeter is approximately equal to**
**(Where D is the diameter of the inlet section and d is the diameter of the throat)**

(a) 2.7 (D-d) (b) 2.2 (D-d)
(c) 2.7 (D-d) (d) 2.2 (D-d)

**Ans. (a) :**




Length of convergent cone = 2.7 (D-d)

dia of throat = $\left(\frac{1}{3}\text{to}\frac{3}{4}\right)$ of D

**108. Greatest shear stress in a thin cylinder is (Where d is the diameter of thin cylinder and t is the thickness of thin cylinder)**

(a) $\frac{pd}{8t}$ (b) $\frac{pd}{2t}$ (c) $\frac{pd}{4t}$ (d) $\frac{pd}{6t}$

**Ans. (a) :** Max shear stress in thin cylinder

$$\tau_{max} = \frac{\sigma_h - \sigma_l}{2} = \frac{\frac{pd}{2t} - \frac{pd}{4t}}{2}$$

$$\tau_{max} = \frac{pd}{8t}$$

$\sigma_h$ = hoop stress
$\sigma_L$ = longitudinal stress

**109. The thin cylinder analysis is generally applicable for a d/t ratio of (Where d is the diameter of the thin cylinder and is the thickness of the thin cylinder)**

(a) $\geq 20$ (b) $\leq 20$
(c) $\geq 10$ (d) $\leq 8$

**Ans. (a) :** A shell is called thin shell if, $\boxed{t \leq \frac{d}{20}}$

**110. Torsional flexibility is the reciprocal of**

(a) Torsional stiffness (b) Torsional rigidity
(c) Polar modulus (d) Twisting moment

**Ans. (a) : Torsional flexibility** $\rightarrow$ It is defined as the angle of twist produced by the unit torque applied. It is reciprocal of torsional stiffness.

Torsional flexibility = $\frac{L}{GJ} = \frac{\theta}{T}$

Torsional stiffness = $\frac{GJ}{L} = \frac{T}{\theta}$

**111. The ratio of diameter of throat to the diameter of the pipe in a Venturi meter may vary from**

(a) 1/3 to 3/4 (b) 1/2 to 1
(c) 1/3 to 1/2 (d) 1/2 t 2/3

**Ans. (a) :** Diameter of throat of venturimeter is 1/3 to 3/4 times dia of pipe.

$$\frac{\text{Dia. of throat}}{\text{Dia. of pipe}} = \frac{1}{3}\text{to}\frac{3}{4}$$

**112. Which of the following equation gives actual discharge through a venturimeter?**
**(Where $a_1$ and $a_2$ are the cross sectional areas at the inlet section and throat of the venturimeter, g is the acceleration due to gravity. H is the venturihead; and $C_d$ is the coefficient of discharge.)**

(a) $C_d \frac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 - a_2^2}}$ (b) $\frac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 - a_2^2}}$
(c) $C_d \frac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 + a_2^2}}$ (d) $\frac{a_1 a_2 \sqrt{2gH}}{2\sqrt{a_1^2 - a_2^2}}$

**Ans. (a) :** $Q_{act} = C_d \frac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 - a_2^2}}$

Where,
$a_1$ & $a_2$ are the cross sectional areas at the inlet section and throat of the venturimeter
g is the acceleration due to gravity.
H is the venturihead;
$C_d$ is the coefficient of discharge.)

**113. If the velocity is zero over half of the cross-sectional area and is uniform over the remaining half, then the momentum correction factor is**

(a) 2 (b) 1
(c) 4/3 (d) 4

**Ans. (a) :** Momentum correction factor ($\beta$)

$$\beta = \frac{\text{Acutal momentum per sec.}}{\text{Average momentum per sec.}}$$

$$\beta = \frac{1}{A}\int_{A/2}^{A}\left(\frac{V}{V/2}\right)^2 dA$$

$$\beta = \frac{4}{A}\int_{A/2}^{A} dA$$

$$\beta = \frac{4}{A} \times \frac{A}{2} = 2$$

**114. Piezometric head in fluid motion is equal to the sum of**

(a) Static head and potential head
(b) Kinetic head and static head
(c) Velocity head and potential head
(d) Kinetic head, static head and potential head

**Ans. (a) :**

$$\underbrace{\frac{P}{\rho g}}_{\text{static head}} + \underbrace{z}_{\text{Potential head}} = \text{Piezometric head}$$

Hydraulic head or piezometric head is specific measurement of liquid pressure above a vertical datum.

**115. Compressibility force in fluid motion is due to**

(a) Elastic property of fluid
(b) Cohesive property of fluid
(c) Adhesive property of fluid
(d) Viscosity of fluid

**Ans. (a) :** Compressibility force in fluid motion is due to elastic property of fluid. It is the property of fluid which unable it to compress.
It is reciprocal of bulk modulus of elasticity.

$$\beta = \frac{1}{K} = -\frac{dV/V}{dP}$$

**116. Flexural rigidity (EI) is expressed in units of**

(a) $kNm^2$ (b) MNm
(c) $kNm^2$ (d) $MNm^4$

**Ans. (a) :** Flexural rigidity = EI

$$\text{Unit of flexural rigidity} = \frac{kN}{m^2} \times m^4 = kNm^2$$

**117. A hinged support can transmit**

(a) two reactive forces
(b) one reactive forces
(c) three reactive forces
(d) four reactive forces

**Ans. (a) :** A hinged support can transmit two reactive forces.
A roller support can transmit one reactive force.
A fixed support can transmit three, reactive force.

**118. Factor of safety is the ratio of**

(a) lower yield and working stresses
(b) ultimate and working stresses
(c) upper yield and working stresses
(d) ultimate and lower yield stresses

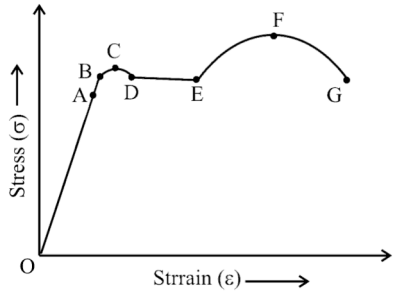
**Ans. (a) :** Factor of safety is the ratio of lower yield stress to the working stress.

$$FOS = \frac{\text{Lower yield stress}}{\text{Working stress}}$$

**119. Strain hardening of a material is**

(a) increases in stress with strain beyond yield stress
(b) increase in strain with stress beyond ultimate stress
(c) increase in hardness as strain in increased
(d) increase in hardness under repeated loading

**Ans. (a) : Stress strain curve for mid steel –**




From the curve,

| | |
|---|---|
| OA = Linear curve, | A = Proportional limit |
| B = Elastic limit, | C = Upper yield point |
| D = Lower yield point, | DE = Plastic region |
| EF = Strain hardening region | F = Ultimate stress point |
| FG = Necking region | G = Fracture point |

Strain hardening in a material increase beyond yield stress.

**120. The Pascal's law states that the**

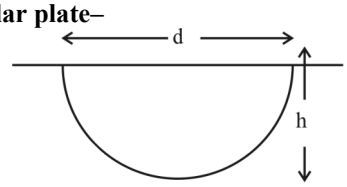
(a) Pressure at any point in a fluid at rest has the same magnitude in all directions
(b) Pressure at any point in a fluid at rest has the dissimilar magnitude in all directions
(c) Pressure at any point in a fluid at rest has the same magnitude in transverse and longitudinal directions only
(d) Pressure at any point in a fluid at rest has the same magnitude in perpendicular direction only

**Ans. (a) : Pascal's Law –** It states that the pressure or intensity of pressure at a point in a static fluid is equal in all directions.

**121. For a vertical semi-circular plate, submerged in a homogenous liquid with its diameter 'd' at the free surface, the depth of centre of pressure from the free surface is**

(a) $3\pi d/32$ (b) $3\pi d/12$ (c) $3\pi d/16$ (d) $\pi d/32$

**Ans. (a) : Depth of centre of pressure for a vertical semi-circular plate–**




$\bar{h}$ = height of centre of gravity
h* = height of centre of pressure

$$h^* = \frac{I_G}{A\bar{h}} + \bar{h} = \frac{I_{base} - Ah^2}{A\bar{h}} + \bar{h}$$

$$h^* = \frac{I_{base}}{A\bar{h}} \qquad A = \frac{\pi}{8}d^4, \ \bar{h} = \frac{2d}{3\pi}$$

$$I_{base} = \frac{1}{2} \times I_{circular} = \frac{\pi d^4}{128}$$

$$h^* = \frac{\frac{\pi d^4}{128}}{\frac{\pi}{8}d^2 \times \frac{2d}{3\pi}} = \frac{3\pi d}{32}$$

**122. Continuity equation for a one dimensional flow through uniform cross sectional area is (Where ρ is the mass density of fluid; and u is the velocity component in x direction; and t is time variable)**

(a) $\frac{\partial\rho}{\partial t}+\frac{\partial\rho u}{\partial x}=0$ (b) $\frac{\partial\rho}{\partial x}+\frac{\partial\rho u}{\partial x}=0$

(c) $\frac{\partial\rho}{\partial t}-\frac{\partial\rho u}{\partial x}=0$ (d) $\frac{\partial\rho}{\partial t}+\frac{\partial\rho u}{\partial y}=0$

**Ans. (a) :** Most general form of continuity equation for three dimensional flow.

$$\frac{\partial\rho}{\partial t}+\frac{\partial}{\partial x}(\rho u)+\frac{\partial}{\partial y}(\rho v)+\frac{\partial}{\partial z}(\rho w)=0$$

For one dimensional flow, flow in y and z direction is zero v = 0, w = 0

$$\frac{\partial\rho}{\partial t}+\frac{\partial}{\partial x}(\rho u)=0$$

**123. Flow of liquid flowing through a long pipe of constant diameter at either increasing or decreasing rate is**

(a) unsteady uniform flow
(b) steady uniform flow
(c) steady non-uniform flow
(d) unsteady non-uniform flow

**Ans. (a) :** Steady and unsteady flow – If flow and fluid properties at any given location does not change with time, flow is called steady otherwise, unsteady.

$$\left.\begin{aligned}\frac{dp}{dt}&=0\\ \frac{dv}{dt}&=0\\ \frac{dp}{dt}&=0\end{aligned}\right\}\text{flow is steady otherwise unsteady.}$$

**124. The pressure at any point in a static mass of fluid varies in vertical direction as indicated by which of the following equation?**
**(where γ is the specific weight ρ is the mass density; and g is the acceleration due to gravity)**

(a) $\frac{dp}{dz}=-\rho g$ (b) $\frac{dp}{dz}=\rho g$

(c) $\frac{dp}{dz}=\gamma g$ (d) $\frac{dp}{dz}=-\gamma g$

**Ans. (a) :** The pressure at any point in static mass of fluid in vertical direction.

$$\frac{dp}{dz}=-\rho g$$

**125. Orthotropic materials have**

(a) different properties in orthogonal directions
(b) different properties in different directions
(c) different strength in different directions
(d) same properties in different direction

**Ans. (a):** Orthotropic materials have different properties in orthogonal directions.

- Isotropic materials have identical properties in all direction.
- Homogeneous materials have same properties at every point throughout the volume.

**126. Rivets in lap joints are subjected to**

(a) axial and shear forces
(b) axial forces only
(c) double shear
(d) single shear alone

**Ans. (a) :** Rivets in lap joint are subjected to axial and shear force.

- It fails in single shear or due to bending stresses.
- Double cover butt joint is free from bending stress and is subjected to double shear.

**127. The maximum shear due to an axial compression of 100 MPa is**

(a) 50 MPa (b) 100 MPa
(c) 150 MPa (d) 75 MPa

**Ans. (a) :** Given,
Axial compression = 100 MPa

$$\tau_{max}=\frac{\sigma}{2}=\frac{100}{2}=50\,MPa$$

**128. Bending moment at a section has the maximum value where**

(a) shear force is zero
(b) the load function has zero value
(c) shear force is zero or changes sign
(d) the load has maximum value

**Ans. (c) :** Bending moment at a section has the maximum value where shear force is zero, or changes sign.

$$\frac{dM}{dx}=V$$

Rate of change of B.M. is equal to shear force and if rate of change is zero then B.M. is at maximum or of constant value.

**129. The diameter of the hole at the gauge point while for making connection for various devices for pressure measurement should be about**

(a) 3 mm to 6 mm (b) 1 mm to 3 mm
(c) 6 mm to 9 mm (d) 9mm to 12 mm

**Ans. (a) :** The diameter of the hole at the gauge point while for making connection for various devices for pressure measurement should be about 3mm to 6 mm.

**130. What is the minimum angle of inclination of an inclined manometer to avoid the force of surface tension?**

(a) $4^0$ (b) $3^0$
(c) $5^0$ (d) $6^0$

**Ans. (a) :** To avoid the force of surface tension in an inclined manometer the minimum angle of inclination is $4^0$.

**131. Which of the following instrument is used to measure the specific gravity of liquid?**

(a) Hydrometer
(b) Densitometer
(c) Differential manometer
(d) Hygrometer

**Ans. (a) :**

| Instrument | Used for Measure |
|---|---|
| Hydrometer | Specific gravity |
| Densitometer | Degree of darkness |
| Differential Manometer | Pressure difference |
| Hygrometer | Humidity |

**132. For manometer, a better liquid combination is one having**

(a) higher surface tension
(b) lower surface tension
(c) high density and high dynamic viscosity
(d) low density and low dynamic viscosity

**Ans. (b) : Property of manometeric fluid–**
- The fluid should have a low vapour pressure & high density.
- It should have low surface tension to avoid the capillary rise.
- The fluid should be immiscible with the working fluid.

**133. Which one of the following with reference to fluid property is true?**

(a) Capillarity is due cohesion and adhesion and surface tension is due to cohesion
(b) Capillarity is due adhesion and surface tension is due to cohesion
(c) Capillarity is due adhesion and surface tension is due to cohesion and adhesion
(d) Capillarity is due cohesion and surface tension is due to adhesion

**Ans. (a) :** The rise or fall of liquid in a small dia. tube when it is immersed in liquid is known as capillarity.
Capillary rise is due to adhesion and capillary fall is due to cohesion.
Hence capillarity is due to cohesion and adhesion whereas surface tension is due to cohesion only.

**134. The ratio of maximum to mean value of shear stress in a circular section is**

(a) 1.33 (b) 1.5 (c) 1.25 (d) 1.75

**Ans. (a) :** $\frac{\tau_{max}}{\tau_{mean}} = \frac{4}{3} = 1.33$

**135. If the moments of inertia of a section about its orthogonal axes are $1 \times 10^6$ and $2 \times 10^6$ and its area is 100 units, its polar moment of inertia is**

(a) $3 \times 10^6$ (b) $3.1 \times 10^6$
(c) $5 \times 10^6$ (d) $4 \times 10^6$

**Ans. (a) :** Given
$I_{xx} = 1 \times 10^6$
$I_{yy} = 2 \times 10^6$
A = 100 units
Polar moment of inertia = $I_{xx} + I_{yy}$
$= 1 \times 10^6 + 2 \times 10^6$
$= 3 \times 10^6$

**136. The curvature of a beam at a section is given by**

(a) $\frac{M}{EI}$ (b) $\frac{\sigma}{y}$
(c) $\frac{\varepsilon}{y}$ (d) $\frac{MI}{E}$

**Ans. (a) :** Bending equation

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\frac{1}{R} = \text{curvature} = \frac{M}{EI}$$

**137. A rectangular bar of width b and height d is being used as a cantilever. The loading is in a plane parallel to the side b. The section modulus is**

(a) $\frac{b^2d}{6}$ (b) $\frac{bd^3}{12}$
(c) $\frac{bd^2}{6}$ (d) $\frac{bd^2}{8}$

**Ans. (a) :** Given,
Width = b
Height = d

$$\text{Section modulus} = \frac{I}{y}$$

$$I = \frac{b^3d}{12}, \quad = y = \frac{b}{2}$$

$$Z = \frac{b^3d}{12} \times \frac{2}{b}$$

$$Z = \frac{b^2d}{6}$$

**138. As compared to uniaxial tension or compression, the strain energy stored in bending is only**

(a) 1/3 (b) 1/8
(c) 1/4 (d) 1/2

**Ans. (a) :** Strain energy due to tension or compression

$$u_1 = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$$

$$u_1 = \frac{1}{2} \times P \times \delta = \frac{P^2L}{2AE} = \frac{\sigma^2}{2E} \times \text{Volume} \ldots\ldots (i)$$

Strain energy due to bending

$$u_2 = \int_0^L \frac{M^2dx}{2E} = \int_0^L \frac{(Wx)^2}{2E}dx = \frac{W^2}{2E}\int_0^L x^2dx$$

$$u_2 = \frac{w^2L^3}{6E} \quad \ldots\ldots\ldots (ii)$$

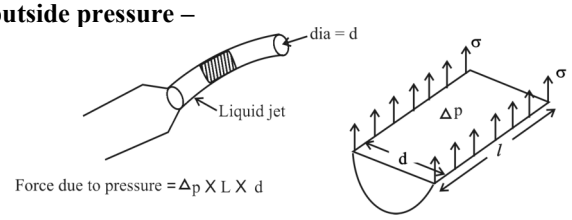
From equation (i) and (ii)

$$\frac{u_1}{u_2} = \frac{1}{3}$$

**139. What is the internal pressure intensity p in excess of the outside pressure intensity for a jet a liquid of radius R and length *l*?**

(a) $\frac{\sigma}{R}$ (b) $\frac{\sigma}{Rl}$
(c) $\frac{\sigma}{2R}$ (d) $\frac{4\sigma}{R}$

**Ans. (a): Pressure intensity inside the liquid above outside pressure –**




Force due to pressure

$= \Delta p \times L \times d$

Force due to surface tension = $\sigma \times 2L$

$= \Delta p \times L \times d = \sigma \times 2L$

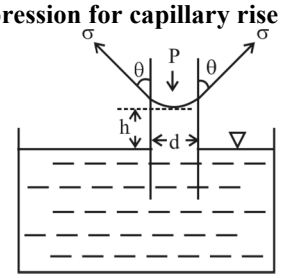
$\because d = 2R \qquad \Delta p = \dfrac{2\sigma}{d} = \dfrac{\sigma}{R}$

**140. The capillary rise of water h, in a glass tube can be determined using**
**(Where σ is the surface is tension; θ is contact or wetting angle; w is specific weight of liquid and R is radius of tube.)**

(a) $h = \dfrac{2\sigma\cos\theta}{Rw}$ (b) $h = \dfrac{\sigma\cos\theta}{Rw}$

(c) $h = \dfrac{4\sigma\cos\theta}{Rw}$ (d) $h = \dfrac{\sigma\cos\theta}{2Rw}$

**Ans. (a) : Expression for capillary rise –**




d = dia of glass tube

h = height of liquid in the tube

σ = surface tension of liquid

θ = angle of contact between liquid and glass tube.

The weight of liquid of height h in the tube

$= (\text{Area of tube} \times h) \times \rho \times g$

$= \dfrac{\pi}{4} d^2 \times h \times \rho g$ ..... (i)

Vertical component of surface tensile force

$= (\sigma \times \text{circumference}) \times \cos\theta$

$= \sigma \times \pi d \times \cos\theta$ ..... (ii)

For equilibrium equating equation (i) and (ii)

$= \dfrac{\pi}{4} d^2 \times h \times \rho g = \sigma \times \pi d \times \cos\theta$

$h = \dfrac{4\sigma\cos\theta}{\rho g d}$

$h = \dfrac{2\sigma\cos\theta}{Rw}$

**141. If mercury is filled in a small circular glass tube, the value of contact angle for mercury and the glass is equal to**

(a) $130^0$ (b) $0^0$ (c) $75^0$ (d) $90^0$

**Ans. (a) :** Value of contact angle –
Mercury and glass = $130^0$
Water and glass = $\theta = 0^0$
Kerosene and glass = $260^0$
Water and silver = $90^0$

**142. How much pressure is required to cause 1% change in the density of water?**

(a) 21 MPa (b) 2.1 MPa
(c) 210 MPa (d) 2100 MPa

**Ans. (a) :** $m = d \times v = \text{Constant}$

$v\Delta d + d\Delta v = 0$

$\dfrac{\Delta v}{v} = -\dfrac{\Delta d}{d} = -\dfrac{1}{100} = -10^{-2}$

$k = -\dfrac{\Delta p}{\left(\dfrac{\Delta v}{v}\right)} \qquad \Delta p = k\left(-\dfrac{\Delta v}{v}\right)$

For water k = 2.1 GPa

$\Delta p = 2.1 \times 10^9 \times 10^{-2}$

$= 2.1 \times 10^7$

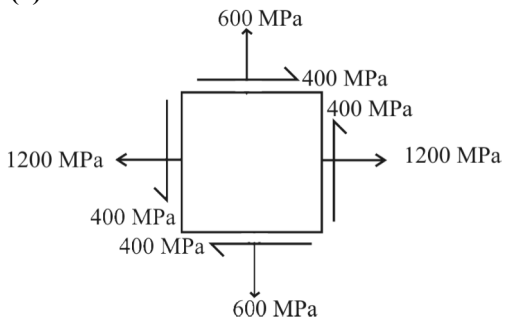
$\Delta p = 21$ MPa

A pressure of about 21 MPa is required to cause 1% change in the density of water.

**143. A body is subjected to tensile stress of 1200 MPa on one plane and another tensile stress of 600 MPa on a plane at right angles to the former. It is also subjected to a shear stress of 400 MPa on the same planes. The maximum shear stress will be**

(a) 500 MPa (b) 1400 MPa
(c) 900 MPa (d) 400 MPa

**Ans. (a) :**




$\tau_{max} = \dfrac{\sigma_1 - \sigma_2}{2}$

$\sigma_1 = \sigma_{major} = \dfrac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$= \dfrac{1200 + 600}{2} + \sqrt{\left(\dfrac{1200 - 600}{2}\right)^2 + 400^2}$

$= 900 + 500 = 1400$ MPa

$\sigma_2 = \sigma_{minor} = \dfrac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right) - \tau^2_{xy}}$

$= 900 - 500 = 400$ MPa

$\tau_{max} = \dfrac{1400 - 400}{2} = 500$ MPa

**144. A shaft of diameter D is the subjected to a twisting moment (T) and a bending moment (M). If the maximum bending stress is equal to maximum shear stress developed, then M is equal to**

(a) T/2 (b) 4T
(c) 2T (d) T

**Ans. (a) :** $\tau_{max} = \dfrac{16T}{\pi D^3}$

$$\sigma_{max} = \frac{32M}{\pi D^3}$$

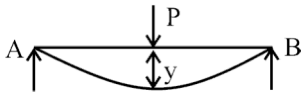
if $\tau_{max} = \sigma_{max}$

$$\frac{16T}{\pi D^3} = \frac{32M}{\pi D^3} \Rightarrow M = \frac{T}{2}$$

**145. If the width of a simply supported beam carrying an isolated load at its centre is doubled, the deflection of the beam at the centre is changed by**

(a) 1/2 (b) 4
(c) 2 (d) 1/8

**Ans. (a) : Deflection of beam carrying point load at centre –**




$$y = \frac{PL^3}{48EI} = \frac{PL^3}{48E} \times \frac{12}{bd^3}$$

$$y \propto \frac{1}{b}$$

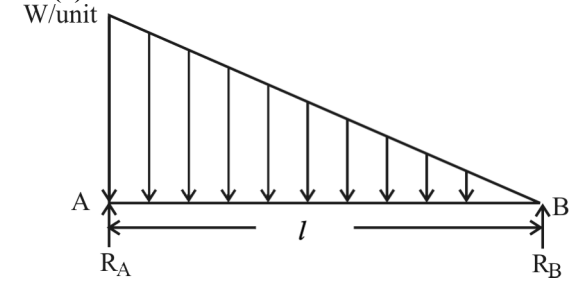
hence, if b' = 2b

$$y' = \frac{1}{2} \times y$$

**146. A simply supported beam with a gradually varying load from zero at 'B' and 'w' power unit length at 'A'. The shear force at 'B' is equal to**

(a) $\dfrac{wl}{6}$ (b) $\dfrac{3wl}{2}$
(c) $wl$ (d) $\dfrac{2wl}{3}$

**Ans. (a) :**




Total load = Area of loading diagram

$$= \frac{1}{2} \times w \times l = \frac{wl}{2}$$

This load will act at the C.G. of loading diagram

i.e $\dfrac{l}{3}$ from support A,

$$\therefore R_A = \frac{wl}{2} \times \frac{2l}{3} \times \frac{1}{2} = \frac{wl}{3}$$

$$R_B = \frac{wl}{6}$$

Shear force at B = $V_B = R_B = \dfrac{wl}{6}$

**147. The neutral axis of the cross-section a beam is that axis at which the bending stress**

(a) zero (b) maximum
(c) minimum (d) infinity

**Ans. (a) :** The neutral axis is an axis in the cross section of a beam or shaft along which there is no longitudinal stresses or strains.

**148. The stress induced in a body, when suddenly loaded, is how many times the stress induced when the same load is applied gradually**

(a) two times (b) four times
(c) equal to (d) one-half times

**Ans. (a) :** The stress induced in a body, when suddenly loaded is 2 times the stress induced when the same load is applied gradually.

**149. Two closely coiled helical springs 'A' and 'B' are equal in all respects but the number of turns of spring 'A' is half that of spring 'B'. The ratio of deflections in spring 'A' to spring 'B' is**

(a) 1/2 (b) 2
(c) 1/8 (d) 1/4

**Ans. (a) :** Deflection of closely coiled helical spring is given by

$$\delta = \frac{64PR^3n}{Gd^4}$$

$\delta \propto n$, $n \rightarrow$ number of turns

$$n_A = \frac{1}{2} n_B$$

$$\frac{\delta_A}{\delta_B} = \frac{\frac{1}{2} n_B}{n_B} = \frac{1}{2}$$

**150. The ratio of bulk modulus to young's modulus for a Poisson's ratio of 0.25 will be**

(a) 2/3 (b) 1/3
(c) 1 (d) 3/2

**Ans. (a) :** $E = 3K(1-2\mu)$

$\mu = 0.25$ (given)

$$\frac{K}{E} = \frac{1}{3(1-2\times0.25)} = \frac{1}{3/2}$$

$$\frac{K}{E} = \frac{2}{3}$$

# Andhra Pradesh Public Service Commission
## Assistant Engineer (Environmental & Civil) Exam., 2016 (P-II)
## Solved Paper

**1. The equation to determine oxygen required for aerobic suspended growth process for the biological degradation of organic matter measured in terms of BOD is (Where Q is the rate of inflow wastewater, V is the volume of the reactor, P is the biomass as VSS wasted per day, $S_{in}$ is the influent substrate concentration and S is the effluent substrate concentration measured in terms of BOD.)**

(a) $Q (S_{in}-S) - 1.42P$ (b) $Q (S_{in}-S) + 1.42P$
(c) $QV (S_{in}-S) - 1.42P$ (d) $Q (S_{in}-S) - 1.42PV$

**Ans. (a) :** The equation to determine oxygen required for aerobic suspended growth process for the biological degradation of organic matter measured in terms of BOD is $Q (S_{in}-S) - 1.42P$
Where,
Q = rate of inflow wastewater
V = volume of the reactor
P = biomass as VSS wasted per day
$S_{in}$ = influent substrate concentration
S = effluent substrate concentration measured in terms of BOD.

**2. Biological reactions in landfills occur in four stages viz., 1. Methanogenesis phase (Steady), 2. Acid Phase, 3. Aerobic & hydrolysis phase and 4. Methanogenesis phase (Unsteady), what is the correct order of these stages?**

(a) 3-2-4-1 (b) 1-2-3-4
(c) 2-3-1-4 (d) 3-2-1-4

**Ans. (a) :** The entire period of refuse stabilization can infect, be divided into five distinct phases–
(i) During the first phase of operation, aerobic bacteria & fungi, which are dominant deplete the available oxygen to effect the oxidation of organic matter.
(ii) In the second phase, anaerobic and facultative bacteria developed and $H_2$ & $CO_2$ gases are evolved through acidogenic activity.
(iii) In the third phase methanogenic bacteria developed.
(iv) In the fourth phase of the decomposition, the methanogenic activity gets stabilized.
(v) In the fifth stage, the methanogenics activity subsides.
Aerobic and hydrolysis phase $\rightarrow$ Acid phase $\rightarrow$ Methogenesis phase (unsteady) $\rightarrow$ Methogenesis phase (steady).

**3. Among the following, what is the ideal C/N ratio for aerobic composting process of solid waste?**

(a) 26 to 30 (b) 5 to 20
(c) 46 to 62 (d) 63 to 76

**Ans.(a):** (i) The ideal C/N ratio for aerobic compositing process of solid waste is 25 to 30.
(ii) Aerobic composting is not only faster but also release no unpleasant odour.
(iii) Solid waste such as winery waste has a high ranging between 35 and 80%.

**4. According to the Government of India standards for noise pollution control, the maximum allowable limit for the noise in dB (A) Leq for residential area during day time is**

(a) 55 (b) 60
(c) 65 (d) 70

**Ans. (a) :** According to the government of India standards for noise pollution control.

| Area/zone | Day time | Night time |
|---|---|---|
| Silence zone | 50 dB | 40 dB |
| Residential area | 55 dB | 45 dB |
| Commercial area | 65 dB | 55 dB |
| Industrial area | 75 dB | 70 dB |

**5. Arrange the following functions associated with an EIA Process in sequential order**
**i) identification ii) prediction iii) scoping iv) impact evaluation and analysis**

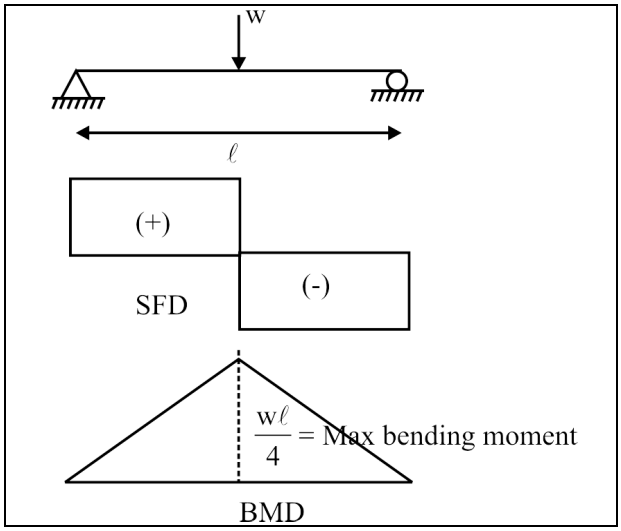
(a) iii,i,ii,iv (b) i,iii,iv,ii
(c) ii,iv,i,iii (d) iv,ii,iii,i

**Ans. (a) :**
**Stages of environmental impact assessment–**
1. Screening
2. Scoping
3. Base line study (collect information on current status of the environment)
4. Impact prediction
5. Impact assessment
6. Mitigation
7. Environment impact statement
8. ETS Review
9. ETA follow up

**6. The point in a beam where the shear force changes its sign is called the point of**

(a) Maximum bending moment
(b) Zero shear force
(c) Zero curvature
(d) Maximum curvature

**Ans. (a) :** The point in the beam where shear force change its sign called paint of max bending moment.
Ex = Assuming simply supported beam point load at centre.




**7. $U_1$ and $U_2$ are the strain energies stored in a prismatic bar due to axial forces $P_1$ and $P_2$ respectively. The strain energy U stored in the same bar due to combined action of $P_1$ and $P_2$ will be**

(a) $U > U_1 + U_2$ (b) $U = U_1 + U_2$

(c) $U = U_1\ U_2$ (d) $U < U_1 + U_2$

**Ans. (a) :**

$$U_1 = \frac{1}{2} \times P_1 \times \delta_1$$

$$= \frac{1}{2} \times P_1 \times \frac{P_1 \times L}{AE}$$

$$U_1 = \frac{1}{2} \times \frac{P_1^2 L}{AE}$$

Similarly for $U_2$

$$U_2 = \frac{1}{2} \frac{P_2^2 L}{AE}$$

And, Strain energy due to combined action of both forces.

$$= \frac{1}{2} \frac{\left(P_1 + P_2\right)^2 L}{AE}$$

$$= \frac{1}{2} \frac{\left(P_1^2 + P_2^2 + 2P_1P_2\right)L}{AE}$$

$$= U_1 + U_2 + \frac{1}{2} \frac{2P_1P_2L}{AE}$$

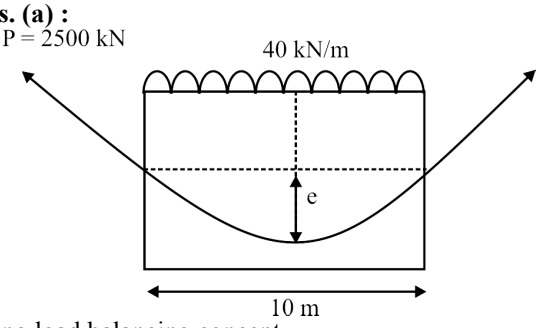
$\therefore \quad U > U_1 + U_2$

**8. If a simply supported concrete beam, prestressed with a force of 2500 kN is designed by load balancing concept for an effective span of 10 m and to carry a total load of 40 kN/m, the central dip of the cable profile should be**

(a) 200 mm (b) 100 mm

(c) 300 mm (d) 400 mm

**Ans. (a) :**




Using load balancing concept–

Central dip, $h = \frac{w\ell^2}{8P}$

We know

W = 40 kN/m

$\ell$ = 10 m

P = 2500 kN

$\therefore h = \frac{40 \times 10 \times 10}{8 \times 2500}$

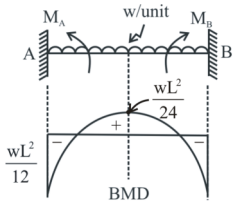
h = 0.2 m  200 mm.

**9. A uniform beam of span *l* is rigidly fixed at both supports. It carries a uniformly distributed load w per unit length. The bending moment at mid-span is**

(a) $wl^2/24$ (b) $wl^2/8$

(c) $wl^2/12$ (d) $wl^2/16$

**Ans. (a) :**




Bending moment at mid span is $\frac{wL^2}{24}$ and at support is $-\frac{wL^2}{12}$

**10. The horizontal thrust due to rise in temperature in a semi-circular two-hinged arch of radius R is proportional to**

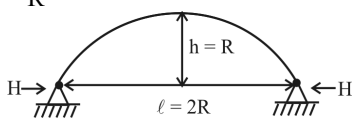
(a) $1/R^2$ (b) R

(c) $R^2$ (d) 1/R

**Ans. (a) :** Due to rise in temperature the horizontal thrust is represented by

$$H = \frac{4EI\alpha T}{\pi R^2}$$

$$H \propto \frac{1}{R^2}$$




- For 2 hinged parabolic arch

$$H = \frac{15}{8} \frac{EI \propto T}{h^2}$$

**11. If the potential infiltration of a watershed having a soil with fair pasture cover, is 10 cm and rainfall is 12 cm, the direct run off is**

(a) 5 cm (b) 3 cm
(c) 2 cm (d) 8 cm

**Ans. (a) :** Direct runoff $= \frac{(P-0.2s)^2}{(P+0.8s)}$

$= \frac{(12-0.2\times10)^2}{12+0.8\times10} = \frac{(10)^2}{20}$

$= \frac{100}{20} = 5\text{cm}$

**12. The average mean velocity of a stream having depth 'h', may be obtained by taking the average of the reading of a current meter at a depth of**

(a) 0.2 h and 0.8 h (b) 0.1 h and 0.9 h
(c) 0.4 and 0.6 h (d) 0.3 h and 0.7 h

**Ans. (a) :** In moderately deep streams the velocity is observed at two points.
(i) At 0.2 times the depth of flow below the free surface ($V_{0.2}$)
(ii) At 0.8 times the depth of flow below the free surface ($V_{0.8}$)
Average velocity in the vertical (v) is taken as–

$$V = \frac{V_{0.2} + V_{0.8}}{2}$$

- For shallow streams of depth upto about 3 m.

$$\bar{V} = V_{0.6}$$

**13. The graph showing the discharge in the stream against the percentage time such discharge is equalled or exceeded is called**

(a) Flow duration curve (b) Double mass curve
(c) Mass curve (d) Flow mass curve

**Ans. (a) :** A flow duration curve of a stream is a plot of discharge against the percent of time the flow was equalled or exceeded. This curve is also known as discharge frequency curve.

- The flow mass curve is a plot of the cumulative discharge volume against time plotted in chronological order.
- The ordinate of mass curve V at any time t is

$$V = \int_{t_0}^{t} Q\,dt$$

**14. A storm with a uniform intensity occurring over a basin for a period of 6 hours produced an effective rainfall of 15 cm and the peak discharge in the corresponding direct run-off hydrograph is 930 $m^3$/see. If another storm of same duration but with an effective rainfall of 7.50 occurs, what is the peak ordinate of the direct rainfall hydrograph produced by it over the same basin?**

(a) 465 $m^3$/sec (b) 930 $m^3$/sec
(c) 2790 $m^3$/sec (d) 1860 $m^3$/sec

**Ans. (a): Case -I** Effective rainfall (n) = 15 cm
Duration (t) = 6 hr
Peak discharge (Q) = 930 $m^3$/sec
Now, for area (A), We know–

$$n = \frac{0.36\times Qt}{A}$$

here, n in cm
Q in $m^3$/sec
t in hr
A in $km^2$

$$15 = \frac{0.36\times930\times6}{A}$$

$A = 133.92\ km^2$

For same basin – i.e,
$A = 133.92\ km^2$
t = 6 hr

Now, $n = \frac{0.36\ Q.t}{A}$

$$Q = \frac{7.5\times133.92}{0.36\times6} = 465\ m^3/\text{sec}$$

**15. The angle of intersection of the two plane mirrors of an optical square is**

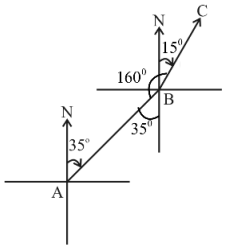
(a) 45º (b) 30º
(c) 60º (d) 90º

**Ans. (a) :** Optical square is used to establish two point at right angle. The principal used is double reflection so angle between two mirrors is $45^0$.

**16. If the fore bearing of a line AB is 35º and the of line BC 15º, then the included angle between the lines is**

(a) 160º (b) 30º
(c) 50º (d) 230º

**Ans. (a) :**




B.B. of AB = F.B. + 180º
= 35º + 180º
B.B. of AB = 215º
included angle between line
AB and BC = 360º – 215º + 15º
= 160º

**17. For which of the following permanent adjustments of theodolite, the spire test is used?**

(a) Adjustment of horizontal axis
(b) Adjustment of plate levels
(c) Adjustment of lien of sight
(d) Adjustment of altitude bubble and vertical index frame

**Ans. (a): Spire test –** It is a test carried out to remove error due to non-perpendicularly of horizontal axis and the vertical axis is surveying instrument.

- Spire test is used for adjustment horizontal axis or trunnion axis. .

**(i) Plate level test–** Make the plate bubbles central to their run when the vertical axis of the theodolite is truly vertical.

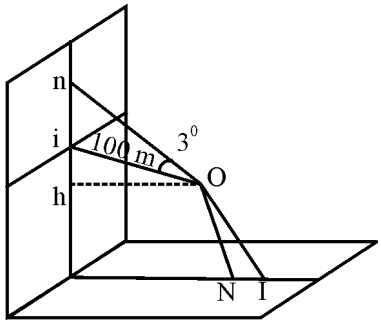
**(ii) Cross hair ring test–** Make the vertical cross hair lie in the plane perpendicular to the horizontal axis.

**(iii) Azimuth test–** Make the line of sight perpendicular to the horizontal axis.

**18. In an aerial photograph, if the photograph has a tilt of 3º and the focal length is 100 mm, then the distance between the perspective center & the plumb point will be**

(a) 100/ cos 3º (b) 100/ tan 3º
(c) 100/ cot 3º (d) 100/ sin 3º

**Ans. (a) :**




Where, NO = distance between the perspective centres and the plumb point.

$$\cos 3^0 = \frac{100}{NO}$$

$$NO = \frac{100}{\cos 3^0}$$

**19. As per the transit rule for balancing of traverse, the correction for latitude of any side of a traverse is given by (where $e_t$ is the total error in latitude)**

(a) (Latitude of that side/arithmetical sum of all latitudes in the traverse)$e_t$
(b) (Length of that side/arithmetical sum of all latitudes in the traverse)$e_t$
(c) (Length of that side/algebraic sum of all latitudes in the traverse)$e_t$
(d) (Latitude of that side/algebraic sum of all latitudes in the traverse)$e_t$

**Ans. (a) : Transit method–** The transit rule may be employed where angular measurements are more precise than that of linear measurements.

Correction to latitude of any side

$$= \text{Total error in latitude} \times \frac{\text{Latitude of that line}}{\text{Arithmetic sum of latitudes}}$$

**20. Which of the following is correct for a Well Graded Sand?**

(a) $C_u > 6$ and $C_c = 1$ to 3
(b) $C_u < 4$ and $C_c = 1$ to 3
(c) $C_u > 4$ and $C_c = 1$ to 3
(d) $C_u < 6$ and $C_c = 1$ to 3

**Ans. (a) :** Coefficient of uniformity ($C_u$) is given as–

$$C_u = \frac{D_{60}}{D_{10}}$$

for well graded sand, $C_u > 6$
for well graded gravel, $C_u > 4$.
Coefficient of curvature ($C_c$) is given as–

$$C_c = \frac{(D_{30})^2}{D_{60} \times D_{10}}$$

for well graded soil, $1 < C_c < 3$.

**21. The ratio of Plasticity Index to Flow Index is termed as**

(a) Toughness Index (b) Activity
(c) Density Index (d) Consistency Index

**Ans. (a) : Toughness Index–** It is defined as the ratio of the plasticity index to the flow index.

$$I_T = \frac{I_P}{I_F}$$

- Toughness index gives us an idea about shear strength of a soil at plastic limit. Two soils, having the same plasticity index value possess toughness which vary in inverse proportion to their flow indices.
- Toughness index generally lies in the range of 0 to 3 for most soils. When $I_T$ is less than '1', the soil is friable (easily crushed) at the plastic limit.
- It indicates the loss of shear strength with increase in moisture.

**22. The Westergard analysis is used for the**

(a) Stratified soils (b) Cohesive soils
(c) Sandy Soils (d) Homogenous soils

**Ans. (a) : Westergard analysis–**
The westergard analysis is used for stratified soil.

**23. Liquefaction occurs in**

(a) Saturated Silty Sand
(b) Saturated Dense Sand
(c) Dry Silty sand
(d) Dry Dense Sand

**Ans. (a) :** The phenomenon when the sand looses it's all shear strength due to oscillatory motion is known as liquefaction. It is commonly observed in sea shore and river beds, in loose saturated fine sand and coarse silts.

**24. For a sandy soil, the angle of internal friction is 30º. If the major principal stress is 50kN/m$^2$ at failure, then the corresponding minor principal stress (in kN/m$^2$) Will be**

(a) 16.66 (b) 12.2
(c) 20.8 (d) 27.2

**Ans. (a) :**

Given, $\sigma_1 = 50$, $\phi = 30^0$

$$\sigma_1 = \sigma_3 \tan^2\left(45 + \frac{\phi}{2}\right) + 2C\tan\left(45 + \frac{\phi}{2}\right)$$

For sandy soil, C = 0

$$\sigma_1 = \sigma_3 \tan^2\left(45 + \frac{\phi}{2}\right)$$

$$50 = \sigma_3 \tan^2\left(45 + \frac{30}{2}\right)$$

$$50 = \sigma_3 \tan^2 60^0 \quad \left(\tan 60^o = \sqrt{3}\right)$$

$$50 = \sigma_3 \times 3$$

$$\sigma_3 = \frac{50}{3} = 16.667$$

Or

$$\frac{\sigma_1}{\sigma_3} = \frac{1+\sin\phi}{1-\sin\phi}$$

$$\sigma_3 = \frac{\sigma_1(1-\sin\phi)}{(1+\sin\phi)} = \frac{50\times(1-0.5)}{1+0.5}$$

$$\sigma_3 = \frac{50}{3} = 16.66$$

**25. The minimum size of the reinforcement bar in RCC column is**

(a) 12 mm (b) 8 mm
(c) 6 mm (d) 10 mm

**Ans. (a) :** As per IS : 456-2000

- Minimum dia of longitudinal bar in column - 12 mm varies from (12 to 50) mm.
- Minimum area of steel - 0.8% of the gross area of column.
- **Maximum area of steel–**

(i) When bar are not lapped - 6% of the gross area of column.
(ii) When bar are lapped - 4% of the gross area of column.

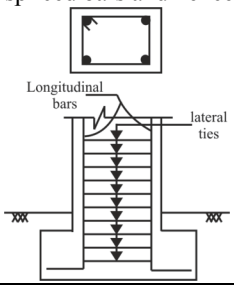
- Maximum distance between longitudinal bar = 300 mm.

**26. Lateral ties in RC columns are provided to resist**

(a) Buckling of longitudinal steel bars
(b) Bending moment
(c) Shear
(d) Both bending moment and shear

**Ans. (a) :** The main functions of lateral reinforcement in RC columns are–

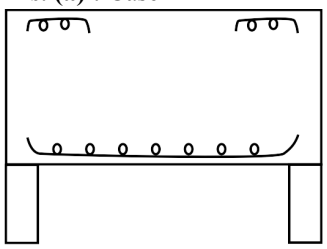
- It prevent longitudinal reinforcement bars from buckling.
- It resist the shear force and hence contributes avoiding shear failure.
- It confines the concrete core to provide sufficient ductility or deformability.
- It restrains the spliced bars and hence prevent their slip

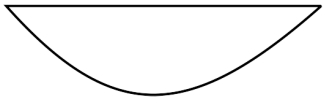



**27. A simply supported RC beam carries uniformly distributed load (ud1) and is referred as beam A. A similar beam is prestressed and carries the same ud1 as the beam A. This beam is referred as beam B. The mid-span deflection of beam A will be**

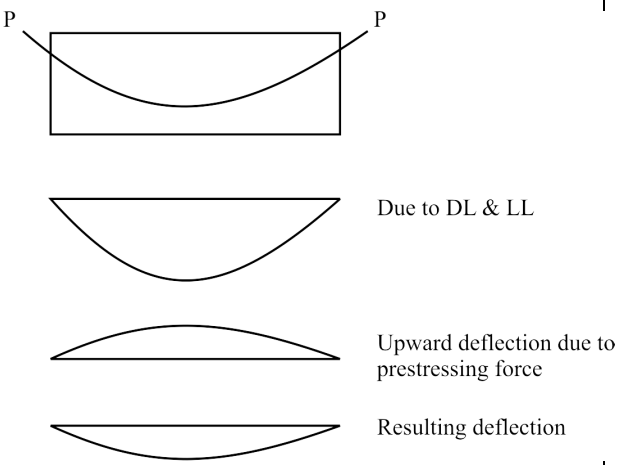
(a) More than that of beam B
(b) Less than that of beam B
(c) The same as that of beam B
(d) Generally less but sometimes more depending upon the magnitude of ud1

**Ans. (a) :** Case-I




Case-II




From above deflection diagram it is clear that deflection in presrtress beam is less than deflection in RCC beam.

**28. Total amount of shrinkage strain for a pretensioned member is**

(a) $3x10^{-4}$ (b) $3x10^{-5}$
(c) $3x10^{-6}$ (d) $3x10^{-7}$

**Ans. (a) :** Shrinkage of concrete–

Loss of stress = $\varepsilon_{ss} \times E_s$

(i) $\varepsilon_{ss}$ = Total residual shrinkage strain
$= 3 \times 10^{-4}$ for pretensioning member

(ii) $\varepsilon_{ss} = \frac{2\times10^{-4}}{\log_{10}(t+2)}$ for post tensioning member.

Whre, $E_s$ = modulus of elasticity of steel
t = Age of concrete at transfer of steel.

**29. Which of the following systems is used for mass production of structural elements?**

(a) Hoyer's system
(b) Freyssinet system
(c) Magnel -Blaton system
(d) Lee-McCall system

**Ans. (a) : Hoyer System-** Hoyer System or long line method is often adopted in pre-tensioning.
• Large scale production
• Two bulk heads or abutments independently anchored to the ground are provided several meters apart, say, 100m. Wires are stretched between the bulk heads moulds are placed enclosing the wires.
• The concrete is now poured so that a number of beams can be produced in one line.

**30. If a vehicle is moving with a speed of V kmph, negotiating a curve with a radius of R meters, and coefficient friction is assumed to be zero, the equilibrium super elevation necessary to counteract the centrifugal force fully is given by**

(a) $\frac{v^2}{127R}$ (b) $\frac{v^2}{12.7R}$
(c) $\frac{v^2}{225R}$ (d) $\frac{v^2}{190R}$

**Ans. (a) :** $e+f=\frac{V^2}{127R}$ ……..(i)
If the entire centrifugal force is counteracted by the superelevation, frictional force will not be called into play.
Thus f = 0, in equation (i) the superelevation then provided is said to be equilibrium superelevation. In such a case, the pressure on the inner and outer wheels will be equal.
So,

$$\boxed{e=\frac{V^2}{127R}}$$

**31. What Bitumen o grade 80/100 means?**

(a) its penetration value lies between 8 to 10 mm
(b) its penetration value is 8 mm
(c) its penetration value is 10 cm
(d) its penetration value lies between 8 to 10 cm

**Ans. (a) :** • The unit of penetration is 1/10 mm.
• 80/100 penetration value means a penetration of 8-10 mm.
• Penetration limits enables bitumen to be classified on the basis of consistency.
• Common grade of bitumen are 30/40, 60/70, 80/100.

**32. The California bearing ratio test is a strength test for**

(a) Sub-grade soil
(b) Cement concrete
(c) Bituminous mixes
(d) Fly ash and cement mix

**Ans. (a) : California bearing ratio method–**
• CBR method of flexible pavement design is based on strength parameter of subgrade soil and subsequent pavement material.
• The total thickness of pavement needed to cover the subgrade of the known CBR value is obtained.
Note– This formula is applicable only when the CBR value of the subgrade soil is less than 12%.

**33. A summit curve is formed at the intersection of a 4% up gradient and 6% down gradient. To prevent a stopping distance of 13m, what is the length of the summit curve?**

(a) 384m (b) 225m
(c) 257m (d) 350m

**Ans. (a) :** Given,
N = 4% – (–6%) = 10% = 0.1
S = 130 m
L > SSD (Let)

$$L=\frac{Ns^2}{4.4}=\frac{0.1\times(130)^2}{4.4}=384\text{ m}$$

**34. In highway material testing, Los Angeles testing machine is used to conduct the**

(a) Abrasion test
(b) Attrition test
(c) Impact test
(d) Crushing strength test

**Ans. (a) : Loss Angels testing machine–**
• It is an abrasion test.
• Loss Angels abrasion testing is done to find the hardness of aggregates.
• It is performed to find the percent wear due to relative rubbing action between the aggregates and steel balls are used as abrasive charge.

**35. Crushing strength of a good building stone should be more than**

(a) 100 MPa (b) 50MPa
(c) 150 MPa (d) 200MPa

**Ans. (a) :** Crushing strength of good building stone.
- A stone should be strong and durable to withstand the disintegration action of weathering
- Compressive strength of building stone in practice range **between 60 to 200 N/mm$^2$**

Crushing strength > 100 MPa (1000 kg/cm$^2$)
- High durability, sufficient hardness (i.e. coefficient of hardness > 14)
- High resistance to wear, good fire resistance, specific gravity more than 2.7
- Toughness index > 13
- Low water absorption (i.e. water absorption after 24 hours less than 5%).

**36. As per IS specifications, the maximum final setting time for ordinary Portland cement should be**

(a) 10 hours (b) 30 minutes
(c) 1 hour (d) 6 hours

**Ans. (a) :**

| Type of cement | Initial setting time (minute) | Final setting time (minute) |
|---|---|---|
| OPC (33, 43, 53) | 30 | 600 |
| PPC | 30 | 600 |
| RHC | 30 | 600 |
| Quick setting cement | 5 | 30 |

**37. The basic purpose of a retarder in concrete is**

(a) to increase the initial setting time of cement paste in concrete
(b) to decrease the initial setting time of cement paste in concrete
(c) to render the concrete more water tight
(d) to improve the workability of concrete mix

**Ans. (a) : Retarder :** Retarders normally increase the setting time and thus delay the setting of cement.

- Retarders increase the compressive strength under freezing and thawing. Calcium sulphate, sugar, starch, cellulose, ammonium, ferrous and ferric chlorides, carbohydrates are few example of retarders.
- To extend the setting time of cement paste in concrete.

**Exp.** Gypsum, sugar, starches, cellulose product, acid or salt acid.

**38. In the brick masonary, the bond produced by laying alternate headers and stretchers in each course in known as**

(a) double Flemish bond
(b) English bond
(c) zigzag bond
(d) single Flemish bond

**Ans. (a) : Double Flemish bond :**

- Each course presents the same appearance both in the face and back of the wall.
- Consists of headers and stretchers alternately in the same course.

**Header bond :** It is the bond in which all the bricks are laid as headers on the faces of walls.

**English Bond :** Alternate course of headers and stretchers.

**Flemish bond :** Each course has alternate header and stretcher.

**39. Mansard roof is a roof which slopes in**

(a) four directions with break in the slope on each side
(b) two directions without break in the slope on each side
(c) two directions with break in the slope on each side
(d) four directions without break in the slope on each side

**Ans. (a) : Mansard roof–**The roof slopes in the four directions, but each slope has a break. It is also called French roof or curb roof.

**Lean to roof –**

- It has slope only one side.

Ex Verandah or single room

**Gable roof –**

- Slopping in two direction without break.

**Hip roof–**

- Sloping in four direction without break.

**Gambrel roof–** Same as gable roof having two sloping surface but these is the break in slope in this case.

**40. When environmental lapse rate exceeds the adiabatic lapse rate, the environmental lapse rate is said to be**

(a) super adiabatic
(b) sub adiabatic
(c) stable atmosphere
(d) neutral

**Ans. (a) : Super adiabatic lapse rate–** When environment (ELR) lapse rate is more than the adiabatic lapse rate (ALR) then ambient lapse rate is super adiabatic (SLR) and the environment is said to be unstable.

**Sub adiabatic lapse rate–** When environment lapse rate is less than the adiabatic lapse rate the environment set to be stable and this prevailing environment lapse rate is called the sub adiabatic lapse rate (as it is less than adiabatic lapse rate)

**Neutral–** ELR = ALR

**41. The moisture content of the sludge is reduced from 90% to 80% in a sludge digestion tank. Then, what is the % reduction in the sludge volume?**

(a) 50% (b) 10%
(c) 25% (d) 60%

**Ans. (a) :** Given,

Moisture content, $P_1 = 90\%$

Moisture content, $P_2 = 80\%$

$V_1 \times (100–P_1) = V_2 \times (100\text{-}P_2)$

$V_1 \times (100–90) = V_2 \times (100\text{-}80)$

$V_1 \times 10 = V_2 \times 20$

$V_1 \times 2V_2$

$$\text{Reduction in volume} = \frac{V_1 - V_2}{V_1} \times 100$$

$$= \frac{2V_2 - V_2}{2V_2} \times 100$$

$$= \frac{1}{2} \times 100 = 50\%$$

**42. Sewage treatment in an oxidation pond is accomplished primarily due to the**

(a) Algal-bacterial symbiosis
(b) Algal photosynthesis only
(c) Bacterial oxidation only
(d) Chemical oxidation only

**Ans. (a) :** Special type of relationship exit between algae and aerobic between aerobic and anaerobic microorganism (M/O).

Algae give oxygen to M/O and M/O decompose the organic matter and get nutrients $CO_3^{2-}$, $NO_3^-$, $SO_4^-$ for algae similarly aerobic M/O give food to anaerobic M/O and anaerobic M/O gives energy to the aerobic M/O. So all these are called symbiosis relationship.

**43. As per CPCB's National Ambient Air Quality standard annual average value for the air pollution parameter $PM_{2.5}$ in residential area is**

(a) $40\mu g/m^3$ (b) $20\mu g/m^3$

(c) $30\mu g/m^3$ (d) $50\mu g/m^3$

**Ans. (a) :** As per National ambient air quality standard 2009–

| Pollutants | Time weighted average | Concentration in ambient air | |
|---|---|---|---|
| | | Industrial, residential, rural and other area | Ecologically sensitive and (notified by central govt.) |
| $PM_{2.5}$ | Annual | 40 | 40 |
| | 24 hours | 60 | 60 |
| $PM_{10}$ | Annual | 60 | 60 |
| | Daily | 100 | 100 |

**44. The percentage of submersion of media without air contact cups in typical rotating biological contactors generally used in the sewage treatment is**

(a) 35-40 (b) 15-25

(c) 25-30 (d) 55-65

**Ans. (a) :** A rotating biological contractor or RBC is a biological fixed film treatment process used in secondary treatment of waste water following primary treatment

- In this case closing space rotatory disc is used as a medium for the growth of biomass layer.
- Unlike trickling filter it is the biomass layer which posses through the waste water in this case.
- This disc are kept immerged in the waste water upto 40% of their size are exposed alternatively to the atmosphere and the waste water.
- The percentage of submersion of media without air contact cups in typical rotating biological contactors generally used in the sewage treatment is 35-40

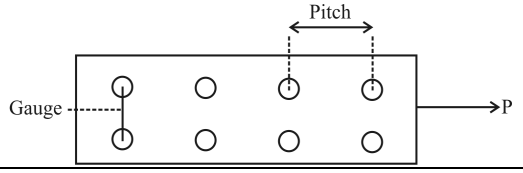
**45. As per IS-800, the minimum pitch of bolts in a row of bolts is recommended as the diameter of the bolt times**

(a) 2.5 (b) 2.0

(c) 3.0 (d) 4.0

**Ans. (a) :** Minimum pitch of rivet, should be 2.5 times of nominal dia i.e, = 2.5 d

**Pitch :** It is the distance between the centre of two consecutive rivets measured in the direction of force applied.




**Maximum pitch–**The tacking rivets are provided at a pitch in line should not more than 32 times the thickness of outside plate or 300 mm (whichever is less)

For tension – minimum (16 t, 200 mm) whichever is less

For compression – Minimum (12 t, 200 mm) whichever is less

**46. The slenderness ratio of single angle single bolted strut should be less than**

(a) 180 (b) 250

(c) 300 (d) 350

**Ans. (a) :** The slenderness ratio of single angle single bolted strut should be less than 180.

- Generally, the strut is a compression member, so maximum slenderness ratio is 180.

**47. The angle of dispersion of a concentrated load on the flange to the web plate of a steel beam is**

(a) 60º with vertical

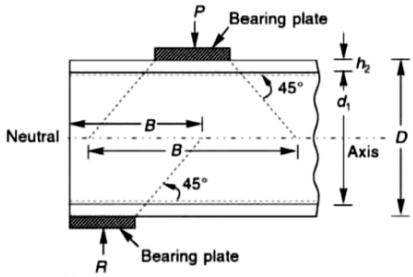
(b) 70º with horizontal

(c) 45º with vertical

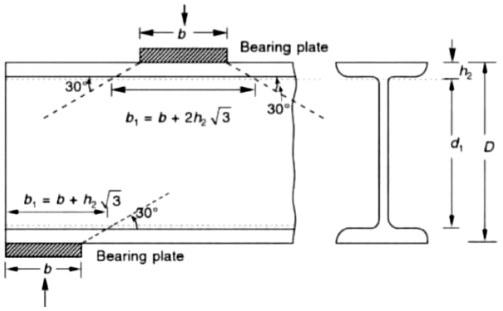
(d) 30º with vertical

**Ans. (a) :** The dispersion of load is $30^0$ to horizontal and $60^0$ to vertical according to crippling criteria and $45^0$ to horizontal and vertical according to buckling criteria.

**For buckling case–**




**For crippling case–**




**48. In a fillet weld, the weakest section is the**

(a) Throat of the fillet

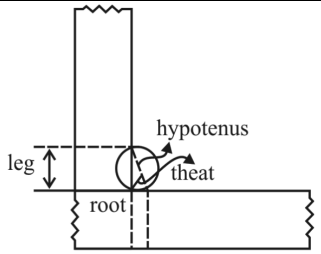
(b) Smaller side of the fillet

(c) Side perpendicular to force

(d) Side parallel to force

**Ans. (a) :** Throat of fillet weld is the weakest section in a fillet weld.

- The theoretical throat of weld is the shortest distance from the root to the hypotenuse of the triangle.



Leg or size of weld

**49. A steel column in a multi-storeyed structure carries a load of 125 kN. It is built up of 2 ISMC 350 channels connected by lacing. The lacing carries a load of**

(a) 3.125 kN (b) 125 kN
(c) 12.5 kN (d) Zero

**Ans. (a) :** Lacing shall be design to resist transverse shear (Vt), at any point in the member, which is equal to at least 2.5% of axial force in the member.

Vt = 2.50% axial force

$$= \frac{2.5 \times 125}{100} = 3.125 \text{ N}$$

**50. Base period of a crop is 120 days and the duty is 864 hectares/cumec on the filed. What is the Delta for this crop?**

(a) 120 cm (b) 200 cm
(c) 7.2 cm (d) 864 cm

**Ans. (a) :** Given,
Duty (D) = 864 hectares/cumec
Base period (B) = 120 days
Delta (Δ) = ?

$$\boxed{\Delta = \frac{8.64\,B}{D}}$$

$$\Delta = \frac{8.64 \times 120}{864}$$

$\Delta = 1.2$ m

$\boxed{\Delta = 120 \text{ cm}}$

**51. Ratio of quantity of water stored in the root zone of the crops to the quantity of the water actually delivered into the field is known as**

(a) Efficiency of water application
(b) Efficiency of water conveyance
(c) Efficiency of water storage
(d) Efficiency of water use

**Ans. (a) : Efficiency of water application–**It is the ratio of the quantity of water stored into the root zone of the crops to the quantity of water actually delivered into the field. It may also be called on farm efficiency, as it takes into consideration the water lost in the farm.

- **Efficiency of water use–**It is the ratio of the water beneficially used, including leaching water, to the quantity of water delivered.

**52. A channel of bed slope 0.0009 carries a discharge of 21 $m^3/s$ when the depth of flow is 1m. If the slope is increased to 0.0001, what is the discharge carried by an exactly similar channel at the same depth of flow?**

(a) 7 $m^3/s$ (b) 21 $m^3/s$
(c) 14 $m^3/s$ (d) 70 $m^3/s$

**Ans. (a) :**
$Q_1 = 21\ m^3/s$, $S_1 = 0.0009$, $S_2 = 0.0001$, $Q_2 = ?$

$$Q = A \times V$$

$$Q = A \times \frac{1}{n} R^{2/3} S^{1/2}$$

$$Q \propto S^{1/2}$$

$d_1 = d_2$

then $$\frac{Q_1}{Q_2} = \frac{S_1^{1/2}}{S_2^{1/2}}$$

$$\frac{21}{Q_2} \Rightarrow \frac{(0.0009)^{1/2}}{(0.0001)^{1/2}}$$

$$Q_2^2 = \frac{21}{9^{1/2}} = \frac{21}{3}$$

$$Q_2 = 7\,m^3/s$$

**53. When high flood level of the drain is below the bottom of the canal, so that drainage of the water flows freely under gravity, the structure is known as**

(a) Aqueduct
(b) Super passage
(c) Syphon aqueduct
(d) inverted syphon

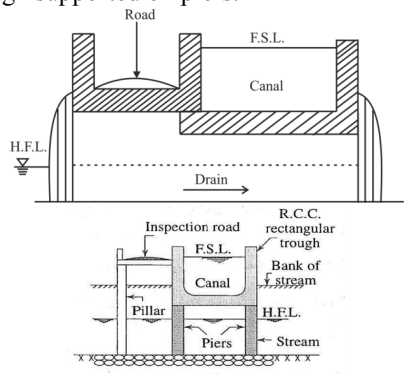
**Ans. (a) : Aqueduct–**In a cross drainage structure, the HFL of the drain is sufficiently below the bottom of the canal, so that the drainage water flow freely under gravity, then such structure is known as aqueduct.

**Type of Aqueduct–**

(i) Arch Aqueduct
(ii) Slab Aqueduct
(iii) Trough Aqueduct
(iv) Syphon Aqueduct
(v) Pipe Aqueduct

When the HFL of the drain is sufficiently below the bottom of the canal such that the drainage water flows freely under gravity, the structure is known as aqueduct.

⇒ Canal water is carried a cross the drainage in a through supported on piers.



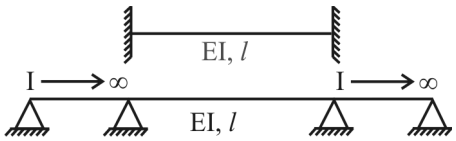

**Super Passage–** When the bed level of the drainage is much above the canal bed level or HFL, then drain is taken above the canal (i.e. just opposite to aqueduct) cross drainage work is called super passage.

**Level crossing–** When the bed level of canal and the stream are approximately the same and quantity of water in canal and stream is not much different the cross drainage work constructed is called level crossing where water of canal and stream is allowed to mix.

**54. Clapeyron's theorem is applied to**

(a) Fixed and continuous beams
(b) Simply supported beam
(c) Fixed beam
(d) Propped cantilever beam

**Ans. (a) :** Clapeyron's theorem of three moment requires at least two span, A fixed end of beam is replaced by the continuous beam can be analyzed with the three moment theorem of infinite stiffness.




$$K = \frac{EI}{\ell}$$

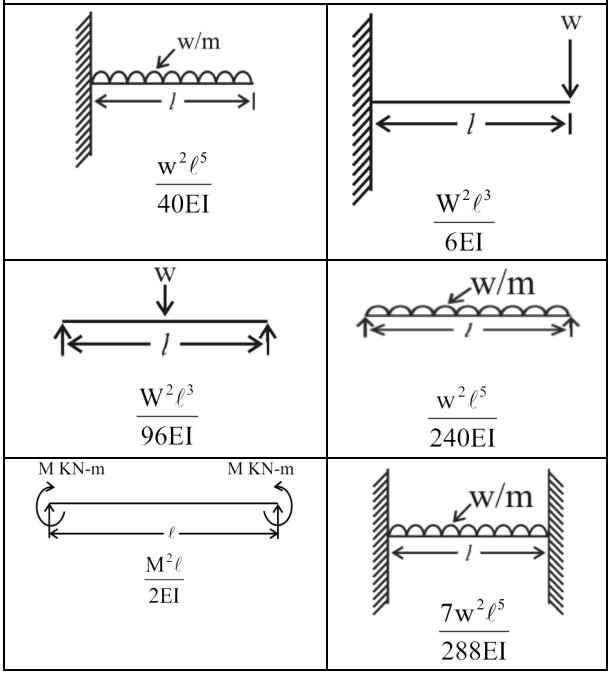
Since, moment of inertia is infinite K(stiffness) will also be infinite.

**55. A simply supported beam of span *l* and flexural rigidity EI is subjected to a unit concentrated load at its mid-span. The strain energy, due to bending, stored in the beam is**

(a) $l^3/(96EI)$ (b) $l^3/(48EI)$
(c) $l^3/(192EI)$ (d) $l^3/(16EI)$

**Ans. (a) :** The strain energy, due to bending, stored in the beam is $l^3/(96EI)$.

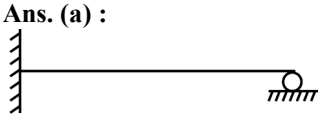
Strain Energy stored in different type of beam–




**56. The Degrees of Freedom of a propped cantilever beam are**

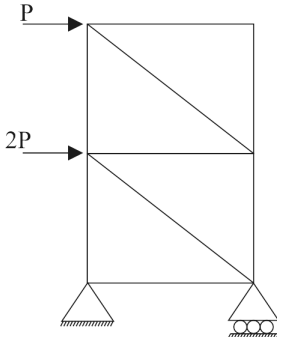
(a) 1 (b) 2
(c) 3 (d) zero

**Ans. (a) :**



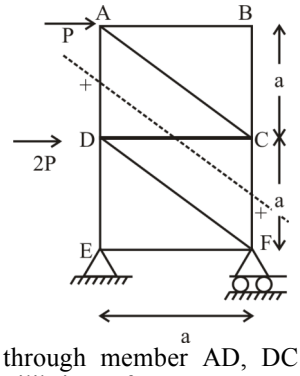
$D_k$ = 3J–Re–M [assume member is inextensible]
= 3 × 2 – (3+1)–1
= 6–5 = 1

**57. If all the vertical and horizontal members of the truss shown in the figure below are of equal length, the force in the middle horizontal member of the truss is**

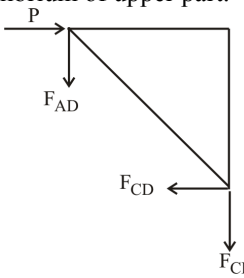



(a) P, tension
(b) Zero
(c) 2P, Compression
(d) P, compression

**Ans. (a) :**




Cutting X-X through member AD, DC and CF and considering equilibrium of upper part.

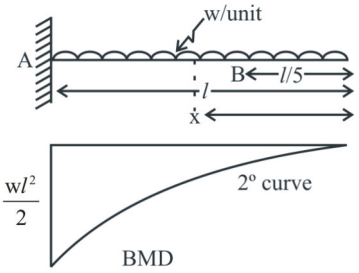



For horizontal equilibrium $\Sigma F_H = 0$, $F_{CD}$ = P Tension

**58. If the bending moment at the fixed end of a cantilever beam of span *l* subjected to uniformly distributed load over its entire span is M, the bending moment at a section *l*/5 from the free end is**

(a) 4% of M (b) 10% of M
(c) 5% of M (d) 0.4% of M

**Ans. (a) :**




$R_A = wl$

$(BM)_{xx} = w \times x \times \frac{x}{2}$

$(BM)_{at\ free\ end} = 0 \quad \because (x=0)$

$(BM)_{at\ B} = w \times \frac{l}{5} \times \frac{\left(\frac{l}{5}\right)}{2} \qquad \left(At\ x = \frac{l}{5}\right)$

$= \frac{wl^2}{25 \times 2}$

$(BM)_{at\ A} = \frac{wl^2}{2} \qquad (x = l)$

$\because$ Bending moment at fixed end is $(M) = \frac{wl^2}{2}$

And B.M at $\frac{l}{5}$ from free end

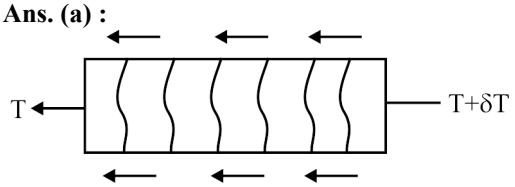
$= \frac{wl^2}{25 \times 2} = \frac{1}{25} \times \frac{wl^2}{2}$

$= \frac{M}{25} = \frac{M}{25} \times 100\% = 4\%\ of\ M$

**59. If L is the lever arm is a reinforced concrete beam, S is the total perimeter of the steel bars and F is the shear force, then the bond stress developed in concrete around the steel reinforcement is equal to**

(a) F/S.L (b) F.S/L
(c) F.L/S (d) F.S.L

**Ans. (a) :**




$$\text{Bond stress} = \frac{\text{Shear force}}{\text{Perimeter} \times \text{lever arm}}$$

$$\tau_{bd} = \frac{F}{S.L}$$

**60. The centre to centre spacing of vertical stirrups, in a rectangular beam, is**

(a) Increased towards the centre of the span of the beam
(b) Decreased towards the centre of the span of the beam
(c) Increased at the ends
(d) Either b or c

**Ans. (a) :** Shear reinforcement shall be provided in any of the following forms–

- Vertical stirrups
- The bent-up bar along with stirrups
- Inclined stirrups
- Shear reinforcement shall be provided to carry a shear force equal to $(V_u - \tau_c.bd)$.

**Note–** As shear strength requirement is more at the supports than centre, the spacing of stirrups increases towards the centre of the beam.

**61. The analysis of slab spanning in one direction is done by assuming it to be a beam of**

(a) 1 m width (b) 1 m length
(c) $1m^2$ area (d) 2 m width

**Ans. (a) :** The analysis of slab spanning in one direction is done by assuming it to be a beam of 1 m width, through the slab width is continuous and is not composed of individual beams of 1 m width. The reinforcement are calculated for 1 m width and the bars are distributed accordingly.

**62. For simply supported slabs spanning in one direction, the ratio of span to overall depth should not exceed**

(a) 30 (b) 15
(c) 20 (d) 50

**Ans. (a) :** For simply supported slabs spanning in one direction, the ratio of span to overall depth should not exceed 30.

If rectangular 2-way slab is supported from all four sides with shorter span upto 3.5 and loading upto $3kN/m^2$ then $\frac{\ell_{eff}}{D}$ ratio should be less than

| Support condition | Fe 250 | Fe415 |
|---|---|---|
| Simply supported beam | 35 | 28 |
| Continuous | 40 | 32 |

**63. A reinforced slab, built monolithically with the supporting columns and is reinforced in two or more directions, without any provision of beams, is called a**

(a) Flat slab (b) Two way slab
(c) Continuous slab (d) Circular slab

**Ans. (a) :** A flat slab is a typical type of construction in which a reinforced slab is build monolithically with supporting columns and is reinforced in two more directions, without any provision of beam.

- The flat slab thus transfers the load directly to the supporting columns suitably spaced below the slab.

**64. Which one of the following is the axis of a gravity dam?**

(a) line of the crown of the dam on the upstream side
(b) line of the crown of the dam on the downstream side
(c) centre line of the top width of the dam
(d) line joining the mid points of the base

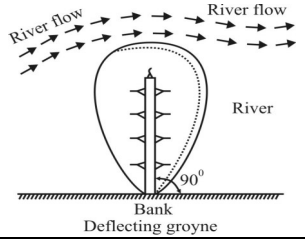
**Ans. (a):** The axis of a dam is taken as the reference line which is defined separately in the plan and in the cross-section of the dam.

- In plan, the axis of dam is defined as the horizontal trace of the u/s edge of the top of the dam and it is also called the base line of dam.
- Axis or base line of dam may be straight, slightly curved u/s or a combination of end curves and central straight line to take the best advantages of topography condition at the site.

**65. A deflecting groyne as**

(a) Small repelling groyne, giving only local protection without repelling the river current
(b) Small attracting groyne
(c) Specially designed curved head groyne
(d) A tree groyne

**Ans. (a) : Deflecting groynes–** A groynes perpendicular to bank or pointing slightly u/s & having a relatively short length lends to only deflect the flow without repelling it and hence it is called a deflecting groynes.




**66. If average rate of infiltration is 5cm/h. What will be the maximum area that can be irrigated by a stream with a discharge of 0.05m$^3$/s?**

(a) 0.36 hectares (b) 36 hectares
(c) 0.864 hectares (d) 0.25 hectares

**Ans. (a) :** Infiltration is 5 cm/h

$Q = 0.05\ m^3/s$

$$\text{Area} = \frac{Q}{\text{Infiltration}} = \frac{0.05}{5\times10^{-2}/3600}$$
$$= 3600\ m^2 = 0.36 \text{ hectare}$$

**67. A horizontal rectangular channel, 9.15 wide carries a flow of 7.65 m$^3$/s when flowing 915 mm deep. The specific energy of the flow is equal to**

(a) 0.9575 m (b) 9.57 m
(c) 7.65 m (d) 1 m

**Ans. (a) :** Given,

$Q = 7.65\ m^3/sec$, $B = 9.15$, $y = 0.915$

$$E = y + \frac{v^2}{2g}$$

$$E = 0.915 + \frac{Q^2}{A^2\times 2g}$$

$$= 0.915 + \frac{7.65^2}{(9.15\times0.915)^2\times2\times9.81}$$

$E = 0.9575$ m

**68. The initial and sequent depth of horizontal hydraulic jump in a rectangular channel are 1 m and 3 m respectively. What is the energy loss in the jump?**

(a) $\frac{2}{3}$m (b) 1m
(c) $\frac{1}{9.81}$m (d) $\frac{3}{2}$m

**Ans. (a) :** Given,

Initial depth $y_1 = 1$ m

Sequent depth $y_2 = 3$ m

$$\text{Energy loss } (E_L) = \frac{(y_2 - y_1)^3}{4y_1.y_2}$$

$$= \frac{(3-1)^3}{4\times3\times1} = \frac{8}{12} = \frac{2}{3}\text{m}$$

**69. The equation of Lemniscates curve is given by (where P is the polar ray of any point, r kis the radius of curvature at that point and $\alpha$ is the polar deflection angle.)**

(a) $r = \frac{P}{3\text{Sin}\,2\alpha}$ (b) $r = \frac{3P}{2\text{Sin}\,2\alpha}$
(c) $r = \frac{2P}{3\text{Sin}\,2\alpha}$ (d) $r = \frac{2P}{3\text{Sin}\,2\alpha}$

**Ans. (a) :**

- The equation of lemniscates curve is given by $r = \frac{P}{3\text{Sin}\,2\alpha}$

where, 'P' is the polar ray of any point
'r' is the radius of curvature
'$\alpha$' is the polar deflection angle.

- A Bernoulli's lemniscates is a special type of transition curve used in modern highways.
- It is specially suited when the angle of deflection between the two tangents is large.

**70. Tellurometer is a device used to measure_____.**

(a) distance (b) angle
(c) bearing (d) altitude

**Ans. (a) :** The tellurometer was designed to yield geodetic accuracy over geodetic distance, but it was also useful for second order work, especially in areas the terrain was rough and temperature extreme.

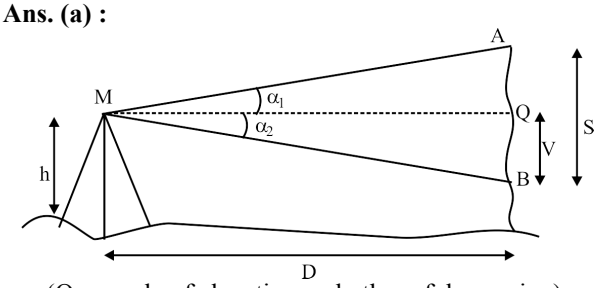
- Tellurometer is a type of EDM (Electro-magnetic distance measurement)
- This instrument comes under the microwave instrument which is a long range instrument, what in the carrier frequency of the ranges 3 to 30 GHz (1 GHz = $10^9$) enable distance measurement upto 100 km.
- Observation can be taken both during day as well as night.

**71. In a tangential method of tachometric survey, if one angle. $\alpha_1$ (positive), is angle of elevation and other angle, $\alpha_2$ (negative) is angle of depression, and the distance between these targets at the staff station is S, then the distance between the instrument axis and the staff station is given by**

(a) $\frac{S}{\tan\alpha_1 + \tan\alpha_2}$ (b) $\frac{S\tan\alpha_2}{\tan\alpha_1 + \tan\alpha_2}$

(c) $\frac{S\tan\alpha_1}{\tan\alpha_1 + \tan\alpha_2}$ (d) $\frac{S\tan\alpha_2}{\tan\alpha_1 + \tan\alpha_2}$

**Ans. (a) :**




(One angle of elevation and other of depression)

$V = D \tan\alpha_2$ ..... (i)

$(S–V) = D \tan\alpha_1$ ..... (ii)

Addition (i) and (ii)

$S = D \tan\alpha_1 + D \tan\alpha_2$

$D = \frac{S}{\tan\alpha_1 + \tan\alpha_2}$

$V = D \tan\alpha_2$

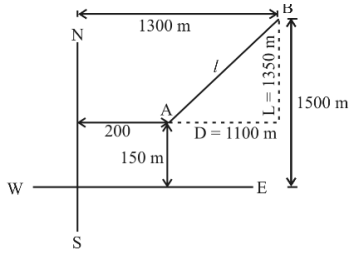
$V = \frac{S\tan\alpha_2}{\tan\alpha_1 + \tan\alpha_2}$

**72. What is the length of the line AB, if the coordinates of A and B are 150N, 200E and 1500N, 1300E respectively?**

(a) 1741.41 m (b) 2229.9 m

(c) 1735.65 m (d) 1984.94 m

**Ans. (a) :**




Departure of line (D) = 1100 m

Latitude of line (L) = 1350 m

Then length of the line

$l = \sqrt{L^2 + D^2}$

$= \sqrt{(1350)^2 + (1100)^2}$

$l = 1741.407 \simeq 1741.41$ m

**73. If the tacheometer is fitted with anallatic lens and the staff is held vertical, upto what vertical angle sloping distances can be taken as horizontal values in stadia tacheoetry, so that the error does not exceed 1 in 350º?**

(a) 3º03'34" (b) 3º10'47"

(c) 3º37'8" (d) 3º57'47"

**Ans. (a) :** True horizontal distance $D = KS\cos^2\theta$

Sloping distance = L = KS

$\frac{\text{Sloping distance}}{\text{Horizontal distance}} = \frac{L}{D} = \frac{KS}{KS\cos^2\theta} = \sec^2\theta$

If the error 1 in 350, we have

$\frac{L}{D} = \frac{350+1}{350} = \frac{351}{350}$

$\sec^2\theta = \frac{351}{350}$

$\sec\theta = \sqrt{\frac{351}{350}}$

$\theta = \sec^{-1}\sqrt{\frac{351}{350}}$

$\boxed{\theta = 3^0 03'34"}$

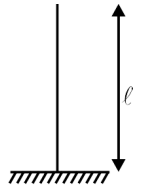
**74. An electric pole 5 m high is fixed into the foundation. It carries a wire at the top and is free to move sideways. The effective length of the pole is**

(a) 10 m (b) 3.25 m

(c) 4 m (d) 5 m

**Ans. (a) :** Effective length of pole = $2\ell$

$= 2 \times 5 = 10$ m




**75. For compression member having the same effective length about any cross-sectional axis, the most preferred section from the point of view of strength is**

(a) a circular tube (b) a box

(c) an I-section (d) a single angle

**Ans. (a) :** For compression member having the same effective length, about any cross sectional axis the most preferred section from the point of view of strength is a circular tube.

**76. The maximum slenderness ratio of compression members carrying loads resulting from dead loads and superimposed loads is**

(a) 180 (b) 100

(c) 150 (d) 200

**Ans. (a) : As per IS 800 – 1984**

| | Type of member | Max slenderness ratio |
|---|---|---|
| (i) | A member carrying compressive loads resulting from dead load and superimposed load | 180 |
| (ii) | A member subject to compressive load resulting from wind/earthquake force provided the deformation of such member does not adversely affect the stress in any part of structure. | 250 |
| (iii) | A member normally carrying tension but subjected to reversal stress due to wind or earthquake force. | 350 |

**77. A channel is designed by Lacey's theory having a mean velocity of one m/s. The silt factor is unity. The hydraulic mean radius will be equal to**

(a) 2.5 m (b) 2 m
(c) 1 m (d) 5 m

**Ans. (a) :** Hydraulic mean radius by Lacey theory–

$$R = \frac{5}{2}\frac{v^2}{f} = 2.5 \times \frac{1^2}{1} = 2.5\,m.$$

**78. The effective span of rolled steel beam is 5m. The permissible deflection in the beam is_____.**

(a) 15.38 mm (b) 16 mm
(c) 20 mm (d) 153.8 mm

**Ans. (a) :** Effective span of rolled steel beam = 5 m = 5000 mm.

The permissible deflection of beam $= \frac{1}{325}$ of span

$$= \frac{1}{325} \times 5000 = 15.38 \text{ mm.}$$

**79. Which one of the following air pollutant is most significant pollutant responsible for metallic corrosion?**

(a) $SO_2$ (b) $NO_2$
(c) Ozone (d) CFC

**Ans. (a) :** Sulphur dioxide ($SO_2$) is most significant pollutant responsible for metallic corrosion.

- In fact most all materials are affected by the deposition of acid, but the degree of damage or intensity may be varied. Some of them are more susceptible to the affect such are carbon, steel, zinc, nickel…. etc.

**80. If P represents the pressure of sound wave and $P_{ref}$ represents the reference pressure, then sound pressure Level, SPL is equal to**

(a) (20) $\log_{10}(P/P_{ref})$ (b) (1/20) $\log_{10}(P/P_{ref})$
(c) (20) $\log_{10}(P_{ref}/P)$ (d) (1/20) $\log_{10}(P_{ref}/P)$

**Ans. (a) :** Sound pressure level ($L_P$)

$$\boxed{L_P = 20\log_{10}\left(\frac{P}{P_{ref}}\right)} \qquad \boxed{P_{ref} = 20\,\mu pa}$$

**81. For every 10 ℓpm of peak flow rate of sewage at 25ºC, the surface area of the septic tank required will be**

(a) $0.90m^2$ (b) $1.90m^2$
(c) $1.50m^2$ (d) $1.20m^2$

**Ans. (a) :** As per IS code 2470 (Part-1) : 1985, as per clause no 3.4.2.1, the surface area of the tank required will be 0.90 $m^2$ for every 10 litre per minute of peak flow rate at a temperature of 25°C.

**82. Which category of projects does not require Environmental Impact Assessment in accordance with the Indian EIA Notification 2006?**

(a) Category $B_2$
(b) Category A
(c) Category $B_1$
(d) None of the given options

**Ans. (a): Category B–** Project undergoes screening process and they are classified into two types–

- Category $B_1$ project (Mandatorily require EIA)
- Category $B_2$ project (Do not require EIA).

Thus, category A projects and category B, projects undergo the complete EIA process where as category $B_2$ project excluded from complete EIA process.

**83. Supporting capacity and assimilative capacity are the components of**

(a) Carrying capacity
(b) Biodiversity holding capacity
(c) Impact mitigation capacity
(d) Carbon capturing capacity

**Ans. (a) :** The concept and premises of carrying capacity are employed as tool for the operationalisation of sustainable development.

- Carrying capacity of a region, comprising its supporitve and assimilative capacities, is defined as the ability to produce desired outputs from a constrained resource base to achieve a higher and more equitable quality of life, while maintaining desired environmental quality and ecological health.

**84. The optimistic, most likely and pessimistic estimates of time for an activity are2 days, 3 days, and 7 days respectively. What is its expected time and variance?**

(a) 3.5 days and 25/36
(b) 5 days and 5/36
(c) 3.5 days and 25/42
(d) 5 days and 5/21

**Ans. (a) :** Given: $t_0 = 2$ days
$t_m = 3$ days
$t_p = 7$ days

$$\text{Expected time } (t_e) = \frac{t_0 + 4t_m + t_p}{6}$$

$$t_e = \frac{2 + 4 \times 3 + 7}{6} = 3.5 \text{ days}$$

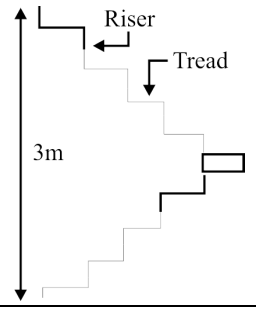
$$\text{Variance } \sigma^2 = \left(\frac{t_p - t_0}{6}\right)^2$$

$$\left(\frac{7-2}{6}\right)^2 = 0.694 = \frac{25}{36}$$

**85. The height between two floors is 3.00m and riser is 150mm. Assuming two flights between the floors, the number of treads will be equal to**

(a) 18 (b) 19
(c) 20 (d) 21

**Ans. (a) :**



Total number of riser = $\frac{3.0}{0.15} = 20$

Number of riser per flight = $\frac{20}{2} = 10$

Total number of tread per flight = 10–1 = 9

Hence total number of tread = 9 × 2 = 18

**86. For an RCC slab, the slump of concrete should be in the range of**

(a) 50-125 mm (b) 0-25 mm

(c) 25-50 mm (d) 25-100 mm

**Ans. (a) : As per IS 456-2000–**

| Type of concrete | Slump (mm) |
|---|---|
| For road construction | 20 to 40 mm |
| Beams and slab | 40 to 50 mm |
| Mass concrete | 25 to 50 mm |
| Normal R.C.C. work | 75 to 150 mm |
| Impermeable work | 75 to 120 mm |
| Concrete to be vibrated | 10 to 25 mm |

**87. Raft foundations are generally used, when the area required for individual footing is more than____.**

(a) 50% of total area (b) 250% of total area

(c) 30% of total area (d) 40% of total area

**Ans. (a) :** The bearing capacity increases with increasing depth width of the foundation and the settlement decreases with the increasing depth of foundation the advantage of mat foundation is two-fold mat foundation is also preferred when the total area of the footing exceed 50% of the total plinth area.

**88. A clay layer 5m thick in field takes 300 days to attain 50% consolidation with condition of double drainage. If the same clay layer is underlain by hard rock then, what will be the time taken to attain 50% consolidation?**

(a) 1200 days (b) 300 days

(c) 600 days (d) 900 days

**Ans. (a) :** Given,

Thickness = 5m

$d_1 = \frac{5}{2} = 2.5$ m due to double drainage

$t_1$ = 300 days

Now, clay layer underlain by hard rock.

Then

$d_2$ = 5m

Now,

$$T_v = C_v \frac{t}{d^2}$$

$t \propto d^2$

$$\frac{t_1}{t_2} = \left(\frac{d_1}{d_2}\right)^2$$

$$\frac{300}{t_2} = \left(\frac{2.5}{5}\right)^2$$

$t_2$ = 1200 days

**89. In a hydraulically most efficient trapezoidal channel section, hydraulics radius is equal to**

(a) Half of depth of flow

(b) One fourth of the depth of flow

(c) The depth of flow

(d) Twice the depth of flow

**Ans. (a) :** In a hydraulically most efficient trapezoidal channel section, hydraulic radius is equal to half of depth of flow.

$$R = \frac{y}{2}$$

$$B = \frac{2y}{\sqrt{3}}, \qquad m = \frac{1}{\sqrt{3}}$$

Where,

$\theta = 60^0$ horizontal

$\theta = 30^0$ vertical

→ For efficient channel section – Wetted perimeter is minimum.

**90. In order to avoid cavitations, the value of Thoma's Cavitation factor for Francis turbine should not be reduced below the critical cavitation factor given by (where $N_s$ is the specific speed.)**

(a) $0.625\left(\frac{N_s}{380.78}\right)^2$ (b) $0.28 + \frac{1}{7.5}\left(\frac{N_s}{444}\right)^2$

(c) $439.22\left(\frac{N_s}{444}\right)^2$ (d) $51.27\left(\frac{N_s}{380.78}\right)^2$

**Ans. (a) :** From is turbine – $\sigma_c = 0.625\left(\frac{N_S}{380.78}\right)^2$

Propeller turbine– $\sigma_c = 0.28 + \left[\frac{1}{7.5}\left(\frac{N_S}{380.78}\right)^3\right]$

**91. Which of the following condition is to be satisfied for a pipe to behave as hydro-dynamically rough pipe?**
**(where $R_e$ is Reynolds number, f is friction factor, R is the radius of the pipe and k is average height of irregularities projecting from the boundary surface)**

(a) $\frac{R_e\sqrt{f}}{R/k} > 400$ (b) $\frac{R_e\sqrt{f}}{R/k} > 100$

(c) $\frac{R_e\sqrt{f}}{R/k} > 200$ (d) $\frac{R_e\sqrt{f}}{R/k} > 300$

**Ans. (a) :** The pipe behaves as hydro-dynamically rough when the parameter.

$\frac{R_e\sqrt{f}}{R/k}$ is more than 400.

- The pipe behaves as hydro-dynamically smooth pipe when the parameter.

$\frac{R_e\sqrt{f}}{R/k}$ is less than 17.

- For $17 < \frac{R_e\sqrt{f}}{R/k} < 400$ the pipe boundary behaves as in transition.

**92. What should be the height of the centrifugal pump from the water surface to avoid cavitation, if atmospheric pressure is 100kPa, vapour pressure is 2.2kPa, effective head of the pump is 50m the head losses in suction pipe 1.5m and a cavitation parameter of 0.12.**

(a) 2.455m (b) 3.955m
(c) 7.5m (d) 4.5m

**Ans. (a) :** $P_a = 100kPa$

$$\Rightarrow P_a = \gamma H_a \Rightarrow H_a = \frac{100\times10^3}{9810} = 10.19m$$

$P_v = 2.2$ kPa

$$\Rightarrow P_v = \gamma H_v \Rightarrow H_v = \frac{2.2\times10^3}{9810} = 0.23m$$

Effective Head, $H_e = 50m$, cavitation factor = 0.12

$H_s = (h_s + h_{fs})$,

$h_{fs}$ - Head loss in suction pipe = 1.5m

$h_s$ - Suction Head = ?

$H_s$ - Total suction Head.

$$\sigma = \frac{H_a - H_v - H_s}{H_e} = \frac{H_a - H_v - (h_s + h_{fs})}{H_e}$$

$$0.12 = \frac{10.19 - 0.23 - (h_s + 1.5)}{50}$$

$$\boxed{h_s = 2.46m}$$

**93. Strouhal number is proportional to the ratio of**

(a) Centrifugal force to inertial force
(b) inertial force to viscous force
(c) inertial force to compressibility force
(d) pressure force to inertial force

**Ans. (a) :**

Strouhal number =

$$\frac{\text{Centrifugal force}\left(\text{inertial force due to local acceleration}\right)}{\text{Inertial force due to convective acceleration}}$$

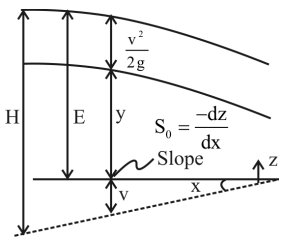
- It is a dimensionless number shown the flow mechanism of oscillating fluids.

**94. The dynamic equation for gradually varied flow in channels is given by (where $\frac{dy}{dx}$ is the slope of water surface with respect to the channel bottom; $S_0$ is the bed slope of the channel; $S_f$ is the slope of the energy line and $f_r$ is the Froude number.)**

(a) $\frac{dy}{dx} = \frac{s_0 - s_f}{1 - f_r^2}$ (b) $\frac{dy}{dx} = \frac{1 - f_r^2}{s_0 - s_f}$

(c) $\frac{dy}{dx} = \frac{s_0 - s_f}{1 + f_r^2}$ (d) $\frac{dy}{dx} = \frac{f_r(s_0 - s_f)}{1 - f_r^2}$

**Ans. (a) :**




Dynamic equation for GVF varies flow in channel given by–

$$\frac{dy}{dx} = \frac{S_o - S_f}{1 - f_r^2}$$

Where,

$S_o$ =bed slope

$S_f$ =Slope of the energy line

$f_r$ = Froude number $\left\{f_r = \frac{Q^2T}{gA^3}\right\}$

**95. What is the seepage velocity of soil having a discharge velocity of $6\times10^{-6}$ m/s and a void ratio of 0.5?**

(a) $18\times10^{-6}\,m/s$ (b) $12\times10^{-6}\,m/s$
(c) $3\times10^{-6}\,m/s$ (d) $6\times10^{-6}\,m/s$

**Ans. (a) :** Given :

Discharge velocity $V_d = 6 \times 10^{-6}$

void ratio (e) = 0.5

porosity $n = \frac{e}{1+e} = \frac{0.5}{1.5} = \frac{1}{3}$

$\therefore$ seepage velocity $V_s = \frac{V_d}{n}$

$$V_s = \frac{6\times10^{-6}}{\left(\frac{1}{3}\right)}$$

$\boxed{V_s = 18\times10^{-7}\text{ m/sec.}}$

**96. If $N_f$, Nd, and H are total number of flow channels, total number of potential drops and total hydraulic head difference respectively, the discharge q through the complete flow is given by (where K is a constant.)**

(a) $q = KH\frac{N_f}{N_d}$ (b) $q = \sqrt{H}\frac{N_f}{N_f}$

(c) $q = \sqrt{H}\frac{N_d}{N_f}$ (d) $q = K\sqrt{H}\frac{N_f}{N_d}$

**Ans. (a) :** Discharge through flow rate is given by–

$$q = KH \times \left(\frac{N_f}{N_D}\right)$$

Where– K = Co-efficient of permeability.

H = Total head loss

$N_F$ = No. of flow channel.

$N_D$ = No. of equipotential drops

here– $\left(\frac{N_f}{N_D}\right)$ = shape factor

$N_f = N_\psi - 1$, $N_\psi$ = No. of flow lines

$N_D = N_\phi - 1$, $N_\phi$ = No. of equipotential lines

**97. The phreatic line is defined as the line within a dam section below which there are**

(a) positive hydrostatic pressure
(b) positive equipotential lines
(c) negative hydrostatic pressure
(d) negative equipotential lines

**Ans. (a) :** A phreatic line is defined as the line within dam section below which there are positive hydrostatic pressure. The pressure on the phreatic line is atmospheric.

**98. Which of the following equation is sued to determine the stability number, If C is cohesion, F is factor of safety, γ is unit weight of soil, and H is the maximum height of embankment?**

(a) $\frac{C}{f\gamma H}$ (b) $\frac{F}{C\gamma H}$

(c) $\frac{H}{C\gamma F}$ (d) $\frac{\gamma F}{CH}$

**Ans. (a) :** Taylor's stability number

$$S_n = \frac{C}{\gamma H F_C}$$

Maximum theoretical value of stability number = 0.5

Maximum practical value = 0.261.

**99. When ends of compression members are not faced for complete bearing, the splices should be designed to transit_____% forces to which they are subjected.**

(a) 100 (b) 50

(c) 25 (d) 75

**Ans. (a) :** • Where the end of compression member are faced for complete bearing over the whole area this should be spliced to hold connected member accurately in position to resist any tension when bending is present.

• When such members are not faced for complete bearing, splices should be designed to transmit all the forces to which these are subjected.

• Splices are designed as short columns.

**100. The minimum width of the pavement of an intermediate lane of a National Highway should be**

(a) 5.7m (b) 4.7m

(c) 6.7m (d) 7.7m

**Ans. (a) :**

| Lane | Width of carriageway |
|---|---|
| Single | 3.75 m |
| Double | 7 m |
| Intermediate | 5.5 m |
| Multi lane (> 2) | 3.5 m per lane |

**101. If E is young's modulus, h is the slab thickness, k is the modulus of sub-grade reaction, and μ is the poisson's ratio of cement, the radius of relative stiffness of cement concrete slab is given by**

(a) $\left[\frac{Eh^3}{12k(1-\mu^2)}\right]^{1/4}$ (b) $\left[\frac{Eh^2}{12k(1+\mu^2)}\right]^{1/2}$

(c) $\left[\frac{Eh^4}{12k(1+\mu^2)}\right]^{1/4}$ (d) $\left[\frac{Eh}{12k(1-\mu^2)}\right]^{1/3}$

**Ans. (a) : Relative stiffeners of slab to subgrade–**

• A certain degree of resistance to slab deflection is offered by the subgrade. This is dependent upon the stiffeners or pressure deformation properties of the subgrade material.

• The resultant deflection of the slab which is also the deformation of subgrade is a direct measure of the magnitude of subgrade pressure. The pressure deformation characteristic of rigid pavement is thus a function of relative stiffeness of slab to that of subgrade. Westergaurd defined this form as the radius of relative stiffeners.

$$\ell = \left[\frac{Eh^3}{12k(1-\mu^2)}\right]^{1/4}$$

**102. If 'V' is the velocity of the vehicle in km/h, 't' is the brake reaction time in seconds and f is the coefficient of friction, the stopping sight distance S of the vehicle is given by**

(a) $0.278Vt + \frac{V^2}{254f}$ (b) $0.278V^2t + \frac{V^2}{0.01f}$

(c) $0.278Vt + \frac{V^2}{0.01f}$ (d) $V^2(0.278t + 0.01f)$

**Ans. (a) :** $SSD = 0.278Vt + \frac{V^2}{254(f \pm S\%)}$

$S = 0$

Then,

$$SSD = 0.278\ Vt + \frac{V^2}{254f}$$

Where, S% is gradient + ve sign for descending gradient SSD = stopping sight distance.

**103. Which one of the following is the Goldbeck's corner formula for determining the thickness of concrete pavements, h? (where is the $\sigma_{max}$ is the stress due to the corner load W)**

(a) $h = \sqrt{\frac{3W}{\sigma_{max}}}$ (b) $h = \sqrt{\frac{2W}{\sigma_{max}}}$

(c) $d = \sqrt{\frac{3W}{2\sigma_{max}}}$ (d) $d = \sqrt{\frac{3W}{4\sigma_{max}}}$

**Ans. (a) :** Goldbleck's formula for stress due to corner load.

$$Sc = \frac{3P}{h^2}$$

$$h^2 = \frac{3P}{Sc} = \frac{3W}{\sigma_{max}} \Rightarrow h = \sqrt{\frac{3W}{\sigma_{max}}}$$

Sc = stress due to corner load (kg/cm$^2$)

P = corner load (kg)

h = thickness of pavement (cm).

**104. In a levelling survey, the height of instrument at a location with a reduced level of 18.185m is observed as 1.275m. For a horizontal line of sight, what is the reading on the staff held at a chajja at a level of 21.150m?**

(a) 1.69m (b) 2.965m
(c) 1.820m (d) 2.965m

**Ans. (a) :** Given, RL = 18.185 m
BS = 1.275 m
Height of instrument (HI) = RL of that point + staff reading of that point
H.I. = 18.185 + 1.275 = 19.46
RL of chhajja is also given which is 21.150 m
**Staff reading for chhajjas–**
= HI – RL
= 19.46 – 21.15 = – 1.69 m or 1.69 (inverted)

**105. A solid shaft is subjected to a torque of 200 N-m. If it is rotating at 150 rpm, the power transmitted by shaft in kW will be**
(a) $\pi$ (b) $\pi/2$
(c) $2\pi$ (d) $2.5\pi$

**Ans. (a) :** $\omega = \frac{2\pi N}{60}$
$= \frac{2\pi \times 150}{60} = 5\pi$
Power transmitted $P = T\omega = 200 \times 5\pi$
$= 1000\pi$
$= \pi$ kW

**106. If $\theta$ is the angle of friction of the soil, the active earth pressure of soil is proportional to**
(a) $\tan^2(45^\circ - \theta/2)$ (b) $\tan^2(45^\circ - \theta)$
(c) $\tan^2(45^\circ + \theta/2)$ (d) $\tan^2(45^\circ + \theta)$

**Ans. (a) :** coefficient of active earth pressure ($k_a$)
$= \frac{1-\sin\phi}{1+\sin\phi} = \tan^2\left(45^\circ - \phi/2\right)$

**107. Within the consolidated process of saturated clay**
(a) a gradual decrease in neutral pressure and a gradual increase in effective pressure takes place and the sum of these two is constant gradual increase in neutral pressure and a gradual decrease in
(b) effective pressure takes place and the sum of these two is constant
(c) both neutral pressure and effective pressure decreases
(d) both neutral pressure and effective pressure increases

**Ans. (a) : :** During consolidation process there will be gradual decrease in neutral pressure and the effective pressure is increase by same amount. The sum of these two (i.e., total stress) remains same.

**108. Terzaghi's formula to determine the ultimate bearing capacity for a strip footing is as follows (where $\gamma$ is the unit weight of the soil, C is unit cohesion, B is the width of the foundation, D is the depth of foundation, $N_c$, $N_q$, $N_\gamma$ are the bearing capacity factors)**
(a) $CN_c + \gamma DN_q + 0.5\gamma N_\gamma B$
(b) $C\gamma N_c + DN_q + 0.5\gamma N_\gamma B$
(c) $CN_c + \gamma DN_q + 0.5N_\gamma B$
(d) $CN_c + DN_q + 0.5\gamma N_\gamma B$

**Ans. (a) :** Terzaghi's formula to determine the ultimate bearing capacity is given as– $CN_c + \gamma DN_q + 0.5\gamma N_\gamma B$.

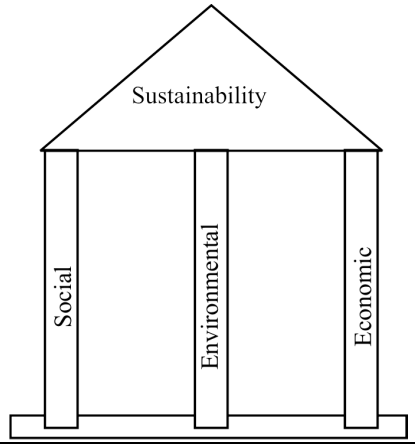
**109. A tri-axial shear test is preferred to direct shear test, because**
(a) It can be performed under all three drainage conditions with complete control
(b) Precise measurement of pore pressure and change in volume during test, is not possible
(c) Stress distribution on the failure plane, is no uniform
(d) All of the given options

**Ans. (a) : Merit and demerit of triaxial test–**
**Merit–**
- There is complete control over the drainage. This can be easily conducted for all three type of drainage condition.
- Pore pressure changes and the volumetric changes can be measured directly.
- The stress distribution on the failure plane is uniform.
- The specimen is free to fail on the weakest plane.
- The test is suitable for accurate research work.

**Demerit–**
- The apparatus is elaborate, costly and bulky
- The drained test takes a longest period as compared with that in direct shear test.
-

**110. The three basic pillars of sustainable development encompasses**
(a) environmental, economical and social attributes
(b) environmental, Industrial and judicial attributes
(c) ecosystem, food security and governance attributes
(d) Ecological governance, and accessibility attributes

**Ans. (a) :** Three basis piller of sustainable development are envrionmental , economic and social attribute.



**111. For the purpose of coastal regulation, coastal regulation Zone 2011 notification states that the coastal zone is**

**i) 500 m on the landward side from high tide line.**

**ii) 500 m on the landward side from low tide line.**

**iii) 12 nautical miles on the seaward side from the high tide line.**

**iv) 12 nautical miles on the seaward side from the low tide line.**

(a) i and iv only (b) ii and iv only

(c) i and iii only (d) ii and iii only

**Ans. (a) :** Coastal regulation zone 2011 notification states that the coastal zone is–

(i) The land area from high tide line (HTL) to 500m on the landward side along the sea front.

(ii) Land area between HTL and low tide line (LTL) which be termed as the intertidal zone.

(iii)The water and the bed area between the LTL to the territorial water limit (12 Nautical mile) in case of sea and the water and the bed area between LTL at the bank to the LTL on the opposite side of the bank, of tidal influenced water bodies.

**112. The sustainable development indicator framework PSIR implies**

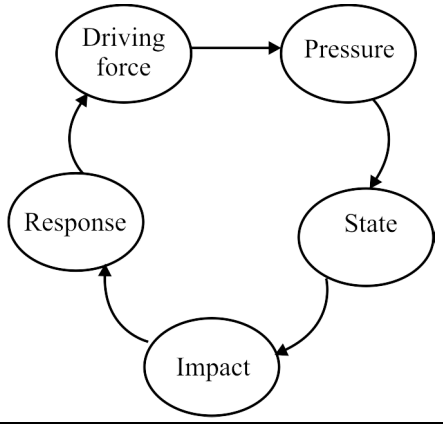
(a) Pressure, State, Impact, Response

(b) Pressure, Society, Industry, Reliability

(c) Pollution, Systems appraoach, Impact, Remedy

(d) Policy, Strategy, Implementation, Response

**Ans. (a) :** PSIR model for indicator of sustainable development.




**113. Which of the following information is provided by ISO 14041?**

(a) Environmental Management: LCA-LC inventory analysis

(b) Environmental Management: LCA-LC principles and guidelines

(c) Environmental Management: LCA-LC impact assessment

(d) Environmental Management: LCA-LC improvement assessment

**Ans. (a) :** ISO 14041 provide information about environmental management - life cycle assessment (LCA) – good & scope definition - inventory analysis.

**114. In carbon trading, which one of following is equivalent to each certified emission reduction unit?**

(a) one metric tonne of $CO_2e$

(b) One metric tonne of Carbon

(c) one metric tonne of green house gases

(d) one metric tonne of $CH_4$

**Ans. (a) :** A certified emission reduction is issued under the condition of the clean development mechanism (article - 12 of the keyoto protocol) and is equal to one meteric ton of $CO_2$ equivalent.

**115. The observed $BOD_5$ of a wastewater is determined as 150 mg/$\ell$ at 20°C. The reaction rate constant is 0.23 per day. What is the $BOD_8$ at 20°C?**

(a) 185mg/$\ell$ (b) 168mg/$\ell$

(c) 220mg/$\ell$ (d) 210mg/$\ell$

**Ans. (a) :** $(BOD)_5 = BOD_u (1-e^{-k\times t})$

$150 = BOD_u (1-e^{-0.23\times 5})$

$150 = (BOD)_u (1-0.3166)$

$(BOD)_u = 219.50$ mg/$l$

$(BOD)_8 = BOD_u (1-e^{-k\times t})$

$(BOD)_8 = 219.50 (1-e^{-0.23\times 8})$

$= 219.50 [1-0.1588]$

$= 184.639$ mg/$l$

$\approx 185$ mg/$l$

**116. Critical deficit of dissolved oxygen due to the discharge of wastewater in a stream that follows the Streeter-phelps model is given by (where $t_c$ is the critic time, $K_1$ and $K_2$ are the deoxygenating and reaeration rate constant respectively and $L_0$ is the ultimate BOD)**

(a) $\frac{k_1}{k_2}L_0e^{-k_1t_c}$ (b) $\frac{k_2}{k_1}L_0e^{-k_1t_c}$

(c) $\frac{k_1}{k_2}L_0e^{-k_2t_c}$ (d) $\frac{k_2}{k_1}L_0e^{-k_2t_c}$

**Ans. (a) :** Streeter – Phelps equation

$$D_t = \frac{K_1L_0}{K_2-K_1}\left[(10)^{-k_1t_c}-(10)^{-k_2t_c}\right]+\left[DO\times(10)^{-k_1t_c}\right]$$

Where

$D_t$ = DO deficit in mg/$l$

$L_o$ = ultimate BOD

$K_1$ = De-oxygenation coefficient

$K_2$ = Re-aeration rate constant

$D_o$ = Initial oxygen deficit of the mix at the mixing point in mg/$l$

• **The critical or max oxygen deficit is given as–**

$$DC = \frac{K_1L_o}{K_2}[10]^{-k_1t_c}$$

Or

$$DC = \frac{K_1}{K_2}L_o[e]^{-k_1t_c}$$

**117. Which one of the following is the Newton's equation for settling velocity of a discrete spherical particle under turbulent settling conditions? (where S is the specific gravity of article, d is the diameter of the particle, $C_d$ is the drag coefficient, and $\mu$ is the dynamic viscosity)**

(a) $\sqrt{3.33g(s-1)d}$ (b) $\sqrt{3.33gC_d(s-1)d}$

(c) $\frac{1}{18}\frac{g}{\mu}(S-1)d$ (d) $\frac{1}{18}\frac{g}{\mu}(S-1)d^2$

**Ans. (a) :** Newton's equation for settling velocity is given as–

$$V_s = \left[\frac{\frac{4}{3}gd(G-1)}{C_D}\right]^{1/2}$$

For turbulent settling–

$C_D = 0.4$ (For $Re > 10^4$)

Putting value of $C_D$ –

$$\boxed{V_S = \sqrt{3.33g(S-1)d}}$$

**118. The principle coagulation mechanism generally occurring in low turbidity-high alkalinity water in water treatment process is**

(a) Sweep coagulation
(b) Interparticle bridging
(c) Ionic layer compression
(d) Adsorption & charge neutralization

**Ans. (a) : Sweep coagulation–** The principle coagulation mechanism generally occurring in low turbidity-high alkalinity water in water treatment process is sweep coagulation.

**119. In split treatment process often used to soften the water with high magnesium hardness, the fraction of total flow bypassed is given by ($Mg_1$ is the magnesium concentration remaining in the fraction of water receiving first stage treatment.)**

(a) $\frac{Mg_f + Mg_l}{Mg_r - Mg_l}$ (b) $\frac{Mg_f - Mg_r}{Mg_r - Mg_l}$

(c) $\frac{Mg_r - Mg_l}{Mg_f - Mg_l}$ (d) $\frac{Mg_f - Mg_l}{Mg_r - Mg_l}$

**Ans. (a) :** In split treatment process after used to soften the water with high magnesium hardness the fraction of total flow–

$$\frac{Mg_f - Mg_1}{Mg_r - Mg_l}$$

Where- $Mg_l$ = fraction of total flow bypass

$Mg_r$ = magnesium concentration in remaining

$Mg_f$ = First stage concentration.

**120. The Carmen-Kozeny equation to compute the friction loss through filter used for water treatment consisting of bed of particles of uniform size $d_p$ is_____. (where f' is the friction factor, L is the depth of the filter, e is the porosity of the bed, $V_s$ is the filtering velocity, $d_p$ is the diameter of the filter media grains, and g is the acceleration due to gravity)**

(a) $\frac{f'L(1-e)V_s^2}{e^3 g d_p}$ (b) $\frac{f'L(1-e)2V_s^2}{e^2 g d_p}$

(c) $\frac{f'L(1-e)}{g d_p}\left(\frac{\rho_m}{\rho_w} - 1\right)$ (d) $\frac{f'Le^3V_s^2}{(1-e)g d_p}$

**Ans. (a) :** Carmen-Kozeny equation $h_f = \frac{f'L(1-e)V_s^2}{e^3 g d_p}$

$$f' = 150\left(\frac{1-e}{R_e}\right) + 1.75$$

**121. The application of hypochlorites for disinfection process in water treatment tends to**

(a) rise the pH and drive the production of less effective $OC1^-$
(b) lower the pH and drive the production of less effective $OC1^-$
(c) lower the pH and drive the production of effective $HOC1^-$
(d) rise the pH and drive the production of effective $HOC1^-$

**Ans. (a) :** Hypochlorites for disinfection process in water treatment tends to rise the ph and drive the production of less effective $OC\ell^-$.

- 100% pure hypochlorite should contain free available chlorine equal to $OC\ell^-$ value of the compound.
- Hypochlorite contains very low amount of chlorine.

**122. The recirculation ratio of standard rate trickling filter is generally kept in the range of**

(a) 0 (b) 1-2
(c) 1-3 (d) 0.2-1

**Ans. (a) :**

| | Standard | High rate | Super high rate |
|---|---|---|---|
| Hydraulic loading (in $m^3/m^2/day$ | 1–4 | 10–14 (including recirculation | 40-200 (including recirculation |
| Organic load in kg $BOD_5/m^3/day$ | 0.08-0.32 | 0.32-1 (excluding recirculation | 0.6-0.8 (excluding recirculation) |
| Depth (m) | 1.5-3.0 | 1.2 m | 4.2 -12m |
| Recirculation ratio $Q_R/Q_0$ | 0 | 0.5-3.0 | 1.4 |

**123. Removal of phosphorus by chemical precipitation through the addition of lime is effective at pH values**

(a) Higher than 9

(b) Lower than 8
(c) Lower than 7
(d) In the range of 6.5 to 7.8

**Ans. (a) : Lime–** Lime is a unique coagulant in cases where phosphours removal is desired and also in instances where it may be desired to remove ammonia. For phosphorous removal, the operating equation.

$$10Ca^{++} + 6PO_4^{3-} + 20H^- \xrightarrow{\text{pH above 9.5}} Ca_{10}(PO_4)_5(OH)_2\,(ppt)$$

**124. In Hardy-Cross method used to design the water distribution system, the correction to be applied to the assumed flows while balancing heads is (where Q is the rate of flow, hf is the head loss, and n is the constant, $\sum h_f$ is the algebraic sum of the head losses around each elementary loop; $\sum(h_f/Q)$ is the arithmetical sum of the ratio of head loss and the rate of flow)**

(a) $\pm\dfrac{\sum h_f}{n\sum(h_f/Q)}$ (b) $\pm\dfrac{\sum h_f}{n\sum(Q/h_f)}$

(c) $\pm\dfrac{\sum h_f}{n\sum(Q/h_f)}$ (d) $\pm\dfrac{\sum Q}{\sum(Q/h_f)}$

**Ans. (a) :** Hardy-Cross method used to design the water distribution system.

The correction to be applied to the assumed flows while balancing heads is $\pm\dfrac{\sum h_f}{n\sum(h_f/Q)}$.

**125. The standard rate anaerobic sludge digester volume can be determined using the equation_____(where $V_1$ is the raw sludge loading rate, $V_2$ is the digested sludge accumulation rate, $t_1$ is the digestion period and t2 is the digested sludge storage period.)**

(a) $\dfrac{(V_1+V_2)t_1}{2}+V_2t_2$ (b) $\dfrac{(V_1+V_2)t_1}{2}+V_1t_1$

(c) $\dfrac{(V_1+V_2)t_2}{2}+V_2t_2$ (d) $\dfrac{(V_1+V_2)t_2}{2}+V_1t_1$

**Ans. (a) :** The total digester volume is the given as–

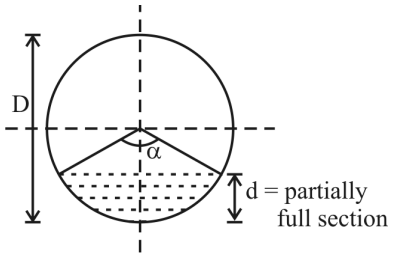
$$V=\left(\frac{V_1+V_2}{2}\right)t_1+V_2t_2$$

Where,
V = volume of the digester, $m^3$
$V_1$ = raw sludge added per day, $m^3/d$
$V_2$ = Volume of digested sludge withdrawn per day, it may be taken as equal to volume of digested sludge produced.
$t_1$ = Digestion period
$t_2$ = number of days for which the digested sludge is stored

**126. The maximum discharge in a circular sewer carrying sewage under gravity will occur when the hydraulic radius is approximately equal to**

(a) 0.29 times the diameter of the channel
(b) 0.95 times the diameter of the channel
(c) 0.75 times the diameter of the channel
(d) Equal to the diameter of the channel

**Ans. (b) :** The maximum discharge is obtained not when sewer is running full, but when the depth is about 0.95 times the full depth and is about 7% greater than that when running full.



Partially full circular sewer section

**127. The Freeman's formula to estimate the quantity of water required to meet the fire demand (Q) of a city with a population of P (in thousands) is_____.**

(a) $1136.5\left(\dfrac{P}{5}+10\right)$lpm

(b) $3861\sqrt{P}\left(1-0.01\sqrt{P}\right)$lpm

(c) $2650\left(\sqrt{P}\right)$lpm

(d) $5663\sqrt{P}$lpm

**Ans. (a) :** Freeman's formula–

$$Q=1136\left(\frac{P}{5}+10\right)\ell pm$$

Where,
P → Population in thousands
Q → Water requirement for fire demand.

Kuichling's formula,

$$Q=3182\sqrt{P}\ \ell/m$$

P → Population in thousands

National board formula

$$Q=4637\sqrt{P}\left(1-0.01\sqrt{P}\right)$$

- Buston formula

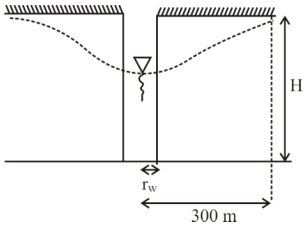
$$Q=5663\sqrt{P}\ \ell/m$$

**128. A well of radius 10cm penetrates fully into a confined aquifer. If the radius of the well is increased to twice its earlier radius with the same radius of influence of 300m, what is the percentage increase in the discharge of the well?**

(a) 9.5% (b) 15.5%
(c) 12.5% (d) 6.02%

**Ans. (a) :**

**Case – 1**




for confined aquifer –

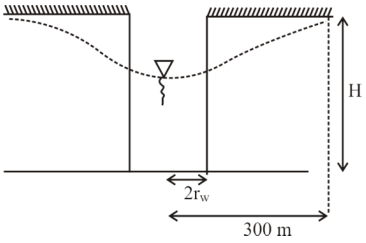
$$Q = \frac{2\pi KHs}{\ell n\left(\frac{R}{r_w}\right)}$$

$$Q \propto \frac{1}{\ell n\frac{R}{r_w}}$$

$$Q_1 \propto \frac{1}{\ell n\frac{30000}{10}}$$

$$Q_1 \propto \frac{1}{\ell n 3000}$$

**Case - 2**




$$Q_2 \propto \frac{1}{\ell n\frac{30000}{20}}$$

$$Q_2 \propto \frac{1}{\ell n 1500}$$

$$\frac{Q_2 - Q_1}{Q_1} \times 100$$

$$\frac{\frac{1}{\ell n 1500} - \frac{1}{\ell n 3000}}{\frac{1}{\ell n 3000}} \times 100\%$$

$$\left(\frac{\ell n 3000}{\ell n 1500} - 1\right) \times 100$$

$$\left(\frac{8.006}{7.313} - 1\right) \times 100$$

$(1.0947–1) \times 100$

$= 0.0947 \times 100$

$= 9.47\% \square 9.5\%$

**129. The mechanism involved in the removal of undesirable constituents such as organic contaminants from water using activated carbon is**

(a) adsorption (b) Absorption

(c) chemical oxidation (d) Ion exchange

**Ans. (a) :** The mechanism involved in the removal of undesirable constituents such as organic contaminants from water using activated carbon is adsorption.

- The process of absorption means that a substance captaincies and transforms energy. The absorbent distributes the material it captures throughout whole and absorbent only distributes it through the surface.

**130. The valve used in a water distribution system to allows water flow in one direction only is**

(a) Reflux valve (b) Butterfly valve

(c) Globe valve (d) Sluice valve

**Ans. (a) :** A valve which allows water to flow only in one direction and prevent it from flowing in the reverse direction is called as reflux valve.

**Check valve or reflux valve or non-returning valve:** In this process some automatic device which allows the water to flow in one direction only.

- This valve is provided in the pipe line which draws water from the pump.
- When the pump is operated, the valve is opened but when the pump is suddenly stopped the valve is automatically closed and the water is prevented from returning to the pump.

**Drain valve :** A valve used to drain off material that has separated from a fluid or gas stream, or one used to empty a process line, vessel or storage tank.

**Pressure relief valve :** The pressure relief valve are also known as relief valve or cut-off valves or safety valve.

- When the pressure of the water suddenly exceeds the permissible pressure due to water hammer phenomenon. Then the valve is opened automatically and the excess pressure is released instantaneously. Thus the pipe line is protected from bursting.

**Air relief valve :** This is also known as pressure relief valves.

- The water flowing through the pipe line always carries same air with it.
- This air tends to accumulate at the summit of pipe line.
- Due to the accumulation of air, a backward pressure is created which causes a blockage to the flow of water. So, the air relief valve is provided at the summit to release the air pressure.

**131. If Q is the discharge in cumecs and D is the economical diameter of the pipe conveying water, According to Lea the relationship between Q and D is**

(a) D= 0.97 to 1.22 $\sqrt{Q}$

(b) D= 0.67 to 0.85 $\sqrt{Q}$

(c) D= 0.57 to 1.30 $\sqrt{Q}$

(d) D= 0.99 to 1.345 $\sqrt{Q}$

**Ans. (a) :** Leas equation is expressed as

$$D = a\sqrt{Q}$$

Where, D is economical diameter in meter, a is coefficient ranging from 0.97 to 1.22 and Q is the amount of water to be pumped in cubic meter per second.

**132. Which one of the following is not a property of hazardous waste?**

(a) carcinogenicity
(b) Ignitability
(c) Corrosiveness
(d) Toxicity

**Ans. (a) : Hazardous waste–** Hazardous waste are classified on the basis of their biological, chemical and physical properties. These properties generate materials that are either toxic, reactive, ignitable, corrosive but not carcinogenicity.

**133. The total energy of a sample from 100kg solid waste is estimated as $14.74 \times 10^5$ kj. What is the energy content on an ash free dry basis, if the moisture and ash content are 21% and 5% respectively?**

(a) 19,919kj/kg (b) 18,658kj/kg
(c) 17,547kj/kg (d) 11,68kj/kg

**Ans. (a) :** 100 kg solid waste = $14.74 \times 10^5$ k.j
Per kg solid waste = $14.74 \times 10^3$ kj/kg
Moisture content = 21%
Ash content = 5%
Energy content on ash free dry basis =

$$\frac{100 \times 14.74 \times 10^3}{100 - 5 - 21} = 19918.91 \text{ kj/kg}$$

$$= 19919 \text{ kj / kgj}$$

**134. For a stationary container system in solid waste collection, if $C_t$ is number of containers emptied per trip, 'uc' is average unloading time per container, $n_p$ is number of container pickup locatins per trip, 'dbc' is the average time spend driving between container locations, what is the pick-up time required per trip?**

(a) $C_t(uc) + (n_p - 1)(dbc)$
(b) $C_t(dbc) + (n_p - 1)(uc)$
(c) $C_t(uc) + (n_p + 1)(dbc)$
(d) $C_t(dbc) + (n_p + 1)(uc)$

**Ans. (a) :** For stationary container system ($P_{scs}$)–

$$P_{scs} = C_t(uc) + (n_p - 1)(dbc)$$

$C_t$ = number of container emptied per trip.
$n_p$ = number of container pick up location per trip
uc = average unloading time per stationary contains for stationary container system.
dbc = average time spent driving between container location.

**135. The composition of biogas is an follows**

(a) Methane, Carbon dioxide, $H_2S$, and $N_2$
(b) Methane, Carbon dioxide, $H_2O$, and $SO_2$
(c) Methane, Carbon monoxide, and $SO_2$
(d) Methane, Carbon monoxide, $SO_2$ and hydrocarbons

**Ans. (a) :** The chemical composition of biogas is follow is–
Methane ($CH_4$) $\Rightarrow$ 50 – 85%, $CO_2 \Rightarrow$ 20 – 35% , $H_2S$ and $N_2$
Biogas is usually made up of around 50-85% methane and carbon dioxide ($CO_2$) is 20-35%.

**136. The splitting of organic substances, through a combination of thermal cracking and condensation reactions, into gaseous, liquid, and solid fractions is termed as**

(a) Pyrolysis (b) Distillation
(c) incineration (d) Wet air oxidation

**Ans. (a) : Pyrolysis–** Also called as destructive distillation.
- It is an endothermic process - requiring continuous input of heat energy.
- Combination of thermal cracking and condensation reaction in absence of oxygen.

**137. The wind rose represents the following characteristics of wind**

(a) wind direction, velocity and frequency
(b) wind direction and frequency
(c) wind direction and velocity
(d) wind velocity and frequency

**Ans. (a) : Wind rose diagram –**
The graphical representation of wind data i.e. direction duration and intensity is called wind rose.
- Wind rose used for orientation of runway.

Type- I Showing direction and duration of wind.
Type-II Showing direction, duration and intensity.

**138. Which one of the following equation relates the efficiency $(\eta)$ of the Electro Static Precipitator to the particle size? (where A is the area of the collection plates, ω is the drift velocity, and Q is the flow rate of the gas stream)**

(a) $\eta = 1 - \exp(-A\omega/Q)$
(b) $\eta = 1 + \exp(-A\omega/Q)$
(c) $\eta = 1 - \exp(-AQ/\omega)$
(d) $\eta = 1 - \exp(-Q\omega/A)$

**Ans. (a) :** The efficiency of removal of particle by an electrostatic precipitates is given by–

$$\eta = 1 - e^{-\left(\frac{\omega A}{Q}\right)}$$

$\eta$ = Fractional collection efficiency
ω = drift velocity (m/min)
A = available collection area ($m^2$)
Q = Volumetric flow rate $m^3$/min

**139. Which one of the following EIA methodologies for environmental impact identification is capable of identifying direct and indirect impacts along with identification of higher order effects and interactions between impacts?**

(a) Network method
(b) Stepped matrix method
(c) Overlay method
(d) Check list method

**Ans. (a) : Network method** is a EIA methodologies for environmental impact identification is capable of identifying direct and indirect impacts along with identification of higher order effects and interactions between impacts.

**140. Which of the following pollutants are responsible for the global climate change?**

(a) $CO_2, CH_4, N_2O, CFC_s$
(b) $SO_2, PAN, CFC_S$
(c) $CO, NH_3, N_2O, CFC_S$
(d) $CO, H_2S, NO_2, PAN$

**Ans. (a) :** The green house gases act like a thermal blanket surrounding the earth as due to the gases the heat remain, within the atmosphere and does not escape out of it.

- The major green house gases are–

(i) carbon di-oxide ($CO_2$) – 57%
(ii) chlorofloro carbon (CFC) – 25%
(iii) methane ($CH_4$) – 12%
(iv) nitrous oxide ($N_2O$) – 6%

**141. The functional aspects of ecosystem deals with the_____**

(a) rate of flow of energy
(b) the condition of existence such as temperature, light etc.
(c) the composition of biological community
(d) the quantity and distribution of non-living material like nutrients, water

**Ans. (a) :** The key functional aspects of ecosystem are energy flow, food chains and food webs, biochemical cycling, ecosystem development and ecosystem regulation and stability.

- Energy flow is the flow of energy through living things with in ecosystem. All living organism can be organised in producer and consumer and those producer and consumer can further be organised into a food chain.

**142. The degree of channels in species composition in a particular environmental gradient is known as**

(a) beta diversity (b) Species diversity
(c) alpha diversity (d) gamma diversity

**Ans. (a) : Beta diversity–**

- It measure the changes in diversity of species from one environment is another.
- In similar terms, at indicate the number of species that are not the same in two different environment.

**143. Herbivore that feeds on the leaves of shrubs or trees is called as**

(a) Folivore (b) Frugivore
(c) Granivore (d) grazzer

**Ans. (a) :** Herbivore that feeds on the leaves of shrubs or trees is called as Folivore.

- Frugivore means an animal that feeds primarily on fruit.
- Granivore means an animal feeding on seeds and grain.
- A grazer is an animal usually a harbibours mammal which eat grass.

**144. The chlorine residue to be maintained in disinfection of a new water distribution system is**

(a) 50mg/1 (b) 2mg/1
(c) 14mg/1 (d) 25mg/1

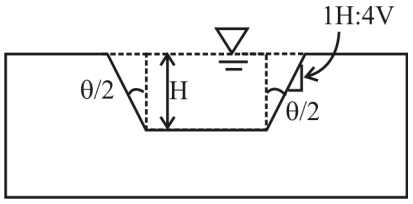
**Ans. (*) :** Chlorine residue of 0.2 mg/*l* is sufficient to safegaurd the water against the future recontamination.

- Above chlorine residue value shall be measured at 10 minutes after adding chlorine.
- Higher chlorine residue will import bad taste in water.

**145. The formula proposed by Cipolletti for discharge over a Cipoletti weir is given by (where L is the length of the weir, n is the number of end contractions, H is the height of water above the crest of the weir and g is the acceleration due to gravity)**

(a) $1.86LH^{3/2}$ (b) $1.86(LH)^{3/2}$
(c) $1.86(L-0.1nH)^{3/2}$ (d) $3.6LH^{3/2}$

**Ans. (a) : Cipolleti weir–** A standard cipoletti weir is trapezoidal in shape. The sides incline outwardly at a slope of 1 horizontal to 4 vertical.




$\tan\frac{\theta}{2}=\frac{1}{4}, \quad \theta = 28^0$

$\theta=\frac{2}{3}C_d\sqrt{2g}\ LH^{3/2}$ Where $C_d = 0.63$

Or $\theta = 1.86\ LH^{3/2}$

**146. If p,V, and A are the pressure, velocity and area of cross section of the pipe which bends through 90º, the stresses developed at the bend is (where $\gamma$ is the specific weight and g is the acceleration due to gravity)**

(a) $\sqrt{2}\left[pA+\frac{\gamma}{g}AV^2\right]$

(b) $\sqrt{3}\left[pA+\frac{\gamma}{g}AV^2\right]$

(c) $\sqrt{5}\left[2pA+\frac{\gamma}{g}AV^3\right]$

(d) $\left(\frac{1}{\sqrt{2}}\right)\left[pA+\frac{\gamma}{g}AV^2\right]$

**Ans. (a) :** If p,V, and A are the pressure, velocity and area of cross section of the pipe which bends through 90º, the stresses developed at the bend is $\sqrt{2}\left[pA+\frac{\gamma}{g}AV^2\right]$.

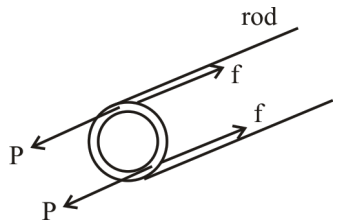
Where,
$\gamma$ = specific weight
g = acceleration due to gravity

**147. If d is the diameter of the pipe, p is the total internal pressure, f is the permissible tensile stress and n is the effectiveness of the joint, and the thickness of the metal pipe is**

(a) $\frac{pd}{2nf}$ (b) $n\frac{pd}{2f}$

(c) $\frac{pf}{nd}$ (d) $\frac{pf}{3nd}$

**Ans. (a) :**




We know, $\frac{Pd}{2t}=\sigma$

Here n is the effectiveness of joint.

then $\frac{Pd}{2tn}=f$

It is the Balow's formula relates the internal pressure that a pipe or cylinder part can withstand to its dimension's and the strength of its material.

**148. A closely coiled helical spring consisting of n turns is carrying an axial load W. If the diameter of the rod of the spring is 'd' and 'R' is the mean radius of the coil the maximum shear stress at any section of the rod is equal to**

(a) $16WR/\pi d^3$ (b) $12WR/\pi d^3$

(c) $16WR^2/\pi d^4$ (d) $16Wd/\pi R^3$

**Ans. (a) :** Turn = n
Axial load = W
d =dia of spring
R = Radius of spring
Coil with maximum shear stress

According to torsion equation = $\frac{\tau}{R}=\frac{G\theta}{l}=\frac{T}{J}$

$$=\frac{\tau}{d/2}=\frac{T}{\frac{\pi d^3}{32}}$$

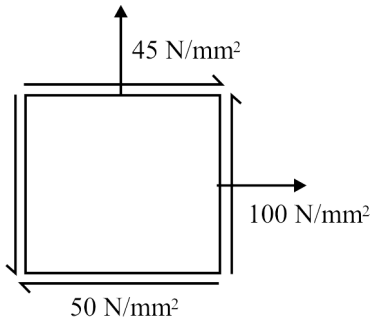
$$\boxed{\tau_{max}=16\,WR/\pi d^3}$$

**149. A rectangular bloc of material is subjected to a tensile stress of 100N/mm$^2$ on one plane and a tensile stress of 45N/ mm$^2$ on a plane at right angle, together with a shear stress of 50N/ mm$^2$ on the same plane. What is the inclination of the principal plane with the plane carrying the tensile stress of 100N/ mm$^2$?**

(a) 61.189º (b) 30.96º

(c) 60.94º (d) 34.6º

**Ans. (a) :**




$$\tan 2\theta_p=\frac{2\tau_{xy}}{\sigma_x-\sigma_y}$$

$$=\frac{2\times 50}{100-45}=\frac{100}{55}$$

$$2\theta_p=\tan^{-1}\left(\frac{100}{55}\right)$$

$$\boxed{2\theta_p=61.189^\circ}$$

**150. What is the slope and downward deflection at the free end of a cantilever beam of span 'L', with a flexural rigidity of EI, subjected to a uniformly distributed load of w per unit run?**

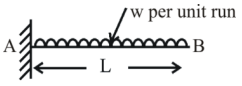
(a) $\text{Slope}=\frac{wL^3}{6EI'}\text{deflection}=\frac{WL^4}{8EI}$

(b) $\text{Slope}=-\frac{wL^2}{2EI'}\text{deflection}=\frac{WL^3}{3EI}$

(c) $\text{Slope}=-\frac{wL^3}{12EI'}\text{deflection}=\frac{WL^4}{16EI}$

(d) $\text{Slope}=\frac{wL^3}{3EI'}\text{deflection}=\frac{WL^4}{8EI}$

**Ans. (a) :**




$$\text{Slope}(\theta)=\frac{wL^3}{6EI}$$

$$\text{Deflection}(\delta)=-\frac{wL^4}{8EI}\text{ (downward)}$$

# Andhra Pradesh Public Service Commission
# Assistant Engineer (Civil, Mains.) Exam, 2016 (P-III)
## Solved Paper

**1. As per Indian standard code 1077, the burnt clay building bricks having compressive strength less than _____ N/mm$^2$ are known as common burnt clay bricks.**
(a) 3.5 (b) 12.5
(c) 30 (d) 40

**Ans. (d) :** As per IS code 1077 : 1962, burnt clay brick having compressive strength less than 40 N/mm$^2$ are known as heavy duty bricks and are used for heavy duty structure such as bridges, piers etc.
• Common burnt clay bricks have minimum strength from 3.5 N/mm$^2$ upto 40 N/mm$^2$.

**2. After 24 hours immersion in cold water, absorption by weight shall not exceed_____ percent of the dry weight of the brick.**
(a) 40 (b) 20
(c) 25 (d) 40

**Ans. (b) :** After 24 hours immersion in cold water, absorption by weight shall not exceed 20% of the dry weight of the brick.

**3. Maximum slenderness ratio allowed as per Indian standard for an unreinforced load bearing wall (using Portland Cement or Portland Pozzolana Cement in mortar) is**
(a) 13 (b) 20
(c) 27 (d) 35

**Ans. (c) :** According to IS 1905 : 1987, Table No. 7 Maximum slenderness ratio for a load bearing wall depends upon number of storey.

| No. of storey | Maximum slenderness ratio | |
|---|---|---|
| | Using portland cement or portland Pozzolana cement in mortar | Using lime mortar |
| Not Exceeding 2 | 27 | 20 |
| Exceeding 2 | 27 | 13 |

**4. The test conducted for the calculation of basic compressive stress of masonry is**
(a) Vibration test (b) Prism test
(c) CBR test (d) Slump cone test

**Ans. (b) :** Prism test is a laboratory test to calculate the compressive strength of a masonry. A masonry prism is an assembly of masonry unit and mortar that is constructed to serve as a test specimen for determining compressive strength.

**5. Efflorescence test is conducted for burnt clay bricks to find out the**
(a) presence of alkaline substance
(b) hardness
(c) soundness
(d) presence of cracks or holes

**Ans. (a) :** Efflorescence test is conducted for burnt clay bricks to find out the presence of alkaline substance.
• For this the bricks are soaked in water at room temperature after the water is evaporated check for presence of white spots (salts) on the brick. It should be < 50% of brick area and for 1st class bricks < 10% of brick area.

**6. The apparatus used for determining the soundness of cement is**
(a) Slump test
(b) Le-Chatelier apparatus
(c) Vicat's needle
(d) UTM

**Ans. (b) : Soundness test :** The purpose of this test is to detect the presence of uncombined lime in cement after setting. It is performed with the help of :

| Autoclave Test | Le-Chatelier's Method |
|---|---|
| • Sensitive to both lime and magnesia | • Unsoundness due to free lime only |
| • Internal mould dimension 25×25×250 | • 100 gm of cement + 0.7 P |
| • Result is given in % | • Result is given in mm |

**7. Cement used for railway sleepers is designated as**
(a) 40-S (b) 53-S
(c) 46-S (d) 48-S

**Ans. (b) :** Cement used for railway sleeper is designated as 53-S. The cement used OPC grade 53 i.e. OPC cement with strength 53 MPa and confirms to BIS specification IS 12269-1987.

**8. The main constituent of cement which is responsible for initial setting of cement is**
(a) Dicalcium silicate
(b) Tricalcium silicate
(c) Tricalcium aluminate
(d) None of the given answers

**Ans. (c) : Tricalcium silicate–** Responsible for one week strength.
**Tricalcium aluminate–** Initial setting, may even cause flash setting if in excess, it does not provide strength.
**Dicalcium silicate–** Responsible for ultimate strength.

**9. For testing compressive strength of hydraulic cement other than masonry cement as per IS 4031-Part-6, the size of cube used is**
(a) 150 mm (b) 100 mm
(c) 70.6 mm (d) 50 mm

**Ans. (c) : Compressive strength test:**
Test specimen = 70.6 mm cube (as per IS 4031 – Part 6)
Surface area = 5000 mm$^2$

• A mixture of cement and sand in the proportion 1:3 by weight is mixed dry.
• Quantity:
1. cement = 185 g
2. sand = 555 g
3. water = $\left(\frac{P}{4}+3\right)\%$

P = Standard consistency of cement
• Three cubes are prepared.
• Cubes are completely filled with mortar.
• Cubes are compacted on vibration table for 2 minutes and cubes are removed from mould and submerged in clean water.
• Average of compressive strength of three cubes at 28 days is taken as avg. compressive strength of cement.
**Note:** Large size specimen (i.e. 150 mm) cubes can not be made since cement shrinks and develops crack.

**10. Plywood has the advantage of**
(a) Greater tensile strength in longer direction
(b) Greater tensile strength in shorter direction
(c) Same tensile strength in both direction
(d) None of the given answers

**Ans. (c) : Advantage of plywood :**
- It has good strength both along as well as across the grains.
- Wood shrinks or swells more across the grains. Since plywood has cross-grained construction, the tendency to shrink or swell is reduced.
- Plywood can be curved into desired shapes.
- It has better splitting resistance due to the grains in adjacent veneers in cross direction.

**11. A queen closer is**
(a) Full brick
(b) Longitudinally $\frac{1}{2}$ bricks
(c) $\frac{3}{4}$ bricks
(d) $\frac{1}{2}$ bricks

**Ans. (b) : Queen closer :** A brick of full length and thickness but half width that is the used at the end of a course next to quoin header.

Queen Closer

**12. The artificial seasoning method that causes timber to become brittle and easy to break is**
(a) Boiling
(b) Chemical seasoning
(c) Electrical seasoning
(d) Kiln seasoning

**Ans. (a): Water seasoning–** Logs of wood immersed in running water with their larger end pointing upstream.
**Boiling–** Timber is immersed in water and it is boiled. This is a very quick method. It affects the strength and elasticity of wood. Boiling seasoning of timber is also achieved by boiling it in water for 3 to 4 hrs. After boiling timber is allowed to dry.
**Electrical Seasoning–** High frequency alternating current is used. It is most rapid method but uneconomical.
**Salt Seasoning–** Timber immersed in solution of suitable salt. It is taken out and seasoned in the ordinary way.

**13. The age of tree can be known by examining**
(a) Cambium layer (b) Annular rings
(c) Medullary rays (d) Heart wood

**Ans. (b) :** The age of a tree can be estimated by the number of concentric rings on a cross-section of its trunk. A tree's age can be determined by counting the annual growth rings in its trunk. Each ring represents one year, and the ring's thickness reveals the relative amount of rainfall that year.

**14. Inadequate compaction during concrete casting results in**
(a) Honey combing
(b) Bleeding
(c) Segregation
(d) Bleeding and Segregation

**Ans. (a) :** When concrete is not vibrated properly it may leave voids called honey combing. The exposed aggregate leaves a honey comb.
• This can be corrected by increasing the sand and cement content of the mix and by proper mixing, placing and compacting honey combing may be caused by leakage of grout or mortar fraction from the concrete at construction or formwork joints.

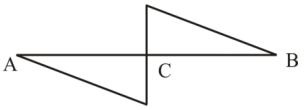
**15. If characteristic compressive strength at 28 days is 40 N/m$^2$ and the standard deviation is 5 N/mm$^2$, the largest strength at 28 day's for concrete mix proportioning**
(a) 40 N/mm$^2$ (b) 35 N/mm$^2$
(c) 25 N/mm$^2$ (d) 48.25 N/mm$^2$

**Ans. (d):** The target mean strength is given as :
$f_m = f_{ck} + (1.65\sigma)$
$= 40 + (1.65\times5)$
$= 48.25$ N/mm$^2$

**16. A statistically indeterminate structure is the one which**
(a) cannot be analysed using equations of statics alone
(b) cannot be analysed at all
(c) is note stable for general loading
(d) can be analysed with equations of statics alone

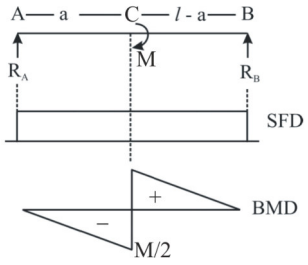
**Ans. (a) :** A statically indeterminate structure can not be analysed by using equation of statics alone, we use compatibility equation also to find all the unknowns.
• When the static equilibrium (force and the moment) are insufficient for determining the internal forces and reaction on that structure.

**17. If the BMD for a simply supported beam is as shown below. the load on the beam will be**




(a) a concentrated load at C
(b) equal and opposite couples at the ends A and B
(c) a uniformly distributed load acting on the entire span
(d) concentrated couple at C

**Ans. (d)** In bending moment diagram-sudden jump indicates couple and straight line means-constant shear force-no load.

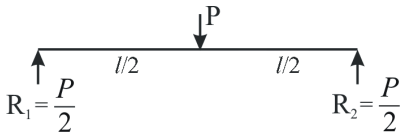



**18. The ratio of maximum deflection to maximum flexural stress in a simply supported beam of span *l* and depth *d* subjected to a concentrated load at mid-span is**

(a) $\frac{l^2}{(6Ed)}$ (b) $\frac{l^2}{(8Ed)}$

(c) $\frac{l^2}{(16Ed)}$ (d) $\frac{l^2}{(60Ed)}$

**Ans. (a) :**




$$\sigma_{max} = \frac{M_{max} y_{max}}{I}$$

We know, max deflection

$$\therefore \quad \delta_{max} = \frac{PL^3}{48EI}$$

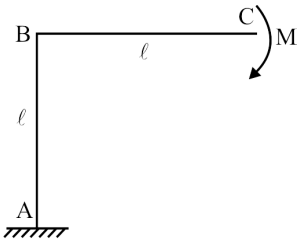
$$M_{max} = \frac{P}{2} \times \frac{l}{2} = \frac{Pl}{4}$$

$$y_{max} = \frac{d}{2}$$

$$\sigma_{max} = \frac{My}{I} = \frac{Pl}{4} \times \frac{d}{2I} = \frac{Pld}{8I}$$

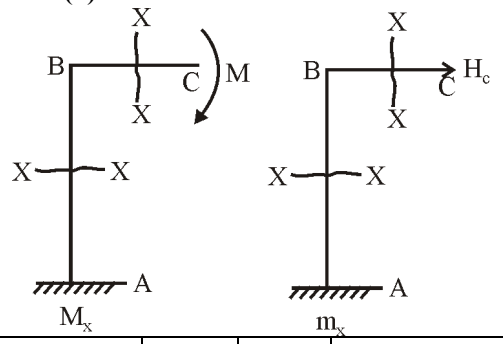
$$\frac{\delta_{max}}{\sigma_{max}} = \frac{Pl^3}{48EI} \times \frac{8I}{Pld} = \frac{l^2}{6Ed}$$

**19. The horizontal deflection at C for the frame loaded and supported as shown below is (EI is constant)**




(a) $M\ell^2 / (4EI)$ (b) $M\ell^2 / (2EI)$
(c) $M\ell^2 / (3EI)$ (d) $2M\ell^2 / (EI)$

**Ans. (b) :**



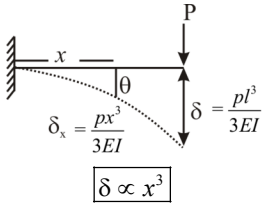

| Member | $M_x$ | $m_x$ | $\delta H_C = \int \frac{M_x m_x}{EI} dx$ |
|---|---|---|---|
| BC | (–M) | 0 | 0 |
| BA | (–M) | -1 × x | $\int_0^{\ell} \frac{-M(-x)}{EI} dx = \frac{M\ell^2}{2EI}$ |

$$\delta H_c = \frac{M\ell^2}{2EI}$$

**20. The influence line for deflection at the free end of a cantilever is**

(a) a triangle with zero ordinate at fixed end and maximum ordinate at free end
(b) a constant line
(c) a cubic parabola with zero ordinate at fixed end and maximum ordinate at free end
(d) a parabola with zero ordinate at fixed end and maximum ordinate at free end

**Ans. (c) :**




$$\boxed{\delta \propto x^3}$$

In a cantilever, the influence line diagram for deflection at the free end is same as elastic Curve of the beam due to unit load placed at free end.

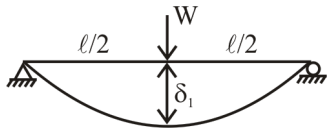
Deflection due to point load placed on cantilever is given as $Px^3/3EI$ is cubic parabola and hence ILD for deflection for a cantilever will be also be cubic parabola.

**21. A prismatic beam simply supported carries a concentrated load W at mid span. If the same beam is fixed at its ends, what load at mid-span can produce the same deflection at mid-span?**

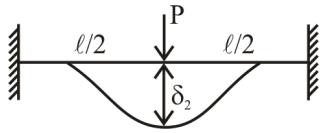
(a) 4 W (b) 2 W

(c) 3 W (d) 5 W

**Ans. (a) :**




When both ends are fixed




As per the question

$\delta_1 = \delta_2$

$$\frac{W\ell^3}{48EI} = \frac{P\ell^3}{192EI}$$

$$\boxed{P = 4W}$$

**22. Two loads of an electrical crane, 50 kN each, spaced at 4 m centre to centre cross a girder of 8 m span. Find the absolute maximum bending moment in the beam**

(a) 1125 kN-m at 3m from support

(b) 112.5 kN-m at 3m from support

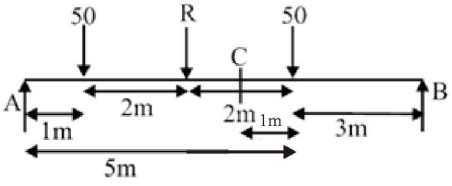
(c) 11.25 kN-m at 3m from support

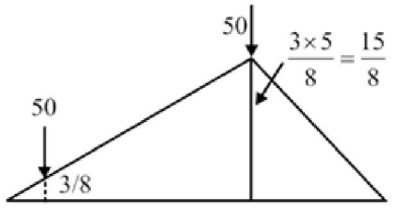
(d) 93.75 kN-m at 3m from support

**Ans. (b) :** Resultant of the forces from left load is-

$$= \frac{50\times 0 + 50\times 4}{50+50} = 2\text{m}$$

Keeping the loads such that 50 kN load and resultant are equidistant from the centre of beam-

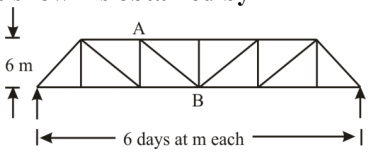



$$(BM)_{abs\,max} = 50\times\frac{3}{8} + 50\times\frac{15}{8}$$

= 112.5 kN-m at 3m from support

**23. Influence line for member force in AB of the truss shown is obtained by**



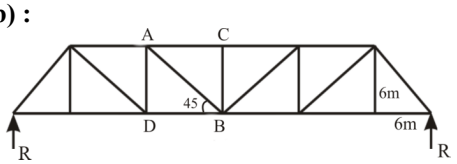

(a) $\sqrt{2}$ times the ordinates of influence line for bending moment at A

(b) $\sqrt{2}$ times the ordinates of influence line for shear in 3rd panel from left

(c) $\sqrt{2}$ times the ordinates of influence line for bending moment at B

(d) $s\sqrt{2}$ times the ordinates of influence line for shear in 3rd panel from right

**Ans. (b) :**




Now use $\sum F_x = 0$ at Joint B

$\therefore \theta = 45^0$

$F_{AB}\cos\theta + F_{DB} = 0$

(Due to symmetry analyze only left portion)

$F_{AB} = -\sqrt{2}F_{BD}$

$\therefore$ i.e., $\sqrt{2}$ force in 3rd panel.

**24. A parabolic arched rib, of span 30 m, is hinged at the springing and crown and is having a central rise of 5 m. If the coefficient of thermal expansion for the arch material is $12 \times 10^{-6}$ per $^0$C , the effect of a temperature rise of 300 C is**

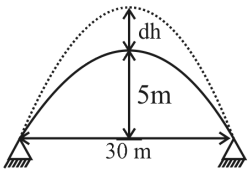
(a) to cause thermal stresses

(b) to cause thermal stresses as well as a central rise of 18 mm

(c) to cause a central rise of 18 mm

(d) to cause no effect on the structure

**Ans. (c) :**




Given, $\alpha = 12\times10^{-6}$ /$^o$C, h = 5m, $\Delta T$ = 30 $^o$C, $l$ = 30m

dh = Rise due to temperature change

$$= \frac{4h^2 + l^2}{4h}\times\alpha\times\Delta T$$

$$= \frac{100+900}{20}\times12\times10^{-6}\times30$$

$$= 0.018\text{m} = 18\text{ mm}$$

**25. A temperature rise in a two hinged symmetric and parabolic arched rib causes**

(a) a uniform bending moment in the rib

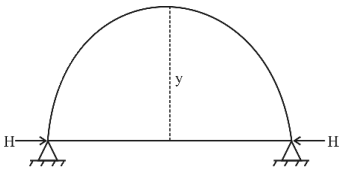
(b) no bending moment in the rib

(c) a maximum bending moment at the crown of the arch

(d) a minimum bending moment at the crown of the arch

**Ans. (c) :** Due to increase in temperature free expansion in 2-hinged arch is not permitted, so horizontal thrust is generated by

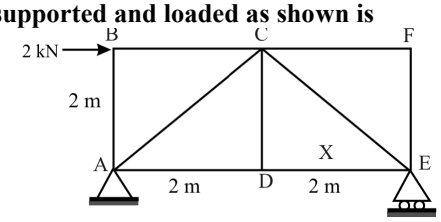
$H = \frac{15}{8}\frac{EI\alpha T}{h^2}$




Moment = Hy

So, maximum B.M. will be at crown as crown has maximum value of y.

**26. The force in the member 'X' of the truss supported and loaded as shown is**




(a) $\sqrt{2}$ kN, Tension (b) $\sqrt{2}$ kN, Compression
(c) 1 kN, Tension (d) 1 kN, Compression

**Ans. (c) :** Members AB, CD, CF and EF are zero member forces

$H_A = 2kN$

balance the moment about roller support E

$V_A \times 4 + 4 = 0$

$V_A = -1$

$V_E = +1$

$F_{AC} \sin\theta = 1$

$F_{AC} = \sqrt{2}$ (Tension)

$F_{AD} = F_{AC} \cos\theta = \sqrt{2} \times \frac{1}{\sqrt{2}} = 1kN$

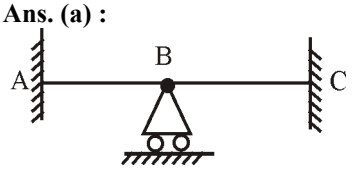
At joint D

$F_{AD} = F_{DE} = 1$ kN (Tension)

**27. A three-span continuous beam is fixed at the ends and supported by unyielding roller supports in between. What is the size of the stiffness matrix?**

(a) 2 × 2 (b) 3 × 3
(c) 1 × 1 (d) 4 × 4

**Ans. (a) :**




$D_s = R\text{-}S = 5\text{–}3 = 2$

**28. Which of the following is displacement method?**

(a) flexibility method
(b) moment distributed method
(c) Kani's method
(d) None of the given answers

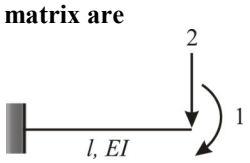
**Ans. (b) : Example of force methods :**

(a) Castigliano's theorem
(b) Strain energy method
(c) Virtual energy method
(d) Clapeyron's three moment equations
(e) Column analogy method
(f) Flexibility matrix method

**Example of displacement method:**

(a) Slope deflection method
(b) Moment distribution method
(c) Stiffness matrix method
(d) Kani's method

**29. For the structure shown. The elements of the flexibility matrix are**




(a) $f_{11} = \frac{l}{(EI)}$; $f_{21} = \frac{l^2}{(2EI)}$; $f_{12} = \frac{l^2}{(2EI)}$;

$f_{22} = \frac{l^3}{(3EI)}$

(b) $f_{11} = \frac{l^3}{(3EI)}$; $f_{21} = \frac{l^2}{(2EI)}$; $f_{12} = \frac{l^2}{(2EI)}$;

$f_{22} = \frac{l}{(EI)}$

(c) $f_{11} = \frac{l}{(EI)}$; $f_{21} = \frac{l^2}{(EI)}$; $f_{12} = \frac{l^2}{(EI)}$;

$f_{22} = \frac{l^3}{(3EI)}$

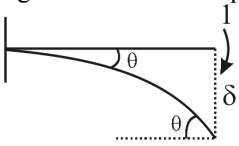
(d) $f_{11} = \frac{l}{(EI)}$; $f_{21} = \frac{l^2}{(2EI)}$; $f_{12} = \frac{l^2}{(2EI)}$;

$f_{22} = \frac{l^3}{(4EI)}$

**Ans. (a) :**

Development of flexibility matrix :

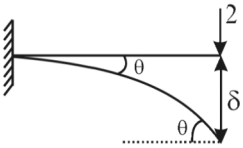
According to moment developed :




$\delta = \frac{ML^2}{2EI} = \frac{L^2}{2EI} = F_{21}$

$\theta = \frac{ML}{EI} = \frac{L}{EI} = F_{11}$

According to force developed :




$\delta = \frac{WL^3}{3EI} = \frac{L^3}{3EI} = F_{22}$

$\theta = \frac{WL^2}{2EI} = \frac{L^2}{2EI} = F_{12}$

**[L = ℓ]**

$$\text{Flexibility matrix} = \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix} = \begin{bmatrix} \frac{\ell}{EI} & \frac{\ell^2}{2EI} \\ \frac{\ell^2}{2EI} & \frac{\ell^3}{3EI} \end{bmatrix}$$

**30. An agricultural land of 437 ha is to irrigate for a particular crop. The base period of the crop is 90 days the total depth of water required by crop is 105 cm, if a rainfall of 15 cm occurs during the base period, the duty of irrigation water is**

(a) 437 ha/cumec (b) 487 ha/cumec
(c) 741 ha/cumec (d) 864 ha/cumec

**Ans. (d) :** Given data,
Area (A) = 437 ha
Base period (B) = 90 days
Total depth of water required for the crop = 105 cm
Rainfall occurring during base period= 15 cm
Delta '$\Delta$' = 105 – 15 = 90 cm

$$\text{Duty of crop} = \frac{864\ B}{\Delta}$$

$$= \frac{864 \times 90}{90} = 864 \text{ ha/cumec}$$

**31. The absolute stiffness of a prismatic member with one end hinged is**

(a) 2 EI/$\ell$ (b) 4 EI/$\ell$
(c) 3 EI/$\ell$ (d) EI/$\ell$

**Ans. (c) :** The absolute stiffness of a prismatic member with one end hinged is 3 EI/$\ell$.

**32. The angle of dispersion of a concentrated load on the flange to the web plate of a steel beam is**

(a) 70 Degrees with horizontal
(b) 60 Degrees with vertical
(c) 45 Degrees with vertical
(d) 30 Degrees with vertical

**Ans. (b) :** The angle of dispersion of a concentrated load on the flange to the web plate of a steel beam is 60 Degrees with vertical and $30^0$ with horizontal.

**33. Sway calculations and non-calculations are carried out in a single operation in**

(a) Kani's method
(b) moment distribution method
(c) unit load method
(d) none of the given answer

**Ans. (a) : Kani Method** given by Dr. Gasper Kani (1947) Kani method is similar to the moment distribution method in that both these method use Gauss-Sidell iteration produce to solve the slope deflection equation, without explicitly writing them down.
However the difference between Kani method and MDM is that Kani's method iterate the member and moment themselves rather than iterating their increment.
- Sway calculations and non-calculations are carried out in a single operation.

**34. The Eddy's theorem is valid for**

(a) Vertical Loads only
(b) Horizontal Loads only
(c) Dynamic Loads only
(d) All loads

**Ans. (a) :** Eddy's theorem is valid only for vertical load.
- For 2-hinged arch, its states that the bending moment at any section of an arch is proportional to the vertical intercept between the linear arch and the centre line of actual arch.

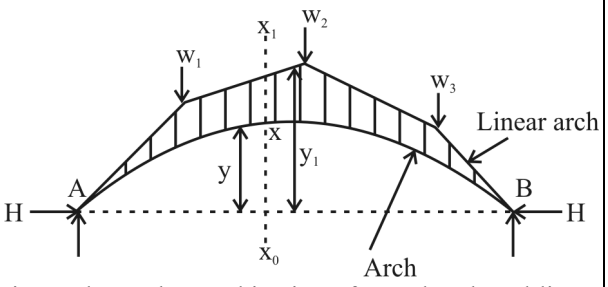



Figure shows the combination of actual arch and linear ach. Let x be the section to determine the bending moment, y and $y_1$ be the rises for actual and linear arch respectively
Bending moment at section $x_0 – x = Hy$
Bending moment at section $x_0 – x_1 = xy_1$
Net bending moment at the overlapped portion of X-section : $H(y_1 – y)$
Therefore, net BM at section X is proportional to the difference in rise i.e. $(y_1 – y)$

**35. As per IS - 875, where access is not provided except for maintenance, live load on roofs, while designing a truss, in respect of its plan area is adopted as**

(a) 100 N/sq.m (b) 400 N/sq.m
(c) 750 N/sq.m (d) 1500 N/sq.m

**Ans : (c)** IS 675 specifies the following live loads to be assumed in the analysis of an industrial building.

| Roof slope | Access | Live load |
|---|---|---|
| $\le 10^0$ | Provided | 1.5 kN/m$^2$ for plain area |
| | Not provided | 0.75 kN/m$^2$ for plain area |
| $\ge 10^0$ | | For roof membrane sheets or purlins 0.75 kN/m$^2$ less 0.02 kN/m$^2$ for every degree increased in slope over $10^0$ $\not<$ 400 N/m$^2$ |

**36. A steel column in a structure carries a load of 125 kN. It is built up of 2 ISMC 350 channels connected by lacing. The lacing carries a load of–**

(a) 125 kN (b) 12.5 kN
(c) 3.125 kN (d) zero

**Ans. (c):** Lacing carries the 2.5% of loading that carries the column structure.

Lacing carries the load $= \frac{125 \times 2.5}{100}$

$= 3.125$ kN

**37. An electric pole 5 m high is fixed into the foundation. It carries a wire at the top and is free to move sideways. The effective length of the pole is**

(a) 3.25 m (b) 4 m

(c) 5 m (d) 10 m

**Ans. (d) :** An electric pole is an example for cantilever end type of column i.e. fixed at one end and loaded at other ends.

OR

Wire does not support to pole by hinge, so pole will be act as cantilever column.

[Effective length ($l_{eff}$) = 2L]

So, $l_{eff} = 2L = 2 \times 5 = 10$ m

**38. The maximum slenderness ratio of compression members carrying loads resulting from dead loads and superimposed loads is**

(a) 100 (b) 180

(c) 150 (d) 200

**Ans. (b) :** As per IS : 800-1964 (Cl.3.7)

Maximum slenderness Ratio ($\lambda$ max)

| Member | $\lambda$ max |
|---|---|
| A member carrying compressive load resulting from dead load and imposed load | 180 |
| A tension member in which reversal of direct stress due to load other than wind and seismic force | 180 |
| A member subjected to compressive forces resulting from wind EQ force provide deformation of such member does not affect stress in any part of the structure | 250 |
| Compression flange of beam | 300 |
| A member normally act as a tie in Roof Truss | 350 |
| Tension member other than Pre-tension | 400 |

**39. The minimum thickness of web plate from corrosion point of view should be–**

(a) 12 mm (b) 6 mm

(c) 3 mm (d) 20 mm

**Ans. (b) :** The minimum thickness of web plate from corrosion point of view should be 6 mm if plate is accessible and it should be 8 mm if it is inaccessible for maintenance work.

**40. For compression member having the same effective length about any cross-sectional axis the most preferred section from the point of view of strength is**

(a) a box (b) an I-section

(c) a circular tube (d) a single angle

**Ans. (a) :** A box section is preferred since it will give maximum moment of inertia. It is also called as the rectangular tube section. I- section has more radius of gyration than other section.

**41. In Pigeaud's coefficient method for the analysis of an interior panel of a T-beam bridge.**

(a) Notation for coefficient as $\alpha x4$ and $\alpha y4$ includes 4 since panel is continuous on all the 4 edges

(b) Poisson's ratio of concrete has no contribution

(c) Applicability is restricted, to the case when wheel load is centrally placed.

(d) Dispersion of load is considered through wearing coat only

**Ans. (c) :** The most commonly used elastic methods are based on Pigeuds or Westergaard's theory.

- It is applicable for slab restrained on all sides, the corners of which are prevented from lifting, adequate provision for torsion being made.
- In Pigeaud's coefficient method for the analysis of an interior panel of a T-beam bridge paplicability is restricted, to the case when wheel load is centrally placed.

**42. The members of a roof truss which carry axial compression are called**

(a) column (b) beam

(c) stanchion (d) strut

**Ans. (d) : Strut :** Strut is part of truss that carry compressive load.

**Column :** Column are not part of truss.

**Beam :** Beam is flexural member.

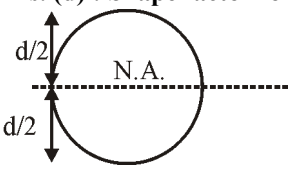
**Stanchion :** It's a bar or post forming a support.

**43. Shape factor for circular section is**

(a) $4/\pi$ (b) $16\ \pi/3$

(c) $20/(3\pi)$ (d) $16/(3\pi)$

**Ans. (d) : Shape factor for circular section–**




$$Z_p = \frac{A}{2}(y_1 + y_2)$$

$$= \frac{\pi d^2}{4}\left(\frac{2d}{3\pi} + \frac{2d}{3\pi}\right)$$

$$Z_p = \frac{d^3}{6}$$

$$Z_e = \frac{I}{y_{max}} = \frac{\frac{\pi d^4}{64}}{\frac{d}{2}} = \frac{\pi d^3}{32}$$

Shape factor S.F. $= \frac{Z_p}{Z_e} = \frac{16}{3\pi}$

**44. As per IS : 800, the minimum pitch of bolts in a row of bolts is recommended as the diameter of the bolt times**

(a) 2 (b) 2.5
(c) 3 (d) 4

**Ans. (b) : Minimum pitch–** Bolt should be placed at a sufficient distance and the minimum pitch is ensured for the following reasons :

- To prevent bearing failure of members between the bolts.
- To prevent efficient installation of bolts

As per IS 800 : 1984,
Minimum pitch = 2.5 d
d = nominal dia of bolt

**Maximum pitch –**

- For tension member $\rightarrow P \not> (16t, 200\ mm)_{min}$
- For compression member $\rightarrow P \not> (12t, 200\ mm)_{min}$
- For rows near the edge $\rightarrow P \not> (100\ mm + 4t, 200\ mm)_{min}$

**45. Loss of stress with time at constant strain in steel is called**

(a) relaxation (b) creep
(c) shrinkage (d) ductility

**Ans. (a) : Relaxation–** Time dependent increase in steel strain at constant stress.
**Creep–** Time dependent deformation due to sustained load.
**Shrinkage–** Total shrinkage of concrete depends upon the constituents of concrete, size of the member and environmental conditions.

**46. In a footing it is usual to assume that the maximum value of transverse bending will occur at a distance, equal to (measured from the face of the column)**

(a) half the effective depth
(b) effective depth
(c) twice the effective depth
(d) none of the given answers

**Ans. (b) :** Maximum bending will occur at affective depth from face of the column in a footing.

**47. The minimum and maximum % of reinforcement in RCC short column are**

(a) 0.8 and 6 (b) 6 and 0.8
(c) 8 and 4 (d) 4 and 6

**Ans. (a) :** According to code IS 456 : 2000 clause 26.5.3.1 (a) the cross-sectional area of longitudinal reinforcement, shall be not less than 0.8% and not more than 6% of the gross cross-sectional area of column.

**48. A simply supported RC beam carries UDL and is referred as beam A. A similar beam is prestressed and carries the same UDL as the beam A This beam is referred as beam B. The mid-span deflection of beam A will be**

(a) more than that of beam B
(b) less than that of beam B
(c) the same as that of beam B
(d) generally less but sometimes more depending upon the magnitude of UDL

**Ans. (a) :** The deflection in PSC beam as compared to RCC beam will be less because a moment due to prestressed cable produces deflection in direction opposite to that of live loads.

**49. As the span of a bridge increase, the impact factor**

(a) decrease
(b) increase
(c) decreases upto a critical value of span and then increase
(d) increase up to a critical value of span and then decrease

**Ans. (a) :** As per IRC class, a loading in connection with design of bridge decks impact factor is inversely proportional span length.

$$I = \frac{A}{B + L} \rightarrow \text{Length}$$

So, span of bridge increases the impact factor decreases.

**50. The neutral axis of the reinforced beam passes through**

(a) centre of gravity of the concrete section
(b) meta-centre of the concrete section
(c) centroid of the transformed section
(d) centroid of the concrete section

**Ans. (c) :** Neutral axis is the axis where the strain in the section is zero. It passes through centroid of the transformed section.

**51. The minimum size of the reinforcement bar in RCC column is**

(a) 8 mm
(b) 6 m
(c) 12 mm
(d) 10 mm

**Ans. (c) :** As per IS 456 in a column the maximum distance between two longitudinal bars should not greater than 300 mm.

- The bars shall not be less than 12 mm is diameter.
- The minimum number of longitudinal bars provided in a column shall be four in rectangular column and six in circular column.

**52. Lateral ties in RC columns are provided to resist**

(a) bending moment
(b) shear
(c) buckling of longitudinal steel bars
(d) both bending moment and shear

**Ans. (c) :** All longitudinal reinforcement in a compression member must be enclosed within transverse reinforcement, comprising either lateral ties (with internal angles not exceeding $135^0$.

**This is required –**

- To prevent the premature buckling of individual bars.
- To confine the concrete in 'core' thus improving ductility and strength.
- To hold the longitudinal bar in position during construction and to provide resistance against shear and torsion, it required.

**53. The section in which concrete is not fully stressed to its maximum permissible value while stress in steel reaches its maximum value, is called**

(a) Under reinforced section
(b) Critical section
(c) Over reinforced section
(d) Balanced section

**Ans. (a) :** In an under reinforced section, the steel yields or tends to yield to its permissible limit before the concrete yield. So, that the structure fails due to steel yielding in under reinforce section.

**54. In a slab, the transverse reinforcement is provided at _____ to the span of the slab.**

(a) 45 degrees (b) 60 degrees
(c) 75 degrees (d) Right angle

**Ans. (d) :** In a slab, transverse reinforcement are provided at right angle to the span of the slab. The transverse reinforcement is provided to take care of temperature stresses.

**55. When the ratio of effective length of the column to its least lateral dimension does not exceed 15, it is termed as**

(a) long column
(b) short column
(c) plain column
(d) none of the given answers

**Ans. (b) :** As per IS Code 456:2000 Slenderness ratio for RC column

$$\lambda = \frac{\text{effective length}}{\text{least lateral dimension}}$$

$\lambda \geq 15 \Rightarrow$ long columns

$\lambda < 15 \Rightarrow$ short columns

**56. What type of stresses are artificially induced by prestressed concrete in a structure before it is loaded?**

(a) Tensile (b) Torsional
(c) Shear (d) Compressive

**Ans. (d) :** Concrete section is strong in compression and weak in tension hence compressive stresses are induced to counter the tensile stresses generated in concrete.

**57. In pre-stressed concrete member, it is advised to use**

(a) low strength concrete only.
(b) high strength concrete only.
(c) low strength concrete but high strength steel.
(d) high strength concrete and high strength steel.

**Ans : (d)** In pre-stressed concrete members, concrete and steel of high grade are used.

- High strength concrete is less liable to shrink and has high modulus of elasticity and smaller creep strain as result loss of pre stress in steel is reduced.
- Use of high strength concrete result in reduction of cross-sectional dimension of pre-stressed concrete hence also reduces weight of structure.
- High tension steel is needed in pre-stressing due to any pre-stress loss like shrinkage and creep all the steel bars will loose their pre-tension simultaneously.

**58. Drops are provided in flat slab to resist primarily**

(a) Bending moment (b) Thrust
(c) Shear (d) Torsion

**Ans. (c) :** Flat slab is without beam or girders and is two way slab. In flat slabs, column are provided with column capital with a drop panel and then flat slab is provided. Drop are provided to resist punching shear from column in the flat slab. Shear section is taken at distance $\frac{d}{2}$ from the column. Minimum thickness of flat slab should be 125 mm.

**59. According to Indian standard the total amount of shrinkage for a pre-tensioned beam is taken as:**

(a) $3 \times 10^{-4}$ (b) $3 \times 10^{-5}$
(c) $3 \times 10^{-6}$ (d) $3 \times 10^{-7}$

**Ans. (a) :** According to Indian standard the total amount of shrinkage,

- for pre-tensioned member is $-3\times10^{-4}$
- for post tensioned member is $-\frac{2\times10^{-4}}{\log_{10}(t+2)}$

here, t=days

**60. The important events located on the activities in a bar chart are known as**

(a) Key events (b) key stones
(c) Milestones (d) Key points

**Ans. (c) :** The important events located on the activities in a bar chart are known as milestones. This is also called mild stone chart.

**61. PERT stands for**

(a) Programme evaluation and research technique
(b) programme examination and review technique
(c) Programming examination and research technique
(d) Programme evaluation and review technique

**Ans. (d) :** PERT stands for programme evaluation and review technique.

- It is event oriented
- It is a tool for planning.
- Three time estimates are made.
- Based on probabilistic approach.
- Cost is directly proportional time hence minimum cost will be achieved C/S to minimum completion time.
- Suitable for new type of project. (which not happened earlier)

**62. Fulkerson's rule stands for**

(a) Planning the events
(b) Scheduling the events
(c) Numbering the events
(d) Controlling the events

**Ans. (c):** Fulkerson's rule is used for numbering the events in a network scientifically.
- Activities are generally identified by node number i.e. the number of events on either end of the activity arrow.
- To make sure that the numbering of events in the networks are in order and not in a random manner, we take the help of Fulkerson's rule.

**63. If the values of $t_o$, $t_m$, $t_p$ are 8, 12 and 18 the value of $t_e$ is**
(a) 12.1 (b) 12.3
(c) 12.6 (d) 13

**Ans. (b) :** Given,
Optimistic time $t_0 = 8$
Most likely time $t_m = 12$
Pessimistic time $t_p = 18$

Expected time, $t_e = \frac{t_0 + 4t_m + t_p}{6}$

$= \frac{8 + 4 \times 12 + 18}{6} = 12.3$

**64. In time estimates PERT follows**
(a) Probabilistic approach
(b) Deterministic approach
(c) Possibility approach
(d) Non- Probabilistic approach

**Ans. (a) : PERT–**
- It is a probabilistic approach of solving technique.
- Event oriented
- It have three time estimates
- The time estimate of activity follows β - distribution curve.
- Use in R & D project
- Concept of slack is represented in PERT
- In PERT critical path by joining critical events are represents.

**65. If the value of $t_0$, $t_p$ of an activity are 4 and 15, what is the variance of the activity?**
(a) 3.22 (b) 3.36
(c) 3.87 (d) 3.92

**Ans. (b) :** Given as,

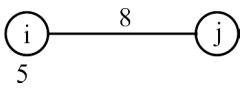
$$\sigma = \left(\frac{t_p - t_o}{6}\right)^2 = \left(\frac{15-4}{6}\right)^2 = 3.36$$

**66. The difference between the total float and free float is known as**
(a) Free float (b) Total float
(c) Independent float (d) interfering float

**Ans. (d) : Float–**Range within which activity time or its finish time may fluctuate without affecting the completion of project.
**1. Total float** = Maximum time available – actual time req.
($F_T$) = LST – EST = LFT –EFT
**2. Free float (FF)–**It is the amount of time by which an activity can be delayed without affecting the EST of the succeeding activity.
**3. Independent float–**It is excess of minimum available time over the required activity duration.
**4. Interfering float–**Different of total float and free float of an activity = Total float – Free float

**67. The duration of an activity i-j is 8. The (EPO) is 5. What is the free float of i-j.**
(a) 5 (b) 13
(c) 10 (d) 8

**Ans. (a) :**




the free float of i-j is 13 – 8 = 5

**68. Acoustics of an auditorium is considered to be excellent when its reverberation time is between**
(a) 0.50 and 1.50 s (b) 50 and 2.00 s
(c) 00 and 3.00 s (d) 00 and 5.00 s

**Ans. (a) :** Acoustics of an auditorium is considered to be excellent when its reverberation time is between 0.50 and 1.50 s.

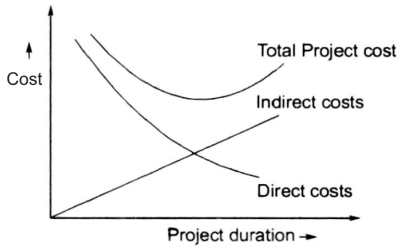
**69. The difference between the latest allowable time and earliest expected of an event is known as**
(a) Float (b) Normal deviate
(c) Free float (d) Slack

**Ans. (d) :** Slack time in PERT analysis is always zero for critical activites.
Slack is the time by which occurrence of the event can be delayed without affecting the schedule completion time of the project.
- Slack time refers to an event

S (slack) = $T_L - T_E$
$T_L$ = Latest allowable occurrence time
$T_E$ = Earliest expected event occurrence time

**70. With the increase of time, the direct costs of the project.**
(a) increase
(b) decrease
(c) remain constant
(d) decrease and the increase

**Ans. (b) :** The reduction in project time normally results in increasing the direct cost and decreasing the indirect cost.




**71. Resource smoothing will be adopted when**
(a) resources are unlimited
(b) resources are limited
(c) resources are constant
(d) in all the cases

**Ans. (a):** Resource smoothing is used when the time constraint takes priority. The objective is to complete the work by the required time through the resources demand. In resource smoothing project duration does not change and the critical path remains unchanged.

**72. A 6 hour storm with intensities of 5, 7, 12, 20, 16 and 2 mm/hr produced a runoff of 30 mm. Then ϕ-index is**

(a) 6.25 mm/hr (b) 5.33 mm/hr
(c) 6.00 mm/hr (d) 5.00 mm/hr

**Ans. (a) :**
Assume ϕ-index is < 2 mm/hr,
Given Runoff = 30 mm

$R = \Sigma (i - \phi)t$

$30 = (5 - \phi)\times1+(7 - \phi)\times1+(20 - \phi)\times1+(16 - \phi)\times1 +(2 - \phi)\times1 + (12-\phi) \times 1$

$30 = 62 - 6 (\phi)$

$6\phi = 32$

$\phi = 5.33$ mm/hr > 2 mm/hr

$\therefore$ Our assumption is wrong

Assuming ϕ-index

$5.33 < \phi < 7$ mm/hrs

$R = \Sigma(i - \phi)t$

$30 = (7-\phi)\times1+(12-\phi)\times1+(20-\phi)\times1+(16-\phi)\times1$

$30 = 55 - 4\phi$

$\phi = 6.25$ mm/hrs

**73. Sprinkler irrigation is not suitable to**

(a) Rice (b) Fodder
(c) Lawn
(d) None of the given answers

**Ans. (a) :** Rice has huge water requirements during its growth, hence sprinkler irrigation is not suitable.
Rice crop will need approximately 10 mm of water per day. Therefore a crop that matures in 100 days will require approximately 1000 mm at water.

**74. For irrigation, water having SAR above 26**

(a) Can be used for all soils and for all crops
(b) Can be used for all soils except fine textured soils
(c) Can be used for all soils if some precautions are taken
(d) is not used for any irrigation

**Ans. (d) : Low Sodium water ($S_1$) – (SAR : 0 to 10)–** suitable fall all type of crops and all types of soils, except for those crops, which are highly, sensitive to sodium.
**Medium Sodium water ($S_2$) - (SAR : 10 to 18) –** Suitable for coarse textured of organic soil with good permeability. Relatively unsuitable in fine textured soils.
**High sodium water ($S_3$) – (SAR : 18 to 26)–**
Harmful for almost all types of soils, requires good drainages high leaching, gypsum additions.
**Very high sodium water ($S_4$) – (SAR above 26 )–**
Unsuitable for irrigation.

**75. As per IS 4987, $N = \left(\frac{C_V}{P}\right)^2$ where $N$ = optimum number of rain gauge stations in a basin, $C_v$ = coefficient of variation of the rainfall values of the existing rain gauge stations, then $P$ is**

(a) Highest discharge at the basin outlet
(b) Annual average rainfall of the basin
(c) Desired degree of percentage error in the estimate of the basin mean rainfall
(d) Highest rainfall recorded in the basin

**Ans. (c) :** Number of rain gauge, $N = \left(\frac{C_V}{P}\right)^2$

P = Desired degree of percentage error in the estimate of the basin mean rainfall.
$C_v$ = Coefficient of variation of the rainfall values of the existing rain gauge stations.

**76. Defective air circulation in plant's root zone is an effect of**

(a) Mixed cropping
(b) Fall in soil moisture content
(c) Water logging
(d) High temperature

**Ans. (c) :** Water logging is the condition where plant's roots are completely submerged in water leaving no space for air circulation, thereby cutting off our supply to roots of the plants.

**77. In the design of impervious apron by Khosla's theory for weirs, the horizontal length of the apron is found by the consideration of**

(a) Velocity of flow over the weir
(b) Permissible exit gradient
(c) Scoring depth
(d) Design discharge

**Ans. (b) :** Horizontal length of apron is based on exit gradient given by Khosla

$$G_E = \frac{H}{d}\frac{1}{\pi\sqrt{\lambda}}$$

$$\alpha = \frac{b}{d}$$

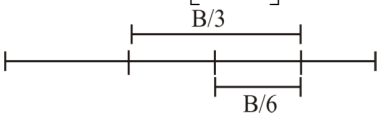
$$\lambda = \frac{1+\sqrt{1+\alpha^2}}{2}$$

**78. For no tension to develop in the gravity dam the resultant of all the external forces should always lie**

(a) At the centre of the base
(b) Within the middle third portion of the base
(c) Within the d/s third portion
(d) With the u/s third portion

**Ans. (b):** The stresses at the base of gravity dam is given as :

$$\sigma_{1/2} = \frac{P}{B}\left[1 \pm \frac{6.e}{B}\right]$$




$$\frac{P}{B}\left[1 - \frac{6.e}{B}\right] > 0$$

$$e < \frac{B}{6}$$

e = Should lie between the middle third region of the dam.

**79. The first watering given to a crop, when the crop is a few centimeters high, is called**
(a) Paleo irrigation
(b) Kor watering
(c) Leaching
(d) None of the given answers

**Ans. (b) :** The first watering is known as kor watering and the depth applied is known as kor depth. The portion of the base period in which kor watering is needed is known as kor period.
• First watering before sowing the crop is called Paleo.

**80. Specific capacity of a confined well**
(a) is constant at all draw downs
(b) increases with increasing discharge
(c) decreases with increasing discharge
(d) is constant at all discharges

**Ans. (c) :** Specific capacity of a confined well decreases with increasing discharges.
The specific capacity of a well is the discharge per unit drawdown in the well and is usually expressed in lpm/m.

**81. Field capacity and optimum moisture content of a soil of density 1.3 g/cc are 28% and 16% respectively. If effective depth of root zone is 70 cm, water available for evapotranspiration is-**
(a) 25.4 cm (b) 22.29 cm
(c) 14.56 cm (d) 10.92 cm

**Ans. (d) :** Depth of evapotranspiration is given as

$$d_a = d_r \times \frac{\gamma_d}{\gamma_w}[FC - OMC]$$

Where, $d_a$ = Available depth
$d_r$ = Root zone depth

$$d_a = 70 \times \frac{1.3}{1}\left[\frac{28-16}{100}\right] = 10.92\text{cm}$$

**82. Flow duration curve is a convenient tool to assess the available at the site**
(a) Firm power (b) Secondary power
(c) Tertiary power (d) Average power

**Ans. (a) : Flow duration curve –**
• The flow duration curve is highly useful in the planning and design of water resources project.
• The flow duration curve also finds use in the design of drainage system and in flood control studies.
• Flow duration curve can be used to estimate what minimum level of flow can be maintained through a reservoir so that minimum demand at all times can be met.

**Firm power–** The power which is guaranteed by the supplier to be available at all time during a period. Flow duration curve is a convenient tool to assess the available at the site.

**83. The best method of estimating runoff is**
(a) Unit hydrograph
(b) Runoff-Coefficient method
(c) Rational formula
(d) Infiltration index method

**Ans. (a) :** Unit hydrograph method is best reliable method to predict floods of a given frequency.
• A large number of methods one proposed to solve this problem and of them probably the most popular and widely used method is the unit hydrograph method.

**84. Blue baby disease may be caused in infants due to drinking water containing higher concentrations of**
(a) Nitrites (b) nitrates
(c) Lead (d) Arsenic

**Ans. (b) :** Nitrates are fully oxidized and hence are stable. Their concentration upto 45 mg/l in water is not harmful for adults but may cause oxygen carrying capacity to decrease in infants. This is known as blue baby disease or methaemoglobinemia.

**85. Sedimentation is the process related to**
(a) Floating material in rivers
(b) Bed material carried by flowing water
(c) Cutting land for channeling
(d) Seepage into embankments

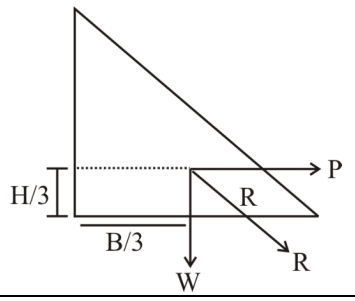
**Ans. (b) :** Sedimentation refers to settling down particle in still water but if the water is in motion, the sediments may flow with it.
Primary treatment of sewage is removal of floating and settleable solids through sedimentation.

**86. If uplift is not considered, base width of elementary profile of gravity dam of height *H* and specific gravity ρ is given by**

(a) $b = \frac{H}{\sqrt{\rho}}$ (b) $b = \frac{H}{\sqrt{\rho - 1}}$

(c) $b = \frac{H}{\sqrt{2\rho}}$ (d) $b = \frac{H}{\sqrt{\rho + 1}}$

**Ans. (a) :**



$\Sigma M_A = 0$

$$P \times \frac{H}{3} - W\frac{B}{3} = 0$$

$$PH = WB$$

$$\frac{\gamma_w H^3}{B} = \frac{\rho\gamma_w B^2 H}{B}$$

$$B = \frac{H}{\sqrt{\rho}}$$

**87. An important hydraulic failure of earth dams is**

(a) Piping
(b) Sloughing
(c) Upstream slope failure due to sudden drawdown
(d) Overtopping

**Ans. (d) :** Important hydraulic failure of earth dams are:
(i) Crushing
(ii) Overtopping
(iii) Sliding

**88. The ratio of rate of change of discharge of canal outlet to the rate of the change of the discharge of the distributing channel is known as**

(a) Proportionality (b) Flexible
(c) Sensitivity (d) Drowning ratio

**Ans. (b) :** • The ratio of rate of change of discharge of canal outlet to the rate of the change of the discharge of the distributing channel is known as flexibility.

$$\text{Flexibility} = \left[\frac{\frac{dq}{q}}{\frac{dQ}{Q}}\right]$$

• A proportional outlet is the one in which the flexibility (F) is equal to unity.

• Drowing ratio between the depth of water level over crest on the downstream and upstream of module.

**89. Three turbines each of capacity 10000 kW are installed at a hydel power station. If the peak load and the average load produced during a certain period are 25000 kW and 15000 kW respectively, then load factor and plant factor are respectively equal to**

(a) 60% and 50% (b) 50% and 60%
(c) 40% and 50% (d) 50% and 40%

**Ans. (a) :** Load factor $= \frac{\text{Average load}}{\text{Peak load}}$

$$= \frac{15000}{25000} \times 100 = 60\%$$

$$\text{Plant factor} = \frac{\text{Average load}}{\text{Installed capacity}}$$

$$= \frac{15000}{3 \times 10000} \times 100 = 50\%$$

**90. In design of alluvial channels, for computing mean velocity of flow, Kennedy used**

(a) Chezy's formula (b) Manning's formula
(c) Kutter's formula (d) Bazin's formula

**Ans. (c) :** For computing mean velocity of flow in design of alluvial channels, Kennedy used Kutter's formula.

$$\text{Kutter's formula } C = \frac{23 + \frac{0.00155}{i} + \frac{1}{N}}{1 + \left(23 + \frac{0.00155}{i}\right)\frac{N}{\sqrt{m}}}$$

N = Kutter's constant

**91. The design period for clear water conveying main is**

(a) 15 years (b) 30 years
(c) 50 years (d) 100 years

**Ans. (b) :** The design period for clear water conveying mains is 30 year.

**92. Slow sand filters require**

(a) Finer sand
(b) Coarser sand media
(c) Medium sand media
(d) Any type of sand media

**Ans. (a) : Slow sand filters–** These filters were constructed of fine sand with an effective size ranges from 0.25 to 0.35 mm with uniformity coefficient ranging from 2 to 3 slow sand filters normally operate at a rate of 1 to 10 million gallons/acre/day. Because of the small size, all of the suspended materials are removed at the filter surface.

**93. A sudden change is the flow velocity or pressure created in water pipe line due to sudden closure of valve is called**

(a) Boiling (b) Implosion
(c) Vibration (d) Water hammer

**Ans. (d) :** A sudden change is the flow velocity or pressure created in water pipe line due to sudden closure of valve is called water hammer.

**94. The minimum dia for public sewer is hilly areas where a steep slopes are prevalent is**

(a) 300 (b) 150
(c) 100 (d) 50

**Ans. (c) :** Minimum diameter for given condition is 100 mm.

**95. Due to incomplete combustion of fuels from petrol engines, the gas librated is**

(a) $CO_2$ (b) CO
(c) $N_2$ (d) He

**Ans. (b) :** Due to incomplete combustion, CO is produced, $CO_2$ is produced if complete combustion is there.

**96. Design period for slow sand filters as suggested by CPHEEO is**

(a) 50 years (b) 30 years
(c) 25 years (d) 10 years

**Ans. (d) :** Design period for slow sand filters as suggested by CPHEEO is 10 year.

**97. Treatment required for raw water containing only carbon dioxide and odourous gases is**

(a) Plain sedimentation and chlorination
(b) Chlorination only
(c) Aeration followed by disinfection
(d) Softening and chlorination

**Ans. (c) :** Disinfection is the least minimum treatment given to water.
Aeration is required to precipitate out the gases and disinfection is provided to free the water from pathogens.

**98. In connection line from water main to building, goose-neck is provided to**

(a) quickly open or stop the flow to building
(b) avoid stresses and strains on the joint due to temperature variations and vibrations
(c) detect leakage in pipe line
(d) to provide additional connections in future

**Ans. (b) :** Goose-neck is a small curved flexible pipe for making connection between ferrule and service pipe. It is made of lead to provide flexibility. It takes care of settlement of service pipe and avoids stresses and strain due to temperature change.
A gooseneck is a $180^0$ pipe fitting at the top of a vertical pipe that prevents entry of water.

**99. Sludge digestion is done by the .....bacteria.**

(a) pathogenic (b) faculative
(c) anaerobic (d) aerobic

**Ans. (b) :** Sludge digestion is done by the faculative bacteria.
The sludge digestion drying is a two-phase process. In the first stage, the dry solid sludge is heated and mixed in a closed tank to enable anaerobic digestion by acid fermentation bacteria.

**100. The sluice valves used for dewatering the pipe line are called**

(a) Scour valve
(b) Safety valve
(c) Sectionalizing valve
(d) Discharge control valve

**Ans. (a) :** The sluice valves used for dewatering the pipe line are called scour valve.

**101. The collapsible soil is associated with**

(a) Dune sands (b) laterite soils
(c) Loess (d) Black cotton soils

**Ans. (c) :** A collapsible soil is also known as loess. It is basically a clayey silt with a metastable structure.

- Collapsible soil shows high strength and stiffness at normal water content condition on the other hand upon wetting it suffers a sudden plastic deformation, i.e., collapse which could be severe in some cases.
- Collapsible soil is also known as loose soil because it collapses at small loading and can't resist stress.
- Loess is a wind deposit soil (Silt). It is generally of uniform gradation, with the particle size between 0.01 – 0.05 mm.
- Loess are compressible in nature when they are in wet form. For Loess the permeability in vertical direction is generally much greater than of horizontal direction.

**102. The probability that a 100 yr flood is equalized or exceeded at once in 100 yr is**

(a) 0.99 (b) 0.64
(c) 0.36 (d) 0.1

**Ans. (b) :** Probability of occurrence

$$P = \frac{1}{\text{Return period (T)}}$$

$$= \frac{1}{100} = 0.01$$

$\therefore$ Probability of non-occurrence

$$q = 1 - 0.01 = 0.99$$

Probability of non occurrence in 100 successive years

$$= (0.99)^{100} = 0.366$$

Probability of occurrence = 1–0.367 = 0.63

**103. The gaseous pollutant such as hydro carbons and carbons monozides can be effectively controlled by**

(a) Combustion or incineration
(b) Electrostatic precipitators
(c) Gravity setting chambers
(d) Fabic fillers

**Ans. (a) : Incineration or combustion units–**

- It's necessary to have the proper combination of four basic elements-oxygen, temperature, fuel and time.
- Combustion as a process is used to purify polluted gases, only when the admixed pollutants are oxidisable to inert gases.
- Hydrocarbons and carbon monoxide can be easily burnt, oxidized and removed.

**104. According to ambient air quality standards in respect of noise (MOEF) mention noise limits for silence zone in day time**

(a) 40 dB (A) Leq (b) 45 dB (A) Leq
(c) 50 dB (A) Leq (d) 75 dB (A) Leq

**Ans. (c) :**

| Category of Area/zone | Day time | Night time |
|---|---|---|
| Industrial area | 75 | 70 |
| Commercial area | 65 | 55 |
| Residential area | 55 | 45 |
| silence zone | 50 | 40 |

**105. Disposal of refuse by Bangalore method involves**

(a) Aerobic method of composting
(b) Anaerobic method of composting
(c) Pulverization
(d) Pyrolysis

**Ans. (b) : Bangloare method–**

- Banglore method is an anaerobic method of composting. In this method, the refuse and right soil are piled up in layers in an underground earthen trench of about 10 m × 1.5 m × 1.5 m.

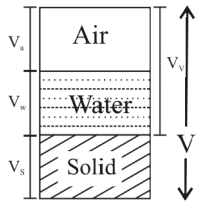
- This mass is covered at its top by layer of earth of about 15 cm depth, and is finally left over for decomposition.
- This method is widely adopted by municipal authorities throughout the country.

**106. Degree of saturation (S) of a soil mass is expressed in terms of volume of water ($V_w$) and volume of voids ($V_v$) as**

(a) $S = \frac{(V_V V_W)}{100}$ (b) $S = 100\left(\frac{V_W}{V_V}\right)$

(c) $S = 100\left(\frac{V_V}{V_W}\right)$ (d) $S = 100(V_V - V_W)$

**Ans. (b) :**




$$\text{Saturation (S)} = \frac{V_W}{V_V} \times 100$$

$$0\% \le S \le 100\%$$

**107. A sample of saturated soil has a water content of 29.6%. If the specific gravity of solids is 2.7, the dry unit weight (in gr/cm$^3$) of the soil is equal to**

(a) 0.5 (b) 0.75

(c) 1 (d) 1.5

**Ans. (d) :** Given,

Water content (w) = 0.296

Specific gravity (G) = 2.7, S = 1

$$wG = Se \Rightarrow e = \frac{wG}{S} = \frac{0.296 \times 2.7}{1} = 0.8$$

$$\frac{\gamma_d}{\gamma_w} = \frac{G}{1+e} = \frac{2.7}{1+0.8}$$

$$\text{Dry unit weight}\left(\gamma_d\right) = 1.5 \text{g/cc}$$

**108. Lacustrine soils according to geological classifications are**

(a) Transported by ice

(b) Transported by wind

(c) Deposited in seas

(d) Deposited in lakes

**Ans. (d) :** Soils carried by water and have been deposited from suspension in still water of lakes are known as lacustrine deposits.

According to the transporting agency, soil are classified as–

**Alluvial soil** →Deposited by river water

**Lacustrine soil** →Deposited by still water like lakes

**Marine soil** →Deposited by sea water

**Aeolian soil** →Transported by wind.

• Residual soils are found at the same location, where they have been formed. Generally, the depth of residual soils varies from 5 to 20m. Residual soils comprise of a wide range of particle sizes, shapes and composition.

**109. If two soils $S_1$ and $S_2$ tested in the laboratory are having liquid limits 38% and 60% and plastic limits I8% and 20% respectively. If natural moisture content for $S_1$ and $S_2$ is 40% and 50% respectively, which soil is better foundation material when remolded.**

(a) $S_1$ (b) $S_2$

(c) $S_1$ and $S_2$ (d) Neither $S_1$ nor $S_2$

**Ans. (b) :** For foundation stability we check the consistency index i.e. in which soil consistency index is more that is better for foundation.

Consistency Index, $I_C = \frac{w_L - w_n}{w_L - w_P}$

$$I_{C_1} = \frac{38-40}{38-18} < 0$$

Soil is in liquid state not suitable for foundation.

$$I_{C_2} = \frac{60-50}{60-20} = 0.25$$

Better soils for foundation.

**110. Neutral stress within a soil mass is also known as**

(a) Pore water pressure

(b) Effective stress

(c) Geostatic stress

(d) Static shear stress

**Ans. (a) :** Pore water pressure ($\mu$) is the pressure due to pore water filling the voids of the soil. Thus

$$\mu = \gamma_w . h$$

$\Rightarrow$ Pure water pressure is also known as neutral pressure or neutral stress, because it can not resist shear stresses.

$\Rightarrow$ Pore water pressure is taken as zero when it is equal to atmospheric pressure because in soil engineering the pressures used are generally gauge pressure and not absolute pressure.

$$\bar{\sigma} = \sigma - \mu$$

For saturated soil = $\bar{\sigma} = \gamma_{sat} . h - \gamma_w . h$

$$\bar{\sigma} = \left(\gamma_{sat} - \gamma_w\right) h$$

$$\bar{\sigma} = \gamma' h$$

**111. The maximum force per unit area that needs to be placed over an expansive soil to prevent value increase in known as**

(a) Pore water pressure

(b) Swelling pressure

(c) Safe bearing capacity

(d) Plastic limit

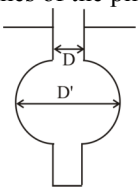
**Ans. (b) :** Swelling pressure $= \frac{\text{Maximum force}}{\text{Area}}$, that needs to be placed over an expansive soils to prevent volume increase.

• When are expensive soil imbibes water from out side, pressure builds up inside the soil. If free swelling of soil is restrained by the placement of a structure over the soil this pressure called the swelling pressure.

**112. The diameter of enlarged bore or bulb for under reamed piles used in expansive soils is generally taken as**

(a) 0.3 – 0.6 times the length of the pile
(b) 1.5 – 1.2 times the pile shaft dia
(c) 2 to 3 times the pile shaft dia
(d) 1000 mm

**Ans. (c) :** For under reamed pile
D' = 2 to 3 times of the pile shaft dia.

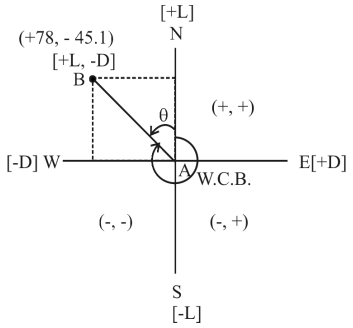



If more than 1 bulb is provided then minimum spacing between them should be 1.5 times the diameter of bulb.

**113. The latitude and departure of a line AB are +78 m and -45.1 m respectively. The whole circle bearing of the line AB is**

(a) $30^o$ (b) $150^o$
(c) $210^o$ (d) $330^o$

**Ans. (d) :**




$\tan\theta = \frac{45.1}{78}$

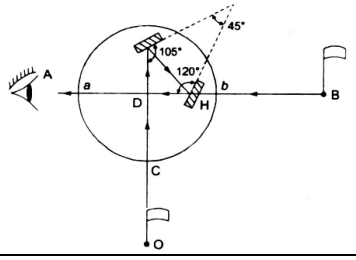
$\theta = \tan^{-1}\left(\frac{45.1}{78}\right) \Rightarrow \theta = 30.0360 \sqcup 30^0$

$\therefore$ W.C.B. $= 360^0 - \theta = 360^0 - 30^0 = 330^0$

**114. What is the angle between two plane mirrors of an optical square**

(a) $30^0$ (b) $45^0$
(c) $60^0$ (d) $90^0$

**Ans. (b) :** Angle between two plane mirrors of an optical square is $45^o$.




**115. The sum of specific yield and specific retention is equal to**

(a) Porosity
(b) Permeability coefficient
(c) Storage coefficient
(d) Saturation percentage

**Ans. (a) :** $S_y + S_R = n$

Where,
$S_y$ = specific yield
$S_R$ = specific retention
n = porosity

**116. A layer of soil having G= 2.67 and e = 0.67 is subjected to an upward head of 1.5 m due to seepage of water. The depth of the soil layer required to provide a factor of safety 2 against piping is**

(a) 1.5 m (b) 2.0 m
(c) 3.0 m (d) 0.75 m

**Ans. (c) :** Given as
G = 2.67
e = 0.67
Critical hydraulic gradient ($i_{cr}$) =

$$\frac{G-1}{1+e} = \frac{2.67-1}{1+0.67} = 1$$

$FOS = \frac{i_{cr}}{i}$, $i = \frac{1.5}{L}$

$$2 = \frac{1}{1.5/L}$$

$= 2 \times 1.5 = 3m$

**117. Boussinesq's solution for the stresses in soil caused by a point load at the surface is based on some assumptions. One of which is**

(a) Soil medium is finite medium
(b) Soil medium is plastic
(c) Soil medium obeys Hook's law
(d) Soil medium is not homogeneous

**Ans. (c) :** Assumptions in the Boussinesq's equation –
(i) It is elastic
(ii) It obey's Hook's law
(iii) Soil medium is semi-infinite.
(iv) It is homogeneous and isotropic.

**118. The failure plane in direct shear test is**

(a) Horizontal
(b) Vertical
(c) Inclined at $45^0$ to the horizontal
(d) Unpredictable

**Ans. (a) : Direct shear test :**

- The soil specimen that is to be tested is confined in a metal box of square cross section that is split into two halves horizontally, a small clearance being maintained between the two halves of the box.
- In this test, drainage cannot be controlled hence rate of loading be such that pore water pressure does not develop i.e. it limit be a drained condition test.
- This test is performed for sand because drainage can not be controlled.
- Effective stress and total stress are same in this test. Volume expansion is measured using dial gauge.

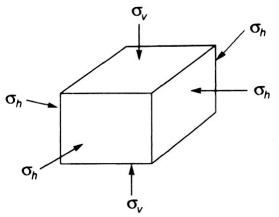
**119. The coefficient of earth pressure at rest is given by-**

(a) $\frac{\mu}{(1-\mu)}$ (b) $\left(\frac{1+\mu}{\mu}\right)$

(c) $\frac{\mu}{(1+\mu)}$ (d) $\frac{(1-\mu)}{\mu}$

**Ans. (a) :** For plain strain condition,

$\varepsilon_x = 0$




Here $\sigma_x = \sigma_y = \sigma_h$ and $\sigma_z = \sigma_v$

$$e_x = \left(\frac{-\sigma_x}{E}\right) - \mu\left(\frac{-\sigma_y}{E}\right) - \mu\left(\frac{-\sigma_z}{E}\right) = 0$$

$\sigma_h(1 - \mu) = \mu\sigma_v$

$$\frac{\sigma_h}{\sigma_v} = k_0 = \left(\frac{\mu}{1-\mu}\right)$$

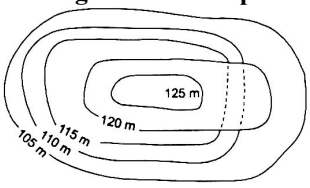
**120. In sands, during earthquakes instantaneous pore pressure are likely to develop leading to sudden and total loss of shearing strength. This phenomenon is known as**

(a) In sands, during earthquakes, instantaneous pore pressures are likely to develop
(b) Quicksand
(c) Liquefaction
(d) Damping

**Ans. (c) :** In sands, during earthquakes, instantaneous pore pressures are likely to develop leading to sudden and total loss of shearing strength. This phenomenon is known as liquefaction.

- It occurs at high frequency of vibration.
- This liquefaction occur in saturated loose sands.
- It occurs during pile driving vibration of machine, explosive blasting, earthquake shake.

**121. The contours given below represent**




(a) Depression (b) Summit
(c) Saddle (d) Over hanging cliff

**Ans. (d) :** Since the higher value contours are inside it is hill but one contour line passes through 2 lesser value contour lines meaning that it is an overhanging cliff. Two contour lines do not intersect each other except in case of overhanging cliff.

**122. Digital elevation model is generated using**

(a) Digitization
(b) Contour interpolation
(c) Hydro graphic analysis
(d) Thiessen polygon method

**Ans. (a) :** Digital elevation model is generated using digitization.
**Digital Elevation Model (DEM)–** DEM is specialized database that represents the relief of a surface between point of known elevation.

**123. Remote sensing work where system has no energy source of its own but depends on external source of energy is called**

(a) Active remote sensing
(b) Passive remote sensing
(c) SLAR
(d) LIDAR

**Ans. (b) :** In passive remote sensing we don't have an energy source of our own but used external source as energy eg. sun.
Remote sensing work where system has no energy source of its own but depends on external source of energy is called passive remote sensing.

**124. In triangulation, the best shape of a triangle from the following is**

(a) Isosceles with base angle $56^0 14'$
(b) Isosceles with base angle $65^0 14'$
(c) Isosceles with base angle $36^0 14'$
(d) Isosceles with base angle $76^0 14'$

**Ans. (a) :** A triangle with such internal angle so that error in angular measurement effects very less the measurement of sides is known as well conditioned triangle. In a well conditioned triangle no angle should be less than 30º and greater than 120º. Mathematically, it is proved that triangle of base angle 56º14' gives rise to minimum error.

**125. The number of GPS Satellites required for GPS receiver to correctly draw 3D map to locate the geo graphical position of the object over the earth surface is**

(a) 1 (b) 2
(c) 3 (d) 4

**Ans. (d) :** GPS (Globle Positioning System)–
- The minimum number of satellites needed for a GPS to determine its position precisely is 4 nos.
- Number of working satellites 24 nos.
- Height of satellites 11500 miles
- Speed 3.9 km/sec

**126. An electronic theodolite combined with EDM is popularly known as**

(a) GPS (b) Total station
(c) Auto Level (d) Compass

**Ans. (b) :** Total station uses the principle of EDM with electronic theodolite.

**127. It is a common practice to design a highway to accommodate the traffic volume corresponding to**

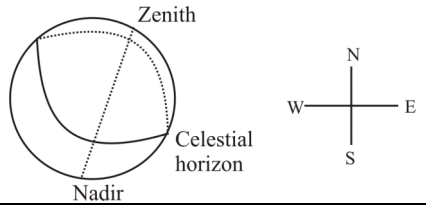
(a) 30th hour (b) peak hour
(c) none (d) 15-min peak period

**Ans. (a):** Thirtieths ($30^{th}$) highest hourly volume is used for design the highway elements and used for design hourly volume.

**128. The poles of the celestial horizon are called**

(a) Zenith and Nadir
(b) North and South poles
(c) Perihelion and aphelion
(d) Perigee and apogee

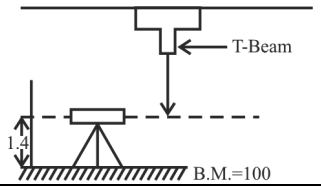
**Ans. (a) :** The poles of the celestial horizon are called zenith and nadir.




**129. The RL of the floor in a building is 100.00 m and the staff reading on the floor is 1.40 m. If the reading on the staff when held inverted with bottom touching the T-Beam of the slab is 3.60 m, then the height of the T-Beam bottom above the floor is equal to**

(a) 2.20 m (b) 2.50 m
(c) 5 m (d) 95 m

**Ans. (c) :** The RL of floor = 100.00 m
Staff reading on the floor = 1.40 m
Staff reading from the bottom of T-beam = 3.6 m
The height of beam bottom from the floor = 3.6 + 1.4
= 5.00 m




**130. An error in observations of either fore or back bearing or both may be due to**

(a) Magnetic declination (b) Dip
(c) Local attraction (d) Inclination

**Ans. (c)** During fore bearing or back bearing calculation magnetic compass is used which may show erroneous result if some metal strip or small magnetic substance nearby. This type of error is known as Local attraction.

**131. In a theodolite, if line of altitude bubble is not parallel to the line of collimation when verniers of vertical circle read zero, the error is known as**

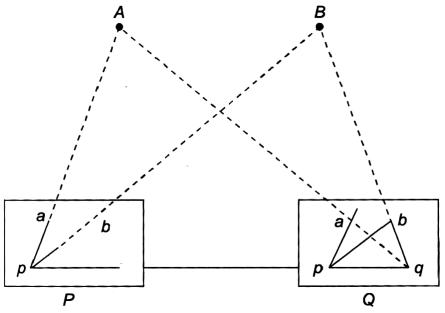
(a) Vertical axis error
(b) Lateral collimation error
(c) Horizontal axis error
(d) Vertical collimation error

**Ans. (d) :** In a theodolite, if line of altitude bubble is not parallel to the line of collimation when vernier of vertical circle read zero, the error is known as vertical collimation error.

**132. The plane table method which is suitable for locating inaccessible points is**

(a) Resection method (b) Intersection method
(c) Radiation method (d) Traversing method

**Ans. (b) :** In intersection method of plane table surveying, the objects or points to be located are obtained at the point of intersection of radial lines drawn from two different stations.




**133. If the stadia interval and the focal length of tacheometer are 12 mm and 240 mm respectively, then the multiplying constant of the tacheometer is equal to**

(a) 0.005 (b) 20
(c) 100 (d) 200

**Ans. (b) :** Multiplying constant = $\dfrac{\text{Focal length}}{\text{Stadia interval}}$

$= \dfrac{f}{i} = \dfrac{240}{12} = 20$

**134. If *L* denotes latitude and D denotes departure, then the direction of closing error is given by**

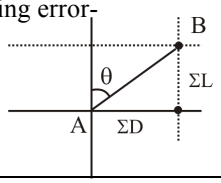
(a) $\tan^{-1}\left(\dfrac{\Sigma D}{\Sigma L}\right)$ (b) $\text{cosec}^{-1}\left(\dfrac{\Sigma D}{\Sigma L}\right)$

(c) $\cos^{-1}\left(\dfrac{\Sigma D}{\Sigma L}\right)$ (d) $\sin^{-1}\left(\dfrac{\Sigma D}{\Sigma L}\right)$

**Ans. (a) :** Direction of closing error-

$\tan\theta = \dfrac{\Sigma D}{\Sigma L}$

$\theta = \tan^{-1}\left(\dfrac{\Sigma D}{\Sigma L}\right)$




**135. In levelling, curvature correction is given by**

(a) $C_C = D^2/(2R)$ (b) $C_C = D^2/(R)$
(c) $C_C = D^2/(4R)$ (d) $C_C = D/(R)$

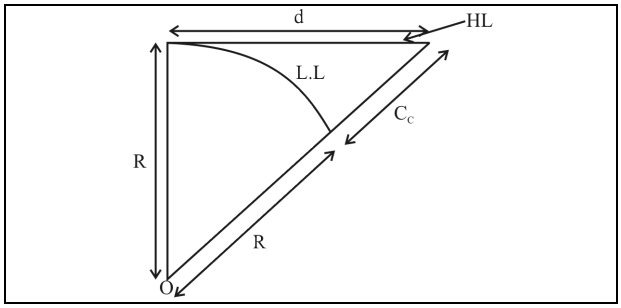
**Ans. (a) :** In leveling, there are three types of correction available-

(i) **curvature correction :** $C_C = \dfrac{-d^2}{2R}$

(ii) **Refraction correction :** $C_R = +\dfrac{1}{7}C_C = \dfrac{1}{7}\times\dfrac{d^2}{2R}$

(iii) **Combined correction due to curvature and Refraction :** $C = -\dfrac{6}{7}\times\dfrac{d^2}{2R}$

Where,
d= horizontal distance between two points, in kilometer
R = Radius of earth in km, which is 6370 km




**136. The absolute minimum radius for a horizontal curve designed for a speed of 80 kmph, given the permissible values of super elevation and coefficient of friction are 0.05 and 0.15 respectively**

(a) 252 m (b) 458 m
(c) 150 m (d) 105 m

**Ans. (a) :** $e + f = \frac{V^2}{127R}$

$$R = \frac{6400}{127 \times 0.2} = 251.96\text{m} \approx 252\text{m}$$

**137. The longitudinal gradient on a road is 1 in 20. If the radius of curve is 76 m, compensated gradient is**

(a) 0.01 (b) 0.015
(c) 0.04 (d) 0.05

**Ans. (c) :**

$$\text{Grade compensation} = \frac{30+R}{R}\% = \frac{30+76}{76} = 1.4\% \Big\} \text{ minimum}$$

$$\text{But can not greater than } \frac{75}{R} = 1\%$$

$$\frac{75}{76} = 0.98\% \approx 1\%$$

Hence grade compensation =1%

$\therefore$ compensated grade = 5 – 1 = 4% = 0.04

**138. The maximum super elevation to be provided on a road curve is 1 in 15. If the rate of change of super elevation is specified as 1 in 120 and the road width including widening is 12 m then the minimum length of the transition curve is (when the outer edge is raised with respect to inner edge)**

(a) 180 m (b) 125 m
(c) 96 m (d) 48 m

**Ans. (c) :** $e_{max} = \frac{1}{15}$, $N = 120$

$W + W_e = 12$ m

length of the transition curve (L) = $eN(W+W_e)$

$$= \frac{1}{15} \times 120 \times 12 = 96\text{m}$$

**139. The gradient which in the normal course must never be exceeded in any part of the road is called as**

(a) Ruling gradient (b) Limiting gradient
(c) Exception gradient (d) Escape gradient

**Ans. (a) : Ruling gradient:** It is the maximum gradient within which the designer attempts to design the vertical profile of a road. It is also known as design gradient.
**Limiting gradient–** The gradient steeper than the ruling gradient, which may be used for a limited road length, is called limiting gradient or maximum gradient.
**Minimum gradient–** The minimum desirable slope essential for effective drainage of rainwater from the purpose is 0.5% if the side drains are lined and 1% if the side drains are unlined.
**Exceptional gradient-** The gradient steeper than the limiting gradient which may be used in short length of road, only in an extraordinary situation is called exceptional gradient.

**140. Which of the following is not considered as design vehicles in IRC-3-1983?**

(a) Single unit truck
(b) Semi trailer combination
(c) Truck trailer combination
(d) passage car

**Ans. (d) :** Passenger car is not considered as design vehicle in IRC : 3-1983.
Any commercial vehicle such as single unit truck, semi trailer combination, and truck trailer combination factor used in design vehicles in IRC : 3 -1983.

**141. Which type of coordinated signal system is not conductive to give continuous movement of all vehicles as given IRC 93-1985?**

(a) Flexible progressive system
(b) Limited progressive system
(c) Simple progressive system
(d) Simultaneous system

**Ans. (d):** Simultaneous system does not give continuous movement.
Flexible progressive system, Limited progressive system & Simple progressive system are conducive to give continuous movement of all vehicles as given in IRC 93-1985.

**142. Bed width (B) and depth (D) of of economical section for a rectangular drains as per IRC guidelines on urban drainage shall be related as**

(a) B = D (b) B = 1.5 D
(c) B = 2D (d) B = 2D/3

**Ans. (c) :** Bed width (B) and depth (D) of economical section for a rectangular drains as per IRC guidelines on urban drainage shall be related as B = 2D

**143. Widening at curves provided to compensate the extra width occupied by a vehicle on the curve due to tracking of the rear wheels is called.**

(a) Mechanical widening
(b) Psychological widening
(c) Super widening
(d) Extra widening

**Ans. (a):** Mechanical widening is provided to cater for off tracking of vehicles.

Its value is given as $W_m = \frac{nl^2}{2R}$

n = number of lane, $\ell$ = length of wheel base

**144. As per IRC code for flexible pavements, designed life for national highways and state highways should be**

(a) 20 years (b) 15 years
(c) 10-15 years (d) <10 years

**Ans. (b) :** As per IRC code for flexible pavements, designed life for national highways and state highways should be 15 years.

**145. The multiplier applied to convert the number of commercial vehicles of different axles loads and axle configuration to the number of standard axle load repetitions is called**

(a) Lane distribution factor
(b) Vehicle damage factor
(c) Safe load factor
(d) Load distribution factor

**Ans. (b) :** The term 'Vehicle Damage Factor' is number of standard axles per commercial vehicle.

Vehicle damage factor –

$$VDF = \frac{V_1\left(\frac{W_1}{W_S}\right)^4 + V_2\left(\frac{W_2}{W_S}\right)^4}{V_1 + V_2 + .....}$$

$W_1$, $W_2$ = Mean values of the various axle load groups

$V_1$, $V_2$ =Traffic volumes

$W_s$ = Standard axle load

**146. Value of camber provided for straight sections of roads in areas of high intensity of rainfall is ____ than/ to that provided for the areas of low intensity of rainfall**

(a) Steeper (b) Milder
(c) Equal
(d) None of the given answers

**Ans. (a) :** Value of camber provided for straight sections of roads in areas of high intensity of rainfall is steeper than to that provided for the areas of low intensity of rainfall.

**147. The safe stopping distance for a two way traffic in a two lane road is 61.4 m, then for a two way traffic in a single lane road when other conditions are unaltered the safe stopping distance will be taken as**

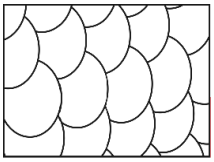
(a) 30.7 m (b) 61.4 m
(c) 92.1 m (d) 122.8 m

**Ans. (d) :** SSD for two way traffic with single lane
= 2 × SSD
= 2 × 61.4 = 122.8 m

**148. Map cracking is the common type of failure in**

(a) Rigid pavements
(b) Cement concrete pavements
(c) Gravel roads
(d) Bituminous surfacing

**Ans. (d) :**



**Alligator cracking**

Above figure shows general pattern of alligator cracking of the bituminous surface. This type of failure occurs due to repeated application of heavy wheel loads resulting fatigue or due to relative movement of pavement layer material.

**149. The most effective and economical method is stabilize block cotton soils is**

(a) Cement stabilization
(b) Lime stabilization with suitable additives
(c) Thermal stabilization
(d) None of the given answer

**Ans. (b) :** Highly unstable plastic and swelling clayey soil such as black cotton soils may be stabilized by hydrated lime.

Normal requirement is 4% to 6% of the soil.

**Lime stabilization–** Lime has been widely used as additive for clay soil or as a binder.

- When clay soils with high plasticity are treated with lime, the plasticity index is decreased and the soil becomes friable and easy to pulverize having less affinity with water.
- Lime is used for stabilization of plastic clayey soils. $Ca^{2+}$ ion migrate to the surface of the clay and displace water and other ions. It reduces the plasticity index of the soil and stabilized the soil.

**150. Equivalent factor of PCU for trucks is**

(a) 0.5 (b) 1
(c) 3 (d) 4

**Ans : (c)** For the study of mixed traffic flow, in common practice passenger car is considered as standard vehicle unit to convert the other vehicle classes and this unit is called passenger car unit or PCV.

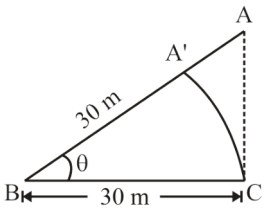
| S.N. | Vehicle | Equivalent factor |
|---|---|---|
| 1 | Passenger car, tempo, auto rickshaw, tractor | 1.0 |
| 2 | Bus, truck, tractor - trailer unit | 3.0 |
| 3 | Motor cycle, scooter pedal cycle | 0.5 |
| 4 | Horse driven vehicle | 4.0 |
| 5 | Small bullock cart and hand cart | 6.0 |

# Telangana State Public Service Commission
## Deputy Engineer Exam, 2015
## Solved Paper

**1. If θ be the angle of slope and the length of chain is 30m, then the correction to be applied per chain length is:**

(a) 30 (1 - sec θ) m (b) 30 (1 - cos θ) m
(c) 30 (sec θ – 1) m (d) 30 (cos θ - 1) m

**Ans. (c) :**



Hypotensusal Allowance

Thus = BA = 30 secθ
BA' = 30 m
Hence- AA' = 30 sec θ – 30
= 30 (secθ – 1) m

**2. Which one of the following angles can be set out with the help of French cross staff?**

(a) 180° (b) 45°
(c) 90° (d) any angle

**Ans. (b/c) : French cross staff-** If consists of a hollow octagonal box. vertical sighting slits are cut in the middle of each face, such that the lines between the centers of opposite slits makes angles of 45º with each other.
■ It is possible, therefore, to out angle of either 45º or 90º with this instrument.

**3. A lamp at the top of a light house is visible just above the horizon from a station at sea level, the distance of the lamp from the station is 30 km, find the height of the light house.**

(a) 2.019 m (b) 20.19 m
(c) 0.57 m (d) 60.57 m

**Ans. (d) :** Height of the light house 'h' $= \left(\frac{d}{3.8553}\right)^2$

$$h = \left(\frac{30}{3.8553}\right)^2$$

$$h = \frac{900}{14.86}$$

$$h = 60.57 \text{ m}$$

**4. The line joining points of equal dip is called**

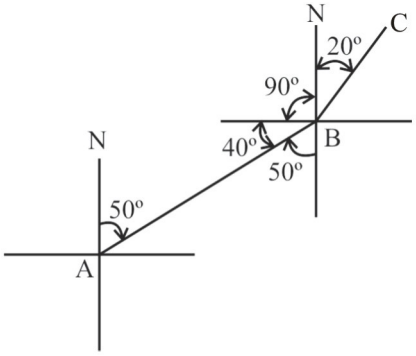
(a) Isoclinic lines (b) Aclinic lines
(c) Isogonic lines (d) Agonic lines

**Ans. (a) : Isoclinic lines-** These are the lines joining points of equal dip or inclination.
**Aclinic line-** A line joining places of zero dip.
**Isogonic line-** These are the lines joining places of equal declination.

**5. If the forebearing of a line AB is 50° and that of line BC is 20°, then the included angle between the lines is**

(a) 70° (b) 220°
(c) 120° (d) 150°

**Ans. (d) :**



Included Angle ∠ABC = 150º

**6. When contours of different elevations each other, it indicates**

(a) Level surface (b) Overhanging cliff
(c) Saddle (d) Vertical cliff

**Ans. (b) : Overhanging cliff-** Two contours of different elevation cross each other in this case.
**Vertical cliff-** Contour lines of different elevations unite to form one line.

**7. In theodolite survey, the telescope is said to be inverted, if the**

(a) Vertical circle is to the right of the observer and the bubble of the telescope is up
(b) Vertical circle is to the right of the observer and the bubble of the telescope is down
(c) Vertical circle is to the left of the observer and the bubble of the telescope is up
(d) Vertical circle is to the left of the observer and the bubble of the telescope is down

**Ans. (b) : Telescope Inverted-** A telescope is said to inverted or reversed when of the vertical circle is to the right and the bubble down.
**Telescope normal-** A telescope is said to be normal or direct when the face of the vertical circle is to the left and the bubble up.

**8. In Geographic information system, line in polygon method is characteristic of**

(a) Buffer operation
(b) Raster overlay
(c) Intersecting operation
(d) Vector overlay

**Ans. (d) :** During vector overlay, map features and associated attributes are integrated to produce a new composite map. Vector overly can be performed on different types of map features. Polygon-on-polygon, line-on-polygon and point-on-polygon.

**9. If the RL of a BM is 100m, the back sight is 1.415m and the foresight is 1.670m, the RL of the forward station is**

(a) 101.670 m (b) 101.415 m
(c) 99.745 m (d) 98.585 m

**Ans. (c) :** Given,
BM = 100 m
BS = 1.415 m
$RL_1$ = 100 + 1.425 = 101.425m
F.B. = 1.670 m
$RL_2$ = 101.415 – 1.670
= 99.745 m

**10. In functional classification of highways, which one of the following highway type have highest mobility and less accessibility**

(a) National Highways
(b) State Highways
(c) Major District roads
(d) Street and Village Roads

**Ans. (a) :**

| Types | Descriptions |
|---|---|
| **Expressway** | • The design speed of expressway should be more than 120 km per hour<br>• Only fast moving vehicles are allowed. |
| **National highways (N.H.)** | • It have highest mobility and less accessibility<br>• These roads connect different states and union territories across India.<br>• An N.H must be continued to the capital of the state. |
| **State highways (S.H.)** | • The state highways are usually roads that link important cities, towns and district headquarters within the state.<br>• Also connect them with national highway or state highways of neighbouring states. |

**11. Webster's equation for computing saturation flow rate in signal design (S = saturation flow rate PCU/hour; W = carriage way width in meters.)**

(a) S = 225 W PCU/hour
(b) S = 525 W PCU/hour
(c) S = 550 W PCU/hour
(d) S = 250 W PCU/hour

**Ans. (b) :** In the absence of data the approximate value of saturation flow is estimated assuming 160 PCU Pe 0.3 meter width of the approach.

Saturation flow (S) $= \frac{160}{0\cdot3} \Rightarrow \frac{1600}{3} = 533\cdot33$

Nearest option (b) is correct.

**12. For the calculation of stopping distance, the longitudinal friction co-efficient values of _____ have been recommended by Indian Roads Congress.**

(a) 0.35 to 0.40 (b) 0.15 to 0.20
(c) 0.40 to 0.45 (d) 0.45 to 0.50

**Ans. (a) :** The longitudinal friction on the high way for calculation of stopping distance in geometrical design is 0.35 to 0.40.
■ Recommended value of the lateral friction coefficient by the Indian Road Congress is 0.15.

**13. In total reaction time of the driver, the time required for the sensations received by the eyes/ears to be transmitted to the brain through the nervous system and spinal chord is called _____.**

(a) Intellection time (b) Emotion time
(c) Volition time (d) Perception time

**Ans. (d) :** Perception time is the time taken by the average motor vehicle driver to comprehend a danger ahead before trying to apply brakes or taking avoiding action.
■ Perception time varies considerably and depends upon many factors such as distance to the object its colour, condition, type and size, alertness and optical ability of the driver, atmospheric visibility, type and condition of the road way etc.
■ Perception time is the time required for the sensations received by the eyes or ears of the driver to be transmitted to the brain through the nervous system and spinal cord.

**14. Calculate the value of lag distance in SSD for a highway with a design speed of 65 kmph?**

(a) 45.18 m (b) 36.14 m
(c) 32.50 m (d) 451.80 m

**Ans. (a) :** Lag distance = 0.278 × V.t
= 0.278 × 65 × 2.5
= 45.175 ⊔ 45.18 m

**15. The psychological widening of pavement is calculated using which one of the following formula (take V = speed of vehicle in kmph; R = radius of the curve in m)**

(a) $\frac{V^2}{9.5\sqrt{R}}$ (b) $\frac{V}{9.5\sqrt{R}}$
(c) $\frac{V}{9.5R}$ (d) $\frac{V^3}{9.5\sqrt{R^2}}$

**Ans. (b) :** Extra widening of pavement–
■ Psychological extra widening

$$W_{ps} = \frac{V}{9.5\sqrt{R}}$$

■ Mechanical extra widening–

$$W_m = \frac{n\ell^2}{2R}$$

**16. Find out the rate of change of centrifugal acceleration for a design speed of 75 kmph, using IRC recommended formula.**

(a) 0.53 m/sec$^3$ (b) 0.53 m/sec$^3$
(c) 0.48 m/sec$^3$ (d) 0.73 m/sec$^3$

**Ans. (b) :** Rate of change of centrifugal acceleration–

$$C = \frac{80}{75 + V} \quad \text{(Given, V = 75 kmph)}$$

$$= \frac{80}{75 + 75}$$

$$C = \frac{80}{150}$$

$$= 0.533 \text{ m/sec}^3$$

**17. Estimate the theoretical capacity of a traffic lane with traffic flow stream speed of 50 kmph (take average centre spacing of vehicles as 10m)**

(a) 500 vehicles/hour/lane
(b) 5000 vehicles/hour/lane
(c) 3000 vehicles/hour/lane
(d) 300 vehicles/hour/lane

**Ans. (b) :** Theoretical capacity

$$C = \frac{1000V}{S} \quad \text{Given, S = 10m}$$

$$C = \frac{1000 \times 50}{10}$$

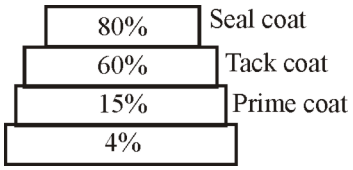
C = 5000 Vehicles/hour/lane

**18. In a flexible pavement, the different materials with the CBR values are available as follows: 80%, 60%, 15% and 4%. Indicate the order (top to bottom) in which the materials are to be placed for making a good pavement**

(a) 4%, 15%, 60%, 80%
(b) 4%, 60%, 80%, 15%
(c) 80%, 60%, 15%, 4%
(d) 4%, 80%, 15%, 60%

**Ans. (c) :** Material having better CBR percentage are considered best for GSB (Granular Sub Base) and subgrades so the material that has more percentage of CBR value is put at the topmost hence the best order to use it is-

CRR value : 80% > 60% > 15% > 4%




**19. Calculate the Equivalent Axle load factor (EALF) for single axle load of 10 tons using fourth power formula**

(a) 2.26 (b) 22.60
(c) 1.23 (d) 11.23

**Ans. (a) :** Single axle load–

$$EALF = \left(\frac{\text{axle load in kg}}{8160}\right)^4$$

$$= \left(\frac{10000}{8160}\right)^4$$

EALF = 2.26

**20. The hardness and toughness properties of a road aggregate will be obtained from**

(a) Aggregate Impact Test
(b) Aggregate crushing test
(c) Los Angeles Abrasion Test
(d) Aggregate Shape test

**Ans. (c) :** The los Angeles abrasion test is a common test method used to indicate aggregate toughness and abrasion (hardness) characteristics.

**21. In the rigid pavement fatigue analysis, the ratio of flexural stress due to load and the flexural strength due to concrete is less than 0.45 indicates**

(a) the allowable number of repetitions of the axle loads is infinity
(b) the allowable number of repetitions of the axle loads is zero
(c) the allowable number of repetitions of the axle loads is 4500
(d) the allowable number of repetitions of the axle loads is 45000

**Ans. (c) :** In the rigid pavement fatigue analysis, the ratio of flexural stress due to load and the flexural strength due to concrete is less than 0.45 indicates the allowable number of repetitions of the axle loads is 4500.

**22. The gauge widths (in meter) for broad, standard and narrow gauges respectively are**

(a) 1.767, 1.650 and 0.760
(b) 1.676, 1.500 and 0.676
(c) 1.676, 1.435 and 0.762
(d) 1.876, 1.656 and 0.800

**Ans. (c) :** Gauge distance for rail gauge–

| Rail gauge | Gauge distance (mm) |
|---|---|
| Broad gauge | 1676 |
| Standard gauge | 1435 |
| Meter gauge | 1000 |
| Narrow gauge | 762 |
| Light gauge | 610 |

**23. For airports serving big aircrafts, ICAO recommends the cross wind component should not exceed _____.**

(a) 25 kmph (b) 35 kmph
(c) 15 kmph (d) 23 kmph

**Ans. (b) :** For airports serving big aircrafts, ICAO recommends the cross wind component should not exceed 35 kmph.

■ Recommendations for cross wind component by FAA-

For small aircraft = (CWC $\not>$ 15 kmph)
For mixed traffic = (CWC $\not>$ 25 kmph)

**24. The monthly mean of average daily temperature for the hottest month of the year is 30°C and the monthly mean of the maximum daily temperature for the same month of the year is 45°C, Find out the airport reference temperature**

(a) 25°C (b) 35°C
(c) 40°C (d) 15°C

**Ans. (b) :** Airport reference temperature (ART)

$= T_a + \frac{T_m - T_a}{3}$

Where, $T_a$ = monthly mean of the average daily temperature for the hottest month of the zear

$T_m$ = monthly mean of the maximum daily temperature for the same month.

$T_a = 30^o$

$T_m = 45^oC$

$ART = 30 + \frac{45-30}{3}$

$ART = 35^oC$

**25. The minimum value of Composite Sleeper Index (CST) prescribed on Indian railways for track sleeper is**

(a) 783 (b) 1352
(c) 1455 (d) 873

**Ans. (a) :** Minimum CSI prescribed on Indian Railway–

| **Types of sleeper** | **Min CSI** |
|---|---|
| Track sleeper | 783 |
| Crossing sleeper | 1352 |
| Bridge sleeper | 1455 |

**26. The formula used for calculation of superelevation on railways is _____ (take G = gauge in meters, V = Speed in kmph; R = radius of curve in meters)**

(a) $\frac{GV^2}{127R}$m (b) $\frac{GV^2}{127R}$cm

(c) $\frac{GV^3}{127R}$m (d) $\frac{GV^2}{225R}$m

**Ans. (a) :** Super elevation/Cant–

$e = \frac{GV^2}{127R}$m

Where, G = Gauge distance in meter
V = Speed in kmph
R = Radius of curve in meter

**27. Cornice and coping is measured in _____**

(a) Running meter & Square meter
(b) Running meter & Running meter
(c) Square meter & Square meter
(d) Square meter & Running meter

**Ans. (b) :**

| **Particulars of item** | **Units of measurements** |
|---|---|
| Cornice and coping | Running meter |
| Earth work in excavation | Cum |
| Lime concrete in foundation | Cum |
| R.C.C. in slab | Cum |
| D.P.C. | Sqm |

**28. In Simpson's formula for areas calculation, the line joining the top of the ordinates is considered as:**

(a) elliptical (b) circular
(c) parabolic (d) straight

**Ans. (c) :** In Simpson's formula for areas calculation the line joining the top of the ordinates is considered as parabolic-

$V = \frac{d}{3}\left[(A_1 + A_n) + 4(A_2 + A_4 + A_6 + .....A_{n-1}) + 2(A_3 + A_5 + A_{n-2})\right]$

$= \frac{d}{3}\left[(A_1 + A_n) + 4(\text{Even area}) + 2(\text{Odd area})\right]$

**29. To obtain the correct volume using the trapezoidal rule, the prismoidal correction should always be**

(a) Multiplied (b) Added
(c) Subtracted (d) Zero

**Ans. (c) :** To obtain the correct volume using the trapezoidal rule the prismoidal correction should always be subtracted.

**Prismoidal formula–**

$V = \frac{L}{6}(A_1 + A_2 + 4A_m)$

**Trapezoidal formula–**

$V = D\left[\frac{A_0 + A_n}{2} + A_1 + A_2 + A_3 + .....A_{n-1}\right]$

**30. In earth work excavation, normally lead and lift is considered for preparation of road estimate**

(a) 30 m and 1.5 m (b) 20 m and 1.5 m
(c) 20 m and 2.0 m (d) 30 m and 2.0 m

**Ans. (a) :** Normally earthwork is estimated for 30 m lead and 1.5 m lift.

**Lead–** It is the average horizontal distance between the centre of excavation to the centre of deposition.

**Lift–** It is the average height through which the earth has to lifted from the source to the place of spreading or heaping.

**31. The unconfined compressive strength of a clay in un-disturbed and disturbed state was found to be 180 kN/sqm and 10 kN/sqm respectively. Based on Sensitivity, the soil may be classified as:**

(a) In-sensitivity
(b) Sensitivity
(c) Quick Clays
(d) Extra Sensitivity Clays

**Ans. (c) :** Sensitivity '$S_t$'

$= \frac{\text{Compressive strength of a clay in un-disturbed}}{\text{Compressive strength of a clay in disturbed}}$

$= \frac{180}{10}$

$S_t = 18$

■ The soil may be classified as quick clays

**32. If $R_1$ and $R_2$ are the radii of curvature of a non-uniform meniscus in two orthogonal planes, the capillary rise is given by:**

(a) $h_c = (\sigma/\gamma_w)\{R_1 + R_2\}$
(b) $h_c = (\sigma/\gamma_w)\{(1/R_1) + (1/R_2)\}$
(c) $h_c = (\sigma/\gamma_w)\{R_1 \times R_2\}$
(d) $h_c = (\sigma/\gamma_w)\{R_1/R_2\}$

**Ans. (b) : Non-uniform meniscus–** If the meniscus is not of uniform curvature but $R_1$ and $R_2$ are the ratio of curvature in two orthogonal plane, the height of capillary rise is given by

$$h_c = \frac{\sigma_s}{\gamma_w}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$h = \frac{4\sigma\cos\theta}{\rho g d}$$

**33. The coefficient of permeability of a soil sample having its void ratio as 0.50 and co-efficient of percolation as $3.00 \times 10^{-4}$ cm/s is:**
(a) $3.00 \times 10^{-4}$ cm/s (b) $1.50 \times 10^{-4}$ cm/s
(c) $6.00 \times 10^{-4}$ cm/s (d) $1.00 \times 10^{-4}$ cm/s

**Ans. (d) :** Void ratio = 0.5

$$\text{Porosity 'n'} = \frac{e}{1+e}$$

$$= \frac{.5}{1+.5}$$

$$= \frac{.5}{1.5} \Rightarrow \frac{5}{15} = \frac{1}{3} = .33$$

n = 0.33
$K_p = 3\times10^{-4}$ cm/s
K = ?
$K = K_p \times n$
$= 3\times10^{-4} \times .33$
$K = 0.99\times10^{-4} \sqcup 1\times10^{-4}$ cm/s

**34. In a Laboratory, to perform IS Heavy Compaction Test, it was required to use a mould of 1400 cc capacity in place of standard 1000 cc capacity mould. All other parameters remaining same, the number of blows to be applied per layer to ensure the designated compaction energy is:**
(a) 56 (b) 25
(c) 35 (d) 50

**Ans. (c) :** Compaction effort 'E' $= \frac{n \times N \times w \times h}{V}$

Where, n =number of layer
N = number of blows
w = weight of hammer
h = height of hammer drop

So, $\left(\frac{n \times N \times w \times h}{V}\right)_1 = \left(\frac{n \times N \times w \times h}{V}\right)_2$

$$\frac{25}{1000} = \frac{N}{1400}$$

$$N = \frac{25\times1400}{1000}$$

N = 35 blow

**35. The magnitude of total primary consolidation of a 6m thick clay with single drainage was estimated as 96 mm. Later it was found that, the medium has double drainage. Then, the magnitude of total primary consolidation settlement will be:**
(a) 48 mm (b) 192 mm
(c) 384 mm (d) 96 mm

**Ans. (a) :** $\left(\frac{\Delta H}{H}\right) = \left(\frac{\Delta e}{1+e_0}\right)$

For single drainage $H_1 = 6$m

For double drainage $H_2 = \frac{6}{2} = 3$m

$$\left(\frac{\Delta H}{H}\right)_1 = \left(\frac{\Delta H}{H}\right)_2$$

$$\frac{96}{6} = \frac{\Delta H_2}{3}$$

$$\Delta H_2 = \frac{96\times3}{6} = 48\,\text{mm}$$

**36. The shear strength of a pure clay specimen when tested in Unconfined compression Test was found to be 100 kPa. If the same specimen was tested in Tri-axial Compression Test, the deviatoric stress at which specimen will undergo failure when the confining stress was 50 kPa, will be:**
(a) 50 kPa (b) 100 kPa
(c) 200 kPa (d) 400 kPa

**Ans. (c) :** $q_u = 2C$
$q_u = 2\times50=100$
$q_u = 100$ kPa
$\sigma = 100$ kPa
Deviatoric stress = 100 + 100 = 200 kPa

**37. A 3 m high retaining wall with vertical face is resisting a moist back fill with horizontal top surface having $\gamma$ = 20 kN/cum. The percentage increase in Total Active Earth Pressure, if the back fill gets submerged with $\gamma_{sat}$ = 22 kN/cum and $Y_W$ = 10 kN/cum, is:**
(a) 20 (b) 40
(c) 60 (d) 100

**Ans. (c) :** $\gamma_{sat} = 22$ kN/cm
$\gamma_w = 10$ kN/cm
Active pressure due to $\gamma$ = 20 kN/cm

$$= \frac{1}{2}k\times20\times3^2$$

Active pressure due to submergence of water

$$= \frac{1}{2}k\times22\times3^2 + \frac{1}{2}\times k\times10\times3^2$$

$$= \frac{1}{2}k\times32\times3^2$$

Ratio of increase in pressure

$$= \frac{P_{sub} - P_{dry}}{P_{dry}} = \frac{32-20}{20} = 0.6$$

Percentage increase = 0.6 × 100 = 60%

**38. The factor of safety of a slope of given inclination of 6 m high constructed using a soil with c = 60 kPa, y = 20 kN/cum and its Taylor's stability number = 0.20, will be:**

(a) 2.50 (b) 5.00
(c) 1.00 (d) 2.00

**Ans. (a) :** Stability number $'S_n' = \frac{C}{\gamma.(F.O.S.)H}$

$$0.2 = \frac{60}{6\times(F.O.S.)\times 20}$$

$$F.O.S. = \frac{60}{20\times 6\times 0.2}$$

FOS = 2.5

**39. According to Boussinesque's theory, under the application of a point load of 100 kN on the surface, the pressure bulb corresponding to an increment in vertical stress of 47.75 kN/sqm will extend to a depth of:**

(a) 0.50 m (b) 1.00 m
(c) 2.00 m (d) 4.00 m

**Ans. (b) :** Boussinesque theory–

q = 100 kN

$\sigma_z$ = 47.75 kN/sqm

$$\sigma_z = \frac{3}{2\pi}\times\frac{q}{Z^2}$$

$$Z^2 = \frac{3\times 100}{2\times 3.14\times 47.75}$$

Z = 1m

**40. The ultimate bearing capacity of a shallow foundation laid on a cohesion-less soil medium was estimated as 200 kN/sqm, when the water table was far below. All other conditions remaining the same, the ultimate bearing capacity of the foundation, when the water table was risen to ground level, is:**

(a) 400 kN/sqm (b) 200 kN/sqm
(c) 100 kN/sqm (d) 50 kN/sqm

**Ans. (c) :** The water table rises to the ground have of a footing resting on cohesionless soil, the bearing capacity approximately reduces to half.

$q_{u2} = 0.5\times q_{u1}$

$= 0.5\times 200$

$= 100\ kN/m^2$

**41. The observations of a Standard Penetration Test are reported as 10/15/20. Assuming correction for overburden is not required and correction for dialatancy only is require, the corrected N-value is:**

(a) 45 (b) 35
(c) 25 (d) 15

**Ans. (c) :** SPT value $N_1$ = 35 (15+20)

Dilatency corrected value $N = 15 + \frac{1}{2}(N_1 - 15)$

N = 25

**42. The ultimate load carrying capacity of an end bearing type of bored cast-in-situ circular pile is estimated as 100 kN All other parameters remaining the same, the capacity of the Pile if the diameter is doubled, is:**

(a) 100 kN (b) 200 kN
(c) 300 kN (d) 400 kN

**Ans. (d) :** Bearing capacity of pile

$Q_u = q_b . A_b$ (For end bearing pile)

$$q_b = 9C,\quad A_b = \frac{\pi}{\ell}d^2$$

$$Q_u = 9C.\frac{\pi}{\ell}d_2^2$$

$$\frac{Q_1}{Q_2} = \left(\frac{d_1}{d_2}\right)^2 \Rightarrow \frac{Q_1}{Q_2} = \left(\frac{d_1}{2d_1}\right)^2$$

$$= \frac{100}{Q_2} = \frac{1}{4}$$

$Q_2$ = 400 kN

**43. Which of the following type of Piles is more appropriate as foundation of structure constructed on Expansive Clays:**

(a) Batter Piles (b) Sheet Piles
(c) Under-reamed Piles (d) Compaction Piles

**Ans. (c) : Under reamed pile foundations-** Buildings built on expansive soils on conventional footing often crack due to the differential ground movements.

The bearing capacity of an under-reamed pile is the sum of the total skin frictional resistance and the total point resistance.

$$Q_{ult} = A_p . NC_p + \alpha\,\overline{C}\,A.$$

Where, $A_p$ = Area of the pile base
N = Bearing capacity factor
$C_p$ = Undisturbed shear strength
$\alpha$ = Reduction factor

**44. The type of Caissons preferred in sites where high upward seepage exists, is:**

(a) Pneumatic Caissons
(b) Open Caissons
(c) Box Caissons
(d) Open caissons and Box caissons

**Ans. (a) :** Pneumatic caissons preferred in sites where high upward seepage exists.

■ These caissons have their top closed and compressed air is used to stop the entry of water into the working chamber.

**Open caissons-** These are concrete or masonry shafts which remain open both at the top and at the bottom during construction.

**45. Coffer Dams are:**

(a) Permanent structures meant for storage of water
(b) Structures built across drains to act as check dams
(c) Temporary structures build to reserve water in side to cure foundation concrete
(d) Temporary structures build to reserve water outside to facilitate construction of foundation

**Ans. (d) : Coffer dams-** Coffer dams are temporary enclosures to keep out water and soil so as to permit dewatering and construction of the permanent facility (structure) in the dry.

**46. The dimensions for Kinematic Viscosity is**

(a) $FL^{-2}T$ (b) $ML^{-1}T^{-1}$

(c) $L^2T^{-2}$ (d) $L^2T^{-1}$

**Ans. (d) :**

| Properties | Dimension |
|---|---|
| Kinematic viscosity | $L^2/T^{-1}$ |
| Vapour pressure | $ML^{-1}T^{-2}$ |
| Surface Tension | $MT^{-2}$ |
| Density | $ML^{-3}$ |

**47. The stream function for a potential flow field is given by $\psi = x^2 - y^2$, corresponding potential function ($\Phi$), assuming zero potential at the origin is**

(a) $x^2 + y^2$ (b) 2xy

(c) $x^2 - y^2$ (d) $x - y$

**Ans. (b) :** The stream function for a potential flow field is given by $\psi = x^2 - y^2$, corresponding potential function ($\Phi$), assuming zero potential at the origin is 2xy.

**48. Water flows through a large size pipe. The stagnation pressure and static pressures measured by a pitot tube are 0.3m and 0.24m of water. The velocity of flow in, m/minute is:**

(a) 1.08 (b) 65.00

(c) 10.8 (d) 0.65

**Ans. (b) :** Velocity $'V' = \sqrt{2g\left(P_{sg} - P_s\right)}$

$= \sqrt{2 \times 9.81(0.3 - 0.24)}$

$V = \sqrt{2 \times 9.81 \times 0.06}$

$= 1.08 \times 60$

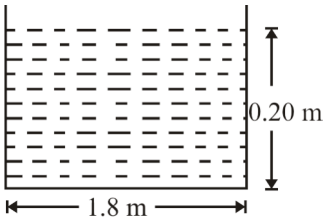
$= 65$ m/minute

**49. A rectangular channel has a width of 1.8 m and carries a discharge of 1.8 $m^3$/sec at a depth of 0.20 m. The specific energy is:**

(a) 1.03 m (b) 1.47 m

(c) 1.87 m (d) 2.0 m

**Ans. (b) :**




Discharge = 1.8 $m^3$/sec

$$\text{Specific energy } 'E' = y + \frac{Q^2}{2gA^2}$$

$$= 0.20 + \frac{(1.8)^2}{2 \times 9.81 \times (.36)^2}$$

$$= 0.20 + \frac{3.2}{2.5427}$$

$E = 1.47$ m

**50. In a catchment there are six rain gauge stations with average depth of rainfall of 92.8 cm and standard deviation of the rainfall is recorded in these rain gauge stations is 30.7 cm. For a 10% degree of error in the measurement of mean rainfall, the optimum number of stations required is:**

(a) 5 nos (b) 6 nos

(c) 10 nos (d) 11 nos

**Ans. (d) :** $\bar{p} = 92.8$ cm

$\sigma = 30.7$ cm

$E = 10\%$

$$C_V = \frac{100\sigma}{\bar{p}}$$

$$= \frac{100 \times 30.7}{92.8}$$

$= 33.08\%$

$$\text{Optimum number of station } 'n' = \left(\frac{C_V}{\varepsilon}\right)^2$$

$$= \left(\frac{33.08}{10}\right)^2$$

$n = 10.94 \sqsubset 11$

**51. The area between the two isohyets 45 cm and 55 cm is 100 $km^2$ and that between 55 cm and 65 cm is 150 $km^2$. What is the average depth of annual precipitation over the basin of 250 $km^2$.**

(a) 50 cm (b) 52 cm

(c) 56 cm (d) 60 cm

**Ans. (c) :** Average depth of annual precipitation

$$= \frac{\left(\frac{45+55}{2}\right) \times 100 + \left(\frac{55+65}{2}\right) \times 150}{100 + 150}$$

$= 56$ cm

**52. The total observed runoff volume during a 4 hr storm with a uniform intensity of 2.8 cm/hr is $25.2 \times 10^6$ $m^3$ from a basin of 280 $km^2$ area. What is the average infiltration rate for the basin**

(a) 3.6 mm/hr (b) 4.8 mm/hr

(c) 5.2 mm/hr (d) 5.5 mm/hr

**Ans. (d) :** The average infiltration rate

$$W = \frac{P - R - \ell_0}{t}$$

$P = 2.8 \times 10 \times 4 = 112$ mm

$$R = \frac{25.2 \times 10^6 \times 10^3}{280 \times 10^6} = 90 \text{ mm}$$

$t_c = 4$ hr

$\ell_0 = 0$

$$W = \frac{112 - 90}{4}$$

$= 5.5$ mm/hr

**53. A volume of $3 \times 10^6$ $m^3$ of groundwater was pumped out from an unconfined aquifer, uniformly over an area of 5 $km^2$. The pumping lowered the water table from initial level of 102 m to 99 m. The specific yield of the aquifer is:**

(a) 0.20 (b) 0.30
(c) 0.40 (d) 0.50

**Ans. (a) :** $\text{Specific field} = \frac{\text{Volume}}{\text{Area} \times \text{WT level}}$

WT level = 102 – 99 = 3m

$\text{Specific field} = \frac{3 \times 10^6}{5 \times 10^6 \times 3}$

$= 0.2$

**54. A hyetograph is a plot of**

(a) Cumulative rainfall Vs time
(b) Rainfall intensity Vs time
(c) Rainfall depth Vs duration
(d) Discharge Vs time

**Ans. (b) : Hyetographs-** A rainfall hyetograph is a plot of rainfall depth or intensity as a function of time.

■ A discharge hydrograph is a graphical representation of flow rate as a function of time at a given location on a river.

**55. The unit of intrinsic permeability is**

(a) cm/day (b) m/day
(c) Darcy/day (d) $cm^2$

**Ans. (d) :** Intrinsic permeability is a property of the porous medium and is independent of the properties of the fluid.

■ The unit of intrinsic permeability is $cm^2$.

**56. If $S_y$ = specific yield and $S_r$ = specific retention then:**

(a) $S_y + S_r$ = void ratio
(b) $S_y + S_r$ = porosity
(c) $S_y + S_r = 1.0$
(d) $Sy + S_r$ = permeability

**Ans. (b) : Specific yield ($S_y$)-** While porosity is a measure of the water bearing capacity of the formation, all this water cannot be drained by gravity or by pumping from wells, as a portion of the water is held in the void spaces by molecular and surface tension forces.

**Specific retention ($S_r$)-** The volume of water retained by molecular and surface tension forces, against the force of gravity, expressed as a percentage of the total volume of the saturated aquifer, is called specific retention '$S_r$' and corresponds to field capacity.

$\boxed{\text{Porosity} = S_y + S_r}$

**57. If duty (D) is 1428 hectare/cumec and base period (B) is 120 days for an irrigated crop, then delta (Δ) in meters is given by:**

(a) 102.8 (b) 0.73
(c) 1.38 (d) 0.01

**Ans. (b) :** Given– duty 'D' = 1428 hectare/cumec
Base period 'B' = 120 days
Delta 'Δ' = ?

$\Delta = \frac{8.64 \times B}{D}$

$= \frac{8.64 \times 120}{1428}$

$\Delta = 0.726 \approx 0.73$ m

**58. Which one of the following equations represents the downstream profile of Ogee spillway with vertical upstream face? (X, Y) are the coordinates of the point on the downstream profile with origin at the crest of the spillway and $H_d$ is the design head.**

(a) $Y/H = -0.5(X/H_d)^{1.85}$
(b) $Y/H_d = -0.5(X/H_d)^{1/185}$
(c) $Y/H_d = -2.0(X/H_d)$
(d) $Y/H_d = -2.0(X/H_d)^{1/185}$

**Ans. (a) :** For a spillway having a vertical v/s face, the d/s crest is given by the equation–

$X^{1.85} = -2\, H_d^{0.85} \times Y$

$X^{1.85} = -2H_d^{0.85}Y \times \frac{H_d}{H_d}$

$X^{1.85} = -2\frac{H_d^{1.85}Y}{H_d}$

$\frac{X^{1.85}}{-2H_d^{1.85}} = \frac{Y}{H_d}$

$\frac{Y}{H_d} = -0.5\left(\frac{X}{H_d}\right)^{1.85}$

**59. For no tension to develop in the gravity dam the eccentricity 'e' of the resultant force should be:**

(a) Less than $b/3$ (b) Less than $b/6$
(c) Less than $b/4$ (d) Less than $b/12$

**Ans. (b) :** ■ No tension should be permitted at any point of the dam under any circumstance.

■ For no tension to develop, the eccentricity should be less than $b/6$. In other words the resultant should always lie within the middle third.

**60. Lacey's equations can be used for the design of**

(a) Unlined channels only
(b) Lined channels only
(c) Both lined and unlined channels
(d) Neither lined nor unlined channels

**Ans. (a) :** ■ Lacey's equations can be used for the design of unlined channels only.

**Design Parameter-**

$\text{Velocity } (V) = \left(\frac{Qf^2}{140}\right)^{1/6}$

Silt factor 'f' = $1.76\sqrt{d_{min}}$

Wetted perimeter 'p' = $4.75\sqrt{Q}$

Hydraulic mean depth, $R = \frac{5}{2}\left(\frac{V^2}{f}\right)$

**61. Syphon aqueduct is a cross drainage work provided to carry canal over a natural drain when**

(a) Canal bed is well above the HFL of the natural drain
(b) Canal bed is at the same level as the bed of the natural drain
(c) Canal bed is below the HFL of the natural drain
(d) Canal bed is below bed of the natural drain

**Ans. (c) :** Syphon aqueduct in which the high flood level of the drain is higher than the bottom of the canal trough.
**Aqueduct-** A hydraulic structure which carries a canal cross and above the drainage.

**62. Poise is the CGS unit of**

(a) Kinematic viscosity (b) Dynamic viscosity
(c) Mass Density (d) Weight Density

**Ans. (b) : Dynamic viscosity-** Viscosity is defined as the property of a fluid which offers resistance to the movement of one layer of fluid over adjacent layer of the fluid.

Dynamic viscosity $\mu = \frac{\tau}{\left(\frac{du}{dy}\right)}$

S.I. unit $= \frac{N-s}{m^2}$

C.G.S. unit = Poise

**63. The velocity gradient is 1000/s. The viscosity is $1.2 \times 10^{-4}$ N-s/m$^2$. The shear stress is**

(a) 0.12 N/m$^2$ (b) $1.2 \times 10^{-7}$ N/m$^2$
(c) 12 N/m$^2$ (d) $12 \times 10^{-5}$ N/m$^2$

**Ans. (a) :** Shear stress $'\tau' = \mu\frac{du}{dy}$

Viscosity $'\mu' = 1.2\times10^{-4}$ N-s/m$^2$

Velocity gradient $\frac{du}{dy} = 1000/s$

$\tau = 1.2\times10^{-4}\times1000$

$\tau = 0.12$ N/m$^2$

**64. If z is measured vertically upwards, dp is given by:**

(a) $dp = \gamma dz$ (b) $dp = \rho dz$
(c) $dp = -\gamma dz$ (d) $dp = -\rho Dz$

**Ans. (c) :** If z is measured vertically upward than it will show the decrease in pressure with height.

$dp = -\gamma dz$

If z is measured vertically downward then it will represent.

$dp = \gamma dz$

**65. If $\psi = 3x^2y - y^3$, the values of u and v are**

(a) $6xy, 3x^2 - 3y^2$ (b) $3x^2 - 3y^2, 6xy$
(c) $(3x^2-3y^2), -6xy$ (d) $3y^2 - 3x^2, 6xy$

**Ans. (c) :** $\psi = 3x^2y - y^3$

$\frac{d\Psi}{dx} = -V = 6xy$

$\frac{d\Psi}{dy} = U = 3x^2 - 3y^2$

So,

$V = -6xy$

$U = 3x^2 - 3y^2$

**66. In a three dimensional motion of a fluid, the component of rotation about the x-axis, $\omega_x$ is**

(a) $\frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right)$ (b) $\frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial y}\right)$
(c) $\frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$ (d) $\frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial w}{\partial y}\right)$

**Ans. (a) :** The rotation of the fluid element about the z axis is the average angular velocity of the two mutually perpendicular line elements

$$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial t} - \frac{\partial u}{\partial y}\right)$$

The rotation of pairs of perpendicular line segment in the yz and xz planes, one can show similar that

$$\omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right)$$

$$\omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)$$

**67. Existence of velocity potential implies that**

(a) Fluid is in continuum
(b) Fluid is irrotational
(c) Fluid is ideal
(d) Fluid is compressible

**Ans. (b) :** If velocity potential exists, the flow will be Irrotational.
Velocity potential function- This function is defined as a function of space and time in a flow such that the negative derivation of this function with respect to any direction given the velocity of fluid in that direction.

**68. Piezometric head is the summation of**

(a) Velocity head and elevation head
(b) Velocity head and pressure head
(c) Pressure head and elevation head
(d) Total head

**Ans. (c) :** Piezometric head $'P_z'$ = Pressure head + Elevation head

$$P_z = \frac{P}{\rho g} + z$$

**69. The Bernoulli's equation is written with usual notation as $p/y + v^2/2g + z$ = constant. In this equation each of the terms represents**

(a) Energy in kg.m/kg mass of fluid
(b) Energy in Nm/kg mass of fluid
(c) Energy in Nm/N weight of fluid
(d) Energy in KW/kg mass of fluid

**Ans. (c) :** Bernoulli's equation $= \frac{P}{\rho g} + \frac{V^2}{2g} + z = C$

■ Bernoulli's equation represents energy per unit weight.

■ Continuity equation is based on conservation of mass principle.

**70. A 50 mm diameter jet having a velocity of 25 m/s strikes a flat stationary plate, the normal of which is inclined at 60° to the axis of the jet. The normal force exerted on the plate in Newtons is:**

(a) 460 N (b) 360 N
(c) 640 N (d) 630 N

**Ans. (*) :** Normal force exerted on the plate–

$P_n = \rho A V^2 \sin\theta$

$= 1000 \times \pi (0.025)^2 \times (25)^2 \times \sin 60^0$

$= 1.9625 \times (25)^2 \times \sin 60^0$

$= 1226.5625 \times \sin 60^0$

$= 1062.23$ N

**71. Reynolds number of a flow is the ratio of**

(a) Gravity forces to viscous forces
(b) Gravity forces to pressure forces
(c) Inertia forces to viscous forces
(d) Viscous forces to pressure forces

**Ans. (c) :**

| Number | Definition |
|---|---|
| ■ Reynolds number | $\frac{\text{Inertia force}}{\text{Viscous force}}$ |
| ■ Mach number | $\sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}}$ |
| ■ Froude number | $\sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}}$ |

**72. Ratio of the average velocity to maximum velocity for steady laminar flow in circular pipe is:**

(a) 1/2 (b) 2/3
(c) 3/2 (d) 2

**Ans. (a) :** Average velocity ($V_{avg}$) for laminar steady flow in circular pipe is given as–

$$V_{avg} = \frac{u_{max}}{2}$$

■ Average velocity ($V_{avg}$) for laminar steady flow between two stationary plate is given as

$$V_{avg} = \frac{3}{2} u_{max}$$

**73. Difference in elevation between TEL and HGL of pipe flow at a point is equal to**

(a) Datum head (b) Velocity head
(c) Pressure head (d) Piezometric head

**Ans. (b) :** ■ HGL always lies below the EGL and the vertical difference between the two at any point is equal to the velocity head $\left(\frac{v^2}{2g}\right)$ at that point.

■ If the diameter of the fluid transmission pipes is uniform throughout then the slope of HGL and EGL are equal which means the two lines run parallel to each other.

**74. Boundary layer separation takes place when**

(a) $\partial p/\partial x > 0$
(b) $\partial v/\partial y \le 0$
(c) $\partial p/\partial x > 0$ & $\partial v/\partial y \le 0$
(d) $\partial p/\partial x < 0$

**Ans. (c) :** Boundary layer separation takes place

When, $\frac{dv}{dx} \le 0$ $\frac{dp}{dx} > 0$

Boundary layer separation takes place whenever an abrupt change in either the magnitude or direction of the fluid velocity is too great for the fluid to keep to a solid surface.

**75. For a uniform flow with a depth of 0.6m and Froude number of 2.0 in a rectangular channel, the specific energy will be:**

(a) 2.4 m (b) 0.8 m
(c) 2.6 m (d) 1.8 m

**Ans. (d) :** ■ Froude number $= \frac{V}{\sqrt{gd}}$

$$2 = \frac{V}{\sqrt{9.81 \times 0.6}}$$

$V = 4.85$ m/s

■ Specific energy (E) $= y + \frac{V^2}{2g}$

$$E = 0.6 + \frac{(4.85)^2}{2 \times 9.81}$$

$= 1.8$ m

**76. A water turbine of expected efficiency 85% operates under 36 m at a flow rate of 10 $m^3/s$. The unit weight of water is 10 $kN/m^3$. The nearest shaft power in MW is**

(a) 3.06 (b) 3060
(c) 3600 (d) 850

**Ans. (a) :** Shaft power $= \frac{\eta Q H W}{1000}$ MW

$$= \frac{0.85 \times 36 \times 10 \times 10}{1000}$$

$= 3.06$ MW

**77. A sedimentation tank 6m wide, 15m long and 3m water depth is treating 2 MLD of water. The surface overflow rate in $lt/hr/m^2$ is**

(a) 858 (b) 926
(c) 1028 (d) 748

**Ans. (b) :** Wide 'B' = 6 m

Long 'L' = 15 m

$Q = 2$ MLD $= 2 \times 10^6$ LD

Surface overflow rate $= \frac{Q}{L \times B}$

$$= \frac{2 \times 10^6}{15 \times 6 \times 24}$$

$= 925.9 \sqcup 926$ *l*t/hr/$m^2$

**78. The efficiency of a sedimentation removal in a continuous sedimentation tank does not depend upon the**

(a) Discharge through the tank
(b) Width of the tank
(c) Length of the tank
(d) Depth of the tank

**Ans. (d) :** In a continuous flow sedimentation tank settling efficiency depends upon–
■ Size, shape and specific gravity of particles, overflow rate. Width and length
■ Temperature, Viscosity
■ The efficiency of sedimentation tank does not depend upon the depth of the tank.

**79. The disinfection efficiency of chlorine in water treatment**

(a) Is not dependent on pH value
(b) Is increased by increase in pH value
(c) Remains constant at all pH values
(d) Is reduced by increase in pH value

**Ans. (d) :** ■ Increasing or decreasing pH reduces efficiency of chlorination.
■ At pH < 5, chlorine doesn't react with water and remains as free chlorine.
■ The disinfection efficiency of chlorine in water treatments is reduced by increased pH value.

**80. A 2% solution of a sewage sample is kept at an incubation temperature of 20°C. If initial DO and final DO values after 5 days of incubation period are 8.5 mg/lit and 5.5 mg/lit respectively, then the BOD will be**

(a) 50 mg/lit (b) 150 mg/lit
(c) 250 mg/lit (d) 350 mg/lit

**Ans. (b) :** $(BOD)_{5@20^o} = (DO_i - DO_f) \times$ Dilution factor

$$= (8.5 - 5.5) \times \frac{100}{2}$$

$$= 150 \text{ mg/lit}$$

**81. If total hardness of water is less than its total alkalinity the non-carbonate hardness will be equal to**

(a) Total alkalinity
(b) total hardness
(c) Total alkalinity - total hardness
(d) Zero

**Ans. (d) :** ■ If total hardness is less than total alkalinity ⇒ Carbonate hardness = Total hardness and non carbonate hardness = zero
■ If total hardness is greater than total alkalinity ⇒ Carbonate hardness = Total alkalinity and non-carbonate hardness = Total hardness carbonate hardness.

**82. If for diluting 25 ml of water sample 175 ml of taste free water is required to be added to make the water sample to just lose its taste then the Flavour Threshold Number (FTN) will be:**

(a) 6 (b) 7
(c) 8 (d) 9

**Ans. (c) :** Given, Diluting = 25 ml
Water sample = 175 ml

$$\text{Flavour threshold number (FTN)} = \frac{\text{Total sample}}{\text{Diluting}}$$

$$= \frac{175 + 25}{25}$$

$$= \frac{200}{25} = 8$$

**83. In network of pipes**

(a) The algebraic sum of discharges around each circuit is zero
(b) The algebraic sum of head losses around each circuit is zero
(c) The elevation of hydraulic grade line is assumed for each junction point
(d) Elementary circuits are replaced by equivalent pipes

**Ans. (b) : In network of pipes-**
■ The algebraic sum of head losses around each circuit is zero.
■ The algebraic sum of the piezometric head drops around each elementary circuit must be zero.
■ The flow entering a junction must be equal to the flow leaving the same junction, that is the law of continuity must be satisfied.

**84. The dispersion of pollutants in atmosphere is maximum when**

(a) Environmental lapse rate is greater than adiabatic lapse rate
(b) Environmental lapse rate is less than adiabatic lapse rate
(c) Environmental lapse rate is equal to adiabatic lapse rate
(d) Maximum mixing depth is equal to zero

**Ans. (a) :** When the environment lapse rate is more than the adiabatic lapse rate, the dispersion of pollutants in atmosphere is maximum.
■ In this condition, a rising parcel of warm lighter air (pollutants) will continue to lift up, whereas a parcel of heavier cooler air will continue to come down.
■ In such circumstances, the environment is unstable and dispersion of pollutants will be rapid due to marked vertical mixing of the air.

**85. Which one of the following pollutant (or) pairs of pollutants is formed due to photochemical reactions**

(a) CO alone (b) $O_3$ and PAN
(c) PAN and $NH_3$ (d) $NH_3$ and CO

**Ans. (b) :** ■ Photochemical smog is produced when pollutants from the combustion of fossil fuels react with sunlight.
■ The energy in the sunlight converts the pollutants into other toxic chemicals.
■ In order for photochemical smog to from, there must be other pollutants in the air, specifically nitrous oxides and other volatile organic compounds.
■ When nitrous oxides and VOC, interact with sunlight, secondary pollutants are formed, such as ozone and peroxyacetyl nitrate.

**86. If a sewer carrying a discharge of 3 cumecs outfalls into a river having a discharge of 10 cumecs and DO equal to 9.1 mg/lt, the resultant DO of the mix will be equal to:**

(a) 5 mg/lt (b) 6 mg/lt
(c) 7 mg/lt (d) 8 mg/lt

**Ans. (c) :** Resultant $DO = \frac{Q_1DO_1 + Q_2DO_2}{Q_1 + Q_2}$

$Q_1$ = 10 cumec, $Q_2$ = 3 cumec
$DO_1$ = 9.1 mg/L
$DO_2$ = 0 mg/L

$$DO = \frac{10 \times 9.1 + 3 \times 0}{10 + 3} = 7 mg/L$$

**87. The natural process under which of the flowing river gets cleaned is known as:**

(a) Oxidation (b) Photosynthesis
(c) Reduction (d) Self-purification

**Ans. (d) :** ■ The natural process under which of the flowing river gets cleaned is known as self purification.
■ The self purification process of water bodies or processes involving biological, chemical and physical processes working simultaneously on biological pollutants, oxidizing them and increasing the amount of dissolved oxygen.

**88. Recirculation in activated sludge process is done to:**

(a) Dilute the incoming sewage
(b) Dampen the effect of the flow variation
(c) Operate the plant continuously
(d) Supply seed materials to the aeration tank

**Ans. (d) :** Recirculation in activated sludge process is done to supply seed materials to the aeration tank.
■ Recirculation is used to remove or lower levels of nitrates, $BOD_5$, TSS and other contaminants in the waste water.

**89. Lower Food to Micro-organic (F/M) ratio in a conventional activated treatment plant will mean:**

(a) Lower BOD removal
(b) Higher BOD removal
(c) No effect on BOD removal
(d) Medium BOD removal

**Ans. (b) :** Food (F) to micro-organism (M)
Ratio- It is also called organic loading

$$\frac{F}{M} = \frac{kg\ of\ BOD\ applied\ per\ day}{kg\ of\ MLSS\ in\ aeration} = \frac{Q.y_1}{V.x_1}$$

■ Lower the F/M ration, higher the BOD removal in plant.

**90. The relative stability of a sewage sample whose dissolved oxygen levels equals the total oxygen required to satisfy its BOD is**

(a) Zero (b) 1%
(c) 100% (d) Infinite

**Ans. (c) : Relative stability-** The ratio of oxygen available in the waste water (as dissolved oxygen) to the total oxygen required to satisfy its first state BOD.

$$\text{Relative stability} = \frac{BOD\ removed}{Total\ BOD} \times 100$$

**91. A rectangular box made with thin uniform plate measures 2000 mm × 1000 mm × 1000 mm. When the box is subjected to certain internal pressure, the dimensions in respective directions have changed by +2.0 mm, –1 mm and +1 mm. The change in the volume of the box is _____.**

(a) $2.0 \times 10^6$ mm$^3$ (b) $1.0 \times 10^5$ mm$^3$
(c) $2.0 \times 10^3$ mm$^3$ (d) $2.0 \times 10^5$ mm$^3$

**Ans. (a) :** $V_1 = 2000 \times 1000 \times 1000$
$= 2 \times 10^9$ mm$^3$
$V_2 = 2002 \times 999 \times 1001$
$= 2001997998$ mm$^3$
Change value $\Delta V = 2001997998 - 2000000000$
$= 1997998$
$= 1.99 \times 10^6 \sqcup 2 \times 10^6$ mm$^3$

**92. A rectangular section of a beam is acted upon by certain amounts of shear force bending moment. Whereas the shear stress varies _____, the variation of bending stress is _____ along the depth.**

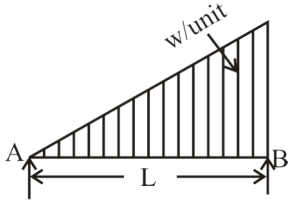
(a) linear with zero value at centroid, linear with zero value at centoid.
(b) parabolic with zero value at centroid, linear with zero value at centoid.
(c) parabolic, with zero value at top & bottom, linear with zero value at middle
(d) linear with zero value at centroid, parabolic with zero value at top & bottom.

**Ans. (c) :** A rectangular section of a beam is acted upon by certain amounts of shear force bending moment whereas the shear stress varies parabolic, with zero value at top and bottom, the variation of bending stress is linear with zero value at middle along the depth.

**93. A simply supported beam of span L, carries a UVL with intensity varying from w/unit downward at left support to w/unit upward at right support. The reactions at left and right supports are respectively:**

(a) wL/4 upward and wL/4 downward
(b) wL/4 downward and wL/4 upward
(c) wL/6 upward and wL/3 downward
(d) wL/8 upward and wL/8 downward

**Ans. (c) :**




■ Reaction at left supported '$R_A$' $= \frac{wL}{6}$ upward

■ Reaction at right supported '$R_B$' $= \frac{wL}{3}$ downward

**94. A cantilever beam of span L, uniform flexural rigidity EI is subjected to a unit couple at its free end. The deflection at the centre of the beam is:**

(a) $L^2/2EI$ (b) $L^2/8EI$
(c) $L^2/4EI$ (d) $L^2/16EI$

**Ans. (b) :** $EI\frac{d^2y}{dx} = M$

$\frac{dy}{dx} = \frac{Mx}{EI} + C$

$y = \frac{Mx^2}{2EI} + C_1x$

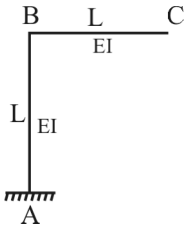
$y = \frac{Mx^2}{2EI} - \frac{Mx^2}{EI} = \frac{-Mx^2}{2EI}$

Let $x = \frac{L}{2}$ then the deflection at the centre will be

$y = -\frac{ML^2}{8EI}$

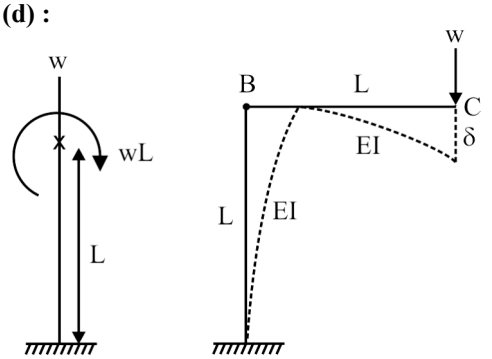
For unit couple $M = 1$ so $y = -\frac{L^2}{8EI}$

***95. For the L bend shown in Fig., the flexural rigidity of both arms AB and BC is EI carries a vertical downward load W at C. The deflection and rotation at B (neglecting axial deformations) are _____ and _____.**




(a) $WL^2/2EI \rightarrow$, $WL/EI$
(b) $2WL^3/EI \downarrow$, $WL^2/3EI$
(c) $WL^3/EI \downarrow$, $WL^2/EI$
(d) $WL^3/2EI \rightarrow$, $WL^2/EI$

**Ans. (d) :**




δ due to w axial will be zero.

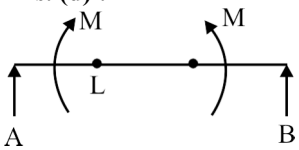
So, $\theta = \frac{ML}{EI} = \frac{w\ell^2}{EI}$

$\delta = \frac{ML^2}{2EI} = \frac{wL^3}{2EI}$

**96. A simply supported beam of span L, carries two couples of magnitude M each acting at both middle third locations of the beam. While one of them is acting clockwise the other is acting counter clockwise. Magnitude of the maximum shear force acting in the beam _____.**

(a) 2 M/L (b) M/L
(c) 1.5 M/L (d) 0

**Ans. (d) :**




$R_A + R_B = 0$

Taking moment at B

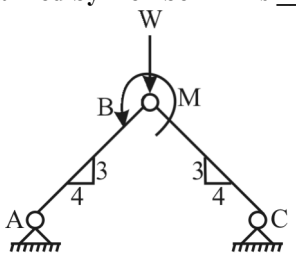
$R_A \times L \times M - M = 0$

$R_A = 0$

$R_B = 0$

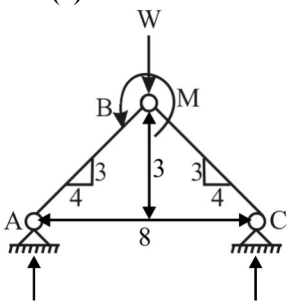
So the shear force will be zero at every points.

**97. The plane truss shown in Fig. carries a point load W and a moment M at the location B. Force carried by member AB is _____.**




(a) 5 W/6 (compressive
(b) W/2 (compressive)
(c) 5 W/6 + M/L (compressive)
(d) W/2 – M/2 (compressive)

**Ans. (a) :**




$\frac{W}{2} = R_A$ $\frac{W}{2} = R_B$

$R_A + R_B = w$

Now considering the point A

$F_{AB}\frac{3}{5} = \frac{w}{2}$

$F_{AB} = \frac{5}{6}w$ compressive

**98. Analysis of trusses is based on the following assumption. The correct combination of assumptions is _____.**

i. All loads act at joints
ii. All joints are smooth friction less pin joints
iii. Truss is made with material which is homogenous, elastic and isotropic.
iv. The axis of all members is straight and their self weights are ignored

(a) i, ii and iv (b) i, ii, iii
(c) i, iv (d) i, iii, iv

**Ans. (a) :** Analysis and Assumptions of truss-
- Self weight of the member is neglected.
- Member of the truss are straight uniform & two force members.
- Truss is loaded at the joints only
- Truss is a pin jointed structure so cannot resist moment.

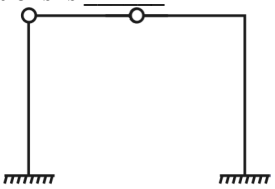
**99. The correct combination of conditions that defines a rigid joint is _____**

i. All members meeting at the joint are rigid.
ii. All deflections and rotations at the joint are zero.
iii. All members meeting at the joint undergo same defections at that joint.
iv. All members meeting at the joint undergo same rotations at the joint.

(a) i, ii (b) i, iii
(c) ii, iv (d) iii, iv

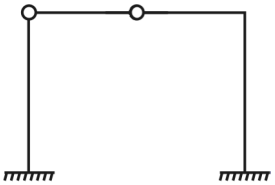
**Ans. (d) : Rigid joint-**
- All members meeting at the joint undergo same defections at that joint.
- All member meeting at the joint undergo same rotations at the joint.
- Rigid joints (moment connections) are designed to transfer axial and shear force in addition to bending moments between the connected members.

**100. The kinematic indeterminacy of the 2-D portal frame shown in Fig. including axial deformations is _____**



(a) 6 (b) 11
(c) 8 (d) 9

**Ans. (b) :**



Kinematic indeterminacy
$D_K$ = 3J–r + number of hinges
= 3 × 5 – 6 + 2
=17 –6 = 11

**101. In the moment distribution method of plane frame analysis, the distribution factor for a member at a joint depends on _____ combination of conditions.**

i. The slenderness ratio of all the members meeting at the joint.
ii. The flexural rigidity of all the members meeting at the joint.
iii. The I/L values of each member meeting at the joint.
iv. Support conditions at the farther ends of members meeting at the joint.

(a) ii, iii (b) i, iii
(c) iii, iv (d) i, iv

**Ans. (c) :** Moment distribution factor for member at joint $= \dfrac{\frac{kL}{EI}}{\sum \frac{kL}{EI}}$

- So the moment distribution factor depend on $\frac{I}{L}$ values of each member connecting at that point.
- Support condition at the farther ends of members meeting at the joints.

**102. A cantilever beam of uniform flexural rigidity with span L and depth D is subjected to temperature $T_1$ on the upper face and $T_2$ on the lower face. If $T_1 < T_2$ and $\alpha$ is the coefficient of linear expansion for the material, the deflection at the free end of the beam is _____.**

(a) $\alpha (T_2 - T_1) L^2/2D$ (upward)
(b) $\alpha (T_2 - T_1) L^2/D$ (upward
(c) $\alpha (T_2 - T_1) L/2D$ (upward)
(d) $\alpha (T_2 - T_1) L^2/2D$ (downward)

**Ans. (a) :** For cantilever beam,

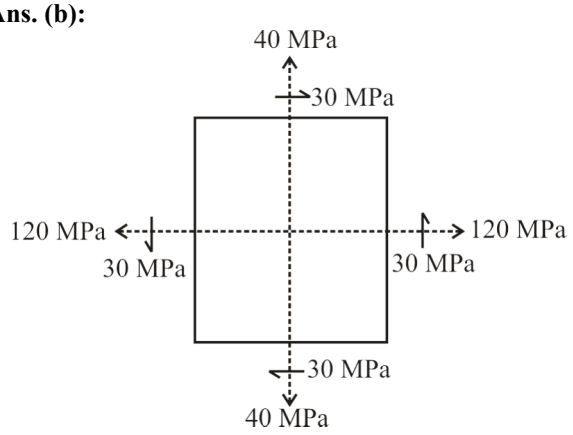
$$\delta_{\text{free end}} = \frac{\alpha (T_2 - T_1) L^2}{2D} (\uparrow)$$

Where,
L = span of beam
$T_1$ = temperature on upper face
$T_2$ = temperature on lower face
D = depth of beam
$\alpha$ = coefficient of linear expansion

**103. At a point in elastic medium normal stresses in two mutually perpendicular directions are 120 MPa, 40 MPa (both tensile) associated with a tangential stress of 30 MPa. The principal stresses at the locations are _____.**

(a) 120 MPa, 40 MPa (both tensile)
(b) 130 MPa, 30 MPa (boh tensile)
(c) 130 MPa, 30 MPa (both compressive)
(d) 130 MPa (tensile), 30 MPa (compressive)

**Ans. (b):**




$$\tan 2\theta = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$$\tan 2\theta = \frac{2\times 30}{120-40} = \frac{3}{4}$$

$2\theta = \tan^{-1}(3/4)$

$2\theta = 36.86^\circ$

Principal stress '$\sigma_n$'

$$= \frac{\sigma_x + \sigma_y}{2} + \left(\frac{\sigma_x - \sigma_y}{2}\right).\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$= \frac{120+40}{2} + \left(\frac{120-40}{2}\right)\cos(36.86^\circ) + 30\sin(36.86^\circ)$$

$= 80 + 40 \times 0.8 + 30 \times 0.599$

$= 80 + 32 + 18$

$\sigma_{n1} = 130$ MPa (Tension)

$$\sigma_{n2} = \frac{\sigma_x + \sigma_y}{2} - \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta - \tau_{xy}\sin 2\theta$$

$= 80–40\times0.8 – 30\times0.599$

$= 80 – 50$

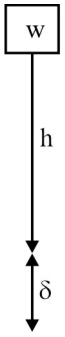
$\sigma_{n2} = 30$ MPa (Tension)

**104. A weight W falls freely on a body from a height of h. If the instantaneous deformation of the body in the direction of weight is δ, the work done by the force is _____.**

(a) $\frac{1}{2}W(h+\delta)$ (b) $W(h+\delta)$

(c) $\frac{1}{2}Wh + W\delta$ (d) $Wh + \frac{1}{2}W\delta$

**Ans. (b) :** The work done by the body if freely falling from height h and deflection is δ, will be $W(h+\delta)$




So the total work done
mg(h+δ) = w(h+δ)

**105. In a beam, where the variation of shear force is a second degree parabola and the variation of loading is _____, the variation of bending moment is _____**

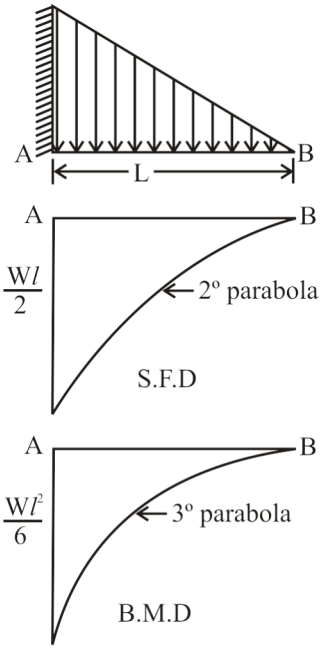
(a) constant, cubic parabola

(b) liner, square parabola

(c) cubic parabola, square parabola

(d) linear, cubic parabola

**Ans. (d) :**




■ In a beam, where the variation of shear force is a second degree parabola and the variation of lading is linear, the variation of bending moment is cubic parabola.

**106. A force of magnitude 5 N moves through a distance of 4mm in a direction, inclined at 60° to the direction of force. The magnitude of the work done by the force is _____.**

(a) $10\sqrt{3}$ N.mm (b) 0 N.mm

(c) 10 N.mm (d) 20 N.mm

**Ans. (c) :** Work = F.d. cosθ

$= 5\times4\times\cos60^\circ$

W = 10 N.mm

**107. The shear centre for an angle section is located _____.**

(a) at the tip of the flange

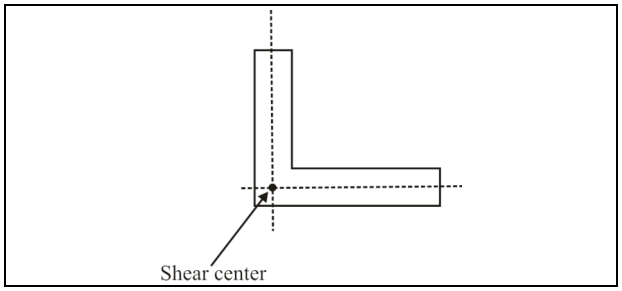
(b) at the intersection of flanges

(c) at the centroid of the angle

(d) at the rigidity centre of the angle

**Ans. (b) : Shear center-** It is a point through which if the external force passes, then the section will be subjected to bending only and there will be no twisting of the section.

■ The shear centre for an angle section is located at the intersection of flanges.




**108. A beam has a circular cross section. If the plane of loading on the beam does not coincide with the centroidal axis of the beam, the member is subjected to _____.**

(a) axial force, shear force and bending moment
(b) torque, shear force
(c) torque, shear force and bending moment
(d) axial force, shear force and torque

**Ans. (c) :** A beam has a circular cross section. If the plane of loading on the beam does not coincide with the centroid axis of the beam, the member is subjected to torque, shear force and bending moment.

**109. The stiffness of a closed coiled spring is more when _____.**

(a) wire diameter, rigidity modulus are more and mean radius, number of turns are lesser
(b) number of turns, rigidity modulus are more and mean radius, number of turns are lesser
(c) number of turns, rigidity modulus are more and mean radius, wire diameter, number of turns are lesser
(d) rigidity modulus is more and wire diameter, mean radius, number of turns are lesser

**Ans. (a) :** ■ The stiffness of a closed coiled spring is more when wire diameter, rigidity modulus are more and mean radius, number of turns are lesser.

■ The primary function of a spring is to deflect or distort under load & to recover it original shape when the load is released.

■ If helix angle is less than or equal to 10º than it is called closed coil spring.

**110. A pressure vessel in the form of a thin cylinder of 1 m diameter and 1 mm plate thickness is subjected to an internal fluid pressure of 0.2 MPa. The maximum shear stress in the material is :**

(a) 50 MPa (b) 0 MPa
(c) 25 MPa (d) 37.5 MPa

**Ans. (a) :** $\tau_{max} = \frac{pd}{4t}$

$d = 1m = 10^3$ mm

$t = 1$ mm

$p = 0.2$ MPa

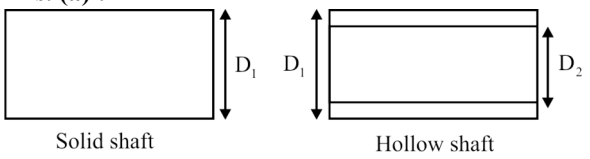
$$\tau_{max} = \frac{0.2\times10^3}{4\times1}$$

$= 50$ MPa

**111. Two circular shafts of same length, weight and material are compared for strength. The first one is a solid shaft and the other is a hollow shaft of outer to inner diameter ratio as 2. The ratio of the torque carrying capacity of the hollow shaft to solid shaft considering the shear stress criterion alone is _____.**

(a) 15/16 (b) 16/17
(c) 16/15 (d) 17/16

**Ans. (a) :**




$$\frac{D_1}{D_2} = 2$$

$$\tau_{solid} = \frac{J_{solid}T_1}{R} = \frac{\frac{\pi}{32}D_1^4T_1}{D^{1/2}}$$

$$\tau_{hollow} = \frac{J_{hollow}T_2}{R} = \frac{\frac{\pi}{32}\left(D_1^4 - D_2^4\right)T_2}{D^{1/2}}$$

$$\tau_{solid} = \tau_{hollow}$$

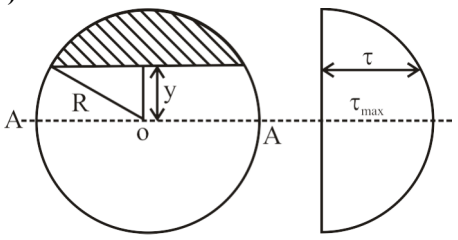
$$\frac{T_2}{T_1} \Rightarrow \frac{D_1^4}{\left(D_1^4 - D_2^4\right)} \Rightarrow \frac{D_1^4}{D_1^4 - \left(\frac{D_1}{2}\right)^4} = \frac{16}{15}$$

$$\frac{T_1}{T_2} = \frac{15}{16}$$

**112. For a circular cross section subjected to shear force, the ratio of minimum shear stress to average shear stress is _____.**

(a) 1.0 (b) 2.0
(c) 1.5 (d) 4/3

**Ans. (d) :**




Shear stress $'\tau' = \frac{FA\bar{y}}{Ib} = \frac{F}{3I}\left(R^2 - y^2\right)$

■ The neutral axis, the shear stress is maximum

$$\tau_{max} = \frac{F}{3I}\times R^2$$

$$I = \frac{\pi D^4}{64} = \frac{\pi R^4}{4}$$

$$\tau_{max} = \frac{F}{3}\times R^2 \times \frac{4}{\pi R^4} = \frac{4}{3}$$

Average shear stress is $\tau_{avg} = \dfrac{\text{Shear force}}{\text{Area}} = \dfrac{F}{\pi R^2}$

$$\tau_{max} = \frac{4}{3} \times \tau_{avg}$$

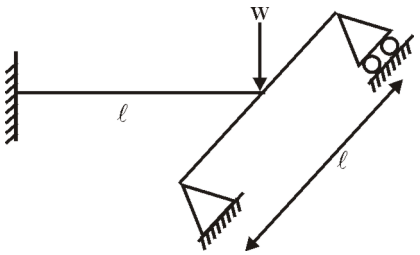
$$\boxed{\frac{\tau_{max}}{\tau_{avg}} = \frac{4}{3}}$$

**113. A cantilever beam of span L and flexural rigidity EI carries a point load W, vertically downwards at its free end. The free end of the beam is resting at the centre of another simply supported beam of span L and flexural rigidity EI. The support reactions for the simply supported beam are _____ and _____.**

(a) W/4, W/4 (b) W/2, W/2

(c) 8W/17, 8W/17 (d) 8W/3, 8W/3

**Ans. (c) :**




$$\frac{w\ell^3}{3EI} - \frac{RL^3}{3EI} = \frac{RL^3}{48EI}$$

$$\frac{w\ell^3}{EI} - \frac{RL^3}{16EI} = \frac{RL^3}{EI}$$

$$w = \frac{R}{16} + R \qquad R = \frac{16w}{17}$$

$$R_A = \frac{8}{17}w, \; R_B = \frac{8}{17}w$$

**114. A simply supported beam of span L and flexural rigidity EI carries a UDL of intensity w/unit run all along its span. The beam is supported at its centre by a linear spring of stiffness k. The force carried by the spring is _____.**

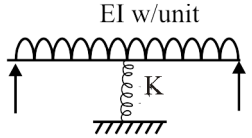
(a) $5\ kwL^4/(kL^3 + 8EI)$

(b) $5\ kwL^4/8(kL^3 + 48EI)$

(c) $5\ wL^4/8(kL^3 + 8EI)$

(d) $5\ wL^4/8(kL^3 + 48EI)$

**Ans. (b) :**




$F = k\delta$

Deflection $\delta = \dfrac{5w\ell^4}{384EI} - \dfrac{F\ell^3}{48EI}$

$$\delta = \frac{5w\ell^4 - 8F\ell^3}{384EI}$$

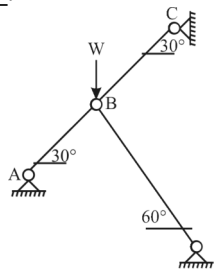
$$\frac{F}{K} = \frac{5w\ell^4 - 8F\ell^3}{384EI}$$

$$\frac{F}{k} = \frac{5w\ell^4}{384EI} - \frac{8F\ell^3}{384EI}$$

$$F\left\{\frac{1}{K} + \frac{8\ell^3}{384EI}\right\} = \frac{5w\ell^4}{384EI}$$

$$F = \frac{5kw\ell^4}{8\left[k\ell^3 + 48EI\right]}$$

**115. The three member plane truss A-B-C-D, shown in Fig. supports a vertical load W at B. The magnitude of the force carried by member BD is _____.**



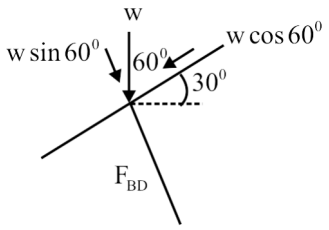

(a) W (compressive)

(b) 0.5 W (compressive)

(c) $\left(\sqrt{3}/2\right)$ W (compressive)

(d) 0.5 W (tensile)

**Ans. (c) :**




$F_{BD} = W \times \sin 60$

$$= \frac{W\sqrt{3}}{2} \text{ (compressive )}$$

**116. The symmetry of flexibility matix is due to ___.**

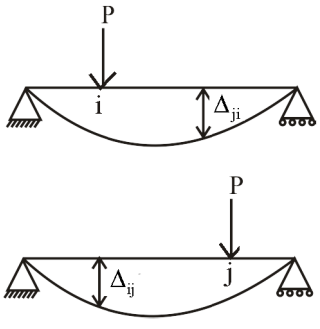
(a) Betti's theorem

(b) Maxwell's reciprocal theorem

(c) Eddy's theorem

(d) Castigliono's theorem

**Ans. (b) : Maxwell's reciprocal theorem :** Symmetry of flexibility matrix-



According to Maxwell's reciprocal theorem-

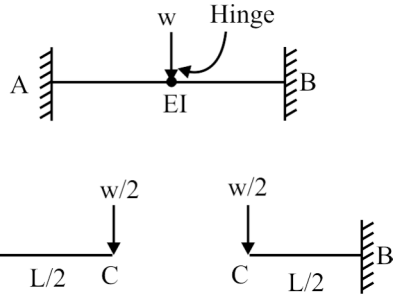
■ The deflection of point j($\Delta_{ji}$) due to a force P applied at point i, is numerically equal to the deflection of point i($\Delta_{ij}$) due to the force P applied at point j, where the deflection are measured along the load direction $\Delta_{ij} = \Delta_{ji}$

**117. A fixed beam of span L and uniform flexural rigidity EI carries a vertical downward load W at its mid span. If a hinge is introduced in the beam at the location of the load, the deflection under the load is _____.**

(a) $WL^3/12EI$ (b) $WL^3/24\ EI$

(c) $WL^3/16EI$ (d) $WL^3/48\ EI$

**Ans. (d) :**

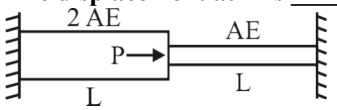



No the deflection will be –

$$\Delta_c = \frac{\frac{w}{2}(L_1)^3}{3EI} = \frac{\frac{w}{2}(L_2)^3}{3EI}$$

$$\frac{wL^3}{2\times 24EI} = \frac{wL^3}{48EI}$$

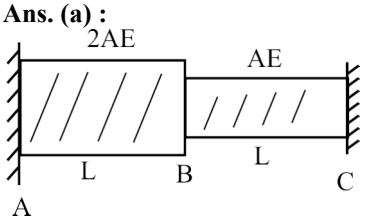
**118. A stepped bar A-B-C of total length 2L carries an axial load P at B as shown in Fig.. Axial rigidity of segment AB is 2 AE and that of BC is AE. The displacement at B is _____.**




(a) PL/3 AE (b) 0

(c) PL/2 AE (d) PL/AE

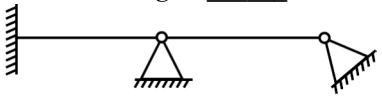
**Ans. (a) :**




$$\Delta_{BC} = \frac{AE}{2AE+AE}\times\frac{PL}{AE} = \frac{PL}{3AE}$$

$$\Delta_{AB} = \frac{2AE}{2AE+AE}\times\frac{PL}{AE} = \frac{2PL}{3AE}$$

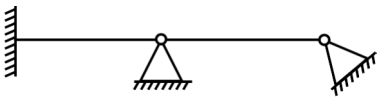
**119. The static indeterminacy for the continuous beam shown in Fig. is _____.**



(a) 6 (b) 3

(c) 2 (d) 4

**Ans. (d) :**



Static indeterminacy for beam will be

= number of total reaction – number of equilibrium equation

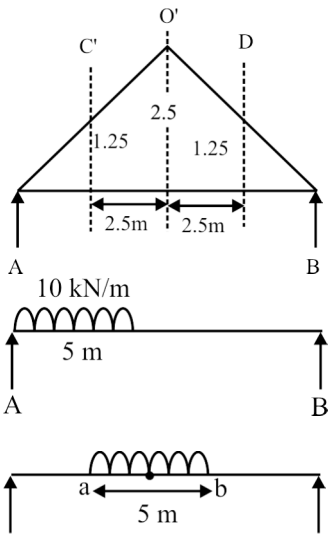
= 7–3 = 4

**120. A simply supported girder of span 10m is traversed by a 5m long UDL segment of intensity 10 kN/m. The magnitude of absolute maximum moment in the beam is ____.**

(a) 93.75 kN.m (b) 98.75 kN.m

(c) 90.25 kN.m (d) 88.75 kN.m

**Ans. (a) :**




Maximum bending moment will be at centre of span when $\left(\frac{a}{b} = \frac{a'}{b'}\right)$

Co-ordinate of maximum absolute bending moment

$$= \frac{ab}{\ell} = \frac{5\times 5}{10} = 2.5$$

So the C'C will be

$$= \frac{2.5}{5}\times 2.5 = 1.25$$

Now the bending moment absolute maximum

Area of deflected co-ordinate X-magnitude of load

$$= 2\times\left(\frac{2.5+1.25}{2}\right)\times 2.5\times 10$$

$$= 10\times 2.5(3.75)$$

$$= 25\times 3.75 = 93.75\ \text{kN-m}$$

**121. For a simply supported girder, EUDLL is defined as the intensity of the UDL acting on full span whose _____.**

(a) intensity is larger than all other loads acting on the span.

(b) moment diagram is above the bending moment moment caused by all loads on the span.

(c) bending moment diagram just envelops bending moment caused by all loads on the span.
(d) intensity is larger than the sum of intensities of all loads on the span.

**Ans. (c) :** For a simply supported girder, EUDL (Equivalent Uniform Distributed Load) is defined as the intensity of the UDL acting on full span whose bending moment diagram just envelops bending moment caused by all loads on the span.

**122. A simply supported beam of span L and uniform flexural rigidity EI is subjected a pres-stressing force of magnitude P, at an eccentricity e below the neutral layer, parallel to the axis of the member. The displacement of the beam at a location L/4 from the support is _____.**

(a) $PeL^2/4EI$ (b) $PeL^2/8EI$
(c) $PeL^2/2EI$ (d) $PeL^2/EI$

**Ans. (b) :**

Moment due to eccentricity = Pe
Than the deflection at the centre will be

$$\Delta_{centre} = \int_{d}^{\ell/2} \frac{Pexdx}{EI} = \frac{Pe\ell^2}{4EI}$$

Then deflection at $\frac{\ell}{4}$ will be from similar triangle rule.

$$\frac{\frac{\ell}{2}}{\frac{\ell}{4}} = \frac{\frac{Pe\ell^2}{4EI}}{\delta'}$$

$$\delta' = \frac{Pe\ell^2}{8EI}$$

**123. For a member subjected to loads on its span, equivalent joint loads are determined in such a way that _____.**

(a) The member end actions caused by equivalent joint loads are same as those caused by loads on the member.
(b) The equivalent joint cause same of nodal displacements as caused by the member loads.
(c) The equivalent joint loads together with member loads ensure the equilibrium of the structure.
(d) The equivalent joint loads together with member loads ensure the compatibility of displacements for the structure.

**Ans. (a) : Equivalent joint loads–** Structural analysis by using flexibility and stiffeners method requires that loads must act only at joints.

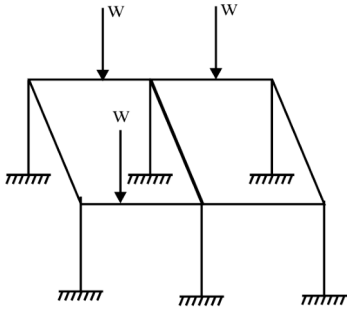
- Thus, loads acting on the members (i.e. loads that are not acting at the joints) must be replaced by equivalent loads acting at joints).
- The loads that are determined from loads on the member are called equivalent joint loads.
- Equivalent joint loads are added to the actual joint loads to get combined joint loads.
- Analysis carried out for combined joint loads.
- Combined joint loads can be evaluated in such a manner that the resulting displacement of the structure are same as the displacements, produced by the actual loads.
- This is achieved through the use of fixed end actions to get equivalent joint loads.

**124. An orthogrid structure is one which satisfies following combinations of conditions _____.**

i. A two dimensional framed structure consisting of members oriented in different directions.
ii. Loading plane perpendicular to the plane of structure.
iii. Member end actions are the axial forces, shear forces and bending moments.
iv. Angle between members is always 90°.
v. Member end actions are the shear forces, torques and bending moments.

(a) i, ii, iii, iv, v (b) i, ii, iii, iv
(c) ii, iii, iv, v (d) i, ii, iv, v

**Ans. (d) :**




**An orthogrid structure–** Orthogrid structures are those have rectangular depending instead of triangular like tress.

- These two dimension formed structure consisting of members oriented is different direction.
- Loading planes are perpendicular to the plane of structure.
- Members end action are the shear forces and bending moments and torques.
- As it has name ortho, so the angle between members are pointed always at $90^0$.

**125. In the stiffness matrix method, the boundary conditions are needed to avert**

(a) divergent solution
(b) singularity
(c) irrational solution
(d) zero displacement vector

**Ans. (b) : Properties of stiffeners matrix–**

- Stiffeners matrix is a square matrix order n × n, where n is equal to number of element in row and column.
- Stiffener matrix is symmetrical matrix hence

$$S_{ij} = S_{ji}$$

- $S_{ij}$ = represents action due to unit positive displacement and while other displacements are zero.
- $S_{ij}$ is principle diagonal element
- Stiffener matrix does exit for unstable structure
- Stiffeners matrix is non-singular matrix [S] is not equal to 0 for stable structure.
- $S_{ij}$ is the action at joint due to unit value of displacement of J point.

**126. In the direct element method of structural analysis, the transformation matrices are needed for following combinations of actions _____.**

i. They facilitate the transformation of element stiffness and element load matrices to global stiffness and global load matrices respectively.
ii. They are useful for determining the displacement matrices.
iii. They facilitate the conversion of resultant displacement matrices into member oriented displacement matrices.
iv. The are useful for determining member end actions.

(a) i, iii (b) ii, iii
(c) iii, iv (d) i, ii

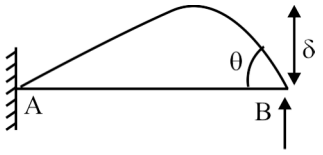
**Ans. (a) :** In the direct element method of structural analysis, the transformation matrices are needed for combinations of actions –
- They facilitate the transformation of element stiffness and element load matrices to global stiffness and global load matrices respectively.
- They facilitate the conversion of resultant displacement matrices into member oriented displacement matrices.

**Direct stiffness method–** It is a matrix method that makes use of the members stiffeners relations for computing member forces and displacement in structure.

**127. For a fixed beam of span L and uniform flexural rigidity EI, with both supports at same level, the support at right rotate by α radians anti clockwise. The support reactions at left and right supports (assuming upward displacement and anti clockwise rotations as positive) respectively are ____.**

(a) $6EI\ \alpha/L^2$, $2EI\ \alpha/L$, $6EI\ \alpha/L^2$, $2EI\ \alpha/L$
(b) $6EI\ \alpha/L^2$, $4EI\ \alpha/L$, $-6EI\ \alpha/L^2$, $2EI\ \alpha/L$
(c) $6EI\ \alpha/L^2$, $2EI\ \alpha/L$, $-6EI\ \alpha/L^2$, $4EI\ \alpha/L$
(d) $6EI\ \alpha/L^2$, $2EI\ \alpha/L$, $-6EI\ \alpha/L^2$, $2EI\ \alpha/L$

**Ans. (c) :**




$$M_{FBA} = \frac{6EI\delta}{\ell^2} = M_{FAB}$$

if fixed having Δ deflection

$$\delta_{AB} = \frac{2EI\delta}{\ell}, \qquad \Delta_{BA} = \frac{4EI\delta}{\ell}$$

$$M_{BA} = M_{AB} = M_{FAB} + \frac{2EI}{L}\left(2\theta_A + \theta_B + \frac{3\delta}{L}\right)$$

$\theta_A = \theta_B = \theta$ (for fixed beam)

$M_{AB} = 0$ if there is no external moment the

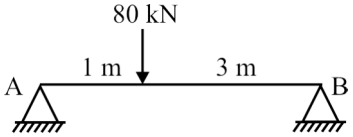
$$M_{FAB} = \frac{2EI}{L}\left(+\frac{3\delta}{L}\right)$$

$$M_{FAB} = \frac{-6EI\delta}{\ell^2} = M_{FBA}$$

**128. Using slope deflection method, the end rotations at A and B for a horizontal member 4m long with flexural rigidity EI are found as EI/1000 and –EI/2000 respectively. The beam carries a 80 kN vertical downward load of 1m from A. The support moments at A and B are (assuming clockwise moments and rotations as positive) taking EI = 5 × $10^4$ kN.m are _____ respectively.**

(a) +30 kN.m –15 kN.m
(b) –45 kN.m +15 kN.m
(c) +7.5 kN.m +15 kN.m
(d) +15 kN.m –15 kN.m

**Ans. (b) :**




$$M_{AB} = M_{F_{AB}} + \frac{2EI}{L}\left(2\theta_A + \theta_B - \frac{3\delta}{L}\right)$$

$$M_{F_{AB}} = -\frac{80\times1\times3^2}{4^2} = -45\ \text{kNm}$$

$$M_{F_{AB}} = -\frac{80\times3\times1^2}{4^2} = 15\ \text{kNm}$$

$$M_{AB} = -45 + \frac{2EI}{4}\left(2\theta_A - \frac{\theta_A}{2}\right) \qquad \left[\theta_B = -\frac{\theta_A}{2}\right]$$

$$= -45 + \frac{EI}{2}\left(\frac{3}{2}\theta_A\right) = -45 + \frac{3}{4}\times\frac{\left(5\times10^4\right)^2}{1000}$$

$$= -43.125 \approx 45\ \text{kNm}$$

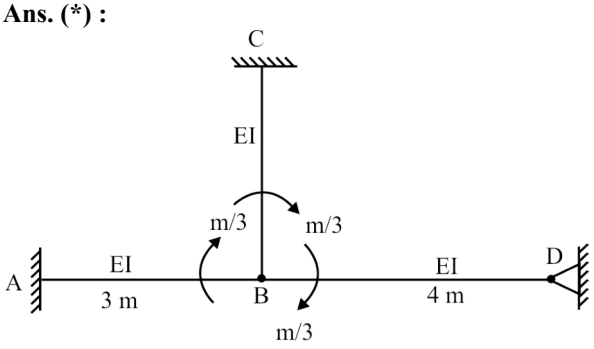
$$M_{BA} = 15 + \frac{2EI}{4}\left(2\theta_B + \theta_A\right)$$

$$M_{BA} = 15 + \frac{2EI}{4}\left(\frac{-2\theta_A}{2} + \theta_A\right) = 15\ \text{kNm}$$

**129. Three members AB, BC and BD 3m, 3m, and 4m long each, with uniform flexural rigidity are rigidly connected at B. If the supports at A and C are fixed while at D it is hinged the distribution factors in directions of BA, BC and BD at the joint B are _____ respectively.**

(a) 3/11, 4/11, 4/11 (b) 4/11, 3/11, 4/11
(c) 4/11, 4/11, 3/11 (d) 1/3, 1/3/ 1/3

**Ans. (*) :**




$$k_{BA} = \frac{4EI}{\ell} = \frac{4EI}{3},\ k_{BD} = \frac{3EI}{\ell} - \frac{3EI}{4}$$

$$k_{BC} = \frac{4EI}{\ell} = \frac{4EI}{3}$$

$$DF_{BA} = \frac{4EI/3}{\frac{4EI}{3}+\frac{4EI}{3}+\frac{3EI}{4}} = \frac{4/3}{41/12} = \frac{16}{41}$$

$$DF_{BC} = \frac{4EI/3}{\frac{4EI}{3}+\frac{4EI}{3}+\frac{3EI}{4}} = \frac{16}{41}$$

$$DF_{BD} = \frac{3EI/4}{\frac{4EI}{3}+\frac{4EI}{3}+\frac{3EI}{4}} = \frac{9}{41}$$

**130. In a three hinged parabolic arch, the third hinge is located at the crown. The beam moment coinciding with H-moment indicates _____**

(a) Arch all along carries only radial shear.
(b) Arch all along carries only beam moment.
(c) Arch all along carries only normal thrust.
(d) Arch all along carries only H-moment.

**Ans. (c) :** In a three hinged parabolic arch, the third hinge is located at the crown. The beam moment coinciding with H-moment indicates arch all along carries only normal thrust.

- Arch is free from bending moment everywhere on span.

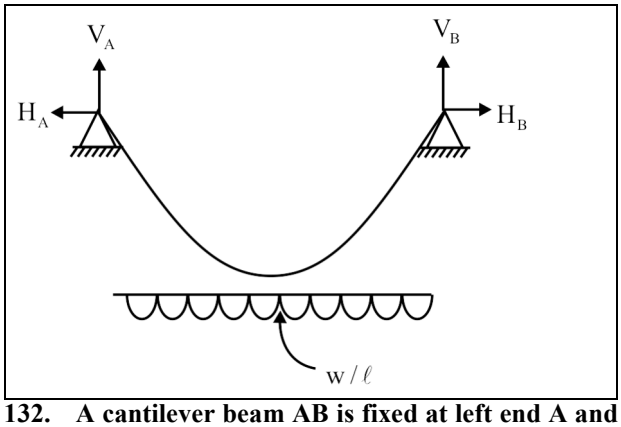
**131. A cable carrying a load of w/unit run is stretched between supports L apart. The supports are at the same level and the central dip is h. The greatest and least tensions in the cable are _____.**

(a) $wL/2,\ wL^2/8h$
(b) $\sqrt{\left((wL/2)^2+\left(wL^2/8h\right)\right)},\ wL^2/16h$
(c) $\sqrt{\left((wL/2)^2+\left(wL^2/8h\right)\right)},\ wL^2/8h$
(d) $\sqrt{\left((wL/2)^2+\left(wL^2/2h\right)\right)},\ wL^2/8h$

**Ans. (c) : Greatest tension in the cable–**

$$\sqrt{H_A^2+V_A^2} = \sqrt{\left(\frac{wL}{2}\right)^2+\frac{wL^2}{8h}}$$

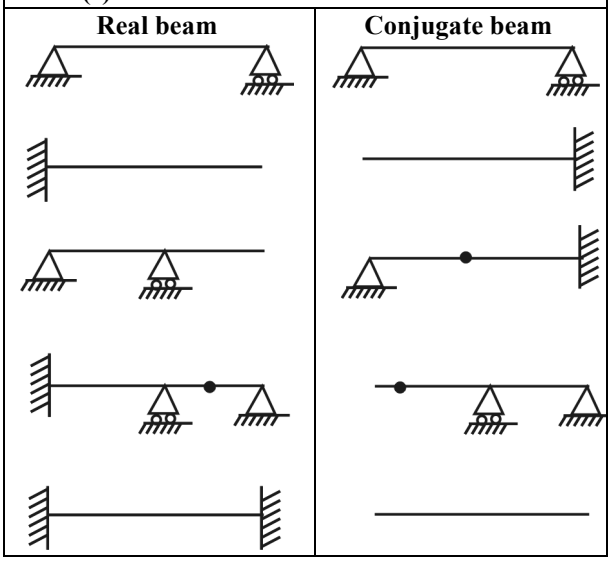
Least tension in the cable, $\frac{wL^2}{8h}$




**132. A cantilever beam AB is fixed at left end A and free at B. The corresponding conjugate beam will supports _____.**

(a) fixed at A and free at B
(b) free at A and simple support at B
(c) free at A and fixed at B
(d) simple supports at both A and B

**Ans. (c) :**




**133. In a steel tension member, it's maximum strength in case of adequately designed bolted connections is governed by _____.**

(a) The slenderness of the member
(b) The tensile strength of net sectional area of the member.
(c) The strength of the bolted connection.
(d) The section modulus of the member

**Ans. (b) :** In a steel tension member, it's maximum strength in case of adequately designed bolted connections is governed by the tensile strength of net sectional area of the member.

The net sectional area of a bolt shall be taken as the area of the threaded part.

- Net area of the bolt at threads

$$A_{nb} = \left(0.78\frac{\pi}{4}d^2\right)$$

**134. The principal reason for adding stiffeners to the web of a steel beam is to _____.**

(a) Increase its moment carrying capacity.
(b) Reduce the deflection of the beam.
(c) Increase the stiffness of the web.
(d) Improve aesthetics.

**Ans. (c) :** The principal reason for adding stiffeners to the web of a steel beam is to increase the stiffness of the web.
Transverse stiffeners are provided to increase buckling resistance of web due to inclined stress due to shear, it is provided vertically along the span.

**135. In the built-up steel columns, the lacing is provided _____.**

i. to keep all individual sections together.
ii. to take-up lateral shear to an extent of 2.5% of axial force.
iii. to increase the bending strength of the section.
iv. to increase the compressive strength of the section.

(a) i, iii (b) ii, iii
(c) iii, iv (d) i, ii

**Ans. (d) :** Flat or angle section are normally used as lacing. The purpose of lacing is to hold the various part of a column straight, parallel, at a correct distance apart and to equalise the stress distribution between its various parts.
The lacing for compression members should be proportional to resist a total transverse shear equal to 2.5% the axial force in the column.

**136. The list of principal components of a plate girder is _____.**

i. Top and bottom flange plates and web plates.
ii. Horizontal stiffeners, Intermediate and bearing stiffeners.
iii. Cleat angles and seat angles.
iv. Web splicing and flange splicing.

(a) i, iii, iv (b) i, ii, iii
(c) i, ii, iv (d) ii, iii, iv

**Ans. (c) :** The list of principal components of a plate girder is –
- Top and bottom flange plates and web plates.
- Horizontal stiffeners, Intermediate and bearing stiffeners.
- Web splicing and flange splicing.

The depth of plate girder is kept as 1/8 to 1/10 of span.
Economical depth of plate girder for riveted–

$$D = 1.1\sqrt{\frac{M}{\sigma_{bt} \times t_w}}$$

**137. The requirements to be satisfied in the design of a gantry girder are that it has to withstand effects due to _____.**

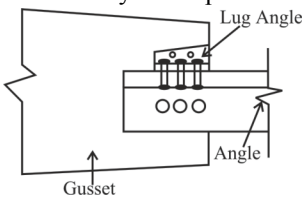
(a) Moving loads, impact effects and fatigue.
(b) Moving loads, longitudinal loads, lateral loads, impact effects and fatigue.
(c) Lateral loads, longitudinal loads, impact effects and fatigue.
(d) Dynamic loads, longitudinal loads, impact effects and fatigue.

**Ans. (b) :** The requirements to be satisfied in the design of a gantry girder are that it has to withstand effects due to moving loads, longitudinal loads, lateral loads, impact effects and fatigue..
Gantry girder are unique in themselves, first it is different from usual becomes in buildings it is generally laterally unsupported except at the columns. These are subjected to force.
1. The reaction from the crane girder acting vertically downwards
2. The longitudinal thrust, due to starting or stopping of crane, acting in the longitudinal direction.
3. The lateral thrust, due to starting or stopping of the crack acting horizontal normal to the gantry girder.

**138. The Lug angle is a member which _____.**

(a) is connected to the main tension member to transfer the tensile force economically to the point.
(b) is connected to the main tension member for erection purpose.
(c) is connected to the main tension member to increase its strength locally.
(d) is connected to the main tension member to increase its stiffness.

**Ans. (a) :** The Lug angle is a member which is connected to the main tension member to transfer the tensile force economically to the point..




**Lug angle–** It is short length of an angle between section used as a joint to connect the outstanding leg of a member, by reducing the length of the joint.
- The channels get twisted symmetrically with regard to its axis parallel to web.
- Twisting takes place about the axis which has less moment or inertia or least radius of gyration.

**139. The expression working out the thickness of slab base, $\sigma_{bs}$ = permissible bending stress in slab base, a, b = longer and shorter projections of the slab base edge to the column member, Posions ratio = 0.25.**

(a) $t = \left((3w/\sigma_{bs})(a - b/4)\right)^{1/2}$
(b) $t = \left((3w/\sigma_{bs})(a^2 - b^2/4)\right)^{1/2}$
(c) $t = \left((3w/\sigma_{bs})(a^2 - b^2)\right)^{1/2}$
(d) $t = \left((3w/\sigma_{bs})(a - b^2/4)\right)^{1/2}$

**Ans. (b):** The thickness of rectangular slab base as per IS 800 : 1984–

$$t = \sqrt{\frac{3w}{\sigma_{bs}}\left(a^2 - \frac{b^2}{4}\right)}$$

Where,

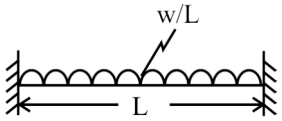
t = thickness of slab base (mm)
w = the pressure or loading on the under side of the base (MPa)
a = the greater projection of the plate beyond the column (mm) = max ($\alpha$, $\beta$)
b = The lesser projection of the plate beyond the column (mm) = min ($\alpha$, $\beta$)

$\sigma_{bs}$ = The permissible bending stress in slab bases.
= 165 MPa for flanged beam
= 185 MPa for solid plate

**140. A fixed beam of span L carries a UDL of intensity w/unit run. If the plastic moment of the beam section is $M_P$, collapse occur when _____ number of plastic hinges are formed and wL = _____.**

(a) 2 and 8 $M_P$/L (b) 3 and 8$M_P$/L
(c) 3 and 16 $M_P$/L (d) 2 and 16 $M_P$/L

**Ans. (c) :**




Degree of static indeterminacy
$D_S = r - S$
$= 4 - 2 = 2$
Number of plastic hinges, $N = D_S + 1$
$N = 2 + 1 = 3$
As per principle of vertical work done–
$w_e = w_i$
$w_u \times \delta = M_p\theta + M_p(\theta + \theta) + M_p\theta$

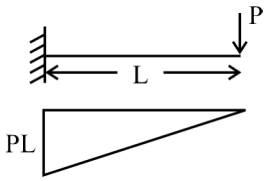
$$\frac{wL}{2} \times \frac{L}{2}\theta = 4M_p\theta$$

$$wL = \frac{16M_p}{L}$$

**141. A cantilever beam of span L carries a point load W at its free end. Plastic moment of the section from support to middle of span is 1.5 $M_P$ and from there to free end is $M_P$. Collapse occurs when W = _____.**

(a) 1.5 $M_P$/L (b) $M_P$/L
(c) 2$M_P$/L (d) 0.75 $M_P$/L

**Ans. (a) :**




$$w = \frac{1.5M_p}{L}$$

Where, $M_p$= Plastic moment = 1.5 $M_p$ .

**142. According to IS: 456-2000, in the design of isolated RCC column footing using RCC with an effective depth d, the critical sections to be checked are _____.**

(a) Bending moment at the face of the column, one-way shear at d/2 away from the face of column and two-way shear at d around the column.
(b) Bending moment at the face of the column, one-way at d away from the face of column and two-way shear at d around the column.
(c) Bending moment at d/2 away from the column, one-way shear at d away from the face of column and two-way shear at d/2 around the column.
(d) Bending moment at the face of the column, one-way shear at d away from the face of column and two-way shear at d/2 around the column.

**Ans. (d) :** The design depth of footing is based on the following three criteria :
(i) Depth based on maximum BM at the face of column–
For illustration consider square footing of width B resting on uniform soil pressure 'p'. The maximum BM is

$$M = \frac{pB(B-b)^2}{8}$$ Where b = column size

The depth is given as $d = \sqrt{\frac{M}{R}}$

Where R is designed constant
(ii) Maximum one way shear at a distance of d from the face of the column.
d – effective depth of footing

$$\tau_v = \frac{V}{bd} < \text{permissible one way shear is footing.}$$

(iii) Based on punching shear stress at a distance of d/2 from face of column, d : effective depth
Punching shear stress

$$= \frac{\text{Net punching force}}{\text{Perimeter} \times d} < \text{permissible punching shear stress}$$

**143. Minimum amount of high strength deformed bar reinforcement used in solid slabs shall not be less than _____ of the total cross sectional area of the slab according to IS: 456-2000.**

(a) 0.15% (b) 0.12%
(c) 0.20% (d) 0.25%

**Ans. (b) :** According to IS code 456:2000
Minimum Reinforcement is either direction of slabs shall not be less than 0.15 percent of cross sectional area in case of mild steel and 0.12 percent of cross sectional area in case of High deformed bar.

**144. When the depth of the web of beam is more than _____mm, side face reinforcement of _____% of the web area is needed to be provided according to IS: 456-2000.**

(a) 600, 0.1% (b) 750, 0.15%
(c) 600, 0.15% (d) 750, 0.1%

**Ans. (d) :** When a beam is subjected to torsion is depth of the beam is more than 450 mm or for beam not subjected to torsion if the depth of web exceed 750 mm then side face reinforcement equal to 0.1% of cross sectional area and is equally distributed on both faces of the beam.

**145. Maximum spacing of vertical shear reinforcement measured along the axis of the RCC beam shall not exceed _____, whichever is lowest.**

(a) 0.75d or 300 mm (b) 0.75d or 400 mm
(c) 0.50d or 250 mm (d) 0.5d or 300 mm

**Ans. (a) :** According to IS 456-2000–
Maximum spacing of shear reinforcement –
(i) 0.75 d for vertical stirrups
(ii) d for inclined stirrups at $45^0$.
(iii) In no case shall be spacing exceed 300 mm
Where,
d = effective depth.

**146. The bearing stress check for column footing in limit state design specified that the value _____ subjected to a maximum of 2 multiplied by bearing stress _____ shall be more than the compressive stress at the base of the column. $A_1$ = supporting area for bearing of footing, which in sloped or stepped footing maybe taken as the area of the lower base of the largest frustum of a pyramid or cone contained wholly within the footing and having for its upper base, the area actually loaded and having side slope of one vertical to two horizontal; and, $A_2$ = loaded area at the column base.**

(a) $\sqrt{(A_1/A_2)}$, 0.30 $f_{ck}$ (b) $\sqrt{(A_1/A_2)}$, 0.65 $f_{ck}$
(c) $\sqrt{(A_1/A_2)}$, 0.45 $f_{ck}$ (d) $\sqrt{(A_1/A_2)}$, 0.60 $f_{ck}$

**Ans. (c) :** The bearing stress check for column footing in limit state design specified that the value $\sqrt{(A_1/A_2)}$, subjected to a maximum of 2 multiplied by bearing stress 0.45 $f_{ck}$ shall be more than the compressive stress at the base of the column.
Where,
$A_1$ = supporting area for bearing of footing, which in sloped or stepped footing maybe taken as the area of the lower base of the largest frustum of a pyramid or cone contained wholly within the footing and having for its upper base, the area actually loaded and having side slope of one vertical to two horizontal.
$A_2$ = loaded area at the column base.
$f_{ck}$ = Characteristic stress

**147. The functionality of a wall, retaining wall and a shear wall in order is _____.**

i. resist predominantly vertical loads
ii. resist lateral loads perpendicular to the plane of wall
iii. resist lateral loads in the plane of wall

(a) ii, iii, i (b) i, ii, iii
(c) iii, ii, i (d) ii, i, iii

**Ans. (b):** The functionality of a wall, retaining wall and a shear wall in order is –
- Resist predominantly vertical loads.
- It resist lateral loads perpendicular to the plane of wall
- Resist lateral loads in the plane of wall.

**148. Splicing of reinforcement in flexure members is taken-up at a location where bending moment is less than _____ the moment of resistance at that section and not more than _____ of bars are spliced at any particular section.**

(a) 75%, 50% (b) 50%, 75%
(c) 25%, 50% (d) 50%, 50%

**Ans. (d) :** Splicing of reinforcement in flexure members is taken-up at a location where bending moment is less than 50% the moment of resistance at that section and not more than 50% of bars are spliced at any particular section.

**149. The loss stress due to creep in steel in a pre-stress problem is given by the formula _____, where $\alpha$ = creep coefficient, $f_c$ = stress in concrete, $E_c$ = modulus of elasticity of concrete and $E_s$ = modulus of elasticity of steel.**

(a) $\alpha(f_c/E_c)\,E_s$ (b) $\alpha(f_c/2E_c)\,E_s$
(c) $\alpha(f_c/E_s)\,E_c$ (d) $2\alpha(f_c/E_c)\,E_s$

**Ans. (a) :** • Loss of stress due to creep

$$\boxed{\phi.m.f_c} = \alpha(f_c/E_c)\,E_s$$

Here,

$$m = \text{modular ratio} = \frac{E_s}{E_c}$$

$f_c$ = original prestress in concrete at the level of steel.

**150. The principal reason for adopting pre-stressing cable profiles in flexure members as parabolic is due to the fact that _____.**

(a) They need to resist both bending moments and shear force in members.
(b) The strength of pre-stressing cables is maximum in parabolic shapes only.
(c) The profile of moment caused by self weight of structure is parabolic and to counter this, the cable profile also needs to be parabolic.
(d) Its a regular practice.

**Ans. (c) :** The principal reason for adopting pre-stressing cable profiles in flexure members as parabolic is due to the fact that the profile of moment caused by self weight of structure is parabolic and to counter this, the cable profile also needs to be parabolic..
Pre-stressed concrete member posses better resistance to shear force due to effect of compressive stresses pressure of eccentric cable profile use of high strength concrete and freedom from cracks, contribute to improve durability under aggressive environmental conditions. It is effective in railways sleeper.

# Telanagana State Public Service Commission
# Assistant Executive Engineer Exam, 2015
## Solved Paper
**Date - 20.09.2015**

**1. The main constituent of cement which is responsible for initial setting of cement is :**
(a) di-calcium silicate
(b) tri-calcium silicate
(c) tri- calcium aluminate
(d) tri- calcium alumino faride

**Ans. (c) :** The main constituent of cement $C_3A$ responsible for initial setting of cement.
**Tricalcium aluminate ($C_3A$) –**
- 5-11% of cement
- It rapidly react with water
- High heat of hydration
- It is responsible for flash set.

**2. Which of the following cement is suitable for use in massive concrete structures such as large dams :**
(a) ordinary portland cement
(b) low heat cement
(c) rapid hardening cement
(d) sulphate resisting cement

**Ans. (b) :** Low heat cement is used in massive construction works like dam, abutments, retaining wall etc. where the rate at which the heat can be lost at the surface is slower than at which the heat is initially generated. Low heat cement has low rate of gain of strength but the ultimate strength is practically the same as that of OPC.

**3. Mean target strength [$f_t$] is given by (Where $f_{ck}$ is 28 day Characteristic compressive strength and 'S' is Standards deviation.)**
(a) $f_t = f_{ck} + 0.65s$ (b) $f_t = f_{ck} - 1.65s$
(c) $f_t = f_{ck} + 1.65s$ (d) $f_t = f_{ck} - 0.65s$

**Ans. (c) :** As per IS 456:2000, the following formula is used to calculate the target mean strength of concrete mix.

Target mean strength $(f_m) = f_{ck} + 1.65 \times \sigma$

Where, $\sigma$ = standard deviation,
$f_{ck}$ = characteristic compressive strength

**4. In which of the following directions, the strength of timber is maximum?**
(a) parallel to grains
(b) $45^0$
(c) perpendicular to grains
(d) same in all directions

**Ans. (a) :**
- Compression strength of the timber section is found to be maximum parallel to the grains (along the longitudinal direction)
- Tensile strength of the timber section is found to be 2 to 4 times greater than its compressive strength parallel to the grains (weak in joint making not using the tension member acts).

**5. Hydraulic lime is obtained by**
(a) burning of limestone
(b) burning of gravel
(c) adding water to quicklime
(d) calcination of pure clay

**Ans.(a):** Hydraulic lime is obtained by burning of kankar.
- Hydraulic lime has proportion of clay 5–30%
- It sets under water.

**6. Soundness of cement is tested by**
(a) Blaine's method
(b) Le chatelier method
(c) Autoclave method
(d) Blains and Lechatelier method

**Ans. (b) :**

| (Cement test) | (Apparatus) |
|---|---|
| (a) Fineness | Nurse and Blains |
| (b) Consistency | Vicat apparatus |
| (c) Soundness | Le-Chatelier apparatus |
| (d) Sp. gravity | Le-chatellier's flask |

**7. Solvent commonly used for oil paints is**
(a) tar (b) petrol
(c) acquaregia (d) turpentine

**Ans. (d) : : Oil paint–** This is the ordinary paint.
- This paint is cheap and easy to apply and it possesses good opacity and low gloss.
- It should not apply during humid and damp weather.

**Composition of oil paint–**
**Base–** White lead, red lead, oxide of iron, zinc white.
**Vehicle–** Linseed oil, tung oil, poppy oil, nut oil.
**Pigment–** Graphite, lamp block, ivory black, indigo, copper sulphate, zinc chrome.
**Solvent–** Turpentine (common), spirit, naptha, petroleum.
**Drier–** Litharge, lead acetate, red lead.
**Adulterants–** Barium sulphate, calcium carbonate, magnesium silicate and silica.

**8. Number of bricks required for one cubic metre of brick masonry is**
(a) 400 (b) 450
(c) 500 (d) 550

**Ans. (c) :** 500 bricks are required to construct 1 $m^3$ brick masonry wall.

$$\text{Number of bricks} = \frac{\text{Volume of wall}}{\text{Volume of 1 brick}}$$

Volume of 1 brick with mortar = 20cm × 10cm × 10cm
$= 0.2 \times 0.1 \times 0.1 = 2 \times 10^{-3}\ m^3$.
Hence, volume of wall = 1 $m^3$

$$\text{Number of bricks} = \frac{1}{2 \times 10^{-3}} = 500$$

**9. Admixtures which cause early setting and hardening of concrete are called**

(a) workability admixtures
(b) accelarators
(c) retarders
(d) air entraining agents

**Ans. (b) :** Admixtures which cause early setting and hardening of concrete are called accelarators.
Accelarators– $CaCl_2$, silicate (sodium silicate) formaldihyde, sodium cloride..... etc.

**10. Which of the following metamorphic rocks has the most weather resisting characteristics?**

(a) Marble (b) Quartzite
(c) Slate (d) Lime stone

**Ans. (b) : Quartzite** is a hard, non-foliated meta morphic rock which was originally sand stone which is converted into quartzite through metamorphic action. Quartzite has a very good weathering resistance, hardness & crushing strength.

**11. The ratio of strength of a solid shaft to hollow shaft for same material, same weight and same length is**

(a) 1.44 (b) 1.0
(c) 0.694 (d) 0.5

**Ans. (*) :** Let assume internal dia and external diameter of hollow shaft respectively D and X
For same length,

$$\frac{\pi}{4}D^2 = \frac{\pi}{4}\left(X^2 - D^2\right)$$

$$2D^2 = X^2$$

$$X = \sqrt{2}D$$

$$Z_S = \frac{I}{y} = \frac{\frac{\pi}{64}\times D^4}{D/2} = \frac{\pi}{32}\times D^3$$

$$Z_H = \frac{I}{y} = \frac{\pi}{64}\times\frac{\left(\sqrt{2}D^4 - D^4\right)}{\frac{\sqrt{2}D\times\sqrt{2}}{2\times\sqrt{2}}}$$

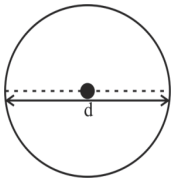
$$\text{Ratio } \frac{Z_S}{Z_H} = \frac{\frac{\pi}{32}\times D^3}{3\sqrt{2}\times\frac{\pi}{64}\times D^3} = \frac{2}{3\sqrt{2}} = 0.4714$$

**12. The ratio of the moment of inertia of a circular plate of diameter same as that of a side of a square plate is**

(a) less than one (b) more than one
(c) equal to one (d) equal to $3\pi/16$

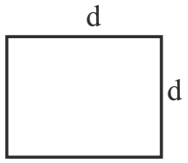
**Ans. (d) :**
Moment of inertia of circular plate ($I_c$)–

$$I_c = \frac{\pi}{64}d^4$$




Moment of inertia of square plate ($I_s$)–

$$I_s = \frac{d^4}{12}$$




$$\text{Ratio, } \frac{I_c}{I_s} = \frac{\frac{\pi}{64}d^4}{\frac{d^4}{12}}$$

$$\boxed{\frac{I_c}{I_s} = \frac{3\pi}{16}}$$

**13. The maximum twisting moment, a shaft can resist, is the product of the permissible shear stress and**

(a) moment of inertia
(b) polar moment of inertia
(c) modulus of rigidly
(d) polar modulus

**Ans. (d) :** For torsion equation,

$$\frac{T}{I_P} = \frac{\tau}{r} = \frac{G\theta}{\ell}$$

$$T = \frac{\tau}{r}\times J$$

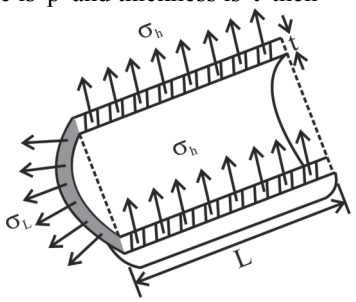
$$T = \tau.\frac{J}{r} = \tau.Z_p$$

$$= q\times Z_P$$

**14. The ratio of longitudinal stress to circumferential stress in the case of a thin cylindrical shell is**

(a) 1.5 (b) 2
(c) 1 (d) 0.5

**Ans. (d) :** Let the diameter of cylindrical shell be 'd' and pressure is 'p' and thickness is 't' then–




$$(\sigma_L) \text{ Longitudinal stress in shell} = \frac{pd}{4t}$$

$$(\sigma_h) \text{ Hoop stress in shell} = \frac{pd}{2t}$$

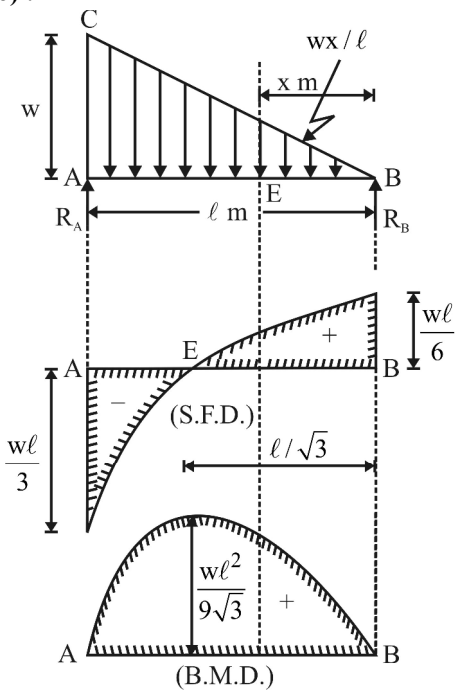
$$\frac{\text{Longitudinal stress}}{\text{Hoop stress}} = \frac{\frac{pd}{4t}}{\frac{pd}{2t}} = \frac{pd}{4t}\times\frac{2t}{pd} = \frac{2}{4} = 0.5$$

**15. The maximum bending moment for a simply supported beam subjected to uniformly varying load with zero intensity at the right end and w per meter length at the left end is**

(a) $w\ell^2/9$ (b) $w\ell^2/9\sqrt{3}$

(c) $w\ell^2/\sqrt{3}$ (d) $w\ell^2/2\sqrt{3}$

**Ans. (b) :**




$\sum M_A = 0$ mes,

$$R_B \times \ell = \left(\frac{1}{2} \times \ell \times W\right)\frac{\ell}{3}$$

$$\boxed{R_B = \frac{W\ell}{6}}$$

$$R_A + R_B = \frac{W\ell}{2}$$

$$\boxed{R_A = \frac{W\ell}{3}}$$

Where shear force change sign, there bending moment is maximum, so distance is where BM is maximum.

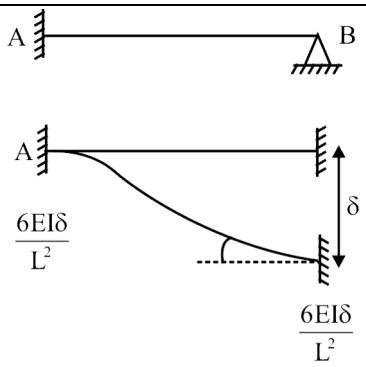
$$\boxed{M_E = \frac{w\ell^2}{9\sqrt{3}} = 0.064\, w\ell^2}$$

**16. The magnitude of fixed end moment for a span if one of the support sinks by δ is**

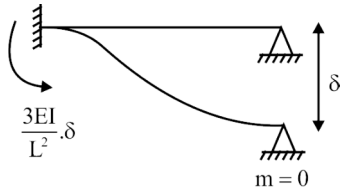
(a) $6EI\delta/L^2$ (b) $3EI\delta/L^2$

(c) $12EI\delta/L^2$ (d) $EI\delta/6L^2$

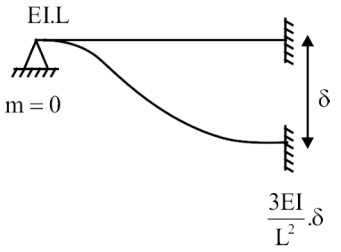
**Ans. (a) :** When both ends of the beam are fixed and one support settles down by the amount.




• When one end is fixed and other has pined are pin support settles down




• Fixed support settle down–

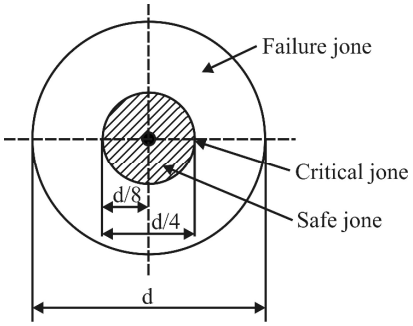



**17. For an eccentric load on a circular section, not to cause any tension in it, the eccentricity of loading should not exceed :**

(a) d/3 (b) d/4

(c) d/6 (d) d/8

**Ans. (d) :**




For an eccentric load on a circular section, not to cause any tension in it, the eccentricity of loading should not exceed d/8.

**For eccentricity–**

$$\frac{Pe.y}{I} \le \frac{P}{A}$$

$$\frac{e.(d/2)}{(\pi d^4/64)} \le \frac{1}{(\pi d^2/4)}$$

$$\boxed{e \le \frac{d}{8}}$$

**18. A spherical ball of volume $10^5$ mm$^3$ is subjected to a hydrostatic pressure of 90 Mpa. If the bulk modulus for the material is 180 kN/mm$^2$, the change in the volume of :**

(a) 50 mm$^3$ (b) 100mm$^3$
(c) 250 mm$^3$ (d) 500 mm$^3$

**Ans. (a) :** Given,
Volume of spherical ball, $V = 10^5$ mm$^3$
Pressure, P = 90 Mpa
Bulk modulus of the material
$K = 180$ kN/mm$^2$ $= 180 \times 10^3$ N/mm$^2$
change in the volume of ball, $\Delta V = ?$

$$K = \frac{\Delta P}{\frac{\Delta V}{V}}$$

$$180\times10^3 = \frac{90\times10^5}{\Delta V}$$

$$\Delta V = \frac{90\times10^5}{180\times10^3}$$

$$\Delta V = 50 mm^3$$

**19. For a column of length 'L', fixed at both ends, and flexural rigidity EI, the critical load is given by :**

(a) $\pi^2 EI/L^2$ (b) $\pi^2 EI/4L^2$
(c) $2\pi^2 EI/L^2$ (d) $\frac{4\pi^2 EI}{L^2}$

**Ans. (d) :** For both end fixed, $\frac{L}{2}$

Critical load = $\frac{4\pi^2 EI}{L^2}$

**20. The deflection at any point of a perfect frame can be obtained by applying a unit load at the joint in :**

(a) vertical direction (b) horizontal direction
(c) inclined direction
(d) the direction in which deflection is required

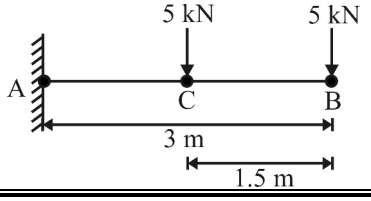
**Ans. (d) :** The deflection at any point of a perfect frame can be obtained by applying a unit load at the joint in the direction in which deflection is required.

**21. A cantilever beam of 3 m long carries a point load of 5 kN at its free end and 5 kN at its middle. The bending moment at the middle of the cantilever beam is :**

(a) 22.5 kN-metre (b) 30.0 kN- metre
(c) 15.0 kN-metre (d) 7.5 kN-metre

**Ans. (d) :** Bending moment from C-
$(B.M)_C = 5 \times 1.5$

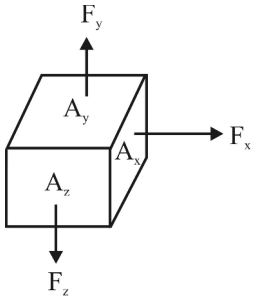
$$\boxed{(B.M)_C = 7.5\,kN-m}$$




**22. When the body is subjected to three mutually perpendicular stresses of equal intensity, the ratio of direct stress to the corresponding volumetric strain is known as :**

(a) modulus of elasticity
(b) modulus of rigidity
(c) bulk modulus
(d) poisson's ratio

**Ans. (c) :**




$$\frac{F_x}{A_x} = \sigma_x, \quad \frac{F_y}{A_y} = \sigma_y, \quad \frac{F_z}{A_z} = \sigma_z$$

$\sigma_x = \sigma_y = \sigma_z$ (Hydrostatics stress)
Initial volume, $V = \ell \times b \times h$

$$V' = (\ell + \Delta\ell)(b + \Delta b)(h + \Delta h)$$

$$\varepsilon_v = \frac{V' - V}{V} = \frac{\Delta V}{V}$$

$$\boxed{K = \frac{\sigma}{\varepsilon_v} = \frac{\text{Direct stress}}{\text{Volumetric strain}}}$$

**23. Which one of the following pairs is not correctly matched?**

(a) Lame's constant : Thick cylinder
(b) Macaulay's method : Deflection of beams
(c) Euler's method : Theory of columns
(d) Eddy's theorem : Torsion of shafts

**Ans. (d) :**

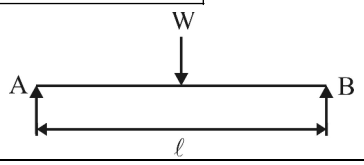
| | | |
|---|---|---|
| Lame's constant | – | Thick cylinder |
| Macaulay's method | – | Deflection of beam |
| Euler's method | – | Theory of column |
| Eddy's theorem | – | Related to arches |

**24. A Simply supported beam of span '$\ell$' carries a unit load at its centre. The strain energy in the beam is :**

(a) $\ell^3/96$ EI (b) $\ell^3/48$ EI
(c) $\ell^3/192$ EI (d) $\ell^2/96$ EI

**Ans. (a) :** A Simply supported beam of span '$\ell$' carries a unit load at its centre, strain energy,

$$\boxed{U = \frac{W^2\ell^3}{96EI} = \frac{\ell^3}{96EI}}$$



**25. The moment required to rotate the near end of a prismatic beam through unit angle, without translation, the far end being fixed is (with usual notations)**

(a) EI/L (b) 2 EI/L

(c) 3 EI/L (d) 4 EI/L

**Ans. (d) : For end is fixed–**

$$K = \frac{4EI}{L}$$

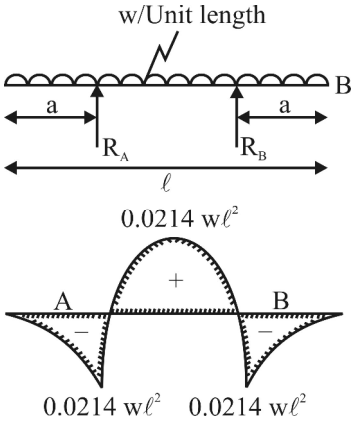
Far end is hinged $K = \frac{3EI}{L}$

**26. A simply supported beam which carries a uniformly distributed load has two equal overhangs. To have maximum B.M. produced in the beam the least possible, the ratio of the length of the overhang to the total length of the beam is :**

(a) 0.207 (b) 0.508

(c) 0.407 (d) 0.307

**Ans. (a) :**




$$R_A = R_B = \frac{w\ell}{2}$$

Hoging B.M. over the support = sagging B.M. at the centre

$$\frac{wa^2}{2} = -w.\frac{\ell}{2}\times\frac{\ell}{4} + \frac{w\ell}{2}\times\left[\frac{\ell}{2} - a\right]$$

$$\frac{wa^2}{2} = -\frac{w\ell^2}{8} + \frac{w\ell^2}{4} - \frac{w\ell a}{2}$$

$$\frac{wa^2}{2} = \frac{w\ell^2}{8} - \frac{w\ell a}{2}$$

$\ell^2 - 4\ell a = 4a^2$ (Add $4a^2$ in both sides)

$$\ell^2 + (2a)^2 - 4\ell a = 8a^2$$

$$(\ell - 2a)^2 = 8a^2$$

$$\ell - 2a = 2\sqrt{2}\,a$$

$$\ell = 2.828a + 2a$$

$$a = 0.207\ell$$

$$\boxed{a/\ell = 0.207}$$

$$\text{Maximum bending moment} = \frac{wa^2}{2} = \frac{w\times(0.207\ell)^2}{2}$$

$$= 0.0214\ w\ell^2$$

**27. A plane carrying normal stress accompanied by no shear stress is called**

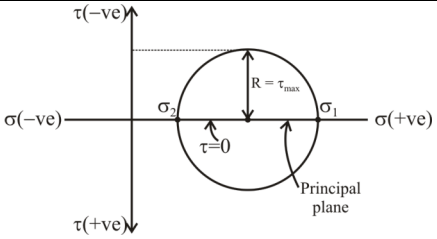
(a) plane of no shear stress

(b) principal plane

(c) normal plane

(d) shear plane

**Ans. (b) :** The shear stress on principal plane is zero but at the maximum shear stress plane principal stresses are not zero.




$$\text{Normal stress } \sigma_n = \sigma_x \cos^2\theta$$

$$\text{Shear stress } q = \frac{\sigma_x}{2}\sin 2\theta$$

**28. The ratio of strengths of solid to hollow shafts, both having outside diameter 'D' and hollow inside diameter 'D/2', in torsion, is :**

(a) 1/16 (b) 1/4

(c) 1/2 (d) 15/16

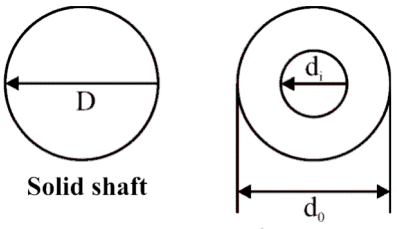
**Ans. (d) :** Given,

Dia of solid shaft = D

External dia of hollow shaft, $d_0 = D$

Internal dia of hollow shaft, $d_i = \frac{D}{2}$

$$\frac{\text{Strength of the hollow shaft}}{\text{Strength of the solid shaft}} = \frac{d_o^4 - d_i^4}{d_o \times D^3}$$



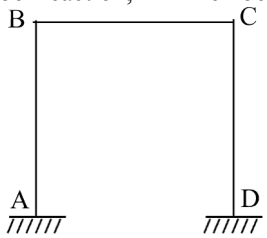

$$\frac{(T.S.M)_H}{(T.S.M)_S} = \frac{D^4 - \left(\frac{D}{2}\right)^4}{D\times D^3} = \frac{16D^4 - D^4}{16D^4} = \frac{15D^4}{16D}$$

$$\boxed{\frac{(T.S.M)_H}{(T.S.M)_S} = \frac{15}{16}}$$

**29. A single–bay, single-storeyed portal frame ABCD has its column ends fixed. If axial deformation is neglected, the kinematic indeterminacy is :**

(a) 3 (b) 2
(c) 6 (d) 4

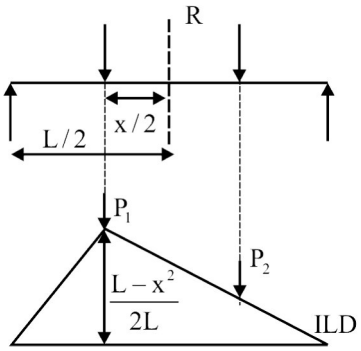
**Ans. (a) :** Axial deformation is not applicable then
$D_k = 3 \times n - R - m = 3 \times 4 - 6 - 3 = 3$
n = Number of joint
R = Number reaction, m = member




**30. The maximum bending moment under a particular point load among a train of point loads crossing a simply supported girder occurs at the location when that load is at :**

(a) at mid span
(b) so placed that load point and the C.G of the train of loads coincides
(c) at one-quarter span
(d) so placed that load point and the C.G of the train of loads equi-distant from the mid span

**Ans. (d) :**




The maximum bending moment under a particular point load among a train of point loads crossing a simply supported girder occurs at the location when that load is at so placed that load point and the C.G of the train of loads equi-distant from the mid span.

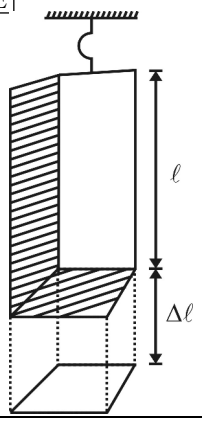
**31. Elongation produced due to self weight in a bar of uniform cross sectional area 'A', length 'ℓ', and weight 'W' having modulus of elasticity 'E', hung vertically at top end is**

(a) $2W\ell/AE$ (b) $2WA/EI$
(c) $W\ell/2AE$ (d) $W\ell/4AE$

**Ans. (c) :** If area of bar = A, Length = ℓ, Weight = W then elongation produce due to self weight-

$$\boxed{\Delta\ell = \frac{\rho\ell^2}{2E}} = \frac{W \times \ell^2}{2A \times \ell \times E} \quad \left[\begin{matrix} W = A \times \ell \times \rho \\ \rho = \dfrac{W}{A \times \ell} \end{matrix}\right]$$

$$\boxed{\Delta\ell = \frac{W\ell}{2AE}}$$




**32. The point of contraflexure in a laterally loaded beam occurs where.**

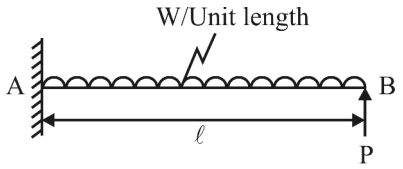
(a) shear force changes its sign
(b) shear force is minimum
(c) shear force is maximum
(d) bending moment changes its sign

**Ans. (d) : The point of contraflexure–**The bending moments of opposite nature always produce curvatures of beams in opposite directions. In a beam if the bending moment changes sign at a point, the point itself having zero bending moment, the beam changes curvature at this point of zero bending moment and this point is called the point of contra flexure.
**Point of inflection–** Point of curve (on deflected shape) changes being concave to convex or vice versa.

**33. Reaction at the level of prop at free end of a cantilever of span 'ℓ' carrying a u.d.l. of 'W' per unit length is**

(a) $W\ell/8$ (b) $W\ell/4$
(c) $3W\ell/8$ (d) $5W\ell/8$

**Ans. (c) :**




$$\frac{P\ell^3}{3EI} = \frac{W\ell^4}{8EI}$$

$$\boxed{P = \frac{3W\ell}{8}}$$

**34. Stress variation across the wall thickness of a thick cylinder is computed by using**

(a) Castigliano theorem
(b) Clapeyron theorem
(c) Maxwell's theorem
(d) Lames theorem

**Ans. (d) :** Analysis of thick cylinders/Lame's theorem is applicable. The theorem is based on that the longitudinal strain is same at all points, i.e., the strain is independent of the radius.
**Assumption–**

(i) Plane section of cylinder, perpendicular to longitudinal axis remain plane under pressure.
(ii) Material is homogenous, isotropic and linear elastic.

**35. The torsional rigidity of a tube of thickness 1.0 mm, diameter 200 mm, and rigidity modulus 100.0 GPa is**

(a) 314.1 $GNmm^2$ (b) 341.1 $GNm^2$
(c) 628.3 $GNm^2$ (d) 157.1 $GNm^2$

**Ans. (d) :** Dia, d = 200 mm = 0.2 m
G = 100 GPa

Torsonal rigidity $= I_P \times G$

$$= \frac{\pi}{32}(0.2)^4 \times 100$$

$$= \frac{3.14}{32} \times 0.2^4 \times 100\,GPa$$

$$= 0.015714$$

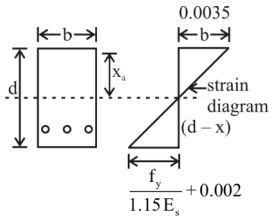
$$= 157.14 \times 10^{-4}\,GNm^2$$

**36. The maximum strain in the tension reinforcement in the section at failure shall not be less than**

(a) 0.002 + (0.87 $f_y$/Es)
(b) 0.0035 + (0.87 $f_y$/Es)
(c) 0.0035 + ($f_y$/1.15 Es)
(d) 0.002 + (0.85 Es/$f_y$)

**Ans. (a) :** The maximum strain in the tension reinforcement in the section at failure shall not be less than–

$$\varepsilon_{su} = \left(\frac{f_y}{1.15 E_s}\right) + 0.002$$




or $$\varepsilon_{su} = \left(\frac{0.87 f_y}{E_s}\right) + 0.002$$

$\varepsilon_{su}$ = Strain in steel at time of failure or limiting strain in steel.
$f_y$ = Characteristic strength of steel
$E_s$ = Modulus of elasticity (steel) = $2 \times 10^5$ $N/mm^2$

**37. Patrial safety factor of the material considered for concrete is**

(a) 1.15 (b) 2.00
(c) 1.50 (d) 0.87

**Ans. (c) : In LSM-**
- Patrial safety factor of the material considered for concrete is 1.50
- Patrial safety factor of the material considered for Steel is 1.15

**In WSM-**
- Safety factor of the material considered for concrete is 3.0
- Safety factor of the material considered for steel is 1.5.

**38. The maximum Deflection which can be allowed in gantry supporting manually operated crane, as per IS : 800–2007 is**

(a) Span/500 (b) Span/550
(c) Span/700 (d) Span/750

**Ans. (a) :** The vertical deflection of a gantry girder should not exceed the values specified below :

| | |
|---|---|
| When the cranes are manually operated | $\frac{L}{500}$ |
| Where the cranes are travelling overhead and operated electrically up to 500 kN | $\frac{L}{750}$ |
| Where the cranes are travelling overhead and operated electrically over 500 kN | $\frac{L}{1000}$ |
| Other moving loads, such as charging cars, etc. Where, L = span of the gantry girder | $\frac{L}{600}$ |

**39. The main function of Column base is to**

(a) transmit the Column load to foundation block
(b) resist the Deflections
(c) resist lateral forces
(d) reduce the effect of vibrations

**Ans. (a) :** Column bases transmit the column land to the concrete or masonry foundation blocks. The column base spreads the load on wider area so that the intensity of bearing pressure on the foundation block is within the bearing strength; there are two types of common bases commonly used in practice.

**40. The members carrying compressive forces in a roof truss are called**

(a) Girders (b) Ties
(c) Struts (d) Purlins

**Ans. (c) :** The members carrying compressive forces in a roof truss are called struts.
- The members carrying tensile forces in a roof truss are called ties.

**41. Battens provided for a compression member shall be designed to carry transverse shear equal to**

(a) 2.5% of axial force in member
(b) 5% of axial force in member
(c) 10% of axial force in member
(d) 25% of axial force in member

**Ans. (a)** Battens should be designed to carry the bending moment and shear arising from the transverse shear force, V, which is 2.5% of the total axial force on the whole compression member.
- The lacing for compression member should be proportioned to resist a total transverse shear V equal to 2.5% of the axial force in the column.

**42. For a singly reinforced over-reinforced section**

1. the lever arm will be less than for a balanced section
2. the maximum stress developed in concrete would be equal to the allowable stress
3. the maximum stress developed in steel would be equal to the allowed stress

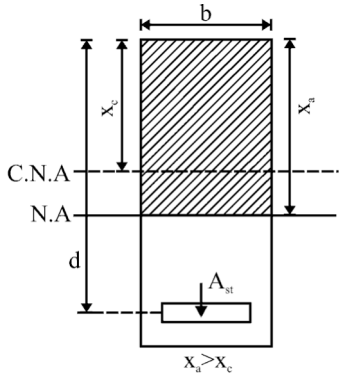
**Of these statements the correct ones are**

(a) 1 and 3 (b) 1 and 2
(c) 2 and 3 (d) 1, 2 and 3

**Ans. (b) :** In the case of the over-reinforced section, concrete yield prior to the steel be stress in concrete reaches its permissible value but stress in steel is less than its permissible stress.

**For a singly reinforced over-reinforced section–**

- The lever arm will be less than for a balanced section.
- The maximum stress developed in concrete would be equal to the allowable stress.



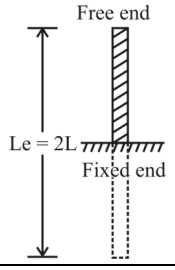

Over reinforcement section

$$\text{Resistance moment of section} = \frac{b x \sigma_{cbc}}{2}\left(d - \frac{x}{3}\right)$$

**43. The effective length of a circular electrical pole of length '$\ell$' and constant diameter erected on ground is, where '$\ell$' is un supported length**

(a) 0.8 $\ell$ (b) 1.2 $\ell$
(c) 1.5 $\ell$ (d) 2.0 $\ell$

**Ans. (d) :** The effective length of a column with given end condition is the length of an equivalent column with one ends fixed and other end free, made up of same material having same cross section, subjected to same crippling load and buckling load as that of given column.




**44. When the column is effectively held in position and restrained against rotation at one end at other end is neither held in position nor restrained against rotation, the effective length of column is 'k' times the unsupported length (L) of the column, where 'k' is equal to**

(a) 1.2 (b) 2.0
(c) 1.5 (d) 0.8

**Ans. (b) :**

| S.N. | Member End resistant | Effective length |
|---|---|---|
| 1. | Effectively held in position at both ends but not restrained against rotation. | 1.00 L |
| 2. | Effectively held in position at both ends and restrained against rotation at one end. | 0.80 L |
| 3. | Effectively held in position and restrained against rotation at both ends. | 0.65 L |
| 4. | Effectively held in position and restrained against rotation at one end, and at the other end partially restrained against rotation but not held in position | 1.50 L |
| 5. | Effectively held in position and restrained against rotation at one end, and other end free in translation and restrained in rotation. | 1.2 L |
| 6. | Effectively held in position at one end but not restrained against rotation and at the other end restrained against rotation but not held in position. | 2.00 L |

**45. Match list-I (column base) with list-II (its application) and select correct answers using the code given below**

| List-I | List-II |
|---|---|
| A. Grillage foundation | 1. Light axial loaded steel column |
| B. Gusseted base | 2. Heavy loaded steel column to be rested on weak, soils |
| C. Slab base | 3. Eccentric loaded steel column |

(a) A-1, B-2, C-3 (b) A-3, B-2, C-1
(c) A-2, B-3, C-1 (d) A-2, B-1, C-3

**Ans. (c) :**

Grillage foundation – Heavy loaded steel column to be rested on weak, soils
Gusseted base – Eccentric loaded steel column
Slab base – Light axial loaded steel column

**46. Which one of the sections is the most efficient for a simply supported gantry girder?**

(a) I-section with equal flanges
(b) I-section with a channel attached to the top flange
(c) I-section with a wide bottom flange
(d) I-section with a heavy plate connected to the bottom flange

**Ans. (b):** The most efficient section for simply supported gantry girder is I-section, the section with a channel attached to the top flange.

- The channel section at the top of the I-section increase the lateral resistance overhead cranes as increases torsional rigidity.

**47. The unit weight of reinforced concrete made with sand and gravel or crushed natural stone aggregate may be taken as (in kN/m$^3$)**

(a) 20 (b) 23

(c) 24 (d) 25

**Ans. (d) :** Unit weight is also as specific weight. Unit weight is the weight of the material per unit volume.

| **Material** | **Unit weight** (in kN/m$^3$) |
|---|---|
| PCC (plain cement concert) | 24 |
| RCC (Reinforced cement concrete) | 25 |
| Portland cement | 14 |
| Cement Mortar | 20 |
| Water | 10 |
| Wet Sand | 11 to 20 |
| Brick masonry | 19.20 |

**48. In an lsolated footing when e > L/6 soil pressure at the base of farther edge from loading point is (with usual notations)**

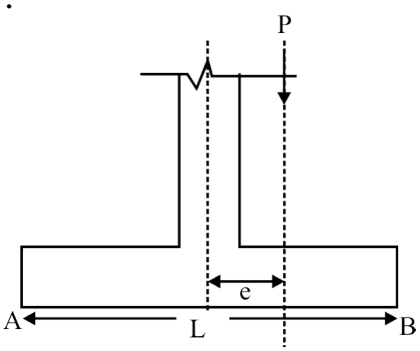
(a) Infinity

(b) Zero

(c) Compressive in Nature

(d) Tensile in Nature

**Ans. (b) :**

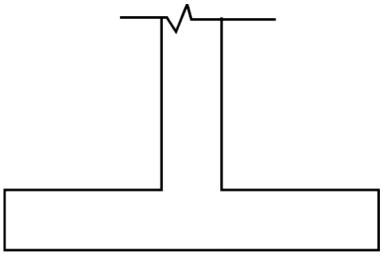


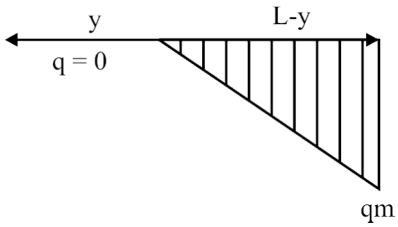

Let length of footing is L and width of footing is 13.

Soil pressure

$$q = \frac{P}{B \times L} \pm \frac{M \times x}{I} \geq 0$$

Where, x = distance from centre of footing to where soil pressure calculated = L/2.






$$I = \frac{B \times L^3}{12}$$

According to question

$$e = \frac{M}{P} > \frac{L}{6}, \qquad 6e > L$$

Soil pressure of father edge ($q_m$) = $\frac{P}{B \times L} - \frac{m \times x}{I}$

$$q_m = \frac{P}{B \times L} - \frac{P \times 6e}{B \times L^2}$$

$6e > L$, it means the value of $\left(\frac{P \times 6e}{B \times L^2}\right)$ is shorter than the value $\frac{P}{B \times L}$, so the soil pressure at further edge is goes into negative, but in soil tension pressure not developed compressive

$q_m$ = zero unit.

**49. Which one of the following is the most critical set for consideration in the design of rolled steel column carrying axial loads?**

(a) percent elongation at yield and net sectional area

(b) critical bending strength and axial yield strength of the material

(c) bucking strength based on the net area of the section and percent elongation at ultimate load

(d) compressive strength based on slenderness ratio and gross sectional area of the section

**Ans. (d) :** Compressive strength based on slenderness ratio and gross sectional area of the section is the most critical set for consideration in the design of rolled steel column carrying axial loads.

**50. Match list-I with list-II and select correct answer :**

| **List-I (Type of stress)** | **List-II (Permissible stress)** |
|---|---|
| A. Bending stress | 1. 0.40 fy |
| B. Bearing stress | 2. 0.45 fy |
| C. Maximum shear stress | 3. 0.75 fy |
| | 4. 0.66 fy |

(a) A-3, B-1, C-2 (b) A-1, B-4, C-3

(c) A-4, B-3, C-2 (d) A-2, B-1, C-3

**Ans. (c) :**

| **Type of stress)** | **Permissible stress)** |
|---|---|
| Bending stress | 0.66 fy |
| Bearing stress | 0.75 fy |
| Maximum shear stress | 0.45 fy |

**51. Compared to mild steel plain bars, high yield strength deformed bars are?**

(a) Less ductile but more strong
(b) More ductile but less strong
(c) More ductile and more strong
(d) Less ductile and less strong

**Ans. (a) :** Compared to mild steel plain bars, high yield strength deformed bars are less ductile but more strong, because increase carbon content the ductility is less and strength increase.

**52. Consider the following statements regarding the working stress design of under reinforced R.C. Section**

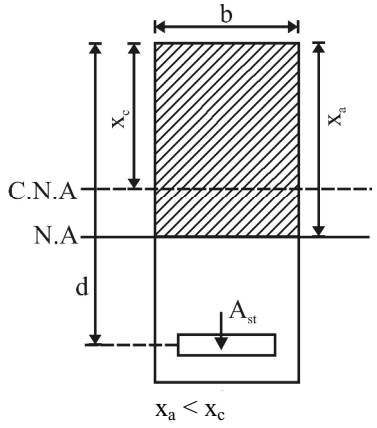
1. The N.A. Depth will be > that of balance section
2. Stress in steel in tension will reach its maximum permissible value
3. The M.R. will be < that of balance section
4. The concrete on tension side is als, to be considered for calculating the M.R. of section

(a) 1 & 2 are correct
(b) 1 & 4 are correct
(c) 3 & 4 are correct
(d) 2 & 3 are correct

**Ans. (d) :** the working stress design of under reinforced R.C. Section–

- Stress in steel in tension will reach its maximum permissible value ($x_a < x_c$)
- The M.R. will be < that of balance section.

$$m_r = \sigma_{st} A_{st}\left(d - \frac{x}{3}\right)$$




$x_a < x_c$

Under reinforcement section

**53. A T-beam behaves as a rectangular beam of width equal to its flange if its 'NA'?**

(a) coincides with centroid of reinforcement
(b) coincides with centroid of T-Section
(c) remains with in the flange
(d) remains in the web

**Ans. (c) :** A T-beam behaves as a rectangular beam of width equal to its flange if its neutral axis remains within the flange.

- In case of I-beam, if depth of the neutral axis lies within the flange region and there is sagging bending moment then only the portion below neutral axis will be in tension and hence, neglected and I-beam will behave like a rectangular beam having with equal to flange width.

**54. In case of a isolated square concrete footing, match the location at which the stress resultant are to be checked?**

P. Bending moment 1. at the face of column
Q. One way shear 2. at d/2 from face of column
R. Punching shear 3. at d from face of column

(a) P-1, Q-2, R-3 (b) P-3, Q-1, R-2
(c) P-1, Q-3, R-2 (d) P-1, Q-2, R-3

**Ans. (c) :** In case of a isolated square concrete footing, match the location at which the stress resultant are to be checked–

- Bending moment : at the face of column
- One way shear : at d from face of column
- Punching shear : at d/2 from face of column.

**55. Match the list-I with list-II and using the codes given below**

| List – I | List - II |
|---|---|
| P. Flexure | 1. Minimum depth of section |
| Q. Shear | 2 Longitudinal steel reinforcement |
| R. Bond | 3. Stirrups |
| S. Deflection | 4. Anchorage in support |

(a) P-3, Q-2, R-1, S-4 (b) P-2, Q-3, R-1, S-4
(c) P-3, Q-2, R-4, S-1 (d) P-2, Q-3, R-4, S-1

**Ans. (d) :** Match the list–

| | |
|---|---|
| Flexure | Longitudinal steel reinforcement |
| Shear | Stirrups |
| Bond | Anchorage in support |
| Deflection | Minimum depth of section |

**56. A two dimensional flow is described by velocity components u = 2x and v = 2y. The discharge between the points (1, 1) and (2, 2) is equal to**

(a) 9 units (b) 8 units
(c) 7 units (d) 6 units

**Ans. (d) :** Given,
Components of velocity, u = 2x, v = –2y
Discharge between point (1, 1) and (2, 2)

$$\vec{A}\vec{V} = \left(u\hat{i} + v\hat{j}\right)\left(\frac{d}{dx}\hat{i} + \frac{d}{dy}\hat{j}\right)$$

$$= \left(2x\hat{i} + -2y\hat{j}\right)\left(\frac{d}{dx}^{(2x)\hat{i}} + \frac{d}{dy}^{(-2y)\hat{j}}\right)$$

$$= \left(2 \times \frac{3}{2}\hat{i} - 2 \times \frac{3}{2}\hat{j}\right)\left(2\hat{i} - 2\hat{j}\right)$$

$$= \left(3\hat{i} - 3\hat{j}\right)\left(2\hat{i} - 2\hat{j}\right)$$

$$= 6\left(\hat{i}\hat{i} \times \hat{j}\hat{j}\right) = 6 \times 1$$

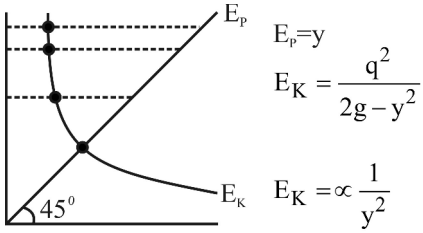
$$\boxed{\vec{A}\vec{V} = 6\,\text{units}}$$

**57. In a rectangular channel, the critical depth is given by**

(a) $\left(\frac{q^2}{g}\right)^{1/2}$ (b) $\left(\frac{q^2}{g}\right)^{1/3}$

(c) $\left(\frac{q^2}{g}\right)^{1/4}$ (d) $\left(\frac{q^3}{g}\right)^{1/3}$

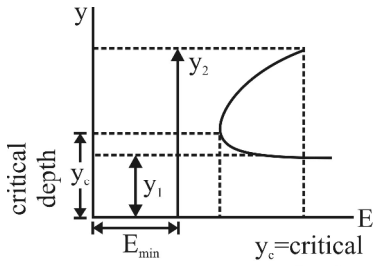
**Ans. (b) :**




$E_P = y$

$E_K = \frac{q^2}{2g - y^2}$

$E_K = \propto \frac{1}{y^2}$




$y_c$=critical depth

= where E will be max

$y_1$ and $y_2$ are alternate depth were E will be equal

$$E = y + \frac{q^2}{2gy}$$

$$\frac{dE}{dy} = 0$$

$$1 - \frac{2q^2}{2gy^2} = 0$$

$$\frac{q^2}{gy^3} = 1$$

$$y^3 = \frac{q^2}{g}$$

$$y = \left(\frac{q^2}{g}\right)^3$$

$$\boxed{y_c = \left(\frac{q^2}{g}\right)^{1/3}} \text{ critical depth}$$

**58. A differential manometer measures**

(a) absolute pressure at a point

(b) local atmospheric pressure

(c) difference in total energy between two points

(d) difference in pressure between two points

**Ans. (d) :**

- A differential manometer – Pressure difference between two points
- Absolute pressure is measured – An aneroid barometer
- A simple manometer – Pressure
- A bourdon gauge – Gauge Pressure.

**59. The condition of "No slip" at rigid boundaries is applicable to**

(a) Flow of Newtonian fluids only

(b) Flow of ideal fluids only

(c) Flow of all real fluids

(d) Flow of all non-newtonian fluids

**Ans. (c) :** The condition of "No slip" at rigid boundaries is applicable to flow of all real fluids.

- At the boundary surface there is no relative motion between the fluid and the boundary. This condition is known as the no slip condition.

**60. The Bernoulli's equation is written with usual notion as $p/W + V^2/2g + z$ = constant. In this equation each of the terms represents**

(a) energy in Kg m/Kg mass of fluid

(b) energy in N.m/Kg mass of fluid

(c) energy in N.m/N weight of fluid

(d) power in kw/kg mass of fluid

**Ans. (c) :** The Bernoulli's equation is written with usual notion as $p/W + V^2/2g + z$ = constant. In this equation each of the terms represents energy in N.m/N weight of fluid.

**61. Pitot tube is used for measurement of**

(a) low pressures

(b) flow of ideal fluids only

(c) velocity at a point

(d) discharge

**Ans. (c) :** The pitot tube is a small open tube bent at right angle it is used to measured the velocity of flow at the required point in a pipe. It is determined by measuring the rise of liquid in a tube.

| | |
|---|---|
| Venturimeter | – Discharge or rate of flow. |
| Orifice & Mouthpiece | – discharge or rate flow |
| Rotameter | – Discharge or rate of flow |
| Pitot tube | – Fluid velocity |
| Elbowmeter | – Discharge or rate of flow |
| Flow nozzle | – Discharge or rate of flow |
| Bendmeter | – Discharge or rate of flow |

**62. The lift force on a body is**

(a) due to buoyant force

(b) the component of the resultant force in the vertical direction

(c) the component of the resultant force in a direction normal to relative velocity

(d) due to drag on the body

**Ans. (c) :** The perpendicular component of the resultant force to the oncoming flow direction is called lift force and the horizontal component of this force parallel to the flow direction is called drag force.

**63. The difference between the total head line and the hydraulic grade line represents**

(a) the velocity head
(b) the piezometric head
(c) the pressure head
(d) the elevation head

**Ans. (a) :** TEL is the line joining the point, representing the value of the total head.

$$TEL = \frac{P}{\rho g} + \frac{V^2}{2g} + z$$

HGL– It is the line joining the points representing the values of pizomentic and elevation head.

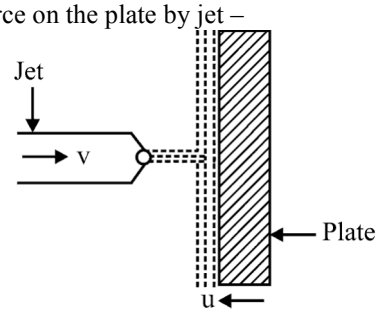
$$HGL = \frac{P}{\rho g} + z$$

• The difference between the total head line and the hydraulic grade line represents the velocity head.

**64. A water jet 0.015m$^2$ in area has a velocity of 15 m/s. If this jet impinges normally on a plate which is moving at a velocity of 5 m/s in the direction of the jet, the force on the plate due to this impact is :**

(a) 3368 N (b) 2246 N
(c) 1497 N (d) 14686 N

**Ans. (c) :** Given,
Area of jet (a) = 0.015 m$^2$
Velocity by jet (v) = 15 m/s
Velocity of let (u) = 5 m/s
Apply force on the plate by jet –




$$F = \frac{Wa(v-u)^2}{g}$$

$$F = \frac{\rho g \times a(v-u)^2}{g} = \rho \times a(v-u)^2$$

$$F = 1000 \times 0.015 \times (15-5)^2$$

$$F = 1000 \times 0.015 \times (10)^2 = 15 \times 100$$

$$\boxed{F = 1500\,N \approx 1497\,N}$$

**65. Which of the following is not a dimension less number**

(a) Darcy Weisbach friction factor f
(b) Coefficient of drag $C_D$
(c) Manning's coefficient N
(d) Coefficient of Velocity $C_V$

**Ans. (c) :** Manning's coefficient–

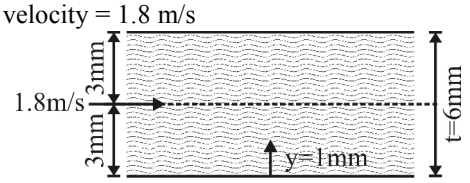
$$\boxed{v = \frac{1}{N} m^{2/3} i^{1/2}} \quad \left(\frac{second}{\sqrt[3]{m}}\right)$$

Dimension of manning coefficient, N = $L^{-0.33}T$

**66. In a laminar flow between two parallel plates with a separation distance of 6 mm, the centre line Velocity is 1.8 m/s. The velocity at a distance of 1 mm from the boundary is**

(a) 0.15 m/s (b) 1.0 m/s
(c) 0.55 m/s (d) 0.75 m/s

**Ans. (b) :** Given,
Distance between two plate (t) = 6 mm
Mean velocity = 1.8 m/s




Mean velocity,

$$U = \frac{1}{8\mu}\left(\frac{-\delta P}{\delta x}\right) t^2$$

$$1.8 = \frac{1}{8\mu}\left(\frac{-\delta P}{\delta x}\right) 6^2$$

$$\left(\frac{-\delta P}{\delta x}\right) = 0.4\mu$$

Velocity of flow to 1 mm from boundary layer–

$$U = \frac{1}{2\mu}\left(\frac{-\delta P}{\delta x}\right)\left(ty - t^2\right)$$

$$U = \frac{1}{2\mu}(0.4\mu)\left(6 - 1^2\right)$$

$$U = \frac{1}{2} \times 0.4 \times 5$$

$$\boxed{U = 1\,m/s}$$

**67. A laminar boundary layer has a velocity distribution given by u/U = y/δ. The displacement thickness δ* for this boundary layer is :**
**(With usual notations)**

(a) δ (b) δ/2
(c) δ/4 (d) δ/16

**Ans. (b) :** $\frac{u}{v} = \frac{y}{\delta}$ and

Displacement thickness = $\int_0^{\delta}\left(1 - \frac{u}{v}\right)dy$

$$= \int_0^{\delta}\left(1 - \frac{y}{\delta}\right)dy$$

$$= \left[y - \frac{y^2}{2\delta}\right]_0^{\delta}$$

$$= \left[\delta - \frac{\delta^2}{2\delta}\right]$$

$$= \frac{\delta}{2}$$

**68. In a hydraulic jump occurring in a horizontal rectangular channel the sequent depths are 0.25 m & 1.25 m. The energy loss in the jump is**

(a) 0.80 m (b) 1.0 m
(c) 1.25 m (d) 1.50 m

**Ans. (a) :** Energy loss $= \dfrac{(y_2 - y_1)^3}{4y_1.y_2}$

$$\frac{(1.25-0.25)^3}{4\times1.25\times0.25} = 0.80 \text{ m}$$

**69. The frictional resistance of a pipe varies approximately with the.....of the liquid**

(a) pressure
(b) velocity
(c) square of velocity
(d) cube of velocity

**Ans. (c) :** Darcy-weishbach equation for friction loss–

$h_f = \dfrac{fLV^2}{2gD}$ {where, f = friction factor}

$h_f \propto V^2$

**70. For subsonic flow, if the area of flow increases**

(a) velocity is constant
(b) velocity increases
(c) velocity decreases
(d) velocity is infinity

**Ans. (c) : Subsonic flow–** When mach no. less than 1 that's called subsonic flow.

$M < 1 \rightarrow$ Subsonic flow
$M = 1 \rightarrow$ Sonic flow
$M > 1 \rightarrow$ Super sonic flow
$M \gg 1 \rightarrow$ Hypersonic ($M > 6$)

**71. The equation of motion for laminar flow of a real fluid is known as**

(a) Euler's equation
(b) Bernoulli's equation
(c) Navier-Stokes equation
(d) Reynold's equation

**Ans. (c) :** Navier-stoke's equation considered pressure and viscous force into account and turbulence and surface tension forces was ignored.

• Navier, stoke's equation represents equation of motion of laminar flow.

**72. The notch angle for maximum discharge over a triangular notch is**

(a) $30^0$ (b) $60^0$
(c) $90^0$ (d) $120^0$

**Ans. (c) :** A v-notch weir also called the triangular notch or weir. The discharge over a triangular weir or notch is given by–

$$Q = \frac{8}{15}C_d\sqrt{2g\tan\frac{\theta}{2}}.H^{5/2}$$

$$\frac{\theta}{2} = 45^0$$

$$\theta = 90^0$$

**73. Square root of the ratio of inertia force of flowing fluid to the gravity force is**

(a) Mach number (b) Weber number
(c) Froude's number (d) Euler's number

**Ans. (c) : Froud number,** $F_r = \sqrt{\dfrac{\text{inertia force}}{\text{gravity force}}} = \dfrac{V}{\sqrt{gL_c}}$

Euler's no = $E_u = \sqrt{\dfrac{\text{Inertia force}}{\text{Pressure force}}} = \dfrac{V}{\sqrt{\dfrac{P}{\rho}}}$

Weber number = $w_n = \sqrt{\dfrac{\text{Inertia force}}{\text{Tension force}}} = \dfrac{V}{\sqrt{\dfrac{\sigma}{\rho\ell}}}$

**74. A centrifugal pump was manufactured to couple directly to a 15 HP electric running at 1500 rpm, delivering 50 lit/min against a total head of 36 m. It is desired to replace the motor by a diesel engine with 1000 rpm and couple it directly to the pump. The head developed by the pump is likely to be**

(a) 41.4 m (b) 29.6 m (c) 20 m (d) 16 m

**Ans. (d) :** Given,
Power, P = 15 H.P.
$N_1$ = 1500 rpm
Discharge Q = 50 $\ell$/min
Total head = 36 m
$N_2$ = 1000 rpm

$$\frac{N}{\sqrt{H}} = \text{Constant}$$

$$\frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$$

$$H_2 = \left(\frac{N_2}{\sqrt{N_1}}\right)^2 \times H_1 = \left(\frac{1000}{1500}\right)^2 \times 36$$

$H_2$ = 16 m

**75. A fluid (specific gravity = 0.9 and $\mu$ = 1.2 Pa.s) flows in a laminar regime between two parallel plates fixed 3 cm apart. If the discharge is 600 $cm^3/s/cm$ width of plate, the shear stress on the boundary, in Pa, is**

(a) 800 (b) 640 (c) 480 (d) 240

**Ans. (a) :** $\Delta P = \dfrac{8\mu v\ell}{t^2}$

$$\Delta P = \frac{8\times1.2\times100\times1}{9\times10^{-4}}$$

$$\Delta P = \frac{48}{9}\times10^4$$

$$\therefore \quad \tau_{max} = \frac{1}{2}\left(\frac{dp}{dx}\right)t \quad (\text{at} = y = 0)$$

$$= \frac{1}{2}\times\frac{48}{9}\times10^4\times3\times10^{-2}$$

$$\boxed{\tau_{max} = 800 \text{ Pascal}}$$

**76. In the laminar flow of a liquid down an inclined plane, the surface velocity is found to be 30 cm/s. The average velocity of the flow, in cm/s is**

(a) 20 (b) 30
(c) 15 (d) 10

**Ans. (a) :** Given,

$V_{max} = 30$ cm/sec

$$V_{avg} = \frac{2}{3} \times V_{max}$$

$$= \frac{2}{3} \times 30 = 20\,m/sec$$

**77. The thickness of laminar sub layer 'd' is given by**
**(with usual notations)**

(a) $11.6u^*/v$ (b) $u^*/(11.6v)$
(c) $11.6v/u^*$ (d) $v/u$

**Ans. (a) :** The laminar sublayer isusually very thin and it thickness 'δ' is found.

$$\delta' = 11.6\frac{V}{u*}$$

$$U* = \sqrt{\frac{\tau_0}{e}} = \text{Shear velocity}$$

**78. In Isentropic flow between two points, the stagnation**

(a) pressure and stagnation temperature may vary
(b) pressure would decrease in the direction of the flow
(c) pressure and stagnation temperature would decrease with an increase in velocity
(d) pressure, stagnation temperature and stagnation density would remain constant throughout the flow

**Ans. (d) :** During / isentropic flow, stagnation pressure stagnation temperature and stagnation remains constant throughout the flow.

- At stagnation point total pressure or temperature is same as stagnation pressure or temperature respectively.

**Note-** Isentropic flow occurs when the change in flow variable is small and gradual, such as the ideal flow through the nozzle.

**79. The ratio of power output of the pump to the power input to the pump is known as**

(a) mechanical efficiency
(b) overall efficiency
(c) manometric efficiency
(d) pump efficiency

**Ans. (b) : Overall efficiency–** It is ratio of power output of the pump to the paver input to the pump.

$$\eta_0 = \frac{wt/m}{100 \times s.p.}$$

or $\eta_0 = \eta_{max} \times \eta_m$

$$\eta_{man} = \frac{\text{Manometric head}}{\text{Head imparted by the impeller power}}$$

$$= \frac{g\,t/m}{Vw_2.u_2}$$

$$\eta_m = \frac{\text{Power at the impeller}}{\text{Power at the shaft}} = \frac{w \times Vw_2.u_2}{1000}$$

**80. The relationship between specific gravity of soil (S), Field capacity (FC) and Permanent wilting print (PWP) exists to compute available water for plant per meter depth is**

(a) AW = S-FC-PWP
(b) AW = S[FC-PWP]
(c) AW = S[PWP-FC]
(d) Relationship does not exist

**Ans. (b) :** Relation between S, Fe & pwp–

Depth of irrigation water $A_w = S\,(F_c - PWP)$

Where, Fc = Field capacity
P.W.P.= Permanent welting point
δ = Special gravity.
$A_w$ = Available water.

**81. The observed run off during 6 h storm with a uniform intensity of 15 mm/h over a basin of area 300 $km^2$ is 21.6 million $m^3$. The average infiltration rate during the storm is**

(a) 3 mm/h (b) 6 mm/h
(c) 12 mm/h (d) 18 mm/h

**Ans. (a) :** Given,

P = 15 mm/h × 6 = 90 mm

$$\text{Run off (R)} = \frac{21.6 \times 10^6 \times 10^3}{300 \times 10^6} = 72 \text{ mm}$$

$t_c$ = 6 hr

$$\frac{90 - 72}{6} = \frac{18}{6} = 3 \text{ mm/hr}$$

**82. The upper limit on the area of the basin for the applicability of unit hydrograph is taken to be**

(a) 100 $km^2$ (b) 2500 $km^2$
(c) 5000 $km^2$ (d) 10000 $km^2$

**Ans. (c) : Limitation of unit hydrograph–**

- Unit hydrograph cannot be used for catchment are greater than 5000 km.
- Unit hydrograph can not be used for catchment less than 200 hec or 2 $km^2$.
- Precipitation must be in the form of rainfall only i.e. not be in the form of show.

**83. A 4-hour unit hydrograph of a drainage basin is triangular in form with a height of 50 $m^3/s$ and a base of 15 hours. The area in $km^2$ of the drainage basin is**

(a) 110 (b) 135
(c) 147 (d) 151

**Ans. (b) :** $\frac{Q_p \times T_B}{2} = A \times \frac{1}{100}$

$$A = \frac{50 \times 15 \times 3600 \times 100}{2 \times 10^6}$$

A = 135 $km^2$

**84. For an annual flood series arranged in decreasing order of magnitude, the return period for a magnitude listed at position m in a total of N entries is**

(a) m/N (b) m/(N + 1)
(c) (N + 1)/m (d) N/(m + 1)

**Ans. (c) : Return period (T) –** It is defined as the average time interval after which a flood of given discharge is equaled or exceed.

$$T = \frac{1}{P}$$

Where,
P = Probability of occurrence exceed of an event with rank (m)

| Method | Probability (P) |
|---|---|
| Weibull method | $\frac{m}{N+1}$ |
| California method | $\frac{m}{N}$ |
| Hozen method | $\frac{m-0.5}{N}$ |

**85. The shape of the phreatic line is**

(a) Parabola
(b) Straight line
(c) Cylindrical
(d) Circular

**Ans. (a) : Phreatic line–** It is the top flow the which follows the path of base parabola. It is stream line. The pressure on this line is atmospheric and below this line is hydrostatic.
• The shape of preatic line is parabola.

**86. If the regime velocity of flow in a Lacey's channel having hydraulic mean radius of 1 m is 0.4 m/s, the silt factor is**

(a) 0.5 (b) 0.6
(c) 0.4 (d) 0.7

**Ans. (c) :** Given,
Regime velocity in channel, V = 0.4 m/s
Hydraulic radius, R = 1 m

$$\text{To, } V = \sqrt{\frac{2}{5}fR}$$

Where f = Silt factor
R = Hydraulic radius

$$V^2 = \frac{2}{5}f.R$$

$$f = \frac{5\times V^2}{2\times R} = \frac{5\times(0.4)^2}{2\times 1}$$

$$= \frac{5\times 0.016}{2}$$

$$\boxed{f = 0.4}$$

**87. A drainage basin is**

(a) The length of long river
(b) A lake or ocean into which river drains
(c) The lower level a river erodes
(d) Total area drained by river and its tributaries

**Ans. (d) :** A drainage basin is total area drained by river and its tributaries.
It is an area of land where water flow rain or snow melt drains downhill into a body of water such as river, lake, method or ocean.

**88. Isohyet is a line joining points having**

(a) Equal evaporation rate
(b) Equal barometric pressure
(c) Equal height above MSL
(d) Equal rainfall depth of given duration

**Ans. (d) :** Isohyet is a line joining point having equal rainfall of given duration.
• Isobar is a lane joining point having equal pressure.
• Isopleths is a line joining point having equal evapotransipiration.

**89. A unit Hydrograph has**

(a) One unit of peak discharge
(b) One unit of rainfall duration
(c) One unit of direct run off
(d) One unit of time base of direct run off

**Ans. (c) :** A hydrograph of direct runoff resulting from 1 cm of effective rainfall applied uniformly over a basin area at a uniform rate during a specified period of tim (D-hr)

• It is given by sherman = $\frac{a}{(t+b)^n}$

Where,
a, b = constant
t = time of concentration

**90. In a sedimentation tank (length L, width B, Depth D) the setting velocity of a particle for a discharge Q is**

(a) Q/BD (b) Q/LD
(c) Q/L (d) Q/BL

**Ans. (d) :** Settling velocity –

$$\boxed{V = \frac{Q}{A} = \frac{Q}{B.L}}$$

Where, Q = discharge
B = width
L = length

**91. When chlorine is added beyond the break-point the process of treating the water is known as**

(a) Plain chlorination
(b) Super chlorination
(c) Post chlorination
(d) Dechlorination

**Ans. (b) :** When chlorine is added beyond the break-point the process of treating the water is known as Super chlorination.
**Plain chloriantion–** Used for clear water.
**Prechlorination–** Chlorine is added before filtration or rather before sedimentation of coagulation.
• Dose require 0.1 to 0.5 ppm
**Post chlorination–** Such does of that residue 0.1 to 0.2 ppm & contact period of 20 min.

**92. A waste water sample of 2 mℓ is made upto 300 ml in BOD bottle with distilled water Initial DO of the sample is 8 mg/ℓ and after 5 days it is 2 mg/ℓ, its BOD is**

(a) 894 mg/ℓ (b) 900 mg/ℓ
(c) 300 mg/ℓ (d) 1200 mg/ℓ

**Ans. (b) :** Dilute water = 300 mℓ
Quantity of residue water = 2 mg
Initial dissolved oxygen = 8 mg/ℓ
dissolved oxygen after two day = 2 mg/ℓ

$$\text{Dilution factor} = \frac{300}{2} = 150$$

Loss of oxygen = 8 mg/ℓ – 2mg/ℓ = 6 mg/ℓ
2 days BOD = loss of oxygen × dilution factor
= 6 × 150 = 900 mg/ℓ

**93. The detention period of a septic tank is of the order of**

(a) 2-6 hours (b) 2-4 hours
(c) 12-36 hours (d) 4-8 hours

**Ans. (c) :** Design parameter of septic tank–
- Detention period = 12-36 hr
- Flow of sewage = 40-40 lpcd
- Rate of accumulation = 30 lpcd/year
- Cleaning period = 6 month to 1 year.

**94. The value of ϕ-index (phi index) for any basin area represents**

(a) separates the value of Rainfall and Runoff in a year
(b) the depth of runoff at a basin
(c) the depth of rainfall in a basin
(d) a value that separates runoff and rainfall intensity for a particular storm

**Ans. (d) :** The value of ϕ-index for any basin area represents a value that separates runoff and rainfall intensity for a particular storm.

$$\phi = \frac{P - R}{t}$$

$$\text{w-index} = \frac{F}{t} = \frac{P - R - S}{t}$$

w-index $\le$ ϕ-index

**95. Which of the option is hygroscopic water**

(a) Water which represents the majority available for plant uptake
(b) Water held tightly as film around individual soil particles and unavailable to plant
(c) Water which ponds up on soil surface
(d) Water which is available to drain through soil by gravity

**Ans. (b) :** Hygroscopic water is absorbed by the particles of dry soil from the atmosphere and is held as a very thin film on the surface of the soil due to adhesion or attraction between the surface of particle and water molecules.
- It is not available for the use of plants.
- Below the permanent welting point, the soil only contains hygroscopic water.

**96. The duty of crop is 1500 hectares in base period is 120 days. The delta of crop is**

(a) 690 mm (b) 860 mm
(c) 1100 mm (d) 1000 mm

**Ans. (a) : Given,**
Duty of crop = 1500 ha
Base period = 120 days
Delta (Δ) = ?

$$\Delta = \frac{8.64\,B}{D}$$

$$= \frac{8.64 \times 120}{150} = \frac{34.56}{5}$$

$$= 0.691\,m$$

$$\boxed{\Delta = 691\,mm}$$

**97. A flood wave in a river is an example of**

(a) steady, non-uniform flow
(b) unsteady, gradually varied flow
(c) steady, spatially varied flow
(d) unsteady, repidly varied flow

**Ans. (b) :** A flood wave in a river is an example of unsteady, gradually varied flow.

For unsteady $\frac{dv}{\partial t} \neq 0$

Non-uniform $\frac{dv}{\partial s} \neq 0$

**98. Seepage through earthen dam can be computed by the following equation (With usual notations)**

(a) $q = Kh[N_d \times N_f]$ (b) $q = Kh[N_d / N_f]$
(c) $q = Kh\sqrt{(N_f / N_d)}$ (d) $q = Kh[N_f / N_d]$

**Ans. (d) :** Seepage through earthen dam can be computed by the following equation (with usual notations) $q = Kh[N_f / N_d]$.

Where,
$N_f$ = Total number of flow channel.
$N_d$ = Total number of equipotential drop.

$[N_f / N_d = \text{Shape factor}]$

**99. Lacey's concept of design of canals is based on the**

(a) lined channels
(b) unlined channels only
(c) both lined and unlined channels
(d) neither lined nor unlined channels

**Ans. (b) :** According to lacey, silt kept in suspension by the vertical component of eddies generated at all point of forces normal to the welted perimeter.
- Lacey's theory is applicable to flow in alluvial river and unlined canal.

**100. Under normal conditions of load and when reservoir is full, the critical stress will be acting on dam at**

(a) Toe (b) Heel
(c) Middle third (d) Centre

**Ans. (a) :** The maximum compressive force in a gravity dam exists at the toe when the reservoir is full due to which toe will be critical section and chances of failure about toe are more.
Maximum compressive force at toe

$$\sigma_{v_{max}} = \frac{\Sigma V}{B}\left(1+\frac{6e}{B}\right)$$

Where,
$\sigma_{vmax}$ = Maximum vertical force at the toe of the dam.

**101. The rate of settling of a particle depends upon the**
(a) viscosity of water
(b) specific gravity of particle
(c) shape and size of particle
(d) viscosity of water, specific gravity and shape and size of particles

**Ans. (d) :** The rate of settling of a particle depend upon the viscosity of water, specific gravity and size of particles.

$$V_s = \frac{(\gamma_s - \gamma_w).d^2}{18\mu}\{Re < 1\}$$

For stones law, D < 0.2 mm

**102. During temperature inversion in atmosphere, air pollutants tend to**
(a) accumulate above inversion layer
(b) accumulate below inversion layer
(c) disperse laterally
(d) disperse vertically

**Ans. (c) :** During temperature inversion is atmosphere air pollutants tend to disperse laterally.
• Negative lapse rate, when the temperature of the environment increase with latitude then the lapse rate becomes inverted or negative.
• From the normal state condition, this condition is known as inversion.

**103. A pollutant undergoes self purification in four distinct zones**
1. Zone of clear water
2. Zone of active decomposition
3. Zone of degradation
4. Zone of recovery

**The correct sequence of these zones is**
(a) 4, 3, 2, 1 (b) 2, 3, 4, 1
(c) 2, 4, 3, 1 (d) 3, 2, 4, 1

**Ans. (d) :** A pollutant undergoes self purification in four distinct zones.
Zone of degradation → Zone of active decomposition → Zone of recovery → zone of clear water.

**104. Activated carbon is used to remove**
(a) Odour and taste (b) Hardness
(c) Iron and manganese (d) Dissolved salts

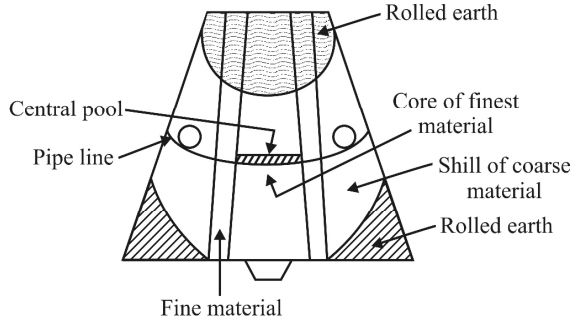
**Ans. (a):** Activated carbon is used to remove odour and taste.
• Intensity of taste and odour is measured by threshold odour number (TON).
• TON allowed is between 1-3.

**105. The main constituents of gas generated during anaerobic digestion of sewage sludge are**
(a) $CO_2$ and $CH_4$ (b) $CH_4$ and Ethane
(c) $CO_2$ and CO (d) $CO_2$ and $N_2$

**Ans. (a) :** The main constituents of gas generates during anaerobic digestion of sewage sludge are $CO_2$ and $CH_4$.
• $CH_4 \rightarrow 70\%$ , $CO_2 \rightarrow 30\%$

**106. Uplift pressure is considered in the analysis of gravity dams**
(a) Only when there is a drainage gallery in the dam
(b) Only when there is tail water
(c) Only where the reservoir is empty
(d) In all situations having water in the reservoir

**Ans. (d) :** Uplift pressure is exerted in all situations when the reservoir has water either in the upstream direction or downstream direction or both.




**107. According to Khosla's theory, the undermining of the floor starts from the**
(a) Tail end (b) Starting end
(c) Intermediate point (d) Foundation bed

**Ans. (a) :** As per Khosla theory, the undermining of the floor started from the fail end because at the tail and head loss due to seepage is maximum and which is turn creates a maximum hydraulic gradient. Therefore, it is essential to have a deep vertical cut off at the down stream end to prevent undermining.

**108. A stream is discharging 400 cumec of water and has a fall of 50 m. The power potential of the hydel station would be**
(a) $1.962 \times 10^5$ KW (b) $19.62 \times 10^5$ KW
(c) 1962 KW (d) $2.616 \times 10^5$ KW

**Ans. (a) :** Given,
Discharge = 400 cumec
Head loss (H) = 50 m
$\because$ $P = W \times H$
or $P = Q \times \rho \times gH$

$$= 400 \times 50 \times 1 \times 10^3 \times 9.8$$
$$= 20,000 \times 10^3 \times 9.8$$
$$= 98 \times 2 \times 10^6 \text{ watt}$$

$= 19.60 \times 10\,KW$

$P = 1.962 \times 10^5$ kW

$P = 1.96 \times 10^5$ KW

**109. Which formula is used to measure velocity of water in water supply main**

(a) Hazen williams
(b) Manning's
(c) Chezy's
(d) Kutters

**Ans. (a) :** Hazen williams formula used to measure velocity of water inwater supply main–

$$V = 0.85C.R^{0.63}S^{0.54}$$

Where,

C= Coefficient of hydraulic capacity
R = Hydraulic radius
S = Slope of energy line.

**110. Acid rain is because of**

(a) $SO_x$ (b) $CO_x$
(c) COH (d) $H_2S$

**Ans. (a) :** Acid rain results when gaseous emission of SOx (70%) and NOx (30%) interact with vapour and sunlight and technical converted to strong acidic compounds ($H_2SO_4$, $HNO_3$)

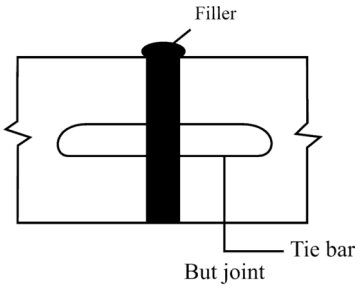
- FOD acid rain $pH \le 5.6$.

**111. The Tie bars in cement concrete pavements are provided across**

(a) Longitudinal joint
(b) Expansion joint
(c) Contraction joint
(d) Warping joint

**Ans. (a) :** The tie bars in cement concrete pavements are provided across longitudinal joint.

- Longitudinal joint provided if the pavement with is more than 4.5 m.
- Tie bars are not load transfer devices but serve as a means to tie two slabs.




**112. In general, the drainage layer facility is provided in which layer of the pavement**

(a) sub-grade (b) sub-base
(c) base course (d) wearing course

**Ans. (b) : Sub-base course–** It is the layer of material beneath the base course and the primary function are to provided structural support, improve drainage, and reduce the instruction of fines form the sub grade in the pavement structure.

**113. In Indian Railways preferabie sleeper density is normally kept as (M being rail length in metres)**

(a) M + 2 to M + 10 (b) M + 2 to M + 7
(c) M + 5 to M + 10 (d) M + 2 to M + 5

**Ans. (b) :** In Indian Railways preferabie sleeper density is normally kept as (M being rail length in metres) M + 2 to M + 7.

**114. A Central Government, semi-official technical body known as Indian Roads Congress (IRC) was formed in the year**

(a) 1943 (b) 1929
(c) 1950 (d) 1934

**Ans. (d) :** A Central Government, semi-official technical body known as Indian Roads Congress (IRC) was formed in the year 1934.

**115. What is the off-tracking while negotiating a horizontal curve on two lane highway with a mean radius of 30 m (take length of wheel base is 6 m)**

(a) 0.66 m (b) 1.0 m
(c) 0.75 m (d) 1.20 m

**Ans. (d) :** Given,
Radius = 30 m, lane n = 2
Length of wheel base = 6 m

$$\text{Off-tracking} = \frac{nl^2}{2R} = \frac{2 \times (6)^2}{2 \times 30} = 1.20 \text{ m}$$

**116. Desire lines are plotted for these surveys**

(a) Origin & destination
(b) Volume
(c) Speed
(d) Accident

**Ans. (a) : Origin-destination study–** It is used to determine travel patterns of traffic called desire line. They are useful when there is substantial charges anticipated in the existing road network.

- The origin and destination study provide the basic data for determining the desired direction of flow. On the basis of these studies, desire lines are draw.

**117. The following formula is used to calculate the Equivalent Axle load Factor (EALF) for single axle load (kg) in vehicle damage factor analysis**

(a) EALF = (axle load/8760)$^4$
(b) EALF = (axle load/8160)$^4$
(c) EALF = (axle load/14968)$^4$
(d) EALF = (axle load/5100)$^4$

**Ans. (b) :** For single axle load–

$$\text{EALF} = \left(\frac{\text{Axel load}}{8160}\right)^4 \quad \{\text{axle load in kg}\}$$

For tandem axle load–

$$\text{EALF} = \left(\frac{\text{Axle load}}{14968}\right)^4$$

**118. The composite sleeper index evolved is from a combination of these properties**

(a) Strength and toughness
(b) Toughness and wear resistance
(c) Strength and Hardness
(d) Toughness and shear resistance

**Ans. (c) : Composite sleeper index–** It is used to measure the mechanical strength of timber.

$$\text{C.S.I.} = \frac{S + 10H}{20}$$

Where,
S = strength
H = Hardness index (measured at 12% moisture)

CSI for – Track sleeper (783)
Crossing sleeper (1352)
Bridge sleeper (1455)

**119. The shape of transition curve suitable for high ways as per IRC is**

(a) Spiral
(b) Lemniscate
(c) Cubic parabola
(d) Parabola

**Ans. (a) : Transition curve–** When a vehicle travelling on a straight road (infinite radius) enters a curve of finite radius. Then, it is suddenly subjected to the centrifugal force which causes shock way.

- For avoiding this, it is necessary to provide a transition curve at the beginning of the circular curve, and gradually reducing the radius to the radius of circular where the curve begins.
- IRS recommends spiral curve for horizontal transition.

**120. Find the compensated gradient at a horizontal curve of radius 50 m with a hill road having ruling gradient of 5 percent is**

(a) 3.5% (b) 4.7%
(c) 3.25% (d) 4.25%

**Ans. (a) :** Given, R = 50 m
Rulling gradient = 5%

$$\text{Grade compensation} = \frac{30 + R}{R}\%$$

$$= \frac{30450}{50} = \frac{80}{50} = 1.6$$

$$\text{or } \frac{75}{R} = \frac{75}{50} = 1.5(\text{take})$$

Then, grade composing, 5 – 1.5 = 3.5%

**121. Which is useful in estimation of highway user revenues, computation of accident rate, establishment of volume trends**

(a) Average Annual Daily Traffic (AADT)
(b) Average Daily Traffic (ADT)
(c) Average Weekday Traffic (AWT)
(d) Peak Hour Traffic Volume (PHTV)

**Ans. (a) : Average annual daily traffic (AADT)–** It count the number of vehicles, passing a point for a year than divide it by 365. It also includes seasonal variation.

**122. The maximum allowable width of any vehicle as per Indian Roads Congress**

(a) 3.5 m (b) 3.75m
(c) 2.5 m (d) 3.80 m

**Ans. (c) :** The maximum allowable width of any vehicle as per Indian road congress is 2.44 m (2.5 m)

- Length = 10.60 m
- Height = 3.8 m (double decker $\rightarrow$ 4.72 m)

**123. The relation between fatigue life and stress ratio (SR) (when 0.45 < = SR < = 0.55) is given to compute number of repetitions of axle loads**

(a) $N = \left[\frac{4.2577}{SR - 0.5325}\right]^{3.258}$

(b) $N = \left[\frac{5.2577}{SR - 0.4325}\right]^{3.268}$

(c) $N = \left[\frac{4.2577}{SR - 0.3325}\right]^{3.268}$

(d) $N = \left[\frac{4.2577}{SR - 0.4325}\right]^{3.268}$

**Ans. (d) :** The relation between fatigue life and stress ratio is to compute the number of repition of axle load.

$$N = \left\{\frac{4.2577}{SR - 0.4325}\right\}^{3.268}$$

Where,

$$SR - [0.45 \le SR \le 0.55]$$

**124. The maximum values of cant deficiency for Broad Gauge and Meter Gauge prescribed by Indian Railways is**

(a) BG = 75 mm & MG = 50 mm
(b) BG = 77 mm & MG = 40 mm
(c) BG = 50 mm & MG = 40 mm
(d) BG = 75 mm & MG = 60 mm

**Ans. (a) : Cant deficiency–** It is difference between the equilibrium can necessary for the maximum permissible speed on a curve and the actual cant provided.

- Higher cant deficiency gives more discomfort to passenger.
- Cant deficiency, BG – 75 mm, MG = 50 mm, NG = 40 mm

**125. As per ICAO, the combined correction for an elevation and temperature should not exceed for runway design**

(a) 45% (b) 35%
(c) 25% (d) 7%

**Ans. (b) :** As per ICAO, the cumulative correction for elevation and temperature together should not be $\not> 35\%$

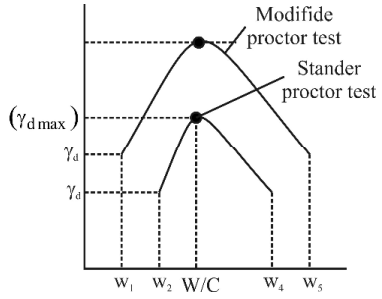
Cumulative correction $\frac{\ell_2 - \ell_1}{\ell} \times 100 \not> 35\%$

If exceeds modify $\ell_2 = 1.35 \times \ell$

**126. What is the effect of increasing the compaction energy on compaction characteristics of a given soil. (with usual notation)**

(a) decreases MDD, increases OMC
(b) decreases OMC, increases MDD
(c) increases OMC and MDD
(d) decreases OMC & MDD

**Ans. (b) :** Effect of increasing the compaction energy on compaction characteristics of a given soil decreases OMC, increases MDD.




Dry density, $\gamma_d = \frac{\gamma_b}{1+w}$

**127. The "Square root of time" fitting method is used to determine**

(a) Compression index ($C_c$)
(b) Coefficient of consolidation ($C_v$)
(c) Coefficient of volume change ($m_v$)
(d) Time factor ($T_v$)

**Ans. (b) : Square root time fitting method–** Used to determine coefficient of consolidation ($C_v$)

- It is also called taylor's method.

$$C_v = \frac{T_{90}.H^2}{t_{90}}$$

**128. In a UU Triaxial compression test on a $\phi = 0$ soil, failure was observed at a deviatoric stress of 60 kN/m$^2$ under a confining stress 50 kN/m$^3$ of the confining stress is increased to 100 kN/m$^2$, the deviatoric stress at failure is**

(a) 60 kN/m$^2$ (b) 110 kN/m$^2$
(c) 120 kN/m$^2$ (d) 160 kN/m$^2$

**Ans. (a) :** Given, $\sigma_d = 60$ kN/m$_2$
Confining stress, $\sigma_c = 100$ kN/m$^3$
$\phi = 0$
$\sigma_1 = \sigma_3 + \sigma_d$
$= 50+60 = 110$ kN/m$^3$

$$\sigma_1 = \sigma_3 \tan^2\left(45+\frac{\phi}{2}\right) + 2C\tan\left(45+\frac{\phi}{2}\right)$$

$$110 = 50\times\tan(45) + 2C\tan 45^0$$

$C = 30$ kN/m$^3$

$$\sigma_1^{'} = \sigma_3^{'}\tan^2\left(45+\frac{\phi}{2}\right) + 2C\tan(45+\phi)$$

$$\sigma_1^{'} = 160\ \text{kN/m}^2$$

$$\sigma_1^{'} = \sigma_3^{'} + \sigma_d^{'}$$

$$\sigma_d^{'} = 160 - 100$$

$$\sigma_d^{'} = 60\ \text{kN/m}^3$$

**129. Which roller is most suitable for the compaction of a gravelly sand mixture with 25% fines?**

(a) Smooth-wheeled roller
(b) Sheepsfoot roller
(c) Vibrating roller
(d) Heavy roller

**Ans. (a) :** Smooth-wheeled roller is most suitable for the compaction of a gravelly sand mixture with 25% fines.

- These are not effective for compacting deep layers of soils are the resulting compaction pressure is low and also cause stratification in deep layers due to non-uniform compaction.

**130. Clover leaf cofferdam belongs to**

(a) braced type (b) cellular type
(c) sheet pile type (d) diaphragm type

**Ans. (b) :** Cofferdam is a temporary structure that is built to keep the water away from the execution sits. So that the structure can be built on the dry surface.

**Type of cofferdams–**
(i) Cellular cofferdam
(ii) Earthen cofferdam
(iii) Braced cofferdam
(iv) Single-walled cofferdam
(v) Rockfill cofferdam,
(vi) Double-walled cofferdam

**131. What is the maximum value of water table correction factor in predicting bearing pressure based on Meyerhof's theory**

(a) 0.5
(b) 0.75
(c) 1.0
(d) Not considered

**Ans. (c) :** Maximum value of the water table correction factor can be 1, when either $Dw_1 = D_f$ or $Dw_f = B$

$$Rw_1 = 0.5\left(1+\frac{Dw_1}{D_f}\right)$$
$$= 0.5\ (1+1) = 1.0$$

**132. The effect of cohesion on the active and passive earth pressures is that**

(a) Both increase
(b) Both decrease
(c) Active earth pressure increases and passive earth pressure decreases
(d) Active earth pressure decreases and passive earth pressure increases

**Ans. (d) :** The effect of cohesion on the active and passive earth pressures is that active earth pressure decreases and passive earth pressure increases .

Active earth pressure $(P_a) = k_a\gamma z - 2c\sqrt{k_a}$

Passive earth pressure $(P_p) = k_a\gamma z + 2c\sqrt{k_a}$

**133. In Terazaghi's theory of bearing capacity of shallow foundation, which of the following zones is assumed to act as if it were part of the following**

(a) Zone of linear shear
(b) Zone of radial shear
(c) Zone of elastic equilibrium
(d) Overburden zone

**Ans. (c) :** In Terazaghi's theory of bearing capacity of shallow foundation, which of the following zones is assumed to act as if it were part of the following zone of elastic equilibrium.

**134. Which of the following type of sampler is preferred to collect an undisturbed sample (UDS) in cohesionless soils**
(a) thin wall tube samplers
(b) split spoon samplers
(c) piston samplers
(d) augers

**Ans. (c) :** Piston samples is preferred to collect an undisturbed sample in cohesionless soil.
• Sampler used to get undisturbed sampler, are Shelby tubes and thin walled sampler, for cohesive soil.

**135. Expansive soils are those which generally consists of**
(a) Silica (b) Feldspar
(c) Mica (d) Montmorillonite

**Ans. (d) :** Expansive soils are those which generally consists of Montmorillonite.

| **Minerals** | **Activity of clay ($A_C$)** |
|---|---|
| Kaolite | 0.38 |
| Illite | 0.92 |
| Montmorillonite | 7.2 |

**136. For a clay, SL = 20%, PL = 40%. The total volume at PL & LL was 1.20 $V_d$ and 1.50 $V_d$ respectively wherein $V_d$ = dry volume. Then, the plasticity index Pl of the soil is**
(a) 10 (b) 20
(c) 30 (d) 40

**Ans. (c) :** Given, SL = 20%
PL = 40%
$V_P = 1.2\ V_d$
$V_L = 1.5\ V_d$

$$\frac{V_L - V_P}{w_L - w_P} = \frac{V_P - V_D}{w_p - w_s}$$

$$\frac{1.5V_d + 0.2V_d}{I_p} = \frac{1.2V_d - V_d}{40 - 20}$$

$I_P = 30\%$

**137. The time required for a consolidating medium with double drainage to undergo 50% of its primary consolidation time was estimated as 8 years. All other conditions remaining same, estimate the time required if the medium has single drainage boundary**
(a) 4 years (b) 8 years
(c) 16 years (d) 32 years

**Ans. (d) :** We known that,

$$T_V = \frac{C_V \times t}{d^2}$$

For Double drainage path $d = \frac{H}{2}$

$$\frac{C_V \times t}{d^2} = \frac{\pi}{4}(U\%)^2$$

$$\Rightarrow \quad \frac{C_V \times 8}{\left(\frac{H}{2}\right)^2} = \frac{\pi}{4}(0.5)^2$$

$$\frac{C_V \times 32}{H^2} = \frac{\pi}{4}(0.5)^2 \qquad \text{.....(i)}$$

II Condition single drainge kesâ efueS d = H

$$\frac{C_V \times t}{H^2} = \frac{\pi}{4}(0.5)^2 \qquad \text{.....(ii)}$$

$$\frac{C_V \times t}{H^2} = \frac{C_V \times 32}{H^2}$$

$$\boxed{t = 32 \text{ years}}$$

**138. The coefficient of earth pressure in "At rest" condition is given by (with usual notations)**
(a) $k_0 = \frac{1-\sin\theta}{1+\sin\theta}$ (b) $k_0 = \frac{1+\sin\theta}{1-\sin\theta}$
(c) $k_0 = \frac{\mu}{1-\mu}$ (d) $k_0 = \frac{1+\cos\theta}{1-\cos\theta}$

**Ans. (c) :** The coefficient of earth pressure in "At rest" condition is given by (with usual notations) $k_0 = \frac{\mu}{1-\mu}$.
For $\phi$-soil, $k_0 = 1 - \sin\phi$

**139. Calculate coefficient of permeability in m/sec of coefficient of consolidation and volume change are 4.8 $mm^2$/min. and $1.02 \times 10^{-3}$ $m^2$/kN.**
(a) $4.8 \times 10^{-8}$ (b) $8.0 \times 10^{-10}$
(c) $1.02 \times 10^{-9}$ (d) $4.8 \times 10^{-10}$

**Ans. (b) :** Given, $C_v = 4.8\ mm^2/min$

Or $4.8 \times \frac{10^{-6}}{60} = 8 \times 10^{-8}\ m^2/sec$

$m_v = 1.02 \times 10^{-3}\ m^2/kN$

$k = C_v.m_v.\gamma$
$= 8 \times 10^{-8} \times 1.02 \times 10^{-3} \times 9.81$
$= 8.0 \times 10^{-10}$ m/sec

**140. Skempton's bearing capacity theory is valid for**
(a) clay soils (b) silt soils
(c) sandy soils (d) gravel soils

**Ans. (a): Skempton's method–** Applicable only for saturated clay to deep foundation also.
$q_{nu} = CNc$

**141. Which one of the following is relatively more accurate estimating method for building**
(a) Service unit
(b) Square meter of floor area
(c) Cubic meter of the building
(d) Elemental bill method

**Ans. (d) : Elemental bill method–** The estimate requires more information other than floor areas. To use this method, drawing with enough information are required i.e. wall elevation, structural frame, roof structure. Ceiling's foundation cross-section floor construction external works, wall and floor finishes.

**142. How much percentage of the estimated cost of the building works are usually provided in the estimate for electrification works**

(a) 10% of the estimated cost of the building
(b) 5% of the estimated cost of the building
(c) 8% of the estimated cost of the building
(d) 12% of the estimated cost of the building

**Ans. (c) :** 8% of the estimated cost of the building of the estimated cost of the building works are usually provided in the estimate for electrification works.
Contingencies work = 3–5%
Plant and tool = 1–1.5%

**143. The optimistic most likely and pessimistic time estimates for an activity are 6, 16 and 22 days, what could be the probabilistic time estimate**

(a) 15.33 days
(b) 30.66 days
(c) 14.66 days
(d) 29.33 days

**Ans. (a) :** Given,
Optimistic time to = 6 day
Most likely time $t_m$ = 16 day
Pessimistic time $t_p$ = 22 day

$$\text{Probabilistic time} = \frac{t_0 + 4t_m + t_b}{6}$$

$$= \frac{6 + (4 \times 16) + 22}{6} = 15.33$$

**144. A document containing detailed description of all the items of work together with their current rates is called**

(a) Analysis of rates
(b) Tender
(c) Schedule of Rates
(d) Cubic rate estimate

**Ans. (c) :** A document containing detailed description of all the items of work together with their current rates is called schedule of Rates.

- The determination of rate per unit of a particular item of work, from the cost of quantities of materials, the cost of labours and their miscellaneous petty expanses required for its completion know the analysis of rates.
- Tender notice giving brief description of work, etc, generally 2–2.5% of the estimate cost is deposited along with tender.

**145. The working from whole to the part is done in surveying in order to ensure that**

(a) plotting is done more quickly
(b) survey work is completed more quickly
(c) errors and mistakes of one portion do not affect the remaining portion
(d) number of errors is minimum

**Ans. (c) :** Accuracy must be maintained throughout the control network and this can be achieved by established different levels of control based on of the fundamental tents of surveying, namely working from the whole to the part.

- The working from whole to the part is done in surveying in order to ensure that errors and mistakes of one portion do not affect the remaining portion.

**146. Match List I with List II and select the correct answer using the codes given below from the lists**

| List-I | List-II |
|---|---|
| A. Correction for sag | 1. Techeometer |
| B. Lease count 30' | 2. Aerial Photograph |
| C. Overlap | 3. Base line |
| D. Additive Constant | 4. Prismatic compass |

(a) A-4, B-3, C-2, D-1 (b) A-3, B-4, C-2, D-1
(c) A-1, B-2, C-3, D-4 (d) A-3, B-4, C-1, D-2

**Ans. (b) : Match the list–**

| | |
|---|---|
| • Correction for sag | Base line |
| • Lease count 30' | Prismatic compass |
| • Overlap | Aerial hotograph |
| • Additive Constant | Techeometer |

**147. Dip of the magnetic needle at magnetic poles is**

(a) $0^0$ (b) $45^0$
(c) $60^0$ (d) $90^0$

**Ans. (d) :** Dip of the magnetic needle at magnetic pole is $90^0$ and at the magnetic equator, the dip needle rests horizontal at an angle of $0^0$.

**148. The datum adopted for Indis is**

(a) MSL at Bombay
(b) MSL at Karachi
(c) MSL at Madras
(d) MSL at Calcutta

**Ans. (a) : Mean sea level–** It is an average level of the surface of various oceans from which heights such as elevations may be measured.

- MSL adopted a survey of India (Dehradoon) for reference is located at mumbai high.

**149. Overturning of the vehicles on a curve can be avoided by using**

(a) Compound curve
(b) Transition curve
(c) Vertical curve
(d) Raverse curve

**Ans. (b) : Transition curve–** It is provided to change the horizontal alignment from straight to circular curve gradually and has a radius which decrease from infinity at the straight end to the desired radius of the circular curve at the other end.

- Overturning of the vehicle on a curve is provided by providing a transition curve.

**150. Which one of the following is an obstacle to chaining but not to ranging?**

(a) River (b) Hill
(c) Building (d) Ground

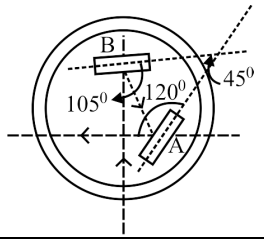
**Ans. (a) :** River is an obstacle to chaining but not to ranging.

- Building is obstacles to both chaining and ranging.

# Telanagana State Public Service Commission
# Assistant Engineer Exam., 2015
## Solved Paper

**1. The working principal of the optical square is based on**
(a) Reflection (b) Refraction
(c) Double reflection (d) Double refraction

**Ans. (c) :** Optical square is based on working principle of double reflection. It is used for setting out right angles and is based on the principle that a ray of light reflected successively from two surface undergoes a deviation of twice the angle between the reflecting surface.




**2. If the magnetic bearing of a line is $48^0$ 24' and the magnetic declination is $5^0$30' East, then the true bearing is**
(a) $42^0$ 54' (b) $37^0$ 24'
(c) $53^0$ 54' (d) $59^0$ 24'

**Ans. (c) :** True bending = Magnetic bearing ± Magnetic declination

- Use plus sign, if declination is in the east and minus sign of declination is in the west.

True bearing = $48^0$ 24' + 50 30'
True bearing = $53^0$54'

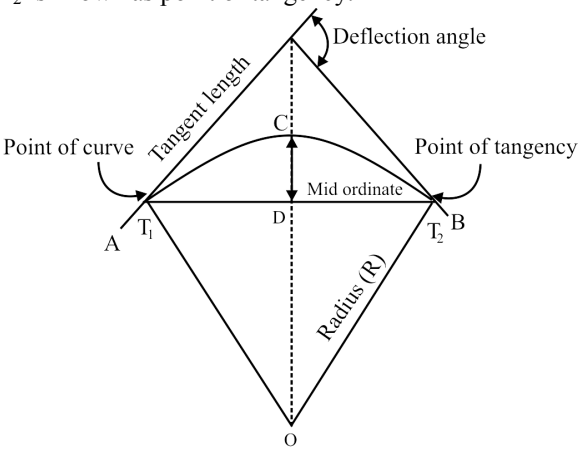
**3. Setting out a simple curve by two theodolite method does not require**
(a) Angular measurement
(b) Linear measurement
(c) Both angular and linear measurement
(d) Any measurement

**Ans. (b) :** Setting out a simple curve by two theodolite method does not require linear measurement because two-theodolite method is most convenient when the ground is undulating, rough and not suitable for linear measurement.

- The angular method are used for setting out curve.
- Rankine method of deflection angle (one-theodolite method.
- Two-theodolite method
- Tacheomatric method.

**4. Point of tangency is the**
(a) Beginning of the curve
(b) End of the curve
(c) Common point where the radius changes
(d) Common point where the radius and direction changes

**Ans. (b) :** End of curve or the last point of the tangent, $T_2$ is known as point of tangency.




**5. In chain surveying, field work is limited to**
(a) Linear measurements only
(b) Both linear and angular measurement
(c) Angular measurements only
(d) Vertical measurement

**Ans. (a) :** The ultimate aim of chain surveying is to make a plan of the area to be surveyed. This require measurement of linear distances (chaining accomplished with chain or tape) between the desire point. It is also a branch of surveying in which only linear measurement are made in the field.

**6. The correction to be applied to each 30 m chain length along slope is**
(a) 30 (1–sec α) m (b) 30 (sec α–1) m
(c) 30 (1–cos α) m (d) 30 (cot α–1) m

**Ans. (c) :** The correction to be applied to chain length along slope is $C_{SL} = L(1-\cos\alpha)$

Where, L = Chain length
$C_{SL}$ = Correction due to slope
For 30 cm chain length correction will be 30 (1–cos α)

**7. The correction for sag is**
(a) Always additive
(b) Always negative
(c) Always zero
(d) Some times additive and some times negative

**Ans. (b) :** Correction due to sag is always negative

Sag correction $\left[C_{sag} = \frac{w^2\ell}{24\ P_m^2}\right]$ (subtractive)

Where,
$P_m$ = Pull applied, W = Total weight of tape = $w\ell$
w = weight per meter length, $\ell$ = length of tape

**8. Which of the following error is not elimiated by the method of repetition for horizontal angle measurement?**

(a) Error due to eccentricity of verniers
(b) Error due to displacement of station signals
(c) Error due to wrong adjustment of line and trunnion axis
(d) Error due to inaccurate graduation

**Ans. (b) :** Repetition method for horizontal angle measurement does not eliminate error due to displacement of station signal.

**Repetition method eliminate different error–**

- Error due to the eccentricity of verniers and centre get eliminated as reading from both the verniers are taken.
- Error due to inaccurate graduations get eliminated as the reading are observed at different part of the circle.
- Error due to lack in the adjustment of line of collimation and the horizontal axis of the instrument get eliminated for considering both face readings.

**9. A triangle is said to be well conditioned when its angles lie between**

(a) $20^0$ and $150^0$ (b) $30^0$ and $120^0$
(c) $15^0$ and $135^0$ (d) $25^0$ and $130^0$

**Ans. (b) :** A triangle is stated to be a well condition triangle when all the angle in it is more than $30^0$ or lesser than $120^0$. A triangle in which an angle of the triangle is less than $30^0$ or greater than $120^0$ is known as ill conditioned triangle.

**10. Which of the following is not used in measuring perpendicular offsets?**

(a) Line ranger (b) Tape
(c) Optical square (d) Cross-staff

**Ans. (a) :** Lime ranger is not used in measuring perpendicular offsets. It is an optical instrument used in surveying for ranging.

Perpendicular offsets are measured by–
(i) using cross staffs
(ii) using optical square
(iii) using prism square
(iv) using tape

**11. Le-Chatelier apparatus is used to determine which of the following properties of cement?**

(a) Soundness (b) Initial setting time
(c) Fineness (d) Compressive strength

**Ans. (a) :** Le-chatelier apparatus and autoclave test are used to find out the soundness of OPC .

**Fineness test :** Method for testing fineness
(i) Sieve method
(ii) Air permeability method
(iii) Wagner turbiditimeter test

**Initials setting time–** Vicat apparatus is used for measuring setting time and consistency of concrete.

**12. The carrier in case of distemper is**

(a) Linseed oil (b) While lead
(c) Poppy oil (d) Water

**Ans. (d) : Distemper–** It is low quality paints

- The coating of distemper are usually thick and they are more brittle than other types of water paints.
- Ingredient of distemper is chalk and water. The chalk is used as base and water is used as carrier in distemper.
- They are less durable than oil paints.

**13. As per NBC 2005, institutional buildings comes under**

(a) Group-A (b) Group-B
(c) Group-C (d) Group-D

**Ans. (c) :** Institutional building comes under group - C as per NBC-2005.
Group A = Residential types of building
Group B = Educational types of building
Group D = Assembly building
Group E = Business building
Group F = Mercantile building
Group G : Industrial building
Group H : Storage building
Group I : Hazardous building

**14. Queen post truss is suitable for spans up to**

(a) 5 to 8 m (b) 12 m
(c) 16 m (d) 24 m

**Ans. (b) : Queen Post truss-** Span length between 8 to 12 meter. It is a tension member in a truss that have span longer opening than a king post truss.

**King post truss–** Span length in the range of 5 to 8 meters. It uses one central supports post, whereas the queen post truss uses two.

**15. Dressing of stone is done**

(a) After seasoning (b) After quarrying
(c) Before use (d) Before seasoning

**Ans. (b) :** The dressing of stone is done immediately after quarrying.

**Dressing of stone–**

- The process of giving a proper size, shape and finish to the roughly broken stone as obtained from the quarry is known as "Dressing of stone".
- Stones obtained from the quarries are very rough and irregular in shape and quite bulky in size and weight.

**16. Low heat cement consists lower percentage of which of the following?**

(a) $C_3A$ (b) $C_3S$
(c) $C_2S$ (d) $C_4S$

**Ans. (a, b) :** Low heat cement consists lower percentage of $C_3A$.

- $C_3S$ in the main contribution to strength of curved concrete in early phases.
- $C_3S$ & $C_3A$ are the main contribution to heat in the early phases.
- $C_3A$ develop approximately 840 Joule/gram and $C_3S$ 500 Joules/gram. Cement used in concrete contain approximately 5 times as much $C_3S$ as $C_3A$. Thus $C_3S$ is the main contributor to the heat development.

**17. Which of the following paints recommended for use on stucco plaster, brick and masonry surface?**

(a) Enamel paints (b) Emulsion paints
(c) Plastic paints (d) Oil paints

**Ans. (b) : Emulsion paints–** Emulsion paints use poly vinyl acetate and polystyrene as binding material and they contain drain like cobalt and manganese. Once applied, enamel paints offer durability, good colour retention and alkali resistance.

- Emulsion points are commonly used for interior walls, ceilings and masonry work.

**Cement Paints–** Cement paint is durable and water proof and its is commonly used in rough internal and external surface.

**Oil Paints–** Oil paints is commonly used in wall, door windows and metal structure and these paint are not suitable for humid condition and it takes time to dry completely.

**18. Gypsum is added to portland cement during its manufacturing so that it may**

(a) Accelerate the setting time
(b) Retard the setting time
(c) Decrease the burning temperature
(d) Facilitate grinding

**Ans. (b) :** Gypsum is called the retarding agent of cement which is mainly used for regulating the setting time of cement and is an indispensable component without gypsum. Cement clinker can condense immediately by mixing with water and release heat.

- Gypsum chemical formula $CaSO_4.2H_2O$.
- Gypsum slows down the setting time of cement and help in controlling the rate of hardening of the cement.

**19. Smith's test is conducted on a sample of stone to find out which of the following parameter?**

(a) Compressive strength
(b) Toughness
(c) Presence of soluble matter
(d) Hardness

**Ans. (c) : Smith's test–** This test is performed to find out the presence of soluble matter.

**Attrition test–** This test is used to find the rate of wear of stone.

**Impact test–** This test is used to find out the toughness index of stone.

**Crushing test–** It can be tested by the universal testing machine in which size sample of 40 × 40 × 40 mm should be placed in water for 72 hours before testing after that load in applied.

- The minimum value of crushing strength of stone should not be less than 100 $N/mm^2$.

**20. Presence of which of the following is responsible for imparting yellow tint to bricks?**

(a) Silica (b) Alumina
(c) Lime (d) Magnesia

**Ans. (d) :** Magnesia in brick earth impart yellow tint to the brick.

**Composition of good brick earth–**

**(i) Silica–**

- A good brick earth should contain 50-60% of silica.
- It prevent cracking, shrinking and wrapping of raw brick.
- Excess of silica make the brick brittle.

**Alumina–** A good brick earth should contain 20-30% of alumina.

- Alumina impart plasticity to the earth so that it can be moulded.
- Excess of alumina lead to shrink and wrap during drying and burning.

**Lime–** It should not be greater than 5% for good bricks.

**Iron Oxide–** It should be range of 5 to 6%

**Magnesia–** A good brick should contain a small quantity of magnesia (about 1%)

- It impart colour to brick.

**21. For a given system of coplanar concurrent forces, If $\sum F_x = 20$ N and $\sum F_y = 20$ N , then**

(a) R = $-20\sqrt{2}$N and $\alpha = 45^0$ with east
(b) R = $20\sqrt{2}$N and $\alpha = 135^0$ with east
(c) R = $20\sqrt{2}$N and $\alpha = 225^0$ with east
(d) R = $-20\sqrt{2}$N and $\alpha = 315^0$ with east

**Ans. (c) :** Given,

$\sum F_x = -20N$

$\sum F_y = -20N$

We know that the resultant $R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$

$R = \sqrt{(-20)^2 + (-20)^2}$

$= \sqrt{400 + 400} = \sqrt{800}$

$R = \sqrt{400 \times 2} = 20\sqrt{2}N$

Direction of resultant

$$\tan\alpha = \frac{\sum F_y}{\sum F_x} = 1$$

$\alpha = 45^0$ or $225^0$ with east

**22. Which of the following cases gives the least moment of inertia, in case of a square?**

(a) M.I. about the base side
(b) M.I. about the top side
(c) M.I. about the diagonal
(d) M.I. about one vertical side

**Ans. (c) :** Base side, vertical side and top side are same in case of square and moment of Inertia in

$I_{base\ side} = I_{C.G.} + A \times h^2$ (distance from base side to C.G.)

Moment of Inertia about diagonal

$I_{C.G.} = \frac{a^4}{12}$ [Diagonal is the axis consider with C.G. of square]

Hence we can say calculate that MOI about diagonal is least as compare to other cases.

**23. In a triangular section of size 'b ×h', if width is reduced to half and height is doubled, then its Ixx is increased by**

(a) 2 times (b) 4 times
(c) 8 times (d) 16 times

**Ans. (b):** Moment of Inertia of triangular section

$$I_{xx} = \frac{bh^3}{36}$$

If, $b = \frac{b'}{2}$ [if width is half]

$h = 2h'$ [If height is double]

$$I'_{xx} = \frac{\frac{b'}{2} \times (2h')^3}{36} = \frac{4b'h^3}{36}$$

$I_{xx} = 4I'_{xx}$

$I_{xx}$ is increased by 4 times.

**24. A structure member is generally designed so that the materials is stressed to**

(a) Yield stress (b) Ultimate stress
(c) Breaking stress (d) Working stress

**Ans. (d) : Working streams–** It is desirable to keep the stress lower than the maximum or ultimate stress at which failure of the material takes place. This stress is known as the working stress or design stress. It is also known as safe or allowable stress.

**Ultimate stress–** The stress which attains its maximum value is known as ultimate stress. It is defined as the largest stress obtained by dividing the largest value of the load reached in a test to the original cross sectional area of the test piece.

**Yield stress–** The stress of which a solid will undergo permanent deformation or plastic flow without a significant increase in the load or external force.

**25. In an RCC column, if $A_s$ = 1000 mm$^2$, $A_c$ = 10000 mm$^2$, $\sigma_c$ = 5N/mm$^2$ and m = 20, then load on the column is**

(a) 75 kN (b) 145 kN
(c) 150 kN (d) 1005 kN

**Ans. (c) :** Load carrying capacity of column is-

$$P = A_s\sigma_s + \sigma_c A_c$$

Transforming the area of steel into area of concrete $[\sigma_s = \sigma_c]$

$$m = \frac{A_c}{A_s}$$

$A_c = mA_s$

Given,

$A_s = 1000$ mm$^2$

$A_c = 1000$ mm$^2$ $\sigma_c = 5$ N/mm$^2$

m =20

$$P = (m \times A_s + A_c)\sigma_{cc}$$

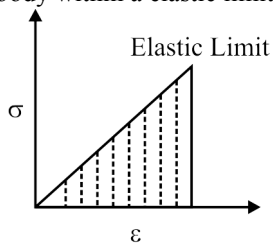
P = (20 × 1000 + 10000 ) × 5

P = 150 kN

**26. If σ and E for a body of volume 2 × 10$^5$ mm$^3$ are 10 N/mm$^2$ and 1 × 10$^5$ N/mm$^2$, Resilience of the body is**

(a) 10 N mm (b) 20 N mm
(c) 100 N mm (d) 200 N mm

**Ans. (c) : Resilience –** Total strain energy stored in a body within a elastic limit.




Where,

σ = stress Resilience = $\left(\frac{1}{2} \times \varepsilon \times \sigma\right)$ * Volume .... (i)

ε = strain

E = modulus of elasticity according to Hook's law

$\sigma \propto \varepsilon$

$\sigma \propto E\varepsilon$

$$\frac{\sigma}{E} = \varepsilon \quad \text{..... (ii)}$$

Putting value of strain in equation (i)

$$\text{Resilience} = \frac{\sigma^2 V}{2E}$$

Given,

$\sigma = 10$

$V = 2 \times 10^5$

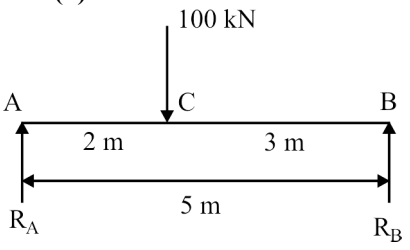
$E = 1 \times 10^5$

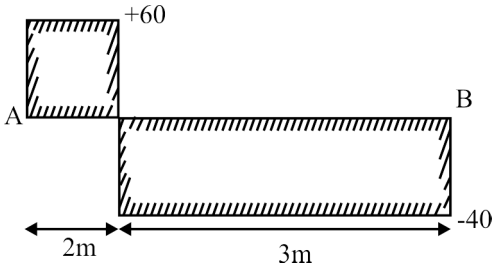
$$\text{Resilience} = \frac{(10)^2 \times 2 \times 10^5}{2 \times 1 \times 10^5} = 100 \text{ N} - \text{mm}$$

**27. If a simply supported beam of 5 m span carries a point load of 100 kN at 2 m from LHS, the maximum bending moment on it is**

(a) 62.5 kNm (b) 120 kNm
(c) 125 kNm (d) 312.5 kNm

**Ans. (b) :**




(a) Beam (Simply supported)

$$R_A = \frac{100 \times 3}{5} = 60 \text{ kN}$$

$R_B = 40$ kN

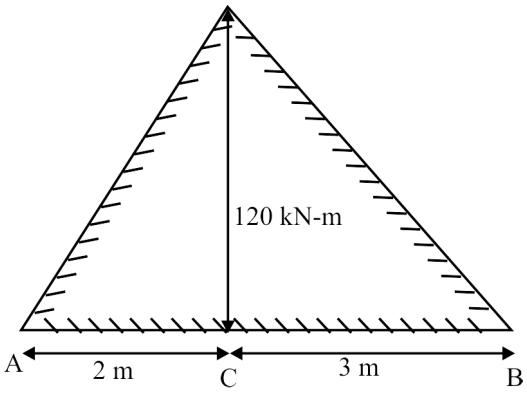
Where $R_A$ & $R_B$ are reaction.

(b) SFD,

S.F at B = –40

S.F. at C = –40+100 = 60

S.F at A = 0




(c) BMD

BM at B = 0

BM at C = 40 × 3 = 120 kn-m

BM at A = 40 × 3 – 60 × 2 = 0

The maximum bending moment on simply supported beam of 5 m span is 120 kn-m

**28. For a solid circular beam of 40 mm dia, the section modulus is**

(a) $\pi \times 10^3$ mm$^3$ (b) $2\pi \times 10^3$ mm$^3$

(c) $4\pi \times 10^3$ mm$^3$ (d) $8\pi \times 10^3$ mm$^3$

**Ans. (b) :** Section modulus of solid circle–

$$Z_{\text{solid cricle}} = \frac{I}{Y}$$

Where,

I = Moment of Inertia about neutral axis.

Y = Distance between neutral axis to extreme fibre

$$I_{\text{solid circle}} = \frac{\pi}{64}D^4$$

$$Y_{\text{solid circle}} = \frac{D}{2}$$

$$Z_{\text{solid circle}} = \frac{\frac{\pi}{64}D^4}{\frac{D}{2}} \Rightarrow Z = \frac{\pi}{32}D^3$$

Given,

Dia of circular beam = 40 mm

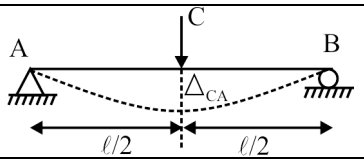
$$Z = \frac{\pi}{32}(40)^3 = 2\pi \times 10^3 \text{ mm}^3$$

**29. If maximum slope of a simply supported beam of span 'L' with midpoint load is $1^0$, then maximum deflection is**

(a) πL/240 (b) πL/270

(c) πL/540 (d) πL/576

**Ans. (c):**

| Simply supported beam | Slope at C w.r.t. A $\theta_{CA}$ | Deflection of C w.r.t. A ($\Delta_{CA}$= $\Delta_{max}$) |
|---|---|---|
|  | $\frac{w\ell^2}{16EI}$ | $\frac{w\ell^3}{48EI}$ |



Given,

Maximum slope = $1^0$

Span of beam = L

$$\theta_{CA} = 1^0$$

$$\frac{w\ell^2}{16EI} = 1 \times \frac{\pi}{180} \quad \text{..... (i) (Changing from degree to retain)}$$

Where,

$\theta_{CA}$ = Slope of beam

$\Delta_{max}$ = Maximum deflection of beam

$\ell$ = span of beam

$$\Delta_{max} \text{ or deflection} = \frac{w\ell^2}{48EI}$$

$$\Delta_{max} = \frac{w\ell^2 \times \ell}{16EI \times 3} \quad \text{..... (ii)}$$

Putting the value of $\frac{w\ell^2}{16EI}$ from equation (i)

$$\Delta_{max} = \frac{\pi}{180} \times \frac{\ell}{3}$$

$$\Delta_{max} = \frac{\pi\ell}{540}$$

**30. Moment area method can be easily adopted for the following case :**

(a) Simply supported beam with eccentric point load

(b) Simply supported beam with point loads & UDL

(c) Simply supported beam with two symmetrically placed unequal point loads

(d) Cantilever with point loads and UDL

**Ans. (d) :** Moment area method can be easily adopted for cantilever with point loads and UDL.

- Moment area method establishes a procedure that utilizes the area of the moment diagram [actually, M/EI diagram] to evaluate the slope or deflection at selected points along the axis of a beam.

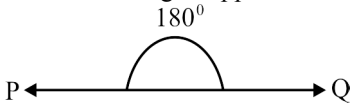
This method is applicable to both prismatic and non-prismatic beams.

**31. The resultant of two collinear forces P and Q, which are acting in opposite direction is**

(a) P + Q (b) P – Q

(c) $\sqrt{P^2 + Q^2}$ (d) $\sqrt{P^2 - Q^2}$

**Ans. (b):** The resultant of two collinear forces P and Q which are acting in opposite direction.




$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$ $\quad [\theta = 180^0]$

$R = \sqrt{P^2 + Q^2 + 2PQ\cos 180}$

$R = \sqrt{P^2 + Q^2 - 2PQ}$

$R\sqrt{(P-Q)^2}$

$R = P - Q$

**32. The force system that is applied to open a bottle cap is called**

(a) Collinear forces
(b) Like parallel forces
(c) Unlike parallel forces
(d) Couple

**Ans. (d) : Couple–** It is a pair of equal parallel forces that are opposite in direction. The only effect of a couple is to produce or prevent the turning of body. The turning effecting or moment of a couple in measured by the product of the magnitude of either force and the perpendicular distance between the action lines of forces.

**Example of couple–**

- Steering wheel of car
- Turning of a screw driver
- Opening of the bottle cap

**33. In a circular section of diameter D, if 'D' is doubled, then its polar M.I. will be increased by**

(a) 4 times (b) 8 times
(c) 16 times (d) 32 times

**Ans. (c) :** The polar moment of inertia of a circular section.

$I_P = \frac{\pi D^4}{64}$ ..... (i)

If D is doubled

$D' = 2D$

$I'_P = \frac{\pi}{64}(2D)^4$

$I'_P = 16\frac{\pi D^4}{64}$

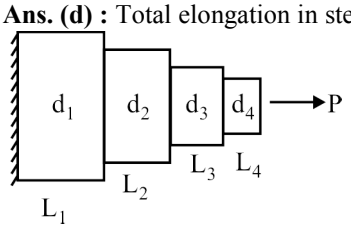
$I'_P = 16\, I_p$

The polar moment of inertia is increased by 16 times of diameter is doubled.

**34. In case of a stepped bar of a material subjected to an axial load, the total elongation is proportional to**

(a) P/E
(b) $[\ell_1 + \ell_2 + ..... + \ell_n]$
(c) $\left[\frac{1}{A_1} + \frac{1}{A_2} + ..... + \frac{1}{A_n}\right]$
(d) $\left[\frac{\ell_1}{A_1} + \frac{\ell_2}{A_2} + ..... + \frac{\ell_n}{A_n}\right]$

**Ans. (d) :** Total elongation in stepped bar




Total elongation of bar = $\Delta L_1 + \Delta L_2 + \Delta L_3 + \Delta L_4$

$\Delta L = \frac{PL_1}{A_1E} + \frac{PL_2}{A_2E} + \frac{PL_3}{A_3E} + \frac{PL_4}{A_4E}$

$\Delta L = \frac{P}{E}\left[\frac{L_1}{A_1} + \frac{L_2}{A_2} + \frac{L_3}{A_3} + \frac{L_4}{A_4}\right]$

$\Delta L \propto \left[\frac{L_1}{A_1} + \frac{L_2}{A_2} + \frac{L_3}{A_3} + \frac{L_4}{A_4}\right]$

Where,
P = axial load
E = Modulus of elasticity
L = Length of bar
A = Area of bar
Here, we can conclude that total elongation in stepped bar in proportional to $\left[\frac{\ell_1}{A_1} + \frac{\ell_2}{A_2} + \frac{\ell_3}{A_3} + \frac{\ell_3}{A_4}\right]$

**35. The unit for modulus of resilience is**

(a) Joule (b) Joules/mm
(c) Joules/mm$^2$ (d) Joules/mm$^3$

**Ans. (d) : Proof resilience–** It is the maximum energy that can be absorbed upto elastic limit without creating a permanent distortion $\mu r = \frac{\sigma^2 V}{2E}$

**Modulus of resilience–** It is the maximum energy that can be absorbed per unit volume without creating permanent distortion.

$\text{Modulus of resilience} = \frac{\text{Proof Resilience}}{\text{Volume}}$

$MOR = \frac{\frac{\sigma^2}{2E}\text{volume}}{\text{Volume}} = \frac{\sigma^2}{2E}$

$MOR = \frac{(kN/mm)^2}{kN/mm^2} = \frac{Joule}{mm^3}$

Unit of modulus of resilience is Joule mm$^{-3}$.

**36. Variation of bending moment under a UDL, in a beam is**

(a) Straight line variation
(b) Parabolic variation
(c) Cubic variation
(d) Zero i.e. horizontal straight line

**Ans. (b) :** In uniformily distributed load the nature of shear force in linear and bending moment is parabolic.

$W_{(x)} = W_{XB}$

$S.F. = \int (x) = wx^1 [\text{Linear}]$

$B.M. = \int S.F.dx = \frac{wx^2}{2} [\text{Parabolic}]$

In presence of distributed load S.F. variation is one order higher than load intensity variation and B.M. variation is one order higher than S.F. variation

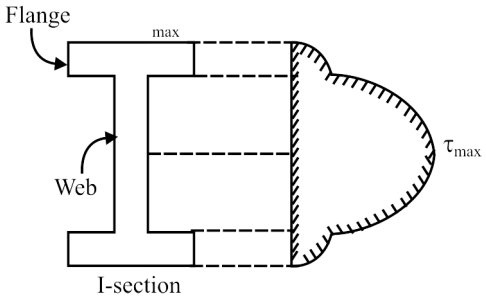
| Load | S.F.D. | B.M.D. |
|---|---|---|
| Moment/ Couple | constant | linear line/straight line |
| Point load | linear/straight line | Inclined line |
| UDL | Inclined line | Parabolic curve |
| UVL | Parabolic curve | Cubic parabola profile. |

**37. In case of an I-beam, major percent of the shear force at a section is resisted by**

(a) Top flange
(b) Bottom flange
(c) Top and bottom flanges together
(d) Web

**Ans. (d) :** I beam major percentage of the shear force at a section in resisted by web.

**Shear stress distribution**




I-section

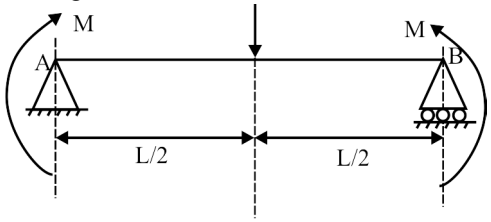
About 85% of the shear force is resisted by the web.
Hence the web resisted the shear force and flange resist the bending moment.

**38. Maximum deflection in a simply supported beam subjected to pure bending is given by**

(a) $\delta = \frac{ML}{2EI}$ (b) $\delta = \frac{ML^2}{8EI}$

(c) $\delta = \frac{ML^2}{12EI}$ (d) $\delta = \frac{4}{48}\frac{ML^2}{EI}$

**Ans. (b) :** Simply supported beam subjected to pure bending-.



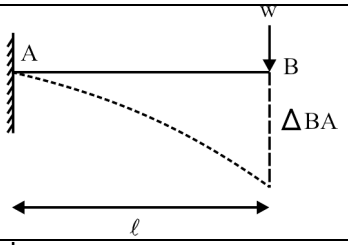
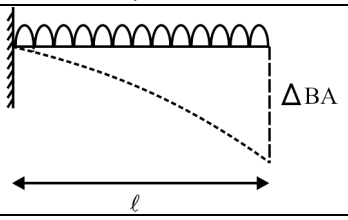

$\theta_{BA} = \theta_{max} = \frac{ML}{2EI}$

$\delta_c = \delta_{max} = \frac{ML^2}{8EI}$

**39. Other parameters being unchanged, if the span of a cantilever carrying end point load is doubled, the maximum slope is increased by**

(a) 2 times (b) 4 times
(c) 8 times (d) 16 times

**Ans. (b) :**

| Cantilever beam | slope of B w.r.t. A, $\theta_{BA}$ | Deflection of B w.r.t. A $\Delta_{BA}$ |
|---|---|---|
|   | $\frac{w\ell^2}{2EI}$ | $\frac{w\ell^3}{3EI}$ |
|   | $\frac{w\ell^3}{6EI}$ | $\frac{w\ell^4}{8EI}$ |

Slope of cantilever carrying end point load = $\frac{w\ell^2}{2EI}$

$\theta_{max_1} = \frac{w\ell^2}{2EI}$

If the span of cantilever beam is doubled

$\theta_{max_2} = \frac{w(2\ell)^2}{2EI}$

$\theta_{max_2} = \frac{4w\ell^2}{2EI}$

$\theta_{max_2} = 4\theta_{max_1}$

**40. The condition for stability of a dam against over turning is**

(a) $\frac{P.h/3}{W(b-\bar{x})} \geq f.s.$ (b) $\frac{W(b-\bar{x})}{P \times \frac{h}{3}} \geq f.s.$

(c) $\frac{P.h/3}{Wb} \geq f.s.$ (d) $\frac{W.b}{P.h/3} \geq f.s.$

**Ans. (b) : Overturning to fhe dam–** The dam does not overturn about the toe point say A the resultant R of the weight of unit length of the dam and the water pressure acting on it must not cut the base line of th dam beyond the toe point A i.e. Z in the limiting case may correspond to the point A and so the eccentricity of the resultant.

$$e = Z - \frac{b}{2} = b - \frac{b}{2} = \frac{b}{2}$$

Over turning moment due to water pressure about the toe point

$$A = P \times \frac{H}{3}$$

Restoring moment due to self weight of the dam about the toe point

$$A = W \times (b - \bar{x})$$

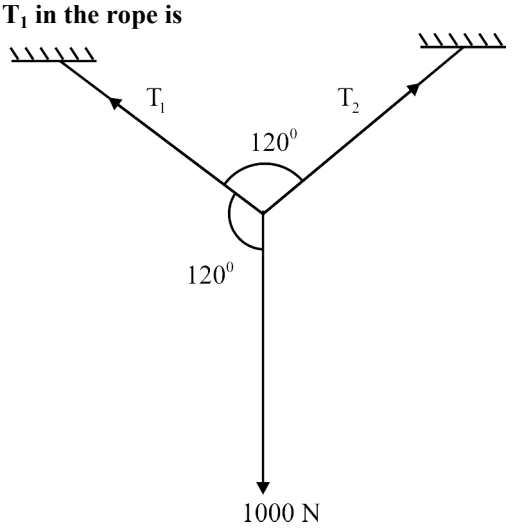
For the dam not to turn about A the condition to be satisfied that

$$W \times (b - \bar{x}) > P \times \frac{H}{3}$$

Factor of safety against overturning = $\dfrac{w \times (b - \bar{x})}{P \times \frac{H}{3}} \geq 1.5$

So for stability of dam $\dfrac{W(b - \bar{x})}{P \times \frac{h}{3}} \geq (\text{f.s.})$

**41. For the force system shown below, the tension $T_1$ in the rope is**




(a) 500 N (b) 866 N
(c) 1000 N (d) 1732 N

**Ans. (c) :** By Lami theorem

$$\frac{T_1}{\sin 120} = \frac{T_2}{\sin 120} = \frac{1000}{\sin 120}$$

$$T_1 = \frac{\sin 120^0 \times 1000}{\sin 120} = 1000\,\text{N}$$

The tension produce in rope $T_1$ is 1000 N.

**42. The horizontal component of a force P acting towards north is**

(a) 0 (b) P
(c) 2P (d) $\infty$

**Ans. (b) :**

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

$\theta = 90^0$, $Q = 0$

$R = P$

**43. The moment of inertia of a semi-circle of diameter D about its base diameter is**

(a) $\frac{\pi}{32}D^4$ (b) $\frac{D^4\pi}{64}$
(c) $\frac{\pi D^4}{128}$ (d) $\frac{\pi D^4}{256}$

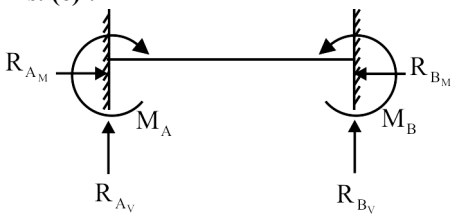
**Ans. (c) :**

| Type of section | Moment of Inertia |
|---|---|
| Rectangle (about it C.G.) | $\frac{bd^3}{12}$ |
| Circle (about it C.G.) | $\frac{\pi d^4}{64}$ |
| Triangle (about it C.G.) | $\frac{bd^3}{36}$ |
| Semicircle about its base diameter | $\frac{\pi D^4}{128}$ |

**44. The number of unknown reactions to be found at a fixed support of a beam, during analysis is/are**

(a) 1 (b) 2
(c) 3 (d) 4

**Ans. (c) :**




For a general system of loading for a fixed beam for each end, the number of unknown are 3 which are horizontal reaction, vertical reaction and moment of the fixed end, so a total of 6 unknown.

We have 3 available equation of equilibrium which is

$$\Sigma F_x = 0,\ \Sigma F_y = 0 \text{ and } \Sigma M_z = 0$$

**45. Effective length of a column of length 'L' with one end fixed and other end hinged is given by**

(a) $L_e = 1$ (b) $L_e = L/2$
(c) $L_e = L/\sqrt{2}$ (d) $L_e = 2L$

**Ans. (c) :**

| End condition | Both end are hinged | One end fixed other free | Both end fixed | One end fixed and other hinged |
|---|---|---|---|---|
| Effective length ($\ell_e$) | L | 2L | $\frac{L}{2}$ | $\frac{L}{\sqrt{2}}$ |

**46. If cohesion > adhesion, then**

(a) Capillary rise occurs
(b) Depression occurs
(c) Remain plane
(d) Either rise or fall

**Ans. (b) :** A concave meniscus occur when the particles of the liquid are more strongly attracted to the container (adhesion) than to each other (cohesion), causing the liquid to climb the walls of the container. This occurs between water and glass.

- A convex meniscus occur when the particle in the liquid have a stronger attraction to each other (cohesion) than to the material of the container.
- When the cohesion between molecule of fluid is greater than the adhesion between fluid and the glass then the free level of fluid in glass tube dipped in the glass vessel will be lower than the surface of the fluid..

**47. For a vacuum pressure of 4.5 m of water, the equivalent absolute pressure is**

(a) 5.83 m of water (b) 14.83 m of water
(c) 12.33 m of water (d) 8.83 m of water

**Ans. (a) :**
Absolute pressure = Gauge pressure + local atmospheric pressure
Gauge pressure can be positive, negative or zero
Vacuum pressure = Negative Gauge Pressure

$P_{absolute} = P_{ATM} + \left(-P_{gauge}\right)$

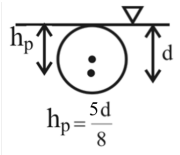
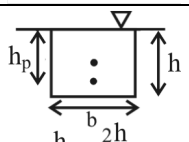
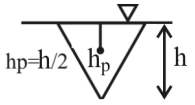
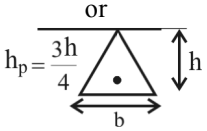
$P_{absolute} = 10.3 – 4.5$

$P_{absolute} = 5.83$ m of water

**48. A vertical triangular area with vortex downward and altitude 'h' has its base on the free surface of the liquid. The centre of pressure below the free surface will be at a depth of**

(a) h/4 (b) h/3
(c) h/2 (d) 2h/3

**Ans.** (c)

| Shape | Centre of Pressure |
|---|---|
| Circular |   $h_p = \frac{5d}{8}$ |
| Square or Rectangular |   $h_p = \frac{2h}{3}$ |
| Triangular |   or   $h_p = \frac{3h}{4}$ |

**49. The velocity at the Reynold's number equal to 2000 is called**

(a) Critical velocity
(b) Lower critical velocity
(c) Higher critical velocity
(d) Uniform velocity

**Ans. (b) : Reynolds number –** It is the ratio of internal forces to viscous forces.
It is dimension less number used to categorize the fluid system in the effect of viscosity is important in controlling the velocity or the flour pattern of fluid.

$Re = R_e = \frac{\rho V d}{\mu}$

Where, $\rho$ = density, v= velocity,
d = diameter, $\mu$= viscosity

**50. To avoid separation, the suitable ratio of throat diameter and pipe diameter in a venturimeter is**

(a) $\frac{1}{4}$ to $\frac{1}{2}$ (b) $\frac{1}{3}$ to $\frac{1}{2}$
(c) $\frac{1}{3}$ to 1 (d) 1 to 4

**Ans. (b) : Venturimeter –** It is a device used for measuring the rate of flow of a fluid flowing through a pipe.

- It is based on principle of Bernaullis equation.
- The angle of convergence is generally, 20-22 degree.
- Length of convergent cone is 2.7 D
- The desirable ratio of throat diameter and pipe diameter lies between $\frac{1}{3}$ to $\frac{3}{4}$
- The most appropriate option is $0.33\left(\frac{1}{3}\right)$ to $0.5\left(\frac{1}{2}\right)$ as the energy losses would minimize of the area difference in minimized.

**51. Due to each and contraction, the crest length is reduced by**

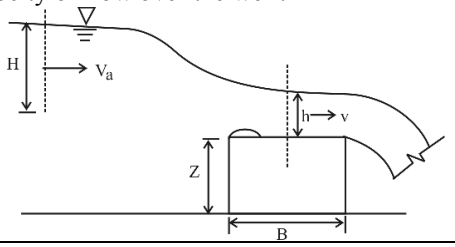
(a) 0.1 L (b) 0.1 H
(c) 0.1 $v^2/2g$ (d) 0.01 L

**Ans. (b) :** Due to each end contraction the crust length in reduced by 0.1 H from each site of length.
$\Rightarrow$ For rectangular, effective length = L – 0.2 H

**52. Condition for broad crested weir is**

(a) 2b > H (b) 2b < H
(c) H < b (d) b = 2H

**Ans. (a) :** A weir having a wide crest is known as broad-crested weir.
H = height of water above crest
b = length of the crest
If 2b > H the weir is known as broad crested weir.
If 2b < H the weir is called as narrow crested weir.
h = head of water at the middle of weir which is constant.
v = velocity of flow over the weir.



Broad crested weir with round upstream end
Discharge,

$$Q_{th} = Lh\sqrt{2g(H-h)}$$

$$Q_{act} = C_d Lh\sqrt{2g(H-h)}$$

The condition for maximum discharge–

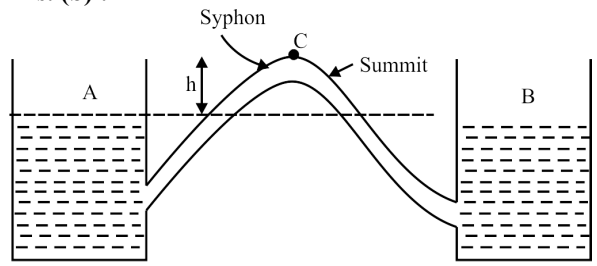
$$\frac{dQ}{dh} = C_d.L\sqrt{2g}\left[\sqrt{H-h} - \frac{h}{2\sqrt{H-h}}\right] = 0$$

$$h = \frac{2}{3}H = 0.6666H = 0.67H$$

**53. To prevent cavitation to occur, the maximum height of the summit of the syphon shall be**

(a) 7.5 m (b) 10.3 m
(c) 15 m (d) 20 m

**Ans. (b) :**




When the pressure is less than atmospheric pressure then vaporization will occur in water, so avoiding boiling construction in siphon pipe because it can harm pipe due to cavitation.

When we increase the elevation, pressure will decrease so there is limiting height where pressure of pipe is equal to atmospheric pressures, it is critical height of pipe till summit.

So, take top of the water level as reference line measured height h from there.

$981 \times 9.81 \times h = 101325$

$h = 10.5$ meter

**54. A turbine is a device which converts**

(a) Hydraulic energy into Mechanical energy
(b) Mechanical energy into hydraulic energy
(c) Kinetic energy into Mechanical energy
(d) Electrical energy into Mechanical energy

**Ans. (a) :** A turbine is a device or machine tht convert hydraulic energy to mechanical energy.

There are different types of turbine–

**Impulse turbine –** If at the inlet of the turbine, the energy available is only kinetic energy the turbine is known as an impulse turbine.

**Tangential flow turbine–** In this type of turbine the water strikes the runner in the direction of the tangent to the wheel.

**Radial flow turbine–** In this type of turbine, the water strikes in the radial direction accordingly.

**It is further classified as –**

**(i) Inward flow turbine–** The flow is inward from the periphery to the centre.

Ex – old Francis turbine

**(ii) Outward flow turbine–** The flow is outward from the centre to periphery.

**55. Francis turbine is a**

(a) Axial flow impulse turbine
(b) Radial flow reaction turbine
(c) Axial flow reaction turbine
(d) Radial flow impulse turbine

**Ans. (b) :** Francis turbine is a inward flow reaction turbine having radial discharge at outlet (mixed flow turbine).

- Specific speed of Francis turbine 60 to 300.
- Flow ratio of Francis turbine 0.15 to 0.30.
- Speed ratio of Francis turbine 0.60 to 0.90.

**56. Capillarity is due to**

(a) Cohesion only
(b) Adhesion only
(c) Both cohesion and adhesion
(d) Viscosity only

**Ans. (c) :** If a tube of the very narrow bore (called capillary) is dipped in a liquid it is found that the liquid in the capillary either ascends or descends relative to the surrounding liquid, this phenomenon is called capillary. Capillary action occur because water molecule bond each other strongly due to forces of cohesion and adhesion.

**57. The position of centre of pressure on a plane surface immersed vertically in a static mass of fluid is**

(a) at the centre of submerged area
(b) railways above the centre of gravity
(c) always below the centre of gravity
(d) has no relation with centre of gravity

**Ans. (c) :** The position of centre of pressure on a plane surface immersed vertically in static mass of fluid in always below the centre of gravity.

Centre of pressure $\Rightarrow h = \bar{x} + \frac{I_G}{A\bar{x}}$

Where, $\bar{x}$ = distance of CG from free water surface

$I_G$ = moment of inertia about C-G axis.

**58. Hydraulic gradient line (HGL) represents sum of**

(a) Pressure head and kinetic head
(b) Kinetic head and datum head
(c) Pressure head, kinetic head and datum head
(d) Pressure head and datum head

**Ans. (d) : Hydraulic gradient Line (H.G.L.) –** Line representing the sum of pressure heat and datum head.

**Total energy line (T.E.L.)–** Line representing the sum of pressure head, datum head and velocity head.

**59. The inlet length of venturimeter**

(a) is equal to the outlet length
(b) is more than the outlet length
(c) is less than the outlet length
(d) has no relation with the outlet length

**Ans. (c) :** The venturimeter is divided into three parts–

**Convergent cone–** It is the region where the cross section emerges into a conical shape for the converging region is attached to the inlet pipe and its cross sectional area decrease from beginning to end.

**Throat–** It is the middle part of the venturimeter and has the lowest cross-sectional area. The length is equal to the diameter of throat. Generally the diameter of throat is 1/4 to 3/4 of the diameter of inlet pipe.

**Diverging cone–** Diverging section in the third part of the device. One side it is attached to the outlet pipe. The diameter of this section in gradually increasing.

The diverging section has an angle of 5 to 15 degree the diverging angle is less than the converging angle due to the length of the diverging cone being larger than the converging cone.

**60. The average value of co-efficient of velocity is**

(a) 0.62 (b) 0.76

(c) 0.84 (d) 0.97

**Ans. (d) : Coefficient of velocity ($C_v$) –** It is defined as the ratio between the actual velocity of a jet liquid at vena contracta and the theoretical velocity of jet.

- The value of $C_v$ varies from 0.95 to 0.99 for different orifices depending on the shop, size of the orifice and on the head under which flow rakes place. Generally the value of $C_v = 0.98$ is taken for sharp edged orifices.

$$C_v = \frac{V}{\sqrt{2gH}}$$

Where, V = actual velocity

$\sqrt{2gH}$ = Theoretical velocity

**Coefficient of contraction ($C_C$) –** It is defined as the ratio of the area of the jet at vena-contracta to the area of the orifice. It is denoted by $C_c$.

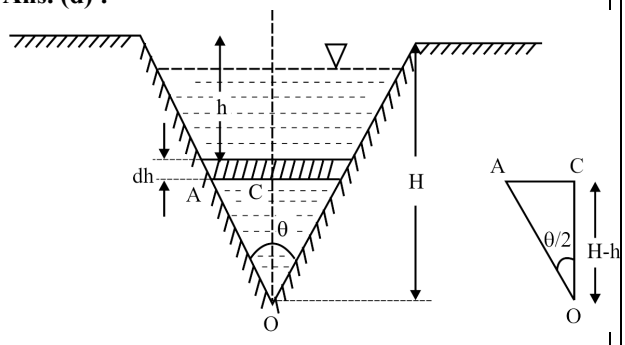
**Coefficient of discharge ($C_d$)–** It is defined as the ratio of the actual discharge from an orifice to the theoretical discharge from the orifice. It is denoted by $C_d$.

**61. The discharge over a right angled V-notch is**

(a) $\frac{8}{15}C_d\sqrt{2gH}$ (b) $\frac{8}{15}C_d\sqrt{2gH^{3/2}}$

(c) $\frac{8}{15}C_d\sqrt{2gH^2}$ (d) $\frac{8}{15}C_d\sqrt{2gH^{5/2}}$

**Ans. (d) :**




H = Head of water above the V-notch.

θ = Angle of notch

$C_d$ = Coefficient of discharge

Total discharge Q is

$$Q = \frac{8}{15}C_d \times \tan\frac{\theta}{2} \times \sqrt{2g} \times H^{5/2}$$

For a right-angled V-notch

$$\theta = 90^0, \quad \tan\frac{\theta}{2} = 1$$

Discharge $Q = \frac{8}{15} \times C_d \times \sqrt{2g} \times H^{5/2}$

$$Q = \frac{8}{15}C_d \times \sqrt{2g} \times H^{5/2}$$

**62. Low discharge are effectively measured by**

(a) Rectangular notch (b) Stepped notch

(c) Trapezoidal notch (d) Triangular notch

**Ans. (d) :** A V-notch weir is also called triangular notch or weir. The discharge over a triangular weir or notch is given by the

$$Q = \frac{8}{15}C_d \tan\frac{\theta}{2}\sqrt{2g}\ H^{5/2}$$

A triangular notch or weir is preferred to a rectangular weir or notch due to following reason –

- The expression for discharge for right - angled V-notch or weir in very simple.
- For measuring the low discharge a triangular notch gives more accurate result.
- In the case of the triangular notch only one reading i.e. H is required for the computation of discharge.

**63. The hydraulic mean depth of a pipe of 1 m diameter flowing full is**

(a) 0.12 m (b) 0.5 m

(c) 0.25 m (d) 2.0 m

**Ans. (c) :** The hydraulic mean depth or also called hydraulic radius is defined as

$$R = \frac{\text{Wetted Area}}{\text{Wetted Perimeter}} = \frac{A}{P}$$

Wetted Area = $A = \frac{\pi}{4}d^2$

Wetted perimeter = $P = \pi d$

Hydraulic Mean depth = $\frac{A}{P} = \frac{\pi d^2}{4\pi d} = \frac{d}{4}$

Given,

$d = 1$ m

For circular pipe

$$R = \frac{A}{P} = \frac{d}{4}$$

$$R = \frac{1}{4} = 0.25\,\text{meter}$$

**64. The discharge through a channel of rectangular section will be maximum if**

(a) Its depth is twice the breadth

(b) Its breadth is twice the depth

(c) Its depth is thrice the breadth

(d) Its breadth is thrice the depth

**Ans. (b) :** The most efficient section is one that gives maximum discharge for a given amount of excavation. For maximum discharge through the rectangular channel the geometric condition can be found by minimizing the wetted perimeter P.

$$P = B + 2d$$

$$P = \frac{Bd}{d} + 2d \qquad [Bd = A]$$

$$P = \frac{A}{d} + 2d$$

Now,

$$\frac{dP}{dd} = \frac{-A}{d^2} + 2$$

For maximum discharge–

$$\frac{dP}{dd} = 0$$

$A = 2d^2$

$Bd = 2d^2$

$B = 2d$

**65. A drop of water maintains its spherical shape on account of its**

(a) Cohesion (b) Adhesion

(c) Viscosity (d) Capillarity

**Ans. (a) :** The cohesive forces between liquid molecules is responsible for the surface tension. The molecule of water on the surface don't have like molecule around it. So the water droplets tends to be pulled into shape of spherical due to cohesive forces.

That's why drop of water is always spherical.

**Surface tension–** It is defined as force per unit length in the plane of the liquid surface at right angles to either side of an imaginary line drawn to surface.

It is tendency of the liquid surface to shrink into a minimum surface area.

**Application of surface tension–**

- A falling drop of water acquire the spherical shape due to surface tension.
- The rise of sap in a tree and the break up of of liquid jet are example of surface tension.

**66. The quantities and unit rate of certain items of work of building are**

**(i) Brick work of 20 $m^3$ and Rs. 2520/- per $m^3$**

**(ii) Plastering of 150 $m^3$ and Rs. 1250/- per $m^3$**

**Calculate the total cost of the items given**

(a) Rs. 50,400 = 00

(b) Rs. 1,87,500 = 00

(c) Rs. 1,37,100 = 00

(d) Rs. 2,37,900 = 00

**Ans. (d) :** Given,

- Brick work of 20 $m^3$ at rate of Rs. 2520 per $m^3$
- Plastering of 150 $m^2$ at rate of Rs. 1250 per $m^2$.

Cost of brickwork = Quantity of brickwork × Rate of brickwork per $m^3$

= 20 × 2520 = 50400

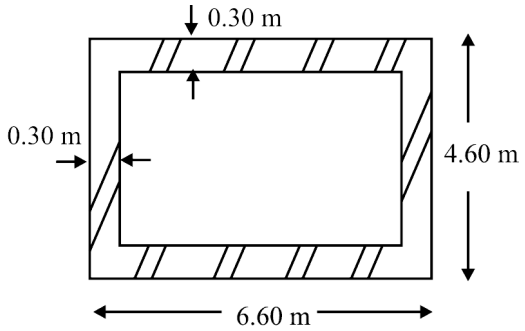
Cost plastering = 150 × 1250 = 187500

Total cost of brick work and plastering

= 50400+ 187500

= 237900 Rs.

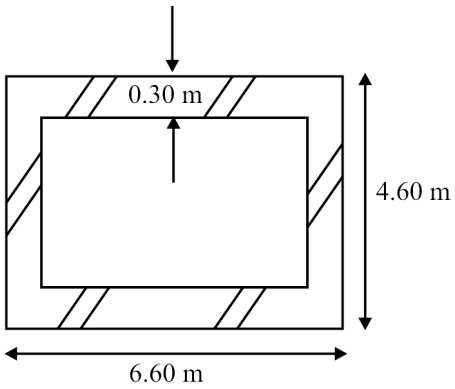
**67. Calculate the quantity of internal plastering for a enclosure as shown below. The height is 3.0 m.**




(a) 10 $m^3$ (b) 20 $m^3$

(c) 60 $m^2$ (d) 60 $m^3$

**Ans. (c) :**




Height of wall = 3 m

Internal length = 6.60 – 0.30 – 0.30 = 6 m

Internal width = 4.60 – 0.30 – 0.30 = 4 m

Total area of plastering = 2 (6×3) +2 ×(4×3)

= 60 $m^2$

The plaster work is measured in $m^2$.

**68. Estimate the cost of brick work for a wall of 4 m long, 3 m high and 30 cm thick. The rate of brick work is Rs. 1,500 = 00 per Cu.m**

(a) Rs. 540,000

(b) Rs. 54,000

(c) Rs. 18,000

(d) Rs. 5,400

**Ans. (d) :** The quantity of brickwork is measured in $m^3$.

Length of wall = L = 4 m

Height of wall = H = 3 m

Thickness of wall = t = 30 cm = 0.30 m

Quantity of brickwork = 4 × 3 × 0.30

= 3.6 m

Total cost of brickwork = 3.6 × 1500 = 5400 Rs.

**69. Calculate the quantity of earth work by mid-ordinate method, for 200 m length of a portion of road in an uniform ground. The heights of banks at the two ends being 1.00 m and 2.00 m. The formation width is 10 m and side slope 2 : 1**

(a) 3800 Cu.m (b) 3900 Cu.m

(c) 3950 Cu.m (d) 4000 Cu.m

**Ans. (b) :** Mid ordinate formula or mean depth formula

$$d_m \text{ or } d_{avg} = \frac{d_1 + d_2}{2}$$

$A = B\, d_{avg} + nd^2_{avg}$

$\boxed{Q = A \times L}$

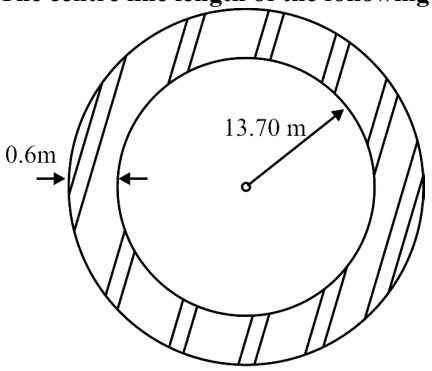
Q = Quantity of work
A = Area of work
L = Length of work
$d_m$ or $d_{avg}$ = Mean depth or height of work
h = slope
B = width
Given,
L = 200 m, $d_1$= 1.0 m, $d_2$=2.0 m, B = 10 m, h = 2 : 1

$$d_m \text{ or } d_{avg} = \frac{1+2}{2} = 1.5 \text{ m}$$

$$A = 10 \times 1.5 + 2(1.5)^2 = Bd_m + nd_m^2$$

Area = 19.5 $m^2$
Quantity = Length × Area = 200 × 19.5 = 3900 $m^3$

**70. The centre line length of the following figure is**

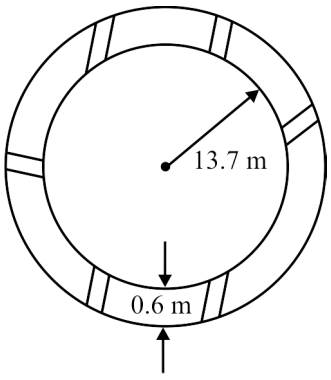



(a) 26.20 πm (b) 28 πm
(c) 28.60 πm (d) 196 πm

**Ans. (b) : Centre line method :** This method consider centre to centre distance between wall.

- Total length of work remains constant from foundation to super structure.
- This method is quick and easy but light chance of error exists.

Given,




Internal radius = 13.7 m

$$\text{Centre radius} = 13.7 + \frac{0.6}{2} = 14$$

Centre line length = 2π × Centre radius
Centre line length = 2 × π × 14
= 28 π

**71. As per IS 3861 : 2002, in the detailed estimate the volumes are worked out to the nearest of**

(a) 0.01 Cu.m (b) 0.05 Cu.m
(c) 0.005 Cu.m (d) 0.001 Cu.m

**Ans. (a) :** As per IS 3861: 2002 in detailed estimate-

- Weight : Should be measured nearest to 1kg or 0.001 tonne.
- Length : It should be worked nearest to 1 cm or 0.01 m
- Area : It should be worked nearest to 100 $cm^2$ or 0.01 $m^2$
- Volume : It should be worked nearest to 1000 $cm^2$ or 0.01 $m^3$

**72. Estimate the rate of metal at site per Cu.m. whose rate at source is Rs. 800/Cu.m (including loading & unloading) and lead of 35 km @a rate of Rs. 20/km.**

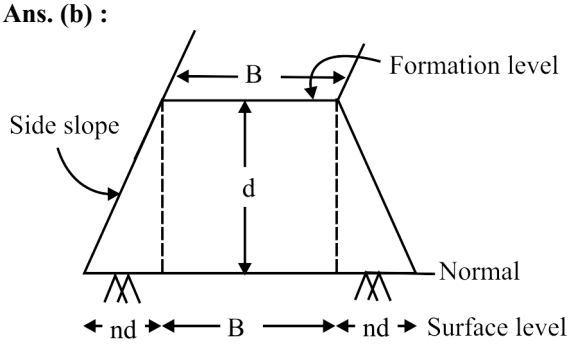
(a) Rs. 800 = 00 (b) Rs. 700 = 00
(c) Rs. 1500 = 00 (d) Rs. 2000 = 00

**Ans. (c) :** Given,
Rate of metal at source = 800 / cu.m
Lead = 35 km
Rate of per km lead = 20 /km
Rate of metal of site = Rate of metal of source + Rate of 35 km lead
Rate of metal at site = 800 + (35 × 20)
Rate of metal at site = 1500

**73. If 'b' is the width of formation, 'd' is the height of the embankment, of length 'L' and side slope in n : 1 for a road, the quantity of earth work is**

(a) (bd+nd) L (b) $(bd+nd^2)$ L
(c) $(b/d - n\sqrt{d})\, L^2$ (d) L/2 (bd) + $nd^2$

**Ans. (b) :**




$$\text{Area} = Bd + 2 \times \frac{1}{2} nd.d$$

Area = Bd + $nd^2$
Quantity = Area × Length
Quantity = (Bd + $nd^2$) × L
Were - B = width of formation
Side slope = n : 1
d = height of embankment
L = length

**74. Abstract estimate is**

(a) Estimation of quantities of various items of work
(b) Estimation of unit rates of various items of work
(c) Estimation of cost of various items of work
(d) Estimation of leads of various items of work

**Ans. (c) : Preliminary or abstract estimate–** It is the least accurate estimate which has very high chance of error.

- This estimate is used to provide cost of project.
- This estimate is prepared from practical knowledge and cost of similar work.

**Cube rate estimate–** It is an approximate estimate and prepared by calculating the cubical content of building by multiplying its length, breadth and height.

**Detailed estimate–** This estimate is most accurate estimate having least chance of error.

- This estimate is prepared for detailing drawing and specification of project with rates of material and labour.

**75. Detailed specification includes**

(a) Rates of various items of work
(b) Measurement taken after execution of work
(c) Quantities of items dumped at site
(d) Qualities and quantities of materials

**Ans. (d) : Detailed specification–** It is a detailed description and expresses the requirement is detail.

**The detailed specification of an item specifies the–**

→ Qualities and quantities of the material
→ The proportion of mortar
→ Workmaship
→ The method of preparation and execution
→ The method of measurement.

**76. The relation between modulus of rupture ($f_{cr}$) and characteristic compressive strength ($f_{ck}$) is given by**

(a) $f_{cr} = 0.7\ f_{ck}$ (b) $f_{cr} = 0.7\ \sqrt{f_{ck}}$
(c) $f_{cr} = 0.75\ f_{ck}$ (d) $f_{cr} = 0.7 / f_{ck}$

**Ans. (b) : Flexural Tensile strength or modulus of rupture ($f_{cr}$) -** It is the minimum strength of concrete at which first tensile crack generate in the beam.

$$f_{cr} = 0.7\sqrt{f_{ck}}$$

Modulus of rupture is also known as flexural tensile strength.

**Characteristic strength of concrete ($f_{ck}$) :** The compressive strength of concrete in which maximum 5% chances of concrete to fail exist and minimum 95% chances of concrete to pass exit.

**77. The approximate value of the total shrinkage strain in concrete for design is taken as**

(a) 0.0001 (b) 0.0003
(c) 0.002 (d) 0.0035

**Ans. (b) :** The volumetric changes of concrete structure due to the loss of moisture by evaporation is known as concrete shrinkage.

- It is a time dependent deformation which depend upon the constituent of concrete, size of member and environmental condition.
- As per IS 456 : 2000 the approximate value of the total shrinkage strain for design may be taken as 0.0003.

**78. A beam of 400 mm effective depth with a neutral axis constant of 0.39, the value of lever arm is**

(a) 250 mm (b) 300 mm
(c) 348 mm (d) 358 mm

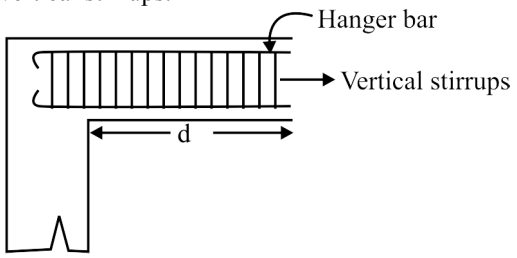
**Ans. (c) : Lever Arm–** It is the vertical distance between compression force and tension force.

$$\text{lever arm} = 400 - \frac{0.39 \times 400}{3} = 348 \text{ mm.}$$

**79. For vertical stirrups, the maximum spacing of shear reinforcement measured along the axis of the member shall not exceed**

(a) 0.75 d (b) 0.40 d
(c) 0.15 d (d) 0.12 d

**Ans. (a) :** Maximum spacing of shear reinforcement in vertical stirrups.




Total shear force = Stress × Area of steel in shear

$V = 0.87\ f_y \times \text{Number of vertical stirrups} \times A_{sv}$

$$V = 0.87 f_y \times \frac{d}{S_v} \times A_{sv}$$

$$\text{Spacing between stirrups} = S_v = \frac{0.87 f_y d A_{sv}}{V}$$

Maximum spacing of vertical stirrups = 0.75d or 300 m (whichever is less)

- As we move from support of the beam to centre the spacing between vertical stirrups increase because the shear force on the beam decreases.

**80. Calculate the pitch of lateral ties for a column of 300 mm square with 20 mm dia longitudinal bar and 8 mm ϕ lateral tie.**

(a) 384 mm (b) 320 mm
(c) 300 mm (d) 280 mm

**Ans. (c) :** As per IS 456 : 2000 –

**Pitch–** The pitch of transverse reinforcement
(i) Least lateral dimension of compressive member
(ii) 16 × Smallest dia of longitudinal reinforcement
(iii) 48 × lateral ties
(iv) 300 mm
Whichever is lesser.
Given,
Longitudinal bar dia = 20 mm
Lateral tie dia = 8 mm

Lateral dimension = 300 $mm^2$
Pitch = 16 × 20 = 320 m
48 × 8 = 384 mm
Least lateral dimension = 300
Whichever is smaller
Hence the pitch of lateral ties shall be kept as 300 mm.

**81. Calculate the strength of fillet weld per 1 mm of 6 mm size with allowable shear stress in the weld 100 MPa**

(a) 700 N (b) 600 N
(c) 424 N (d) 420 N

**Ans. (d) :** Allowable shear strength of the weld = Throat thickness × Length of the (L) weld × Allowable shear stress of the weld
Throat thickness = 0.7 ×1 = 0.7
Length of the weld = 6 mm
Allowable stress shear = 100 MPa
Strength of weld = 0.7 × 6 × 100 = 420 N

**82. The section modulus and the plastic modulus of a section are Z and P respectively. Then its shape factor is given by**

(a) Z/P (b) P/Z
(c) (P-Z)/P (d) (P-Z)/Z

**Ans. (b) :** Shape factor = $\frac{\text{Plastic modulus}}{\text{Elastic modulus}}$

$$S = \frac{P}{Z}$$

Where, P is plastic modulus
Z is elastic modulus
Sis shape factor

**83. The lacing of a compression member is designed to resist a total transverse shear 'V' equal to**

(a) 1.25% of the axial force in the member
(b) 1.5% of the axial force in the member
(c) 2.0% of the axial force in the member
(d) 2.5% of the axial force in the member

**Ans. (d) :** The lacing of a compression member is designed to resist a total transverse shear 'V' equal to 2.5% of the total force in the member.
• According to IS 800 : 2007. Minimum width of lacing bar = 3 times the diameter of bolt.
• Minimum thickness is 1/40th of the effective length for single lacing and 1/60th for double lacing.
• The slenderness ratio for lacing bar shall not exceed 145.

**84. In case of limit state, the maximum strain in concrete at the outermost compression fibre in bending is**

(a) 0.35 (b) 0.035
(c) 0.0035 (d) 0.002

**Ans. (c) :** In case of limit state, the maximum strain in concert at the outermost compression fibre in bending is 0.0035.
• Maximum compressive strain in case of direct compression in case member subjected to axial compression = 0.002.
• Tensile strength of the concrete is ignored during the design for the limit state of collapse in flexure.

**85. In the designation of concrete mix, the letter 'M' and the number stands for**

(a) Mix and characteristic compressive strength of 100 mm cube at 28 days
(b) Mix and characteristic compressive strength of 75 mm cube at 28 days
(c) Mix and characteristic compressive strength of 150 mm cube at 28 days
(d) Mix and characteristic compressive strength of 125 mm cube at 28 days

**Ans. (c) :** In the designation of a concrete mix, the letter 'M' and the number stands for mix and characteristic compressive strength of 150 mm cube at 28 days.
• The compressive strength of concrete is determined by the cube test. As per IS 456 : 2000, it can be found out by testing standard - sized concrete cubes (15 cm × 15 cm × 15 cm) under compression testing machine after 7 and 28 days taking an average of three sample.
• The characteristic compressive strength is defined as the strength of concrete below which not more than 5% of the test results are expected to fail.

**86. The size of the rectangular section is fixed and the moment of resistance of a singly reinforced section is less than design moment. The beam shall be designed as**

(a) Under-reinforced (b) Over-reinforced
(c) Doubly reinforced (d) Compressive failure

**Ans. (c) :** The size of the rectangular section is fixed and the normal of resistance of a singly reinforced section is less than design moment then the beam shall be designed as doubly reinforced section.
• When reversal of stresses condition exist the beam shall be designed as doubly reinforced.
• In this beam the reinforcement to provided along tension sdte and compression side. The function of compression reinforcement is to resist compressive force.

**87. In M20 and M25, the number is characteristic compressive strength in $N/mm^2$ at_____ days.**

(a) 7 (b) 14
(c) 21 (d) 28

**Ans. (d) :** In M20 & M25, the number is characteristic compressive strength in $N/mm^2$ at 28 days.

**88. The grade of concrete generally not used in the reinforced concrete is**

(a) M 40 (b) M 25
(c) M 20 (d) M 10

**Ans. (d) :** The grade of concrete generally not used in the reinforced concrete in M10.
• As per IS : 456-2000, the code has recommended that minimum grade of concrete shall not be less than M20 in reinforced concrete work for design mix and M15 for nominal mix.

**89. If the given bending moment is greater than moment of resistance of balanced section, then the section is**

(a) Balanced section
(b) Under-reinforced section
(c) Over-reinforced section
(d) Critical section

**Ans. (c) : Over-reinforced system–** In this section concrete reaches its maximum value earlier as compare to steel.

- The failure takes place along compression side and the failure is known as brittle failure.
- If the given bending moment is greater than moment of resistance of balanced section, then the section is over-reinforced section.

$$(x_u > x_{ulim})$$

- This section does not give warning before failure.

**90. Modulus of elasticity of steel shall be taken as**

(a) 200 kN/mm$^2$ (b) $2 \times 10^3$ N/mm$^2$
(c) $2 \times 10^4$ N/mm$^2$ (d) $2 \times 10^2$ N/mm$^2$

**Ans. (a) :** Modulus of elasticity of steel shall be taken as 200 kN/mm$^2$ or ($2 \times 10^5$ N/mm$^2$)

- The ratio of longitudinal stress to longitudinal strain is called as modulus of elasticity (E)

$$E = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}}$$

**91. The area between the isohyets 0.45 and 0.55 m is 100 sq.km and between 0.55 and 0.60 m is of 150 sq. km. The average depth of annual precipitation over the above basin of 250 sq.km will be**

(a) 0.50 m (b) 0.55 m
(c) 0.56 m (d) 0.60 m

**Ans. (c) :** Average depth =

$$\frac{A_1h_1 + A_2h_2 + A_3h_3 .....A_nh_n}{A_1 + A_2 + A_3 + .....A_n}$$

$$h_1 = \frac{45+55}{2} = 50 \text{ cm.}$$

$$h_2 = \frac{55+65}{2} = 60\text{cm.}$$

$$\text{Average depth} = \frac{50\times100 + 60\times150}{250}$$

Average depth = 56 cm = 0.56 m

**92. Duty should clearly state**

(a) Time of measurement of water
(b) Place of measurement of water
(c) Method of measurement of water
(d) Maximum rainfall

**Ans. (b) :** Duty should clearly state place of measurement of water.

**Delta–**The total depth of water (in cm) required by a crop to come to maturity is called its delta.

**Duty–**Usually defined as area irrigated using 1 cumec of water applied to land continuously for entire base period of the crop. (hectares/cumec)

Total depth of water applied over an irrigated land at different watering throughout entire base period. It is denoted by $\Delta$ and expressed in cm or m.

$$\boxed{\Delta = \frac{8.64}{D}.B}$$

where,

$\Delta$ = Meters
B = Base period in days
D = hec/cumec

**93. If the catchment area is 100 sq. km. and the value of constant is 12.47, then the maximum flood discharge as per Dicken's formula is :**

(a) $Q = 12.47 \times 100^{2/3}$ (b) $Q = 100 \times 12.47^{3/4}$
(c) $Q = 100 \times 12.47^{2/3}$ (d) $Q = 12.47 \times 100^{3/4}$

**Ans. (d) :** Given,
A = 100 sq. mm.
C = 12.47

$Q = C.A^{3/4}$

Where, Q = maximum/peak flood discharge
C = constant
A = Catchment area

$Q = 12.47 \times 10^{3/4}$

$\rightarrow$ In 1985, Dickens made the first attempt in India to obtain an empirical formula for determining the maximum flood discharge Q(m$^3$/sec) in a river.

**94. The escape of earth from underneath the foundation of a weir along with the percolated water results in a phenomenon called**

(a) Piping (b) Creep
(c) Uplift (d) Scour

**Ans. (a) :** The escape of earth from underneath the foundation of a weir along with the percolated water results in a phenomenon called piping.

- Escape is a structure constructed on an irrigation channel for the disposal of surplus water from the channel. It is also known as surplus water escape.

**95. The net free board is**

(a) Difference of level between MWL and FRL
(b) Difference between top of demand MWL
(c) Difference between top of dam and FRL
(d) Difference between of of dam and bed level

**Ans. (a) : Net free board–** It is the difference of level b/w M.W.L. and the top of the dam.

**Dead Storage–** It is the of the stored water in the reservoir basin which is not available for the use.

**Live Storage–** It is the difference b/w gross storage end dead storage. It is amount of water available from the F.R.L. to the still of lowest sluice.

**96. A canal which is aligned at right angles to the contour is called**

(a) Contour canal (b) Watershed canal
(c) Ridge canal (d) Side slope canal

**Ans. (d) : Side slope canal–** A canal which is aligned at right angles to the contour is called side slope canal.
**Water shed canal–** The canal which aligned along any natural watershed is called a watershed canal.
**Ridge canal–** A canal aligned along the ridgeline or watershed line of an area is called ridge canal.
**Contour canal–** Aligned nearly parallel to the contour of the country.
→ Can irrigate areas only on one side.
→ Cross drainage works are required.

**97. The sill of the notches in a canal is kept is**
(a) Bed level of downstream channel
(b) FSL of upstream channel
(c) FSL of down-stream channel
(d) Bed level of upstream channel

**Ans. (d) :** The sill of notches in a canal is kept at bed level of upstream channel.

**98. The process of making unfertile barren land as fertile land is called**
(a) Soil conservation (b) Land reclamation
(c) Gully erosion (d) Afforestation

**Ans. (b) :** The process of making unfertile barren land as fertile land is called 'Land reclamation'.
→ The best method of reclaiming alkaline soils is by addition of gypsum to the soil and leaching.
→ Acidic soils are reclaimed by suing limestone a soil amendment.

**99. Correct statement from the following is**
(a) Crop period is slightly more than base period
(b) Base period is slightly more than crop period
(c) Crop period is equal to the base period
(d) Crop period and base period are expressed in hours

**Ans. (a) : Crop period–** The time period that elapses from the instant of its sowing to the instant of its harvesting is called the crop-period.
**Base period–** The time between the first watering of a crop at the time of its sowing to it's last watering before harvesting is called the base period or the base of the crop.
→ Crop period is slightly more than the base period, but for all practical purposes, they are taken as one and the same thing and generally expressed in days.

**100. Pick out the factor which does not affect runoff**
(a) Shape of catchment
(b) Existence of vegetation
(c) Type of soil
(d) Existence of building

**Ans. (*) : Factor affecting runoff-** The factor affecting run-off as follows–
- Precipitation
- Size and shape of catchment
- Geographical characteristics of the catchment
- Meteorological characteristics of the catchment
- Effect of drainage net
- Other factors– Cultivation is an area, contour, bonding, ploughing, deforestation, urbanization.

**101. The major resisting force in a gravity dam is**
(a) Water pressure (b) Wave pressure
(c) Self weight of dam (d) Uplift pressure

**Ans. (c) :** In a gravity dam, the major resisting force is self-weight of the dam which resists external forces (such as, water pressure, uplift pressure, wind pressure etc.)
- The main overturning force in the gravity dam is the water pressure.
- In case of gravity dam, the factor of safety against overturning should not be less than 1.50.

**102. Dead storage in a reservoir is provided**
(a) To meet emergency needs
(b) To mitigate floods
(c) To accommodate the sill-trapped in the reservoir
(d) To increase the useful life period

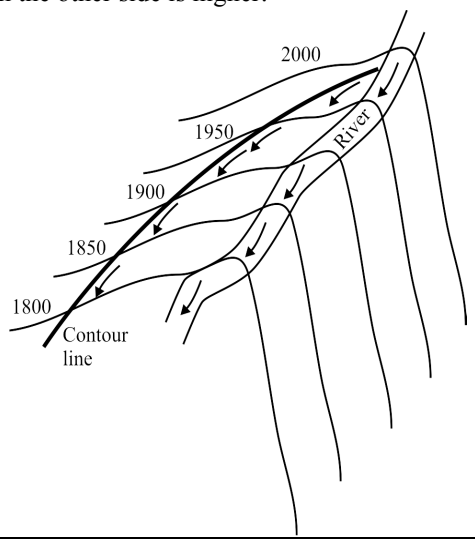
**Ans. (c) :** The dead storage in a reservoir is provided to accommodate the silt-trapped in the reservoir.
- The water stored in the reservoir below the minimum pool level is called dead storage.
- The storage of water in a reservoir up to top level of sluice gates is called line storage or useful storage.
- The volume of water stored between the full reservoir level and minimum pool level is called the live storage.

**103. A surplus weir of an earthen dam with stepped apron is classified as**
(a) Type - A (b) Type - B
(c) Type - C (d) Type - D

**Ans. (d) :** A surplus weir of an earthen dam with stepped apron is classified as type-D.
- A filter is provided at the downstream toe of an earthen dam to collect and drain out the seepage water.

**104. Generally irrigation can be practiced on one side of a**
(a) Contour canal (b) Side slope canal
(c) Ridge canal (d) Branch canal

**Ans. (a) : Contour canal–** Contour canal usually runs almost parallel to natural drainage.
- A contour canal irrigates only one side because the area on the other side is higher.

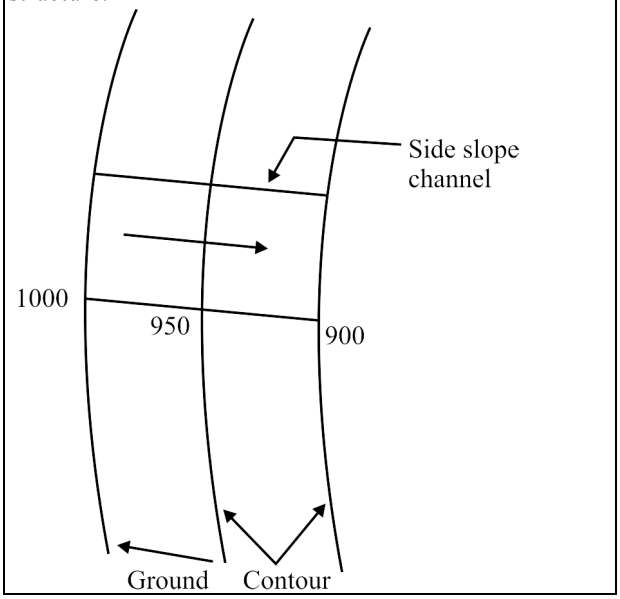


**105. A canal which will not intercept any natural drainage is called as**

(a) Contour canal (b) Side slope canal
(c) Ridge canal (d) Branch canal

**Ans. (b) : Slide slope canal–** A side slope canal is that which is aligned at right angles to the contours along the side slopes.

- Since such a canal runs parallel to the natural drainage flow, it usually does not intercept drainage channels, thus, avoiding the construction of cross-drainage structure.




**106. The method used to remove the salinity of the soil is**

(a) Flood irrigation (b) Sprinkler irrigation
(c) Leaching (d) Surface irrigation

**Ans. (c) : Leaching–** The method used to remove the salinity of the soil is leaching.

- Leaching is the process by which excess salt in the soil is pushed down by standing water above the soil.
- Salinity in soils make soil infertile.

**107. The average Δ of rice crop is nearer to**

(a) 400 mm (b) 800 mm
(c) 1200 mm (d) 1600 mm

**Ans. (c) :** The average Δ for certain important crops in India-

| Crop | Delta on field (Δ) |
|---|---|
| Sugarcane | 120 cm |
| Rice | 120 cm (1200 mm) |
| Tobacco | 75 cm |
| Cotton | 50 cm |
| Wheat | 40 cm |
| Maize | 25 cm |
| Peas | 15 cm |
| Vegetable | 45 cm |

**Delta (Δ) -** The total depth of water (in cm) required by a crop of come to maturity is called its delta (Δ)

**108. Water loss through the leaves of plants is termed as**

(a) Precipitation (b) Infiltration
(c) Transpiration (d) Surface evaporation

**Ans. (c) : Transpiration –** Transpiration is the evaporation of water from plants occurring at the leaves while their stomata are open for the passing of $CO_2$ and $O_2$ during photosynthesis.

→ Water loss through the leaves of plants is termed as transpiration.

**Precipitation–**

- It is condensed water vapours that fall to the earth.
- Most precipitation occurs as rain, but also snow, hail fog drip, grouped and steel.

**Evaporation–** Evaporation is the process in which a liquid changes to the gaseous state at the free surface, below the boiling point through the transfer of heat energy.

**109. A divide wall is constructed for the purpose of**

(a) Controlling seepage
(b) Scouring silt
(c) Creating a still pond
(d) Providing a fish passage

**Ans. (c) : Divide wall–** Divided wall is a long masonry or concrete wall constructed at right angle to axis of the weir to separate the under sluices from the rest of weir.

**Function of divide wall–**

- It control the eddy current or cross current in front of the canal head.
- It forms a still water rocket in front of the canal head so that the suspended silt can settle and scoured later through scouring sluices from time to time.
- It provides at straight approach in front of the canal head.

**110. A rock toe filter in an earthen dam is provided on**

(a) upstream end of the bund
(b) under the base of the bund
(c) downstream end of the bund
(d) at the centre of the bund along the length

**Ans. (c) :** A rock toe filter is an earthen dam is provided on downstream end of the bund.

- The rock toe consists of stone of size usually varying from 15 to 20 cm.
- Toe filter generally, consists of three layers of fine sand, coarse sand, and gravel; as per the filler criteria requirements. The height of the rock toe is generally kept between 25 to 35% of the reservoir head.

**111. The formula $P_n = P\left(1+\frac{r}{100}\right)^n$ is used for forecasting population by**

(a) Arithmetical increase method
(b) Geometrical increase method
(c) Incremental increases method
(d) Graphical method

**Ans.(b): Geometric Increase Method–** In this method, the per decade percentage increase or percentage growth rate (r) is assumed to be constant.

- For new younger cities expanding at faster rate, this method is better.
- It is also called uniform increase method & give highest value of forecasted.

$$P_n = P_0\left(1+\frac{r}{100}\right)^n$$

Where,

$P_o$ = Population at the end of last known centre.

r = Avg% growth rate per decade.

$r = r = \frac{r_1 + r_2 + r_3 .....r_n}{n}$ or $r = \left(r_1 \times r_2 \times r_3 .....r_n\right)^{1/n}$

(ii) Arithmetical increase method–

$$P_n = P_0 + n\bar{x}$$

$P_n$ = Population after decades from present.

n = Number of decades, $P_o$ = present population,

$\bar{x}$ = Arithmetic mean of population increase in known decades.

Incremental increase method–

$$P_n = P_0 + n\bar{x} + \frac{n(n+1)}{2}\bar{y}$$

$\bar{x}$ = Avg. Increase of population of known decades

$\bar{y}$ = Avg. of increment increase of the known decades

**112. The geological formation which yields sufficient quantity of water is called**

(a) Aquifuge (b) Aquiclude
(c) Aquifer (d) Aquitard

**Ans. (c) : Aquifer–** In aquifer is a saturated formation of earth material from which water is yielded in sufficient quantity, due to the high permeability.

- Unconsolidated deposits of sand and gravel are good for aquifer formation.

**Aquiclude–** It contains a large amount of water in pores, but extraction of water is very difficult. Ex-clay.

**Aquitard–** Aquitard form by that material through which the only seepage possible but extraction of water is not so easy as in aquifer. ex - silt clay.

**Aquifuge–** It is neither porous, not permeable, these are not inter-connected opening, so it does not allow it to pass through it.

**113. Hardness is expressed in mg/*l* as**

(a) $Ca(HOC_3)_2$ (b) $CaCO_3$
(c) $Ca(OH)_2$ (d) $CaSO_4$

**Ans. (b) : Hardness–** It is defined as concentration of multivalent metallic ions most abundant in natural water are calcium and magnesium.

- Hardness can be divided in two parts i.e. carbonate, hardness.
- Hardness is expressed in mg/$\ell$ as $CaCO_3$.
- It is measured by using spectrophotometric techniques.

**114. Super chlorination is done**

(a) in day to day practice
(b) during an epidemic
(c) during winter
(d) during summer

**Ans. (b) : Super chloration–** When excess chlorine (5 to 15 mg/$\ell$) is added in water during epidemic such that it gives a residual of 1 to 2 mg/litre beyond break point is called super chlorination.

**115. The normal temperature of sewage when compared to that of water is generally**

(a) Lower (b) Higher
(c) Same (d) Has no relation

**Ans. (b) :** The normal temperature of sewage is generally slightly higher than the temperature of water. The average temperature of sewage in India is $20^0$C, which is near about ideal temperature for the biological activities. However, when the temperature is more, the dissolved oxygen content (D.O.) of sewage gets reduced.

**116. The detention period in primary sedimentation tank in a sewage treatment plant is**

(a) 1 to 3 hrs (b) 4 to 8 hrs
(c) 8 to 12 hrs (d) 12 to 18 hrs

**Ans. (a) :** The detention period in primary sedimentation tank in a sewage treatment plant is 1-3 hourse.

- The detention time (f) of a setting tank may be defined as the average theoretical time required for the sewage to flow through the stuck. It is thus the time that would be required for the flow of sewage to fill the tank if there was no outcome.

**117. The upper most layer of atmosphere is called?**

(a) Stratosphere (b) Mesosphere
(c) Ionosphere (d) Exosphere

**Ans. (d) :** Exosphere is the uppermost layer of the atmosphere.

Above the stratosphere is the mesosphere. It extends upward to height of about 95 km above our planet.

- The stratosphere is a layer of earth's atmosphere. It is the second layer of the atmosphere as you go upward.

**118. The base of cone of depression is called**

(a) Circle of influence (b) Radius of influence
(c) Draw down (d) Specific yield

**Ans. (b) :** Pumping from a well in a water table aquifer lower the water table near the well. The area is known as cone of depression. The land area above a cone of depression, is called the area of influence.

- The base of cone of depression is called radius of influence.

**119. Presence of chlorides and sulphates of calcium and magnesium in water causes**

(a) Acidity
(b) Temporary hardness
(c) Permanent hardness
(d) Softness

**Ans. (c) :** Sulphate, chloride and nitrate of calcium and magnesium gives permanent hardness. It is also called non-carbonate hardness. This hardness cannot be removed by simple boiling. It requires softening techniques.

**120. Odour and taste is controlled by**

(a) Disinfection (b) Aeration
(c) Coagulation (d) Sods-lime process

**Ans. (b) :** Intensity of taste and odour is measured by TON (Threshold odour number).

- Taste and odour are caused by dissolve gases like $H_2S$ mecaptaus, methane, organic matter derived from certain dead or living micro organism, decomposing organic matter etc.
- Odour and taste is controlled by Aeration.
- Odour is generally measured by an instrument known as osmoscope.

**121. Distribution layout in which the mains, sub-mains and branches are interconnected with each other is**

(a) Tree-System (b) Grid Iron system
(c) Radial system (d) Dead end system

**Ans. (b) : Tree system / Dead End system**–

- This system is easy to design and simple.
- In this system free circulation of water is not possible.
- In this system the water can reach only from one direction. In case of maintenance if supply is disturbed the water can not be passed from any other direction.
- This system contain high wastage of water because at dead end the water has to be removed.

**Grid Iron system–** In this system submains are connected with each other using branch pipe or lateral and also called as reticular system.

- More than one route are available for flow of water and there is is no dead end exit in this system.
- This system is used for planned cities and require large number of pipe and valves.

**Radial system–** In this system each area is divided into number of zones and water is supplied radially to distribution pipes and thus also known as zonal distribution.

- This method is the most efficient method used for distribution of water due to high pressure head.

**122. The valves which are used to remove the sediment in a pipe line is called.**

(a) Drain valve (b) Scour valve
(c) Air valve (d) Sluice valve

**Ans. (b,a) : Gate valve/Sluice valve :** This valve is provided at the summit of the pipe. This valve regulate the flow of water through the pipe.

**Air valve–** Air valve are of two types

**(i) Air relief valve–** These valve are used to release air present in a pipe.

**(ii) Air inlet valve–** These valve are used to allow the inlet movement of air.

**Drain valve/Scour valve–** This valve are used to remove excess water from pipe.

- This valve is provided at the bottom most point of pipe.
- This valve allow periodic flushing of the lines to remove sediment and to allow the line to be drained for maintenance and repair work.

**123. The deplorable aspect of conservancy system is:**

(a) Recurring cost is high
(b) Vehicles are required to carry night soil
(c) Vast areas for disposal are necessary
(d) Human element is involved in collection and transportation of human waste

**Ans. (d) :** The deplorable aspect of conservancy system is human element is involved in collection and transportation of human waste.

**Conservancy system –**

- Very cheap in initial cost.
- The aesthetic appearance of the city cannot be improved.
- No technical persons required.

**124. The settleble faecal and other organic solids are removed in**

(a) Activated sludge process
(b) Trickling filter
(c) Primary sedimentation tank
(d) Secondary sedimentation tank

**Ans. (c) : Primary setting tank–** This is a simple setting tank, normally rectangular or circular in shape. This unit is provided in the domestic waste water treatment system mainly to remove a large portion of suspended material, which is primarily organic and inorganic in natures.

**Secondary sedimentation tank–** This tank is used to remove the settleable solid remaining in the effluent after the biological treatment process and used to reduce the organic matter. It is an essential unit of producing a clear effluent.

**Trickling filter–** This is used to remove organic matter from waste water. This is an aerobic treatment system that utilizes micro organism attached to a medium to remove organic matter from waste water.

**125. A water borne disease is**

(a) Malaria (b) Plague
(c) Dysentry (d) Encephalitis

**Ans. (c) : Dysentry–** It is waterborne disease characteristics by sewer diarrhea as well as blood or mucus in the stool. It is transmitted through the ingestion of food or water that has been contaminated.

**Malaria–** It is disease caused by the bites of infected female anophetes mosquitoes.

**Plague–** It is a disease caused by the bacterium, Yersinia Pestes.

**Encephalities–** It is disease caused often due to virus.

**126. The permissible turbidity in mg/*l* for potable water is**

(a) 30-40 (b) 20-30

(c) 10-15 (d) 5-10

**Ans. (d) : Turbidity–** It represent resistance to passage of light.

**Source of turbidity–** Suspended and colloidal solid of silt, clay and algae.

**Measurement of turbidity–**

(i) Turbidity rod method

(ii) Jackson turbidimeter

(iii) Nephlo turbidimeter

(iv) Boylis turbidimeter

**Permissible turbidity in potable water –** 5 to 10 mg/$\ell$

**127. Over flow pipes are provided in service reservoirs at**

(a) Floor level

(b) Top of the reservoir

(c) Full reservoir level

(d) Mid level

**Ans. (c) : Service Reservoir–** It is used in distribution to provide storage to meet fluctuation in demand of water, to provide storage for firefighting and emergencies such as breakdown, repair etc and to stabilize pressure in distribution system.

The various accessories provided for service reservoirs are-

(i) Inlet pipe for entry of water

(ii) Outlet pipe for the exit of water

(iii) Overflow pipe are provided at full reservoir level to exit of water above full supply level.

(iv) Manhole in top cover or roof of reservoir

(v) Washout pipe for removing water after cleaning of reservoir.

**128. For ease in the design of sewers, the following are made used of**

(a) Thumb rules

(b) Nomograms

(c) Emperical formulae

(d) Hydraulic formulae

**Ans. (d) :** For the design of sewer hydraulic formula are used.

•The equation used for sewer that are flowing in the presence of gravity is Manning's equation –

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Where,

V= Velocity of the flow in sewer in m/sec.

$$R = \frac{\text{Area}}{\text{Wetted Perimeter}}$$

S= Slope of the sewer

n = Roughness coefficient for pipes (0.013-0.15)

**129. The hydraulic mean depth of a circular sewer running half full is equal to**

(a) d/2 (b) d/4

(c) d/6 (d) d/8

**Ans. (b) :** Hydraulic mean depth of sewer running half fall

Area of the cross section (A) = $\frac{\pi}{4} \times d^2$

Wetted perimeter (P) = $\pi D$

Hydraulic mean depth (R) = $\frac{A}{P} = \frac{(\pi/4)D^2}{\pi D} = \frac{D}{4}$

**130. The animals which feed directly on living plant or plant remains are called**

(a) Carnivores (b) Annivores

(c) Herbivores (d) Macrophytes

**Ans. (c) : Herbivores–** These are the animal that eat only plants or plants remain, ex-cow, goat.

**Carnivores–** These are the animal that eats only meat, ex-lion, tiger.

**Omnivores–** These are the animals that eat both plants and meat ex-bears, dogs.

**Macrophytes–** These are the aquatic plant growing in or near water.

**131. A common problem in hill roads is**

(a) Skid (b) Earthquake

(c) Slip (d) Land slide

**Ans. (d) : Land slide–** It is defined as the movement of a mass of rock or earth down a slope.

• The land slides are a problem which is caused due to earth quake and land slide ultimately leading to the disintegration of rock and they cause upward and downward and this is the common problem in hell roads.

**Skid–** When a vehicle suddenly slide forward and distance travelled in longitudinal direction is greater than the circumferential distance.

**Slip–** It occur when wheel revolves more than the corresponding longitudinal movement or the distance travelled in a circumferential direction is greater than longitudinal distance.

**132. Calculate the rise of crown with respect to edges for a cement concrete road of 10.00 m width and situated in areas of heavy rainfall**

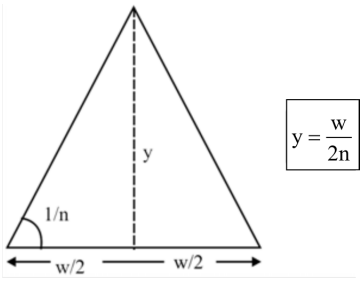
(a) 0.01 m (b) 0.05 m

(c) 0.07 m (d) 0.10 m

**Ans. (d):** Given,

Width of road= 10 m

- For cement concrete road having heavy rainfall camber or slope is $\frac{1}{50}$.




$y = \frac{w}{2n}$

Water, y = rise of crown

w = width of road

h = slope or camber

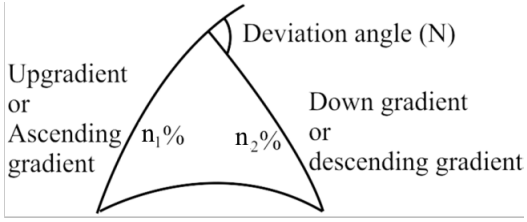
$$y = \frac{10}{2 \times 50} = 0.1 \text{ m}$$

**133. When an ascending gradient 1 in 100 meets with a descending gradient of 1 in 20, calculate the deviation angle (N).**

(a) 1/600 (b) 11/600

(c) 15/600 (d) 30/600

**Ans. (*): Deviation angle–** This is known as the direction of change of two gradient angles.




$$N = n_1\% - (-n_2\%)$$

$$N = n_1\% + n_2\%$$

$$N = \frac{1}{100} + \frac{1}{20} = \frac{6}{100} = \frac{3}{50}$$

**134. A pavement has a horizontal curve of 1000 m for a design speed of 75 kmph. The super elevation is**

(a) 1 in 70 (b) 1 in 60

(c) 1 in 40 (d) 1 in 50

**Ans. (c) : Super elevation–** Raising of outer edge with respect to inner edge of counteract centrifugal force and to reduce tendency of vehicle to over turn.

- IRC has suggested to provide superelevation to counter act centrifugal force due to 75% of design speed by neglecting lateral friction.

$$e + f = \frac{v^2}{127R} \Rightarrow e = \frac{(0.75V)^2}{127R} \quad [f=0]$$

$$e = \frac{V^2}{225R}$$

Where, V= Design speed

e = Super elevation

R = Radius of curve

Given,

V = 75 km/hr

R = 1000

$$e = \frac{(75)^2}{225 \times 1000} = \frac{1}{40} \text{ or 1 in 40}$$

**135. A soil has a volume of 100 $cm^3$ and mass of 200 gm. on over drying for 24 hrs. The mass is reduced to 100 gm. Calculate the dry density.**

(a) 9.81 $kN/m^3$ (b) 14.72 $kN/m^3$

(c) 19.62 $kN/m^3$ (d) 24.53 $kN/m^3$

**Ans. (a) :** Dry density = $\frac{\text{Mass of solid}}{\text{Volume of soil}}$

$$\gamma_{\text{dry density}} = \frac{100}{100} = 1 \text{ g/cm}^3$$

$$\text{Dry weight density}(\gamma)(\text{kn/m}^3) = \gamma_{\text{dry density}} \times g(\text{g/cm}^3)$$

$$\gamma_d = 1 \times 9.81$$

$$\gamma_d = 9.81 \text{ kN/m}^3$$

**136. A soil deposit having water content 15%, specific gravity 2.5 and voids ratio 0.5, calculate degree of saturation.**

(a) 50% (b) 70%

(c) 75% (d) 90%

**Ans. (c) :** From relation between properties of soil.

We know,

Se = wG

Where,

S = degree of saturation

e = void ratio

W = water content

G = Specific gravity

Given,-

G = 2.5

e = 0.5

w = 15%

$S \times 0.5 = 15 \times 2.5$

S = 75%

**137. Exceptional gradient in plains as per IRC**

(a) 1 in 90 (b) 1 in 15

(c) 1 in 40 (d) 1 in 30

**Ans. (b) : Gradient–** Gradual rise or fall along the length o a road is called gradient.

| Terrain | Rulling gradient | Limiting gradient | Exceptional gradient |
|---|---|---|---|
| Plain and rolling terrain | 3.3% or $\frac{1}{30}$ | 5% or $\frac{1}{20}$ | 6.7% or $\frac{1}{15}$ |

| | | | |
|---|---|---|---|
| Moutaneous and steeper terrain elevation more than 3000 m above mean sea level | 5% or $\frac{1}{20}$ | 6% or $\frac{1}{17}$ | 7% or $\frac{1}{14}$ |
| Steeper train elevation not more than 3000 m above MSL. | 6% or $\frac{1}{17}$ | 7% or $\frac{1}{14}$ | 8% or $\frac{1}{12.5}$ |

**138. As per IS soil classification, inorganic soils with low compressibility are represented by**

(a) MH (b) SL

(c) ML (d) OL

**Ans. (c):**

**MH** = High plastic Silt
**CH** = High plastic Inorganic clay
**OL** = Low plastic organic clay
**ML** = Low plastic silt
**CI** = Inorganic clay of medium plasticity
**MI** = Inorganic silt of medium plasticity
**CL** = Low plastic Inorganic clay
**OH** = High plastic organic clay

**139. As per IS soil classification, clays, organic soils are represented by-**

(a) MH (b) SL

(c) ML (d) OH

**Ans. (d) : Coarse grained soil :**

- **Sand** – (0.075-4.75) mm
  - Coarse sand – (4.75 mm - 2 mm)
  - Medium sand – (0.425 – 2 mm)
  - Fine sand – (0.075 – 0.425 mm)

**Fine grained soil :**
- → Silt – (0.002 – 0.075mm)
- → Clay – < 0.002

**Peat soil :** Organic soil with fibrous aggregate's formed from vegetable matter in excess moisture, highly compressible not suitable for foundation.

**140. As per IS soil classification gravelly soils, more than half of the grains larger than IS Sieve 480 are represented by**

(a) GM (b) GL

(c) GS (d) SM

**Ans. (a) :**

| Major soil group (x) | Descriptor (y) | Possible symbols ( $x\bar{y}$ ) |
|---|---|---|
| Gravel (G) More than 50% particle retained over 4.75 mm | Well graded (W) Poorly graded (P) | GW, GP, GM, GC |
| Sand (S) more than 50% particle passing through 4.75 mm sieve | Silt (M) Clays (C) | SW, SP, SM, SC |

**141. A soil sample has a porosity of 50%, calculate voids ratio**

(a) 3.0 (b) 1.0

(c) 0.5 (d) 0.33

**Ans. (b) :** Relation between porosity and void ratio.

$$e = \frac{n}{1-n}$$

where, e = void ratio

n= porosity

Given,

$n = 50\% = 0.50$

$$e = \frac{0.50}{1-0.50} = 1$$

So,

The void ratio of soil is 1.0

**142. A soil has a volume of 100 $cm^3$ an mass of 200 gr. on over drying for 24 hrs, the mass is reduced to 150 gr. Calculate the water content.**

(a) 30% (b) 33%

(c) 50% (d) 55%

**Ans. (b) :** Relation between dry density , bulk density and water content is given as–

$$\rho_d = \frac{\rho_{bulk}}{1+w}$$

where,

w = water content

$\rho_d$ = dry density = $\frac{M_{soil\ after\ dry}}{Volume}$

$\rho_{bulk}$ = bulk density = $\frac{M_{soil}}{Volume}$

Given,

Volume (V) = 100 $cm^3$

Mass of soil = 200 gm

Mass of soil after drying = 150 gm

$$\rho_d = \frac{150}{100} = 1.5\ g/cm^3$$

$$\rho_{bulk} = \frac{200}{100} = 2\ g/cm^3$$

$$1.5 = \frac{2}{1+w}$$

$$w = 33.33\% \text{ or } \frac{100}{3}$$

So, water content will be 33.00 %

**143. A soil has a volumeof 100 $cm^3$ and mass of 200 gr. on over drying for 24 hrs, the mass is reduced to 150 gr. Calculate the bulk density**

(a) 9.81 kN/m$^3$
(b) 14.72 kN/m$^3$
(c) 19.62 kN/m$^3$
(d) 24.53 kN/m$^3$

**Ans. (c) :** $\rho_{bulk} = \frac{M_{bulk}}{Volume}$

$\rho_{bulk}$ = Bulk density of soil

$M_{bulk}$ = Mass of soil carrying air and water void

V = Volume of soil

$$\rho_{bulk} = \frac{200}{100} = 2g/cm^3$$

$$\gamma_{bulk}\left(\text{in kN/m}^3\right) = \rho_{bulk}\left(\text{in g/cm}^3\right) \times g$$

$$\gamma_{bulk} = 2 \times 9.81 = 19.62 kN/m^3$$

$\gamma_{bulk}$ = Bulk weight density of soil

**144. Layout of centre line of the highway on the ground is called as**
(a) Setting out (b) Stake out
(c) Alignment (d) Base line

**Ans. (c) : Alignment –** The position or the layout of the centre line of the highway on the ground is called the alignment. Alignment survey is widely provided for safety and protection purpose of different route and highway. The alignment survey include horizontal alignment, cross section angle and vertical alignment of roads. The process of alignment is available as most important and crucial process for road safety.

**145. Alternate routes for a highway project are suggested by the study by**
(a) Political map
(b) Traffic map
(c) Topographic map
(d) Road map

**Ans. (c) :** Alternate route of highway project are suggested by the study of topographic map. It is map characterized by large scale detail and quantitative representation of relief features usually using contour lines. It show both natural and artificial feature of earth. These map depict in detail of ground relief (land forms and terrain), drainage (lakes and river), forest cover, administrative areas and transportation routes and facilities (including road and railway) and other man made feature.

**146. IRC committee was appointed by the Government with M.P. Jayakar as chairman in**
(a) 1920 (b) 1925
(c) 1926 (d) 1927

**Ans. (d) : Jayakar committee (1927) –**

- road development should be considered as a matter of national interest
- Tax on petrol should be levied to collect for road development work. Central Road fund was formed in 1929
- A research organisation should be instituted to carryout research and development work.
- CRRI was formed in 1950
- A semi- official technical body should be formed to act as advisory body on various aspect of road.
- IRC was formed in 1934

**147. In urban areas, when the volume of cycle traffic in high, minimum width provided for the cycle track is**
(a) 3.5 m (b) 3.0 m
(c) 2.0 m (d) 1.5 m

**Ans. (c) :** These are provided in urban areas when the cycle volume is very high. The minimum width is 2 m, which can be increased by 1 meter for each additional cycle lane.

**148. To divide the traffic moving in opposite direction_____ is provided.**
(a) kerb
(b) footpath
(c) median
(d) drive way

**Ans. (c) : Median or traffic separator–** The main function of a traffic separator is to prevent head-on collision between vehicle moving in opposite direction on adjacent lanes.

**149. Pycnometer is used**
(a) to measure only specific gravity
(b) to measure only water content
(c) to measure specific gravity and water content
(d) to measure specific area

**Ans. (c) : Pycnometer:-** Pycnometer is used to determine the specific gravity and water content of the soil sample.

For water content, $w = \left[\left(\frac{W_2 - W_1}{W_3 - W_4}\right)\left(\frac{G-1}{G}\right) - 1\right]$

Specific gravity, $G = \frac{(W_2 - W_1)}{(W_2 - W_1) - (W_3 - W_4)}$

**150. Water formed transported soil is**
(a) Loess
(b) Glacier
(c) Alluvial
(d) Marine

**Ans. (c) : Alluvial soil :** Alluvium is loose, unconsolidated soil.
- It comes by erosion and reshaped by water is some forms and get redeposited.
- It has variety of materials, gravel, silt, sand, clay etc

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer (Observers) Exam, 2013
## Solved Paper

**1. The boundary layer thickness in turbulent flow varies as**

(a) $X^{1/7}$ (b) $X^{1/2}$

(c) $X^{4/5}$ (d) $X^{3/5}$

Where X is the distance from the leading edge

**Ans. (c) :** On laminar region $\delta \times \sqrt{x}$, while in turbulent region $\delta \propto x^{4/5}$, So $\delta$ increases more rapidly in turbulent region than in laminar region.

**2. Surge wave in a rectangular channel is an example of**

**A. steady flow** **B. unsteady flow**

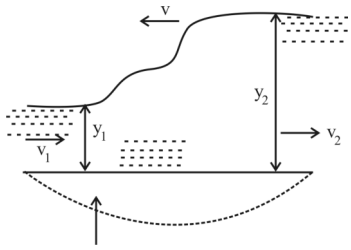
**C. uniform flow** **D. non- uniform flow**

**The correct answer is**

(a) both A and C (b) both B and C

(c) both A and D (d) both B and D

**Ans. (d) :**




A surge is a moving wave front which results in an abrupt change of the depth of flow. It is a rapidly varies unsteady flow condition.

- Surge wave in a rectangular channel is an example of unsteady flow and non-uniform flow.
- It is produced by a rapid change in the rate of flow, for example, by the rapid opening or closure of a control gate in a channel.

**3. If f is friction factor, then Chezy's coefficient is proportional to**

(a) f (b) $\sqrt{f}$

(c) $\frac{1}{f}$ (d) $\frac{1}{\sqrt{f}}$

**Ans. (d) :** Chezy's coefficient (c) = $\sqrt{\frac{8g}{f}}$

$c \propto \frac{1}{\sqrt{f}}$ f = Friction factor

**4. For any channel section, the specific energy increases with**

**A. increase in depth of sub- critical flow**

**B. increase in depth of super- critical flow**

**C. decrease in depth of sub- critical flow**

**D. decrease in depth of super- critical flow**

**The correct answer is**

(a) both A and B (b) both A and D

(c) both B and C (d) both C and D

**Ans. (b) :** For any channel section the specific energy increase with increase in depth of sub-critical flow, and decrease in depth of super-critical slow.

- Critical depth – Minimum specific energy
- Normal depth – Uniform flow
- Alternate depth – Same specific energy
- Conjugate depth – Same specific force

**5. The mild slope profile $M_2$ occurs for depth**

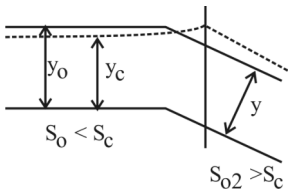
(a) above normal but below critical

(b) above critical but below normal

(c) above critical and below critical

(d) above normal and above critical

**Ans. (b) :**




The mild slope profile $M_2$ occurs for depth above critical but below normal.

$\boxed{y_0 > y > y_c}$ and $S_o < S_c$

Critical slope –

$\boxed{y_0 = y = y_c}$ and $S_o = S_c$

Steep slope –

$\boxed{y_c = y_o}$ and $S_c = S_o$

**6. In a gradually varied flow, if $= \frac{dy}{dx}$ is positive, then $= \frac{dE}{dx}$ will be**

(a) always zero (b) positive if $y > y_c$

(c) negative if $y > y_c$ (d) always negative

**Ans. (b) :** Gradually varied flow applies for steady and non uniform flow where depth of flow changes over large length. s

If $\frac{dy}{dx} > 0$, then $\frac{dE}{dx} > 0$ If $y > y_c$

**7. $S_2$ profile can occur at**

(a) a break in slope from mild to steep

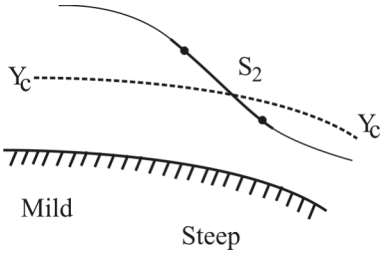
(b) a break in slope from steep to mild

(c) the downstream of a sluice gate on a steep slope

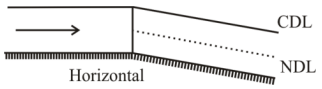
(d) a sudden drop in bed in a steep slope

**Ans. (a):** Steep slope ($y_c > y_0$)

- Supercritical flow at normal depth.
- $S_2$ profile can occur at a break in slope from mild to slope.




**8. Figure shown below represents gradually varied flow in an open channel with a break in bed slope**

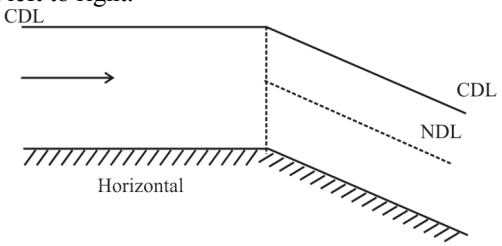



**types of water surface profiles occurring from left to right are**

(a) $H_3, S_3$ (b) $H_2, S_2$

(c) $H_2, M_2$ (d) $H_3, M_2$

**Ans. (b) :** The $H_2$, $S_2$ of water surface profiles occuring from left to right.




**Type of channel slope -**

(1) mild slope ($y_n > y_c$)

(2) Steep slope ($y_c > y_n$)

(3) Critical slope ($y_c = y_n$)

**9. The critical velocity for a flow of q $m^3/s/m$ width of a wide rectangular channel is given by**

(a) $\left[\frac{q^2}{g}\right]^{1/3}$ (b) $(qg)^{1/3}$

(c) $\sqrt{qg}$ (d) None of the above

**Ans. (b) :** Critical depth $Y_c = \sqrt[3]{\frac{q^2}{g}}$

and Critical Velocity $V_c = \sqrt{g.y_c} = \left[g \times \sqrt[3]{\frac{q^2}{g}}\right]^{\frac{1}{2}}$

$$= \left[\left(\frac{q^2 \times g^3}{g}\right)^{\frac{1}{3}}\right]^{\frac{1}{2}}$$

$$= (q.g)^{1/3}$$

**10. In the most efficient trapezoidal section, which of the following is not true?**

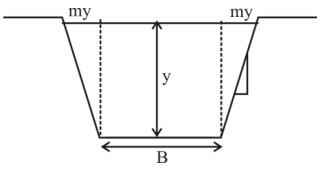
(a) Top width is twice the length of the sloping side

(b) Hydraulic mean depth is half the depth of flow

(c) Shape is half of regular hexagon

(d) Depth must be equal to top width

**Ans. (d) :**




**Trapezoidal section–**

- Most efficient trapezoidal section
- Top width is twice length of the sloping side
- Depth of section $y = \frac{B}{2}$

$$\left\{B = \frac{2y}{\sqrt{3}},\ m = \frac{1}{\sqrt{3}}\right\}$$

- Hydraulic mean radius $R = \frac{y}{2}$
- Shape is half of regular hexagon.

**11. Consider the following measuring structures:**

(1) submerged broad- created weir

(2) free broad- created weir

(3) free sharp- created weir

(4) free ogee spillway

The correct sequence of the increasing order of the discharge coeffieinent of these structures is

(a) 1, 2, 4, 3 (b) 2, 1, 3, 4

(c) 2, 1, 4, 3 (d) 1, 2, 3, 4

**Ans. (d) :** The classification of weir is–

$\frac{H_1}{B_w} \le 0.1$ long crested weir

$0.1 < \frac{H_1}{B_w} < 0.35$ Broad-crusted weir

$0.35 = \frac{H_1}{B_w} < 1.5$ Narrow crested weir

$\frac{H_1}{B_w} \ge 1.5$ Sharp crested weir

**Note-** Submerged weir have lesser coefficient of discharge compared to free weir.

For ogge spillway $C_d = 1.19 \times C_{ds}$

**12. Which of the following equations are used for the derivation of the differential equation for water surface profile in open channel flow?**

A. continuity equation

B. energy equation

C. momentum equation

**Select the correct answer using the code given below:**

(a) A, B, and C (b) Only A and C
(c) Only A and B (d) Only B and C

**Ans. (c) :** Continuity equation and energy equation are used for the derivation of the differential equation for water surface profile in open channel flow.

- The continuity is statement of the law of conservation of matter in open-channel flows since we deal with in-compressible fluids. This equation is relatively simple and much more for the case of steady flow. $\rho AV = \rho A_1 V_1$
- The energy equation in the form of the Bernoulli's equation is used. According to this equation, the total energy at a downstream section differs from the total energy at the upstream section by an amount equal to the loss of energy between the sections.

$$H = z + \frac{P}{\rho g} + \frac{v^2}{2g}$$

**13. Consider the following statements is respect of specific energy of flow in an open channel of fixed width:**

**A. There is only one specific energy curve for a given channel**

**B. Alternate depths are depths of flow at which the specific energy is the same**

**C. Critical flow occurs when the specific energy is minimum**

(a) Only A and B (b) Only A and C
(c) Only B and C (d) A, B and C

**Ans. (c) :** Alternate depths are depths of flow at which the specific energy is the same.

- Critical flow occurs when the specific energy is minimum.

**14. The chlorine demand of a water sample was found to be 0.5mg/*l*. The amount of bleaching powder containing 30% available chlorine to be added to treat one litre of such a water sample is**

(a) 1.67 mg (b) 1.92 mg
(c) 1.01 mg (d) 1.36 mg

**Ans. (a) :** Total amount of chlorine required = Chlorine demand × Quantity of water to be treated.

Chlorine demand = 0.5 mg/$\ell$

Quantity of water to treated = 1 $\ell$

Chlorine required = 0.5 × 1 = 0.5 mg

Amount of bleaching powder = $\frac{0.5 \times 100}{30}$ = 1.67 mg

**15. The populations of a town in three consecutive years are 5000, 7000 and 8400 respectively. The populations of the town in the fourth consecutive year according to geometrical increase method is**

(a) 9500 (b) 9800
(c) 10920 (d) 10100

**Ans. (c) :**

| Population | Increase in population | Percentage increase in population |
|---|---|---|
| 5000 | 2000 | $r_1 = \frac{2000}{5000} \times 100 = 40\%$ |
| 7000 | 1400 | $r_2 = \frac{1400}{7000} \times 100 = 20\%$ |
| 8400 | | |

$$r = \frac{r_1 + r_2}{2} = \frac{40 + 20}{2} = 30\%$$

We know that,

$$P_n = P_0\left(1 + \frac{r}{100}\right)^n$$

$n = 1$

$r = 30$

$$P_n = 8400\left(1 + \frac{30}{100}\right)^1$$

$P_n = 10920$

**16. The rate of filtration of pressure filters is**

(a) less than that of slow sand filters
(b) in between the filtration rate of slow sand filters and rapid sand filters
(c) greater than that of rapid sand filters
(d) equal to that of slow and filters

**Ans. (c) :**

- The rate of filtration of pressure filters is greater than that of rapid sand filters and slow sand filters.
- The filtration rate of slow sand filter is 100-200 *l*/m$^2$/hr
- The filtration rate of rapid sand filter is 3000 to 6000 *l*/m$^2$/hr
- The filtration rate of pressure filter is 6000 to 15000 *l*/m$^2$/hr

**17. When the coefficient of rugosity is increased from 0.01 to 0.02, the gradient of a pipe of a given diameter to carry the same flow at the same velocity will be**

(a) increased by 4 times
(b) increased by 2 times
(c) decreased by $\sqrt{2}$ times
(d) decreased by 4 times

**Ans. (a) :** $Q = A\frac{1}{n}R^{2/3}S^{1/2}$

$n \propto S^{1/2}$

$$\frac{n_2}{n_1} = \left(\frac{S_2}{S_1}\right)^{1/2}$$

$$\frac{0.02}{0.01} = \left(\frac{S_2}{S_1}\right)^{1/2}$$

$\because$ $S_2 = 4\,S_1$

**18. The water supply to a house begins with the connection of a service pipe with the municipal water mains. the connection comprises**

**A. Stop cock**

**B. goose neck**

**C. ferrule**

**D. water meter**

**The correct sequence of these connections is**

(a) A, B, C, D (b) C, A, B, D

(c) C, B, A, D (d) A, B, D, C

**Ans. (c) :** Correct sequence–

Ferrule → goose neck → service pipe → stop cock → water meter.

**Ferrule–** Mode up of gun metal, size of dia (10-50 mm).

**Gooseneck –** Length 75 m, generally mode up of lead.

**Service pipe –** Made up of G.I. Pipe having dia (12-50) mm.

**19. Consider the following statements:**

**While deciding to locate an intake structure for a city situated on a river bank, intake for water supply should be located**

**A. in deep water**

**B. sufficiently away from shore lines**

**C. upstream of the populated city**

**D. near navigational channel**

(a) A, B and D (b) A, B and C

(c) B, C and D (d) A, C and D

**Ans. (b) :** While deciding to locate an intake structure for a city situated on a river bank, intake for water supply should be located given as below.

1. It should not be located on inner bank but outer back. It should not be located on the curve of the river.
2. Site should be located such that maximum water may be intake.
3. It should be located in deep water.
4. It should be located sufficiently away from the shore line.
5. It should be located upstream of the populated city.
6. It should not be near the navigational channel.

**20. For a flow of 5.7 MLD and a detention time of 2 hours, the surface area of a rectangular sedimentation tanks to remove all particles having settling velocity of 0.33 mm $g^1$ is**

(a) 20 $m^2$ (b) 100 $m^2$

(c) 400 $m^2$ (d) 200 $m^2$

**Ans. (d) :**

Given, flow = 5.7 MLD (Million litres per day)

$= 5.7 \times 10^6$ lit/day

Detention period $t_d$ = 2 hours.

Setting velocity = 0.33 mm/sec or 0.00033 m/sec

Flow Q = $\frac{5.7\times10^3}{24}$ $m^3$/hour = 237.5 $m^3$/hour

Settlement within detention period

$= 237.5 \times 2 \times 60 \times 60 = 2.376$ (m)

Flow within detention period = $237.5 \times 2$

$= 475$ $m^3$/hour

Area (A) = $\frac{\text{flow}}{\text{settlement}} = \frac{475}{2.376} = 199.91$ $m^3$ □ 200$m^2$

**21. If the total hardness of water is greater than its total alkalinity, the carbonate hardness will be equal to**

(a) total alkalinity

(b) total hardness

(c) total hardness- total alkalinity

(d) non- carbonate hardness

**Ans. (a) :** Total hardness is the sum of carbonate and non-corbonate hardness.

**Case-1 : If total hardness is greater than total alkalinity.**

Carbonate hardness (C.H.)= Total alkalinity (T.A.)

**Case-2 : If total hardness is less than total alkalinity .**

Then, carbonate hardness = Total hardness and non-corbonate hardness = 0

**22. Uniformity coefficient of filter sand is given**

(a) $\frac{D_{50,}}{D_5}$ (b) $\frac{D_{50,}}{D_{10}}$

(c) $\frac{D_{60,}}{D_6}$ (d) $\frac{D_{60,}}{D_{10}}$

**Ans. (d) : Uniformity coefficient ($c_u$) –**

$$C_u = \frac{D_{60}}{D_{10}}$$

- It is a measure of particle size range. It is always greater than or equal to 1.
- For a well graded gravel, $C_u$ must be greater than 4.
- Where as for a well graded sand $C_u$ must be greater than 6.

**23. When two identical centrifugal pumps are operating in series on a common rising main, then**

(a) The pressure in the rising main will be nearly doubled, while the discharge will remain the same

(b) the discharge will be nearly doubled, while the pressure remains the same

(c) discharge as well as pressure in the rising main will be doubled

(d) discharge as well as pressure in the rising main will increase but not become double.

**Ans. (a) When pump working (identical pump) in series-**

- When two identical centrifugal pump working in series, discharge is identical and head produce it doubled and if they are working in parallel, discharge is doubled and head remains the same.

Two identical centrifugal pump are operating in series then pressure in the rising main will be nearly doubled, while discharge will remains constant -
Head (H) = $H_1 + H_2 + H_3$
Discharge $(Q) = Q_1 = Q_2 = Q_3$ or Q = constant
If pump operating in parallel then
Discharge (Q) = $Q_1 + Q_2 + Q_3$
and Head H = $H_1 = H_2 = H_3$ or Head = constant

**24. Which of the following is not correctly matched?**
(a) Aerator - Excess $CO_2$ and $H_2S$
(b) Rapid sand filter - Suspended matter
(c) Sedimentation tank - settleable and colloidal matter
(d) Slow sand filter - Hardness

**Ans. (d) : Slow sand filter** – Used to remove fine particle and flocs and also removes organic matter and bacteria.
**Aeration** – Excess $CO_2$ and $H_2S$
**Rapid sand filter** – Suspended matter
**Sedimentation tank** – Settleable and colloidal matter
**Lime-soda process** – hardness

**25. The dissolved oxygen level in natural unpolluted water, at normal temperatures is found to be the order of**
(a) 10 mg/*l* (b) 5 mg/*l*
(c) 50 mg/*l* (d) 100 mg/*l*

**Ans. (a) :** Oxygen content present in water is called dissolved oxygen. Minimum level of oxygen content of survival of aquatic life in water is 4 mg/lit.
• In river streams dissolved oxygen present is 10 mg/lit and large water bodies it is 8 gm/lit. The dissolved oxygen level in natural unpolluted water at normal temperature is 10 mg/lit.

**26. Most suitable section of sewer in separate sewage system is**
(a) rectangular section
(b) Circular section
(c) Standard form of egg shaped sewer
(d) modified egg shaped section

**Ans. (b)** Circular sewer sections are most widely used, which are preferably provided with separate sewerage system.
• The circular sections are generally preferred to all other shapes because it can be manufactured most easily and utilizes the minimum quantities of materials and the cheapest and most economical.

**27. The correct relation between theoretical oxygen demand (TOD), biochemical oxygen demand (BOD) and chemical oxygen demand (COD) is given by**
(a) TOD > BOD > COD
(b) TOD > COD > BOD
(c) BOD > COD > TOD
(d) COD > BOD > TOD

**Ans. (b) :** The quality of sewage water in terms of BOD, COD and TOD is :
TOD > COD > BOD
• COD is the measure of oxygen being consumed during the oxidation of oxidizable organic matter in presence of a strong oxidizing agent.
• The amount of oxygen required to oxidize he biodegradable organics is called B.O.D.
• If the chemical formula and the quantity of all organic matter present in the sewage is known, the extract amount of oxygen required to oxidize then can be calculated. This is called (T.O.D.).

**28. In a BOD test, 1.0mL of raw sewage was diluted to 100 mL. and the dissolved oxygen concentration of diluted sample at the beginning was 6 ppm and it was 4 ppm at the end of 5 day incubation at $20^0$C. The BOD of raw sewage will be**
(a) 100 ppm (b) 200 ppm
(c) 300 ppm (d) 400 ppm

**Ans. (b) :** Given, diluted sample = 1 in 100 = 1%
Final BOD = 4 ppm
Initial BOD = 6 ppm
$\therefore$ dilution factor = $\frac{100}{1}$
$BOD_5$ = (Initial BOD – Final BOD) × Dilution factor
= [6 – 4] × 100
= 2 × 100
= 200 ppm

**29. Dissolved oxygen in streams is**
(a) maximum at noon
(b) minimum at noon
(c) minimum at midnight
(d) same throughout the day

**Ans. (a) :** If temperature of sewage is more then dissolved oxygen content will be less.
• So dissolved oxygen in stream is maximum at noon.
• The minimum D.O. content should be 4 ppm for survival of aquatic things & saturation D.O. in normal river water is 9.2 ppm.

**30. One litre of sewage, when allowed to settle for 30 minutes gives volume of 27 $cm^3$. If the dry weight of this sludge is 3 grams, then its sludge volume index will be**
(a) 9 (b) 24
(c) 30 (d) 81

**Ans. (a):** Sludge volume index (SVI)

Sludge volume index = $\frac{V_{ob}}{X_{ob}}$

$V_{ob}$ = Settled sludge volume (m*l*/lit)

$X_{ob}$ = Concentration of suspended solids in the mixed liquor in mg/litre

Given,

$V_{ob} = 27\ cm^3$/litre $\Rightarrow$ 27 ml/litre

The concentration of suspended solids in mixed liquor $X_{ob} = 3$ gm/litre

$$SVI = \frac{V_{ob}}{X_{ob}}$$

$$\frac{27}{3} = 9\ ml/gm$$

**31. The forces, whose lines of action lie in the same plane and are meeting at one point, are known as**

(a) coplanar concurrent forces
(b) coplanar non- concurrent forces
(c) non- coplanar concurrent forces
(d) none of the above

**Ans. (a) :** The force, whose lines of action lie in the same plane and are meeting at one paint, are known as coplanar concurrent forces.

**32. A forces P, of high magnitude acts on a body for a small interval of time $\Delta t$. The product of P and $\Delta t$ is called**

(a) implusive force
(b) kinetic energy of the body
(c) impulse
(d) none of the above

**Ans. (c) :** A force P of high magnitude acts on a body for a small internal of time $\Delta t$. the product of P and $\Delta t$ is called as impulse.

$$F_0 = P \times \Delta t$$

**33. When two bodies of mass M and 2M are connected by a light inextensible string and pass over a smooth pulley, than**

(a) tension in both sides of the string will be equal
(b) tension in one side of the string is two times the tension in the other side of the string
(c) tension in one side of the string is half the tension in the other side of the string
(d) none of the above

**Ans. (a) :** When two bodies of mass M and 2M are connected by a light inextensible string and pass over a smooth pulley, than tension in both sides of the string will be equal.

**Note–** Two masses $M_1$ and $M_2$ ($M_1 > M_2$) are connected by a light, in extensible string which passes over a smooth pulley. When released, mass m, will go down and the mass $m_2$ will move up and their common acceleration if $f = \left(\frac{M_1 - M_2}{M_1 + M_2}\right)g$ and the tension in the string is $T = \frac{2M_1 \times M_2}{M_1 + M_2}.g$

**34. A body is moving with a velocity of 10m/s. The time required to stop the body within a distances of 5 m is equal to**

(a) 3 seconds (b) 5 seconds
(c) 1 seconds (d) 0.5 seconds

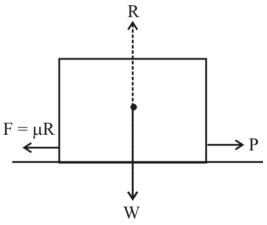
**Ans. (c) :** Given,

Velocity (v) = 10 m/s
Distance (d) = 5 m
Time (t) = ?
u = 10 m/s
v = 0
$\because$ $v^2 = u^2 + 2as$
$0 = (10)^2 + 2a \times 5$
$10a = -100$
$a = -10\ m/sec^2$
$\therefore$ $v = u + at$
$0 = 10 - 10t$
$10t = 10$
$t = 1$ sec.

**35. A body of weight W is resting on a horizontal plane. A force P is applied parallel to the plane to move the body. The value of P necessary to move the body against the resistance of friction is**

(a) $\frac{W}{\tan\phi}$ (b) $W\sin\phi$
(c) $W\tan\phi$ (d) $W\cos\phi$

**Ans. (c) :**




Here, the coefficient of friction is

$\mu = \tan\theta$

and $mg = W$

The minimum force,

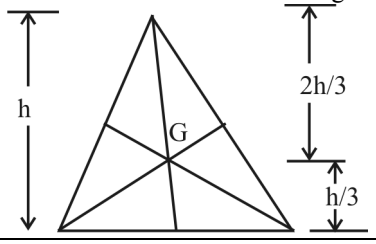
$P = F = \mu mg$

Friction $(F) = W\tan\theta$

**36. The centre of gravity of a triangle lies at the point of concurrence of**

(a) the right bisectors of the angles of the triangle
(b) the medians of the triangle
(c) the altitudes from the vertices on the opposite side
(d) none of the above

**Ans. (b):** The centre of gravity (G) of triangle lies at point where the three medians of the triangle intersect.




**37. The moment of inertia of a solid sphere of mass M and radius R is given by**

(a) $\frac{MR^2}{12}$ (b) $\frac{2}{5}MR^2$

(c) $\frac{MR^2}{36}$ (d) $\frac{3}{5}MR^2$

**Ans. (b) :** Moment of inertia of solid sphere of mass M and radius R about an axis passing through the centre of mass

$$\boxed{I_{xx\,or\,G} = \frac{2}{5}MR^2}$$

**38. The gravitational acceleration at a place is 6 times the value or gravitational acceleration on Earth. The weight of the body at that place will be**

(a) one- sixth of the total weight on Earth
(b) same as an Earth
(c) six times the weight on Earth
(d) none of the above

**Ans. (c) :** Gravitational acceleration at a place (g) is six times the value of gravitational accelerated on earth.

$g = 6g_{Earth}$

$mg = 6 \times mg_{Earth}$

$$\boxed{W = 6W_{Earth}}$$

**39. The velocity of a particle moving with SHM is maximum, when it is at**

(a) extreme position
(b) its means position
(c) a point between its mean and extreme position
(d) none of the above

**Ans. (b) :** The velocity of a moving body with simple harmonic motion at any instant is given by-

$$v = \omega\sqrt{r^2 - x^2}$$

The velocity is maximum at the mean position i.e. when $x = 0$

**40. The maximum velocity of a particle moving with SHM is 2 m/s and maximum acceleration is 20 m/s$^2$. The frequency of the motion is equal to**

(a) $\frac{20}{\pi}$ (b) $\frac{10}{\pi}$

(c) $\frac{5}{\pi}$ (d) $\frac{1}{\pi}$

**Ans. (c) :** Given,
maximum velocity (v) = 2m/s,
maximum acceleration (a) = 20 m/s$^2$
the frequency of the motion (f) = ?
maximum velocity (v) = $\omega A$
maximum acceleration ($a_{max}$) = $A \times \omega^2 = v \times \omega$

$$\omega = \frac{a_{max}}{v}$$

$$\therefore\ \omega = \frac{20}{2} = 10\,\text{rad/s}$$

$$\therefore \text{ frequency of oscillation } (f) = \frac{\omega}{2\pi}$$

$$= \frac{10}{2\pi} = \frac{5}{\pi}$$

**41. If a body is moving in a circular path, a force comes into play which acts along the radius of the circular path and is directed towards the centre of the path. This force is called**

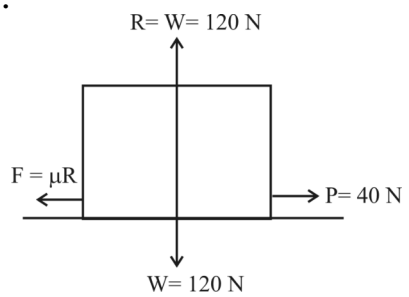
(a) centrifugal force (b) centripetal force
(c) shear force (d) none of the above

**Ans. (b) :** If a body is moving in a circular path, a force comes into a play which acts along the radius of the circular path and is directed towards the centre of the path. This force is known as centripetal force.

**42. A horizontal force of 40 N is applied on a body of weight 120 N, Placed on a horizontal plane, If the body is just on the point of motion, the angle of friction would be.**

(a) $20^0$ (b) $18.4^0$
(c) $10^0$ (d) $25^0$

**Ans. (b) :**




Given,
Horizontal free (P) = 40 N
Weight of body (W) = 120 N
Angle of friction (θ) = ?
We know that-
Coefficient of friction (μ) = tan θ ……. (i)

$$F = \mu R$$

$$\mu = \frac{F}{R} = \frac{40}{120}$$

$$\mu = \frac{1}{3}$$

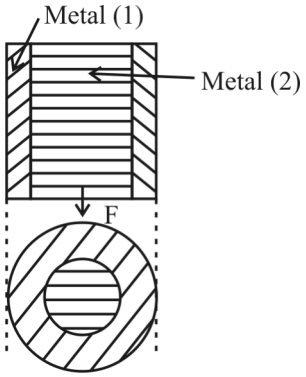
From equation (i)
$\tan\theta = 1/3$
$\theta = \tan^{-1}(1/3) = 18.4$

$$\boxed{\theta \approx 16.7^0}$$

**43. In a composite bar, the load distribution among different materiatls of which it is made is based on the assumption that all the materials will have**

(a) equal areas
(b) same Young's modulus
(c) same strain
(d) same stress

**Ans. (c) :** In case of composite bar total force, $F = F_1 + F_2$ and elongation due to load F,




$\Delta\ell_1 = \Delta\ell_2 = \Delta\ell$

So, $\varepsilon_1 = \varepsilon_2 = \varepsilon$

So, in this case strain will be same.

**44. A strut is a member which**

(a) connects two joints
(b) is subjected to axial tension primarily
(c) is subjected to axial compressive force predominantly
(d) is subjected to bending moment and shear force along with any axial force

**Ans. (c) :** A strut is a member which is subjected to axial compressive force predominantly.

- Smaller in size than column.
- It is pushed from both ends.

**45. A plate 100 mm wide, 10 mm thick is having a hole of diameter 10 mm, aymmetrical about the axis of the plate. The plate is subjected to a force of 9 kN. The maximum stress on a section passing through the centre of the hole will be**

(a) 10 N mm$^{-2}$ (b) >10 N mm$^{-2}$
(c) <10 N mm$^{-2}$ (d) 9 N mm$^{-2}$

**Ans. (b) :** Given,
Force = 9000 N
Thickness = 10 mm
Diameter = 10 mm
Width (b) = 100 mm

$\sigma_{max} = 3 \times \sigma_{nominal}$

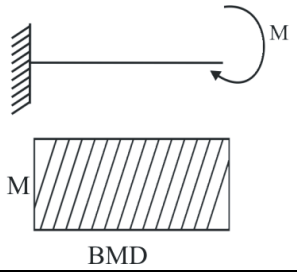
$$\sigma_{nominal} = \frac{F}{(b-d)t} = \frac{9000}{(100-10)\times t} = 9\ N/mm^2$$

$\sigma_{max} = 3 \times 9 = 27\ N/mm^2$

**46. A cantilever of length 'l' is subjected to bending moment M at the free end. The SF diagram will be**

(a) a triangle with maximum ordinate at fixed end
(b) a rectangle with ordinate M
(c) a parabola with maximum ordinate of $\frac{Ml^2}{2}$ at the fixed end
(d) no shear force at all

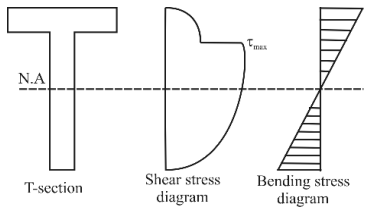
**Ans. (d) :** The bending moment diagram of cantilever beam subjected to moment at the free end will be rectangular and no any shear force occurs.




**47. In case of T- section, maximum bending stress will occurs at**

(a) Neutral axis
(b) extreme fibre in the flange
(c) extreme fibre in the web
(d) junction of web and flange

**Ans. (c) :**




Bending stress is zero at neutral axis and maximum value at the extreme fibers at the top and bottom of the beam.

Therefore maximum bending stress occurs in T-section at extreme fibers in the web.

**939. A cantilever AB is subjected to a concentrated load at the free end. The slope and deflection at the free end are respectively $\frac{WL^2}{2EI}$ and $\frac{WL^3}{3EI}$ If the same load is applied at mid- span point, the deflection at the free- end will be**

(a) $\frac{5}{384}\frac{WL^3}{EI}$ (b) $\frac{5}{48}\frac{WL^3}{EI}$
(c) $\frac{WL^3}{6EI}$ (d) $\frac{WL^3}{16EI}$

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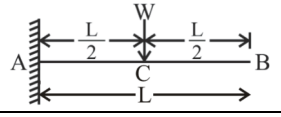
**Ans. (b) :**



Deflection $(\delta)_B$ = Deflection $(\delta)_c + \frac{L}{2} \times \text{slope}(\theta)_c$

$$= \frac{W \times \left(\frac{L}{2}\right)^3}{3EI} + \frac{L}{2} \times \frac{W\left(\frac{L}{2}\right)^2}{2EI}$$

$$= \frac{WL^3}{24EI} + \frac{WL^2}{16EI}$$

$$= \frac{2WL^3 + 3WL^3}{48EI} = \frac{5}{48}\frac{WL^3}{EI}$$

**49. Polar moment of inertia of a hollow circular section with external diameter D and internal diameter d is equal to**

(a) $\frac{\pi}{64}\left(D^4 - d^4\right)$ (b) $\frac{\pi}{64}\left(D^3 - d^3\right)$

(c) $\frac{\pi}{32}\left(D^4 - d^4\right)$ (d) $\frac{\pi}{32}\left(D^3 - d^3\right)$

**Ans. (c) :** A hollow circular shaft of outer dia D and internal dia d.

Polar moment of inertia = $\frac{I_{xx} + I_{yy}}{2}$

For circular section, $I_p = \frac{\pi}{32}\left(D^4 - d^4\right)$

**50. A thin cylinder of diameter 100 mm and thickness 5 mm is subjected to an internal fluid pressure of 10 N mm$^{-2}$. The hoop stress is**

(a) 100N mm$^{-2}$ (b) 10 N mm$^{-2}$

(c) 50 N mm$^{-2}$ (d) 5000 N mm$^{-2}$

**Ans. (a) :** Hoop stress = $\frac{Pd}{2t}$

P = 10 Nmm$^{-2}$

d = 100 mm

t = 5 mm

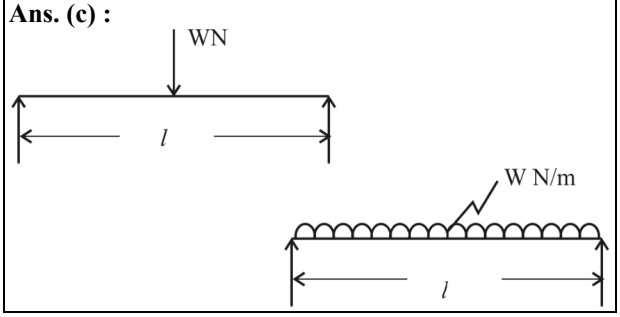
$\sigma_h = \frac{10 \times 100}{5 \times 2} = 100 \text{ N mm}^{-2}$

**51. A simply supported beam of span L carries a UDL of w N m$^{-1}$ what is the magnitude of concentrated load to be supplied at the centre of this beam which would produce the same deflection as the UDL**

(a) $\frac{3}{8}wL$ (b) $\frac{wL}{2}$

(c) $\frac{5}{8}wL$ (d) $\frac{7}{8}wL$

**Ans. (c) :**




Deflection $\delta_{(u.d.l)} = \frac{5}{384}\frac{wl^4}{EI}$.....(i)

Deflection $[\delta_{(Point\ load)}] = \frac{1}{48}\frac{Wl^3}{EI}$.....(ii)

$\because$ From equation (i) = (ii)

$$\frac{1}{48}\frac{Wl^3}{EI} = \frac{5}{384}\frac{wl^4}{EI}$$

$$\boxed{W = \frac{5}{8}wl}$$

Here, W = Point load, w = u.d.*l*

**52. Three beams made of same material (but of different sections) of same space are subjected to same maximum bending moment. then which section will have the maximum weight per unit length?**

(a) circular (b) square

(c) rectangular (d) none of the above

**Ans. (a) :** Three beams made of same material (but of different sections) of same space are subjected to same maximum bending moment. then circular section will have the maximum weight per unit length.

**53. A spring has a stiffness of k. if it is cut into two equal portions, the stiffness of the cut portion of the spring is**

(a) k (b) $\frac{k}{2}$

(c) 2k (d) $\frac{k}{4}$

**Ans. (c) :** Stiffness of spring (K) = $\frac{cd^4}{64R^3n}$

$$K \propto \frac{1}{n}$$

where, n = Number of coils

c = modulus of rigidity

d = diameter of wire

R = mean radius of coil

$$\frac{K_1}{K} = \frac{n}{n/2}$$

$$\frac{K_1}{K} = \frac{2n}{n}$$

$$\boxed{K_1 = 2K}$$

**54. which of the following relations is correct?**

(a) factor of safety = crippling load- safe load

(b) factor of safety = crippling load/ safe load

(c) factor of safety = crippling load× safe load

(d) factor of safety = safe load/ crippling load

**Ans. (b) :** It is the load to which a column is actually subjected to and is well below the buckling load is obtained by dividing the buckling load by a suitable factor of safety.

Factor of safety = $\frac{\text{Crippling load}}{\text{Safe load}}$

**55. The maximum eccentricity of the vertical load on a short column of external diameter D and internal diameter d, for no tensions to occur, is**

(a) $\frac{8D}{D^2+d^2}$ (b) $\frac{D^2+d^2}{8D}$

(c) $\frac{D^2+d^2}{D}$ (d) $\frac{D^2+d^2}{8D}$

**Ans. (b) :** For no tension minimum combination stress (direct and bending) thrust be greater or equal to zero.

$$\frac{P}{A}-\frac{My}{I}\geq 0$$

$$\frac{P}{A}-\frac{P.e.D/2}{\frac{\pi}{64}(D^4-d^4)}\geq 0$$

$$\frac{4\times P}{\pi(D^2-d^2)}\geq \frac{P.e.D/2}{\frac{\pi}{64}(D^4-d^4)}$$

$$\frac{4\times\frac{\pi}{64}(D^4-d^4)}{\pi(D^2-d^2)\cdot\frac{D}{2}}\geq e$$

$$\frac{(D^4-d^4)}{8D(D^2-d^2)}\geq e$$

$$\frac{(D^2-d^2)(D^2+d^2)}{8D(D^2-d^2)}\geq e$$

$$\boxed{e\leq\frac{(D^2+d^2)}{8D}}$$

**56. A solid circular shaft of diameter D is subjected to an axial load to produce a stress of p N-cm$^{-2}$ If the same axial load is applied to a hollow circular shaft of external diameter D and internal diameter D/2, the corresponding stress produced would be**

(a) $\frac{p}{2}$ Ncm$^{-2}$ (b) $\frac{p}{4}$ Ncm$^{-2}$

(c) $\frac{3p}{2}$ Ncm$^{-2}$ (d) $\frac{4p}{3}$ Ncm$^{-2}$

**Ans. (d) :** Given, For solid shaft,

dia = D

($p_s$) stress = p N/cm$^2$

For hollow shaft,

dia $\Rightarrow$ inner = D/2

outer = D

stress $p_H$ = ?

$$p_s=\frac{p}{A}=\frac{p\times 4}{\pi D^2}$$

$[p_s = p = 4p/\pi D^2]$ .....(i)

$$p_H=\frac{p}{A}=\frac{4p}{\pi(D^2-D^2/4)}=\frac{4p\times 4}{\pi\times 3D^2}$$

$$p_H=\frac{4}{3}\times\left(\frac{4p}{\pi D^2}\right)$$ (from equation (i))

$$\boxed{p_H=\frac{4}{3}p}\,N/cm^2$$

**57. A body is subjected to an axial tensile stress p. Then the normal stress on any oblique plane inclined at an angle θ to the cross section of the body is given by**

(a) $P_n = P\sin\theta$ (b) $P_n = P\cos\theta$

(c) $P_n = P\sin^2\theta$ (d) $P_n = P\cos^2\theta$

**Ans. (d) :** $\sigma_n=\frac{\sigma_x+\sigma_y}{2}+\frac{\sigma_x+\sigma_y}{2}\cos 2\theta$

$\sigma_y = 0,\ \sigma_x = \sigma$

$$\sigma_n=\frac{\sigma}{2}+\frac{\sigma}{2}\cos 2\theta=\frac{\sigma}{2}[1+\cos 2\theta]$$

$$\sigma_n=\frac{\sigma}{2}[1+2\cos^2\theta-1]$$

$$\boxed{\sigma_n=\sigma\cos^2\theta}$$

$$\sigma_n=\frac{P_n}{A}$$

$$P_n=A.\sigma\cos^2\theta$$

$$\boxed{P_n=P\cos^2\theta}$$

**58. To what category do gravel and sand belong?**

(a) cohesionless soils (b) cohesive soils

(c) marine soils (d) expansive soils

**Ans. (a) :** Cohesionless soil is gravel, sand, etc.

Sand : 0.075 – 9.75 mm

Gravel : 4.75– 80 mm

**59. Which of the following statement is true?**

(a) In a dry soil, all the voids are filled with air

(b) In a saturated soil, all the voids are filled with water

(c) In a partly saturated soil, the voids are occupied by both air and water

(d) All the above

**Ans. (d)** In a dry soil all the voids are filled with air.

$\boxed{V_v = V_a}$

Total volume ($V_T$) = $V_a$ + $V_s$ $\{\because V_T = V_V + V_S\}$

In a standard soil all the voids are filled with water.

In partially saturated soil voids are occupied by both air and water.

$V_v = V_a + V_w$

$V_T = V_a + V_w + V_s$

**60. If the temperature at the time of test is more than that of calibration of the hydrometers, the hydrometer reading will be**

(a) less (b) more

(c) same (d) none

**Ans. (a) :** Hydrometer analysis is based on the principle of Stoke's law.

- Results form the test show the grain size distribution for soil finer than the number.
- Hydrometer is used for measuring the relative density of liquid based on concept of buoyancy.
- Hydrometer reading will be less if temperature test is more than calibration of the hydrometer.

**61. If the overland flow from the critical point to the mouth of a drain is 12 km, and the difference in levels between these points is 14.5 m, then the inlet time would be of the order of**

(a) 2 hours (b) 4 hours
(c) 6 hours (d) 8 hours

**Ans. (c) :** Given,
Flow path length ($L_L$) = 12 km
Flow path slope (different level) = 14.5 m

Inlet time in hour $T_c=\left[\frac{0.885\ L_c^3}{H}\right]^{0.385}$ (Kirpitch's formula)

$$T_c=\left[\frac{0.885\times(12)^3}{14.5}\right]^{0.385}$$

$T_c$ = 6 hour

**62. Which one of the following is correctly matched?**

(a) Trickling filter- attached growth
(b) Oxidation ditch- extended aeration
(c) Oxidation pond- symbiotic
(d) All of these

**Ans. (d) :** Ponding due to logging of the filter (trickling filter), odour due to anaerobic decomposition and fly nuisance takes place in trickling filters.

**Oxidation ditch–** The oxidation ditch is a modified form of extended aeration activated sludge process.

- It has maximum efficiency of the BOD removal.
- It is suitable for small communities or small industries.
- It is best suitable for fluctuation in quality and quantity.
- No primary sedimentation and sludge treatment is needed.
- It is cheap in construction, operation and maintenance.

**Oxidation pond–**

- The waste water treatment initiated by symbiotic relationship between algae and bacteria.
- Oxidation pond or waste stabilization ponds or lagoons are relatively shallow body of water contained in an earthen basin of controlled slope. They are designed and constructed to treat sewage and biodegradable industrial wastes.
- Ponds require low initial cost end low operation cost.
- It is required more land.

**63. The following reacting takes place during anaerobic digestion of organics.**

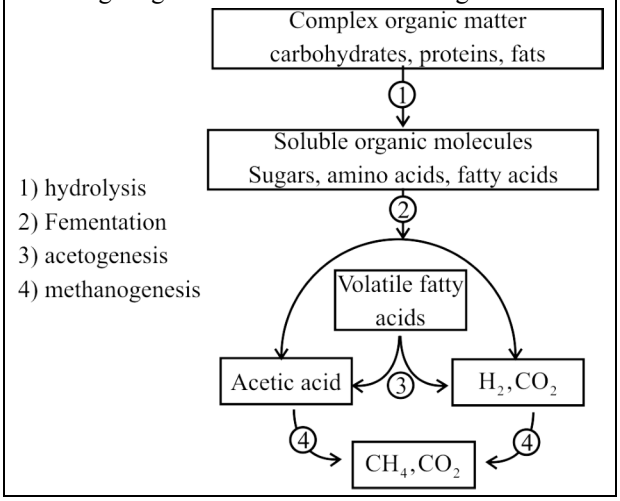
(A) methane production
(B) alkaline fermentation
(C) acid fermentation
(D) acid regression

The correct sequence of these reactions is

(a) C, D, B, A (b) D, C, B, A
(c) C, D, A, B (d) D, C, A, B

**Ans. (a) : Anaerobic decomposition or anaerobic digestion–** Anaerobic digestion is a sequence of processes by which microorganisms break down biodegradable material in the absence of oxygen. The process is used for industrial or domestic purposes to manage waste or to produce fuels.

The full process of anaerobic digestion is shown in the following diagram for better understanding.




**64. The advantage of aerobic digestion over anaerobic digestion is**

(a) lower BOD concentration is supernatant liquor
(b) production of sludge with excellent dewstering
(c) lesser capital cost
(d) all of these

**Ans. (d) :** Advantage of aerobic digestion over anaerobic digestion–

- Lower BOD concentration is supernatant liquor.
- Production of sludge with excellent dewatering.
- Lesser capital cost.

**65. Soluble organics in domestic wastewaters include**

(a) carbohydrates (b) proteins
(c) lipids (d) all the above

**Ans. (d) :** Soluble organics in domestic wastewaters include carbohydrates, proteins, lipids and other organic materials such as detergents, pesticides, fat, oil and grease, colouring solvents etc.

**66. The settling velocity of a spherical body in still water is given by**

(a) Lacey's formula
(b) Darcy's law
(c) Hazen- William's formula
(d) Stoke's law

**Ans. (d) Stoke's law of settling–**The gravitation air force pulling the particle down while the drag force of the fluid resisting this sinking or frictional resistance of a quit sedimentation fluid upon a sinking spherical particle.

$$V_s = \frac{(G-1)gd^2}{18\nu}$$

$\therefore$ Where,

$V_s$ = velocity of settlement of particle in m/sec

d = dia of the particle in meter

G = Specific gravity of the particle

**67. When the times of earliest finish and the latest finish are equal in an operation, the operation is called**

(a) critical (b) float

(c) free float (d) all the above

**Ans. (a) :** When the times of earliest finish and the latest finish are equal in an operation, the operation is called critical.

**68. If for an activity optimistic time is 1 day, pessimistic time is 8 days and most likely time is 3 days, then the expected time is**

(a) 3.5 days (b) 3.6 days

(c) 3.75 days (d) 4 days

**Ans. (a) :** Given,

Active optimistic time = 1 day

Optimistic time = 8 day

Most likely time = 3 day

Expected time,

$$t_e = \frac{1+4\times3+8}{6} = 3.5 \text{ days}$$

**69. Float is**

(a) the earliest time by which an activity may commence

(b) latest time by which an activity can by delayed without hampering the completion of project

(c) difference of latest allowable time and earliest expected time

(d) difference of time available and time required for an activity

**Ans. (b) :** Float is latest time by which an activity can be delayed without hampering the completion of project.

**70. Free float is mainly used to identify the activities which can be delayed without affecting the total float of**

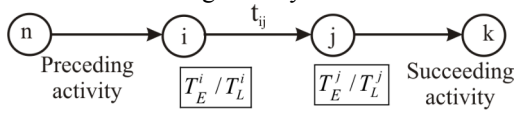
(a) preceding activity

(b) succeeding activity

(c) either the preceding or succeeding activity

(d) none of the above

**Ans. (b) :** Free float is mainly used to identify the activities which can be delayed without affecting the total float of succeeding activity.




i = Tail event

j = Head event

$$F_T = \left(T_L^j - T_E^i\right) - t_{ij}$$

**71. the direct cost of project with respect to normal times is**

(a) minimum (b) maximum

(c) zero (d) infinite

**Ans. (a)** The direct cost of a project with respect to normal time is minimum.

- The goal of any scheduling problem is essentially to find the project duration that minimizes sum of direct costs and indirect cost or in other words, finding the optimum point in a time-cost trade-off.

**72. Updating may result in**

(a) change of critical path

(b) decrease of project completion time

(c) increase of project completion time

(d) all of the above

**Ans. (d) :** Once the scheduled plan has been prepared and execution has started, a check or control over their progress of work has to be exercised in order to complete the work by target or schedule date for this the process of incorporation of change and replanning or rescheduling is done which is called as updating. It has following important point.

(i) Any major change in duration of an activity updating is mandatory.

(ii) For critical activities update is mandatory.

(iii) Any activity is neither added nor deleted.

(iv) Critical part may change during updating and hence critical activities may also get change.

**73. Activity having negative float is called**

(a) critical activity

(b) supercritical activity

(c) sub critical activity

(d) all of the above

**Ans. (b) :** Supercritical activity is a critical activity that has negative float (means its now behind schedule and that it needs to be immediately finished.

**74. A contractor has two options:**

**I. Invest his money in project A or B**

**II. Invest his money in project B. If he decides to invest in A, for every rupee invested, he is assured of doubling his money in ten years. If he decides to invest in B, he is assured of making his money 1.5 times in 5 years. If the contractor values his money at 10% interest rates, he**

(a) should invest in neither of the two projects

(b) could invest in either of the two projects

(c) should invest in project A

(d) should invest in project B

**Ans. (a) :** A contractor should not invest his money either of the two projects.

**75. Consider the following operation:**

**A. drilling**

**B. blasting**

**C. Mucking**

**D. placing steel**

**E. placing concrete**

**The correct sequence of these operation is tunnel constructions is**

(a) A, B, D, C, E (b) A, C, B, D, E
(c) A, B, C, D, E (d) A, C, D, B, E

**Ans. (c) :** • Sequence of operations in tunnel construction is–

Drilling → Blasting→ Mucking → Placing steel → Placing concrete

• Construction of tunnel is done in small - small sections.

• Each section is drilled to a certain length of depth using a pre-engineered drilled pattern.

**76. Functional Organization system of working was introduced by**

(a) F.W. Taylor (b) Henry Gantt
(c) M.R. Walker (d) J.E. Kelly

**Ans. (a) :** A functional organization is one in which work is organized on basis of specialization. Thus, expert staff personnel without taking the route of formal command – chain.

This form of organization was devised by F.W. Taylor the father of Scientific Management.

**77. Consider the following statements:**

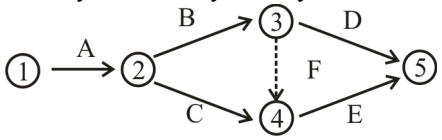
**A: A dummy job takes zero time to perform.**

**R: It is used solely to illustrate precedence relationship of these statements.**

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

**Ans. (a) :** Dummy activity is a artificial activity represented by dashed arrows which neither consumes any time or resources.

Activity F is dummy activity




A dummy job takes zero time to perform because it used solely to illustrate precedence relationship of these statements.

**78. Capital recovery factor considers**

(a) developing a predetermined sum at a future date at a uniform rate
(b) consuming a predetermined sum till a future date at a uniform rate
(c) consuming a predetermined sum till a future date at a uniformly varying rate
(d) consuming a predetermined sum till a future date at uniform classes

**Ans. (b) : Capital recovery factor–**

$$\Rightarrow A/P = \left[\frac{i(1+i)^n}{(1+i)^{n-1}}\right]$$

• Capital recovery factor is the ratio of a constant (uniform) annuity to the present value of receiving that annuity for a given length of time using an interest rate.

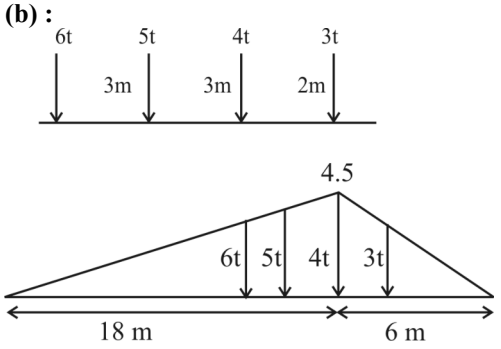
• When n is the number of annuities received. This is related to the annuity formula which gives the present value in term of the annuity, the interest rate and the number of annuities.

• If n = 1 then CRF reduces to 1 + i. As n goes to infinity the CRF goes to i.

**79. A number of wheel loads 3t, 4t, 5t and 6t spaced 2m, 3m and 3 m respectively move on a simply supported beam AB of span 24 m with the 3t load leading from left to right. To find the maximum bending moment at 18 m from A, the load that must be placed at the section is**

(a) 3t (b) 4t (c) 5t (d) 6t

**Ans. (b) :**



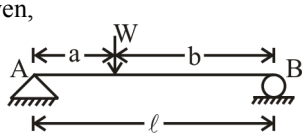

$$\text{Maximum ordinate} = \frac{18\times 6}{18+6} = 4.5$$

So, the load that must be placed at the section is 4t.

**80. A load W is moving from left to right support on a simply supported beam to span $\ell$. The maximum bending moment at 0.4 $\ell$ from the left support is**

(a) 0.16 W$\ell$ (b) 0.20 W$\ell$
(c) 0.24 W$\ell$ (d) 0.25 W$\ell$

**Ans. (c) :** Given,




W is a moving load from left to right.

At, 0.4 $\ell$, $BM_{max}$ = ?

In this case of beam,

$$BM_{max} = \frac{Wab}{a+b}$$

$$= \frac{W\times 0.4\ell\times 0.6\ell}{0.4\ell+0.6\ell} \qquad \left(\because a = 0.4\ell,\ b = \ell - 0.4\ell = 0.6\ell\right)$$

$$\boxed{BM_{max} = 0.24W\ell}$$

**81. Which of the following does not fall under the category of displacement method?**

(a) Method of consistent deformation
(b) Equilibrium method
(c) Moment distribution method
(d) Stiffness method

**Ans.(a): Displacement method–** Kanis method, Moment distribution method, Slope deflection method, Stiffness method, equilibrium method.
**Force method–** Method of consistent deformation, strain energy method, 3 moment method / clapey ron's method, flexibility matrix, column analogy etc.

**82. The ratio of design discharge to surface area of a sedimentation take is known as its**
(a) surface loading
(b) overflow velocity
(c) overflow rate
(d) all of the above

**Ans. (c) : Over flow rate :** It is also called surface loading or surface overflow rate.

$$\text{Over flow rate } (V_o) = \frac{\text{Discharge}}{\text{Surface area}}$$

$$\boxed{(V_o) = \frac{Q}{BL}}$$

Overflow rate for pain sedimentation
– 500 – 750 lit/hr/$m^2$
Overflow rate for coagulation with sedimentation
– 1000 – 1250 lit/hr/$m^2$

**83. Fixed end of fixed beam of span L settles by an amount δ. The ratio of the slope and deflection at the centre of the beam will be**
(a) 0.5 L (b) 1.5 L
(c) 0.75 L (d) 0.667 L

**Ans. (b) :** $K = \frac{M}{\theta} = \frac{P}{\delta}$

$$\theta = \frac{\frac{6EI\Delta}{\ell^2}}{\frac{12EI\Delta}{\ell^3}}$$

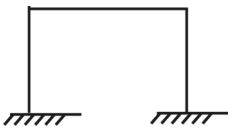
$$\theta = 0.5\ell$$

$$\delta = \frac{\frac{4EI\Delta}{\ell^3}}{\frac{12EI\Delta}{\ell^3}} = \frac{1}{3}$$

$$\frac{\theta}{\delta} = \frac{0.5}{1/3}\ell = 1.5\ \ell$$

**84. The moment distribution method is best suited for**
(a) indeterminate pin jointed truss
(b) rigid frames
(c) space frames
(d) trussed beam

**Ans. (b) :** The moment distribution method is best for rigid frames.



Rigid frames are non-sway frames which has all the fixed support.

**85. The number of unknowns to be determined in the stiffness method is equal to**
(a) static indeterminacy
(b) kinematic indeterminacy
(c) sum of static and kinematic indeterminacy
(d) none of the above

**Ans. (b) : Displacement / stiffness method–** The unknowns are takes as joint displacements.
- To find unknown displacements, joint equilibrium conditions are written.
- Joint equilibrium conditions needed = number of unknowns degree of kinematic indeterminacy.
- This method is suitable when $D_K < D_S$

**86. It can be stated in general that if a statically indeterminate structure has k redundancy than it will become a statically determinate structure if the number or plastic hinges formed is equal to**
(a) k (b) k+1
(c) k-1 (d) none of the above

**Ans. (b) :** The number of plastic hinges required for total collapse of structure is one more than the order of statically indeterminacy
Plastic hinges = k + 1
k = order of indeterminacy

**87. Corrosion in concrete sewers is caused by**
(a) septic conditions (b) dissolved oxygen
(c) chlorine (d) nitrogen

**Ans. (a) :**
- Corrosion in concrete sewers in caused by septic conditions.
- Corrosion in concrete sewers is mainly caused by hydrogen sulphide gas which is formed due to anaerobic decomposition of sewage. During anaerobic decomposition of sewage acids, alcohols and gases are formed and hydrogen sulphide is one of the formed gases.

**88. A two- hinged arch (parabolic) is subjected to a temperature rise of $t^0$ C, and then horizontal thrust at support will.**
(a) increase (b) decrease
(c) remain same (d) increase or decrease

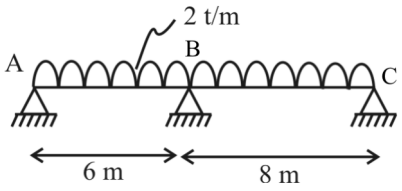
**Ans. (a) :** H is two hinged arch due to rise in temperature can be represented by

$$H = \frac{\alpha T \ell EI}{\int_0^\ell y^2 dx}$$

The H increased due to rise in temperature.

**89. A two span continuous beam ABC is simply supported at A and C and is continuous over support B. Span AB= 6 m, BC =8 m. The beam carries a UDL of 2 t/m over both spans. EI is constant for the entire beam. The fixed end moment at B is span BA or BC would be**
(a) 12 t m (b) 9 t m
(c) 8 t m (d) 6 t m

**Ans. (b):**




$$\overline{M}_{BA} = \frac{2\times 6^2}{12} = 6 \text{ t m}$$

$$\overline{M}_{BC} = -\frac{2\times 8^2}{12} = -10.6$$

Slope deflection equation for BA

$$M_{BA} = 6 + \frac{2EI}{6}(2\theta_B) \quad \text{.....(i)}$$

Slope deflection equation for BC

$$M_{BC} = -10.6 + \frac{2EI}{8}(2\theta_B) \quad \text{.....(ii)}$$

$$M_{BA} + M_{BC} = 0$$

$$4.6 = \frac{2EI}{6}(2\theta_B) + \frac{2EI}{8}(2\theta_B)$$

$$\frac{4.6}{EI} = \frac{2}{3}\theta_B + \frac{\theta_B}{2}$$

$$\theta_B = \frac{4.6\times 6}{7\times EI}$$

$$\theta_B = \frac{3.942}{EI}$$

$$M_{BA} = 6 + \frac{2EI}{6}\times\frac{3.942}{EI}\times 2$$

$$M_{BC} = 9 \text{ t m}$$

**90. Given the x- component of the velocity u= 6×y - 2×$^2$, then y- component of the flow v is given by**

(a) $6y^2 - 4\times y$ (b) $6\times y + 2\times^2$

(c) $5\times^2 - 2\times y$ (d) $4xy - 3y^2$

**Ans. (d) :** Continuity equation

$$\frac{du}{dx} + \frac{dv}{dy} = 0$$

$u = 6xy - 2x^2$

$$\frac{du}{dx} = 6y - 4x$$

$$6y - 4x + \frac{dv}{dy} = 0$$

$$\frac{dv}{dy} = 4x - 6y$$

$$v = 4xy - \frac{6y^2}{2}$$

$$v = 4xy - 3y^2$$

**91. If S= strength of timber and H= hardness of timber at 12% moisture content for a timber sleeper, then its " composite sleeper index" is**

(a) $\frac{S+H}{2}$ (b) $\frac{S+10H}{11}$

(c) $\frac{S+9H}{10}$ (d) $\frac{S+10H}{20}$

**Ans. (d) :** • Composite sleeper index is the measure of strength and hardness of timber.

• Minimum CSI for track sleeper (783), Crossing sleeper (1352) & bridge sleeper (1455)

$$CSI = \frac{S+10H}{20}$$

S = Strength index of timber at 12% moisture

H = Hardness index of timber.

**92. Sleeper density is**

(a) number of sleepers per kM length

(b) number of sleepers per rail length

(c) density of the material of the sleeper

(d) minimum distance between any two neighbouring sleepers

**Ans. (b) : Sleeper density–** Sleeper density is number of sleeper per rail length.

It is denoted by N + x

N = rail length (13 for BG)

x = varies b/w 3 to 7

for BG sleeper density in N + 5 (18 sleepers/rail)

**93. The outer and inner wheels though cast monolithic cover different distances on a curve for the same number of revolutions because of**

(a) adzing of sleepers (b) coning of wheels

(c) widening of gauge (d) tilting of rails

**Ans. (b) : Coning of wheels**– Coning of wheel is done at slope of 1 in 20.

• On curve track outer wheel has to travel greater length than inner wheel, the coning of wheel help the outer wheel tread to move up and down relatively to curve without any jerk and disturbance to inner wheel movement.

**94. Switch is**

(a) a tongue rail

(b) a stock rail

(c) combination of both tongue and stock rails

(d) tongue and stock rail combination but separated by flange way

**Ans. (c) : Switch–** It is the combination of stock rail & tongue rail forming a switch.

**Points–** A set of switches is known as points.

**Turnout–** The combination of points & crossing is called turnout.

**95. Flat type of marshalling yard is preferred to the others when**

(a) space available is small and restricted

(b) less number of engines are available

(c) goods to be transported include perishables

(d) pneumatic retarders are provided to bring the bogies to rest quickly.

**Ans. (a) : Flat yard–** It is a type of marshalling yard which is preferred when space available is small and restricted.

• A flat yard has not hump, and relies on locomotives for all car movements.

**96. Home signal is located not more than- from the start of points or switches.**

(a) 1 m (b) 860 m
(c) 180 m (d) 580 m

**Ans. (c) : Home signal–** Due to its location at the door of station, it is termed as home signal.

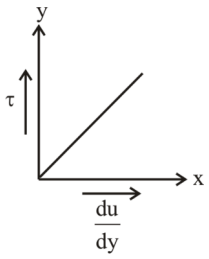
- The home signal has bracketed arms which shows which line is to be used.
- It is located not more than 180 m from the start of points or switches.

**97. Newton's law of viscosity states that**

(a) shear stress is directly proportional to the velocity
(b) shear stress is directly proportional to velocity gradient
(c) shear stress is directly proportional to the viscosity
(d) shear stress is directly proportional to the viscosity of fluid

**Ans : (b)** Shear stress is directly proportional to velocity gradient.

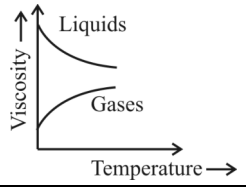
$$\tau = \mu \frac{du}{dy}$$




**98. The increase of temperature**

(a) increase the viscosity of a liquid
(b) decrease the viscosity of a gas
(c) decrease the viscosity of a liquid
(d) increase the viscosity of a fluid.

**Ans. (c) :** The increase of temperature decrease the viscosity of a liquid.




**99. For submerged curved surface, the vertical component of the hydrostatic force is**

(a) mass of the liquid supported by the curved surfaces
(b) weight of the liquid supported by the curved surface
(c) the force of the projected area of the curved surface on vertical place
(d) none of the above

**Ans. (b) :** For submerged curved surface, the vertical component of the hydrostatic force is weight of the liquid supported by the curved surface.

**100. The condition of stable equilibrium for a floating body is**

(a) the metacentre M coincides with the centre of gravity G of the body
(b) the metacentre M lies below the centre of gravity G of the body
(c) the metacentre M lies above the centre of gravity G of the body
(d) the centre of buoyancy B is above the centre of gravity G of the body

**Ans. (c) : Stable equilibrium–** When a body is given a small angular displacement, and then it return back to it's original position due to the internal forces then it is said that the body is in stable equilibrium. For floating body to be in stable equilibrium- metacentre must be above centre of gravity.

| | Submerged body | Floating body |
|---|---|---|
| Stable equilibrium | G below B | M above G |
| Unstable equilibrium | G above B | M below G |
| Neutral | G & B coincide | M & G coincide |

B = Centre of buoyancy
G = Centre of gravity
M = meta centre

**101. The gases are considered incompressible when the Mach number**

(a) is equal to 1.0 (b) is equal to 0.5
(c) is more than 0.3 (d) is less than 0.2

**Ans. (d) :** The gases are considered incompressible when the Mach number is less than 0.2.

- When apart from inertial force, compressibility force predominant, we use mach law.
- Compressibility force are predominant when mach number $\geq 0.3$.
- When mach number $< 0.3$ compressibility effect can be neglected.

**102. For a floating body, the buoyant force passes through the**

(a) centre of gravity of the body
(b) centre of gravity of the submerged portion of the body
(c) metacentre of the body
(d) centroid of the liquid displaced by the body

**Ans. (d) :** Centre of buoyancy is the centroid of the volume of displaced liquid. It is applicable in both floating as well as submerged bodies.

**103. Fluid statics deals with the following forces**

(a) viscous and gravity forces
(b) viscous and pressure forces
(c) gravity and pressure forces
(d) surface tension and gravity forces

**Ans. (c) :** Fluid statics deals with gravity and pressure forces.

- A static fluid can have positive normal struss and zero shear stress.
- The fluid is in rest. i.e. the velocity gradient is zero, therefore no shear force, will be acting on it, the only force present is due to gravity and pressure of the fluid i.e. normal /force only.

**104. The necessary condition for the flow to be uniform is that**

(a) the velocity is constant in the flow field with respect to space
(b) the velocity is constant at a point with respect to time
(c) the velocity changes at a point with respect to time
(d) the velocity changes in the flow field with respect to space

**Ans. (a) : Uniform flow–** If the velocity is constant in the flow field with respect to space or the flow is same at any instant of time at all points is called uniform flow.

For uniform flow $\boxed{\frac{\partial v}{\partial s} = 0}$

**105. The velocity components in x and y directions in terms of stream function Ψ are.**

(a) $u = \frac{\partial \psi}{\partial \chi}; v = \frac{\partial \psi}{\partial y}$ (b) $u = \frac{\partial \psi}{\partial \chi}; v = \frac{\partial \psi}{\partial y}$

(c) $u = \frac{\partial \psi}{\partial y}; v = \frac{\partial \psi}{\partial \chi}$ (d) $u = -\frac{\partial \psi}{\partial y}; v = \frac{\partial \psi}{\partial \chi}$

**Ans. (d) :** Stream line function ($\psi$)

$$u = \frac{-\partial \psi}{\partial y}; v = \frac{\partial \psi}{\partial x}$$

If stream function satisfies Laplace equation then it is case of irrotational flow.

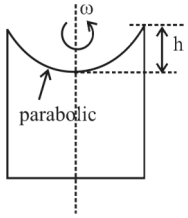
• The stream function ($\psi$) is defined as a scalar function of space and time, such that its partial derivative w.r.t. any direction gives the velocity component at right angle to this direction.

**106. For a forced vortex flow, the height of paraboloid formed is equal to**

(a) $\frac{p}{\gamma} + \frac{V^2}{2g}$ (b) $\frac{V^2}{2g}$

(c) $\frac{V^2}{r \times 2g}$ (d) $\frac{\omega r^2}{2g}$

**Ans. (b) :** For a forced vertex flow, the height of paraboloid form is equal to $\frac{v^2}{2g}$




$$h = \frac{v^2}{2g}$$

$v = r\omega$

$$h = \frac{\omega^2 r^2}{2g}$$

Volume of parabolid $= \frac{1}{2}\left(\pi r^2\right)\left(\frac{\omega^2 R^2}{2g}\right)$

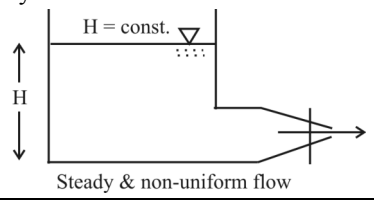
**Note :** for no spelling case, rise above original water level = fall below original water level.

**107. Flow at a constant rate through a tapering pipe is**

**i. steady flow**
**ii. uniform flow**
**iii. unsteady flow**
**iv. non- uniform flow**
**The correct answer is**

(a) (i) and (ii) (b) (i) and (iv)
(c) (ii) and (iii) (d) (ii) and (iv)

**Ans. (b) :** Flow at a constant rate through a tapering pipe is steady and non-uniform flow.




Steady & non-uniform flow

**108. In a steady flow, the temporal acceleration of any fluid particle**

(a) can be zero
(b) is never zero
(c) is always zero
(d) is independent of coordinates

**Ans. (a) :**

| Type of flow | Convective Acceleration | Temporal Acceleration |
|---|---|---|
| Steady & uniform | 0 | 0 |
| Steady & non-uniform | Exists | 0 |
| Unsteady & uniform | 0 | Exists |
| Unsteady & non-uniform | Exists | Exists |

**109. The velocity potential function for source varies with distance r as**

(a) $\frac{1}{r}$ (b) $\frac{1}{r^2}$

(c) r (d) $l$ n r

**Ans. (d) :** A source or sink represents a purely radial flow

So, $V_\theta = 0$

The corresponding velocity potential can be obtained by Integrating, the equations.

$$\frac{\partial \phi}{\partial r} = \frac{q}{2\pi r} \qquad \frac{1}{r}\frac{\partial \phi}{\partial \theta} = 0$$

it follows that

$$\phi = \frac{q}{2\pi} \ln r$$

$\boxed{\phi \propto \ln r}$

• If m is positive, the flow is radially outward and the flow is considered to be a source flow.
• If m is negative, the flow is towards the origin and the flow is considered to be a sink flow.
• The flowrate, m is the strength of the source or sink.

**110. Which one of the following velocity fields represents a possible fluid flow?**
(A) U=X, V=Y (B) $U=X^2$; $V=Y^2$
(C) U=xy; $v=x^2 y^2$ (D) U= x; v=-y

**Ans. (d) :** Flow is possible only if velocity fields follow continuity equation–

$$\frac{du}{dx}+\frac{dv}{dy}=0$$

$$\Rightarrow \frac{d(x)}{dx}+\frac{d(-y)}{dy}$$

$$\Rightarrow 1-1=0$$

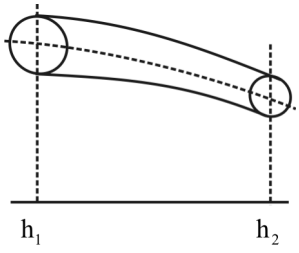
only option (d) satisfies it.

**111. Consider the following parameters related to fluid flow:**
**1. Vorticity, 2. Velocity potential**
**3. Stream function**
**Which of these parameters exist both in rotational and irrotational flows?**
(a) 1 & 2 (b) 2 & 3
(c) 1 & 3 (d) 1, 2 & 3

**Ans. (c)** Vorticity and stream function both follow rotational & irrotational parameters
• if vorticity $(\xi)$ at a point is zero than that is irrotational.
• Velocity potential function does not exist in rotational flow.
• Vorticity exists in both rotational and irrotational flow but irrotational flow its value is zero.

**112. In the most general form of Bernoulli's equation $z+\frac{P}{\gamma}+\frac{V^2}{\rho g}$ = constant, each term represents**
(a) energy per unit mass
(b) energy per unit weight
(c) energy per unit volume
(d) energy per unit area

**Ans. (b):** The Bernoulli's equation written in conventional form represents total energy per unit weight.
Bernoulli's equation states conservation of energy.




$$P_1+\frac{1}{2}\rho v_1^2+\rho g h_1=P_2+\frac{1}{2}\rho v_2^2+\rho g h_2$$

Divide by ρg –

$$\frac{P_1}{\rho g}+\frac{v_1^2}{2g}+h_1=\frac{P_2}{\rho g}+\frac{v_2^2}{2g}+h_2$$

$$\boxed{\frac{P}{\rho g}+h+\frac{v^2}{2g}=\text{ constant}}$$

where,

$\frac{P}{\rho g}$ = Pressured energy per unit weight of fluid or pressure head.

$\frac{v^2}{zg}$ = Kinetic energy per unit weight or kinetic head.

z = Potential energy per unit weight or potential head.

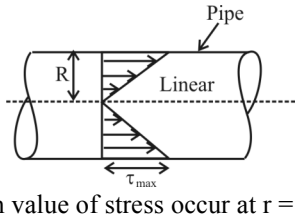
**113. The locus of elevations that water will raise in a series of pitot tubes is called**
(a) energy grade line (b) hydraulic grade line
(c) pressure head (d) velocity head

**Ans. (a) :** The locus of elevations that water will raise in a series of pitot tubes is called energy grade line.
• Hydraulic gradient line (HGL)– It joins piezometric head $\left(\frac{P}{\gamma}+z\right)$ at various points.
• Total energy line (TEL) – It joins total head of various points $\left(\frac{P}{\gamma}+z+\frac{V^2}{2g}\right)$
• Energy gradient line = $\frac{V^2}{2g}$

**114. Which of the following statements is correct in respect of steady is minor flow through a circular pipe ?**
(a) shear stress is zero at the centre
(b) Velocity is maximum at the centre
(c) Hydraulic gradient varies directly with the velocity
(d) All the above

**Ans. (d) :** Laminar flow through circular pipe–
(i) Shear stress (τ) is zero at the centre.

$$\tau=\left(-\frac{\delta p}{\delta x}\right).\frac{r}{2}$$

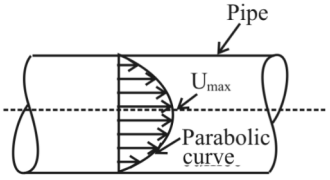



The maximum value of stress occur at r = R
(ii) Velocity is maximum at the centre,

$$v=\frac{1}{4\mu}\left(\frac{\delta p}{dx}\right)\left(R^2-r^2\right)$$

At centre r = 0 thus

$$u_{max}=\frac{1}{4\mu}\left(-\frac{\delta p}{\delta x}\right)R^2$$




(iii) Hydraulic gradient varies directly with the velocity.

$$\frac{fLv^2}{2gd} = h_L$$

$$\frac{h_L}{L} = \frac{fv^2}{2gd} \qquad \text{where, } \frac{h_L}{L} = i$$

$$i \propto v^2$$

**115. A jet of water issues from a 5 cm diameter nozzle, held vertically upwards, at a velocity of 20 m/s. If air resistance consumes 10% of the initial energy of the jet, find the height it would reach above the nozzle.**

(a) 18.35 m (b) 19.14 m
(c) 19.92 m (d) 20.00 m

**Ans. (a) :** Nozzle dia (d) = 5 cm
Initial velocity (u) = 20 m/s
Air resistance = 10%
Height (h) = ?
$V^2 = u^2 - 2gh$
V = 0, u = 20 m/s
$0 = (20)^2 - 2 \times 9.81 \times h$
h = 20.38 m
Actual height of air resistance is 10%
Then H = h × 90%

$$H = \frac{20.38 \times 90}{100} = 18.35 \text{ m}$$

**1375. For laminar flow is a pipe of circular cross section, the Darcy's a friction f is**

(a) indirectly proportional to Reynolds number and independent of pipe wall roughness
(b) directly proportional to pipe wall roughness and independent of Reynolds number
(c) inversely proportional to Reynolds numbers and directly proportional to pipe wall roughness
(d) inversely proportional to Reynolds number and directly proportional to pipe wall roughness

**Ans. (a) :** For laminar flow is a pipe of circular cross section, the Darcy's friction factor is inversely proportional to Reynolds numbers.

- The laminar flow in a pipe does not depend on the roughness of the pipe.

Friction factor is the only function of Reynolds number in laminar flow.

$$f = g(R_e)$$

- For turbulent flow $f = g(R_e, k)$

We know that,

$$f = \frac{64}{R_e}$$

So, $$f \propto \frac{1}{R_e}$$

And, $$h_f = \frac{f\ell v^2}{2gd}$$

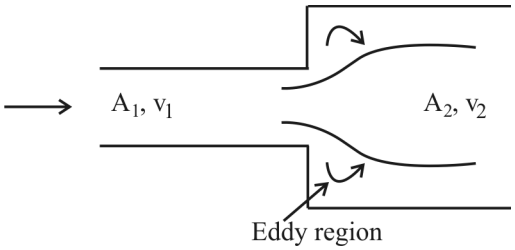
So, $$\boxed{f \propto h_f}$$

**117. The loss of energy due to sudden enlargement is given by**

(a) $\frac{v_2^2}{2g}\left(\frac{A_2}{A_1} - 1\right)^2$ (b) $\left(\frac{v_1 - v_2}{2g}\right)^2$

(c) $\frac{v_2^2 - v_1^2}{2g}$ (d) $\frac{v_1^2}{2g}\left(1 - \frac{A_2}{A_1}\right)^2$

**Ans. (a) :** The loss of energy due to sudden enlargement is given by $\frac{v_2^2}{2g}\left(\frac{A_2}{A_1} - 1\right)^2$.




We know that,
Loss of energy due to sudden enlargement,

$$h_{se} = \frac{(v_1 - v_2)^2}{2g}$$

$$h_{se} = \frac{v_2^2}{2g}\left(\frac{v_1}{v_2} - 1\right)^2$$

$$\left\{\begin{array}{l} \because A_1 v_1 = A_2 v_2 \\ \therefore \frac{v_1}{v_2} = \frac{A_2}{A_1} \end{array}\right\}$$

$$\therefore \quad h_{se} = \frac{v_2^2}{2g}\left(\frac{A_2}{A_1} - 1\right)^2$$

**118. If the speed of pressure wave is $v_0$ and pipe length is L, rapid closure occurs when time of closure is**

(a) less than $\frac{2L}{v_0}$ (b) greater than $\frac{2L}{v_0}$

(c) less than $\frac{L}{v_0}$ (d) zero

**Ans. (a) :** Water hammer is a phenomenon that can occur in any piping system where valves are used to control the flow of liquids or stream. Water hammer is the result of a pressure surge or high pressure shockwave that propagates through a piping system when a fluid in motion is force to change direction or stop abruptly.

- The valve closure is gradual if $\left[ t > \frac{2L}{v_0} \right]$
- The valve closure is sudden if $\left[ t < \frac{2L}{v_0} \right]$

Where- L = Length of pipe
T = Time in sec
$v_0$ = Velocity of pressure wave

**119. If the physical quantities involved in a fluid flow phenomenon are discharge Q, diameter D, acceleration due to gravity g, dynamic viscosity μ and mass density ρ, then the number of π parameters needed to express the function f (Q, D, g, μ,ρ) =0 are**

(a) 2 (b) 3 (c) 4 (d) 5

**Ans. (a) :**

| | |
|---|---|
| $Q = m^3/sec$ | dimension $[M^0L^3T^{-1}]$ |
| $d = m$ | dimension $[M^0LT^0]$ |
| $g = m/sec^2$ | dimension $[M^0L^1T^{-2}]$ |
| $\mu = kg/m\text{-}sec$ | dimension $[M^1L^{-1}T^{-1}]$ |
| $\rho = kg/m^3$ | dimension $[M^1L^{-3}T^0]$ |

$\int (Q, D, g, \mu, \rho) = 0$

Total number of variables (n) = 5
Number of fundamental dimensions [M,L,T] (m) = 3
Number of dimensionless π-terms = n-m = 5-3 = 2

**120. Which of the following quantities is dimensionless?**

(a) $\frac{\rho F}{\mu}$ (b) $\frac{\mu^2 \rho}{F}$

(c) $\frac{\rho F}{\mu^2}$ (d) $\frac{\mu}{\rho^2 F}$

**Ans. (c) :** Dimensional representation of

$\rho = ML^{-3}$

$\mu = ML^{-1}T^{-1}$

$F = MLT^{-2}$

From option (iii)

$$\frac{\rho F}{\mu^2} = \frac{ML^{-3} \times MLT^{-2}}{\left(ML^{-1}T^{-1}\right)^2}$$

$$\frac{M^2L^{-2}T^{-2}}{M^2L^{-2}T^{-2}} = 0 \text{ (dimensionless)}$$

**121. Two soils A and B are having a flow index of 9 and 6 respectively. Then which soil has better strength as a function of WC?**

(a) A (b) B
(c) Both A and B (d) None

**Ans. (b) :** Flow index $I_f = \frac{w_2 - w_1}{\log_{10}\left(\frac{N_2}{N_1}\right)}$

A soil, $I_f = 9$
B soil, If = 6
Strength, $A_{I_f} > B_{I_f}$

$\text{Strength} \propto \frac{1}{I_f}$

**122. In case of stratified soils, the permeabilities, $K_x$ and $k_2$, along and across stratification are related as**

(a) $K_x < K_2$ (b) $K_x = K_2$
(c) $K_x > K_2$ (d) None of the above

**Ans. (c) :** Permeabilitiy of stratified soil deposit.
Flow parallel to planes of stratification

$$K_x = \frac{K_1H_1 + ..... + K_nH_n}{H_1 + H_2 ..... H_n}$$

Flow normal to the plane of stratification

$$K_y = \frac{H_1 + ..... + H_n}{\frac{H_1}{K_1} + \frac{H_2}{K_2} + \frac{H_n}{K_n}}$$

Always $K_x > k_y$

**123. The hydrostatic pressure on the phreatic line with a dam section is**

(a) greater than atmospheric pressure
(b) less than atmospheric pressure
(c) equal to atmospheric pressure
(d) none of the above

**Ans. (c) :** Phreatic line is the top flow line which separate the saturated soil mass from unsaturated soil mass. The pressure on the phreatic line is atmospheric. Pressure below phreatic line is hydrostatic.

**124. The seepage velocity $v_s$ and the Darey's velocity are related as**

(a) $v = \frac{v_s}{n}$ (b) $v_s = \frac{v}{n}$

(c) $v_d = v.n$ (d) $v_s = \frac{n}{v}$

**Ans. (b) :** Seepage velocity and Darcy velocity are related as–

$$\boxed{V_S = \frac{V}{n}}$$

$V_S \rightarrow$ Seepage velocity/actual velocity
$V \rightarrow$ Darcy velocity/average velocity/discharge velocity
$n \rightarrow$ Porosity

**125. The unit of coefficient of consolidation is**

(a) $cm^2/cm$ (b) cm /sec
(c) $cm^2/sec$ (d) gram $cm^2/sec$

**Ans. (c) : Coefficient of consolidation ($C_v$)–**
The coefficient of consolidation required to estimate the amount of settlement due to consolidation of homogeneous saturated soil deposit. It is given by the equation.

$$C_v = \frac{K}{m_v \times \gamma_w}$$

Unit of $(C_v) = cm^2/sec$

**126. Secondary consolidation of soil is due to the expulsion of**

(a) absorbed water only (b) adsorbed water only
(c) both (1) and (2) (d) none of the above

**Ans. (b):** Secondary consolidation occurs at constant effective stress.

- Secondary consolidation occurs due to the readjustment of soil Skeleton after a disturbance during primary consolidation.
- Rate of secondary consolidation is controlled by the viscous adsorbed layer surrounding clay particle in soil.

**127. For the same soil, increase in compaction effort**

(a) does not affect OMC
(b) increases OMC
(c) decreases OMC
(d) decreases more density

**Ans. (c) :** For the same soil, increase in compaction effort decreases OMC.

$$\gamma_d = \frac{G\gamma_w}{1+e}$$

$$\gamma_d = \frac{\gamma_t}{1+w}$$

**128. For calculating the stress distribution in soils, Boussinesq assumed the point load to exist**

(a) below the ground level
(b) below water table
(c) at the ground level
(d) at water table

**Ans. (c) : Assumptions made in Boussinesq's theory–**
(i) Soil is homogeneous, semi-infinite, elastic and isotropic.
(ii) Self weight is neglected.
(iii) Soil is initially unstressed
(iv) Hook's law is valid
(v) Any change in volume of soil due to application of load is neglected.
(vi) Top surface of soil is free from shear stress and is subjected to only point load.

**129. Two soils A and B are tested in the laboratory for the consistency limits, The results are**

| | A | B |
|---|---|---|
| **Liquid limit** | **40%** | **60%** |
| **Plastic limit** | **20%** | **25%** |

**Which soil is more plastic ?**

(a) A
(b) B
(c) both A and B
(d) none of the above

**Ans. (b) :** For sample, A–
L.L = 40% P.L = 20%
Plasticity index ($I_P$) = L.L – P.L = 40 – 20
$I_P$ = 20%
For sample, B–
L.L = 60% P.L = 25%
Plasticity index ($I_P$) = L.L – P.L
= 60 – 25 = 35%
$I_P$ of B > $I_P$ of A.
So, sample B is more plastic.

**130. With reference to shrinkage and swelling, the most active clay mineral is**

(a) Kaolinite
(b) Montmorillonite
(c) Illite
(d) none of the above

**Ans. (b) :** With reference to shrinkage and swelling, the most active clay mineral is montmorillonite.

**Activity of minerals**

| | |
|---|---|
| Kaolinite | 0.38 |
| Illite | 0.90 |
| Montmorillonite | 7.2 |

**131. Due to raise in temperature, the unit weight and viscosity of the permeant fluid are reduced to 90% and 60% respectively. If other things remain constant, the coefficient of permeability.**

(a) increases by 33.3%
(b) decreases by 33.3%
(c) increases by 50%
(d) increases by 25%

**Ans. (c) :** According Kozeny Carman equation

$$k = \frac{1}{C}\times\frac{1}{S^2}.\frac{\gamma_w}{\mu}.\frac{e^3}{1+e}$$

$$k = \frac{\gamma_w}{\mu} \quad \text{..... (i)}$$

According to question,

$$k_2 = \frac{0.9\gamma_w}{0.6\mu} \quad \text{..... (ii)}$$

(ii) ÷ (i)

$$\frac{k_2}{k_1} = \frac{0.9\gamma_w}{0.6\mu} / \frac{\gamma_w}{\mu}$$

$$k_2 = 1.5k$$

% increased of permeability

$$= \frac{k_2 - k_1}{k_1} = \frac{1.5k_1 - k_1}{k_1}\times 100 = 50\%$$

Hence, the coefficient of permeability ($k_2$) increased by 50%.

**132. A clay deposit suffers a total settlement of 10 cm with one way drainage, and then with two way drainage, it suffers a total settlement of**

(a) 20 cm (b) 5 cm
(c) 10 cm (d) Nil

**Ans. (c) :** Total settlement does not depend on drainage path. The settlement due to double drain will quick compare to single drain but total settlement will be same.

**133. For better strength and stability, the coarse grained and fine grained soils are respectively compacted as**

(a) dry of OMC and wet. of OMC
(b) wet of OMC and wet. of OMC
(c) wet of OMC and dry of OMC
(d) dry of OMC and dry of OMC

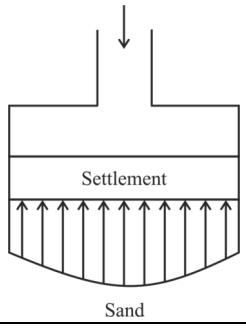
**Ans. (a) :**

- For better strength and stability the fine grained soils are compacted wet of OMC.
- For better strength and stability the coarse grained soils are compacted dry of OMC.
- Soils compacted dry of optimum have more swelling and more shear strength.
- Soils compacted wet of optimum have low swelling and low shear strength.

**134. The maximum contact pressure for a rigid footing on a cohesionless soil will be at**

(a) edges
(b) centre
(c) between centre and edge
(d) none of the above

**Ans. (b) :** The maximum contact pressure for a rigid footing on granular soil like sand will be at centre and minimum contact pressure will be at edges.




**135. Standard penetration test is suitable for**

(a) cohesionless soils (b) clayey soils
(c) silty soils (d) loamy soils

**Ans. (a) :** Standard penetration test is suitable for cohesionless soil only. In this the split spoon sampler is allowed to penetrate into the soil by applying impact load of 63.5 kg hammer having a free fall of 75 cm.

**136. The maximum value of $N_c$ for a strip footing according to Meyerhof's when $D_f$ /b ratio is greater than 2.5 is**

(a) 5.14 (b) 7.5
(c) 8.25 (d) 9.0

**Ans. (c) : Meyerhof's Nc value–**
Strip footings : $N_c = 5.5\ (1 + 0.25D_f/b)$
With a limiting value of 8.25 for $N_c$ for $D_f/b > 2.5$
Square or circular footing : $N_c = 6.2\ (1+0.32\ D_f/b)$
With Limiting value of 9 for $N_c$ for $D_f/b > 2.5$

**137. Width of a traffic lane is**

(a) 5.50 m (b) 3.75 m
(c) 7.50 m (d) 7.00 m

**Ans. (b) :** Width of a traffic lane is 3.75 m.

- The width of traffic lane is dependent upon the width and operational characteristics of vehicles, the speed, composition and volume of the traffic and the location of barrier curbs.

**138. One of the natural factors influencing camber is**

(a) type of material used for wearing course
(b) topography of the area
(c) nature of subsoil met with
(d) amount of rainfall

**Ans. (d) :** One of the natural forces influencing camber is amount of rainfall.

- The amount of camber to be provided depends upon the smoothness of the surface and the intensity of rainfall.

**139. While negotiating a curve**

(a) wheels of both axles tread the same path
(b) front wheels follow a less radius than the near wheels
(c) rear wheels follow a less radius than the front wheels
(d) depending on right had or left hand curve it varies

**Ans. (c) :** While negotiating a curve rear wheels follow a less radius than the front wheels.

**140. Find the compensated gradient at horizontal curve of radius 60 m with a ruling gradient of 6%**

(a) 1.52 % (b) 4.75 %
(c) 1.5 % (d) 5.0%

**Ans. (b) :** As per IRC,

$$\text{Grade compensation (GC)} = \left.\begin{matrix}\dfrac{30+R}{R} \\ \dfrac{75}{R}\end{matrix}\right\}\min$$

$$= \left.\begin{matrix}\dfrac{30+60}{60} \\ \dfrac{75}{60}\end{matrix}\right\} = \left.\begin{matrix}1.5 \\ 1.25\end{matrix}\right\} = 1.25$$

Compensated Gradient = 6% – 1.25% = 4.75%

**141. A pavement is classified as flexible pavement or rigid pavement based on its.**

(a) base course (b) wearing course
(c) sub- grade (d) sub- base

**Ans. (b) :** • Pavement are classified as flexible and rigid on the basis of wearing coarse.
**Flexible pavement–** In flexible pavement wearing course as binding surface of flexibility even in case of rigid pavement wearing course as slab which is considered as liquid plate very low flexibility, other lower layers are foundation material.
On the other hand, rigid pavement and flexible pavement are changeable only wearing course and other formation course.

**142. Cement stabilization is best suited for**

(a) sandy soils
(b) clays rich in sulphates
(c) clayey soils
(d) soils rich in lime

**Ans. (a) :** The cementing action is due to result of the chemical reaction of cement with siliceous soil and by binding action of the individual particles through cement.

- It is used for coarse grained soil.

**143. The best road suited to pneumatic and iron wheeled traffic is**

(a) WBM road
(b) earthen road
(c) bitumen bound macadam road
(d) cement concrete road

**Ans. (d) :** The best road suited to pneumatic and iron wheeled traffic is cement concrete road.

- Roads in India carry both pneumatic and iron wheel traffic. The design chart as presented are on the basis of pneumatic tyre traffic only.

**144. Maximum equivalent single wheel load as per LRC is**

(a) 8160 kg (b) 2040 kg
(c) 4080 kg (d) 1020 kg

**Ans. (c) : Design wheel load–**

| Type of load | Flexible pavement | Rigid pavement |
|---|---|---|
| Maximum legal axle load | 8170 kg | 10.2 t |
| Maximum equivalent single wheel load | 4080 kg | 5.1 t |
| Maximum tandem axle load | 14500 kg | 19 t |

**Maximum wheel load–**

- Total load influences the thickness requirements of pavements.
- Design of pavement is passed on $98^{th}$ percentile of axle load.

**145. The safe speed on a highway is**

(a) $50^{th}$ percentile speed
(b) $98^{th}$ percentile speed
(c) $85^{th}$ percentile speed
(d) $75^{th}$ percentile speed

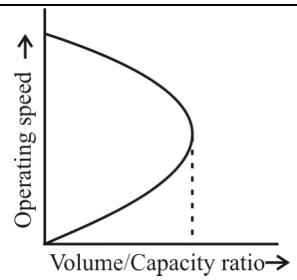
**Ans. (c) :** The safe speed on a highway is $85^{th}$ percentile speed (upper speed limit).
$15^{th}$ lower speed limit.
$98^{th}$ percentile design speed
$50^{th}$ percentile average speed

**146. The range of volume to capacity ratio theoretically**

(a) 0 to 1.0 (b) 0.5 to 1.0
(c) 0.5 to 1.5 (d) 0 to 1.5

**Ans. (a) : Volume to capacity ratio–**

- It is the measurement of the operating capacity of roadway or intersection where the number of vehicles passing through is divided by the number of vehicles that would theoretically pass through when at capacity.




- Range of volume to capacity ratio theoretically is 0 to 1.

**147. Grade separation has the advantage of**

(a) easy right turn
(b) no aped restriction
(c) number of points of conflicts are minimum
(d) occupying the least space

**Ans. (c) : Grade separation–** A grade separation is a crossing of a highway and a railway, two highway or a pedestrian walkway and a highway at different levels without interruption.
**Advantages–** It provides maximum facility to crossing traffic and avoid accidents while crossing.

- They provide increased safety for turning traffic.
- They help to increase speed limits.
- Reduces number of conflict to nil.

**Disadvantage–**They are space intensive i.e it takes a lot of space.

- They are costly to build.

**148. Cost of construction of a railway track is proportional to**

(a) $\sqrt{\text{Gauge}}$ (b) Gauge
(c) $\text{Gauge}^2$ (d) $\text{Gauge}^3$

**Ans. (b) : Cost of track–** The cost of railway track is directly proportional to the width of its gauge.

- Hence, if the fund available is not sufficient to construct a standard gauge, a metre gauge or narrow gauge is preferred rather than to have no railway at all.

**149. The side slope that is quite common in cutting is**

(a) 2H : 1V (b) 1.5 H : 1V
(c) 1 H : 1V (d) 1H : 1V

**Ans. (b) :** Stability of earthwork depends mainly on two factors namely cohesion and friction.
**Cohesion–** Achieved temporary stability by cohesion.
**Friction–** Permanent stability achieved by friction.

- The cutting slope is canal work is 1:1 and filling 1.5 :1.
- In common or over steep it can be taken 1.5 : 1 in cutting.

**150. The minimum depth of ballast cushion recommended for truck routes of Broad Gauge is**

(a) 150 mm (b) 250 mm
(c) 200 mm (d) 300 mm

**Ans. (c) :**

| Particulars | Broad gauge | Metre Gauge |
|---|---|---|
| Width of Ballast | 3.35 m | 2.25 m |
| Depth of ballast | 20 to 25 cm | 15 to 20 cm |
| Quantity of stone ballast per m length | 1.036 $m^3$ | 0.71 $m^3$ |

# Andhra Pradesh Public Service Commission
# Assistant Engineer Exam, 2013
## Solved Paper

**1. Triangle law of forces is derived from ____**

(a) Parallelogram law of forces
(b) Law of transmissibility of forces
(c) Newton's law of gravitation
(d) All the above

**Ans. (a) :** Derived triangle law of forces from fundamental law parallelogram law of forces. Triangle law of forces may be stated as 'If two forces acting on a body are represented are after another by the sides of a triangle, their resultant is represented by the closing side of the triangle taken from first point tot he last point.

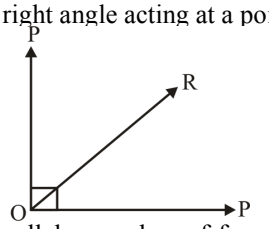
- The resultant force of given system of forces, may also be found out by the following laws.
- Triangle law of forces (for two forces)
- Polygon law of forces (for more forces)

**Polygon law of forces–** If a number of forces acting simultaneously on a body, represented in magnitude and direction, by the sides of a polygon taken in order, then the resultant of all these forces may be represented, in magnitude and direction by the closing side of the polygon, taken in opposite order.

**2. The resultant of two forces each equal to P/4 and acting at right angles is ____**

(a) $\frac{P}{2}$ (b) $\frac{P}{\sqrt{2}}$ (c) $\sqrt{2}P$ (d) 2P

**Ans. (*) :**
Two force 'P' at right angle acting at a point 'O'.




According to Parallelogram law of forces if two forces acting at a point are represented in magnitude and direction by the two sides of parallelogram, then the resultant of these two force is represented in magnitude and direction by the diagonal of parallelogram passing through the same point.

$$R = \sqrt{\left(\frac{p}{4}\right)^2 + \left(\frac{p}{4}\right)^2 + 2\frac{p}{4}\times\frac{p}{4}\cos 90^0}$$

$$= \sqrt{\frac{P^2}{16} + \frac{P^2}{16} + 0}$$

$$= \sqrt{2\frac{p^2}{16}}$$

$$= \sqrt{\frac{p^2}{8}}$$

$$R = \frac{P}{2\sqrt{2}}$$

**3. What is the torque transmitted by a solid shaft of diameter (D), when subjected to a shear stress (τ)?**

(a) $\frac{\pi}{16}\tau D^2$ (b) $\frac{\pi}{16}\tau D^3$
(c) $\frac{\pi}{32}\tau D^2$ (d) $\frac{\pi}{32}\tau D^3$

**Ans.(b):** Torque transmitted by a solid shaft of diameter (D), when subjected to a shear stress (τ) is equal to $\frac{\pi}{16}\times\tau\times D^3$.

Explanation-

$$\boxed{\frac{\tau}{r} = \frac{T}{I_p} = \frac{G\theta}{L}}$$

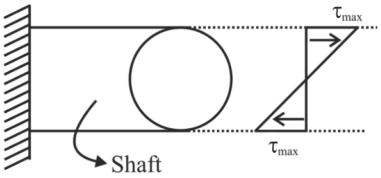
$$\frac{\tau}{r} = \frac{T}{I_p} \Rightarrow T = \frac{\tau}{r}\times I_p$$

$$T = \frac{\tau}{D/2}\times\frac{\pi D^4}{32} = \frac{\tau.\pi D^3}{16}$$

**4. When a solid shaft is subjected to torsion, the shear stress induced in the shaft at its centre is**

(a) Zero (b) Minimum
(c) Maximum (d) Average

**Ans. (a) :** The shear stress is zero at its centre in shaft under torsion is-




$$\tau_{max} = \frac{T}{z_p} \text{ or } \frac{G\theta r}{\ell}$$

$$\tau \propto r$$

$$\left[\frac{\tau_{max}}{R} = \frac{\tau}{r}\right] \qquad [\tau\downarrow \text{Then } r\downarrow]$$

**5. Strain energy stored in a hollow circular shaft of external diameter D and internal diameter d when subjected to a shearing stress τ is equal to**

(a) $\frac{\tau^2}{2C}\left[\frac{D^2-d^2}{D}\right]$ × Volume of shaft
(b) $\frac{\tau^2}{2C}\left[\frac{D^2+d^2}{D}\right]$ × Volume of shaft
(c) $\frac{\tau^2}{4C}\left[\frac{D^2-d^2}{D}\right]$ × Volume of shaft
(d) $\frac{\tau^2}{4C}\left[\frac{D^2+d^2}{D}\right]$ × Volume of shaft

**Ans. (d): For hollow shaft–**

Strain energy, $\boxed{U = \frac{\tau^2}{4C}\frac{D^2+d^2}{D^2} \times \text{Volume of shaft}}$

For solid circular shaft due to torsion strain energy is

$$\boxed{U = \frac{\tau^2}{4C} \times \text{Volume of shaft}}$$

Where,

$\tau$ – Shear stress due to torsion

C – Modulus of rigidity.

**6. Find the maximum shear stress induced in a solid circular shaft of diameter 200 mm when the shaft transmits 190 kW power at 200 rpm.**

(a) 4.67 N/mm$^2$ (b) 5.78 N/mm$^2$

(c) 5.14 N/mm$^2$ (d) 6.12 N/mm$^2$

**Ans. (b) :** $\frac{\tau}{r} = \frac{T}{J}$ $\qquad P = \frac{2\pi NT^3}{60}$

$\frac{\tau}{100} = \frac{T}{J}$ $\qquad 190\times10^3 = \frac{2\times\pi\times200\times T}{60}$

T = 9071.83 N-m

$$\tau = \frac{9071.83\times100}{\frac{\pi}{32}\times(200)^2} = 5.77 \text{ N/mm}^2 \approx 5.78 \text{ N/mm}^2$$

**7. Euler's buckling formula is applicable for columns**

(a) Subjected to eccentric loads

(b) Having initial curvature

(c) Initially straight and subjected to only axial loads

(d) All of the above

**Ans. (c) :** The Euler's buckling formula is applicable –

- When material is isotropic, homogenous & linear elastic in which hook's law is applicable.
- Before loading column is perfectly straight
- The load is perfectly axial.
- Euler formula holds good only for long columns.
- For short or long columns Rankine's formula is used.
- The plain c/s before the application of load remains plain even after the application.
- The failure of column is assumed to be due to buckling.

**8. A cantilever of length of span *l* caries a uniformly distributed load of w per unit length over its entire span. If its span is halved, then its slope will become**

(a) Half (b) One-fourth

(c) One-eight (d) One-sixteenth

**Ans. (c) :**

$$\delta = \frac{w\ell^3}{6EI} = \frac{w\left(\frac{\ell}{2}\right)^3}{6EI}$$

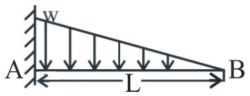
$$= \frac{w\ell^3}{8\times6EI} = \frac{1}{8}\left(\frac{w\ell^3}{6EI}\right)$$

**9. A cantilever beam of span is carrying a triangular load of zero intensity at its free end to w per unit length at its fixed end. The deflection at its free end will be**

(a) Wl$^4$/30EI (b) Wl$^4$/24EI

(c) Wl$^4$/16EI (d) Wl$^4$/12EI

**Ans. (a) :**




**slope at the free end ,** $\theta_B = \frac{wL^3}{24EI}$

**Maximum Deflection,** $\delta_B = \frac{wL^4}{30EI}$

**10. The shear force at any section of the conjugate beam is equal to the slope of the elastic curve at the corresponding section of the actual beam.**

(a) Mohr's theorem I

(b) Mohr's theorem II

(c) Conjugate beam theorem I

(d) Conjugate beam theorem II

**Ans. (c) :** The slope and deflection at any section of a beam by conjugate beam method is given by –

**Conjugate beam theorem I-**

The slope at any section of the given beam is equal to the shear force at the corresponding section of the conjugate beam.

**Conjugate beam theorem II-**

The deflection at any section, for the given beam is equal to the bending moment at corresponding section of the conjugate beam.

**11. If the actual beam has both ends fixed, then the ends of the conjugate beam will be**

(a) Fixed at both ends

(b) Free at both ends

(c) Fixed at one end, free at other end

(d) Hinged at one end, free at other end

**Ans. (b) :**

- A fixed end for the real beam become free end for the conjugate beam.
- If the actual beam has both ends fixed, then the ends of the conjugate beam will be free at both ends.
- The point of zero shear for conjugate beam corresponding to a point zero slope for real beam.
- The point of maximum moment for conjugate beam corresponding point of maximum deflection for real beam.

| S.N. | Real Beam | Conjugate Beam |
|---|---|---|
| (i) | Slope at any point in the real beam | Becomes shear force at that point in conjugate beam. |
| (ii) | Deflection at any point | Becomes moment at that point |
| (iii) | Fixed end | Becomes free end |
| (iv) | Free end | Becomes fixed end |
| (v) | Internal support | Becomes internal hinged |

**12. All the members in a planar truss are ____ force members.**

(a) Two (b) Three
(c) Four (d) None of these

**Ans. (a) :** A truss is a structure where member are pin connected at their ends, and loads act only at the joint, due to this configuration, all member is a truss are two force member meaning that they carry only tension or compression.

**13. Forces at the ends of the member act ___ member in a plane truss.**

(a) Perpendicular to
(b) Along the axis of
(c) Inclined to
(d) Inclined at 30º to

**Ans. (b) : As per assumption the analysis of truss–**
- Member of the truss are pin-connected to each other
- Loads act only at joint
- Member of the truss are not subjected to bending moment
- Member of the truss are rigid.
- Self weight of the members are neglible.
- All member are of uniform c/s

**14. All the loadings in plane truss are applied at __**

(a) Any point in the member
(b) Joints
(c) Mid points
(d) Base

**Ans. (b) : As per assumption the analysis of truss–**
- Member of the truss are pin-connected to each other
- Loads act only at joint
- Member of the truss are not subjected to bending moment
- Member of the truss are rigid.
- Self weight of the members are neglible.
- All member are of uniform c/s

**15. Relation between number of members (m) and joints (j) in a plane truss is**

(a) m-3 = 4(j-3) (b) m-2 = 4(j-2)
(c) m-3 = 2(j-3) (d) m-2 = 2(j-2)

**Ans. (c) :** For plane truss-

$m = 2j - 3$

$m-3 = 2j-3-3$

$m-3 = 2j-6$

$m-3 = 2[j-3]$

**16. An ideal flow of any fluid must fulfill the following:**

(a) Newton's law of viscosity
(b) Newton's law of motion
(c) Boundary layer theory
(d) Continuity equation

**Ans. (d) :** An ideal flow of liquid follows continuity equation, not Newton's law of viscosity.

Newtons law of viscosity

$\tau \propto \frac{du}{dy}$

Continuity equation, $a_1v_1 = a_2v_2$

An ideal flow must satisfy the following requirement.
- The flow in irrotational.
- The continuity equation should be satisfied.
- Newton's secured law of motion applies at each instant and at every point.
- These should be no separation between fluid and solid boundaries and the fluid should not penetrate the boundaries.

**17. The stress-strain relation of the Newtonian fluids is**

(a) Linear (b) Parabolic
(c) Hyperbolic (d) Involutic

**Ans. (a) :** For viscous fluid the stress is related to earth of strain and that relation. Depends on the nature of the fluid.

For Newtonian fluids the stress - strain relation is linear, while for non-Newtonian fluids a non-linear relation is most applicable.

Liquids which follow Newton's law of viscosity are known as Newtonian's fluids.

**Newton's law of viscosity–**

$$\boxed{\tau = \mu.\frac{du}{dy}}$$

$\tau$ = shear stress

$\mu$ = dynamic viscosity

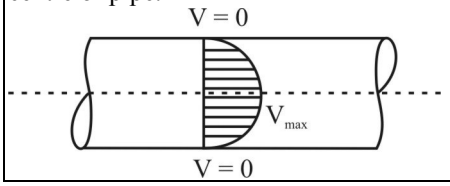
$\frac{du}{dy}$ = velocity gradient

- For Newtonian fluid, coefficient of viscosity remains constant.

**18. The velocity distribution in a laminar flow through a circular pipe follows as**

(a) Parabolic (b) Linear law
(c) Logarithmic law (d) None of these

**Ans. (a) :** Velocity distribution for laminar flow through a circular pipe varies parabolic variation.
- Mean velocity occurs at a distance of 0.707 R from centre of pipe.




**19. The units of kinematic viscosity are**

(a) $kg/m^2.sec$ (b) $kg.sec/m^2$
(c) $m/kg.sec$ (d) $m^2/sec$

**Ans. (d) :** In C.G.S. the units of kinematic viscosity is stokes.

In m.k.s the units of kinematic viscosity is = $\frac{m^2}{sec.}$

$1\frac{m^2}{sec.} = 10^4$ stokes

**20. Froude's number is defined as the ratio of**

(a) Inertia force to viscous force
(b) Inertia force to gravity force
(c) Inertia force to elastic force
(d) Inertia force to pressure

**Ans. (b) :** Froude's number = $\sqrt{\frac{\text{Inertia force}}{\text{gravity force}}}$

Froude number = $\frac{V}{\sqrt{gL}}$

F < 1 – sub critical flow
F = 1 – critical flow
F > 1 – super critical flow

**21. Differential manometers are used for measuring**

(a) Velocity at a point in a fluid
(b) Pressure at a point in a fluid
(c) Pressure difference between two points
(d) All of the above

**Ans. (c) :** A differential manometer is used to measure pressure difference between two points in a pipe.

• Differential manometers are used to measure the difference of pressures between two points in a pipe or in two different pipes. There are two type differential manometers.

1) U-tube differential manometer.
2) Inverted u-tube differential manometer.

**22. A rotameter is a device used to measure**

(a) Velocity of fluid in pipes
(b) Velocity of gauges
(c) Vortex flow
(d) Flow of fluids

**Ans. (d) : Rotameter –** A rotameter is a flow meter that measures volumetric flow of liquids and gases.

• The technique for measuring flow is accomplished by a freely moving float finding equilibrium in tapered tube.

• The flow rate is then read from either a scale next to the tube or a scale on the tube.

**23. Surface tension is a phenomenon due to**

(a) Cohesion only
(b) Viscous force
(c) Adhesion between liquid and solid molecules
(d) Difference in magnitude between the force due to adhesion and cohesion

**Ans. (a) :** Surface tension is defined as the tensile force acting on the surface of a liquid in contact with a gas or on surface between two immiscible liquids such that the contact surface behaves like a membrane under tension. Surface tension of a fluid is depend on cohesion only.

**24. A measure of the effect of compressibility in fluid flow is the magnitude of dimensionless parameter known as**

(a) Reynolds number (b) Mach number
(c) Weber number (d) Froude number

**Ans. (b) :** Mach number is defined as the square root of the inertia force of a fluid to the elastic force.

Mach number, $M = \sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}} = \frac{V}{\sqrt{K/\rho}} = \frac{V}{C}$

$\because \sqrt{K/\rho} = c$

When apart from inertial force, compressibility force are dominant, we use mach law. Compressibility force are pre-dominant when mach number $\geq$ 0.3 i.e speed of movement $\geq$ 0.3 × velocity of sound)
Mach number < 1 is called subsonic
Mach number > 1 is called super-sonic
Mach number = 1 is called sonic
Mach number > 6 hyper sonic

**25. For air flow at room temperature to be incompressible, the fluid velocity must not exceed**

(a) 100 m/sec (b) 70 m/sec
(c) 50 m/sec (d) 25 m/sec

**Ans. (a) :** The magnitude of the compressibility effect can be judged with the flow velocity.

• For air, when the flow velocity is 100 m/s or less. The air is treated as an in compressible fluid. (Mach no. < 0.3)

**26. Mercury is used in manometers on account of**

(a) Its high density
(b) Negligible capillarity effect
(c) Very low vapour pressure
(d) Its low compressibility

**Ans. (c) :** For large pressure difference, mercury is used in differential manometer.

**Mercury is used in manometer for–**

• It has high density.

• Mercury has low vapour pressure, so it does not evaporate easily.

• It is not freezing point is much lower than that of water.

Option (c) is most appropriate.

**27. Which of the following fluids has the highest compressibility?**

(a) Water
(b) Air at 1 atm. pressure
(c) A gas at 5 atm. pressure
(d) All of (a), (b) & (c)

**Ans. (c) :** The physical state of matter which can be easily compressed is the gaseous state. They are very easy to compress compared to solid and liquid because gas has more inter molecular attraction between the molecules.

**28. The pressure gradient in the horizontal direction in a static fluid is represented by**

(a) $\frac{\partial p}{\partial z} = -\gamma$ (b) $\frac{\partial p}{\partial z} = \gamma$

(c) $\frac{\partial p}{\partial x} = 0$ (d) $\frac{\partial p}{\partial z} = 0$

**Ans. (c) :** $\frac{\partial p}{\partial x} = 0$, In a fluid at a rest, the only external force acting is that of gravity, the presence gradient is zero along any horizontal line, that is planes normal to the direction of gravity.

**29. Flow of a fluid in a pipe takes place from**

(a) Higher level to lower level
(b) Higher pressure to lower pressure
(c) Higher energy to lower energy
(d) None of these

**Ans. (c) :** In case of turbulent flow of a fluid through a circular tube, as compared to the case of laminar flow, at the same flow rate the maximum velocity is lower, shear stress at the wall is higher and the pressure drop across a given length in higher.

| **Laminar flow** | **Turbulent flow** |
|---|---|
| Flow is smooth | Flow is randomly |
| Energy loss ∝ velocity | Energy loss∝(velocity high ) |
| Velocity is small | Velocity high |
| Re < 2000 | Re > 4000 |
| High viscosity | Low viscosity |

**30. The point through which the resultant hydrostatic force acts is called**

(a) Metacentre (b) Centre of pressure
(c) Centre of buoyancy (d) Centre of gravity

**Ans. (b) : Centre of pressure ($C_p$)–** The centre of pressure is the point where the total sum of a pressure field acts on a body. Causing a force to act through that point.

It is always below the centre of gravity / centroid.

Depth of centre of pressure, $\bar{h} = \frac{I_G \sin^2\theta}{A\bar{x}} + \bar{x}$

**31. An oil of specific gravity 0.7 and pressure 0.14 kgf/cm$^2$ will have the height of oil as**

(a) 70 cm of oil (b) 2m of oil
(c) 20 cm of oil (d) 80 cm of oil

**Ans. (b) :** 1 kgf/cm$^2$ = 9.81 ×10$^4$ N/m$^2$

0.14 kgf/cm$^2$

x = 0.14 × 9.81 × 104 N/m$^2$

P= ρgh

$h = \frac{0.14\times 9.81\times 10^4\ \text{N/m}^2}{0.7\times 10^3\times 9.81}$

h = 2 m of oil

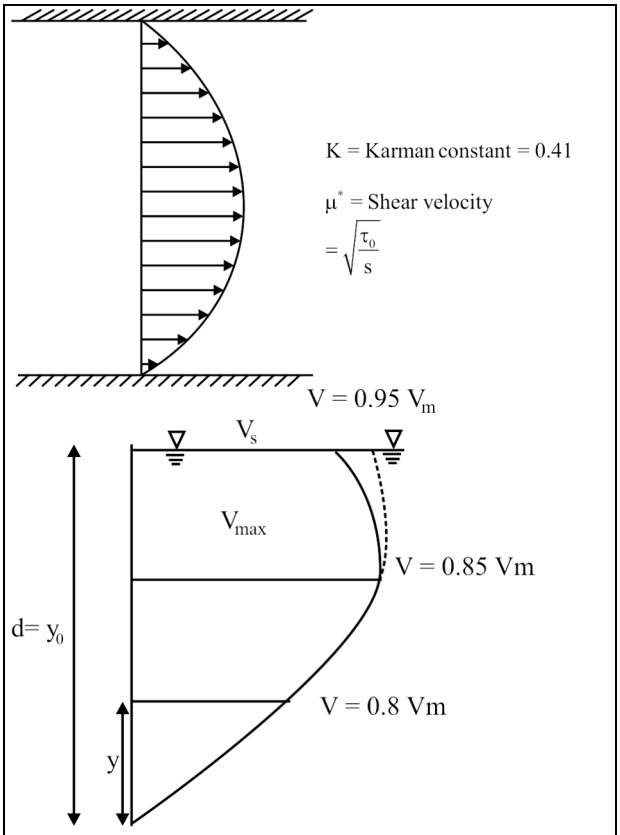
**32. The mean velocity in open channels can be estimated from the known velocity at the free surface. It is approximately equal to**

(a) 0.88 (b) 0.75
(c) 0.65 (d) 1.10

**Ans. (a) : Velocity distribution–** A typical velocity profile at a section in a plane normal to the direction of flow it observed can be described by logarithmic distribution or power law distribution .

Velocity is zero at the boundaries and gradually increases with increase in distance from the boundary.

- It has been observed that maximum velocity of flow occurs at a curtain distance below the free surface.




Average velocity is also related to surface velocity as

$V_{avg} = KVs$

K = 0.8 – 0.45 = 0.88

**33. The critical depth is the depth of flow at which**

(a) The specific energy is maximum
(b) The unit discharge 'q' is minimum
(c) The specific energy is minimum
(d) The Froude number is greater than unity

**Ans. (c) :** specific energy is minimum is called critical depth ($y_c$).

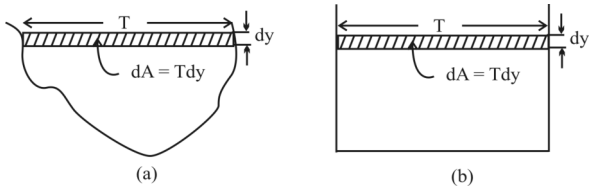
**Criteria for minimum specific energy–**

Value of the specific energy (E) for a given discharge is minimum for critical flow.

$$\because\ E = y + \frac{Q^2}{2gA^2}$$

Differentiating above equation with respect to y for critical flow;

$$\frac{dE}{dy} = 1 - \frac{Q^2}{gA^3}\times\frac{dA}{dy} = 0 \quad \ldots\ldots\ldots\ldots \text{(i)}$$




**Irregular and rectangular channel**

Since, $T = \frac{dA}{dy}$

Thus, equation (i) can be represented as

$$1-\frac{Q^2T}{gA^3}=0$$

To denote critical conditions use

$T_c$, $A_c$, $V_c$ and $y_c$

So, $\frac{Q^2T_c}{gA_c^3}=1;\frac{Q^2}{g}=\frac{A_c^3}{T}$ { Condition for critical flow / Or Minimum specific energy

Or,

$$\frac{v_c^2}{g\left(\frac{A_c}{T_c}\right)}=1 \qquad \text{..... (ii)}$$

Since, hydraulic depth (D) = $\frac{A}{T}$

Putting this value in above equation (ii)

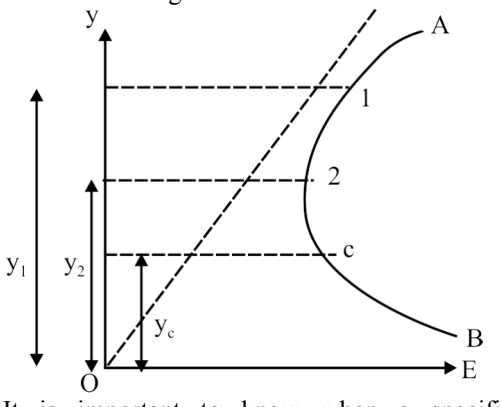
$$\frac{v^2}{gD_c}=1 \text{ or } \frac{v_c}{\sqrt{gD_c}}=1$$

Therefore, $F_r=\frac{v}{\sqrt{gD}}$

**34. In a subcritical flow, as the specific energy in a channel is decreased, the depth of flow**

(a) Also decrease
(b) Increase
(c) Does not vary
(d) Approaches to a minimum value which corresponds to the minimum specific energy

**Ans. (a/d) :** In the supercritical flow, when the specific energy decrease, the depth of flow above channel bottom has to decrease i.e. $y_2$ has to decrease and the free surface has to drop down at the hump. Along with the decease in depth $y_2$, their will be an increase on kinetic head. $\frac{V^2}{2g}$




It is important to know when a specific energy approaches its minimum value as well as the corresponding depth and velocity when the specific energy is at its minimum ($E_{min}$).

**35. In case of uniform flow in the channel, the water surface slope $\frac{dy}{dx}$ is equal to**

(a) 1 (b) ∞
(c) 0 (d) 1000

**Ans. (c):** In uniform flow with spatially constant Q, it must also be true that depth and width are constant, so "uniform flow" implies dy/dx = 0 and dw/dx = 0.1 and, since the depth does not change, "uniform flow" implies that the water-surface slope is identical to the channel slope.

**According to Newton's second law**

$$F_1=\int_0^H \rho g y dy=\frac{1}{2}\rho g h_1^2, F_2=\frac{\rho g h_2^2}{2}$$

$$F_1-F_2=\frac{\rho g h_1^2}{2}-\frac{\rho g h_2^2}{2}$$

$$\boxed{F_1-F_2=0}$$

**36. A control section is one where a definite relationship between depth of flow and discharge exists. A critical depth section is a control section. Which of the following cases represents control sections?**

**i. Hydraulic jump**
**ii. Flow under a sluice gate**
**iii. Flow over a small hump**
**iv. Overflow spillway**
**v. Change of slope from mild to steep**
**vi. Supercritical flow in channel**

(a) ii, iv, v (b) iv, v, vi
(c) v, iv, iii (d) None of these

**Ans. (a) :** Control sections–
Steady flow in open channels is generally uniform, provided that the relevant portion of the channel is far from particular sections. Where flow conditions are not connected with parameter of the open channel flow along the major portion of the channel. In those particular cross section, called control sections. The water depth is deformed either solely by the channel (Q) discharge by external conditions.

- In OCF flow control section is a section where for a given discharge the depth of flow is know or it can be controlled.
- Hydraulic jump, flow over a small hump and change of slope from mild to steep, is not given the relation between discharges the depth of flow.

**37. Mach number is a measure of compressibility effects in fluid flow. Indicate up to what mach number can a fluid flow be considered incompressible.**

(a) 1 (b) 0.1
(c) 0.3 (d) 0.8

**Ans. (c) :** The following are the assumptions made in the derivation of Bernoulli's equation :

- The flow is incompressible
- The fluid is ideal, i.e. viscosity is zero
- The flow is steady
- The flow is irrotational
- Streamline flow
- Incompressible flow density is taken as constant in derivation of the Bernoulli's equation. The flow is incompressible for liquids and also by gases at Mach number less than about 0.3.

**38. An isometric flow is one which is**

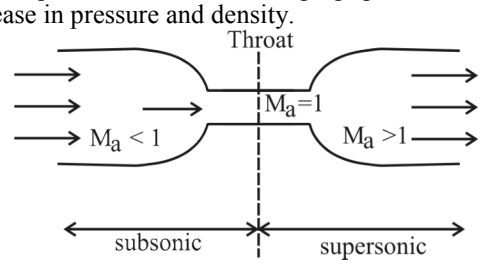
(a) Adiabatic
(b) Isothermal
(c) Isothermal and reversible
(d) Adiabatic and irreversible

**Ans. (c) :** An isometric flow is one is Isothermal and reversible.

**39. A shock wave which occurs in a supersonic flow represents a region in which**

(a) There is no change in pressure, temperature and density
(b) There is sudden change in pressure, temperature and density
(c) A zone of silence exists
(d) Velocity is zero

**Ans. (b) :** Sometimes, the oblique shock waves remains attached to the body. This occur when a super-sonic, flow is turned to itself through an angle, when the flow is forced to change direction, suddenly, as in a sharp concave corner of a passage, the flow takes turn abruptly, causing the formation of oblique shock.
In a supersonic flow a diverging passes result in decrease in pressure and density.




Mach number = $\sqrt{\frac{\text{inertia force}}{\text{elastic force}}}$ or $\left(\frac{v}{c}\right)$

**40. The sonic velocity in a fluid medium is directly proportional to**

(a) Square root of pressure
(b) Square root of temperature
(c) Density
(d) Mach number

**Ans. (a) : Sonic velocity is given by-**

(i) $C = \sqrt{\frac{k}{\rho}}$ in terms of bulk modulus for fluid medium.

**For gas medium–**

(ii) $C = \sqrt{RT}$ for isothermal process

(iii) $C = \sqrt{\gamma RT}$ for adiabatic process

**41. A fluid jet of cross-sectional area 'A' and velocity 'v' strikes a flat plate moving with a velocity 'u'. The fluid mass striking it for a second is**

(a) ρAv (b) ρA(v – u)
(c) ρA(u – v) (d) ρA(v + u)

**Ans. (b) :** Mass flow rate in moving = $\delta a(v-\mu) = \rho aVr$
Mass flow rate in static = ρav
Where,
v= Absolute velocity of jet of water at inlet/exit
u = Absolute velocity of vane / plate at inlet/exit
Vr = Relative velocity of jet of water at inlet/exit

**42. The dynamic force exerted by a fluid of cross section area 'A' and velocity 'v' on a moving flat plate held at right angles to the direction of jet is**

(a) ρAvu (b) ρA(v – u)u
(c) $\rho A(v-u)^2$ (d) ρA(v + u)u

**Ans. (c) :**
Assume
Vane surface smooth-

$$vr_2 = vr_1$$

$$\dot{m} = s_a vr_1$$

In x-direction-

$$F_x = \dot{m} v_1 - \left[\frac{\dot{m}}{2}(u_2) + \frac{\dot{m}}{2}(u_2)\right]$$

$$F_x = \dot{m}[v_1 - u_1]$$

$$= s_a (v_1 - u_1)(v_1 - u_1)$$

$$F_x = s_a (v_1 - u_1)^2$$

In y-direction

$$F_y = 0 - \left[\frac{\dot{m}}{2} vr_2 - \frac{\dot{m}}{2} vr_2\right]$$

$$F_y = 0$$

**43. Turbines give best performance (i.e. work at peak efficiency) when they are operated at full or design load. The performance of many turbines deteriorates considerably at part loads. Which of the following turbines is best suited for operation at part loads?**

(a) Pelton turbine (b) Francis turbine
(c) Propeller turbine (d) Kaplan turbine

**Ans. (d) : Partload operation–** In general the efficiency at part-loads in less than normal for the, sake, of economy the turbine, should always seen with the maximum possible efficiency to get more revenue.
When the turbine has to run at part or overload condition. Driaz turbine is employed .

**44. The power which appears in the expansion for the specific speed is**

(a) The water speed is
(b) The horse power developed by the runner
(c) The shaft horse power (brake horse power)
(d) The power input to the turbine

**Ans. (c) :** The power which appears in the expansion for the specific speed is the shaft horse power (brake horse power).

**45. The power obtainable from an impulse turbine is proportional to the number of nozzles used. A pelton turbine with six nozzles has a specific speed of 8.1. The specific speed per nozzle is**

(a) 1.35 (b) 3.3
(c) 2 (d) 8.1

**Ans. (b):** Specific speed –

$$N_{s_1} = \frac{N\sqrt{P}}{(H)^{5/4}} \text{ for single jet}$$

$$N_{s_2} = \frac{N\sqrt{nP}}{(H)^{5/4}} \text{ for multiple jet}$$

$$N_{s_2} = \sqrt{n}\,\frac{N\sqrt{P}}{(H)^{5/4}}$$

$$N_{s_2} = \sqrt{n}\,(N_{s_1})$$

$$8.1 = \sqrt{6}N_{s_1}$$

$$N_{s_1} = \frac{8.1}{\sqrt{6}} = 3.3$$

**46. The difference between the power obtained from the turbine shaft and the power supplied by water at its entry to the turbine is equal to the**

(a) Hydraulic losses
(b) Mechanical losses
(c) Hydraulic and mechanical losses
(d) Mechanical and volumetric losses

**Ans. (c) : Hydraulic losses–**

• Shock or eddy losses at the entrance to and exit from the impeller.

**Mechanical losses–**

- Losses due to disc friction between the impeller and the liquid which fills the clearance spaces between the impeller and casing.
- Losses pertaining to friction of the main bearing and glands.

**Leakage loss**–The loss of energy due to leakage of liquid is known as leakage loss.

**47. Cavitation, being related to occurrence of low pressure in flow, would definitely occur if**

(a) The pressure anywhere in the flow approaches the atmospheric value
(b) The pressure anywhere in the flow falls to a value very close to the vapour pressure of the liquid at the prevailing temperature
(c) The pressure is lower than the vapour pressure
(d) None of these

**Ans. (b) : Cavitation–**Formation of vapour bubbles within a liquid at low-pressure regions that occur in places where the liquid has been accelerated to high velocities, as in the operation of centrifugal pumps, water turbines, and marine propellers.

**48. The velocity with which the water approaches a notch is called**

(a) Velocity of flow
(b) Velocity of approach
(c) Velocity of whirl
(d) None of these

**Ans. (b) :** Velocity of approach is defined as the velocity with which the water approaches or reaches the weir or notch before it flows over it.

Thus if $V_a$ is the velocity of approach, then an additional head $h_a$ equal to $\frac{V_a^2}{2g}$ due to velocity of approach, is acting on the water flowing over the notch. Then initial height of water over the notch becomes $(H+h_a)$ and final height becomes equal to $h_a$. Then all the formulae are changed taking into consideration of velocity of approach.

• The velocity of approach, $V_a$ is determined by floating the discharge over the notch or weir neglecting velocity of approach. Then dividing the discharge by the cross-sectional area of the channel on the upstream side of the weir on notch, the velocity of approach is obtained.

$$V_a = \frac{Q}{\text{Area of channel}}$$

• This velocity of approach is used to find an additional head $h_a = \frac{V_a^2}{2g}$.

**49. Power transmitted through pipes will be maximum when load lost due to friction is equal to**

(a) $\frac{1}{2}$ total heat at the inlet of the pipe
(b) $\frac{1}{4}$ total heat at the inlet of the pipe
(c) total heat at the inlet of the pipe
(d) $\frac{1}{3}$ total heat at the inlet of the pipe

**Ans. (d) :** The power transmitted through the pipe is maximum, when the head lost due to friction in the pipe is equal to 1/3 of the total supply head.

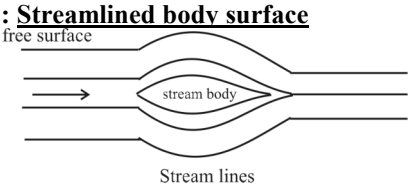
• The maximum efficiency of transmission through a pipe is 66.67%

where, $h_f$ = head lost due to friction
H = total supply head.

**50. A body is called a stream lined body when it is placed in a flow and the surface of the body**

(a) Coincides with the stream lines
(b) Does not coincide with the stream lines
(c) Is perpendicular to the stream lines
(d) Is inclined to the stream lines

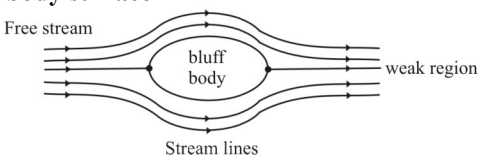
**Ans. (a) : <u>Streamlined body surface</u>**




**Streamlined body surface –**

Stream line body coincide with the streamline when placed in flow, the body is called a streamlined body

**Bluff body surface –**




Bluff body surface does not coincide with stream lines.

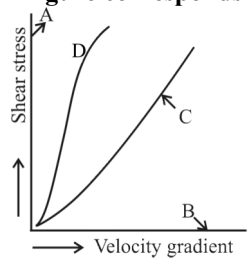
**51. The entrance length or length of establishment of flow is**

(a) The pipe length inside the reservoir
(b) The initial length in which the flow develops fully such that the velocity profile does not change downstream
(c) The length in which the boundary layer remains uniform
(d) The length of pipe from its entrance in which the flow may be assumed irrotational

**Ans. (b) :** The entrance length or length of establishment of flow is the initial length in which the flow develops fully such that the velocity profile does not change downstream.

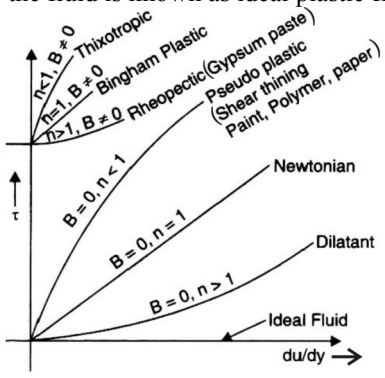
• When a fluid enters a tube, at the entrance of a tube, there is length where the flow is not yet established. The shape of the velocity profile evolves until it reaches the established profile (Hagen-Poiscessille profile for laminar flow), the evaluation of the profile is due to the progressive increase of the boundary layer at the wall, when the boundary layer merge at the centre axis, the flow is established.

**52. Curve-D in figure corresponds to**




(a) Ideal fluid
(b) Newtonian fluid
(c) Non-Newtonian fluid
(d) Non-ideal fluid

**Ans. (c) :** A fluid is which shear stress is more than the yield value and shear stress is proportional to the rate of shear strain, the fluid is known as ideal plastic fluid.




**Real fluid–** A real fluid is one which has viscosity, surface tension and compressibility in addition to the density.

The real fluids are actually available in nature.

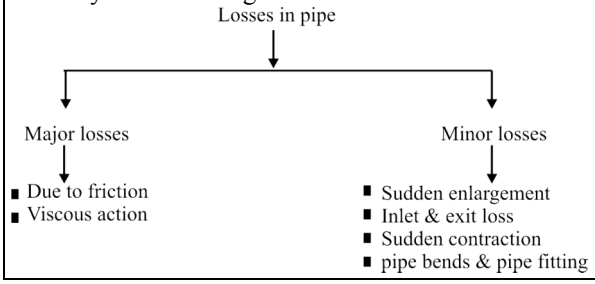
**Bingham or ideal plastic–** An ideal plastic (or bingham plastic) has a definite yield stress and a constant linear relation between shear stress and the rate of angular deformation.

e.g. Sewage sludge, drilling muds etc.

**53. The energy loss in pipe lines is due to**

(a) Viscous action
(b) Surface roughness only
(c) Friction offered by pipe walls as well as by the viscous action
(d) Turbulent shear stress alone

**Ans. (c) :** Pipe friction losses for circular cylindrical pipes depend on the flow velocity, pipes size, fluid viscosity and wall roughness.




**54. In flow through pipe bends, the pressure on inner and outer radii**

(a) Do not vary and are the same as at the centre of pipe
(b) Vary, it being more on the inner one
(c) Are different; pressure increase in radius and is; therefore, more at the outer radius
(d) Stand at the same level increasing gradually towards the pipe centre

**Ans. (c) :** In flow through pipe bends, the pressure on inner and outer radii are different; pressure increase in radius and is; therefore, more at the outer radius.

**55. Vorticity is given by ____ times the rotation.**

(a) 1 (b) 2
(c) 3 (d) 4

**Ans. (b) :**

Circulation $(\Gamma)$ = vorticity × Area under closed loop

so vorticity = $\frac{\text{circulation}}{\text{Area under closed loop}}$

In another way vorticity = 2 × angular velocity

**56. The condition for maximum transmission of power through a pipe line is that one-third of the available head must be consumed in friction. The corresponding efficiency of pipe line is**

(a) 66.67% (b) 33.3%
(c) 100% (d) $\sqrt{2}/33$

**Ans. (a) :** The power transmitted through the pipe is maximum, when the head lost due to friction in the pipe is equal to 1/3 of the total supply head.

• The maximum efficiency of transmission through a pipe is 66.67%

where, $h_f$ = head lost due to friction

H = total supply head.

**57. In order to have a continue flow through a siphon, no portion of the pipe should be higher than**

(a) 10m (b) 10.33m
(c) 5.5m (d) 7.75m

**Ans. (d) : Siphon–**

- Inverted siphon is appurtenant that is used to over come obstruction in sewer line due to road, canal and railway lines.
- The maximum vacuum at the summit of a siphon is 7.4 m of water.
- Provision of inverted siphon is made when a sewer line crosses a river.
- Siphon is a long bent which is used to carry water from a reservoir at higher elevation to another reservoir at a lower elevation.
- In order to have a continuous, flow through a siphon, no portion of pipe should be higher than 7.75 m above HGL.

**58. Water hammer is a phenomenon which is caused by**

(a) Sudden opening of a valve in a pipe line
(b) Sudden closing of a valve in a pipe line
(c) Incompressibility of fluid
(d) The pipe material being elastic

**Ans. (b) :** Water hammer is a phenomenon that can occur in any piping system where valves are used to control the flow of liquids or stream. Water hammer is the result of a pressure surge or high pressure shockwave that propagates through a piping system when a fluid in motion is force to change direction or stop abruptly.

- The valve closure is gradual if $\left[t > \frac{2L}{C}\right]$
- The valve closure is sudden if $\left[t < \frac{2L}{C}\right]$

Where- L = Length of pipe
T = Time in sec
C = Velocity of pressure wave

**59. A surge tank is a device connected to the penstock pipe line and installed as close to the powerhouse as the topography of the area permits. The main function of the surge tank is to**

(a) Restrict the water hammer effects to a small length of penstock
(b) Provide a free water surface near the turbines
(c) Act as a reservoir
(d) Protect the penstock pipe line from bursting

**Ans. (a) :** It the pumping system consists of a penstock under a very high pressure, a surge tank is usually required at the upstream of the power house to reduce the water hammer. Surge tank provides a free reservoir surface just upstream of the turbine in tank to absorb and compensate water hammer effect.

**60. The difference in pressure head, measured by a mercury water differential manometer for a 20 cm difference of mercury level will be**

(a) 2.72 m (b) 2.52 m
(c) 2.0 m (d) 0.2 m

**Ans. (b) :** Differential manometer reading = 0.20 m.

$0.20\left[13.6-1\right] = 0.20\times 12.6 = 2.52\,\text{m}$

**61. The hydrostatic pressure on a plane surface of area 'A' is given by**

(a) $0.5\gamma Ah_c$ (b) $\gamma Ah_c$
(c) $2\gamma Ah_c$ (d) $1.5\gamma Ah_c$

**where $h_c$ = depth of centroid of the area below the liquid surface.**

**Ans. (b) : Total pressure–** It is defined as the force exerted by a static fluid on a surface either plane or curved when the fluid comes in contact with the surface. This force always acts normal to the surface.

$\boxed{P = wA\bar{h}}$

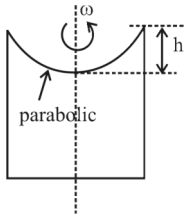
Where,
$\bar{h}$ = Centre of pressure
W = Specific weight of liquid ($\rho g$)
A = Area of surface
P = Total pressure

| Case | Force | Center of pressure (h) |
|---|---|---|
| Horizontal Surface | $wA\bar{x}$ | $h = \bar{x}$ |
| Vertical Surface | $wA\bar{x}$ | $h = \bar{x} + \frac{I_G}{A\bar{x}}$ |
| Inclined Surface | $wA\bar{x}$ | $h = \bar{x} + \frac{I_G}{A\bar{x}}\sin^2\theta$ |

**62. When a static liquid is subjected to uniform rotation in a container, the free surface assumes a shape of:**

(a) A cone
(b) A circular cylinder
(c) A paraboloid of revolution
(d) An ellipsoid of revolution

**Ans. (c) :** For a forced vertex flow, the height of paraboloid form is equal to $\frac{v^2}{2g}$




$h = \frac{v^2}{2g}$

$v = r\omega$

$h = \frac{\omega^2 r^2}{2g}$

Volume of parabolid $= \frac{1}{2}\left(\pi r^2\right)\left(\frac{\omega^2 R^2}{2g}\right)$

Note: for no spelling case, rise above original water level = fall below original water level.

**63. A circular plane 1 m in diameter is submerged vertically in water such that its upper edge is 8 m below the free surface of water. The total hydrostatic pressure force on one side of the plate is**

(a) 6.7 kN (b) 65.4 kN
(c) 45.0 kN (d) 77.0 kN

**Ans. (b) :**

$F = \delta g \bar{h}.A$

$\bar{h} = (y + D/2) \Rightarrow 8 + \frac{1}{2} = 8.5$

$A = \frac{\pi}{4} \times D^2 \Rightarrow \frac{\pi}{4} \times 1^2 = 0.785$

$F = 1000 \times 9.81 \times 8.5 \times 0.785$

F = 65.45 kN

**64. The flow in a river during the period of heavy rainfall is**

(a) Steady, uniform, two-dimensional
(b) Unsteady, uniform, three-dimensional
(c) Unsteady, non-uniform and three-dimensional
(d) Steady, non-uniform and three dimensional

**Ans. (c) :** The flow in a river during period of heavy rainfall is unsteady non-uniform and 3-D.

**65. A water supply pipe line changes its alignment through a bend. When the flow in the pipe line is increased by operating a valve, the flow in the bend is classified as**

(a) Unsteady, uniform flow
(b) Unsteady, non-uniform flow
(c) Steady, uniform flow
(d) Steady, non-uniform flow

**Ans. (b) :**
- Flow through a tapering pipe at either decreasing or increasing flow rate– unsteady non-uniform flow. Ex- flow through pipe bend.
- Flow through a tapering pipe at a constant flow rate – steady non-uniform flow.
- Flow through a long constant diameter pipe at either decreasing or increasing flow rate–unsteady-uniform flow.
- Free vortex flow is one in which the fluid mass rotates without any external impressed contact force. The whole fluid mass rotates either due to fluid pressure itself or the gravity or due to rotation previously imparted.

**66. The continuity equation in fluid mechanics is a mathematical statement embodying the principle of**

(a) Conservation of energy
(b) Conservation of mass
(c) Conservation of momentum
(d) None of these

**Ans. (b) :** Equation of continuity in case of steady incompressible flow–

$$\frac{\partial \rho}{\partial t} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Continuity equation based on the conservation of mass continuity equation = $\boxed{A_1V_1 = A_2V_2}$

**67. The concept of stream function which is based on the principle of continuity is applicable to**

(a) Three dimensional flow
(b) Two-dimensional flow only
(c) Uniform flow cases only
(d) Irrotational flow only

**Ans. (b) :** The concept of stream function is applicable for 2-D flow field & velocity potential function is applicable for 3-D flow field.

**68. The velocity potential function for a source varies with distance 'r' as**

(a) $\frac{1}{r}$ (b) $\frac{1}{r^2}$ (c) e' (d) ln.r

**Ans. (d) :** A source or sink represents a purely radial flow

So, $V_\theta = 0$

The corresponding velocity potential can be obtained by Integrating, the equations.

$\frac{\partial \phi}{\partial r} = \frac{m}{2\pi r}$ $\qquad \frac{1}{r}\frac{\partial \phi}{\partial \theta} = 0$

it follows that

$\phi = \frac{m}{2\pi} \ln r$

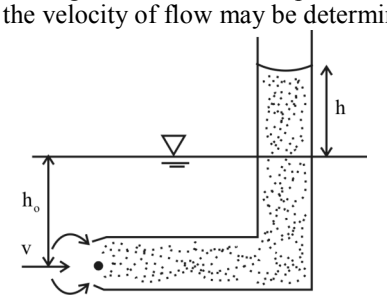
$\boxed{\phi \propto \ln r}$

- If m is positive, the flow is radially outward and the flow is considered to be a source flow.
- If m is negative, the flow is towards the origin and the flow is considered to be a sink flow.
- The flowrate, m is the strength of the source or sink.

**69. A stagnation point is a point**

(a) Where the pressure reduced to zero
(b) Where the total energy is zero
(c) Where the velocity of flow reduces to zero
(d) Where the total energy is maximum

**Ans. (c) : Stagnation point–** The velocity of flow at a particular point is reduced to zero, which is known as stagnation point, the pressure there is increased due to the conversion of kinetic energy into pressure energy, and by measuring the increase in the pressure energy at this point, the velocity of flow may be determined.




$V_{th} = \sqrt{2gh}$

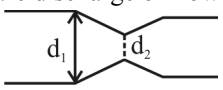
$V_{th} = \sqrt{2g\left(\frac{P_s - P_0}{\rho g}\right)}$

$h = \left(\frac{P_s - P_0}{\rho g}\right)$

**70. A venturimeter is a device based on the Bernoulli principle and is used for measuring**

(a) Piezometric head (b) Velocity head
(c) Flow rate (d) Total energy

**Ans. (c) : Venturimeter–** Venturimeter is a device used to measure discharge or flow rate in pipe.




$$d_2 = \frac{1}{2} \text{ to } \frac{1}{3}\ d_1$$

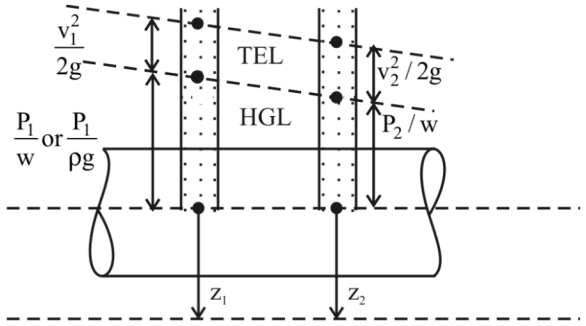
- Angle of conversion = $20^o$ to 22
- Angle of diversion = $5^o$ to $7^o$ (It should be not greater than $7^o$ to avoid flow separation)
- For venturimeter ($C_d$) = 0.94 to 0.98

$$Q = \frac{A_1 A_2 \sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}}$$

**71. The total energy line is always higher than the hydraulic grade line. The vertical distance between the two represents**

(a) The datum head
(b) The pressure head
(c) The velocity head
(d) The piezometric head

**Ans. (c) :** The total energy line is always higher than the hydraulic grade line, and the vertical distance between these represents the velocity head.




$\frac{P}{\rho g} + \frac{v^2}{2g} + z$ = Total energy head at various points.

$\frac{P}{\rho g} + z$ = Piezometirc head at various point.

Total energy line will be horizontal when the fluid is Ideal. Where as TEL will be sloping downward when the fluid is real.

**72. While applying the momentum equation the net force that acts over the control volume is required to be evaluated. Which of the following forces are generally considered in determining the net force?**

A. Gravity force
B. Viscous forces
C. Pressure force
D. Boundary resistance
E. Elastic force
F. Surface tension force

(a) A, B, C (b) A, E, F
(c) A, D, E (d) A, C, D

**Ans. (d) :** The momentum equation the net force that acts over the control volume is required to be evaluated is Gravity force , Pressure force, Boundary resistance.

**73. The boundary layer exists in**

(a) Flow of ideal fluids
(b) Flow of real fluids
(c) Only pipe flow
(d) Only flow over flat surfaces

**Ans. (b) :** The boundary layer concept is valid for infinitely large medium of real fluid and not for ideal fluid.

• When a real fluid flows past a solid surface, the fluid particle adheres to the boundary due to no-slip condition i.e. if boundary is stationary, fluid on it will be stationary and if boundary is moving, fluid attained to it moves with the same velocity.

**74. The existence of boundary layer is on account of**

(a) Fluid density (b) Gravitational effect
(c) Fluid viscosity (d) Surface tension

**Ans. (c) :** In a viscous fluid the velocity field must the no slip boundary condition at the solid surface confining the fluid one can satisfy the no slip. Boundary condition by adding viscous modes to the intentional wave. This leads tot he existence of boundary layer where viscous effect are confined. The concept of boundary layer widely used in classical hydrodynamics. It the width of the boundary layer is much less then the fluid dimension, the pressure of viscous modes (viscosity) can be taken into account by an effective boundary condition.

**75. Laminar flow through a circular tube was studied experimentally by**

(a) Newton
(b) Pascal
(c) Hagen and Poiseuille
(d) Prandtl

**Ans. (c) : Hagen-Poiseuille Flow–**

Shear stress $\tau = \frac{-\partial P}{\partial x} \cdot \frac{r}{2}$ (Linear distribution)

Velocity $u = \frac{-1}{4\mu}\left(\frac{\partial P}{\partial x}\right)[R^2 - r^2]$ (Parabolic distribution)

Where, $\frac{\partial P}{\partial x}$ = Pressure gradient

**76. The lower limit of the critical Reynolds number below which all disturbances (or source of turbulence) in pipe flow are damped out by viscous action has a value approximately equal to**

(a) 1 (b) 500
(c) 1000 (d) 2000

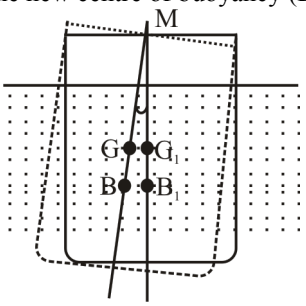
**Ans. (d) :** The lower limit of the critical Reynolds number below which all disturbances in pipe flow are damped out by viscous action has a value approximately equal to 2000.

- If Reynolds number increases the disturbance increases and due to instability disturbance is amplified.
- If there is lower $R_e$ value then viscous force damp the disturbance.

**77. The point, about which a floating body starts oscillating when the body is tilted, is called**

(a) Centre of pressure (b) Centre of buoyancy
(c) Centre of gravity (d) Metacentre

**Ans. (d) :** Metacentre may be defined as the intersection of the line passing through the original centre of buoyancy (B) and c.g. (G) of the body and the vertical line through the new centre of buoyancy ($B_1$)




**78. In a laminar flow through a circular tube, the Darcy-Weibach friction factor (f) depends only on the Reynolds number (R), and the two are related by**

(a) $f = \frac{1}{R}$ (b) $f = \frac{16}{R}$
(c) $f = \frac{0.316}{R^{1/4}}$ (d) $f = \frac{64}{R}$

**Ans. (d) :** Darcy formula–

$$\text{Frictional losses} = \frac{f.L.V^2}{2gD}$$

f = friction factor
f' = coefficient of friction

$$f' = \frac{f}{4}$$

If $(Re)_D$ <2000, laminar flow

$$\boxed{f = \frac{64}{(Re)_D}} \qquad \boxed{f' = \frac{16}{(Re)_D}}$$

$$f = \frac{0.3164}{(Re)^{1/4}} \text{ for turbulent flow} \qquad \because f = 4f'$$

**79. According to the Darcy's law is valid for velocities 'u' less than**

(a) 1 m/s (b) 10 cm/s
(c) 3 to 4.5 mm/s (d) 5 to 10 mm/s

**Ans. (c) :** According to the Darcy's law is valid for velocities 'u' less than 3 to 4.5 mm/s.

**80. The discharge in $m^3/s$ for laminar flow through a pipe of diameter 0.04 m having a cetnre line velocity of 1.5 m/s is**

(a) $3\pi/59$ (b) $3\pi/2500$
(c) $3\pi/5000$ (d) $3\pi/10000$

**Ans. (d) : :** $V_{max} = 2V_{avg}$

$V_{max} = 1.5$ m/s $\quad V_{avg} = \frac{1.5}{2}$ m/s

$$Q = A\bar{V} = AV_{avg}$$

$$Q = \frac{\pi}{4}d^2 \times V_{avg}$$

$$= \frac{\pi}{4}(0.04)^2 \times \frac{1.5}{2} = \frac{1.5\pi}{5000}$$

$$Q = \frac{3\pi}{10000} \text{ m}^3/\text{sec}$$

**81. Which of the following devices are usually used in measuring of pipe flow?**

(a) Mouthpiece (b) Cipolletti weir
(c) Venturimeter (d) Pitot tube

**Ans. (c) :**

| Device | Measurement |
|---|---|
| Venturimeter | rate of flow (discharge) |
| Flow nozzle | rate of flow |
| Orifice meter | rate of flow |
| Bend meter | rate of flow |
| Rotameter | rate of flow |
| Pitot tube | velocity |
| Hot wire anemometer | air and gas velocity |
| Current meter | velocity in open channels |
| Venturi-flume | rate of flow in open channels |

**82. Arrange the following flow measuring devices in the decreasing order of head loss caused by them.**

A. nozzle meter B. Venturimeter
C. orifice meter
(a) B, A, C (b) A, B, C
(c) C, A, B (d) C, B, A

**Ans. (c) :** coefficient of discharge ($C_d$)–
$C_d \rightarrow 0.95$ to 0.98 for venturimeter
$C_d \rightarrow 0.93$ to 0.98 for nozzle meter
$C_d \rightarrow 0.62$ to 0.65 for orifice meter
So decreasing order is C, A, B

**83. Arrange the hydraulic coefficients of orifice in increasing order of magnitude:**

A. $C_v$ B. $C_c$
C. $C_d$
(a) A, B, C (b) C, B, A
(c) B, C, A (d) C, A, B

**Ans. (b) :** Hydraulic coefficients of orifice meter–
$C_C$ (Coefficient of contraction) $\rightarrow 0.611$
$C_d$ (Coefficient of discharge velocity) $\rightarrow 0.64$ to 0.76
$C_V$ (Coefficient of velocity) $\rightarrow 0.98$

**84. The discharge through a sharp-crested rectangular weir is given by (with usual notations)**

(a) $C_dB\sqrt{2g}H$ (b) $\frac{2}{3}C_dB\sqrt{2g}H$
(c) $\frac{2}{3}C_dB\sqrt{2g}H^{3/2}$ (d) $C_dB\sqrt{2g}H^{3/2}$

**Ans. (c) :** $Q_{act} = \frac{2}{3} C_d B \sqrt{2g} H^{3/2}$

$C_d = 0.62$

H = depth of water above crest level.

**85. A mouthpiece and an orifice, both of the same diameter 'd' are discharging under the same head, H. The discharge through the mouthpiece will be**

(a) The same as that through the orifice
(b) More than discharge or orifice
(c) Less than that through the orifice
(d) None of these

**Ans. (b) :** The orifice is a small opening in the wall or base of a vessel through which the fluid flows and mouthpiece is an attachment in a form of a small tube or pipe fixed to the orifice. It's length is usually two and three times the diameter of orifice. It is used to increase the amount of discharge

**86. The head over a 90° V-notch weir increase from 0.15 to 0.30m. The ratio of the net discharge to the original discharge is**

(a) 1.414 (b) 2.00
(c) 4.000 (d) 5.657

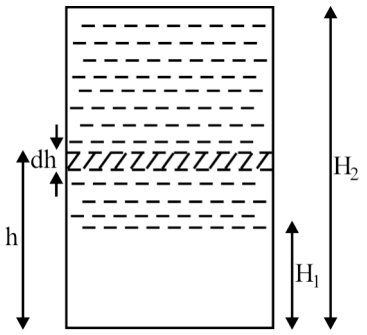
**Ans. (d) :** $\theta = \frac{8}{15} Cd\sqrt{2g} \tan\left(\frac{90}{2}\right) H^{5/2}$

$\frac{(0.3)^{5/2}}{(0.15)^{5/2}} = (2)^{5/2} = 4\sqrt{2} = 5.657$

**87. The time taken for a tank (filled to a height 'h' above its flat base) to empty through an orifice in the base varies as the flowing power of 'h':**

(a) 1 (b) $\frac{1}{2}$
(c) $-\frac{1}{2}$ (d) –1

**Ans. (b) :** Time of emptying a task through an orifice at its Bottom.




$$T = \frac{2A\left[\sqrt{H_1} - \sqrt{H_2}\right]}{C_d a\sqrt{2g}}$$

For empty the tank completely $H_2 = 0$

$$T = \frac{2A\sqrt{H_1}}{C_d . a . \sqrt{2g}}$$

**88. Uniform flow in open channels is characterised by**

(a) A change in depth of flow
(b) A constant discharge passing down the channel
(c) A constant depth of flow
(d) No variation of velocity at flow

**Ans. (c) :** Uniform flow in open channel are characterized by constant water depth (h) and constant mean velocity (V) along the flow direction (s).

$$\frac{\partial h}{\partial s} = 0, \ \frac{\partial v}{\partial s} = 0$$

Uniform flow can occur only in cylindrical river beds, with constant bed slope and constant discharge.

**89. The Chezy's constant 'c' which appears in the Chezy's equation for mean velocity in open channels**

(a) Is a dimensionless constant
(b) Has a constant value of different types of channels
(c) Had dimensions $L^{1/2}T^{-1}$
(d) None of these

**Ans. (c) :** The chezy equation can be used to calculate mean flow velocity in open channel.

Chezy's constant is given by

$$C = \sqrt{\frac{8g}{f}}, \ f = \text{friction factor}$$

It's nit is $\sqrt{\frac{m}{s}}$

So, the dimension of chezy coefficient $L^{1/2}T^{-1}$

**90. The maximum velocity in open channels occurs**

(a) At the mid depth
(b) At the free surface
(c) A little below the free surface
(d) Near the channel bottom

**Ans. (c) :** The maximum velocity in open channels occur a little below the free surface.

- Current meter is used for velocity measurement in open channels flow.

**91. The following data is related to a screw jack:**
**Pitch of the treaded screw = 8 mm**
**Diameter of the thread screw = 40 mm**
**Coefficient of the friction between screw and nut 0.1, Load = 20 kN**
**Assuming that the load rotates with the screw, determine the Efficiency**

(a) 0.683
(b) 0.386
(c) 0.836
(d) 0.638

**Ans. (b) :**

$$\eta = \frac{1-\sin\phi}{1+\sin\phi} = \frac{\tan\alpha}{\tan(\alpha+\phi)}$$

$$\tan\alpha = \frac{P}{\pi d} = \frac{8}{40\times\pi} = 0.06366$$

$\alpha = 3.6426^0$

$\phi = 5.71$

$\tan\phi = 0.1$

$\phi = 5.71$

$$\eta = \frac{\tan\alpha}{\tan(\alpha+\phi)} = \frac{0.06366}{\tan(5.71+3.6426)}$$

$\eta = 38.64\%$

**92. A load of 15 kN is raised by means of a screw jack. The mean diameter of the square threaded screw is 42 mm and the pitch is 10 mm. A force of 120 N is applied at the end of lever to raise the load. Determine the length of the load.**

(a) 218.7 (b) 318.7
(c) 418.7 (d) 518.7

**Ans. (d) :** W = 15 kN = 15000 N

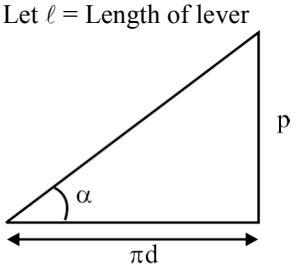
$d_m$ = 42 mm

p = 10 mm

$P_1$= 120 N

$\mu$ = 0.12

Let $\ell$ = Length of lever




We know that

$$\tan\alpha = \frac{p}{\pi d_m}$$

$$\tan\alpha = \frac{10}{\pi\times 4L}$$

$\alpha = 4.334^0$

$\tan\phi = \mu$

$\tan\phi = 0.12$

$\phi = 6.84^0$

Effort required at the circumference of the screw to raise the load is given by

$P = w \tan(\alpha+\phi)$

$P = 15 \times 10^3 \tan(4.334 + 6.843)$

$P = 15 \times 10^3 \times 0.1976$

P = 2.964 N

Torque required overcoming the friction–

$$T = P\times\frac{d_m}{2}$$

$$T = 2964\times\frac{42}{2}$$

T = 62244 N-mm

Torque applied at the end of the lever,

$T = P_1 \times \ell$

$T = 120 \times \ell$

Required torque and applied torque should be equal

$62244 = 120\times\ell$

$$\ell = \frac{62244}{120}$$

$\ell$ = 518.7 mm

**93. A load of 15 kN is raised by means of a screw jack. The mean diameter of the square threaded screw is 42 mm and the pitch is 10 mm. A force of 120 N is applied at the end of lever to raise the load. Determine the mechanical advantage**

(a) 25 (b) 125
(c) 625 (d) 0

**Ans. (b) :** $W = 15\,kN$, P = 120

$$M.A. = \frac{15000}{120} = 125$$

**94. A shaft runs at 80 rpm and drives another shaft at 150 rpm through the belt drive. The diameter of the driving pulley is 600 mm. Determine the diameter of the driven pulley taking the belt thickness as 5 mm.**

(a) 320 mm (b) 317.7 mm
(c) 304.8 mm (d) 304.9 mm

**Ans. (b) :** Given data,

$N_1$= 80 rpm, $N_2$= 150 rpm

$D_1$ = 600 mm,

t = 5 mm

$$\text{Velocity ratio (V.R.)} = \frac{N_2}{N_1} = \frac{D_1+t}{D_2+t}$$

$$= \frac{150}{80} = \frac{600+5}{D_2+5}$$

$D_2$ = 317.66 = 317.7 mm

**95. A shaft runs at 80 rpm and drives another shaft at 150 rpm through the belt drive. The diameter of the driving pulley is 600 mm. Determine the diameter of the driven pulley taking the belt thickness as 5 mm and a total slip of 4%.**

(a) 320 mm (b) 317.7 mm
(c) 304.8 mm (d) 304.9 mm

**Ans. (c) :** $N_1$ = 80 rpm , $N_2$ = 150 rpm, $D_1$ = 600 mm

Total slip = 4%

$$\text{Velocity ratio } \left(\frac{N_2}{N_1}\right) = \frac{D_1+t}{D_2+t}\times\left[1-\frac{S}{100}\right]$$

$$\frac{N_2}{N_1} = \frac{D_1+t}{D_2+t}\times\left[1-\frac{s}{100}\right]$$

$$\frac{150}{80} = \frac{600+5}{D_2+5}\times\left[1-\frac{4}{100}\right]$$

Diameter of driven pulley $D_2 = 304.76 \approx 304.8$ mm

**96. A shaft runs at 80 rpm and drives another shaft at 150 rpm through the belt drive. The diameter of the driving pulley is 600 mm. Determine the diameter of the driven pulley taking the belt thickness as 5 mm and a slip of 2% on each pulley.**

(a) 320 mm (b) 317.7 mm
(c) 304.8 mm (d) 304.9 mm

**Ans. (*) :** $N_1 = 80$ rpm $N_2 = 150$ rpm

$D_1 = 600$ mm, $t = 5$ mm, $D_2 = ?$

$S = 2\%$

$$\text{Velocity ratio} = \frac{N_2}{N_1} = \frac{D_1 + t}{D_2 + t} \times \left[1 - \frac{S}{100}\right]$$

$$\frac{150}{80} = \frac{600 + 5}{D_2 + 5} \times \left[1 - \frac{2}{100}\right]$$

$= 311.21 = 312.22$

**97. A bar of copper and steel form a composite system which is heated to a temperature of 45°C. The stress in the copper bar will be____**

(a) Tensile stress (b) Compressive stress
(c) Bursting stress (d) Shear stress

**Ans. (b) :** $\because \alpha_{cu} > \alpha_{steel}$

So, in composite system of both metal after heating them, the type of stress induced in copper is compressive and in steel is tensile.

**98. A cylindrical pipe of diameter 1.5m and thickness 1.5 cm is subjected to an internal fluid pressure of 1.2N/mm$^2$. The hoop stress developed in the pipe is ____**

(a) 30N/mm$^2$ (b) 60N/mm$^2$
(c) 90N/mm$^2$ (d) 15N/mm$^2$

**Ans. (b) :** $$\text{Hoop stress} = \frac{p.d}{2t}$$

$$= \frac{1.2 \times 1500}{2 \times 15} = 60 \text{ N/mm}^2$$

**99. A cylinder of internal diameter 2.5m and of thickness 5 cm contains a gas. If the tensile stress in the material is not to exceed 80 N/mm$^2$, the internal pressure of the gas should be____**

(a) 3.2N/mm$^2$ (b) 6.4N/mm$^2$
(c) 7.2N/mm$^2$ (d) 1.6N/mm$^2$

**Ans. (a) :** $$\text{Stress in gas} = \frac{p.d}{2t}$$

$$80 = \frac{p \times 2.5 \times 1000}{2 \times 5 \times 10}$$

$p = 3.2 \text{ N/mm}^2$

**100. A thin cylinder of internal diameter 1.25m contains a fluid at an internal pressure of 2N/mm$^2$. The longitudinal and circumferential stresses are not to exceed 30 N/mm$^2$ and circumferential stresses are not to exceed 30N/mm$^2$ and 45N/mm$^2$ respectively. The maximum thickness of the cylinder is ____**

(a) 2.08 cm (b) 2.22 cm
(c) 2.77 cm (d) 2.80 cm

**Ans. (c) :** $$\text{Hoop stress} = \frac{pd}{2t}$$

$$t > \frac{pd}{2 \times \text{hoop stress}} = \frac{2 \times 1.2}{2 \times 45}$$

$t > 2.66 \text{ cm} \approx 2.77$

$$\text{Longitudinal stress} = \frac{Pd}{4t}$$

$$t > \frac{pd}{4 \times \text{longitudinal stress}} = \frac{2 \times 1.2}{4 \times 30}$$

$t > 2$ cm

Hence adopt maximum value so the answer is 2.66 = 2.77 cm.

**101. A water main 80 cm diameter contains water at a pressure head of 100m. If the weight density of water is 9810 M/m$^3$ and the permissible stress is 20N/mm$^2$, then the thickness of the metal required for the water main is _____**

(a) 0.5 cm (b) 1 cm
(c) 2 cm (d) 4 cm

**Ans. (c) :** Local pressure in a pipe hoop strut is applicable

$$\sigma_h = -\frac{pD}{2t}$$

$$20 = \frac{0.981 \times 800}{2 \times t}$$

$$t = \frac{0.981 \times 800}{2 \times 20} = 19.62 \text{ mm} = 1.962 \approx 2 \text{ cm}$$

**102. A cylinder of thickness 1.5 cm has to withstand maximum internal pressure of 1.5 N/mm$^2$. If the ultimate tensile stress in the material of the cylinder is 300 N/mm$^2$, factor of safety 3.0 and joint efficiency 80%, the diameter of the cylinder is ____**

(a) 160 m (b) 160 cm
(c) 160 mm (d) 160 km

**Ans. (b) :**

$$\frac{\text{Tensile stress}}{\text{F.O.S.}} = \frac{pd}{2t \times \eta}$$

$$\frac{300}{3} = \frac{1.5 \times d}{2 \times 1.5 \times 10 \times 0.8}$$

$d = 1600$ mm $= 160$ cm

**103. A hollow cylindrical drum 600 mm in diameter and 3m long, has a shell thickness of 10mm. If the drum is subjected to an internal air pressure of 3 N/mm$^2$, the increase in its volume is (Take E = 2 × 10$^5$ N/mm$^2$ and Poisson's ratio = 0.3 for the material)**

(a) 792623000 mm$^3$
(b) 692723000 mm$^3$
(c) 962723000 mm$^3$
(d) 862793000 mm$^3$

**Ans. (*) :** $\varepsilon_v = \frac{pd}{4tE}(5-4\mu)$

$\Delta v = \frac{pd}{4tE}(5-4\mu) \times \text{area} \times \text{length}$

$= \frac{3\times 600}{4\times 10\times 2\times 10^5}\left[5-(4\times 0.3)\right]\times \frac{\pi}{4}\times 600^2 \times 3000$

$\Delta V = 724869 \text{ mm}^3$

**104. A steel plate of width 120 mm and of thickness 20 mm is bent into a circular of radius 10m. The maximum stress induced is ____ (Take E = $2 \times 10^5$ N/mm$^2$)**

(a) 100N/mm$^2$ (b) 200N/mm$^2$
(c) 400N/mm$^2$ (d) 600N/mm$^2$

**Ans. (b) :** $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$

Equation of pure bending

Bending stress, $\sigma = \frac{E}{R}.y$

$\sigma = \frac{2\times 10^5}{10000}\times \frac{20}{2}$

$\sigma_{max} = 200 \text{ N/mm}^2$

**105. A steel place of width 120 mm and thickness 20 mm is bent into a circular arc of radius 10m. The bending moment which will produce the maximum stress is given by ____ (Take E = $2\times 10^5$ N/mm$^2$)**

(a) 1.0 kN-m (b) 1.6 kN-m
(c) 1.2 kN-m (d) 1.4 kN-m

**Ans. (b) :** Equation of pure bending

$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R} \Rightarrow M = \frac{E}{R}\times I$

$M = \frac{2\times 10^5}{10000}\times \frac{120\times 20^3}{12}$

$M = 16 \times 10^5$ N-mm

$M = \frac{16\times 10^5}{10^3\times 10^3}\text{kN-m}$

M = 1.6 kN-m

**106. A rectangular beam 200 mm deep and 300 mm wide is simply supported over a span of 8m. What uniformly distributed load per metre may the beam carry, if the bending stress is not to exceed 120 N/mm$^2$?**

(a) 3 kN-m (b) 30 kN-m
(c) 300 kN-m (d) 0.3 kN-m

**Ans. (b) :** $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$

$I = \frac{bd^3}{12}$, $y_{max} = \frac{d}{2}$ $\quad M = \frac{w\ell^2}{8}$

$z = \frac{I}{y_{max}} = \frac{bd^2}{6}$ $\quad M = \frac{w\times(8)^2}{8}$

$M = 8\,w$

$z = \frac{300\times 200^2}{6}$ $\quad M = 8000$ w-Nmm

$z = 2 \times 10^6$ mm$^4$

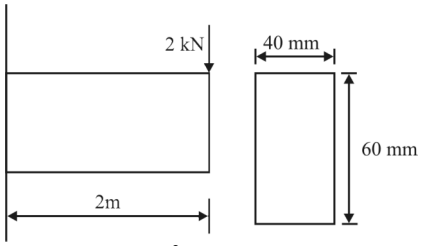
$M = \sigma_{max}.z$

$8000\,w = 2\times 10^6 \times 120$

w= 30000 N/m

w = 30 kN/m

**107. A cantilever of length 2 m fails when a load of 2 kN is applied at the free end. If the section of the beam is 40mm × 60 mm, the stress at failure will be**




(a) 1666.7 N/mm$^2$ (b) 166.67 N/mm$^2$
(c) 16.667 N/mm$^2$ (d) 1.6667 N/mm$^2$

**Ans. (b) :** $I = \frac{bd^3}{12} = \frac{40\times 60^3}{12} = 7.2\times 10^5 \text{ mm}^4$

M = W L = 2 ×2 = 4 kN-m

$\frac{M}{I} = \frac{\sigma}{y}$

$\sigma_{max} = \frac{M}{I}y$

$= \frac{4\times 10^6 \text{ N-mm}}{7.2\times 10^5}\times \frac{60}{2} = 166.67 \text{ N/mm}^2$

**108. The maximum stress induced in a cast iron pipe of external diameter 40 mm, internal diameter 40 mm, internal diameter 20 mm and of length 4m when the pipe is supported at its ends and carries a point load of 80N at its centre, is**

(a) 1.358 N/mm$^2$ (b) 13.58 N/mm$^2$
(c) 135.8 N/mm$^2$ (d) 1358 N/mm$^2$

**Ans. (b) :** $D_0$ = 40 mm

$D_i$ = 20 mm

L = 4000 mm

W = 80 N

$M_{max} = \frac{WL}{4} = \frac{80\times 4000}{4}$

$= 8 \times 10^4$ N-mm

$I = \frac{\pi}{64}\left(D^4 - d^4\right)$

$= \frac{\pi}{64}\left(40^4 - 20^4\right)$

$= 117809.7 \text{ mm}^4$

$\sigma_{max} = \frac{M}{I}\times y_{max}$

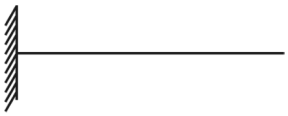
$= \frac{8\times 10^4 \times \frac{40}{2}}{117809.7}$

$\sigma_{max} = 13.58 \text{ N/mm}^2$

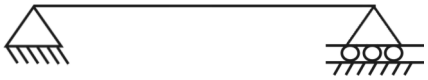
**109. A beam which is fixed at one end and free at the other end is known as**

(a) Cantilever beam
(b) Simply supported beam
(c) Overhanging beam
(d) Roller beam

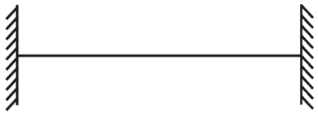
**Ans. (a) : Cantilever beam :** Cantilever beam is type of beam which is fixed at one end and free at the other end.



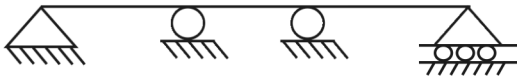
**Simply supported beam :** A type of beam which is supported at both ends is called as simply supported beam. It is the one of the simplest structural elements in existence.



**Fixed beam :** The type of beam which is fixed at both ends is called as fixed beam.



**Continuous beam :** A beam having more than two supports is known as continuous beam.



**110. Thin cylinders are frequently required to operate under pressure upto**

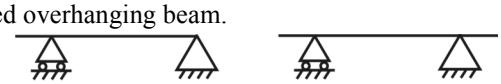
(a) 5 MN/m$^2$ (b) 15 MN/$^2$
(c) 30 MN/m$^2$ (d) 250 MN/m$^2$

**Ans. (c) :** In this cylinder, the stress may be assumed uniformly distributed over the all thickness boiler, tanks, steam pipes, water pipes are usually considered as thin cylinder. Thin cylinder or frequently required to operate under pressure upto 30 MN/m$^2$ or more, for high pressure such as 250 MN/m$^2$ or more, thick cylinder are used

**111. If the end portion of a beam is extended beyond the support, such a beam is known as ___**

(a) Cantilever beam
(b) Simply supported beam
(c) Overhanging beam
(d) Roller beam

**Ans. (c) :** A beam extending beyond the support is called overhanging beam.



(One side overhang) (Both side overhang)

**112. In a case of a circular section, the maximum shear stress is ____ percent more than the mean shear stress**

(a) 10 (b) 20
(c) 33.33 (d) 66.66

**Ans. (c) :** In case of circular section

$$\tau_{max} = \frac{4}{3}\tau_{mean}$$

$$\% = \frac{\frac{4}{3}\tau_{mean} - \tau_{mean}}{\tau_{mean}} \times 100$$

$$= \frac{\frac{4}{3} - 1}{1} \times 100$$

$$= \frac{1}{3} \times 100 = 33.33\%$$

**113. The shear force ____ at a section where there is a vertical point load.**

(a) Remains constant (b) Change gradually
(c) Changes suddenly (d) Is zero

**Ans. (c) :** S.F. at a section is the resultant vertical force to the right or left of the section.

- The S.F. changes suddenly at a section where there is a vertical point load.
- The S.F. between way two vertical loads remains constant.

**114. The shear force between any two vertical loads**

(a) Remains constant (b) Change gradually
(c) Changes suddenly (d) Is zero

**Ans. (a) :**
- The S.F. between way two vertical loads remains constant.
- S.F. at a section is the resultant vertical force to the right or left of the section.
- The S.F. changes suddenly at a section where there is a vertical point load.

**115. A square section with side 'x' of beam is subjected to a shear force 'S'. The magnitude of shear stress at the top edge of the square is**

(a) $\frac{1.5S}{x^2}$ (b) $\frac{S}{x^2}$
(c) $\frac{0.5S}{x^2}$ (d) zero

**Ans. (d) :** $\tau = \text{shear stress} = \frac{5A\bar{y}}{I.b}$

$A\bar{y}$ = is the moment of the sectional area above that point above the NA

So, in this question, this is no area above the top edge of the square so $A\bar{y}$ is zero i.e.

$\boxed{\tau = 0}$

**116. The diameter of kernel of a circular section is**

(a) $\frac{d}{2}$ (b) $\frac{d}{3}$

(c) $\frac{d}{4}$ (d) $\frac{d}{\sqrt{2}}$

**Ans. (c) :** $\sigma_{max} = 0 = \frac{P}{A} + \frac{M_x}{I_x}(-y)$

$$0 = \frac{P}{\frac{\pi}{4}d^2} + \frac{P_e}{\frac{A}{64}d^4}\left(-\frac{d}{2}\right)$$

$$e_y = \frac{d}{8}$$

i.e. for a maximum eccentricity of d/8 will be no tensile stress in the section.

So the diameter of the core will be

$$= \frac{d}{8} + \frac{d}{8} = \frac{d}{4}$$

**117. Bending moment is ____ at a section where shear force is zero after changing its sign.**

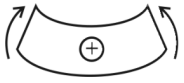
(a) Maximum
(b) Minimum
(c) Zero
(d) Either maximum or minimum

**Ans. (d) :**

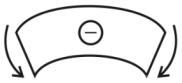
- A point where shear force diagram changes sign bending moment will be maximum or minimum compare to neighbourhood.
- **Bending moment :**
- Bending moment at any section is the internal reaction due to all the transverse forces either from left side or from right side of that section.
- It is equal to algebraic sum of moments at that section either from left or from right side of that section.

Bending moment is different from twisting moment.

Its sign conventions is–



Sagging



Hogging

**118. A cantilever of length *l* is carrying a uniformly distributed load of 'w' per unit run for a distance of 'b' from fixed end. The slope of the free end is given as**

(a) $\frac{wb^3}{6EI}$ (b) $\frac{wb^3}{8EI}$

(c) $\frac{wb^3}{12EI}$ (d) $\frac{wb^3}{24EI}$

**Ans. (a) :** A cantilever of length *l* is carrying a uniformly distributed load of 'w' per unit run for a distance of 'b' from fixed end. The slope of the free end is given as $\frac{wb^3}{6EI}$.

**119. When a beam is subjected to a couple at a section, then bending moment ____ at the section.**

(a) Changes suddenly
(b) Changes gradually
(c) Remains unaltered
(d) Is zero

**Ans. (a) : Beams subjected to coupled –** Sometimes, if a beam is subjected to a couple (may be clockwise or counterclockwise) at a section, then the magnitude of the couple must be considered while calculating the support reactions. While calculating the shear force, one need not have to consider the couple (couple is nothing but two unlike parallel non-collinear force acting at the section.) While calculating the bending moment at that section, there will be jump or drop in the magnitude of the moment to the tune of the magnitude of the couple at that section according to the sense of the couple.

**120. The radius of gyration of a solid sphere of radius 'r' is equal to ____**

(a) 0.8 r (b) 0.5 r
(c) 0.4 r (d) 10.3 r

**Ans. (*) : Radius of gyration of a solid sphere–**

Moment of inertia for a solid sphere having radius R and mass M is

$$I = \frac{2}{3}MR^2$$

If K is radius of gyration of the solid sphere, then

$$I = MK^2$$

Also, $K = \sqrt{\frac{I}{M}} = \sqrt{\frac{2MR^2}{5M}} = \sqrt{\frac{2}{5}}R = 0.63\ R$

**121. The ratio of shear modulus to the elastic modulus for a Poisson's ratio of 0.4 will be**

(a) 5/6 (b) 6/5
(c) 3/5 (d) 1/2

**Ans. (*) :** $\frac{G}{E} = ?, \mu = 0.4$

$E = 2G(1+\mu)$

$$\frac{E}{G} = 2(1+\mu)$$

$$\frac{E}{G} = 2(1+0.4)$$

$$\frac{E}{G} = 2.8 = \frac{28}{10}$$

$$\frac{G}{E} = \frac{5}{14}$$

**122. The maximum strain energy stored in a body without permanent deformation is**

(a) Resilience
(b) Proof Resilience
(c) Modulus of Resilience
(d) Impact Resilience

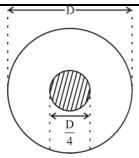
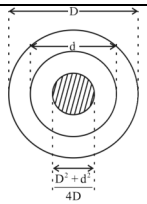
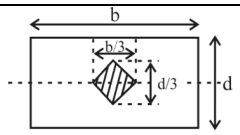
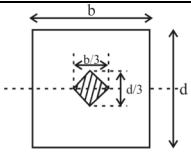
**Ans. (b) : Proof resilience** the maximum energy stored in the body without permanent deformation( ie elastic limit) is known as proof resilience.

$$\text{Proof Resilience} = \frac{p^2}{2E} \times \text{Volume}$$

**123. The shape of the kern area of a rectangular section is**

(a) Circle (b) Square
(c) Rectangle (d) Parallelogram

**Ans. (d)**

| Section | Dimension of core/kern | Shape of core or kern |
|---|---|---|
|  Solid circular | dia of kern = D/4 | Circular |
|  Hollow circular | dia of kern = $\frac{D^2+d^2}{4D}$ | Circular |
|  Rectangle | b/3 × d/3 | Rhombus (Parallogram ) |
|  Square | d/3 × d/3 (b = d) | Square |

**124. A train starts from rest on a curved track of radius 800 m. Its speed increases uniformly and after 3 minutes it is 72 km/hr. The tangential acceleration after 2 minutes would be**

(a) $\frac{1}{9}\text{m/sec}^2$ (b) $\frac{2}{9}\text{m/sec}^2$
(c) $\frac{1}{2}\text{m/sec}^2$ (d) $\frac{1}{3}\text{m/sec}^2$

**Ans. (*) :** The linear of equation of motion

$v_f = v_i + at$

$v_f$ = 72 km/hr = 20 m/s

20 m/s = 0 + a (120)

$a = \frac{1}{6}\ \text{m/s}^2$

**125. The maximum stress induced in a body if the load 'P' is applied suddenly upon an area of cross section 'A' is**

(a) P/A (b) 2P/A
(c) 3 P/A (d) 4 P/A

**Ans. (b) :** When the load is applied all of sudden and not stepwise is called suddenly applied load. Now let the load W is applied all of sudden and maximum stress thus produced by $\sigma_{su}$; the extension being the same as $\delta\ell$.

Then, Energy stored = External work done

$$W \times \delta\ell = \frac{1}{2}\sigma_{su}.A \times \delta\ell$$

**126. The maximum stress induced in a body if the load 'P' is applied with impact upon an area of cross section 'A' is (Where h = Height through which the load falls, E = Modulus of rigidity, L = Length of the body)**

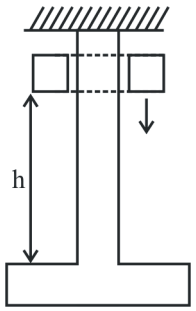
(a) $\frac{P}{A}\left(1+\sqrt{1+\frac{2AEh}{PL}}\right)$

(b) $\frac{P}{A}\left(2+\sqrt{1+\frac{2AEh}{PL}}\right)$

(c) $\frac{P}{A}\left(1+\sqrt{2+\frac{2AEh}{PL}}\right)$

(d) $\frac{P}{A}\left(2+\sqrt{2+\frac{2AEh}{PL}}\right)$

**Ans. (a) :**




$\sigma_{impact} = \sigma_{static} \times \text{I.F}$

$\text{Impact power} = 1+\sqrt{1+\frac{2h}{\delta_{static}}}$

$\therefore$ Used only when load is fallen from certain height or suddenly applied.

$$\delta_{static} = \frac{WL}{AE}$$

$$\sigma_{impact} = \left[1+\sqrt{1+\frac{2hAE}{WL}}\right] \times \sigma_{static}$$

$\sigma_{impact}$ is inversely proportional to the length of the rod. So, the axial stress in the rod can be reduced by increasing the length of the rod.

**127. Poisson's ratio for a cast iron is ____**

(a) 0.27 (b) 0.31
(c) 0.33 (d) 0.36

**Ans. (a):** It is the ratio of lateral strain to the longitudinal strain.

Mathematically, $\mu = \frac{-\text{Lateral strain}}{\text{Longitudinal strain}}$

Stainless steel : 0.30 - 0.31
Steel : 0.27- 0.30
Cast iron : 0.21 - 0.26
Cork : 0.0
Rubber = 0.5
Aluminium = 0.2

**128. The tensile longitudinal stress produces**
(a) Compressively longitudinal strain
(b) Tensile longitudinal strain
(c) Shear strain
(d) Tensile lateral strain

**Ans. (b) :** A strain occurs in the perpendicular direction of the applied load are called transverse normal strain or lateral strains.
- So, pulling on it in the x-direction causes it to shrink in the y-direction.
- Or we can say that tensile stress results in tensile strain in a linear direction and compressive strains in lateral direction.

**129. Rivets are made normally of ____ material.**
(a) Brittle (b) Hard
(c) Ductile (d) Malleable

**Ans. (c) :** Rivets are usually made from ductile materials, such as carbon steel, aluminium and brass.

**130. The depth of weld in case of butt weld is ___ the thickness of the plate**
(a) Less than (b) More than
(c) Two times (d) Equal to

**Ans. (d) :** The throat thickness in case of a butt weld joint is equal to the thickness of the plate whereas in case of fillet weld or lap weld the throat thickness is equal to 0.707 $t_1$ where t = size of weld.

**131. An angle section welded to a plate is an example of**
(a) Symmetrical welded joint
(b) Unsymmetrical welded joint
(c) Axi-symmetrical welded joint
(d) Symmetrical welded section

**Ans. (b) :** An angle section welded to a plate is an example of unsymmetrical welded section.
The length of the welds at the top and bottom of an unsymmetrical welded sections should be so proportional that the sum of the resisting moment of the welds about the centre of the gravity axis is zero.

**132. In case of thin walled cylinders, the ratio of longitudinal stress to hoop stress**
(a) 2 (b) $\frac{1}{2}$
(c) 4 (d) $\frac{1}{4}$

**Ans. (b) :** Hoop stress = $\frac{Pd}{4t}$

Longitudinal stress = $\frac{Pd}{4t}$

$$\frac{\text{longitudinal stress}}{\text{Hoop stress}} = \frac{\frac{Pd}{4t}}{\frac{Pd}{2t}} = \frac{1}{2}$$

**133. If the value of Young's modulus of elasticity is zero, it implies that the material is**
(a) High elastic (b) Plastic
(c) Compressible (d) Incompressible

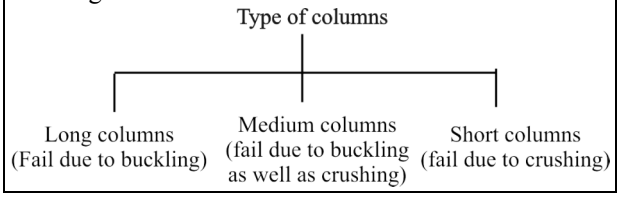
**Ans. (b) :** When young's modulus of elasticity E = 0 then for a given stress

$$\varepsilon_\ell = \frac{\sigma}{E} = \infty$$

Means material will have plastic behaviour.

**134. All short columns fail due to**
(a) Crippling (b) Buckling
(c) Crushing (d) Twisting

**Ans. (c) : Column** is defined as a vertical structural member which is fixed at both ends and is subjected to an axial compressive load.
**Strut** is defined as a structural member subjected to an axial compressive load.
All columns are struts but vice-versa is not true.
- As the length of structure, chances of it failing by buckling are more.




**135. For a beam of span 'L' subjected to a couple 'M' at the centre, the shear force at the left support is _____**
(a) $\frac{4M}{L}$ (b) $\frac{2M}{L}$
(c) $\frac{M}{L}$ (d) Zero

**Ans. (c) :** $\sum M_A = 0$

$$P_B \times L + M = 0 \Rightarrow R_B = -\frac{M}{L}$$

$$R_A = \frac{M}{L}$$

$$V_x = R_A \text{ so } V_A = \frac{M}{L}$$

**136. The load at which the column just buckles is called**
(a) Breaking load (b) Permissible load
(c) Crippling load (d) Ultimate load

**Ans. (c) :** The load at which the column just buckles is known as buckling load or critical load or crippling load.

**Column** is defined as a vertical structural member which is fixed at both ends and is subjected to an axial compressive load.

**Strut** is defined as a structural member subjected to an axial compressive load.

All columns are struts but vice-versa is not true.

- As the length of structure, chances of it failing by buckling are more.

**137. The Crippling load (P) for a column by Euler's formula when both ends are hinged is (where E = Young's Modulus of the material of the column, L = Actual Length of the column, I = Least moment of inertia of the column)**

(a) $\frac{\pi^2 EI}{L^2}$ (b) $\frac{\pi^2 EI}{4L^2}$

(c) $\frac{4\pi^2 EI}{L^2}$ (d) $\frac{2\pi^2 EI}{L^2}$

**Ans. (a) : Euler's formula–** Assumptions

The self weight of column is neglected

Crushing effect is neglected

Flexural rigidity is uniform

Load applied is truly axial

Length is very large compared to cross-section

$P_e \propto f\left[E, I_{min}, \text{end condition}, L^2\right]$

$P_e = \frac{\pi^2 EI_{min}}{L_e^2}$

$P_e$ = Euler's buckling load.

I min : min ($I_{xx}$ and $I_{yy}$),

Le = effective length of column = $\alpha L$

L = actual length of column.

$\alpha$ = length fixity coefficient

$\alpha = 1$ in this question. $L_e = L$

So, $P_e = \frac{\pi^2 EI_{min}}{L_e^2} = \frac{\pi^2 EI_{min}}{L^2}$

**138. A uniform beam of effective length L, fixed at one end and loaded at the centre, will have maximum deflection at**

(a) Free end (b) $\frac{L}{\sqrt{3}}$ from the end

(c) $\frac{L}{\sqrt{2}}$ from free end (d) $\frac{L}{\sqrt{5}}$ from free end

**Ans. (a) :**

$\delta_B = \frac{P\ell^3}{3EI} \Rightarrow \frac{W \times \left(\frac{L}{2}\right)^3}{3EI} = \frac{WL^3}{24EI}$

$\theta_B = \frac{P\ell^3}{2EI} \Rightarrow \frac{W \times (L/2)^2}{2EI} = \frac{WL^2}{8EI}$

$\delta_{max}$ at C = $\delta_B + \theta_B \times \frac{L}{2}$

$= \frac{WL^3}{24EI} + \frac{WL^2}{8EI} \times \frac{L}{2}$

$= \frac{2WL^2}{24EI} + \frac{WL^3}{6EI}$

$= \frac{2WL^3 + 3WL^3}{48}$

$\delta_{max} C = \frac{5WL^3}{48EI}$ at free end

**139. The crippling load (P) for a column by Euler's formula when both ends are fixed is (where E = Young's Modulus of the material of the column, L = Actual Length of the column, I = Least moment of inertia of the column).**

(a) $\frac{\pi^2 EI}{L^2}$ (b) $\frac{\pi^2 EI}{4L^2}$

(c) $\frac{4\pi^2 EI}{L^2}$ (d) $\frac{2\pi^2 EI}{L^2}$

**Ans. (c) : Euler's formula–** Assumptions

The self weight of column is neglected

Crushing effect is neglected

Flexural rigidity is uniform

Load applied is truly axial

Length is very large compared to cross-section

$P_e \propto f\left[E, I_{min}, \text{end condition}, L^2\right]$

$P_e = \frac{\pi^2 EI_{min}}{L_e^2}$

$P_e$ = Euler's buckling load.

I min : min ($I_{xx}$ and $I_{yy}$),

Le = effective length of column = $\alpha L$

L = actual length of column.

$\alpha$ = length fixity coefficient

From the given condition,

$\alpha = \frac{1}{2}$, $L_e = \frac{L}{2}$

So, $P_e = \frac{\pi^2 EI_{min}}{L_e^2} = \frac{\pi^2 EI_{min}}{\left(\frac{L}{2}\right)^2} = \frac{4\pi^2 E I_{min}}{L^2}$

**140. The Crippling load (P) for a column by Euler's formula when one end is fixed and the other end is hinged (where E = Young's Modulus of the material of the column, L = Actual Length of the column, I = Least moment of inertia of the column).**

(a) $\frac{\pi^2 EI}{L^2}$ (b) $\frac{\pi^2 EI}{4L^2}$

(c) $\frac{4\pi^2 EI}{L^2}$ (d) $\frac{2\pi^2 EI}{L^2}$

**Ans. (d): Euler's formula–** Assumptions
The self weight of column is neglected
Crushing effect is neglected
Flexural rigidity is uniform
Load applied is truly axial
Length is very large compared to cross-section

$$P_e \propto f\left[E, I_{min}, \text{end condition}, L^2\right]$$

$$P_e = \frac{\pi^2 E I_{min}}{L_e^2}$$

$P_e$ = Euler's buckling load.
I min : min ($I_{xx}$ and $I_{yy}$),
Le = effective length of column = $\alpha L$
L = actual length of column.
$\alpha$ = length fixity coefficient

From the given condition,

$$\alpha = \frac{1}{\sqrt{2}},\ L_e = \frac{L}{\sqrt{2}}$$

$$\text{So, } P_e = \frac{\pi^2 E I_{min}}{L_e^2} = \frac{\pi^2 E I_{min}}{\left(\frac{L}{\sqrt{2}}\right)^2} = \frac{2\pi^2 E I_{min}}{L^2}$$

**141. The value of Poisson's ratio of cork is ____**
(a) 0.0 (b) 0.1
(c) 0.2 (d) 0.3

**Ans. (a) :** $\boxed{\mu = 0}$ for cork

The near zero poission's ratio for cork makes it an ideal material as a bottle stopper. This is because cork almost does not expand even when compressed on either side, in contract, rubber, stopper will expand laterally when exposed so axial compressive.

**142. A body is to be moved up and inclined plane by applying a force parallel to the plane surface. It is found that a force of 3 kN is required to just move it up to the plane when the angle of inclination is 10º whereas the force needed increases to 4 kN when the angle of inclination increases 15º. Find the weight of the body.**
(a) W = 10950 N (b) W = 12950 N
(c) W = 13950 N (d) W = 11950 N

**Ans. (d) :** Equation from the given condition.

$$\mu W\cos\theta + W\sin\theta = P_1 \quad \text{..... (i)}$$

$$\mu W\cos\theta + W\sin\theta = P_2 \quad \text{..... (ii)}$$

$$\mu W\cos 10 + W\sin 10 = 3 \quad \text{..... (iii)}$$

$$\mu W\cos 15 + W\sin 15 = 4 \quad \text{..... (iv)}$$

From the equation (iii) and (iv)

$$\mu W(0.087) = 0.0818$$

$$\mu W = 0.94 \quad \text{..... (v)}$$

Put the value of equation (v) in equation (iii)

$$0.94 \times \cos 10 + W\sin 10 = 3$$

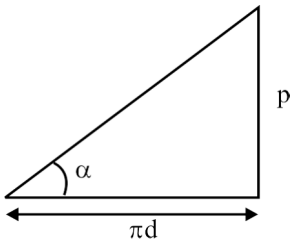
$$W = 11.93 \text{ kN} = 11950 \text{ N}$$

**143. A square threaded bolt with a core diameter of 25mm and a pitch of 10mm is tightened by screwing a nut. The mean diameter of the bearing surface of the nut is 60 mm. the coefficient of friction for the nut and bolt is 0.12 and for the nut and the bearing surface it is 0.15. Determine the force required at the end of 400 mm long spanner if the load on the bolt is 12 kN.**
(a) 112.7 N (b) 212.7 N
(c) 312.7 N (d) 500 N

**Ans. (b) :** 212.7 N

**144. A load of 15 kN is raised by means of a screw jack. The mean diameter of the square threaded screw is 42 mm and the pitch is 10 mm. A force of 120 N is applied at the end of a lever to raise the load. What is the efficiency?**
(a) 0.384 (b) 0.483
(c) 0.843 (d) 0.834

**Ans. (a) :**




W = 15 kN = 15000 N
$d_m$ = 42 mm
p = 10 mm
$p_1$ = 120 N
$\mu$= 0.12
$\ell$ = length of level
We know that

$$\tan\alpha = \frac{p}{\pi d_m}$$

$$\tan\alpha = \frac{10}{\pi \times 42}$$

$$\alpha = 4.334^0$$

$\tan\phi = \mu$
$\tan\phi = 0.12$
$\phi = 6.84^0$
Effort required at the circumference of the screw to raise the load is given by–
$P = w \tan(\alpha+\phi)$
$P = 15 \times 10^3 \tan(4.334 + 6.843)$
$P = 15 \times 10^3 \times 0.1976$
P = 2964 N
Torque required to overcome the friction is–

$$T = P = \frac{d_m}{2}$$

$$T = 2964 \times \frac{42}{2}$$

= 62244 N-mm

Torque applied at the end of the lever,

$$T = P_1 \times \ell$$

$$T = 120 \times \ell$$

Required torque and applied torque should be equal

$62244 = 120 \times \ell$

$\ell = 518.7$ mm

Mechanical advantage, $MA = \frac{W}{P_1}$

$$MA = \frac{15000}{120} = 125$$

Velocity ratio, VR= $\frac{2\pi\ell}{p}$

$$VR = \frac{2\pi \times 518.7}{10}$$

$VR = 325.9$

Efficiency is given by –

$$\eta = \frac{MA}{VR}$$

$$= \frac{125}{325.9} = 0.38355 = 0.384$$

**145. The deformation of any structure takes place in such a manner that the work of deformation is a minimum. This is known as**

(a) Principle of least work
(b) Law of minimum energy
(c) Principle of failure of structure
(d) None of these

**Ans. (a) : Principle of least work–** In the absence of prescribed displacement a linearly elastic system undergo compatible deformation if and only if the strain energy is a minimum with respect to arbitrary changes of the statically admissible forus.

- When using the principle of least work, the system's complementary strain energy its strain energy must be expressed in terms of the statically admissible forus rather than deformation.

**146. A body is to be moved up an inclined plane by applying a force parallel to the plane surface. It is found that a force of 3 kN is required to just move it up the plane when the single of inclination is 10º whereas the force needed increases to 4 kN when the angle of inclination increases to 15º. Find the coefficient of friction.**

(a) 0.786 (b) 0.0786
(c) 7.86 (d) 0.00786

**Ans. (b) :** Equation from the given condition.

$\mu W\cos\theta + W\sin\theta = P_1$ ..... (i)

$\mu W\cos\theta + W\sin\theta = P_2$ ..... (ii)

$\mu W\cos 10 + W\sin 10 = 3$ ..... (iii)

$\mu W\cos 15 + W\sin 15 = 4$ ..... (iv)

From the equation (iii) and (iv)

$\mu W(0.087) = 0.0818$

$\mu W = 0.94$ ..... (v)

Put the value of equation (v) in equation (iii)

$0.94 \times \cos 10 + W\sin 10 = 3$

$W = 11.93$ kN $= 11950$ N

Put the value of W in (iii) equation

$\mu \times 11.93\cos 10 + 11.93\sin 10 = 3$

$\mu = 0.0786$

**147. A rope of weight 0.50 kg/m hinges from a drum for a height of 6m. The work down in winding up the rope will be**

(a) 3 kg.m (b) 6 kg.m
(c) 9 kg.m (d) 12 kg.m

**Ans. (*) :**

Weight = 6 × 0.5 kg/m = 3 kg

work done = Weight × displacement

= 3 × 6 = 18 kg-m

**148. Find the moment of inertia of a triangle about the base AB where AB = b and height = h.**

(a) $\frac{bh^3}{12}$ (b) $\frac{hb^3}{12}$
(c) $\frac{bh^3}{3}$ (d) $\frac{bh^3}{4}$

**Ans. (a) :** Moment of inertia of a triangle with base b & height h

$$I_{base} = \frac{bh^3}{12}$$

MOI about the centroidal axis of triangle

$$I_{centroid} = \frac{bh^3}{36}$$

**149. What is the centroid of a rectangle with length = a and width = b?**

(a) (a/2, b/2) (b) (a/3, b/3)
(c) (a/4, b/4) (d) (a/5, b/5)

**Ans. (a) :** Centroid of rectangle lies where two diagonal intersect each other i.e. $\frac{a}{2}$ & $\frac{b}{2}$

**150. If a number of forces are acting at a point, their resultant will be inclined at an angle θ with the horizontal, such that**

(a) $\tan\theta = \Sigma H / \Sigma V$ (b) $\tan\theta = \Sigma H \times \Sigma V$
(c) $\tan\theta = \Sigma V / \Sigma H$ (d) $\tan\theta = \sqrt{\Sigma H / \Sigma V}$

**Ans. (c) :** If a number of forces are acting at a point, their resultant will be inclined at an angle θ with the horizontal, such that– $\tan\theta = \Sigma V / \Sigma H$ .

- The value of the angle θ will vary depending upon the value of ΣV and ΣH as discussed below.
- When ΣV is positive, the resultant makes an angle between $0^0$ to $180^0$ but when ΣV is negative, the resultant makes an angle between $180^0$ and $360^0$.
- When ΣH is positive, the resultant makes and angle between $0^0$ to $90^0$ or $270^0$ to $360^0$. But when ΣH is negative, the resultant makes an angle between $90^0$ to $270^0$.

# Andhra Pradesh Public Service Commission
## Assistant Executive Engineer (Common, Civil/Mech.) Exam, 2012 (P-I)
## Solved Paper

**1. The ratio between the change in volume and orginal volume of the body is called**

(a) tensile strain
(b) compressive strain
(c) volumetric strain
(d) shear strain

**Ans. (c)** Volumetric strain $= \frac{\text{change in volume}}{\text{original volume}}$

$$\boxed{e_v = \frac{\Delta V}{V}}$$

**2. The ratio between tensile stress and tensile strain or compressive stress to compressive strain is termed as**

(a) modulus of rigidity
(b) modulus of elasticity
(c) bulk modulus
(d) modulus of sub grade reaction

**Ans. (b) :**

Modulus of elasticity $= \frac{\text{Normal stress}}{\text{Normal strain}}$, $E = \frac{\sigma}{\varepsilon}$

**3. Relation between E, K and C is given by**

(a) $E = \frac{3K + C}{6KC}$ (b) $E = \frac{6KC}{K + 3C}$

(c) $E = \frac{3KC}{3K + C}$ (d) $E = \frac{9KC}{3K + C}$

**Ans. (d) :** Young's modulus (E) can be expressed using Bulk modulus (K) and Poisson's ratio (μ) as-

$E = 3K (1 – 2\mu)$ ... (1)

Similarly, Young's modulus (E) can also expressed using rigidity modulus (C) and Poisson's ratio (μ) as-

$E = 2C (1 + \mu)$ ... (2)

Combining the above two equation and solving then to eliminate Poisson's ratio we can get relationship b/w Young's modulus (E), Bulk modulus (K) and Modulus of rigidity (C) as-

$$E = \frac{9KC}{3K + C}$$

**4. The elongation of a conical bar due to its self weight is**

(a) $\frac{\gamma \ell^2}{6E}$ (b) $\frac{\gamma \ell^2}{2E}$

(c) $\frac{\gamma^2 \ell}{2E}$ (d) $\frac{\gamma \ell^2}{E}$

where γ = unit weight of the material

**Ans. (a)** Elongation of conical bar due to self weight.

$$\Delta = \frac{\gamma \ell^2}{6E} = \frac{1}{3} \times \text{Deflection of prismatic bar}$$

γ= Specific weight or unit weight

$\ell$ = length of bar

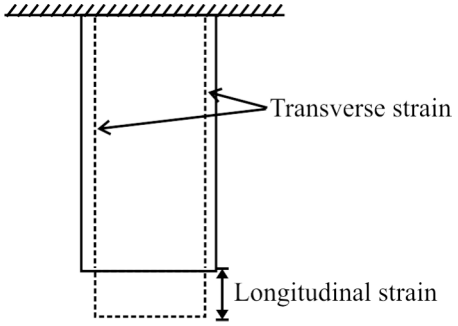
E = Modulus of Elasticity.

**5. Strain in a direction at right angles to the direction of applied force is known as**

(a) shear strain
(b) lateral strain
(c) longitudinal strain
(d) volumetric strain

**Ans. (b) Lateral strain –** The dimensions of the body change in all direction at right angles to its line of application the strains thus produced are called lateral or secondary or transverse strain.

**Longitudinal strain–** It body is subjected to a load, its length change ratio of this change in length to the original length is known as linear or primary strain.




**6. Factor of safety is defined as the ratio of**

(a) $\frac{\text{working stress}}{\text{ultimate stress}}$ (b) $\frac{\text{ultimate load}}{\text{design safe load}}$

(c) $\frac{\text{ultimate stress}}{\text{working stress}}$ (d) $\frac{\text{design safe load}}{\text{ultimate load}}$

**Ans. (c) :** Factor of safety is defined for brittle material is the ratio of ultimate stress to the working stress.

• Factor of safety is defined for ductile material is the ratio of yield strength to the working stress.

$$\text{Factor of safety} = \frac{\text{uttimate stress}}{\text{working stress}}$$

**7. The strain due to a temperature change in a simple bar is**

(a) α T (b) α/ T

(c) T/ α (d) α + T

**Ans. (a) Thermal stress and strain –**

$$\sigma_{th} = E\alpha T$$

$$\Delta = L\,\alpha T$$

$$\text{Strain} = \frac{L\alpha T}{L} = \alpha T$$

$\Delta$ = Change in length

T = Temperature Change

$\alpha$ = Coefficient of thermal expansion.

$\sigma$ = Thermal stress

**8. The ratio of total elongation of a bar of uniform cross- section produced under its own weight to the elongation produced by and external load equal to the weight of the bar is**

(a) 1 (b) 2

(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

**Ans. (c)** Axial elongation $(\Delta_1)$ of bar due to self weight

$$\Delta_1 = \frac{PL}{2AE}$$

- Axial elongation $(\Delta_2)$ of bar due to external load

$$(\Delta_2) = \frac{PL}{AE}$$

$$\frac{\Delta_1}{\Delta_2} = \frac{PL}{2AE} \Big/ \frac{PL}{AE} = \frac{1}{2}$$

**9. Two bars A and B are of equal length but B has an area half that of A and bar A has Young's modulus double that of B. When a load 'P' is applied to the two bars, the ratio of deformation between A and B is**

(a) $\frac{1}{2}$ (b) 1

(c) 2 (d) $\frac{1}{4}$

**Ans. (d) :** Deformation due to load, $\delta = \frac{PL}{AE}$

$$\delta \propto \frac{1}{AE} \qquad \{PL = \text{constant}\}$$

$$\frac{\delta_A}{\delta_B} = \frac{1}{2AE} \div \frac{1}{\frac{A}{2}E}$$

$$= \frac{1}{2AE} \div \frac{2}{AE} = \frac{1}{2AE} \times \frac{AE}{2} = \frac{1}{4}$$

**10. The elongation of beam of length '$\ell$' and cross-sectional area 'A' subjected to a load 'P' and deflection is $8\ell$. If the modulus of elasticity is halved, the new elongation will be**

(a) $\frac{8\ell}{4}$ (b) $2(8\ell)$

(c) $8\ell$ (d) $\sqrt{28\ell}$

**Ans. (b)** $\delta_1 = 8\ell$

$$\delta_1 = \frac{P\ell}{AE} = 8\ell, \frac{P}{AE} = 8$$

$$\delta_2 = \frac{P\ell}{AE/2} = 2\frac{P\ell}{AE} = 2(8\ell)$$

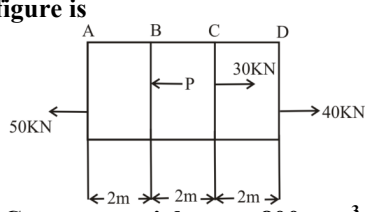
**11. A 16 mm diameter central hole is bored out of a steel rod of 40 mm diameter and length 1.6 m. The tensile strength because of this operation.**

(a) increase (b) remains constant

(c) decreases (d) None of these

**Ans. (c) :** A 16 mm diameter central hole is bored out of a steel rod of 40 mm diameter and length 1.6 m. The tensile strength because of this operation decreases.

**12. The force p for equilibrium of the bar show in the figure is**

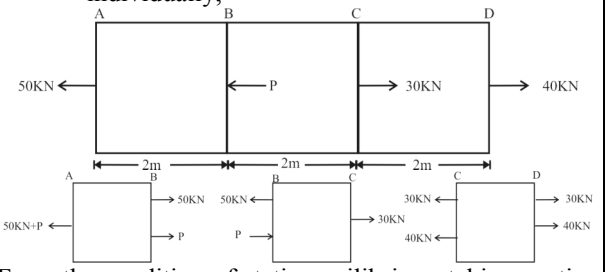



**Cross- sartorial area= 800 mm$^3$**

(a) 60 kN (b) 40 kN

(c) 120 kN (d) 20 kN

**Ans. (d) :** Consider $FBD_s$ of Portion AB,BC&CD individually,




From the condition of static equilibrium, taking portion BC–

$$+50 - P = 30$$

$$P = 20\text{kN}$$

**13. The relationship between Young's modulus and shear modulus when $\frac{1}{m} = 0$, is**

(a) E= 2C (b) E= 3C

(c) E= 2C+1 (d) C= 2E

**Ans. (a)**

$$E = 2C\left(1 + \frac{1}{m}\right)$$

$$E = 3K\left(1 - \frac{2}{m}\right)$$

If $\frac{1}{m} = 0$ then

E = 2C, E = 3K

**14. If a rigidly connected bar of steel and copper is cooled, the copper bar will be subjected to**

(a) compression (b) shear

(c) tension (d) None of these

**Ans. (c)** If a rigidly connected bar of steel and copper is cooled, the copper bar will be subjected to tension,
Because $\alpha_{copper} > \alpha_{steel}$

- If composite bar is heated, the copper bar will be under compression.

**15. The force required to punch a 10 mm thick, if the shear strength of mild steel is 360 MPa is**

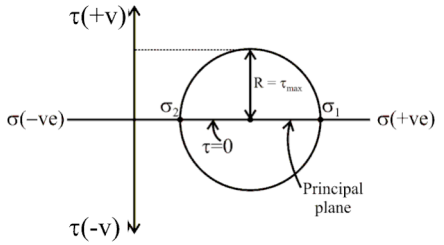
(a) $9\pi$kN (b) $36\pi$kN
(c) $18\pi$kN (d) 2.25kN

**Ans. (b) :** $P = \tau.A$
$= 360\ MPa \times \pi\ 10 \times 10\ mm^2$
$= 36\ \pi\ kN$

**16. The shear stress on principal plane is**

(a) minimum (b) maximum
(c) zero (d) infinity

**Ans. (c) :** The shear stress on principal plane is zero but at the maximum shear stress plane principal stresses are not zero.




**17. If a body carries two unlike principal stresses, the maximum shear stress is given by**

(a) sum of the principal stresses
(b) difference of the principal stresses
(c) half the difference of the principal stresses
(d) half the sum of the principal stresses

**Ans. (d)** If a body carries two unlike principal stresses the maximum shear stress is given by half the sum of the principal stresses.

Given, $\sigma_{P_1} = +\sigma_1$

$\sigma_{P_2} = -\sigma_2$

$$\tau_{max} = \frac{\sigma_{P_1} - \sigma_{P_2}}{2} = \frac{+\sigma_1 - (-\sigma_2)}{2} = \frac{\sigma_1 + \sigma_2}{2}$$

**18. The radius of Mohr's circle for two unlike principal stresses of magnitude $\sigma$ is**

(a) $\frac{\sigma}{2}$ (b) $\sigma$
(c) $\frac{\sigma}{4}$ (d) Zero

**Ans. (b)** Given, $\sigma_{P_1} = +\sigma$

$\sigma_{P_2} = -\sigma$

$$R = \tau_{max} = \frac{\sigma_{P_1} - \sigma_{P_2}}{2}$$

$$= \frac{+\sigma - (-\sigma)}{2} = \frac{2\sigma}{2} = \sigma$$

**19. A solid circular shaft is subjected to a maximum shearing stress of 140 MPa The magnitude of maximum normal stress developed in the shaft is**

(a) 140 $MP_a$ (b) 80 $MP_a$
(c) 70 $MP_a$ (d) 60 $MP_a$

**Ans. (a)** Shaft is under pure shear state. Hence,

$$\sigma_{max} = \tau_{max} = \frac{16T}{\pi D^3} = 140\,MPa$$

**20. If the principal stresses at a point in a strained body are $\sigma_x$ and $\sigma_y(\sigma_x > \sigma_y)$, then the resultant stress on a plane carrying the maximum shear stress is equal to**

(a) $\sqrt{\sigma_x^2 + \sigma_y^2}$ (b) $\sqrt{\sigma_x^2 - \sigma_y^2}$
(c) $\sqrt{\frac{\sigma_x^2 + \sigma_y^2}{2}}$ (d) $\sqrt{\frac{\sigma_x^2 - \sigma_y^2}{2}}$

**Ans. (c)** Principal stress on the plane of maximum shear stress $= \frac{\sigma_x + \sigma_y}{2}$

and $\tau = \frac{\sigma_x - \sigma_y}{2}$

Hence, $\sigma_R = \sqrt{\left(\frac{\sigma_x + \sigma_y}{2}\right)^2 + \left(\frac{\sigma_x - \sigma_y}{2}\right)^2}$

$$\sigma_R = \sqrt{\frac{1}{4}\left(\sigma_x^{\,2} + \sigma_y^{\,2} + 2\sigma_x\sigma_y + \sigma_x^{\,2} + \sigma_y^{\,2} - 2\sigma_x\sigma_y\right)}$$

$$\boxed{\sigma_R = \sqrt{\frac{\sigma_x^{\,2} + \sigma_y^{\,2}}{2}}}$$

**21. A body is subjected to two normal stresses 20 $kN/m^2$ (tensile) and 10 $kN/m^2$ (compressive) acting mutually perpendicular to each other. The maximum shear stress is**

(a) 30 $kN/m^2$ (b) 5 $kN/m^2$
(c) 15 $kN/m^2$ (d) 10 $kN/m^2$

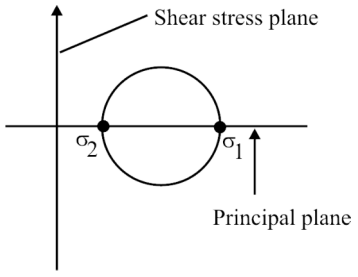
**Ans. (c)** For two mutually perpendicular stress

$$\tau_{max} = \frac{\sigma_x - \sigma_y}{2} = \frac{20 - (-10)}{2} = 15\,KN/m^2$$

**22. Principal planes will be free of**

(a) Normal stress
(b) shear stress
(c) both normal and shear stresses
(d) None of these

**Ans. (b) :** Principal planes will be free of both normal and shear stresses.




$\sigma_1$ & $\sigma_2$ = normal stress at the principal

**23. On two perpendicular planes there are normal stresses, $\sigma_1$ and $\sigma_2$ shear stress q. If $q^2$, $\sigma_1$, $\sigma_2$ the major and minor principal stresses respectively are**

(a) $\sigma_1+\sigma_2$ and zero (b) $\sigma_1+\sigma_2$ and $\sigma_1+\sigma_2$

(c) zero and $\sigma_1-\sigma_2$ (d) $\sigma_1-\sigma_2$ and $\sigma_1+\sigma_2$

**Ans. (a) :** $\sigma_x = p_x, \sigma_y = p_y, \tau_{xy} = q = \sqrt{p_x p_y}$

$$\sigma_{1,2} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{1,2} = \frac{p_x+p_y}{2} \pm \sqrt{\left(\frac{p_x-p_y}{2}\right)^2 + \left(\sqrt{p_x p_y}\right)^2}$$

$$\sigma_{1,2} = \frac{p_x+p_y}{2} \pm \sqrt{\frac{p_x^2}{4} - \frac{2p_x p_y}{4} + \frac{p_y^2}{4} + p_x p_y}$$

$$\sigma_{1,2} = \frac{p_x+p_y}{2} \pm \sqrt{\frac{1}{4}\left(p_x^2 + 2p_x p_y + p_y^2\right)}$$

$$\sigma_{1,2} = \frac{p_x+p_y}{2} \pm \frac{p_x+p_y}{2}$$

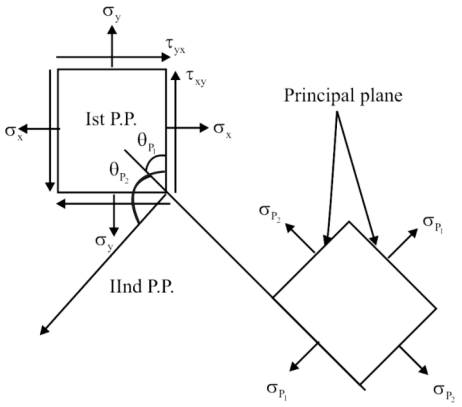
$\sigma_1 = p_x + p_y, \sigma_2 = 0$

**24. Angel between the principal planes is**

(a) $270^o$ (b) $180^o$

(c) $90^o$ (d) $45^o$

**Ans. (c) :** Angel between the principal planes is $90^0$.




Location of principal plane

$$\tan 2\theta_{P_1} = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$\theta_{P_1} \rightarrow$ Location of first principal plane

$\theta_{P_2} \rightarrow \theta_{P_1} + 90^0$ location

$\theta_{P_3} \rightarrow$ location of III[rd] principal plane perpendicular to $\theta_{P_1}$ and $\theta_{P_2}$ principal plane in z-direction.

**25. For a two- dimensional stress system the coordinates of the centre of Mohr's circle are**

(a) $\left[\frac{\sigma_x-\sigma_y}{2}, 0\right]$ (b) $\left[\frac{\sigma_x+\sigma_y}{2}, 0\right]$

(c) $\left[0, \frac{\sigma_x-\sigma_y}{2}\right]$ (d) $\left[0, \frac{\sigma_x+\sigma_y}{2}\right]$

**Ans. (b)** For a two - dimensionl stress system the Co-ordinates of the centre of Mohr's circle are $\left[\frac{\sigma_x+\sigma_y}{2}, 0\right]$

The general equation of mohr's circles

$$\left(\sigma_0 - \frac{\sigma_x+\sigma_y}{2}\right)^2 + \tau_0^2 = \left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2$$

Centre $= \left(\frac{\sigma_x+\sigma_y}{2}, 0\right)$

Radius $= \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$

**26. For a maximum bending moment shear force at that section should be**

(a) zero (b) maximum

(c) minimum (d) None of these

**Ans. (a) :** For a maximum bending moment shear force at that section should be zero.

- If shear force is zero at any section the BM is either maximum or minimum at that section.
- If shear fore is changes its sign is +ve to -ve then BM is maximum at that section.
- It shear force is change the sign -ve to +ve then BM is minimum at that section.

**27. For uniform stress force throughout the span of a simply supported beam. It should carry**

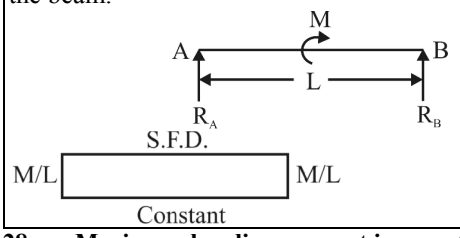
(a) a concentrated lead at the mid-span

(b) a couple anywhere in the section

(c) udl over its entire span

(d) two concentrated loads equally spaced

**Ans. (b) :** If a couple is applied anywhere in simply supported beam then S.F will be constant throughout the beam.

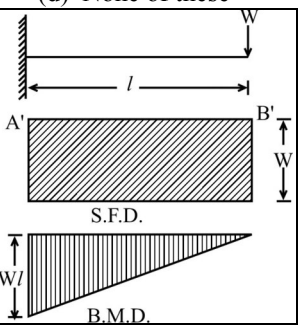



**28. Maximum bending moment in a cantilever carrying a concentrated load at the free end occurs**

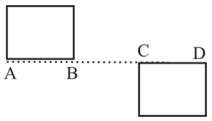
(a) at the fixed end (b) at the free end

(c) at the mid- span (d) None of these

**Ans. (a)**
Maximum bending moment in a cantilever beam carrying a concentrated load at the free end occurs at the fixed end.



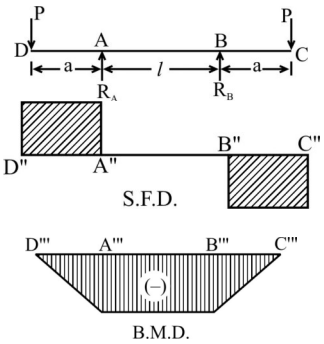
**29. The given figure shows the shear force diagram for the beam ABCD**




**Bending moment in the portion BC of the beam**

(a) is zero
(b) varies linearly from B to C
(c) parabolic variation between B and C
(d) is a non- zero constant

**Ans. (d)**




Bending moment in the portion AB of the beam is a non-zero constant.

$\because \dfrac{dM}{dx} = V = 0 \left(\text{for AB portion}\right)$

$\Rightarrow \boxed{M = \text{constant}}$

**30. At the point of contra flexure in a beam**

(a) B.M. is zero
(b) B.M. changes its sign
(c) S.F. is zero
(d) Both S.F and B.M. change sign

**Ans. (b) : The point of contraflexure–**The bending moments of opposite nature always produce curvatures of beams in opposite directions. In a beam if the bending moment changes sign at a point, the point itself having zero bending moment, the beam changes curvature at this point of zero bending moment and this point is called the point of contra flexure.

**Point of inflection–** Point of curve (on deflected shape) changes being concave to convex or vice versa.

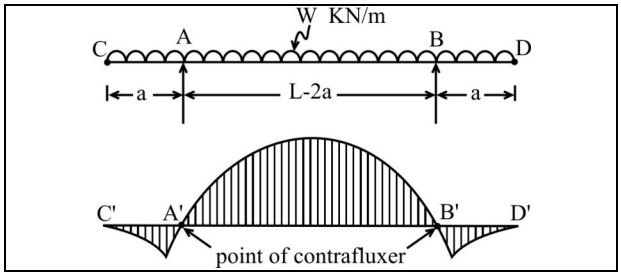
• The point at which bending moment changes its sign (from +ve to -ve and vice-versa) is known as point of contra flexure.

**31. In a double overbalancing beam carrying udl throughout its length, the number of points of contra flexure are**

(a) 1 (b) 2
(c) zero (d) 3

**Ans. (b)** In a double overhanging beam carrying UDL throughout its length, the number of points of contraflexure are two.

**Points of contraflexure:–** Where the bending moment will change sign from negative to positive or vice versa. Such a point, where the bending moment change sign, is known as a point of contraflexure.

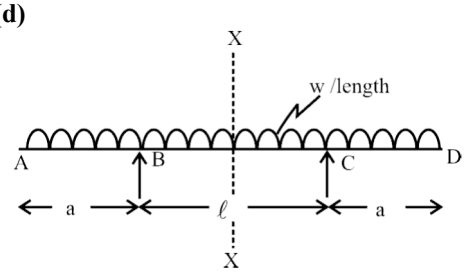



**32. A beam ABCD is simply supported at B and C. the simply supported span BC= $\ell$. The over hangs are each 'a' If the beam carries a udl of w/metre over the entire length, the maximum BM at the centre is**

(a) $\dfrac{w\ell^2}{8} - \dfrac{wa^2}{4}$ (b) $\dfrac{wa^2}{4}$

(c) $\dfrac{wa^2}{2}$ (d) $\dfrac{w\ell^2}{8} - \dfrac{wa^2}{2}$

**Ans. (d)**



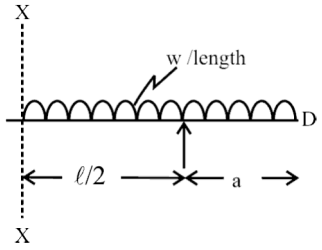

$\sum f_y = 0$

$$R_B + R_C = w\left(\ell + 2a\right) \quad \text{..... (i)}$$

By the symmetrical section,

$$R_B + R_C = \frac{w\left(\ell + 2a\right)}{2}$$

BM taken at the centre XX,




$$+M_{xx} - \frac{w\left(\ell + 2a\right)}{2} \times \frac{\ell}{2} + w\left(a + \frac{\ell}{2}\right) \times \frac{\left(\frac{\ell}{2} + a\right)}{2} = 0$$

$$M_{xx} = \frac{w\ell^2}{4} + \frac{wa\ell}{2} - \frac{wa^2}{2} - \frac{w\ell^2}{8} - \frac{wa\ell}{2}$$

$$M_{xx} = \frac{w\ell^2}{8} - \frac{wa^2}{2}$$

**33. Rate of change of shear force is equal to**

(a) Bending moment
(b) Intensity of loading
(c) Maximum deflection
(d) Slope

**Ans. (b)** (i) Rate of change of Shear force (V) is equal to intensity of loading

$$\frac{dV}{dx} = W$$

W = load per unit length.

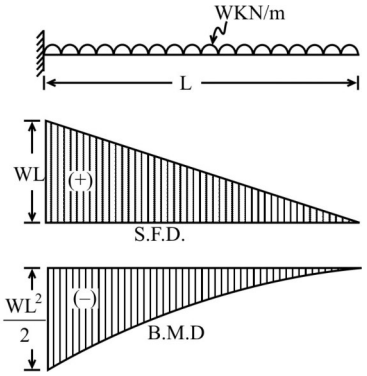
(ii) Rate of change of bending moment along the length of beam is equal to shear force.

$$\frac{dM}{dx} = V_x$$

**34. A cantilever is subjected to udl throughout the length. If the maximum shear force is 200 kN and maximum bending moment is 400 kN-m, the span, "L" of the beam is metres is**

(a) 3 (b) 2 (c) 4 (d) 8

**Ans. (c)**




Maximum shear force = 200kN

WL = 200kN ……..(i)

Maximum Bending moment

$\frac{WL^2}{2}$ =400kN …..(ii)

for equation (i) and (ii)

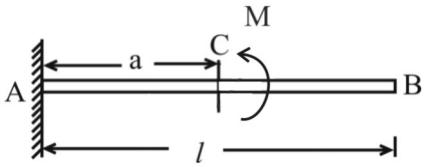
$$\frac{\frac{WL^2}{2}}{WL} = \frac{400}{200}$$

$\frac{L}{2} = 2 \Rightarrow \boxed{L = 4\ m}$

**35. A cantilever beam AB of length $\ell$ is subjected to an anticlockwise couple of 'M' at a section C, distance 'a' from support, Then the maximum shear force is equal to**

(a) M (b) $\frac{M}{a}$ (c) Zero (d) Ma

**Ans. (c)**




A cantilever beam AB of length $l$ is subjected to an anticlockwise couple of M at a section C, distance 'a' from support, then the maximum shear force is equal to zero.

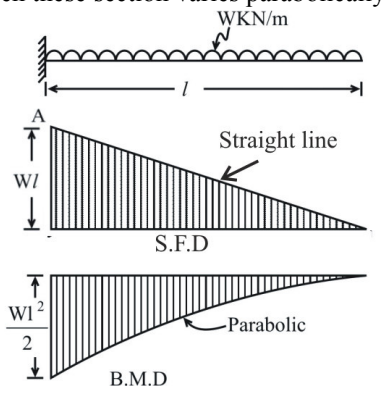
**Note–** Since there is no transverse loading hence there will be no shear force.

**36. If SFD between two sections varies linearly, BM between these sections varies**

(a) linearly (b) parabolically

(c) constant (d) None of these

**Ans. (b) :** If SFD between two section varies linearly, BM between these section varies parabolically




**Note–** $\frac{dM}{dx} = V$ (Linear variation given)

$dM = Vdx$ .....(i)

Integration of above equation will yield quadratic equation hence variation is parabolic.

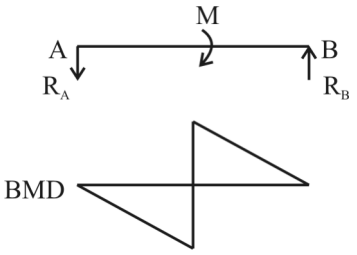
**37. At section of a beam sudden change is BM indicates the action of**

(a) point load (b) couple

(c) point load or couple (d) udl

**Ans. (b) :** At section of a beam sudden change in BM indicates the action of couple.

**Ex.–**




**38. In a double equal overhang beam, for maximum BM to be as small as possible, the supports must be placed at a distance 'x' from the ends of the beam. The value of x is**

(a) 0.5 $\ell$ (b) 0.207 $\ell$

(c) 0.53 $\ell$ (d) 0.7 $\ell$

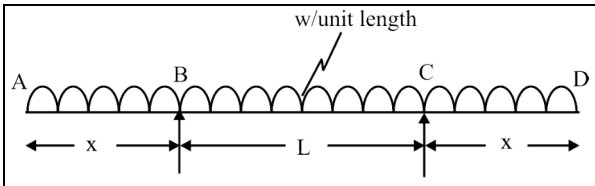
**Ans. (b) :** The condition of maximum bending moment to be minimum will occur when maximum sagging bending moment is equal to maximum hogging moment and that time will be most economical beam.

Maximum sagging bending moment = Maximum hogging bending moment

$$\frac{w}{8}\left(L^2 - 4x^2\right) = \frac{wx^2}{2}$$

$$L = 2\sqrt{2}x$$

$$x = \frac{L}{2\sqrt{2}}$$




$$\% \text{ of over hangs on each side} = \frac{x}{2x + 2\sqrt{2}x} \times 100$$

$$= 20.7\%$$

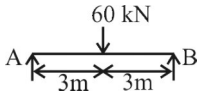
Value of x is = 0.207 L

**39. A freely supported beam of span 6 m is subjected to a point load of 60kN at mid span. The maximum BM is equal to**

(a) 300 kN-m (b) 180 kN-m

(c) 90 kN-m (d) 270 kN-m

**Ans. (c)**




$R_A + R_B = 60$ kN

$\Sigma M_B = 0$

$R_A \times 6 = 60 \times 3$

$R_A = 30$ kN

$R_B = 30$ kN

Max. bending moment = $R_A \times 3$

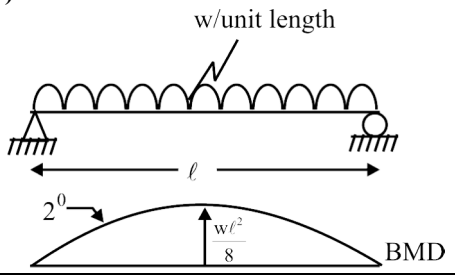
$= 30 \times 3 = 90$ kN-m

**40. If a freely supported beam is subjected to udl throughout the span, the shape of the BMD is**

(a) rectangle (b) straight line

(c) equilateral triangle (d) parabola

**Ans. (d) :**




**41. Section modulus of a beam is defined as**

(a) IY (b) $\frac{Y}{I}$

(c) $\frac{I}{Y_{max}}$ (d) $Y^2I$

**Ans. (c) Section modulus:–**

$\frac{M}{I} = \frac{\sigma}{y}$ OR $M = \sigma \times \frac{I}{y}$

$M = \sigma_{max} \times \frac{I}{y_{max}} = \sigma_{max} \times Z$

Where $Z = \frac{I}{Y_{max}}$. The term Z is known as modulus of section or section modulus.

**42. A beam of uniform strength is one which as same**

(a) bending moment throughout the section

(b) shearing force throughout the section

(c) deflection throughout the baem

(d) Bending stress at every section

**Ans. (d) : Beam of uniform strength–**

- In common practice, a beam of constant cross-section through the length is provided. This gives rise of constant section modulus and has constant or uniform moment of resistance.
- Since the actual moment generally varies along the length, such a section is not economical, under heavy loads, a beam of variable section is designed to have uniform strength, and there can be two such configurations.

(i) Beam of uniform width,

(ii) Beam of uniform depth

We shall investigate both the configurations for the case of beam with a concentrated load at the mid span, it is to be noted that a beam of uniform strength has the same maximum bending stress all along the length.

**43. Neutral axis of a beam is the axis at which**

(a) the shear force is zero

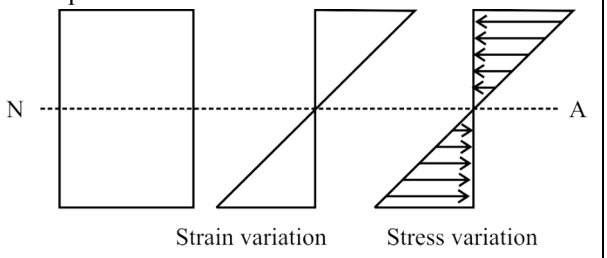
(b) the section modulus is zero

(c) the bending stress is maximum

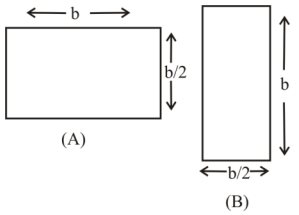
(d) the bending stress is zero

**Ans. (d) :** The neutral axis is the basis through a beam where the stress is zero, that is there is neither tension nor compression.

- On applying load on beam, upper portion of NA goes to compression and bottom on is in tension.




**44. A beam cross-section is used in two different orientations as shown in figure:**

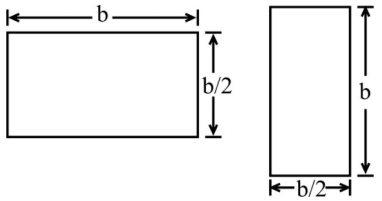



**Bending moments applied in both cases are same, The maximum bending stresses induced in cases (A) and (B) are related as**

(a) $\sigma_A = \sigma_B$ (b) $\sigma_A = 2\sigma_B$

(c) $\sigma_A = \frac{\sigma B}{2}$ (d) $\sigma_A = \frac{\sigma B}{4}$

**Ans. (b)**




(A) (B)

**For first case –**

$$I_A = \frac{1}{12} \times b\left(\frac{b}{2}\right)^3 = \frac{b^4}{96}$$

$$\frac{M}{I_A} = \frac{\sigma_A}{y} \Rightarrow \frac{M \times 12 \times 8}{4 \times b^3}$$

$$\boxed{\sigma_A = \frac{24M}{b^3}} \text{ ..... (i)}$$

For second case

$$I_B = \left(\frac{b}{2}\right) \times (b)^3 \times \frac{1}{12}$$

$$I_B = \frac{b^4}{24}$$

$$\frac{M}{I_B} = \frac{\sigma_B}{y}$$

$$\frac{M}{\frac{b^4}{12 \times 2}} = \frac{\sigma_B}{\frac{b}{2}}$$

$$\boxed{\sigma_B = \frac{12M}{b^3}} \text{ ..... (ii)}$$

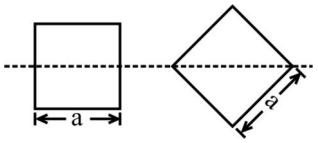
For equation first and second $\boxed{\sigma_A = 2\sigma_B}$

**45. The ratio of flexural strength of a square section with its two sides horizontal to its diagonal horizontal is**

(a) $\sqrt{2}$ (b) 2

(c) $2\sqrt{2}$ (d) $\sqrt{\frac{2}{5}}$

**Ans. (a)**




a = Side of the square beam

$M_1$ = Moment of resistance of Section 1

$M_2$ = Moment of resistance of Section 2

$$Z_1 = \frac{bd^2}{6} = \frac{a \times a^2}{6} = \frac{a^3}{6} \text{ .....(i)}$$

$$I_2 = 2 \times \frac{bh^3}{12} = 2 \times \frac{a\sqrt{2} \times \left(\frac{a}{\sqrt{2}}\right)^3}{12} = \frac{a^4}{12}$$

$$y_{max} = \frac{a}{\sqrt{2}}$$

$$Z_2 = \frac{I}{y_{max}} = \frac{\frac{a^4}{12}}{\frac{a}{\sqrt{2}}} = \frac{a^3}{6\sqrt{2}}$$

Moment of resistance of a section is directly proportional to their section modulus.

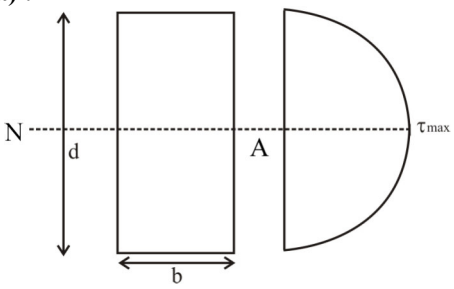
$$\frac{M_1}{M_2} = \frac{Z_1}{Z_2} = \frac{\frac{a^3}{6}}{\frac{a^3}{6\sqrt{2}}} = \sqrt{2}$$

**46. The ratio of maximum shear stress to the average shear stress in case of a rectangular beam is equal to**

(a) 1.5 (b) 2.0

(c) 2.5 (d) 3

**Ans. (a) :**




$$\tau = \frac{6S}{bd^3}\left(\frac{d^2}{4} - y^2\right)$$

Where S = shear force

for $\tau_{max}$, $\frac{d\tau}{dy} = 0 \Rightarrow y = 0$

$$\tau_{max} = \frac{6S}{bd^3} \times \frac{d^2}{4} = \frac{3}{2}\frac{S}{bd}$$

we know

$$\frac{S}{bd} = \tau_{avg}$$

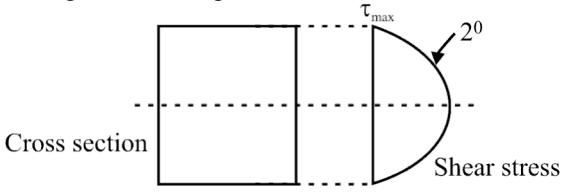
$$\frac{\tau_{max}}{\tau_{avg}} = 1.5$$

**47. The nature of distribution of horizontal shear stress in a rectangular beam is**

(a) linear (b) parabolic

(c) hyperbolic (d) elliptic

**Ans. (b) :** The nature of distribution of shear stress in a rectangular beam is parabolic.




$$\tau = \frac{V}{2I}\left[\frac{d^2}{4} - y^2\right]$$

$\tau \propto -y^2$ (Parabolic variation)

(i) Rectangular section:

$$\tau = \frac{6S}{bd^3}\left(\frac{d^2}{4} - y^2\right) \quad \text{..... Parabolic}$$

(ii) Circular section:

$$\tau = \frac{4S}{3\pi R^4}\left(R^2 - y^2\right) \quad \text{..... Parabolic}$$

(ii) Triangular section:

$$\tau = \frac{12S}{bh^3}\left(hy - y^2\right) \quad \text{..... Parabolic}$$

where S= shear force.

**48. Section modulus of a circular section about an axis through its centre of gravity is**

(a) $\frac{\pi}{32}d^3$ (b) $\frac{\pi}{16}d^3$

(c) $\frac{\pi}{8}d^3$ (d) $\frac{\pi}{64}d^3$

**Ans. (a) :** Section modulus (z) = I/y

I = Moment of Inertia $= \frac{\pi d^4}{64}$

$$y = \frac{d}{2} \Rightarrow Z = \frac{I}{y} = \frac{(\pi/64)d^4}{d/2}$$

$$\boxed{z = \frac{\pi d^3}{32}}$$

**49. A steel plate 50 mm wide and 100 mm thicke is to be bent into a circular arc of radius 10 m. If $E=2\times10^5$ N/mm$^2$, then the maximum bending stress induced will be**

(a) 200 N/mm$^2$ (b) 100 N/mm$^2$

(c) 10,000 N/mm$^2$ (d) 1000 N/mm$^2$

**Ans. (d)** b=50mm, d = 100mm, R = 10m

$E= 2\times10^5$ N/mm$^2$

**Bending Equation –**

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

y = 50 mm

$$\frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{E}{R} \times y$$

$$\sigma = \frac{2\times10^5}{10\times10^3} \times 50$$

$$\boxed{\sigma = 1000 N/mm^2}$$

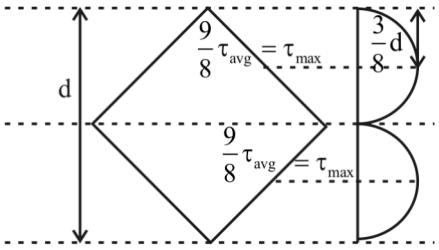
**50. A beam of square section is placed with one diagonal placed horizontally. The maximum shear stress occurs at**

(a) the N.A. (b) $\frac{3}{8}$d from top

(c) $\frac{3}{8}$d from N.A. (d) at the extreme fibres

where d= depth of the beam

**Ans. (b)** $\frac{3}{8}$ d from top




**51. Torsional rigidity of a shaft is given by**

(a) $\frac{T}{L}$ (b) $\frac{T}{J}$

(c) $\frac{T}{E}$ (d) $\frac{T}{r}$

**Ans. (*) :** We know that torsion equation

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}$$

$$\frac{T}{J} = \frac{G\theta}{L}$$

$$\boxed{\frac{T}{\theta} = \frac{GJ}{L}}$$

- Stiffness$(K) = \frac{T}{\theta}$

**52. Shear stress for a circular shaft due to torque varies**

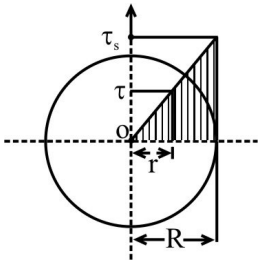
(a) from surface to centre parabolically

(b) from surface to centre linearly

(c) from centre to surface parabolically

(d) from centre to surface linearly

**Ans. (d)** Shear stress for a circular shaft due to torque varies from centre to surface linearly.




$$\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{r}$$

The above relation states that the intensity of shear stress at any point in the cross-section of a shaft subjected to pure torsion is proportional to its distance from the centre.

**53. If two shafts of the same length, one of which is hollow, transmit equal torque and have equal maximum stress, then they should have equal**

(a) angle of twist

(b) polar modulus of section

(c) polar moment of inertia

(d) diameter

**Ans. (b) :** Given, $T_H = T_S$

$(\tau_{max})_H = (\tau_{max})_S$

$$\frac{T}{J} = \frac{\tau}{r}$$

$$\frac{T}{\tau} = \frac{J}{r}$$

$$Z_p = \frac{J}{r_{max}}$$

Where,

$Z_p$= Polar modulus of section.

**54. A circular shaft subjected to torsion undergoes a twist of $1^0$ in a length of 1.2 m. If the maximum shear stress induced is 100 MPa and the rigidity modulus is $0.8\times 10^5$ MPa, the radius of the shaft in mm should be**

(a) $\frac{270}{\pi}$ (b) $\frac{\pi}{270}$ (c) $\frac{180}{\pi}$ (d) $\frac{\pi}{180}$

**Ans. (a)** Twist angle $(\theta) = 1^0$

Length of shaft = 1.2 m

Maximum shear stress = 100 MPa

Modulus of rigidity $(G) = 0.8\times 10^5$ MPa.

Torsion equation $\frac{T}{J} = \frac{G\theta}{\ell} = \frac{\tau}{r}$

$$\frac{100}{r} = \frac{0.8\times 10^5 \times \pi}{180\times 1.2\times 10^3}$$

$$\boxed{r = \frac{270}{\pi}\text{mm}}$$

**55. Two shafts are of same length and same material. The diameter and maximum shear stress of the second shaft is twice that of the first shaft. Then the ratio of power developed between the first and second shaft is**

(a) 16 (b) $\frac{16}{3\sqrt{3}}$ (c) $\frac{1}{16}$ (d) $\frac{3}{16}$

**Ans. (c)** Power develop (P) = $T\omega$

Case 1st:-

$$T = \frac{\pi d^3 \tau}{16}$$

$$P_1 = \frac{\pi d^3 \tau}{16}\times\omega$$

Case 2nd :-

$$P_2 = \frac{\pi(2d)^3 \times 2\tau}{16}\times\omega$$

assume, both case angular speed same then

Ratio of power developed $\left(\frac{P_1}{P_2}\right) = \frac{1}{16}$

$$\boxed{\frac{P_1}{P_2} = \frac{1}{16}}$$

**56. The differential equation which gives the relation between BM, slope and deflection of beam is**

(a) $EI\frac{d^2y}{dx^2} = \frac{M}{I}$ (b) $\frac{d^2y}{dx^2} = M$

(c) $EI\frac{d^2y}{dx^2} = M$ (d) $EI\frac{dy}{dx} = \frac{M}{F}$

**Ans. (c) :** Bending Moment (M) = $EI\frac{d^2y}{dx^2}$

Shear force (V) = $EI\frac{d^3y}{dx^3}$

Load intensity (w) = $EI\frac{d^4y}{dx^4}$

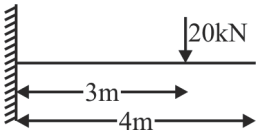
$y \rightarrow$ deflection

$\frac{dy}{dx}$ = slope

**57. A rolled steel beam having a span of 4 m carries a point load of 20 kN at 3 m from the left support, If the moment of inertia of the section is $12\times10^7$ mm$^4$ and E= 200 kN/mm$^2$, then the deflection of the beam under the point load is equal to**

(a) 25 mm (b) 7.5 mm (c) 13.33 (d) 50 mm

**Ans. (b)**




$l$ = 4m

a = 3m

E = 200 kN/mm$^2$

I = $12\times10^7$ mm$^4$

The deflection of the beam under the point load.

$$= \frac{Wa^3}{3EI} = \frac{(20\times1000)\times(3\times10^3)^3}{3\times200\times10^3\times12\times10^7}$$

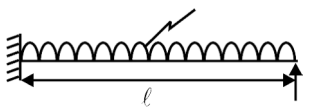
$$= \frac{2\times10^4\times27\times10^9}{3\times2\times12\times10^5\times10^7} = \frac{270}{36} = 7.5 \text{ mm}$$

**58. A cantilever of length '$\ell$' carries a udl of a w per unit run, over then whole length. If the free end be supported over a rigid prop. The reaction of the prop will be**

(a) $\frac{2w\ell}{8}$ (b) $\frac{5w\ell}{8}$

(c) $\frac{3w\ell}{8}$ (d) $\frac{7w\ell}{8}$

**Ans. (c) :**




$$\frac{R\ell^3}{3EI} = \frac{w\ell^4}{8EI}$$

$$R = \frac{3}{8}w\ell$$

**59. Radius of curvature of the beam is equal to**

(a) $\frac{ME}{I}$ (b) $\frac{M}{EI}$ (c) $\frac{EI}{M}$ (d) $\frac{MI}{E}$

**Ans. (c)** Bending equation

$$\frac{M}{I}=\frac{\sigma}{y}=\frac{E}{R.}$$

$$\frac{M}{I}=\frac{E}{R.}$$

$$\text{Radius of curvature of beam } (R)=\frac{EI}{M}$$

Where,

EI = Flexural rigidity

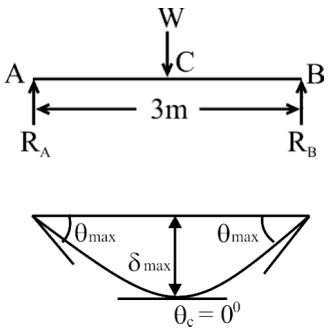
M = Bending moment

**60. A simply supported beam span 3 m is subjected to a central point load of 5 kN Then the slope at the mid span is equal to**

(a) $\frac{25}{24EI}$ (b) $\frac{256}{EI}$

(c) $\frac{40}{48EI}$ (d) Zero

**Ans. (d)**




$$\text{Slope}=\frac{w\ell^2}{16EI},\ \text{Deflection}=\frac{w\ell^3}{48EI}$$

W = 5 KN (Central point load.)

$l$ = 3 m (beam span)

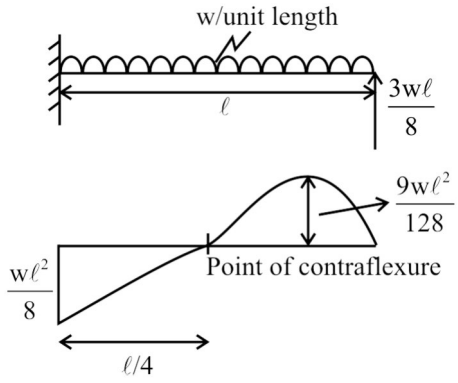
Slope at the mid span = 0

Deflection at the mid span = Maximum.

**61. In a propped cantilever beam, the number of points of contra flexure is**

(a) 1 (b) 2 (c) 3 (d) 4

**Ans. (a)** In a propped cantilever beam, the number of point of contra-flexure is one.




**Contraflexure–** A point where BMD changes the sign is termed as "point of contraflexure" at the same point in the deflected curve shape is termed as "point of inflection".

**62. A fixed beam 'AB' 6m long carries a vertical load 90 kN at 2 m from 'A' The fixed end moments at 'A' and 'B' are**

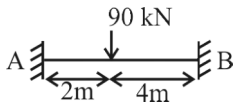
(a) 40 kNm, 80 kN-m

(b) 40 kNm, 120 kN-m

(c) 80 kNm, 40 kN-m

(d) 120 kNm, 80 kN-m

**Ans. (c)**




$$M_{AB}=\frac{Pab^2}{L^2}=\frac{90\times2\times16}{36}=80\,kN-m$$

$$M_{BA}=\frac{Pa^2b}{L^2}=\frac{90\times4\times4}{36}=40\,kN-m$$

**63. In a fixed beam, at the fixed ends**

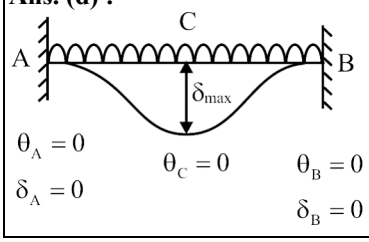
(a) slope is zero and deflection is maximum

(b) slope is maximum and deflection is zero

(c) both slope and deflection are maximum

(d) slope and deflection are zero

**Ans. (d) :**

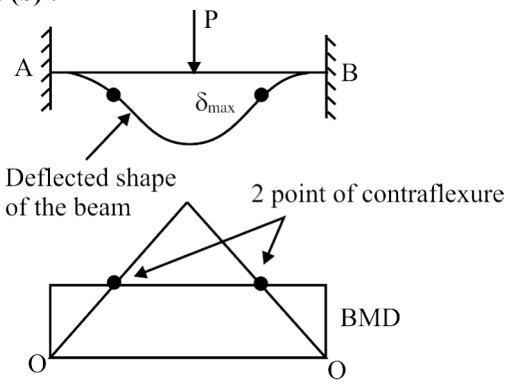



**64. If a fixed beam is subjected to a point load at mid span, total number or points of contraflexure are**

(a) 1 (b) 2

(c) 3 (d) zero

**Ans. (b) :**



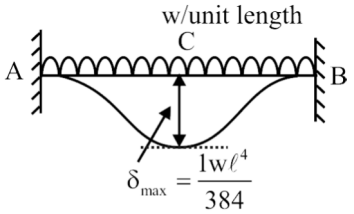

• Point of contraflexure, where bending moment diagram changes it's sign.

**65. A beam of length, $\ell$, fixed at both ends carries a uniformly distributed load of w per unit length. If EI is the flexural rigidity, then the maximum deflection in the beam is**

(a) $\frac{w\ell^4}{192EI}$ (b) $\frac{w\ell^4}{24EI}$

(c) $\frac{w\ell^4}{384EI}$ (d) $\frac{w\ell^4}{12EI}$

**Ans. (c) :**




$\delta_{max} = \frac{1w\ell^4}{384}$

$\theta_c = 0^0$

**66. Slenderness ratio of a column may be defined as the ratio of its effective length to the**

(a) radius of column
(b) minimum radius of gyration
(c) maximum radius o gyration
(d) area of the cross- section

**Ans. (b) :** Slenderness ratio $(\lambda) = \frac{l_{eff}}{r_{min}}$

$\ell_{eff}$ = Effective length of column.
$r_{min}$ = least radius of gyration

**67. The crippling load of a column with one end fixed and other end hinged is**

(a) $\sqrt{2}$ time that of a both ends hinged column
(b) Two times that of a both ends hinged column
(c) four times that of a both ends hinged column
(d) Eight times that of a both ends hinged column

**Ans. (b)**

| End Conditions | Effective length |
|---|---|
| Both end hinged | L |
| One fixed other free | 2L |
| Both end fixed | L/2 |
| One end fixed and other hinged | $L/\sqrt{2}$ |

(i) Crippling load of a column with one end fixed and other end hinged.

$$P_1 = \frac{\pi^2 EI}{(L/\sqrt{2})^2} = \frac{2\pi^2 EI}{L^2} \quad \ldots\ldots\ldots\ldots (i)$$

(ii) Crippling load of a column both end hinged

$$P_2 = \frac{\pi^2 EI}{L^2} \quad \ldots\ldots\ldots\ldots (ii)$$

$$P_2 = 2 \times P_1$$

**68. The formula given by I.S. code in calculating allowable stress for the design of eccentrically loaded columns is base on**

(a) Johnson's parabolic formula
(b) Straight line formula
(c) Perry's formula
(d) Secant formula

**Ans. (c) :** The formula given by I.S. code in calculating allowable stress for the design of eccentrically loaded columns is base on perry's formula.

**69. The Rankine constant ($\alpha$) in Rankine's formula is equal to**

(a) $\frac{\pi^2 E}{\sigma_C}$ (b) $\frac{\sigma_C}{\pi^2 E}$

(c) $\frac{\pi^2}{\sigma_C E}$ (d) $\frac{E\sigma_C}{\pi^2}$

**Ans. (b) Rankine's formula :-**

$$\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_e}$$

$P_R$ = Rankine load

$P_c = \sigma_c \times A$ = crushing load

$$P_R = \frac{\sigma_c \times A}{1 + \frac{\sigma_c}{\pi^2 E}\left(\frac{Le}{K}\right)^2}$$

A = Area of column

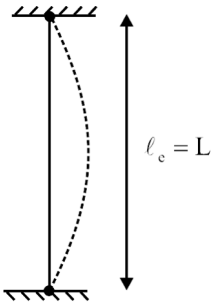
$\alpha = \frac{\sigma_c}{\pi^2 E}$ = Rankin's Constant.

**70. When both ends of the column are pinned, then the formula for crippling load (P) is equal to**

(a) $P = \frac{\pi^2 EI}{\ell^2}$ (b) $P = \frac{4\pi^2 EI}{\ell^2}$

(c) $P = \frac{2\pi^2 EI}{\ell^2}$ (d) $P = \frac{\pi^2 EI}{\ell^2}$

**Ans. (a) :** $P_{cr} = \frac{\pi^2 EI}{\ell_e^2}$




$P_{cr} = \frac{\pi^2 EI}{L^2}$ or $\frac{\pi^2 EI}{\ell_e^2}$

**71. In Rankine's formula, the material constant for mild steel is**

(a) $\frac{1}{9000}$ (b) $\frac{1}{5000}$

(c) $\frac{1}{1600}$ (d) $\frac{1}{7500}$

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**Ans. (d)**

| Material | Value of 'a' |
|---|---|
| Wrought iron | 1/9000 |
| Cast iron | 1/1600 |
| Mild Steel | 1/7500 |
| Timber | 1/750 |

**72. If the flexural rigidity of the column is doubled, then the strength of the column is increased by**

(a) 16 (b) 8

(c) 2 (d) 4

**Ans. (c)** Case 1st :-

$$\text{Strength of column } (P_1) = \frac{\pi^2 EI}{L^2}$$

Case 2st :-

Flexural rigidity of column is doubled.

$$\text{strength of column } (P_2) = \frac{\pi^2 \times 2EI}{L^2}$$

$$P_2 = 2 \times P_1$$

In the flexural rigidity of the column is doubled, then the strength of the column is increased by two times.

**73. The diameter of the core for no tension in a column of diameter 120 mm is**

(a) 30 mm (b) 15 mm

(c) 40 mm (d) 20

**Ans. (a) :** Diameter of the core for no tension.

$$\frac{D}{4} \Rightarrow \frac{120}{4} = 30\,\text{mm}$$

**74. The least radius of gyration for solid circular column is**

(a) d (b) $\frac{d}{2}$

(c) $\frac{d}{4}$ (d) $\frac{d}{3}$

**Ans. (c) :** For circular section $I = \frac{\pi}{64} d^4, A = \frac{\pi}{4} d^2$

$$\text{Radius of gyration } = k = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{\pi}{64} d^4}{\frac{\pi}{4} d^2}} = \frac{d}{4}$$

**75. In a mild steel tube 4 m long, the flexural rigidity of the tube is $1.2\times 10^{10}$ N/mm$^2$. The tube is used as a strut with both ends hinged. The crippling load in kN is given by**

(a) 14.80 (b) 7.40

(c) 29.60 (d) 1.85

**Ans. (b)** Tube length = 4 m

Flexural rigidity (E×I) = $1.2 \times 10^4$ N – m$^2$

End condition = Both ends hinged

Effective length = $l$

$$\text{Crippling load } = \frac{\pi^2 EI}{\ell^2}$$

$$P_e = \frac{\pi^2 \times 1.2 \times 10^4}{4^2}$$

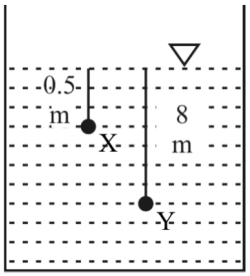
$$\boxed{P_e = 7.40 \text{ kN}}$$

**76. The ratio of pressures between two points X and Y located respectively at depths of 0.5 m and 8 m below water level in a tanks is**

(a) 1: $\sqrt{2}$ (b) 1 : 2

(c) 1 : 8 (d) 1 : 16

**Ans. (d) :**




$P = \rho g h$

$P \propto h$

$$\frac{P_x}{P_y} = \frac{0.5}{8} = \frac{1}{16}$$

1 : 16

**77. The hydrostatic pressure in kgf exerted on one side of an angular area enclosed by concentric circles of radii 2 m and 1m, and having its concentric 4 m below water surface is**

(a) $10000\pi$ (b) $11000\pi$

(c) $12000\pi$ (d) $24000\pi$

**Ans. (c) :** Area = $\frac{\pi}{4}\left(4^2 - 2^2\right)$

$$\text{Area} = \frac{12\pi}{4}\text{m}^2$$

$\bar{x} = 4$m

Hydraulic pressure (P) $= \rho g A \bar{x}$

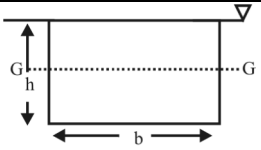
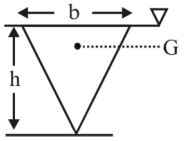
$$= 1000 \times \frac{12\pi}{4} \times 4 = 12000\ \pi\text{N}$$

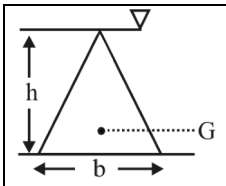
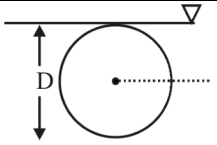
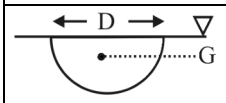
**78. The depth of the centre of pressure of a vertical semi- circular plane of diameter 'd' submerged in a liquid, with diameter located at the free surface is**

(a) $\frac{\pi d}{12}$ (b) $\frac{\pi d}{32}$

(c) $\frac{\pi d}{64}$ (d) $\frac{3\pi d}{32}$

**Ans. (d)**

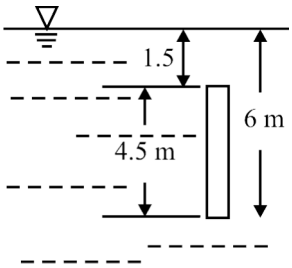
| Surface | C.G. ($\bar{x}$) | C.P ($\bar{h}$) |
|---|---|---|
|   | $\frac{h}{2}$ | $\frac{2h}{3}$ |
|   | $\frac{h}{3}$ | $\frac{h}{2}$ |

| | | |
|---|---|---|
|  | $\frac{2h}{3}$ | $\frac{3h}{4}$ |
|  | $\frac{D}{2}$ | $\frac{5D}{8}$ |
|  | $\frac{2D}{3\pi} = \frac{4r}{3\pi}$ | $\frac{3\pi D}{32}$ |

**79. A vertical rectangular plane surface is submerged in water with its top surface and bottom surface at 1.5 m and 6.0 m below the free water surface. The depth of centre of pressure below the free water surface is**

(a) 40.0 m (b) 4.5 m
(c) 4.375 m (d) 4.2 m

**Ans. (d) :**




$$\bar{h}_{cp} = \bar{h} + \frac{I_{GG}\sin_2\theta}{A\bar{h}} \left\{\theta = 90^0\right.$$

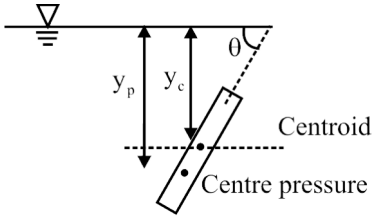
$$= 3.75 + \frac{1\times(4.5)^2}{12\times4.5\times1\times3.75}$$

$= 4.2$ m

**80. In an inclined plane submerged in water, the centre of pressure is located**

(a) at the concentric
(b) below the concentric
(c) above the concentric
(d) anywhere is the plane

**Ans. (b) :** centre of pressure is located below the concentric.

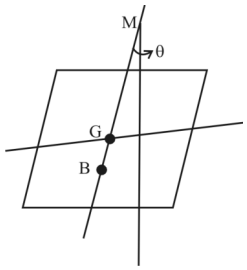



Here,
$y_c$ = Centroid from the liquid surface
$y_p$ = Centre of pressure from liquid surface.

**81. When a body floating in a liquid is given a small angular displacement, it stands oscillating about a point known as**

(a) centre of pressure (b) centre of gravity
(c) centre of buoyancy (d) metacentre

**Ans. (d) :** If the body is tilled by angle θ in clockwise direction the amount of liquid displaced became more on right side of centre line and becomes less on left side of center line . So centre of buoyancy shifts to the right to position B. This line cuts the original center line at MM is called meta centre.




• Thus meta center is the point of intersection of lines of action of buoyant force before and after rotation.

**82. A vertical triangular plane area, submerged in water, with one side coinciding the free surface vertex downward, with altitude 'h' has the centre of pressure below the free surface by**

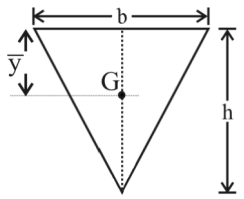
(a) $\frac{h}{4}$ (b) $\frac{h}{3}$
(c) $\frac{2h}{3}$ (d) $\frac{h}{2}$

**Ans. (d) :** Depth of center of pressure

$$h^* = \frac{I_G.\sin^2\theta}{A\bar{y}} + \bar{y} \qquad \bar{y} = \frac{h}{3}$$

$\theta = 90^0$

$$h^* = \frac{I_G}{A\bar{y}} + \bar{y}$$




$$h^* = \frac{\frac{bh^3}{36}}{\left(\frac{1}{2}bh\right)\left(\frac{h}{3}\right)} + \frac{h}{3}$$

$$h^* = \frac{h}{3} + \frac{h}{6}$$

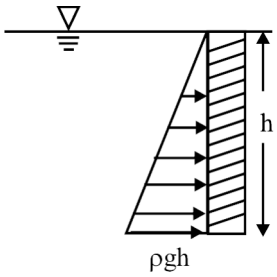
$$\boxed{h^* = \frac{h}{2}}$$

**83. A vertical wall is subjected to a pressure due to a liquid on one of its sides. the total pressure on the wall per unit length is**

(a) wh (b) $\frac{wh^2}{2}$

(c) $\frac{wh}{2}$ (d) $\frac{wh^2}{3}$

**Ans. (b) :**




Total pressure

$= \frac{1}{2} \times \rho g h \times h$

$= \frac{1}{2} \rho g h^2$

$= \frac{1}{2} wh^2$

**84. A vertical gate closes a horizontal tunnel 5 m high and 3 m wide running full with water, The pressure at the bottom of the gate is 196.2 kN/m$^2$. the total pressure on the gate is**

(a) 2.0 MN (b) 2.575 MN

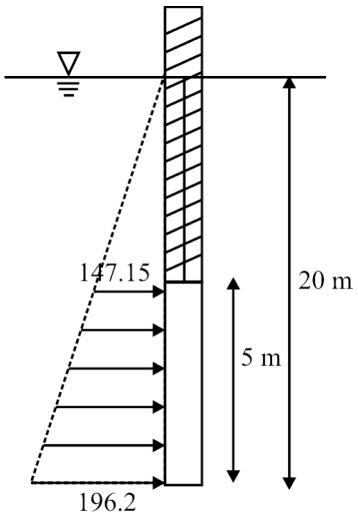
(c) 5.525 MN (d) 1.75 MN

**Ans. (b) :**

$P = \rho g h$

$h = \frac{P}{\rho g}$

$h = \frac{196.2}{9.81} = 20 \text{ m}$




Total pressure on gate = $\frac{147.15 + 196.2}{2} \times 5 \times 3$

= 2.575 MN

**85. A body floating in a liquid is said to be in neutral equilibrium, if its metacentre**

(a) coincides with its centre of gravity

(b) lies above its centre of gravity

(c) lies blow its centre of gravity

(d) lies below the centre of buoyancy and centre of gravity

**Ans. (a) :** If a body when given a small angular displacement, occupied a new position and remains at rest in this new position its is said to possess a neutral equilibrium.

• For neutral equilibrium, the position of meta centre M coincide with the centre of gravity G.

**86. A flow whose stream line is represented by a curve is called**

(a) coincides with its centre of gravity

(b) lies above its centre of gravity

(c) lines below its centre of gravity

(d) lines below the centre of buoyancy and centre of gravity

**Ans. (b)** A flow whose stream line is represented by a curve is called two dimensional flow.

(i) A flow whose stream line is represented by a line is called one dimensional flow.

$u = f(x), v = 0 \text{ and } w = 0$

(ii) for a steady two dimensional flow, the fluid parameter are function of two space co-ordinates.

$u = f_1(x,y)$, $v = f_2(x,y)$ and $w = 0$

(iii) For a steady three dimensional flow, the fluid parameter are function of three space co-ordinates.

$u = f_1(x,y,z), v = f_2(x,y,z) \text{ and } w = f_3(x,y,z)$

**87. The Bernoulli constants for points lying on the same stream line and those which lie on other stream line will have the same value, if the flow is**

(a) incompressible (b) steady

(c) irrotational (d) uniform

**Ans. (c) :** The Bernoulli constants for points lying on the same stream line and those which lie on other stream line will have the same value, if the flow is irrotational.

**88. The Beronoulli's equation written in conventional form represents total energy per unit of a certain quantity, Identify this quantity.**

(a) energy per unit volume

(b) energy per unit mass

(c) energy per unit weight

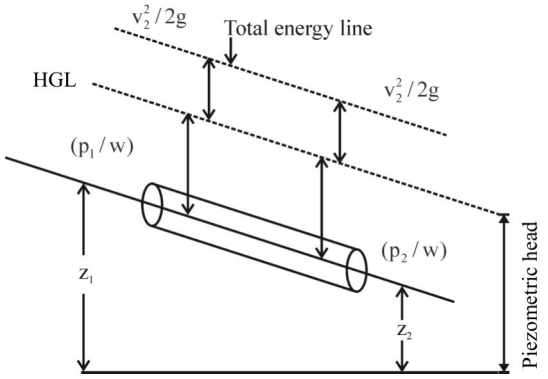
(d) energy per unit specific weight

**Ans. (c) :** The Beronoulli's equation written in conventional form represents total energy per unit of a certain quantity, Identify this quantity energy per unit weight.

**89. The total energy line is always higher than the hydraulic grade line, and the vertical distance between the two represents**

(a) the datum head
(b) the pressure head
(c) the velocity head
(d) the piezometric head

**Ans. (c) :**




Total energy line $= \left(z + \frac{v^2}{2g} + \frac{p}{w}\right)$

Piezometric Head $= \left(z + \frac{p}{w}\right)$

**90. The total energy represented by the Bernoulli's equation $\left(\frac{v^2}{2g} + \frac{p}{y} + z\right)$ has the units**

(a) N-m/m (b) N-m/N
(c) N-m$^2$/s (d) N-m/s

**Ans. (b) :** The total energy represented by the Bernoulli's equation $\left(\frac{v^2}{2g} + \frac{p}{y} + z\right)$ has the units N-m/N.

$\frac{V^2}{2g}$ = Velocity head

$\frac{P}{y}$ = Pressure head

Z = Datum head

**91. The kinetic energy correction factor a is a measure of effect of non- uniform distribution of velocity which is caused on account of viscous and other resistance It is expressed by**

(a) $\frac{1}{A}\int_A \left(\frac{u}{V}\right)^2 dA$ (b) $\frac{1}{A}\int_A \left(\frac{u}{V}\right) dA$

(c) $\frac{1}{A}\int_A \left(\frac{u}{V}\right)^2 dA$ (d) $\frac{1}{A}\int_A \left(\frac{u}{V}\right)^3 dA$

**Ans. (d)** Kinetic energy correction factor $(\alpha)$ :-

$$\alpha = \frac{\text{K.E / sec based on actual velocity}}{\text{K.E / sec based on average velocity}}$$

$$\alpha = \frac{1}{A}\int_A \left(\frac{v}{V}\right)^3 .dA$$

Kinetic energy correction factor for laminar flow through a circular pipe is 2 and for turbulent flow it is 1.02 to 1.15

**92. A stagnation point is a point where**

(a) pressure is zero
(b) total energy is zero
(c) total energy is maximum
(d) velocity of flow reduces to zero

**Ans. (d) :** A stagnation point is a point where velocity of flow reduces to zero.
Pitot tube is based on principal of conversion of kinetic head into pressure head. The point at which velocity reduces to zero is called stagnation point.

**93. Cavitation in fluid flow occurs when**

(a) the total energy suddenly increases
(b) total energy decrease suddenly
(c) velocity head reduces to zero
(d) Pressure of flow decreases to a value close to its vapour pressure

**Ans. (d) :** Cavitation is defined as the phenomenon of formation of vapour bubbles of a flowing liquid in a region where the pressure of the liquid falls below its vapour pressure.

**94. The momentum correction factor β is used to account for**

(a) change in pressure
(b) change in mass rate of flow
(c) change in total energy
(d) non- uniform distribution of velocities at inlet and outlet sections

**Ans. (d) Momentum correction factor $(\beta)$–** It is defined as the ratio of momentum/sec based on actual velocity to the momentum/sec based on average velocity.

$$\beta = \frac{1}{AV^2}\int u^2 dA$$

V = Average velocity
u = local velocity at distance r
For laminar flow $(\beta) = 1.33$
For uniform flow $(\beta) = 1$
Momentum correction factor is less for turbulent flow, because there is not much variation in velocity across the depth in turbulent flow.

**95. The change in moment of momentum of fluid due to flow along a curved path results is**

(a) a dynamic force which passes through the centre of curvature
(b) a torque
(c) a change in energy
(d) a change in pressure

**Ans. (b) :** The change in moment of momentum of fluid due to flow along a curved path results in a torque.

$$\boxed{\frac{d}{dt}(mvr) = T}$$

**96. The velocity head representing the kinetic energy per unit weight of fluid is denoted by**

(a) V2 (b) $\frac{V^2}{2}$

(c) $\sqrt{2gh}$ (d) $\frac{V^2}{2g}$

**Ans. (d) :** $Z+\frac{P}{\rho g}+\frac{V^2}{2g}=c$

Z = Datum head

$P/\rho g$ = Pressure head

$V^2/2g$ = velocity head

$$\text{Velocity head} = \frac{\text{Kinetic energy}}{\text{weight}}$$

**97. A Prandtl type pitot tube is used to measure the**

(a) velocity of flow at the required point in a pipe

(b) pressure difference between two points in a pipe

(c) total pressure of liquid flowing in a pipe

(d) discharge through a pipe

**Ans. (a)** A prandtl type pitot tube is used to measure the velocity of flow at the required points in a pipe.

**98. While using the pitot-tube, it must ensured that its alignment is such that**

(a) its horizontal leg is at right angles to the direction of flow

(b) its opening faces the downstream direction

(c) its opening faces upstream and the horizontal leg is perfectly aligned with the direction of flow

(d) the horizontal leg be inclined at $45^0$ in plan

**Ans. (c)** While using the pitot-tube, it must ensured that its alignment in such that its opening faces upstream and the horizontal leg is perfectly aligned with the direction of flow.

**99. The coefficient of discharge '$C_d$' of a venturimeter lies within the limits**

(a) 0.7-0.9 (b) 06.-0.8

(c) 0.75- 0.95 (d) 0.95- 0.99

**Ans. (d) :**

| Instrument | Use | $C_d$ |
|---|---|---|
| Venturimeter | Discharge measurement | 0.95-0.99 |
| Orifice meter | Discharge measurement | 0.65 |
| Pitot tube | Velocity measurement | 0.98-0.99 ($C_v$) |

**100. When the venturimeter is inclined, then for a given flow it will give the reading us**

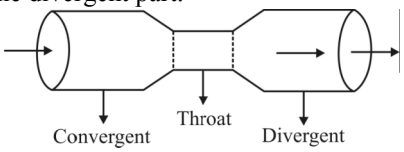
(a) same (b) more

(c) less (d) no relation

**Ans. (a) :** The venturimeter is inclined, then for a given flow it will give the reading us same.

**Venturimeter–**

- A venturimeter is a device used for measuring the rate of flow of a fluid of a liquid flowing through a pipe.
- The venturimeter always have a smaller convergent portion and larger divergent portion.
- This is done to ensures a rapid converging passage and a gradual diverging passage in the direction of flow to avoid the loss of energy due to separation.
- In coarse of flow through the converging part, the velocity increases in the direction of flow according to the principle of continuity, while the pressure decreases according to Bernoulli's theorem.
- The velocity reaches its maximum value and pressure reaction its minimum value at the throat.
- Subsequently a decrease in the velocity and an increase in the pressure take place in course of flow through the divergent part.




$Q = AV$

$\left[A \propto \frac{1}{V}\right]$

**101. The velocity of liquid flowing through the divergent portion of a venturimeter**

(a) remains constant (b) decreases

(c) increases (d) no relationship

**Ans. (b) :** The velocity of liquid flowing through the divergent portion of a venturimeter decrease because cross-section area gradually increases.

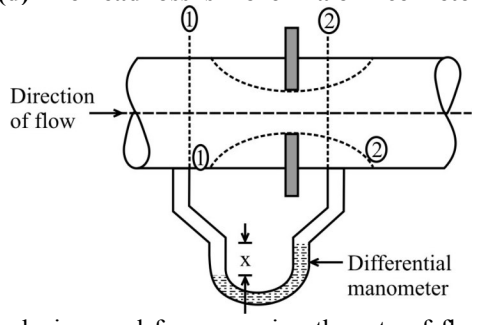
$$Q = AV, \quad V \propto \frac{1}{A}$$

**102. The head loss is more in a**

(a) nozzle meter (b) venturimeter

(c) inclined venturimeter (d) orifice meter

**Ans. (d) The head loss is more in a orifice meter :–**




It is a device used for measuring the rate of flow of a fluid through a pipe.

$$\text{Discharge}(Q) = \frac{C_d a_0 a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_0^2}}$$

$C_d$ = Co-efficient of discharge for orifice meter

The co-efficient of discharge for orifice meter is much smaller than that for a venturimeter.

**103. If successive measurements with a prandtl-pitot tube indicate that the tip piezometer reading varies only across the flow, and the side piezometer reading varies only in the direction of flow, the flow is evidently**

(a) uniform, irrotational
(b) non- unifrom, rotational
(c) uniform, rotational
(d) non- uniform, irrotational

**Ans. (b) :** For flow measurement with prandtl-pitot tube showed that tip reading varies only across the flow while the side opening varied only along the flow. The type of flow is Non Uniform rotational flow.

**104. A mechanical device, which has rotation elements, the speed of rotation of which is function of velocity of flow, and which is used to measure the velocity of liquid in open channels, is**

(a) Anemometer (b) Orifice meter
(c) Current meter (d) Rotameter

**Ans. (c) Current meter–** The current meter is a mechanical device used to measure. The velocity of water in river and in open channel.

- The current meter consist of hallow hemisphere or cups mounted on spokes. So as to cause rotation about a shaft perpendicular to the direction of flow.

A mechanical device, which has rotation elements, the speed of rotation of which is a function of velocity of flow, and which is used measure the velocity of liquid in open channels is current meter.

**105. A mouthpiece and an orifice, both of the same diameter 'd', are discharging under the same head 'H'. The discharge through the mouthpiece will be**

(a) the same as that of the orifice
(b) less than that of the orifice
(c) more than that of the orifice
(d) no relationship

**Ans. (c) :** A mouthpiece and an orifice, both of the same diameter 'd', are discharging under the same head 'H'. The discharge through the mouthpiece will be more than that of the orifice.

$Q = C_d a\sqrt{2gh}$, $C_d = 0.82$

**106. As compared to a rectangular weir, a triangular weir measures low discharges more accurately, the discharge through the latter being**

(a) $\frac{2}{3}C_d\sqrt{2g}\tan\theta.H^{5/2}$

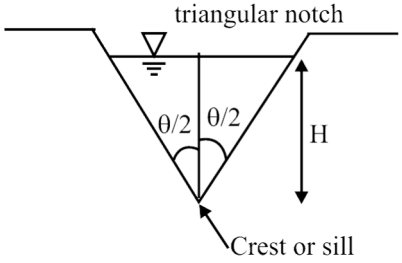
(b) $\frac{8}{15}C_d\sqrt{2g}\tan\frac{\theta}{2}.H^{5/2}$

(c) $\frac{8}{15}C_d\sqrt{2g}\tan\theta.H^{3/2}$

(d) $\frac{2}{3}C_d\sqrt{2g}\tan\theta.H^{3/2}$

**Ans. (b) :**

V-Notch or triangular weir or triangular notch




$Q = \frac{8}{15}C_d\sqrt{2g}\tan\frac{\theta}{2}H^{5/2}$

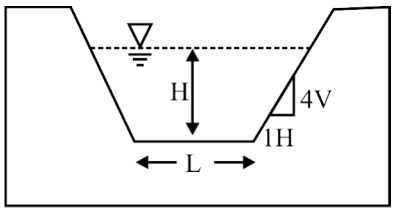
% error in discharge measurement =

$$100\times\frac{dQ}{Q} = \frac{5}{2}\frac{dH}{H}\times 100$$

**107. A Cippoletti weir is a**

(a) rectangular weir with sharp edges
(b) high triangular notch
(c) trapezoidal notch with $45^0$ slopes
(d) trapezoidal notch with sides inclined at 1 H : 4V

**Ans. (d) :** A Cippoletti weir is a trapezoidal notch with sides inclined at 1 H : 4V.



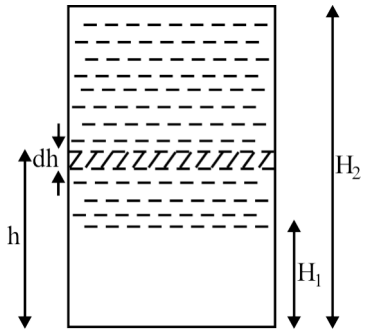

$Q = \frac{2}{3}C_d L\sqrt{2g}H^{3/2}$

$C_d = 0.63$

**108. The time taken for a tank, filled to a height 'h' above its flat base, to empty through an orifice in the base varies as the following power of 'h**

(a) 1 (b) $\frac{1}{2}$
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

**Ans. (b) :** Time of emptying a task through an orifice at its Bottom.



$$T = \frac{2A\left[\sqrt{H_1} - \sqrt{H_2}\right]}{C_d a\sqrt{2g}}$$

For empty the tank completely $H_2 = 0$

$$T = \frac{2A\sqrt{H_1}}{C_d.a.\sqrt{2g}}$$

$$T = \frac{2A}{a\sqrt{2g}}\sqrt{h}$$

$$T \propto h^{1/2}$$

**109. The equation of state for a perfect gas is**

(a) $\frac{P}{V} = RT$ (b) $\frac{P}{T} = \frac{V}{R}$

(c) $PV = RT$ (d) $\frac{P}{R}T$

**Ans. (c) :** The equation of state for a perfect gas is $PV = RT$.

**110. If the compression or expansion of a gas takes place in such a way that the gas neither gives heat nor takes heat from its surrounding , the process is said to be**

(a) Isothermal (b) Adiabatic

(c) Isobaric (d) None of these

**Ans. (b) :** If the compression or expansion of a gas takes place in such a way that the gas neither gives heat nor takes heat from its surrounding, the process is said to be adiabatic.

**111. For an adiabatic process,**

(a) $PV$ = a constant (b) $PV^n$ = a constant

(c) $\frac{P}{pK}$ = a constant (d) $\frac{P}{K^p}$ = a constant

**Ans. (c) :** For an adiabatic process,

$$\frac{P}{pK} = \text{a constant}$$

**112. Momentum equation is completely independent of**

(a) compressibility effects

(b) frictional effects

(c) viscous effects

(d) momentum flax

**Ans. (c)** Momentum equation is completely dependent of

(i) Compressibility effects

(ii) Frictional effects

(iii) Momentum flux

Von Karman momentum equation for boundary layer flow– $\frac{\tau_0}{\rho U^2} = \frac{d\theta}{dx}$

**219. The velocity of elastic pressure wave in a fluid medium is equal to**

(a) sonic velocity

(b) half of sonic velocity

(c) square root of sonic velocity

(d) Bulk Modulus

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**Ans. (a)**

Mach number

$$= \frac{\text{Velocity of fluid or body moving in fluid}}{\text{Velocity of sound in the fluid}}$$

The velocity of elastic pressure wave in a fluid medium is equal to sonic velocity. (i.e mach number = 1)

**114. Mach number is given by**

(a) $\frac{\text{Acoustic speed}}{\text{Stream speed}}$

(b) $\frac{\text{Stream speed}}{\text{Acoustic speed}}$

(c) product of gas constant and temperature

(d) half of Bulk Modulus

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**Ans. (b)** Mach number = $\frac{\text{Stream speed}}{\text{Acoustic Speed}}$

Subsonic flow :- $M < 1$

Sonic flow :- $M = 1$

Supersonic flow : - $M > 1$

Hypersonic flow : - $M > 6$

**115. Mach cone is possible in**

(a) Stationary fluids (b) Subsonic flow

(c) Transonic flow (d) Supersonic flow

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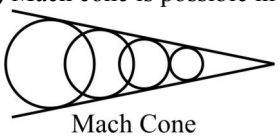
**Ans. (d)** Mach cone is possible in Supersonic flow.



Mach Cone

Mach angle $\theta = \sin^{-1}\left(\frac{1}{M}\right)$

**116. The range of Mach number for a subsonic flow is**

(a) $0 < M < 1$ (b) $0.3 < M < 1$

(c) $0.8 < M < 1.2$ (d) $M > 1$

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**Ans. (b)**

(i) Subsonic flow - $M < 1$

(ii) Sonic flow - $M = 1$

(iii) Supersonic flow - $M > 1$

(iv) Hypersonic flow - $M > 6$

M = Mach number

**117. The differential form of continuity equation for one dimensional steady flow compressible fluids with usual terms is**

(a) $\frac{d\rho}{\rho} + \frac{dA}{A} = 0$ (b) $\frac{dA}{A} = \frac{d\rho}{\rho} + \frac{dV}{V}$

(c) $\frac{dA}{\rho} = \frac{dV}{V} + \frac{d\rho}{\rho}$ (d) $\frac{dA}{A} = \frac{-d\rho}{\rho} - \frac{dV}{V}$

**Ans. (d) Continuity equation :-** This is based on law of conservation of mass which states that matter cannot be created nor destroyed.

- For one-dimensional steady flow.

the mass per second = $\rho AV$.

$\rho$ = mass density. A = Area of cross - section, V = velocity

$$\rho AV = \text{Constomt}$$

$$d(\rho AV) = 0$$

$$\rho d(AV) + AVd\rho = 0$$

$$\rho[A.dV + V.dA] + AVd\rho = 0$$

$$\rho AdV + \rho VdA + AVd\rho = 0$$

dividing by $\rho AV$

$$\boxed{\frac{dV}{V} + \frac{dA}{A} + \frac{d\rho}{\rho} = 0}$$

**118. For flow in a nozzle discharging from a tank, "choking" condition occurs, when the flow at the nozzle exit is**

(a) subsonic (b) supersonic
(c) critical (d) transonic

**Ans. (c)** For flow in a nozzle discharging from a tap choking condition occurs, when the flow the nozzle exit is critical.

For maximum flow through orifice or nozzle fitted to the tank, pressure ratio $= \frac{P_2}{P_1}; n = 0.528$.

**119. Effect of compressibility of a fluid can be neglected if Mach number is**

(a) equal to 1
(b) greater than 1
(c) less than 1 but grate than 0.4
(d) less than 0.4

**Ans. (d) :** Effect of compressibility of a fluid can be neglected if Mach number is less than 0.4

Subsonic flow - $M < 1$
Sonic flow - $M = 1$
Supersonic flow - $M > 1$
Hypersonic flow - $M > 6$
M = Mach number

**120. Laminar flow through a circular tube studied experimentally by**

(a) Newton (b) Pascal
(c) Hagen and Poiseuille (d) Prandtl

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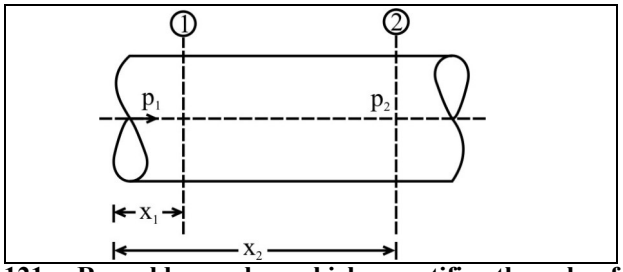
**Ans. (c)** Laminar flow through a circular tube studies experimentally by Hagen and poiseuille.

**Hagen poiseuille formula :-**

$$\frac{p_1 - p_2}{\rho g} = H_L = \frac{32\mu\bar{u}L}{\rho g D^2}.$$

$H_L$ = loss of pressure head




**121. Reynolds number which quantifies the role of viscous effect expressed as**

(a) $\frac{v}{\sqrt{gd}}$ (b) $\rho\frac{vd}{\mu}$
(c) $\rho\frac{v^2L}{\sigma}$ (d) $v\sqrt{\frac{\rho}{E}}$

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**Ans. (b)** Reynolds Number ($R_e$)

$$R_e = \frac{\text{Inertia force}}{\text{Viscous force}}$$

$$R_e = \frac{\rho vd}{\mu}$$

Here $\rho$= Density
d = Characteristics length
v = Average velocity

**122. The lower limit of the critical Reynolds number below which all disturbances in pipe flow are damped out by viscous action has a value approximately equal to**

(a) 1 (b) 500
(c) 1000 (d) 2000

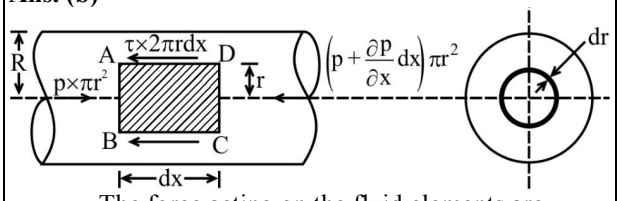
**Ans. (d)** The limiting values of Reynolds Number corresponding to which flow is laminar is given by

| Flow condition | Pipe flow | Open channel flow |
|---|---|---|
| Laminar flow | $R_e \le 2000$ | $R_e \le 500$ |
| Transitional flow | $2000 < R_e < 4000$ | $500 < R_e < 2000$ |
| Turbulent flow | $R_e > 4000$ | $R_e > 2000$ |

**123. The shear stress distribution in pipe flow is expressed by**

(a) $\tau = \frac{1}{r}\left(\frac{dp}{dx}\right)$ (b) $\tau = \left(\frac{dp}{dx}\frac{r}{2}\right)$
(c) $\tau = 2r\left(\frac{dp}{dx}\right)$ (d) $\tau = \frac{r2}{L}\left(\frac{dp}{dx}\right)$

**Ans. (b)**




The force acting on the fluid elements are

(i) The pressure force $p \times \pi r^2$ on face AB

(ii) The pressure force $\left(p+\frac{\partial p}{\partial x}.dx\right)\pi r^2$ on force CD

(iii) The shear force $\tau\times 2\pi r.dx$ on the surface of fluid elements.

$$p\times\pi r^2-\left(p+\frac{\partial p}{\partial x}.dx\right)\pi r^2-\tau\times 2\pi r.dx=0$$

$$\tau=\frac{-\partial p}{\partial x}\left(\frac{r}{2}\right)$$

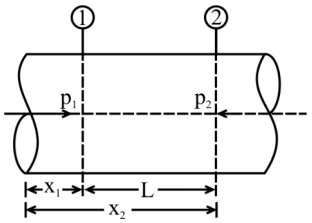
$$\tau=-\left(\frac{\partial p}{\partial x}\right)\frac{r}{2}$$

**124. The Hagen – Poiseuille equation which governs the velocity distribution in laminar flow through pipes may be expressed as**

(a) $Q=\frac{\pi d^4\Delta p}{128\mu L}$ (b) $Q=\frac{\pi d^2\Delta p}{128\mu L}$

(c) $Q=\frac{128\mu L}{\pi d^2\Delta p}$ (d) $Q=\frac{128\mu L}{\pi d^4\Delta p}$

**Ans. (a)**




Drop pressure for a given length (L) of a pipe :-

$$\frac{p_1-p_2}{\rho g}=\frac{32\mu\bar{u}L}{\rho gd^2}$$

$$\frac{p_1-p_2}{\rho g}=\frac{32\mu\bar{u}L}{\rho gd^2}$$

$$\frac{p_1-p_2}{\rho g}=\frac{32\mu\bar{u}L}{\rho gd^2}=\frac{128\mu QL}{\rho g\pi d^4}\left(\bar{u}=\frac{Q}{A}\right)$$

$$p_1-p_2=\frac{128\mu QL}{\pi d^4}$$

$$\boxed{Q=\frac{\Delta p\pi d^4}{128\mu L}}$$

**125. The pressure drop per unit length of pipe in laminar flow is equal to**

(a) $\frac{d^2}{32\mu V}$ (b) $\frac{32\mu VL}{\gamma d^2}$

(c) $\frac{32\mu V}{d^2}$ (d) $\frac{8\mu V}{d^2}$

**Ans. (c)** $\frac{p_1-p_2}{\rho g}=\frac{32\mu\bar{u}L}{\rho gd^2}$

$$p_1-p_2=\frac{32\mu\bar{u}L}{d^2}$$

For per unit pressure drop, L = 1

$$p_1-p_2=\frac{32\mu\bar{u}L}{d^2}\text{ OR }\frac{32\mu v}{d^2}$$

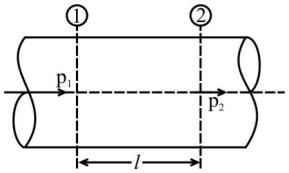
$\bar{u}=v=$ average velocity

**126. In laminar flow through a circular tube, the Darcy-Weisbach friction factor is related to the Reynolds number by**

(a) $f=\left(\frac{1}{R}\right)$ (b) $f=\left(\frac{16}{R}\right)$

(c) $f=\left(\frac{64}{R}\right)$ (d) $f=\left(\frac{0.316}{R^{1/4}}\right)$

**Ans. (c)**




Darcy- Weishbach equation

$$\frac{p_1-p_2}{\rho g}=h_f=\frac{f\,lv^2}{2gd}=\frac{32\mu vl}{\rho gd^2}$$

$h_f$ = friction loss in pipe.

$$\frac{flv^2}{2gd}=\frac{32\mu vl}{\rho gd^2}$$

$$f=\frac{64}{\left(\frac{\rho vd}{\mu}\right)}$$

$\frac{\rho vd}{\mu}=R_e$ (Raynold's Number)

$$\boxed{f=\frac{64}{Re}}$$

**127. The discharge in ($m^3$/s) for laminar flow through a pipe of diameter 0.04 m bearing a centerline velocity of 1.5 m/s is**

(a) $\frac{3\pi}{59}$ (b) $\frac{3\pi}{2500}$

(c) $\frac{3\pi}{5000}$ (d) $\frac{3\pi}{10000}$

**Ans. (d)**

$V=\bar{u}$

$u_{max}=1.5$ m/sec

$u_{max}=2V$

$V=\frac{1.5}{2}=0.75$ m/sec

Pipe diameter (d) = 0.04 m

Maximum velocity = 1.5 m/sec.

Average velocity = 0.75 m/sec.

Discharge (Q) = A × V

$$=\frac{\pi}{4}(0.04)^2\times 0.75$$

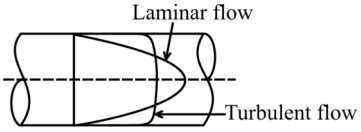
$$Q=\frac{3\pi}{10000}m^3/\text{sec}$$

**128. The most essential feature of a turbulent flow is**
(a) Large discharge
(b) High velocity
(c) Velocity and pressure at a point exhibit irregular fluctuations of high frequency
(d) Velocity at a point remains constant with time

**Ans. (c)** The most essential feature of a turbulent flow is velocity and pressure at a point exhibit irregular fluctuation of high frequency.
• Turbulent flow is that type of flow in which the fluid particles move in zig-zag way. Due to the movement of fluid particles in zig-zag way, the eddies formation take place which are responsible for high energy loss.

**1020. The velocity distribution in turbulent flow follows a**
(a) parabolic law (b) logarithmic law
(c) liner law (d) hyperbolic law

**Ans. (b)**




Velocity distribution in laminar and turbulent flow. The velocity distribution in turbulent flow follows a logarithmic law.

**1021. Steel and cast iron pipes carrying fluid under pressure are regarded as hydraulically smooth when**
(a) the boundary surface is relatively smooth
(b) the roughness projections are of low height
(c) the roughness elements are completely covered by the laminar sub-layer
(d) the laminar layer is thin as compared to the a average height of roughness elements

**Ans. (c)** Steel and cast iron pipes carrying fluid under pressure are regarded as hydraulically smooth when the roughness elements are completely covered by the laminar sub layer.

**131. The Darcy-Weisbach friction factor 'f' which is a direct measure of resistance to flow in pipes is dependent on**
(a) roughness height, diameter and velocity
(b) relative roughness, diameter and viscosity
(c) relative roughness velocity and viscosity
(d) relative height, diameter, velocity and kinematic viscosity

**Ans. (d)** Friction factor (f) = $\frac{64}{R_e}$

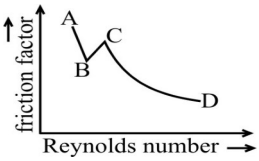
For laminar flow.

$$f = \frac{64 \times \mu}{\rho v d}, \quad f = \frac{64\nu}{VD} \qquad \left(\because \nu = \frac{\mu}{\rho}\right)$$

**Friction factor depend upon :-**
**(i)** Diameter
**(ii)** Velocity
**(iii)** Kinematic viscosity
**(iv)** Relative height.

**132. The parameters which determine the friction factor for turbulent flow in a rough pipe are**
(a) Froude number and relative roughness
(b) Froude number and Mach number
(c) Reynolds number and relative roughness
(d) Mach number and relative roughness

**Ans. (c) Friction factor :** Friction factor is dimensional terms it is depends two terms.




(i) Reynolds number
(ii) Relative Roughness of pipe surface

$$f = \phi(R_e, k/d)$$

**133. In case of turbulent flow of a fluid through a circular tube as compared to the case of laminar flow, at the same flow rate, the maximum velocity is _____shear stress at the wall is_____ and the pressure drop across a given length is____.**
(a) higher, higher, higher
(b) higher, lower, lower
(c) lower, higher, higher
(d) lower, higher, lower

**Ans. (c)** In case of turbulent flow of a fluid through a circular tube, as compared to the case of laminar flow, at the same flow rate the maximum velocity is lower, shear stress at the wall is higher and the pressure drop across a given length in higher.

| Laminar flow | Turbulent flow |
|---|---|
| (i) Flow is smooth | (i) Flow is Randomly |
| (ii) Energy loss $\propto$ velocity | (ii) Energy loss $\propto$ $(\text{velocity})^2$ |
| (iii) velocity is small | (iii) velocity high |
| (iv) $R_e < 2000$ | (iv) $Re > 4000$ |
| (v) High viscosity | (v) Low viscosity |

**134. The hydraulic efficiency of an impulse turbine is maximum, when the velocity of wheel is ..... that of the jet velocity.**
(a) one- fourth (b) one- half
(c) three- fourth (d) double

**Ans. (b) :** Hydraulic efficiency,

$$\eta_{hyd} = \frac{\text{Runner power}}{\text{Kinetic energy per second}}$$

$$\eta_{max.hyd} = \left(\frac{1 + k\cos\phi}{2}\right)$$

When wheel velocity = $\frac{\text{jet velocity}}{2}$

**135. The number of buckets on the periphery of a Pelton wheel is given by**

(a) $\frac{D}{2d}+5$ (b) $\frac{D}{2d}+10$

(c) $\frac{D}{2d}+15$ (d) $\frac{D}{2d}+20$

**Ans. (c) :** The number of bucket 'z' to be attached to the periphery of the wheel should be such that the water jet is fully utilized so that no water escapes without drinking the buckets.

$z = 0.5\ m + 15 \text{ or } \frac{D}{2d}+15 \quad \left\{ m = \frac{D}{d} \approx 11 \text{ to } 15 \right\}$

**136. In a reaction turbine, the draft tube is used**

(a) to run the turbine full
(b) to prevent air to enter the turbine
(c) to increase the effective head of water
(d) to transport water to downstream

**Ans. (c) :** Reaction turbine operate with their runner completely flooded and a draft tube is used to take the water exiting, from the turbine to the tail water.

- The draft tube maintains a column of water from the tail water and recovers energy from the velocity of the water existing the turbine.
- By the use of a draft the head of a reaction turbine extends from the surface of the upper reservoir to the surface of the tail water.

**137. In an inward flow reaction turbine**

(a) the water flows parallel to the axis of the wheel
(b) the water enters at the centre of the wheel and from there flows towards the outer periphery of the wheel
(c) the water enters the wheel at outer periphery and then flows towards the centre of the wheel
(d) the flow of water is partly radial and partly axial

**Ans. (c) :** Inward flow reaction turbine, in which water enters at the outer periphery and flows inward towards the centre of the turbine.

- Outward flow reaction turbine in which water enters at the inner periphery and flows outward towards to outer periphery.

**138. In a Kaplan turbine runner, the number of blades is generally**

(a) 2 to 4 (b) 4 to 8
(c) 8 to 16 (d) 16 to 24

**Ans. (b) :**

| Turbine | No. of vanes/blades |
|---|---|
| Pelton | $15\frac{M}{2}\left\{ \text{jet ratio}(m) = \frac{D}{d} = 11 \text{ to } 15 \right\}$ = 11 to 15 |
| Francis | 25 to 30 |
| Kaplan | 3 to8 |

**139. The power developed by a turbine is**

(a) Directly proportional to $H^{1/2}$
(b) Inversely proportional to $H^{1/2}$
(c) Directly proportional to $H^{3/2}$
(d) Inversely proportional to $H^{3/2}$

**Ans. (c) :** $N_u = \frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$

$N \propto H^{1/2}$

Speed $\propto H^2$ ..... (i)

$Q_u = \frac{Q_1}{\sqrt{H_1}} = \frac{Q_2}{\sqrt{H_2}}$

$Q \propto H^{1/2}$

Discharge $\propto H^2$.....(ii)

$P_u = \frac{P_1}{H_1^{3/2}} = \frac{P_2}{H_2^{3/2}}$

Power $\propto H^{3/2}$ ..... (iii)

**140. The specific speed of a turbine is given by**

(a) $\frac{N\sqrt{P}}{H^{3/2}}$ (b) $\frac{N\sqrt{P}}{H^{5/4}}$

(c) $\frac{N\sqrt{P}}{H^{2}}$ (d) $\frac{N\sqrt{P}}{H^{3}}$

**Ans. (b) :** $N_s = \frac{N\sqrt{P}}{H^{5/4}}$

$H \rightarrow$ in meter

$P \rightarrow$ kw (in SI)

mhp (in mks)

mhp $\rightarrow$ Metric horse power

1 mhp = 735.5 watt

**141. The turbine to be used for 450 m head of water is**

(a) Pelton wheel (b) Francis turbine
(c) Kaplan turbine (d) None of these

**Ans. (a) :**

| Type of turbine | Discharge | Head (m) | Example |
|---|---|---|---|
| High head | Low | >300 | Pelton turbine |
| Medium head | Medium | 60–300 | Francis turbine |
| Low head | High | < 60 | Kaplan and Propeller turbine |

**142. The cavitation is a hydranlic machine**

(a) cause noise and vibration of various parts
(b) makes the surface rough
(c) reduces the discharge of a turbine
(d) cause sudden drop in power output and efficiency

**Ans. (a) : Effects of Cavitation –**

- The metallic surface are damaged and cavities are formed on the surface.
- Due to sudden collapse of vapour bubble, considerable noise and vibration are produced.
- The efficiency of a turbine decrease due to cavitation.

**143. The specific speed of a turbine is speed of an imaginary turbine. identical with the given turbine, which**

(a) delivers unit discharge under unit load
(b) delivers unit discharge under unit speed
(c) develops unit H.P. under unit head
(d) develops unit H.P. under unit speed

**Ans. (c) :** The specific speed of a turbine is speed of an imaginary turbine. identical with the given turbine, which develops unit H.P. under unit head.

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

Here, $N_s$ = specific speed
H = head of water
P = Power
N = No. of rotation (in rpm)

**144. In a centrifugal pump, the liquid enters the pump**

(a) at the centre (b) at the top
(c) at the bottom (d) from sides

**Ans. (a) :** In practically all centrifugal pumps the liquid enters the pump at the centre through the eye and is a given a rotary motion in the pump chamber by the rotation of a number of blades (the impeller).

- The rotation of the impeller in a true centrifugal pump does two things to the liquid.

**145. Multistage centrifugal pumps are used to**

(a) give high discharge
(b) pump viscous fluids
(c) produce high heads
(d) None of these

**Ans. (c) :** Multistage centrifugal pumps are used to produced high head.

**Multistage centrifugal pumps :** If a centrifugal pump consists of two or more impeller, the pump is called a multistage centrifugal pump. The impellers may be mounted on the same shaft or on different shaft.

A multistage pump is having the following two important functions.

- To produced a high head.
- To discharge a large quantity of liquid

If a high head is to be developed, the impellers are connected in series, while for discharge large quantity of liquid, the impeller are connected in parallel.

**146. Theoretical power required to drive a reciprocal pump is**

(a) $\frac{WQH_s}{60}$ (b) $\frac{WQH_s}{75}$
(c) $\frac{WQH_d}{60}$ (d) $\frac{WQ(H_s + H_d)}{75}$

**Ans. (d) :** Theoretical power required to drive a reciprocal pump is $\frac{\rho gQH}{75}$

$$P = \frac{\rho gQH}{75}$$

$$P = \frac{WQH}{75}$$

$H = H_s + H_d$

**147. The specific speed of a centrifugal pump is given by**

(a) $\frac{N\sqrt{Q}}{H^{2/3}}$
(b) $\frac{N\sqrt{Q}}{H}$
(c) $\frac{N\sqrt{Q}}{H^{3/4}}$
(d) $\frac{N\sqrt{Q}}{H^{5/4}}$

**Ans. (c) : Specific speed–** It is the speed of a geometrical similar pump delivering unit discharge against unit head.

$$N_s = \frac{N\sqrt{Q}}{H^{3/4}}$$

**148. For centrifugal pump impeler, the maximum value of the vane exit angle is**

(a) $10^0$ to $15^0$
(b) $15^0$ to $20^0$
(c) $20^0$ to $25^0$
(d) $25^0$ to $30^0$

**Ans. (d) :** For centrifugal pump impeller, the maximum value of the vane exit angle is $25^0$ to $30^0$.

**149. Which of the following pumps is preferred for flood control and irrigation applications?**

(a) Centrifugal pump
(b) Mixed flow pump
(c) Axial flow pump
(d) Reciprocation pump

**Ans. (c) :** Axial flow pump is preferred for flood control and irrigation applications.

**150. In order to avoid cavitation in centrifugal pumps**

(a) the suction pressure should be high
(b) the delivery pressure should be high
(c) the suction pressure should be low
(d) the delivery pressure should be low

**Ans. (c) :** In order to avoid cavitation in centrifugal pumps the suction pressure should be low.

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 2012 (P-II)
# Solved Paper

**1. The limiting value of cant gradient for all gauges is**

(a) 1 in 360 (b) 1 in 720
(c) 1 in 1000 (d) 1 in 2000

**Ans. (b) :** Limiting value of various parameters concerning curves–

| Parameter | Limiting value | |
|---|---|---|
| | **BG** | **MG** |
| Maximum degree | $10^o$ | $16^o$ for MG and $40^o$ for NG |
| Maximum cant | Group A, B and C - 65 mm<br>Group D and E - 140 mm | 90 mm (100 mm with special permission of chief) engineer |
| Maximum cant deficiency | In normal cases – 75 mm<br>(in special cases, 100 mm for group A and B routes with a nominated rolling stock and with permission of the chief engineer) | 50 mm dia |
| Cant excess maximum cant gradient | 1 in 720 (in exceptional cases, 1 in 360 with permission of (E) | 65 mm<br>1 in 720 |
| Date of change of cant or cant deficiency | desirable -35 mm/ sec<br>maximum - 55 mm/sec | Desirable 35 mm/sec<br>maximum 35 mm/sec |
| Minimum cant deficiency in turnout | 75 mm | 50 mm |

**2. If 'R' is the radius of a circular curve, then the versine on a chord of length 'C' is given by**

(a) $\frac{C^2}{4R}$ (b) $\frac{C}{8R}$
(c) $\frac{C^2}{8R}$ (d) $\frac{C}{4R}$

**Ans. (c) :** length of chord $C = 2R\sin\frac{\Delta}{2}$

$$\sin\frac{\Delta}{2} = \frac{C}{2R} \text{.....(i)}$$

Versine on chord $= R\left[1-\cos\frac{\Delta}{2}\right]$

$$\cos^2\frac{\Delta}{2} = 1-\frac{C^2}{4R^2}$$

Versine on chord $\Rightarrow R\left[1--\sqrt{1-\frac{C^2}{4R^2}}\right]$

$$= R\left[1-\left(1-\frac{C^2}{4R^2}\right)^{1/2}\right]$$

$$= R\left[1-1+\frac{C^2}{8R^2}\right]$$

Expansion by binomial theorem versine on chord $= \frac{C^2}{8R}$

**3. The correct relation between curve lead (CL), switch lead (SL) and lead of crossing (L) is given by**

(a) CL = L – SL (b) SL = L + CL
(c) L = CL – SL (d) L = (CL + SL)/2

**Ans. (c) :** The relation between curve lead (CL), switch lead (SL) and lead of crossing (L); L = CL – SL

**4. The grade compensation on a 4º curve on a broad gauge railway track is**

(a) 0.20% (b) 0.16%
(c) 0.12% (d) 0.08%

**Ans. (b) :** Given : 4º curve
on broad gauge railway track, 0.04% per degree of curve (or) 70/R whichever is minimum.
Grade compensation = 0.04×4
= 0.16%

**5. A treadle bar is used for**

(a) interlocking points and signal
(b) setting points and crossings
(c) setting marshalling yard signals
(d) track maintenance

**Ans. (a) :** A treadle bar is used for interlocking points and signal.
• It is mechanical or electrical device to detect where the train wheel passed a particular location or not.

**6. The background color of the informatory sign board is**

(a) red (b) yellow
(c) green (d) white

**Ans. (c & d) :**
- The colour chosen for informatory or guide signs depend on the classification on the road. For national highway and state highway the informatory signs shall be of green background with the borders, legends and word massages in white colour.
- For all other categories of roads such as MDR, ODR and VR these signs shall have white background with border, legends and word message in black colour.

**7. Limiting values of Poisson's ratio are**

(a) -1 and 0.5 (b) 1 and -0.5

(c) 1 and -0.5 (d) 0 and 0.5

**Ans. (a) :** The limiting values of μ poison's ratio are –1 to 0.5. in other words 0.5 is the maximum value of the Poisson ratio.

$$\text{Poisson's ratio} = \frac{-\text{Lateral strain}}{\text{Longitudinal strain}}$$

**8. A beam of square cross-section with side 100 mm is placed with one diagonal, vertical. If the shear force acting on the section is 10 kN, the maximum shear stress is**

(a) $1\frac{N}{mm^2}$ (b) $1.125\frac{N}{mm^2}$

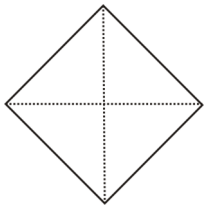
(c) $2\frac{N}{mm^2}$ (d) $2.25\frac{N}{mm^2}$

**Ans. (b):** Given:

Shear force = 10 kN

P = 10 kN = 10000N

Area (A) = 100 × 100 = 10000 $mm^2$



Max. shear stress $(z_{max}) = \frac{9}{8} z_{avg.}$

$$z_{max} = \frac{9}{8} \times \frac{P}{A}$$

$$= \frac{9}{8} \times \frac{10000}{10000}$$

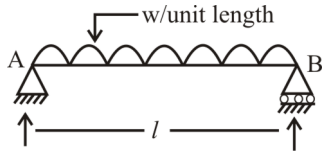
$$= \frac{9}{8} = 1.125 \frac{N}{mm^2}$$

**9. Slope at the end of the simply supported beam of span *l* with uniformly distributed load w/unit length over the entire span is given by**

(a) $\frac{wl^2}{16EI}$ (b) $\frac{wl^3}{16EI}$

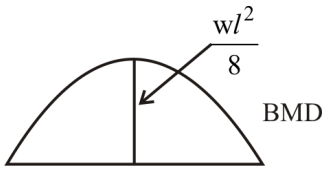
(c) $\frac{wl^3}{24EI}$ (d) $\frac{wl^2}{24EI}$

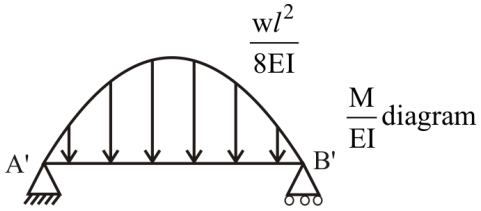
**Ans. (c) :**




$$R_A = \frac{w\ell}{2} \qquad R_B = \frac{w\ell}{2}$$

By conjugate method




$\theta_A = \theta_B$ = Area under moment curve.

$$\theta_A = \theta_B = \frac{1}{2} \times \frac{2}{3} \times l \times \frac{wl^2}{8EI}$$

$$\theta_B = \frac{wl^3}{24EI}$$

& Deflection = $\frac{5}{384} \frac{w\ell^4}{EI}$

**10. A linear arch has**

(a) normal thrust only

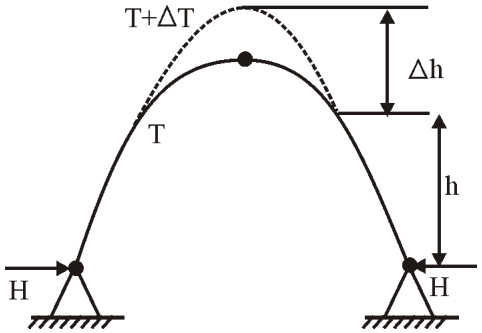
(b) shear force only

(c) normal thrust and shear force

(d) None of these

**Ans. (a) :** Linear arch is a compression member in which shear force and bending moment is zero throughout the length and it has only normal thrust.

Under this condition the arch is called funicular shape of arch because of no bending or shear force in arch.




$$\Delta h = \frac{L^2 + 4h^2}{4h} \propto \Delta t$$

**11. If a circular shaft is subjected to a torque "T" and moment 'M', the ratio of maximum bending stress and maximum shear stress is**

(a) $\frac{2M}{T}$ (b) $\frac{M}{2T}$

(c) $\frac{M}{T}$ (d) $\frac{2T}{M}$

**Ans. (a):** Given:
T = Torque
M = Moment
from bending equation

$$\sigma = \frac{M}{I} \times y$$

$$\sigma_{max} = \frac{M}{\frac{\pi}{64}d^4} \times \left(\frac{d}{2}\right)$$

$$\sigma_{max} = \frac{M}{\frac{\pi}{32}d^3}$$

from torsion eqn.

$$\frac{T}{J} = \frac{\tau}{R}$$

$$\tau_{max} = \frac{T}{\left(\frac{\pi}{32}d^4\right)} \times \left(\frac{d}{2}\right)$$

$$\tau_{max} = \frac{1}{2} \times \frac{T}{\left(\frac{\pi}{32}d^3\right)}$$

$$\therefore \frac{\sigma_{max}}{\tau_{max}} = \frac{2M}{T}$$

**12. If the diameter of a shaft subjected to torque alone is doubled, then horse power P can be increased to**

(a) 16P (b) 8P
(c) 4P (d) 2P

**Ans. (b) :** We know that, $P = \frac{2\pi NT}{60}$ and $\tau = \frac{16T}{\pi d^3}$

By both equation –

$$P = \frac{2\pi N}{60} \cdot \frac{\tau \pi d^3}{16} \qquad P_1 = P$$

So , $P \propto d^3$

$$\frac{P_2}{P_1} = \left(\frac{d_2}{d_1}\right)^3 \Rightarrow \frac{P_2}{P_1} = \left(\frac{2d}{d}\right)^3$$

$$P_2 = 8P_1 \Rightarrow P_2 = 8P$$

**13. A shaft turns at 150 rpm under a torque of 1500 Nm. Power transmitted is**

(a) $15\pi$ kW (b) $10\pi$ kW
(c) $7.5\pi$ kW (d) $5\pi$ kW

**Ans. (c) :** Given:
N = 150 rpm, T = 1500 Nm

$$\text{Power} = \frac{2\pi NT}{60}$$

$$= (2\pi \times 150 \times 1500) / (60 \times 100)$$

Power = $7.5\pi$ kW

**14. In a particular material, if the modulus of rigidity is equal to the bulk modulus, then the Poisson's ratio will be**

(a) $\frac{1}{8}$ (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) 1

**Ans. (a) :** Given:
G = K
E = 2G (1 + μ) .....(i)
E = 3K (1 – 2μ) .....(ii)
eq$^n$ (i) ÷ eq$^n$ (ii)

$$1 = \frac{2G(1+\mu)}{3K(1-2\mu)}$$

$$\mu = \frac{1}{8}$$

**15. The number of independent equations to be satisfied for static equilibrium of a plane structure is**

(a) 1 (b) 2 (c) 3 (d) 4

**Ans. (c) :** The number of independent equation to be satisfied for static equilibrium of a plane structure is three and for space structure is six.

**16. Castigliano's first theorem is applicable**

(a) for statically determinate structures only
(b) when the system behaves elastically
(c) only when principle of superposition is valid
(d) None of the above

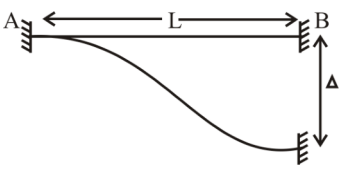
**Ans. (c) :** Condition for validity of super-position principle–

- Material should obey the Hook's law.
- Structure is not subjected to any temperature change.
- The structure is not analyzed for any support settlement.
- The castigliano's theorem is valid only when super position principle is valid.

**17. If one end of the prismatic beam AB with fixed ends is given a transverse displacement 'Δ' without any rotation, then the transverse reactions at A or B due to displacement is**

(a) $\frac{6EI\Delta}{l^2}$ (b) $\frac{6EI\Delta}{l^3}$
(c) $\frac{12EI\Delta}{l^2}$ (d) $\frac{12EI\Delta}{l^3}$

**Ans. (d) :**




At point A $M_{AB} = M_{BA} = \frac{6EI\Delta}{L^2}$

$-M_{AB} + R_B \times L = M_{BA}$

$$R_B = \frac{2M_{AB}}{L}$$

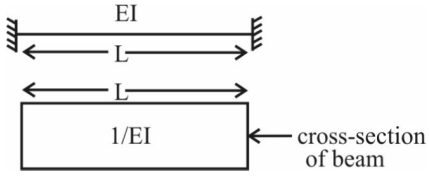
$$R_B = \frac{12EI\Delta}{L^3}$$

**18. In column analogy method, the area of analogous column for a fixed beam of span 'L' and flexural rigidity EI is taken as**

(a) $\frac{L}{EI}$ (b) $\frac{L}{2EI}$

(c) $\frac{L}{3EI}$ (d) $\frac{L}{4EI}$

**Ans. (a) :**




Area of cross-section beam $= L \times \frac{1}{EI}$

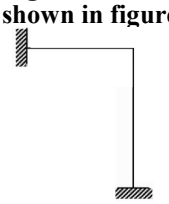
$= \frac{L}{EI}$

**19. The three moments equation is applicable only when**

(a) the beam is prismatic
(b) there is no settlement of supports
(c) there is no discontinuity such as hinges with the span
(d) the spans are equal

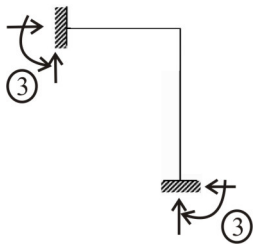
**Ans. (c) : There moment method's applicable –**
(1) To any type of C/S not limited to prismatic cross section.
(2) In case of support settlement, only extra moments generated due to support settlement is incorporated into the three moment equation.
(3) If there is no discontinuity within the span because discontinuity within span cannot produce a valid $\frac{M}{EI}$ diagram for the span.
(4) For different span length, not necessarily the spans should be equal.

**20. What is the degree of static indeterminacy of the structure shown in figure?**



(a) 1 (b) 2
(c) 3 (d) 4

**Ans. (c) :**




Degree of static indeterminacy = R – E
= 6 – 3 = 3

**21. The strain energy of a structure due to bending is given by**

(a) $\int \frac{M^2 dx}{EI}$ (b) $\frac{1}{2}\int \frac{M^2 dx}{EI}$

(c) $\int \frac{2M^2 dx}{EI}$ (d) $\frac{1}{3}\int \frac{M^2 dx}{2EI}$

**Ans. (b) :** $\sigma = \frac{M}{I} \times y$

for small element of beam strain energy

$$U = \frac{\sigma^2}{2E} \times V$$

$$U = \frac{1}{2E}\left(\frac{M}{I}.y\right)^2 da.dx$$

$$U_{total} = \int \frac{1}{2E}\left(\frac{M}{I}.y\right)^2 da.dx$$

$$= \frac{1}{2E}\frac{M^2}{I^2}.dx \int y^2.dx$$

$$= \frac{1}{2E}.\frac{M^2}{I^2}.dx.I \qquad I = \int y^2.da$$

$$U_{total} = \frac{M^2}{2EI}.dx$$

for entire span of beam, strain energy

$$U = \int_o^l \frac{M^2}{2EI}.dx$$

**22. In the displacement method of structural analysis, the basic unknowns are**

(a) displacements
(b) force
(c) displacements and force
(d) None of the above

**Ans. (a) :** Displacement method as also called equilibrium / stiffness method displacement ($\Delta$, $\theta$) are considered as unknown.
Example are :
- Slope deflection method
- Moment distribution method
- Kani's method
- Equilibrium method
- Stiffness matrix method

**23. The deformation of a spring produced by a unit load is called**

(a) Stiffness (b) Flexibility
(c) Unit strain (d) None of these

**Ans. (b) :** Flexibility is the inverse of stiffness–

$$\left[\text{stiffness} \propto \frac{1}{\text{flexibility}}\right]$$

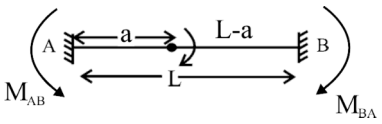
- Stiffness of a spring is defined as force required to cause unit deflection.
- Flexibility of a spring is defined as deflection produced by an unit load.

**24. A fixed beam AB of span L is subjected to a clockwise moment M at a distance 'a' from end 'A'. Fixed end moment at end 'A' will be**

(a) $\frac{M}{L^2}(L-a)(L-3a)$ (b) $\frac{M}{L^2}a(2L-3a)$

(c) $\frac{M}{L^2}(L-a)$ (d) $\frac{M}{L^2}(L-a)(2L-a)$

**Ans. (a) :**



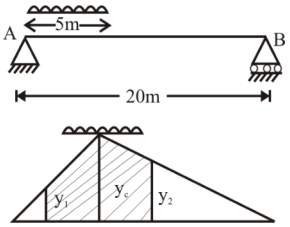

$$M_{AB} = +\frac{M}{L^2}(L-a)[L-3a]$$

$$M_{BA} = -\frac{M}{L^2}a[2L-3a]$$

**25. The absolute maximum bending moment in a simply supported beam of span 20 m due to a moving udl of 4 t/m spanning over 5 m is**

(a) 87.5 t-m at the support
(b) 87.5 t-m near the mid point
(c) 3.5 t-m at the mid point
(d) 87.5 t-m at the mid point

**Ans. (d) :**




$$y_c = \frac{Z(L-Z)}{L} = \frac{10(20-10)}{20}$$

$y_c = 5$

$\frac{x}{5-x} = \frac{10}{10}$ $\qquad y_1 = \frac{10-2.5}{10}\times 5$

$x = 2.5$ m $\qquad y_1 = 3.75\text{m} = y_2$

$$\text{Maximum moment} = 4\times\left[\frac{3.75+5}{2}\times 2.5\right]\times 2$$

= 87.5 t–m at mid point

**26. The degree of static indeterminacy of a rigid jointed space frame is**

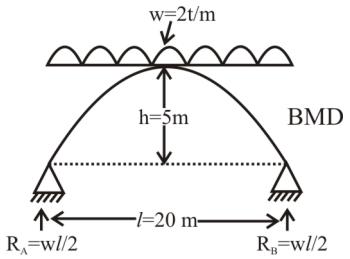
(a) m + r – 2j (b) m + r – 3j
(c) 3m + r – 3j (d) 6m + r – 6j

**Ans. (d) :** The degree of static indeterminacy of a rigid-jointed space frame is 6 m + r – 6j.
Where,
m, r and j are unknown as member forces,
r is unknown as reaction components
and j is number of joints–

**27. Asymmetrical parabolic arch of span 20 m and rise 5 m is hinged at the springings. If supports uniformly distributed load of 2 tonnes per meter run of the span. The horizontal thrust in tonnes at each of the springings is**

(a) 8 (b) 16 (c) 20 (d) Zero

**Ans. (c) :**




$$\sum M_c = 0$$

$$R_A \times \frac{l}{2} - \frac{wl}{2}\times\frac{l}{4} = H\times h$$

$$\frac{wl^2}{4} - \frac{wl^2}{8} = H\times h$$

formula of horizontal thrust $H = \frac{wl^2}{8h}$

$$H = \frac{2\times 20^2}{8\times 5}$$

H = 20 tonnes

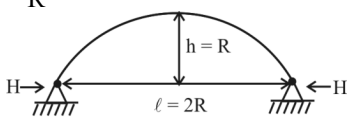
**28. The horizontal thrust due to rise in temperature in a semi-circular two hinged arch of radius R is proportional to**

(a) R (b) $R^2$
(c) $\frac{1}{R}$ (d) $\frac{1}{R^2}$

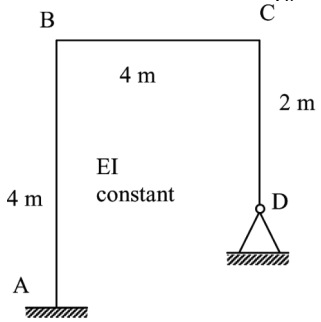
**Ans. (d) :** Due to rise in temperature the horizontal thrust is represented by

$$H = \frac{4EI\alpha T}{\pi R^2}$$

$$H \propto \frac{1}{R^2}$$

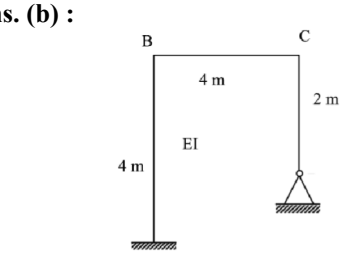


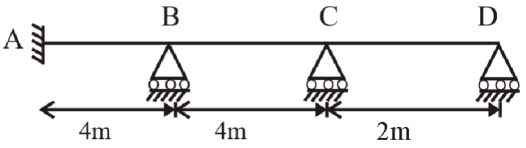

**29. In the frame shown in the figure, the support 'D' settles by 'δ'. The fixed end moment in the horizontal member of the frame will be (other symbols have the usual meaning)**




(a) $\frac{EI\delta}{8}$ (b) $\frac{6EI\delta}{16}$
(c) $\frac{3EI\delta}{16}$ (d) $\frac{EI\delta}{16}$

**Ans. (b) :**



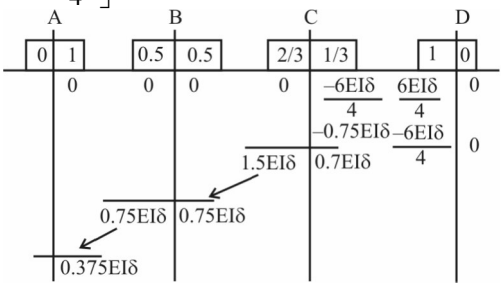




$$M_{CD} = \frac{6EI\delta}{L^2} = \frac{6EI\delta}{4}$$

$$M_{DC} = \frac{-6EI\delta}{4}$$

$$\text{At B}\begin{cases} BC = \dfrac{4EI}{4} = 0.5 \\ BA = \dfrac{4EI}{4} = 0.5 \end{cases} 2EI$$

$$\text{B}\begin{cases} CD = \dfrac{4EI}{2} = .33 \\ CB = \dfrac{4EI}{4} = 0.67 \end{cases} 3EI$$

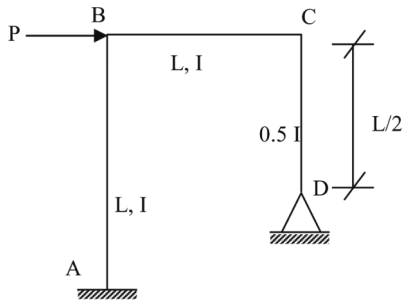



$M_{FED} = 0.375EI\delta$

or

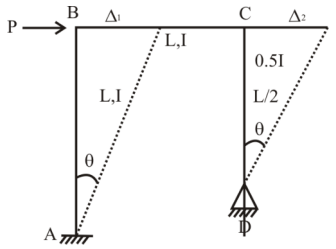
$$M_{FED} = \frac{6}{16}EI\delta$$

**30. Figure shows a portal frame with one end fixed and other hinged. The ratio of the fixed end moments $\frac{M_{BA}}{M_{CD}}$ due to side sway will be**




(a) 1.0 (b) 2.0
(c) 2.5 (d) 3.0

**Ans. (a):**




$$M_{BA} = \frac{2I}{L^2}$$

$$M_{CD} = \frac{I}{L^2}$$

$$= \frac{0.5I}{\left(\frac{L}{2}\right)^2} = \frac{4I}{2\times L^2} = \frac{2I}{L^2}$$

$$\frac{M_{BA}}{M_{CD}} = \frac{2I/L^2}{2I/L^2} = 1$$

**31. The amount of irrigation water required to meet the evapotranspiration needs of the crop during its full growth is called**

(a) effective rainfall (b) consumptive use
(c) consumptive irrigation requirement
(d) net irrigation requirement

**Ans. (c) : Consumptive irrigation requirement–** Consumptive use of water by a crop is expressed at the depth of water per unit area for a specified period, such as days, months or seasons.

The value of the consumptive use of water is needed to calculate the irrigation requirement of the crop.

CIR = consumptive use – effective rainfall

**Effective rainfall / useful rainfall–**

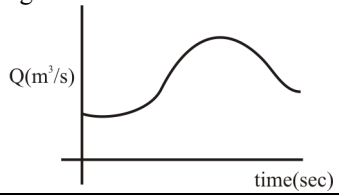
A portion of rain falling over a crop during the growing period of the crop is utilized to fulfill its evapotranspiration requirement. Thereby it reduce irrigation requirement of the crop.

- The portion of rainfall that is effectively used for the growth of crop is termed as useful rainfall.

**32. Hydrograph is the graphical representation of**

(a) runoff and time
(b) surface runoff and time
(c) ground water flow and time
(d) rainfall and time

**Ans. (a) :** Hydrograph is the graphical representation of runoff discharge v/s time.




**33. Cyclonic precipitation is caused by lifting of an air mass due to**

(a) Pressure difference
(b) Temperature difference
(c) Natural topographical barriers
(d) None of these

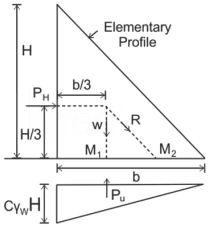
**Ans. (a):** Cyclonic precipitation is caused by lifting of an air mass due to pressure difference.
**Orographic precipitation :** This type of precipitation occur due to presence of mountain barriers because they can not move forward, they rise upward and undergo cooling, condensation and precipitation.
**Frontal Precipitation :** This precipitation occurs due to conflict between two air masses of different temperature and densities.

**34. If it rains between 2 PM and 3 PM and the entire basin area just starts contributing water at 3 PM to the outlet, then the time of concentration will be**
(a) 15 minutes (b) 20 minutes
(c) 30 minutes (d) 60 minutes

**Ans. (d) :** Time of concentration is the time required for an entire area/watershed to contribute to runoff
Toc = 1 hour = 60 minutes

**35. The elementary profile of a dam is**
(a) a rectangle
(b) a trapezoidal
(c) an equilateral triangle
(d) a right angled triangle

**Ans. (d) :** The elementary profile of a dam subjected only to external water pressure on the upstream side will be a high angled triangle, having zero width at water level and a base width (B) at bottom i.e. where the hydrostatic pressure is maximum.

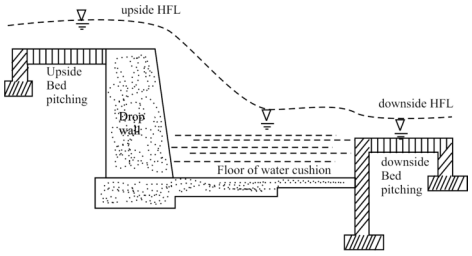



**36. In a chute spillway, the flow is usually**
(a) uniform (b) sub critical
(c) critical (d) supercritical

**Ans. (d) :** Spillway is a structure constructed at a dam sill, for effectively disposing of the surplus water from upstream to downstream
• The frequency of flow over spillway, depend upon.
→ Runoff characteristics of the drainage area
→ Reservoir storage.
In a chute spillway, the flow is usually supercritical having high velocity of flow.

**37. Vertical drop fall is satisfactory for a height upto**
(a) 0.5 m (b) 1.5 m
(c) 3.5 m (d) 5.0 m

**Ans. (b) :** Vertical drop fall is satisfactory for height upto 1.5m because these dissipates excess energy.
Sarda fall is an example of vertical fall.
**Sarda type canal fall–** Sarda fall as vertical drop fall consist of single vertical drop which allows the upstream water to fall with sudden impact on downstream, the down stream acts like cushion for the upstream water and dissipate extra energy.



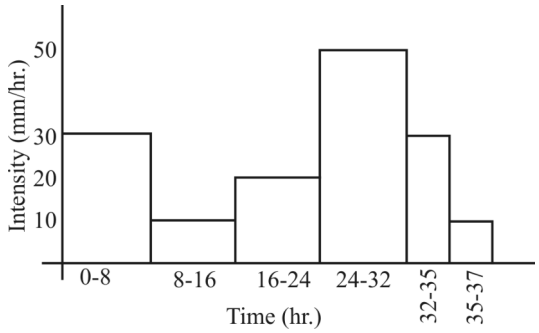

• This canal fall were constructed on the Sarda canal in Uttar Pradesh, So called as Sarda fall.

**38. A land is known as water logged**
(a) when the permanent wilting point is reached
(b) when the gravity drainage has ceased
(c) when capillary fringe reaches the root zone of plants
(d) none of these

**Ans. (c) :** An agricultural land is said to be waterlogged when its productivity or fertility is affected by high water table. The depth of water-table at which it tends to make the soil water-logged and harmful to the growth and subsistence of plant life depends upon the height of capillary fringe.
**The main cause of water logged–**
• Over irrigation of land
• Seepage losses from canal
• Surface spreading of waste water
• Excessive rainfall
• Poor land drainage
• Poor permeability of the soil system etc.

**39. A hyetograph is a graphical representation of**
(a) rainfall intensity and time
(b) rainfall depth and time
(c) discharge and time
(d) cumulative rainfall and time

**Ans. (a) :** Hyetograph is a plot of rainfall intensity vs time. It is derived from the mass curve and is usually represented as a bar chart.




**40. The peak of a 4-hour flood hydrograph is 240 $m^3$/sec. If the rainfall excess is 80 mm and base flow which is constant is 40 $m^3$/sec, then the peak of 4-hour unit hydrograph will be**
(a) 20 $m^3$/sec (b) 25 $m^3$/sec
(c) 30 $m^3$/sec (d) 35 $m^3$/sec

**Ans. (b) :** For 4- hr flood hydrograph
$Q_{peak}$ = 240 $m^3$/sec, base flow = 40$m^3$/sec
Peak of 4- hr direct runoff = 240 – 40
= 200$m^3$/sec

Peak of 4- hour unit hydrograph $= \frac{200 m^3/sec}{8cm}$
$= 25\ m^3/sec$

**41. If 'p' is the precipitation, 'a' is the area represented by a rain gauge, and 'n' is the number of rain gauges in a catchment area, then the weighted mean rainfall is**

(a) $\frac{\sum ap^3}{\sum a^2}$ (b) $\frac{\sum ap}{n}$

(c) $\frac{\sum ap}{\sum a}$ (d) $\frac{\sum ap^5}{\sum a^3}$

**Ans. (c) :** Weighted mean Rainfall
$= \frac{p_1a_1 + p_2a_2 + p_3a_3}{a_1 + a_2 + a_3}$
p = precipitation
a = area

**42. For medium silt whose average grain size is 0.16 mm, Lacey's silt factor is likely to be**

(a) 0.30 (b) 0.45
(c) 0.70 (d) 1.32

**Ans. (c) :** Lacey's silt factor $= 1.75\sqrt{d_{mm}}$
$= 1.75\sqrt{0.16} = 0.7$

**43. According to Indian standards, the number of raingauge stations for an area of 5200 km$^2$ in plains should be**

(a) 10 (b) 15
(c) 20 (d) 30

**Ans. (a) :** As per IS
In plain area, 1 raingauge is required for 520 km$^2$
No. of raingauge required $= \frac{Area}{520}$
$= \frac{5200}{520} = 10 units$

**44. The maximum permissible limit for fluoride in drinking water is**

(a) 0.1 mg/litre (b) 1.5 mg/litre
(c) 5 mg/litre (d) 10 mg/litre

**Ans. (b) :** Maximum permissible limit for fluoride in water is 1.5 gm/liter.

**45. Standard EDTA (ethylene-diamine tetra acetic acid) solution is used to determine the**

(a) hardness of water
(b) turbidity of water
(c) dissolved oxygen in water
(d) residual chlorine in water

**Ans. (a) : EDTA** (Ethylene Diamine Tetra Acetic Acid)–
- It is used to determine hardness of water.
- It define the amount of calcium carbonate in various solid materials.
- EDTA stands for Ethylenediamine Tetraacetic Acid.
- Titrant : 0.01 M EDTA
- Indicator = EBT

**46. Turbidity is measured on**

(a) Standard silica scale
(b) Standard cobalt scale
(c) Standard platinum scale
(d) Platinum cobalt scale

**Ans. (a) :** Turbidity is a measure of the degree to which the water loses its transparency due to presence of suspended particular.
It is measured in following units–
(a) STU (Silica turbidity unit)
(b) JTU (Jackson turbidity unit)
(c) NTU (Nephlometric turbidity unit)
(d) FTU (Formazine turbidity unit)

**47. The length of rectangular sedimentation tank should not be more than**

(a) B (b) 2B
(c) 4B (d) 8B

**Ans. (c) :** Length of sedimentation tank should not be more than 4 times the width (B)

**48. Orthotolidine test is used for determination of**

(a) dissolved oxygen
(b) residual chlorine
(c) biochemical oxygen demand
(d) None of these

**Ans. (b) : Orthotolidine test**–This is carried out to find the presence of both free chlorine and chloromines in water on addition of the reagent (orthotolidine in 10% HCL).
- Both free and combined C*l* can be determind.
- Reaction with free C*l* is fast.
- After 30 sec examine for a blue colour
- Blue green colour within 30 sec means positive test.

**Test for the presence of chlorine residual in water–**
- Orthotolidine (O.T.) Test.
- Starch – Iodide Test
- Orthotolidine - Arsenite (O.T.A.) test
- DPD (Di ethyle phenyline daimine)

**49. The suitable layout of distribution system for a city with roads of rectangular pattern is**

(a) grid iron system (b) dead end system
(c) ring system (d) radial system

**Ans. (a) : Grid iron system–** It is suitable for cities with rectangular layout where the water mains and branches are laid in rectangles.
The system has following advantages–
- There is free circulation of water without any stagnation or sediment deposit.
- Due to interconnection, water is available at each end point with min loss of head.
- Enough water is available in the street for fire fighting.

**50. A sewer that receives the discharge of a number of house sewers is called**

(a) house sewer (b) lateral sewer
(c) intercepting sewer (d) submain sewer

**Ans. (b) :** Types of sewer in typical collection system

| Types of sewer | Purpose |
|---|---|
| **Lateral or branch sewer** | Lateral sewers from the first element of a waste water collection system and are usually in streets or special easements. They are used to collect waste water from one or more building sewers and convey it to a main sewers. |
| **Building or house sewer** | Building sewers, sometimes called building connections, connect to the building plumbing and are used to convey waste water from the building to lateral or branch sewers, or any other except another building sewer. The distance from the foundation wall to where the sewer begins depends on the local building regulators |
| **Main sewer** | Main sewer are used to convey waste water from one or more lateral sewer to trunk sewer or to intercepting sewer. |
| **Trunk sewer** | Trunk sewers are large sewers that are used to convey |
| **Intercepting sewer** | Trunk sewers are large sewers that are used to convey waste-water from main sewer to treatment or other disposal facilities or too large intercepting sewers. |
| **Intercepting sewer** | Intercepting sewers are large sewers that are used to intercept a number of a main or trunk sewers and convey the waste water to treatment or other disposal facilities. |

**51. The correct relation between theoretical oxygen demand (TOD), biochemical oxygen demand (BOD) and chemical oxygen demand (COD) is given by**

(a) TOD > BOD > COD
(b) TOD > COD > BOD
(c) BOD > COD > TOD
(d) COD > BOD > TOD

**Ans. (b) : Theorectical oxygen demand (TOD)–** It is the oxygen demand than can be worked out the oretically.

**Chemical oxygen demand (COD)–** It is chemical oxidation with potassium permagnate or potassium dichromate in an acid solution.

COD represents both biologically active and inactive organic matter.

**Bio-chemical oxygen demand (BOD)–** BOD gives biologically active organic matter only.

- Generally TOD > COD > BOD > TOC

**52. In a BOD test 1.0 ml of raw sewage was diluted to 100 ml and the dissolved oxygen concentration of diluted sample at the beginning was 6 ppm and it was 4 ppm at the end of 5-day incubation at 20°C. The BOD of raw sewage will be**

(a) 100 ppm (b) 200 ppm
(c) 300 ppm (d) 400 ppm

**Ans. (b) :** $BOD_5 = [\text{initial D.O.} - \text{final D.O}] \times \text{D.F}$

$$BOD_5 = [6 - 4] \times \frac{100}{1}$$

$BOD_5 = 200$ ppm

**53. The minimum dissolved oxygen which should always be present in water in order to save the aquatic life is**

(a) 1 ppm (b) 4 ppm
(c) 10 ppm (d) 40 ppm

**Ans. (b) :** Oxygen content present in water is called dissolved oxygen. Minimum level of oxygen content of survival of aquatic life in water is 4 mg/lit.

In river streams dissolved oxygen present is 10 mg/lit and large water bodies it is 8 gm/lit. The dissolved oxygen level in natural unpolluted water at normal temperature is 10 mg/lit.

**54. Temporary hardness in water is caused by the presence of**

(a) Bicarbonates of Ca and Mg
(b) Sulphates of Ca and Mg
(c) Chlorides of Ca and Mg
(d) Nitrates of Ca and Mg

**Ans. (a) :** Carbonate hardness (temporary hardness) is removed by boiling it is caused due to presence of bicarbonate of calcium or maganism in water i.e., calcium hardness is not a health hazard, because a in zeolite process when permanent hardness is removed than precipitate of Calcium Maganism is formed, which is not to much harmful.

- Hardness water consume more soap because hard water contains dissolved salt of calcium, magnesium and other heavy metals and it does not form lather with soap readily to the reaction show below.

$$2RCOONa + Ca_2 + \longrightarrow (RCOO)_2\,Ca + 2Na^+$$

**55. Blue baby disease (methemoglobinemia) in children is caused by the presences of excess**

(a) chlorides (b) nitrates
(c) fluorides (d) lead

**Ans. (b) :**

- Blue baby disease in children is due to presence of excess amount of nitrates in water.
- The permissible value of nitrate in water should be 45 mg/*l*.
- Nitrates have been defined as the non-objectionable final end product in aerobic treatment of sewage, yet its concentration in potable water are controlled, because larger concentrations (above 45 ppm) may cause nitrate poisoning in infants.

- This happens because there exists lower acidity in the intestines of the infants, which permits growth of nitrate reducing bacteria, which converts nitrates to nitrites, the nitrites, when absorbed in blood, prove very harmful, because they have greater affinity for hemoglobin than does oxygen and thus, nitrite replace oxygen in the blood complex. This diseases, popularly called blue baby disease or technically as methaemoglobinemia.

**56. Two samples of water, A and B have pH values of 4.4 and 6.4 respectively. How many times more acidic is sample A than sample B ?**
(a) 0 (b) 50
(c) 100 (d) 200

**Ans. (c) :** $(pH)_{\pi} = 4.4, \quad (pH)_B = 6.4$

$pH = -\log_{10}[H^+]$

$[H^+]_A = 10^{-4.4}, \quad [H^+]_B = 10^{-6.4}$

$$\frac{[H^+]_A}{[H^+]_B} = \frac{10^{-4.4}}{10^{-6.4}} = 100$$

$\therefore$ Sample A is 100 times more acidic than B

**57. Dechlorination of water is achieved by adding**
(a) Sodium thiosulphate (b) Sodium sulphate
(c) Sodium biosulphate (d) None of these

**Ans. (a): Dechlorination–** When the chlorine residue is high, excess chlorine will be remove by dechlorinating agent.
**Dechlorinating agent–**
(i) Sodium thiosulphate ($Na_2S_2O_3$)
(ii) Sulphur dioxide ($SO_2$)
(iii) Sodium sulphate ($Na_2SO_3$)
(iv) Activated carbon.

**58. The efficiency of a sedimentation tank does not depend upon**
(a) depth of the tank (b) detention time
(c) length of the tank
(d) horizontal velocity of water

**Ans. (a) :** As per IS 10313 : 1982, the factor does effluence the sedimentation process in a sedimentation tank is–
(i) Inlet and outlet arrangement
(ii) Surface overflow
(iii) Size, shape and weight of particle.
**Note–**
- pH value of water not a factor of sedimentation process is sedimentation tank.
- The depth of tank will not have any effect on the efficiency of tank.

**59. Which one of the following would contain water with the maximum amount of turbidity?**
(a) lakes (b) oceans
(c) rivers (d) wells

**Ans. (c) :** In rivers, water contain with the maximum amount of turbidity.
- Due to availability of large amount of suspended matter, water becomes turbid. The turbidity depends upon the linear and concentration of particles present in water.
- In oceans due to turbulence the suspended impurities comes to the shore and in case of lakes and well they settle. But rivers have high turbidity in comparison to other sources of water.

**60. The waste stabilization ponds can be**
(a) aerobic (b) anaerobic
(c) facultative (d) all of the above

**Ans. (d) : Oxidation pond–**
- The waste water treatment initiated by symbiotic relationship between algae and bacteria.
- Oxidation pond or waste stabilization ponds or lagoons are relatively shallow body of water contained in an earthen basin of controlled slope. They are designed and constructed to treat sewage and biodegradable industrial wastes.
- Ponds require low initial cost end low operation cost.
- It is required more land.

Stabilization ponds are $\rightarrow$ Aerobic / Anaerobic / Facultative

**61. For a slab supported on its four edges with corners held down and loaded uniformly, the Marcus correction factor to the moments obtained by Grashoff Rankine's theory**
(a) is always less than 1
(b) is always greater than 1
(c) is equal to 1
(d) None of the above

**Ans. (a) :** Marcus correction factor

$$C = \left[1 - \frac{5}{6}\frac{r^2}{1+r^4}\right] \qquad r = \frac{l_y}{l_x}$$

C is always less than 1

**62. The permissible diagonal tensile stress in the reinforced brick work is**
(a) zero
(b) about 0.1 $N/mm^2$
(c) 0.3 $N/mm^2$ to 0.7 $N/mm^2$
(d) None of the above

**Ans. (b) :** Reinforced brick work is very weak against diagonal tension. The permissible shear stress may vary between 0.07 to 0.14 $N/mm^2$.
The average value of 0.1 $N/mm^2$ may be adopted for design purposes.
- Normally, shear stress should not exceed this value in exceptional cases, if the diagonal tension exceeded this value, suitable shear reinforcement should be provided. In such cases however, the value of shear stress should not exceed 0.3 $N/mm^2$ otherwise section should be redesigned.

**63. The limits of percentage P of longitudinal reinforcement in a column are given by**
(a) 0.15 % to 2% (b) 0.8% to 4%
(c) 0.8% to 6% (d) 0.15% to 6%

**Ans. (c) :** According to IS 456 : 2000 (Cl.26.5.3.1)
The cross-sectional area of longitudinal reinforcement, shall be not less than 0.8 percent nor more than 6 percent of the gross cross-sectional area of the column.

Minimum steel area = 0.8% of gross section area
Maximum steel Area

$$= \begin{cases} 6\% \text{ of gross Area; in non Lapped} \\ 4\% \text{ of gross Area; in Lapped case} \end{cases}$$

**64. In a pile of length *l*, the points of suspension from the ends for lifting it are located at**

(a) 0.207 *l* (b) 0.25 *l*
(c) 0.293 *l* (d) 0.333 *l*

**Ans. (a) :** A pile of length L has the distance of points of suspension from either end is 0.207L to ensure the hogging bending moment at two point of suspension equals the sagging moment at its center.

**65. The partial safety factor for steel as per IS 456 – 1978 is taken as**

(a) 1.15 (b) 1.35
(c) 1.50 (d) 1.65

**Ans. (a) :** As per IS code 456 : 200

Partial safety factor for $\begin{cases} \text{steel} = 1.15 \\ \text{concrete} = 1.5 \end{cases}$

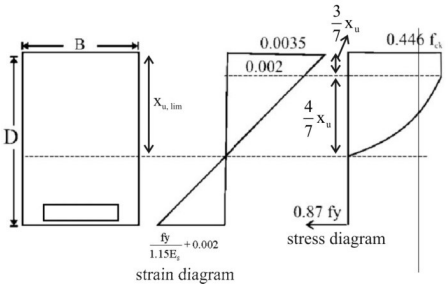
**66. According to IS 456 – 1978, the maximum strain in concrete at the outermost compression fibre in the limit state design of flexural members is**

(a) 0.0020 (b) 0.0035
(c) 0.0050 (d) 0.0065

**Ans. (b) :** The maximum strain in concrete in compression in bending is taken as 0.0035 where as maximum strain in tension reinforcement is taken as

$$\frac{f_y}{1.15E_s} + 0.002$$

Maximum strain when concrete under compression = 0.002 and when flexural maximum strain is taken 0.0035 in outer most compression fiber.




So, By IS: 456-2000

Maximum strain in concrete = 0.0035 and that of in steel is $0.002+\frac{f_y}{1.15E_s}$

**67. In a spherical dome subjected to concentrated load at crown or uniformly distributed load, the meridional force is always**

(a) Zero (b) Tensile
(c) Compressive (d) None of these

**Ans. (c):** The primary stresses induced in the dome are (i) a compressive force acting along the meridian, called the meridional thrust and (ii) a force compressive or tensile in nature, acting along a horizontal circle on the dome called the hoop tension or hoop compression.

Hence, a spherical dome subjected to concentrated load at crown or uniformly distributed load, the meridional force is always compressive.

- Meridional thrust is maximum at support and hoop force is maximum at crown.
- Radial bar are provided for meridional thrust while circular rings are provided for circumferential/hoop force.

$T_1$ = Meridonal thrust $\left(\frac{wR}{1+\cos\theta}\right)$

$T_2$ = Circumferential force = $wR\left(\frac{\cos\theta - 1}{1+\cos\theta}\right)$

Let–

t = thickness of done generally varies in the range of 75 to 100 mm.

h = rise of dome = $\frac{1}{5}$ to $\frac{1}{6}$ dia meter of tank (D) span

R = radius of dome

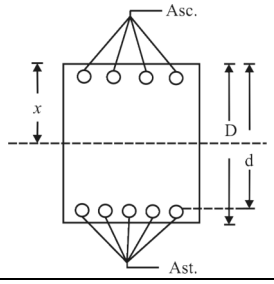
w = uniformly distributed load per unit SA.

**68. In a doubly reinforced rectangular beam, the allowable stress in compression steel is**

(a) equal to the permissible stress in the tension steel
(b) more than permissible stress in the tension steel
(c) less than permissible stress in the tension steel
(d) None of these

**Ans. (c) :** In doubly reinforced rectangular beam, the allowable stress in compression steel is less than permissible stress in the tension steel

- In doubly reinforced beam we provide some steel in compression zone to reduce the depth of beam so compressive steel is under stressed.




**69. Loss of stress with time at constant strain in steel is called**

(a) relaxation (b) creep
(c) ductility (d) shrinkage

**Ans. (a) :** Relaxation is the loss of stress in steel with time at constant strain.

Loss due to relaxation of strands is a long term pre-stress loss in a pre-stressed member.

• Early attempts to produce pressed concrete was not successful due to loss of pre-stress transferred concrete after few years.

• Pre-stress loss is nothing but the reduction of initial applied pre-stress to an effective value.

• In other words, loss in pre-stress is the difference between initial pre-stress and the effective pre-stress that remains in a member.

Relaxation of steel is defined as the decrease in stress with time under constant strain. Due to the relaxation of steel, the prestress in the tendon is reduced with time.

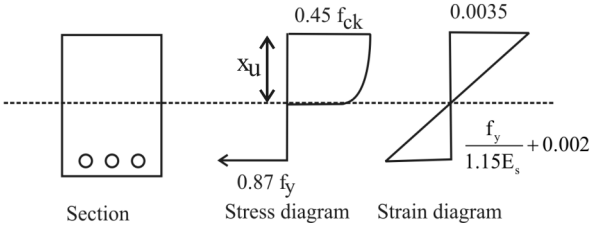
| Initial stress | Relaxation loss (MPa) |
|---|---|
| $0.5\ f_p$ | 0 |
| $0.6\ f_p$ | 35 |
| $0.7\ f_p$ | 70 |
| $0.8\ f_p$ | 90 |

**70. In the limit state design of concrete structures, the strain distribution is assumed to be**

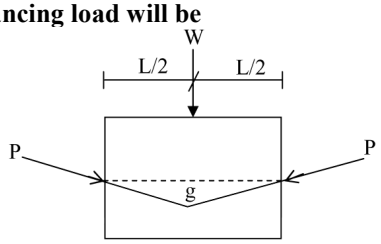
(a) linear (b) non-linear
(c) parabolic (d) rectangular

**Ans. (a) :** In LSM, it is assumed that plane section remain plane before and after bending i.e. strain variation is linear.

In the limit state design of the concrete structure the strain distribution is assumed to be linear. Limit state design is the one in which the load and resistance factor design are used in the engineering of structural.




**71. If 'P' is the prestressing force applied at a maximum eccentricity 'g' at mid span (figure), to balance the concentrated load 'W', the balancing load will be**




(a) 2.5 Pg/L (b) 3.0 Pg/L
(c) 3.5 Pg/L (d) 4.0 Pg/L

**Ans. (d) :** $\tan\theta = \sin\theta = \frac{g}{\ell/2} = \frac{2g}{\ell}\{\theta \text{ is small}\}$

From load balancing concept $2P\sin\theta = w$

$$\frac{2P \times 2g}{\ell} = w$$

$$w = \frac{4Pg}{\ell}$$

**72. Yield line theory results in**

(a) elastic solution
(b) upper bound solution
(c) lower bound solution
(d) unique solution

**Ans. (b) :** The yield line analysis is an upper bound method in which the predicted failure load of a slab for given moment of resistance (capacity) may be higher than the true value. Thus, the solution of the upper bound method (yield line analysis) may result into unsafe design if the lowest mechanism.

• Lower bound theorem or static represents the lower limit to the true ultimate load and has maximum factor of safety. The static theorem satisfies the equilibrium and yield condition.

**73. For ultimate load design of prestressed concrete girders used for bridges, combination of load factors used is (here L.L. and D.L. are line load and dead load respectively)**

(a) 1.5 D.L. + 2.5 L.L. (b) 1.0 D.L. + 2.0 L.L.
(c) 1.0 D.L. + 2.5 L.L. (d) 2.0 D.L. + 2.0 L.L.

**Ans. (d) :** In ultimate load design concept for prestressed concrete girder and bridges, the load factor taken of both L.L and D.L is 2

Ultimate load =2.0 D.L. + 2.0 L.L

**74. A reinforced concrete slab is 75 mm thick. The maximum size of reinforcement bar that can be used is**

(a) 12 mm diameter (b) 10 mm diameter
(c) 8 mm diameter (d) 6 mm diameter

**Ans. (c) :** Maximum size of reinforcement

$$\not< \frac{\text{Depth of slab}}{8}$$

$$= \frac{75}{8} \approx 9.375$$

= 8 mm diameter

**75. In the design of two way slab restrained at all edges, torsional reinforcement required is**

(a) 0.75 times the area of steel provided at mid span in the same direction
(b) 0.375 times the area of steel provided at mid span in the same direction
(c) 0.375 times the area of steel provided in the shorter span
(d) None of these

**Ans. (a) :** Torsional reinforcement should be 0.75 times area of steel provided at mid span as per annex D clause D-1.8, of IS 456 : 2000.

- Torsional reinforcement is provided in the form of a grid or mesh both at the top and bottom of the slab.
- 0.75 $A_{st}$ if the both the meeting edges are restrained.
- 0.375 $A_{st}$ if one of the two meeting edges, one in continuous and other discontinuous.

**76. PERT technique of network analysis is mainly useful for**

(a) small projects
(b) large and complex projects
(c) research and development projects
(d) deterministic activities

**Ans. (c):** PERT technique of network analysis is mainly useful for research and development projects.

- It is probabilistic approach and it also considers uncertainties of activity duration.

**77. In PERT analysis, the time estimates of activities and probability of their occurrence follow**

(a) Normal distribution curve
(b) Poission's distribution curve
(c) β - distribution curve
(d) Binomial distribution curve

**Ans. (c) :** In PERT analysis, the time estimate of activities and probability of their occurrence follow β-distribution.

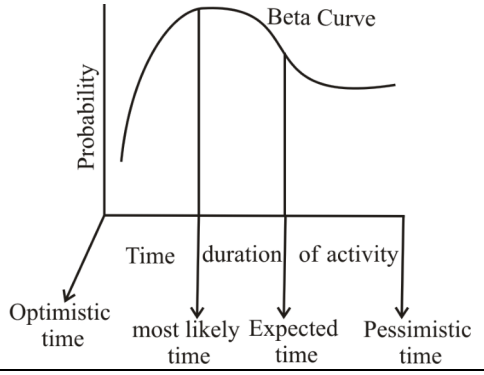
The Program Evaluation and Review Technique (PERT) is a project scheduling technique to analyze and represent the task involved in completing a given project.

Malcolm's PERT network, henceforth referred to as classical PERT network, assumes that all activities durations are independent random variables, having approximately beta distributions parameterized by three times estimates :

(a) Optimistic time
(b) Pessimistic time
(c) Most likely time.

The project duration follows a normal distribution with the following parameter :

(a) Expected project duration
(b) Expected variance




**78. Critical path**

(a) is always longest (b) is always shortest
(c) may be longest (d) may be shortest

**Ans. (a) : Critical path**-Critical path represents the longest path of the network from the initial event to the final event.

- Critical path consist of only critical activities.
- Critical path is shortest path in terms of time in which project can be completed as early as possible.

**79. The time by which a particular activity can be delayed without affecting the preceding and succeeding activities is known as**

(a) free float (b) Total float
(c) Independent float (d) Interfering float

**Ans. (c): Independent float :** The time by which a particular activity can be delayed without affecting preceding and succeeding activities is known as independent float.

**Total float :** Total float is the time by which starting/finishing of an activity can be delayed without affecting the scheduled completion time of the project.

$F_T = LST - EST$

**80. Economic saving of time results by crashing**

(a) Cheapest critical activity
(b) Cheapest non – critical activity
(c) Costliest critical activity
(d) Costliest non – critical activity

**Ans. (a) : Crashing–**The process of reduction of total project duration along the path (time wise) of the network i.e. along the critical path to obtain the optimum project cost and optimum duration is called as crashing.

- Crashing is a methodology or procedure to obtain project cost and optimum duration.
- Crashing of the activity having the least cost slope along the critical path of the network.

**81. Slack time refers to**

(a) An activity
(b) An event
(c) Both event and activity
(d) None of the above

**Ans. (b) : Slack–**The difference between the two times of an activity indicates the range between which the occurrence time of an event can vary.

- Slack may be simply defined as the difference between the latest allowable time and the earliest expected time of an event.

$$S = T_L - T_E$$

where S is the slack for any event.

**82. A tractor has purchase price of Rs. 4.7 lakhs and could save the organization an amount of Rupees one lakh per year on operating costs. The salvage value after the amortization period is 10% of the purchased price. The capital recovery period will be**

(a) 3.7 years (b) 4.2 years
(c) 5 years (d) 7.8 years

**Ans. (b) :** Given:
Purchase price = Rs. 4.7 Lakhs
Save amount = ₹ 1 Lakhs/year on operating cost
Salvage value = 10% purchase price

$$\text{Salvage value} = \frac{10}{100} \times 4.7 \text{ Lakhs}$$

= ₹ 0.47 Lakhs

$S = P - (I \times T)$
$47000 = 470000 - (100000 \times T)$
$T = 4.23$ year

**83. Site order book is used for recording**

(a) Instructions of the executive engineer
(b) Construction measurements
(c) Requisition of plants and equipments
(d) Indents for materials to be ordered

**Ans. (a):** Site order book is used for recording the instructions of the executive engineer or higher authorities.

**84. The system of organization introduced by F.W. Taylor is known as**

(a) Effective organization
(b) Functional organization
(c) Line organization
(d) Line and staff organization

**Ans. (b) :** Functional Organization is suggested by F.W. Taylor. In functional organization different experts (specialists) guide the subordinates in organization. Workers or subordinates thus receive order from several specialists or officers and are responsible for their performance to different experts.

**85. The original cost of an equipment is Rs. 10,000. Its salvage value at the end of its total useful life of live years is Rs. 1,000. Its book value at the end of two years of its useful life (as per straight line method of evaluation of epreciation) will be**

(a) Rs. 8,800 (b) Rs. 7,600
(c) Rs. 6,400 (d) Rs. 5,000

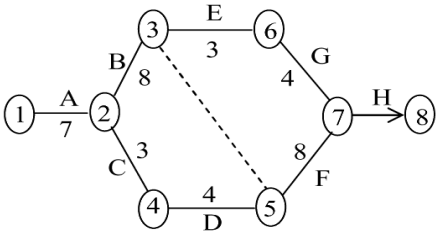
**Ans. (c) :** P = ₹ 10000 time = 2 year
S = ₹ 1000
$S = P(1-I)^t$
$1000 = 10000\ (1-I)^2$
I = 68.37%
$I = \frac{68.37 \times 10000}{100} = ₹\ 6837$

**86. Grader is used mainly for**

(a) Trimming and finishing
(b) Shaping and trimming
(c) Finishing and shaping
(d) Finishing, shaping and trimming

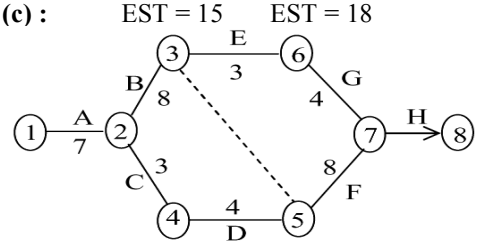
**Ans. (d) :** Grader is a equipment used for :
(A) Levelling (B) shaping (C) finishing
(D) Mixing gravel (E) Flattening & smoothing
(F) Trimming slopes

**87. The flow net of the activities of a project are shown in the network given in figure indicating the duration of the activities along their arrows. The critical path of the activities is along.**




(a) $1 \to 2 \to 4 \to 5 \to 7 \to 8$
(b) $1 \to 2 \to 3 \to 6 \to 7 \to 8$
(c) $1 \to 2 \to 3 \to 5 \to 7 \to 8$
(d) $1 \to 2 \to 4 \to 5 \to 3 \to 6 \to 7 \to 8$

**Ans. (c) :** EST = 15 EST = 18




EST = 10 EST = 15
Critical path = longest path of Network
= 1 – 2 – 3 – 5 – 7 – 8

**88. Capital recovery factor at 15% p.a. discrete compounding for 4 years is 0.35. Rs. 10,000 deposited now at 15% p.a. discrete compounding will yield an amount 'X' at the end of each 4 – year – period in perpetuity. The value of 'X' is**

(a) Rs. 7,500 (b) Rs. 6,666
(c) Rs. 6,000 (d) Rs. 9,000

**Ans. (a) : :** Given,
C.R.F. = 0.35, P = 10,000, i = 0.15
A = 10000 × 0.35 = 3500 Rs.
now,

$X = A \times (CRF) - P$ Given $CRF = \frac{(1+i)^4 - 1}{i}$

$$= 3500 \times \left[\frac{(1+0.15)^4 - 1}{0.15}\right] - 10000$$

$= 3500 \times 5 - 10000$
X = 7500 Rs.

**89. A machine costs Rs. 16,000 by constant rate of declining balance method of depreciation, its salvage value after an expected life of 3 years is Rs. 2,000. The rate of depreciation is**

(a) 0.25 (b) 0.30
(c) 0.40 (d) 0.50

**Ans. (d) :** Machine cost = ₹ 16,000
Salvage value = ₹ 2000
time = 3 year

$$\text{Rate of depreciation} = 1 - \sqrt[t]{\frac{\text{salvage value}}{\text{cost}}}$$

$$= 1 - \sqrt[3]{\frac{2000}{16000}}$$

$= 1 - 0.5$
Rate of depreciation = 0.5

**90. The relation between 'D' in hectares/cumec, depth of water 'Δ' in meters and the base period B in days is given by**

(a) $\Delta = \frac{1.98B}{D}$ (b) $\Delta = \frac{8.64B}{D}$
(c) $\Delta = \frac{5.68B}{D}$ (d) $\Delta = \frac{8.64D}{D}$

**Ans. (b):** Relation b/w duty, Delta and base period

$$\Delta = \frac{8.64B}{D}$$

D = Duty (hectares/cumec)
$\Delta$ = Depth of water in meter
B = Base period (day)

**91. If P is the percentage of water required for normal consistency, water to be added for determination of initial setting time is**

(a) 0.75P (b) 0.85P
(c) 0.085P (d) 0.075P

**Ans. (b) :** Standard consistency or normal consistency test is conducted to find out the quantity of water to be mixed with cement to form a paste of normal consistency i.e. the paste of standard solidity.

- This test is done before the setting time test to fix the quantity of water to be mixed in the cement.
- The amount of water used to conduct the initial setting time test of cement which has a standard consistency of **0.85 W.**
- For soundness test = 0.78W
- For compression strength test = $\frac{W}{4} + 3$
- For tensile strength test = $\frac{W}{5} + 4$

**92. According to IS 399 – 1963, the weight of the timber is specified at**

(a) 10% moisture content
(b) 12% moisture content
(c) 8% moisture content
(d) 14% moisture content

**Ans. (b) :** Weight and strength properties of timber are generally influenced by moisture content.
Therefore the strength values and weight obtained are usually standardized at 12% moisture content.
Freshly cut tree (timer) contains moisture and sap of 100 to 200% of its weight.

**93. Due to attack of dry rot, the timber**

(a) Shrinks (b) Cracks
(c) Reduces to powder (d) Reduces its weight

**Ans. (c) :** The dry rot occurs where there is no free circulation of air. The other cause are unseasoned wood and improper storage the most favourable conditions for the rapid growth of fungus responsible for dry rot are absence of sunlight, dampness, presence of sap, stagnant air and warmth.

**94. Plywood is made from**

(a) Bamboo fibre (b) Teak wood only
(c) Common timber (d) Asbestos sheets

**Ans. (c) :** Plywood is made from common timber.
Plywood is specially processed type of wood, which consist of essentially on odd number of plies (or thin sheets called veneers) glued together in such a manner that the grain of adjacent layer is perpendicular to each other.

**95. For a 50 kg bag of cement, water required is**

(a) 22.5 liters (b) 20.5 liters
(c) 18.5 liters (d) 23.5 liters

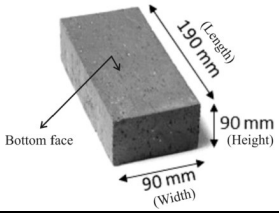
**Ans. (a):** Generally the water of cement ratio is considered as = 0.45. So for a 50 kg cement bag.
The amount of water is required is
$0.45 \times 50 = 22.5$ litres.

**96. The standard size of a brick is**

(a) 19.5 cm × 9.5 cm × 9.5 cm
(b) 18 cm × 8 cm × 8 cm
(c) 19 cm × 9 cm × 9 cm
(d) 18.5 cm × 8.5 × 8.5 cm

**Ans. (c) : Size and weight of bricks-** For India,

- Standard size of bricks is 19 cm × 9 cm × 9 cm
- Nominal size (with mortar) is 20 cm × 10 cm × 10 cm
- The commonly adopted nominal size of traditional bricks is 23 cm × 11.4 cm × 7.6 cm
- It is found that the weight of 1 $m^3$ of bricks earth is about 1800 kg hence the average weight of a brick will be about 3 to 3.50 Kg.




**97. Initial setting cement is caused due to**

(a) 3 $CaO.SiO_2$ (b) 2 $CaO.SiO_2$
(c) 3 $CaO.Al_2O_3$ (d) 4 $CaO.Al_2O_3.Fe_2O_3$

**Ans.(c):** Tricalcium Aluminate ($C_3A$) – It is responsible for the initial setting time, it has high heat of hydration and has greater tendency to volume changing causing cracking.

**98. Clay and silt content in a good brick earth must be at least**

(a) 25% (b) 30%
(c) 40% (d) 50%

**Ans. (d) :** A good brick earth should contain about 50-60% of silica. Presence of this constituent prevents cracking, shrinkage and warping of raw bricks. It thus imparts uniform shape to the bricks. According to IS-2117 the proportions of various ingredients in a good bricks earth are as follows

| | | |
|---|---|---|
| Silica ($SiO_2$) | : | 50 – 60% |
| Alumina | : | 20 – 30% |
| Iron oxide | : | 3 – 8% |
| Lime | : | 2 – 5% |
| Na, K, Mg | : | 1 – 3% |
| Alkalies | : | Less than 10% |

Silt is granular material of a size between sand and clay. A good bricks earth should contains about not less than 50% of clay and silt.
Addition of sandy loam is effective in controlling the drying behavior of plastic soil.
- Its composition should be –

| | |
|---|---|
| Clay (<2 μ) | 8–10% |
| Silt (2–20 μ) | 30–50% |
| Sand (>20 μ) | 40–60% |

**99. For construction of structures under water the type of lime used is**

(a) Pure lime (b) Fat lime
(c) Quick lime (d) Hydraulic lime

**Ans. (d) : Hydraulic lime–** It is different in chemical composition from fat lime in that it contains a definite amount of clay in addition to CaO, clay content in hydraulic lime may range from 10 to 30% by weight. This clay and lime composition gives the hydraulic lime a property of hydraulicity.

**100. Cement is said to be of good quality if**

(a) It is not smooth when rubbed in between fingers
(b) It's colour is greenish gray
(c) A handful of cement thrown into a bucket of water does not float
(d) None of the above

**Ans. (c) :** Cement is said to be of good quality if (As per Physical test IS : 4031)- the sample under test.

1. The given sample of cement under test must posses uniform gray colour.
2. It feel smooth when rubbed in between the fingers.
3. It be free from presence of air set lumps.
4. It sink and not float when thrown in bucket of water.
5. It feel cool inside cement then hand is put in it.
6. A thin paste of given sample of cement should feel sticky in between the fingers.

**101. The most valuable timber may be obtained from**

(a) Sal (b) Re Sander
(c) Sandle (d) Teak

**Ans. (d) :** Teak is the most valuable timber used for making furnitures.

- It is moderately hard, durable and fire resistance.
- It is not attacked by white ant and dry rot.
- It is used for house construction, railway carriage, flooring, structural works, ship building etc.

**102. Soundness test of cement determines**

(a) Durability (b) Tensile strength
(c) Quality of free lime (d) Initial setting

**Ans. (c) :** In the soundness test of cement, the amount of excess lime is determined.

- Unsoundness due to lime is found using "Le-chatlier's test" which consists of split cylinder of diameter and height of 30 mm and indicator arms of length 165 mm.

**103. A badly mixed cement concrete results in**

(a) Bleeding (b) Honeycombing
(c) Segregations (d) None of above

**Ans. (b) :** Honey combing occurs when there is an improper concrete mix which is usually a short-term deterioration.

**Honey combing–** The consist of areas in the concrete that are composed of large aggregates and an insufficient amount of sand and cement which results in void amount the large aggregates. Isolated pockets of honey combing can occurs when a stiff mix (1 to 3 slump) is being used without sufficient consolidation.

- Voids between aggregates from a defects known as honey combing.

**104. Inner part of a timber log surrounding the pith is called**

(a) Annual ring (b) Cambium layer
(c) Medullary sheath (d) Heart wood

**Ans. (d) :** The innermost layer is known as pith and surrounding pith is heartwood.

- **Heart wood** - Inner of a timber log surrounding the pith. It is called as heart wood.
  Dead central wood of trees, Heart wood gives a strong and firm support to the tree with the growth of tree, the cells in the inner older portion of trunk gradually become inactive and lifeless, but do not decay. This portion of the trunk is called heart wood.
- **Sapwood** - It assists in the life process of tree by staring up starch and conducting sap. The cells in the sap wood are active.

**105. Portland Pozzolana cement possesses**

(a) Lower heat of hydration
(b) Water tightness
(c) Lower shrinkage on drying
(d) All the above

**Ans. (d) :**

- Portland Pozzolana cement posses lower heat of hydration.
- It possess lower shrinkage on drying.
- It has water tightness ( marine work)

Portland Pozzolana cement contain 20-30% Pozzolana do not have binding property (non-cementious material)

**106. If P, Y and Z are the weights of the cement, fine aggregates, and coarse aggregates respectively and W/C is the water cement ratio, the minimum quantity of water to be added to first batch, is obtained by the equation.**

(a) $0.1\ P + 0.3\ Y + 0.1\ Z = (W/C) \times P$
(b) $0.2\ P + 0.5\ Y + 0.1\ Z = (W/C) \times P$
(c) $0.3\ P + 0.1\ Y + 0.01\ Z = (W/C) \times P$
(d) $0.5\ P + 0.3\ Y + 0.01\ Z = (W/C) \times P$

**Ans. (c) :** The minimum quantity of water added to first batch is obtain by equation–

$0.3\ P + 0.1\ Y + 0.01\ Z = (W/C) \times P$

30% of cement content + 10% of fine aggregate +1% of coarse aggregate

Where,

P = Quantity of weight of the cement
Y = Quantity of weight of the fine aggregates
Z = Quantity of weight of the coarse aggregates

**107. The main advantage of steel member, is**

(a) Its high strength (b) Its long service life
(c) Its water tightness (d) All the above

**Ans. (d) : Advantages of steel member :**

- **(i)** Steel, being a ductile material, does not fail suddenly, but give visible evidence of impending failure by large defection.

(ii) Structural steel are tough, ie, they have both strength and ductility, thus, steel member subjected to large deformations during fabrication and erection will not fracture.
(iii) The properties of steel mostly do not change with time.
(iv) Steel has the highest scrap value amongst all building materials
(v) Steel is ultimate recyclable materials.
(vi) Steel has high strength
(vii) Steel is gas and water tight
(viii) Steel has long service life

**108. As per ISI, rolled steel beam sections are classified into**

(a) Three series (b) Four series
(c) Five series (d) Six series

**Ans. (c) :** As per Indian standards rolled steel I-section are classified into five series.
**(i) ISWB -** Indian standard wide flange beam has high moment of inertia about y-axis as compared to other beam sections.
**(ii) ISLB -** Indian standard light beams are generally used as purlins in roof or as secondary beams where loading is less.
**(iii) ISMB -** Indian standard medium weight in frames, floor beams etc with high moment of inertia about x-axis.
**(iv) ISJB** - Indian standard Junior beam.
**(v) ISCS -** Indian standard column section

**109. With a percentage increase of carbon in steel, decreases**

(a) Hardness (b) Ductility
(c) Brittleness (d) Strength

**Ans. (b) :** With a percentage increase of carbon in steel, ductility decreases because of increment in hardness.

**110. The critical load for a column of length '$l$' hinged at both ends and having flexural rigidity EI, is given by**

(a) $P_c = \frac{\pi^2 EI}{l^3}$ (b) $P_c = \frac{\pi^2 EI}{l^2}$

(c) $P_c = \frac{\pi (EI)^2}{l^2}$ (d) None of these

**Ans. (b) :** Critical load for a column hinged at both ends

$\Rightarrow P_c = \frac{\pi^2 EI}{l^2}$

EI = flexural rigidity
$l$ = length of column.

**111. Slenderness ratio of compression member is**

(a) $\frac{\text{Moment of Inertia}}{\text{Radius of gyration}}$

(b) $\frac{\text{Effective length}}{\text{Radius of gyration}}$

(c) $\frac{\text{Radius of gyration}}{\text{Area of cross-section}}$

(d) $\frac{\text{Moment of Inertia}}{\text{Area of cross -section}}$

**Ans. (b):** Slenderness ratio of compression member

$= \frac{\text{Effective length}}{\text{Radius of gyration}}$

$\lambda = \frac{l}{r}$

**112. The most economical section for a column is**

(a) Hexagonal (b) Rectangular
(c) Tubular section (d) Solid round

**Ans. (c) :** The most economical section for a column is tubular section.
Tubular section $\rightarrow$ Moment of area (maximum).

- Tubular section has greater second moment of area. Tubular section is best suited for smaller compression member subjected to smaller loads.
- It has same radius of gyration in all direction.

**113. If W and L are the total superimposed load and the span of a plate girder in meters, the approximate self weight of the girder, is taken as**

(a) $M = \frac{WL}{300}$ (b) $M = \frac{WL}{100}$

(c) $M = \frac{WL}{400}$ (d) $M = \frac{WL}{200}$

**Ans. (a) :** Self weight of girder (M) $= \frac{WL}{300}$

W = total superimposed load
L = span
Self weight of plate girder may be taken between wL/200 to wL/400 depending upon joint conditions.

| **Condition** | **Self weight** |
|---|---|
| Riveted/bolted plate girder | $\frac{wL}{300}$ kN/m |
| Welded plate girder | $\frac{wL}{400}$ kN/m |

**114. According to IS 800 – 71, the minimum thickness of vertically stiffened web plate shall not be less than**

(a) $\frac{d}{95}$ (b) $\frac{d}{250}$

(c) $\frac{d}{225}$ (d) $\frac{d}{200}$

**Ans. (d) :** If $\frac{d_1}{t_w} > 200$, vertical stiffeners are provided to prevent buckling of web due to diagonal compression.

**115. Stiffeners are used in a plate girder**

(a) To reduce compressive strength
(b) To reduce the shear stress
(c) To avoid the buckling of web plate
(d) To take the bearing stress

**Ans. (c) :** Stiffeners are used in plate girders to avoid the buckling of web plates by providing resistance to longitudinal and transverse shear.

**116. The minimum pitch of rivet holes of diameter 'd' should not be less than**

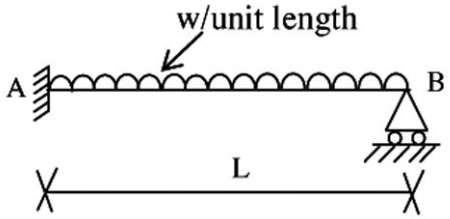
(a) 1.5d (b) 2.5d

(c) d (d) 2d

**Ans. (b) :** Minimum pitch of rivet holes $\geq 2.5d$

d : diameter of rivet

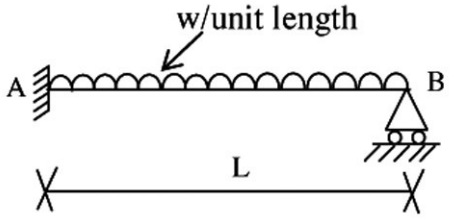
**117. In propped cantilever loaded as shown in figure, the plastic hinge will form at**




(a) 0.414 L from B (b) 0.414 L from A

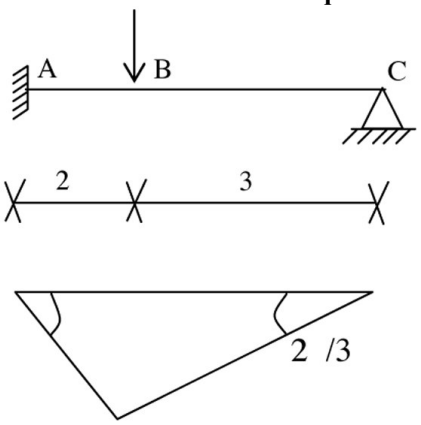
(c) L/2 from B (d) L/2 from A

**Ans. (a) :**




Whenever, a beam is supported as fixed from one end and rolled at other end, the plastic hinge will always be formed at 0.414 times span of beam from roller support i.e. from (B)

**118. A uniform beam shown in figure has the plastic moment Mp for span AB and 0.9 Mp for span BC. The correct virtual work equation is**



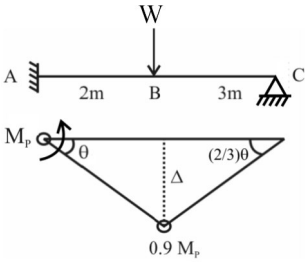

(a) $M_P.\theta + M_P\left(\theta + \frac{2\theta}{3}\right) = W.2\theta$

(b) $M_P\theta + M_P\theta + 0.9M_P\frac{2\theta}{3} = W.2\theta$

(c) $M_P.\theta + 0.9M_P\left(\theta + \frac{2\theta}{3}\right) = W.2\theta$

(d) $M_P\theta + 0.9M_P\left(\theta + \frac{2\theta}{3} + \frac{2\theta}{3}\right) = W.2\theta$

**Ans. (c):**




• If at 2 meter length θ is the slope then for 3 meter the slope will be

$\Delta = 2\theta$

$\Delta = 3x'$, where x' is slope at C.

$2\theta = 3x'$

$x' = \frac{2\theta}{3}$

So,

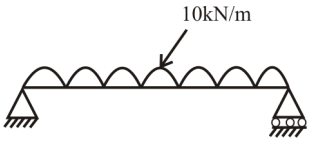
$M_p\theta + 0.9M_p\left(\theta + \frac{2}{3}\theta\right) = W.2\theta$

**119. A steel beam of rectangular cross – section is clamped at both ends. Plastic deformation is just observed when the udl on the beam is 10 kN/m. At the instant of collapse, the load on the beam will be**

(a) 10 kN/m (b) 15 kN/m

(c) 20 kN/m (d) 30 kN/m

**Ans. (b) :**




When the load is such that maximum moment is equal to yield moment ($M_y$), the beam start deforming. The load for M is given as

$W_1 = 10$ kN/m

The collapse load will be 4/3 of given load (as given condition)

$\therefore$ Collapse load $= \frac{4}{3} \times 10$

$= 13.33$ kN/m

So, as given option, collapse load will be 15 kN/m (nearest)

**120. One of the criteria for the effective width of flange of T – beam is $b_f = \frac{l_0}{6} + b_w + 6D_f$ In the above formula, $l_0$ signifies**

(a) Effective span of T – beam

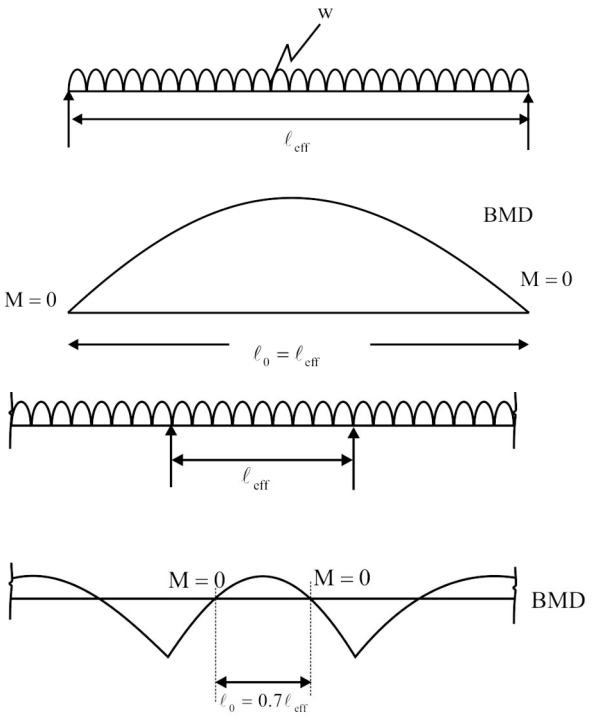
(b) distance between points of zero moments in the beam

(c) Clear span of the beam

(d) Distance between points of maximum moments in the beam

**Ans. (b):** $b_f = \frac{l_o}{6} + b_w + 6D_f$ (effective width of flange of T beam)

- This effective width is calculated on the basis of distance between points of zero moment in the beam.
- For simply supported beam, $\ell_0 = \ell_{eff}\,(\text{span})$
- For fixed beam or continuous beam, $\ell_0 = 0.7\ell_{eff}\,(\text{span})$




**121. The trap used for a water closet is called**
(a) Gully trap (b) Anti – siphon trap
(c) Intercepting trap (d) P – trap

**Ans. (d) :** Traps may be defined as fittings, placed at the ends of the soil pipes or sullage pipes to prevent passage of foul gases from the pipes to the outside. Generally P-traps are used for water closet systems.

**122. A soil has a bulk density of 22 kN/m$^2$ and water content 10%. The dry density of soil is**
(a) 18. 6 kN/m$^2$ (b) 20. 0 kN/m$^2$
(c) 22. 0 kN/m$^2$ (d) 23. 2 kN/m$^2$

**Ans. (b) :** Bulk density of soil $\gamma = 22$ kN/m$^2$

$w = 10\%$

dry bulk density $\gamma_d = \frac{\gamma}{1+w}$

$\gamma_d = \frac{22}{1+0.1}$

$\gamma_d = 20\text{kN/m}^2$

**123. Which of the following is a measure of particle size range?**
(a) Effective size
(b) Uniformity co – efficient
(c) Co – efficient of curvature
(d) None of the above

**Ans. (b) :** Uniformity coefficient is a measure of particle size range.

Coefficient of uniformity ($C_u$) = $\frac{D_{60}}{D_{10}}$

$D_{10}$ = effective size of particle

$D_{10}$ = particle size such that 10% of the soil is finer than this size

$D_{60}$ = particle size such that 60% of the soil is finer than this size

**124. If the plasticity index of a soil mass is zero the soil is**
(a) Sand (b) Silt
(c) Clay (d) Clayey silt

**Ans. (a) :** "If the plasticity index of a soil is zero the soil is sand."

$\boxed{I_P = W_L - W_P}$

if, $W_L = W_P$

$I_P = W_L - W_L = 0$

if, $W_P > W_L$

$I_P = W_L - W_P$

= zero (because $I_P$ is can't be –ve)

Where,

$I_P$ = plasticity index

$W_L$ = liquid limit

$W_P$ = plastic limit

In the case of sandy soils, plastic limit should be determined first. When plastic limit cannot be determined, the plasticity index is reported as NP(non-plastic). When the plastic limit is equal to or greater than the liquid limit, the plasticity index is reported as zero.

**125. According to IS – classification, the range of silt size particles is**
(a) 4.75 mm to 2.00 mm
(b) 2.00 mm to 0.425 mm
(c) 0.425 mm to 0.075 mm
(d) 0.075 mm to 0.002 mm

**Ans. (d) :** Range of silt size particle = 0.075mm to 0.002 mm

**126. Which one of the following clays behaves like dense sand ?**
(a) Over – consolidated clay with a high over – consolidation ratio
(b) Over – consolidated clay with a low over – consolidation ratio
(c) Normally consolidated clay
(d) Under - consolidated clay

**Ans. (a) :** Over-consolidated clay with a high over consolidation ratio behaves like dense sand

over consolidation ratio $= \frac{\text{Pre consolidation pressure}}{\text{current effective stress}}$

**127. Relationship between dry density $r_d$, percentage of air voids $\eta_a$, water content w and specific gravity G of any soil is**

(a) $\gamma_d = \frac{(1+\eta_a)}{1+wG}$ (b) $\gamma_d = \frac{(1+\eta_a)G\gamma_w}{1-wG}$

(c) $\gamma_d = \frac{(1-\eta_a)G\gamma_w}{1+wG}$ (d) $\gamma_d = \frac{(1-\eta_a)G\gamma_w}{1-wG}$

**Ans. (c) :** $\gamma_d = \frac{(1-\eta_a)G\gamma_w}{1+wG}$ Se = Gw

$\eta_a$ = % air void
G = specific gravity
$\gamma_w$ = unit wt. of water
w = water content

**128. The ratio of the undisturbed shear strength to the remoulded shear strength, in cohesive soils under undrained conditions is**

(a) Zero (b) 1
(c) Greater than 1 (d) Between 0 and 1

**Ans. (c) :** $\frac{\text{undisturbed shear stress}}{\text{Remoulded shear strength}} > 1$ (for cohesive soil)

because, on remolding due to internal friction reduction, shear strength reduces.

**129. The critical height of unsupported vertical cut in a cohesive soil is given by**

(a) $\frac{4c}{\gamma}\tan\left(45^\circ+\frac{\phi}{2}\right)$ (b) $\frac{2c}{\gamma}\tan\left(45^\circ+\frac{\phi}{2}\right)$

(c) $\frac{4c}{\gamma}\cot\left(45^\circ+\frac{\phi}{2}\right)$ (d) $\frac{2c}{\gamma}\cot\left(45^\circ+\frac{\phi}{2}\right)$

**Ans. (a) :** Critical height of unsupported vertical cut at cohesive soil $= \frac{4c}{\gamma}\tan\left(45^\circ+\phi/2\right)$

c = soil cohesion
$\gamma$ = unit wt. of soil
$\phi$ = internal friction

**130. A plate load test is useful to estimate**

(a) Bearing capacity of foundation
(b) Settlement of foundation
(c) Both (1) and (2)
(d) None of these

**Ans. (c) :** A plate load test is used to –
- Bearing capacity of foundation
- Settlement of foundation

taking empirical formula as relation b/w plates and foundation

**131. The correct increasing order of the surface areas of the given soils is**

(a) Silt, sand, colloids, clay
(b) Sand, silt, colloids, clay
(c) Sand, silt, clay colloids
(d) Clay, silt, sand, colloids

**Ans. (c) :** Surface area→ sand < silt < clay < colloids. As size by particle decreases, surface area increases.

**132. For a given soil sample,**
**$C_c$ = Co - efficient of gradation**
**$C_u$ = Co - efficient of uniformity**
**$D_{10}$ = effective size**
**$D_{30}$ = diameter through which 30 percent of the total soil mass is passing.**
**If $C_c$ = 1.0, $C_u$ = 4.0, then the value of $D_{30}$ / $D_{10}$ would be**

(a) 2.00 (b) 1.75
(c) 1.50 (d) 1.25

**Ans. (a) :** $C_u = \frac{D_{60}}{D_{10}}$ , $C_c = \frac{D_{30}^2}{D_{60}.D_{10}}$

$u = \frac{D_{60}}{D_{10}}$ , $1 = \frac{D_{30}^2}{D_{60}.D_{10}}$

$D_{60} = 4D_{10}$ $D_{30}^2 = 4D_{10}^2$

$\frac{D_{30}}{D_{10}} = 2$

**133. The total settlement of a compressible soil stratum 2 m deep and having a coefficient of volume compressibility of 0.02 $cm^2$/kg under a pressure increment of 2 $kg/cm^2$ will be**

(a) 2 cm (b) 4 cm
(c) 8 cm (d) 10 cm

**Ans. (c) :** Coefficient of volume compressibility
$= 0.02\ cm^2/kg$

$\Delta p = 2kg/cm^2$

soil depth = 2m

Total settlement of soil $\Rightarrow$

$\Delta h = Hm_v \Delta p$

$\Delta h = 2\times0.02\times2$

$\Delta h = 0.08$ m

$\Delta h = 8$ cm

**134. The determination of ultimate bearing capacity on eccentrically loaded square footing depends upon the concept of useful**

(a) square (b) width
(c) triangle (d) circle

**Ans. (b) :** Ultimate bearing capacity (square footing)

$= cN_c + qN_q + \frac{1}{2}\gamma\ BN_\gamma$

$= cN_c + \gamma D_\delta N_q + \frac{1}{2}\gamma\ BN_\gamma$

depends upon width of footing

**135. In consolidation testing, curve fitting method is used to determine**

(a) Compression index
(b) Swelling index
(c) Co-efficient of consolidation
(d) None of these

**Ans. (c) :** In consolidation test, curve fitting method is used to determine coefficient of consolidation.

**136. Westergaard's analysis for stress distribution beneath loaded areas is applicable to**

(a) sandy soils (b) clayey soils
(c) stratified soils (d) silty soils

**Ans. (c) :** Westergaard's analysis for stress distribution is applicable for stratified soil.
Assumptions in Westergard's analysis–

(i) The soil is elastic and semi-infinite and used for stratified soils.
(ii) Soil is composed of numerous closely spaced horizontal layers of negligible thickness of a finite rigid material.
(iii) The rigid material permits only the downward deformation of mass in which horizontal deformation is zero.

Werstergard's equation $(\sigma_{zw}) = \frac{Q}{\pi z^2}\frac{1}{\left(1+2\left(\frac{r}{z}\right)^2\right)^{3/2}}$

**137. If the true bearing of a line AB is 269º 30' then the azimuth of the line AB is**

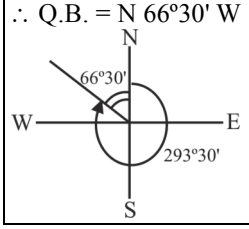
(a) 0º 30' (b) 89º 30'
(c) 90º 30' (d) 269º 30'

**Ans. (c) :** Line AB true bearing = $269^0 30'$
Hence line AB azimuth = $360^0 00'$ – true bearing
$= (360^0\text{-}00') - (269^0 30')$
$= \boxed{90^0 30'}$

**138. In the quadrantal bearing system, a whole circle bearing of 293° 30' can be expressed as**

(a) W 23° 30' N (b) N 66° 30' W
(c) S 113° 30' N (d) N 23° 30' W

**Ans. (b) :** W.C.B. = 293º30'
Q.B. = 360º – 293º30' = 66º30'
∴ Q.B. = N 66º30' W




**139. Which of the following methods of contouring is most suitable for a hilly terrain?**

(a) Direct method
(b) Square method
(c) Cross-sections method
(d) Tacheometric method

**Ans. (d) :** There are two methods of contouring:–
**(a) Direct method**
**(i) Vertical control –** Location of point on the contour.
(ii) Horizontal control
**(b) Indirect method**
**(i) Tacheometric method :** Suitable for contouring of very steep hill.
**(ii) Square (or) grid method :** Suitable for plain or faulty sloping area.
**(iii) Cross–section method:** Suitable for a type of land having very long strip but narrow width. (e.g. road, canal etc.)

**140. If a 30 m length can be taped with a precision of ± 0.01 m, then the standard error in measuring 1.08 km with the same precision will be**

(a) ± 0.54 m (b) ± 0.45 m
(c) ± 0.36 m (d) ± 0.06 m

**Ans. (c) :** No. of times tape used $= \frac{1.08\times1000}{30}$
$= 36$
Error in measurement = $36 \times (\pm 0.01)$
$= \pm 0.36$ m

**141. If in triangle ABC, b=300 m and angle ∠ABC = 60°, then the radius of the circular curve passing through the points A, B and C will be**

(a) 86.6 m (b) 100.0 m
(c) 173.2 m (d) 300.6 m

**Ans. (c) :** Now, AC is a chord of length 300 m having tangent angle 60º,
$\Delta = 180º - 60º$
$\Delta = 120º$
$b = 2R \sin \Delta/2$
$300 = 2R \sin (120º/2)$
$R = 173.21$ m

**142. Which one of the following specifications for length of base line refers to the "Third order triangulation system"?**

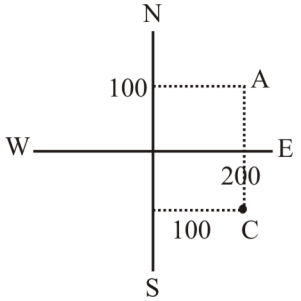
(a) 0.5 to 3 km (b) 1.5 to 5 km
(c) 5 to 15 km (d) 10 to 20 km

**Ans. (d) :** According to books:
'Third order triangulation system' has length of base line = 10 to 20 km

**143. If the co-ordinates of A are 100 N and 200 E and those of C are 100 S and 200 E, then the length AC is**

(a) 400.00 (b) 282.84
(c) 244.94 (d) 200.00

**Ans. (d) :**




AC = AE + EC
From the given geometry length of AC = 200

**144. The maximum limit of water absorption for aggregate suitable for road construction is**

(a) 0.4% (b) 0.6%
(c) 0.8% (d) 1.0%

**Ans. (b) :** Rock specimen having more than 0.6 percent water absorption are considered unsatisfactory unless found acceptable based on strength test. However slightly higher value of porosity may be acceptable for aggregates used in bituminous pavement construction if the aggregates are other-wise found suitable.

**145. Which of the following represents hardest grade of bitumen?**

(a) 30/40 (b) 60/70

(c) 80/100 (d) 100/120

**Ans. (a) :** 30/40 Bitumen is hardest because the penetration of 30/40 bitumen is 3-4 mm which is least.

$$\text{penetration} = \frac{1}{10} \times \text{Reading scale}$$

$$= \frac{1}{10} \times (30 \text{ to } 40)$$

= (3 to 4) mm

- The hardest grade of bitumen is 30/40
- The unit of penetration is 1/10 mm.
- Thus 80/100 grade bitumen means a penetration of 8-10 mm.
- Common grades are 30/40, 60/70 and 80/100.

**146. For the construction of water bound macadam roads, the correct sequence of operations after spreading the coarse aggregate is**

(a) dry rolling, wet rolling, application of screening and application of filler

(b) dry rolling, application of filler, wet rolling and application of screening

(c) dry rolling, application of screening, wet rolling and application of filler

(d) dry rolling, application of screening, application of filler and wet rolling

**Ans. (b) :** For construction of water bound macadam road steps are

dry rolling

↓

application of filler

↓

wet rolling and application of screening

In WBM mainly stone dust is used as binding material.

- In WBM the voids are filled with screening and binding material with the assistance of water.
- WBM is the sub-base layer of flexible pavement.
- Thickness of each compacted layer of WBM ranges from 10.0 cm to 7.5 cm (descending on size of aggregate and gradation).

**147. The thickness of bituminous carpet varies from**

(a) 20 to 25 mm

(b) 50 to 75 mm

(c) 75 to 100 mm

(d) 100 to 120 mm

**Ans. (a) :** Bitumen carpet having thickness of 20 to 25 mm.

**148. Expansion joints in cement concrete pavements are provided at an interval of**

(a) 10 m

(b) 15 m

(c) 18 m to 21 m

(d) 25 m to 30 m

**Ans. (d) :** Cement concrete pavements, expansion joints should be at an interval of 25m to 30m

As per IS 6509 : 1485 clause 5.3.1.2

| Type of slab | Width of slab (m) | Spacing for expansion joint | Spacing for contraction joint |
|---|---|---|---|
| RCC | 0.25 | 51 m | 17 m |
| | 0.20 | 45 m | 14 m |
| | 0.15 | 36 m | 13 m |
| | 0.15 | 30 m | 7.5 m |
| PCC | 0.20 & above | 36 m | 4.5 m |
| | 0.15 | 27 m | 4.5 m |
| | 0.10 | 27 m | 4.5 m |

As per IRC the maximum specify speeding of contraction joints in rigid joints in rigid pavements is 4.5 m.

**149. A summit curve is formed at the intersection of a 3% up gradient and 5% down gradient to provide a stopping distance of 128 m. The length of summit curve needed will be**

(a) 271 m (b) 298 m

(c) 322 m (d) 340 m

**Ans. (b) :** Deviation angle, $n_1 = 3\%, n_2 = 5\%$

$N = n_1 - (-n_2) = n_1 + n_2$

$N = 3\% + 5\% = 8\%$

$L > SSD, \quad S = 128$ m.

$$\text{Length of summit curve } L = \frac{NS^2}{4.4} = \frac{8}{100} \times \frac{(128)^2}{4.4}$$

L = 298 m.

**150. Which one of the following binders is recommended for a wet and cold climate**

(a) 80/100 penetration asphalt

(b) tar

(c) cut back

(d) emulsion

**Ans. (d) : Bitumen emulsion–**

- Emulsion in the two phase system consisting of two immiscible liquid.
- The bitumen content in emulsion range from 40–60% and remaining portion of water.
- The average size of bitumen particle in this emulsion is about 2 mm.
- These are generally used in maintenance, patch repair work and even in wet weather condition

# Andhra Pradesh Public Service Commission
## Assistant Engineer (Public Health & Municipal Engineer) Exam, 2012
## Solved Paper

**1. For the same discharge in a pipe with laminar flow, reduction of radius to half will increase the pressure gradient by a factor of**

(a) 2 (b) 4
(c) 8 (d) 16

**Ans. (d) :** According to Hagen-Poiseuille Equation

Discharge (Q) $= \frac{\pi}{8\mu}\left(\frac{-\partial P}{\partial x}\right)R^4$; where R = Radius of pipe

For same discharge–

Pressure gradient $\left(\frac{-\partial P}{\partial x}\right) \propto \frac{1}{R^4}$

where insight shows the pressure decreases in the pipe along the length

Let, Pressure gradient $\left(\frac{-\partial P}{\partial x}\right) = P$

$\therefore \frac{P_1}{P_2} = \left(\frac{R_2}{R_1}\right)^4$

Given, $R_1 = R$ and $R_2 = R/2$

$\therefore \frac{P_1}{P_2} = \left(\frac{R/2}{R}\right)^4 = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$

$P_2 = 16\, P_1$

$\Rightarrow$ Factor of pressure gradient $\frac{P_2}{P_1} = 16$

**2. A pipe is said to be a siphon if**

(a) it can operate without external power
(b) it can discharge a liquid at a fast rate
(c) it has sub-atmospheric pressure in it
(d) the exit is at a lower level than the inlet

**Ans. (c) :** A pipe is said to be a siphon if it has sub-atmospheric pressure in it.
- Siphon is a long bent pipe which is used to transfer liquid from a reservoir at a higher elevation in another reservoir at a lower level when the two reservoirs are separated by a hill or high level ground as shown in figure–

**3. In supersonic flow, a diverging passage results in**

(a) increase in velocity and pressure
(b) decrease in pressure and density
(c) increase in velocity and density
(d) decrease in velocity and pressure

**Ans. (b) :** In supersonic flow, a diverging passage results in increasing divergent cross-section area (Example : Divergent Nozzle, increase in velocity and density.

**4. Which one of the following is dimensionless?**

(a) $\frac{\partial P}{\partial x}\cdot\frac{D^4}{\mu Q^2}$ (b) $\frac{\partial P}{\partial x}\cdot\frac{D^3}{\mu Q}$
(c) $\frac{\partial P}{\partial x}\cdot\frac{D^4}{\mu Q}$ (d) $\frac{\partial P}{\partial x}\cdot\frac{\mu Q}{D^4}$

**Ans. (c) :** $\frac{\partial P}{\partial x}\times\frac{D^4}{\mu Q}$ is dimension less.

Now, dimension of $\frac{\partial P}{\partial x} = [ML^{-2}T^{-2}]$, $D^4 = [L^4]$

$\mu = [ML^{-1}T^{-1}]$ $Q = [L^3T^{-1}]$

$\frac{\partial P}{\partial x}\cdot\frac{D^4}{\mu Q} = \frac{\left[ML^{-2}T^{-2}\right]\times\left[L^4\right]}{\left[ML^{-1}T^{-1}\right]\times\left[L^3T^{-1}\right]} = [M^0L^0T^0]$

**5. Assertion (A) : In a supersonic flow, the velocity increases as the area increases.**
**Reason (R) : Conservation of masses, momentum and energy require that dA/A should increase with dM/M for M greater than unity.**

(a) A is true; R is true, and it explains A
(b) A is true; R is true, but it does not explain A
(c) A is true; R is false
(d) A is false; R is true

**Ans. (a) :** In the supersonic flow, velocity of flow exceed the velocity of sound. For supersonic flow divergent portion of the nozzle is used and Mach number is greater than unity. (M>1). Hence R is true, A is true, and R explains A.

**6. How does the sonic velocity in air at STP condition change with the altitude in the standard atmosphere?**

(a) 340.3 m/s (b) 170.15 m/s
(c) 0 (d) 680.6 m/s

**Ans. (a) :** The velocity of air at STP condition–

$$C = \sqrt{\gamma RT}$$

Where, $\gamma = 1.4$
R = 287 J/kg
K and T = 288.15 K

So, $C = \sqrt{1.4\times 287\times 288.15}$
C = 340.3 m/s

**7. Match List I with List II and select the correct answer using the codes given below the Lists:**

| List I | List II |
|---|---|
| **A. Oscillating jump** | **i. less than 2** |
| **B. Strong jump** | **ii. between 2 and 3** |
| **C. Weak jump** | **iii. around 4** |
| **D. Undular jump** | **iv. around 10** |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | iii | iv | ii | i |
| (b) | i | ii | iii | iv |
| (c) | iv | iii | i | ii |
| (d) | ii | i | iii | iv |

**Ans. (a) :**
(a) Oscillating jump → around 4, (2.5 – 4.5)
(b) Strong jump → around 10, ( ≥9 )
(c) Weak jump → between 2 and 3, (1.7 – 2.5)
(d) Undular jump → less than 2, (1 – 1.7)

**8. A rectangular channel of 4 m width conveys water at 8 $m^3$/s under critical condition. Specified energy for this flow is**
(a) 1.1123 m (b) 1.4830 m
(c) 0.3703 m (d) 0.7416 m

**Ans. (a) :** Given, Width of channel (b) = 4 m
Discharge (Q) = 8 $m^3$/sec

$$\therefore \text{Critical depth } (h_c) = \left(\frac{q^2}{g}\right)^{1/3} = \left[\frac{2^2}{9.81}\right]^{1/3}$$

$h_c = 0.74153$ m
Therefore, Minimum specific energy is related with critical depth by relation –

$$E_{min} = \frac{3}{2}h_e = \frac{3}{2}\times 0.74153$$

$E_{min} = 1.1123$ m

**9. Match List I with List II and select the correct answer using the codes given below the Lists:**

| List I | List II |
|---|---|
| **A. a sphere** | **i. 0.2** |
| **B. an aerofoil** | **ii. 0.3** |
| **C. a bluff body** | **iii. 1.6** |
| **D. a racing car** | **iv. 0.1** |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | i | iv | iii | ii |
| (b) | i | iv | ii | iii |
| (c) | iv | i | iii | ii |
| d) | iv | i | ii | iii |

**Ans. (a) :** (a) Asphere → 0.2
(b) Air aerofoil → 0.1
(c) A bluff body → 1.6
(d) A racing car → 0.3

**10. A trip wire is mounted near the leading edge of a body**
(a) to increase the lift
(b) to decrease the lift
(c) to advance the point of separation
(d) to delay the point of separation

**Ans. (d) :** To delay the point of separation, A tripwire is mounted near the leading edge of body.

**11. The flow in a capillary tube is laminar because**
(a) the capillary tube is made of glass
(b) the surface tension and capillarity promote laminarisation
(c) only inviscid liquids can flow through a capillarity tube
(d) the diameter of the capillarity tube is very small

**Ans. (d) :** The flow in a capillary tube is laminar flow because the diameter of the capillary tube is very small. If the diameter of the capillary tube is large then there will be turbulence in the capillary tube and laminar flow not occurs.

**12. Impingement of a jet on a flat plate may be idealised by**
(a) $\psi = xy$ (b) $\psi = x^2 - y^2$
(c) $\psi = x^2 + y^2$ (d) $\psi = x/y$

**Ans. (a) :** Impingement of a jet on a flat plate may be idealized by equation $\boxed{\psi = xy}$

**13. Coefficients of velocity, contraction and discharge arranged in increasing order of value are**
(a) $C_v$, $C_C$, $C_d$ (b) $C_d$, $C_C$, $C_v$
(c) $C_v$, $C_d$, $C_C$ (d) $C_C$, $C_d$, $C_v$

**Ans. (b) :** $C_v$ = coefficient of velocity
$C_c$ = coefficient of contraction
$C_d$ = coefficient of discharge
For an orifice –
$C_v$ = 0.95 to 0.99
$C_c$ = 0.61 to 0.69
$C_d$ = 0.61 o 0.65
Hence, Arranging in increasing order –
$C_d < C_c < C_v$ or $C_d, C_c, C_v$

**14. In a converging steady flow, there is**
(a) no acceleration
(b) no temporal acceleration
(c) only convective acceleration
(d) convective and temporal acceleration

**Ans. (c) :** In a converging steady flow, there is change in magnitude w.r.t. space. Therefore, only convective acceleration is exist and tampered acceleration is zero because there is no any nominal component.

**15. A solid body sinks in a fluid when**
(a) the specific gravity of its material is greater than unity
(b) the buoyancy force does not pass through the metacentre
(c) the weight of the fluid displaced is less than the weight of the body
(d) the metacentre lies below the C.G.

**Ans. (c) :** When a solid body is emerged in the fluid which weight is more than the weight of displaced fluid then the solid body sinks in the fluid.

**16. Select the correct statement.**
(a) Viscosity of gas increases with temperature.
(b) Density of gas increases with temperature.
(c) Surface tension of liquid increases with temperature.
(d) Bulk modulus is independent of temperature.

**Ans. (a) :** Viscosity of gas increases with temperature. in case of gases the cohesive forces are small and molecule momentum transfer predominates. With the increase in temperature, molecular momentum transfer increases and hence viscosity increases.
For gas; $\mu = \mu_0 + \alpha t + \beta t^2$
where; $\alpha$ and $\beta$ are constant for gas
$\mu_0$ = viscosity of gas at $0^0$C in poise
$\mu$ = viscosity of gas at $t^0$C in poise

**17. Measurement of velocity without any obstruction to the flow in a pipe may be made by**
(a) pilot-static probe
(b) hot-wire anemometer
(c) hot-film anemometer
(d) laser-doppler anemometer

**Ans. (d)** Lasser-doppler anemometer used in measurement of velocity without any obstruction to the flow in a pipe.

**18. Schlieren flow visualisation technique operates by employing**
(a) density variation in flow
(b) density gradients in flow
(c) second derivatives of density
(d) higher derivatives of density

**Ans. (b) :** Schlieren is a device which indicates the density gradient $\frac{d\rho}{dy}$ in flow.
The illumination or contrast on a screen is dependent the focal length of a lens, adjustment of a knife edge the density gradient in fluid flow.

**19. The integral momentum equation requires the assumption that.**
(a) the flow is uniform
(b) the flow is unidirectional
(c) the fluid is incompressible
(d) the flow is steady

**Ans. (d) :** The integral momentum equation requires the assumption that the flow is steady.

**20. The critical angle of attack of an aerofoil is that where**
(a) the lift becomes zero
(b) the drag becomes zero
(c) the drag begins to rise
(d) the lift begins to drop

**Ans. (d) :** The critical angle of attack of an aerafoil is that where the lift begins to drop.

**21. The height through which water rises by capillary action in a glass tube of 2 mm bore is the surface tension at the prevailing temperature is 0.075 g/cm, is**
(a) 1.5 cm (b) 3 cm
(c) 0.75 cm (d) 10.3 cm

**Ans. (a) :** Given,
Glass tube bere (d) = 2 mm = $2 \times 10^{-3}$ m
Surface tension ($\sigma$)= 0.075 g/cm
= 0.07354 Newton/meter

$$= 0.07354 \ \frac{kg - \frac{m}{s^2}}{m}$$

Let,
h= height of rise water due to capillary action

$$h = = \frac{4\sigma\cos\phi}{\rho g d}$$

for water glass tube, $\theta = 0^0$ and $\rho = 1000$ kg/m$^3$

$$h = \frac{4\sigma}{\rho g d} = \frac{4 \times 0.07354 \ \frac{kg - m/sec^2}{m}}{1000\frac{kg}{m^3} \times 9.81\frac{m}{sec^2} \times 2 \times 10^{-3} m}$$

h = 0.01499 m
= 1.499 cm ≈ 1.5 cm

**22. Calculate the maximum allowable discharge of water through a venturimeter throat 5 cm, fitted in a 10 cm diameter line with its inlet at an open channel. Assume $C_d$ = 0.95.**
(a) 0.0224 m$^3$/s (b) 0.0448 m$^3$/s
(c) 0.8 m$^3$/s (d) 0.04 m$^3$/s

**Ans. (a) :** Question incomplete.

**23. There is a pin bole of diameter 0.1 mm at the bottom of a glass vessel containing mercury. What depth of mercury can be put in the vessel without any leakage occurring? Take $\sigma$ = 0.55 N/m**
(a) h = 0.165 cos θ m (b) h = 0.23 cos θ m
(c) h = 0 m (d) h = 10.3 m

**Ans. (a) :** Given :
diameter of hole (d) = 0.1 mm = $0.1 \times 10^{-3}$ m
surface tension ($\sigma$) = 0.55 N/m
Let, θ = Angle of contact for mercury and glass

$$h = \frac{4\sigma\cos\theta}{\rho g D} \qquad \therefore \rho_{mercury} = 13600 \text{ kg/m}^3$$

$$= = \frac{4 \times 0.55 \cos\theta}{13600 \times 9.81 \times 0.1 \times 10^{-3}}$$

= 0.1648 cos θ meter
= 0.1648 cos θ meter
h ≈ 0.165 cos θ m

**24. Match List I with List II and select the correct answer using the codes given below the Lists:**

| List I | List II |
|---|---|
| A. Kaplan turbine | i. works at atmospheric pressure |
| B. Pelton wheel | ii. high-part load efficiency |
| C. Axial flow pumps | iii. pressure head recovery |
| D. Draft tube | iv. high value of $N_S$ |

Code:

| | A | B | C | D |
|---|---|---|---|---|
| (a) | ii | i | iv | iii |
| (b) | i | ii | iv | iii |
| (c) | ii | i | iii | iv |
| (d) | i | ii | iii | iv |

**Ans. (a) :**
Kaplan turbine → high-part load efficiency
Pelton wheel → work at atmospheric pressure
Axial flow pump → high value of specific speed (Ns)
Draft tube → pressure head recovery

**25. A dimensionless group formed with the variables ρ, w, μ and D is**
(a) $\rho w\mu/D^2$ (b) $\rho wD^2/\mu$
(c) $\mu D^2\rho w$ (d) $\rho w\mu D$

**Ans. (b)** Dimension of each variables are–
$\rho = kg/m^3 = ML^{-3}$
$w = 1/second = T^{-1}$
$\mu = N\text{-}sec/m^2 = ML^{-1}T^{-1}$
$D = meter = L$

$$\therefore \frac{\rho wD^2}{\mu} = \frac{\left[ML^{-3}\right]\times\left[T^{-1}\right]\times\left[L^2\right]}{\left[ML^{-1}T^{-1}\right]}$$
$$= [M^0L^0T^0]$$
This is the dimensionless with the variables

**26. An air stream with a velocity of 300 m/s at 10 kN/m² vacuum and temperature of 320 K passes through a section where its gauge is 20 kN/m². Compute its stagnation properties (M, T, P).**
(a) 0.84, 365K, 142 kN/m²
(b) 0.96, 385K, 284 kN/m²
(c) 0.42, 225K, 71kN/m²
(d) None of the above

**Ans. (a) :** From data,
$P_1 = 100 - 10 = 90$ kN/m²
$\gamma = 1.4$,
$u_1 = 300$ m/s, $R = 287$ J/kg
$T_1 = 320$ K,
$C_P = \gamma/(\gamma - 1)$

$$R = \frac{1.4\times 287}{0.4} = 1004.5$$

$$C_1 = \sqrt{\gamma RT_1} = \sqrt{1.4\times 287\times 320} = 358.57 \text{ m/s}$$

$$M = \frac{u_1}{C_1} = \frac{300}{358.57} = 0.8366 \cong \boxed{0.84}$$

$$T = T_1\left(1+\frac{(\gamma-1)}{2}\right)M_1^2$$
$= 320\,(1 + 0.2 \times 0.84^2) = 364.79$ K $\cong \boxed{365\text{ K}}$
$P = 90 \times 1.14^{3/5} = 142.36 \cong \boxed{142\text{ kN/m}^2}$

**27. A 3 meter wide rectangular channel flowing at its normal depth of 0.8 m carries a discharge of 5.5 m³/s. The bed slope of the channel is**
(a) Steep (b) Critical
(c) Mild (d) Adverse

**Ans. (c) :** Given,
$Q = 5.5$ m³/sec.
$y_n = 0.8$ m

$$y_c = \left(\frac{q^2}{g}\right)^{1/3}$$
$$= \left(\frac{\left(\frac{5.5}{3}\right)^2}{9.81}\right)^{1/3}$$
$y_c = 0.70$
$y_n > y_c$
So, slope is mild

**28. The head loss in a sudden pipe expansion from area $A_1$ to area $A_2$ and from velocity $u_1$ to velocity $u_2$ is given by**
(a) $(1-A_1/A_2)^2.\frac{u_1^2}{2g}$ (b) $(1-A_1/A_2)^2.\frac{u_2^2}{2g}$
(c) $(1-A_2/A_1)^2.\frac{u_2^2}{2g}$ (d) $(1-A_2/A_1)^2.\frac{u_1^2}{2g}$

**Ans. (a) :** Head loss in a sudden pipe expansion = $\left(1-\frac{A_1}{A_2}\right)^2\times\frac{u_1^2}{2g}$

**29. Compare the cost of pumping the same fluid at the same flow rate through a 150 mm pipe and through 200 mm pipe both having the same roughness factor = 0.03 (negligible minor loss).**
(a) 4.2 times (b) 2.1 times
(c) 8.4 times (d) 10 times

**Ans. (a) :** Flow rate = Q
$d_1 = 150$ mm, $d_2 = 200$ mm

$$\text{Head loss, } h = \frac{flQ^2}{12d^5}$$

The ratio of head loss in two pipes–

$$\frac{h_1}{h_2} = \frac{flQ^2}{12d_1^5}\times\frac{12d_2^5}{flQ^2}$$

$$\frac{h_1}{h_2} = \frac{d_2^5}{d_1^5} = \left(\frac{0.2}{0.15}\right)^5 = 4.21$$

The cost of pumping varies as the power required for pumping, which in turn depends directly upon the friction head loss. The cost of pumping the same fluid at same discharge through 150 mm pipe is, therefore, 4.2 times the cost of pumping it through a 200 mm pipe.

**30. A 2 m diameter cylinder rotates at 1800 r.p.m. in a stream ($\rho = 1.225$ kg/m³) of 25 m/s. Estimate the lift per unit length of the cylinder.**
(a) L = 11.55 kN (b) L = 9.22 kN
(c) L = 24 kN (d) L = 5.99 kN

**Ans. (a) :**

**31. A thin cylinder of radius r and thickness t when subjected to an internal hydrostatic pressure 'p' causes a radial displacement 'u'. Then the tangential strain caused is**
(a) $\frac{du}{dr}$ (b) $\frac{1}{r}.\frac{du}{dt}$
(c) $\frac{u}{r}$ (d) $\frac{2r}{r}$

**Ans. (a) :** Given : r = radius of the thin cylinder
u = radial displacement
∴ Tangential strain =

$$= \frac{\text{change in radial displacement}(du)}{\text{Change in radius}(dr)}$$

$$= \frac{du}{dr}$$

**32. A body having weight of 1000 N is dropped from a height of 10 cm over a close coiled helical spring of stiffness 200 N/cm. The resulting deflection of spring is nearly**
(a) 5 cm (b) 16 cm
(c) 35 cm (d) 100 cm

**Ans. (b) :** We known,
mg = 1000N, n = 10 cm, k = 200 N/cm

$$mg(h + x) = \frac{1}{2}kx^2$$

$$1000[10 + x] = \frac{1}{2}[200]x^2$$

$$100 + 10x - x^2 = 0$$

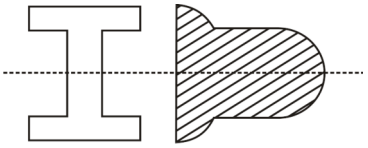
∴ x = 16.18
x ≈ 16cm

**33. Which of the following stresses are associated with the design of pins in bushed pin type flexible coupling?**
(i) Bearing stress (ii) Bending stress
(iii) Axial tensile (iv) Transverse shear
**Code:**
(a) (i), (iii) and (iv) (b) (ii), (iii) and (iv)
(c) (i), (ii) and (iii) (d) (i), (ii) and (iv)

**Ans. (d) :** Stresses are associated with the design of pins in bushed pin type flexible coupling is –
1- Bearing stress
2- Bending stress
3- Transverse shear

**34. In a beam of I section, the maximum shear force is carried by**
(a) the upper flange (b) the web
(c) the lower flange (d) Any of these

**Ans. (b) :** In a beam of I-section, the maximum shear force is carried by the web.
In a I-section of a beam subjected to transverse shear force the maximum shear stress is developed at the centre of the web.



■ The maximum shear stress is at the neutral axis. So, max. shear stress is carried by the web portion in I-beam

**35. Autofrettage is the method of**
(a) joining thick cylinders
(b) calculating stresses in thick cylinders
(c) prestressing thick cylinders
(d) increasing the life of thick cylinders

**Ans. (c) :** Autofrettage is the method of prestressing thick cylinders.

**36. During tensile-testing of a specimen using a UTM, the parameters actually measured include.**
(a) True stress and true strain
(b) Poisson's ratio and Young's modulus
(c) Engineering stress and Engineering strain
(d) Load and elongation

**Ans. (d) :** From the Universal testing machine (UTM) during tensile testing of a specimen load and elongation measured.
A universal testing machine is a machine that is used to perform standard tensile and compressive tests on materials and structures.

**37. In the formulation of Lewis equation for toothed gearing, it is assumed that tangential tooth load '$F_1$' acts on the**
(a) pitch point
(b) tip of the tooth
(c) root of the tooth
(d) whole face of the tooth

**Ans. (a)** Lewis equation for tooth gearing, it is assumed that tangential tooth load $F_1$ acts on the pitch point.
Where,
P = power transmitted in watts
v = pitch line velocity in m/s
$C_s$ = service factor

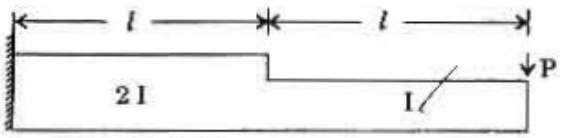
**38. In a thick cylinder pressurized from inside, the hoop stress is maximum at**
(a) the centre of the wall thickness
(b) the outer radius
(c) the inner radius
(d) both inner and outer radii

**Ans. (c) :** When a thick cylinder pressurized from inside, the hoop stress is maximum at the inner radius.

**39. Which one of the following features improves the fatigue strength of a metallic material?**
(a) Increasing the temperature
(b) Searching the surface
(c) Over stressing
(d) Under stressing

**Ans. (d) :** When we applying the stress at the surface of metallic material, then the fatigue strength of material improves.

**40.**




**$I = 375\times10^{-6}$ m$^4$**
***l* = 0.5 m**
**E = 200 GPa**
**Determine the stiffness of the beam shown in the above fig.**
(a) $12 \times 10^{10}$ N/m (b) $10 \times 10^{10}$ N/m
(c) $4 \times 10^{10}$ N/m (d) $8 \times 10^{10}$ N/m

**Ans. (a) :** Given ; $I = 375 \times 10^{-6}\ m^4$

$l = 0.5m$

$E = 200\ GPa = 200 \times 10^9\ Pa$

$\therefore$ Bending moment $(M_x) = P.x$

$$U = \int_0^l \frac{(px)^2}{2EI}.dx + \int_0^{2l} \frac{(px)^2}{2EI(2I)}.dx$$

$$= \left[\frac{P^2x^3}{3\times 2EI}\right]_0^l + \left[\frac{P^2x^3}{3\times 2E\times 2I}\right]_l^{2l}$$

$$= \frac{P^2l^3}{6EI} + \frac{P^2\left(8l^3 - l^3\right)}{12\ EI} = \frac{P^2l^3}{6EI} + \frac{7P^2l^3}{12EI}$$

$$U = \frac{9P^2l^3}{12EI} = \frac{3P^2l^3}{4EI} \qquad \text{..... (i)}$$

$$\because U = \frac{1}{2}P\times\delta = \frac{1}{2}\frac{P^2}{K} \qquad \text{.....(ii)}$$

From equation (1) and (2)

$$\frac{1}{2}\frac{P^2}{K} = \frac{3P^2l^3}{4EI} \Rightarrow K = \frac{2EI}{3l^3}$$

$$K = \frac{2\times 200\times 10^9 \times 375\times 10^{-6}}{3\times(0.5)^3} = 4\times 10^8\ N/m$$

No any option matching with the solution.

**41. Cermets are**

(a) metals for high temperature use with ceramic like properties

(b) ceramics with metallic strength and luster

(c) coated tool materials

(d) metal-ceramics composites

**Ans. (d) :** Composites material are composed of ceramic and metallic materials. A cermet is ideally designed to have the optimal properties of both a ceramic and metal, such as the ability to under plastic deformation.

**42. Circumferential and longitudinal strains in a cylindrical boiler under steam pressure are $\varepsilon_1$ and $\varepsilon_2$ respectively. Change in volume of the boiler cylinder per unit volume will be**

(a) $\varepsilon_1 + 2\varepsilon_2$ (b) $\varepsilon_1\varepsilon_2^2$

(c) $2\varepsilon_1 + \varepsilon_2$ (d) $\varepsilon_1^2\varepsilon_2$

**Ans. (c) :** Given, for cylindrical boiler steam pressure

$\varepsilon_1$= Circumferential strains

$\varepsilon_2$= Longitudinal strains

$$\frac{\delta V}{V} = ?$$

Where; $\delta V$ = change in volume

V = initial volume of cylinder.

For a cylinder, volume $(V) = \frac{\pi}{4}d^2L$

$\therefore$ Change in the volume $(\delta_V) = \frac{\pi}{4}d^2.dL + 2\frac{\pi}{4}Ld.\delta d$

$$\frac{\delta V}{V} = \frac{\delta L}{L} + 2.\frac{\delta d}{d}$$

$$= 2\varepsilon_1 + \varepsilon_2 \qquad \varepsilon_1 = \frac{\delta d}{d},\ \varepsilon_2 = \frac{\delta \ell}{\ell}$$

Hence; Change in the volume of boiler cylinder per unit volume = $2\varepsilon_1+\varepsilon_1$

**43. If two shafts of the same length, one of which is hollow, transmit equal torque and have equal maximum stress, then they should have equal**

(a) polar moment of inertia

(b) diameter

(c) polar modulus of section

(d) angle of twist

**Ans. (c) :** Given; solid and hallow shafts have the same length let us assume shafts have the same materials.

Given, torque applied to both shafts is same, and the maximum shear stress is equal.

$\therefore$ Torque applied in both the case is same. (Given)

ie, $T_S = T_H \qquad \because T = \frac{\pi}{16}d_3.\tau$

$$\tau.\frac{\pi}{16}D_s^3 = \tau\frac{\pi}{16}\left[\frac{D_H^4 - d_H^4}{D_H}\right]$$

$$\frac{\pi}{16}D_S^3 = \frac{\pi}{16}\left[\frac{D_H^4 - d_H^4}{D_H}\right]$$

Where;

$\frac{\pi}{16}D_S^3$ = Polar section modulus of solid shaft

& $\frac{\pi}{16}\left[\frac{D_H^4 - d_H^4}{D_H}\right]$ Polar section modulus of hollow shaft

Hence, polar section modulus in both the case is also same.

**44. A plane stressed element is subjected to the state of stress given by $\sigma_x = \tau_{xy} = 100\ kgf/cm^2$ and $\sigma_y = 0$. Maximum shear stress in the element is equal to**

(a) $50\sqrt{3}\ kgf/cm^2$ (b) $100\ kgf/cm^2$

(c) $50\sqrt{5}\ kgf/cm^2$ (d) $150\ kgf/cm^2$

**Ans. (c) :** Given ; $\sigma_x = \tau_{xy} = 100\ kgf/cm^2$

& $\sigma_y = 0$

Let, Maximum shear stress = $\tau_{max}$

We know that in Mohr circle maximum stress $(\tau_{max})$ is equal to the radius (R) of Mohr circle.

Hence,

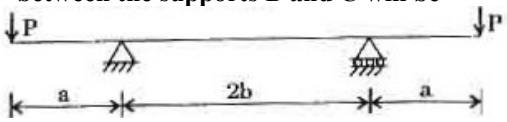
$$\tau_{max} = R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{\left(\frac{100-0}{2}\right)^2 + (100)^2} = \sqrt{(50)^2 + (100)^2}$$

$$= 50\sqrt{1+4}$$

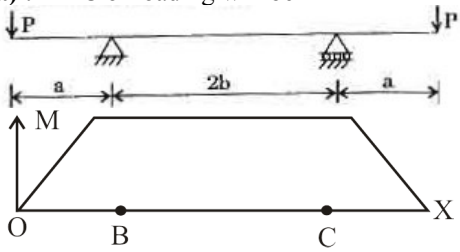
$$= 50\sqrt{5}\ kgf/cm^2$$

**45. For the beam shown is the fig. the elastic curve between the supports B and C will be**




(a) circular (b) parabolic
(c) elliptic (d) a straight line

**Ans. (a) :** BMC of loading will be




Between B and C, BMD is straight line
BM = constant

Now, $\frac{M}{I} = \frac{E}{R}$

$\therefore \frac{1}{R} = \frac{M}{EI} = \text{constant}$

R = constant means curve of constant radius
Hence circular

**46. Dynamic viscosity (μ) has the dimensions as**

(a) $MLT^{-2}$ (b) $ML^{-1}T^{-1}$
(c) $ML^{-1}T^{-2}$ (d) $M^{-1}L^{-1}T^{-1}$

**Ans. (b) :** Dimensions of Dynamic viscosity (μ) = ?

We know that SI unit of $\mu = \frac{\text{Newton second}}{m^2} = \text{N-s/m}^2$

$\therefore$ dimension of $\mu = [ML^{-1}T^{-1}]$

**47. Atmospheric pressure held in terms of water column is**

(a) 7.5 m (b) 8.5 m
(c) 9.81 m (d) 10.30 m

**Ans. (d) :** We know that –

Pressure head (h) = $\frac{P}{\rho g}$ in meter

Let,

P = Pa = Atmospheric pressure = 1.013 bar
$= 1.013 \times 10^5$ N/m$^2$
and g = 9.81 m/s$^2$

Atmospheric pressure in water column ($h_w$) = $\frac{P_a}{\rho g}$

$h_w = \frac{1.013\times10^5}{1000\times9.81} = \frac{101.3}{9.81} = 10.32$

$h_w$ = 10.30 m

**48. The flow rate through a circular pipe is measured by**

**(i) pitot-tube (ii) venturimeter**
**(iii) orificemeter (iv) None of the above**
**Code:**

(a) (i) (b) (i), (ii) and (iii)
(c) (ii) and (iii) (d) All of the above

**Ans. (c):** Venturimeter and orificemeter measure the rate of flow (ie, discharge) in a circular pipe. While pitot-tube measure the velocity

**49. Hydrostatic law of pressure is given by**

(a) $\frac{\partial P}{\partial z} = \rho g$ (b) $\frac{\partial P}{\partial z} = 0$

(c) $\frac{\partial P}{\partial z} = z$ (d) $\frac{\partial P}{\partial z} = c$

**Ans. (a) : Hydrostatic law of pressure :** Pressure in a static mass of fluid increasing in the vertically downward direction at a rate is equal to the specific weight of fluids. It is the hydrostatic law.

$\therefore \frac{\partial P}{\partial Z} = w$

Where, specific weight (w) = ρg

$\therefore \frac{\partial P}{\partial Z} = \rho g$

**50. Study of fluid at rest is known as**

(a) Kinematics (b) Dynamics
(c) Statics (d) None of the above

**Ans. (c) :** Study of fluid at rest is known as statics.
The branch of physics which deals with the study of fluids at rest is called hydrostatics. Hydro stands for liquids and statics is for fluid at rest.

**51. The rate of flow through a V-notch varies**

(a) H (b) $\sqrt{H}$
(c) $H^{3/2}$ (d) $H^{5/2}$

**Ans. (d) :** The rate of flow i.e, discharge through a v-notch (Q) = ?

$\therefore$ Rate of flow (Q) = $\frac{8}{15} C_d . \tan\frac{\phi}{2} . \sqrt{2g} . H^{5/2}$

$\therefore$ $\boxed{Q \propto H^{5/2}}$

**52. Notch is a device used for measuring**

(a) rate of flow through pipes
(b) rate of flow through a small channel
(c) velocity through a pipe
(d) velocity through a small channel

**Ans. (b) :** A notch is a device used for measuring the rate of flow of a liquid through a small channel or a tank. It may be defined as an opening in the side of a tank or a small channel in such a way that the liquid surface in the tank or channel is below the top edge of the opening.

**53. For supersonic flow, if the area of flow increases then**

(a) velocity decreases (b) velocity increases
(c) velocity is constant (d) None of the above

**Ans. (b) :** For supersonic flow, if the area of flow is increases then the velocity of flow is increases. In the divergent portion of the Nozzle, for supersonic flow, when the area of the Nozzle increases than flow velocity increases.

**54. Chezy's formula is given by**

(a) $v = i\sqrt{mC}$ (b) $v = C\sqrt{mi}$

(c) $v = m\sqrt{Ci}$ (d) None of the above

**Ans. (b) :** Chezy's formula : This is used for loss of head due to friction in pipes. Which is –

$\boxed{V = C\sqrt{mi}}$

Where, V = mean velocity of flow

m = hydraulic mean depth or hydraulic radius = $\frac{A}{P}$;

A = cross section area of pipe

P = wetted perimeter of pipe

C = Chezy's constant

i = loss of head per unit length of pipe = $\frac{h_f}{L}$

$\frac{h_f}{L}$ = loss of head due to friction in pipe.

L = Length of pipe.

**55. The square root of the ratio of inertia force to gravity force is called**

(a) Reynolds number (b) Froude number

(c) Mach number (d) Euler number

**Ans. (b) :** The square root of the ratio of inertia force to gravity force is called Froude number.

Froude Number ($F_r$)= $\sqrt{\frac{F_i}{F_g}}$;

Where; $F_i$ = Inertia force

$F_g$ = Gravity force

Or $\frac{V}{\sqrt{gL}}$

**56. Muschel curve means**

(a) curve at constant head

(b) curve at constant speed

(c) curve at constant efficiency

(d) None of the above

**Ans. (c) :** Muschel curve mean the curve at constant efficiency or Iso-efficiency curve.

- The constant efficiency curves or Muschel curves help to determine the range of pump operation for a particular efficiency.
- Operating characteristic curve called constant speed curve.
- Main characteristics curve called constant head curve.

**57. The hydraulic mean depth is given by Where A- Area, P- watted perimeter**

(a) $\frac{P}{A}$ (b) $\frac{P^2}{A}$

(c) $\frac{A}{P}$ (d) $\sqrt{\frac{A}{P}}$

**Ans. (c) :** Hydraulic mean depth (m) = $\frac{A}{P}$

Where; A = Area of cross-section of pipe

P = Wetted perimeter of pipe

The ratio of $\frac{A}{P} = \left(\frac{\text{Area of flow}}{\text{Wetted perimetre}}\right)$ is called hydraulic mean depth or hydraulic radius and is denoted by 'm'.

**58. Cavitation will take place if the pressure of the following fluid at any points is**

(a) more than vapour pressure of the fluid

(b) equal to vapour pressure of the fluid

(c) less than vapour pressure of the fluid

(d) None of the above

**Ans. (c) :** If the pressure of the following fluid at any points is less than vapour pressure of the fluid.

Due to the cavitation, erosion occurs in the runner blade of turbine etc.

**59. The speed ratio for Pelton wheel varies from**

(a) 0.45 to 0.50 (b) 0.6 to 0.7

(c) 0.3 to 0.4 (d) 0.8 to 0.9

**Ans. (a) :** Speed ratio ($K_u$) of a Pelton wheel turbine is defined as the ratio of the tangential velocity of wheel u to the theoretical velocity of the jet.

$K_u = \frac{u}{\sqrt{2gh}}$

where;

$u = \frac{\pi DN}{60}$ N = speed in rpm

Speed ratio for pelton wheel various from = 0.45 to 0.50

**60. Maximum efficiency of power transmission through pipe is**

(a) 50% (b) 66.67%

(c) 75% (d) 100%

**Ans. (b) :** Maximum efficiency of power transmission

$(\eta_{max}) = \frac{\text{flow head at out let}}{\text{flow head at inlet}} \times 100 = \frac{H_{outlet}}{H_{inlet}} \times 100$

$(\eta_{max}) = \frac{H - h_f}{H} \times 100$

$= \left(\frac{H - H/3}{H}\right) \times 100$

$= \frac{2}{3} \times 100$

$(\eta_{max})$ = 66.67%

where;

H = head at inlet

$h_f$ = loss of head due to friction in pipe = $\frac{H}{3}$ (we know)

**61. One number of elastic constants for a completely anisotropic elastic material which follows Hooke's law is**

(a) 3 (b) 4

(c) 21 (d) 25

**Ans. (c) :** The number of elastic constants for a completely anisotropic elastic material which follow the Hooke's law is 21.

**62. A close-coiled helical spring absorbs 80 N mm of energy while extending by 4 mm. The stiffness of the spring is**

(a) 5 N/mm (b) 10 N/mm
(c) 16 N/mm (d) 20 N/mm

**Ans. (b) :** For a closed - coiled helical spring–

Absorbe energy (V) = $\frac{1}{2}$W.δ ..... (i)

Given; δ = deflection of coil under the load W
= 4 mm
U = Absorbs energy
= 80 N-mm

$\therefore$ $W = \frac{2V}{\delta}$ [From equation (i)]

$W = \frac{2\times 80\ N-mm}{4mm} = 40\ N$

$\therefore$ Stiffness of spring (K) = $\frac{W}{\delta} = \frac{40N}{4mm} = 10$ N/mm

**63. Match List I with List II and select the correct answer using the codes given below the lists:**

| List I | List II |
|---|---|
| **A. Both ends hinged** | **(i) L** |
| **B. One end fixed and other end free** | **(ii) $\sqrt{2}$ L** |
| **C. One end fixed and the other pin-joined** | **(iii) L/2** |
| **D. Both ends fixed** | **(iv) 2L** |

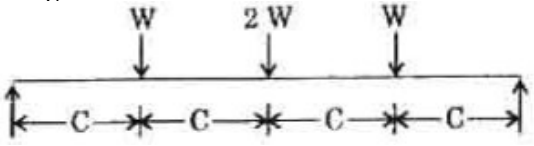
**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | i | iv | ii | iii |
| (b) | i | iii | ii | iv |
| (c) | iii | i | ii | iv |
| (d) | iii | i | iv | ii |

**Ans. (a) :**

A. Both ends hinged – L
B. One end fixed and other end free – 2L
C. One end fixed and the other pin-joined – $L/\sqrt{2}$
D. Both ends fixed – L/2

**64. A simply supported beam is loaded as shown in fig. The maximum shear force in the beam will be**




(a) Zero (b) W
(c) 2 W (d) 4 W

**Ans. (c) :** A simply supported beam–
From the figure –
Reaction at A and B in the beam is $R_A$ and $R_B$ respectively due to symmetrical loading.
$R_A = R_B$ .....(i)
& $R_A + R_B = 4$ W .....(ii)
$\therefore$ from equation (i) and (ii)
$R_A = R_B = 2$ W
therefore, from SFD; maximum shear force = 2W

**65. The equivalent bending moment under combined action of bending moment M and torque T is**

(a) $\sqrt{M^2, T^2}$ (b) $\frac{1}{2}\sqrt{M^2+T^2}$
(c) $M+\sqrt{M^2+T^2}$ (d) $\frac{1}{2}\left[M+\sqrt{M^2+T^2}\right]$

**Ans. (d) :** Combined effect of bonding and Torsion torque :

Equivalent bending moment ($M_e$)

$= \frac{1}{2}\left[M+\sqrt{M^2+T^2}\right]$

Where; M = Bending moment
T = Torque or Twisting moment

**66. Young's modulus of elasticity and Poisson's ratio of a material are $1.25 \times 10^5$ MPa and 0.34 respectively. The modulus of rigidity of the material is**

(a) $0.4025 \times 10^5$MPa (b) $0.4664 \times 10^5$MPa
(c) $0.8375 \times 10^5$MPa (d) $0.9469 \times 10^5$MPa

**Ans. (b) :**
Given : Young's Modulus of elasticity (E)
$= 1.25 \times 10^5$ MPa
Poisson's ratio (μ) = 0.34
Let, Modulus of rigidity = G
$E = 2G(1+\mu)$

$\Rightarrow G = \frac{E}{2\times(1+\mu)} = \frac{1.25\times10^5\ MPa}{2\times(1+0.34)} = \frac{1.25\times10^5}{2\times1.34}$

$G = 0.4664 \times 10^5$ MPa

**67. A solid circular shaft is subjected to a maximum shearing stress of 140 MPa. The magnitude of the maximum normal stress developed in the shaft is**

(a) 140 MPa (b) 80 MPa
(c) 70 MPa (d) 60 MPa

**Ans. (a)** $\tau_{max} = \frac{\sigma_x - \sigma_y}{2}$

If $\sigma_x = -\sigma_y$

$\tau_{max} = \frac{\sigma_x - (-\sigma_x)}{2} = \frac{2\sigma_x}{2}$

$\sigma_x = \tau_{max} = 140$ MPa

**68. The ratio of circumferential stress to longitudinal stress in a thin cylinder subjected to internal hydrostatic pressure is**

(a) 1/2 (b) 1
(c) 2 (d) 4

**Ans. (c) :**

$\frac{\text{Circumferential stress}}{\text{Longitudinal stress}} = \frac{\sigma_C}{\sigma_L} = ?$

When a thin cylinder subjected to internal hydrostatic pressure then there will be in the cylinder circumferential stress, Longitudinal stress and longitudinal stress is acting. But in case in thin cylinder, tangential stress is negligible.

We know that;

$$\sigma_C = \frac{Pd}{2t}$$

$$\sigma_L = \frac{Pd}{4t}$$

Where P = Internal pressure
d = Dia of cylinder
t = thickness of cylinder

therefore;

$$\frac{\sigma_C}{\sigma_L} = \frac{Pd/2t}{Pd/4t} = 2$$

**69. The independent elastic constants for a homogeneous and isotropic material are**

(a) E, G, K, ν (b) E, G, K
(c) E, G, ν (d) E, G

**Ans. (a) :** E, G, K and ν are the independent elastic constant for a homogeneous and isotropic material.

**70. The ratio of the deformation of a bar due to its own weight, to the deformation due to axial load equal to its weight, is**

(a) 1 (b) 1/2 (c) 2 (d) 4

**Ans. (b) :** We know that -
Deformation of a bar due to its own weight

$$(\Delta_1) = \frac{WL}{2AE}$$

Deformation due to axial load

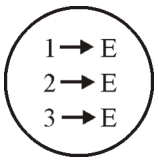
$$(\Delta_2) = \frac{WL}{AE}$$

$$\frac{\Delta_1}{\Delta_2} = \frac{\frac{wL}{2AE}}{\frac{wL}{AE}} = \frac{1}{2}$$

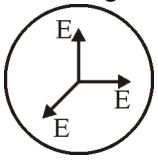
**71. A material having identical properties in all directions, is called**

(a) elastic (b) homogeneous
(c) isotropic (d) Any of these

**Ans. (c)** • Material is termed to be homogenous if it exhibit same elastic property at any point in given direction.




• Material is termed to be isotropic if it same elastic properties in all direction at a given point.




**72. The stresses in a thick cylinder subjected to uniform pressure vary proportional to**

(a) r (b) 1/r (c) $r^2$ (d) $1/r^2$

**Ans. (d) :** From the Lame's equation for thick cylinder–

$$\sigma_r = A - \frac{B}{r^2}(\text{compressive})$$

and $\sigma_c = \frac{B}{r^2} + A\ (\text{Tensile})$

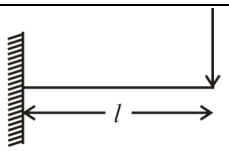
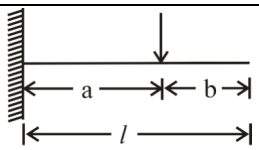
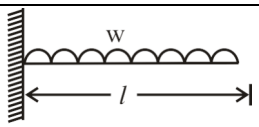
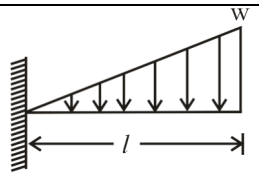
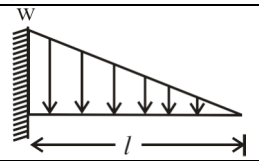
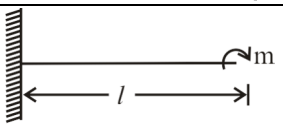
where; $\sigma_r$ = radial stress
$\sigma_c$ = circumferential stress
A & B is constant
if A = 0; then

$$\left\{\begin{matrix}\sigma_r \propto \frac{1}{r_2} \\ \sigma_C \propto \frac{1}{r_2}\end{matrix}\right\}$$

**73. Maximum deflection for a cantilever of span L loaded at the free end by P is given by**

(a) $\frac{PL^2}{3EI}$ (b) $\frac{PL^2}{6EI}$
(c) $\frac{PL^3}{8EI}$ (d) $\frac{PL^3}{3EI}$

**Ans. (d) :**

| Beam and load cases | Maximum beam deflection |
|---|---|
|   | $\delta_{max} = \frac{Pl^3}{3EI}$ |
|   | $\delta_{max} = \frac{Pa^2(3l-a)}{6EI}$ |
|   | $\delta_{max} = \frac{wl^4}{8EI}$ |
|   | $\delta_{max} = \frac{11wl^4}{120EI}$ |
|   | $\delta_{max} = \frac{wl^4}{30EI}$ |
|   | $\delta_{max} = \frac{ml^2}{2EI}$ |

**74. The ratio of lateral strain to linear strain is known as**

(a) Elastic limit (b) Poisson's ratio
(c) Elasticity (d) Rigidity

**Ans. (b):** The ratio of lateral strain to linear strain is know as Poisson's ratio. It is represented by 'μ'

$$\mu = \frac{\text{Lateral strain}}{\text{Linear strain}}$$

**75. Polar moment of inertia of a circular area is**

(a) $\frac{\pi}{32}d^4$ (b) $\frac{\pi}{64}d^4$

(c) $\frac{\pi}{4}d^4$ (d) $\frac{\pi}{4}d^3$

**Ans. (a) :** d = Dia of circular area

For a circular area –

Moment of Inertia $(I_{xx}) = \frac{\pi}{64}d^4$

Moment of Inertia $(I_{yy}) = \frac{\pi}{64}d^4$

$\therefore$ Polar moment of inertia $(I_{zz}) = I_{xx} + I_{yy}$

$$= \frac{\pi}{64}d^4 + \frac{\pi}{64}d^4$$

$$I_{zz} = \frac{\pi}{32}d^4$$

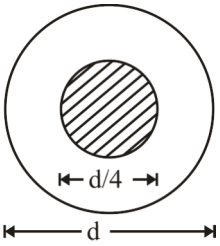
**76. The diameter of kernel of a circular section of diameter 'd' is**

(a) d/2 (b) d/3

(c) $d/\sqrt{2}$ (d) d/4

**Ans. (d) : Kern-** Kern is a zone (or area) in which an axial arbitrary load can be applied without inducing tensile stress in the section.

■ In circular section, kern is $\frac{d}{4}$ in diameter.




**77. The flexural rigidity of a beam is**

(a) EI (b) E/I

(c) I/E (d) $E/I^2$

**Ans. (a) :** The flexural rigidity of a beam is the product of the elastic constant (E) and Moment of inertia of the beam.

i.e, flexural rigidity of a beam = EI

**78. One kgf/cm$^2$ when converted into SI units is**

(a) 0.0981 MPa (b) 0.981 Pa

(c) $10^4$ Pa (d) 1 Pa

**Ans. (a) :** In SI units –

1 kgf/cm$^2$ = 0.0981 MPa

**79. Point of contraflexure is where**

(a) B.M. is maximum (b) S.F. is maximum

(c) S.F. is zero (d) B.M. is zero

**Ans. (d) :** At the point of contraflexure bending moment is zero. i.e, bending moment (B.M.) changes sign.

Point of contraflexure is the point where bending moment changes its sign from positive to negative or vice-versa.

■ At a point of contraflexure the value of bending moment is zero.

**80. The equivalent length of a column fixed at both ends is**

(a) 0.7 *l* (b) 0.5 *l*

(c) *l* (d) 2 *l*

**Ans. (b) :** Let,

The equivalent length of column fixed at both the end = $L_e$

$$L_e = \frac{l}{2} = 0.5l$$

where;

*l* = Length of fixed column

**81. If two springs with stiffness $K_1$ and $K_2$ are connected in series, then stiffness of the composite spring is given by**

(a) $K_1 + K_2$ (b) $\frac{1}{K_1} + \frac{1}{K_2}$

(c) $\frac{1}{K_1} - \frac{1}{K_2}$ (d) $K_1 - K_2$

**Ans. (b) :**

Given,

$K_1$ and $K_2$ are the stiffness of two springs which is connected in series.

Let,

K = equivalent stiffness of springs

$$\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2}$$

**82. If $\sigma_1$ and $\sigma_2$ are principal stresses, the shear stress on the principal planes is given by**

(a) $\frac{\sigma_1 - \sigma_2}{2}$ (b) 0

(c) $\frac{\sigma_1 + \sigma_2}{2}$ (d) $\sigma_1 - \sigma_2$

**Ans. (b) :** Principal plane are those planes where shear stress ($\tau_s$) is zero. If principal stresses $\sigma_1$ and $\sigma_2$ are given

i.e, $\boxed{\tau_s = 0}$

**83. The reaction at the prop in a propped cantilever beam subjected to u.d.*l*. is**

(a) $\frac{Wl}{4}$ (b) $\frac{3Wl}{8}$

(c) $\frac{5Wl}{8}$ (d) $\frac{6Wl}{7}$

**Ans. (b) :** Draw a propped cantilever beam subjected to U.D.L.

Let, W = Uniform distributed load

$l$ = Length of cantilever beam

R = Reaction at the prop

Deflection at point B due to UDL in cantilever beam in downward direction

$$\delta_B = \frac{Wl^4}{8EI}(\downarrow)$$

Deflection at point B due to reaction (R) in upward direction

$$\delta'_B = \frac{Rl^3}{3EI}(\uparrow)$$

$\therefore$ equilibrium condition $\delta'_B = \delta_B$

$$\Rightarrow \frac{Rl^3}{3EI} = \frac{Wl4}{8EI}$$

$$R = \frac{3Wl}{8}$$

**84. The ratio of maximum shear stress to average shear stress is 1.5 in a beam of**

(a) Circle (b) Rectangle

(c) Triangle (d) Any cross-section

**Ans. (b) :** The maximum shear stress developed in a beam of rectangular cross-section is

$\tau_{max} = 1.5\ \tau_{avg}$

$$\frac{\tau_{max}}{\tau_{avg}} = 1.5$$

where;

$\tau_{max}$ = Maximum shear stress

$\tau_{avg}$ = Average shear stress = $\frac{\text{Load}}{\text{Area}}$

**85. A solid circular shaft is subjected to pure torsion. The ratio of maximum shear to maximum normal stress at any point would be**

(a) 1:1 (b) 1:2

(c) 2:1 (d) 2:3

**Ans. (a) :** D = Dia of solid circular shaft

T = Torque or twisting moment

Maximum shear stress ($\tau_{max}$) = $\frac{16T}{\pi D^3}$ .....(i)

Maximum normal stress in case of shaft subjected to combined bending and torsion.

$$(\sigma_b)_{max} = \frac{16T}{\pi D^3}\left[M + \sqrt{M^2 + T^2}\right]$$

In case of pure torsion;

Bending moment (M) = 0

$(\sigma_b)_{max}$ = Maximum normal stress

$$\therefore (\sigma_b)_{max} = \frac{16T}{\pi D^3} \quad \text{..... (ii)}$$

$$\frac{\tau_{max}}{(\sigma_b)_{max}} = \frac{16T/\pi D^3}{16T/\pi D^3} = \frac{1}{1}$$

$$\tau_{max} : (\sigma_b)_{max} = 1:1$$

**86. The unit of elastic modulus is the same as those of**

(a) stress, strain and pressure

(b) strain, shear modulus and pressure

(c) shear modulus, stress and force per area

(d) strain, shear modulus and force

**Ans. (c) :** Let, E = Elastic modulus

$$\because E = \frac{\sigma}{\varepsilon}$$

where; $\sigma$ = stress ($N/m^2$)

$\varepsilon$ = linear strain

Unit of elastic constant ($\varepsilon$) = $N/m^2$

Therefore, the unit of shear modulus, stress and force per area is same as the unit of elastic modulus.

**87. The work done per unit volume in elongating a body by a uniaxial force is**

(a) stress/strain (b) stress × strain

(c) $\frac{1}{2}$stress × strain (d) None of the above

**Ans. (c) :**

$$\frac{\text{workdone}}{\text{volume}} = \frac{1}{2}\text{stress} \times \text{strain} = \frac{1}{2}\sigma.\delta$$

**88. A solid thick cylinder is subjected to an external hydrostatic pressure 'P'. The state of stress in the material of the cylinder is represented as**

(a) P → □ ← P (b) ↑P, P → □ ← P, ↓P

(c) ↓P, P → □ ← P, ↑P (d) P →, P → □ ← P, ← P

**Ans. (c) :** A solid thick cylinder is subjected to an external hydrostatic pressure 'P'. the state of stress represented by –

↓P, P → □ ← P, ↑P

Radial stress $\sigma_r$ is compressive for thick cylinder.

Circumferential stress $\sigma_c$ is tensile for thick cylinder.

**89. Consider the following statements:**

**A splined shaft is used for**

**(i) transmitting power**

**(ii) holding a flywheel rigidly in position**

**(iii) moving axially the gear wheels mounted on it.**

**(iv) mounting v-belt pulleys on it**

**Code:**

(a) (ii) and (iii) are correct

(b) (i) and (iv) are correct

(c) (ii) and (iv) are correct

(d) (i) and (iii) are correct

**Ans. (d) :** Splines are used when there is requirement of relative axial motion between shaft land hub. The gear shifting mechanism in automobile uses their mechanism.

**90. Two close-coiled springs are made from a small diameter wire. One wound on 2.5 cm diameter core and the other on 1.25 cm diameter core. If each spring had 'n' coils, then the ratio of their spring constant would be**

(a) 1/16 (b) 1/8
(c) 1/4 (d) 1/2

**Ans. (b) :** Stiffness 'k' $= \frac{Gd^4}{64R^3N}$

$$\frac{k_1}{k_2} = \left(\frac{D_2}{D_1}\right)^3 = \left(\frac{2.5}{1.25}\right)^3$$

$$= \frac{1}{8}$$

**91. To measure ____strain, strain rosettes are used**

(a) Linear (b) Shear
(c) Volumetric (d) Any of the above

**Ans. (a) :** Strain rosettes method is used to measure Linear strain.
**Strain gauge Rosettes-** A gauge made of very fine wire is used per measurement of surface strains in sample.

**92. A localised compressive stress at the area of contact between two members is known as**

(a) Shear (b) Crushing
(c) Bending (d) Tensile

**Ans. (b) :** A localised compressive stress at the area of contact between two members is known as crushing.

**93. Moment of inertia of a semicircle about its XX-axis is given by**

(a) $0.22\ r^3$ (b) $0.11\ r^4$
(c) $0.14\ r^4$ (d) $0.2\ r^4$

**Ans. (b) :** Let,
r = radius of semi-circle
LM = base of the semi-circle or lamina
moment of inertia of a circular lamina out a diameter

$$LM = \frac{\pi r^4}{4} = \frac{\pi d^4}{64}$$

Moment of inertia of the semi-circle about LM

$$I_{LM} = \frac{\pi r^4}{8}$$

Let XX be centroidal axis parallel to the base LM
Let h be the distance between the axis XX and L

$$h = \frac{4r}{3\pi}$$

Applying parallel axis theorem
$I_{LM} = I_{XX} + Al^2$

$$\frac{\pi r^4}{8} = I_{XX} + \frac{\pi r^2}{2} \times \left(\frac{4r}{3\pi}\right)^2$$

$$I_{XX} = \frac{\pi r^4}{8} - \frac{8\pi r^4}{9\pi^2}$$

$I_{XX} = 0.11\ r^4$

**94. The point of contraflexure is also called as**

(a) the point of inflexion (b) a virtual hinge
(c) Either of the above (d) None of the above

**Ans. (c):** The point of contraflexure is also called as the point of inflection or virtual hinge. Because at the point of contraflexure bending moment changes sign and bending moment is zero.

**95. Which of the following are the statically determinate beams?**

(a) Cantilevers (b) SSB
(c) Overhanging beams (d) All of the above

**Ans. (d) :** A beam is said to be statically determined if all its reaction components can be calculated by applying three condition of statical equilibrium.
i.e, $\Sigma V = 0$, $\Sigma H = 0$ and $\Sigma M = 0$
Ex. Cantilever beam, simply supported beam (SSB) and overhanging beams etc.

**96. The strength of the beam mainly depends on**

(a) Bending moment (b) C.G. of the section
(c) Section modulus (d) Its weight

**Ans. (c) :** The beam is stronger when section modulus is more the strength of the beam depends on section modulus.
■ The strength of the beam also depends on the material, size and shape of cross section.

Section modulus, $\sigma_b = \frac{M}{Z}$

There fore, $Z = \frac{M}{\sigma_b}$

Z = Section modulus

**97. In case of rectangular section**

(a) $\tau_{max} = \frac{1}{2}\tau_{mean}$ (b) $\tau_{max} = \tau_{mean}$

(c) $\tau_{max} = \frac{3}{2}\tau_{mean}$ (d) $\tau_{max} = \frac{5}{2}\tau_{mean}$

**Ans. (c) :** In case of rectangular section –

$$\tau_{max} = \frac{3}{2}\tau_{mean}$$

where ; $\tau_{max}$ = Maximum shear stress
$\tau_{mean}$ = Mean shear stress

**98. A square section with side 'X' of a beam is subjected to a shear force 'S'. The magnitude of shear stress at the top edge of the square is**

(a) $\frac{1.5S}{X^2}$ (b) $\frac{S}{X^2}$

(c) $\frac{0.5S}{X^2}$ (d) Zero

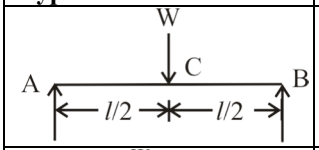
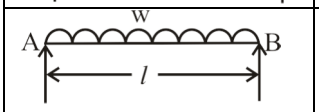
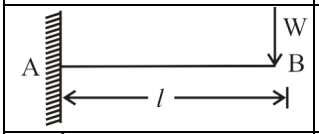
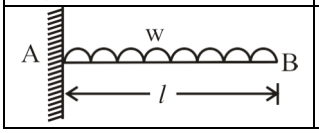
**Ans. (d) :** At the top edge of the square, the magnitude of the shear stress should be zero. Because shear stress is a surface phenomina

**99. A simply supported beam of span '*l*' is carrying point load 'W' at the midspan What is the deflection at the centre of the beam?**

(a) $\frac{Wl}{48\,EI}$ (b) $\frac{Wl^3}{48\,EI}$

(c) $\frac{5Wl^3}{348\,EI}$ (d) $\frac{11Wl^3}{120\,EI}$

**Ans. (b) :**

| Type of beam | Slope | Deflection |
|---|---|---|
|   | $\theta_B = \frac{Wl^2}{16EI}$ | $y_c = \frac{Wl^3}{48EI}$ |
|   | $\theta_B = \frac{wl^3}{24EI}$ | $y_c = \frac{5}{384}\frac{wl^4}{EI}$ |
|   | $\theta_B = \frac{Wl^2}{2EI}$ | $y_B = \frac{Wl^3}{3EI}$ |
|   | $\theta_B = \frac{wl^3}{6EI}$ | $y_B = \frac{wl^4}{8EI}$ |

**100. A beam length 4 m, fixed at both ends carries a point load of 120 kN at the centre. If EI for the beam is 2000 kN m$^2$, deflection at the centre of beam is**

(a) 1.0 mm (b) 20 mm

(c) 5.0 mm (d) 10.0 mm

**Ans. (b)** Let, $\delta_c$ = deflection at the centre (c) of beam is fixed at both ends $= \frac{wl^3}{192EI}$

Given,

$EI = 2000$ kN-m$^2$, $w = 120$ kN, $l = 4$m

$$= \frac{120\ \text{kN} \times (4)^2\ \text{m}^3}{192 \times 2000\ \text{kN} - \text{m}^2} = 0.02\,\text{m}$$

$= 20$ mm

**101. Thin cylinders are frequently required to operate under pressure upto**

(a) 5 MN/m$^2$ (b) 15 MN/m$^2$

(c) 30 MN/m$^2$ (d) 250 MN/m$^2$

**Ans. (c) :** In thin cylinder the stress may be assumed uniformly distributed over the wall thickness.

■ Thin cylinders are frequently required to operate under pressures up to 30 MN/m$^2$ or more, for high pressures such as 250 MN/m$^2$ or more thick cylinders are used.

**102. The bursting pressure for a cold drawn seamless steel tubing of 60 mm inside diameter with 2 mm wall thickness is (the ultimate strength of steel is 380 MN/m$^2$)**

(a) 25.33 MN/m$^2$ (b) 24.33 MN/m$^2$

(c) 26.33 MN/m$^2$ (d) 50.66 MN/m$^2$

**Ans. (a) :** For thin cylinder

$\sigma_C = \frac{Pd}{2t}$

P = bursting pressure = ?

d = Inside diameter of steel tube = 60 mm

t = wall thickness of seamless steel tube = 2 mm

$\sigma_c$ = circumferential stress or brushing stress = ultimate strength of steel = 380 MN/m$^2$

$$P = \frac{\sigma_c \times 2t}{d} = \frac{380\ \frac{MN}{m^2} \times 2 \times 2mm}{60\,mm}$$

P = 25.33 MN/m$^2$

**103. Pressure vessels are made of**

(a) non-ferrous materials

(b) sheet metal (steel)

(c) cast iron

(d) Any of the above

**Ans. (d) :** Pressure vesseles are made of cost iron, shear metal (steel) and non-ferrous materials.

**104. _____riveting is used in structural units**

(a) Chain (b) Zig-zag

(c) Diamond (d) None of the above

**Ans. (c):** In the structural units, Diamond riveting is used.

**Diamond Riveting-** All the rivets are arranged symmetrically about the centre line of plate.

The strength of plate in tearing in diamond riveting at the section can be computed.

$P_t\,(b - d).\ t \times \sigma_t$

Where-

b = width of the plate

d = gross dia of the rivet

t = thickness of plate

**105. In shafts with keyways, the allowable stresses are usually_____ of the value given.**

(a) 25% (b) 50%

(c) 75% (d) 95%

**Ans. (c) :** In shafts with keyway, the allowable stress are usually 75%

**106. For the same material, length and given torque, a hollow shaft weighs_____a solid shaft.**

(a) less than (b) more than

(c) equal to (d) None of the above

**Ans. (a) :** For the same material, length and given torque a hollow shaft weights less than a solid shaft.

**107. Wahl's correction factor (K) is given by the relation where S = Spring Index**

(a) $K = \frac{3S-1}{3S-4} + \frac{0.615}{S}$ (b) $K = \frac{4S-1}{4S-4} + \frac{0.615}{S}$

(c) $K = \frac{5S-1}{5S-4} + \frac{0.615}{S}$ (d) $K = \frac{6S-1}{6S-4} + \frac{0.615}{S}$

**Ans. (b) :** Wahl's correction factor (k) $= \frac{4S-1}{4S-4} + \frac{0.615}{S}$

where;

$S = \frac{D}{d}$ = spring index

D = mean diameter of the coil

d = diameter of the wire of the coil

**108. In case of laminated springs, the load at which the plates become straight is called.**

(a) Working load (b) Safe load
(c) Proof load (d) None of the above

**Ans. (c) :** In case of laminated springs, the load at which the plates become straight is called proof load.
Proof load is the limit of the elastic range of the bolt. Designing for the use of bolts according to proof load can be help prevent plastic deformation as long as bolt is never tensioned beyond its specified proof load.

**109. The stress due to suddenly applied load is _____ times that of gradually applied load.**

(a) 2 (b) 3
(c) 4 (d) 5

**Ans. (a) :** Let,

$\sigma$ = Stress due to gradually applied load = $\frac{P}{A}$

$\sigma_{imp.}$ = Impact stress due to suddenly applied load

$$\therefore\ \sigma_{imp} = \sigma \times \left[1 + \sqrt{1 + \frac{2h}{\delta_{static}}}\right]$$

where ;

$\delta_{static}$ = static deflection

for suddenly applied load; h = 0

$$\sigma_{imp} = \sigma \times \left[1 + \sqrt{1+0}\right] = \sigma \times 2$$

$$\sigma_{imp} = 2\sigma$$

Therefore, the stress due to suddenly applied load is 2 times that of gradually applied load.

**110. Strain energy (U) caused by bending is given by the relation**

(a) $U = \int \frac{M}{2EI} dx$ (b) $U = \int \frac{M^2}{2EI} dx$

(c) $U = \int \frac{M^2}{EI} dx$ (d) $U = \int \frac{M^2}{3EI} dx$

**Ans. (b) :** Strain energy energy is the energy stored in body due to its elastic deformation.
The most general formula-

$$U = \frac{1}{2} \times f \times \delta$$

f = applied force
$\delta$ = deformation
(1) Strain energy due to bending is-

$$U \int \frac{M^2}{2EI} dx$$

(2) Strain energy due to torsion is–

$$U = \int \frac{T^2 L}{2GJ} dx$$

**111. The safe compressive load on a hollow cast iron column (one end fixed other hinged) of 150 mm external diameter, 100 mm internal diameter and 10 m length is (Use Euler's formula with a factor of safety of 5, and E = 95 GN/m$^2$)**

(a) 74.8 kN (b) 149.6 kN
(c) 37.4 kN (d) 299.2 kN

**Ans. (a) :** For safe compressive load – Applying Euler's formula–

$$P_e = \frac{\pi^2 E I_{min}}{l_e^2}$$

Given; E = 95 GN/m$^2$ = 95 × 10$^9$ N/m$^2$
$l_e$ = Effective length of the column,
For one end fixed and one end hinged column

$l_e = L/\sqrt{2}$ ; L = length of column = 10 m (given)

For hollow cast iron column

$$I_{min} = \frac{\pi}{64}\left[D^4 - d^4\right];$$

D = External diameter
= 150 mm = 0.15 m
d = Internal diameter
= 100 mm = 0.1 m

$$I_{min} = \frac{\pi}{64} \times \left[(0.15)^4 - (0.1)^4\right]$$

= 1.994 × 10$^{-5}$ m$^4$

If $l_e = L/\sqrt{2}$

$$\therefore P_e = \frac{2\pi^2 E I_{min}}{L^2}$$

$$= \frac{2\pi^2 \times 95 \times 10^9 \times 1.994 \times 10^{-5}}{(10)^2}$$

= 373919.83 N/m$^2$

$$\text{Safe. compressive load} = \frac{P_e}{FOS}$$

$$= \frac{373919.83}{5}$$

= 74783.96 N

$P_e$ = 74.8 kN

**112. The radius of gyration of a circular column of diameter 'd' is**

(a) d/4 (b) d/2
(c) d$^2$/4 (d) d$^2$/16

**Ans. (a) :** We know- $I = Ak^2$

$$k = \sqrt{\frac{I}{A}}$$

For solid circle-

$$I = \frac{\pi d^4}{64}$$

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{\pi}{64}d^4}{\frac{\pi}{4}d^2}} = \sqrt{\frac{1}{16}d^2} = \frac{d}{4}$$

**113. Euler's formula is applicable for**

(a) short column (b) long column
(c) medium column (d) All of the above

**Ans. (b) :** Euler's formula–

$$P_e = \frac{\pi^2 EI}{L_e^2}$$

Where- $P_e$ = Buckling load
I = Min. moment of inertia about centroidal axis.
$L_e$ = Effective length
■ It is applicable for long column. Effect of crushing is neglected.

**114. The ratio of equivalent length of the column to the maximum radius of gyration is called**
(a) Poisson's ratio (b) Buckling factor
(c) Factor of safety (d) None of the above

**Ans. (b) :** Buckling factor is define as the ratio of equivalent length ($L_e$) of column to the maximum radius of gyration. It is also called slenderness ratio (S)

$$S = \frac{L_e}{K}$$

Where, K = radius of gyration

**115. A member under tension is called**
(a) strut (b) tie
(c) strut-tie (d) column

**Ans. (b) :** A member, carrying a tension is known as a tension member. It is also known as a tie.

**116. A perfect frame should satisfy the relation**
(a) m = 2j – 3 (b) m = 2j – 4
(c) m = 3j – 2 (d) m = 3j – 3

**Ans. (a) :** A perfect frame should satisfy the relation–
m = (2J – 3)
Where,
M = No. of members in frame
J = No. of joints in the frame
■ A frame in which the number of members is less than (2J – 3), is known as deficient frame.
■ A frame, in which the number of members is more than (2J – 3) is known as redundant frame.

**117. Which one of the following properties is more sensitive to increase in strain rate?**
(a) Yield strength (b) Proportional limit
(c) Elastic limit (d) Tensile strength

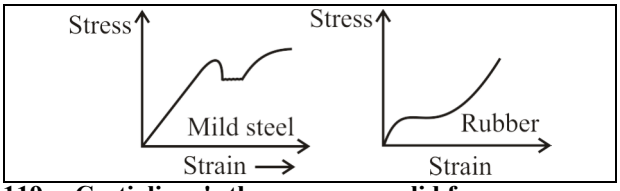
**Ans. (a) :** The yield strength is found to be more sensitive to strain rate than ultimate strength.
■ Strain rate will increase when the change in length is faster with time and it depends on yield strength there fore yield strength is more sensitive to change in strain rate.

**118. Which one of the following materials is highly elastic?**
(a) Rubber (b) Brass
(c) Steel (d) Glass

**Ans. (c) :** Modulus of elasticity or young's modulus is the slope of stress-strain curve in the elastic region

$$E = \frac{\sigma}{\varepsilon}$$

■ Steel is a highly elastic material.
■ Rubber is a brittle material.




**119. Castigliano's theorems are valid for**
(a) elastic structure (b) truss
(c) beam (d) linear structure

**Ans. (a) :** Castigliano's first theorems is applicable to linearly or non-linearly elastic structures which the temperature is constant and the supports are unyielding.
■ Castigliano's is applicable to linearly elastic structures with constant temperature and unyielding supports.
■ These theorems are applicable to any structure for which the force deformation relations are linear.

**120. If diameter of a long column is reduced by 20%, the percentage of reduction is Euler buckling load is**
(a) 4 (b) 36
(c) 49 (d) 59

**Ans. (d) :** Let, $D_1$ = initial diameter of column
$D_2$ = Reduced diameter of column = 0.8 $D_1$

$$\because P_e = \frac{\pi^2 E I_{min}}{L_e^2}$$

$$\because \frac{(Pe)_2}{(Pe)_1} = \frac{I_2}{I_1} = \frac{(0.8D_1)^4}{D_1^4}$$

$$\frac{(Pe)_2}{(Pe)_1} \approx 0.4096 \approx 0.41$$

$(Pe)_2 = 0.41\ (Pe)_1$
$\therefore$ Percentage reduction = 59%

**121. An oil of specific gravity 0.7 and pressure 0.14 kgf/cm$^2$ will have the height of oil as**
(a) 70 cm of oil (b) 2 m of oil
(c) 20 cm of oil (d) 80 cm of oil

**Ans. (b) :** Given specific gravity of oil = 0.7
Pressure (P) = 0.14 kgf/cm$^2$
$P = \rho gh$

$$0.14 \times 9.81 \times 10^4\ N/m^2 = (1000 \times 0.7) \times 9.81 \times h$$

$$h = \frac{1.4}{0.7}$$

h = 2m of water

**122. The term $v^2/2g$ is known as**
(a) kinetic energy
(b) pressure energy
(c) kinetic energy/unit weight
(d) None of the above

**Ans. (c) :** The beam $\frac{V^2}{2g}$ is known as kinetic energy permit useful or kinetic head in meters.

**123. Bernoulli's theorem deals with the law of conservation of**

(a) mass (b) momentum
(c) energy (d) None of the above

**Ans. (c) :** Bernoulli's theorem deals with the law of conservation of energy.

**124. For a submerged body, if the centre of buoyancy coincides with the centre of gravity, the equilibrium is called**
(a) stable (b) unstable
(c) neutral (d) None of the above

**Ans. (c) :** For a submerged body, if the centre of buoyancy coincides with centre of gyration, then the body is in the Neutral equilibrium.

**125. If the fluid particles move in a zig-zag way, the flow is called**
(a) unsteady (b) non-uniform
(c) turbulent (d) incompressible

**Ans. (c) :** IP the fluid particles move in a zig-zag way then the flow is called turbulent flow i.e, the direction of fluid particles in randomly.

**126. Compressibility is equal to**
(a) $\frac{dV/V}{dp}$ (b) $\frac{dp}{-(dV/V)}$
(c) $\frac{dp}{d\rho}$ (d) $\sqrt{\frac{dp}{d\rho}}$

**Ans. (a) :** Compressibility (β) It is inverse of bulk modulus of elasticity (K).

$$\beta = \frac{1}{K} = \frac{dv/v}{dp}$$

Where, $\frac{dV}{V}$ = change in volume per unit volume

dP = degree in the pressure

**127. A current meter is a device used for measuring**
(a) velocity (b) viscosity
(c) current (d) pressure

**Ans. (a) :** Current meter is the device that is used to measure the velocity of the flow of fluid.
The relation between velocity and the number of revolutions completed by current meter is-

$V = a \times N + b$

Where,
V = Velocity
N = No. of revolutions done by current meter in 1 second
a, b = Current meter constant

**128. The discharge through a single-acting reciprocating pump is**
(a) $Q = \frac{ALN}{60}$ (b) $Q = \frac{2ALN}{60}$
(c) Q = ALN (d) Q = 2ALN

**Ans. (a) :** Discharge through the reciprocating pump-
■ The discharge through a single acting reciprocating is given as

$$Q = V \times \frac{N}{60}$$

Q = (Area × length of stoke) × N/60

$$Q = (A \times L) \times \frac{N}{60}$$

$$Q = \frac{ALN}{60}$$

**129. The boundary layer takes place**
(a) for ideal fluid
(b) for pipe flow only
(c) for real fluid
(d) for flow over flat plates only

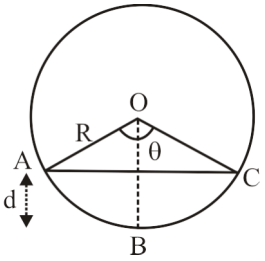
**Ans. (c) :** In flowing fluids the boundary layer is the area in the immediate vicinity of a solid wall where the velocity rises asymptotically from the value at the wall to the value of the main flow which is not influenced by friction.

**130. The discharge through a rectangular channel is maximum when**
(a) m = d/3 (b) m = d/2
(c) m = 2d (d) m = 3d/2

**Ans. (b) :** Maximum discharge through a rectangular channel if – $m = \frac{d}{2}$
where; m = hydraulic mean depth.

**131. For a circular channel, the wetted perimeter is given by**
(a) $\frac{R\theta}{2}$ (b) 3Rθ
(c) 2Rθ (d) Rθ

**Ans. (c) :**




For a circular channel, let wetted perimeter = P
In arc ABC
$\therefore \boxed{P = 2R\theta}$

**132. The discharge through a trapezoidal channel is maximum when**
(a) half of top width = sloping side
(b) top width = half of sloping side
(c) top width = 1.5 × sloping side
(d) None of the above

**Ans. (a) :** The discharge through a trapezoidal channel is maximum when–
half of top width = sloping side

**133. The difference in pressure head, measured a mercury water differential manometer for 20 cm difference of mercury will be**
(a) 2.72 m (b) 2.52 m
(c) 2.0 m (d) 0.2 m

**Ans. (b):** Let, x = difference of mercury level
= 20 cm = 0.2 m

$\therefore$ The difference in pressure load $\frac{P_1}{\rho g}-\frac{P_2}{\rho g}$

$$\Rightarrow \frac{P_1-P_2}{\rho g}=h=x\left(\frac{S_m}{S}-1\right)$$

where,
$S_m$ = specific gravity mercury = 13.6
S = sq. specific gravity of water = 1

$$\therefore \frac{P_1}{\rho g}-\frac{P_2}{\rho g}=0.2\times\left(\frac{13.6}{1}-1\right)=2.52\text{ m}$$

**134. Shear strain rate is given by**

(a) $\frac{1}{2}\left(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}\right)$

(b) $\frac{1}{2}\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)$

(c) $\frac{1}{2}\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}$

(d) $\frac{1}{2}\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}$

**Ans. (b) :** Angular deformation– It is the average change in the angle contained by two adjacent sides.
It is given by–

Shear strain rate = $\frac{1}{2}\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)$

**135. The value of momentum correction factor for the viscous flow through a circular pipe is**

(a) 1.33 (b) 1.50
(c) 2.0 (d) 1.25

**Ans. (a) :** The momentum correction factor ($\beta$) for turbulent flow $\beta$ is slightly higher than one near 1.2.
1. For laminar, $\beta = 1.33$
2. Kinetic energy factor-
for laminar = $\alpha = 2$

**136. A known as hydro dynamically smooth when**

(a) $\frac{k}{\delta'}=0.3$ (b) $\frac{k}{\delta'}>0.3$

(c) $\frac{k}{\delta'}<0.25$ (d) $\frac{k}{\delta'}=6.0$

**Ans. (c) :** Hydrodynamically smooth when,

$$\frac{k}{\delta}<0.25$$

Boundary transition condition, when

$$0.25<\frac{k}{\delta}<6$$

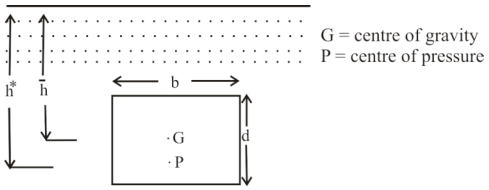
Hydrodynamically rough when-

$$\frac{k}{\delta}>6$$

**137. The resultant hydrostatic force acts through a point known as**

(a) centre of gravity
(b) centre of buoyancy
(c) centre of pressure
(d) None of the above

**Ans. (c) :** Centre of pressure is defined as the point of application of the total pressure on the surface.
The point of action of total Hydrostatic force on the submerged surface is called the centre of pressure.




$$h^*=\bar{h}+\frac{I_G\sin^2\theta}{A\bar{h}}$$

**138. The loss of pressure head for the laminar flow through pipe varies**

(a) as the square of velocity
(b) directly as the velocity
(c) as the inverse of the velocity
(d) None of the above

**Ans. (b) :** The loss of pressure head for the laminar flow through pipe varies directly as the velocity.

i.e., $\frac{P_1-P_2}{\rho g}=h_f=\frac{32\mu\bar{u}L}{D^2}$

Loss of pressure head ($h_f$) $\propto \bar{u}$

**139. Geometric similarity between model and prototype means**

(a) the similarity discharge
(b) the similarity of linear dimensions
(c) the similarity of motion
(d) the similarity of forces

**Ans. (b) :** Geometric similarly between model and prototype means the similarity of linear dimensions.
**Geometric Similarity:-** When the ratio of all corresponding linear dimensions is the model and proto type are equal.

**140. The work saved by fitting an air vessel to a single acting reciprocating pump is**

(a) 39.2% (b) 89.4%
(c) 48.8% (d) 92.3%

**Ans. (*) :** 84.4% work saved by fitting on air vessel to a single acting reciprocating pump.
• Work saved in a double acting reciprocating pump = 39.2%

**141. The thickness of laminar boundary layer at a distance x from the leading edge over a flat plates varies as**

(a) $x^{4/5}$ (b) $x^{1/2}$
(c) $x^{1/5}$ (d) $x^{3/5}$

**Ans. (b) :** Let,
$\delta$ = thickness of laminar boundary layer at a distance x from loading edge.
Blasius's equation

$$\delta = \frac{4.91x}{\sqrt{R_{ex}}}$$

$$= \frac{491 \times x}{\sqrt{\rho \frac{Vx}{u}}}$$

$$\delta \propto x^{1/2}$$

$$\because R_{ex} = \frac{\rho V x}{\mu}$$

**142. If the surface tension at the air- water interface is 0.073 N/m, estimate the pressure difference between inside and outside an air bubble of 0.01 mm.**
(a) 159.7 kN/m$^2$ (b) 159.7 MN/m$^2$
(c) 319.4 kN/m$^2$ (d) 0

**Ans. (*) :** Given:-
$\sigma$ = 0.073 N/m.
d = 0.01 mm = $0.01 \times 10^{-3}$ m.
$\therefore$
Pressure inside droplet

$$p = \frac{4\sigma}{r}$$

$$p = \frac{4 \times 0.073 \times 10^3}{0.01/2}$$

$$p = 58.4 \text{ kN/m}^2$$

**143. Piezometric head of a fluid is defined as**
(a) the sum of absolute pressure head and datum
(b) the stagnation pressure head
(c) the sum of stagnation head and datum head
(d) the sum of gauge pressure head and datum head

**Ans. (d) :** Piezometric head of a fluid is defined as the sum of gauge pressure head and datum head

$$\text{Piezometric head} = \frac{P}{\rho g} + z$$

**144. The pressure at a point in a fluid is not equal in all directions if**
(a) the fluid is at rest
(b) there are shear stresses
(c) the fluid is accelerated
(d) the fluid is rotated at a constant speed

**Ans. (b) :** The pressure at a point in a fluid is not equal in all direction if there are shear stresses because when we applying the shear on the fluid, then in the fluid continuous deformation occurs.

**145. Assertion (A) : Pressure is equal in all directions at a point in an ideal fluid flow.**
**Reason (R) : Pascal's law is valid for all cases where shear stresses are zero.**
(a) A is true; R is true, and it explains A
(b) A is true; R is true, but it does not explain A
(c) A is true; R is false
(d) A is false; R is true

**Ans. (a) : Pascal's law :** The state that pressure at any point in a fluid at rest has a same magnetite in all directions.
In other words if a certain pressure is applied at any point in the fluid at rest then the pressure is transmitted equally in all the directions and to every other point in the fluid.

**146. Observation of a flow net enables us to**
(a) determine the velocity at all points
(b) estimate the pressure at all points
(c) estimate the velocity variations
(d) determine the energy loss in flow

**Ans. (c) :** Application of flow net:-
(i) Determination of seepage
(ii) Determination of Hydrostatic pressure
(iii) Determination of seepage pressure
(iv) Determination of exit gradient.

**147. The assumption not made in the derivation of Bernoulli's equation is**
(a) inviscid flow (b) steady flow
(c) two-dimensional flow
(d) uniform flow

**Ans. (d) : Bernulli equation:-** It is the integration of Euler is equation of motion a long a stream line under steady incompressible flow conditions.
**Assumption:-**
(i) Along stream line.
(ii) Effect of friction is negligible.
(iii) Steady and incompressible.

$$\text{Total head} = \frac{p}{\rho g} + \frac{V^2}{2g} + Z$$

The assumption not made in the derivation of Bernoulli's equation is uniform flow.

**148. Venturimeter (V), flow nozzle (N) and orificemeter (O) arranged in increasing order of co-efficient of discharge are**
(a) V, N, O (b) N, O, V
(c) O, N, V (d) O, V, N

**Ans. (c) :** O < N < V
Increasing order of coefficient of discharge for venturimeter (V), flow nozzle (N) and orifice meter (O).

**149. According to the King's Law for a constant temperature hot wire anemometer**
(a) $E \propto U$ (b) $E^2 \propto U$
(c) $E^2 \propto \sqrt{U}$ (d) $E \propto U^2$

**Ans. (c) :** According to the King's law – $E^2 \propto \sqrt{U}$

**150. The dividing streamline for a uniform flow superimposed over a two-dimensional droplet is**
(a) a straight line (b) a circle
(c) a sphere (d) an ellipse

**Ans. (b) :** The dividing stream line for a uniform flow super imposed over a two - dimensional droplet is a circle.

# Telangana State Public Service Commission
# Assistant Executive Engineer Exam, 2011 (P-I)
# Solved Paper

**1. The net head (H) on the turbine is given by**

(a) H= gross head + head lost due to friction

(b) H= gross head - head lost due to friction

(c) H= gross head $+\frac{v^2}{2g}-$ head lost due to friction

(d) H= gross head $-\frac{v^2}{2g}-$ Head lost due to friction

**Ans. (b) :** Net head is also called effective head and is defined as the head available at the inlet of the turbine.

$$\boxed{H = H_g - h_f}$$

Where, Net head $\rightarrow H$

Gross head $\rightarrow H_g$

Head lost due to friction $\rightarrow h_f$

**2. Mechanical efficiency of a turbine is the ratio of**

(a) Power at the inlet to the power at the shaft of the turbine

(b) Power at the shaft to the power given to the runner

(c) Power at the shaft to the power at the inlet of the turbine

(d) None of the above

**Ans. (b) :** Mechanical Efficiency ($\eta_m$)

$$\boxed{\eta_m = \frac{\text{Power at the shaft of the turbine}}{\text{Power delivered by water to runner}}}$$

$$\boxed{\eta_m = \frac{S.P}{R.P}}$$

Overall efficiency ($\eta_0$)

$$\boxed{\eta_0 = \frac{\text{Power available at the shaft of turbine}}{\text{Power supplied at the inlet of turbine}}}$$

$$\boxed{\eta_0 = \frac{\text{Shaft Power (S.P)}}{\text{Water power (W.P)}}}$$

Hydraulic efficiency ($\eta_m$)

$$\boxed{\eta_h = \frac{\text{Power delivered to runer}}{\text{Power supplied inlet}} = \frac{R.P}{W.P}}$$

Volumetric efficiency ($\eta_v$)–

$$\boxed{\eta_v = \frac{\text{Volume of water actually strinking the runner}}{\text{Volume of water supplied to the turbine}}}$$

$$\boxed{\eta_0 = \eta_m \times \eta_h}$$

**3. The relation between the hydralic efficiency ($\eta_h$), mechanical efficiency ($\eta_m$) and overall efficiency ($\eta_o$) is**

(a) $\eta_h = \eta_o \times \eta_m$ (b) $\eta_o = \eta_h \times \eta_m$

(c) $\eta_o = \frac{\eta_m}{\eta_h}$ (d) None of the above

**Ans. (b) :** The relation between the hydralic efficiency ($\eta_h$), mechanical efficiency ($\eta_m$) and overall effieiency is $\boxed{(\eta_o) = \eta_m \times \eta_h}$.

$$\eta_o = \frac{\text{Power available at the shaft of the turbine}}{\text{Power supplied at the inlet of the turbine}} = \frac{S.P.}{W.P.}$$

**4. A turbine is called impulse turbine if at the inlet of the turbine**

(a) Total energy is only kinetic energy

(b) Total energy is only pressure energy

(c) Total energy is the sum of kinetic energy and pressure energy

(d) None of the above

**Ans. (a) :** • In an impulse turbine all the available energy of water is converted to kinetic energy.

• Where as in reaction turbine, the available water is pressure and kinetic both.

• Girard turbine is an example of impulse turbine.

**5. A turbine is called reaction turbine if at the inlet of the turbine the total energy is**

(a) Kinetic energy only

(b) kinetic and pressure energy

(c) Pressure energy only

(d) Mechanical energy

**Ans. (b) :** A turbine is called reaction turbine if at the inlet of the turbine the total energy is kinetic and pressure energy.

All blades are completely sub-merged in water.

**Reaction turbine :**

• It is useful for low head and high discharge.

• Degree of reaction are not zero, pressure drop takes place.

• Example of reaction turbines are Francis Turbine, Kaplan turbine, Scotch turbine and Propeller turbine.

**6. Tick mark the correct statement**

(a) Pelton wheel is a reaction turbine

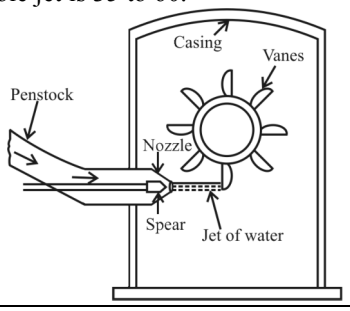
(b) Pelton wheel is a radial flow turbine

(c) Pelton wheel is an impulse turbine

(d) None of the above

**Ans. (c) : Pelton turbine–** The pelton wheel turbine is a tangential flow impulse turbine used for high heads (> 250 m) of water. The energy available at the inlet of the turbine is only kinetic energy.

• Specific speed of pelton wheel for single jet 10 to 35 and for double jet is 35 to 60.




**7. Francis turbine is**

(a) An impulse turbine
(b) A radial flow impulse turbine
(c) An axial flow turbine
(d) A reaction radial flow turbine

**Ans. (d) :** Francis turbine is a reaction radial flow turbine.
Francis turbine is a inward flow reaction turbine having radial discharge at outlet (mixed flow turbine).
• Specific speed of Francis turbine 60 to 300.
• Flow ratio of Francis turbine 0.15 to 0.30.
• Speed ratio of Francis turbine 0.60 to 0.90.

| $N_S$ | Type | $\beta_1$ | $\alpha_1$ |
|---|---|---|---|
| 40–120 | slow | $90^0$-$120^0$ | $15^0$-$25^0$ |
| 120–200 | medium | $90^0$ | $25^0$-$32^0$ |
| >120 | fast | $45^0$-$90^0$ | $32^0$-$40^0$ |

**8. Kaplan turbine is**

(a) An impulse turbine
(b) A radial flow impulse turbine
(c) An axial flow reaction turbine
(d) A radial flow reaction turbine

**Ans. (a) :** Kaplan turbine is an axial flow reaction turbine used for low head and high flow rate or discharge.
**Other reaction turbine**–
(1) Fourneyron turbine
(2) Jonval turbine
(3) Thomson's turbine.
In Kaplan turbine we have double control guide vane which controls flow and inlet angle and individual blades can also be rotated about their respective axis.

**9. Flow ratio is defined as the ration of**

(a) Velocity of flow at inlet to the velocity given by $\sqrt{2gh}$
(b) Velocity of runner at inlet to the velocity of flow at inlet
(c) Velocity of runner of the velocity given by $\sqrt{2gh}$
(d) None of the above

**Ans. (a) :** Flow ratio is the ratio of the velocity of flow at inlet to the theoretical jet velocity

$$K_f = \frac{V_{f_1}}{\sqrt{2gH}}$$

• The value of $K_f$ varies from 0.15 to 0.30.

**10. The discharge through Francis turbine is**

(a) $Q = \pi DBVf$
(b) $Q = \frac{\pi}{4} d^2 \times \sqrt{2GH}$
(c) $Q = \frac{\pi}{4}\left(D_A^2 - D_B^2\right)$
(d) None of the above

**Ans. (a) :** The discharge through Francis turbine is $\boxed{Q = \pi DBV_f}$.
• The discharge through Francis turbine is radial.

**11. The discharge through Kaplan turbine is**

(a) $Q = \pi DBV_f$
(b) $Q = \frac{\pi}{4} d^2 \times \sqrt{2GH}$
(c) $Q = \frac{\pi}{4}\left(D_a^2 - D_b^2\right)$
(d) $Q = 0.9\pi DBV_f$

**Ans. (c) :** Discharge through the turbine–

$$Q = V_f \times \frac{\pi}{4}\left(D_a^2 - D_b^2\right)$$

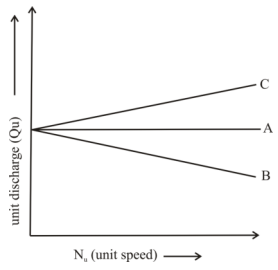
D – Dia of turbine
$D_b$ – Dia of the boss
$V_f$ – Velocity of flow at inlet

**12. Draft tube is used for discharging water from the exit of**

(a) An impulse turbine
(b) A kaplan turbine
(c) A pelton wheel
(d) reciprocation pump

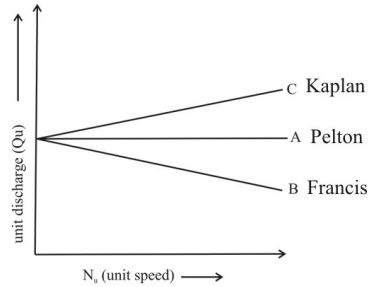
**Ans. (b) :** Draft tube is used for discharging water from the exit of a reaction turbine (Kaplan).
• In reaction turbine, draft tube is used to convert the kinetic by a gradual expansion of the flow cross-section.

**13. the unit discharge $(Q_u)$ and unit speed $(N_u)$ curves for different turbines are shown in the adjacent figure 1. curve A is for**




(a) Francis turbine
(b) Kaplan turbine
(c) Pelton turbine
(d) Propeller turbine

**Ans. (c):**

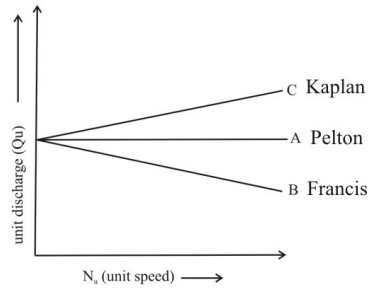



According to figure curve A is for Pelton turbine, curve B is for Francis turbine and curve C is for Kaplan turbine.

**14. Curve B is Figure 1 is for**

(a) Francis turbine
(b) Kalpan turbine
(c) Pelton turbine
(d) Propeller turbine

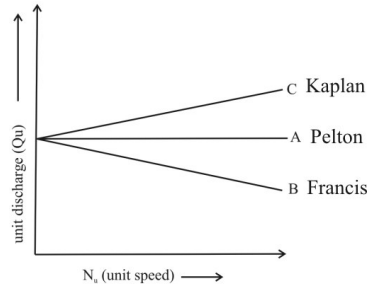
**Ans. (a) :**




**15. Curve C in Figure 1 is for**

(a) Francis turbine
(b) Kaplan turbine
(c) Pelton turbine
(d) Propeller turbine

**Ans. (b) :**




**16. Tick mark the correct statement**

(a) Curves at constant speed are called main characteristic curves
(b) Curves at constant head are called main characteristics
(c) Curves at constant efficiency are called operating characteristic curves
(d) Curves at constant efficiency are called main characteristic curves

**Ans. (b) :** Characteristics curves are used for predicting the behavior and performance of a pump when it is working under different flow rate, head and speed.

The following are the importants characteristics curves for pumps–
1. Main characteristics curves,
2 Operating characteristics curves, and
3. Constant efficiency or Muschel curves.

- The main characteristics of a centrifugal pump consist of variation of head (manometeric head, $H_m$), power and discharge with respect to speed. For plotting curves of manometric head versus speed, discharge is kept constant. For plotting curves of discharge versus speed, manometric head is kept constant. And for plotting curves of power versus speed, the manometric head and discharge are kept constant.
- If the speed is kept constant the variation of manometric head power and efficiency with respect to discharge gives the operating characteristics curve of pump.

**17. Main characteristic curves of a turbine means**

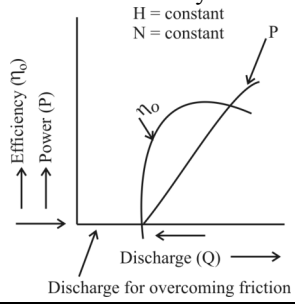
(a) Curves at constant speed
(b) Curves at constant efficiency
(c) Curves at constant head
(d) None of the above

**Ans. (c) :** The characteristic curve obtained all the following –
(a) Constant head curves or main characteristics curves.
(b) The constant head at a constant gate opening and speed is varied by changing the load on the turbine.
(c) Constant speed curve or operating characteristic curves.
(d) Constant efficiency curve or muchel curves.

**18. Operating characteristic curves of a turbine means**

(a) Curves drawn at constant speed
(b) Curves drawn at constant efficiency
(c) Curves drawn at constant head
(d) None of the above

**Ans. (a) :** Operating characteristic curves– Operating characteristic curves are plotted when the speed on the turbine is constant. The head may or not remain same.




**19. the work done by the impeller of a centrifugal pump on water per second per unit weight of water is given by**

(a) $\frac{1}{g}V_{w_1}u_1$ (b) $\frac{1}{g}V_{w_2}u_2$

(c) $\frac{1}{g}\left(V_{w_2}u_2 - V_{w_1}u_1\right)$ (d) None of the above

**Ans. (b):** Work done per second by the impeller on the liquid $= -\frac{w}{g}\left(V_{w_1}u_1 - V_{w_2}u_2\right)$

w = weight of liquid per second

$u_1$ = tangential velocity at the exit point of the impeller

u = Tangential velocity of the impeller at the entrance.

Work done per unit weight of liquid $= \frac{1}{g}\left(V_{w_2}u_2\right)$

$\left\{V_{w_1} = 0\right\}$

**20. The manometric head of a centrifugal pump is**

(a) Pressure head at the outlet of pump-pressure head at inlet

(b) Total head at inlet-total head at outlet

(c) Total head at outlet - total head at inlet

(d) none of the above

**Ans. (c) : Manometric head**–The head against which a centrifugal pump has to work is known as the manometric head.

- $H_{mano}$ = Head imparted by the impeller to liquid – loss of head in the pump.
- $H_{mano}$ = Total head at outlet of the pump – Total head at inlet of the pump.
- $H_{mano}$ = static + losses in pipes + $\frac{V_d^2}{2g}$

**21. Mechanical efficiency of a centrifugal pump is**

(a) Power at the impeller/ H.H.P.

(b) S.H.P./ power at the impeller

(c) Power possessed by water/power at the impeller

(d) Power possessed by water / S.H.P.

**Ans. (d) : Mechanical efficiency ($\eta_m$)**–The ratio of the power delivered by the impeller to the power input to the pump shaft is known as mechanical efficiency.

$$\eta_m = \frac{\text{Power delivered by the impeller to the liquid}}{\text{Power input to the pump shaft}}$$

**22. Specific speed of a pump ($N_s$) is given by the expression**

(a) $N_s = \frac{N\sqrt{Q}}{H_m^{5/4}}$ (b) $N_s = \frac{N\sqrt{P}}{H_m^{3/4}}$

(c) $N_s = \frac{N\sqrt{Q}}{H_m^{3/4}}$ (d) $N_s = \frac{N\sqrt{Q}}{H_m^{5/4}}$

**Ans. (c) :** Specific speed of a pump ($N_s$) is given by the expression $N_s = \frac{N\sqrt{Q}}{H_m^{3/4}}$

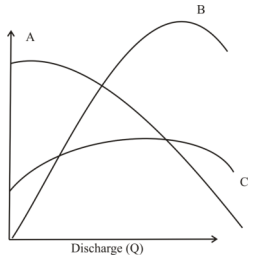
**Specific speed ($N_s$)** of any turbine is the speed in rpm of a turbine, geometrically similar to the actual turbine but of such a size that the under corresponding conditions it will develop unit power when working under unit head.

$$\boxed{N_S = \frac{N\sqrt{P}}{H^{5/4}}}$$

**Note**– Specific speed is not a dimension less parameter, its unit is

$\left[F^{1/2}L^{-3/4}T^{-3/2}\right]$

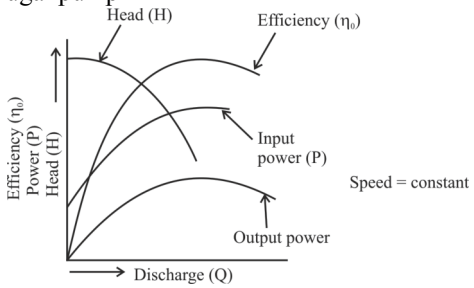
**23. The operating characteristic curves of a centrifugal pump are shown in Fig. 2 curve A is for**




(a) Head (b) Efficiency

(c) Power (d) Voltage

**Ans. (a) :** Operating characteristics curves of centrifugal pump–




Performance of a centrifugal pump under variable rotational speeds. It is a parabolic curve, which shows that for a given rotational speed, the manometric head decreases with the discharge.

**24. The curve C in fig. 2 is for**

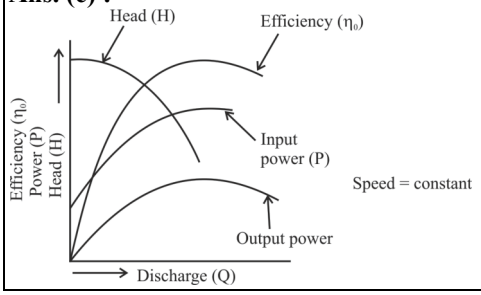
(a) head

(b) efficiency

(c) power

(d) velocity

**Ans. (c) :**




**25. In a centrifugal pump regulating valve is provided on the**

(a) Casing (b) Delivery pipe

(c) suction pipe (d) Impeller

**Ans. (b) :** A check/regulating valve is provided in the delivery pipe near the pump, in order to protect the pump from water hammer and also to regulate the discharge from the pump.

**26. The discharge of a centrifugal pump is**

(a) $Q = \pi DV_f$ (b) $Q = \pi bV_f$

(c) $Q = \pi DbV_f$ (d) $Q = \pi DbV_f$

Where D= diameter of impeller at inlet
b= width of impeller of inlet
$V_f$ velocity of flow at inlet

**Ans. (c) :** The discharge of a centrifugal pump is same as that of a reaction turbine. A centrifugal pump lifting water from a lower level to a higher level.

Q = Area × velocity of flow

$\boxed{Q = \pi DbV_f}$

**27. A centrifugal pump will start delivering liquid only when the pressure rise in the impeller is equal to the**

(a) Kinetic head (b) Velocity head

(c) Manometric head (d) Static head

**Ans. (c) :** A centrifugal pump will start delivering liquid only when the head developed by it is equal to the manometric head.

$\boxed{\text{Pressure head} \geq H_m}$

**28. For centrifugal pump impeller, the maximum value of the vane exit angle is**

(a) $10^o$ to $10^0$ (b) $15^o$ to $20^0$

(c) $20^o$ to $25^0$ (d) $25^o$ to $30^0$

**Ans. (c) :** For centrifugal pump impeller, the maximum value of the vane exist angle is$20^o$ to $25^{0.}$

**29. If the net positive suction head (NPSH) requirement for the pump is not satisfied, then**

(a) No flow will take place
(b) Cavitation will be formed
(c) Efficiency will be low
(d) Excessive power will be consumed

**Ans. (b) : NPSH**–Net positive suction head may be defined "as the difference between the net inlet head and the head corresponding to the vapour pressure of the liquid.

- NPSH has significance only when cavitating liquids are handled.
- NPSH is a parameter that can be used to check cavitation in pump.

**30. The centrifugal pump preferred for a specific speed between 80 to 160 r.p.m. is**

(a) Slow speed with radial flow at outlet
(b) Medium speed with radial flow at outlet
(c) High speed with radial flow at outlet
(d) High speed with mixed flow at outlet

**Ans. (d) :**

| Specific speed | Type of centrifugal pump |
|---|---|
| 10 to 30 | Slow speed pump with radial flow at outlet |
| 30 to 50 | Medium speed pump with radial flow at outlet |
| 50 to 80 | High speed pump with radial flow at outlet |
| 80 to 160 | High speed pump with mixed flow at outlet |
| 160 to 500 | High speed pump with axial flow at outlet |
| Above 500 | Very high speed pump |

**31. The number or contra flexure points that occur in a fixed ended beam and subjected to concentrated load is**

(a) 0 (b) 2

(c) 1 (d) 3

**Ans. (b) :** The number of contra flexure points that occur in a fixed ended beam and subjected to concentrated load is two.

| Beam and loading | No. of Points of contraflexure |
|---|---|
| (i) Simply supported beam with UDL over part of the structure | Zero |
| (ii) An overhanging beam with loading only over supported span not on overhangs | Zero |
| (iii) Fixed beam subjected to concentrated load/UDL. | Two |
| (iv) Cantilever beam subjected to UDL with zero load intensity at free end | Zero |
| (v) Propped cantilever beam for any load | one |

**32. The deflection of a beam depends mainly on**

(a) Total stresses and boundary conditions
(b) Total external loads
(c) Fixity of joints
(d) Cumulative strains and boundary conditions

**Ans. (b) :** The deflection of a beam depends mainly on total external loads.

Deflection in bar due to axial loading,

$$\Delta L = \frac{PL}{AE}$$

$$\text{Strain energy } (U) = \frac{1}{2} P \times \Delta L$$

$$= \frac{1}{2} \times P \times \frac{PL}{AE}$$

$$\therefore \quad \boxed{U = \frac{P^2 L}{2AE}}$$

**33. Maximum deflection of a cantilever beam subjected to a concentrated load at free end is**

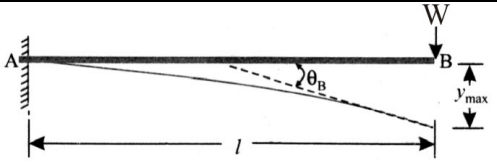
(a) $\frac{WL^3}{3EI}$ (b) $\frac{WL^3}{8EI}$

(c) $\frac{WL^3}{48EI}$ (d) $\frac{WL^3}{36EI}$

Were W= total load, L= span, EI= flexural rigidity

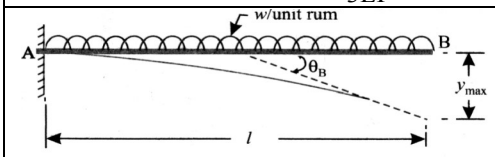
**Ans. (a) :**

**Slope and deflection for cantilever beam**



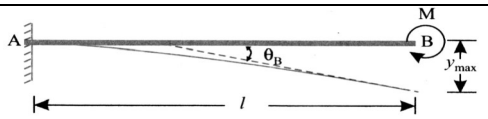
Slope, $\theta_B = -\frac{W\ell^2}{2EI}$

Maximum deflection, $y_{max} = \frac{-W\ell^3}{3EI}$



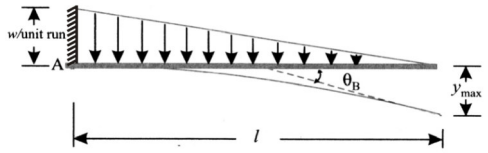

Slope, $\theta_B = \frac{-w\ell^3}{6EI}$

Maximum deflection, $y_{max} = -\frac{w\ell^4}{8EI}$




Slope, $\theta_B = -\frac{M\ell}{EI}$

Maximum deflection, $y_{max} = -\frac{M\ell^2}{2EI}$




Slope, $\theta_B = -\frac{w\ell^3}{24EI}$

Maximum deflection, $y_{max} = \left[-\frac{w\ell^4}{30EI}\right]$

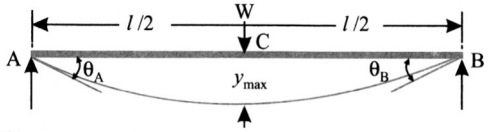
**34. Maximum deflection of a simply supported beam subjected to uniformly distributed load over entire span is**

(a) $\frac{WL^3}{3EI}$ (b) $\frac{WL^3}{8EI}$

(c) $\frac{WL^3}{48EI}$ (d) $\frac{5WL^3}{384EI}$

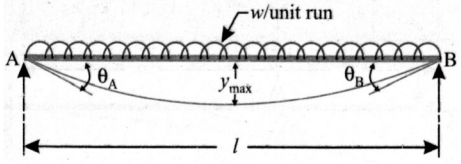
Were W= total load,

**Ans. (d) :**




Slope, $\theta_A = -\frac{w\ell^2}{16EI}$ $\theta_B = +\frac{w\ell^2}{16EI}$

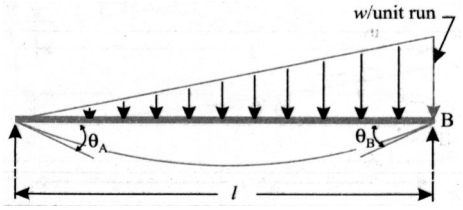
Maximum deflection, $y_{max} = y_C = -\frac{w\ell^3}{48EI}$




Slope, $\theta_A = \frac{-w\ell^3}{24EI} = \frac{-W\ell^2}{24EI}$ $\theta_B = \frac{+w\ell^3}{24EI}$ or $\left(\frac{-W\ell^2}{24EI}\right)$

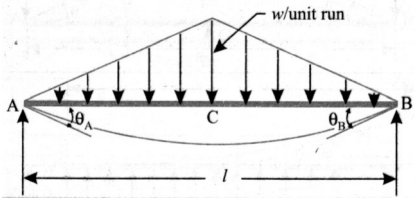
Maximum deflection, $y_{max} = \frac{-5w\ell^4}{384EI}$ OR

$\left[= -\frac{5W\ell^3}{384EI}\right]$ where, W = w$\ell$




Slope, $\theta_A = \frac{-7w\ell^3}{360EI}$ $\theta_B = \frac{w\ell^3}{45EI}$

Maximum deflection, $y_{max} = -\frac{-2.5w\ell^4}{384EI}$ (at x = 0.519 *l* from A)




Slope, $\theta_A = \frac{-5w\ell^3}{192EI}$ $\theta_B = \frac{+5w\ell^3}{192EI}$

Maximum deflection, $y_{max} = y_C = \left(\frac{-w\ell^4}{120EI}\right)$

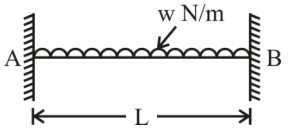
**35. Maximum deflection of a beam caused by uniformly distributed load acting on a beam with both the ends fixed is**

(a) $\frac{WL^3}{15EI}$ (b) $\frac{11WL^3}{60EI}$

(c) $\frac{WL^3}{60EI}$ (d) $\frac{WL^3}{384EI}$

Were W= total load, L= span, EI= flexural rigidity

**Ans. (d) :**




Maximum deflection $(\delta) = \frac{wL^4}{384EI} = \frac{WL^3}{384EI}$

**36. Maximum deflection caused by a concentrated load at mid point of a fixed ends beam is**

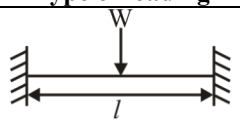
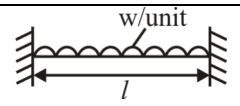
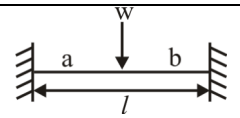
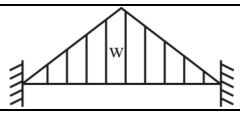
(a) $\frac{WL^3}{185EI}$ (b) $\frac{WL^3}{192EI}$

(c) $\frac{5WL^3}{384EI}$ (d) $\frac{WL^3}{384EI}$

Were W= total load, L= span, EI= flexural rigidity

**Ans. (b) :**

**Slopes and Deflections for Different Loadings on fixed beam**

| Type of loading | Slope | Deflection |
|---|---|---|
|   | 0 | $\Delta = \frac{WL^3}{192EI}$ |
|   | 0 | $\Delta = \frac{WL^4}{384EI}$ |
|   | 0 | $\Delta = \frac{Wa^3b^3}{3EIL^3}$ |
|   | 0 | $\Delta = \frac{7wl^4}{3840EI}$ |

**37. The maximum deflection caused in a propped cantilever beam subjected to a uniformly distributed load is**

(a) $\frac{WL^3}{185EI}$ (b) $\frac{WL^3}{192EI}$

(c) $\frac{5WL^3}{384EI}$ (d) $\frac{WL^3}{384EI}$

Were W= total load, L= span, EI= flexural rigidity

**Ans. (a) :**

$R_1 = \frac{5}{8}wL,\ R_2 = \frac{3}{8}wL$

$M = \frac{w}{8}\left(4x^2 + 5x - \ell^2\right)$

$M_1 = \frac{-w\ell^2}{8}$

$y = \frac{wx^2}{48EI} \times (1-x) \times (2x-3)$

$y_{max} = \frac{w\ell^4}{185EI} = \frac{W\ell^3}{185EI}$

**38. The relation between rate of loading and deflection is**

(a) Rate of loading= $EI\frac{d^4y}{dx^4}$

(b) Rate of loading= $EI\frac{d^3y}{dx^4}$

(c) Rate of loading= $EI\frac{dy}{dx}$

(d) Rate of loading= $EI\frac{d^2y}{dx^2}$

**Ans. (a) :** As we know,

$EI\frac{d^2y}{dx^2} = M$

Then, $EI\frac{d^3y}{dx^3} = \frac{dM}{dx} = F$

& $EI\frac{d^4y}{dx^4} = \frac{d^2M}{dx^2} = \frac{dF}{dx} = w$

So, relation between loading (w) and deflection (y),

Rate of loading = $EI\frac{d^4y}{dx^4}$

**39. The bending moment and displacement are related as**

(a) $B.M. = EI\frac{d^3y}{dx^3}$ (b) $B.M. = EI\frac{dy}{dx}$

(c) $B.M. = EI\frac{d^2y}{dx^2}$ (d) $B.M. = EI.y$

**Ans. (c) :** $Slope = \frac{dy}{dx}$

$BM = EI\frac{d^2y}{dx^2}$

$Shear\ force = EI\frac{d^3y}{dx^3}$

$Load\ intensity = EI\frac{d^4y}{dx^4}$

**40. A rigid prop is one which**

(a) Permits 50% of free deflection, that would have occurred if the prop were not there

(b) Does not permity any displacement perpendicular to plane of prop

(c) Does not offer any reaction

(d) Supports the entire load and relives all other supports completely

**Ans. (b) :** A rigid prop is one which does not permity any displacement perpendicular to plane of prop

For a prop tgere is no change in length of prop due to the force in it.

**41. Torsional rigidity is**

(a) EI (b) Z

(c) NJ (d) $\frac{N\theta}{l}$

**Ans. (c) : Torsional Rigidity–** The minimum force required to deform an object by twisting through a unit angle.

Torsional rigidity is best defined as the torque required for unit rotation for a unit length of the bar.

Torsional equation $\Rightarrow \left[\frac{T}{J} = \frac{\sigma}{R} = \frac{G\theta}{\ell}\right]$

Where, $T \rightarrow$ Torque (N-m)

$\tau \rightarrow$ Shear stress ($N/m^2$)

$J \rightarrow$ Polar moment of inertia ($m^4$)

$R \rightarrow$ Radius of shaft

$G \rightarrow$ Modulus of rigidity ($N/m^2$)

$\theta \rightarrow$ Angle of twist (radius)

$\theta = \frac{TL}{GJ} \Rightarrow T = \frac{GJ\theta}{L}$

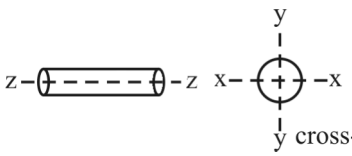
• Torque required to produce a twist 1 radian per unit length of the shaft called torsional rigidity.
$\theta = 1$ unit
$L = 1$ unit.
$T = GJ$
Here, $G = N$
$T = NJ$

**42. Polar moment of inertia of a circular section with radius, R is**

(a) $\frac{\pi R^3}{32}$ (b) $\frac{\pi R^4}{32}$

(c) $\frac{\pi R^4}{16}$ (d) $\frac{\pi R^4}{2}$

**Ans. (d) :** Polar moment of inertia of a solid circular section–




Polar moment of Inertia : M.I about polar axis (J)
"Z - Z" : Polar axis

$J = I_{ZZ} = I_{XX} + I_{YY}$

$= \frac{\pi D^4}{64} + \frac{\pi D^4}{64}$

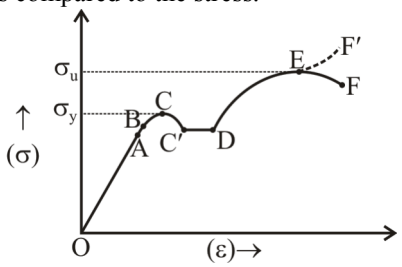
$= \frac{2\pi D^4}{64}$

$J = I_{ZZ} = \frac{\pi D^4}{32} = \frac{\pi}{2} R^4$

**43. In a tensile test, when the material is stressed beyond elastic limit, the tensile strain------- as compared to the stress**

(a) decreases slowly
(b) increases slowly
(c) decreases more quickly
(d) increases more quickly

**Ans. (d) :** In a tensile test, when the material is stressed beyond elastic limit, the tensile strain increases more quickly as compared to the stress.




From the stress-strain diagram, B ⇒ elastic limit.
Beyond elastic limit strain increases more quickly.

**44. A tensile test is performed on a round bar. After fracture it has been found that the diameter remains approximately same at fracture. The material under test was**

(a) Mild steel (b) Cat iron
(c) Glass (d) Copper

**Ans.(c):** The diameter of glass remain and approximately same after fracture under tensile testing due to very low value of Poisson's ratio.
$\mu_{glass} = 0.02$ to $0.03$
$\mu_{cast\ iron} = 0.23$ to $0.27$
$\mu_{copper} = 0.33$ to $0.36$
$\mu_{mild\ steel} = 0.25$ to $0.33$
Low value of $\mu$ it means low value of lateral strain. So, change in dia. $\Delta d$ is very low.

**45. Factor of safety is defined as the ratio of**

(a) Ultimate stress to working stress
(b) Working stress to ultimate stress
(c) Breaking stress to ultimate stress
(d) Ultimate stress to breaking stress

**Ans. (a) :** Factor of safety is defined for brittle material is the ratio of ultimate stress to the working stress.
• Factor of safety is defined for ductile material is the ratio of yield strength to the working stress.

**46. The hydrostatic law states that rate of increase of pressure in a vertical direction**

(a) is equal to density of fluid
(b) is equal to specific weight of fluid
(c) is equal to weight of fluid
(d) none of the above

**Ans. (b) :** A hydrostatics low state that rate of increase of pressure in a vertically downward direction in liquid is equal to weight density of the fluid and is given as–

$\boxed{P = \rho gh}$

where, $\rho g$ = Specific weight.

**47. Fluid statics deals with the following forces**

(a) Viscous and gravity forces
(b) Viscous and pressure forces
(c) Gravity and pressure forces
(d) Surface tension and gravity forces

**Ans. (c) :** Fluid static deals with fluids at rest.
• A fluid at rest has no shear stress. Any force developed is only due to normal stresses i.e. pressure.
• Fluid statics deals with gravity and pressure forces.

**48. The hydrostatic pressure on a plane surface is equal to**

(a) $wA\bar{h}$ (b) $wA\bar{h}\sin^2\theta$

(c) $\frac{wA\bar{h}}{2}$ (d) $wA\bar{h}\sin\theta$

**Ans. (a) : Total pressure–** It is defined as the force exerted by a static fluid on a surface either plane or curved when the fluid comes in contact with the surface. This force always acts normal to the surface.

$\boxed{P = wA\bar{h}}$

Where,
$\bar{h}$ = Centre of pressure
W = Specific weight of liquid ($\rho g$)
A = Area of surface
P = Total pressure

| Case | Force | Center of pressure (h) |
|---|---|---|
| Horizontal Surface | $wA\bar{x}$ | $h = \bar{x}$ |
| Vertical Surface | $wA\bar{x}$ | $h = \bar{x} + \frac{I_G}{A\bar{x}}$ |
| Inclined Surface | $wA\bar{x}$ | $h = \bar{x} + \frac{I_G}{A\bar{x}}\sin^2\theta$ |

**49. For submerged curved surface, the vertical component of the curved surface, the vertical component of the hydrostatic force is**

(a) Mass of the liquid supported by the curved surface
(b) Weight of liquid supported by the curved surface
(c) The force of the projected area of the curved surface on a vertical plane
(d) None of the above

**Ans. (b) :** For submerged curved surface, the vertical component of the hydrostatic force is weight of the liquid supported by the curved surface.

**50. The centre of pressure for a vertically immersed surface lies at a distance equal to..... the centre of gravity**

(a) $\frac{I_G}{A\bar{x}}$ below (b) $\frac{I_G}{A\bar{x}}$ above
(c) $\frac{Ax}{A_x}$ below (d) $\frac{A_x}{I_{\bar{G}}}$ above

**Ans. (a)** The centre of pressure for a vertically immersed surface lies at a distance equal to $\frac{I_G}{A\bar{x}}$ below the centre of gravity.

The point in the immersed body through which the resultant pressure of the liquid may be taken to act is known as centre of pressure.

$$\bar{h}_{c.p.} = \bar{h} + \frac{I_{GG}.\sin^2\theta}{A\bar{h}}$$

Where,

$\bar{h}_{cp} \rightarrow$ height of centre pressure

$\bar{h} \rightarrow$ height of centre of gravity Act

$I_{GG} \rightarrow$ moment of inertia

$A \rightarrow$ Area of plane

For horizontal plane

$\theta = 0^0$, $\bar{h}_{cp} = \bar{h}$

For vertical plane

$\theta = 90^0$

$$\bar{h}_{c.p.} = \bar{h} + \frac{I_{GG}}{A\bar{h}}$$

**Note :** As the depth of the surface increases $\bar{h}$ increase & $\left(\bar{h}_{c.p.} = \bar{h}\right)$ decrease (It means centre of pressure shift towards centre of gravity).

**51. When a vertical wall is subjected to pressures due to liquid on both sides, the resultant pressure is the------ of the two pressures.**

(a) Sum (b) Difference
(c) Arithmetic mean (d) Geometric mean

**Ans. (b) :** When a vertical wall is subjected pressures due to liquid on both sides, the resultant pressure is the difference of the two pressure. Because the pressure is acting in opposite direction. So we calculate net pressure in one direction as–

$$P = P_1 - P_2$$

**571. In a lock gate, the reaction between two gates is**

(a) $\frac{\rho}{\sin\alpha}$ (b) $\frac{2\rho}{\sin\alpha}$
(c) $\frac{\rho}{2\sin\alpha}$ (d) $\frac{\rho}{\sin\alpha/2}$

Where =ρ resultant pressure on the lock gate, α= inclination of the gate with the normal to the side of the lock

**Ans. (c) :** The lock gates are provided in navigation chambers or river for navigation. there are two sets of gates, one set on either side of the chamber. In a lock gate, the reaction between the two gates (R) is given by

$$R = \frac{\rho}{2\sin\alpha}$$

Where,

ρ = Resultant water pressure on the lock gate.

α = Inclination of the gate with the normal to the side of the lock.

**53. When a body, floating in a liquid, is given a small angular displacement, it starts oscillating about a point known as**

(a) Centre of pressure (b) Centre of gravity
(c) Centre of buoyancy (d) metacentre

**Ans. (d) :** When a body, floating in a liquid is a small angular displacement, it starts oscillating about a point known as metacentre.

**54. The metacentric height is the distance between**

(a) Centre of gravity of the floating body and the centre of buoyancy
(b) Centre of gravity of floating body and the meta centre
(c) Meta centre and centre of buoyancy
(d) Original centre of buoyancy and new centre of buoyancy

**Ans. (b) Metacentric height –** It is distance between the centre of gravity of the body and the metacentre.

$$GM = \frac{I}{V} - BG \text{ (V is volume displaced)}$$

GM = Metacentric height

Metacentre height for rolling condition will be less then metacentric height for pitching condition.

**For stable condition –** Metacentric height > 0 and metacentre above centre of gravity.

**Unstable condition –** Metacentric height < 0

Metacentre < centre of gravity

**55. A body floating in a liquid is said to be in neutral equilabrium, if its meta centre**

(a) Coincides with its centre of gravity
(b) Lies above its centre of gravity
(c) Lies below its centre of gravity
(d) Lies between the centre of buoyancy and centre of gravity

**Ans. (a) :** Floating body

| | |
|---|---|
| Stable equilibrium | M above G |
| Unstable equilibrium | M below G |
| Neutral equilibrium | M & G coincide |

**56. The meta centric height of a ship is 0.6m and the radius of gyration is 4 m. The time of rolling of a ship is**

(a) 4.1 s (b) 5.2 s
(c) 10.4 s (d) 14.1 s

**Ans. (c) :** Given,
least radius of gyration, (k) = 4m
metacentric height GM = 0.6m

Time of rolling $T = 2\pi\sqrt{\frac{k^2}{GM.g}}$

$$T = 2\pi\sqrt{\frac{4^2}{0.6\times 9.81}}$$

$$T = 10.4\,\text{sec}$$

**57. One cubic metre of water weighs**

(a) 100 litres (b) 250 litres
(c) 500 litres (d) 1000 litres

**Ans. (d) :** 1 $m^3$ = 1000 litres
1 litre = $1 \times 10^{-3}$ $m^3$

**58. The water pressure per metre length on a vertical masonry wall of dam is**

(a) WH/2 (b) WH
(c) $WH^2/2$ (d) $WH^2/4$

Where W= Specific weight of the liquid
H= Height of the liquid

**Ans. (c) : Water pressure–** This force is exerted by the weight of the water stored on the upstream side of the dam. It is estimated from the rules of hydrostatic pressure distribution.

- The water pressure per metre length on a vertical masonry wall of dam is $WH^2/2$.

**839. Bernoulli's theorem deals with the principle of conservation of–**

(a) Principle of conservation of mass flow
(b) Principle of conservation of energy
(c) Principle of conservation of momentum
(d) Both the principle of conservation of energy and principle of conservation of mass flow

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**Ans. (b) :** Bernoulli's equation states conservation of energy.
Bernoulli's equation is obtained by integrating the Euler's equation of motion as–

$$\int\frac{dp}{\rho}+\int gdz+\int \gamma dv = \text{constant}$$

If flow is incompressible, P is constant and

$$\therefore \frac{P}{\rho}+gz+\frac{v^2}{2}=\text{constant}$$

$$\frac{P}{\rho g}+z+\frac{v^2}{2g}=\text{ constant}$$

where,

$\frac{P}{\rho g}$ = Pressured energy per unit weight of fluid or pressure head.

$\frac{v^2}{zg}$ = Kinetic energy per unit weight or kinetic head.

z = Potential energy per unit weight or potential head.

**60. The Bernoulli's equation can take the from**

(a) $\frac{p_1}{\rho_1}+\frac{\upsilon_1^2}{2g}+z_1\frac{p_2}{\rho_2}+\frac{\upsilon_2^2}{2g}+z_2$

(b) $\frac{p_1}{\rho_1 g}+\frac{\upsilon_1^2}{2}+z_1=\frac{p_2}{\rho_2 g}+\frac{\upsilon_2^2}{2}+z_2$

(c) $\frac{p_1}{\rho_1 g}+\frac{\upsilon_1^2}{2g}+gz_1=\frac{p_2}{\rho_2 g}+\frac{\upsilon_2^2}{2g}+gz_2$

(d) $\frac{p_1}{\rho_1 q}+\frac{\upsilon_1^2}{2g}+gz_1=\frac{p_2}{\rho_2 g}+\frac{\upsilon_2^2}{2g}+z_2$

**Ans: (b)** As per Bernoulli's equation

$$\frac{P}{\rho g}+Z+\frac{V^2}{2g}=\text{constant head}$$

Where,

$\frac{P}{\rho g}$ = Pressure head or static head

Z = elevation head or datum

$\frac{V^2}{2g}$ = Velocity head or dynamic head or kinetic head

$\frac{P}{\rho g}+z$ = Piezometric head

$\frac{P}{\rho g}+\frac{V^2}{2g}$ = stagnation head.

**61. An orthotropic material has**

(a) Non- homogeneous properties
(b) Inelastic properties
(c) different properties in three perpendicular directions
(d) Same properties in orthogonal directions

**Ans. (c) : Orthotropic material–** Material having property, that differ along three mutually orthogonal direction. They are a subset of anisotropic materials because their properties change when measured from different directions.
**Anisotropic materials–** Properties such as young modulus, change with the direction along the object → Wood and composite material. .

**62. Bending stress on a beam section is zero at**

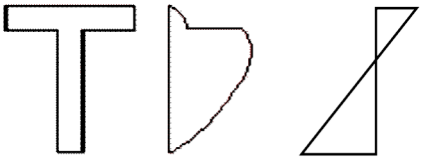
(a) Depends on the shape of the beam
(b) Top fibre

(c) Bottom fibre
(d) Centroid of the section

**Ans. (d) :** Bending stress on a beam section is zero at the centroid at the section (If stress is within the limit of elastic limit, centroid will be the neutral axis.).
We know that,
Bending stress = $\frac{M}{Z}$
Where, Z = Second moment of inertia
$Z = \frac{I}{Y}$, Y = D/2
At neutral axis, Z = 0
So, bending stress = 0

**63. Bending stress on a T- beam section is maximum at**
(a) Top fibre
(b) Centroid
(c) Bottom fibre
(d) Depends on the moment

**Ans. (c) :** Bending stress on a T- beam section is maximum at bottom fibre.




**64. Ratio of maximum bending stress in the flange to that in the web or an I-section at any distance along the length of the beam is always**
(a) Less than one
(b) Equal to one
(c) More than one
(d) No exact relation exists

**Ans. (c) :** Ratio of maximum bending stress in the flange to that in the web or an I-section at any distance along the length of the beam is always more than one.

**65. An equal flange I- section of 300 mm depth and 200 mm wide flanges is having a uniform flange width of 100 mm at the top and bottom segments. It is subjected to a bending moment of 100 kN-m. The bending stress at top most fibre is (in MPa)**
(a) 135 (b) 75
(c) 35 (d) 15

**Ans. (*) :** Data discrepancy.

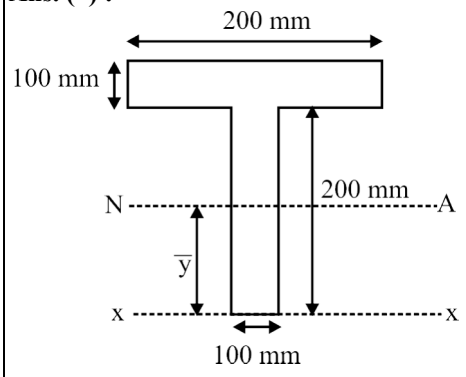
**66. A circular section of 200 mm diameter is subjected to a bending moment of 10 kN-m. The maximum tensile stress caused in the section (in MPa) is**
(a) 120 (b) 100
(c) 65 (d) 40

**Ans. (*) :** Given,
Dia (D) = 200 mm
B.M. (M) = 10 kN-m

$$\sigma = \frac{M}{I} y$$
$$= \frac{10 \times 10^6}{\frac{\pi}{64}(200)^4} \times 100$$
= 12.7 MPa

**67. A T- section with 300 mm depth has 200× 100 mm flange and 100 × 200 mm web projection. It is subjected to a bending mement of 100 KN-m. The approximate bending stress at top fibre is (in MPa)**
(a) 115 (b) 95
(c) 65 (d) 35

**Ans. (*) :**




$$\bar{y} = \frac{200 \times 100 \times \left(\frac{200}{2}\right) + 100 \times 200 \times \left(200 + \frac{100}{2}\right)}{2 \times (200 \times 100)}$$

$$I_{xx} = \frac{100 \times (200)^3}{3} + \frac{200 \times (100)^3}{12} + 200 \times 100 \times 250^2$$
$$= 1.533 \times 10^9 \text{ mm}^4$$

$I_{NA} = I_{xx} - Axy^2$

$$= 1.533 \times 10^9 - 2 \times (200 \times 100) \times (175)^2 = 3.08 \times 10^8 \text{ mm}^4$$

$$\sigma_{\text{top fibre}} = \frac{My_{top}}{I_{NA}} = \frac{100 \times 10^6 \times (300 - 175)}{3.08 \times 10^8}$$
$$= 40.58 \text{ N/mm}^2$$

**68. Which of the following are the most efficient in carrying bending moments?**
(a) Rectangular sections
(b) Circular sections
(c) I- sections
(d) T- sections

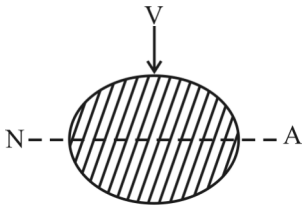
**Ans. (c) :** The section modulus of the I-section is more. It is very efficient form for carrying both bending and other loads in the plane of web.

**69. The ratio of maximum shear stress to average shear stress in the case of rectangular section is**
(a) $\frac{4}{3}$ (b) $\frac{3}{2}$
(c) 1 (d) 2

**Ans. (b):** The ratio of maximum shear stress to average shear stress in the case of rectangular section is 3/2.

For circular section, the maximum shear stress = $\frac{4}{3} \times$ Average shear stress

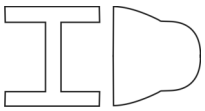
$$\frac{\text{Maximum shear stress}}{\text{Average shear stress}} = \frac{4}{3}$$




**70. In an I-section all most all shear force is taken by**

(a) Web
(b) Top flange
(c) Bottom flange
(d) Half the depth of each flange

**Ans. (a) :** In an I-section all most all shear force is taken by Web.



Symmetrical I-section the intensity of shear stress is maximum at the centroid of the section.

**71. If a shaft is subjected to pure twisting moment, an element on the surface is subjected to**

(a) normal tensile stress
(b) normal compressive stress
(c) Pure shear stress
(d) Bending stress

**Ans. (c) :** Pure shear stress is produced when shaft is subjected to pure twisting (torsion). The shear stress due to twisting moment is zero at the axis of the shaft.

**72. The maximum shear stress produce in a shaft is $5N/mm^2$. The shaft is of 40 mm diameter. The value of twisting moment is**

(a) 628 N-m (b) 62.8 N-m
(c) 125.6 N-m (d) 1256 N-m

**Ans. (b) :** Given,

$\tau = 5\,N/mm^2$

D = 40 mm

Shear stress $(\tau) = \frac{16T}{\pi D^3}$

$$5 = \frac{16T}{\pi \times (40)^3}$$

T = 62.8 N-m

**73. The power transmitted in KW by a shaft roating at speed of n rpm, transmitting a mean torque of T (in kg-m) is given by**

(a) $\pi nT/3060$ (b) $2\pi nT/4500$
(c) $\pi nT$ (d) $3060\pi nT$

**Ans. (a) :** Speed = n (rpm); Torque (T) (kg-m)

Power (P) = ? (in kw) $T = T \times 9.81$ N-m

$P = T\omega$

$$= \frac{T \times 2\pi n}{60} \times \frac{1}{1000} kW$$

$$= \frac{T \times 9.81 \times 2\pi \times n}{60000}$$

$$= \frac{\pi nT}{\frac{60000}{2 \times 9.81}} = \frac{\pi nT}{3058.10}$$

$$\boxed{P \approx \frac{\pi nT}{3060} kW}$$

**74. How does they Young's modulus vary with the temperature?**

(a) Will increase (b) Will not change
(c) Will decrease (d) None of the above

**Ans. (a,c) :** Young's modulus of material decreases with increase in temperature.

Thermal strain $\Delta_{th} = \alpha.\Delta T.E$

$$E \propto \frac{1}{\Delta T}$$

If temperature increases then young's modulus will decrease.

If temperature decreases then young's modulus will increase.

**75. If $\alpha$ denotes co- efficient of linear expansion T is the rise in temperature, the thermal stress is given by**

(a) $ET\alpha$ (b) $\frac{E\alpha}{T}$
(c) $\frac{ET}{\alpha}$ (d) $\frac{\alpha}{ET}$

**Ans. (a)** If $\alpha$ denotes co- efficient of linear expansion T is the rise in temperature, the thermal stress is given by $ET\alpha$

**76. Two shafts 'A' and 'B' are made up of same material. The shaft A is of diameter 'D' and shaft B is of diameter D/2. the strength of shaft B is..... as that of shaft 'A'**

(a) One- eighth (b) one- fourth
(c) one - half (d) four times

**Ans. (a) :** Given,

Shaft A $\rightarrow$ dia = $D_A$ Strength of shaft A = $T_A$

Shaft B $\rightarrow$ dia = $D_B$ Strength of shaft B = $T_B$

$$T = \frac{\pi D^3 \cdot \tau}{16} \Rightarrow T \propto D^3$$

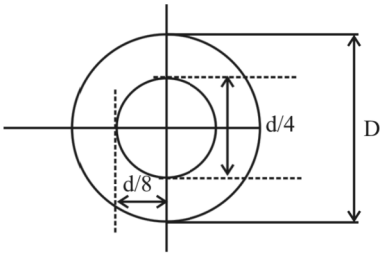
$$\frac{T_A}{T_B} = \left(\frac{D_A}{D_B}\right)^3 = \left(\frac{D \times 2}{D}\right)^3 = 8$$

$$\boxed{T_B = \frac{1}{8}(T_A)}$$

**77. The limit of eccentricity for no- tensile conditions for column of circular section of diameter 'd' is**

(a) d/4 (b) d/8
(c) d/12 (d) d/16

**Ans. (b) :** The limit of eccentricity for no- tensile conditions for column of circular section of diameter 'd' is d/8.




$e = \frac{D^2 + d^2}{8D}$, for hallow circular column.

**78. Compression members always tend to buckle in the direction of the**

(a) Axis of load
(b) Perpendicular to axis of load
(c) Minimum cross section
(d) Least radius of gyration

**Ans. (d) :** Compression members always tend to buckle in the direction of the least radius of gyration

**79. According to Euler's column theory, the cripping load for a column of length l, fixed at both ends is---- the crippling load for a similar column hinged at both the ends.**

(a) equal to (b) two times
(c) four times (d) eight times

**Ans. (c) :** According to Euler's column theory, the crippling load for a column of length l, fixed at both ends is four times the crippling load for a similar column hinged at both the ends.

**80. According to Euler's column theory, the cripping load of a column is given by $p=\pi^2EI/cI^2$. If this equation, the value of c for a column with both ends hinged, is**

(a) 1/4 (b) l/2
(c) 1 (d) 2

**Ans. (c) :** According to Euler's column theory, the cripping load of a column is given by $p=\pi^2EI/cI^2$. If this equation, the value of c for a column with both ends hinged, is 1.

**81. A column of length T with both ends fixed may be considered as equivalent to a column of length ..... with one end fixed and the other end free**

(a) *l*/8 (b) *l*/4
(c) *l*/2 (d) *l*

**Ans. (b) :** A column of length L with both ends fixed may be considered as equivalent to a column of length with one end fixed and the other end free *l*/4.

**82. A vertical column has two moments of inertia (i.e. $I_{xx}$ and $I_{yy}$). The column will tend to buckle in the direction of the**

(a) axis of load
(b) perpendicular to axis of load
(c) maximum moment of inertia
(d) minimum moment of inertia

**Ans. (d) :** Vertical column has two moment of inertia ($I_{xx}$ and $I_{yy}$) under an axial load the column will tend to buckle in the direction of the minimum moment of inertia.

$$P_c = \frac{\pi^2 EI}{L_{eff}^2}$$

$P_c \propto I$

where, E = modulus of elasticity
I = moment of inertia
$L_{eff}$ = effective length

$$I_{xx} = \frac{bh^3}{12}, I_{yy} = \frac{hb^3}{12}$$

So, $I_{xx} < I_{yy}$
$P_{cx} < P_{cy}$

**83. The Rankine's constant for a mild steel column with both ends hinged is**

(a) 1/750 (b) 1/1600
(c) 1/7500 (d) 1/9000

**Ans. (c) :** The Rankine's constant for a mild steel column with both ends hinged is 1/7500.

For cast iron = $\frac{1}{1600}$

For strong timber = $\frac{1}{750}$

**84. In case of eccentrically loaded struts------ is preferred.**

(a) solid section (b) Hollow section
(c) Composite section (d) Reinforced section

**Ans. (c) :** A structural member subjected to an axial compressive stress is called strut. In case of eccentrically loaded struts, a composing section is preferred.
Strut may be horizontal, vertical or inclined.

**85. For all end conditions as per secant formula the maximum compressive stress in the column is given by**

(a) $\frac{\rho}{A}\left\{1+\frac{ey_c}{r^2}\sec\frac{L_E}{2}\sqrt{\frac{\rho}{EI}}\right\}$

(b) $\frac{\rho}{A}\left\{1+\frac{ey_c}{r^4}\sec\frac{L_E}{2}\sqrt{\frac{\rho}{EI}}\right\}$

(c) $\frac{\rho}{A}\left\{1+\frac{ey_c^2}{r^2}\sec\frac{L_E}{2}\sqrt{\frac{\rho}{EI}}\right\}$

(d) $\frac{\rho}{A}\left\{1+\frac{ey_c^4}{r^4}\sec\frac{L_E}{2}\sqrt{\frac{\rho}{EI}}\right\}$

**Ans. (a) :** Secant formula was adopted in the earlier IS 800 prior to 1984.
For all end conditions as per secant formula the maximum compressive stress in the column is given by $\frac{\rho}{A}\left\{1+\frac{ey_c}{r^2}\sec\frac{L_E}{2}\sqrt{\frac{\rho}{EI}}\right\}$.

**86. The slope of an elastic curve of a beam cab be related as**

(a) $\int\frac{M}{EI}dx$ (b) $\int\frac{Mx}{EI}dx$

(c) $\int\frac{Vx}{EI}dx$ (d) $\int\frac{w}{EI}dx$

where M= bending moment, EI= flexural rigidity, V= shear force, w=load intensity and x= distance of the point.

**Ans. (a) :** As we know,

$$EI\frac{d^2y}{dx^2}=M$$

$$\boxed{\frac{dy}{dx}=\int\frac{M\cdot dx}{EI}}$$

So, the slope of an elastic curve of a beam can be related as $\int\frac{M}{EI}\cdot dx$.

**87. The elastic curve profile of a beam can be obtained from**

(a) $\int\frac{w}{EI}dx$ (b) $\iint\frac{wx}{EI}dx$

(c) $\int\frac{wx}{EI}dx$ (d) $\iiint\frac{wx}{EI}dx$

**Ans. (b) :** Elastic curve $=\frac{M}{EI}$

And we know,

$$\frac{d^2M}{dx^2}=w(x)$$

$$M=\iint w(x)dx$$

or, $$\frac{M}{EI}=\iint\frac{w(x)}{EI}dx$$

So, elastic curve profile $=\iint\frac{w(x)}{EI}dx$

**88. Maximum or minimum deflection of a beam occurs at**

(a) zero bending moment location of the beam
(b) zero shear force location of the beam
(c) zero slope location of the beam
(d) point of contra flexure of the beam

**Ans. (c) :** If y is the deflection of beam, then we know, y will be maximum or minimum, if $\frac{dy}{dx}=0$.
So, maximum or minimum deflection of a beam occurs at zero slope location of the beam.

**89. The following happens at the point of contra flexure in a beam**

(a) deflection changes its nature
(b) beam develops crack
(c) radius of curvature changes its sign and nature
(d) bending moment is zero

**Ans. (c) :** The point of contra-flexure in beam is where radius of curvature changes its sign and nature.

| Beam and loading | No. of Points of contraflexure |
|---|---|
| (i) Simply supported beam with UDL over part of the structure | Zero |
| (ii) An overhanging beam with loading only over supported span not on overhangs | Zero |
| (iii) Fixed beam subjected to concentrated load/UDL. | Two |
| (iv) Cantilever beam subjected to UDL with zero load intensity at free end | Zero |
| (v) Propped cantilever beam for any load | one |

**90. The number or points of contra flexure that occur in a cantilever beam subjected to uniformly distributed load is**

(a) 0 (b) 1
(c) 2 (d) 3

**Ans. (a) :**

| Beam and loading | No. of Points of contraflexure |
|---|---|
| (i) Simply supported beam with UDL over part of the structure | Zero |
| (ii) An overhanging beam with loading only over supported span not on overhangs | Zero |
| (iii) Fixed beam subjected to concentrated load/UDL. | Two |
| (iv) Cantilever beam subjected to UDL with zero load intensity at free end | Zero |
| (v) Propped cantilever beam for any load | one |

**91. The percentage reduction in area of a member from a direct tension test indicates**

(a) ductility (b) elasticity
(c) malleability (d) brittleness

**Ans. (a) : Ductility–** It is the property which permits a material to be drawn out longitudinally to a reduced section under the action of tensile force.
**Malleability–** It is the property of a material which permits the material to be extended in all direction without rupture.
**Brittleness –** It is lack of ductility. Brittleness implies that it can not be drawn out by tension to smaller section.
In brittle material failure take place under load without significant deformation.

**92. The standard gauge length over which extension is to be measured to determine percentage elongation of a specimen with initial cross section of $a_0$ is**

(a) $6.56\sqrt{a_0}$ (b) $5.65\sqrt{a_0}$

(c) $6.65\sqrt{a_0}$ (d) $a_0\sqrt{6.56}$

**Ans. (b) :** The standard gauge length over which extension is to be measured to determine percentage elongation of a specimen with initial cross section of $a_0$ is $5.65\sqrt{a_0}$ .

**93. Modulus of rigidity is the ratio of**

(a) shear stress to shear strain
(b) normal stress to normal strain
(c) poisson's ratio to ultimate strength in compression
(d) lateral stress to lateral strain

**Ans. (a) :** The ratio of shear stress to shear strain is known as Modulus of rigidity.

$$\tau = \phi G$$

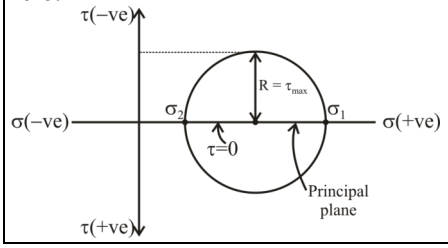
or $$\boxed{G = \frac{\tau}{\phi}}$$

where, G = modulus of rigidity ($N/mm^2$)
$\tau$ = shear stress
$\phi$ = shear strain.

**94. A principal plane is a plane which carries**

(a) maximum shear stress
(b) to give stress of higher magnitude acting
(c) no shear stress
(d) plane inclined at $45^0$ to X-axis

**Ans. (c) :** On principal plane, only normal stresses act, no shear stress exist.
The shear stress on principal plane is zero but at the maximum shear stress plane principal stresses are not zero.




**95. If an element of a specimen of brittle material is subjected to shear stress, the crack propagation if occurs, will be inclined to the sides of the element at**

(a) $45^0$ (b) $90^0$
(c) $30^0$ (d) $60^0$

**Ans. (a) :** If the element of a specimen of brittle material is subjected to shear stress, the crack occurs inclined to the sides of the element at $45^0$.

- Mechanism leading to brittle failure is the maximum shear stress (tresca yield criteria).
- The yield plane for maximum shear stresses at $45^0$ relative to the torsion axis.
- Maximum shear stress should be less than or equal to maximum shear stress under uniaxial loading.

**96. Bulk modulus, Young;s modulus and Poisson's ratio are connected by the relation**

(a) $E = \dfrac{E}{3\left(1-\frac{2}{3}\right)}$ (b) $E = \dfrac{E}{3\left(1+\frac{2}{3}\right)}$

(c) $E = 2K\left(1+\frac{1}{m}\right)$ (d) $E = 3K\left(1-\frac{2}{m}\right)$

**Ans. (d) :** Bulk modulus, Young;s modulus and Poisson's ratio are connected by the relation $E = 3K\left(1-\frac{2}{m}\right)$.

**Relation between K, $\mu$ and G –**

$E = 2G(1+\mu)$ .....(1)

& $E = 3K(1-2\mu)$ .....(2)

From equation (1) & (2)

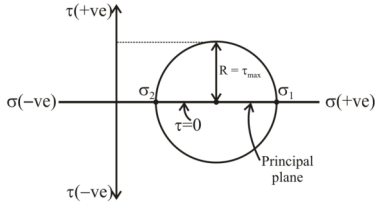
$$\frac{(1-2\mu)}{(1+\mu)} = \frac{2G}{3K}$$

or $$\boxed{G = \frac{3K(1-2\mu)}{2(1+\mu)}}$$

**97. The maximum shear stress from a Mohr's circle is given by**

(a) diameter of the circle
(b) the distance of centre from the origin
(c) the distance of farthest point on the Mohr's circle from the origin
(d) the radius of the circle

**Ans. (d) :** Maximum shear stress from a Mohr's Circle is given by the radius of the Mohr's circle.

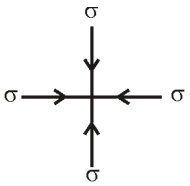



$\tau_{max} = \dfrac{\sigma_1 - \sigma_2}{2} = R$ $\tau_{max}$ = Radius of Mohr's circle

**98. An element is subjected to two equal and like stresses $\sigma$, on two mutually perpendicular planes. The shape of the Mohr's circle will be**

(a) a circle of radius $2\sigma$ (b) a circle of radius $\sigma$
(c) a circle of radius $\dfrac{\sigma}{2}$ (d) a point

**Ans. (d) :**




$$\text{Radius} = \frac{\sigma_1 - \sigma_2}{2} = \frac{\sigma - \sigma}{2} = 0$$

The point Mohr's Circle will be getting here.
It is case of hydrostatic loading.

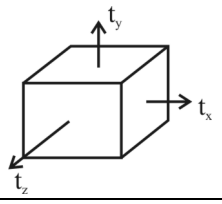
**99. If the strains in three principal directions are $t_x$, $t_y$ and $t_z$ the volumetric strain will be**

(a) $\frac{1}{3}(t_x + t_y + t_z)$ (b) $(t_x + t_y + t_z)$

(c) $3(t_x + t_y + t_z)$ (d) $(t_x + t_y + t_z)^2$

**Ans. (b) :** Volumetric strain ($\varepsilon_v$)–

$$\varepsilon_v = t_x + t_y + t_z$$




**100. The maximum shear stress will be equal to**

(a) one half of the algebraic difference of maximum and minimum stresses at that point

(b) the intensity of shear stress acting on the major principal plane

(c) the shear stress acting on the planes in the direction of two principal directions

(d) none of these

**Ans. (a) :** The maximum shear stress will be equal to one half of the algebraic difference of maximum and minimum stress at that point.

$$\tau_{max} = \frac{1}{2}(\sigma_1 - \sigma_2)$$

Where, $\sigma_2$ = Maximum stress

$\sigma_1$ = Minimum stress

**101. If the Poisson's ratio is 0.25, the ratio of modulus of rigidity to the Young's modulus is**

(a) 2 (b) 0.4

(c) 2.5 (d) 4.0

**Ans. (d) :** Using relation between E, Poisson's ratio ($\mu$) and modulus of rigidity (G).

$$E = 2G(1+\mu)$$

$$\frac{E}{G} = 2(1+\mu)$$

$$\frac{G}{E} = \frac{1}{2(1+\mu)}$$

$$\frac{G}{E} = \frac{1}{2(1+0.25)}$$

$$\frac{G}{E} = \frac{1}{2\times 1.25}$$

$$\frac{G}{E} = \frac{1}{2.5}$$

$$\frac{G}{E} = 0.40$$

**102. Modular ratio of two materials is the ratio of**

(a) linear stress to liner strain

(b) shear stress to shear strain

(c) their modulus of elasticitics

(d) their modulus of rigiduties

**Ans. (c) :** Modular ratio of two materials is the ratio of their modulus of elasticitics.

The modular ratio of two materials is defined as $\frac{E_1}{E_2}$.

**103. A steel bar of 5 mm is heated from $15^0$ C to $40^0$ C and it is free to expand. the bar will induce**

(a) no stress (b) shear stress

(c) tensile stress (d) compressive stress

**Ans. (a) :** A steel bar of 5 mm is heated from $15^0$ C to $40^0$ C and it is free to expand. the bar will induce no stress.

Stress ($\sigma$) = $\alpha \times \Delta T \times E$

**104. If the modulus of elasticity for a given material is twice its modulus of rigidity (c), then the bulk modulus is equal to**

(a) 2c (b) 3c

(c) $\frac{2c}{3}$ (d) $\frac{3c}{2}$

**Ans. (c) :** $E = \frac{9KC}{3K + C}$ $(\because E = 2C)$

$$2C = \frac{9KC}{3K + C}$$

$$6KC + 2C^2 = 9KC$$

$$2C^2 = 3KC$$

$$2C = 3K$$

$$K = \frac{2C}{3}$$

**105. A body is subjected to tensile stress of 1200 MPa on one plane and another tensile stress of 600 MPa on a plane at right angles to the former. it is also subjected to a shear stress of 400 MPa on the same planes. The maximum normal stress will be**

(a) 400 MPa (b) 500 MPa

(c) 900 MPa (d) 1400 MPa

**Ans. (d) :**

$$(\sigma_n)_{max} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{1200 + 600}{2} + \sqrt{\left(\frac{1200 - 600}{2}\right)^2 + 400^2}$$

$$= 900 + \sqrt{300^2 + 400^2}$$

$$= 900 + \sqrt{90000 + 160000}$$

$$= 900 + \sqrt{250000} = 900 + 500$$

So, $(\sigma_n)_{max}$ = 1400 MPa

**106. Shear force is usually associated with**

(a) Bending moment (b) Torsional moment

(c) Normal thrust (d) Axial thrust

**Ans. (a) :** Rate of change of bending moment along the length of a beam is equal to shear force.

$$-\frac{dM}{dx} = S$$

So, Shear force is usually associated with bending moment.

**107. Shear force and transverse load on a beam are relatedby**

(a) $V = \sum wdx$ (b) V=wL

(c) V=wydx (d) $V = \frac{dw}{dx}$

**Ans. (a) :** Rate of change of shear force along the length of a beam is equal to transverse load.

$$\boxed{\frac{dV}{dx} = w}, \text{ or } V = \sum w.dx$$

**108. Shear force and the displacement of the beam are approximately related by**

(a) $V = EI\frac{d^2y}{dx^2}$ (b) $V = EI\frac{d^3y}{dx^3}$

(c) $V = EI\frac{d^4y}{dx^4}$ (d) Not related to all

**Ans. (b) :** Given, shear force = V

Displacement = y

As we know,

$$EI\frac{d^2y}{dx^2} = M$$

$$EI\frac{d^3y}{dx^3} = \frac{dM}{dx} = V$$

So, $\boxed{V = EI\frac{d^3y}{dx^3}}$

**109. Approximate relation between the bending moment, M and the intensity of transverse load, w on a beam is**

(a) $M = \frac{wL^2}{8}$ (b) $M = \iint wdx$

(c) $M = \frac{d^2w}{dx^2}$ (d) $M = w(x-a)x$

**Ans. (b) :** Relation between bending moment (M) and intensity of transverse load (w) is $M = \iint w.dx$

- Intensity of loading = slope of shear force. $\left(\frac{dV}{dx} = W\right)$
- Slope of B.M.D. at any section = S.F. at that section

$$\frac{dM}{dx} = V$$

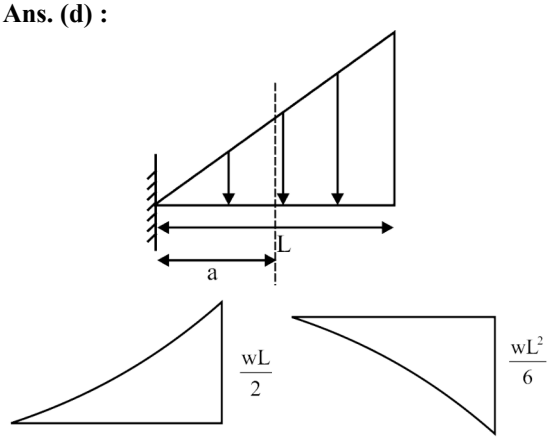
**110. Maximum bending moment caused by a hydrostatic type of load acting over a segment 'a' from the fixed end with zero intensity at the support on a cantilever beam is**

(a) $\frac{-Wa}{2}$ (b) $\frac{-Wa}{2}(L+a)$

(c) $\frac{-Wa}{3}$ (d) $-2\frac{Wa}{3}$

Where W= total load, L= span and BM is + ve if it caused compression on load face

**Ans. (d) :**




Maximum bending moment caused by hydrostatic type of loading $= -\frac{1}{2} \times a \times \frac{wa}{L} \times \frac{2a}{3} = -\frac{2wa}{3}$

**111. Variation of bending moment in a beam along a segment where no external load is present in**

(a) Zero (b) Linear

(c) Parabolic

(d) Depends on the beam span

**Ans. (b) :** When there is no external load is present the variation in bending moment is linear.

Here, $R_A = R_B = \frac{w}{2}$

B.M. at section x from support A

$B.M. = R_A \times x$ (linear variation)

$B.M. = \frac{wx}{2}$

At $x = \frac{\ell}{2}$, $B.M._{(max)} = \frac{w\ell}{4}$

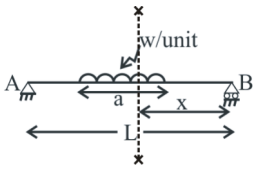
**112. Maximum bending moment on a simply supported beam caused by a uniformly distributed load over a segment of length 'a' spread symmetrically about the mid span is**

(a) $\frac{Wa}{4}$ (b) $wa\left(1-\frac{a}{2L}\right)$

(c) $\frac{Wa}{2}$ (d) $\frac{W}{8}(2L-a)$

Where W= total load, L= span

**Ans. (d) :**




$R_A + R_B = wa$

Moment about 'A'

$R_A \times 0 - R_B \times L + wa(L/2) = 0$

$R_A = R_B = wa/2$

$\because$ Maximum bending moment occurs where shear force is zero.

$\therefore$ Maximum bending moment will be at $x = \frac{L}{2}$

So, At mid point,

$$BM_{max} = R_B \times \frac{L}{2} - w \times \frac{a}{2} \times \frac{a}{4}$$

$$\frac{waL}{4} - \frac{wa^2}{8} \qquad \{W = wa$$

$$= \frac{W}{8}(2L - a)$$

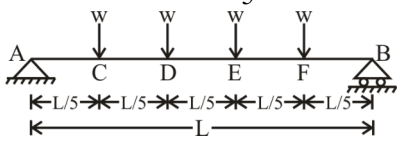
**113. Maximum bending moment caused by four equal loads each are spaced apart at intervals of a on a simply supported beam is**

(a) $\frac{WL}{6}$ (b) $\frac{5WL}{36}$

(c) $\frac{WL}{8}$ (d) $\frac{3WL}{20}$

Where L= span and W= total of the loads.

**Ans. (d) :** As per question, $a = \frac{L}{5}$




$R_A + R_B = 4w$

So, $R_A = R_B = 2w$ (Due to symmetry of load)

$BM_B = 0,\ BM_F = 2w \times \frac{L}{5} = \frac{2wL}{5}$

$BM_E = 2w \times \frac{2L}{5} - \frac{w \times L}{5} = \frac{3wL}{5}$

$BM_D = 2w \times \frac{3L}{5} - \frac{w \times 2L}{5} - \frac{wL}{5} = \frac{3wL}{5}$

$BM_C = 2w \times \frac{4L}{5} - \frac{w \times 3L}{5} - \frac{w \times 2L}{5} - \frac{w \times L}{5} = \frac{2wL}{5}$

$BM_A = 2w \times \frac{5L}{5} - \frac{w \times 4L}{5} - \frac{w \times 3L}{5} - \frac{w \times 2L}{5} - \frac{wL}{5} = 0$

So, maximum B.M. or designed BM, $= \frac{3wL}{5}$

According to the question,

Total load = W

So, $w = \frac{W}{4}$

Then, $BM_{max} = \frac{3 \times \frac{W}{4} L}{5} = \frac{3WL}{20}$

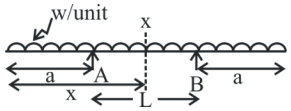
**114. Maximum positive bending moment caused in an overhang beam subjected to a uniformly distributed load over entire span is**

(a) $w\left\{\frac{L}{8} - \frac{a}{4}\right\}$ (b) $w\left\{\frac{L^2}{8} - \frac{a^2}{2}\right\}$

(c) $\frac{wa}{2}$ (d) $\frac{-wa}{4}$

Where L= distance between the supports.
a= overhang span on either side (each)
and w= intensity of load.

**Ans. (b) :**




$R_A + R_B = w(L+2a)$

Moment about 'A'

$$R_A \times 0 - R_B \times L + w(L+a) \times \left(\frac{L+a}{2}\right) - w \times a \times \frac{a}{2} = 0$$

$$R_A = R_B = \frac{w}{2}(L+2a)$$

$$(BM)_x = \frac{-wx^2}{2} + \frac{w(L+2a)}{2} \times (x-a)$$

Maximum positive B.M. occurs at the mid span,

So, at $x = a + \frac{L}{2}$

$$(BM)_{max} = -\frac{w}{2}\left(a + \frac{L}{2}\right)^2 + \frac{w}{2}(L+2a)\left(a + \frac{L}{2} - a\right)$$

$$\boxed{(BM)_{max} = w\left[\frac{L^2}{8} - \frac{a^2}{2}\right]}$$

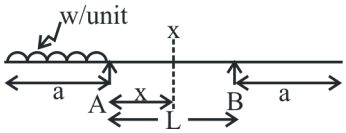
**115. Bending moment at mid span caused by a uniformly distributed load over one overhang of a double overhang beam is**

(a) $W\left\{\frac{L}{8} - \frac{a}{4}\right\}$ (b) $W\left\{\frac{L}{8} - \frac{a}{2}\right\}$

(c) $\frac{-a}{2}$ (d) $\frac{-Wa}{4}$

Where L= distance between supports
a = overhang span on either side (each)
and w= intensity of the load

**Ans. (d) :**




$R_A + R_B = wa$

Moment about 'A'

$$R_A \times 0 - R_B \times L - wa \times \frac{a}{2} = 0$$

$$R_B = \frac{-wa^2}{2L}$$

$$R_A = \frac{wa}{2L}(2L+a)$$

$$(BM)x\text{-}x = -wa \times \left(x + \frac{a}{2}\right) + R_A \times x$$

$$\left(\text{at } x = \frac{L}{2}\right)(BM)_{\text{at centre}} = -wa \times \left(\frac{L}{2} + \frac{a}{2}\right) + \frac{wa}{2L}(2L+a) \times \frac{L}{2}$$

$$= -\frac{wa^2}{4}$$

$$\boxed{(BM)_{\text{at centre}} = -\frac{Wa}{4}} \qquad (\because W = wa)$$

**116. The slope of curve of S.F. diagram at any section will be equal to**

(a) the slope of loading at that section
(b) the ordinate of loading diagram from end to that section
(c) the area of loading diagram from end to that section
(d) the bending moment at that section

**Ans. (b) :** The slope of the curve of SFD at any section will be equal to the ordinate of loading diagram at that section.

$$\boxed{\frac{dS}{dx} = W}$$

The slope of shear force diagram at any point is equal to the intensity of the applied distributed loading. If the distributed loading is a curve of degree n, the shear force will be a curve of degree (n+1).

**117. The slope of the curve of B. M. diagram at any section will be equal to**

(a) the slope of loading diagram at that section
(b) the slope of S.F. diagram at that section
(c) the ordinate of S.F. diagram at that section
(d) the area of S.F. diagram starting from any one end

**Ans. (c) :** The slope of the curve of B.M. diagram at any section will be equal to the ordinate of S.F. diagram at that section.

i.e., $\boxed{\frac{dM}{dx} = S.F.}$

**118. Which of the following end conditions permits the displacement in any direction and also rotation?**

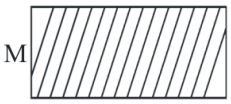
(a) fixed end (b) hinged end
(c) free end (d) roller end

**Ans.(d):** Roller end conditions permits the displacement in any direction and also rotation.
Fixed end condition does not permit the displacement in any direction as well as rotation.

**119. A cantilever of length 'l' is subjected to bending moment M at the free and. The SF diagram will be**

(a) a traiangle with maximum ordinate at fixed end
(b) a rectangle with ordinate M
(c) a parabola with maximum ordinate of $\frac{ML^2}{2}$ at the fixed end
(d) no shear force at all

**Ans. (d) :** The bending moment diagram of cantilever beam subjected to moment at the free end will be rectangular and no any shear force occurs.




BMD

**120. A rectangular section 100×200 mm is subjected to a moment of 20 kN-m. The maximum bending stress is**

(a) 30 N/mm$^2$ (b) $\frac{5}{6}$ N/mm$^2$
(c) 10,000 N/mm$^2$ (d) 300 N/mm$^2$

**Ans. (a) :** $\sigma_b = \frac{M}{z} = \frac{6M}{bd^2}$

$$\frac{6\times 20\times 10^3\times 10^3}{100\times(200)^2} = 30 \text{ N/mm}^2$$

**121. The pressure variation along a radial direction for vertex flow along a horizontal plane is given as**

(a) $\frac{\partial p}{\partial r} = -\rho\frac{u^2}{r}$ (b) $\frac{\partial p}{\partial r} = \rho\frac{u}{r^2}$
(c) $\frac{\partial p}{\partial r} = \rho\frac{u^2}{r}$ (d) None of the above

**Ans. (c):** For forced vortex flow total pressure derivatives.

$$dp = \frac{\rho u^2}{r}dr - \rho g dz$$

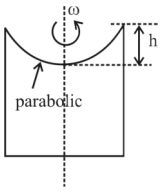
Pressure variation in radial direction $\frac{dp}{dr} = \frac{\rho u^2}{r}$

Pressure variation in vertical direction $\frac{dp}{dz} = -\rho g$

**122. For a forced vertex flow the height of paraboloid formed is equal to**

(a) $\frac{P}{w} + \frac{v^2}{2g}$ (b) $\frac{v^2}{2g}$
(c) $\frac{v^2}{r^2\times 2g}$ (d) $\frac{wr^2}{2g}$

**Ans. (b) :** For a forced vertex flow, the height of paraboloid form is equal to $\frac{v^2}{2g}$




$h = \frac{v^2}{2g}$ $\quad v = r\omega$

$h = \frac{\omega^2 r^2}{2g}$

Volume of paraboloid $= \frac{1}{2}\left(\pi r^2\right)\left(\frac{\omega^2 R^2}{2g}\right)$

Note: for no spelling case, rise above original water level = fall below original water level.

**123. Euler's equation in the differential form for the motion of liquids is given by**

(a) $\frac{p}{w} + gdz + vdv = 0$

(b) $\frac{dp}{\rho} - g.dz + vdv = 0$

(c) $\rho dp + gdz + udv = 0$

(d) $\rho dp - gdz + udv = 0$

**Ans.(a):** Euler's equation of motion along the streamline–

$$\frac{dp}{\rho} + gdz + vdv = 0$$

Assumption –
(i) Flow is laminar
(ii) Flow is irrotational
(iii) Flow is inviscid

**124. Venturimeter is used to**

(a) measure the velocity of a flowing liquid
(b) measure the pressure of a flowing liquid
(c) measure the discharge of a liquid flowing in a pipe
(d) measure the pressure difference of liquid flowing between two points in a pipe line

**Ans. (c) : Venturimeter–** A venturimeter is used to measure the flow of fluid or discharge in the pipe. A venturimeter may also be used to increase the velocity of any type fluid in a pipe at any particular point.

$$Q_{actual} = \frac{C_d A_1 A_2}{\sqrt{(A_1^2 - A_2^2)}} \times \sqrt{2gh}$$

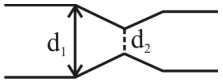
where,
$Q_{actual}$ = Actual discharge
$C_d$ = Coefficient of discharge varies between 0.96 and 0.98
$A_1$ = Area of flow cut off on entering venturimeter
$A_2$ = Area of flow cut on the throat

**125. In order to avoid tendency of separation at throat in a venturimeter, the ratio of the diameter at throat to the diameter of pipe should be**

(a) $\frac{1}{16}$ to $\frac{1}{8}$ (b) $\frac{1}{8}$ to $\frac{1}{4}$

(c) $\frac{1}{4}$ to $\frac{1}{3}$ (d) $\frac{1}{3}$ to $\frac{1}{2}$

**Ans. (d) : Venturimeter–** Venturimeter is a device used to measure discharge in pipe.




$$d_2 = \frac{1}{2} \text{ to } \frac{1}{3}\ d_1$$

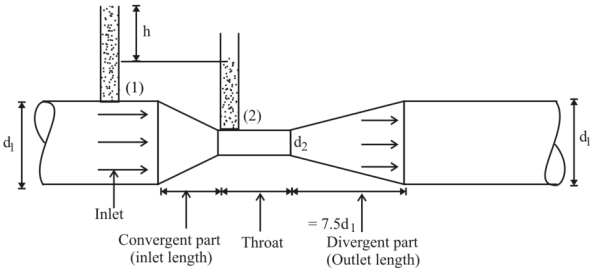
- The ratio of the diameter at throat to the diameter of pipe usually take 1 : 3.
- Angle of convergent = $20^o$-$23^o$
- Angle of divergent = $6^o$-$7^o$ (It should be not greater than $7^o$ to avoid flow separation)
- For venturimeter ($C_d$) = 0.94 to 0.98

$$Q = Cd \frac{A_1 A_2 \sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}}$$

**126. In order to measure the flow with a venturimeter, it is installed in**

(a) horizontal line
(b) inclined line with flow upwards
(c) inclined line with flow downwards
(d) any direction and in any location

**Ans. (d) :** In order to measure the flow with a venturimeter. It is installed in any direction and in any location. It is use to find discharge from a large diameter pipe reduction in area leads to increase in velocity and decrease in pressure, this pressure decrease is rated and used in Bernoulli to calculate discharge.




$$Q_{actual} = c_d \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gh}$$

$$c_d = \sqrt{\frac{h - h_L}{h}} \approx 0.98$$

**127. A pitot tube is used to measure the**

(a) velocity of flow at the required point in a pipe
(b) pressure difference between two points in a pipe
(c) total pressure of liquid flowing in a pine
(d) discharge through a pipe

**Ans. (a)** Pitot tube is a simple L shape structure by the pitot tube velocity of the fluid flowing through the open drain is determining. The flow of fluid in an open drain is determined by the following formula.

$V = \sqrt{2gh}$

Where, V – Velocity of liquid
h – The height of the fluid in the tube above the surface.

**128. In case of a forced vertex**

(a) velocity decreases with radius
(b) velocity increases with radius
(c) the fluid rotates as a composite solid
(d) the motion of fluid varies at randon

**Ans. (b) :** For free vortex flow $v = \frac{C}{r}$

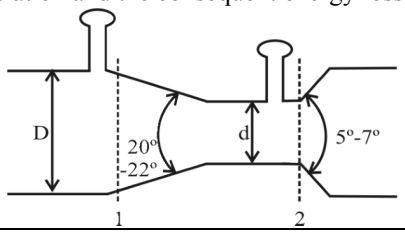
For forced vortex flow, $v = r\omega$

- In case of a forced vortex velocity increases with radius.
- Forced vortex flow is one in which the fluid mass is made to rotate by means of some external agency.

• E.g. rotation of liquid in a vertical cylinder.
• Free vortex flow is one in which the fluid mass rotates without any external impressed contact force.
• E.g. A whirpool in a river.

**129. In a venturimeter**
(a) Length of divergent cone is equal to the length of the convergent cone.
(b) Length of divergent cone is half that of convergent cone
(c) Divergent cone is kept shorter than convergent cone
(d) Divergent cone is kept longer than convergent cone

**Ans. (d) :** In a venturimeter divergent cone is kept longer than convergent cone. To avoid the possibility of flow separation and the consequent energy loss.




**130. A current meter is used to measure**
(a) pressure (b) viscosity
(c) velocity
(d) electrical conductivity

**Ans. (c) : Current meter–** The current meter is used to measure water velocity of predetermined points (subsection) along a marked line. Suspended cableway or bridge across a river or stream. The depth of the water is also measured at each point.

**131. the discharge through a rectangular weir varies as**
(a) H (b) $H^{1/2}$ (c) $H^{3/2}$ (d) $H^{6/2}$

**Ans. (c) :** The discharge through a rectangular weir varies as $H^{3/2}$
The discharge through the rectangular weir.

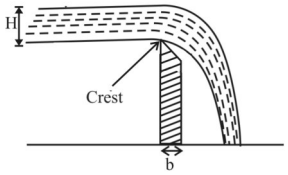
$$Q = \frac{2}{3} C_d L \sqrt{2g} H^{3/2}$$

$$Q \propto H^{3/2}$$

Where, Q = Discharge
L = Length of weir
H = Depth of water above crest level

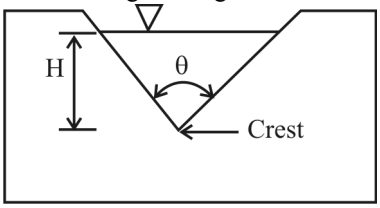
**132. The upper surface of a weir over which water flows is known as**
(a) Nappe (b) Crest (c) Sill (d) Vein

**Ans. (b) : Crest or sill–** The bottom edge of a notch or a top of weir over which the water flows is know as the sill or crest. Its height above the bottom of the tank or channel is know as crest height.




**133. The discharge through a V- notch weir varies as**
(a) $H\frac{1}{2}$ (b) H
(c) $H^{-5/2}$ (d) $H^{5/2}$

**Ans. (d) :** The discharge through the V notch weir




$$Q = \frac{8}{15} C_d \sqrt{2g} \tan\left(\frac{\theta}{2}\right) H^{5/2}$$

$$Q \propto H^{5/2}$$

$C_d = 0.52$ (Coefficient of discharge)

**134. The horizontal to vertical side slope in case of cippoletti weir is**
(a) 1 : 1 (b) 1 : 2
(c) 1 : 3 (d) 1 : 4

**Ans. (d) : Cippoletti weir -** The cippoletti weir a trapezoidal weir, having side slope horizontal to 4 vertical.
Francis formula's for cippoletti weir is -

$$Q = 1.86\, LH^{3/2}$$

and General equation $Q = \frac{2}{3} C_d . L \sqrt{2g} \left(H_1^{3/2} - H_2^{3/2}\right)$

Discharge over an ogee weir remain the same as that as cippoletti weir.
The crest and sides of the weir plate are placed for enough from the bottom and sides of the approach channel to produce full contraction.

**135. According to equation of continuity**
(a) $w_1 a_1 = w_2 a_2$ (b) $w_1 u_1 = w_2 u_2$
(c) $a_1 u_1 = a_2 u_2$ (d) $w_1 u_1 = w_2 / u_2$

**Ans. (c) :** Continuity equation $a_1u_1 = a_2u_2$
Continuity equation for compressible fluid–

$$\rho_1 a_1 u_1 = \rho_2 a_2 u_2$$

• The continuity equation is based on the principle of conservation of mass. It states as follows– "If no fluid is added or removal from the pipe in any length then the mass passing across different sections shall be some."

**136. In one dimensional flow, the flow**
(a) is steady and uniform
(b) takes place in straight line
(c) takes place in curve
(d) takes place in one direction

**Ans. (d) :** The flow in which the velocity is the function of time and one space co-ordinate (x) only is called one dimensional flow or Its involve zero transverse component of flow.
Experiment flow through the pipe is consider as a one dimensional flow u = f(x) v = 0, w = 0

**137. The total energy of a liquid particle in motion is equal to**

(a) pressure energy+ kinetic energy+ potential energy
(b) pressure energy- (kinetic energy + potential energy)
(c) potential energy - (pressure energy+ kinetic energy)
(d) kinetic energy- (pressure energy+ potential energy)

**Ans. (a) :** The total energy of a liquid particle in motion is equal to–
Pressure energy + kinetic energy + potential energy
• The line representing the sum of pressure head ($P/\gamma$), datum head (2) and velocity head ($v^2/2g$) with respect to some reference line is known as total energy line (TEL).

**138. A structure used to dam up a stream or river over which water flows is called**

(a) orifice (b) notch
(c) weir (d) dam

**Ans. (c) : Weir–** It is a concrete or masonry structure, placed in open channel over which the flow occurs. It is generally in the form of vertical wall, with a sharp edge at the top, running all the way across the open channel.
• Its measured the discharged.
• Weir is a low dam built across a river to raise the level of water upstream or regulate the its flow.

**139. According to Darcy's formula, the loss of head**

(a) $\frac{flu^2}{2gd}$ (b) $\frac{flu^2}{gd}$ (c) $\frac{3flu^2}{2gd}$ (d) $\frac{4flu^2}{2gd}$

Where $f$ = Darcy's coefficient
$l$ = length of pipe
u = velocity of liquid in pipe
d = diameter of pipe

**Ans. (a) :** According to Darcy's formula, the loss of head due to friction in the pipe is $\frac{flu^2}{2gd}$
Where f = Darcy's coefficient,
$l$ = length of pipe,
u = velocity of liquid in pipe,
d = diameter of pipe)

**140. The velocity at which the flow changes from laminar flow to turbulent flow is called**

(a) critical velocity
(b) velocity of approach
(c) sub- sonic velocity
(d) supper sonic- velocity

**Ans. (a) :** The velocity at which the flow changes from laminar flow to turbulent flow is called critical velocity.

**141. Reynold's number is the ratio of the inertia forces to the**

(a) surface tension force (b) viscous force
(c) gravity force (d) elastic force

**Ans. (b) :** Reynold's number is the ratio of inertia force to viscous force

$$Re = \frac{F_i}{F_v} = \frac{\rho V^2 L^2}{\mu VL}$$

$$\boxed{Re = \frac{\rho VL}{\mu}}$$

V = velocity in pipe, $\rho$ = density
$\mu$ = dynamic viscosity, L = Length of pipe

$$\text{Froude No.} = \sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}} = \frac{V}{\sqrt{gL}}$$

**142. The flow in a pipe is turbulent when Reynold's number is**

(a) less than 2000 (b) between 2000 and 2800
(c) more than 2800 (d) none of these

**Ans. (d) :** The flow in a pipe is turbulent when Reynold's number is more than 4000 and the flow is laminar when Reynold's number's is less than 2000.

$$\text{Reynold's number } (R_e) = \frac{\rho vd}{\mu}$$

Where, $\rho$ = Density,
d = Diameter of the pipe
v = Velocity of the flow

**143. The loss of pressure head in case of laminar flow is proportional to**

(a) velocity (b) $(\text{velocity})^2$
(c) $(\text{velocity})^3$ (d) $(\text{velocity})^4$

**Ans. (a) : For pipe–** The loss of pressure head in case of laminar flow is given by–

$$h_L = \frac{32\mu \bar{U} L}{\rho g d^2}$$

Parallel plate –

$$h_L = \frac{32\mu \bar{U} L}{\rho g t^2} \qquad \because h_L \propto \bar{U}$$

Where, $\mu$ = viscosity,
L = length of pipe/plate
$\bar{U}$ = Average velocity
d = diameter of pipe
Thus, the loss of the pressure head is proportional to the velocity of flow in laminar flow.

**144. The Reynold's number of a ship is ------ to its velocity and length**

(a) directly proportional
(b) inversely proportional
(c) partly directly and partly inversely proportional
(d) not related

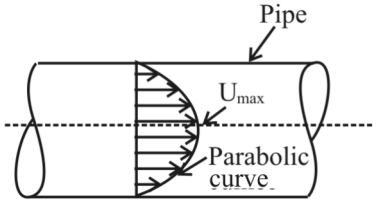
**Ans. (a) :** $R = \frac{\rho VD}{\mu}$

$R \propto D \rightarrow$ Length or we can say its linear dimension.
$R \propto V$
So, Reynolds's number of ship is directly proportional to its velocity and length.

**145. For a laminar flow through circular pipes**

(a) the maximum velocity = 1.5 time the average velocity
(b) the maximum velocity =2 time the averages velocity
(c) the maximum velocity = 2.5 time the average velocity
(d) none of the above

**Ans. (b) :** For a laminar flow through circular pipes the maximum velocity is two times the averages velocity.




$\therefore \boxed{U_{max} = 2 \times v_{avg}}$

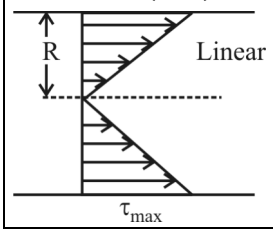
or $\boxed{v_{avg} = \frac{U_{max}}{2}}$

$$V_{av} = \frac{U_{max}}{2} = \frac{1}{32\mu}\left(\frac{-dP}{dx}\right)D^2$$

**146. for a laminar flow through a pipe, the shear stress over the cross section**
(a) varies inversely as the distance from the centre of the pipe
(b) varies directly as the distance from the surface of the pipe
(c) varies directly as the distance from the centre of pipe
(d) remains constant over the cross section

**Ans. (c) :** For a laminar flow through a pipe, the shear stress over the cross section varies directly as the distance from the centre of pipe.
Shear stress ($\tau$) distribution

$$\tau = \left(\frac{-\partial p}{\partial x}\right)\frac{R}{2}$$




**147. The specific speed of a hydraulic turbines is defined as the speed of the turbine which**
(a) produces unit horse power at unit head
(b) produces unit horse power at unit discharge
(c) delivers unit discharge at unit head
(d) delivers unit discharge at unit power

**Ans. (a) : Specific speed of turbine–** The specific speed of a turbine is the speed at which the turbine will run when developing unit power under a unit head.

$$N_s = \frac{N\sqrt{P}}{H^{5/4}} \qquad (\text{Dimension } F^{1/2}, L^{-3/4}, T^{-3/2},)$$

- Specific speed of pelton turbine (single jet) = 10 to 35 and double jet 35 to 60.
- For Francis turbine = 60 to 300.
- For Kaplan turbine = 300 to 1000.

The dimensional formula for the specific speed of a turbine $M^{\frac{1}{2}}L^{\frac{-1}{4}}T^{\frac{-5}{2}}$.

**148. A turbine is a device which converts**
(a) hydraulic energy into mechanical energy
(b) mechanical energy into hydralic energy
(c) kinetic energy into mechanical energy
(d) electrical energy into mechanical energy

**Ans. (a) :** A turbine is a device which converts hydraulic energy into mechanical energy
This energy obtained is known as hydroelectric power which is one of the cheapest forms of energy generation.

Pump $\xrightarrow{\text{Converts}}$ Mechanical energy into hydraulic energy

Turbine $\xrightarrow{\text{Converts}}$ Hydraulic energy into mechanical energy

**149. A pump is defined as device which converts**
(a) hydralic energy into mechanical energy
(b) mechanical energy into hydraulic energy
(c) kinetic energy into mechanical energy
(d) none of the above

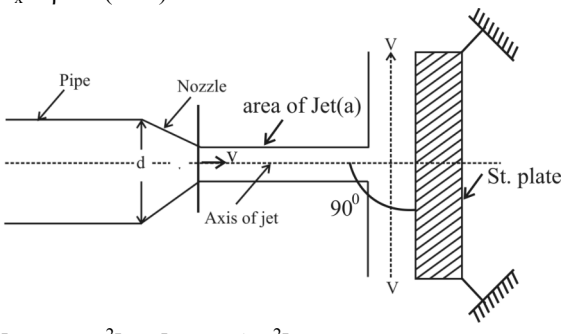
**Ans. (b) :** A pump is defined as device which converts mechanical energy into hydraulic energy.

Pump $\xrightarrow{\text{Converts}}$ Mechanical energy into hydraulic energy

Turbine $\xrightarrow{\text{Converts}}$ Hydraulic energy into mechanical energy

**150. The force exerted by a jet of water on a stationary vertical plate in the direction of jet is given b**
(a) $f_x = \rho AV^2 \sin^2\theta$
(b) $f_x = \rho AV^2(1+\cos\theta)$
(c) $f_x = \rho AV^2$
(d) none of the above

**Ans. (c) :** Force exerted on a stationary flat plate held normal to the jet–
The force exerted by the jet on the plate (assuming it smooth) in the direction of jet (x-direction),
$F_x$ = Rate of change momentum (in the direction of forxce)
$F_x$ = (Initial momentum – Final momentum) ……. Impluse – momentum principle.
$F_x$ = (mass/sec) × [velocity of jet before striking the plate– Velocity of jet after striking plate]
$F_x = \rho aV\,(V-0)$




$[F_x = \rho aV^2]$ or $[F_x = \rho AV^2]$
Where, $\rho$ = mass density of liquid

$$A = \text{Area of jet} = \frac{\pi}{4}d^2$$

# Telangana State Public Service Commission
# Assistant Executive Engineer Exam, 2011 (P-II)
# Solved Paper

**1. Magnitude of capillary rise is more in**
(a) silts (b) sands
(c) clays (d) gravels

**Ans. (c)** Finer the particle size higher will be the capillary rise, because in fine grained soil the diameter of capillary tube (formed by voids) is small.
As we know–

$$\boxed{h = \frac{4\sigma\cos\theta}{\rho g d}} \Rightarrow \left(h \propto \frac{1}{d}\right)$$

**2. The permeability of a soil deposit in-situ can be best obtained by**
(a) falling head permeameter
(b) constant head permeameter
(c) pumping test
(d) none of the above

**Ans. (c) :** The permeability of soil in the field is determine by two method.
**i) Pumping out test–** It is useful for homogenous coarse grained soil and large project.
• For unconfined aquifer,

$$k = \frac{2.303q}{\pi\left(h_2^2 - h_1^2\right)}\log_{10}\left(\frac{r_2}{r_1}\right)$$

• For confined aquifer, $\boxed{k = \frac{2.303q}{2\pi D\left(h_2 - h_1\right)}\log_{10}\left(\frac{r_2}{r_1}\right)}$

**ii) Pumping in test–** It is more economical but less reliable than pumping out test as it gives coefficient of permeability of stream which is close to the whole. It is use for small project.

**3. Consolidation is a process of involving**
(a) sudden compression of soil
(b) tilting and failure of structure
(c) abnormal sinking of foundation
(d) gradual expulsion of pore water

**Ans. (d) :** Consolidation is a process in which gradual expulsion of pore water
• Consolidation settlement is function of effective stress not function of total stress.

**4. Vane shear test is used for**
(a) measuring shear strength of cohesive soil
(b) measuring void ratio of sandy soils
(c) measuring bearing capacity of soils
(d) all the above

**Ans. (a)** Vane shear test is used for measuring shear strength of cohesive soil.
For Lab – Diameter of vane (D) = 20 mm
Height and vane (H) = 10 mm

$$\text{shear strength } (\tau) = \frac{T}{\pi D^2\left(\frac{H}{2} + \frac{D}{6}\right)}$$
[for top and bottom (both)]

$$\text{shear strength } (\tau) = \frac{T}{\pi D^2\left(\frac{H}{2} + \frac{D}{12}\right)}$$ [for only bottom]

**5. Shearing strength of cohesionless soil depends upon**
(a) dry density (b) void ratio
(c) loading rate (d) normal stress

**Ans. (d) :** Shearing strength of cohesionless soil depends upon normal stress.

**6. Shear strength of soil is determined by the equation**
(a) $c = s + \sigma \tan\phi$ (b) $\sigma = c + s \tan\phi$
(c) $s = c + \sigma \tan\phi$ (d) $s = \sigma + c \tan\phi$

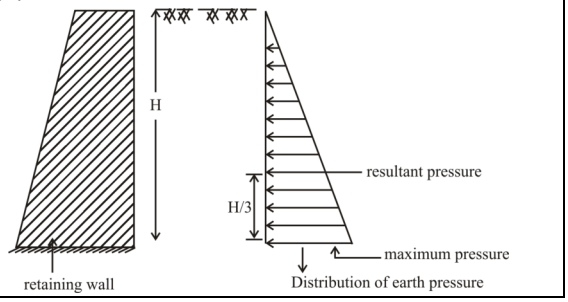
**Ans. (c) : Coulomb's equation–**
$S = (c + \sigma_n \tan\phi)$
S = shear stress on failure plane
C= cohesion of soil
$\phi$ = angle of internal friction.
According to coulomb law–
• Material fails essentially by shear.
• The critical shear stress causing failure depends upon the properties of the material as well as on normal stress on the failure plane.
• The ultimate strength of the materials is determined by the stress on the potential failure plane.

**7. In Rankine's earth pressure theory, the following assumption is made**
(a) wall face is rough and vertical
(b) back fill is cohesive soil
(c) wall face is smooth and vertical
(d) elastic equilibrium is satisfied

**Ans. (c) : Rankine earth pressure theory assumptions –**
• Soil is homogenous, isotropic, elastic, semi-infinite.
• The ground surface is planar, which may be horizontal or inclined.
• The base of the wall in contact with the backfill is vertical and smooth.
• Soil is in state of plastic equilibrium.
• Rupture surface is planner.

**8. Distribution of earth pressure with depth is**
(a) parabolic
(b) hydrostatic
(c) non- linear but increasing
(d) decreasing and increasing

**Ans.(b):** Distribution of earth pressure with depth is hydrostatic at the top zero linearly increasing with depth.




**9. For normally consolidated clay deposits stability analysis of slopes by the following method is appropriate**
(a) friction circle method
(b) swedish circular arc method
(c) slices method
(d) none of the above

**Ans. (*) : Swedish method–** The assumption of a circular slip surface and its application in sweden.
The method is known as the Swedish circle method or the method of slices.
• Fellinious method is used for purely cohesive soil.
**Friction circle method–** It is based on the assumption that resultant force R on the rupture surface is tangential to a circle of radius r= R sin ϕ which is concentric with the trail slip circle.
• Friction is assumed to be fully moblised.

**10. Which of the following requires greatest deformation**
(a) local shear failure
(b) general shear failure
(c) composite shear failure
(d) rigid failure

**Ans. : (a)** Local shear failure involves rupture of the soil only immediately below the footing. There is soil bulging on both sides of the footing, but the bulging is not as significant as in general shear. A local shear failure occurs for soils that are in a loose or soft soil in case of shallow foundation.

**11. Factor of safety of embankments shall be**
(a) atleast 1.0 (b) altleast 1.5
(c) atleast 2.0 (d) atleast 2.5

**Ans. (b) :** Short and long term safety factor for embankment stability should be 1.3 and 1.5 respectively.

**12. The maximum size of the plate for plate load test is**
(a) 30 cmsq. (b) 45 cmsq.
(c) 60 cmsq. (d) 75 cmsq.

**Ans. (d) :** The maximum size of the plate for plate load test is 75 cmsq.
Minimum – 30 $cm^2$
Thickness of steel plate should be greater than 25 mm.

**13. Contribution due to skin friction is maximum in**
(a) sands (b) soft clays
(c) piles resting on rock (d) none of the above

**Ans. (a) :** Contribution due to skin friction is maximum in sand.
• For soft clays contribution due to skin friction is minimum.

**14. Caissons are structures used for**
(a) underpinning
(b) lifting and transporting foundating parts
(c) dewatering while laying foundations
(d) constructing foundations in proper position

**Ans. (d) :** • In geotechnical engineering, a caisson is a water tight retaining structure used.
• To work on the foundations of a bridge pier, for the construction of concrete dam, or for the repair of ships.

**15. An advantage of plane table survey is**
(a) it is a tropical instrument
(b) it has many accessories
(c) plotting is done out=door
(d) chance to miss details are less

**Ans. (d) : Advantage of plane table survey–**
• It is less costly than theodolite or any other type of survey.
• No great skill is required
• Method is rapid than other type of survey
**Disadvantage of plane table survey–** It does not give very accurate results to other survey.

**16. A leveling station is place where**
(a) the level is setup
(b) the level staff is held
(c) both B.S. and F.S. are taken
(d) temporary adjustments are done

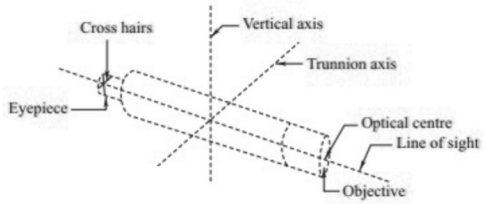
**Ans. (b) :** A station point is a point whose elevation is to be determine or a point which is to be established at a known elevation.
• Therefore, this is the point on which the staff is held and not the point on which the instrument is setup.

**17. Reciprocal leveling eliminates**
(a) collimation error
(b) collimation, curvature and refraction error
(c) curvature and refraction error
(d) collimation and curvature error fully and refraction

**Ans. (d) :** Reciprocal levelling is used to determine the correct difference of elevations of two points which are quite apart and when it is not possible to setup the instrument between the points to balance B.S. and F.S.
• Reciprocal levelling eliminates error due to –
(i) Curvature error
(ii) Refraction error
(iii) Inclination of line of collimination

**18. In a theodolite the line of collimations is**
(a) parallel to axis of plate levels
(b) parallel to the vertical axis
(c) perpendicular to the trunnion axis
(d) parallel to the horizontal axis

**Ans. (c) :** In a theodolite the line of collimations is perpendicular to the trunnion axis–




**The line of sight or line of collimation –** It is the line passing through the intersection of the horizontal and vertical cross-hairs and the optical centre of the object glass and its continuation.

**19. Lime suitable for making mortar of good strength**

(a) hydraulic lime (b) fat lime
(c) lean lime (d) none of the above

**Ans. (a) : Hydraulic lime–** Lime contains small quantities of Silica, Alumina, Iron oxide in chemical combination. with calcium oxide component.
- It has the property to set and harden under water.

**20. The ingredient which gives the desired colour to a paint is called**

(a) base (b) pigment
(c) vehicle (d) solvent

**Ans. (b) :** Pigments are used to hide the surface imperfections and to impart the desired colour.

| Pigment | Colour |
|---|---|
| Zinc oxide | White |
| Copper sulphate | Green |
| Burnt siena | Brown |
| Prussian blue, Indigo | Blue |
| Ivory black | Black |
| Red lead | Red |

**21. The quantity of partition walls and honeycomb walls are worked out is**

(a) m (b) $m^2$
(c) $m^3$ (d) lump-sum

**Ans. (b) :** Honeycomb brick wall shall be measured in square meters.

| S N | Particulars of items | unit of measurement in mks. |
|---|---|---|
| 1. | Honey- comb brick work thickness specified | sq. m. |
| 2. | cement concrete (c.c.) | cu. m. |
| 4. | Sun dried brickwork | cu. m. |
| 5. | Thin partition wall | sq. m. |
| 6. | half brick work with or without reinforcement | sq. m. |
| 7. | Jack arch roofing including top finishing | sq. m. |
| 8. | Cast iron pipe dia specified | meter |

**22. If valuable properties are found during excavation, it becomes the property of**

(a) contractor (b) laborers
(c) owner of the site (d) government

**Ans. (d) :** If valuable properties are found during excavation, it becomes the property of government.

**23. A deep and big depression on a road is called**

(a) longitudinal rut (b) cross rut
(c) pot hole (d) crack

**Ans. (a) :** A rut is a depression or groove worn into a road on path by the travel of wheels or skis.

**24. Grade separation had the advantage of**

(a) easy right turn
(b) no speed restriction
(c) number of points of conflicts is nil
(d) occupying the least space

**Ans. (c) : Grade separation–**
A grade separation is a crossing of a highway and a railway, two highway or a pedestrian walkway and a highway at different levels without interruption.

**Advantages–**
- It provides maximum facility to crossing traffic and avoid accidents while crossing.
- They provide increased safety for turning traffic.
- They help to increase speed limits.
- Reduces number of conflict to nil.

**Disadvantage–**
- They are space intensive i.e it takes a lot of space.
- They are costly to build.

**25. Unguarded level crossing comes under**

(a) warning sign (b) informatory sign
(c) regulater sign (d) route marker sign

**Ans. (a) : Unguarded level crossing–**It comes under warning sign.

- Sign indicates that there is a railway crossing not manned by personnel.
- It has to be crossed by driver himself very cautiously.

**Classification of traffic signs**

**(1) Mandatory/Regulatory sign**
- Give way, STOP, compulsory left, speed limit are example of regulatory signs.
- It's violation is a legal offence.

**(2) Cautionary/Warning sign**
- Men at work, major road ahead, hair pin bend are example of warning sign.
- Warn road users of existence of certain hazardous conditions.

**(3) Informatory sign**
- Petrol pump, hospital direction sign are example of informatory sign.
- It is used for guiding the road users.

**26. Positive float for any activity means that the activity is**

(a) supercritical (b) sub- critical
(c) critical (d) none of the above

**Ans. (b) :** Positive float for any activity means that the activity is sub-critical.
Negative float for any activity means that the activity is supercritical.

**27. The total float for any activity along the critical path in CPM is**

(a) positive (b) negative
(c) zero (d) any value

**Ans. (c) :** The total float for any activity along the critical path in CPM is zero.
A CPM network comprises least total cost durations for activities. In CPM cost is not proportional to time. Hence time cost models are prepared to find the minimum project cost and corresponding time.

**28. Which of the following is a dummy activity**

(a) excavation of the foundations
(b) await the arrival of concrete material
(c) both (1) and (2)
(d) none of these

**Ans. (b) :** Dummy activity await the arrival of concrete material. Excavation of the foundations is an activity.

**29. One of the most important Benefits of a GIS is the ability to spatially interrelate multiple types of information stemming from a....**

(a) range of sources (b) range of date
(c) source of resolution (d) none of the above

**Ans. (a) :** One of the most important benefit of a GIS is the ability to spatially inter related multiple type of information to spatially inter related multiple type of information stemming from a range of source.
The system contain data from a range of source maps.
(a) That have been gecoded on a cell by all basis toform a series of layer.
(b) All in geographic registration. The analyst can then manipulate and overlay the information contained.

**30. The instrument used to reproduce plans to a different scale is called**

(a) Planimeter (b) clinometer
(c) ghat tracer (d) pantograph

**Ans. (d) :** The instrument used to reproduce plans to a different scale is called pentagraph.
• Planimeter is an instrument used in surveying to compute the area of any given plan.

**31. As per elastic theory of design the factor of safety in the ratio of**

(a) working stress to yield stress
(b) Yield stress to working stress
(c) Ultimate strength to yield stress
(d) Ultimate load to load at yield

**Ans. (b) : In elastic theory of design–**

$$\text{Factor of safety} = \frac{\text{yield stress}}{\text{working stress}}$$

**32. The stress due to temperature change in a member depends on**

(a) length of the member
(b) area of the cross section
(c) supporting conditions at the two ends
(d) none of the above

**Ans. (d) :** The stress due to temperature change in a member do not depends upon properties of cross-section like length and area and supporting conditions at the two ends. They only depends upon properties of the material and temperature change.
Thermal stress ($\sigma$) = $E\alpha\Delta T$

**33. The percentage elongation of a material frame a direct tensile test indicates**

(a) ductility (b) strength
(c) yield stress (d) ultimate strength

**Ans. (a) :** The percentage elongation of a material frame a direct tensile test indicates ductility.
Both Percentage elongation and Reduction of area are the conventional and relevant measures of ductility obtained from tensile test.

**34. An element" of a specimen of brittle material is subjected to shear stress the crack propagation if occurs will be inclined to the sides of the element at**

(a) $45^0$ (b) $90^0$ (c) $30^0$ (d) $60^0$

**Ans. (a) :** If the element of a specimen of brittle material is subjected to shear stress, the crack occurs inclined to the sides of the element at $45^0$.
• Mechanism leading to brittle failure is the maximum shear stress (tresca yield criteria).
• The yield plane for maximum shear stresses at $45^0$ relative to the torsion axis.
• Maximum shear stress should be less than or equal to maximum shear stress under uniaxial loading.

**35. A plate 100mm wide, 10 mm thick is having a hole of diameter 10 mm, symmetrical about the axis of the plate. The plate so subjected to a force of 6 kN. The maximum stress on a section passing through centre of the hole will be**

(a) 10 N/ mm$^2$ (b) > 10 N/ mm$^2$
(c) < 10 N/ mm$^2$ (d) 9 N/ mm$^2$

**Ans. (b) :** Given,
F = 4000 N, d = 10 mm
b = 100 mm, t = 0 mm

$$\sigma_{max} = 3 \times \sigma_{nominal}$$

$$\sigma_{nominal} = \frac{F}{(b-d)t}$$

$$= \frac{9000}{(100-10)\times 10} = 9\,N/mm^2$$

$$\sigma_{max} = 3 \times 9 = 27\ N/mm^2$$

**36. The ratio of strengths of a rivet in a double shear to that in a single shear will be**

(a) 1 (b) $\frac{1}{2}$
(c) 2 (d) none of these

**Ans. (c) :** Strength of rivet in single shear

$$T_1 = 1 \times \frac{\pi}{4} d^2 \times \tau_v f$$

Strength of rivet in double shear

$$T_2 = 2 \times \frac{\pi}{4} d^2 \times \tau_v f$$

Now, $\frac{T_2}{T_1} = \frac{2 \times \frac{\pi}{4} d^2 \times \tau_v f}{1 \times \frac{\pi}{4} d^2 \times \tau_v f}$, $\frac{T_2}{T_1} = 2$

**37. If the poisson's ratio of a material is 0.25 the ratio of modulus of rigidity to the young's modulus is**

(a) 2 (b) 0.4 (c) 2.5 (d) 4

**Ans. (b) :** Using relation between E, Poisson's ratio ($\mu$) and modulus of rigidity (G).

$$E = 2G(1+\mu)$$

$$\frac{E}{G} = 2(1+\mu)$$

$$\frac{G}{E} = \frac{1}{2(1+\mu)}$$

$$\frac{G}{E} = \frac{1}{2(1+0.25)}$$

$$\frac{G}{E} = \frac{1}{2\times 1.25}$$

$$\frac{G}{E} = \frac{1}{2.5}$$

$$\boxed{\frac{G}{E} = 0.40}$$

**38. If 'K' is the neutral axis constant for a singly reinforced member, the lever arm constant is given by**

(a) $1-\frac{K}{3}$ (b) $1+\frac{K}{3}$ (c) $1-\frac{K}{6}$ (d) $1+\frac{K}{6}$

**Ans. (a) :** While design of balanced section

$\boxed{BM = Q.Bd^2}$ for balance section.

$Q = \frac{1}{2}cjk$ $\boxed{j = 1-\frac{K}{3}}$ $\boxed{K = \frac{mc}{t+mc}}$

K= neutral axis constant

j = lever arm constant

**39. The maximum allowable percentage of tension reinforcement in RCC beams is**

(a) $\frac{85}{f_y}$ (b) 0.4

(c) 4 (d) $\frac{40su}{f_y d}$

**Ans. (c) : Tension reinforcement in RCC beam –**

**(a) Maximum reinforcement -** The maximum area of tension reinforcement shall not exceed 0.04 bD. (i.e. 4%)

Where,

D– total depth of the beam.

**(b) Minimum reinforcement -** The minimum area of tension reinforcement shall not less than that given by the following.

$$\boxed{A_{st} = \frac{0.85\ bd}{f_y}}$$

Where,

$A_{st}$ – minimum area of tension reinforcement

b – breadth of beam

d – effective depth, and

$f_y$ – characteristic strength of reinforcement in $N/m^2$.

**40. The effective width of flange in RCC T. beams should be restricted to**

(a) $\frac{l_0}{6}+b_w+6D_f$ (b) $\frac{l_0}{12}+b_w+3D_f$

(c) $\frac{l_0}{3}$ (d) Centre to centre distance of the beams

**Ans. (a) :** The effective width of flange in RCC T. beams should be restricted to $\frac{l_0}{6}+b_w+6D_f$.

**For isolated T-beam–**

(i) $b_f = \left[\frac{\ell_o}{\left(\frac{\ell_o}{b}\right)+4}+b_w\right]$

(ii) $b_f = b$

Whichever is less–

L-beam (i), $b_f = \frac{0.5l_0}{\left(\frac{l_0}{b}\right)+4}+b_w$

(ii) $b_f \le b$ whichever is less

where, $b_f$ = effective width of flange

$b_w$ = width of the web

b = actual width of the flange

$l_0$ = distance between points of zero moments in the beam.

**41. The minimum distance over which a bar should be extended (curtailment) beyond which it is no longer needed for bending in RCC beam is**

(a) effective depth of beam or 12 times the diameter of the bar whichever is greater

(b) 48 times the diameter of the bar

(c) 1.5 times the depth or 24 times the diameter of the bar whichever is greater

(d) $\frac{\phi\sigma_s}{4\tau bd}$

**Ans. (a) :** As Per IS 456 : 2000 –

- For curtailment, reinforcement shall extend beyond the point at which it is no longer required to resist flexures a distance equal to the effective depth of the member or 12 mm times the bar diameter whichever is greater except at simple support or end of cantilever.

**42. In uncracked reinforced concrete columns the actual bending compressive stress is**

(a) $\frac{M}{Bjd^2}$ (b) $\frac{M}{Z}$

(c) $\frac{P}{A_c+1.5Ma_{SC}}$ (d) $\frac{M}{A_c+1.5Ma_{SC}}$

**Ans. (b) :** $\frac{M}{I} = \frac{\sigma_c}{y}$

$$\frac{\sigma_c}{y} = \frac{M}{I}\times y \qquad \sigma_c = \frac{M}{I/y} \Rightarrow \boxed{\sigma_c = \frac{M}{Z}}$$

**43. The minimum thickness of unstiffened out standing legs of mild steel compression member should be**

(a) 10 mm (b) 16 mm

(c) $\frac{B}{14}$ (d) $\frac{B}{16}$

**Ans. (d) :** The minimum thickness of unstiffened out standing legs of mild steel compression member should be $\frac{B}{16}$.

**44. The effective height of bearing stiffener is**

(a) more than 'd' (b) less than 'd'

(c) 0.7d (d) 2d

**Ans. (c) : Bearing stiffeners-** Bearing stiffeners are used to transfer concentrated loads on the girder and heavy reaction at supports to the full depth of web.

- The effective height of bearing stiffener is taken between 0.7 and 0.75 times the depth of the web.

$l_{eff} = 0.7\ d$

**45. The allowable bending tension in M30 grade concrete of RCC water tanks is (in N/mm$^2$)**

(a) 1.2 (b) 1.5

(c) 1.7 (d) 2.0

**Ans. (d) :** Permissible concrete stresses in calculation relating to resistance to cracking of an RC water tank–

| S.No. | Grade of concrete | Permissible concrete stress N/mm$^2$ | |
|---|---|---|---|
| | | Direct tension | Tension due to bending |
| (i) | M 25 | 1.3 | 1.8 |
| (ii) | M 30 | 1.5 | 2.0 |
| (iii) | M 35 | 1.6 | 2.2 |
| (iv) | M 40 | 1.8 | 2.4 |

**46. The pre-stressing is most efficient when applied to**

(a) struts (b) ties

(c) pipes (d) panels

**Ans. (d) :** Prestressed concrete is very economical for long span structure such as girders, roofs upto span 30–50 m.

- The pre-stressing is most efficient when applied to panels.

**47. Circular prestressing is best suited to**

(a) circular beams in buildings

(b) pipes

(c) railway sleepers

(d) columns

**Ans. (b) :** Circular prestressing is generally used for circular liquid retaining structures such as water tanks, pressure conduits etc.

- Circular prestressing is advantageous in pipes and tanks because they can reduced the hoop stress by auto fringing process.

**48. Linear prestressing is used in**

(a) beams (b) columns only

(c) water tanks (d) pipes

**Ans. (a) :** Linear prestressing is used to include all other types of prestressing, where the cables may be either straight or curved, but not wound in circles around a circular structure.

Linear prestressing is used in beams.

**49. A load- bearing wall is designed to carry**

(a) vertical loads

(b) live loads+ self weight

(c) imposed loads

(d) live loads/ wind loads

**Ans. (a) :** Load bearing wall : Transfer loads all the way to the foundation or other suitable from member.

- A wall directly above the beam is called a load bearing wall if it is designed to carry the vertical load.

**50. A masonry wall is said to be a short wall if the slenderness ratio is less than**

(a) 8 (b) 12

(c) 15 (d) 20

**Ans. (a) :** A masonry wall is said to be a short wall if the slenderness ratio is less than 8.

**51. Dams located in seismically active regions, the design engineer should consider the following**

(a) ultimate load design factor

(b) limit state load design factor

(c) seismic design factor

(d) none of the above

**Ans. (c) :** Seismic design factor should be considered while dams located in seismically active regions.

**52. The resultant hydrostatic force acts through a point known as**

(a) centre of gravity (b) centre of buoyancy

(c) centre of pressure (d) none of the above

**Ans. (c) :** The resultant hydrostatic force acts through a point known as centre of pressure.

**53. Continuity equation can take the form**

(a) $A_1V_1 = A_2V_2$ (b) $P_1A_1 = P_2A_2$

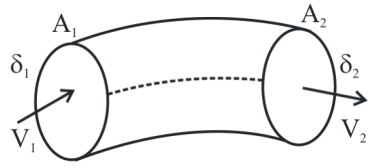
(c) $P_1A_1 = P_2A_2V_2$ (d) None of the above

**Ans. (a) :** One-dimensional continuity

$$\dot{M}_{in} = \dot{M}_{out}$$

where, $\dot{M}$ = mass rate

$\therefore \delta_1A_1V_1 = \delta_2A_2V_2$




If density of fluid is constant, i.e.

$P_1 = \delta_2 = \delta$

Then $A_1V_1 = A_2V_2$

**54. The pressure variation along the radial direction for vortex flow along a horizontal plane is given as**

(a) $\partial P/\partial r = -\rho u^2/r$ (b) $\partial P/\partial r = -\rho u/r^2$

(c) $\partial P/\partial r = -\rho u^2/r$ (d) none of the above

**Ans.(c):**For forced vortex flow total pressure derivatives.

$$dp = \frac{\rho u^2}{r} dr - \rho g dz$$

Pressure variation in radial direction $\frac{dp}{dr} = \frac{\rho u^2}{r}$

Pressure variation in vertical direction $\frac{dp}{dz} = -\rho g$

**55. The flow rate through a circular pipe is measured by**

(a) pitot- tube (b) venturimeter
(c) volt meter (d) none of the above

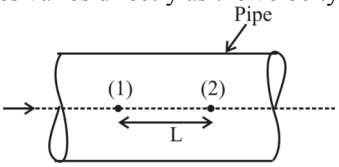
**Ans. (b) :** The flow rate through a circular pipe is measured by venturimeter.

- A venturimeter is a device used for measuring the rate of flow of a fluid flowing through a pipe.
- Venturimeter is a type of flow meter that works on the principle of Bernoullis equation.

**56. Losses of pressure head for the laminar flow through pipes varies**

(a) as the square of velocity
(b) directly as the velocity
(c) as the inverse of the velocity
(d) none of the above

**Ans. (b) :** Losses of pressure head for the laminar flow through pipes varies directly as the velocity.




$$P_1 - P_2 = \frac{32\mu VL}{D^2}$$

$$\boxed{h_f = \frac{32\mu VL}{\gamma D^2}}$$

**57. Kinematie similarity between model and prototype means**

(a) the similarity of forces
(b) the similarity of shape
(c) the similarity of motion
(d) the similarity of discharge

**Ans. (c) : Kinematic similarity** - Kinematic similarity between model and prototype means the similarity or motion .

Kinematic similarity requires that the model and prototype have the same length scale and same time scale ratio.

- Kinematic parameters are velocity, acceleration, discharge.
- In the model and prototype the ratio of velocity as well as acceleration must be same.

Kinematic similarity contains parameters like velocity, accelerator and discharge.

- At all corresponding points in the model and prototype the ratio of velocity as well as acceleration must be same (both in magnitude and direction).

Velocity ratio = $\frac{v_m}{v_p} = v_r = \frac{L_r}{T_r}$

Acceleration ratio = $\frac{a_m}{a_p} = ar = \frac{L_r}{T_r^2}$

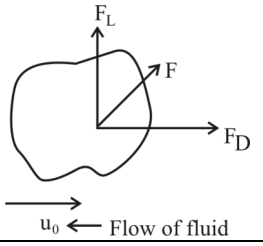
Discharge ratio = $\frac{Q_m}{Q_p} = Q_r = \frac{L_r^3}{T_r}$

**58. Drag is defined as the force exerted by a flowing fluid on a solid body**

(a) in the direction of flow
(b) perpendicular to the direction of flow
(c) in the direction which is at an angle of $45^0$ to the direction of flow
(d) none of the above

**Ans. (a) :** The force acting on the body in direction of flow by the fluid is called as the drag force. It is denoted by $F_D$.

- The other force acting on the body perpendicular to the direction of fluid flow is called as the lift force $F_L$.




**59. Kinematic viscosity ($\nu$) is equal to**

(a) $\mu \times \rho$ (b) $\mu / \rho$
(c) $\rho \times \mu$ (d) none

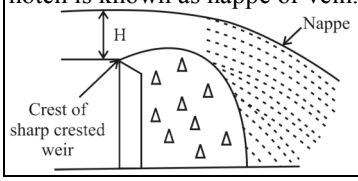
**Ans. (b) :** $\nu = \frac{\mu}{\rho}$

S.I. unit $m^2$/sec
CGS stoke's

**60. Notch is a device used for measuring**

(a) rate of flow through pipes
(b) rate of flow through a small channel
(c) velocity through pipe
(d) velocity through a small channel

**Ans. (b) :** Notch is a device used for measuring the rate of flow of a liquid through a small channel. The bottom of the notch over which the water flows is known as crest and the thin sheet of water flowing through the notch is known as nappe or vein.




**61. Statically indeterminate structures are opted inspite of being indeterminate because**

(a) They can be made of good material
(b) They can be designed to have larger dimensions
(c) higher loads can be supported with less consumption of material
(d) they can be easily analysed

**Ans. (c):** Statically indeterminate structures are opted inspite of being indeterminate because higher loads can be supported with less consumption of material.

**62. The force method in structural analysis starts with**
(a) Equilibrium field
(b) compatible deformations
(c) Equilibrium and compatibility conditions
(d) Force deformation relation only

**Ans. (c) :** Force method is called as flexibility method or consistent deformation method.
- to calculate the force compatibility the original structure is required.
- Once we obtain the reaction, redundant force, we can simply use the equilibrium equation to get the rest of the unknowns
- Used for indeterminate structure.

**63. The slope deflection method in structural analysis falls in the category of**
(a) Force method
(b) Flexibility method
(c) Consistent deformation method
(d) Stiffness method

**Ans. (d) : Example of force methods :**
(a) Castigliano's theorem
(b) Strain energy method
(c) Virtual energy method
(d) Clapeyron's three moment equations
(e) Column analogy method
(f) Flexibility matrix method
**Example of displacement method :**
(a) Slope deflection method
(b) Moment distribution method
(c) Stiffness matrix method
(d) Kani's method

**64. Horizontal stiffener in plate girders are used to**
(a) Increase the bending strength of the web
(b) Increase the shear capacity of the web
(c) Prevent local buckling of the flange
(d) Prevent local buckling of the web

**Ans. (d) :** Horizontal stiffness in plate girders are used to prevent local buckling of the web.

**65. For finding forces in members or truss by the graphical method, the force diagram is suggested by**
(a) Mohr (b) Timoshenko
(c) Maxwell (d) Betti

**Ans. (c) :** The cremona diagram also known as the cremona-maxwell method, is a graphical method used in statics of trusses to determine the forces in members.

**66. Minimum number of equilibrium equations to be considered at a joint in the analysis of plane truss are**
(a) 0 (b) 1
(c) 2 (d) None of the above

**Ans. (c) :** Since in truss all the joints are in equilibrium, we can write two equilibrium equation for each joints. Therefore if the structure is statically determinate we can write the relation.

$2j = m + R$
j = number of joint
m = number of member
R = number of unknown reaction

**67. The ultimate tensile stress of cold twisted bars should be grater than (in N/mm$^2$)**
(a) 580 (b) 490
(c) 420 (d) 260

**Ans. (c) :** The ultimate tensile stress of cold twisted bars should be grater than 420 in N/mm$^2$.

**68. Minimum percentage of elongation of high yield strength bars should be about**
(a) 10 (b) 15
(c) 20 (d) 25

**Ans. (a) :** Minimum percentage of elongation of high yield strength bars should be about 10.
For mild steel = 15%

**69. The minimum quantity of cement content needed in one m$^3$ of a reinforce concrete which is exposed to sea weather conditions is (in kg)**
(a) 200 (b) 250
(c) 300 (d) 350

**Ans. (c) :** The minimum quantity of cement content needed in one m$^3$ of a reinforce concrete which is exposed to sea weather conditions is 300 in kg.

**70. The approximate allowable axial compressive stress under earthquake load condition in reinforced concrete is**
(a) 0.25 fck (b) 0.33 fck
(c) 0.44 fck (d) 0.3 fck

**Ans. (c) :** As per IS code 456 : 2000
**Clause B- 2.3 Increase in permissible stress –**
When stresses due to wind (or earthquake) temperature & shrinkage effect are combined with those due to dead live & impact load, the stress specified in table no 21, 22 & 23 may be exceeded up a limit of $33\frac{1}{3}$ percent. Winds seismic force need not be considered as acting simultaneously.

**71. According to IS 456-2000, the maximum cement content that can be used in concrete work in (kg/m$^2$)**
(a) 250 (b) 350 (c) 450 (d) 550

**Ans. (c) :** As per clause 8.2.4.2 the maximum cement content that can be used in concrete work in 450 kg/m$^2$

**72. The ratio of the allowable bond stress in deformed bars to that of plain bars is**
(a) 1.15 (b) 1.40 (c) 1.66 (d) 1.20

**Ans. (c) :** Bond stress $(\tau_{bd}) = \frac{\phi\sigma_{st}}{4L_d}$ in plain bar

$(1.60)\tau_{bd} = \frac{\phi\sigma_{st}}{4L_d}$ is deformed bar

Ratio of bond stress = $\frac{\text{in case of deformed bar}}{\text{in case of plain bar}}$

$= \frac{1.60\ \tau_{bd}}{\tau_{bd}} = 1.60 \sqcup 1.66$

The values specified for '$\tau_{bd}$' (desigh bond stress), As per Clause 26.2.1.1 of IS 456:2000 are 1.2 MPa, 1.4 MPa, 1.5 MPa, 1.7 MPa and 1.9 MPa for the concrete grades M–20, M–25, M–30, M–35 & M–40 and above respectively.
For plain bar in tension, with an increase of 60% for deformed bars in tension. And further a increase of 25% for bars in compression.
• i.e, the ratio of allowable bond stress is deformed bar to that of plain bar is 1.6.

**73. The minimum thickness of a web stiffened in both the directions should be**
(a) 6 mm (b) L/425
(c) d/250 (d) h/250

**Ans. (c) :** The minimum thickness of a web stiffened in both the direction should be $\frac{d}{250}$ If it is stiffened in both by horizontal and vertical stiffeners.
If $\frac{d}{t_w} > 400$ then the section must be redesigned.
Where, d = stiffener spacing.

**74. The bending resistance of steel beams is governed mostly by**
(a) Moment of Inertia
(b) Modulus of the section
(c) Radius of gyration
(d) Area of cross-section

**Ans. (b) : Moment of resistance –** It is the bending moment which a beam can resist.
Moment of resistance = $z \times \sigma_{bc}$ or $\sigma_{bt}$
• Bending resistance of steel beam is governed mostly by modulus of the section.
• More the section modulus of beam have more bending resistance.

**75. The maximum spacing of the intermediate vertical stiffeners in plate girders should be**
(a) 3 d (b) 1.5 d
(c) d (d) 0.5 d

**Ans. (b) :** Maximum spacing of the intermediate vertical stiffeners in plate girders should be 1.5 d.
• Stiffeners are transverse plates used in plate girders to avoid local web buckling and are spaced 1.5 times the distance between flange angles.

**76. The effective length of an angle member in welded joint truss is equal to**
(a) L (b) 0.85 L
(c) 0.7 L (d) 0.65 L

**Ans. (c) :** The effective length of an angle member in welded joint truss is equal to 0.7 L.

**77. The workability of concrete can be improved by the addition of the following admixture**
(a) Calcium carbonate (b) Calcium sulphate
(c) Hydrated lime (d) Hydrogen peroxide

**Ans. (c) :** The workability of concrete can be improved by the addition of the hydrated lime.

**78. The work ability of concrete can be improved by-**
(a) More sand
(b) More cement
(c) More fine aggregate
(d) Fineness of coarse aggregate

**Ans. (b) :** The work ability of concrete can be improved by more cement.

**79. The approximate ratio of the direct tensile strength to the compressive strength of concrete is**
(a) 0.25 (b) 0.10
(c) 0.05 (d) 0.01

**Ans.(b):** Tensile strength of concrete is approximately 10–15% of compressive strength.

**80. The shrinkage of concrete is directly proportional to**
(a) water content at the time of mixing
(b) sand content
(c) coarse aggregate content
(d) aggregate to cement ratio

**Ans. (a) :** The drying shrinkage increases with an increase in cement content or an increase in water content, shrinkage is greater with richer mixes (more of cement) and also with aluminious cement other things being equal, shrinkage of concrete is almost directly proportional to the amount of water in the mix.

**81. Most of the shrinkage in concrete takes place in**
(a) 28 days of curing
(b) one month of drying
(c) 6 months of drying
(d) 12 months of drying

**Ans. (a) :** The rate of shrinkage decreases with time. It has been observed that 14 to 34% of 20 years shrinkage takes place in two week's time. Where 40 to 70% in 3 months and 66 to 80% in one year.

**82. The minimum compressive strength of perforated bricks should be about (in $N/mm^2$)**
(a) 4 (b) 7
(c) 10 (d) 15

**Ans. (b) :** The minimum compressive strength of perforated bricks should be about 7 $N/mm^2$).
Water absorption should not be greater than 15%.

**83. The maximum percentage of silt content soil for making good tiles is about**
(a) 10 (b) 20
(c) 30 (d) 40

**Ans. (c) :** The maximum percentage of silt content in soil for making good tiles is 30.
• The maximum percentage of sand content soil for making good tiles is 40.

**84. Good quality building stones should not contain soluble salts more than**
(a) 1 % (b) 5 %
(c) 10 % (d) 12 %

**Ans. (b) :** Good quality building stones should not contain soluble salts more than 5%.
• Good quality of building stones should not contain soluble salts.

• Building stones used for the force work of the building should have fine, compact texture.
• Stone used in construction should be strong and durable to with stand the disintegrating action of wheatear.

**85. The flakiness of aggregate has the following effect on the strength of the concrete in which it is used**
(a) increase the strength
(b) Does not effect the strength
(c) Decreases the strength
(d) Decreases the soundness

**Ans. (c) :** • Flaky and elongated aggregates leads to the formation of poor mix, which has low strength and low workability as poor is the inter particle locking and higher is the area to be lubricated in this case.

**86. The approximate temperature in kiln in which granules of cement material are exposed to is**
(a) 800 to $1000^0$C (b) 1000 to $1200^0$C
(c) 1300 to $1500^0$C (d) 1500 to $1700^0$C

**Ans. (c) :** A cement kiln, a huge in dined rotating furnace, is a continuous stream process vessel, in which the feed and fuel are held in a dynamic balance. As the raw materials – limestone, clay & shale – tumble towards the $1500^0$C flame, the chemical reactions transform them into clinker.

**87. The approximate percentage of tricalcium aluminate in Portland cement is**
(a) 40 (b) 30 (c) 20 (d) 10

**Ans. (d) :** Bogue's compounds –

| **Name** | **Percentage** |
|---|---|
| $C_3S$ | 30-50% |
| $C_2S$ | 20-45% |
| $C_3A$ | 8-12% |
| $C_4AF$ | 6-10% |

**88. The development of strength of cement and its fineness are related as**
(a) Inversely proportional
(b) Directly proportional
(c) Not related
(d) Randomly connected

**Ans. (b) Factor affecting Fineness of cement–**
- Faster reaction
- Higher heat of hydration
- Large particle do not react with water completely
- Higher shrinkage indicate higher fineness
- Reduce bleeding with higher fineness
- Reduced durability with higher fineness

$$\text{strength of cement} \propto \text{Fineness}$$

**89. The following material contains pozzolanic properties**
(a) Diatomaceous clay (b) Black cotton clay
(c) Dolomite (d) Lime stone

**Ans. (a) :** Diatomaceous clay has a pozzolanic properties and other three have not pozzolanic properties.

**90. The setting time of cement can be increased by the addition of**
(a) Calcium chloride (b) Hydrogen peroxide
(c) Gypsum (d) sodium

**Ans. (c) :** The setting time of cement can be increased by the addition of gypsum.

**91. Timber can be made reasonable fire resistant**
(a) by soaking it in ammonium sulphate
(b) by applying tar paint
(c) by pumping creosote oil in to timber under high pressure
(d) none of the above

**Ans. (a) :** Timber can be made reasonable fire resistant by soaking it in ammonium sulphate, borax and boric acid.

**92. Most economical of the methods or sawing wood is**
(a) radial sawing (b) tangential sawing
(c) quarter sawing (d) flat sawing

**Ans. (c) : Quarter sawing –** Most economical method Used for high class work.
Wastage of timber is maximum.

**93. Ply wood is identified by**
(a) volume (b) weight
(c) thickness (d) area

**Ans. (c) :** Ply wood is identified by thickness.
The plywoods broads are prepared from thin layer of wood or veneers.

**94. Bogue's compounds are**
(a) $C_3S$ (b) $C_2S$
(c) $C_4AF$ (d) All (1),(2) &(3)

**Ans. (d) :** Bouue's compounds are

| | **% by mass in cement** |
|---|---|
| $C_3S$ | 30-50 |
| $C_2S$ | 20-45 |
| $C_3A$ | 8-12 |
| $C_4AF$ | 6-10 |

**95. Compressive strength of cement essentially depends up on**
(a) W/C ratio (b) Fineness of cement
(c) A/C ratio (d) (1),(2) & (3)

**Ans. (d):** Compressive strength of cement depend upon.
(i) Water - cement ratio.
(ii) Fineness of cement
(iii) Aggregate cement ratio
Temperature $27 \pm 2^0$C

**96. The constituent responsible for setting of hydraulic lime under water is**
(a) silica (b) clay
(c) calcium oxide (d) carbon dioxide

**Ans. (b) :** The constituent responsible for setting of hydraulic lime under water is clay.
• It contains small proportion of clay 5–30% and iron oxide in chemical combination with the calcium oxide content [CaO + MgO 7–80% with MgO less than 5%]

**97. The initial setting time of the ordinary Portland cement should not be less than**
(a) 15 minutes (b) 30 minutes
(c) 45 minutes (d) one hour

**Ans. (b) :** The initial setting time of the ordinary Portland cement should not be less than 30 minutes.
Final setting time 10 hrs.

**98. The process of keeping concrete wet to enable it to attain full strength is known as**

(a) curing (b) wetting
(c) 30 drenching (d) quenching

**Ans. (a) Curing :** It is process of preventing the loss of moisture from the concrete. Curing period for rapid hardening cement 3 to 7 days and 7 days for OPC.

**99. PPC is a type of cement where in**

(a) Fly ash is added
(b) silica fume is added
(c) GGBS is added
(d) Fly ash & silica fume are added

**Ans. (a) : PPC [IS 1489, Part-I] –**

- It is manufactured by grinding Portland cement clinker and pozzolana [usually fly ash 10-25% by mass of PPC].
- It has low heat evolution and is used in the place of mass concrete.

**100. The relationship between compressive strength and water cement ratio is called**

(a) Abram's law (b) Coloumb's law
(c) James law (d) Bernaullis law

**Ans. (a) :** The relationship between compressive strength and water cement ratio is called Abram's law. This law states that the strength of a concrete mix is inversely related tot he mass ratio of water to cement.

**101. Which one of the following is not a type of brick bond**

(a) English Bond (b) Flemish Bond
(c) French Bond (d) Both (1) and (2)

**Ans. (c) :** Various types of bonds generally found in brick wall –

(a) Stretcher bond (b) Header bond
(c) Facing bond (d) English bond
(e) Flemish bond (f) Dutch bond
(g) Ranking bond (h) Zig-zag bond
(i) Garden wall bond

So, French bond is not any type of bond.

**102. One of the following is not a NDT for concrete**

(a) UPV (b) Rebound Hammer
(c) Profometer (d) Core extraction

**Ans. (d) :**Profometa test is a non-destructive testing technique used to detect location and size of reinforced and concrete cover quickly and accurately.

**103. Swelling of the brick is known as**

(a) bulking (b) bladdening
(c) bloating (d) none of the above

**Ans. (c) Bloating :** This defect observed as spongy swell mass over the surface of burned bricks is caused due to the presence of excess carbonaceous matter and sulphur in brick-clay.

**104. A mortar prepared by mixing wood powder or saw dust to cement mortar or lime mortar is known as**

(a) cinder mortar (b) wood mortar
(c) light mortar (d) gauged mortar

**Ans. (c) :** A mortar prepared by mixing wood powder or saw dust to cement mortar or lime mortar is known as light weight mortar.

**105. The coefficient of linear expansion of concrete is**

(a) less than that of steel
(b) more than that of steel
(c) almost same as that of steel
(d) many times more than that of steel

**Ans. (c):**The linear expansion of concrete is - $1.2 \times 10^{-5}$/°C

The linear expansion of steel is - $1.1 \times 10^{-5}$/°C

almost same as that of steel so option (c) is correct.

**106. The analysis of a structure is**

(a) deciding the material of the member
(b) deciding the dimension of the member
(c) calculating the magnitude and nature of various straining actions at salient points of the structure
(d) planning of the structure

**Ans. (c) :** The analysis of a structure is calculating the magnitude and nature of various straining actions at salient points of the structure.

**107. A beam is a structural member predominantly subjected to**

(a) transverse loads (b) axial loads
(c) twisting moment (d) none of the above

**Ans. (a) :** A beam is a structural member predominantly subjected to transverse loads.

**108. In a coplanar force system, the number of unknown forces that can be found by the principles of statics is**

(a) 3 (b) 2 (c) 1 (d) 0

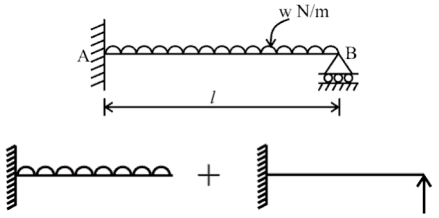
**Ans. (b) : Coplanar forces–** The forces, whose lines of action lie on the same plane are known as coplanar forces.

- In a coplanar force system, the number of unknown forces that can be found by the principle of statics is 2

**109. A cantilever beam subjected to a uniformly distributed load of intensity w/m run is propped by a rigid prop to the same level of fixed support, The reaction in the prop is**

(a) $\frac{3}{8}wl$ (b) $\frac{wl}{2}$ (c) $wl$ (d) $\frac{5}{8}wl$

**Ans. (a) :** A propped cantilever beam A as fixed and B is propped. Deflection at B = 0.




So, deflection at B for cantilever beam with UDL is equal to deflection at B for cantilever beam with upward reaction.

So, $\frac{wl^4}{8EI} = (R_b)\frac{l^3}{3EI}$ ...(1)

$R_A + R_B = wl$ ...(2)

By equation (1)

$\boxed{R_B = \frac{3wl}{8}}$ $\boxed{R_A = \frac{5wl}{8}}$

**110. A pin jointed frame with number of joints *j*, and number of members *n* will be a perfect frame, if**

(a) n= (2j+3) (b) n> (2j-3)

(c) n< (2j -3) (d) n= (2j -3)

**Ans. (d) :** A pin jointed frame with number of joints *j*, and number of members *n* will be a perfect frame, if n= (2j -3).

Pin jointed frame is designed in such a way that member of frame always carries only axial forces.

Hence equation of equilibrium available at truss joint.

2 Nos. ⟶ Plane pin jointed frame [$\Sigma F_x = 0, \Sigma F_y = 0$]

3 Nos. ⟶ Space pin jointed frame [$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0$]

**111. If a fixed beam is subjected to udl throughout the span, the point of contraflexure will occur at**

(a) L/2

(b) at two fixed supports

(c) 0.21 L, L from each of the supports

(d) 0. 667 L from each of the supports

**Ans. (c) :** In a fixed beam is subjected to udl throughout the span, the point of contraflexure will occur at 0.21 L, L from each of the supports.

**112. In a rigid jointed frame, the joints are considered**

(a) to rotate only as a whole

(b) not to rotate at all

(c) that 50% of members rotate in clock wise direction and 50% in anti-clock wise direction

(d) none of the above

**Ans. (a) Rigid jointed frame–** They are structure which has been rigidity jointed.

- Rigid joints are designed to transfer axial and shear forces in addition to bending moment between the connected members.
- In rigid jointed frame, the joints are considered to rotate as whole.
- In rigid jointed frame, the member has zero rotation with relative to other connected member or reaction at that joint.

**113. If *Ki* is the stiffness of $i^{th}$ member at a joint, the distribution factor for the member is**

(a) $\frac{Ki}{\sum Ki}$ (b) $\sum Ki$

(c) $Ki$ (d) $\left(\sum Ki - Ki\right)$

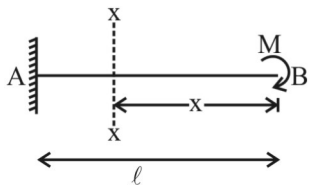
**Ans. (a) :** If *Ki* is the stiffness of $i^{th}$ member at a joint, the distribution factor for the member is $\frac{Ki}{\sum Ki}$.

**114. A cantilever AB of length L has a moment *'M'* applied at free end. The deflection at the free end 'B' is given as**

(a) $\frac{M^2L}{EI}$ (b) $\frac{ML^2}{2EI}$

(c) $\frac{ML}{2EI}$ (d) $\frac{ML^3}{2EI}$

**Ans. (b) :**




$$EI\frac{d^2y}{dx^2} = -M_{xx}$$

$$EI\frac{dy}{dx} = -Mx + C_1 \quad .....(1)$$

$$EI \cdot y = -M\frac{x^2}{2} + C_1x + C_2 \quad .....(2)$$

At $x = l, \frac{dy}{dx} = 0$ | $x = \ell,\ y = 0$

So, $C_1 = Ml$ | $0 = -\frac{Ml^2}{2} + Ml \times l + C_2$

$C_2 = -\frac{Ml^2}{2}$

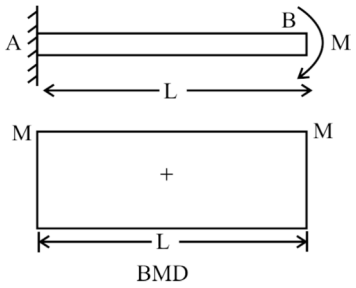
From equation (2) –

$x = 0$

$$EI \times y_B = 0 + 0 - \frac{Ml^2}{2}$$

So, $y_B = \frac{Ml^2}{2EI}$ (In downward direction)

**Other Method–**




BMD

Mohr's second theorem,

$$\delta = \frac{A\bar{x}}{EI}$$

$$= \frac{ML}{EI} \times \frac{1}{2} = \frac{ML^2}{2EI}$$

**115. The amount of deflection of a beam subjected to some type of loading depends upon**

(a) cross - section (b) bending moment

(c) either (2) & (2) (d) both (1) & (2)

**Ans. (d) :** The amount of deflection of a beam subjected to some type of loading depends upon cross section and bending moment.

**116. In a rigid jointed frame, the rotating of various members meeting at the joint will be**

(a) equal

(b) proportional to the length of the member

(c) proportional to the stiffness

(d) proportional to the respective moment of inertia

**Ans. (a): Rigid jointed frame–**

- They are structure which has been rigidity jointed.
- Rigid joints are designed to transfer axial and shear forces in addition to bending moment between the connected members.
- In rigid jointed frame, the joints are considered to rotate as whole.
- In rigid jointed frame, the member have zero rotation with relative to other connected member or reaction at that joint.

$$M \propto \frac{EI\theta}{L}$$

$$M = K\,\theta$$

$$K = \frac{EI}{L}$$

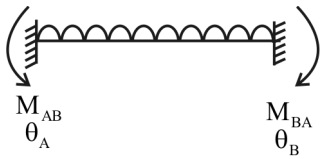
So, $L \propto \theta$

The rotation of various members meeting at joint will be indirectly proportional to the respective moment of inertia.

**117. In slope deflection method, the unknown rotations at various joints are determined by considering**

(a) the equilibrium of the joint
(b) the rigidity of joint
(c) the equilibrium of structure
(d) none of the above

**Ans. (a) :**




In the slope deflection method, the unknown rotations and displacement are determined by considering moment equilibrium condition at joint.

$$M_{AB} = \bar{M}_{AB} + \frac{2EI}{\ell}\left(\theta_A + \theta_B - \frac{3\Delta}{\ell}\right) \text{____(i)}$$

$$M_{BA} = \bar{M}_{BA} + \frac{2EI}{\ell}\left(2\theta_B + \theta_A - \frac{3\Delta}{\ell}\right) \text{_____(ii)}$$

Solving eq (i) & (ii) we get known rotation values ($\theta_A$ & $\theta_B$)

**118. A portal frame having single bay, single storey configuration can be analysed by using**

(a) Slope deflection method
(b) Kanis method
(c) Moment distribution method
(d) All (1), (2) & (3)

**Ans. (d) :** A portal frame having single bay, single storey configuration can be analysed by using Slope deflection method, Kanis method and Moment distribution method.

**119. In a slope deflection method, the end moments for any member are expressed**

(a) as zero
(b) in terms of the unknown end rotations
(c) as equal
(d) none of the above

**Ans. (b) :** The slope deflection method is a structural analysis method for beams and frames.

- In this method, the end moments for any member are expressed as unknown end rotations.
- The slope deflection equations can also be written using the stiffness factor, $k = \frac{I_{ab}}{L_{ab}}$

And the chord rotation $(\psi) = \frac{\Delta}{L_{ab}}$.

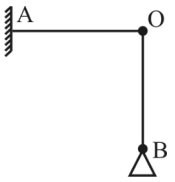
$$\rightarrow M_{AB} = \bar{M}_{AB} + \frac{2EI}{\ell}\left(\theta_A + \theta_B - \frac{3\Delta}{\ell}\right) \text{____(i)}$$

$$\rightarrow M_{BA} = \bar{M}_{BA} + \frac{2EI}{\ell}\left(2\theta_B + \theta_A - \frac{3\Delta}{\ell}\right) \text{_____(ii)}$$

**120. The ratio of stiffness of a member when far end is hinged to that of the member when far end is fixed is**

(a) 1 (b) 2 (c) $\frac{3}{4}$ (d) $\frac{4}{3}$

**Ans. (c) :**




$$\boxed{\frac{K_2}{K_1} = ?}$$

$$K_1 = \frac{4EI}{\ell} \text{ (For end fixed)}$$

$$K_2 = \frac{3EI}{\ell} \text{ (For end hinged)}$$

$$\frac{K_1}{K_2} = \frac{\frac{4EI}{\ell}}{\frac{3EI}{\ell}} = \boxed{\frac{4}{3}}$$

$$\boxed{\frac{K_2}{K_1} = \frac{3}{4}}$$

**121. If the froude number in open channel flow is equal to 1.0 the flow in called**

(a) critical flow (b) streaming flow
(c) shooting flow (d) none of the above

**Ans. (a) :** If the froude number in open channel flow is equal to 1.0 the flow in called critical flow.
Froude number greater than 1 is shooting flow.

**122. Kaplan turbine is**

(a) an impulse turbine
(b) a radial flow impulse turbine
(c) an axial flow reaction turbine
(d) a radial flow reaction turbine

**Ans. (c) :** Kaplan turbine is an axial flow reaction turbine, it is also a low head.
Kaplan turbine works on the principle of axial flow reaction. In axial flow turbines, the water flows through the runner along the direction parallel to the axis of

rotation of the runner. The guide vanes direct the water to the runner blades. The vanes are adjustable and can adjust itself according to the requirement of flow rate.

**123. Hydraulic ram is a pump which works**
(a) on a principle of water hammer
(b) on the principle of centrifugal action
(c) on the principle of reciprocationg action
(d) none of the above

**Ans. (a) :** A hydraulic ram or hydram, is a cyclic water pump powered by hydropower, used to lift water without electric motor. It uses the water hammer effect to develop pressure.

**124. Rainfall hyetograph shows the variation of**
(a) cumulative rainfall with time
(b) rainfall intensity with time
(c) rainfall depth over an area
(d) rainfall intensity with the cumulative rainfall

**Ans. (b)** Rainfall hyetograph is the relation between the intensity of rain fall with time and can be graphically presented.

| Type of curve | Meaning |
|---|---|
| Mass curve | Cumulative rainfall vs time |
| Double mass curve | Cumulative rain fall of nearby station vs cumulative rainfall of concerned station |
| Hydrograph | Discharge vs time |
| Hyetograph | Rainfall intensity vs time |

**125. The most accurate method of finding the average depth of rainfall over an area is**
(a) thiessen polygon
(b) Isohyetal method
(c) Arithmetic mean method
(d) Any of the above

**Ans. (b) : Isohyetal method–** This method of finding average rainfall is best provided that there are large number of stations. In this method topographic influence are taken into account.
This method requires large number of rain gauge. So it is cost intensive and it required to draw new isohyete every time the rainfall pattern changes, so it is cumbersome process.

$$P_{avg} = \frac{\varepsilon P_{ij} A_{ij}}{A}$$

$P_{ij} = \frac{(P_i + P_j)}{2}$, $A_{ij}$ is area between two successive isohyets $P_i$ and $P_j$.
A is total area of the catchment.

**126. The word unit in the unit hydrograph denotes**
(a) the unit depth of run off
(b) unit duration of the storm
(c) unit base period of the hydro graph
(d) arbitrary

**Ans. (a) :** A unit hydrograph is defined as the direct runoff hydrograph resulting from 1 unit depth (1 cm) of rainfall excess occurring uniformly over the basin at a uniform rate for specific duration of 'n' hrs.
.It is based on two assumption.
(i) Time invariance and
(ii) Linear response

**127. The 4h unit hydrograph of a basin can be approximated as a triangle with base period of 48 h and a peak ordinate of 200 $m^3/S$. The area of the basin would be**
(a) 1728 $km^2$ (b) 3456 $km^2$
(c) 864 $km^2$ (d) 5184 $km^2$

**Ans. (a) :** For unit hydrograph–
Runoff depth (n) = 1 cm
So,

$$n = \frac{\text{Area of hydrograph}}{\text{Area of catchement}}$$

$$1\times10^{-2} = \frac{\frac{1}{2}\times(48\times60\times60)\times200}{A}$$

$$A = \frac{1}{2}\times48\times60\times60\times200\times100$$

$A = 1728 \times 10^6\ m^2$
$A = 1728\ km^2$.

**128. Another name for an unconfined aquifer is**
(a) Leaky aquifer (b) perched aquifer
(c) artesian aquifer (d) water table aquifer

**Ans. (d) :** An unconfined aquifer (also known as water table aquifer) is one in which a free water surface exists. Recharge of this aquifer takes place through infiltration of precipitation from the ground surface.

**129. The trap efficiency of a reservoir is a function of**
(a) age of the reservoir
(b) the reservoir capacity
(c) total inflow
(d) reservoir capacity/ total inflow

**Ans. (d) : Trap efficiency–** It is defined as the percentage of the sediment deposited in the reservoir even inspite of taking precautions & measures to control it's deposition. Therefore, trap efficiency ($\eta$)

$$\eta = \frac{\text{Total sediment deposited}}{\text{Total sediment flowing}}$$

The ratio of the reservoir capacity to the total in flow of water in it, is known as capacity in flow ratio. It is very important factor, because the trap efficiency ($\eta$) has been found to be function of capacity inflow ratio, i.e,–

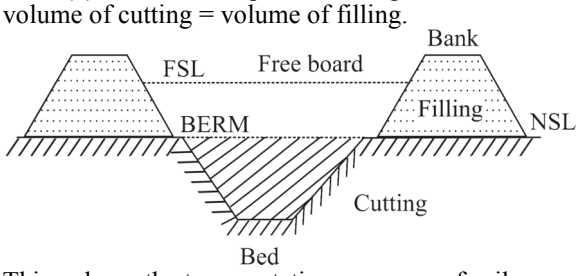
$$\eta = f\left(\frac{\text{capacity}}{\text{in flow}}\right)$$

**130. A climatic region lacking enough water for agriculture without artificial irrigation is called**
(a) arid zone (b) dry zone
(c) desert zone (d) none of the above

**Ans. (a) :** A climatic region lacking enough water for agriculture without irrigation is called as arid zone.

**131. The balanced depth of cutting of a canal is one in which**
(a) the volume of the cutting is equal to the volume of embankment
(b) the volume of cutting is less than the volume of embankment
(c) these are not cutting but only embankment
(d) there is only cutting and no embankment

**Ans. (a):** Balanced depth of cutting of canal is where volume of cutting = volume of filling.




This reduces the transportation expenses of soil.

**132. The main reason for the sitting up of a canal is**
(a) in adequate slope
(b) non- regime section
(c) defective head regulator
(d) all the above

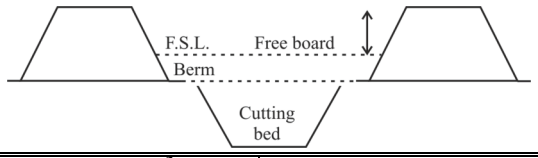
**Ans. (d) :** The main reason for the sitting up of a canal is – in adequate slope, non- regime section, defective head regulator.

**133. The pipe which carries water under pressure from reservoir to turbine is called**
(a) scroll case pipe (b) draft tube
(c) pen stock (d) none of the above

**Ans. (c) :** The pipe which carries water under pressure form reservoir to turbine is called pen stock.
**Penstock–** These are pipes of large diameters which carry water under pressure from the storage reservoir to the turbine. Penstock are usually made of steel, wood or reinforced concrete.

**134. The difference between the level of the top of the bank the F.S.L. is know as**
(a) safe margin depths (b) beam
(c) free board (d) none of the above

**Ans. (c) :** The difference in level between the top of a bank and full supply level in a canal is called Free board.
♦ The margin b/w FSL and bank level is known as free board. The amount of freeboard depends upon the size of the channel.




| Discharge ($m^3/s$) | Free board (m) |
|---|---|
| 1 to 5 | 0.50 |
| 5 to 10 | 0.60 |
| 10 to 30 | 0.75 |
| 30 to 150 | 0.90 |

**135. The power which is available in water resources engineering on and above the firm power is known as**
(a) surplus power (b) secondary power
(c) grace power (d) none of the above

**Ans. (b) :** The power which is available in water resources engineering on and above the firm power is known as secondary power.

**136. The ratio of inertial forces to gravity forces is called**
(a) Gravity number (b) Reynolds number
(c) Weber number (d) Froude number

**Ans. (d) :** The ratio of inertial forces to gravity forces is called Froude number.
The ratio of inertial forces to viscous forces is called Renyold's number.
The ratio of inertial forces to surface tension is called Froude number.

**137. In a homogeneous earth dam without d/s horizontal filter, portion of the d/s face through which seepage flow comes out is**
(a) stream line
(b) equipotential line
(c) neither a stream line nor an equipotential line
(d) both (1) and (2)

**Ans. (d) :** In a homogeneous earth dam without d/s horizontal filter, portion of the d/s face through which seepage flow comes out is due to the stream line and equipotential line both.
The flow net take place where the stream lines and equipotential lines cut each other in a small cross section.

**138. The difference is levels between top of capillary zone and the water table is**
(a) zero (b) negative
(c) positive (d) none of the above

**Ans. (b) :** Capillary zone is always above the water table, so the level difference between top of capillary zone and the water table is negative.

**139. The irrigation canals are generally aligned along**
(a) ridge line (b) straight line
(c) colour line (d) valley line

**Ans. (a)**
- Watershed canal runs perpendicular to contour line.
- Irrigation canal are generally aligned along the water shed line.
- It irrigating the both side of are so that CCA is large for irrigation.
- The water shed line does not cross the drainage line hence cross drainage work are not required.

**140. The length of water way required for the drain at a cross -drainage work is obtained as**
(a) $L = 475\sqrt{Q}$ j (b) $L = 47.5\sqrt{Q}$
(c) $L = 4.75\sqrt{Q}$ (d) $L = 0.475\sqrt{Q}$

**Ans. (c) :** An approximate value of required waterway for the drain may be obtained by using Lacey's equation.
$L$ or $P = 4.75\sqrt{Q}$
P = wetted perimeter
Q = total discharge in cumec

**141. The water is unsuitable of irrigation when the boron content exceeds the limit of**
(a) 0.5 ppm (b) 1 ppm (c) 1.3 ppm (d) 2 ppm

**Ans. (d) : Boron content in water–** Boron unlike sodium is an essential elements of plant growth. Chloride is also essential but in such small quantity that its frequently said non essential. The boron quantity if exceeds to its requirement then it become toxic for plant.
For some crop 0.2 mg/l boron water in essential but 1 to 2 mg/l may be toxic.

**142. A natural method of disposal of sewage is**
(a) sewage irrigation
(b) septic tank treatment
(c) composting
(d) aerated lagooning

**Ans. (c) : Disposal of refuse by composting–** This decomposition can be effected either under aerobic condition or under anaerobic condition or both. The final end product is a manure called compost or humus, as fertilizer for forms. Compositing is considered to be as aerobic process.
• In India, the composting is practiced in rural areas on the mixture of night soil and refuse.
Two methods, which are generally adopted here are–
(i) Indore process (ii) Bangalore process.

**143. The activity that may not affect ecological balance is**
(a) growing crops (b) developing gardens
(c) industries (d) use of pesticides

**Ans. (b) :** Due to development of garden, ecological balance remains unaffected.

**144. An agent responsible for the transportation of soil is**
(a) Wind (b) Water
(c) Gravity (d) All the above

**Ans. (d) :** Soil can be transported from one place to another due to weathering.
There are various method or agents, throw which soil is being transported such as. Water, wind, gravty and glaciers.

**145. Soil is considered as**
(a) single phase system (b) two phase system
(c) three phase system (d) none of the above

**Ans. (c) :** Soil is considered as three phase system with solid, air and water.
• Dry soil is considered as two phase system with solid and air.
• Fully saturated soil is considered as two phase system with a solid and water.

**146. A moist sample of soil weights 24 grams in a tin lid. The tin lid above weight 14 gram. Over dry weight to thin an sample is 22 gram. What is the water content of the soil?**
(a) 12.5 % (b) 37.5 %
(c) 25 % (d) 50 %

**Ans. (c) :** Given,
Weight of moist soil sample with tin = 24 gm
Tin weight = 14 gm
Weight of dry soil with tin = 22 gm
Then water content, $w = \frac{W_w}{W_s} \times 100\%$

$$w = \left(\frac{24-22}{22-14}\right) \times 100$$

$$w = \frac{2}{8} \times 100$$

$w = 25\%$

**147. To obtain grain size distribution, hydrometer analysis is appropriate for**
(a) Peats (b) Sands and gravels
(c) Silt and clays (d) All soils

**Ans. (c) : Sedimentation analysis–**
• The soil particle less than 75μ size can be further analyzed for the distribution of the various grain size of the order of silt and clay by sedimentation analysis as wet analysis.
**Type of sedimentation analysis–**
**(a) Pipette method –**
• It is a laboratory method
• The method consists in drowning off 10 ml samples of soil suspension by means of the sampling pipette from a standard depth of 10 cm at various time intervals after the start of sedimentation.
**(b) Hydrometer analysis –**
The hydrometer method differs from the pipette analysis in that the weight of solid per ml in the suspension at the chosen depth at chosen instants of time are obtained in directly by reading the specific gravity of the soil suspension with the aid of a hydrometer.

**148. A soil has a bulk density of 2.4 g/cm$^3$ and water content of 20% what is its dry density**
(a) 1 g/cm$^3$ (b) 1.2 g/cm$^3$
(c) 1.4 g/cm$^3$ (d) 1.6 g/cm$^3$

**Ans. (*) :** Given, Bulk density $\gamma_b = 2.4$ g/cm$^3$
Water content w = 20%
Dry density $\gamma_d = \frac{\gamma_b}{1+w} = \frac{2.4}{1.20} = 2.0\,g/cm^3$

**149. Plasticity index is obtained as the difference between**
(a) liquid limit and shrinkage limit
(b) shrinkage limit and plastic limit
(c) liquid limit and plastic limit
(d) none of the above

**Ans. (c) :** Plasticity index ($I_P$) = $w_L - w_P$
$w_L$ = Liquid limit
$w_P$ = plastic limit
$I_P$ = Plasticity index
• Is $w_p >> w_L$ then $I_P$ would be zero

| $I_P$ | | Soil description |
|---|---|---|
| 0 | – | Non plastic |
| 1 - 5 | – | Slight plastic |
| 5 - 10 | – | Low plastic |
| 10 - 20 | – | Medium plastic |
| 20 - 40 | – | High plastic |
| > 40 | – | Very high plastic |

**150. Capillary force is dependent on**
(a) pore pressure (b) water content
(c) depth to water (d) surface tension of water

**Ans. (d) :** The capillary force of soil moisture depends upon various factors such as surface tension of water, pressure in water relation to atmospheric pressure and the size and conformation of soil pores.

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 2004 (P-I)
## Solved Paper

**1. The Principle of superposition is valid for**
(a) Elastic materials
(b) Linear elastic materials
(c) Only metals
(d) Homogenous and isotropic materials

**Ans. (b) :** According to the principle of superposition, for a linearly elastic material, the load effects caused by two or more loadings are the sum of the load effects caused by each loading separately. Note that the principle is limited to:
(i) Linear elastic material only.
(ii) Structures undergoing small deformations only.

**2. The maximum deflection of a fixed beam of 4.0 m span and EI = 20.0 MNm$^2$ is subjected to a distributed load of 6.0 kN/m is?**
(a) 1.00 m (b) 0.20 m
(c) 0.20 mm (d) 10.00 mm

**Ans. (c) :** The maximum deflection of a fixed beam with uniformly distributed load is given by

$$S = \frac{WL^4}{384\,EI}$$

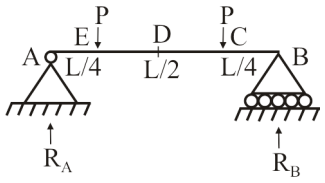
$$= \frac{6\times(4)^4\times10^3}{384\times20\times10^6}$$

$$= 0.0002 \text{ m}$$

$$= 0.2 \text{ mm}$$

**3. A simply supported beam of span $\ell$ carries two concentrated load of magnitude P at each of the quarter span sections; the bending moment at the mid-span section is**
(a) P/2 (b) P$\ell$/4
(c) 2P$\ell$ (d) P$\ell$

**Ans. (b) :**




$R_A = R_B = P$ (Symmetric loading)
$M_D = ?$
Bending moment at mid section

$$M_{mid} = \frac{P.L}{2} - \frac{P.L}{4}$$

$$M_D = \frac{P.L}{4}$$

Bending moment at any section is algebraic sum of moment either section of beam.

**4. The bulk modulus of a material is less than its Young's modulus the Poisson's ratio for the material is**
(a) 0.45 (b) 0.15 (c) 0.35 (d) 0.50

**Ans. (c) :** $K < E$ ..... (i)
We know that

$$E = 3K(1-2\mu) \quad \text{..... (ii)}$$

From (i) and (ii)

$$K < 3K(1-2\mu)$$

$$\frac{1}{3} < (1-2\mu)$$

$$2\mu \le \frac{2}{3}$$

$$\mu < \frac{1}{3}$$

**5. A propeller shaft transmits 540.0 kW at 720 rpm; the torque developed in the shaft is**
(a) 65.65 kNm (b) 53.71 kNm
(c) 7162.0 Nm (d) 7.162 Nm

**Ans. (c) :** Power (P) = 540 kW
Speed (N) = 720 rpm

$$\text{Power (P)} = \frac{2\pi NT}{60}$$

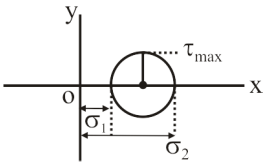
$$540 = \frac{2\times3.14\times720\times T}{60}$$

$$T = 7.1619 \text{ kN-m}$$

$$T = 7161 \text{ N-m}$$

**6. The maximum shear stress from Moh'r circle, is given by the**
(a) diameter of the circle
(b) distance of the centre from the origin
(c) radius of the circle
(d) distance of the farthest point on the circle from the origin

**Ans. (c) :** Maximum shear stress in mohr's circle




$\tau_{max}$ = radius of circle

$$\tau_{max} = \frac{\sigma_2 - \sigma_1}{2}$$

**7. A solid circular shaft of diameter D is subjected to a torque T; the maximum tensile stress induced in the shaft is**
(a) zero (b) 16T/ ($\pi D^3$)
(c) 32T/($\pi D^3$) (d) 64 T/($\pi D^3$)

**Ans. (b):** The maximum shear stress developed on the surface of the shaft subjected to torque T is given by

Equivalent bending moment $M_{eq} = \frac{1}{2}\left(M + \sqrt{M^2 + T^2}\right)$

$$M_{eq} = \frac{1}{2}\sqrt{T^2} \Rightarrow M = 0$$

$$M_{eq} = \frac{T}{2}$$

From bending equation :-

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

Bending Stress $\sigma = \frac{M.y}{I}$

$$= \frac{\frac{T}{2}}{\frac{\pi}{64}d^4} \times \frac{d}{2} \Rightarrow \frac{d}{2} \times \frac{64}{\pi d^4} \times \frac{d}{2}$$

$$\boxed{\sigma = \frac{16T}{\pi d^3}}$$

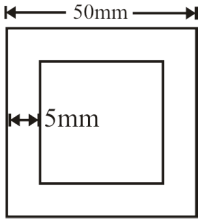
**8. A hollow beam of square section with external dimensions of 50.0 mm and thickness 5.0 mm can sustain a stress of 100.0 MPa. Its capacity in flexure is**

(a) 1.54 KNm (b) 1.23 KNm
(c) 0.72 KNm (d) 7.16 KNm

**Ans. (b) :** Given,

$\sigma$ = 100 MPa = N/mm$^2$

y = 50/2 = 25mm




$$I_{NA} = \frac{A^4 - B^4}{12} \quad \left[\begin{array}{l}\text{Where, A = Side of outer square} \\ \text{B = Side of inner square}\end{array}\right]$$

$$I = \left(\frac{50^4 - 40^4}{12}\right) mm^4$$

$$M = \frac{\sigma.I_{NA}}{y} \quad [\text{From Bending equation}]$$

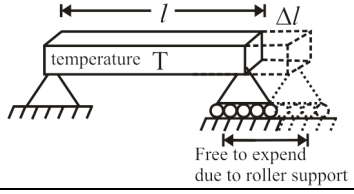
$$M = \frac{100 \times (50^4 - 40^4)}{12 \times 25 \times 10^6} kN - m$$

$$\boxed{M = 1.23 \text{ kN-m}}$$

**9. A beam of span *l* with one end on roller support and the other end hinged, when subjected to a uniform rise in temperature T develops**

(a) compressive stresses (b) flexural stresses
(c) tensile stresses (d) zero stresses

**Ans.(d):** When body is free to expand no thermal stresses will induced in a member




**10. The modulus of elasticity of high carbon steels is about**

(a) 200.0 GPa (b) 200.0 Mpa
(c) 80.0 GPa (d) 120.0 GPa

**Ans. (a) :**

| Material | Young's modulus(GPa) |
|---|---|
| Aluminium | 70 |
| Brass | 91 |
| Copper | 110 |
| Steel | 200 |
| Glass | 55 |

**11. Newton is the force exerted, under gravity by a body of mass**

(a) 0.100 kg (b) 250.0 kg
(c) 102.0 kg (d) 1.0 kg

**Ans. (d) :** F = Mg

1N = 1 kg × g

$\boxed{M = 1 \text{ kg}}$

**12. Pascal in SI units is a unit of**

(a) force (b) stress
(c) mass (d) length

**Ans. (b) :** The Pascal is the SI unit of stress or pressure.

1 Pa = 1 N/m$^2$

**13. Poisson's ratio of aluminium is usually adopted as**

(a) 0.22 (b) 0.14
(c) 0.30 (d) 0.45

**Ans. (c) :** Poisson's ratio for various material are

| Materials | Poisson's Ratio |
|---|---|
| Cast Iron | 0.21-0.26 |
| Steel | 0.27-0.30 |
| Aluminium | 0.31 |
| Rubber | 0.5 |
| Cork | 0.0 |

**14. Complementary and applied shear stresses are equal in**

(a) flexure only (b) punching shear only
(c) direct shear only (d) All of the above

**Ans. (d) :** Complementary shear stresses are equal irrespective of flexure punching shear or direct shear.

**15. Shear strain is the**

(a) Ratio of the change in the angle to the original angle
(b) ratio of lateral and longitudinal strains
(c) ratio of longitudinal and lateral strains
(d) change in a right angle

**Ans. (a) :** The ratio of change in angle to the original angle is called shear strain.
The ratio of lateral strain to the longitudinal strain is called Poisson's ratio.

**16. The coefficient of discharge $C_d$ for flow through an orifice, the coefficient of velocity $C_v$ and the coefficient of contraction $C_e$ are related as**

(a) $C_d = C_c + C_v$ (b) $C_d = C_c + C_v$

(c) $C_d = C_c + C_v$ (d) $C_d = C_c + C_v$

**Ans. (*) :** Relation between coefficient of discharge ($C_d$), coefficient of contraction ($C_c$) and coefficient of velocity ($C_v$)

$$C_d = C_c \times C_v$$

**17. Reynold's number is the ratio of**

(a) gravity force to inertia force

(b) inertia force to viscous force

(c) elastic force to gravity force

(d) gravity force to elastic force

**Ans. (b) :** The Reynolds number is defined as the ratio of inertial forces to viscous forces.

$$\text{Reynold's number} = \frac{\text{Inertia force}}{\text{viscous force}} = \frac{\rho v D}{\mu}$$

**18. Manometers measure**

(a) low pressure (b) high pressures

(c) differential pressures (d) velocity of flow

**Ans. (c) :** Manometers are devices in which columns of a suitable liquid are used to measure the difference in pressure between two points or between a certain point and the atmosphere.

**19. The minimum angle of inclination of a manometer to avoid the effects of surface tension is**

(a) 2" (b) 3"

(c) 4" (d) 5"

**Ans. (*) :** Minimum inclination angle of manometer to avoid the effects of surface tension = 4°

**20. Surface tension in fluids is measured in**

(a) MPa (b) N/mm

(c) kPa (d) Nmm

**Ans. (b) :** Surface tension is the tension of the surface film of a liquid caused by the attraction of the particles in the surface layer by the bulk of the liquid, which tends to minimize surface area.

Mathematically, surface tension can be expressed as follows

$$\sigma = \frac{\text{Force}}{\text{Length}}\left(\text{N/m}\right) = \frac{\text{Surface Energy}}{\text{Surface area}}\left(\text{J/m}^2\right)$$

**21. The depth of the centre of pressure of a vertical rectangular plate of depth 3.0 m and width 4.0 m when the water surface coincides with the top of the gate is**

(a) 1.0 m (b) 2.0 m

(c) 1.5 m (d) 3.0 m

**Ans. (b):** Given b= 4 m and h = 3m, $\bar{h} = \frac{h}{2} = \frac{3}{2}$ Distance of centre of pressure from the free surface of liquid

$$h_{C.P.} = \frac{I_G \sin^2\theta}{A\bar{h}} + \bar{h}$$

For vertical plane surface $\theta = 90^\circ$ and $I_G = \frac{bh^3}{12}$; A = bh

$$h_{C.P.} = \frac{(bh^3/12)}{b \times h \times (h/2)} + (h/2)$$

$$= \frac{2h}{12} + \frac{h}{2}$$

$$= \frac{2h}{3} \qquad \text{(Putting h = 3 m)}$$

$$= \frac{2 \times 3}{3}$$

$$= 2 \text{ m}$$

**22. A body is in stable equilibrium, if its metacentre is**

(a) at the centroid

(b) above the centroid

(c) below the centroid

(d) at the mid-depth

**Ans. (b) :** (i) Stable Equilibrium: The metacentre is above the centre of gravity of body. (MG = +ve)

(ii) Unstable equilibrium: The metacentre is below the centre of gravity of the body. (MG = –ve)

(iii) Neutral equilibrium: The metacentre and the centre of gravity coincides with each other (MG = 0)

**23. Gauge pressure is**

(a) absolute pressure + atmospheric pressure

(b) absolute pressure - vacuum pressure

(c) atmospheric pressure + vacuum pressure

(d) absolute pressure - atmospheric pressure

**Ans. (d) :** We know that,

Absolute pressure = Atmospheric pressure + Gauge pressure

Absolute pressure = Atmospheric pressure – Vacuum pressure.

**Note:–** Negative gauge pressure = Vacuum pressure

**24. A line whose tangent at any point indicates the direction of flow of fluid particles at that point is known as**

(a) stream line (b) potential line

(c) phreatic line (d) velocity line

**Ans. (a) :** Streamline is an imaginary line or series of imaginary lines in a flow field, such that a tangent to this line at any point at any instant represents the direction of the instantaneous velocity vector at the point.

**25. Steady flow occurs when the**

(a) conditions of flow change steadily the time

(b) conditions of flow remain constant with time

(c) conditions of flow remains the same everywhere

(d) velocity vector is constant at a point

**Ans. (b) :** A flow that is not a function of time is called a steady flow. Steady state flow refers to the condition where the fluid properties at a point in the system do not change over time.

**26. As the temperature is increased, the viscosity of water**

(a) increases (b) decreases

(c) remains constant (d) varies with pressure

**Ans. (b):** As the temperature is increased, the viscosity of water decreased.

**27. If the velocities of fluid particles vary from point to point in magnitude and direction, and with time, the flow is said to be**

(a) laminar (b) turbulent
(c) uniform (d) non-uniform

**Ans. (b) :** The flow in which the particles of a fluid attain such velocities that vary from point to point in magnitude and direction as well as from instant to instant, is known as turbulent flow.

**28. Non-uniform steady flow occurs in a**

(a) long tube with increasing flow rate
(b) pipe of tapering diameter under decreasing flow rate
(c) long pipe with decreases flow rate
(d) pipe of tapering diameter with constant flow

**Ans. (d) :** Non-uniform steady flow occurs in a pipe of tapering diameter with constant flow rate. In this flow, the stream changes from point to point in flow direction but does not change with respect to time.

**29. An ideal flow of fluid satisfies**

(a) Newton's law of viscosity
(b) Darcy law
(c) Continuity equation
(d) The condition of zero velocity at the boundaries

**Ans. (c) :** In ideal flow of any fluid must satisfy continuity equation. The equation based on the principal of conservation of mass is called the continuity equation.

**30. In a flow net, the distance between two successive stream lines at two successive sections are 10.0 mm and 5.0 mm. If the velocity of the first section is 1.0 m/s, that at the second section is**

(a) 1.0 m/s (b) 0.5 m/s
(c) 1.5 m/s (d) 2.0 m/s

**Ans. (d) :** let consider 1 unit width here so

$A_1 = 10 \times 1 = 10\ mm^2$ $v_1 = 1$ m/s

$A_2 = 5 \times 1 = 5\ mm^2$

By continuity equation $A_1v_1 = A_2v_2$

$10 \times 1 = 5 \times v_2$

$v_2 = 2$ m/s

**31. Water flows through a convergent mouth piece of 40.0 mm diameter under a head of 3.0 m. If the maximum vacuum pressure is 9.0 m of water; the maximum divergence to avoid separation of flow is**

(a) 60.0 mm (b) 56.6 mm
(c) 40.0 mm (d) 34.4 mm

**Ans. (b) :** Given,

Diameter of convergent = 40 mm

Head of water = 3 m

Vacuum pressure = 9 m

Dia of divergent = ?

To avoid flow separation min$^m$ dia of divergent:–

$$\frac{A_{divergent}}{A_{convergent}} = \sqrt{1+\frac{P_{vacuum}}{H_{water}}}$$

Where, A = Cross section area

$$\frac{\frac{\pi}{4}D^2_{divergent}}{\frac{\pi}{4}D^2_{convergent}} = \sqrt{1+\frac{9m}{3m}} \quad \left[\because P_{vacuum} = P_{atm} - P_{abs}\right]$$

$$D^2_{divergent} = 40^2 \times 2\,mm^2$$

$$D_{divergent} = 40\sqrt{2}\ mm^2$$

$$\boxed{D_{divergent} = 56.5\,mm}$$

**32. Cavitation is caused by**

(a) low pressure (b) high pressure
(c) low velocity (d) high velocity

**Ans. (a) :** Cavitation occurs in fluid flow systems where the local static pressure are below the fluids vapours pressure i.e. low pressure.

**33. Discharge through a submerged orifice is proportional to the**

(a) difference in the elevations of water level.
(b) square root of the difference in the elevations of water level
(c) square of the difference in the elevations of water level
(d) reciprocal of the difference in the elevation water levels

**Ans. (b) :** The rate of discharge through the orifice is given by

$$Q = a_0\ C_d\sqrt{2gH}$$

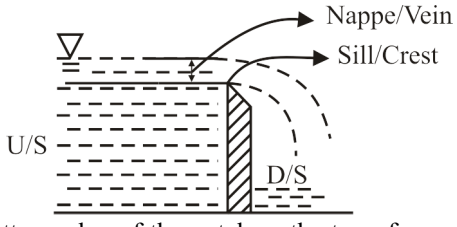
Where, $C_d$ is the coefficient of discharge h is the difference in elevation of water surface.

**34. The surface of the wier over which water flows is known as**

(a) vein (b) nappe
(c) sill (d) soffit

**Ans. (c) :** A weir may be defined as a structure constructed across a river or canal to store water on the upstream side. Water flows over the crest of weir.

The sheet of water flowing through a notch or over a weir is known as nappe or vein.




The bottom edge of the notch or the top of a weir over which water flows is known as sill or crest.

**35. When the Mach number is approaching unity from a value of 1.75, the flow is said to be**

(a) subsonic (b) supersonic
(c) transonic (d) sonic

**Ans. (d):**

| Sr. No. | Mach No. | Nature of flow |
|---|---|---|
| 0.1 | M < 0.8 | Sub-sonic |
| 0.2 | 0.8 –1.3 | Transonic |
| 0.3 | = 1 | Sonic |
| 0.4 | 1.3 – 5 | Super sonic |
| 0.5 | 5 – 10 | Hyper sonic. |
| 0.6 | 10 – 25 | High hyper sonic |
| 0.7 | > 25 | Re-entry speed |

**36. When water enters the turbine at high pressure and low velocity, it is of the type**

(a) radial flow (b) axial flow
(c) impulse (d) reaction

**Ans. (c) :** Impulse turbines operate at high pressure head and low discharge

| Sr. No. | Head | Turbine |
|---|---|---|
| 1. | High (150 – 2000) m | Pelton |
| 2. | Medium (30 – 150) m | Francis |
| 3. | Low < 30 m | Kaplan |

**37. The ratio of percentage error in the discharge over a rectangular notch, and the percentage error in the measurement of head is**

(a) 0.50 (b) 0.67
(c) 1.50 (d) 0.75

**Ans. (c) :** Discharge through a rectangular notch

$$Q = \frac{2}{3} C_d B \sqrt{2g}\, H^{3/2}$$

Where, Q = Discharge
$C_d$ = Coefficient of discharge
B = Width
H = Head of water
g = acceleration due to gravity

Differentiating w.r.t. H

$$\frac{dQ}{dH} = \frac{2}{3} C_d B \sqrt{2g} \left(\frac{3}{2} H^{1/2}\right)$$

$$dQ = \frac{2}{3} C_d B \sqrt{2g} \left(\frac{3}{2} H^{1/2}\right) dH \quad -(\text{Divided by } Q)$$

$$\frac{dQ}{Q} = \frac{\frac{2}{3} C_d B \sqrt{2g} \left(\frac{3}{2} H^{1/2}\right) dH}{\frac{2}{3} C_d B \sqrt{2g} \left(H^{3/2}\right)}$$

$$\boxed{\frac{dQ}{Q} = \frac{3}{2} \frac{dH}{H}}$$

$$\text{Error} = \frac{3}{2} = 1.5$$

**38. If H is the height of the liquid above the sill, the effect of end contraction according to Francis formula is**

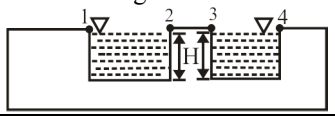
(a) 0.1 H (b) 0.2 H
(c) 0.3 H (d) 0.5 H

**Ans. (a):** If H is the height of the liquid above the sill, the effect of end contraction according to Francis formula is 0.1H

According to Francis formula, the discharge over a rectangular weir is –

$$Q = \frac{2}{3} C_d (L - 0.1 n H) H^{3/2}$$

Where, n = no. of end contraction
L = Effective length




**39. In a long pipe, the head loss is negligible at the**

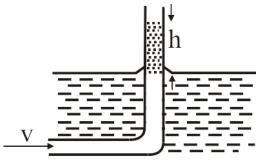
(a) entrance (b) outlet
(c) entrance and outlet (d) bends

**Ans. (c) :** For a long pipe, the head loss at the entrance and outlet both are ignored.

**40. Flow of water in pipes of diameter more than 3.0 m can be measured by**

(a) pitot tube (b) venturimeter
(c) piezometer (d) rotameter

**Ans. (a) :** Flow of water in pipes of diameter more than 3.0m can be measured by pitot tube. Pitot tube is a device used for calculating the velocity of flow at any point in a pipe or a channel.




Pitote tube

**Venturimeter:–** Used for measuring rates of flow in both incompressible and compressible fluids.

- Energy and heat losses are insignificant.
- Coefficient of discharge is 0.98

**Rotameter:–** It is work as a constant pressure drop variable area meter that measure the flow of fluid in closed tube.

**41. Enthalpy is**

(a) kinetic energy (b) specific energy
(c) molecular energy (d) internal energy

**Ans. (d) :** Enthalpy is a thermodynamics property of a substance and is defined as the sum of its internal energy and the product of its pressure and volume.

**42. When pipes are attached end to end, the**

(a) head loss is the same in all the pipes
(b) discharge is the same in all the pipes
(c) velocity of flow is the same in all the pipes
(d) frictional losses are reduced

**Ans. (b) :** When pipes are attached end to end (in series) then discharge is same in all the pipes.
(ii) When pipes are attached in parallel then head loss is same in all the pipes.

**43. If the specific speed of a pump is 5000, the type of pump is**

(a) slow speed radial flow
(b) medium speed radial flow

(c) very high speed mixed flow
(d) high speed axial flow

**Ans. (d) :** Specific speed of centrifugal pump is about to (500 – 1000) RPM with axial flow.

**44. Water is to be pumped of a rate of 6.0 cubic metres per minute to a height of 12.0 m; if the head loss is estimated as 5.0 m, the minimum rating of the pump required is**
(a) 1.70 KW (b) 17.0 KW
(c) 170.0 KW (d) 12.5 KW

**Ans. (b) :** Given,
$Q = 6\ m^3/min = 0.1\ m^3/s$
$H = 12m, h_f = 5m$
$\rho = 1000\ kg/m^3, g = 10\ m/s^2$
Power required to run pump
$= \rho \times Q \times g = (H + h_f)$
$= 1000 \times 0.1 \times 10 \times (12 + 5)$
$= 17\ kW$

**45. A centrifugal pump discharge 40.0 cumecs of water at 1500 rpm for a head of 25.0 m; its specific speed is**
(a) 848.5 (b) 14.1
(c) 1200.0 (d) 169.7

**Ans. (a) :** Given,
$Q = 40\ m^3/s$, $N = 1500$ rpm
$H = 25$ m
Specific speed of a centrifugal pump is given by

$$N_s = \frac{N\sqrt{Q}}{H^{\frac{3}{4}}}$$

where; Q = Discharge
H = Head, N = Normal Speed

$$= \frac{1500\sqrt{40}}{(25)^{\frac{3}{4}}}$$

$= 848.5$ rpm

**46. In a centrifugal pump, the flow of water leaving a size that is**
(a) rectilinear (b) radial
(c) free vortex (d) forced vortex

**Ans. (d) :** In a centrifugal pump casing, the flow of water leaving the impeller is free vortex.
The centrifugal pump works on the principle of forced vortex flow which means that when a certain mass of liquid is rotated by an external torque, the rise in pressure head of the rotating liquid takes place.

**47. Specific speed of pump is the speed of a unit of such a size that it**
(a) delivers unit discharge at unit head
(b) delivers unit discharge at unit power
(c) consumes unit power per unit discharge
(d) produces unit power per unit head

**Ans. (a) :** The specific speed of a centrifugal pump is defined as the speed of a geometrically similar pump which would deliver one cubic meter of liquid per second against a head of one meter or we can say that it delivers unit discharge at unit head.

**48. The head developed by a centrifugal pump depends upon**
(a) suction, lift and friction losses
(b) impeller diameter and speed
(c) fluid density
(d) type of casing

**Ans. (b) :** The centrifugal pumps are the ones that are used for the purpose of transporting the fluids. This is done by doing the transportation of rotational kinetic energy into a different form of energy that is referred as the hydrodynamic energy flow.

**49. While starting an axial flow pump, its delivery valve should be**
(a) closed (b) opened gradually
(c) open or closed (d) open

**Ans. (d) :** Axial flow pumps have poor suction capability so require more starting power. They are started against an open valve to minimize the starting power required.

**50. The flow of water is tangential in**
(a) Pelton wheel
(b) Franics turbine
(c) Kaplan and Fracis turbines
(d) Derias turbines

**Ans. (a) :** A pelton turbine is a type of impulse turbine in which the flow enters the wheel tangentially.

**51. The optimum value of the vane exit angle for a centrifugal pump impeller is**
(a) $10^0$-$15^0$ (b) $20^0$-$25^0$
(c) $30^0$-$45^0$ (d) $1^0$-$10^0$

**Ans. (b) :** The optimum value of the value of the vane exit angle for a centrifugal pump impeller is 20-25.

**52. The liquid enters in centrifugal pump**
(a) from the sides (b) at the top
(c) at the bottom (d) at the centre

**Ans. (d) :** In a centrifugal pump, the liquid enters the pump from the centre.
Rotation of a centrifugal pump impeller causes the liquid it contains to move outwards from the centre to beyond the circumference of the impeller. The revolving liquid is impelled by centrifugal effect.

**53. Medium specific speed implies a**
(a) Propeller turbine (b) Francis turbine
(c) Impulse turbine (d) Kaplan turbine

**Ans. (b) :** Following are the range of specific speed of different turbines.

| Turbine | Specific speed |
|---|---|
| Pelton (Impulse Turbine) | 10 -35 |
| Francis | 60-300 |
| Kaplan (propeller turbine) | > 300 |

**54. Run away speed of a turbine indicates**
(a) full load speed
(b) the speed at which the turbine will be damaged
(c) the speed at which the turbine runs freely without load
(d) the maximum permissible speed

**Ans. (c):** Runaway Speed of a water turbine indicates the speed at which the turbine runs freely without load. The turbine will be designed to survive the mechanical forces of this speed.

**55. The unit discharge through a turbine running at 4500 rpm and discharging 400.0 cumec at 120. head is**

(a) 36.52 (b) 715.24
(c) 3.33 (d) 6.00

**Ans. (a) :** Given

$N = 4500$ rpm $\quad Q = 400 m^3/s$

$H = 120$ m

Unit discharge $Q_u = \dfrac{Q}{\sqrt{H}}$

$= \dfrac{400}{\sqrt{120}} = 36.52$

**56. A pelton wheel is**

(a) axial flow impulse turbine
(b) radial flow impulse turbine
(c) tangential flow impulse turbine
(d) radial flow reaction turbine

**Ans. (c) :** Pelton turbine is a tangential flow impulse turbine that transforms the potential energy of water into kinetic energy in a form of a water jet, which impacts and drives a pelton runner.

**57. The ratio of the speed of a pelton wheel and of the water jet for maximum efficiency is**

(a) 1.0 (b) 0.5
(c) 2.0 (d) 1.5

**Ans. (b) :** For maximum efficiency–

- The ratio of bucket speed to jet speed is 0.5.
- The speed ratio varies from 0.43 – 0.48
- The jet ratio is 12.

No. of bucket = 15 + 0.5m

**58. The most susceptible part of a turbine to cavitation is**

(a) penstock
(b) guide blade
(c) inlet of the draft tube
(d) blade inlet

**Ans. (c) :** The most susceptible part of a turbine to cavitations is inlet of the draft tube.

**59. The maximum number of jets that can be employed in an impulse turbine without jet interference is**

(a) four (b) six
(c) eight (d) twelve

**Ans. (b) :** The maximum number of jets that can be employed in an impulse turbine without jet interference is six.

**60. The most suitable type of turbine for low heads and high discharge is**

(a) Pelton (b) Francis
(c) Kaplan (d) Propeller

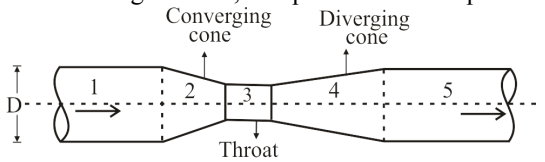
**Ans. (c) :** The most suitable type of turbine for low heads and high discharge is kaplan turbine. It is a propeller type of reaction turbine.

| Turbine | Pelton wheel | Francis | Kaplan |
|---|---|---|---|
| Head | High > 150 m | Medium 30 - 150 m | Low < 30 m |
| Discharge | Low | Medium | High |
| Energy used | Kinetic | Kinetic | Kinetic + Potential |

**61. In a venturimeter; the divergent cone is**

(a) shorter than the convergent cone
(b) equal in length to the convergent cone
(c) longer than the convergent cone
(d) often shorten than the convergent cone

**Ans. (c) :** In a venturimeter, the divergent cone is longer than the convergent cone, this prevents flow separation.




**Venturimeter consists of five parts :–**

1. Cylindrical inlet section
2. Conical convergent section
3. Cylindrical throat
4. Conical divergent outlet
5. Cylindrical outlet section

- Convergent angle :– $(21° \pm 2°)$
- Divergent angle :– $(5°–7°)$

$$Q = \frac{a_1 . a_2}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gh}$$

$$C_d = \frac{Q_{act}}{Q_{th}} = \sqrt{\frac{h - h_f}{h}} \approx 0.98$$

**62. In order to avoid the tendency of separation of fluid flow, the most suitable ratio of the diameter of the throat and the pipe is**

(a) 1/4 to 1/3 (b) 1/3 to 1/2
(c) 1/2 to 3/4 (d) 3/4 to 1.0

**Ans. (b) :** In order to avoid the tendency of separation of fluid flow, the most suitable ratio of the diameter of the throat and the pipe is 1/3 to 1/2.

**Throat of venturimeter:–**

Length = Dia. of inlet (D)

Diameter = generally $\left(\frac{1}{3} - \frac{3}{4}\right) D$

**63. Metacentre of body submerged in a fluid is the point through which the**

(a) body oscillates
(b) buoyant force acts
(c) fluid pressure acts
(d) body mass acts

**Ans. (a) : Metacentre**:– A body submerged in a fluid is the point through which the body oscillates when it is tilted by a small angle.

"It is defined as the point at which the line of action of the force of buoyancy will meet the normal axis of the body when the body is given as small angular displacement

1.0 $GM_{pitching} > GM_{rolling}$

2.0 GM↑ , Stability↑

**64. The flow in pipe is laminar, if the Reynold's number is**

(a) less than 2000 (b) 2000-2500

(c) 2500-3000 (d) more than 3000

**Ans. (a) :** Reynold's number can be defined for pipe flow is

$$Re = \frac{\rho VD}{\mu}$$

Where, ρ = Density of fluid

V = Velocity of fluid in pipe

D = Diameter of pipe

μ = Dynamic viscosity of fluid

If $R_e \le 2000$ – Laminar flow will occur in pipe flow

If $2000 < R_e < 4000$- Transition will occur in pipe flow

If $R_e \ge 4000$ – Turbulent flow will occur in pipe flow

**65. The equation of continuity of flow is based on the principle of the conservation of**

(a) force (b) work

(c) mass (d) momentum

**Ans. (C) :** The equation of continuity of flow is based on the principle of the conservation of mass.

Thus, for a fluid flowing through the pipe at all the cross-section, the quantity of fluid flowing per second is constant.

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Where, ρ = Density of fluid

A = Cross-section area of pipe at two different location

V = Velocity of the pipe at the measured cross-section

**66. The velocity of flow at the centre of a smooth pipe is**

(a) minimum

(b) maximum

(c) of the mean value of flow velocity

(d) the same as that the boundary

**Ans. (b) :** The common application of laminar flow would be in the smooth flow of a viscous liquid through a tube or pipe.

In that case, the velocity of flow varies from zero at the walls to a maximum along the centerline of the vessel.

**67. The flow is called rotational, if the velocity of flow normal to the plane is equal to**

(a) half the angular velocity

(b) the angular velocity

(c) twice the angular velocity

(d) three times the angular velocity

**Ans. (c) :** The flow is called rotational, if the velocity of flow normal to the plane is equal to twice the angular velocity.

Vorticity = 2 × Rotational component.

**68. The equation for irrotational flow was formulated by**

(a) Cauchy-Riemann (b) Reynold

(c) Bernoulli (d) Laplace

**Ans. (d) :** For an Irrotational and incompressible flow, the velocity potential and the stream function each obey Laplace's equation i.e.

$$\nabla^2\phi = 0$$

$$\nabla^2\Psi = 0$$

***69. If the orthogonal component of velocity of flow in a two-dimensional problem vary linearly along the respective axes, the flow is**

(a) circular arcs (b) parabolic arcs

(c) hyperbolic arcs (d) elliptical arcs

**Ans. () :**

**70. The point at which the velocity of particles is zero is known as**

(a) critical point (b) neutral point

(c) sub-critical point (d) stagnation point

**Ans. (d) :** The point at which the velocity of particle's is zero is known as stagnation point. Stagnation points occur at places where the fluid is brought to a state of rest by an object. They usually exist at the surface of objects.

**71. Energy equation is applicable generally to**

(a) steady flow (b) laminar flow

(c) turbulent flow (d) uniform flow

**Ans. (a) :** Energy equation usually applicable to steady flow. The steady flow energy equation (SFEE) is used for open systems to determine the total energy flows.

• This equation holds good for non-uniform flow also.

**72. Piezometers measure very**

(a) low pressure (b) high pressure

(c) high velocity (d) low velocity

**Ans. (a) : Piezometers:–** It is the simplest form of manometer used for measuring low gauge pressure of flowing liquid in terms of the height of liquid column.

**Limitations:–** Gas pressure cannot be measured by means of piezometer because a gas forms no free atmosphere surface. [One and of piezometer is open to the atmosphere.]

• It can't be used when large pressure in the lighten liquid are to be measured.

**73. The stream line function for a two -dimensional flow is given by 2xy; the velocity of flow at a point (2.0, 2.0) is**

(a) 4.00 (b) 2.83

(c) 5.66 (d) 1.41

**Ans. (c) :** Given, Stream function Ψ = 2ry

$$u = -\frac{\partial\Psi}{\partial y} = -2x$$

$$v = -\frac{\partial\Psi}{\partial x} = 2y$$

Velocity $\vec{V} = u\hat{i} + v\hat{j}$

$$\vec{V} = -2x\hat{i} + 2y\hat{j}$$

Velocity at point (2, 2) i.e. at x = 2 and y = 2

$$\vec{V} = -4\hat{i} + 4\hat{j}$$

$$\left|\vec{V}\right| = \sqrt{(4)^2 + (4)^2} = 4\sqrt{2}$$

$$= 5.66$$

**74. The point at which the resultant pressure acts on an immersed body is known as**

(a) centroid (b) centre of depth
(c) centre of pressure (d) metacentre

**Ans. (c) :** The point of action of total hydrostatic force on a submerged surface is called the centre of pressure. It is the point at which the resultant pressure on a immersed surface acts.

**75. The velocity of flow at vena contracta, when the head is 1.0 m is**

(a) 2.21 m/s (b) 1.41 m/s
(c) 1.00 m/s (d) 4.43 m/s

**Ans. (d) :** Velocity at vena contracta $= \sqrt{2gH}$

$$= \sqrt{2 \times 9.61 \times 1}$$

$$= 4.42 \text{ m/s}$$

**76. For fluid flow between two parallel plates, the velocity is**

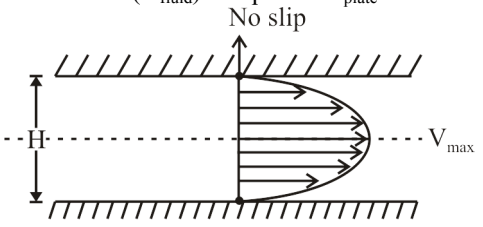
(a) uniform under steady state conditions
(b) maximum at the mid-depth of flow
(c) maximum at the plate boundaries
(d) uniform under turbulent flow conditions

**Ans. (b) :** Velocity between two parallel palates is parabolic with maximum at the centre (mid depth) and minimum (Zero) at the boundary/plate surface due to **No Slip** condition(if plates are stationary)

**No Slip condition** – Velocity of fluid particle at the surface of plate is equal to the velocity of plate due to its viscosity.

If plate is stationary v = 0

If plate velocity is equal to $V_{plate}$ then velocity of fluid particle at surface ($V_{fluid}$) is equal to $V_{plate}$




$\boxed{V_{max} = 1.5\ V_{avg}}$

**77. The path of fluid jet flow from a nozzle exposed to atmosphere is a**

(a) straight line (b) circular arc
(c) parabolic arc (d) hyperbolic arc

**Ans. (c) :** The path of fluid jet flow from a nozzle exposed to atmosphere is parabolic.

**78. The ratio of the maximum height to which is jet inclined at θ rises to the total horizontal path of flow is**

(a) 0.25 sin θ (b) 0.25 cos θ
(c) 0.25 tan θ (d) 0.25 cot θ

**Ans. (c):** The ratio of the maximum height to which a jet inclined at θ rises to its total horizontal path is 0.25 tanθ.

**79. When the whole fluid mass rotates either due to gravity or rotation imparted previously, the flow is known as**

(a) free vortex
(b) forced vortex
(c) rotational vortex
(d) non-potential vortex

**Ans. (a) : Free Vortex:–** No-external torque applied

- Over all flow energy remains constant
- Irrotational flow – (Particles of fluid does not rotate)
- Fluid mass rotates due to the conservation of angular momentum.
- $V \propto \frac{1}{r}$ $\quad \begin{bmatrix} v = \text{velocity} \\ r = \text{radius} \end{bmatrix}$
- The point where velocity is infinite (r = 0) is known as singular point.
- Bernoullie's equation can be applied

**Example:–**

- Whirlpool in a river
- Flow in sink or bathtub or wash basin
- Flow in centrifugal pump casing and
- Flow around the circular bend in a pipe

**Force vortex:–** Tangential velocity ∝ radii

$\boxed{V = r.\omega}$

- Profile – parabolic
- Flow – rotational

$\boxed{\text{Vorticity} = 2\omega}$

**Example:–** Flow inside the impeller of centrifugal pump

**80. Froude number is the ratio of inertia force and**

(a) viscous force (b) surface tension
(c) elastic force (d) gravitational force

**Ans. (d) :** Froude number is the ratio of inertia force and gravitation force.

**Non-Dimensional number:–**

1. Reynold's number $R_e = \frac{\text{inertia force}}{\text{viscous force}} = \frac{\rho VD}{\mu}$

2. Froude's number $F_r = \sqrt{\frac{\text{inertia force}}{\text{gravitational force}}} = \frac{V}{\sqrt{Lg}}$

3. Euler's number $E_u = \sqrt{\frac{\text{inertia force}}{\text{pressure force}}} = \frac{V}{\sqrt{p/\rho}}$

4. Weber's number $W_e = \sqrt{\frac{\text{inertia force}}{\text{Surface tension force}}} = \frac{V}{\sqrt{\sigma/\rho L}}$

5. Mach's number $M = \sqrt{\frac{\text{inertia force}}{\text{elastic force}}} = \frac{V}{C}$

**81. The value of kinetic energy corrective factor for laminar flow through a circular pipe is**

(a) 0.50 (b) 1.25
(c) 1.50 (d) 2.00

**Ans. (d) : Kinetic energy correction factor ($\alpha$)**:– It is defined as the ratio of kinetic energy/second based on actual velocity to the kinetic energy/second based on average velocity.

$$\alpha = \frac{1}{A}\int\left(\frac{u}{V}\right)^3 dA$$

(i) For laminar flow in circular pipe is 2.
(ii) For laminar flow between parallel plates is 1.543.

**Momentum correction factor():-**It is defined as the ratio of momentum of the flow per second based on actual velocity to the momentum of the flow per second based on average velocity across a section

$$\beta = \frac{1}{AV^2}\int u^2.dA$$

For laminar low through a circular pipe. $\beta = \frac{4}{3} = 1.33$

**82. If the velocities of flow due to sudden enlargement in a pipe are $v_1$ and $v_2$, the loss of head is**

(a) $\left(v_2^2 - v_1^2\right)/2g$ (b) $\left(v_1 - v_2\right)^2/2g$
(c) $\left(v_1^2 - v_2^2\right)/2g$ (d) $\left(v_1^2 - v_2^2\right)/2g$

**Ans. (b) :** Loss due to expansion $= \frac{\left(v_1 - v_2\right)^2}{2g} = K\frac{V_1^2}{2g}$

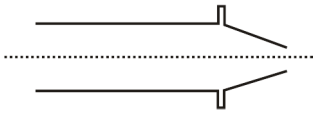
Losses in pipe flow generally expressed in terms of smaller diameter pipe

**83. The shape of fire hose nozzle is**

(a) cylindrical
(b) divergent
(c) convergent
(d) convergent-divergent

**Ans. (c) :** The hose nozzle is cylindrical in shape with continuous decrease in its size i.e. the convergent shape.

Fire Hose Nozzle



The general purpose of a hose nozzle is to supply the flow of water at high velocity and reduced pressure.

**84. For a short cylinderical external mouthpiece, vena contracta occurs at a distance of, from the outlet of the orifice**

(a) diameter of the orifice
(b) half of the diameter
(c) one-third of the diameter
(d) one-fourth of the diameter

**Ans. (d):** One-fourth of the diameter

| Device | Vena contracta |
|---|---|
| Orifices | D/2 |
| Mouthpiece (Internal) | D/2 |
| Mouthpiece (external) | D/4 |

**85. If a venturimeter is inclined from horizontal position its reading**

(a) increases (b) decreases
(c) remains the same (d) changes

**Ans. (c) :** Inclination doesn't cause any effect on the reading of a venturimeter.
Venturimeter is a device used for measuring the rate of flow of a fluid flowing through a pipe.

**86. Centrifugal pumps are adopted for**

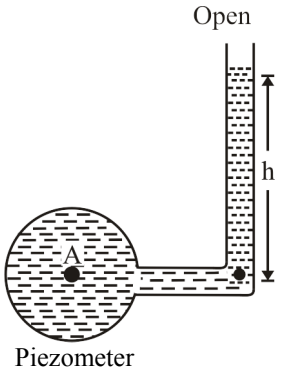
(a) large head and low discharge
(b) low head and low discharge
(c) large head and large discharge
(d) low head and large discharge

**Ans. (d) :** Centrifugal pumps are adopted for low head and large discharge whereas reciprocating pumps are adopted for low discharge and large head.

**87. Piezometers measure**

(a) velocity head (b) static pressure
(c) total pressure (d) discharge

**Ans. (b) :** For measurement of static pressure in a pipe, piezometer is used.




Piezometer

**88. For the maximum discharge, the angle of a triangular notch should be**

(a) $30^o$ (b) $60^o$
(c) $90^o$ (d) $120^o$

**Ans. (c) :** For maximum discharge over a triangular notch, the angle of the notch should be 90º.

**89. An orifice is termed small, if the ratio of water head to the diameter is more than**

(a) 2 (b) 3
(c) 4 (d) 4.5

**Ans. (d) :** Discharge through orifice

$Q = a.C_d\sqrt{2gh}$ $C_d = (0.60 - 0.65)$

**Classification:–**
1. Small orifice H > 5D
2. Large orifice H ≤ 5D

Where,
H = head of water
D = depth of orifice

- Used for measuring rate of flow
- Diameter of orifice is kept half of the diameter of pipe
- Economical and less space required for fitting.
- Head loss is more in orifice meter as compare to venturimeter

**90. In a lock gate, with the gates inclined at $\theta$ to the channel walls, the reactive force between the gates sustaining pressure P is**

(a) P/sin $\theta$ (b) 0.5 P cosec $\theta$
(c) P cos $\theta$ (d) P cosec ($\theta$/2)

**Ans. (b):** The lock gates are provided in navigation chambers to change a water level in a canal or river in navigation. There are two sets of gates, one set on either side of the chamber. In a lock gate, the reaction between the two gates is given by

$$R = \frac{P}{2\sin\theta}$$ Where, P = resultant pressure

R = reaction

$R = 0.5\ P\ \text{coses}\ \theta$

**91. The kern of a section of size 300.0 × 600.0 mm is a**

(a) rectangle of size 50.0 × 100.0 mm
(b) rectangle of size 100.0 × 200.0 mm
(c) parallelogram of size 100.0 × 200.0 mm
(d) parallelogram of size 50.0 × 100.0 mm

**Ans. () :** Kern is that are through which of load passes then there will be no resultant anywhere in the cross section

| | Cross section | Shape of kern |
|---|---|---|
| 1. | Rectangle | Rhombus |
| 2. | I-Section | Rhombus |
| 3. | Square | Rhombus |
| 4. | Solid circular | Circular |
| 5. | Hollow circular | Circular |

**92. The ratio of the polar moment of inertia of a circular section to that of its second moment of area about its diameter is**

(a) 1.0 (b) 2.0
(c) $\sqrt{2}$ (d) 0.50

**Ans. (b) :** Polar moment of inertia of circular section

$$J_{circular} = \frac{\pi}{32}d^4$$

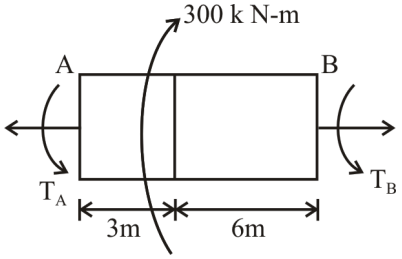
$2^{nd}$ moment of area = moment of inertia $(I) = \frac{\pi}{64}d^4$

$$\frac{J_{circular}}{I_{circular}} = \frac{\frac{\pi}{32}d^4}{\frac{\pi}{64}d^4} = 2$$

**93. A shaft of 9.0 m length is subjected to a torque of 300.0 kNm at a distance of 3.0 m from one end. The maximum torque in the shaft is**

(a) 50.0 kNm (b) 100.0 kNm
(c) 300.0 kNm (d) 200. kNm

**Ans. (d) :**




$$T_A = \frac{300\times 6}{9} = 200\ \text{kN}$$

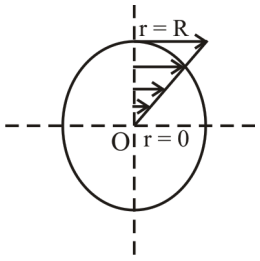
$$T_B = \frac{300\times 3}{9} = 100\ \text{kN}$$

**94. In a circular shaft subjected to torsion the shear stress varies**

(a) linearly from zero at the centre of the section to the maximum value at its periphery
(b) linearly from zero at the centre of the section to the maximum value at the centre of the section
(c) linearly from the minimum value at the centre to the maximum value at the periphery
(d) parabolically along the depth

**Ans. (a) :**
In a circular shaft subjected to torsion the shear stress varies linearly from zero at the centre of the section to the maximum value at its periphery.

**OR**

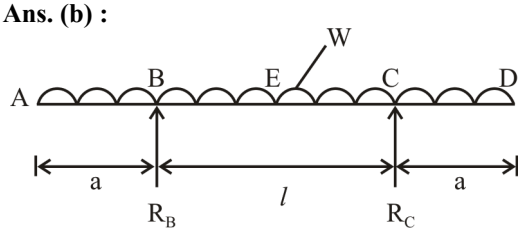



Here the maximum shear stress occurs on the outer surface of the shaft where r = R and at the centre the shear stress is zero.

**95. A beam with length $\ell$ between the simple supports overhange by both sides. The mid-span bending moment in the beam is zero under uniformly distributed load covering the entire span, if**

(a) $\ell = a$ (b) $\ell = 2a$
(c) $\ell = 3a$ (d) $\ell = 4a$

**Ans. (b) :**




$R_B + R_C = W\ (l + 2a)$

$$R_B = R_C = \frac{W(l+2a)}{2}$$

Take moment about point E

$$R_B \times \frac{l}{2} = W\left(\frac{l}{2}+a\right)\left(\frac{l}{2}+a\right)\times\frac{1}{2}$$

$$\frac{W}{2}(l+2a)\times\frac{l}{2} = W\left(\frac{l}{2}+a\right)^2\times\frac{1}{2}$$

$$W\left(\frac{l}{2}+a\right)\times\frac{l}{2} = W\left(\frac{l}{2}+a\right)^2\times\frac{1}{2}$$

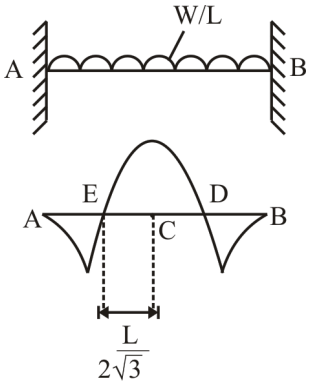
$$l = \frac{l}{2}+a$$

$$l = 2a$$

**96. Points of inflextion in a fixed beam of span $\ell$ with uniformly distributed load over the entire span occur at**

(a) 0.207 *l* from each end
(b) 0.211 *l* from each end
(c) 0.222 *l* from each end
(d) 0.333 *l* from each end

**Ans. (b) :**




Point of contraflexure of beam from mid span is $\frac{L}{2\sqrt{3}}$

Contraflexure from each side.

$$\frac{L}{2}-\frac{L}{2\sqrt{3}} \Rightarrow \frac{L}{2}\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right)$$

$$\Rightarrow L\frac{\sqrt{3}-1}{\sqrt{3}} \text{ from each side.}$$

$$= 0.211L$$

**Point of inflection:-** Where B.M. changes its curveture.
**Point of Contraflexure:-** Where B.M. changes its sign.
**Point of vertual hinge:-** Point in a beam with internal hinge. (B.M = 0 at hinge)
**Note-** every point of contraflexure is a point of inflection but every point of inflection is not a point of contracflexure.

**97. If the depth of a beam is doubled, the maximum deflection for a given loading will be**

(a) halved (b) doubled
(c) reduced by 4 (d) reduced to 1/8

**Ans. (d) :** The central deflection in a simply supported beam due to point load,

$$\Delta_C = \frac{Wl^3}{48EI}, \qquad \Rightarrow \Delta \propto \frac{1}{b}$$

$$\Delta_C \propto \frac{1}{I}, \qquad \Rightarrow \Delta_C \propto l$$

$$\propto \frac{12}{bd^3}$$

$$\propto \frac{1}{d^3}$$

As the depth of the beam is double

$$\frac{(\Delta_C)}{(\Delta_C)} = \left(\frac{d_1}{d_2}\right)^3 = \left(\frac{d_1}{2d_1}\right)^3 = \frac{1}{8}$$

$$\frac{(\Delta_C)_2}{(\Delta_C)_1} = \frac{1}{8}$$

**98. The slenderness ratio of a free standing column of length 3.0 m and 30.0 mm square section is**

(a) 100.6 (b) 200.8
(c) 692.8 (d) 346.4

**Ans. (c) :** Given,

$l = 3$ m

$b = 30$ mm $= 0.03$ m

$$I = \frac{b^4}{12} = \frac{(0.03)^4}{12}$$

$$A = b^2 = (0.03)^2$$

$$\text{Slenderness Ratio } (\lambda) = \frac{L_e}{K}$$

$$\text{Radius of Gyration (K)} = \sqrt{\frac{I_{min}}{A}} = \sqrt{\frac{a^4}{12}\times\frac{1}{a^2}}$$

$$= \sqrt{\frac{30^4}{12\times 30^2}}$$

$$= 8.660$$

$\boxed{L_{eff} = 2L}$ ∵Column is vertically free

$$\lambda = \frac{2\times 3\times 1000}{8.660}$$

$$\lambda = 692.8$$

**Note-** $\text{Slenderness Ratio} = \frac{\text{Effective length}}{\text{Least leteral dimension}}$

This formula is mostly used for RCC.
According to this formula,

$$= \frac{2\times 3000}{30} = 200$$

No option according to this formula.

**99. The point of contraflexure occurs where**

(a) bending moment is minimum
(b) bending moment changes sign
(c) shear force changes sign
(d) shear force is minimum

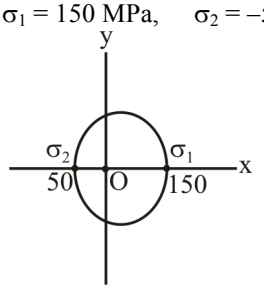
**Ans. (b) :** The point (other than extreme ends of a beam) in a beam at which B.M. is zero, and its sign changes.

**100. The principal stresses at a point are + 150.0 MPa and – 50.0 MPa; the radius of the corresponding Mohr's circle will be**

(a) 100.0 (b) 200.0
(c) 50.0 (d) 75.0

**Ans. (a):** Given,

$\sigma_1 = 150$ MPa, $\sigma_2 = -50$ MPa




Radius of Mohr's circle $(R) = \dfrac{\sigma_1 - \sigma_2}{2} = \dfrac{150 - (-50)}{2}$

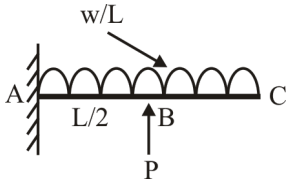
$= 100$ MPa

**101. A cantilever of span $\ell$ subjected to uniformly distributed gravity loading w/m is subjected to an upward force P at a distance $\ell/2$ from the free end. If the deflection at the free end of the beam is zero, ($w\ell/p$) is**

(a) 1.50 (b) 2.00

(c) 0.67 (d) 0.83

**Ans. (d):**




Deflection at free end due to u.d.l

$$\delta_1 = \frac{wl^4}{8EI}$$

Deflection at free end due to vertical load

$$\delta_2 = \frac{P(l/2)3}{3EI} + \frac{P(l/2)^2}{2EI} \times \left(\frac{l}{2}\right)$$

According to the given condition deflection at free end is equals to zero. It means all deflection due to udl = deeflection due to point load 'P' (upward)

$$\frac{Wl^4}{8EI} = \frac{Pl^3}{24EI} + \frac{Pl^3}{16EI}$$

$$\frac{Wl^4}{8EI} = \frac{5Pl^3}{48EI}$$

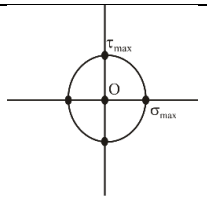
$$\frac{Wl}{P} = \frac{5}{6} = 0.83$$

**102. The ratio of the maximum shear stress to the maximum longitudinal stress in a shaft subjected to uniform torque is**

(a) 2.0 (b) 1.0

(c) 1.5 (d) 0.375

**Ans. (b):** The ratio of maximum shear stress to the maximum longitudinal stress in a shaft subjected to uniform torque is 1




$$\frac{\tau_{max}}{\sigma_{max}} = 1$$

**103. Torsional rigidity of shaft is an index of the**

(a) power transmitted

(b) shear stress sustained

(c) deformability

(d) failure load

**Ans. (c):** From torsion equation

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{l}$$

$$GJ = \frac{TL}{\theta}$$

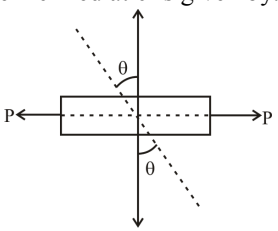
From the above equation we can conclude that torsional rigidity of the shaft is expressed as the torque required to produce a twist of one radian per unit length of shaft. So, GJ is index of deformability.

**104. A shaft of sectional area A is subjected to an axial force P; the normal stress on a plane inclined at θ to the shaft axis is**

(a) $P \cos 2\theta/A$ (b) $P \cos^2\theta/A$

(c) $P \sin 2\theta/A$ (d) $P \sin^2\theta/A$

**Ans. (a):** For a bi-axial state of stress normal stress at any plane inclined at θ is given by.




**Maximum normal stress on oblique plane:-**

$$\left[\sigma_n = \frac{\sigma_x + \sigma_y}{2} + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta\right]$$

**Maximum shear on oblique plane:-**

$$\left[\tau_n = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta\right]$$

Given,

$$\sigma_x = \frac{P}{A}, \qquad \sigma_y = 0$$

$$\Rightarrow \quad \sigma_n = \frac{\frac{P}{A} + 0}{2} + \left(\frac{\frac{P}{A} - 0}{2}\right)\cos 2\theta + \tau_{xy} \quad [\because \tau_{xy} = 0]$$

$$\Rightarrow \quad \sigma_n = \frac{P}{2A} + \frac{P}{2A}\cos 2\theta + 0$$

$$\Rightarrow \quad \sigma_n = \frac{P}{2A}(1+\cos 2\theta) \quad [\because \cos 2\theta = 2\cos^2\theta - 1]$$

$$\boxed{\sigma_n = \frac{P}{A}\cos^2\theta}$$

**105. On a principal plane**

(a) shear stress is zero but not the normal stress
(b) normal stress is zero but not the shear stress
(c) normal and shear stresses are zero
(d) shear stress is maximum

**Ans. (a):** On principal plane shear stress is zero but normal stress is minimum and maximum known as minor principal stress and major principal stress respectively.

**106. In a truss member, in order to avoid shear failure, the shear strength of the material should be at least**

(a) equal to tensile strength
(b) half that in tension
(c) twice the tensile strength
(d) one third that in tension

**Ans. (c):** Twice the tensile strength

- Ductile materials are weak in shear so that to avoid shear failure; the shear strength of ductile material should be at least twice of tensile strength.

**107. A mild steel bar of 250.0 mm length is subjected to an axial force of 20.0 kN. If the Young's modulus is 200.0 GPs, allowable stress is 100.0 MPa and allowable deformation is 0.10 mm, its cross-sections should be at least**

(a) 200.0 $mm^2$ (b) 500.0 $mm^2$
(c) 250.0 $mm^2$ (d) 400.0 $mm^2$

**Ans. (c):** Elongation in the bar when an axial load (P) is applied is given by

$$\Delta = \frac{PL}{AE}$$

Given, L = 250 mm

$P = 20\ kN = 20\times10^3\ N$

$E = 200\ GPa = 200\times10^3\ N/mm^2$

$\Delta = 0.1\ mm$

$\Delta = df$

$$\Delta \le \frac{PL}{AE} \qquad [\because \text{allowable } \Delta = .1\ mm]$$

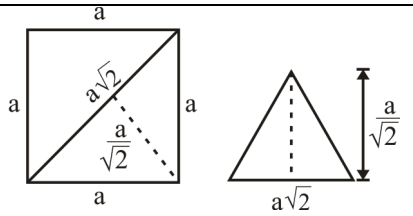
$$0.1 \le \frac{20\times10^3\times250}{A\times200\times10^2}$$

$A \ge 250\ mm^2$

**108. The moment of inertia of square section of size 10.0 mm about a diagonal is**

(a) 1666.7 $mm^4$ (b) 833.33 $mm^4$
(c) 333.33 $mm^4$ (d) 833.33 $mm^4$

**Ans. (b):** Moment of inertia of the square about the diagonal




$$I_{square} = 2\left(I_{triangle}\right)base$$

$$= 2\times\frac{bh^3}{12}$$

$$= 2\times\frac{\left(a\sqrt{2}\right)\left(\frac{a}{\sqrt{2}}\right)^3}{12}$$

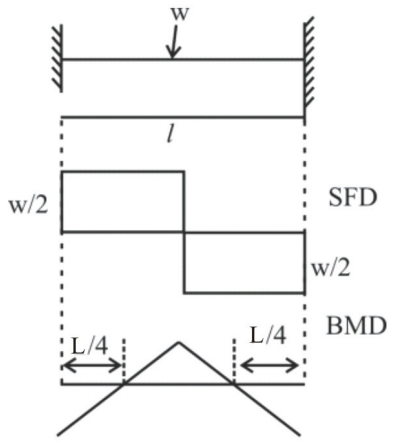
$$= \frac{a^4}{12} \qquad [\because a = 10\ mm]$$

$$= \frac{10^4}{12} = 833.33\ mm^4$$

**109. In a fixed beam with a central point load, the bending moment is zero at**

(a) mid span section
(b) quarter span sections
(c) one third span sections
(d) a distance of 0.211 times the span from the supports

**Ans. (b):**




The bending moment at centre is

$$M_{max} = \frac{wL}{8}$$

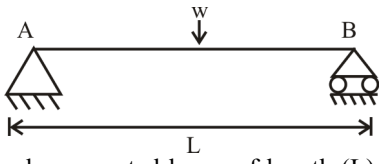
From figure, the zero bending moment in a beam fixed at both ends with a central load w is at quarter span section.

**110. A central concentrated load causes a maximum bending moment M in a simply supported beam. If the same load is distributed uniformly over the entire span, the maximum bending moment is**

(a) M (b) 2M
(c) M/2 (d) M/4

**Ans. (c):** Maximum bending moment for a simply supported beam with central concentrated load w is given by

$$(BM)_{max} = \frac{wL}{4} = M$$



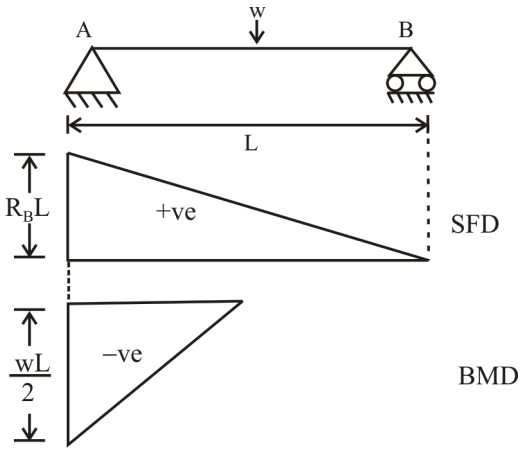

For a simply supported beam of length (L) loaded with uniformly distributed load (w), the maximum bending moment is given by

$$(BM)_{max} = \frac{wL}{8} = \frac{wL}{4\times 2}$$

$$= \frac{M}{2} \qquad \left[\because \frac{wL}{4} = M\right]$$

**111. The maximum bending moment in a propped cantilever subjected to a mid-span concentrated load occurs**
(a) between the prop and mid-span section
(b) under the load
(c) at the fixed end
(d) between the fixed end and mid-span section

**Ans. (c):**




**112. The reaction at a hinged support of a beams in**
(a) normal to the beam axis
(b) normal to supports
(c) a function of beam parameters for the given load
(d) a function of the applied loads

**Ans. (d):** The reaction at a hinged support of a beams is a function of the applied load.

**113. The rate of change of shear force at a sections in a beam is the**
(a) rate of loading at section
(b) bending moment at the section
(c) deflection at the section
(d) loading at the section

**Ans. (a):** The relation between shear force (v) and loading rate (w) is

$$\frac{dV}{dx} = w \text{ (loading rate)}$$

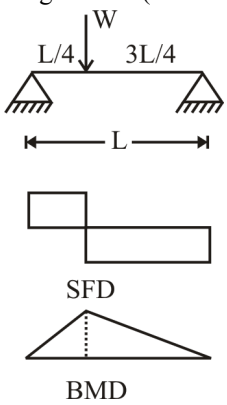
$$\frac{dM}{dx} = V$$

$$\frac{d^2M}{dx^2} = w \text{ (loading rate)}$$

It means the rate of change of shear force at a section in a beam is the loading rate at the section.

**114. A simply supported beam carries a concentrated load at a quarter, span section. Failures in bending occurs at its**
(a) mid-span section
(b) supports section
(c) both the quarter span sections
(d) loaded section

**Ans. (d):** Maximum bending moment in SSB is located at the loading section (load is maximum)




$BM_{max}$ = at loading section.

**115. The ratio of the moment capacity of a square section with its sides horizontal to that of the section with its diagonal horizontal is**
(a) 2.0 (b) 0.577
(c) 1.414 (d) 0.707

**Ans. (c):** Moment capacity $\propto$ Z [section modulus]

$$M = \sigma.Z$$

$$\Rightarrow \quad Z = \frac{I}{y_{max}}$$

$$Z_{square} = \frac{a^3}{6}$$

$$Z_{digonal\ horizontal} = \frac{a^3}{6\sqrt{2}}$$

$$\frac{Z_{square}}{Z_{digonal\ horizontal}} = \frac{\frac{a^3}{6}}{\frac{a^3}{6\sqrt{2}}}$$

$$\frac{Z_{square}}{Z_{digonal\ horizontal}} = \sqrt{2} = 1.414$$

**116. Modulus of section is the ratio of its**

(a) moment of inertia and extreme fibre distance from the neutral axis

(b) moment of inertia and extreme fibre distance from the soffit

(c) bending moment and moment of inertia

(d) moment of inertia and bending moment

**Ans. (a):** Section modulus is the ratio of moment of inertia of the beam cross-section about the neutral axis to the distance of extreme fibre from the neutral axis.

**117. Compression members fail in buckling about**

(a) major principle axis

(b) minor principle axis

(c) neutral axis

(d) centroid axis

**Ans. (b) :** Moment of inertia is inversely proportional to bending moment so column will buckle in the direction of the axis which has least MOI. Buckling takes place about minimum radius of gyration or least moment of inertia.

**118. Torsional rigidity of a shaft of diameter D is proportional to**

(a) $D^4$ (b) $D^3$

(c) $D^2$ (d) D

**Ans. (a):** Torsional rigidity is the amount of torque required to produce unit twist.

$$\frac{T}{J}=\frac{G\times\theta}{L}$$

$$GJ=\frac{TL}{\theta}=\frac{T}{\theta/L}$$

When unit angle of twist is produced in unit length, we have,

$$GJ = T$$

Thus the term GJ may be looked as torque required to produce unit angle of twist in unit length and is called as torsional rigidity.

$$\text{Torsional rigidity} = G\times\frac{\pi}{32}d^4$$

$$\text{Torsional rigidity} \propto d^4$$

**119. The ratio of the strength of a solid shaft and hollow shaft of the same external diameter and internal diameter half of the external diameter is**

(a) 0.94 (b) 1.02

(c) 1.07 (d) 0.53

**Ans. (c):** The ratio of the strength of a solid shaft ($T_s$) and hollow shaft ($T_H$) is given by

$T \propto J$ (Polar moment of inertia)

$$\frac{J_S}{J_H}=\frac{\frac{\pi}{32}D^4}{\frac{\pi}{32}\left(D^4-d^4\right)} \qquad \left[\because d=\left(\frac{D}{2}\right)\right]$$

$$\frac{J_S}{J_H}=\frac{D^4}{\left[D^4-\left(\frac{D}{2}\right)^4\right]}$$

$$\frac{J_S}{J_H}=\frac{D^4}{\frac{15D^4}{16}}$$

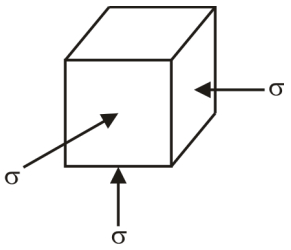
$$\frac{J_S}{J_H}=\frac{16}{15}=1.066$$

**120. A solid cube is subjected to equal compressive stress on all the faces; the ratio of the volumetric strain to linear strain is**

(a) 0.30 (b) 0.50

(c) 1.00 (d) 3.0

**Ans. (d):** If $\in_x$, $\in_y$ and $\in_2$, are the linear strain in x, y and z direction then volumetric strain is given by




$$\in_v = \in_x = \in_y + \in_z$$

Here $\in_x = \in_y = \in_z = \in$

$$\in_v = 3\in$$

**121. A material with large deformations at failure is termed**

(a) brittle (b) elastic

(c) ductile (d) elasto-plastic

**Ans. (c):** The material in which large deformation is possible before the absolute failure or rupture is termed as ductile.

**122. A 30.0 m long wire of 28.5 $mm^2$ sectional area stretches by 25.0 mm under an axial pull of 4.45 kN. Its Young's modulus of elasticity is**

(a) 187.368 GPa

(b) 187.37 GPa

(c) 0.130 GPa

(d) 130 GPa

**Ans. (b):** Given,

L = 30 m = $30\times10^3$ mm

A = 28.5 mm2, P = 4.45 kN

$\Delta$ = 25 mm = 4450 N

$$\Delta=\frac{PL}{AE}$$

$$E=\frac{PL}{A\Delta}=\frac{4450\times30\times10^3}{28.5\times25}$$

$= 187368 \text{ N/mm}^2$

= 187.368 GPa

**123. A mild steel bar of 20.0 mm diameter is enclosed in a brass tube of the same length and internal radius 12.5 mm and external diameter 30.0 mm. If the Young's modulus of elasticity is 200.0 GPa for steel and 100.0 GPa for brass, the ratio of stress developed in the steel bar to the in the brass tube under axial force will be**

(a) 0.50 (b) 1.00
(c) 2.00 (d) 1.45

**Ans. (c):** If bars are enclosed
Strain will same

$$\frac{P_1L_1}{A_1E_1} = \frac{P_2L_2}{A_2E_2} \quad [\because L_1 = L_2]$$

$$\frac{\sigma_1}{E_1} = \frac{\sigma_2}{E_2}$$

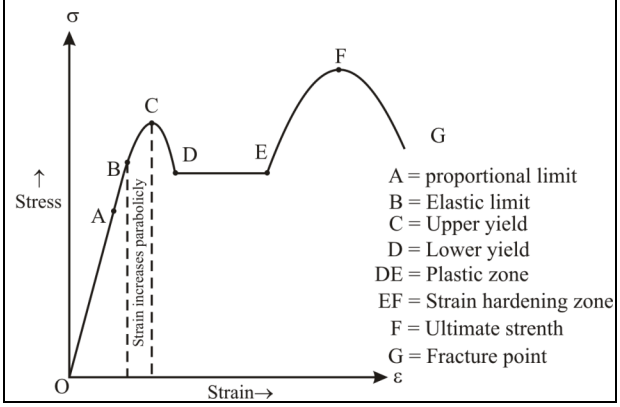
$$\frac{\sigma_1}{\sigma_2} = \frac{E_1}{E_2}$$

$$\frac{\sigma_{steel}}{\sigma_{Brass}} = \frac{E_{steel}}{E_{Brass}} = \frac{200}{100} = 2$$

**124. In a mild steel bar subjected to an axial pull, the strain after elastic limit**

(a) increase slowly (b) increase suddenly
(c) decreases suddenly (d) is zero

**Ans. (b) :**




**125. The strain energy in a bar of 20.0 mm diameter, 1.0 m length, and Young's modulus 200.0 GPa under an axial force of 100.0 kN is**

(a) 0.796 N-mm (b) 0.796 kN-m
(c) 0.080 N-m (d) 0.080 kN-m

**Ans. (d):** Given, d = 20 mm, L = 1000 mm
$E = 200\times10^3$ N/mm$^2$,
$P = 100$ N $= 100\times10^3$ N

$$\text{Strain Energy } U = \frac{P^2L}{2AE}$$

$$= \frac{\left(100\times10^3\right)^2\times1000}{2\times\frac{\pi}{4}\times(20)^2\times200\times10^3}$$

$$= 0.0795\times10^6 \text{ N-mm}$$

$$\approx 0.080 \text{ kN-m}$$

**126. The principal strains in a body are + 800.0, + 400.0 and -1200.0 in micro units. The volumetric strain in micro units is**

(a) 2400.0 (b) 1200.0
(c) zero (d) 4800.0

**Ans. (c):** Volumetric Strain = 800+400–1200
= 0

**127. A steel bar of 20.0 mm diameter and 4.50 m length is heated to $40^0$C above the ambient temperature, and clamped at the end. If the clamps have yielding by 0.50 mm on cooling and the coefficient of thermal expansion is 10.0 × $10^{-6}$/ºC, the strain in the bar is**

(a) 0.0003 tensile (b) 0.0003 compressive
(c) 0.0004 tensile (d) 0.0004 compressive

**Ans. (c):** The change in length due to thermal expansion

$$\Delta L = \alpha\, L\, \Delta T$$

$$\text{Strain } \frac{\Delta L}{L} = \varepsilon = \alpha\, \Delta T$$

$$= 10\times10^{-6}\times40$$

$$= 0.0004 \text{ tensile}$$

**Note-** When bar cools, it exert a pull on clamp so the strain will be tensile.

**128. Maximum bending moment occurs in a simply supported beam, where the**

(a) load is maximum
(b) shear force is maximum
(c) load changes sign
(d) shear force changes sign

**Ans. (d):** Maximum bending moment occurs in a simply supported beam, where the share force changes sign.

**129. A body is subjected to equal tensile stresses on orthogonal planes; the maximum shear stress in the body is**

(a) half the applied stress
(b) twice the applied stress
(c) equal to the applied stress
(d) zero

**Ans. (d) :** Maximum shear stress on oblique plane,

$$\tau_{max} = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$[\because \sigma_x = \sigma_y = \sigma;\ \tau_{xy} = 0]$$

$$\tau_{max} = -\left(\frac{\sigma - \sigma}{2}\right)\sin 2\theta + 0$$

$$\tau_{max} = 0$$

**130. A fixed beam of span '$\ell$' is subjected to a uniformly distributed has w/m over its entire length; the maximum deflection in the beam is**

(a) 0.0026 w$l^3$/(EI) (b) 0.0130 w$l^4$/(EI)
(c) 0.0026 w$l^4$/(EI) (d) 0.0130 w$l^3$/(EI)

**Ans. (c):** The maximum deflection of a fixed beam with uniformly distributed load of w/m over its entire length is given by

$$\Delta = \frac{wl^4}{384EI}$$
$$= 0.0026\ wl^4/EI$$

**131. If the shear force diagram of a simply supported beam is parabolic, the beam is subjected to**

(a) uniformly distributed load
(b) a series of concentrated load
(c) linearly varying load
(d) moments applied at the supports

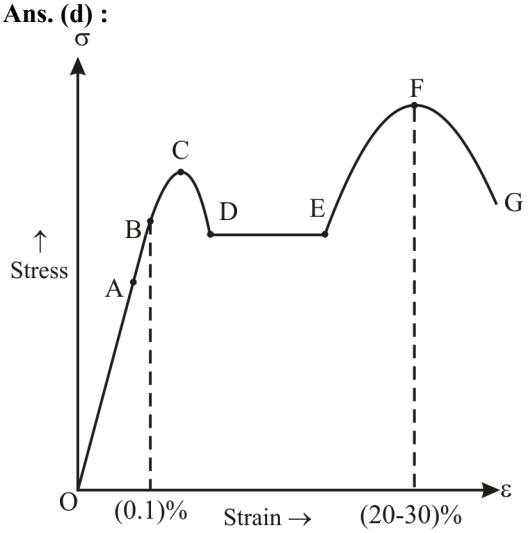
**Ans. (c):**

| Load | Shear force | B.M |
|---|---|---|
| (1). Point | Rectangle | Linear |
| (2). UDL | Linear | Parabolic |
| (3). UVL | Parabolic | Cubic |

**132. The ratio of the strain at fracture to that at yield for mild steel is approximately**

(a) 10.0 (b) 100.0
(c) 1000.0 (d) 250.0

**Ans. (d) :**



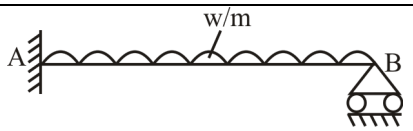

Strain at fracture = 20 – 30% ≈ 25%
Strain at yield point = 0.1%

$$\frac{\varepsilon_f}{\varepsilon_y} = \frac{25\%}{0.1\%} = 250$$

**133. A propped cantilever of span $\ell$ with distributed load w/m over the entire length is subjected to design bending moment**

(a) $-(w\ell^2/8)$ and shear force $w\ell/2$
(b) $+(w\ell^2/2)$ and shear force $3w\ell/8$
(c) $-(w\ell^2/2)$ and shear force $3w\ell/8$
(d) $-(w\ell^2/8)$ and shear force $5w\ell/8$

**Ans. (d):**




Shear force at A = $\frac{5wL}{8}$

Shear force at B = $-\frac{2wL}{8}$

Bending moment at A = $-\frac{wL^2}{8}$

**134. Euler-Bernoulli theory is valid for**

(a) T-section
(b) L-sections
(c) only steel members
(d) only concrete members

**Ans. () :** j

**135. If the rigidity modulus of a material is 40.0 GPa and the Young's modulus is 100.0 GPa, its bulk modulus is**

(a) 200.0 GPa
(b) 66.7 GPa
(c) 140.0 GPa
(d) 70.0 GPa

**Ans. (b):** Given G = 40 GPa , E = 100 GPa, K = ?
Relation between E, G and K is given by

$$E = \frac{9KG}{3K+G}$$
$$100 = \frac{9\times K\times 40}{3K+40}$$
$$K = 66.67\ GPa$$

**136. The law of proportionally of stress and strain in materials was formulated by**

(a) Thomas Young (b) Poisson
(c) Mohr (d) Robert Hooke

**Ans. (d):** Hooke's law states that within the elastic limit the stress applied on a body is directly proportional to strain produced.

**137. The principal planes at a point in body, subjected to external forces, are inclined to each other at**

(a) 180º (b) 60º
(c) 90º (d) 120º

**Ans. (c):** The principal planes at a point in body, subjected to external forces, are inclined to each other at 90º.

**138. Bending moment at a section in a beam is the**

(a) Integral of the load function
(b) Differential of the load function
(c) Integral of the shear force function
(d) Differential of the shear force function

**Ans. (c):** The slope of bending moment diagram gives shean force i.e

$V = dM/dx$

or $M = \int v\,dx$

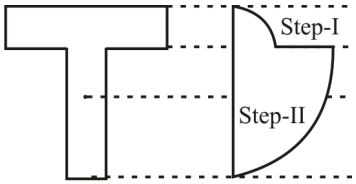
Where V is the shear force and M is the bending moment.

Hence Bending moment at a section in a beam is the integral of shear force function.

**139. The variation of shear stress over the cross section of a beam of T-section is**

(a) Rectangular
(b) Parabolic
(c) Parabolic with a step
(d) Parabolic with two steps

**Ans. (d):** Shear stress variation is parabolic below neutral axis and normal stress is linear below neutral axis.




Shear stress variation

**140. The torsional ridigity of a tube of 200.0 mm diameter and 1.0 mm thickness, and rigidity modulus 100.0 GPa is**

(a) 314.1 GN $mm^2$
(b) 314.1 GN $mm^2$
(c) 157.1 GN $mm^2$
(d) 628.8 GN $mm^2$

**Ans. (d):** d = 200 mm, G = 100 GPa = $100\times10^3$ MPa

r = 100 mm t = 1 mm

A = Perimeter of tube (t) thickness

$= (2\pi r) \times t$

$= 2 \times 3.14 \times 100 \times 1$

$= 628\ mm^2$

$J = A \times r^2 = 628 \times 100^2\ mm^4$

Torsional rigidity of the material is given by $T = GJ$

Here, G is the modulus of rigidity

J is the polar moment inertia

$T = 100 \times 10^3 \times 628 \times (100)^2$

$= 628 \times 10^9\ Nmm^2$

= 628 GN-mm

**141. The assumption that sections plane before bending remain plane after bending in a beam implies that the strains very linearly along the**

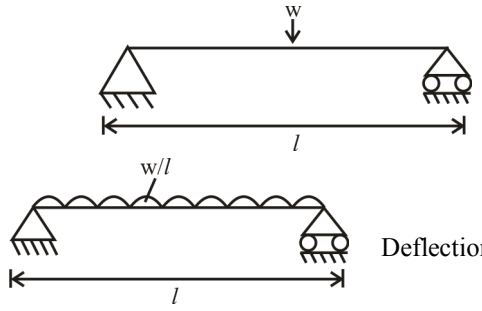
(a) span
(b) depth
(c) width
(d) width and span

**Ans. (b):** Plane section before bending remains plane after bending, this assumption means that strain at any point on cross section is directly proportional to its distance from the neutral axis or strain diagram being linear.

**142. The ratio of the maximum deflections in a simply supported beam subjected to a concentrated load and in that subjected to the same load distributed over the entire span is**

(a) 1.0 (b) 0.625
(c) 0.500 (d) 1.600

**Ans. (d):**




Deflection $= \frac{5wl^4}{48EI}$

Here $w = wl$

Required ratio $= \frac{wl^3/48EI}{wl^3/384EI}$

$= \frac{384}{48\times5}$

$= 1.6$

**143. The theory of simple bending assumes that**

(a) the young's modulus is the same everywhere in the beam
(b) the strains are inversely proportional to the distance from the neutral axis
(c) the strains are proportional to the distance from the beam soffit
(d) the Young's modulus varies linearly with distance from the neutral axis

**Ans. (a) :** Following are the assumptions made in the theory of simple bending

(i) The material of the beam is homogenous and isotropic.

(ii) The beam is subjected to pure bending and the effect of shear is neglected.

(iii) Strain is proportional to the distance from the neutral axis

(iv) Young's modulus assumed to be same in tension as well as compression.

**144. The section modulus of a rectangular section of depth d and breadth b is**

(a) $bd^2/6$ (b) $bd^3/6$
(c) $bd^3/12$ (d) $db^2/6$

**Ans. (a):** Rectangular section

Moment of Inertia

$$I = \frac{bd^3}{12}$$

$$Y_{max} = \frac{d}{2}$$

Section modulus, $z = \frac{I}{y_{max}}$

$$= \frac{bd^3/12}{d/2}$$

$$= \frac{bd^2}{6}$$

**145. The ratio of the moment of inertia of a square plate to that of the circular plate of the same depth is**

(a) 1.563 (b) 1.571

(c) 1.698 (d) 3.141

**Ans. (c):** Moment of inertia of square plate = $\frac{d^4}{12}$

Moment of inertia of circular plate = $\frac{\pi d^4}{64}$

$$\text{Required Ratio} = \frac{d^4/12}{\pi d^4/64}$$

$$= \frac{64}{12\pi}$$

$$= 1.698$$

**146. The radius of gyration of a section to proportional to the square root of the**

(a) depth of the section

(b) inverse of its area

(c) inverse of its second moment of area

(d) inverse of its first moment of area

**Ans. (b):** The radius of gyration of a section is proportional to the square root of the moment of inertia divided by area of the section.

$$\text{Radius of Gyration (K)} = \sqrt{\frac{I}{A}} \Rightarrow K \propto \frac{1}{\sqrt{A}}$$

**147. The maximum deflection in a cantilever of span *l* subjected to a moment M at its free end is**

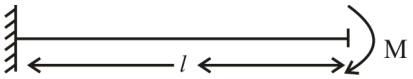
(a) $Ml^2/(2\,EI)$

(b) $Ml/(2\,EI)$

(c) $Ml^2/(3\,EI)$

(d) $Ml/(3\,EI)$

**Ans. (a):**




$$\text{Maximum Deflection} = \frac{Ml^2}{2EI}$$

**148. The shear force, diagram for a cantilever subjected to uniformly distributed load over its entire span is**

(a) rectangular

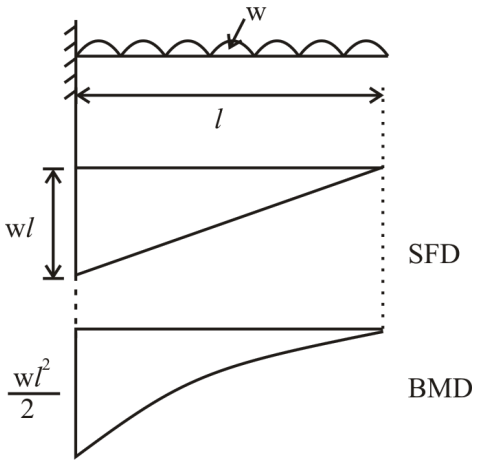
(b) triangular

(c) parabolic

(d) trapezoidal

**Ans. (b):** Cantilever beam carries a

(b) uniformly distributed load




**149. The equivalent length of a column with its fixed ends $\ell$ apart is**

(a) $\ell$ (b) $\ell/2$

(c) $\ell/\sqrt{2}$ (d) $2\ell$

**Ans. (b):**

| End Condition | Effective length |
|---|---|
| Both ends hinged | L |
| Both ends fixed | L/2 |
| One end fixed other free | 2L |
| One end fixed & other hinged | $L/\sqrt{2}$ |

**150. The ratio of the effective length of a column and its minimum radius of gyration is known as**

(a) buckling factor

(b) toughness ratio

(c) slenderness ratio

(d) slenderness factor

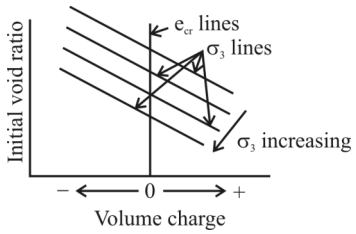
**Ans. (c):** The ratio of the effective length of a column and its minimum radius of gyration known as slenderness ratio.

$$SR\ (\lambda) = \frac{L_{eff}}{\text{Min. Radius of gyration (K)}}$$

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 2004 (P-II)
## Solved Paper

**1. Critical void ratio of soil**
(a) increases with confining pressure
(b) decreases with confining pressure
(c) is not affected by the confining pressure
(d) depends upon the oil type but is independent of confining pressure

**Ans. (b) :** At critical void ratio there is no change in volume. It decreases with confining pressure if increases.




**2. if the porosity of a soil sample is 20 percent, its void ratio is**
(a) 0.167 (b) 0.205
(c) 0.80 (d) 0.25

**Ans. (d) :** Given,
Porosity, n = 20%
Void ratio, $e = \frac{n}{1-n} \Rightarrow e = \frac{0.20}{0.80} = 0.25$

**3. On a retaining wall, the friction between wall and soil**
(a) increase the wall pressure
(b) reduces the wall ressure
(c) induces skin friction
(d) has no effect on wall ressure

**Ans. (a) :**

$$p_a = \frac{1}{2}\gamma H^2 \times \left[ \frac{\sec\theta.\cos[\phi-\theta]}{\sqrt{\cos(\theta+\delta)} + \sqrt{\frac{\sin(\phi+\delta)\sin(\phi-\beta)}{\cos(\beta-\theta)}}} \right]$$

$\beta$ = Surcharge angle
$\theta$ = Angle made by inclined face of the wall with vertical.
$\delta$ = angle of friction between wall and soil.
$\phi$ = friction angle of soil.
• So angle of friction between wall and soil is directly proportional to the wall pressure.

**4. In order to estimate foundation settlement, the required parameter is**
(a) coefficient of consolidation
(b) coefficient of permeability
(c) coefficient of compressibility
(d) time factor

**Ans. (a) :** In order to estimate foundation settlement, the required parameter is coefficient of consolidation. Permeability is the capacity to permit water through its void.

**5. Skirted foundation is preferred over flat footing due to**
(a) increased bearing capacity
(b) compaction of sand due to skirts
(c) reduced compaction of soil due to construction of skirts
(d) improved drainage

**Ans. (a) :** In Skirted foundation there is vertical or inclined wall surrounds one or more sides of soil mass beneath the footing. It is one of the recognized bearing capacity improvement techniques.

**6. Standard penetration test measures**
(a) shear strength of soil
(b) angle of internal friction
(c) density index of sand deposit
(d) depth of hard stratum

**Ans. (a) :** • Standard Penetration Test measures shear strength of soil.
- It is specially used for cohesionless soil. It is used to determine the relative density and the angle of shearing resistance of cohesionless soils it can also be used to determine the unconfined compressive strength of cohesive soil.
- It can also be used to determine the unconfined compressive strength of cohesive soil.

**7. Plate load test results are plotted on log-log scale because**
(a) of large quantities involved
(b) of ease of plotting
(c) it yields a straight line
(d) failure point is indicated clearly

**Ans. (d) :** Plate load test results are plotted on log-log scale so that the failure point is indicated clearly. Plate load test is used to determine ultimate bearing capacity and settlement.

**8. In the case of under- reamed piles, if the swelling index of the soil increases with depth, it is advantageous to increase the**
(a) bulb diameter (b) length of the pile
(c) pile reinforcement (d) all the above

**Ans. (a) :** In the case of undreamed piles, if the swelling index of the soil increases with depth, it is advantageous to increase the bulb diameter.

**9. Negative skin friction is not developed in a pile driven through a**

(a) clay layer undergoing consolidation
(b) cohesive layer below a compressible layer
(c) clay layer with low water table level
(d) soft cohesive layer to a firm stratum

**Ans. (d) :** Negative skin friction is not developed in a pile driven through a soft cohesive layer to a firm stratum.

**10. Allowable settlement has the maximum value in the case of**

(a) isolated footings on sand
(b) isolated footings on clay
(c) rafts on sand
(d) rafts on clay

**Ans. (d) :**

| **Foundation/Footing** | **Total settlement (mm)** |
|---|---|
| Isolated footing on clay | 65 |
| Isolated footing on sand | 40 |
| Raft footing on clay | 65–100 |
| Raft footing on sand | 40–65 |

Allowable settlement has the maximum value in the case of rafts on clay (i.e. 65–100 mm).

**11. Mohr- Coulomb law relates soil properties to the**

(a) shear and normal stresses on failure planes
(b) passive earth pressure on retaining walls
(c) safe bearing capacity
(d) seepage through the soil medium

**Ans. (a) :** Mohr - Coulomb law relates soil properties with shear and normal stresses on failure planes.

$$S = C + \sigma_n \tan \phi$$

Where,

S = shear stress
C = cohesion
$\sigma_n$ = normal stress
$\phi$ = angle of internal friction

**12. Bearing capacity of soil**

(a) increases with the friction angle but decreases with soil density
(b) decrease with friction angle and decreases with flow factor
(c) decreases with flow factor but increases with depth
(d) increases with flow factor

**Ans. (c) :** Bearing capacity of soil decreases with flow factor but increases with depth.

**13. Effective stress on soil**

(a) increases with the degree of saturation
(b) decrease with pore pressure
(c) decreases with foundation depth
(d) increases with pore pressure

**Ans. (b) :** Effective stress on soil decrease with increases in pore pressure. As we know–

$$\text{Effective stress} = \text{Total stress} - \text{Pore water pressure}$$

**14. Rankine's theory of earth pressure is valid for**

(a) water tanks (b) pavements
(c) retaining walls with cohesive soils
(d) rataining walls with non- cohesive soils

**Ans. (d) :** Rankine's earth pressure theory is applied initially cohesionless soil only how ever it was extended to include cohesive soil also later on.

- This theory is also been extended to satisfied partially or fully submerged soil.

**15. Toughness index of soils is the ratio of**

(a) liquid limit to plastic limit
(b) plastic limit to liquid limit
(c) plasticity index to plastic limit
(d) plasticity index to flow index

**Ans. (d) : Toughness Index–** It is defined as the ratio of the plasticity index to the flow index.

$$I_T = \frac{I_P}{I_F}$$

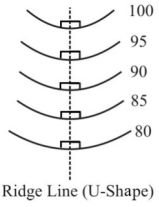
- Toughness index gives us an idea about shear strength of a soil at plastic limit. Two soils, having the same plasticity index value possess toughness which vary in inverse proportion to their flow indices.
- Toughness index generally lies in the range of 0 to 3 for most soils. When $I_T$ is less than '1', the soil is friable (easily crushed) at the plastic limit.
- It indicates the loss of shear strength with increase in moisture.

**16. A contour and ridge line intersect at**

(a) $30^0$ (b) $45^0$
(c) $60^0$ (d) $90^0$

**Ans. (d) :** • Angle of intersection of contour line and ridge line be 90º.

- Pointing towards lower elevation
- U shape contour.

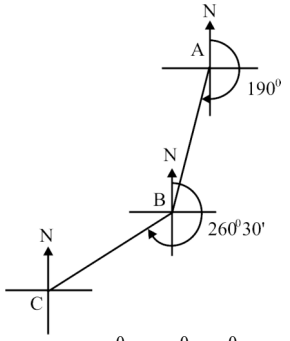


Ridge Line (U-Shape)

**17. The bearing of a line AB is $190^0$, and that of BC is $260^0 30'$; the included angel ABC is**

(a) $210^0 15'$ (b) $59^0 30'$
(c) $70^0 30'$ (d) $90^0 30'$

**Ans. (*) :**




Included angle ABC = $10^0 + 90^0 + 9^0 30' = 109^0 30'$

**18. If s is the sum of the three angles of a spherical triangel, the spherical excess equals.**

(a) $s- 90^0$ (b) $s- 180^0$

(c) $180^0 -s$ (d) $270^0 -s$

**Ans. (b) : Spherical excess–** The spherical excess of a spherical triangle is the value by which the sum of the three angles of the triangle exceeds $180^0$.

Thus, spherical excess, $E = (A + B + C – 180^0)$

where, $A + B + C = s$

$$\boxed{E = s – 180^0}$$

**19. The moon rotates round the sun once in very**

(a) 29 days (b) 29.35 days

(c) 29.53 days (d) 30 days

**Ans. (c) :** The moon takes about one month to orbit earth (27.3 day to completed a revolution, but 29.5 day to change from new moon to new moon).

- It takes a little more than two additional days for sunlight to hit the moon in the same way it did on day zero. That's why it takes 29.5 days to get from new moon to new moon.

**20. If H is the flying height of an object above the mean ground level, and f is the principal distance of a vertical photograph, the mean scale of the photograph is**

(a) H.f (b) H/f

(c) f/H (d) f+H

**Ans. (c) :** If H is the flying height of an object above the mean ground level, and f is the principal distance of a vertical photograph, the mean scale of the photograph is f/H.

Scale of horizontal photograph

$$\text{scale} = \frac{\text{photo distance}}{\text{ground distance}}$$

$$\text{scale} = \frac{x}{x} = \frac{y}{Y} = \frac{f}{H}$$

Scale of vertical photograph

$$\text{Scale} = \frac{x}{X} = \frac{y}{Y} = \frac{f}{H - h}$$

A = Co-ordinate (X, Y) on photo plate

a = (x,y) on negative plate i.e, image plate.

f = focal length

H = Flying height

n = Elevation of ground level

**21. Under Indian conditions, water bound Macadam roads are considered suitable, if the daily traffic does not exceed**

(a) 2000 t (b) 2500 t

(c) 3000 t (d) 3500 t

**Ans. (a) :** Water bound macadam surface is suitable for about 1000 tonnes of traffic per lane per day.

**22. The convexity provided to the carriage way between the crown and the edge of the pavement is known as**

(a) super elevation (b) camber

(c) crown (d) chord

**Ans. (b) :** The convexity is provided to the carriage way between the crown and the edge of the pavement, is known as camber.

Road are not flat but are designed and constructed to assist in the drainage of water.

**23. The polar axis of earth is about**

(a) 12713 168 m (b) 15 286 563 m

(c) 9 989 366 m (d) 131 236 786 m

**Ans. (a) :** The earth is a oblate spheroid of revolutions, the length of its polar axis (12,713,800 metres) being somewhat less than that of its equatorial axis (12,756,750 meters). Thus, the polar axis is shorter than equatorial axis by 42.95 km. Relative to the diameter of the earth this less than 0.34 percent.

**24. If the design speed of vehicles is 100.0 kmph, the change of radial acceleration in $m/s^3$ is**

(a) 0.38 (b) 0.35

(c) 0.42 (d) 0.44

**Ans. (d) :** Radial acceleration (c) = $\frac{80}{75 + v}$

$$C = \frac{80}{75 + 100} = 0.45\ m/s^3 \approx 0.44\ m/s^3$$

**25. The maximum super elevation for hill road should not exceed**

(a) 0.005 (b) 0.050

(c) 0.800 (d) 0.100

**Ans. (d) :** Max SE limit as per IRC

| **(Super elevation) maximum** | **Terrain** |
|---|---|
| $\frac{1}{15} = 7\%$ | Plain and rolling terrain |
| $\frac{1}{10} = 10\%$ | For hill roads |
| $\frac{1}{25} = 4\%$ | For urban road. |

**26. The area of a plot to be surveyed in about 1200.0 $km^2$; the most suitable method is**

(a) compass (b) tachometric

(c) geodetic (d) plane table

**Ans. (c) : Compass–** In compass surveying horizontal angles are measured with the help of a magnetic compass.

**Tacheometric–** In tacheometric surveying horizontal angles, horizontal distances and vertical distance are measured with tacheometer.

**Geodetic survey–** Large areas of earth's surface are involved and the curvature of earth is taken into account.

**Plane table–** Graphical method, plotting and measurement both taken same time.

**27. The length of the sewer line between A and B laid at a gradient 1 in 125 is 250.0m. If the reduced level of the invert at A is 205.75 m and the height of the sight rail at B is**

(a) 280.75 m (b) 202.75 m

(c) 206.75 m (d) 211.75 m

**Ans. (c) :** Given, length between A and B = 250 m
Gradient = 1 in 125
RL of the invert at A = 205.50 m
Height of boning rod = 3 m

$$\text{RL of invert at B} = 205.75 - \left(\frac{250}{125}\right)$$

$$= 203.75$$

RL of sight rail height at B = 203.75 + 3 = 206.75

**28. The perpendicular offset from the tangent of a 500.0m curve at a distance of 20.0 m from the commencement of the curve is approximately**

(a) 0.20 m (b) 0.40 m
(c) 0.22 m (d) 0.30 m

**Ans. (*) :** Perpendicular offset $(y) = \frac{L^2}{6R}$

L= 20 m
R= 500m

$$y = \frac{L^2}{6R}$$

$$= \frac{20^2}{6 \times 500}$$

=0.133m

**29. The super elevation required for a curve of 100.0 m radius and a design speed of 50.0 kmph on a hill road is**

(a) 0.11 m (b) 0.08 m
(c) 0.22 m (d) 0.30 m

**Ans. (a) :** Given,
Design speed (v) = 50 kmph
Radius (R) = 100 m

$$e = \frac{v^2}{225R} = \frac{(50)^2}{225 \times 100} = 0.11 \text{ m}$$

**30. Radiation intersection traversing and resection are the methods adopted in**

(a) levelling (b) plane table survey
(c) theodolite survey (d) geodetic survey

**Ans. (b) : Plane table survey–** Plane table surveying is a graphical method of surveying in which field work and plotting are done simultaneously.
The method of surveying with a plane table include–
1. Radiation 2. Traversing 3. Intersection
4. Resection.

**31. The maximum diameter of bar in a 200.0 mm thick concrete slab is**

(a) 12.0 mm (b) 16.0 mm
(c) 20.0 mm (d) 25.0 mm

**Ans. (d) :** Maximum diameter used in slab should not exceed 1/8 of the total thickness of slab. If slab thickness is 200 mm, maximum diameter which can be use is $\frac{200}{8} = 25$ mm .

**32. The breakage of wires in any pre-stressed concrete member during tensioning as a percentage of total number of wires should not exceed.**

(a) 1.0 (b) 1.5
(c) 2.0 (d) 2.5

**Ans. (d) :** The breakage of wires in any one prestressed concrete member shall not exceed 2.5 percent during tensioning. If the breakages are more than 2.5 percent. It shall be examined and corrective actions taken.

**33. The minimum compressive strength of 100 mm cubes of grout in prestressing tendons should not be less than**

(a) 17.0 MPa at 7 days
(b) 17.0 MPa at 28 days
(c) 22.0 MPa at 7 days
(d) 22.0 MPa at 28 days

**Ans. (a) :** The water-cement ratio shall be low as possible consistent with workability and shall not exceed 0.45, then the compressive strength of 100 mm cubes of the grout shall be not less than 170 kg/cm$^2$ or 17 MPa at 7 days.

**34. The minimum longitudinal reinforcement in post- tensioned concrete units as a percentage of cross sectional area should not be less than**

(a) 0.12 (b) 0.15
(c) 0.20 (d) 0.25

**Ans. (c) :** A minimum longitudinal reinforcement of 0.2 percent of the total concrete area shall be provided in all cases except in the case of pretensioned units of small section. This reinforcement may be reduced to 0.15 percent. In the case of high yield strength deformed reinforcement.

**35. The minimum number of bars in a circular R.C. column is**

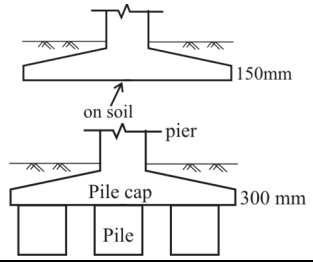
(a) 4 (b) 6
(c) 8 (d) 10

**Ans. (b) :** The minimum number of bars in column–
For circular-6 Nos.
For square-4 Nos.

**36. The thickness of a footing at the edge should not be less than**

(a) 100.0 mm (b) 150.0 mm
(c) 200.0 mm (d) 250.0 mm

**Ans. (b) :** The minimum thickness at the edge of reinforced and plain concrete footings shall be at least 150 mm for footings on soils and at least 300 mm above the top of piles for footings on piles, as per the IS 456-2000 clause No. 34.12




**37. Prestressing of concrete members is effective in**

(a) columns (b) walls
(c) railway sleepers (d) struts

**Ans. (c) :** Pre-stressed concrete member posses better resistance to shear force due to effect of compressive stresses pressure of eccentric cable profile use of high strength concrete and freedom from cracks, contribute to improve durability under aggressive environmental conditions. It is effective in railways sleeper.

**38. The effective flange width of T- beams spaced at 3.5 m over 15.0m span with 100.0 mm thick flange and 300.0 mm wide webs is**

(a) 3.50 m (b) 3.40 m
(c) 4.65 m (d) 2.90 m

**Ans. (b) :** Given,
Thickness of flange ($D_f$) = 100 mm
breadth of flange ($b_w$) = 300 mm
$l_0$ = 15 m = 15000 mm

$$b_f = \frac{l_0}{6} + b_w + 6D_f = \frac{15000}{6} + 300 + 6\times 100$$

$$= \frac{15000}{6} + 900$$

$b_f$ = 3.4 m

**39. Working stress design of R.C. members assumes**

(a) isotropic material
(b) parabolic stress strain curve
(c) linear elastic material
(d) homogeneous and isotopic material

**Ans. (c) :** The WSM follow the elastic theory of reinforced concrete section. In simple words. It follows the Hook's law. The WSM will follow the assumption that the structural material will behave linearly elastic till a point.

**Features of WSM–**

- In this method, the stress will be limited to this point value so that the structure will remain same or in other words the stress are limited under this safe stress value. This stress or load value is called as the service or working loads.
- The permissible stress for steel and concrete can be obtained by dividing the characteristic strength of the material by the factor of safety.
- This is done to restrict the working stress acting on the material under the action of service loads to be within the linear elastic phase of material.

**40. The maximum permissible compressive strain in an R.C. column when the minimum strain at one edge is 0.0010 should not be**

(a) less than 0.00350
(b) more than 0.00250
(c) less than 0.00275
(d) more than 0.00275

**Ans. (a) :** • The maximum permissible compressive strain in an RCC column should not be less than 0.0035, when the minimum strain at one edge is 0.002.

• Maximum compressive strain in concrete under axial loading at the limit state of collapse in compressible is specified of 0.0035 by the code.

**41. Mobilisation advance is given to a contractor**

(a) for payment of advance to staff
(b) for purchase of equipment
(c) for commencement of work as a loan
(d) for commencement of work on finalisation of contract

**Ans. (d)** Mobilization advance is given to a contractor physical and manpower resource for transfer to a construction site until the completion of the contracts.

• In civil construction projects it is known as mobilization advance.

• It is 10% to 15% of total contract value.

• The prerequisite for the issue of advance is that contractor has to provide a guarantee in the shape of bank or insurance equal to the amount being issued to the contractor.

• M.C. is deducted from bill's in equal installments.

**42. A sinking fund is**

(a) raised to meet maintenance costs
(b) the sum payable to the municipal corporation
(c) the fund for rebuilding a structure when its useful life is over
(d) The amount kept in reserve for providing additional structures and modifications

**Ans. (c) :** Sinking fund is the fund for rebuilding a structure when its economic life is over.

**Sinking Fund–**A certain amount of gross rent is set aside annually as sinking fund to accumulate the total cost of construction when the life of the building is over. This annual sinking fund is also taken as outgoings

$$(S) = \frac{S_n \times R}{(1+R)^n - 1}$$

$S_n$ = Sinking fund to be accumulated in 'n' years
R = Rate of interest in decimal
S = Yearly installment of sinking fund

**43. A logic splitter is used in critical path net work when**

(a) each of the two activities follow each of the two prior activities
(b) each of the two activities which, in turn, follow two other separate activities each individually also follow yet a third activity
(c) two activities follow one activity, and one of them follows yet another activity
(d) two concurrent activities are followed and preceded by the same activity

**Ans. (d) :** A logic splitter is used in critical path net work when two concurrent activities are followed and preceded by the same activity.

**44. A CPM network comprises**

(a) activity sequencing without resource availability
(b) least direct cost durations for activities
(c) least total cost durations for activities
(d) activity sequencing considering resource availability for each activity separately

**Ans. (c) :** A CPM network comprises least total cost durations for activities. In CPM cost is not proportional to time. Hence time cost models are prepared to find the minimum project cost and corresponding time.

**45. An order book is maintained at site to record**

(a) instructions to engineer
(b) measurements of construction work
(c) requisition of equipment
(d) indents for materials

**Ans. (a) :** In civil engineering works on site should have site order book.

- It has technical terms and prescribed work recorded in formats that is followed at site.
- It has the comments and prescriptions of higher officials and authorities.

**46. Crashing of the project means**

(a) reducing the time of project by opening more resources
(b) reducing the cost of project by delaying the time of completion
(c) reducing the project size
(d) abandoning the project

**Ans. (a) : Crashing :** The process of reduction of the total project duration along the longest path ( time wise) of the network (away the critical path to obtain the optimum project cost & optimum duration).

**47. Pertaining to burst event, the latest permissible occurrence time is the**

(a) Maximum LST of the bursting activities
(b) Maximum EST of the bursting activities
(c) Minimum LST of the bursting activities
(d) Minimum EST of the bursting activities

**Ans. (a) : Latest start time (LST)–**

- Latest start time for an activity is the latest time by which an activity can be start without delaying the completion of the project.
- For no-delay condition to be fulfilled it should be naturally equal to the latest finish time (LFT) minus the activity duration.

LST=LFT–Activity duration.

**48. The time by which the completion of an activity can be delayed without affecting the start of succeeding activities is known ad**

(a) total float
(b) interfering float
(c) independent float
(d) free float

**Ans. (c) Total float**– The amount of time that can be delayed from its early start date without delaying the project finish date.

**Free float**– The amount of time that an activity can be delayed without delaying the early start date of any successor activity.

- Subsequent tasks ('free float')
- Project completion date ('total float')

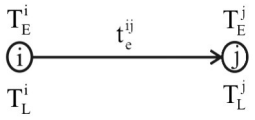
**Free Float ($F_F$)–** $\boxed{F_F = T_E^j - T_E^i - t_e^{ij}}$ or $\boxed{F_F = F_T - S_j}$

where. $S_j$ = Head event slack

$\boxed{F_T = T_L^j - T_E^i - t_e^{ij}} - F_T \rightarrow$ total float

$T_E^i$ = earliest start time

$T_L^j$ = Latest finish time




**49. An activity of a project requires**

(a) events (b) resources
(c) time and resources (d) equipment

**Ans. (c) :** An activity of a project requires time and resources.

- A task or activity represents an identical unit of work that is performed by a single individual or group of workers, with the clear beginning and end.
- For example, in constructing a house, the project is divided into task and sub-tasks such as making the house, electric connection, painting, etc.

**50. Bar charts are suitable for**

(a) major projects only
(b) minor projects only
(c) medium projects only
(d) any project

**Ans. (b) : Bar charts :**

- Bar charts are suitable for minor projects only.
- It is a graphical representation of activity v/s time.
- It is also being referred as Gantt chart.

**51. Symon's rain gauge is**

(a) recording type
(b) non- recording type
(c) indicative type
(d) cumulative recording type

**Ans. (b) :** Symon's rain-gauge is most common type of non-automatic rain-gauge and is used by metrological department of Government of India. It consider of cylindrical vessel 127 mm (or 5") in diameter with a base enlarged to 210mm diameter.

**52. with continuous rains, the infiltration capacity of soil**

(a) decreases (b) increases
(c) becomes steady (d) becomes uniform

**Ans. (d) :** Infiltration is the movement of water through soil into the ground water table. The max rate at which can be observed by soil is called ' Infiltration capacity". This capacity is minimum at the beginning of the storm and it generally increases so as to attain a constant value.

**53. Large run off is produced by drainage basin of the shape of**

(a) fan (b) cone
(c) fern (d) ball

**Ans. (a):** The shape of the basin influences the time taken for water from the remote parts of the catchment to arrive at the outlet. Thus, the occurrence of the peak and hence the shape of the hydrograph are affected by the basin shape. Fan-shaped, i.e., nearly semi-circular shaped catchment gives high peak and narrow hydrographs while elongated catchments give broad and low peaked hydrographs.

**54. Rehbock's empirical relationship to obtain discharge is valid for**

(a) rectangular weirs (b) triangular weirs
(c) cippoletti weirs (d) parabolic weirs

**Ans. (a) :** Rehbock's empirical relationship to obtain discharge is valid for rectangular weirs.

**55. Dicken's formula for maximum follow discharge Q a river basin of area A is of the form $Q= CA^n$, where C is a constant and n is**

(a) 0.50 (b) 0.66
(c) 1.10 (d) 0.75

**Ans. (d) :** Dicken's formula is suitable for north and central India, and is given as –

$$Q_p = C_D.A^{3/4}$$

$Q_p$ in $m^3$/sec
A is area in $km^2$

$$n = \frac{3}{4}$$

$C_D$ = Dicken's constant lies between 6 to 30.

**56. When the rate of demand is lower than the natural stream flow, the mass curve will be**

(a) flatter than the demand line
(b) equal to the demand line
(c) steeper than the demand line
(d) fluctuating about the demand line

**Ans. (c) :** If the rate of demand is lower than the natural stream flow, the mass curve will be steeper than the demand line.

**57. River Ganga is a/an**

(a) normal river (b) perennial river
(c) is pluvial river (d) torrential river

**Ans. (b) : River Ganga–** This is himalyan river

- Take off from himalyas and flow through alluvial plains.
- These river are almost pereminal and can give dependable yields throughout the year.
- These river carry huge sediment along with ganga there are many other himalyan river as– Indus, Jhelum, Chenab, Ravi, Beas, Sutlet, Kosi, Brahmaputra, etc.
- They derive their water from rain during monsoon and winter and from melting snow during summer.

**58. A hydrograph produced by continuous effective rainfall at constant rate for indefinite period is known as**

(a) H- hydrograph (b) B- hydrograph
(c) S- hydrograph (d) M- hydrograph

**Ans. (c) :** The S-curve, also known as S-hydrograph is a hydrograph produced by a continuous effective rainfall at a constant rate for an infinite period. It is a curve obtained by summation of an infinite series of D-hr unit hydrograph spaced D-hr apart.

**59. An instantaneous unit hydrograph is formed when the duration of effective precipitation is**

(a) infinitesimal (b) exponential
(c) infinite (d) asymptotic

**Ans. (a) :** If time gets reduces for same rainfall the intensity will go on increasing. It will be maximum when $D \to 0$, hydrograph for which rainfall ratio is very small & intensity is very high. Then such hydrograph is known as "instantaneous unit hydrograph.

**60. The requirement of water is the highest for**

(a) livestock (b) irrigation
(c) domestic needs (d) toughness modulus

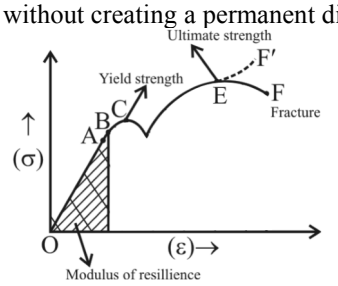
**Ans. (b) :** • The requirement of water is highest for irrigational need.

- The 70% of total fresh water goes for the irrigation apart from drinking, sanitary, industrial and other uses.

**61. The maximum strain energy stored in a material at its elastic limit per unit volume is known as**

(a) resilience (b) proof resilience
(c) modulus of resilience (d) toughness modules

**Ans. (b) : Proof resilience :** It is defined as the maximum strain energy that can be absorbed upto the elastic limit, without creating a permanent distortion.




Area of shaded portion = modulus of resilience

$$\sigma = \frac{\sigma_B^2}{2E}$$

= strain energy stored per unit volume upto elastic limit
Proof resilience = Modulus of resilience × volume

**62. The stiffness of a beam with the far end simply supported is**

(a) 4 EI/*l* (b) 3 EI/*l*
(c) 0.75 EI/*l* (d) EI/*l*

**Ans. (b) :**

- Far end hinged : Beam hinged at both ends then stiffness (K) = $\frac{3EI}{L}$
- Far end fixed : Beam hinged at near and fixed at far

$$K = \frac{4EI}{L}$$

Where, E = young modulus, I = moment of inertia
L = length of member

**63. In a linear elastic structural system**

(a) stiffness is directly proportional to its flexibility

(b) stiffness is inversely proportional to its flexibility

(c) stiffiness in equal to its flexibility

(d) stiffness and flexibility are not related

**Ans. (b) : Flexibility–**The flexibility of a structure is defined as the displacement caused by a unit force.

$$f = \frac{\delta}{P}$$

**Stiffness–**The stiffness is defined as the force required to produce a unit displacement.

$$K = \frac{1}{f} = \frac{P}{\delta}$$

So, Stiffness is inversely proportional to flexibility.

**64. A beam with both ends fixed and a shear hinge at the midspan section can be analysed by**

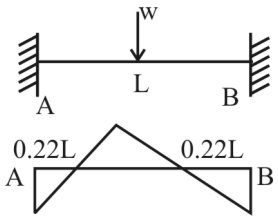
(a) Kani's method

(b) Hardy cross method

(c) Slop- deflection method

(d) Clapeyron's method

**Ans. (d) :** The shear hinge problem can be solved by the clapeyron's theorem.




If a fixed beam is subjected to point load at mid then the two contraflexure points are arise at the distance of 0.221 L from both ends.

•Clapeyron's method is used to solve continuous beam.

**65. The maximum sagging moment in a propped cantilever under uniformly distributed load occurs at a distance of**

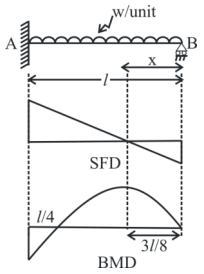
(a) 3*l*/8 from the simple support

(b) 3*l*/8 from the fixed support

(c) 5*l*/8 from the simple support

(d) 5*l*/8 from the fixed support

**Ans. (a) :**




Deflection, $\delta_B = 0$

$$\frac{wl^4}{8EI} - \frac{R_B l^3}{3EI} = 0$$

$$\frac{R_B l^3}{3EI} = \frac{wl^4}{8EI}$$

$$R_B = \frac{3}{8}wl$$

$$(SF)_{at\ x-x} = R_B - wx$$

$$= \frac{3wl}{8} - wx$$

Maximum sagging moment is the point where shear force is zero.

$$\therefore \quad \frac{3wl}{8} - wx = 0$$

$$\frac{3wl}{8} = wx$$

$$\boxed{x = \frac{3}{8}l}$$

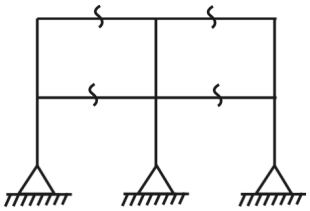
Therefore, maximum sagging moment at $\frac{3}{8}l$ from the simple support.

**66. The degree of redundancy of a two storey two bay rigid jointed frame with hinged supports is**

(a) 12 (b) 9

(c) 10 (d) 2

**Ans. (b) :**



Degree of redundancy or indeterminacy for frame is

$D_S = 3C - R - \sum(m' - 1)$

C= Number of loop

R= Number of constraint reaction to make joint fix

m' = Number of member meeting at hinge

C = 4

R = 3

m = 0

$D_S = 12-3$

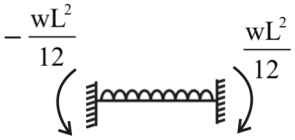
$D_S = 9$

**67. A continuous beam of two equal spans *l*, and end supports fixed supports uniformly distributed load w/m over one span; the value of numerically largest support moment is**

(a) $wl^2/12$ (b) $5wl^2/48$

(c) $3wl^2/8$ (d) $wl^2/8$

**Ans. (a) :** From the standard results




Fixed end moment under UDL = $\frac{wL^2}{12}$

**68. Shape factor of a section in plastic analysis is the ratio of**

(a) cross sectional area to its perimeter

(b) plastic and elastic moment capacities

(c) elastic moment capacity and yield stress
(d) plastic moment capacity and yield stress

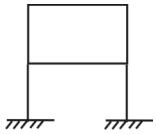
**Ans. (b) :** In plastic analysis the shape factor may be defined as the ratio of the plastic moment and the yield moment of the section.

$$S = \frac{M_P}{M_y} = \frac{f_y Z_p}{f_y Z_e} = \frac{Z_p}{Z_e}$$

Where,
$Z_p$ = Plastic section modulus
$Z_e$ = Elastic section modulus

**69. A single bay two storey rigid jointed frame may develop at collapse a maximum of**
(a) 7 hinges (b) 4 hinges
(c) 3 hinges (d) 8 hinges

**Ans. (a) :**



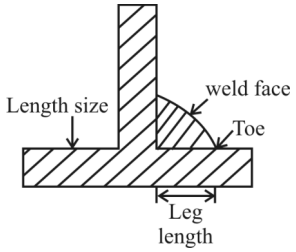
R = 6, m = 6 J = 6
$D_S = (3m+R) - 3J = 18 + 6 - 18 = 6$
Total plastic hinges = $D_s + 1 = 6 + 1 = 7$

**70. Engesser's theorem is applicable to structures with**
(a) geometrical non- linearity only
(b) linear stress- strain behavior
(c) material non-linearity only
(d) any non- linearity

**Ans. (b) : Engesser's Theorem**– Engesser theorem states that the partial derivative the complementary strain energy with respect to any load acting on a linear or non-linear elastic structure is equal to the displacement of the load point of the structure in the direction of the load acting at that point.

**71. The effective length of a fillet weld is taken as the**
(a) length of the weld
(b) length of the weld less twice the weld size
(c) length of the weld less the weld size
(d) length of the weld less three times the weld size

**Ans. (b) :** Effective length of fillet weld,
$\ell$ = over all length – 2s
s = size of weld
• Effective length of fillet weld is the length for which the specified size and throat thickness of weld exist.
• Effective throat thickness = k × S

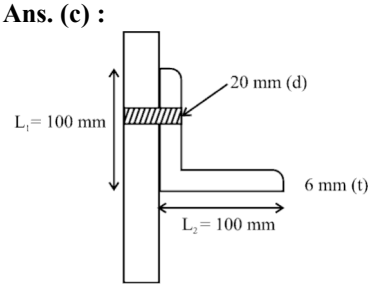



**72. In a welded joint, the side fillet is subjected to**
(a) shear stresses (b) tensile stresses
(c) bending stresses (d) compressive stresses

**Ans. (a) : Side fillet weld–** In a side convex fillet weld subjected to shear stress along the weld failure occurs down the throat of the weld.
A fillet weld is placed on the sides or end of the base metal and it is subjected to shear along with tension or compression and usually bending.

**73. The effective area of ISA 100×100× 6 connected to a gusset plate through one leg with 20 mm diameter rivets is**
(a) 1044.0 $mm^2$ (b) 936.7 $mm^2$
(c) 885.3 $mm^2$ (d) 1164.0 $mm^2$

**Ans. (c) :**




Net effective area
$A_{net} = A_1 + KA_2$

$$K = \frac{3A_1}{3A_1 + A_2}$$

$$A_1 = \left(L_1 - d - \frac{t}{2}\right) \times t = \left(100 - 20 - \frac{6}{2}\right) \times 6 = 462 \text{ mm}^2$$

$$A_2 = \left(L_2 - \frac{t}{2}\right) \times t = \left(100 - \frac{6}{2}\right) \times 6 = 582 \text{ mm}^2$$

$$K = \frac{3 \times 462}{3 \times 462 + 582} = 0.704$$

$A_{net} = 462 + 0.704 \times 582 = 871.73$ $mm^2$
(Nearest option 885.3 $mm^2$)

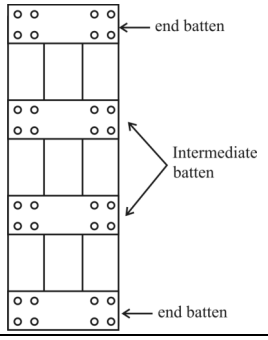
**74. Bearing stiffeners in plate girders**
(a) eliminate web buckling
(b) eliminate flange buckling
(c) transfer loads
(d) reduce deflections

**Ans. (a) :** Bearing stiffeners are applied in plate girder to prevent the bend of web and to provide the stability to web.
• Bearing stiffeners are used to transfer concentrated loads on the girder & heavy reactions at support to the full depth of the web. Thus are required the web has insufficient strength for any of the limit states of web yielding web crippling or sides way web buckling. Although the web can be proportional to resist directly any applied concentrated loads.
• If stiffeners are used to resist the full concentrated load, the limit states of web yielding, web crippling & sideway web buckling are not required to be checked.

**75. The minimum number of battens in a compression steel member is**

(a) 4 (b) 6 (c) 8 (d) 3

**Ans. (a) :** There should be minimum four batten in a compressive steel member for 3 bays, within it actual length. Battens provided at ends are called end batten and others intermediate battens.




**76. The effective length of a column fixed at one end, and restrained against rotation but not held in position at the other end as a ratio of the actual length is**

(a) 1.0 (b) 1.2

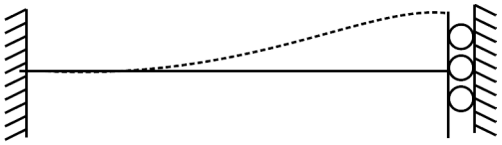
(c) 1.5 (d) 2.0

**Ans. (b) :** The effective length of column fixed at one end and restrained against rotation but not held in position at the other end as a ratio of actual length is –

Length = L

Effective length = 1.2 L

$$\therefore \frac{1.2\ L}{L} = 1.2$$

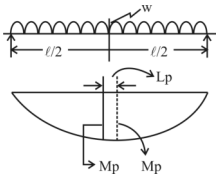


**77. The length of elasto-plastic region at failure of a simply supported beam of rectangular section subjected to uniformly distributed load over its span *l* is**

(a) $l/\sqrt{3}$ (b) $l(2\sqrt{3})$

(c) $l/2$ (d) $l/3$

**Ans. (a) :**




The length of elasto-plastic zone $(L_p) = L\sqrt{\left(1-\frac{1}{S.F.}\right)}$

$$L_p = \ell\sqrt{\left(1-\frac{1}{1.5}\right)}$$

$$L_p = \ell\sqrt{\left(1-\frac{1}{1.5}\right)} \qquad \boxed{L_p = \frac{\ell}{\sqrt{3}}}$$

**78. The maximum pitch or staggered rivets in compression members of 10.0 mm thick elements is**

(a) 120.0 mm (b) 180.0 mm

(c) 200.0 mm (d) 300.0 mm

**Ans. (b) :** Maximum pitch in compression member– In case of staggered rivets increased 60%.

$$\text{When staggered} = \min\begin{cases}18t\\300\,mm\end{cases}$$

$$= \min\begin{cases}18\times 10 = 180mm\\300\,mm\end{cases} = 180mm$$

In tension when rivet are in staggered form then minimum pitch $= \min\begin{cases}24t\\300\,mm\end{cases}$

**79. The allowable bending stress in tension in mild steel sections is**

(a) 157.5 MPa (b) 197.0 MPa

(c) 150.0 MPa (d) 165.0 MPa

**Ans. (d) :** The allowable bending stress in mild steel.

$f_y = 250$

Bending stress = $0.66\ f_y$

$= 0.66 \times 250$

$= 165\ N/mm^2$ or MPa

**80. The effective length of a cantilever beam ϕ projecting 3.0 m with partial restraint against torsion at the support and free at the end is**

(a) 2.25 m (b) 3.00 m

(c) 6.00 m (d) 9.00 m

**Ans. (a) :** The effective length of a cantilever beam ϕ projecting 3.0 m with partial restraint against torsion at the support and free at the end is $0.75\ l = 0.75 \times 3 = 2.25$ m.

| | End connections | Effective length "ℓ" |
|---|---|---|
| (i) | Built-in at the support, free at end | 0.85 L |
| (ii) | Built-in at the support, restrained against torsion at the end by cross member. | 0.75 L |
| (iii) | Built-in at the support, restrained against lateral deflection and torsion at the end by continuous, cross members over several bearing. | 0.50 L |
| (iv) | Continuous and unrestrained against torsion at the support and free at the end | 3 L |
| (v) | Continuous and partially restrained against torsion at the support and free at end | 2 L |
| (vi) | Continuous at the support, restrained against torsion at the support and free at the end | L |

**81. The inclination of a lacing bar with the axis of a compression member should not be**

(a) more than $30^0$ (b) less than $40^0$

(c) less than $45^0$ (d) more than $75^0$

**Ans. (b):** The angle of inclination of the lacing with the longitudinal axis of the column should be between $40^0$ to $70^0$.

**82. The maximum permissible shear stress in structural member of mild steel is**

(a) 125.0 MPa (b) 100.0 MPa
(c) 150.0 MPa (d) 112.5 MPa

**Ans. (d) :** The maximum permissible shear stress–
(a) Maximum 0.45 $f_y$
(b) Average 0.40 $f_y$
$f_y = 250$
Maximum = 0.45 $f_y$
$= 0.45 \times 250 = 112.5$ N/mm$^2$ or MPa

**83. If angle purlins are adopted, the slope of the roof in degrees should not exceed**

(a) 25 (b) 30
(c) 45 (d) 10

**Ans. (b) : IS 800-2007** provides general design procedure for angle purlins procedure for angle Purlins confirming the roof slope not exceeding $30^0$ based on a minimum live load of 750 N/m$^2$

**84. The minimum thickness of main steel members directly exposed to weather, and not accessible for repainting is**

(a) 8.0 mm (b) 6.0 mm
(c) 4.5 mm (d) 10.0 mm

**Ans. (a) : Specification for steel member–**
- When the steel work is directly exposed to weather and is fully accessible for cleaning and re-painting the thickness shall not be less than 6 mm.
- When the steel work directly exposed to weather and is not accessible for cleaning and repainting the thickness shall not be less than 8 mm.
- When the steel work is not directly exposed to weather, the thickness of steel in main members shall not be less than 6 mm.
- When the steel work is directly exposed to weather the thickness of steel in secondary member shall not be less than 8 mm.

**85. The battens of a column carrying an axial load of 2.5 MN, with two parallel planes battens spaced as 1.0 m and 600.0 m transverse length should be designed for transverse shear force of**

(a) 62.5 kN (b) 50.0 kN
(c) 31.25 kN (d) 25.0 kN

**Ans. (a) : As per IS 800 : 2007–** Batten should be designed to carry the bending moment and shear from the transverse shear force, which is 2.5% of the total axial force.

$$V = \frac{2.5}{100} \times \text{axial force}$$

$$= \frac{2.5}{100} \times 2.5 \times 10^6 \text{ N} = 6.25 \times 10^4 \text{ N} = 62.5 \text{ kN}$$

**86. The minimum factor of safety against sliding a masonry wall is**

(a) 1.50 (b) 2.00
(c) 2.50 (d) 3.00

**Ans. (a) :** The minimum factor of safety against sliding a masonry wall is 1.50.

**87. The nominal maximum permissible size aggregates in a 120.0 mm thick concrete is**

(a) 20.0 mm (b) 25.0 mm
(c) 30.0 mm (d) 40.0 mm

**Ans. (c) :** As per IS 456 : 2000, clause 5.3.3
The nominal maximum size of coarse aggregate should be as large as possible within the limits specified but.
In no case greater than 1/4th of the minimum thickness of the member provided that the concrete can be placed without difficulty. So as to surround all reinforcement the roughly and fill the corners of the form.

$$\text{Nominal maximum size of aggregate} = \frac{120.0 \text{ mm}}{4}$$

$$= 30 \text{ mm}$$

**88. The maximum cement content in concrete should not exceed**

(a) 350.0 kg/m$^3$ (b) 450.0 kg/m$^3$
(c) 500.0 kg/m$^3$ (d) 550.0 kg/m$^3$

**Ans. (b) :** IS 456:2000 (Grade of designation)

| | |
|---|---|
| • Grade of designation | M25 |
| • Type of cement | PPC confirming to IS 1489 (Part 1) |
| • Maximum nominal size of aggregate | 20 mm |
| • Minimum cement content | 300 kg |
| • Maximum cement content | 450 kg |
| • Max. water-cement ratio | 0.50 |
| • Workability (Slump) | 60 mm |
| • Type of aggregate | Crushed angular aggregate |

**89. The minimum anchorage length of 12 mm diameter bars of Fe 415 grade steel in M grade concrete with a design bond strength 2.40 MPa and a $90^0$ standard bend at the end is**

(a) 355.3 mm (b) 451.3 mm
(c) 282.1 mm (d) 186.1 mm

**Ans. (d) :** Given data
$\phi = 12$ mm
$f_y = 415$ N/mm$^2$, $\sigma_{sv} = 0.87 fy = 361.05$ N/mm$^2$
$\tau_{bd} = 2.40$ N/mm
$L_d = ?$
•In LSM

$$L_d = \frac{\phi \times \sigma_s}{6.4 \times \tau_{bd}}$$

$$L_d = \frac{12 \times 361.05}{6.4 \times 2.4}$$

$$L_d = 282.07 \text{ mm}$$

Since $90^0$ bend is provided at the end.
$90^0$ bend is equivalent to – 8d.
Hence $(L_d)$required = 282.1– 8d
$= 282.1 - 8 \times 12$
$= 186.10$ mm

**90. The maximum permissible area compression reinforcement in reinforce concrete beame as a percentage cross- sectional area is**

(a) 0.12 (b) 2.00
(c) 4.00 (d) 6.00

**Ans. (c) : According to IS-456:2000,**
clause 26.05.1.2 compression reinforcement,
The maximum area of compression reinforcement shall not exceed 0.04 bD. Compression reinforcement in beams.

**91. The type of lime used for structures submerged in water is**

(a) quick lime (b) hydraulic lime
(c) fat lime (d) anhydrous lime

**Ans. (b) : Hydraulic lime**–The lime in which the percentage of impurities is 5 to 30 percent. It is called hydraulic lime.
There are three types of this, which is based on impurities.

| Type of lime | Impurities Percentage |
|---|---|
| • Feebly hydraulic lime | 5 to 10 |
| • Moderately hydraulic lime | 10 to 20 |
| • Eminently hydraulic lime | 20 to 30 |

Hydraulic lime are used in masonry work and hydraulic structures.

**92. timber is seasoned to**

(a) strengthen the cells
(b) facilitate uniform shrinkage and minimise warping
(c) reduce moisture content below the fibre saturation point
(d) prevent cracking and defects

**Ans. (c) : Seasoning of timber :**

- Seasoning is the process of reducing the moisture content (drying) of timber in order to prevent the timber from possible formation and making it suitable for use.
- Increase strength durability and workability.
- Reduce the tendency to split and decay.

**93. Plywood is mansufactured from**

(a) bamboo (b) teak wood
(c) waste wood (d) structural timber

**Ans. (b) :**

- Plywood is manufactured from teak wood.
- A wood panel glued under- pressure from an odd number (usually 3 to 13) of layrs of veneers is known as plywood.
- Normally the alternate plies are oriented at $30^0$ or $60^0$ in star plywood.
- High grade plywood is superior to most metals in strength - to - weight ratio.

**94. White decayed spots in timber are known as**

(a) doatiness (b) druxiness
(c) upsets (d) foxiness

**Ans. (b) : Druxiness :** It is indicated by white decayed spots which are concealed by healthy wood. They are probably formed for the access of fungi.
**Foxiness :** It is indicated by red or yellow tinge in wood or reddish brown stains or spots round the pith of tree discolouring the timber.
**Upsets :** These are also known as the rupture and they indicate the wood fibres which are injured by crushing or compression.

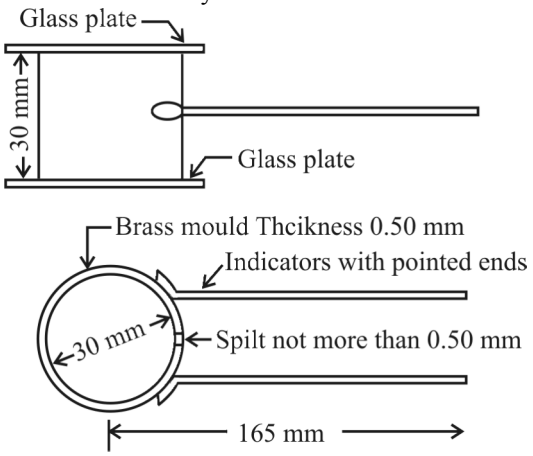
**95. Le chatelier's apparatus determines, in cements, the**

(a) setting time (b) soundness
(c) tensile strength (d) creep

**Ans. (b) : Soundness–** Soundness refer to the ability of cement of maintain constant volume.

- Carried out to detect the presence of uncombined lime in cement.
- Determined by Le chatelier apparatus or auto clave test
- Expansion more than 10 mm indicates un soundness of cement.

**Le-Chatelier method –** Apparatus gives unsoundness due to excess lime only.




**Autoclave test –** Test for expansion due to excess both for lime and magnesia.

**96. Refractory bricks resist**

(a) high temperatures
(b) high chemical contamination
(c) dampness
(d) high compressive loads

**Ans. (a) : Refractory brick –** A refractory brick is built primarily to withstand high temperature but will also usually have a low thermal conductivity for greater energy efficiency.
• It is a block of ceramic material used in lining furnace, kilns, fire boxes and fireplace.

**97. Long term development of strength by cement concrete is due to**

(a) tri calcium silicate
(b) di calcium sillicate
(c) tetra calcium aluminate
(d) tri calcium alumino ferrite

**Ans. (b): Dicalcium silicate ($C_2S$)–**

- It undergoes hydrated within a year or so after the addition of water into the cement hence is responsible for the ultimate or progressive strength in cement.
- It also increases the resistance of cement against the attack of chemicals and acids.
- it in any construction progressive strength is required proportion of $C_2S$ is increased.
- It has been found that hydration of $C_2S$ produces comparatively lower calcium hydroxide $Ca(OH)_2$ than $C_3S$

**98. Slump of concerete indicates its**

(a) strength (b) water content
(c) workability (d) durability

**Ans. (c) :** Slump of concerete indicates its workability
The main purposed of a slump test is usually to indicate whether the correct amount of water has been used.

**99. The age of a tree can be estimated by**

(a) radius of the stem
(b) height of the tree
(c) number of branches
(d) number of annular rings

**Ans. (d) :** The age of a tree can be estimated by number of annular rings

**100. Ultrasonic pulse velocity method indicates the**

(a) length of the member
(b) density of concrete
(c) increase durability
(d) increase workability

**Ans. (b) :** Ultra sonic pulse method indicates the–
1. Evaluate the quality and homogeneity of concrete materials.
2. Predict the strength of concrete.
3. Evaluate dynamic modulus of elasticity of concrete.
4. Estimate the depth of cracks in concrete.
5. Detect internal flows, cracks, honeycombing and poor patches.

**101. Super plasticisers, when added to concrete,**

(a) increase the strengths
(b) reduce permeability
(c) increase durability
(d) increase workability

**Ans. (d) : Super plasticizers :** These admixture are chemically distinct from normal plasticizers and although their action is basically the same.
Air entraining agents : It is also enhance the workability.

**102. The standard rate of loading for testing concrete is**

(a) 1.4 MPa per second
(b) 2.8 MPa per minute
(c) 10.0 MPa per second
(d) 3.6 MPa per minute

**Ans. (a) :** As per, IS 456 : 2000 Compression testing machine should be able to apply gradual load of 14 $N/mm^2$/minute till the specimen is crushed.

**103. Fly ash when added to concrete**

(a) acts as aggregates (b) acts as pozzolana
(c) improves appearance (d) reduces setting time

**Ans. (b) : Pozzolana–**These are siliceous material which are themselves inactive but react in the presence of water with lime to form compounds having cementitious properties.
**Example–** Lime, flyash, burnt clay, blast furnace slag etc.
- The use of flyash as an admixture in concrete reduces the segregation and bleeding. It increase the workability of concrete and has no significant effect on drying shrinkage of concrete.

**104. Water used for mixing concerete should be**

(a) slightly acidic (b) free from bacteria
(c) distilled (d) potable

**Ans. (d) :** While use of potable water is generally safe, water not fit for drinking may also be satisfactorily used in making concrete, as a fule, any water pH 0.6 to 0.8 which does not taste saline or brackish is suitable for use.

**105. The proportion of coarse aggregates to fine aggregates in structural concrete is usually**

(a) 0.50- 1.25 (b) 1.25-4.50
(c) 1.50-2.50 (d) 0.25-0.75

**Ans. (b) :** The proportion of coarse aggregates to fine aggregates in structural concrete is usually in the range of 1.25-4.50.

**106. Hardy Cross method of analysis is based on**

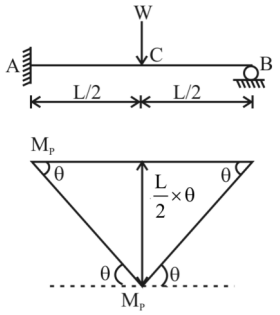
(a) Slope displacement method
(b) Moment area method
(c) conjugate beam method
(d) Virtual work method

**Ans. (b) :** → This Moment Distribution Method (MDM) suggested by Prof. Hardy Cross in 1930.
→ Moment distribution method is a displacement method of analysis of kinematic indeterminate structure based on stiffness approach 1.
→ MDM is a successive method, where degree of accuracy can be obtained by repeated iteration.
→ It also known as hardy cross method of successive approximation.
→ Moment distribution method is also called as lock and release methods.

**107. A beam of span *l* and moment capacity Mp is fixed at one end and simply supported at the other end. The maximum mid-span load the beam can support is**

(a) $4M_p/l$ (b) $6M_p/l$
(c) $3M_p/l$ (d) $M_p/l$

**Ans. (b) :**




Number of plastic hinge (N) = 2
Degree of indeterminacy ($D_S$) = 3 – 2 = 1
Plastic hinge required to form mechanism (n)

$D_S + 1 = 1 + 1 = 2$

By the principle of virtual work done–
External work done = Internal work done.

$$W\times\left(\frac{L}{2}\times\theta\right)=M_P(\theta)+M_P(\theta+\theta)$$

$$\boxed{W_P=\frac{6M_P}{L}}$$

**108. The extensional stiffness of a member is the**
(a) moment per unit rotation
(b) moment per unit extension
(c) axial force per unit elongation
(d) elongation per unit force

**Ans. (c) :** The extensional stiffness of a member is the axial force per unit elongation.

$$P=K\delta$$

or $$K=\frac{P}{\delta}$$

Here, K= extensional stiffness
P=Axial force
δ=unit elongation

**109. the number of points of contra flexure in a propped cantilever is**
(a) one for uniformly distributed load
(b) two for uniformly distributed load
(c) one for any load
(d) two for any load

**Ans. (c) :** The number of point of contraflexure in a propped cantilever beam is one for any load.

| Beam and loading | No. of Points of contraflexure |
|---|---|
| (i) Simply supported beam with UDL over part of the structure | Zero |
| (ii) An overhanging beam with loading only over supported span not on overhangs | Zero |
| (iii) Fixed beam subjected to concentrated load/UDL. | Two |
| (iv) Cantilever beam subjected to UDL with zero load intensity at free end | Zero |
| (v) Propped cantilever beam for any load | one |

**110. Mueller Breslau Principle is applicable to**
(a) only determinate structures
(b) any stable structure
(c) only beams
(d) beams but not to arches

**Ans. (b) :**
Mueller Breslau is applicable to find the ILD of indeterminate structure as well as determinate.
- Mueller Breslau Principle is applicable to any stable structure.

**111. In a cantilever of unit length the ratio of slope to vertical displacement at the free end for a load at the free end is**
(a) 2/3 (b) 3/2
(c) 7/9 (d) 9/7

**Ans. (b) :** Slope (θ) = $\frac{P\ell^2}{2EI}$

Deflection (δ) = $\frac{P\ell^3}{3EI}$

Given,
$\ell = 1$ unit

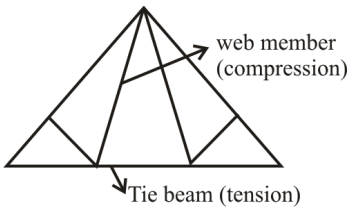
$$\frac{\theta}{\delta}=\frac{P\ell^2}{2EI}\div\frac{P\ell^3}{3EI}=\frac{3}{2}$$

**112. The influence lines are**
(a) always straight line segments
(b) curved in determinate beams
(c) straight line segments for indeterminate frames
(d) curved for indeterminate structures

**Ans. (d) :** The influence lines are always straight line segments for determinate structure whereas it can be curved for indeterminate structure.

**113. The maximum compressive force in a French roof truss with all the joints loaded occurs in the**
(a) mid-span members
(b) lower chord members
(c) members connected to the supports
(d) web members

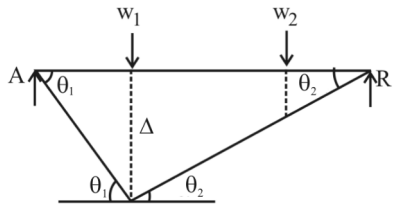
**Ans. (d) :**




(French roof truss)

**114. In a simply supported prismatic beam under symmetric two point loading**
(a) plastic hinges form under the loads
(b) plastic hinges from under one of the loads
(c) plastic hinges from at the supports and under the loads
(d) plastic zone extends from one load to the other

**Ans. (b):** In a simply supported prismatic beam under symmetric two point loading plastic hinges from under one of the loads.

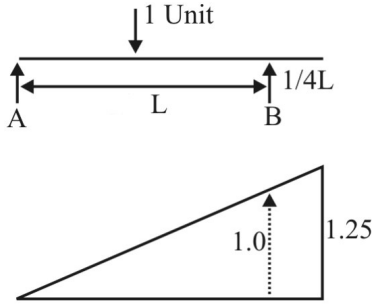



$W_1 > W_2$

**115. A simply supported beam of span *l* has an overhang of one quarter of its span over the right support. The influence line for the right support is**

(a) straight line with maximum ordinate 1.25
(b) straight line with maximum ordinate 1.00
(c) curved with maximum ordinate 1.0
(d) curved with maximum ordinate 1.25

**Ans. (a) :**




$$\left(1+\frac{e}{\ell}\right)=1+\frac{\frac{1}{4}L}{L}=1.25$$

- ILD for the load reaction at B when load is at A will be Zero.
- ILD for the reaction at B when load is at B is 1. And when the load is at the edge of overhang the reaction at the B is 1.25.

**116. Willot- mohr diagram is used to determine joint displacements in**

(a) beams (b) frames
(c) trusses (d) arches

**Ans. (c) : Truss deflection–**

- The Willot diagram, Although graphical methods have been superseded by other methods, ever a sketch of Willot-Mohr diagram can provide an insight into how a truss deflects, or can form a basis for checking results computed by other means.
- In a Willot diagram, deflections are assumed to be small enough for second order effects to be neglected, and are plotted all on the same diagram.

**117. Poisson's ratio for cork is**

(a) 0.50 (b) 0.00
(c) 0.30 (d) 0.15

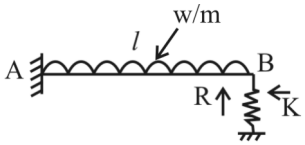
**Ans. (b) :** Poisson's ratio for cork is 0.00

| **Type of material** | **Poisson's ratio (μ)** |
|---|---|
| Cork | 0 |
| Concrete | 0.1 – 0.2 |
| Cast iron | 0.23 – 0.27 |
| Mild steel | 0.25 – 0.33 |
| Rubber | 0.45 – 0.5 |

**118. A cantilever of span *l* is propped by a spring of stiffness k at the free end at the same level as the fixed support. If the beam is loaded uniformly w/m, the prop reaction is**

(a) $(3wl/8)+k$
(b) $(3wl/8)--k$
(c) $wl(3/8)/(1+3EI/)(wl^3)\}]$
(d) $wl(3/8)/\{1-3EI/(wl^3)\}]$

**Ans. (c) :**




Deflection in beam at C = contribution in spring

$$\frac{wl^4}{8EI}-\frac{Rl^3}{3EI}=\frac{R}{K}$$

$$\frac{wl^4}{8EI}=\frac{R}{K}+\frac{Rl^3}{3EI}$$

$$R\left(\frac{1}{K}+\frac{l^3}{3EI}\right)=\frac{wl^4}{8EI}$$

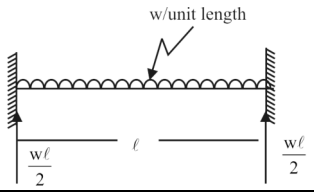
$$R=\frac{\frac{wl^4}{8EI}}{\left(\frac{l^3}{3EI}+\frac{1}{K}\right)}=\frac{\frac{wl^4}{8EI}\times\frac{3EI}{l^3}}{\left(1+\frac{3EI}{l^3K}\right)}$$

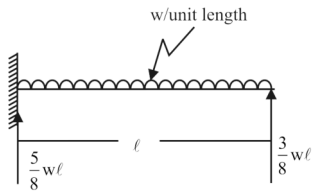
$$\boxed{R=\frac{\frac{3wl}{8}}{1+\frac{3EI}{l^3K}}}$$

**119. The ratio of the load carrying capacity of a fixed beam to that of a propped cantilever of the same span subjected to uniformly distributed lad over the entire span with reference to shear force is**

(a) 1.25 (b) 0.80
(c) 1.33 (d) 0.75

**Ans. (c) :** j






Ratio of the fixed beam
to that of propped cantilever

beam w.r.t shear force= $\frac{w\ell/2}{3w\ell/8}$

$=\frac{4}{3}$

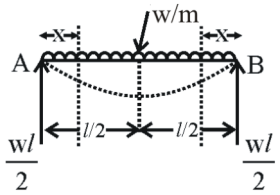
$=1.33$

**120. A simply supported beam of span *l* carries a uniformly distributed load w/m over the entire span; the strain energy of the beam is given by**

(a) $\frac{w^2l^2}{84EI}$ (b) $\frac{w^2l^5}{184EI}$

(c) $\frac{w^2l^5}{240EI}$ (d) $\frac{w^2l^5}{384EI}$

**Ans. (c) :**




$$\because M_x = \frac{wl}{2}x - \frac{wx^2}{2}$$

$$U_{AB} = U_{AC} + U_{BC}$$

$$= \int_0^{l/2} \frac{M_x^2}{2EI}dx + \int_0^{l/2} \frac{M_x^2}{2EI}dx$$

$$U_{AB} = 2\int_0^{l/2} \frac{M_x^2}{2EI}.dx$$

$$= \frac{2}{2EI}\int_0^{l/2}\left(\frac{wl}{2}.x - \frac{wx^2}{2}\right)^2 dx$$

$$\boxed{U_{AB} = \frac{w^2l^5}{240EI}}$$

**121. Irrigation water should contain**

(a) high sodium (b) low sodium

(c) medium sodium (d) any amount of sodium

**Ans. (b) : Irrigation water–** Irrigation water should contain generally less than 5% of the total exchangeable cations.

- If the percentage increases to about 10% or more, the aggregation of soils grains break down.
- The soil becomes less permeable and of pores fills. It starts crusting when dry and its pH increases towards that of an alkaline soil.
- High sodium soils are, therefore plastic sticky when wet, and are prone to form cloggs and they crust on drying.

$$SAR = \frac{Na^+}{\sqrt{\frac{Ca^{++} + Mg^{++}}{2}}}$$

**122. The duty or crops requiring 1.08 m of water over 120 days is**

(a) 960.0 hectares/cu m

(b) 960.0 cu. m/ hectare

(c) 1200.0 hectares/cu m

(d) 1200.0 cu m / hectare

**Ans. (a)**

$\Delta = 1.08$ m

B = 120 day

$$D = \frac{8.64 \times B}{\Delta} = \frac{8.64 \times 120}{1.08} = 960 \text{ hec/cumec}$$

**123. Kharif season generally extends from**

(a) January - July (b) March- August

(c) September- April (d) June- October

**Ans. (d) :** • Kharif season generally extended from June to October.

- It is sown at the beginning of the first rains during the advent of the south – west mansoon.
- Rice, Ragi, groundnut, cotton, maize, bajra all the type are the kharif crops.

**124. The formula generally applicable to the catchment areas in South India is due to**

(a) Dickens (b) Ryve

(c) Inglis (d) Creager

**Ans. (b) : Emprical formula for flood peak–**

(a) Dickens formula–

$$Q_p = C_D A^{3/4}$$

Dickens constant ($C_D$) = 6 to 30

Used in central and northern parts of country.

(b) Inglis formula–

$$Q_p = \frac{124A}{\sqrt{A + 10.4}}$$

Used in western ghats of Maharashtra.

(c) Ryves formula–

$$Q_p = C_R A^{2/3}$$

$C_R$ = Ryves formula

$Q_P$ in $m^3$/sec

A in $km^2$

**125. The formation that does not contain any water is known as**

(a) aquifer (b) aquifuge

(c) artesian well (d) spring

**Ans. (b) : Aquifuge–**

• An aquifuge is a formation that neither contains nor transmits significant quantities of water,

• Unfractured crystalline rocks would fall into this category.

**126. A detention dam stores water**

(a) and releases water at a safe rate
(b) and diverts to another stream
(c) permanently
(d) raises groundwater level

**Ans. (d) :** Detention dams is built to catch surface runoff and stream water flow to regulate the water flow in areas below dam. It is used to reduced damage caused by flooding or to manage the flow rate through a channel.

**127. A surplussing work is also known as**

(a) reservoir (b) dam
(c) barrage (d) spillway

**Ans. (d) :** A spillway is a waterway provided to dispose off surplussing flood water from a reservoir after it has been filled to its maximum capacity .
Spillway are provided for almost all the dams and these acts as safety values for the dams.
• A spillway may be located either within the body of the dam or at one ends of the dam or entirely away from the dam in a natural saddle as on independent structure.

**128. Canal lining is essential to**

(a) store water (b) release water
(c) minimise water losses
(d) minimise silting

**Ans. (c) :** • Canal lining is essential to minimise water losses due to seepage that cause the water logging and that leads to the growth of unwanted water beeds.
• Canal lining reduces the repair cost and increases the life of the canal.

**129. An aqueduct is constructed when the high flood level is**

(a) below the canal level
(b) same level as the canal
(c) above the canal level
(d) above the road level

**Ans. (a) :** An aqueduct is the structure constructed in cross drainage work which carries the to bridge in which instead of road or a railway, a canal is carried over the natural drain.
• The drain flows at atmospheric pressure under the work.
• Generally an inspection road is provided along with the through.
• HFL (Highest flood level) of canal is much above the HFL of drain.

**130. A canal fall**

(a) releases water to crops
(b) releases silt
(c) increases water flow
(d) lowers water level

**Ans. (d) : Canal Falls–**
• A canal fall is a hydraulic structure constructed across a canal to lower down its water level.
• This is achieved by negotiating the change in bed elevation of the canal necessitated by difference in ground slope and canal slope.
• The necessity of a fall a rises when the available ground slope exceeds the design bed slope of the canal and which condition sis rather usual.

**131. The average domestic consumption under normal conditions in Indian cities per day per persons is**

(a) 105 *l* (b) 135 *l*
(c) 500 *l* (d) 150 *l*

**Ans. (b) :**

| S.N. | Description | Amount of water in lit per head per day |
|---|---|---|
| 1. | Bathing | 55 |
| 2. | Washing of clothes | 20 |
| 3. | Flushing of W.C. | 30 |
| 4. | Washing the house | 10 |
| 5. | Washing of utensils | 10 |
| 6. | Cooking | 5 |
| 7. | Drinking | 5 |
| | Total | 135 litres |

• As per IS 1172 - 1993 recommends as per capita water consumption of 135 litres per day.

**132. Water losses in a supply system are assumed as**

(a) 15 percent (b) 7.5 percent
(c) 10 percent (d) 22.5 percent

**Ans. (a) :** Water losses in supply system are assumed as 15%.
• Total water loss is calculated by subtracting the amount of water consumed from the amount of water produced.
• It is important to differentiate between total water loss and leakage.
• Losses comprise leakage from pipes, joints and fittings.

**133. For the least effect on the water table, the location of tubewells should be one per every**

(a) 0.50 sq km (b) 0.75 sq km
(c) 1.50 sq km (d) 2.50 sq km

**Ans. (c) :** For the least effect on the water table, the location of tube wells should be dug one in every 1.50 $km^2$.
• The water table marks the boundary between that available water and the dry surface.
• Ground water is impacted by precipitation, irrigation and ground cover.
• It also may be affected by land use and tidles.

**134. If a 2 percent solution of sewage sample is incubated for 5 days at $20^0$ C and the depletion of oxygen was found to be 5 ppm, the B.O.D of the sewage is**

(a) 200 ppm (b) 225 ppm
(c) 250 ppm (d) 300 ppm

**Ans. (c) :** Depletion of oxygen = 5 ppm

$$\text{Dilution factor} = \frac{100}{2} = 50$$

B.O.D. at 5 days = $5 \times 50$
= 250 ppm

**135. If the length, breadth and depth of water in a rectangular sedimentation tank of discharge Q are *l*,b and d, respectively, the settling velocity is**

(a) Q/ *l* (b) Q/b

(c) Q/(*ld)* (d) Q/(*lb)*

**Ans. (d) :**

Settling velocity = Q/(*lb)*

Because settling velocity don't depends on the depth of water.

Settling velocity depends on the plan area of the settling basin.

**136. Lead caulked joints are used for laying**

(a) stoneware pipes (b) earthenware pipes

(c) C.I. pipes (d) G.I. pipes

**Ans. (b) :** Lead caulked joints are used for laying earthen ware pipes.

Earthenware is fired at 1900 degrees Fahrenheit, much lower than stoneware.

**137. To lift water from a tube well with water level at 45.0 m below general ground level, the most suitable pump is**

(a) centrifugal pump

(b) reciprocating pump

(c) deep well turbine pump

(d) centrifugal and reciprocating pump

**Ans. (c) :**

**Characteristics of Pumps Frequently Employed in Wells**

| Type of pump | Practical Suction Lift' | Usual well-Pumping Depth | Usual pressure Heads |
|---|---|---|---|
| Reciprocation shallow well | 6–7m | 6–7m | 30–60m |
| Deep well | 6–7m | Up to 180m | Up to 180m above cylinder |
| Centrifugal shallow well straight centrifugal (single stage) | 6m max | 3–6m | 30–45m |
| Regenerative vane turbine type (single impeller) | 8m max | 8m | 30–60m |
| deep well Vertical line shaft turbine (multistage) | Impellers submerged | 15–90m | 30–250m |

**138. In activated sludge process, the sludge is mixed thoroughly with activated or**

(a) treated sludge (b) untreated sludge

(c) raw sludge (d) oxidised sludge

**Ans. (c) :** In activated sludge process, the sludge is mixed thoroughly with activated or raw sludge.

- It is a biological treatment process in which biologically active mass, called activated sludge is continuously mixed with the (sewage flow) in an aeration basin in the presence of oxygen.

**139. If the percentage decrease in the volume of sludge is 50, the percentage reduction in the moisture content is about**

(a) 90-100 (b) 80-90

(c) 70-80 (d) 60-70

**Ans. (b) :** $\frac{V_2}{V_1} = \left(\frac{100 - P_1}{100 - P_2}\right)$

$\frac{V_2}{V_1} = 0.5$ (Given)

Now,

$= \frac{100 - P_1}{100 - P_2} = \frac{100 - 90}{100 - 80} = 0.5$ (putting value from option)

So, $P_1 = 90$, $P_2 = 80$

**140. The highest yield of water can be expected from**

(a) gravity springs (b) surface springs

(c) artesian wells (d) aquifuge

**Ans. (c) :** The highest yield of water can be expected from artesian wells.

- Artesian well from which water flows under natural pressure without pumping.
- Gravity spring result where water moves from the water table aquifer through permeable materials to the land surface, or where the land surface intersects the water-table.
- Gravity spring are normally low-yielding source of ground water.

**141. Perched aquifers are found**

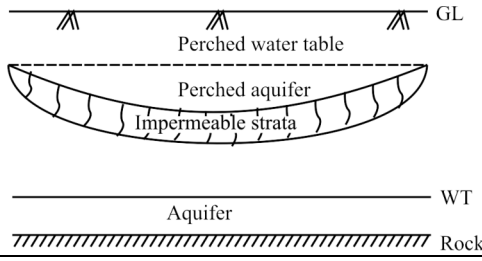
(a) at ground surface

(b) below ground surface but above water table

(c) below ground water table

(d) at ground water table level

**Ans. (b) :** Perched aquifer is a special type of unconfined aquifers and occur where a ground water body is separated from the main ground water by a relatively impermeable strata of some aerial extent and by the zone of aeration above the main body of ground water.




**142. Underground water is obtained from**

(a) rivers (b) lakes

(c) reservoirs (d) springs

**Ans. (d) :** Source of underground water is spring.

- A river is a natural flowing watercourse usually freshwater, flowing towards an ocean, sea, lake or another river.

- A reservoir is most commonly an enlarged natural or artificial lake created using a dam to store water.
- A lake is an area filled with water, localized in a basin, surrounded by land.

**143. The property of the ground that allows water to pass through is know as**

(a) perviousness (b) porosity
(c) permeability (d) transmissivity

**Ans. (c) : Permeability–** It is the property of the soil by virtu of which it allows the flow of fluid through it.

- It is also termed as hydraulic conductivity.
- Permeability of coarse grain soil is more than that of fine grain soil.

**144. The maximum permissible colour for domestic water supply on cobalt scale is**

(a) 5 ppm (b) 10 ppm
(c) 12 ppm (d) 15 ppm

**Ans. (*) :** Maximum permissible colour for domestic supplies is 20 ppm based on cobalt scale.

- For public supplies, the colour number on cobalt scale should not be exceed 20 and should be preferably less than 10.

**145. The maximum permissible fluorine content in water for domestic supplies should be**

(a) 0.15 ppm (b) 1.5 ppm
(c) 15 ppm (d) 150 ppm

**Ans. (b) : The permissible limits of various compounds are as follows–**

| Parameters | Permissible Limit | Cause for rejection |
|---|---|---|
| Total suspended solids | 500 | 2000 |
| Turbidity (NTU) | 1 | 10 |
| Colour (TCU) | 5 | 25 |
| Taste & odour (TON) | 1 | 3 |
| Total dissolved solids (mg | 500 | 2000 |
| Alkalinity | 200 | 600 |
| pH | 7-8.5 | <6.5&>9.2 |
| Hardness (mg/L) | 200 | 600 |
| Chloride content (mg/L) | 250 | 1000 |
| Free ammonia (mg/L) | 0.15 | 0.15 |
| Organic ammonia (mg/L) | 0.3 | 0.3 |
| Nitrite (mg/L) | 0 | 0 |
| Nitrate (mg/L) | 45 | 45 |
| Fluoride content (mg/L) | 1 | 1.5 |
| Iron as Fe (mg/L) | 0.1 | 1.0 |
| Sulphate (mg/L) | 200 | 400 |
| Calcium (mg/L) | 75 | 200 |

**146. The normal values of over time is *l*/h/m$^2$for plain sedimentation tank using coagulants is**

(a) 750-1000 (b) 1000-1250
(c) 1250-1500 (d) 1500-2000

**Ans. (b) :** Overflow rate 500–750 *l*/h/m$^2$ for plain sedimentation 1000–1250 *l*/h/m$^2$ for sedimentation with coagulation.

- Detention time is 3–4 hours for plain sedimentation tank and 2–2.5 hrs. for sedimentation with coagulation.

Q × detention time = volume of tank.

**147. Elutrition is the process of**

(a) sludge digestion
(b) filtration
(c) sedimentation
(d) washing digested sludge

**Ans. (d) :** Before the sludge is dewatered by vacuum filtration method its generally elutriated. It is the process of washing the sludge water, to remove the organic and fatty acid from it. Sometimes, before the dewatering, the sludge is thickened or conditioned and then elutriated. During sludge digestion process, the volatile acid, alcohol and organic acid are formed, which is not removed, will interfere with coagulation process during dewatering by vacuum filtration. If elutriation is done before dewatering, it will reduce the quantity of coagulants considerabaly.

**148. Non- disposal of solid waste may cause the spread of**

(a) malaria
(b) typhoid
(c) rodent related diseases
(d) dysentry

**Ans. (c) :** Non-disposal of solid waste may cause the spread of rodents related disease such as plague, diarrhea, bassillary dysentery, cholera etc.

**149. A scour valve**

(a) removes silt in a pipe
(b) releases the accumulated air
(c) controls water flow
(d) improves water quality

**Ans. (a) : Scour valve/Blow off valve/Drain valve–**

- Scour valves are also known as wash-out valves.
- These are similar to the sluice valves, but the function is different.
- Scour valves are provided at the dead end of the pipe line.
- The function of this valve is to remove the sand, silt etc.

**150. A rain sanitary sewer is constructed to carry**

(a) sewage (b) storm sewage
(c) surface water (d) ground water

**Ans. (b) :** A rain sanitary sewer is constructed to carry storm rainfall-runoff which is known as storm sewage.

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 2003
## Solved Paper

**1. The minimum compressive strength of building bricks is**

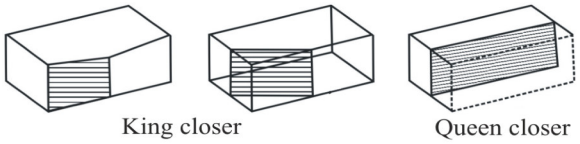
(a) 3.5 MPa (b) 5.0 MPa
(c) 10.0 MPa (d) 15.0 MPa

**Ans. (a) :** The minimum crushing/compressive strengths of burnt bricks tested flat-wise prescribed are :
(i) Common building bricks 3.5 MPa,
(ii) Second class bricks 7 MPa
(iii) First class bricks 10.5 MPa.

**2. A brick with the width at one end cut to half is known as**

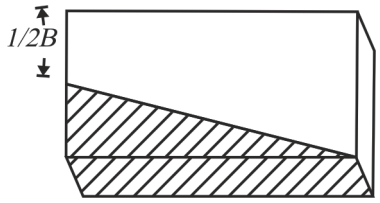
(a) beveled closer (b) king closer
(c) mitred closer (d) queen closer

**Ans. (b) : King closer –** If a brick is cut in such a way that the width of one end becomes half that a full brick, while the width at the other end is equal to the full width, then its is called as king closer.

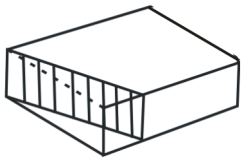



**Queen closer–** When a brick is cut along its length, making it two equal halves then it is called queen closer.

**Beveled closer–** It is a form of king closer in which the whole length of the brick (i.e. stretcher face) is chamfered on beveled in such a way that half width is maintained at one end and full width is maintained at the other end.




**Mitred closer–** It is a portion of a brick whose one end is cut splayed or mitred for full width. The angle of splay may vary from $45^0$ to $60^0$.



**3. The weight of timber is specified, as per the Indian standards, at a moisture content of**

(a) 8 percent (b) 10 percent
(c) 12 percent (d) 14 percent

**Ans. (c) :** Seasoning is the process of reducing the moisture content by drying under controlled conditions as nearly as possible at uniform rate from all parts of timber in order to prevent the timber from possible fermentation and making it suitable for use.
The objectives of seasoning are following—

- Reduce the shrinkage and warping after placement in structure.
- Increase strength, durability and workability.
- Reduce the weight.
- Reduce the tendency to split and decay.
- Remove sap from the timber.
- The moisture content in a well seasoned timber is about 10 - 12%.

**4. A concrete mix of good workability should have a minimum water cement ratio of**

(a) 0.20 (b) 0.40
(c) 0.60 (d) 0.80

**Ans. (b) :** • minimum w/c ratio = 0.4
• normal for ordinary concrete w/c = 0.6 to 0.7
• specified if a higher quality concrete, w/c = 0.4
Particle range of w/c ratio = 0.3 to 0.8

**5. Initial setting time of concrete is governed by the content of its**

(a) tricalcium silicate
(b) dicalcium silicate
(c) tricalcium aluminate
(d) tetracalcium alumino ferrite

**Ans. (c) :**

| Chemical compound | Avg. % | Function |
|---|---|---|
| Tricalcium sillicate ($C_3S$) | 40 | It is responsible for 28 days strength that is early maximum strength. |
| Dicalcium silicate($C_2S$) | 32 | It gives ultimate strength. |
| Tricalcium aluminate($C_3A$) | 10 | It react rapidly with water and is responsible for flash setting. |
| Tetra calcium aluminoferrite ($C_4AF$) | 8 | It is responsible for flash set but generates less heat. |

**6. The percentage of water for normal consistency is**

(a) 5 - 15 (b) 10 - 25
(c) 15 - 25 (d) 25 - 35

**Ans. (d):** The percentage of water for normal consistency is 20-25%

**Standard consistency –** The standard consistency of a cement paste is defined as that consistency which will permit a vicat plunger having 10 mm dia and 50 mm length to penetrate to a depth of 33-35 mm from the top of the mould.

**7. Inadequate compaction of concrete results in**

(a) segregation
(b) bleeding
(c) bleeding and segregation
(d) honey combing

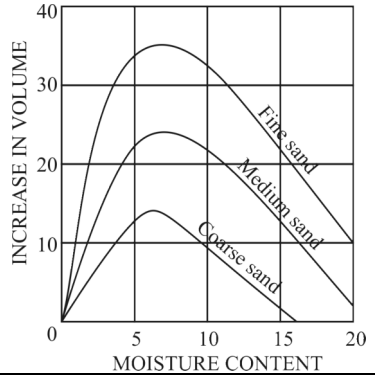
**Ans. (d) :** Honey combing in concrete caused by mainly–

- Inadequate compaction
- Poor mixing of concreting
- Gaps in form work which allowing some of the mortar to drain out of the concrete.

**8. For the same moisture content, bulking will be the highest in**

(a) gravelly sand
(b) coarse sands
(c) medium grained sands
(d) fine grained sands

**Ans. (d) :** Bulking of sand is maximum in fine sand for same moisture content. Bulking will be maximum at moisture content of 4-6%.




**9. Admixtures are added to improve the performance of**

(a) cement during manufacture
(b) cement at the site
(c) concrete during mixing
(d) reinforcement

**Ans. (c) :**

- Admixture are chemical compounds other than water, aggregate hydraulic cement and additives like pozzolana and fiber reinforcement used as an ingredient of concrete or mortar.
- It is immediately added before or during mixing to modify one or more properties of concrete in plastic and hardened state.
- Admixture are added in concrete to give it certain improved qualities for changing different physical properties in its fresh & hardened stages.

**10. Schmidt rebound hammer estimates the**

(a) hardness of aggregates
(b) setting time of concrete
(c) strength of concrete
(d) composition of concrete

**Ans. (a) : Rebound hammer test (Schmidt hammer test)**

- This test measures toughness (strength) of concrete.
- This is a non-destructive test of concrete.
- It is a surface hardness tester for which an empirical correlation has been established between strength and rebound number.
- It is suitable for both field and laboratory work.
- It weight about 1.8 kg

**11. The number of standard bricks per cubic metre is approximately**

(a) 300 (b) 200
(c) 800 (d) 500

**Ans. (d) :**

Nominal Size of brick = 20 cm × 10 cm × 10 cm

Volume of one brick = 20 cm × 10 cm × 10 cm = 0.002 $m^3$

The number of bricks in 1 $m^3$ brick works.

$$\text{Number of bricks} = \frac{1\text{m}^3}{0.002\text{m}^3} = 500$$

**12. The maximum permissible water absorption of building bricks as a percentage of weight is**

(a) 1.0 (b) 5.0
(c) 10.0 (d) 25.0

**Ans. (d) :** The maximum permissible water absorption of building bricks as a percentage of weight is 25%.

**13. An increase of 1.0 percent in water cement ratio**

(a) increases concrete strength by 10 percent
(b) reduces concrete strength by 4 percent
(c) reduces concrete strength by 10 percent
(d) increases concrete strength by 4 percent

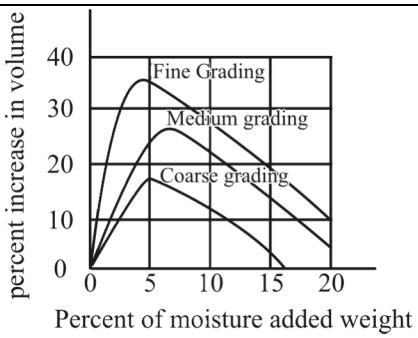
**Ans. (b) :** An increase of 1.0 percent in water cement ratio strength reduces by 4%.

- An increase of water cement ratio the strength of concrete reduces. Due to inter particle lubrication fluidity of concrete increases with water and lower the heat of hydration.

**14. Bulking of sand is caused by**

(a) surface absorption
(b) surface adsorption
(c) surface tension
(d) surface texture

**Ans. (c) : Bulking of sand–**The increase in the volume of sand due to increase in moisture content is known as bulking of sand. A film of water is created around the sand particles which forces the particles to get a side from each other and thus the volume is increased, bulking of sand depends on the quantity of moisture in the sand and also the size of the particles. Four to six percent of the increase in moisture in the sand can increase the volume of sand up to 20 to 40 percent.




**15. Standard size of cubes for testing the strength of cement is**

(a) 150.0 mm (b) 200.0 mm
(c) 70.7 mm (d) 110.5 mm

**Ans. (c) :** • Cement is tested for compressive strength, by this test, the quality and the quantity of concrete can be controlled and degree of adulteration can be checked.
• Material for each three specimen cube is mixed separately.
• Three specimen cubes are prepared of size 70.6 mm (having face area of about 5000 $mm^2$)
• Compressive strength is taken to be the average of the results of the three cubes.

**16. The sum of rotational factors of all the members of a joint in Kani's method of analysis is**

(a) +10 (b) –1.5
(c) –0.5 (d) –0.6

**Ans. (c) :** The sum of the rotational factors of all the members at a joint in Kain's methods of analysis is taken as –0.5.
• Kani's method is also known as rotation contribution method.
• Sum of the distribution factor at a joint of all the member is taken as 1.

**17. A truss with one support on a roller bearing and the other hinged with 10 joints and 21 members is**

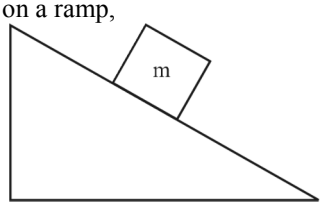
(a) statically determinate
(b) unstable
(c) statically indeterminate to the fourth degree
(d) statically indeterminate to the second degree

**Ans. (c) :** roller bearing reactions + hinged bearing reactions = 3
m = 21
j = 10
So static indeterminacy in truss $= m + r - 2j$
$= 21 + 3 - 2 \times 10$
$= 24 - 20$
$= 4$
$\boxed{D_s = 4}$

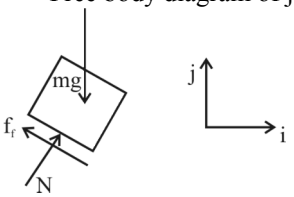
**18. A free body diagram is**

(a) the structure with supports removal
(b) an unstable structure
(c) the structure with loads removed
(d) a part of the structure with internal forces imposed

**Ans. (d) : Free body diagram (FBD) :** A free body diagram is a graphical illustration used to visualize the applied internal forces, moments and resulting reactions on a body in a given condition, they depict a body or connected bodies with all the applied forces and moments and reactions, which act on the body.
Like :
A block on a ramp,




Free body diagram of just the block




**19. Compatibility conditions satisfy**

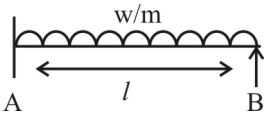
(a) equilibrium conditions of the structure
(b) displacement conditions
(c) equilibrium condition of the free body
(d) symmetry conditions of the structure

**Ans. (b) : Compatibility equations–**
A structure apart from satisfying equilibrium conditions should also satisfy all the compatibility conditions.
• These conditions require that the displacement and rotations be continuous throughout the structure and compatible with the nature of supports conditions.

**20. A propped cantilever is subjected to a uniformly distributed load w/m over its span $\ell$. The fixed end moment is**

(a) $-w\ell^2/12$ (b) $-w\ell^2/24$
(c) $+w\ell/8$ (d) $-w\ell^2/8$

**Ans. (d) :**




We know that at the end point deflection will be zero
So

$$\frac{w\ell^4}{8EI} - \frac{R_b\ell^3}{3EI} = 0$$

$$R_B = \frac{3w\ell}{8}, R_A = \frac{5w\ell}{8}$$

So, $$M_A = \frac{w\ell^2}{2} - \frac{3w\ell^2}{8} = \left(\frac{w\ell^2}{8}\right)$$

**21. The shear force at a section in the conjugate beam corresponds in the real beam to**

(a) shear force at the corresponding section
(b) deflection at the corresponding section
(c) slope at the corresponding section
(d) bending moment at the conjugate section

**Ans. (c) : Conjugate beam method –**

Conjugate beam method is the modified moment area method. This method based on the construction of conjugate beam, defined as an imaginary beam of length equal to that of real beam and loaded with an elastic weight $\left(\frac{M}{EI}\right)$.

**Conjugate beam theorem I –** Shear force at any section of the conjugate beam is equal to the slope of elastic curve at the corresponding section of the actual beam.

**Conjugate beam theorem II –** Bending moment at any section of the conjugate beam is equal to the deflection of elastic curve at the corresponding section of the actual beam.

**22. The number of joint equilibrium equations available for the analysis of trusses is**

(a) 3 (b) 1
(c) 2 (d) zero

**Ans. (c) :** The number of equilibrium equation of 2 – D truss is taken equals to 2

$\sum f_x = 0$

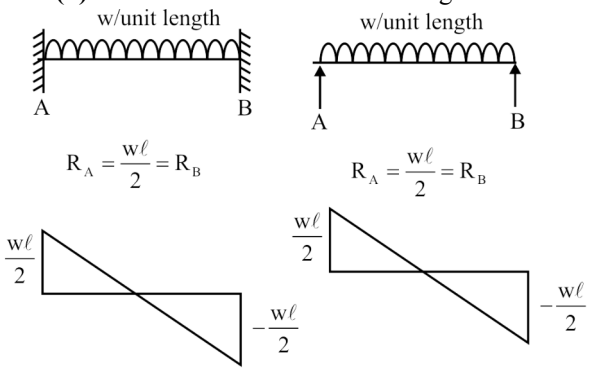
$\sum f_y = 0$

All the joints do not carry any moment, only axial forces are considered in the members.

**23. The ratio of the strength of a fixed beam to that of a simply supported beam of the same span under uniformly distributed load on the entire spans with reference to shear to**

(a) 3.0 (b) 2.0
(c) 0.5 (d) 1.0

**Ans. (d) :** On the basis of shear stress diagram–




both having same shear stress value and diagonal.

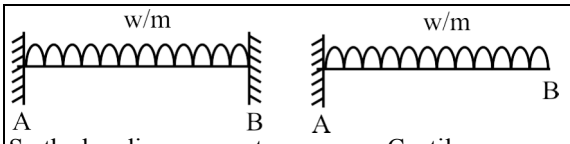
So,

$$\frac{V_{A\,fix}}{V_{A\,simply}} = \frac{w\ell/2}{w\ell/2} = 1$$

**24. The ratio of the load carrying capacity of a fixed beam to that of a cantilever of the same span subjected to uniformly distributed load over the entire spans with reference to the bending moment is**

(a) 6.0 (b) 4.5
(c) 3.0 (d) 1.0

**Ans. (a) :**




So the bending moment

$$M_{AB\ fix} = \frac{-w'\ell^2}{12}$$

Cantilever moment

$$M_{AB\ cant} = \frac{-w''\ell^2}{12}$$

$$\frac{M_{AB(fix)}}{M_{AB(cant)}} = \frac{\frac{-w'\ell^2}{12}}{\frac{-w''\ell^2}{12}}$$

Let $M_{AB\ fix} = M_{AB\ cant}$

$$\frac{12}{2} = \frac{w'}{w''} = \frac{6}{1} = 6$$

$$\frac{w'}{w''} = 6$$

**25. The theorem of three moments was formulated by**

(a) Thomas Young (b) Williot-Mohr
(c) J.C. Maxwell (d) Clapeyron

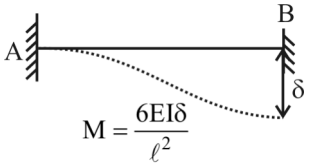
**Ans. (d) : Theorem of three moment**–Structural analysis clapeyron's theorem three moments is a relationship among the bending moments at three consecutive supports of a horizontal beam.

- It is applicable only when there is no discontinuity such as hinges within the span.
- It is not applicable to trusses and frames.
- It is not applicable to fixed supports.

**26. If one end of a fixed beam of span $\ell$ and flexural rigidity EI sinks by $\delta$, the beam is subjected to**

(a) sagging moment ($6EI\ \delta/\ell^2$)
(b) hogging moment ($6EI\ \delta/\ell^2$)
(c) sagging moment at the lower support and hogging moment at thee other support of magnitude ($6EI\ \delta/\ell^2$)
(d) hogging moment at the lower support and saging moment at the other support of magnitude ($6EI\ \delta/\ell^2$)

**Ans. (c) :** If one end of a fixed beam of span $\ell$ and flexural rigidity EI sinks by $\delta$, the beam is subjected to sagging moment at the lower support and hogging moment at thee other support of magnitude ($6EI\ \delta/\ell^2$)

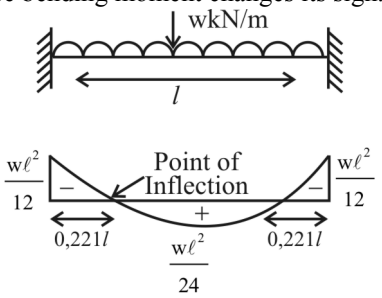



**27. In a fixed beam subjected to uniformly distributed load throughout the span, points of inflection occur at**

(a) midspan section
(b) fixed ends
(c) 0.211 $\ell$ from the supports

(d) 0.202 $\ell$ from the supports

**Ans. (c) :** Point of enflection or contraflexure is the point where bending moment changes its sign.




At the point of inflection bending moment found to be zero.

**28. At a rigid joint in a frame all the members have**

(a) same rotation but different displacements
(b) same displacements but different rotations
(c) different displacements and rotation
(d) same displacements and rotations

**Ans. (a) :** In a rigid jointed frame, the points are considered to rotate as a whole. In these members, the applied load is transmitted into axial, shear and bending effects. Rigit joints are designed to transfer Axial as well as shear in addition to bending moments b/w connected member.

**29. The slope deflection method formulates**

(a) equilibrium conditions only
(b) compatibility conditions only
(c) both equilibrium and compatibility conditions
(d) either equilibrium or compatibility conditions

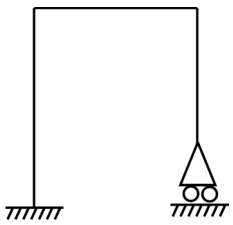
**Ans. (a) :**

| Force methods | Displacement |
|---|---|
| • Governing equation - compatibility equation<br>• Type of indeterminacy - Static Indeterminacy<br>• Force displacement relations : Flexibility matrix | • Governing equation - equilibrium equation<br>• Type of indeterminacy - Kinematic indeterminacy<br>• Force displacement relation stiffness matrix |

**30. A single bay single storey portal frame has one roller support while the other end is fixed. Its statical and kinematic degrees of redundancy are, respectively**

(a) 2 and 3 (b) 1 and 5
(c) 5 and 3 (d) 3 and 1

**Ans. (b) :**

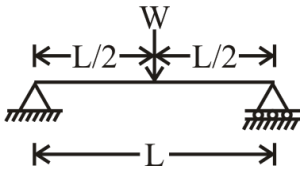


DS $\Rightarrow$3 × no. of loop – no. of reaction to make joint fix
= 3 × 1 – 2 = 1
DK = 3 × no. of joint – no. of reaction
– no. of inextensible members
= 3 × 4 – 4 – 3 = 5

**31. A simply supported beam of span '$\ell$' and flexural rigidity EI carries a unit load at its midspan section the strain energy of the beam is**

(a) $\frac{\ell^3}{48EI}$ (b) $\frac{\ell^3}{192EI}$
(c) $\frac{\ell^3}{96EI}$ (d) $\frac{\ell^3}{24EI}$

**Ans. (c) :**




Strain energy due to bending :

$$U = \int \frac{M_x^2 dx}{2EI}$$

The deflection under unit point load in simply supported beam:

$$\delta = \frac{WL^3}{48EI} = \frac{L^3}{48EI}$$

$\therefore$ Strain energy : $U = \frac{1}{2} \times w \times \delta$

[$\because$ W = 1 unit]

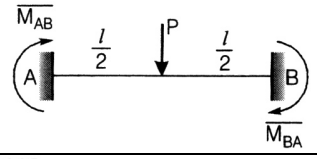
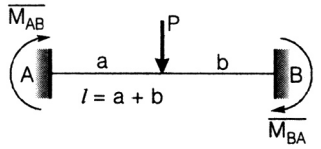
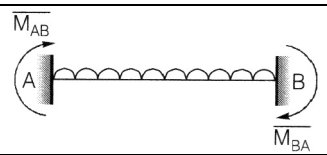
$$U = \frac{1}{2} \times 1 \times \frac{L^3}{48EI}$$

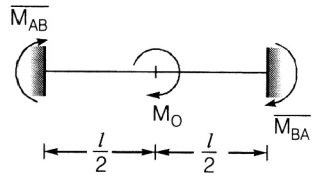
$$= \frac{L^3}{96EI}$$

**32. A fixed beam is subjected to a moment M at the midspan section; the fixed end moments are**

(a) M (b) M/2
(c) M/4 (d) M/3

**Ans. (c) :** Fixed end moment of both side of fixed and continuous beam

| Fixed Beam | $\overline{M_{AB}}$ | $\overline{M_{BA}}$ |
|---|---|---|
|  | $\frac{-P\ell}{8}$ | $\frac{P\ell}{8}$ |
|  | $\frac{-Pab^2}{\ell^2}$ | $\frac{Pa^2b}{\ell^2}$ |
|  | $\frac{-w\ell^2}{12}$ | $\frac{w\ell^2}{12}$ |

| | | |
|---|---|---|
|  | $\frac{M_O}{4}$ | $\frac{M_O}{4}$ |

**33. The ratio of the maximum deflection of a fixed beam under a point load at midspan to that of the beam with the same load distributed uniformly over the entire span is**

(a) 2 (b) 3

(c) 4 (d) 6

**Ans. (a) :**

$$\Delta \text{uniformly} = \frac{w\ell^3}{384EI}$$

$$\Delta \text{Point} = \frac{w\ell^3}{196EI}$$

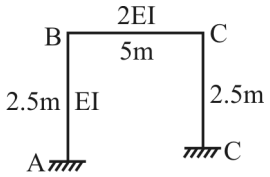
$$\frac{\Delta \text{Point}}{\Delta \text{Uniformly}} = \frac{\frac{w\ell^3}{196EI}}{\frac{w\ell^3}{384EI}} = \frac{384}{196} = 2$$

**34. A closed frame of span 5.0 m and height 2.5 m has the flexural rigidities of the horizontal members twice that of the vertical members. The distribution factors at a joint of the frame for Hardy Cross Method are**

(a) 0.5 and 0.5 (b) -0.25 and -0.25

(c) -0.36 and -0.64 (d) 0.33 and 0.67

**Ans. (d) :**




At point B

$$K_{BC} = \frac{8EI}{\ell}$$

$$K_{BA} = \frac{4(EI)}{\ell}$$

$$\sum K = K_{BC} + K_{BA} = \frac{8EI}{\ell} + \frac{4EI}{\ell} = \frac{12EI}{\ell}$$

$$M_{BC} = \frac{K_{BC}}{\sum K} = \frac{8EI/\ell}{\frac{12EI}{\ell}} = \frac{8}{12} = \frac{2}{3} = 0.67$$

$$M_{BA} = \frac{\frac{4EI}{\ell}}{\frac{12EI}{\ell}} = \frac{4}{12} = 0.34$$

**35. A Pratt truss has ten joints and 21 members. If the truss is simply supported, it is**

(a) unstable

(b) statically determinate

(c) statically indeterminate to the first degree

(d) statically indeterminate to the fourth degree

**Ans. (d) :** For simply supported Pratt truss

m = 21

j = 10

r = 3

$D_s = m + r - 2j$

$21 + 3 - 2 \times 10 = 4$

So the truss is indeterminate by fourth degree.

**36. A portal frame with one column inclined and the other column vertical can be analysed by**

(a) Hardy Cross method but not Kani's method

(b) Kani's method but not by Hardy cross method

(c) Kani's and Hardy Cross methods

(d) Neither Kani's nor Hardy Cross methods

**Ans. (a) :** Portal frame with one column inclined and the other column vertical can be solved by Handy-cross method.

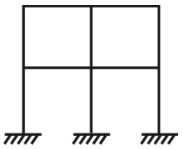
Kanis method is used to solve indeterminate frame and beam, non sway and deflected frame.

**37. The degree of freedom of a two bay two storey rigid jointed frame are**

(a) 27 (b) 18

(c) 6 (d) 9

**Ans. (b) :**



$D_K = 3J - r \rightarrow$ if member's are extensible

$D_K = 3J - r - m \rightarrow$ if member are in extensible

$= 3 \times 9 - 9$ (for extensible member)

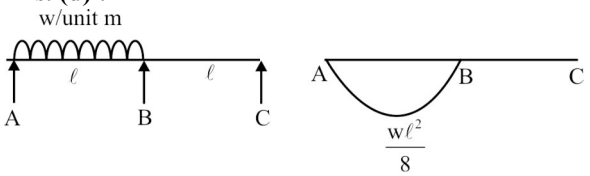
$= 27 - 9 = 18$

**38. A beam is continuous over three simple supports, and is subjected to uniformly distributed load w/m over one span. If the spans are of equal length l, the moment at the middle support is**

(a) $-w\ell^2/12$ (b) $-w\ell^2/2$

(c) $-w\ell^2/8$ (d) $-w\ell^2/16$

**Ans. (d) :**




$$m_a\ell_1 + 2m_b(\ell_1 + \ell_2) + m_c\ell_2 + \frac{6A_1\bar{x}_1}{\ell_1} + \frac{6A_2\bar{x}_2}{\ell_2} = 0$$

$m_a = 0$

$m_c = 0$

$$2m_b(\ell_1 + \ell_2) = \frac{-6\frac{2}{3} \times \frac{w\ell^2}{8} \times \frac{\ell}{2}}{\ell} - 0$$

$$2m_b(2\ell) = \frac{-2 \times 2w\ell^2}{16}$$

$$m_b = -\frac{w\ell^2}{16}$$

**39. The shape factor of a standard rolled steel joint is about**

(a) 1.50 (b) 1.73
(c) 1.95 (d) 1.21

**Ans. (d) :** The shape factor of standard rolled steel joint is about 1.10-1.20
It is a function of cross-section or shape and is represented by 'f'
As we know that

$$\text{Shape factor} = \frac{\text{Plastic section modulus}}{\text{section modulus}}$$

$$f = \frac{MP}{My} = \frac{f_y z_p}{f_y z_y} = \frac{z_p}{z_y}$$

**40. Plastic modulus of a section is the ratio of**

(a) plastic and elastic moment capacities
(b) plastic moment capacity and yield strength
(c) elastic moment capacity and yield strength
(d) plastic moment capacity and working stress

**Ans. (b) :** Plastic modulus of the section is denoted by $z_p$
It is the ratio of the plastic moment capacity and yield strength

$$z_p = \frac{m_p}{\sigma_y}$$

- The ratio of plastic failure load to elastic failure load is called as shape factor.

**41. Slenderness ratio of lacing bars should not be**

(a) greater than 180
(b) less than 145
(c) greater than 145
(d) less than 180

**Ans. (b) : According to IS 800 : 2007**

- The slenderness ratio $\frac{\ell_{eff}}{r}$ of the lacing bars should not exceed 145.
- The angle of inclination of the lacing with the longitudinal axis of the column should be between $40^0$ to $70^0$.
- The minimum thickness of lacing bars.

$t = \frac{\ell}{40}$ for single lacing

$t = \frac{\ell}{60}$ for double lacing riveted or welded at intersection.

**42. Intermediate vertical stiffeners are provided in plate girders, if the depth to web thickness ratio is more than**

(a) 45 (b) 85
(c) 100 (d) 200

**Ans. (b) :** If $\frac{d}{t_w} < 67 \Rightarrow$ unstiffened girder can be designed i.e. no stiffener required.

- If $85\,\varepsilon < \frac{d}{t_w} < 200\,\varepsilon \Rightarrow$ Vertical stiffeners provided ($C_1$ & $C_2$)
- If $200\,\varepsilon < \frac{d}{t_w} < 250\,\varepsilon \Rightarrow$ Vertical stiffener along with longitudinal stiffness at 0.2 d may be provided.
- If $250\,\varepsilon < \frac{d}{t_w} < 345\,\varepsilon \Rightarrow$ Vertical stiffeners along with two longitudinal stiffener at 0.2 d and 0.5 d respectively may be provided

**Calculation :**

$$\frac{d}{t_w} = \frac{1000}{6} = 166.67$$

as $85 < \text{and } \frac{d}{t_w} < 200$

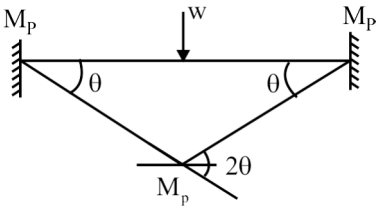
Therefore, intermediate vertical stiffness are to be provided.

**43. The ratio of the plastic failure load to the elastic failure load of a fixed beam of prismatic profile is**

(a) 1.0 (b) 1.5
(c) 3.0 (d) 2.0

**Ans. (a) :** Let we take rectangular section of fixed beam and it is having a point load at centre so the bending moment at the centre will be $M = \frac{w\ell}{8}$

So, the plastic moment will be




$$\frac{L}{2} w \times \theta = M_p\theta + 2M_p\theta + M_p\theta$$

$$\frac{wL}{8} = M_p$$

and $m = \frac{wL}{8}$

$$\frac{M_p}{M} = \frac{wL/8}{wL/8} = 1$$

**44. For a welded joint, the edge distance of a slot should be more than**

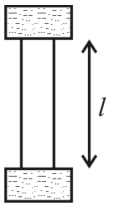
(a) twice the plate thickness or 25 mm

(b) 1.5 times the plate thickness or 20 mm
(c) twice the plate thickness or 15 mm
(d) 1.5 times the plate thickness 15 mm

**Ans. (a) :** For a welded joint the edge distance between the edge of the part and edge of the slot or hole, or between adjacent slots or holes, should be not less than twice the thickness(t) and not less than 25mm for holes.
For slot weld the diameter of hole should not be less than three times the thickness or 25mm, whichever is greater.
Corners at the enclosed ends or slots should be rounded with a radius not less than 1.5 times the thickness or 12mm which ever is greater.

**45. In a case of compression steel members with both ends fixed the effective length as a ratio of actual length is**
(a) 0.50 (b) 1.00
(c) 0.65 (d) 1.20

**Ans. (c) :**




The ratio of the actual lengths to effective lengths of both ends fixed compression steel member is taken 0.65 L

| End Restraints | Effective Lengths |
|---|---|
| (a) Effectively held in position at both ends but not restrained against rotation. | 1.00 L |
| (b) Effectively held in position at both ends and restrained against rotation at one end. | 0.80 L |
| (c) Effectively held in position and restrained against rotation at both ends. | 0.65 L |
| (d) Effectively held in position and restrained against rotation at one end, and at the other end restrained against rotation but not held in position. | 1.20 L |
| (e) Effectively held in position and restrained against rotation at one end, and at the other end partially restrained against rotation but not held in position. | 1.50 L |
| (f) Effectively held in position at one end but not restrained against rotation, and at the other end restrained against rotation but not held in position. | 2.00 L |

**46. The ratio of the distance between the innermost connecting lines of rivets to the thickness of a batten in a steel column should not be**
(a) more than 45 (b) less than 45
(c) less than 60 (d) more than 50

**Ans. (c) :** If single lacing system is adopted the thickness t of the lacing flat $\not<\frac{1}{40}$ of the lengths between the inner ends rivets or weld. if double lacing system is adopted, the thickness t of the lacing flat $\not<\frac{1}{60}$ of the length between the inner end rivets of welds.

**47. The length of elastoplastic region at failure of a simply supported beam of rectangular section subjected to a central point load over a span $\ell$ is**
(a) $\ell/\sqrt{3}$ (b) $2\ell/3$
(c) $\ell/3$ (d) $\ell/\left(2\sqrt{3}\right)$

**Ans. (c) :** The length of elasto plastic region at failure of a simply. Supported beam of rectangular section subjected to central load even span *l* is

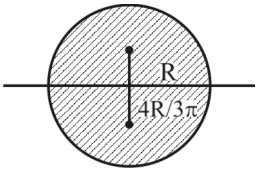
$$\ell_p \Rightarrow \ell\left(1-\frac{1}{\text{shape factor}}\right)$$

$$\ell\left(1-\frac{1}{3/2}\right)$$

$$\ell/3$$

**48. Plastic modulus of a circular section of radius R is**
(a) $3\,R^3/4$ (b) $4\,R^3/3$
(c) $\pi R^3/4$ (d) $\pi R^3/3$

**Ans. (b) :** Plastic modulus of a circular section




Plastic modulus of the circular section

$$\frac{\pi R^2}{2}\times\frac{4R}{3\pi}+\frac{\pi r^2}{2}\times\frac{4r}{3\pi}\Rightarrow\frac{4R^3}{3}$$

**49. The Euler load of a section of area A, slenderness ratio $\lambda$, moment of inertia I, Young's modulus E and equivalent length $\ell$ is given by**
(a) $\pi^2 EI/\lambda^2$
(b) $\pi^2 EAs/\ell^2$
(c) $\pi^2 EAs/\lambda^2$
(d) $4\pi^2 EI/\ell^2$

**Ans. (c) :** The Euler load of a section of Area A, slenderness $\lambda$ moment of inertia I, young's modulus E and equivalent length *l* is given by

$$P_C=\frac{\pi^2 EI}{\ell^2} \qquad r=\sqrt{\frac{I}{A}} \text{ and } \lambda=\frac{\ell}{\sqrt{\frac{I}{A}}}$$

So, $= \frac{\pi^2 EA}{\lambda^2}$ $\lambda^2 = \frac{\ell^2 A}{I}$

**50. The maximum shear stress in a steel beam of yield strength $f_y$ is**

(a) 0.45 $f_y$ as per plastic design
(b) 0.55 $f_y$ as per elastic design
(c) 0.75 $f_y$ as per plastic design
(d) 0.45 $f_y$ as per elastic design

**Ans. (d) : As per elastic design:-** Maximum permissible Axial stress in compression and tension is given by

$$\sigma_{ac/at} = 0.6f_y$$

maximum permissible average shear stress is given by

$$\tau_{v\,avg} = 0.40f_y$$

maximum permissible maximum shear stress is given by

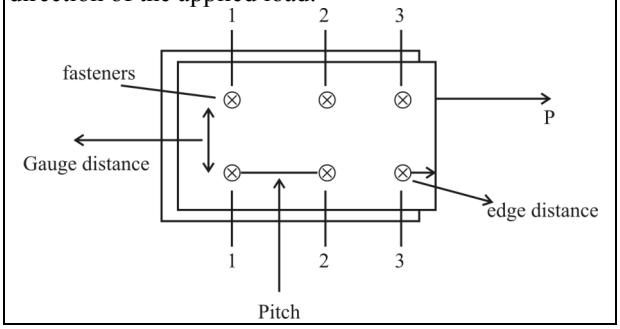
$$\tau_{v\,max} = 0.45f_y$$

maximum permissible bending stress in tension and compression is used in design $\sigma_{bt} = 0.66f_y$

**51. Gauge of rivets is the spacing between**

(a) adjacent fasteners
(b) adjacent parallel lines of fasteners
(c) end fastener and plate edge
(d) fasteners in a line

**Ans. (d) :** Gauge of rivets is the spacing between adjacent parallel lines of fasteners perpendicular to the direction of the applied load.




**52. Factor of safety is the ratio of**

(a) yield stress and design load
(b) upper yield stress and working load
(c) lower yield stress and working stress
(d) fractures stress and working stress

**Ans. (b) :** Factor of safety is the ratio of yield stress (upper yield stress) and working stress.
But in L.S.M. the factor of safety is taken for plastic region.
Here the FOS will be desired for each loads by considering load combination and strength and serviceability requirements.
Hence it will be

$$\text{Working load} = \frac{\text{Corresponding characteristics load}}{\text{Partial safety factor}}$$

**53. The minimum edge distance for a fastener hole of 19.5 mm diameter, as Per IS : 800 : 1984, is**

(a) 25.0 mm
(b) 29.0 mm for machine flame cut plates
(c) 32.0 mm for machine flame cut plates
(d) 39.0 mm

**Ans. (d) :** The minimum edge distance and end distance.
= 1.5 × gross dia of rivets (machine cut)
= 1.7 × gross dia of rivets (for hand driven element)
So, for 19.5 mm diameter of hole
edge distance = 1.5 × 19.5 = 29.25 mm (machine cut)
= 1.7 × 19.5 = 33.15 mm (for hand driven)
But for analysis and design purpose, end distance will be = 2.0 × gross dia of rivet
= 2 × 19.5
= 39.00
So, (d) Answer is correct.

**54. The maximum permissible shear stress in hand driven rivets is**

(a) 80.0 MPa (b) 100.0 MPa
(c) 50.0 MPa (d) 250.0 MPa

**Ans. (a) :** Connection

| N/mm² | Axial Tension | Shearing | Bearing |
|---|---|---|---|
| Power shop rivets | 100 | 100 | 300 |
| Power driven field rivets | 90 | 90 | 270 |
| Hand driven field rivets | 80 | 80 | 250 |

**55. Bearing stiffeners in a plate girder**

(a) prevent shear failure
(b) eliminate local buckling
(c) reduce deflections
(d) transfer concentrated loads

**Ans. (d) :** Bearing stiffeners in plate girder used to transfer concentrated loads at the ends, on the supports to full depth of the web
A bearing stiffener at the support is called as end bearing stiffener.
Bearing stiffeners should be placed
(i) Tight with the web
(ii) Straight (it is not crimped or joggled)
(iii) In pairs of two or four angles, symmetrically placed on both side of the web for a riveted place girder and two or four plates for a welded plate girder.

**56. M 120 grade concrete indicates a compressive strength of 120.0 MPa in**

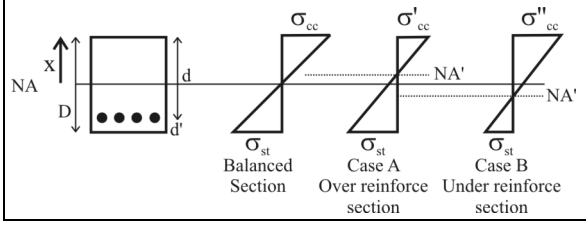
(a) 200 mm cubes at 28 days
(b) 150 mm cubes at 28 days
(c) 150 mm cubes at 28 days
(d) 200 mm cubes at 56 days

**Ans. (b):** In M120, M denotes the mix and 120 the strength after 28 days period of casting in a 150 mm size of cube and strength is expressed in MPa or N/mm$^2$
There are various grades of concretes in which

(1) M10, M15 and M20 comes in ordinary concrete
(2) M25, M30, M35, M40, M50 and M55 comes in standard concrete and
(3) M60, M65, M70, M75, M80, comes in high strength concrete.

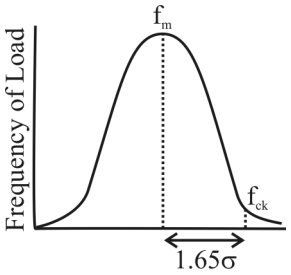
**57. In a balanced section the neutral axis**
(a) moves down as the steel area is reduced
(b) moves up as the steel area is increased
(c) remains the same as steel area is changed
(d) moves up as the grade of concrete is increased

**Ans. (a) :** In a balanced section steel and concrete fails at the same stress level and bears same loads at failure.
Neutral axis remains at the same places during balanced state.
**Case (A)** If the grade of concrete (or Area of concrete) reduced and quantity of steel (Area of steel) increased NA will shift upward of balanced section N.A.
**Case (B)** If the grade of concrete is increased and the quantity of steel is reduced the N.A. shifts downward direction.




**58. Characteristic load on a structure has a probability of being**
(a) exceeded 95 percent
(b) not exceeded 5 percent
(c) exceeded 5 percent
(d) not exceeded 99 percent

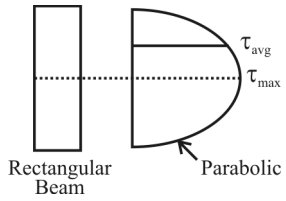
**Ans. (b) :**




The strength of concrete should not be less than 5% of characteristics strength, so the probability of characteristics strength should be 95%.
Design load Ld = Characteristics load × Partial factor of safety.

**59. The distribution of shear stress in a rectangular R.C. beam is**
(a) parabolic
(b) rectangular
(c) triangular
(d) parabolic upto neutral axis and rectangular upto tension steel

**Ans. (a) :**




The distribution of shear stress in a rectangular R.C Beam is in parabolic form
$\tau_{max} = 3/2\ \tau_{avg}$
At the upside and downside end layer of beam the shear stress is found to be zero.

**60. Crack width in a reinforced concrete member should not exceed**
(a) 0.20 mm in mild environment
(b) 0.02 mm in aggressive environment
(c) 0.30 mm in mild environment
(d) 0.20 mm in serve environment

**Ans. (c) :** Crack width as per IS 456 : 2000 in a reinforced concrete member should not be more than 0.3 mm for mild exposure conditions.
For moderate exposure condition a lower limit of crack width of 0.2 mm is prescribed for structures exposed to continuous moisture conditions or which are in direct contact of soil or the ground water.

**61. Flexural tensile strength of M25 grade concrete is**
(a) 3.5 MPa (b) 5.0 MPa
(c) 2.5 MPa (d) 1.5 MPa

**Ans. (a) :** Flexural tensile strength of M25 grade concrete is taken =
$= 0.7\sqrt{f_{ck}}$
$= 0.7\sqrt{25} = 3.5\ N/mm^2$ or MPa

**62. The effective depth of a continuous beam of 9.0 m span should not be**
(a) less than 50.0 mm
(b) more than 846.2 mm
(c) less than 900.0 mm
(d) less than 346.2 mm

**Ans. (d) :** Effective depth of continuous beam of 9.0 m span should be less than

$$\frac{\ell}{d} \le 26 \qquad \frac{9000}{d} \le 26$$

$$\frac{9000}{26} \ge d$$

$$d \le 346.2mm$$

**63. The maximum spacing of 12 mm diameter bars in a reinforced concrete slab of 350.0 mm effective depth is**
(a) 350.0 mm (b) 120.0 mm
(c) 240.0 mm (d) 300.0 mm

**Ans. (b):** Dia of bar = 12m
as per the IS code 456 : 2000
depth of the slab should be = 8 × dia of bar

$= 12 \times 8 = 96$

So, total depth of slab will be = effective depth + clear cover + dia of bar/2

= 96 + 20 + 12/2 = 122 mm ⊔ 120 mm

**64. In working stress method of design, the capacity of a column of 300.0 × 600.0 mms section and 5.0 m effective length is reduced by**

(a) zero percent (b) 9.7 percent

(c) 0.12 percent (d) 0.80 percent

**Ans. (b) :** Working stress method of design

section = 300.0 × 600.0

$l_{eff}$ = 5.0 m

$$Cr = 1.25 - \frac{\ell_{eff}}{48b} = 1.25 - \frac{5000}{48 \times 300}$$

$$= 0.9027$$

So, percentage reduction in length

$$= \frac{(5 - 5 \times 0.9027)}{5} \times 100$$

$$\boxed{9.72\%}$$

**65. The minimum area of tension reinforcement in a beam for Fe 415 grade steel is**

(a) 0.20 percent (b) 0.15 percent

(c) 0.12 percent (d) 0.80 percent

**Ans. (a) :** The minimum area of tension steel in beam for fe 415 grade steel.

$$\frac{A_{st}}{b_d} = \frac{0.85}{f_y}$$

$$A_{st} = \frac{0.85bd \times 100}{415}$$

$$\boxed{A_{st} = 0.2048bd}$$

**66. Rankine's theory assumes that the soil behind a retaining wall is**

(a) dry and cohesive

(b) dry and non-cohesive

(c) granular and moist

(d) moist but non-cohesive

**Ans. (b) : Rankine's theory:-**

- The soil is cohesionless in Rankine's theory

But later it was extended to cohesive & submerged soil.

- For cantilever & counter fort wall, we use Rankines theory.
- Back surface is taken smooth and frictionless.
- Soil is semi-infinite, homogeneous isotropic, dry & cohesionless.
- The backfill soil is Horizontal.
- Rupture surface is a plane surface which may be horizontal/inclined.

**67. The maximum spacing of stirrups inclined at $45^0$ to the axis of beam of 342.0 mm effective depth is**

(a) 171.0 mm (b) 256.5 mm

(c) 300.0 mm (d) 342.0 mm

**Ans. (c) :** Maximum spacing of stirrup inclined at 45° to the axis of beam of 342.0 mm

$$\left.\begin{array}{l} 0.75\ d\ (\text{Vertical Stirrup}) \\ d\ (\text{for inclined}) \\ 300\ \text{mm} \end{array}\right\} \text{whichever is minimum}$$

So, 300 will be right answer.

**68. The effective width of the flange of T-beams or span 10.0 m, web thickness 300.00 mm, flange thickness 120.00 mm and spaced at 3.0 mm is**

(a) 2.69 m (b) 2.50 m

(c) 3.00 m (d) 2.39 m

**Ans. (a) :** $l_0 = 10$ m. $b_w = 0.3$, $D_f = 0.12$

$$b_f = \frac{l_0}{6} + b_w + 6D_f$$

$$= \frac{10}{6} + 0.3 + 6 \times .12$$

$$= 2.68\text{m}$$

**69. Nominal concrete cover in a beam of section 300.0 × 600.00 mm, span 12.0 m and reinforced with 20 mm diameter bars should not be less than**

(a) 20.0 mm in mild exposure

(b) 30.0 mm in mild exposure

(c) 40.0 mm in very severe exposure

(d) 30.0 mm in very severe exposure

**Ans. (a) :**

| Environmental condition | Minimum grade of concrete | Nominal cover |
|---|---|---|
| Mild | M 20 | 20 mm |
| Moderate | M 25 | 30 mm |
| Severe | M 30 | 45 mm |
| Very severe | M 35 | 50 mm |
| Extreme | M 40 | 75 mm |

So, cover of 20.0 mm in mild exposure condition is provided.

**70. Freyssinet's method of pre-stressing is suitable for**

(a) pre-tensioning beams

(b) railway sleepers

(c) post-tensioning beams

(d) pre-fabricated beams

**Ans. (c) :** Freyssinet's method is used for pre-stressing in post-tensioning beams.

- Wires stands are used as tendons.
- It is used for medium to large range of force.
- Arrangement of tendons in ducts it done Annular, spaced by helical wire care.
- Anchoring is of concrete wedge type.

**71. The first method developed for project planning is**

(a) PERT (b) CPM

(c) Bar chart (d) Milestone

**Ans. (c):** The first method developed for project planning is bar chart.

Features of bar chart are:
1. It is a pictorial chart.
2. It has two co-ordinates axes, the horizontal co-ordinate represents the elapsed time and vertical co-ordinates represent the job or activity to be performed.
3. The beginning end ends bar represents starting and finishing time of particular activity respectively.
4. The length of bar shows the time required for completion.

**72. Critical activity has**
(a) maximum float (b) minimum float
(c) zero float (d) mean float

**Ans. (c) :** Critical activity has zero float.
**Critical activity –** It has zero slack at head event as well as at tail event. The sequence of critical activities in a network is called the critical path.
So, the critical path has zero total float or delay.

**73. The estimated time required to perform activity is known as**
(a) dummy (b) float
(c) duration (d) event

**Ans. (c) :** The estimated time required to perform activity is term as duration of an activity.
Activity may contain, excavation, casting of beam and erection of machines etc.
- Activity lies between two nodes.
- Length of arrow is insignificant in an activity.
- An activity can start only when tail event completed.
- An activity cannot occur more then once in the network.

**74. Critical path moves along the activity having a total float of**
(a) positive magnitude (b) negative magnitude
(c) zero (d) maximum value

**Ans. (c) :** Critical path moves along the activity having a total float of zero.
- If float is zero float event, is called as critical event and this represents that the project is on schedule.
- If float is less than zero i.e. negative float and such events are called as super-critical event and found to be behind the schedule.
- If float is greater than zero i.e. positive float, such event called sub-critical events and these represents that the project is ahead of schedule.

**75. The time which results in the least possible direct cost of an activity is known as**
(a) normal time (b) crash time
(c) slow time (d) standard time

**Ans. (a) :** The time which results in the least possible direct cost of an activity is known as normal time.
**Crash cost–** This is the direct cost corresponding to the completion of the activity within crash time. If the time decreases the direct cost increase but after a limit increase in direct cost will not let any change in crashing duration whether infinite resources are allocated for the activity.

**76. The duration of various activities of a project are shown on a bar chart by**
(a) vertical lines (b) horizontal lines
(c) inclined lines (d) discontinuous lines

**Ans. (b) :** In the bar chart or Gantt chart method at horizontal axis duration and at vertical axis activity is represented.
- Job can be concurrent or can be started are after other.

So, some bars can run parallel or overlap each other or may run serially.

**77. A milestone chart shows**
(a) delays in the jobs
(b) jobs ahead of schedule
(c) interdependence of jobs
(d) jobs completed

**Ans. (d) :** Each activity there are certain key events which works the completion of certain portion of the main activity, such event are called milestone.
By mile stone charts, the relationship between two specific shear stones on an activity is revealed but inter depending among milestones between different activities is not indicated.

**78. Cost-benefit analysis indicates**
(a) total cost of project
(b) variation in costs in the project
(c) expenditure involved in the project
(d) viability of a project

**Ans. (d) : Cost–** Benefit analysis is process for evaluating the merits of a particular project in a systematic way is terms of cost and benefits of the project.
- Cost is the value of money that has been used upto producing something and hence is not available for use anymore.
- Benefits are the monetary value of desirable consequence of economic polices and decisions.
- The conceptual and theoretical frame work of CBA is derived from welfare economics.
- It is an economic evaluation technique that measures all the positive (beneficial) and negative (costly) consequences of an intervention or program in monetary terms.

**79. Bar charts are preferred because they**
(a) are simple
(b) indicate interdependence of activities
(c) distinguish between critical and non-critical activities
(d) reflect uncertainty in duration

**Ans. (a):** Bar charts are preferred because they are simple.
- It is unable to depict inter dependencies of various activities clearly.
- Bar chart cannot distinguish between critical and non-critical activities and hence resource smoothening and resource leveling can not be done
- It is not useful in the projects where there are uncertainties indetermination of estimation of time required for completion of various activities such as R&D projects.

**80. The A-O-N (activity on Node network system) is preferred over A-O-A system because**

(a) it allows efficient use of dummy activities
(b) it does not contain activity times
(c) it does not allow dummy activates
(d) it allows revisions and modification easily

**Ans. (c) :** No activities are represented on nods.
- There is no place for events in AON.
- In AON there is no place for dummies as the relationship is being developed with help of arrows itself.
- It is comparatively easy to modify.

**81. Hydrology deals with**
(a) rain water
(b) river water
(c) sea water
(d) surface and underground water

**Ans. (d) :** Hydrology deals with surface and underground water.
Hydro mean water and logy means science.
It deals with the occurrence and movement of water, the physical and chemical properties of water and its remain ship with the living and material components of the environment.
Therefore , it is the steady of water moving through the earths surface into the atmosphere and back in its various forms.
- It includes the precipitation evaporation, surface runoff, stream flows in catchment, ground water etc.

**82. Surface run-off is the quantity of water that**
(a) is absorbed by soil
(b) is intercepted by building and roads
(c) fills ground depressions
(d) reaches streams

**Ans. (d) :** Surface run-off is the quantity of water that reaches streams channels.
- When rain occurs, and the water flows over the ground and reaches to join nearly channel flow then the amount of water that reaches is known as surface runoff.
- When the water flows after rainfall some of the quantity get trapped by building, vegetation and some of the water fills in the depression present in the ground. This amount of water is calculated as lost water and the remaining of the water is absorbed by the soil.

**83. Relative humidity is the ratio of actual vapour pressure to the saturated vapour pressure**
(a) at the same temperature
(b) at the same pressure
(c) at the same volume
(d) in the atmosphere

**Ans. (a) :** Relative humidity is the ratio of actual vapour pressure to the saturated vapour pressure at the same temperature.
- It is denoted by $\phi \left[ RH(\phi) = \frac{m_v}{m_{vs}} \right]$
- It feels about the amount of water vapour in the air compared to how much vapour could be hold at same temperature.
- If the water vapour content stays the same and the temperature drops, the relative humidity increases.

**84. The standard height of a rain gauge is**
(a) 100.0 mm (b) 200.0 mm
(c) 300.0 mm (d) 500.0 mm

**Ans. (c) :** The standard height rain gauge – (IS : 4986 : 1968)
- In India non-recording type rain gauge that is used mostly, is synopsis rain gauge.
- It has weight upto 30.5 cm (approximately 1 ft.) and dia of 12.7 cm
- It is also called 5-inch rain gauge. It is suitable for measuring rainfall 70.1 mm of depth.

**85. Isohyets are the lines joining the points of equal**
(a) pressure (b) height
(c) humidity (d) rainfall

**Ans. (d) :** Isohyets are the lines joining the points of equal rainfall.
- It is the contour lines drawn on maps that connects places or points that received the same amount of rainfall or precipitation at a given period of time.
- The mainly used isohyets represent area that received a similar amount of rain or equal density of drain during particular storm.
- It is used to represent or estimate various parameter of rainfall like the minimum, maximum and average rainfall of a particular region.

**86. If P and A are perimeter and area of drainage basin, respectively, its compactness coefficient is**
(a) $P^2/(2\pi A)$ (b) $P/(2\pi A)$
(c) $P/2(\sqrt{\pi A})$ (d) $P^3/(\pi A)$

**Ans. (c) :** If P and A are the perimeter and area of drainage basin, then its compactness coefficient.
Can be written by

$$K_c = \frac{P}{2\sqrt{\pi A}} = 0.28\frac{P}{\sqrt{A}}$$

This index represents the shape of the surface of the basin, according to its delimitation and its influence of runoff and the hydrograph resulting from the precipitation.
Form factor, This is one of the parameter that explains the elongation of the basin. It is defined as the ratio of the basin area to its length.

$$F_f = \frac{A}{L^2}$$

$F_f$ = Form factor, $A \rightarrow$ Area of the basin in $km^2$
$L \rightarrow$ Length of the basin in km.

**87. The time of overland flow is affected by**
(a) slope of the basin but its length
(b) ground surface but not the slope
(c) length of flow path but not the area
(d) slope and length of path

**Ans. (c):** The time of overland flow is affected by land slope and length of path.

- It depends on the velocity of flow, type of ground, and length of flow etc.
- It is also called as sheet flow; It is a phase of runoff when water flows as a sheet on plain land.

**88. The distance of the critical point is 20.0 km and the difference in elevation is 193 m from a survey of India map; the overland flow time is**

(a) 2 hours (b) 3 hours
(c) 2 hours 30 minutes (d) 4 hours

**Ans. (d) :** By Kirpich equation –

$$T_c \text{ (in minute)} = 0.01947 \times (L^{0.77}) \times \left(\frac{h}{L}\right)^{-0.385}$$

h = 193 m, L = 20000 m

$$T_c = 0.01947 \times (20000)^{0.77} \times \left(\frac{193}{20000}\right)^{-0.385}$$

= 238.296 minute
= 3.97 hour $\approx$ 4 hour

**89. The quantity of water retained by subsoil against gravity is known as**

(a) yield (b) porosity
(c) specific yield (d) specific retention

**Ans. (d) :** The quantity of water retained by subsoil against gravity is known as specific retention.

- Its depends on the porosity of the soil.

$$n = \frac{v_v}{v}$$

If n > 25% $\rightarrow$ Sufficient amount of ground water will in sole.

5% < n < 25% $\rightarrow$ Moderate amount of ground water will be in sole.

n < 5% $\rightarrow$ Less amount of ground water will be in soil.

**90. The best unit period of a unit hydrograph is equal to the basin lag divided by**

(a) 2 (b) 3
(c) 4 (d) 5

**Ans. (c) :** By experience it is found that the duration of unit hydrograph should not exceed $\frac{1}{5}$ to $\frac{1}{3}$ basin lag.

- For catchment of sizes large than 250 $km^2$ the duration of 6 is generally satisfactory.
- But the $\frac{1}{4}$ basin lag is best choice.

**91. The zone below water table is known as the zone of**

(a) saturation (b) aeration
(c) separation (d) hydration

**Ans. (a) :** The zone below water table is known as the saturated zone and the water below the water table is called ground water table.

- Around 65% of irrigation in India is from ground water.
- It is found mainly between soil pore spaces and in the fractures of rock formations.
- Ground water is accessed by drilling well into aquifer and pumping water out.
- Ground water stored between layers of hard rock below the water table is called as aquifer.

**92. Water in the zone of saturation is**

(a) under pressure
(b) at atmospheric pressure
(c) vacuum pressure
(d) at atmospheric temperature

**Ans. (b) :** Water in the zone of saturation is called as ground water and this remains under the hydrostatic pressure and above that atmospheric pressure exits.

**93. Ground water is in the zone of**

(a) saturation (b) high pressure
(c) capillary water (d) dykes

**Ans. (a) :** Ground water is found in the saturation zone under the earth surface.

**94. isopiestic lines are the contours representing**

(a) water table (b) piezometric heads
(c) piezometric surface (d) rainfall

**Ans. (c) : Isopiestic lines –** These are the lines representing the contour of same piezometric surface.

- Piezometric surface is an imaginary surface to which ground water rises under hydrostatic pressure in wells are springs.
- Lines joining the points of equal rainfall depths called isphyeteal lines.

**Isobar$\rightarrow$** It is a line on a map connecting points having the same atmospheric pressure at a given time.

**95. If h is the loss due to friction in a pipe, the losses in strainer and bends may be taken as**

(a) 0.01 h (b) 0.45 h
(c) 0.20 h (d) 0.25 h

**Ans. (d) :** If h is the loss in a pipe
Then losses in strainer and bends is taken as = 0.25 h

$h_f = \frac{f\ell v^2}{2dg} \rightarrow$ Friction loss

**96. Hydraulic grade line**

(a) remains above the conduit line
(b) remains below the conduit line
(c) remains parallel to the conduit line
(d) may be above or below the conduit line

**Ans. (d) : Hydraulic grade line–** The line which point the piezometric head $\frac{P}{\gamma} + z$ at various points in a flow is known as hydraulic grade line.

**TEL–** The line which joins the total energy $\left(\frac{P}{\gamma} + \frac{v^2}{2g} + z\right)$ at various points in a flow is known as a TEL.

- Distance between TEL and HGL gives KE head or velocity head.
- HGL may rise or fall.

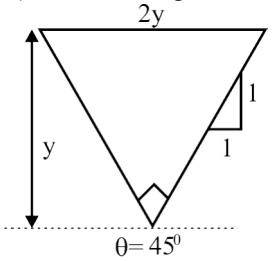
• Total energy lines always falls only in the direction of flow.
• HGL may be above or below the centre line of conduit.
• If HGL is above the centre of pipe line the pressure is positive and if HGL is below the centre of the pipeline, the pressure is negative.

**97. In a long pipe, the head loss is ignored**
(a) at the entrance
(b) at the outlet
(c) due to friction
(d) at the inlet and outlet

**Ans. (c) :** In a long pipe, the head loss due to friction are more prominent then entry and exit so the minor losses are ignored in the pipe having long length. But if length is short then friction loss is ignored and losses due to bend, exit and entry are considered.

**98. The most economical profile of a triangular open channel is**
(a) equilateral triangle
(b) right angle with equal sides
(c) isosceles triangle with $45^0$ angle
(d) triangle with $30^0$ angle

**Ans. (c) :** The most economical profile of a triangular. Open channel is (isosceles triangle with $45^0$ angle.




$$R = \frac{A}{R} = \frac{y^2}{2\sqrt{2}y}$$

$$R = \frac{y}{2\sqrt{2}}$$

**99. The equation for viscous flow was first formulated by**
(a) Bernoulli (b) Chezy
(c) Newton (d) Pascal

**Ans. (c) :** The equation for viscous flow was first formulated by Newton.
• He formulated the shear produced betweens the layer with the charge of velocity and vertical gap between layer $\left[\tau = \mu \frac{du}{dy}\right]$ $\mu \rightarrow$ viscosity
• Upper layer which is moving faster tries to draw the lower slowly moving layers along with it.

**100. The Thiessen polygon method is adopted to compute over an area**
(a) mean depth of rainfall
(b) maximum depth of rainfall
(c) instantaneous depth of rainfall
(d) minimum yield of ground water

**Ans. (a) :** The Thiessen polygon method is adopted to compute over an area mean depth of rainfall.

$$\overline{P} = \frac{P_1A_1 + P_2A_2 + .....P_nA_n}{A_1 + A_2 + A_3.....A_n}$$

This is common method of weighing the rain gauge observation according to the area. This method is also called weighted mean method.

• It assumes that rainfall at any point within the polygon is the same at that of the nearest gauge.
For this method, all gauge in and around, the basin are considered.

**101. Effluent irrigation indicates**
(a) effluents created due to irrigation
(b) irrigation by treated sewage
(c) irrigation by untreated sewage
(d) excessive irrigation

**Ans. (b) : Effluent irrigation–** This is also called as irrigation by treated sewage.
• Properly treated sewage effluent becomes on alternative source of irrigation water.
• In effluent water total salt concentration and electrical conductivity of the effluent must be low.
• The concentration of toxic ions, such as heavy metal, boron and Ct should be low and appropriate measures should be taken.
• Concentration of $Ca^{++}$, $mg^{++}$ and $Na^{+}$ should be in appropriate measure as per the crop susceptibility.

**102. The permissible limit of fluorides in drinking water is**
(a) 1.0 ppm (b) 3.0 ppm
(c) 5.0 ppm (d) 10.0 ppm

**Ans. (a) :** If present in small concentrations, fluoride may be useful to human beings. Fluorine upto 1.2 ppm will prevent dental cavity.
- Excess value > 1.5 to 2 mg/*l* result in discoloration of teeth called mottling of teeth.
  {Infants are affected not adult diseases called flourosis}
- > 5mg/*l* cause deformation of bone called bone flourness.

Acceptable limit = 1 mg/*l*
Cause for rejection = 1.5 mg/*l*

**103. A tube-well that supplies water to a surface canal during non-monsoon season is known as**
(a) direct irrigation tube-well
(b) stand-by tube-well
(c) augmentation tube-well
(d) indirect irrigation tube-well

**Ans. (d) :** Indirect irrigation tube wells are used to supply water to a surface canal during arid or non-monsoon seasons.
Due to scarcity of water and non perennial river the canal is supplied water through indirect irrigation tube wells.

**104. The type of bacteria which can survive with on without free oxygen is known as**

(a) aerobic (b) anaerobic
(c) facultative (d) coliformic

**Ans. (c) : Facultative bacteria–** These are the type of the bacteria those can survive in aerobic as well as anaerobic condition both.

• There kind of bacteria generate ATP (Adenosine triphasphate) through aerobic respiration when present in the oxygen - enriched environment, however in oxygen deficient environment too.

**105. Electrical conductivity of water**

(a) decrease with total dissolved solids (TDS)
(b) increases with TDS
(c) decreases initially and then increase
(d) increases initially and then decrease

**Ans. (b) :** Electrical conductivity of water increases with TDS.

A rapid estimation of the total dissolved solids (TDS) content of a water supply can be obtained by specific conductance measurements i.e the electrical conductivity of water.

→ Special conductance measurements indicate the capacity of a sample to carry electric current, which in turn is related to the concentration of ionized substances in the water.

→ Most dissolved inorganic substances in water supply are in ionized form, such measurement can give a practical estimate of the TDS content of a water supply.

$\left(\text{EC in } \frac{\mu\text{mho}}{\text{cm}} \text{ at } 25^{\circ}\text{C}\right) \times 0.65 =$ Total Dissolved solid in mg/*l*.

**106. Which of the following pairs is correctly matched?**

(a) Lime soda process fluoride removal
(b) Nalgonda technique - softening
(c) Aeration - coagulation
(d) Ozonation - disinfection

**Ans. (d) : Ozonation–** It is used to disinfection the water and helps in killing bacteria.

**Lime soda process** is used for hardness removal to about 50 ppm, used per softening of water.

• It also reduces the corrosion of pipes due to increased akalinity.

**Aeration–** To remove dissolve gas as iron and manganese as precipitate.

• Removal of 90% $CO_2$ to 99% $H_2S$.

Cascade aerator– Removes 20-45% $CO_2$ and 35% $H_2S$

Nalgonda techniques– Used to deflouridation (removal of fluorine).

**107. The chloride demand of water sample was found to be 0.20 mg/ℓ. The amount of bleaching powder with 30 percent available chlorine per kilo litre of water is**

(a) 0.67 g (b) 0.67 mg
(c) 0.67 kg (d) 0.06 mg

**Ans. (b) :** The chloride demand of water sample = 0.20 mg/l

Amount of bleaching powder with 30% available chlorine per kilolitre water will be.

$$\text{Total water} \times \frac{\text{Total amount of Chlorine per liter}}{\text{Percentage available chlorine in compound}}$$

$$\frac{0.20 \times 1}{0.30} \Rightarrow 0.67 \text{ mg}$$

**108. E, coli is determined to examine the quality of water because**

(a) it is pathogenic
(b) it indicates viral contamination
(c) it indicates pathogenic bacterial pressure
(d) it is a unique indicator organism for sewage pollution

**Ans. (b)** E-colli is determined to examine the quality of water because it is facultative bacteria that contaminate the water.

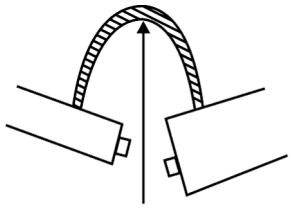
**109. A goose neck is used in service pipes in order to**

(a) control discharge
(b) prevent damage due to unequal settlement
(c) control pressure in service pipe
(d) increase service pipe capacity

**Ans. (b) :** A goose neck is used in service pipes in order to prevent damage due to unequal settlement of pipe.

This is shaped connecting pipe main with sub-main or service.

it is flexible pipe of 75 m length forming a flexible connection between pipes.



Flexible pipe (Goose neck)

**110. One litre of sewage when allowed to settle for 30 minutes yields a sludge volume of 30.0 cc. If the dry mass of the sludge is 3.0 g, the sludge volume index is**

(a) 10.0 (b) 27.0
(c) 30.5 (d) 81.0

**Ans. (c) :**

One litre of sewage form sludge volume =30.00 CC.

If dry mass of sludge = 3.0 g

So the sludge volume index =

$$\frac{\text{V m}\ell/\text{L}}{\text{m(g)/L}} = \frac{\text{Settled sludge volume}}{\text{Suspended solid concentration}} \times 100$$

$$\frac{30 \text{ mL} \times 1000}{3.0 \times 1000} = 30 \text{ mL/g}$$

**111. Under natural flow conditions, a polluted river will contain**

(a) more dissolved oxygen in summer than in winter
(b) more dissolved oxygen in winter than summer
(c) about the same dissolved oxygen in summer and winter
(d) the least amount of dissolved oxygen during floads

**Ans. (b) :** At normal temperature D.O level found to be 10 mg/litre
- D.O. content depends on the water temperature.
- It helps to aquatic life to do respiration into water.
- If the temperature of sewage is more, the D.O. content is young to be less and vice, versa.
- So in summer DO in water is found to be lower than winter.
- It is determined by the wrinkle's test.

**112. Elutriation is a process of**
(a) sludge digestion
(b) filtration
(c) sedimentation
(d) washing digested sludge

**Ans. (d) : Elutriation–** Elutriation is the process of separation of the particles based in their size, shape and density, using a stream of gas or liquid flowing in a direction usually opposite to the direction of sedimentation.
- It is the process to washing digested sludge with fresh water plant effluent or other waste water to remove fine particulate or change of alkalinity reducing the demand for conditioning and improving settling/filtering characteristics of solids.
- Smaller particle have slower terminal velocity due to its lights weight (1 μm) and they rise to the top.
- Elutriation is a process of sludge conditioning in which sludge is washed either by fresh water or plant effluent.

**113. Symbiosis, the beneficial association of and bacteria for treatment of waste water is used in**
(a) activated sludge unit
(b) rotating biological disc
(c) aerobic digester
(d) oxidation pond

**Ans. (d) :** Symbiosis, the beneficial association of bacteria for treatment of waste water is used in oxidation pond.
- In this process the relationship between algae and bacteria where algae in the presence of sunlight produces oxygen by photosynthesis which is used by bacteria for oxidizing the waste organic matter which are required by algae to grow and continue producing oxygen.

**114. The best method of disposal of refuse to ensure complete destruction of pathogenic bacteria is**
(a) land disposal (b) incineration
(c) pulverization (d) sea disposal

**Ans. (b):** Land fill is done for the non-biodegradable waste-
- I and fill helps to prevent the contamination of soil and water from waste and helps to reduce odor and pests.
- In incineration process the waste is burst in furnace till ash.
- Combusted waste changes into water and carbondioxide.
- Reduces waste 20 to 300 of its volume.
- It is done in such a way that does not release extreme amount of heat.
- This is best method of disposal of refuge to ensure complete destruction of pathogenic bacteria.

**115. Water can be dechlorinated by adding**
(a) sodium thiosulphate
(b) sodium sulphate
(c) sodium hexametaphosphate
(d) sodium bisulphate

**Ans. (a) :** Dechlorination agent used are
Sodium thiosoluphate, activated carbonl, sodium sulphate, and sodium bisulphit, sulphure dioxide.

**Dechlorination–** When the chlorine residue is high, excess chlorine will be remove by dechlorinating agent.
**Dechlorinating agent–**
(i) Sodium thiosulphate ($Na_2S_2O_3$)
(ii) Sulphur dioxide ($SO_2$)
(iii) Sodium sulphate ($Na_2SO_3$)
(iv) Activated carbon.

**116. Sewage sickness indicates**
(a) toxicity of sewage interfering with treatment
(b) destruction of aquatic flora and fauna
(c) reduction in the waste purifying capacity of soil
(d) clogging of soil pores and septic conditions

**Ans. (d) : Sewage sickness–** when sewage is continuously applied on a piece of land the pores or void in the soil get clogged and the free circulation of air will be prevented this is know as sewage sickness.

**117. Area method of land filling is more suitable when**
(a) the area is unsuitable for excavation of trenches
(b) adequate depth of cover material is available at the site
(c) natural or artificial depressions exist in the vicinity
(d) water table is near the ground surface

**Ans. () :** There are various types of land fills (sanitary land fills) –
- Trench method • Area method, • Ramp method

**Area method –** Best used in areas where natural depression exists as in quartzite, ravines and valleys. The waste is put in the natural depression and compacted. A layer of earth is given on top and compacted. Process is repeated till the depression is filled up.

**118. The function of a sluice valve is to**

(a) control flow through pipe lines
(b) release accumulated air
(c) remove silt accumulated in the pipes
(d) reduce water hammer effects

**Ans. (a) :** Sluice valve is also known as gate valve or shut off valve.

- These valves are provided to regulate the flow of water through the pipe and are essential to divide the main line into several section.
- These valve are usually placed at the summit of the pressure conduit where pressure is low

**119. Ringlemann's scale**

(a) measures CO
(b) measures $SO_2$
(c) grades density of smoke
(d) grades automobile exhaust

**Ans. (c) : Ringlemann's scale –** It is used to measure density of smoke.

- This scale was developed by a French professor of agricultural engineering in 1888. He also developed Ringlemann effect.

**120. Eutrophication of water bodies is caused by**

(a) discharge of toxic substances
(b) excessive discharge of nutrients
(c) excessive discharge of suspended solids
(d) excessive discharge of chlorides

**Ans. (b) : Eutrophication–** It is the excessive richness of nutrients in a water body of water, generally due to run off from the land, which causes a dense growth of plant life.

**121. Clayey minerals are identified from non-clayey minerals by**

(a) specific surface (b) atterberg's limits
(c) permeability (d) lattice structure

**Ans. () :** Clayey minerals are identified from non-clayey mineral by their size difference and mineralogy. Their are the various minerals like, silica, alumina or magnesia or both and water.

**122. Cohesionless soil is**

(a) sand (b) clay
(c) clay and silt (d) sandy clay

**Ans. () :** Cohesionless soil is soil that have no clay content in soil mass.
Sand is considered as cohesionless.
Silt has very less cohesive value.
Collidal are most cohesive soil in all soils.

**123. Specific gravity of sand is about**

(a) 1.6 (b) 2.0
(c) 2.2 (d) 2.7

**Ans. (d) :**

- Specific gravity of cement = 3.15
- Specific gravity of manufactured sand = 2.73
- Specific gravity of coarse aggregate = 2.6
- Specific gravity of wood = 1.54
- Specific gravity of mercury = 13.6

**124. Liquid and plastic limits are determined for**

(a) sandy soils (b) clay soils
(c) silty soils (d) gravel

**Ans. (b) :** Liquid and plastic limits are determined for the clay soils. The water contents at which the soil changes from one state to the other all known as consistency limits or atterbergs limits.

- Clay soil have high liquid limit and plastic limit.
- Clay are the fine grained soil and exist in ..... states, namely, liquid, plastic, semi-solid and solid state.

**125. The maximum shear stress occurs on a fillment on a plane at an angle with the horizontal of**

(a) $30^0$ (b) $45^0$
(c) $60^0$ (d) $90^0$

**Ans. (b) :** The maximum shear stress occurs on a filament on a plane at an angle with the horizontal of $45^0$.

- The maximum shear stress $\tau_{max}$ is numerically equal $t_0(\sigma_1 - \sigma_3)/2$ and it occurs on a plane inclined at $45^0$.

**126. Triaxial test is preferred to direct shear test because**

(a) it can be performed with complete control under drainage in all the three directions
(b) pore pressure can be measured precisely and volume change is not possible during the test
(c) stress distribution is uniform on failure plane
(d) pore pressure does not change during the tests

**Ans. (d) :** Triaxial test is preferred over the direct shear test because it can be performed with complete control under drainage in all the three direction.

- Plain of failure in direct shear test is predefined and failure plain is always horizontal and shear plain.
- Triaxial test is controlled test in which expulsion of pore water is done under controlled way during test.

**127. For the base failure of a slope, the depth factor is**

(a) 1.0 (b) greater than 1.0
(c) less than 1.0 (d) greater than 1.5

**Ans. (b) :** Rotational failure are of 3 types –
(i) face failure or slope failure
(ii) Toe failure
(iii) Base failure

**Face or slope failure** occurs when soil above the toe contents weak stratum. In this case, the failure plane intersects the slope above toe, $D_f < 1$.

- Toe failure occur when failure plane passes through tow of slope . For toe failure, depth of foundation, $D_f = 1$.
- Base failure occurs when there is a weak soil stratum under the toe and failure plane passes through base of slope. For base failure, depth of foundation, $D_f = 1$

**Base Failure**

**128. If C is the consolidation coefficient, t is the time and d is the drainage path of one dimensional consolidation, the time factor T is given by**

(a) $T = C.t/d^2$ (b) $T = t^2/(d^2C)$

(c) $T = C/(dt^2)$ (d) $T = C^3/(d^2t)$

**Ans. (a) :** If C is the consolidation coefficient, t is the time and d is the drainage path of one dimensional consolidation, the time factor T is given by

$$T_V = \frac{C_V.t}{d^2}$$

Time factor– It is dimensionless parameter.

$$C_V = \frac{T_V.d^2}{T}$$

where,

$C_V$ = Coefficient of consolidation
d = length of drainage path
$T_V$ = Time factor
t = time

**129. The minimum depth of footings in sandy soils is about**

(a) 0.10 - 0.50 m (b) 0.50 - 1.00 m
(c) 0.80 - 1.00 m (d) 0.50 - 1.90 m

**Ans. (c) :** Minimum depth of foundation in sandy soil is kept around 0.8 to 1 m deep bended the soil where hard strata of sandy soil mix with either clay, loam and gravel is found above ground water table.

• For medium plastic soil minimum foundation. depth is kept 1 m and 0.9 m for clayey soil beneath the soil where hard strata of soil is found.

**130. A pile penetrates by 5.0 mm with last blow of a drop hammer of 1800 kg mass and free fall of 1.0 m; the capacity of the pile as per Engineering News formula is**

(a) 1000.0 kN (b) 500.0 kN
(c) 200.0 kN (d) 100.0 kN

**Ans. (c) :** Engineering new formula.

$$Q_H = \frac{whn_n}{S + C}$$

C = 2.54 cm
w = 1800 kg
h = 1.0 m
S = 5.0 mm
$n_n$ = 0.7 to 0.85 for vertical drop hammer.

$$QH = \frac{1800 \times 1 \times 0.7}{(5 + 2.54) \times 10^{-3}}$$

$= 202.91 \times 10^3$
$= 202.91$ kN

**131. The clay soil group that does not swell under wet conditions is**

(a) kaolinite (b) ollite
(c) vermicilitel (d) montrorillinte

**Ans. (a) :** The clay soil group that does not swell under wet conditions is kaolinite.

• Illite swells less than montomorillonite. However, swelling is more than in kaolinite.

**132. The equation for direct shear $\tau = c + \sigma \tan \phi$ was formulated by**

(a) rankine (b) mohr
(c) coulomb (d) culmann

**Ans. (c) :** The equation for direct shear $\tau = c + \sigma \tan \phi$ was formulated by coulomb.

When pore water pressure is also inexistence than

$$\tau = c' + \bar{\sigma} \tan \phi'$$

c' and $\phi'$ are the cohesion intercept and angle of shearing resistance in terms of the effective stresses.

**133. Under-reamed piles are suitable**

(a) in clay soils (b) in sandy soils
(c) for precasting (d) for steel sections

**Ans. (a) :** Undereamed piles are the bored pile and constructed by drilling deep ends the ground and filling it with concrete.

• Under reamed biles are used in claysoils and provided bulb of increased area for better load support.

**134. Standard penetration test is performed to determine**

(a) soil depth (b) nature of soil
(c) bearing capacity (d) ground water depth

**Ans. (c) :** Standard penetration test is performed to determine the shear strength of the cohessionless soil.

• It determines the relative density and angle of shearing resistance of sand.

• Used also to determine the unconfined compressive strength of cohesive soils. standard penetration number is taken equal to the number of blows required for 300 mm penetration beyond the seating drive of 150 mm.

• If blows number for 150 mm drives exceeds 50, it is taken as refusal and test is discontinued.

**135. The depth of influence of loaded area indicates the depth**

(a) upto which the soil should be excavated
(b) below which the soil is stressed to 50 percent
(c) below which the soil is stressed to 10 percent
(d) below which hard soil exists

**Ans. (c) :** • Isobars are useful for determining the effect of the load or vertical stress at various point.

• The zone within which the stress have a significant effect on the settlement of structures is known as the pressure bulb. It is generally assumed that isobar of 0.10 forms a pressure bulb.

So the depth of influence of loaded area indicates the depth below which soil is stressed to 10 percent.

**136. The process of photography from ground is known as**

(a) aerial surveying
(b) terrestrial surveying
(c) telescopic surveying
(d) astronomical surveying

**Ans. (b):** The process of photography from ground is known as terrestrial surveying.

- But if the photography from air is taken then the term will be called as axial photography, or aerial photo grammarte.

**137. The parallel formed by a plane passing through the centre of spheroid and normal to its axis is known as**

(a) equator (b) meridian
(c) polar radius (d) longitude

**Ans. (a) :** The parallel formed by a plane passing through the centre of spheroid and normal to its axis is known as equator.

- When the earth's axis of rotation extended in both the directions meets the celestial sphere at celestial north and celestial south called as poles of the earth.

**138. The reduced representation of ground contour in horizontal projection is known as**

(a) isotype (b) profile
(c) isocline (d) plan

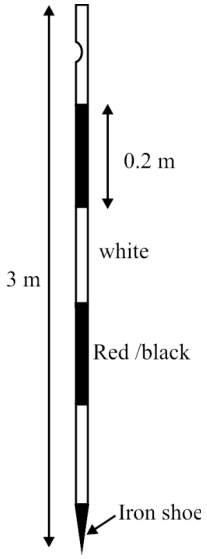
**Ans. (c) :** The reduce representation of ground contour in horizontal projection is known as isocline.

- It is line joining the equal projection or elevation.

**139. Offset rods are useful to**

(a) measure distances
(b) test chains or tapes
(c) mark stations
(d) measure the offsets

**Ans. (d) : Offset rods–** It is mainly used for measuring offsets nearby. It is similar to ranging rod and has a length of 3 meter.




**140. The smaller angle made by a survey line with the true meridian is known as**

(a) true bearing (b) azimuth
(c) magnetic bearing (d) grid meridian

**Ans. (b) :** The smaller angle made by a survey line with the true meridian is known as azimuth.

- It is generally observed from north.
- It depends on the reference meridian and it can also called geodetic, astronomic, magnetic, gird.

**141. Graphic triangulation method of plane table surveying is also known as**

(a) intersection method
(b) radiation method
(c) resection method
(d) traversing method

**Ans. (a) : Intersection method –** It is used when the base stations are visible accessible but other station are only visible but not accessible. It is most suitable for hilly area and also known as graphical triangulation.

**142. Systematic error in surveying**

(a) are cumulative and can be determined
(b) cannot be measured
(c) are compensatory
(d) can be eliminated during survey

**Ans. (a) : Systematic error–**

- This error arises from the source that act in a similar manner on observations.
- Systematic errors are cumulative in nature.
- The error caused by systematic error are eliminated by applying corrections.

**143. A quick and accurate method of solving three point problem in resection plane table surveying is**

(a) Besel's graphical method
(b) Tracing paper method
(c) Lehmann's method
(d) Mechanical method

**Ans. (c) : Trial and error method or Lehman's method–** This method is very commonly used in field measurement as it is very accurate. The position of a plane table is estimated by judgment. This triangle is reduced to point by trial and error.
This is the quickest and most accurate method to solve three-point problem in resection of plane tabling.

**144. Nagpur road plan recommended for road system**

(a) star and circle pattern
(b) star and grid pattern
(c) star and block pattern
(d) star and hexagonal pattern

**Ans. (b) : Nagpur Road plan (1943–63)–** In this first 20 year road development plan popularly known as the Nagpur road plan, all road where classified into five categories and 20 year development programme from period 1943–63.

- The Nagpur road plan formula where prepared on the basis of star and grid pattern.

• The total road length 5,32,700 km with a density of 16 km of road length per 100 $km^2$.
• It is design star and grid pattern.
• This plan divide into 5 categories road (NH, SH, MDR, ODR, VR)

**145. An endoscope measures**
(a) running speed
(b) time-mean speed
(c) spot speed
(d) average speed

**Ans. (c) :** Spot speed study on highway is conducted by endoscope.
• The spot speed may be obtained either by finding the running speed of vehicles over a short distance of less than 50 metre or by finding the instantaneous speed while crossing a section, depending upon the method used.
• It is the simple, cheap and easy to use.

**146. Traffic conflicts likely to occur at a rotary intersection are**
(a) crossing, merging and diverging
(b) crossing and diverging
(c) crossing and merging
(d) merging and diverging

**Ans. (d) :** The traffic conflicts that may occur in a rotary intersection are merging and diverging.

**147. The gradient on a highway is limited by**
(a) maximum tractive effort for short distance
(b) minimum tractive effort on the whole gradient
(c) alignment design
(d) drainage requirements

**Ans. (d) : Ans. (c) :** Where topography of a place compels adopting steeper gradients than ruling gradients, limiting gradients are used in view of enormous increase in cost in constructing roads with gentler gradients. However the length of continuous grade line steeper than ruling value should be limited. On rolling terrain and on hill roads, it may be frequently necessary to exceed ruling gradient and adopt limiting gradient, but care should be taken to separate such stretches of steep gradients by providing either a level road of a road with easier grade

| Terrain | Limiting gradient |
|---|---|
| Plain or rolling | 5% |
| Mountainous terrain and steep terrain, elevation > 3000 m | 6% |
| Steep terrain < 3000 m | 7% |

**148. The effect of impact on the design of rigid pavement is accounted by**
(a) increasing the thickness required for static loading
(b) providing flexible base course
(c) reducing the allowable flexural stress in concrete
(d) increasing the stresses under static loading

**Ans. (d) : CBR test–**
• The CBR method limitations gives the total thickness required of the pavement above a subgrade and this thickness value would remain same respective of the quality of material used in component layer.
• It surcharge weights are used to stimulate the effect of overlaying pavement.
• The component of material should be judiciously chosen for durability and economy.
• Rigid pavements posses note worthy flexural strength or flexural rigidity.
• Rigid pavement transfer the load through slab action but not grain to grain as in case of flexible pavement.
• The effect of impact on the design of rigid pavement is accounted for adopting an increased stress relative to that produced by static wheel load.

**149. A vehicle travelling on a dry, level pavements at 80.0 kmph stops at a distance of 70.0 m after applying brakes; the coefficient of friction developed is**
(a) 0.36 (b) 0.37
(c) 0.38 (d) 0.39

**Ans. (a) :** Given,
Stopping sight distance, SSD = 70 m
speed of vehicle, V = 80 kmph
We know that

$$SSD = \frac{V^2}{254f}$$

$$70 = \frac{80^2}{254 \times f}$$

$$f = \frac{80^2}{70 \times 254}$$

$$\boxed{f = 0.35995 \approx 0.36}$$

**150. The minimum radius of the horizontal curves designated for a speed of 100.0 kmph, super elevation of 0.08 and coefficient of friction 0.15 is**
(a) 312 m (b) 322 m
(c) 333 m (d) 342 m

**Ans. (d) :** Given v = 100 km/h = 27.78 m/sec,
e= 0.08, f = 0.15

we know that superelevation, $e + f = \frac{v^2}{Rg}$ or

$$R = \frac{v^2}{g(e+f)}$$

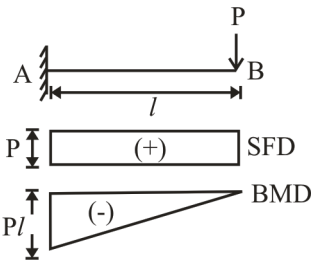
$$R = \frac{(27.78)^2}{9.81(0.08+0.15)}$$

$$\boxed{R = 342.032\text{m} \approx 342\text{m}}$$

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 1993
## Solved Paper

**1. A cantilever beam is loaded by a point load at the free end. It is likely to fall by B.M.**
(a) under the load (b) at mid span
(c) at support (d) at any section

**Ans. (c) :**




The above figure shows a cantileveler beam which is loaded by a point load ρ at the free and.
Shear force and Bending moment diagram is also shown in figure.
Here maximum bending moment (M = P$\ell$)
Occurs at the fixed end and constant shear force (P) throughout the length.
So maximum stress will be at the fixed end and failure will occur at that point.
figure

**2. In a beam under loading**
(a) SF is maximum when BM is maximum
(b) SF is zero when BM is zero
(c) SF is zero when BM is maximum
(d) SF is maximum when BM is zero

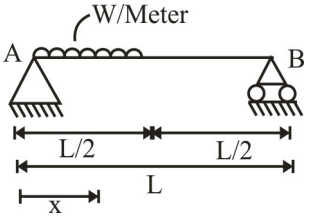
**Ans. (c) :** Bending moment will be maximum when shear force changes its sign i.e zero.

**3. Where the intensity of loading varies linearly, the B.M. will vary**
(a) Linearly (b) Cubically
(c) Parabolically (d) Constant rate

**Ans. (b) :** In uniformly distributed load, the nature of shear force is linear and bending moment is parabolic.
• In uniformly varying load, the nature of shear force is parabolic and bending moment is cubic.

**4. The left half of a simply supported beam of span "L" is loaded with u.d.l. of w/metre run. The point of maximum B.M. will occur at**
(a) Mid span
(b) 2L/3 from left support
(c) 3L/8 from left support
(d) 5L/8 from left support

**Ans. (c) :**




$$R_A + R_B = \frac{WL}{2}$$

Take moment about point A

$$R_B \times L = W \times \frac{L}{2} \times \frac{L}{4}$$

$$R_B = \frac{WL}{8}$$

$$R_A = \frac{3WL}{8}$$

For maximum bending moment, Shear force at

$$X = R_A - w \times \mathbf{x} = \mathbf{0}$$

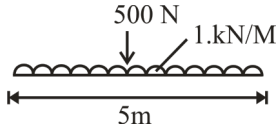
$$w \times \mathbf{x} = R_A$$

$$w \times x = \frac{3wL}{8}$$

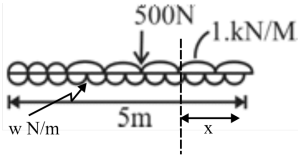
$$x = \frac{3L}{8}$$ (from left-support)

**5. A wooden log of length 5 m is floating in water. Its self weight is 1 kN/m. If a man of weight 500 N is standing at the mid span, the maximum B.M. is**
(a) 0 (b) 625 Nm
(c) 250 Nm (d) 312.5 Nm

**Ans. (d) :**




Assume buoyancy force exerted in the UDL in the upward direction




$w \times 5 = 500 + 1000 \times 5$
$w = 1100$ N/m

Bending moment

$$M_{x\text{-}x} = \frac{1100 \times x^2}{2} - \frac{1000 \times x^2}{2}$$

$$= \frac{100x^2}{2}$$

Maximum BM at x = 2.5 at mid – span

$$M_{xx} = \frac{100 \times (2.5)^2}{2}$$

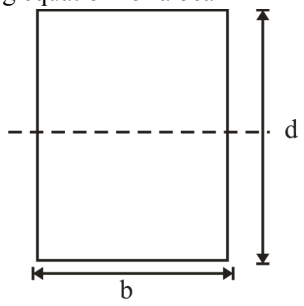
= 312.5 Nm

**6. For a rectangular section, keeping breadth "b" constant, the depth "d" for uniform strength beam will have relation with bending moment 'M' as**

(a) $d \propto M^2$ (b) $d \propto \overline{M}$

(c) $b \propto \sqrt{M}$ (d) $b \propto M^2$

**Ans. (c) :** Bending equation for a beam




as $\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$

$M = \sigma \times z$

$= \sigma \times bd^2/6$

$M \propto b$

$M \propto d^2$

$d \propto \sqrt{M}$

**7. The beam of uniform strength is one in which**

(a) bending stress varies uniformly throughout the length of the beam

(b) bending stress is constant throughout the depth of the beam

(c) bending stress is constant throughout the length of the beam

(d) none

**Ans. (c) :** A beam of uniform strength is one in which bending stress developed at every cross – section of the beam remains the same along the longitudinal axis.

**8. The dia of the circular cross section is 400 mm. The breadth of strongest beam is ____ mm.**

(a) $400\sqrt{3}$ (b) 200

(c) $400\sqrt{6}$ (d) $400/\sqrt{3}$

**Ans. (d) :** Strongest beam is the one for which section modulus Z is maximum.

Let rectangular section is of width b and depth d,

Diameter of cylindrical log, D = 400 mm

$b^2 + d^2 = D^2$

$$z = \frac{bd^2}{6} = \frac{b\left(D^2 - b^2\right)}{6} = \frac{bD^2 - b^3}{6}$$

for maximum z ;

$$\frac{dz}{db} = 0$$

$$\frac{D^2 - 3b^2}{6} = 0$$

$$b = D/\sqrt{3} = 400/\sqrt{3}$$

**9. The ratio of maximum shear stress developed in a beam of rectangular section that of the average shear stress is _____**

(a) 4/3 (b) 3/4

(c) 3/2 (d) 2/3

**Ans. (c) :** The ratio of maximum and average shear stresses on a rectangular section is 3/2

**10. A wooden beam of rectangular cross section 100 mm × 200 mm is formed by giving two identical beams of square cross section. If the safe shear strength of the glue is 3 N/mm$^2$, the safe shear straight**

(a) 30 kN (b) 40 kN

(c) 50 kN (d) 60 kN

**Ans. (b) :** Maximum shear stress

$$\tau_{max} = \frac{3}{2}\frac{F}{bh}$$

$$3 = \frac{3}{2} \times \frac{F}{100 \times 200}$$

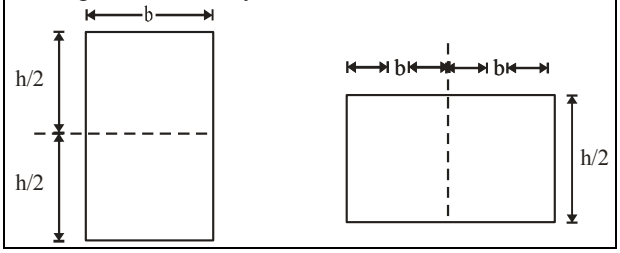
F = 40 KN

**11. If a beam is cut in halves horizontally and the two halves are laid side by side, it can in comparison to the original beam carry.**

(a) the same load (b) double the load

(c) half the load (d) one-fourth the load

**Ans. (c) :** If a beam is cut in halves horizontally and the two halves are laid side by side it can in comparison to the original beam carry half the load.




**12. Curvature in deflection of beams is given by**

(a) dy/dx (b) $d^2y/dx^2$

(c) $d^3y/dx^3$ (d) $d^4y/dx^4$

**Ans. (b) :** From bending equation

$$\text{Radius of curvature (R)} = \frac{EI}{M} = \frac{1}{\frac{d^{2y}}{dx^2}}$$

$$M = EI\ \frac{d^2y}{dx^2}$$

**13. When a beam deflects into a circular arc, the expression for slope is given by ..... radians**

(a) L/2R (b) L/8R

(c) $L^2/8R$ (d) $L^2/2R$

**Ans. (c) :** When a beam deflects into a circular arc, the expression for slope is given by $L^2/8R$ radians.

**14. Deflection at the free end of cantilever beam of span L loaded with concentrated load at the free end is**

(a) $PL^2/2EI$ (b) $PL^2/3EI$

(c) $PL^3/3EI$ (d) $PL^4/8EI$

**Ans. (c) :** Deflection is beam = Bending moment in Conjugate beam

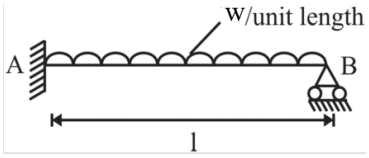
$$= \frac{1}{2} \times \frac{PL}{EI} \times L \times \frac{2L}{3} = \frac{PL^3}{3EI}$$

**15. A propped cantilever of span L carries a UDL of w per unit length. If prop is at free end prop reaction is**

(a) 1/2wL (b) 5/8wL

(c) 3/8wL (d) 3/4wL

**Ans. (c) :**




Deflection of end B = 0

(Downward deflection due to uniformly distributed load) – ( Upward deflection due to $R_B$) = 0

$$\frac{wl^4}{8EI} - \frac{R_B \times l^3}{3EI} = 0$$

$$R_B = \frac{3}{8} wl$$

**16. A rectangular beam simply supported at the ends and carrying point load W at the centre causes a deflection $\Delta_1$. If the width of the beam is doubled, the deflection at the centre under the same load will be.**

(a) $\Delta_1$ (b) $\Delta_1/4$

(c) $2\Delta_1$ (d) $\Delta_1/2$

**Ans. (d) :** Deflection at the center due to central conjugate load on a simply supported beam

$$\Delta = \frac{WL^3}{48EI}$$

Moment of inertia for rectangular beam

$$I = \frac{bd^3}{12}$$

$$\Delta \propto \frac{1}{b}$$

$$\frac{\Delta_2}{\Delta_1} = \frac{b_2}{b_1}$$

$$\frac{\Delta_2}{\Delta_1} = \frac{b}{2b}$$

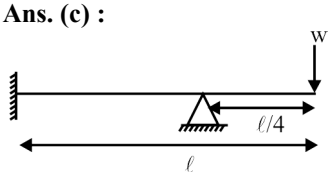
$$\Delta_2 = \frac{\Delta_1}{2}$$

**17. A cantilever beam of span L is subjected to point load W at the free end. It is propped at the distance of 1/4 from free end. The prop reaction is**

(a) W (b) W/2

(c) 3W/2 (d) None

**Ans. (c) :**




Net deflection at B

$$\frac{R_B\left(\frac{3\ell}{4}\right)^3}{3EI} = \frac{w\left(\frac{\ell}{4}\right)\left(\frac{3\ell}{4}\right)^2}{2EI} + \frac{w\left(\frac{3\ell}{4}\right)^3}{3EI}$$

$$\frac{27R_A}{64 \times 3} = \frac{9w}{64 \times 2} + \frac{27w}{64 \times 3}$$

$$(R_A - w) \times \frac{27}{64 \times 3} = \frac{9w}{64 \times 2}$$

$$R_A - w = \frac{w}{2}$$

$$R_A = \frac{w}{2} + w$$

$$R_A = \frac{3w}{2}$$

**18. If the cylindered pipe is subjected to internal fluid pressure the nature of hoop stress is**

(a) shear (b) compression

(c) Tension (d) any one

**Ans. (c) :** When cylindrical pipe is subjected to internal fluid pressure the hoop stress is tensile in nature and when cylindrical pipe is subjected to external fluid pressure the hoop stress is compressive in nature.

**19. In a thin cylinder, circumferential stress is x time longitudinal stress, where x is**

(a) 0.5 (b) 1

(c) 2 (d) 0.2

**Ans. (c) :** Circumferential or hoop stress

$$(\sigma_n) = \frac{Pd}{2t}$$

$$\text{longitudinal stress } (\sigma_l) = \frac{Pd}{4t}$$

$$\frac{\text{Circumferential stress}}{\text{longitudinal stress}} = \frac{pd/2t}{pd/4t} = 2$$

$$\sigma_n = 2\sigma_l$$

**20. A cylindrical shell 3 meters long. 1 metre in diameter, thickness of metal 10 mm is subjected to a internal pressure of 1.5 N/mm$^2$. It E = 200 kN/mm$^2$ and 1/m= 0.3, the volumetric strain produced in the cylinder is**

(a) 0.000075 (b) 0.000319

(c) 0.0007125 (d) 0.0006375

**Ans. (c) :** Given $l = 3m = 3000$ mm

$d = 1m = 1000$ mm

$t = 10$ mm

$p = 1.5$ N/mm$^2$

$E = 200$ kN/mm$^2 = 200 \times 10^3$ N/mm$^2$

$$\frac{1}{m} = \mu = 0.3$$

Volumetric strain $\epsilon_v = \frac{\delta v}{v} = \frac{pd}{4tE}(5 - 4\mu)$

$$= \frac{1.5 \times 1000}{4 \times 10 \times 200 \times 10^3}(5 - 4 \times 0.3)$$

$$= 0.0007125$$

**21. When a shaft is subjected to a twisting moment, every cross section of the shaft will be under**

(a) Tensile stress (b) Compressive stress
(c) Shear stress (d) Bending stress

**Ans. (c) :** Twisting moment impart an angular displacement of one and cross-section with respect to the other end and it will setup shear stresses an any cross section of bar perpendicular to its axis.

**22. In pure torsion of hallow circular shafts, the maximum shear stress occurs at ----- of the cross section.**

(a) the axis (b) the outer surface
(c) the inner surface (d) every point

**Ans. (b) :** The shear stress distribution is linear.
The shear stress of a circular shaft is maximum at the outer surface of the shaft.

**23. The torque required in a circular shaft rotating at 100 r.p.m to produce I h.p. is _____ kg. cm.**

(a) 35.8 (b) 7.16
(c) 716 (d) 358

**Ans. (c) :** Given $P = 1$ HP $= 746$ W

$N = 100$ rpm

$$\text{Power (p)} = \frac{2\pi NT}{4500}$$

$$1 = \frac{2 \times 3.14 \times 100 \times T}{4500}$$

$T = 7.1656$ kg m

$= 716.56$ k cm

**24. If closely coiled helical spring is subjected to torsion, the type of stress developed inthe spring material is**

(a) tensile (b) compressive
(c) shear
(d) varies with type of material

**Ans. (c) :** If closely coiled helical spring is subjected to torsion, the type of stress developed in the spring material is shear.

**25. In a closely coilde helical spring subjected to a axial load, other quantities remaining the same, if the wire diameter is doubled, then the stiffness of the spring when compared to the original one will become.**

(a) twice (b) four times
(c) eight times (d) sixteen times

**Ans. (d) :** Stiffness of the spring is given by

$$k = \frac{W}{\delta} = \frac{Gd^4}{8D^3n}$$

D = Mean diameter of the spring
d = wire diameter
n = number of active wills
G = modulus of rigidify

$k \propto d^4$

$d^1 = 2d$

$$\frac{k^1}{k} = \frac{(2d)^4}{d^4} = 16$$

$k^1 = 16\ k$

**26. Long column is a column whose slenerness ratio is greater than ..... or whose length is more than ..... times the least lateral dimension.**

(a) 112;40 (b) 80;20
(c) 120;30 (d) 32;8

**Ans. (b) :** long column are those with a slenderness ratio greater than 100 for ductile materials and greater than 80 for cast iron or whose length is more 20 times the length dimension.

**27. For a column fixed at both ends, the effective length of the column as a fraction of unsupported length according to Euler's formula is equal to**

(a) $1/\sqrt{2}$ (b) 2
(c) 1 (d) 1/2

**Ans. (d) :**

| **End Condition** | **Effective length column** |
|---|---|
| Both end hinged | $l$ |
| Both end fixed | $l/2$ |
| One fixed and other hinged | $l/\sqrt{2}$ |
| One fixed and other free | $2l$ |

**28. The slenderness ratio of column is ZERO, when**

(a) Its length is exactly equal to radius of gyration Its length is half the radius of gyration
(b) Its length is half the radius of gyration
(c) Its lengths is supported all sides throughout its full length.
(d) None

**Ans. (c) :** When the when is supported throughout its length, then the effective length of a column will be zero, so the slenderness ratio also become zero.
Slenderness ratio is defined as the ratio of effective length and ratio of gyration.

**29. A shaft column of square cross section 2 m wide is loaded with a point load of 8 kN at one of the corner. The stress at the corner in Kn/m$^2$ is**

(a) 14 (b) 2
(c) 12 (d) 8

**Ans. (b) :** Given,

$A = 2 \times 2 = 4m^2$

$P = 8$ kN

$\text{Stress} = \frac{\text{Load}}{\text{Area}} = \frac{8}{4} = 2kN/m^2$

**30. Relation between number of joints J and number of members 'm' in pin joined plane determinate truss is**

(a) $m = 2j - 1$ (b) $m = 2j - 3$

(c) $m = 2j - 5$ (d) $j = 2m - 3$

**Ans. (b) :** (i) if $m = 2j - 3$ then system is statically determinate

(ii) if $m < 2j - 3$ then system is collapsed

(iii) if $m > 2j - 3$ then system is statically indeterminate.

Where m = number of member

j = number of joints

**31. The number of members in a frame is 10, and the no. of joints is 9. It is an example of**

(a) Perfect frame (b) Deficit frame

(c) Redundant frame (d) None

**Ans. (b) :** Given $m = 10$ , $j = 9$

(i) for perfect frame

$m = 2j - 3$

(ii) for imperfect frame

$m \neq 2j - 3$

(iii) For redundant frame

$m > 2j - 3$

(iv) for deficient frame

$m < 2j - 3$

$10 < 2 \times 9 - 3 = 10 < 15$

**32. If a rigid body is a equilibrium under the action of three forces, they must be**

(a) Equal (b) Non-concurrent

(c) Concurrent (d) Collinear

**Ans. (c) :** If a body in equilibrium is acted upon by three forces, then the resultant of any two forces must be equal, opposite and collinear with the third force.

If a three force member is in equilibrium and the forces are not parallel, they must be concurrent. Therefore the lines of action of three forces acting on such a member must intersect at a common point; any single force is, therefore. the equilibrium of the other two forces.

**33. Parallelogram law of forces states that if two forces acting simultaneously at a point be represented in magnitude and direction by two adjacent sides of a parallelogram, their resultant may be represented in magnitude and direction by**

(a) Longer side of the other two sides

(b) Shorter side of the other two sides

(c) Diagonal of the parallelogram which does not pass through their point of intersection

(d) Diagonal of the parallelogram which passes through their points of intersection

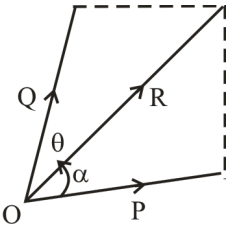
**Ans. (d) :** The law of parallelogram of two forces states that if two vectors acting on a particle at the same time are represented in magnitude and direction by two adjacent sides of a parallelogram drawn form point their resultant is represented by he diagonal of the parallelogram drawn from the same point.

**34. The resultant of two forces acting at right angles is $\sqrt{34}$ N and acting at $60^0$ is 7N. The forces are**

(a) N and 4N (b) 3N and $\sqrt{5}$ N

(c) $\sqrt{3}$ N and $\sqrt{5}$ N (d) 3N and 5N

**Ans. (d) :** Let P & Q are two forces and R is the resultant of a both forces. Resultant forces R is given by




$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$

$R^2 = P^2 + Q^2 + 2PQ\cos\theta$

Case (i) = $\theta = 90^\circ$

$\left(\sqrt{34}\right)^2 = P^2 + Q^2 + 2PQ\ Cos90^\circ$

case (ii) $\theta = 60^\circ$

$(7)^2 = P^2 + Q^2 + 2PQ\ \cos 60^\circ$

$P^2 + Q^2 + PQ = 49$ ..... (ii)

On solving equation (i) and (ii)

$P = 3N$ , $Q = 5N$

**35. Limiting friction is the**

(a) Minimum static friction

(b) Minimum sliding friction

(c) Maximum static friction

(d) Friction that limit the velocity of a body

**Ans. (c) :** The maximum static friction that a body can exert on the other body in contact with it, is called limiting friction, this limiting friction is proportional to normal contact force between the two bodies.

$f_{max} = \mu_s N$

Where $f_{max}$ = maximum possible static friction

N = normal force

**36. Angle of friction is the angle**

(a) made by the surface with the horizontal

(b) limiting friction and normal reaction

(c) limiting friction and the resultant of limiting friction and normal reaction

(d) normal reaction and the resultant of limiting friction and normal reaction

**Ans. (d) :** Angle of friction is the angle between normal reaction and the resultant of normal reaction and the limiting friction.

**37. For an ideal machine**

(a) mechanical advantage > velocity ratio

(b) mechanical advantage < velocity ratio

(c) mechanical advantage = velocity ratio

(d) mechanical advantage = velocity ratio/2

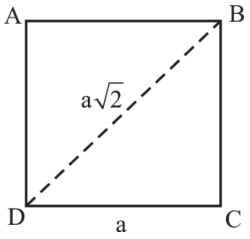
**Ans. (c) :** For an ideals machine, mechanical advantage is equal to the velocity ratio. for a practical machine, mechanical advantage is less than the velocity ratio as efficiency is less than one.

**38. In which of the following the centric lies outside the section?**
(a) Symmetrical I-section
(b) Tee-section
(c) Unsymmetrical I-section
(d) Angle section

**Ans. (d) :** The centroid of a shape represent the point at which the area of the section is evenly distributed.
It is also defined as a point where the whole body mass can be concentrated for evaluating the physical and numerical quantities.
Angle section is the section in which centriod lies outside the section.

**39. The moment of inertia about a diagonal of a square plate of side a is**
(a) $a^4/12$ (b) $a^4/16$
(c) $a^4/6\sqrt{2}$ (d) None

**Ans. (a) :** Moment of inertia of the square about the diagonal




$I_{square} = \frac{a^4}{12}$ figure

**40. When the diameter of the circular section is doubled, its radius of gyration**
(a) in reduced to half
(b) increased by 8 times
(c) is doubled
(d) increases by three times

**Ans. (c) :** Radius of gyration $k = \sqrt{\frac{I}{A}}$

$$I = I = \frac{\pi}{64} d^4,\ A = \frac{\pi}{4} d^2$$

$$k = \sqrt{\frac{\frac{\pi}{64} d^4}{\frac{\pi}{4} d^2}} = k = \sqrt{\frac{d^2}{16}} \Rightarrow K \propto d$$

$k^1 = 2k$

**41. The diameter 'd' (in mm) of a rivet to suit the thickness 't' (in mm) of a plate may be determined by the Unwin formula**
(a) $d = \sqrt{5}t - 0.2$ (b) $d = 1.5t + 4$
(c) $d = 6\sqrt{t}$ (d) $d = 1.9\sqrt{t}$

**Ans. (c) :** To calculate the size of the rivet, Unwin's formula is used

$d = 6.04\sqrt{t}$

**42. The weakest plane in a fillet weld is**
(a) A side parallel to the force
(b) A side normal to the force
(c) The one along the throat
(d) The one normal to the throat

**Ans. (c) :** A fillet weld is used to fill in the edges of the plates created by corner, lap and tee joints . it is adopted when two members are to be welded at $90^\circ$ to each other. for a given amount of weld material, the fillet weld is not as strong as due to the heavy Uncentration of the stress. for fillet weld, the weakest plane is along the throat section.

**43. The effective length of the fillet weld of length L and size S is given by**
(a) $L = \sqrt{2}S$ (b) $L - 2S$
(c) $L - S\sqrt{2}$ (d) L

**Ans. (b) :** Effective length of the fillet weld of length l and size S is given by L-2S

**44. The material which show same properties in all direction is called**
(a) Homogenous (b) Elastic
(c) Isotropic (d) Orthotropic

**Ans. (c) :** Isotropic material means a material having identical values of a property in all direction.
A material is said to be homogenous if the materials elastic properties are same at all the points or throughout the body.

**45. The highest possible value of Poisson's ratio for isotropic material is**
(a) 0.1 (b) 0.3
(c) 0.5 (d) 1.5

**Ans. (c) :** The engineering range of poisson's ratio
$0 \le \mu \le 0.5$ While the theoretical range is $-1 \le \mu \le 0.5$
The value of poisson's ratio for elastic material cannot be more than 0.5

**46. In terms of bulk modulus (K) and modulus of rigidity (C), the poissons ratio can be expressed as**
(a) $\frac{3K - 4C}{6K + 4C}$ (b) $\frac{3K + 4C}{6K - 4C}$
(c) $\frac{3K - 2C}{6K + 2C}$ (d) $\frac{3K + 2C}{6K - 2C}$

**Ans. (c) :** We know that
$E = 3K(1 - 2\mu)$ .....(i)
$E = 2C(1 + \mu)$ .....(ii)
From equation (i) and (ii)
$$\mu = \frac{3K - 2C}{6K + 2C}$$

**47. A rod of length L and area of cross-section A, whose material has a modulus of elasticity E and a coefficient of liner thermal expansion 'α' is subjected to a charge of temperature ΔT. The change in length of the rod is**
(a) $\alpha\Delta(T)$ (b) $\alpha(\Delta T)L/E$
(c) $\alpha(\Delta T)L$ (d) $\alpha L/AE$

**Ans. (c) :** Strain developed in the rod is given by

$$\in = \alpha\,\Delta T$$

$$\frac{\Delta L}{L} = \alpha \Delta T$$

$$\Delta L = \alpha(\Delta T)L$$

**48. The maximum stress produced in a bar when a load is applied suddenly is 40 N/mm$^2$. If the same load is applied gradually, the stress produced is**

(a) 10 N/mm$^2$ (b) 20 N/mm$^2$
(c) 40 N/mm$^2$ (d) 80 N/mm$^2$

**Ans. (b) :** Given $\sigma_s = 40 N/mm^2$

$$\sigma_{sudden} = 2\sigma_{gradual}$$

$$\sigma_{gradual} = 40/2 = 20 N/mm^2$$

**49. The Youngs modulus of a certain material is 2 ×10$^5$ N/mm$^2$ and its Modulus of Rigidity is 0.8×10$^5$ N/mm$^2$. The value of Poissons ratio is**

(a) 4 (b) 2.5 (c) 1 (d) 0.25

**Ans. (d) :** Given

$$E = 2\times10^5\ N/mm^2$$

$$G = 0.8\times10^5\ N/mm^2$$

$$E = 2G(1+\mu)$$

$$2\times10^5 = 2\times0.8\times10^5(1+\mu)$$

$$\mu = 0.25$$

**50. Stress in concrete is 5 N/mm$^2$. Ratio of Young' modulus of steel to concrete is 15. Stress in steel**

(a) 20 N/mm$^2$ (b) 1 N/mm$^2$
(c) 75 N/mm$^2$ (d) 0.25 N/mm$^2$

**Ans. (d) :** Given

Modular ratio $M = \frac{E_s}{E_c} = 15$

Stress in steel = modular ratio × stress in concrete.

= 15 × 5 = 75 N/mm$^2$

**51. An iron needle placed on water surface in a tank floats on the water although its weight is more. This is due to**

(a) Specific gravity (b) Cohesion
(c) Compressibility (d) Surface tension

**Ans. (c) :** An iron needle placed on water surface in a tank floats on the water although its weight is more, this is due to surface tension of water, which does not allow the needle to sink.

**52. The dynamic viscosity of a fluid is 0.139 kgf sec/m$^2$. If the specific gravity of the fluid is 0.95, its kinematic viscosity is**

(a) 0.1463 m$^2$/sec (b) 0.001435 m$^2$/sec
(c) 0.000146 m$^2$/sec (d) 0.5 m$^2$/sec

**Ans. (c) :** Given

$\mu$ = 0.139 kgf sec/m$^2$

$\delta = 0.95 \times 1000 = 950$ kg/m$^3$

Kinematics viscosity = $\frac{\mu}{\delta} = \frac{0.139}{950}$

= 0.000146 m$^2$/sec

**53. Coefficient of viscosity has the dimensions of**

(a) $FL^{-2}T$ (b) $FL^{-1}T^{-1}$
(c) $FLT^{-2}$ (d) $FL^2/T$

**Ans. (a) :** Shear stress in fluid is given by

$$\tau = \mu\frac{dv}{dx}$$

$$\frac{F}{A} = \mu\frac{dv}{dx}$$

$$\frac{[F]}{[L^2]} = \mu\frac{[LT^{-1}]}{[L]}$$

$$\mu = [FL^{-2}T]$$

**54. The height of water in capillary tube**

(a) increases with increase in diameter
(b) increase with decrease in diameter
(c) decrease with an increases in surface tension
(d) increases with an increase in specific weight

**Ans. (c) :** The capillary rise or depression in a small diameter tube is inversely proportional to the diameter, which is given by expression

$$h = \frac{4\mu\cos\theta}{\delta g d} \Rightarrow h \propto \frac{1}{d}$$

**55. The intensity of pressure developed by surface tension of 0.0075 kg/m in a droplet of water 6 ×10$^{-3}$ mm diameter is**

(a) 0.01 kg/cm$^2$ (b) 0.04 kg/cm$^2$
(c) 0.005 kg/cm$^2$ (d) 0.05 kg/cm$^2$

**Ans. (*) :** Given,

$\sigma$ = 0.0075 kg/m, d = 6 × 10$^{-3}$ mm

The intensity of pressure developed by surface tension ($\sigma$) in a droplet of water of diameter (d) is given by

$$\Delta P = \frac{4\sigma}{d}$$

$$= \frac{4\times0.0075}{6\times10^{-3}\times10^{-3}}$$

= 5000 kg/m$^2$ = 0.5 kg/cm$^2$

**56. An ideal fluid is**

(a) One which obeys Newton's law of viscosity
(b) Frictionless and incompressible
(c) Very viscous
(d) Frictionless and compressible

**Ans. (b) :** An ideal fluid is both incompressible, frictionless and non viscous. This means that on ideal liquid is a non-viscous liquid that contains zero tangible forces among the layer's of the liquids in relative motion.

**57. The pressure in metres of oil of specific gravity 0.8, equivalent to 80 m of water is**

(a) 64 (b) 80
(c) 100 (d) 88

**Ans. (c) :** Given,

Specific gravity = 0.8

$h_w$ = 80 m

Pressure = $\frac{80}{0.8} = 100$

**58. Select the correct statement**

(a) Absolute pressure = Gauge pressure - atmospheric pressure

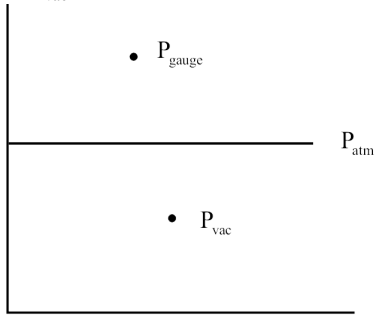
(b) Gauge pressure = Absolute pressure – Atmospheric pressure

(c) Absolute pressure = Atmospheric pressure + Vacuum pressure

(d) Gauge pressure = Atmospheric pressure - vacuum pressure

**Ans. (b) :** $P_{abs} = P_{atm} + P_{gauge}$

$P_{abs} = P_{atm} - P_{vac}$




Absolute pressure

**59. A mercury water manometer has a gauge difference of 0.6 m. The difference in pressure, measured in metres of water, is**

(a) 0.6 (b) 7.58

(c) 9.16 (d) 7.86

**Ans. (b) :** For manometer, the head different can be expressed

$$h = \left(\frac{S_m}{S_w} - 1\right) \times x$$

$$= \left(\frac{1.36}{1} - 1\right) \times 0.6 = 7.56 \text{ m}$$

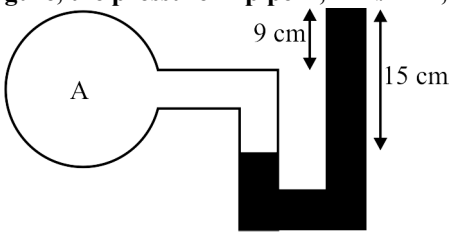
**60. In the inverted U-tube the manometer liquid is ----- the flowing liquid.**

(a) heavier than (b) lighter than

(c) same as (d) none of these

**Ans. (b) :** In the inverted U-tube differential manometer, specific gravity of manometric fluid used is lighter than relative to the fluid flowing in the pipes as the manometric fluid is at the top.

**61. Water is flowing through two different pipes to which an inverted differential manometer having an oil of specific gravity 0.8 is connected. The pressure in the pipe A is 2m of water. The manometric ridings as shown in figure, the pressure in pipe B, in N/mm$^2$, is**




(a) 1.896 (b) 1.475

(c) 1.86 (d) 1.52

**Ans. (c) :**

Pressure head at A = $\frac{P_A}{\rho g} = 2\,\text{m of water}$

$P_A = \rho g \times 2 = 1000 \times 9.81 \times 2$

$= 19620\,\text{N/m}^2$

Pressure below X-X in left limb

$= P_A - \rho_l g h_1$

$= 19620 - 1000 \times 9.81 \times 0.3 = 16677\ \text{N/m}^2$

Pressure below X-X in right limb

$= P_B - 1000 \times 9.81 \times 0.1 - 800 \times 9.81 \times 0.12$

$= P_B - 1922.76$

Equating both,

$P_B = 16677 + 1922.76$

$= 18599.76\ \text{N/m}^2$

$P_B = 1.85 \sqcup 1.86\ \text{N/cm}^2$

**62. The horizontal component of force on a curved surface is equal to the**

(a) weight of liquid vertically above the curved surface

(b) weight of liquid retained by the curved surface

(c) product of pressure at its centroid and area

(d) force on a vertical projection of the curved surface

**Ans. (d) : Horizontal component–** The horizontal component of the total pressure force is equal to the total pressure force on the projected are of the curved surface in the vertical plane.

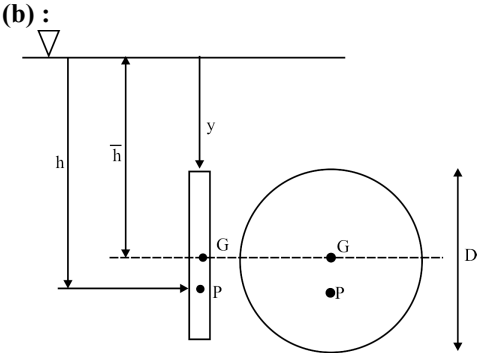
**Vertical component–** The vertical component is equal to the weight of liquid supported by the curved surface upto the free liquid surface.

**63. A circular plate 1 m diameter is submerged vertically in water such that its upper edge is 8 m below free surface of water. The total hydrostatic pressure force on the side of the plate is**

(a) 6.7 kN (b) 65.4 kN

(c) 45.0 kN (d) 77.0 kN

**Ans. (b) :**




D = 1 m, y = 8 m, $F = \rho g \bar{h}.A$

$= \rho.g\left(y + \frac{D}{2}\right).\frac{\pi}{4}D^2$

$= 1000 \times 9.81\left(8 + \frac{1}{2}\right) \times \frac{\pi}{4} \times (1)^2 = 65.4\ \text{kN}$

**64. When the velocity of flow of fluid does not change from point to point at any given time, the flow is defined as**

(a) Steady flow (b) Laminar flow
(c) Irrotational flow (d) Uniform flow

**Ans. (a) : Steady flow–** When the fluid properties do not change with respect to time, then the flow is known as steady flow.
**Uniform flow –** It is defined as the type of flow in which the velocity at any given time does not change with respect to space.

**65. The discharge in $m^3$/sec for laminar flow through a pipe of diameter 0.04 m having a centre line velocity of 1.5 m /sec is**

(a) $\frac{3\pi}{50}$ (b) $\frac{3\pi}{2500}$
(c) $\frac{3\pi}{5000}$ (d) $\frac{3\pi}{10000}$

**Ans. (d) :** For laminar flow, maximum velocity occurs at centre.
$U_{max}$ = Centre – Line velocity = 1.5 m/s

For laminar flow $U_{avg} = \frac{U_{max}}{2} = \frac{1.5}{2} = 0.75\ m/s$

Discharge (Q) = Area × Average velocity

$$= \frac{\pi}{4} \times (0.04)^2 \times 0.75$$

$$= \frac{3\pi}{10000}\ m^3/s$$

**66. The pressure at a point in a water is 196.2 $kN/m^2$. The velocity at that point is 0.5m and the point is 5 m above the datum. The total head at the point is .....**

(a) 25.0 m (b) 25.5 m
(c) 15.0 m (d) 5.25 m

**Ans. (b) :** Given,
P = 196.2 $kN/m^2$, $V_1$= 0.5 m/s, $Z_1$ = 5 m

Total head = $\frac{P}{\rho g} + \frac{V_1^2}{2g} + Z_1$

$$= \frac{196.2 \times 10^3}{1000 \times 9.81} + \frac{(0.5)^2}{2 \times 9.81} + 5$$

= 20 + 0.0127 + 5 = 25.01 m

**67. Bernoulli's equation is based on the principle of conservation of**

(a) Momentum (b) Mass
(c) Velocity (d) Energy

**Ans. (d) :** Bernoulli's equation is based on the principle of conservation of energy applied to a liquid in motion.

**68. The head loss in the venturimeter between inlet and throat is given by the formula**

(a) $h_f = (1 - C_d)H$ (b) $h_f = (1 - \sqrt{C_d})H$
(c) $h_f = (1 - C_d^2)H$ (d) $h_f = (1 - C_d)^{1/3} H$

**Ans. (c) :** We know that, $Q_{theoretical} = Q_{actual}$

$$C_d = \sqrt{\frac{h - h_L}{h}}$$

$$C_d^2 = 1 - \frac{h_L}{h}$$

$$\frac{h_L}{h} = 1 - C_d^2$$

$$h_L = (1 - C_d^2)h$$

**69. If the venturimeter is discharging into the atmosphere, the pressure at the throat will be ..... atmospheric pressure.**

(a) Less than (b) More than
(c) Equal to (d) None

**Ans. (b) :** If the venturimeter is discharging into the atmosphere, the pressure at the throat will be less than atmospheric pressure.
• The velocity reaches its maximum value and pressure reaches its minimum value at the throat.

**70. The total head in a flow is the sum of**

(a) Piezometric head and datum head
(b) Piezometeric head and pressure head
(c) Piezometeric head and velocity head
(d) Piezometric head, velocity head and datum head

**Ans. (c) : Piezometric head :** The sum of pressure head and datum is known as piezometric head. It is given by–

$$\frac{P}{\rho g} + Z$$

$\frac{V^2}{2g}$ = Velocity head

$\frac{P}{Pg}$ = Pressure head

Z = Datum head

**71. Vena contracta is a point at which**

(a) The velocity of flow is maximum
(b) The velocity of flow is average
(c) The velocity of flow is minimum
(d) The velocity of flow is constant for any head

**Ans. (a) :** An orifice meter is a thin plate with a centrally located hole is inserted into the flow passage.
The flow contracts suddenly as the fluid passes through the hole. The flow continues to contracts a short distance downstream the hole.
The region of the smallest cross-section is known as vena contracta, which is developed downstream the orifice. At vena contracta the kinetic energy of the fluid flow is maximum.

**72. The coefficient of velocity for an orifice is 0.96. The head loss at the orifice while discharging under a head of 3.0 m is**

(a) 5.765 m (b) 3.255 m
(c) 0.23 m (d) 0.196 m

**Ans. (c) :** Given,
$C_v$ = 0.96, h = 3 m
Head loss at the orifice

$$H_F = h(1 - C_V^2)$$

$$= 3\ [1 - (0.96)^2]$$

$$= 0.2352\ m$$

**73. A small orifice is provided at the bottom of a tank with liquid. For it 25 seconds is taken to draw the liquid from 49 cm to 25 cm. Taken the time required to draw the liquid from 36 to 16 cm is**

(a) 25 sec (b) 15 sec
(c) 37.5 sec (d) 50 sec

**Ans. (a) :** $\frac{t_1}{t_2} \propto \left(\sqrt{H_1} - \sqrt{H_2}\right)$

$$\frac{25}{t_2} = \frac{\left(\sqrt{49} - \sqrt{25}\right)}{\sqrt{36} - \sqrt{16}}$$

$t_2 = 25$ sec

**74. An external cylindrical mouthpiece is discharging water under a constant head of 10 m. The absolute pressure head at vena contracta is**

(a) 19.2 m (b) 0.3 m
(c) 1.1 m (d) 5.85 m

**Ans. (b) :** absolute pressure head at vena contracta is 10.3–10 = 0.3 m

**75. The discharge over a triangular notch is proportional to**

(a) $H^{1/2}$ (b) $H^{3/2}$
(c) $H^{5/2}$ (d) $H^{1/3}$

**Ans. (c) :** The discharge through a rectangular notch is,

$$Q = \frac{2}{3} C_d L \sqrt{2g} H^{3/2}$$

The discharge through a triangular notch is,

$$Q = \frac{8}{15} C_d \tan\frac{\phi}{2} \sqrt{2g} H^{5/2}$$

Where,
Q = Discharge of fluid
$C_d$ = Coefficient of discharge
θ = Notch angle
H = Height of water above the notch

**76. A triangular notch is preferred to a rectangular notch, because**

(a) it gives more accurate result for low discharges
(b) It has its coefficient of discharge greater than that of rectangular notch
(c) It measurement wide ranges of flow
(d) End contraction do not play a significant role

**Ans. (a) :** A triangular notch is preferred to a rectangular notch due to the following reasons.
(i) For measuring the low discharge, a triangular notch gives more accurate results than a rectangular.
(ii) In the case of the triangular notch, only one reading i.e. H is required for the computation of discharge.

**77. The head over a $90^0$ V notch increases from 0.15 m to 0.3m. The ratio of the new discharge to the original discharge is**

(a) 1.414 (b) 2.000
(c) 4.000 (d) 5.657

**Ans. (d):** Ratio of new discharge to the original discharge

$$\frac{Q_2}{Q_1} = \left(\frac{H_2}{H_1}\right)^{5/2} = \left(\frac{0.3}{0.15}\right)^{5/2} = 5.657$$

**78. A cippoletti weir has a side slope of**

(a) 1 vertical; 4 horizontal
(b) 1 vertical; 2 horizontal
(c) 1 horizontal; 4 vertical
(d) 1 horizontal; 2 vertical

**Ans. (c) :** A type of contracted weir that is related to the rectangular sharp-crested weir is cippoletti weir, which has a trapezoidal cross-section with side slopes 1 : 4 (H:V).

**79. A rectangular weir is 9 m long and discharges 900 litres. The discharge is decreased to 600 litres, keeping the head constant, % decrease in the length of the weir is**

(a) 15.2 (b) 33.53
(c) 66.67 (d) 34.43

**Ans. (c) :** Given,
$L_1 = 9$ m, $Q_1 = 900$ litre
$Q_2 = 300$ litre
We know that discharge for rectangular notch

$$Q = \frac{2}{3} C_d L \sqrt{2g} H^{3/2}$$

$$Q \propto L$$

$$\frac{L_2}{L_1} = \frac{Q_1}{Q_2}$$

$$\frac{L_2}{9} = \frac{300}{900}$$

$L_2 = 3$ m

Percentage decrease in length $= \frac{L_1 - L_2}{L_1}$

$$= \frac{9-3}{9} = \frac{2}{3} = 66.67\%$$

**80. In a sutro weir, if H is the Head over the crest, the discharge is proportional to**

(a) $\sqrt{H}$ (b) $H^{3/2}$
(c) $H^{5/2}$ (d) H

**Ans. (d) :** The discharge equation of weir is generally expressed as Q is directly proportional to $H^n$. A weir with n = 1 is called a proportional weir or sutro weir.

**81. A siphon pipe is**

(a) any bent pipe
(b) that part of pipe above hydraulic gradient line
(c) that part of pipe below hydraulic gradient line
(d) useful for pumping water to high level

**Ans. (b) :** A siphon is a long-bent pipe used to connect two reservoir at different levels through a high ridge. The highest point of the syphon is called the summit.
Or
A pipe which rises above its hydraulic grade line has negative pressure and is known as syhpon.

**82. At a point along a horizontal water pipe line, the cross sectional area of the pipe is suddenly doubled. If the velocity of head before the sudden expansion is 2 m, the loss of head at pipe expansion is**

(a) 0.1 m (b) 0.25 m
(c) 0.05 m (d) 1.0 m

**Ans. (c) :** The head loss due to sudden enlargment in a pipeline is given by

$$h_L = \frac{(V_1 - V_2)^2}{2g} = \frac{V_1^2}{2g}\left(1 - \frac{A_1}{A_2}\right)^2$$

$$h_L = \frac{(2)^2}{2\times 10} \times \left(1 - \frac{1}{2}\right)^2$$

$$= \frac{4}{2\times 10} \times \frac{1}{4} = 0.05 \text{ m}$$

**83. The head of water supplied to a nozzle is 15 m. The loss of head due to friction at maximum power transmission is**

(a) 3 m (b) 1.5 m
(c) 5 m (d) 10 m

**Ans. (c) :** The power transmitted through the pipe is maximum when the head loss due to friction in the pipe is 1/3 of the total supply head.

$$h_f = \frac{H}{3} = \frac{15}{3} = 5\text{m}$$

**84. A liquid flows in a 25 cm diameter pipe at a Reynolds number of 1600. The coefficient of friction 'f' is equal to**

(a) 0.0123 (b) 0.001
(c) 0.01 (d) 0.016

**Ans. (c) :** Given, Re = 1600

$$\text{Coefficient of friction} = \frac{16}{\text{Re}}$$

$$= \frac{16}{1600} = 0.01$$

**85. The hydraulic mean depth of a pipe of 4 m diameter flowing half full is**

(a) 2 m (b) 1 m
(c) 0.5 m (d) 0.25 m

**Ans. (b) :** The hydraulic mean depth or also called hydraulic radius is defined as

$$R = \frac{\text{Wetted Area}}{\text{Wetted Perimeter}} = \frac{A}{P}$$

$$= \frac{\frac{\pi}{4}d^2}{\pi d} = \frac{d}{4}$$

$$= \frac{4}{4} = 1 \text{ m}$$

**86. Two pipes from a reservoir are connected in parallel, then**

(a) The total head loss is the sum of the loss of head in each pipe
(b) The total discharge is the discharge in each pipe
(c) The total head loss is the loss of head in each pipe
(d) The total head loss is the reciprocal or the sum of loss of head in each pipe

**Ans. (c) : Parallel connection–** When two pipes are connected in parallel, then the head loss between the two pipes is same and the discharge of each pipe adds up to give a larger discharge.

$$H_1 = H_2 \text{ and } Q = Q_1 + Q_2$$

**87. The hydraulic grade line is**

(a) Always above the energy grade line
(b) The velocity head below the energy grade line
(c) Always above the closed conduit
(d) Always sloping downwards in the direction of flow

**Ans. (d) :** For the pipe of constant diameter, the hydraulic gradient line is always sloping down in the direction of flow and is parallel to the total energy line and the difference between both the lines represents velocity head. In a flow of fluid through pipes, the hydraulic gradient line (HGL) is always placed below the total energy line.

**88. A jet of water of c.s. area 0.1962 $m^2$ strikes a flat plate normally with a velocity of 15 m/sec. The horse power of the jet is**

(a) 33750 (b) 450
(c) 30 (d) 900

**Ans. (d) :** $P = \rho A V^2 \times V$

$= 1000 \times 0.1962 \times 15^3$

$= 887.03 \approx 900$ HP

**89. In two point method of measuring velocity with a current meter, the velocities are measured at depths**

(a) 0.25 d and 0.75 d (b) 0.2 d and 0.8 d
(c) 0.1 d and 0.7 d (d) 0.15 d and 0.75 d

**Ans. (b) :** In two point method of measuring velocity with a current meter, the velocities are measured at depths 0.2 d and 0.8 d.

**90. For laminar flow through pipes, the kinetic energy correction factor is**

(a) less than 1 (b) 1.03
(c) 1.33 (d) 2.0

**Ans. (d) :** (i) kinetic energy correction factor ($\alpha$) ; It is defined as the ratio of kinetic energy per second based on actual velocity to the kinetic energy per second based on average velocity.

$$\alpha = \frac{1}{A}\int_A \left(\frac{u}{V}\right)^3 dA$$

Where,

A= area, V= average velocity
u = local velocity at distance r

(ii) In circular pipe laminar flow $\alpha = 2$

**91. The expression $\frac{\rho v d}{\mu}$ represents**

(a) Froude number (b) Mach number
(c) Reynolds number (d) Weber number

**Ans. (c): Reynolds number** – The Reynolds number is the ratio of inertial force to viscous forces within a fluid which is subjected to relative internal movement due to different fluid velocities.

$$Re = \frac{\rho uL}{\mu} = \frac{uL}{\upsilon}$$

where, $\rho$ = density, u = flow velocity
$\mu$ = dynamic viscosity
$\upsilon$ = Kinematic viscosity

- Laminar flow occurs at low reynold's number.
- Turbulent flow occurs at high reynold's number.

**92. The jet of water impinging on the runner of a Pelton wheel has a velocity of 60 m/sec. For maximum efficiency of the turbine, the velocity of runner is**

(a) 120 m/sec (b) 60 m/sec
(c) 30 m/sec (d) 15 m/sec

**Ans. (c) :** For maximum efficiency of the turbine, the velocity of runner is V/2 = 30 m/sec.

**93. A hydraulic turbine has an output of 1024 h.p. when it works under a head of 16m and runs at 100 r.p.m. The turbine is**

(a) Kaplan (b) Francis (c) Pelton (d) Propeller

**Ans. (b) :** Given, N= 100 rpm, P = 1024 × 746 W
= 763.904 kW
H =16 m

$$\text{Specific speed of turbine } (N_S) = \frac{N\sqrt{P}}{H^{5/4}}$$

$$= \frac{100\sqrt{763.904}}{(16)^{5/4}} = 86.37$$

Range of specific speed of francis turbine in between 51-255.

**94. The resultant upward pressure of the fluid on an immersed body is called**

(a) Buoyancy (b) Upthrust
(c) Metacentre (d) Centre of pressure

**Ans. (a) :** Buoyancy is an upward force exerted by a fluid that opposes the weight of an immersed object.

**95. At a hydro-electric power plant the head is 10 m and the load is variable. The type of turbined most suitable is**

(a) Pelton wheel (b) Francis turbine
(c) Kaplan turbine (d) Thomson turbine

**Ans. (c) :** In kaplan turbine very low head is required, generally less than 30m. In Francis turbine the medium head is required generally in the range of 30-500 m.

**96. Penstocks are made up of**

(a) Galvanized iron (b) Cast iron
(c) Lime concrete (d) Steel

**Ans. (b) :** Penstocks are channels which transport water from reservoir to turbines which are usually made up of cast iron or concrete.

**97. Curved vanes are preferred to flat vanes in hydraulic machines because of**

(a) low cost of maintenance
(b) higher hydraulic efficiency can be obtained
(c) convenience of moulding and machining
(d) It is possible to use high velocity jet

**Ans. (b):** The using curved vanes over flat vanes in a hydraulic turbine is higher hydraulic efficiency can be obtained.

**98. The specific speed of a pump is defined as the speed of a pump**

(a) of unit size with unit discharge at unit head
(b) of such a size that it requires unit power for unit head
(c) of such a size that it delivers unit discharge at unit head
(d) of such a size that is delivers unit discharge at unit power

**Ans. (c) :** Specific speed of a pump is defined as the speed of unit of such a size that it delivers unit discharge at unit head.
Specific speed of pump is given by

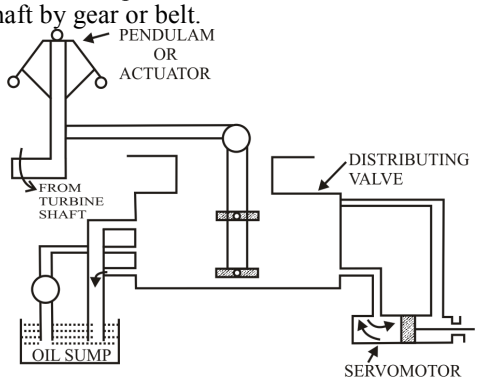
$$N_S = \frac{N\sqrt{Q}}{H^{3/4}}$$

**99. The common type of governor used in modern turbines is ..... governor.**

(a) needle (b) centrifugal
(c) diffuser (d) oil pressure

**Ans. (d) :** Component parts for an oil pressure governor in modern turbines are–

- Servomotor–Relay cylinder.
- Control value or distribution valve.
- Oil sump.
- Oil pump, driven by belt connected to turbine shaft.
- A mechanism which supplies oil sump with the relay valve and relay valve with Servomotor by oil supply pipes.
- Actuator or pendulum, driven from turbine main shaft by gear or belt.




- Due to oil pressure, the piston of servomotor or relay cylinder moves.
- Oil pump is used to pressurize the oil.

**100. Draft tube is a pipe which has**

(a) A gradually increasing cross sectional area
(b) A gradually decreasing cross sectional area
(c) Constant area
(d) perforated linings

**Ans. (a) :** The draft tube is a conduit that connects the runner exit to the tailrace where the water is being finally discharged from the turbine. It is used with reaction turbines only. A draft tube is made divergent (area increases as we go towards the outlet of the draft tube). So as to reduce the velocity at the outlet to a minimum.

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 1992
## Solved Paper (Held on 1993)

**1. The working stress for mild steel specimen is calculated factor of safety on**

(a) Breaking stress (b) Ultimate stress
(c) Yield point stress (d) None of these

**Ans. (c) :** Working stress for mild steel specimen is factor of safety on yield point stress.

$$\text{Factor of safety, FOS} = \frac{\text{ultimate / yield stress}}{\text{working stress}}$$

**2. Cup and the cone fractures usually indicates, the material as**

(a) Elastic (b) Brittle
(c) Ductile (d) Plastic

**Ans. (c) :** Ductile material strong in tension, therefore failure is due to shear, the plane of failure is at 45° from the axis of shaft, such failure is known as cup and cone failure.

**3. Poisson's ratio is defined as ratio of**

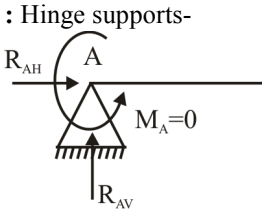
(a) Stress to strain
(b) Stress to Young's modulus
(c) Lateral strain to longitudinal strain
(d) Rigidity modulus to bulk modulus

**Ans. (c) :** $\text{Poisson's ratio} = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

**4. In plane structure a hinge support has**

(a) one reaction (b) two reactions
(c) three reactions (d) no reactions

**Ans. (b) :** Hinge supports-




Bending moment, $M_A = 0$
Axial component, $R_A$
Vertical component, $R_V$

**5. Stress-strain relation of a material for structural analysis should be**

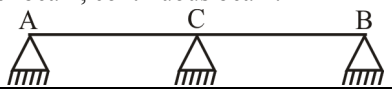
(a) Elastic only
(b) Linear only
(c) Elastic and linear
(d) Elastic and non-linear

**Ans. (a) :** The relation between stress and strain is that They are directly proportional to each other upto an elastic limit.
According to Hooke's law, the strain in a solid is proportional to the applied stress and this should be within the elastic limit.

**6. A simply supported beam with a hinge at mid-span is**

(a) Statically determinate
(b) Statically indeterminate
(c) Unstable
(d) Determinate and unstable

**Ans. (b) :** Statically Indeterminate beam is defined as the total number of reaction in a beam is greater than number of useful static equilibrium equations.
**Example-** Fixed beam or Built in beam, propped cantilever beam, continuous beam.




**7. Principle of superposition is valid in a**

(a) Statically determinate structure
(b) Statically indeterminate structure
(c) Structure which obeys Hooke's law
(d) Structure which undergoes small deformation

**Ans. (c) :** Principle of superposition is valid in a conditions-
The material must behave in a linear elastic manner, so, that Hooke's law is valid and therefore the load will be proportional to displacement i.e. stress-strain relationship is linear.

**8. Shear force at any section is the algebraic sum of the following force on the either side.**

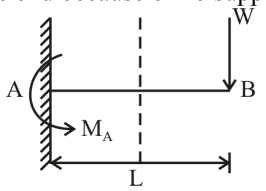
(a) Axial forces (b) Transverse forces
(c) Moments (d) None of these

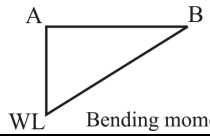
**Ans. (b) :** Shear force at any section can be defined as the algebraic sum of all the lateral component of the forces acting on either side of the cross section.
Or
Algebraic sum of transverse forces on the either side.

**9. Bending moment in a cantilever beam is zero at**

(a) A simple support (b) A fixed support
(c) Moments (d) None of these

**Ans. (d) :** Bending moment is zero in cantilever beam at its free end because of no support.




Bending moment diagram

**10. The ratio of maximum bending moment in a cantilever beam to that of a simply supported beam of same span and uniformly distributed load at**

(a) 1 : 4 (b) 1: 0.25
(c) 1 : 2 (d) None of these

**Ans. (d) :** Bending moment of a contilever beam

$(BM)_{contilever} = WL$

Bending moment of a simply supported beam with uniformly distributed loading.

$$(BM)_{SSB} = \frac{WL}{8} \qquad [\because W = wL]$$

$$\therefore \quad \text{Ratio} = \frac{(BM)_{cantilever}}{(BM)_{SSB}} = \frac{WL}{\frac{WL}{8}}$$

$$= \frac{8}{1}$$

$$= 8 : 1$$

**11. Between point loads the shear varies**

(a) Constant (b) Linearly
(c) Parabolically (d) Cubically

**Ans. (a) :** If we have two point load on the beam then shear force diagram is constant at all sections.
But it changes its magnitude suddenly.

- Between two concentrated loads, the shear force diagram for any part of the beam is a straight line.
- The shear force diagram for uniformly distributed load is linear.
- The shear force diagram for the uniformly varying load is parabolic.

**12. The bending moment diag. Due to uniformly varying load is**

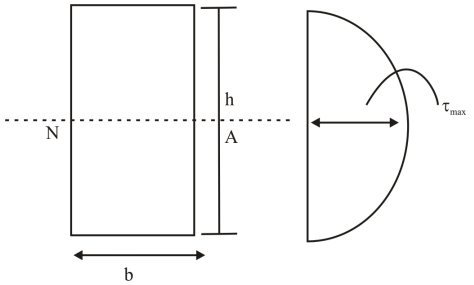
(a) Parabolic (b) Elliptic
(c) Hyperbolic (d) Cubic

**Ans. (d) :** The shape of bending moment diagram due to a uniformly varying load is a cubic parabola.

**13. At a given rectangular section the shear stress varies along the depth as**

(a) Constant (b) Linearly
(c) Parabolically (d) None of these

**Ans. (c) :** The shear stress distribution over a rectangular cross section of a beam follows parabolic path.




$$\tau_{max} = \frac{3}{2}\tau_{av}$$

$\tau_{av}$ = Average shear stress

**14. Shear stress in a beam with rectangular cross-section is maximum at**

(a) Extreme top fibre (b) Neutral axis
(c) Extreme bottom fibre (d) Mid-depth

**Ans. (b) :** Shear stress distribution of a rectangular cross-section is define as the shear stress is distributed parabolically across the rectangular section, shear stress maximum at neutral axis and will be zero at extreme end fiber.

**15. The ratio of maximum shear stress to mean stress in triangular cross-section is**

(a) 1.5 (b)1.33 (c) 1.125 (d) 1.25

**Ans. (a) :**

$$\text{Maximum shear stress} = \frac{3}{2} \times \text{mean shear stress}$$

$$\frac{\text{max . shear stress}}{\text{mean shear stress}} = \frac{3}{2}$$

$$= 1.5$$

**16. Bending stress at a section is maximum at**

(a) Mid-depth
(b) Extreme top and bottom fibre
(c) Centroid axis
(d) None of these

**Ans. (b) :** Bending stress is perpendicular to cross-section of beam. Bending stress varies linearly over the depth of beam. At extreme fibre bending stress is maximum. Bending stress is zero at neutral axis.

**17. Section modulus of circular cross-section of diameter D is**

(a) $\pi D^2/64$ (b) $\pi D^3/32$
(c) $\pi D^2/4$ (d) $\pi D^4/32$

**Ans. (b) :** Section modulus, $Z_{NA} = \frac{I}{Y_{max}} = \frac{\frac{\pi D^4}{64}}{\frac{D}{2}}$

$$= \frac{\pi D^4}{64} \times \frac{2}{D}$$

$$Z_{NA} = \frac{\pi D^3}{32}$$

**18. The most efficient section to resist bending stress is**

(a) Rectangular section (b) I-section
(c) T-section (d) Circular section

**Ans. (b) :** Section modulus of the I-section is more than that of the rectangular cross-section for the same depth. I-section is most efficient in carrying bending moment.

**19. For no tension in columns of diameter D the eccentricity of compressive load must be**

(a) D/4 (b) D/6 (c) D/8 (d) D/12

**Ans. (c) :** For no tension condition-

$$\sigma_b \le \sigma_a$$

$$\frac{P.e}{\frac{\pi d^3}{32}} \le \frac{4P}{\pi d^2}$$

$$e \le \frac{D}{8}$$

**20. The ratio of Euler's load for a column with both ends hinged to that of with both ends fixed with same length is**

(a) 1 : 2 (b) 1 : 4

(c) 1 : $\sqrt{2}$ (d) 4 : 1

**Ans. (b) :** Euler's load for a column with both ends hinged, $l_{eff} = l$

Euler's load for a column with both ends fixed,

$$l_e = \frac{l}{2}$$

$$(P_e)_{fixed} = \frac{\pi^2 E I_{min}}{(l_{eff})^2} = \frac{\pi^2 E I_{min}}{\left(\frac{l}{2}\right)^2}$$

$$= \frac{4\pi^2 E I_{min}}{l^2}$$

$$\text{Ratio} = \frac{(P_e)_{Hinged}}{(P_e)_{fixed}} = \frac{\frac{\pi^2 E I_{min}}{l^2}}{\frac{4\pi^2 E I_{min}}{l^2}} = \frac{1}{4}$$

$$= 1 : 4$$

**21. The nature of combined stress in a beam depends on the nature of**

(a) T only (b) M only

(c) T and M both (d) None of these

**Ans. (c) :**

**Equivalent torque** : The torque which acting alone would produce maximum shear stress equal the maximum. Shear stress caused by combined bending and torsion.

$$T_e = \sqrt{M^2 + T^2}$$

**Equivalent moment** : The moment which acting alone would produce maximum principal equal the 'maximum principal stress' caused by combined Bending and twisting.

$$M_e = \frac{1}{2}\left(M + \sqrt{M^2 + T^2}\right)$$

**22. The number of independent stress components at a point in a dimension is**

(a) 3 (b) 6

(c) 9 (d) 12

**Ans. (b) :** State of stress at a point in the cube can be categorized into six independent normal shear stress components.

Thus, $\sigma_x$, $\sigma_y$, $\sigma_z$ represents the three normal stress, and $\tau_{xy}$, $\tau_{zy}$, $\tau_{zx}$ represents the three shear stress.

**23. The principal planes at a point are inclined at**

(a) 45° (b) 60°

(c) 90° (d) 180°

**Ans. (c) :** Principal stresses occur on mutually perpendicular planes. Shear stresses are zero on principal planes. Planes of maximum shear stress inclined at 45° to the principal planes.

**24. Maximum shear stress is equal to**

(a) Sum of principal stresses

(b) Half of the difference of principal stresses

(c) Difference of principal stresses

(d) None of these

**Ans. (b) :** Maximum shear stress is equal to one half of the difference of the principal stresses.

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

Where,

$\tau_{max}$ = maximum shear stress

$\sigma_1$ = maximum principal stress

$\sigma_2$ = minimum principal stress

**25. Mohr's circle can be used to determine**

(a) Principle stresses

(b) Principal stresses and strains

(c) Principal strains only

(d) None of these

**Ans. (b) :** Mohr's circle is used to determine the relationships between normal and shear stresses acting on any inclined plane at a point.

It's helpful in finding maximum and minimum principal stresses, maximum shear stress and strain.

**26. The deflection of elastic curve of a beam is given by**

(a) $\iint M.dx / EI$ (b) $\iint W.dx / EI$

(c) $\iint V.dx / EI$ (d) $\iint M_n.dx / EI$

**Ans. (a) :** Deflection of elastic curve of a beam

$$EI.\, Y_{max.} = \iint M.dx$$

$$Y_{max.} = \frac{\int\int M.dx}{EI}$$

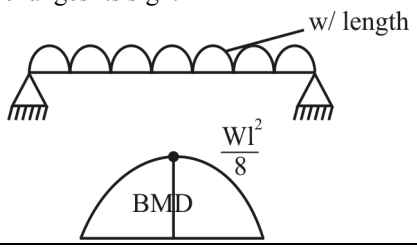
**27. The number of points of inflection in a simply supported beam with u.d.l is**

(a) 0 (b) 1

(c) 2 (d) 3

**Ans. (a) :** The points of inflection are points of zero bending moment that is where the bean changes its curvature from sagging to hogging.

A point of contra-flexure is location where the bending moment changes its sign.

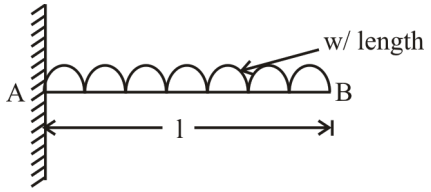



**28. The maximum deflection at the free end of a cantilever subjected to u.d.l is**

(a) $W\ell^2/16EI$ (b) $W\ell^4/6EI$

(c) $W\ell^4/8EI$ (d) $(5/384)\times W\ell^2/EI$

**Ans. (c):**




$$\text{Slope, } \theta = \frac{Wl^3}{6EI}$$

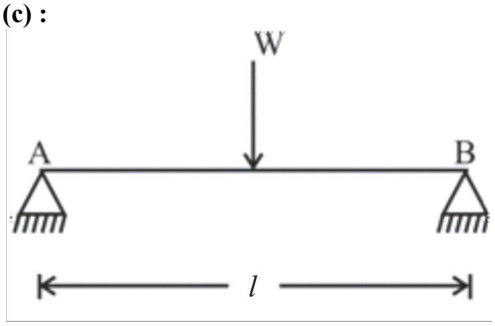
$$\text{Max. deflection, } \delta_{max} = \frac{Wl^4}{8EI}$$

**29. The maximum slope in a simply supported beam subjected to a concentrated load at its mid-span is**

(a) $W\ell^3/16EI$ (b) $(5/384)\times W\ell^2/EI$

(c) $W\ell^2/16EI$ (d) $(5/384)\times W\ell^2/EI$

**Ans. (c) :**




$$\text{Slope, } \theta = \frac{Wl^2}{16EI}$$

$$\text{Max. deflection, } \delta_{max} = \frac{Wl^3}{48EI}$$

**30. The unit load method is based on**

(a) Principle of superposition

(b) Maxwell's reciprocal theorem

(c) Principle of virtual work

(d) Newton's laws of equilibrium

**Ans. (c) :** The virtual work load method is also known as unit-load method or virtual force method used to find the deflection and slope at a point of a beam.

**31. The relationship between the three moduli of an elastic material is given by**

(a) E = 9KG/(G+3K) (b) G = K + E

(c) K = 2 G/ (3K+G) (d) None of these

**Ans. (a) :** $E = \frac{9KG}{3K+G}$

**32. Pick the correct bending equation :**

(a) (M/I) = (y/f) = (E/R)

(b) (M/I) = (f/y) = (E/R)

(c) (M/R) = (y/E) = (f/I)

(d) (R/I ) = (E/M) = (y/f)

**Ans. (b) :** Bending equation-

$$\frac{M}{I} = \frac{\sigma}{Y} = \frac{E}{R}$$

**33. A beam of uniform strength has**

(a) Uniform cross-section

(b) Uniform bending moment

(c) Uniform stress across the section

(d) Same maximum across at all section

**Ans. (c) :** A beam is said to be uniform strength, if bending stress developed at every cross-section of the beam remains the same along longitudinal axis.

Two methods to make a beam of uniform strength-

(i) Varying the width of the beam and keeping depth constant.

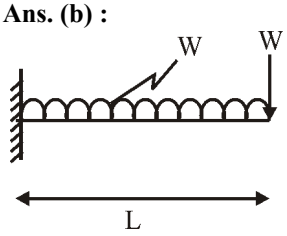
(ii) Varying the depth of the beam and keeping width constant.

**34. A cantilever beam of span L carries a concentrated load of W at the free end also a total load of W over entire length. The maximum bending moment is**

(a) WL (b) 1.5 WL

(c) 0.25 WL (d) W + 0.25 WL

**Ans. (b) :**




$$\text{Bending moment, } M = Wx + W\left(x - \frac{L}{2}\right)$$

At point, x = 0

$$M = Wx_0 + W\left(0 - \frac{L}{2}\right)$$

$$= -\frac{WL}{2} = -0.5WL$$

Maximum Bending moment,

$$(BM)_{max} = Wx + W\left(x - \frac{L}{2}\right)$$

At, x = L

$$(BM)_{max} = W \times L + W\left(L - \frac{L}{2}\right)$$

$$= WL + \frac{WL}{2}$$

$$= \frac{3WL}{2}$$

$$= 1.5\ WL$$

**35. The strain along longitudinal direction is 0.25, if the Poisson's ratio is 0.2, the lateral strain is**

(a) 0.050 (b) 0.50

(c) 0.125 (d) 1.25

**Ans. (a) :** Given,

Strain along longitudinal, $E_{lg} = 0.25$

Poisson's ratio $\mu = 0.2$

lateral strain, $E_{lateral}$ = ?

Poisson's ratio, $\mu = \dfrac{\varepsilon_{lateral}}{\varepsilon_{long}}$

$$0.20 = \frac{\varepsilon_{lateral}}{0.25}$$

$$\varepsilon_{lateral} = 0.25 \times 0.2$$

$$\varepsilon_{lateral} = 0.050$$

**36. The transverse load and shear in a beam has the relation**

(a) W = dV/dx (b) V = dW/dx

(c) V = W.I. (d) $V = d^2W/dx^2$

**Ans. (a) :** Transverse load, $W = \dfrac{dv}{dx}$

Shear force, $V = \dfrac{dM}{dx}$

**37. The shear stress on the principal plane is**

(a) Maximum (b) Zero

(c) Minimum

(d) Equal to one of the principal stress

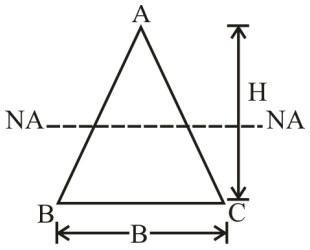
**Ans. (b) :** The shear stresses are zero on principal planes. Principal stress occurs on mutually perpendicular planes. Planes of maximum shear stress at 45° to the principal planes.

**38. The moment of inertia of a triangular lamina of base width B and height H about its base is**

(a) $BH^3/36$ (b) $BH^3/12$

(c) $BH^3/6$ (d) None of the above

**Ans. (b) :** Moment of inertia of a triangular lamina.




$$I_{NA} = \frac{BH^3}{36}$$

$$I_{base} = \frac{BH^3}{12}$$

**39. When a circular shaft is a subjected to a state of pure torsion the cross-section of the shaft is subjected to**

(a) Shear stress (b) Bending stress

(c) Bending and shear (d) None of the above

**Ans. (a) :** When a circular shaft is a subjected to a state of pure torsion the cross-section of the shaft is subjected to shear stress.

**40. Euler theory is applicable to**

(a) Short columns (b) Long columns

(c) Short and long columns

(d) None of the above

**Ans. (b) :** Euler's theory is valid for long column only.

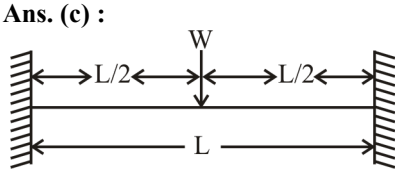
- The columns whose slenderness ratio is more than 80 – long column.
- Slenderness ratio < 80 – short column
- For short or long columns Rankine's formula is used.

**41. The mid-span deflection of a fixed beam due to concentrated load at mid-span is**

(a) $W\ell^3/48EI$ (b) $(5/385)W\ell^4/EI$

(c) $W\ell^3/192EI$ (d) $W\ell^3/384EI$

**Ans. (c) :**




Slope, $\theta = 0$

Deflection, $\delta_{max} = \dfrac{WL^3}{192EI}$

**42. In double overhanging beam for the positive bending moment to be equal to negative bending moment the position of supports must be at**

(a) 0.25 $\ell$ (b) 0.207 $\ell$

(c) 0.33 $\ell$ (d) None of these

**Ans. (c) :**

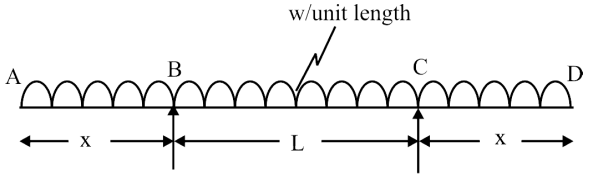
$$BM_{+ve} = \frac{w}{8}\left(L^2 - 4a^2\right)$$

$$BM_{-ve} = \frac{wa^2}{2}$$

Equating,

$$\frac{w}{8}\left(L^2 - 4a^2\right) = \frac{wa^2}{2}$$

$a = 0.35\ L$




**43. The torsion equation of the form**

(a) $T/J = f_s / R = q/r = N\theta/\ell$

(b) $T/R = f/R = g/j = N\ell/\theta$

(c) $T/I = f/y = E/R = q\theta/N$

(d) None of the above

**Ans. (a) :** Torsion equation–

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

$T/J = f_s / R = q/r = N\theta/\ell$

**44. For efficient transmission of power the cross-section of shaft must be**

(a) Rectangular (b) Circular

(c) Elliptical (d) Hollow circular

**Ans. (d) :** A shaft is a rotating element which is used to transmit power from one place to another. Transmission of power the cross-section of shaft must be circular for efficient transmission of power the cross-section of shaft must be Hollow circular.

**45. Bulk modulus is the ratio of**

(a) stress and strain

(b) shear stress and strain
(c) stress and volumetric strain
(d) lateral strain and longitudinal strain

**Ans. (c) :** Bulk modulus is defined as the ratio of volumetric stress to volumetric strain.

$$\text{Bulk modulus (k)} = \frac{\text{Volumetric stress}\,(\sigma)}{\text{Volumetric strain}\,(\varepsilon_v)}$$

**46. Gauge length is the ratio of**
(a) length of specimen between grips
(b) it is 8 times dia of specimen
(c) distance over which change in length is measured
(d) distance between 2 points in the specimen

**Ans. (a) :** Gauge length is the reference length used in the elongation calculations. Multiplying the width and thickness within the gauge length determines the initial cross-section area before testing. Grips tightly clamp the edges of the specimen at opposite ends.

**47. When a bar is fixed at the two ends and subjected to temperature raise the nature of stress developed in the bar is**
(a) Tensile (b) Compressive
(c) Shear (d) None of the above

**Ans. (b) :** When a bar is fixed at the two ends and subjected to temperature raise the nature of stress developed in the bar is compressive stress.

**48. Under uniformly varying load the bending moment varies**
(a) Linearly (b) Parabolical
(c) Cubic parabolically (d) Elliptical

**Ans. (c) :** In uniformly varying load the nature of shear force is parabolic and bending moment is cubic.

**49. The shear stress on principal plane is**
(a) Minimum
(b) Zero
(c) Minimum normal stress
(d) None of the above

**Ans. (b) :** Shear stress on principal plane is Zero.

**50. If poisson's ratio is 0.3, its rigidity modulus is**
(a) 0.77E (b) 0.385E
(c) 0.6E (d) 2.4E

**Ans. (b) :** Given, poission's ratio ($\mu$) = 0.3
Modules of rigidity (G) = ?

$\therefore$ $E = 2G(1+\mu)$
$E = 2G(1+0.3)$
$E = 2G \times 1.3$
$E = 2.6\,G$
$G = \frac{1}{2.6}E$
$G = 0.3846\,E$

Modulus of rigidity (G) = 0.385 E

**51. The first moment of an area 'A' about the y axis in the plane of the area is given by**
(a) $\int Axda$ (b) $\bar{y}A$
(c) $\bar{x}dA$ (d) $\int^{-} Ayda$

**Ans. (a) :**
First moment of an area 'A' about the y axis, Ax or $Ax_{max}$.
First moment of an area 'A' about the x axis, Ay or $Ay_{max}$.

**52. The centre of pressure is**
(a) At the centroid of sub-merged area
(b) The centroid of the pressure prism
(c) Independent of the orientation of the area
(d) A point on the line of action of the resultant force

**Ans. (d) :** Centre of pressure is defined as the point where the total sum of a pressure field resultant in an equivalent force acting through that point.

**or**

The centre of pressure is a point on the line of action of the resultant force.

**53. The pressure in a tank is 50 N/cm$^2$, The equivalent pressure head of oil with specific gravity 0.9 is**
(a) 0.5663 m (b) 5.663 m
(c) 56.63 m (d) 566.3 m

**Ans. (c) :** Given, pressure (P) = 50 N/cm$^2$
Oil specific gravity (s) = 0.9

$$\text{Specific gravity of oil (s)} = \frac{\rho_{oil}}{\rho_{standard\ fluid}}$$

$\rho_{oil} = 0.9 \times 1000$
$\rho_{oil} = 900\ kg/m^3$

$\therefore$ Pressure (P) = $\rho gh$
$50 \times 10^4 = 900 \times 9.81 \times h$

$$h = \frac{50 \times 10^4}{900 \times 9.81}$$

$h = 56.63$ m

**54. A circular area of diameter'd' is vertical and submerged in a liquid. Its upper edge is at liquid surface. Its center of pressure is at a depth of**
(a) d/2 (b) d/4
(c) 3d/4 (d) 5d/8

**Ans. (d) :** For circular section–

$$\text{Centre of pressure (C.P.)} = \bar{x} + \frac{I_{C.G.}\sin^2\theta}{A\bar{x}}$$

$$\text{C.P.} = \frac{d}{2} + \frac{\frac{\pi d^4}{64}}{\frac{\pi d^2}{4} \times \frac{d}{2}} \qquad [\because \theta = 90^\circ]$$

$$\text{C.P.} = \frac{d}{2} + \frac{\pi d^4}{64} \times \frac{8}{\pi d^3}$$

$$\text{C.P.} = \frac{d}{2} + \frac{d}{8} = \frac{4d+d}{8}$$

$$\text{Centre of pressure (C.P.)} = \frac{5d}{8}$$

**55. A body floats in a stable equilibrium**

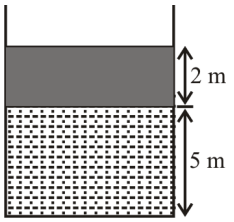
(a) When its metacentric height is zero
(b) When the metacentre is below the centre of gravity
(c) When I/V is positive
(d) When the metacenter is above center of gravity

**Ans. (d) :** A body floats in stable equilibrium when the meta centre is above centre of gravity.

**56. An open tank contains 5m of water covered with 2m of oil whose specific weight is 8 kN/m$^3$. The pressure at the interface is**

(a) 16 kN/m$^2$ (b) 65 kN/m$^2$
(c) 16 N/m$^2$ (d) 65 N/m$^2$

**Ans. (a) :**




Given,

Height of water ($h_w$) = 5 m

Height of oil ($h_o$) = 2 m

Specific weight of oil (w) = $\rho g$ = 8 kN/m$^3$

$\therefore$ Pressure at the interface of two fluids–

$P = \rho gh$

$P = w \times h$

$P = 8 \times 2$

$P = 16$ kN/m$^2$

**57. Which of the following velocity fields does not satisfy conservation of mass for in compressible plane flow?**

(a) u = x, v = -y (b) u = y, v = x
(c) u = 2x, v = 2y (d) u = -2x, v = +2y

**Ans. (c) :** Continuity equation for a flow of an in compressible fluid is-

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

For option check–

(a) $u = x, \quad v = -y$

$$\frac{\partial u}{\partial x}=1, \quad \frac{\partial v}{\partial y}=-1$$

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

$$1 + (-1) = 0$$

(b) $$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

$$\frac{\partial}{\partial x}(y)+\frac{\partial}{\partial y}(x)=0$$

$$0 + 0 = 0$$

(c) $$\frac{\partial}{\partial x}(2x)+\frac{\partial}{\partial y}(2y)$$

$$= 2 + 2$$

$$= 4$$

$$4 \neq 0$$

(d) $$\frac{\partial}{\partial x}(-2x)+\frac{\partial}{\partial y}(2y)=0$$

$$-2 + 2 = 0$$

$$0 = 0$$

**58. The condition of irrotationality of flow**

(a) $\delta^2\psi/\delta^2 x - \delta^2\psi/\delta^2 y = 0$
(b) $\delta\psi/\delta x + \delta\psi/\delta y = 0$
(c) $\delta^2\psi/\delta^2 x - \delta^2\psi/\delta^2 y = 0$
(d) $\delta^2\psi/\delta x^2 + \delta^2\psi/\delta y^2 = 0$

**Ans. (d) :** According to the irrotationality condition–

$$\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}=0$$

or $$\frac{\partial^2\phi}{\partial x^2}-\frac{\partial^2\phi}{\partial y^2}=0$$

$$\delta^2\psi/\delta x^2 + \delta^2\psi/\delta y^2 = 0$$

**59. Flow-net can be drawn, if**

(a) Flow is turbulent (b) Fluid is ideal
(c) Fluid is unsteady (d) Fluid is viscous

**Ans. (b) :** A flow-net is a grid obtained by drawing a series of equipotential lines and stream lines. Flow net is very useful two dimensional, irrotational flow.
The flow should be steady.

**60. Orifice meter is used for measuring**

(a) Pressure (b) Velocity
(c) Volume (d) Volume rate of flow

**Ans. (d) :** Orifice meter is used for measuring volumetric rate of flow.

**61. A venturimeter laid 45$^0$ to horizontal is connected to a differential mercury manometer. The deflection shown is 'h' cm, if the venturimeter is made horizontal, the deflection in the manometer will be**

(a) $h/\sqrt{2}$ (b) $h/\sqrt{2}$
(c) h (d) $\sqrt{2}/h$

**Ans. (c) :** Manometer reading of a verturimeter does not depends on inclination of venturimeter with the horizontal.

- The angle of divergence = 5° – 7°
- If should be not greater than 7° avoid flow separation.

**62. The coefficient of discharge of a flow nozzle meter is**

(a) Greater than a venturimeter
(b) Lower than a orificemeter
(c) Greater than both the venturimeter and orificemeter
(d) Lower than a venturimeter but greater than orificemeter

**Ans. (d):** Coefficient of discharge for nozzle meter lies between venture meter and orifice meter. Coefficient of discharge for nozzle meter is lower than venturimeter and greater than orifice meter.

**63. Pitot tube can be used to measure velocity**

(a) Only in pipe flow
(b) Only in open channel flow
(c) Only in compressible flow
(d) In all the three types started above

**Ans. (d) :** Pitot tube is a device used for calculating the velocity of flow at any point in a open channel, pipe flow and compressible flow.

**64. A cippoletti weir is one**

(a) Which is a rectangular weir with end contraction
(b) Which can be treated as rectangular weir without any end contractions
(c) Which is a combination of a rectangular and triangular weir
(d) Which is a combination of a rectangular and circular weir

**Ans. (c) :** A cippoletti weir is trapezoidal weir. The slopes of cipoletti weir are adjusted in such a way that, the reduction in discharge due to end contraction in rectangular weir. The increase in discharge due to triangular portion.

**65. The ratio of inertial forces to pressure forces yield**

(a) Reynold's number (b) Mach number
(c) Euler number (d) Cauchy number

**Ans. (c) :** Euler number $= \sqrt{\dfrac{\text{Inertia force}}{\text{Pressure force}}}$

**66. Bernoulli's equation for compressible fluid flow is**

(a) The same as that of incompressible flow
(b) Not at all same as that of incompressible flow
(c) The same as incompressible flow except pressure energy term
(d) The same as incompressible flow except kinetic energy term

**Ans. (c) :** Bermoulli's equation for incompressible flow

$$\frac{dP}{\rho} + Vdv + gdz = \text{constant}$$

For compressible flow ρ (Density is not constant)

$$\int \frac{dP}{\rho} + \int Vdv + \int gdz = \text{constant}$$

(Same as incompressible flow except of pressure energy term because ρ is variable in case of compressible flow.)

**67. A coloured dye is injected at a point in water flow. The resulting visible line is a**

(a) Stream line (b) Streak line
(c) path line (d) None of the above

**Ans. (b) :** A coloured dye is injected at a point in water flow. The resulting visible line is a streak line.

- Streak line indicates the instantaneous position of particles of fluid passing through a point.

**68. In forced vortex flow, velocity V and radial distance r are related as**

(a) $V \alpha r$ (b) $V \alpha 1/r$
(c) $V \alpha 1/t^2$ (d) $V \alpha r^2$

**Ans. (a) :** Forced vortex flow is defined as that type of vortex flow in which external torque is required to rotate the fluid mass.

$$v = r\omega \quad \Rightarrow \quad \omega = \frac{v}{r} = \text{constant}$$

$$v \propto r$$

**69. The non-dimensional parameter connected with cavitations is**

(a) Mach number (b) Cauchy number
(c) Thoma number (d) Strouhal number

**Ans. (c) :** Thoma's cavitation parameter (σ)

$$\sigma = \frac{NPSH}{H} = \frac{\left(\frac{P_A}{\rho g} - \frac{P_V}{\rho g} - z - h_f\right)}{H}$$

It is the ratio of net positive suction head (NPSH) to the total head

- NPSH must be positive for preventing the cavitation.
- If NPSH reaches zero then liquid starts boiling and cavitation starts.

Mach number- $(M) = \sqrt{\dfrac{\text{Inertia force}}{\text{elastic force}}}$

- Used for flow condition

| Sr. no | Mach no. | Type of flow |
|---|---|---|
| 1 | $< .8$ | subcritical |
| 2 | $.8 - 1.3$ | transition |
| 3 | $= 1$ | Sonic |
| 4 | $1.3 - 5$ | Supersonic |
| 5 | $> 5$ | Hyper sonic |

**70. The lower critical Reynold's number for pipe flow is**

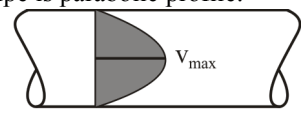
(a) 5000 (b) 2000
(c) 3000 (d) 3200

**Ans. (b) :** Lower critical Reynold's number for pipe flow is 2000.

**71. The velocity distributed for Laminar flow through closed pipe is**

(a) Linear (b) Elliptical
(c) Parabolic (d) Trapezoidal

**Ans. (c) :** Velocity distributed for laminar flow through closed pipe is parabolic profile.




(Velocity distribution laminar)

**72. For laminar flow through circular pipe the ratio of average velocity to maximum velocity is**

(a) 0.75 (b) 0.50

(c) 0.25 (d) 1.50

**Ans. (b) :** Maximum velocity, $V_{max} = 2\ V_{avg.}$

$$\therefore \quad \text{Ratio} = \frac{\text{Average velocity}}{\text{Maximum velocity}}$$

$$\text{Ratio} = \frac{V_{avg.}}{V_{max}} = \frac{1}{2} = 0.5$$

**73. For a flow between two reservoirs connected by a series pipeline, the head loss due to friction is proportional to**

(a) Q (b) 1/Q

(c) $Q^2$ (d) $1/Q^2$

**Ans. (c) :** For a flow between two reservoirs connected by series pipeline–

$$h_f = \frac{8fLQ^2}{\pi^2 g D^5} \Rightarrow h_f \propto Q^2$$

Where,

f = Friction factor

L = Length of pipe

D = Dia. of pipe

Q = Flow rate

$h_f$ = Head loss due to friction

When two pipes are connected in series then the discharge in each pipe is same, but the head loss in each pipe.

Head loss (H) = $h_{f1} + h_{f2} + \ldots\ldots h_{fn}$

Discharge, $Q_1 = Q_2 = \ldots\ldots Q_n$

**74. Loss of head in pump's foot value....**

(a) $0.5\ v^2/2g$ (b) $v^2/2g$

(c) $1.5\ v^2/2g$ (d) $v^2/g$

**Ans. (b) :** Loss of head in pipe's foot volve–

$$h_f = \frac{v^2}{2g}$$

**75. A jet of water 5 cm is diameter impinges on a flat plate normally with 18 m/sec velocity. If the plate is moving in the same direction at 12 m/sec force exerted on the plate is**

(a) 5547 N (b) 565.5 N

(c) 5.65 N (d) 554.7 N

**Ans. (*) :** Given,

Dia of plate = 5 cm = $5 \times 10^{-2}$ m

Velocity of plate = 12 m/sec

Velocity of jet = 18 m/sec

Force exerted by jet-

$$(F) = \rho\ A\ (V_{jet} - V_{plate})^2$$

$$F = 1000\ kg/m^3 \times \frac{\pi}{4} \times (5 \times 10^{-2})^2\ m^2 \times (18-12)^2\ m/sec^2$$

F = 70.68 N

Option given by board (d)

**76. A turbine develops 1600 kW while running at 360 rpm under 8 m head. The turbine shall be**

(a) Kaplan turbine (b) Francis turbine

(c) Pelton wheel

(d) Turbo impulse wheel

**Ans. (a) :** Pressure (P) = 1600 kW

N = 360 rpm

Head (H) = 8 m

$$\therefore \text{ Specific speed } (N_s) = \frac{N\sqrt{p}}{H^{5/4}}$$

$$N_s = \frac{360 \times \sqrt{1600}}{8^{5/4}}$$

$$N_s = 1070.28$$

$N_s > 1000$ —— Kaplan Turbine

**77. A turbine develops 600 kW under a head of 30 m at 450 rpm speed. The speed and power at 24 m head is**

(a) 1070 rpm, 429 kW

(b) 403 rpm, 429 kW

(c) 429 rpm, 403 kW

(d) 1070 rpm, 403 kW

**Ans. (b) :** Given,

Power ($P_1$) = 600 kW

Head ($h_1$) = 30 m

Head ($h_2$) = 24 m

Normal speed ($N_1$) = 450 rpm

Normal speed ($N_2$) = ?

Power ($P_2$) = ?

1. Unit speed under unit head

$$N_{unit} = \frac{N}{\sqrt{H}}\text{rpm} \Rightarrow \frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$$

$$\frac{450}{\sqrt{30}} = \frac{N_2}{\sqrt{24}} \Rightarrow N_2 = 403 \text{ rpm}$$

2. Unit power under the unit head

$$P_{unit} = \frac{P}{H^{3/2}} \Rightarrow \frac{P_1}{H_1^{3/2}} = \frac{P_2}{H_2^{3/2}}$$

$$\frac{600}{30^{3/2}} = \frac{P_2}{24^{3/2}} \Rightarrow P_2 = 429 \text{ kW}$$

**78. A draft tube is a part of the installation of a**

(a) Propeller turbine

(b) Pelton turbine

(c) Turbo impulse wheel

(d) All the above turbines

**Ans. (a) :** Draft tube is used only in reaction turbines as there is a change in pressure energy involved

**Purpose of draft tube:–**

**1.** It permits a negative or suction head to be established at runner exit to install the turbine above the tail race level without head loss.

**2.** Converts proportion of velocity energy rejected from runner into useful pressure energy.

• Reactions Turbins are – Kaplan, Francis, Propeller.

**79. The function of a governor in a turbine is to**

(a) Give maximum overall efficiency

(b) Give maximum hydraulic efficiency

(c) Control the speed of the runner

(d) Control the flow through the runner

**Ans.(c):** Governor is the main controller of the hydraulic turbine. The governor varies the water flow through the turbine "to control its speed or power output".

**80. Run away speed of a turbine is the one at**

(a) Maximum efficiency (b) Optimum efficiency
(c) Zero load (d) Minimum input

**Ans. (c) : Run away speed**:– The runway speed of a turbine is a speed at full flow and no shaft load. It is the maximum speed that the turbine can theoretically attain.

**81. In a Pelton wheel the depth of the bucket is approximately**

(a) 0.5 diameter of jet (b) 0.95 diameter of jet
(c) 1.5 diameter of jet (d) 1.90 diameter of jet

**Ans. (c) :** The depth of the bucket for a pelton wheel is generally 1.2 times the dia. of the jet and width is 5 times of the dia. of jet.

**No. of bucket on the periphery** = $\left(15+\frac{D}{2d}\right)$

**Where :** D = Pitch dia.
d = dia. of jet

**82. Tubular turbines are low head, low power**

(a) Kaplan turbines (b) Propeller turbines
(c) Francis turbines (d) Pelton turbines

**Ans. (a) :** Kaplan turbines are Tubular turbines low head, low power.

**83. Turbine constants are**

(a) Input power, output power
(b) Input power, flow through runner, output power
(c) Unit power, unit speed, specific speed
(d) Unit power, unit speed, unit discharge

**Ans. (d) :** 1. Unit discharge :–

$$Q_{unit}=\frac{Q}{\sqrt{H}}\Rightarrow\frac{Q_1}{\sqrt{H_1}}=\frac{Q_2}{\sqrt{H_2}}$$

2. Unit Speed :–

$$N_{unit}=\frac{N}{\sqrt{H}}\text{ rpm}\Rightarrow\frac{N_1}{\sqrt{H_1}}=\frac{N_2}{\sqrt{H_2}}$$

3. Unit Power :–

$$P_{unit}=\frac{P}{H^{3/2}}\Rightarrow\frac{P_1}{H_1^{3/2}}=\frac{P_2}{H_2^{3/2}}$$

**84. The motion that helps in the working of a centrifugal pump is**

(a) Radial flow (b) Free vortex flow
(c) Forced vortex flow (d) Axial flow

**Ans. (c) : Centrifugal pump** :– It works on the principle of "forced vortex" where external torque is procided to the impeller by the means of the prime mover in volute casing the external torque is absent therefore it becomes "free vortex flow"

**85. A centrifugal pump converts**

(a) Electrical energy to mechanical energy
(b) Kinetic energy to pressure energy
(c) Pressure energy to kinetic energy
(d) Mechanical energy to electrical energy

**Ans. (b):** Centrifugal pump converts Kinetic energy to pressure energy.
The vanes of centrifugal pumps are curved backward to prevent overloading of the impeller motor and have better stabilizing of power.
The impeller in the centrifugal pump is immersed in water and imparts a centrifugal force which moves out the water particles radially. The impellers are known to have backward curved vanes which have less than 90º of blade.

- The centrifugal pump converts mechanical energy into hydraulic energy.
- The efficiency of centrifugal pump range from 75 to 93%.

**86. A centrifugal pump when used to work with compressible fluid flow it is called**

(a) A diffusor (b) A nozzle converter
(c) A compressor (d) A pressure converter

**Ans. (c) :** A centrifugal pump when used to work with compressible fluid flow it is called compressor.

**87. In a centrifugal pump the work done by the impeller is**

(a) The manometric head
(b) Greater than manometeric head
(c) Less than manometric head
(d) The manometric head less losses

**Ans. (b) :** In a centrifugal pump the work done by the impeller is Greater than manometeric head.

**88. A turbine pump is**

(a) A diffuser vane pump
(b) A single volute pump
(c) A double volute pump
(d) A variable speed pump

**Ans. (a) :** A turbine pump is a diffuser vane pump.

**89. A multi-stage centrifugal pump is equivalent to**

(a) Centrifugal pumps connected in parallel
(b) Centrifugal pumps connected in series
(c) Centrifugal pumps connected in series first and coupled to parallel pumps
(d) Centrifugal pumps connected in parallel first and coupled to series connected.

**Ans. (b) :** Multistage centrifugal pumps have multiple liquid chamber that are connected in series in order to increases head.

**90. If N is speed of rotation of impeller and Q is the discharge delivered, then the manometeric head can be expressed as**

(a) $H_m = AN^2 + BNQ^2 + CQ$
(b) $H_m = AN^2 + BN^2Q + CQ^2$
(c) $H_m = AN^2 + BNQ + CQ^2$
(d) $H_m = AN^2 + BNQ^2 + CQ^3$

**Ans. (b) :** If N is speed of rotation of impeller and Q is the discharge delivered, then the manometeric head can be expressed as :– $H_m = AN^2 + BN^2Q + CQ^2$

**91. In a centrifugal pump, cavitation occurs generally**

(a) In the volue chamber (b) In the impeller
(c) In the shaft (d) In the casing

**Ans. (b): NPSH**–Net positive suction head may be defined "as the difference between the net inlet head and the head corresponding to the vapour pressure of the liquid.

- NPSH has significance only when cavitating liquids are handled.
- NPSH is a parameter that can be used to check cavitation in pump.

**92. An isentropic flow is**

(a) An accelerating adiabatic flow
(b) A retarding adiabatic flow
(c) A reversible adiabatic flow
(d) An irreversible adiabatic flow

**Ans. (c) : Isentropic flow** :– It is a fluid flow that is both reversible and adiabatic. i.e. no heat is added to the flow and no energy transformation takes place due to friction or dissipative effects.

**93. The continuity equation for one dimensional compressible fluid flow is**

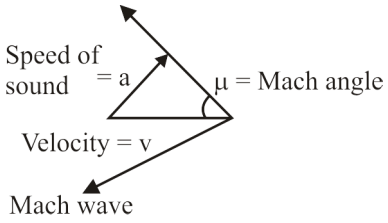
(a) $d(\rho AV) = 1$ (b) $d(\rho AV) = 0$
(c) $\rho d(AV) = 1$ (d) $\rho d(AV) = 0$

**Ans. (b) :** The continuity equation for one dimensional compressible fluid flow is:– $d(\rho AV) = 0$
in compressible flow $\rho$ is variable.

**94. Mach angle in a compressible fluid flow is formed for a flows with mach number**

(a) Less than one (b) Equal to one
(c) Greater than one (d) Very less than one

**Ans. (c) :** Greater than one




$\sin\mu = \frac{a}{v}$ Mach number = M > 1.0

$\text{Mach angle} = \mu = \sin^{-1}\frac{1}{M}$

**95. Flow through a diffuser in a compressible flow will have**

(a) Rise in temperature
(b) Fall in temperature
(c) Fall in temperature and pressure
(d) Fall in temperature and rise in pressure

**Ans. (d) : Diffuser**:– It s a device for reducing the velocity and increasing the static pressure of fluid passing through a system.

**96. A normal shock wave is**

(a) An infinite pressure wave
(b) A mild finite pressure wave
(c) A steep finite pressure wave
(d) A steep infinite pressure wave

**Ans. (c) : Shock wave :–** It is nothing but a steep finite pressure wave.
"If shock wave is perpendicular to the flow direction is called **Normal shock wave** ".

**97. If Q is heat transfer, E is energy and W is work, then energy equation for compressible flow is**

(a) $dE = dQ + dW$ (b) $dW = dQ + dE$
(c) $dE = dQ - dW$ (d) $dW = dE + dQ$

**Ans. (c) :** According to thermodynamics second law.

$E = Q - W$

Energy equation for compressible flow

$dE = dQ - dW$

Where,

Q = Heat transfer, W = Work
E = Energy

**98. For an isentropic air flow the stagnation temperature is given by**

(a) $T_0 = T(1+0.4M^2)$ (b) $T_0 = T(1-0.4M^2)$
(c) $T_0 = T(1+0.2M^2)$ (d) $T_0 = T(1-0.2M^2)$

**Ans. (c) :** Stagnation temperature

$$T_\circ = T\left(1+\frac{\gamma-1}{2}M^2\right)$$

For air $\gamma = 1.4$

$T_0 = T(1+0.2M^2)$

**99. A hot wire anemometer works with**

(a) Constant voltage and constant temperature
(b) Constant voltage or constant temperature
(c) Constant temperature and constant current
(d) Constant temperature or constant current

**Ans. (d) :** A hot wire anemometer works with constant temperature or constant current.

**100. For isentropic flow through a convergent divergent nozzle the area ratio is given for air,**

(a) $\frac{A}{A_1} = 1/M\left[\left(5-M^2\right)/6\right]^3$

(b) $\frac{A}{A_1} = 1/M\left[\left(5+M^2\right)/6\right]^3$

(c) $\frac{A}{A_1} = 1/M\left[\left(M^2-6\right)/6\right]^3$

(d) $\frac{A}{A_1} = M\left[\left(M^2-5\right)/6\right]^3$

**Ans. (b) :** Area ratio for ideal gas :-

$$\frac{A}{A_1} = \frac{1}{M}\left(\frac{1+\frac{\gamma-1}{2}M^2}{\frac{\gamma+1}{2}}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

For air $\gamma = 1.4$

Putting $\gamma = 1.4$

$$\frac{A}{A_1} = \frac{1}{M}\left[\frac{\left(5+M^2\right)}{6}\right]^3$$

Where, M = Mach no.

**Isentropic Nozzle flow:–** It describes the movement of a gas or liquid through a narrowing opening without increase or decrease in entropy.

# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 1987
# Solved Paper

**1. The ratio of maximum shear to the mean shear stress in case of a beam of circular cross section is**

(a) 4 : 3 (b) 3 : 2
(c) 5 : 4 (d) 2 : 1

**Ans. (a) :** $\tau_{max} = \frac{4}{3}\tau_{avg}$ for circular cross section.

$$\frac{\tau_{max}}{\tau_{avg}} = \frac{4}{3} \Rightarrow 4:3$$

**2. For punching a hold of 20 mm dia in a plate 20 mm thick, if the shear stress is 30 kg/sq. mm. the force required in kg would be**

(a) 10000 (b) 18850
(c) 28850 (d) 37699

**Ans. (d) :** Punching shear force = Hole perimetre × thickness × shear stress

$F_{PS} = \pi d \times t \times \tau$

$= \pi \times 20 \times 20 \times 30$ kg./sq.mm

$F_{PS} = 37699.11$ kg

**3. If E & K be the Young's modulus and Bulk modulus of a material the ratio E/K is**

(a) 1 + (2/m) (b) 1 – (2/m)
(c) 1 + (1/2m) (d) 1 – (3/m)

**Ans. (b) :** $E = 3K(1 - 2\mu)$

$$E = 3K\left(1 - \frac{2}{m}\right)$$

$$\frac{E}{K} = 3\left(1 - \frac{2}{m}\right)$$

$\because \mu = \frac{1}{m} =$ poisson's

E = young's mod uolus

K = Bulk mod ulus

$$\frac{E}{K} \propto \left(1 - \frac{2}{m}\right)$$

**4. A short cast iron column carries a load of 80 MT. If the original diameter is 8 cm and $E_s = 1 \times 10^6$ kg/sq.cm and Poisson's ratio = 0.25 then the increase in dia. of column in cm would be**

(a) 0.00318 (b) 0.00216
(c) 0.00164 (d) 0.00288

**Ans. (a) :** Given,

D = 8 cm

$E = 1 \times 10^6$ kg/cm$^2$

$P = 80$ MT $= 80 \times 10^3$ kg

$\Delta D = ?$ in cm

$\mu = 0.25$

$$E = \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{\sigma}{\varepsilon_{long}}$$

(i) $\varepsilon_{long} = \frac{\sigma}{E} = \frac{P}{AE} = \frac{80 \times 10^3 \text{ kg}}{\frac{\pi}{4} 8^2 \times 10^6 \text{ kg/cm}^2} = 0.001591$

(ii) $\mu = -\frac{\text{Lateral strain}}{\text{Longitudinal strain}} = 0.25 = \frac{\text{Lateral strain}}{0.001591}$

(iii) Lateral strain = 0.0003978, Lateral strain $= \frac{\Delta D}{D}$

$$\frac{\Delta D}{D} = 0.0003978$$

$\Delta D = 0.0003978 \times 8$ cm

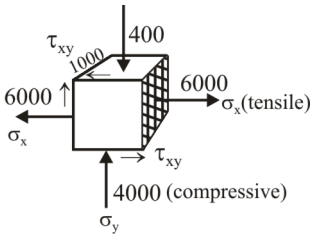
$\Delta D = 0.003183$

**5. An element has a tensile stress of 6000 Ksc and compressive stress of 4000 Ksc acting on two mutually perpendicular planes, and two equal shear stresses of 1000 Ksc on these plane the maximum (tensile) principal stress would be in Ksc**

(a) 5,200
(b) 4,125
(c) 8,600
(d) 6,095

**Ans. (d) :** Principal stress $\sigma_{max}/\sigma_{min}$

$$= \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$




Given,

$\sigma_x = +6000$ksc

$\sigma_y = -4000$ksc

$\tau_{xy} = 1000$ ksc

$$= \frac{6000 + (-4000)}{2} \pm \sqrt{\left(\frac{6000 - (-4000)}{2}\right)^2 + 1000^2}$$

$= 1000 \pm \sqrt{26000000}$

$= 1000 + 5099 = 6099$ Ksc.

**6. For a column fixed at one end and free at the other, the Euler's crippling load is**

(a) $\pi^2$ EI/ $L^2$ (b) $2\pi^2$ EI/ $L^2$

(c) $\pi^2$ EI/$4L^2$ (d) $\pi^2$ EI/$2L^2$

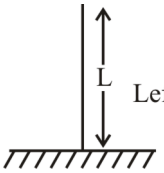
**Ans. (c) :**

Euler's crippling load = $\frac{\pi^2 \, EI_{min}}{(\ell_{eff})^2}$

$$P_{cr} = \frac{\pi^2 EI_{min}}{(2L)^2} \qquad [\because \ell_{eff} = 2L]$$

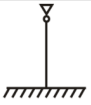
$$P_{cr} = \frac{\pi^2 EI_{min}}{4L^2}$$

Condition




Left = 2L

Additional:-

| End Conditions | | | | Respresentation | Effective Length |
|---|---|---|---|---|---|
| One End | | Other End | | | |
| Translation | Rotation | Translation | Rotation | | |
| Restrained | Restrained | Free | Free | | 2.0 L |
| Free | Restrained | Free | Restrained | | 2.0 L |
| Restrained | Free | Restrained | Free | | 1.0 L |
| Restrained | Restrained | Restrained | Free |  | 0.8 L |
| Rest | Rest | Rest | Rest | | 0.65 L |

**7. Column of given length, cross-section area and material, have different buckling loads for different end conditions, the strongest column is one for which**

(a) one end is fixed and the other hinged

(b) both ends hinged

(c) one end fixed and other free

(d) both ends fixed

**Ans. (d) :** Explanation:-

Euler load (P) $\propto \frac{1}{\text{(effective length)}}$

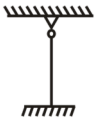
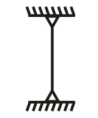
P↑ $\ell_{eff}$ ↓

For fixed beam $L_{eff} = \left(\frac{\ell}{2}\right)$

$$P \propto \frac{1}{\left(\frac{l}{2}\right)^2}$$

$$P \propto \frac{1}{l^2}$$ [4 times strong]

**Additional**

| | **Conditions** | **Effective length** |
|---|---|---|
| 1. |  | Left = $\frac{L}{\sqrt{2}}$ |
| 2. |  | Left = L |
| 3. | | Left = 2L |
| 4. | | Left = $\frac{L}{2}$ |

$$P = \frac{\pi^2 \, EI_{min}}{(Left)^2}$$

$$P = \frac{\pi^2 \, EI}{\left(\frac{L}{2}\right)^2}$$

$$P = \frac{4\pi^2 \, EI}{L^2}$$ For fixed beam

**8. If a member is subjected to a tensile force 'P' having its normal cross section 'A', the resulting shear stress in an oblique plane inclined at an angle 'θ' to its transverse plane will be**

(a) P $\sin^2\theta$/A (b) P $\sin 2\theta$/A

(c) P $\cos \theta$/2A (d) P $\cos^2\theta$/A

**Ans. (*) :**

Normal stress on oblique plane ($\sigma_n$)

$$= \left(\frac{\sigma_x + \sigma_y}{2}\right) + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

Shear stress on oblique plane ($\tau_n$)

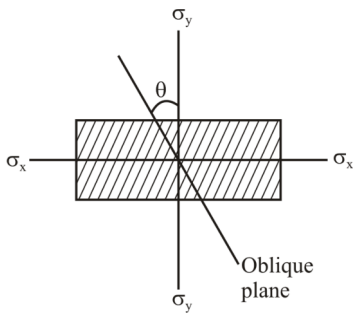
$$= -\left(\frac{\sigma_x - \sigma_y}{2}\right) \sin 2\theta + \tau_{xy} \cos 2\theta$$

Given,

Tensile force = P

Cross-section = A

$\sigma_x = \frac{P}{A},\ \sigma_y = 0$ (uni axial loading)




$$\tau_n = -\left(\frac{P}{2A}\right)\sin 2\theta + \tau_x \cos 2\theta$$

$$\tau_n = \frac{P}{2A}\sin 2\theta$$

**9. Modulus of rigidity is given by**

(a) Lateral strain / longitudinal strain
(b) Compressive stress / strain
(c) Tensile stress / tensile strain
(d) Shear stress / shear strain

**Ans. (d) :**

- Modulus of Rigidity G = $\frac{\text{Shear stress}}{\text{Shear strain}}$
- Poisson's Ratio $\mu$ = $-\frac{\text{Lateral strain}}{\text{longitudinal strain}}$
- Young's modulus E = $\frac{\text{Linear stress}}{\text{Corresponding strain}}$
- Bulk modulus K/N = $\frac{\text{Volumetric stress}}{\text{Volumetric strain}}$

$0.5 > \mu > -1$ Theoretically

**Note-** Strain is a dimensionless parameter Thus, it has No units

**10. A shaft 10 cm dia. and 2 m. long is subjected to torque of 800 kg.m. The maximum shear stress, in kg/sq.cm. developed is**

(a) 308.3 (b) 407.4
(c) 128.6 (d) 216.2

**Ans. (b) :** Given,

D = 10 cm
l = 2 m
T = 800 kg .m
$\tau$ = ?

$$\frac{T}{J} = \frac{\tau}{R} \Rightarrow \tau = \frac{TR}{J}$$

$$\tau = \frac{800 \times \frac{10}{2} \times 10^2}{\frac{\pi}{32} \times (10)^4} = 407.4 \text{ kg/ cm}^2$$

**11. A shaft transmits 80 H.P. at 60 rpm. If the maximum torque is 30% greater than the mean, the maximum torque is kg.m is**

(a) 1800 / $\pi$ (b) 3200 / $\pi$
(c) 1920 / $\pi$ (d) 3879 / $\pi$

**Ans. (d) :** Given,

P = 80 H.P = 80 × 746 watt = 59680 watt
N = 60 R.P.M
$T_{max}$ = ?
$T_{max} = 1.3\ T_{mean}$

$$P = \frac{2\pi N T_{mean}}{60}$$

$$80 \times 746 = \frac{2\pi \times 60 \times T_{mean}}{60}$$

$$T_{max} = \frac{1.3 \times 29840}{\pi} \text{N} - \text{m}$$

$$= \frac{3879.2}{\pi} \text{kg} - \text{m} \qquad [\because 1\text{N} = 0.1 \text{ kg}]$$

$$T_{max} = \frac{3879}{\pi}$$

**12. The ratio of moments of resistance of a solid circular shaft of dia. 'D' and a hallow shaft of external dia. 'D' and internal dia 'd' is**

(a) $D^4/\left(D^4 - d^4\right)$ (b) $D^3/\left(D^3 - d^4\right)$
(c) $\left(D^4 - d^4\right)/D^4$ (d) $\left(D^3 - d^3\right)/D^3$

**Ans. (a) :**

Moment of Inertia of solid shaft = $\frac{\pi}{64}D^4 \Rightarrow Z_S \frac{\pi}{32}D^3$

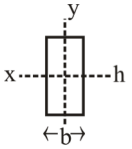
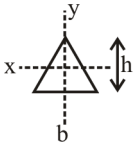
Moment of Inertia of Hollow shaft = $\frac{\pi}{64} \times \left(D^4 - d^4\right)$

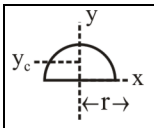
$$\Rightarrow Z_H = \frac{\pi}{32}\left(\frac{D^4 - d^4}{D}\right)$$

$M_r \propto Z$

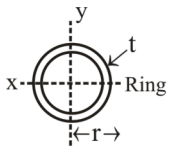
$$\frac{m_1}{m_2} = \frac{z_1}{z_2} = \frac{\frac{\pi}{32}}{\frac{\pi}{32}\left(\frac{D^4 - d^4}{D}\right)} = \frac{D^4}{D^4 - d^4}$$

**Additional**

| Shape | MOI $_{NA}$ |
|---|---|
|   | $I_{NA} = \frac{bh^3}{12}$ |
|   | $I_{NA} = \frac{bh^3}{36}$ ; $I_{base} = \frac{bh^3}{12}$ |




$I_{xx} = I_{yy} = \frac{\pi r^4}{8}$ $y_C = \frac{4r}{3\pi}$

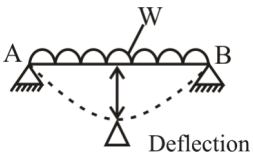



$I = 2\pi r^3 t$

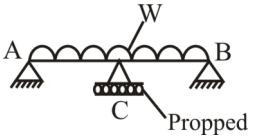
**13. A prismatic beam of length L is simply supported at its ends and subjected to a total U.d.l. of W spread over its entire span. It is then propped at its centre to neutralize the deflection. The net B.M. at its centre will be**

(a) WL (b) WL/3
(c) WL/24 (d) WL/32

**Ans. (d) :**




(i) $\Delta_1 = \frac{5WL^3}{384EI}$ $(W = W.L)$




(ii) Case:2 To avoid deflection in upward direction due to reaction Rc

$$\Delta_1 = \frac{R_c L^3}{48EI}$$

(iii) $\Delta_1 = \Delta_2$

$$= \frac{5WL^3}{384EI} = \frac{R_c L^3}{48EI}$$

$$R_C = \frac{5}{8}W$$

(iv) $R_A + R_B + R_C = W$

$$R_A + R_B = W - \frac{5}{8}W = \frac{3W}{8}$$

$$R_A = R_B = \frac{3}{16}W \qquad \because R_A = R_B$$

(v) B.m. at 'C'

$$BM_C = R_A \times \frac{L}{2} - \frac{W}{2} \times \frac{L}{4}$$

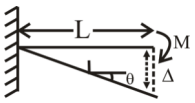
$$= \frac{3}{16} \times \frac{WL}{2} - \frac{WL}{8}$$

$$BM_C = \frac{3WL - 4WL}{32} = \frac{-WL}{32}$$ (–) clockwise moment

**14. A cantilever of flexural rigidity EI and span L is subjected to a moment M at its free end. The slope at its free end would be**

(a) ML/2EI (b) $ML^2/EI$
(c) ML/EI (d) ML/4EI

**Ans. (c) :**




$$\theta_{max} = \frac{ML}{EI}$$

$$\Delta = \frac{ML^2}{2EI}$$

$\Rightarrow$ Relation between θ and Δ

$$\Delta = \theta \times \frac{L}{2}$$

Where,

M = Moment
L = Length/span of beam
EI = Flexural rigidity

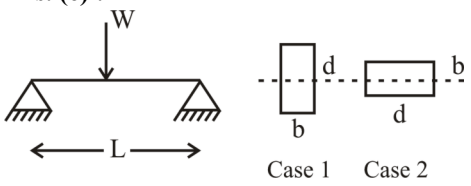
Note-

- iff EI ↑ Δ↓
- AE → Axial rigidity
- GJ → Torsional rigidity

**15. A simply supported beam of width 'b' and depth 'd' subjected to a load W at its centre causing deflection 'y' at that point. If the beam be turned such that its width is 'd' and depth is 'b' and be subjected to same load at same point, the Central deflection would be**

(a) (d/b)y (b) $(dy/b)^2$ (c) $(d/b)^2 y$ (d) (b/d)y

**Ans. (c) :**




- Deflection due to point load (δ) $y/\delta = \frac{WL^3}{48EI}$

$$y/\delta \propto \frac{I}{L}$$

$\delta.I$ = constant

$\delta_1 I_1 = \delta_2 I_2$

$\because \delta_1 = \frac{bd^3}{12}$, $\delta_2 = \frac{db^3}{12}$

$$\frac{\delta_1}{\delta_2} = \frac{b^2}{d^2}$$

$$\delta_2 = \left(\frac{d}{b}\right)^2 \delta_1 \Rightarrow \delta_2 = \left(\frac{d}{b}\right)^2 y_1$$

- δ = y = Δ = deflection.

**16. In boiler plate the principal stresses across two perpendicular planes are 600 Ksc and 300 Ksc both tensile. Poisson's ratio is 0.3. The stress in Ksc, which acting alone would produce the same maximum strain is**

(a) 510 (b) 430
(c) 620 (d) 300

**Ans. (a) :** Maximum principal strain theory

Stress produce maximum strain = $\sigma_1 - \mu\sigma_2 - \mu\sigma_3$

Where, $\sigma_1, \sigma_2, \sigma_3$ = principal stress

Given,

$\sigma_1 = 600$ Ksc

$\sigma_2 = 300$ Ksc

Poisson's ration = $\mu = 0.3$

$\sigma = 600 \text{ Ksc} - 0.3 \times 300 \text{ Ksc}$

$= 600 - 90$ Ksc

$\sigma = 510$ Ksc

**17. The equivalent length of a column fixed at one end and free at the other is**

(a) 0.5 L (b) 0.7 L
(c) L (d) 2 L

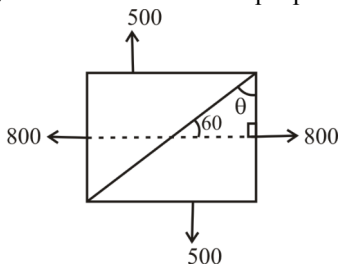
**Ans. (d) :** Equivalent length (effective length) of a column fixed at one end and free at other is = 2L

| Sr. | End condition | | Effective length |
|---|---|---|---|
| 1. | Both end fixed | — | L/2 |
| 2. | One end fixed other free | — | 2L |
| 3. | Both end hinged | — | L |
| 4. | One end fixed other hinged | — | $\frac{L}{\sqrt{2}}$ |

**18. A steel plate is subjected to tensile stresses of 800 and 500 Ksc at right angles to each other. The normal stress in Ksc on a plane inclined at 60 degree to the 800 Ksc stress would be**

(a) 600 (b) 360
(c) 840 (d) 725

**Ans. (d) :** Normal stress at oblique plane




$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\sigma_n = \frac{800 + 500}{2} + \left(\frac{800 - 500}{2}\right)\cos 2 \times 30^\circ + \sin 60^\circ$$

$\sigma_n = 650 + 75$

$\sigma_n = 725$ Ksc

**19. Materials which exhibit same elastic properties in all directions are called**

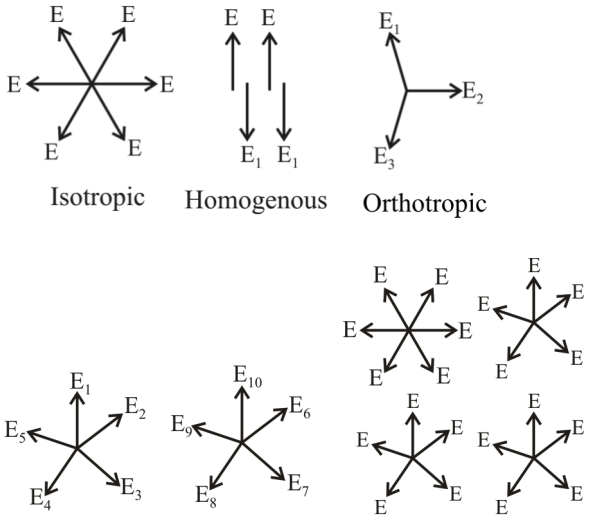
(a) Homogenous (b) Isentropic
(c) Allotropic (d) Isotropic

**Ans. (d) : Isotropic-** same properties in all direction at a point.

**Homogenous-** Same properties in same direction at all point.

**Orthotropic-** Different properties in orthogonal direction at a point.

**Anisotropic-** Different properties in all direction.




**20. A steel shat 5 cm. dia. and 50 cm long is subject to a twisting couple of 11,000 kg. cm. If the angle of twist be 0.6 degree modulus of rigidity of material of shaft in kg/sq.cm.**

(a) $0.762 \times 10^6$ (b) $0.936 \times 10^6$
(c) $0.856 \times 10^6$ (d) $0.892 \times 10^6$

**Ans. (c) :** Given,

D = 5 cm

L = 50 cm

T = 11000 kg. cm

$$\theta = 0.6^\circ = \frac{\pi \times 0.6}{180}$$

From Torsion equation,

$$\frac{T}{J} = \frac{G\theta}{L}$$

$$G = \frac{T \times L}{J \times \theta} = \frac{11000 \times 50}{\frac{\pi}{32} \times 5^4 \times \frac{\pi}{180} \times 0.6}$$

$G = 0.856 \times 10^6$ kg/cm$^2$

**21. For a given stress, the moment of resistance of a beam of square section placed with two sides horizontal, is K times the moment of resistance of the same beam section placed with a diagonal horizontal, where K is**

(a) $\sqrt{3}$ (b) $\sqrt{2}$
(c) 1.5 (d) 1.8

**Ans. (b) :** For given stress σ

$M_r \propto Z$

$\frac{M_{r_1}}{M_{r_2}} = \frac{Z_1}{Z_2}$ $\quad \because Z$ = section modulus

M = MOR

$Z_1$ of 2 side horizontal section $= \frac{a^3}{6}$

$Z_1$ of Diagonal horizontal $= \frac{a^3}{6\sqrt{2}}$

$\frac{M_{r_1}}{M_{r_2}} = \frac{a^3}{6} \times \frac{6\sqrt{2}}{a^3} = \sqrt{2}$

$\frac{M_{r_1}}{M_{r_2}} = \sqrt{2}$

**22. Two beams carrying identical loads, simply supported have same width, but beam A has double the depth of beam B. The elastic strength of beam A is K times that of beam B, where K is**

(a) 2 (b) 4

(c) 8 (d) 6

**Ans. (c) :** For same load

$M \propto z$

$\frac{M_1}{M_2} = \frac{Z_1}{Z_2}$

Beam (A) $Z_A = \frac{b(2d)^2}{6}$; Beam (B) $Z_B = \frac{bd^2}{6}$

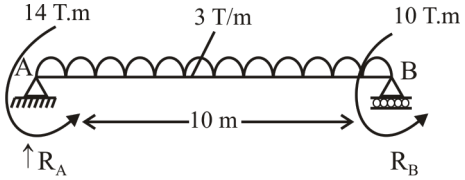
$\frac{M_A}{M_B} = \frac{Z_A}{Z_B} = \frac{4bd^2}{6} \times \frac{6}{bd^2} = 4$

**23. A horizontal beam is simply supported at its ends and carries a,u.d.l. of 3 T/m between supports places 10 m. apart, counter clockwise moments of 14 and 10 T.m. are applied at the left and right supports respectively. The shear force is tone at support A is**

(a) 14 (b) 17.4

(c) 13.6 (d) 15

**Ans. (b) :**




$R_A + R_B = 30T$ .....(i)

Moment about point 'B'

$R_A \times 10 + 14$ T.m $+ 3 \times 10 \times 5$ T.m $+ 10$ Tm $= 0$

$R_A = 17.4$ T $\Rightarrow R_B = 12.6$T

$\because R_A = 17.4$ T

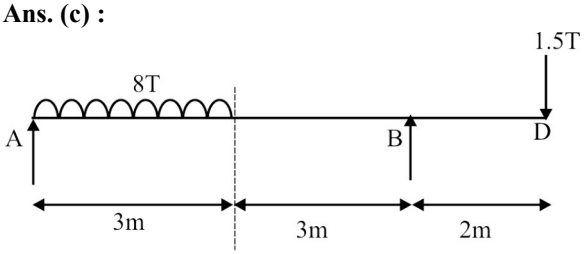
So, shear force at 'A' is equals to = 17.4 T

**24. A beam ABD is 8 m long and is simply supported at A and at a point B such than AB = 6 m. A point load of 1.5 T acts at D, and a total u.d.l. of 8T acts over half of AB. The shear force at A in Tone would be**

(a) 6 (b) 8

(c) 5.5 (d) 7.5

**Ans. (c) :**




$R_A + R_B = 8 + 1.5 = 9.5$ T

$R_A \times 0 + R_B \times 6 = 8 \times 1.5 + 1.5 \times 8$

$R_B \times 6 = 12 + 12$

$R_B = 4$T

$R_A = 9.5 - 4$

$R_A = 5.5$ T

**25. A fixed beam of span L carries a concentrated load W at mid point. The beam has flexural rigidity EI the deflection at the mid-point of beam is**

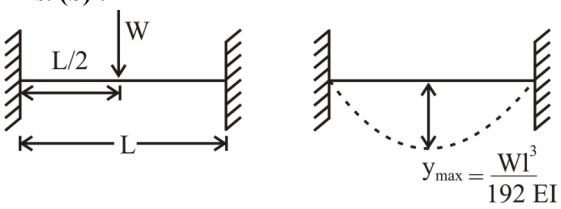
(a) $WL^3/128$ EI

(b) $WL^3/192$ EI

(c) $WL^2/64$ EI

(d) $WL^3/384$ EI

**Ans. (b) :**




Flexural rigidity = EIs

• Fixed beam is 4 times stronger than simply supported beam in case of point load and 5 times in case of udl.

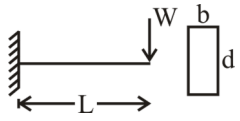
| $y_{max}$ | | Load |
|---|---|---|
| SSB | Fixed | |
| $\frac{Wl^3}{48EI}$ | $\frac{Wl^3}{192EI}$ | Point load |
| $\frac{5Wl^4}{384EI}$ | $\frac{Wl^4}{384EI}$ | Udl |

**26. A cantilever beam of rectangular cross-section carries a point load W at its free end. If the depth of beam load W at its free end, if the depth of beam is doubled, and the load halved, the deflection of the free end, as compared to its original value, will be**

(a) 1/2 (b) 1/4

(c) 1/8 (d) 1/16

**Ans. (d) :**



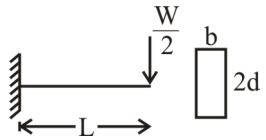

(i)

$$y_{max} = \frac{WL^3}{3EI}$$

$$= \frac{WL^3}{3E\frac{bd^3}{12}}$$

$$y_{max} = \frac{12.WL^3}{3\,E\,bd^3}$$

$$y_1 = \frac{4WL^3}{Ebd^3}$$




(ii)

$$y_{max} = \frac{WL^3}{3E\frac{b(2d)^3}{12}}$$

$$y_2 = \frac{12\times\frac{W}{2}L^3}{8\times 3E\,bd^3}$$

$$y_2 = \frac{6\,WL^3}{8\times 3E\,bd^3}$$

$$y_2 = \frac{1}{4}\frac{WL^3}{Ebd^3}$$

(iii) $\frac{y_2}{y_1} = \frac{1}{4}\frac{WL^3}{Ebd^3}\times\frac{Ebd^3}{4WL^3}$

$$\frac{y_2}{y_1} = \frac{1}{16}$$

**27. Two beam of cross-section circular and square have the same length, same allowable bending stress, and the same moment of resistance. The 'wl' of the beam with circular section is K times that of the square section, where K is**

(a) 0.852 (b) 1.118

(c) 1.856 (d) 2.00

**Ans. (b) :** w.l of circular beam = weight of Beam

$z = \frac{m}{\sigma}$ = constant $\because$ m, σ, L is same for both beam

$$Z_C = Z_S$$

$$\frac{\pi}{32}d^3 = \frac{a^3}{6}$$

$$\boxed{a = 0.838d}$$

(iii) $\frac{w_e}{w_s} - \frac{\text{weight of circular beam}}{\text{Weight of square beam}}$

$$K = \frac{\lambda\, A_c \times L}{\lambda \times A_s \times L} \qquad \because L, \lambda = \text{some}$$

$$K = \frac{\pi}{4}\frac{d^2}{a} \Rightarrow \frac{\pi\times d^2}{4(0.838d)^2}$$

K = 1.1184

**28. Two shafts are made of same material and same outer dia. with one having a circular section with inner dia. half that of outer dia. the torque that can be transmitted by the solid section is N time that of the hollow section, where N is**

(a) 8/7 (b) 17/16

(c) 1.5 (d) 15/16

**Ans. (d) :** $\frac{T_s}{T_H} = \frac{Z_s}{Z_H} = \frac{\frac{\pi}{32}D^3}{\frac{\pi}{32}\left(\frac{D^4 - d^4}{D}\right)} = \frac{D^4}{D^4 - d^4} \because d = \frac{D}{2}$

$$\frac{D^4}{D^4 - \left(\frac{D}{2}\right)^4} = \frac{D^4}{D^4 - \frac{D}{16}^4} = \frac{D^4}{\frac{15}{16}D^4} = \frac{16}{15}$$

$$\boxed{\frac{T_s}{T_H} = \frac{15}{16}}$$

$$\boxed{N = \frac{15}{16}}$$

**29. A shaft is simultaneously subject to a torque T and a bending moment M. The ratio of maximum bending stress and maximum shearing stress is**

(a) M/T (b) T/M

(c) 2M/T (d) 2T/M

**Ans. (c) :** Maximum bending stress.

$$\sigma_{max} = \frac{M}{Z} = \frac{M}{\frac{\pi}{32}d^3} = \frac{32\,M}{\pi d^3}$$

Maximum shear stress given by Torque

$$\tau_{max} = \frac{T.R}{J} = \frac{T}{\frac{\pi}{32}d^4 \times \frac{d}{2}} = \frac{16T}{\pi d^3}$$

$$\frac{\sigma_{max}}{\tau_{max}} = \frac{32\,M\times\pi d^3}{\pi d^3 \times 16T} = \frac{2M}{T}$$

Additional

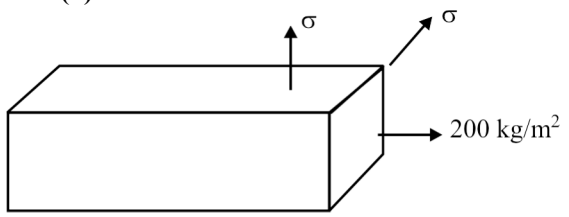
# Equivalent moment, $M_{eq} = \frac{1}{2}\left[M + \sqrt{M^2 + T^2}\right]$

# Equivalent Torque, $T_{eq} = \left[\sqrt{M^2 + T^2}\right]$

**30. A bar of rectangular section is tension under an axial stress of 200 kg/sq.m. If Poission's ratio is 1/3 for the material, the stress to be applied to the side faces, to prevent any change in cross section dimensions, is kg/sq.cm. is**

(a) 80 (b) 100
(c) 120 (d) 96

**Ans. (b) :**




$\varepsilon_y = 0$

$= \frac{\sigma}{E} - \frac{\mu\sigma}{E} - \frac{200\mu}{E} = 0$

$\sigma(1-\mu) = 200\mu$

$\sigma\left(1-\frac{1}{3}\right) = 200 \times \frac{1}{3}$

$\sigma = 100$ kg/m$^2$

**31. A column hinged at both ends is 4 m long and has cross-sectional area 145.3 sq.cm. and radius of gyration 6.25 cm. Assuming $f_c$ = 5500 kg/sq.cm. and $\alpha$ = 1/1600. The rankin's crippling load in tons would be**

(a) 224.8 (b) 248.4
(c) 316.8 (d) 296.6

**Ans. (a) :** Effective length of column is equal to 'L'

$\because$ Both end hinged.

L = 4 m = 400 cm

Radius of gyration K = 6.25 cm

$f_c = \sigma_c$ = 5500 kg/sq.cm.

A = 145.3 cm$^2$

$\alpha = \frac{1}{1600}$

Slenderness ratio $\lambda = \frac{\ell_e}{K}$

$= \frac{400}{6.25} = 64$

Rankine load=

$\frac{\sigma_c A}{1+\alpha\lambda^2} = \frac{5500 \times 145.3}{1+\frac{1}{1600}\times(64)^2} = 224,480$ kg.

Load in Tons = $\frac{224480}{1000} = \boxed{224.48 \text{ Tons}}$

**32. The ratio of effective length of column to its minimum radius of gyration is known as**

(a) Bulking factor (b) Slenderness ratio
(c) Crippling factor (d) Section modular

**Ans. (b): Slenderness ratio–** It is the ratio of effective length of column to its minimum radius of gyration.

$\lambda = \frac{L_{effective}}{K}$ Where K = min. radius of gyration.

I = moment of inertia.

$K = \sqrt{\frac{I_{min}}{A}}$

**33. The relation between the three elastic constants is**

(a) E = 3KC/(K+C) (b) E = 6KC/(3K+C)
(c) E = 9KC/(3K+C) (d) E = 3KC/(3K+C)

**Ans. (c) :**

- $E = 26(1+\mu)$
- $E = 3K(1-2\mu)$
- $E = \frac{9KG}{3K+G}$
- $\frac{1}{E} = \frac{1}{9K} + \frac{1}{36}$

Where E = young's modulus
K = bulk modulus
$\mu$ = Poisson's Ratio.
N/C/G = Shear modulus.

**34. A steel road of sectional area 3 sq. cm. connects two parallel walls 6 m apart and when the rod is heated to 100 degrees c, the nuts at the ends are tightened up. The pull exerted by the bar on cooling to 20 degrees C with ends not yielding and assuming.**
**$\alpha$ = 0.000012 / C, E = 2.1 × 10$^6$ Ksc would be in kg.**

(a) 4056 (b) 3482
(c) 6048 (d) 7218

**Ans. (c) :** Thermal stress $\sigma = E \times \Delta t \times \alpha$

Given Area = 3 cm$^2$

$E = 2.1 \times 10^6$ kg/cm$^2$

$\Delta t = (100 - 20)\,c^\circ$

L = 6000 mm

$\alpha = 0.000012^\circ C^{-1}$

$\sigma = 2.1 \times 10^6 \times 0.000012 \times (100 - 20)$

$= 2016$ kg/cm$^2$

Total full applied $P = \sigma \times A$

$= 2016 \times 3$cm$^2$

$\boxed{P = 6048 \text{ kg}}$

**35. A cast iron column 5 m long is of hollow circular section with internal dia 25 cm and external dia 30 cm. The axial load in Tonnes which develops a compressive stress of 800 kg/sq. cm. in the column is**

(a) 96.4 (b) 172.8
(c) 148 (d) 120.6

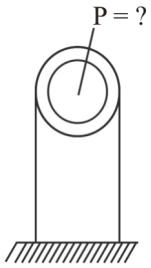
**Ans. (b) :** Given,

Length of column = 5 m

Internal dia. meter = 25 cm

External dia. meter = 30 cm

Compressive Stress = 800 kg/cm$^2$




Load = Stress × Area'

$$P = \frac{800\,Kg}{cm^2} \times \frac{\pi}{4}\left(30^2 - 25^2\right)cm^2$$

= 1727875 kg.

$$\boxed{P_{Tones} = 172.787 \text{ Tons}}$$

**36. When a shaft is subjected to purer twisting, the type of stress developed in the shaft is**

(a) Bending stress (b) Axial stress

(c) Shear stress (d) Normal stress

**Ans. (c) :** When a shaft is subjected to pure twisting moment, shear stress developed in the shaft.

$$\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{R}$$

Where T = Torque

J = Polar moment of Inertia

G = Shear modulus N/mm$^2$

$\theta$ = Angle of twist in Radian

L = length of shaft

$\tau$ = Shear stress.

R = Radius of shaft.

**37. Young's modulus of material is K times its rigidity modulus, when K is**

(a) 2(1–1/m) (b) 3(1+1/m)

(c) 2(1+1/m) (d) 3(1–1/m)

**Ans. (c) :** Relation between young's modulus 'E' and Rigidity modulus 'G'

$E = 2G(1+\mu)$

$$\frac{E}{G} = 2(1+\mu)$$ $\mu$ is also denoted by $\frac{1}{m}$

$$\frac{E}{G} = 2\left(1+\frac{1}{m}\right)$$

**38. Poisson's ratio is the ratio of**

(a) Lateral stress/Lateral strain

(b) Lateral stress/Linear strain

(c) Lateral strain /Linear strain

(d) Linear stress/Lateral strain

**Ans. (c) :** Poisson's Ratio = $-\frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

(–) sign denotes that change in volume is opposite in nature in both lateral direction and longitudinal direction.

Practically $\mu$ varies $0 \le \mu \le 0.5$

Theoratically $-1 \le \mu \le .5$

(–1) value of Poisson's ratio for 'Auxetic material' or Biological materials.

**39. A shaft running at 150 rpm is subject to torque of 150 kg.m. The H.P. transmitted by the shaft is**

(a) $\pi$ (b) $10\pi$

(c) $\pi^2$ (d) $1/\pi$

**Ans. (b) :** Given,

N = 150 RPM

T = 150 kg m = 1500 N.m

P = ?

$$\text{Power } P = \frac{2\pi NT}{60}$$

$$P = \frac{2 \times \pi \times 150 \times 1500 N - m}{60 \times 746 Watt\,(1HP)}$$

P □ 10π H.P

**40. In a shaft subject to pure twist, the shear stress at a section is maximum at**

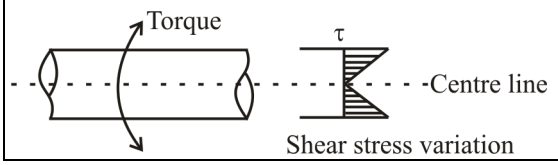
(a) Centre of section

(b) Mid-radius

(c) Surface

(d) 3/8 radius from centre

**Ans. (c) :** When a shaft is subjected to pure twist the shear stress is maximum at the surface of shaft.




**41. A cantilever has length of 2.5 m. It is of T section with $I_x$ = 2127 cm$^4$ and fibre distance (tensile) from neutral axis is 7 cm. if the maximum allowable tensile stress is 300 kgs/sq. cm. then the maximum u.d.l. that can be applied in kg/m is**

(a) 368 (b) 2409

(c) 485 (d) 292

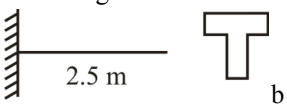
**Ans. (d) :** Given, length = 2.5 m = 250 cm

$I_x$ = 2127 cm$^4$

$y_{max}$ = 7 cm

$\sigma$ = 300 kg/cm$^2$

u.d.l in kg/m



From bending Equation

$$M = \sigma \times \frac{I_x}{y_{max}} \qquad BM = \frac{Wl^2}{2} = \frac{WL}{2}$$

$$\frac{Wl}{2} = \frac{300 \times 2127}{7} \qquad \because W = Wl$$

$$W = \frac{2 \times 300 \times 2127}{7 \times 250\,cm}$$

$W = 729.25$ kg

$Wl = 729.25$ kg

$$W = \frac{729.25}{2.5\,m}\,kg = 291.70\,kg/m$$

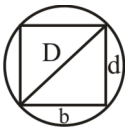
Ud$l$ = 291.7 kg/m  292 kg/m

**42. The width of the strongest beam of rectangular section than can be cut out of a cylindrical log of wood whose dia. is 30 cm. would be (in cm).**

(a) 10.4 (b) 15.6
(c) 20.2 (d) 17.3

**Ans. (d) :** $D^2 = b^2 + d^2$

$d^2 = D^2 - b^2$




Section modulus $Z = \frac{bd^2}{6}$

$$Z = \frac{b\left(D^2 - b^2\right)}{6}$$

$$Z = \frac{bD^2 - b^3}{6} = \text{For maximum value } \frac{dz}{ab} = 0$$

$$\frac{dz}{db} = \frac{D^2 - 3b^2}{6} = 0$$

$3b^2 = D^2$

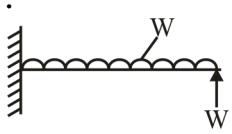
$$b = \frac{D}{\sqrt{3}} \Rightarrow \frac{30}{\sqrt{3}} = 17.32 \text{ cm}$$

$$d = \sqrt{\frac{2}{3}}D$$

**43. A cantilever carries a total u.d.l. of 'W' over its entire length, and a force W acts as its free end upwards. The max. deflection at the free end is**

(a) zero
(b) $5WL^3/24$ EI upward
(c) $5\ WL^3/\ 24$ EI downward
(d) None of these

**Ans. (b) :**




Net deflection = deflection due to udl – deflection due to upward point

$$\Delta = \frac{WL^3}{8EI} - \frac{WL^3}{3EI} \qquad \because W = wl \text{ for ud}l$$

$$\Delta = \frac{3WL^3 - 8WL^3}{24EI} = -\frac{5WL^3}{24EI}$$ (–) upward deflection.

**44. The property of material by which it can be beaten or rolled in to plates is called**

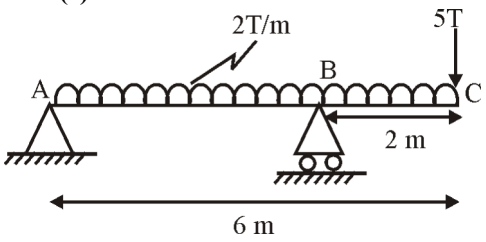
(a) Malleability (b) Ductility
(c) Plasticity (d) Elasticity

**Ans. (a) : Malleability–** Property of material by which it can be beaten into thin sheets.
**Ductility–** large deformation before failure
elongation > 5% Ductile material
elongation < 5% brittle material
**Elasticity–** Property by which material can gain its original shape after removal of loading up to a limit.

**45. A prismatic beam ABC is simply supported at points A and spaced 6 m apart and has an overhand BC of 2 m. A u.d.l. of 2T/m acts over entire length AC and a point load of 5T acts at C. The hogging moment in T-m at B would be**

(a) 15 (b) 18
(c) 14 (d) 12

**Ans. (c) :**




$$R_B = \frac{16(4) + 40}{6} = 17.33 \text{ T}$$

$R_A = 3.67$ T

$$M = 3.67 \times 2\frac{x^2}{2}\Big|_B + 17.33(x - 6)$$

$M_{(hogging)_{max}} = $ BM at B

$= 3.67 \times 6 - 6^2 = -13.98 \approx 14$ T-m

**46. A bar of certain material of section 4 $cm^2$ is subjected to a pull of 16 T. Thereby the extension is 0.01 cm in length of 20 cm. If Poisson's ratio is 1/4, the rigidity Modulus of the material in Kg/Sq. cm is**

(a) $2 \times 10^6$ (b) $1.6 \times 10^6$
(c) $1.3 \times 10^6$ (d) $3.2 \times 10^6$

**Ans. (d) :** Given, A = 4 $cm^2$

f = 16 T = $16 \times 10^3$ kg

$\Delta L = 0.01$ cm

L = 20 cm

$$\mu = \frac{1}{4} = 0.25$$

G = ?

Elongation Formula

$$\Delta L = \frac{PL}{AE}$$

$$\frac{PL}{\Delta L.A} = \frac{16000 \times 20}{0.01 \times 4\,cm^2}$$

$E = 8\times10^6$ kg/cm$^2$

$E = 2G(1+\mu)$

$8\times10^6 = 2\times G(1+0.25)$

$G = 3.2\times10^6$ kg/cm$^2$

**47. Along the principal plane subject to the maximum principal stress there would be**

(a) Maximum shear stress
(b) Minimum shear stress
(c) No shear stress
(d) None of the above

**Ans. (c) :** Along the principle plane subjected to maximum principal stress, no shear stress acts.

• Principal plane is the plane of maximum normal stress at which shear stress is equal to zero.

**48. A hollow cylindrical column of cross. Sectional area 1.7 sq. Cm. and I = 1766 cm$^4$, is 6 m long. If E= 0.8 ×10$^6$ kg/sq. cm. the Euler's crushing load in kg. is**

(a) 30,500 (b) 28,600
(c) 42,000 (d) 38,700

**Ans. (d) :** $P_e = \frac{\pi^2 EI_{min}}{(L_{eff})^2} = \frac{\pi^2 \times 0.8\times10^6 \times 1766}{(600)^2}$

$P_e = 38727.14 = 38{,}700$ kg

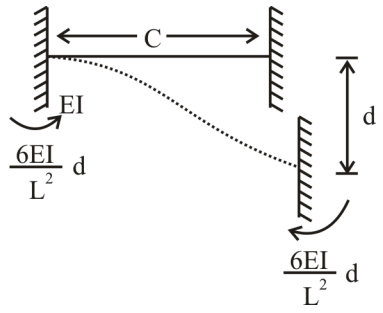
• Euler Theory is valid for long column only.

**49. In a fixed beam of flexural rigidity EI and span L, if one of the supports sinks by 'd', the moment consequent of this is**

(a) $6EI/d^2/L^2$ (b) $6EI^2d/L$
(c) $6EId/L^2$ (d) $6EI^2d^2/L^2$

**Ans. (c) :**

(1)

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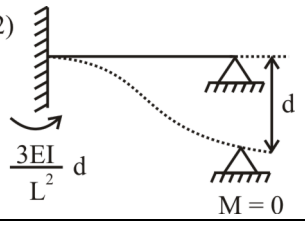
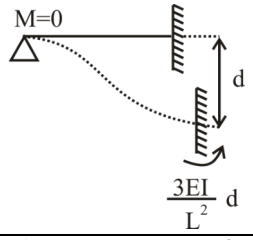



(3)




**50. A short column of external dia D and internal dia 'd' carries an eccentric load W. The greatest eccentricity which the load can have without producing tension in the column, is**

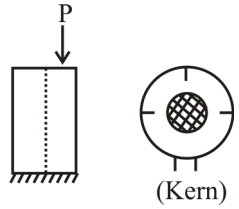
(a) (D+d)/8 (b) $(D^2+d^2)/8$
(c) $(D+d)^2/8d^2$ (d) $(D^2+d^2)/8D$

**Ans. (d) :** Bending Equation

$$\sigma_b \le \frac{M}{Z}$$

$$\frac{P}{A} \ge \frac{P.e}{Z}$$

$$\frac{e}{Z} \le \frac{1}{A}$$




$$e \le \frac{\frac{\pi}{32}\frac{(D^4-d^4)}{D}}{\frac{\pi}{4}(D^2-d^2)}$$

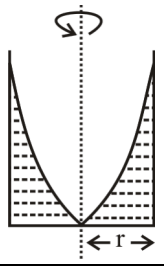
$$e \le \frac{1}{8}D \times \frac{(D^2+d^2)(D^2-d^2)}{(D^2-d^2)}$$

$$e \le \frac{D^2+d^2}{8D}$$

**51. When a liquid rotates at constant angular velocity about a vertical axis, the pressure**

(a) Increase linearly as the radial distance
(b) Varies inversely as the altitude along any vertical times
(c) Varies as square of the radial distance
(d) Decreases as square of the radial distance

**Ans. (c) :** The pressure varies as square of the radial distance.



$$P = \frac{\rho\omega^2 R^2}{2}$$

$$P \propto R^2$$

$$H = \frac{\omega^2 R^2}{2g}$$ Height of parabolied

**52. A triangular channel section will be most economical when each of the sloping sides make with vertical an angle of**

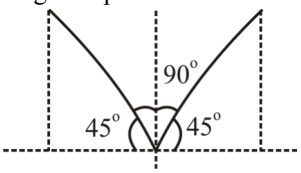
(a) 30 degrees (b) 60 degrees

(c) 40 degrees (d) 45 degrees

**Ans. (d) :** Most economical triangular section

$\theta$ for most economical section = $90^o$

Angle slop with vertical side = $45^o$




**53. A Francis turbines is an**

(a) Inward flow reaction turbine

(b) Inward flow impulse turbine

(c) Outward flow reaction turbines

(d) Outward flow impulse turbine

**Ans. (a) :** Francis turbine is an Inward flow reaction turbine.

Head – Medium (30 m – 300 m)

Specific speed = (60– 300)

- Modified Francis turbine is a mixed flow turbine.

**54. A right circular cylinder open at top, is filled with liquid and rotated about its vertical axis at such speed that half the liquid splits out. The pressure at the centre of bottom is**

(a) Zero

(b) 1/8 of that when cylinder was full

(c) 1/4 of that when cylinder was full

(d) 1/2 of that when cylinder was full

**Ans. (a) :** A cylinder rotating about the vertical axis forced vortex flow the free surface is paraboloid.

- Which has half the volume of cylinder of same height when half the liquid spills out, the centre of the bottom will be exposed and pressure will be zero.

**55. Water hammer in pipes occurs when**

(a) excessive leakage occurs

(b) pipe is hit with hammer

(c) flow of liquid through pipe is suddenly stopped by closing the valve

(d) pipe bursts under excessive fluid pressure

**Ans. (c) : Water hammer–** When flow of liquid through pipe is suddenly stopped by closing the valve. Sudden closure of valve in a pipe car carrying flowing liquid destroys the momentum of flowing liquid and sets up a high pressure wave.

Water hammer pressure = PVC

For Rigid Pipe $C = \sqrt{\frac{K}{\rho}}$

Where,

K = bulk modulus

$\rho$ = density

**56. The value of friction factor for laminar flow of liquid in terms of Reynolds number R, is given by**

(a) 64/R (b) 32/R

(c) 24/R (d) 48/R

**Ans. (a) :** Friction factor for Laminar flow 'f' = 4f friction coefficient

$$f = \frac{64}{Re}$$ Re < 2000 – Laminar flow in pipe.

4000 > Re > 2000 – Transition flow

Re > 4000 – Turbulent

- $R_e$ < 2000 – Laminar – flow through pipe
- R < 500 – Laminar flow through open channel
- Re < 1000 – Laminar flow through parallel plate.

**57. In a free Cylinder vortex of water, the tangential velocity at a radius of 12 cm. from axis of rotation is 7.2 m/sec and intensity of pressure is 2.5 kg/sq.m. The velocity at a radius of 24 cm. from the axis (in m/sec.) is**

(a) 2.4 (b) 3.2

(c) 3.6 (d) 4.8

**Ans. (c) : For free vortex motion–**

mvr = constant

here,

$$\frac{v_1}{v_2} = \frac{r_2}{r_1}$$

$$v_2 = \frac{7.2 \times 0.12}{0.24}$$

$v_2$= 3.6 m/s

**58. The depth of the Centre of pressure of a vertical Semi Circular Plane Submerged in a liquid with dia 'd' at the free Surface is**

(a) $\pi$ d/16 (b) $\pi$ d/32

(c) $\pi$ d/12 (d) 3$\pi$d/32

**Ans. (d) :** $\bar{y} = \frac{4r}{3\pi} = \frac{2d}{3\pi}$

$$I_{xx} = I_{cc} + Ay^2$$

$$I_{cc} = I_{xx} - Ay^2$$

$$= \frac{1}{2}\left[\left(\frac{\pi}{64}d^4\right) - \frac{\pi}{4}d^2 \times \frac{\pi}{4}d^2 \times \left(\frac{2d}{3\pi}\right)^2\right]$$

$$Cp = \bar{y} + \frac{I_{cc}\sin\theta}{A\bar{y}} \qquad \theta = 90^o$$

$$= \bar{y} + \frac{I_{xx} - Ay^2}{A_{\bar{y}}}$$

$$= \bar{y} + \frac{I_{xx} - Ay^2}{A_{\bar{y}}}$$

$$= \bar{y} + \frac{I_{xx}}{A_{\bar{y}}} - \bar{y}$$

$$\text{C.p.} = \frac{I_{xx}}{A\bar{y}} = \frac{\frac{\pi}{128}d^4}{\frac{\pi}{8}d^2 \times \frac{2d}{3\pi}} = \frac{3\pi d}{32}$$

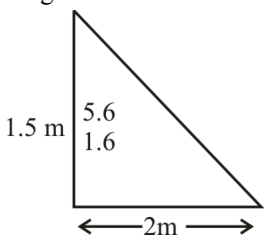
**59. A Cubical Tank of side 2 m is filled with glycerine of Sp. Gr. 1.6 to a depth of 1.5 m. If the tank is accelerated vertically upward at 5 m/sec$^2$, the force acting on the side of the tank would be in (Kg.)**

(a) 2448.6 (b) 4800.0
(c) 6425.2 (d) 5434.5

**Ans. (d) :** Given,
Side of cubical tank = 2m
Specific gravity of glycerine = 1.6

$\rho_{gl} = 1600$ kg/m$^3$
$a = 5$ m/sec$^2$
$h = 1.5$ m
$P = \rho gh$
$g' = (g+a)$
$g' = 9.81+5$
$= 14.81$
$P = 35544$ N/m$^2$

Pressure diagram



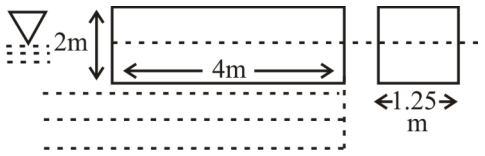

$$A = \frac{1}{2} \times 1.5 \times 2$$

$= 1.5$ m$^2$
Force = Pressure × Area
$F = P \times A$
$F = 35544$ N/m$^2 \times 1.5$ m$^2$
$F = 53{,}316$ N $\quad \because 1$ kgf $= 9.81$ N
$F = 5434.86$ kgf

**60. A wooden block of rectangular section 4m long, 1.25 wide and 2 m deep floats horizontally in Sea water of Sp. Wt. 1025 kg/cu.m. If the Sp. Gr. of wood is 0.64, the depth of block under water would be (in m.)**

(a) 2.4 (b) 1.8
(c) 0.8 (d) 1.25

**Ans. (d) :**



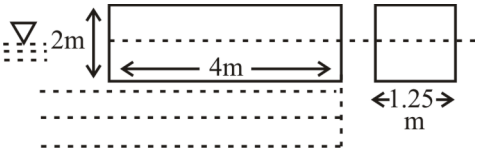

$$\frac{\text{Density of Material}}{\text{Density of water (or any liquid)}} \times D = \text{Submurged part}$$

Where D = Depth of block

$$\text{Submerged part} = \frac{640}{1025} \times \text{Depth of block}$$

$$= \frac{640}{1025} \times 2\text{m}$$

$= 1.24\ 8$m




depth of block under water = 1.248 m ≈ 1.25 m

**61. For flow through pipes the head loss by Darcy-Weisbach equation is equal to**

(a) $fLV/2gd$ (b) $fLV^2/gd$
(c) $fLV^2/2gd$ (d) $fL^2V/2gd$

**Ans. (c) :** The Darcy-Weishbach equation is one of the most commonly used equations for estimating frictional pressure drops in internal flow.

$fLV^2/2gd$

**62. A certain liquid has a volume of 1 litre when compressed at 1 MN/Sq.m. and a volume of 995 C.C. when compressed to 2 MN/sq. The Bulk Modulus of Elasticity would be (in MPa)**

(a) 10 (b) 1000
(c) 200 (d) 150

**Ans. (c) :** Given,
$v = 1$ liter
$P_1 = 1$ MN/m$^2$
$P_2 = 2$mn/m$^2$
$V_2 = 995$ cc
$\Delta V = 1000–995$
$= 5$ CC
$\Delta P = (2–1)$ MN/m$^2$

$$\text{Bulk Modulus} = \frac{\Delta P}{\frac{\Delta V}{V}}$$

$$K = \frac{1 \times 10^6 \text{ N/m}^2}{\frac{5}{1000}}$$

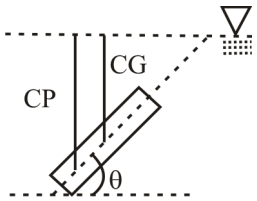
$$K = \frac{1000 \times 10^6}{5} N \times 10^{-6} mm^{-2}$$

$K = 200$ Mpa $\quad \because 1\ MN/m^2 = 1\ MPa$

**63. On an inclined plane, the centre of pressure is located**

(a) above its centroid (b) below centroid
(c) at centroid (d) anywhere

**Ans. (b) :** $Cp = \bar{n} + \frac{I_G \sin\theta}{A\bar{n}}$




Centre of pressure = Centroid + $\frac{I_G \sin\theta}{A\bar{n}}$

C.P > CG

On inclined plane CP is located below centroid.

**64. Thrust on a plane area is equal to area multiplied by intensity of pressure at centroid if**

(a) Area is horizontal (b) Area is inclined
(c) Area is vertical (d) All the above

**Ans. (d) :** Thrust on plane area = Pressure Intensity × Area.

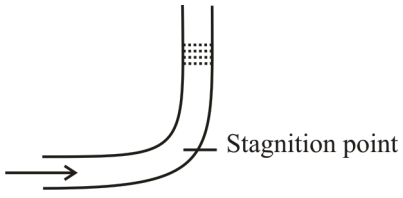
- Area can be horizontal inclined or vertical.

**65. A pitot tube is used to measure**

(a) Discharge in a flowing stream
(b) Pressure in a static fluid
(c) Velocity in a flowing fluid
(d) Dynamic pressure

**Ans. (c) : Pitot tube:–** Used to measure fluid flow velocity.

$V = \sqrt{2gh} \qquad \because H = \frac{V^2}{2g}$




$V_{actual} = C_v\sqrt{2gh}$

$C_v = 0.98$ = Coefficient of velocity.

- Pitot static tub is known as prandtl tube $C_v = 0.99$
- Based on principle of conversion of kinetic energy into pressure energy in term head.

**66. If the specific speed of a turbine is 6, then the turbine should be**

(a) kaplan (b) francis
(c) pelton wheel (d) thompson

**Ans. (c):**

| Flow | Energy | Head | Specific speed | Example |
|---|---|---|---|---|
| Tangential | Impulse | High >300 m | 0-60 | Pelton wheel |
| Radial | Reaction | medium 30-300 m | 60-300 | Fransis |
| Axial | Reaction | Low < 30 m | 300-600 | Propeller |
| | | | 600-1000 or more | Kaplan |

Specific speed Ns = $N_S = \frac{N\sqrt{P}}{H^{5/4}}$

**67. Mannings formula is used to determine friction head loss in**

(a) pipes running full
(b) pipes running partially
(c) open channel
(d) irregular sections

**Ans. (c) :** Mannings Formula 'V' = $\frac{1}{N} \times R^{2/3} \times S_o^{1/2}$

Where:– N = Roughness coefficient
$S_0$ = Slope of channel
V = Velocity
R = Hydraulic mean depth

| Channel | N |
|---|---|
| Cement | 0.011 |
| Wood | 0.015 |
| Concrete | 0.012 |
| Gravel | 0.025 |
| Earthen | 0.05 |

**68. The discharge through a triangular notch under a constant head of 2.5 cm, if angle of the notch is 120 degrees, assuming $C_d = 0.62$, is (in cum/sec)**

(a) 0.180 (b) 0.079
(c) 0.098 (d) 0.667

**Ans. (b) :** Discharge through triangular notch.

$$Q = \frac{8}{15} \times C_d\sqrt{2g}\tan\frac{\theta}{2}H^{5/2}$$

$$= \frac{8}{15} \times 0.62 \times \sqrt{2 \times 9.81}\tan\frac{120^o}{2}(0.25)^{5/2}$$

$Q = 0.0792\ m^3/sec$

**69. Water is to be pumped out a deep well under a total head of 95m. A number of indentical pumps of design speed 1000 rpm and specific speed 900 rpm with a rated capacity of 150 litres/sec are available. The number of pumps needed would be**

(a) 2 (b) 4
(c) 6 (d) 3

**Ans. (d):** Specific speed of pump

$$N_s = \frac{N\sqrt{Q}}{H_m^{3/4}}$$

$$H_m^{3/4} = \frac{1000\sqrt{150}}{900}$$

$$H_m^{3/4} = 13.608 \Rightarrow H_m = 32.489$$

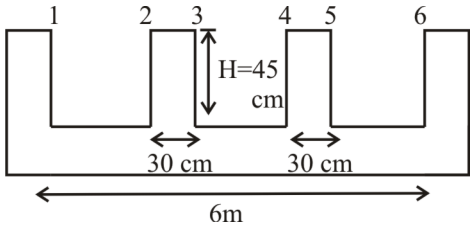
$$\text{No. of pump} = \frac{\text{Total head}}{\text{Head of single pump}} = \frac{95}{32.489} = 2.9 \square 3$$

3 pump are in series to get 95 m of total head.

**70. A rectangular weir 6 m long is divided into 3 bays, by a 2 vertical post, each 30 cm wide. If the head is 45 cm, the discharge in Cu m /sec over the weir, using Francis formula (considering and contraction) is**

(a) 2.25 (b) 2.83
(c) 2.58 (d) 2.96

**Ans. (b) :**




No. of end contraction = 6

Length of weir = 6 –2×0.3

= 5.4 m

Effective length of channel due to end contraction.

$L_{eff} = L_{actual} - 0.1\ nH$

n = No of contraction points

$= 5.4 - .1 \times 0.45 \times 6$

$L_{eff} = 5.13$ m

$$Q = \frac{2}{3} C_d \sqrt{2g}\ L_{eff} \times H^{3/2}$$

$$= \frac{2}{3} \times 0.62 \times \sqrt{2 \times 9.81} \times 5.13 \times 0.45^{1.5}$$

$Q = 2.835\ m^3/sec$

**71. A tank 6 m long, 2.5 wide, 2 m deep, is fully filled with oil. If it is accelerate in the direction of its length at 1.5 m/sec$^2$, the volume of soil spilled in litres, would be**

(a) 6880 (b) 460
(c) 5460 (d) 6400

**Ans. (a) :** Given,

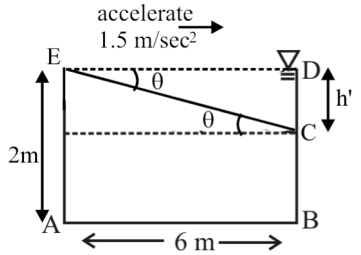
Dimension of tank

Length (L) = 6 m,

Depth (D) = 2m

Width (B) = 2.5 m

Acceleration = 1.5 m/sec$^2$




At the time of spilling depth of oil at A = 2m and at B it will be reduced by h' and ΔCDE is the volume spilled out.

$$\tan\theta = \frac{a}{g} = \frac{1.5}{9.81} = 0.1529$$

In ΔCDE

$$\tan\theta = \frac{h'}{ED} = \frac{h'}{6}$$

$h' = 6 \times 0.1529 = 0.9174$ {tan θ = 0.1529 }

Volume of oil spilled out

V = Area of ΔCDE × Width

$$V = \frac{1}{2} \times 6 \times 0.9174 \times 2.5$$

$= 6.8805\ m^3$

V = 6880.5 litre.

**72. A ship weights 13,000 T, when the ship's boats at a mean distance of 10 m from the centre, are filled with water; weighting 60 T, the angle of displacement of plumb line is $2^0 16'$. Given tangent of this angle as 0.396, the metacentri heights is (in metres)**

(a) 1.16 (b) 1.36
(c) 0.96 (d) 1.6

**Ans. (a) :** 1.16

**73. A vertical gate close horizontal tunnel 5 m height and 3 m wide running full with water. The pressure at the bottom of the gate is $2 \times 10^4$ kg/sq.m. The total pressure on the gate would be in kg.**

(a) 192.000 (b) 250,000
(c) 262,500 (d) 225,000

**Ans. (c) :** 262,500

**74. The discharge through a venturimeter, with inlet and throat sectional area A and B is given by the formula**

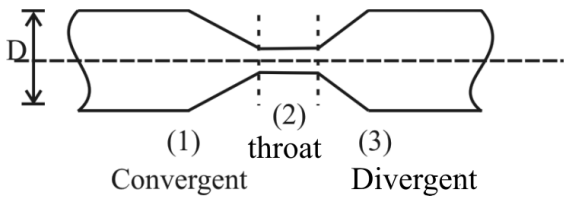
(a) $Q_{AB} = \frac{\sqrt{2gh}}{\left(A^2 - B^2\right)}$

(b) $Q_{AB} = \frac{\sqrt{gh}}{\left(A^2 - B^2\right)}$

(c) $Q_{AB} = \frac{\sqrt{gh}}{\left(A - B\right)^2}$

(d) $Q_{AB} = \frac{\sqrt{2gh}}{\left(A^2 - B^2\right)}$

**Ans. (d):** Venturimeter




Convergent angle = 20 ± 2

Divergent angle = $6^o$–$7^o$

Throat = $\left(\frac{1}{3}-\frac{1}{2}\right)D$

Principle – Bernoulli equation

Used for measuring flow rate of fluid through pip.

$$Q=\frac{a_1a_2\sqrt{2gh}}{\sqrt{a_1^2-a_2^2}}$$

$\frac{a_1}{a_2}$ = Area ratio

$C_d=\sqrt{\frac{h-h_2}{h}}=0.98$ Coefficient of discharge

If venturimeter, is contracted such that $C_C = 1$ then it is termed as Nozzle meter and also used for calculating discharge.

**75. If the shaft HP of the motor of a centrifugal pump be P, the overall efficiency of the pump is**

(a) $wQH^2/P$

(b) $wQ^2H/75P$

(c) $wQH/75P$

(d) $wQH/125P$

**Ans. (c) :** Overall efficiency '$\eta$' = It is defined as a ratio of the power output of the pump to the power input to the pump.

$$\eta=\frac{\text{water power}}{\text{Shaft power}}=\frac{wQH}{P}$$

if shaft power in HP

Then, $\eta=\frac{wQH}{75P}$

**76. Hydraulic jump occurs when**

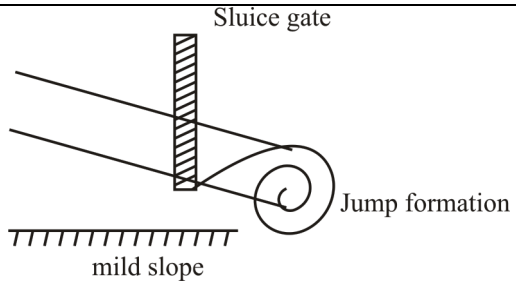
(a) Flow is Sub-Critical

(b) Flow in Super-Critical

(c) Flow in Sub-critical and adequate downstream depth is available

(d) Flow is super-critical and adequate downstream depth is available

**Ans. (d): Hydraulic Jump:–** Hydraulic jump occurs when there is a transition from supercritical flow to subcritical flow and it's an example of rapidly varied flow where flow properties change in short reach.




When bed slope of channel changes from steep of mild.

| Sr. | Hydroulic jump | Froude No. | Nature of Jump |
|---|---|---|---|
| 01. | Undular Jump | 1–1.7 | Weak |
| 02. | Weak Jump | 1.7–2.5 | Weak |
| 03. | Oscillating Jump | 2.5–4.5 | Oscillating |
| 04. | Steady Jump | 4.5–9.0 | – |
| 05. | Strong Jump | 9.0 | Strange |

**77. Reynold's Number of pipe flow is gives by**

(a) $VD/\nu$ (b) $VD\mu/g$

(c) $VDP/\mu$ (d) $VD/\mu$

**Ans. (a) :**

Reynold's Number for pipe flow = $\frac{\rho VD}{\mu}=\frac{VD}{\nu}$

Kinematic viscosity $\nu=\frac{\mu}{\rho}$

**78. When a venturimeter is used in an inclined position it will show**

(a) Less reading (b) More readings

(c) Same reading (d) None

**Ans. (c) :** When a venturimeter is in inclined position it will show same reading. No effect of inclination or reading.

**79. A kaplan turbine is suitable for**

(a) High head, low discharge

(b) Low head, high discharge

(c) High head, high discharge

(d) Low head, low discharge

**Ans. (b) : Kaplan turbine:–**

- Axial flow reaction turbine.
- Low head (ranging 10 to 70 m), high discharge.
- Specific speed > 1000 RPM
- Adjustable blades
- Very high efficiency.

**80. Uniform flow is said to occur in a fluid when**

(a) Rate of change of velocity is zero

(b) At every point, the velocity vector is identical in magnitude and direction, at any given instant

(c) Conditions do not change with time at any point

(d) Changes in transverse directions are zero

**Ans. (b) : Uniform and Non uniform flow:–** The flow of a fluid in which each particle moves along its line of flow with constant speed and in which the cross section of each stream tube remains unchanged is known as uniform flow otherwise non – uniform flow.

• **Steady flow and unsteady flow:–**

At any location the flow and fluid properties (i.e. density, viscosity etc) do not change with time then it is steady flow otherwise unsteady.

• **Rotational and irrotational flow:–** When fluid particles rotate about their mass centre during movement flow is said to be rotational otherwise irrotational.

Free vortex– Irrotational flow.

Force vortex– Rotational flow.

**81. If total head at the nozzle of a pipe is 37.5 m and discharge is 1 cumec, the HP generated is**

(a) 550 (b) 500

(c) 400 (d) 450

**Ans. (b) :** Power Power $'P' = \frac{\rho QH}{75}$ in HP.

Where:–

Q= Discharge

H = head

ρ = density of water

Given, H = 37.5 m

$Q = 1\ m^3$ sec

P = ?

$\rho = 10^3\ kg/m^3$

$$P = \frac{1000 \times 1 \times 37.5}{75}$$

P = 500 HP

Shaft power SP = Tω

$$\omega = \frac{2\pi N}{60} \text{rod/sec}$$

**82. Specific speed of a centrifugal pump is**

(a) $\left(N\sqrt{Q}\right)/H$ (b) $\left(N\sqrt{Q}\right)/H^2$

(c) $\left(N\sqrt{H}\right)/Q$ (d) None of the above

**Ans. (d) :**

Specific speed of pump $'N_s' = \frac{N\sqrt{Q}}{H_m^{3/4}} \Rightarrow \left[L^{3/4}T^{-3/2}\right]$

Where: $N_S$= Specifics speed in r.p.m

N = Normal speed r.p.m

Q = Dishcarge in litre/sec

$H_m$ = head in m

Specific speed of Turbine $N_s = \frac{N\sqrt{P}}{H_m^{5/4}}$

Where: P = Power

**83. The power transmitted through a pipe is maximum when the loss of head due to friction is K times the head supplied where K is**

(a) 1/4 (b) 1/2

(c) 1/3 (d) 2/3

**Ans. (c) :** Power transmitted through a pipeline is maximum when the head lost due to friction in the pipe is equal to one-third of the total supply head.

i.e., $H_f = \frac{H}{3}$

where, H = Total supply head,

$H_f$= Head loss due to friction.

Power transmitted at outlet of the pipe–

$$P = \gamma Q\left(H - H_f\right)$$

$$P = \gamma \times \frac{\pi d^2 v}{4}\left(H - H_f\right)$$

$$P = \gamma\left(\frac{\pi}{4}d^2 \times V\right)\left(H - \frac{f\ell V^2}{2gd}\right)$$

Condition for maximum transmission of power

$$\frac{dp}{dv} = \gamma \times \frac{\pi}{4}d^2\left(H - \frac{3f\ell V^2}{2gd}\right)$$

$$\frac{dP}{dv} = 0$$

$$\frac{dP}{dv} = H - 3 \times \frac{f\ell v^2}{2gd} = 0$$

$$H = 3H_f$$

$$H_f = \frac{H}{3}$$

**84. The maximum continuous power available from a hydel plant under the most adverse hydraulic conditions, is known as**

(a) Base power

(b) Firm power

(c) Primary power

(d) Installed capacity

**Ans. (b) : Firm power:–** The net amount of power which is continuously available from a plant without any break on a guaranteed basis. The consumers can always be sure of getting this power and this power is available under the most adverse hydraulic conditions.

**85. The purpose of a Surge tank in a pipe line is to**

(a) Minimize friction loss is pipe

(b) Smooth flow of water

(c) Prevent occurrence of hydraulic jump

(d) Relieve pressure due to water hammer

**Ans. (d) : Surge tank:–** It is a device or chamber introduced within a hydropower water conveyance system having a rather long pressure conduit to absorb the excess pressure rise in case of sudden valve closure.

**Water hammer:** When the water flowing in a long pipe is suddenly brought to rest by closing the valve or by any similar case there will be a sudden rise in pressure due to the momentum of the moving water being destroyed. The phenomenon of the sudden rise in pressure in the pipe is known as water hammer.

**86. Reynold's number is the ratio of**

(a) Viscous to inertia forces
(b) Elastic to pressure force
(c) Gravity to inertia force
(d) None of the above

**Ans. (d) :**

- Reynold's numbers:- $R_e = \frac{\text{Inertia force}}{\text{Viscous force}}$
- Euler's numbers:- $E = \sqrt{\frac{\text{Inertia force}}{\text{Pressure force}}}$
- Froude numbers:- $F_r = \sqrt{\frac{\text{Inertia force}}{\text{Gravitational force}}}$
- Weber numbers:- $W = \sqrt{\frac{\text{Inertia force}}{\text{Surface tension force}}}$
- Mach numbers:- $M = \sqrt{\frac{\text{Inertia force}}{\text{Eelastic force}}}$

**87. A metallic piece of sp. gr. 3.4 floats in mercury of sp. gr. 13.6. The fraction of its volume under mercury is**

(a) 0.50 (b) 0.90
(c) 0.25 (d) 0.75

**Ans. (c) :** Given,
Specific gravity of metallic piece = 3.4
then $\rho_{metal} = 3400$
Specific gravity of mercury = 13.6
then $\rho_{mercury} = 13600$
Applying the equilibrium equation in the vertical direction.
Weight of metal piece (w) = buoyancy force

$$mg = F_B$$

$$\rho_{metal} V_{metal}\, g = \rho_{mercury} V_{mercury}\, g$$

$$3400 \times V_{metal} = 13600 \times V_{mercury}$$

$$\frac{V_{mercury}}{V_{metal}} = \frac{3400}{13600} = 0.25$$

**88. Bernoulli's equation assumes that**

(a) Fluid is non-viscous
(b) Fluid is homogenous
(c) Flow is steady and along streamline
(d) All of the above

**Ans. (d) : Bernoulli's Equation:** It is the integration of Eular's equation of motion along a stream line under steady incompressible flow condition.

**Assumption:**

- Fluid is non-viscous
- Flow is steady, Incompressible and irrotational
- Fluid is Ideal – homogeneous

Hydrostatic pressure head

- $\frac{\rho}{\rho g} + \frac{v^2}{2g} + z = \text{constant}$

static head dynamic head.

$\left(\frac{\rho}{\rho g} + z\right)$ – Piezometric pressure head

$\left(\frac{\rho}{\rho g} + \frac{v^2}{2g}\right)$ – Stagnation pressure head

$\left(\frac{\rho}{\rho g} + \frac{v^2}{2g} + z\right)$ – Total pressure head

**89. The water level in a tank is 50 m above the centre of circular hole of dia 2.5 cm. The velocity of water coming out through the hole in m/sec, is**

(a) 31.4 (b) 31.2
(c) 31.1 (d) 31.3

**Ans. (d) :** Given, H = 50 m, d = 2.5 cm, v = ?
Velocity of water coming out through hole $v = \sqrt{2gh}$

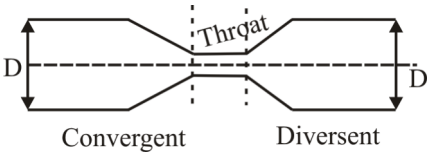
$$v = \sqrt{2 \times 9.81 \times 50}$$

$$v = 31.320 \text{ m/sec}$$

**90. The Divergent cone in a venturimeter is kept**

(a) Equal in length to the Convergent cone
(b) Longer than convergent cone
(c) Shorter than convergent cone
(d) None of the above

**Ans. (b) : Venturimeter:–**




Convergent pipe is shorter than the divergent pipe to avoid boundary layer separation.

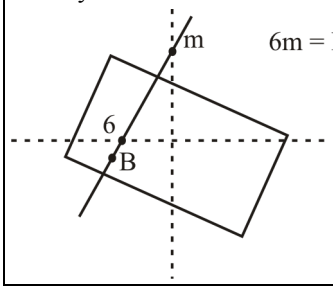
- Convergent angle $\approx 20^0$
- Divergent angle ($5^o$–$7^o$)

**91. Centre of buoyancy is**

(a) The point of intersection of buoyant force and centre line of body
(b) Centre of gravity of the body
(c) Centroid of displaced volume of fluid
(d) Mid-point between centre of gravity and metacentre

**Ans. (c) : Centre of buoyancy:–** It is defined as the centre of gravity of the displaced fluid.
**Centroid:–** Geometric centre of the of the body.

**Metacentre:–**The point of intersection of lines of action of buoyant force before and after rotation.




**92. Full load is supplied by a turbine shaft when dia. of jet issuing form nozzle is 15 cm. If the load suddenly drops of 36% of full load, the dia. of the jet to be regulated is**

(a) 7 cm
(b) 10 cm
(c) 6 cm
(d) 9 cm

**Ans. (d) :** Given,
Diameter ($D_1$) = 15 cm
Assume initial power ($P_1$) be P
Final power ($P_2$) drops of 36% of the full load = 0.36 P
Where
P = Power
ρ = density of fluid
g = acceleration due to gravity
Q = Discharge
h = head

$P = \rho gQh$

$$Q = AV = \frac{\pi d^2}{4} \times V$$

$$P \propto Q \propto D^2$$

$$\frac{P_1}{P_2} = \left(\frac{D_1}{D_2}\right)^2$$

$$\frac{P}{0.36P} = \left(\frac{15}{D_2}\right)^2$$

$$D_2 = \sqrt{15^2 \times 0.36}$$

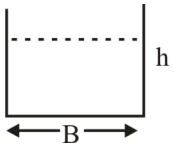
$D_2$ = 9 cm

**93. In case of rectangular notch, the ratio of % error in discharge to the % error in the measurement of head is**

(a) 1/2 (b) 3/2
(c) 3/4 (d) 2/3

**Ans. (b) :** Discharge through ractangular notch

$$Q = \frac{2}{3} C_d B\sqrt{2g}\, H^{3/2}$$ diff. w.r.t. H.




$$\frac{dQ}{dH} = \frac{2}{3} C_d B\sqrt{2g}\,\frac{3}{2} H^{1/2}$$ divided by Q.

$$\frac{dQ}{Q} = \frac{\frac{2}{3} C_d B\sqrt{2g}\,\frac{3}{2} H^{1/2} dH}{\frac{2}{3} C_d B\sqrt{2g} H^{3/2}}$$

$$= \frac{dQ}{Q} = \frac{3}{2}\frac{dH}{H} = \frac{3}{2}$$

**94. A floating body attains stable equilibrium if its metacentre is**

(a) Above the centroid
(b) Below the centroid
(c) At centroid
(d) Anywhere

**Ans. (a) : Types of equilibrium :**

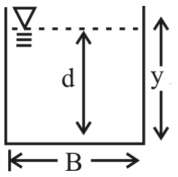
| | **Submerged body** | **Floating body** |
|---|---|---|
| Stable | G below B | M above G; GM = +ve |
| Unstable | G above B | M below G; GM = (–ve) |
| Neutral | G & B coincide | M & G coincide; GM = 0 |

G = Centroid
M = Metacentre
B = Centre of buoyancy
GM = Metacentric height

**95. A rectangular channel section will be most economical when depth of flow is equal to**

(a) Bottom width
(b) Three-fourth bottom width
(c) Half bottom width
(d) Two-third bottom width

**Ans. (c) : Most economical rectangular section–**
Most economical section is the one whose wetted perimeter is minimum for the given value of discharge.

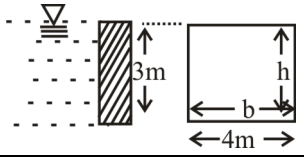



d =B/2
d/b = 1:2

**96. For a vertical rectangular gate 4 m wide and 3m high, with water surface at top of the gate, the depth of centre of pressure in meters, is**

(a) 1.0 (b) 2.0
(c) 1.5 (d) 2.5

**Ans. (b) :**



$$C_p = \bar{x} + \frac{I_G \sin^2\theta}{A\bar{x}}$$

Where,

$\bar{x}$ = depth of center of gravity from a free surface

A = Area

$I_G$ = Moment of inertia centre of gravity

$$C_p = \frac{h}{2} + \frac{bh^3/12}{bh \times \frac{h}{2}} \quad \text{(for vertical plane surface } \theta = 90^0\text{)}$$

$$C_p = \frac{h}{2} + \frac{h}{6} = \frac{2h}{3} = \frac{2\times 3}{3} = 2\,m$$

**97. Gauge pressure is equal to**

(a) Absolute pressure – atmospheric pressure

(b) Absolute pressure + atmospheric pressure

(c) Atmospheric pressure – absolute pressure

(d) None of the above

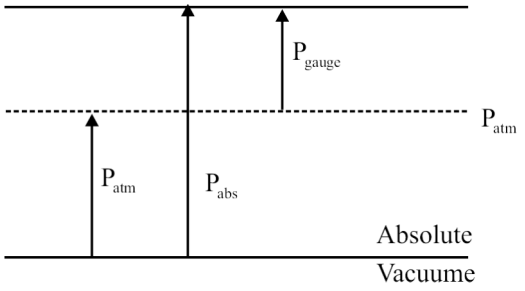
**Ans. (a) :** Gauge pressure:– It is the pressure wrt atmospheric pressure as datum.

It can be +ve, –ve, and zero also.

**Absolute pressure–** It is the pressure wrt absolute zero or complete vacuum. It is the actual pressure and measured by "Aneroid Barometer"

**Atmospheric pressure:–** Pressure exerted by atmospherics.

- $P_{abs} = P_{atm} - P_{vacuum}$
- $P_{abs} = P_{atm} + P_{gauge}$




**98. If a water tank partially filled with water is being carried on a truck moving with constant horizontal acceleration, the level of the liquid will**

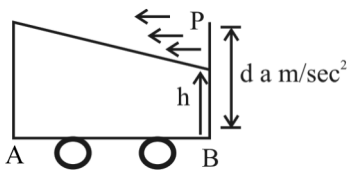
(a) Rise on the front side of tank

(b) Remains same

(c) Rise on backside and fall in front side

(d) Fall on backside of tank

**Ans. (c) :**




When a tank partially filled with water is being carried on a truck moving with constant horizontal carried on a truck moving with constant horizontal acceleration the level of liquid will rises on back side and fall in front side.

$$\tan\theta = \frac{a}{g}$$

Height of liquid at B = h – d tanθ, Less than original

Height of liquid at A = h+ d tan θ → more.

**99. Euler's equation of motion of liquids assumes that**

(a) Fluid is viscous

(b) Fluid is homogeneous and in compressible

(c) Flow of liquid is unsteady

(d) Velocity of flow is non-uniform

**Ans. (b) :** Euler's Equation of motion of liquid assumption– Based on Newton's 2nd law of motion

- Fluid is homogeneous
- Fluid is incompressible
- Fluid is non – viscous
- Flow is continuous, steady and along the stream line.
- The Velocity of the flow is uniform over the section.
- No energy or force (except gravity and pressure forces) is involved in the flow.

**100. A turbine develops 10140 HP under a head of 24.7 m at 180 rpm is specific speed in rpm is**

(a) 247

(b) 336

(c) 284

(d) 304

**Ans. (c) :** Given,

P = 10140 HP

= 10140×746 W

= 7564.44 kW

N = 180 r.p.m

Head (H) = 24.7 m

Specific speed of turbine

$$N_S = \frac{N\sqrt{P}}{H^{5/4}}$$

$$N_s = \frac{180 \times \sqrt{7564.44}\ \text{kw}}{(24.7)^{5/4}}$$

$N_s$ = 284.3 RPM

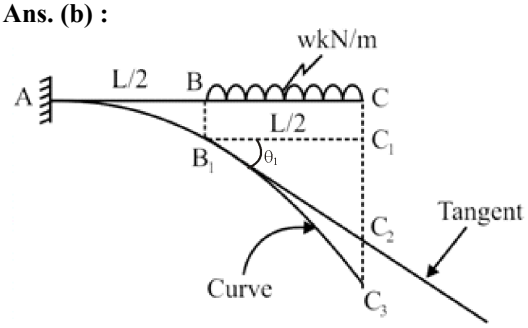
# Andhra Pradesh Public Service Commission
# Assistant Executive Engineer Exam, 1984
## Solved Paper

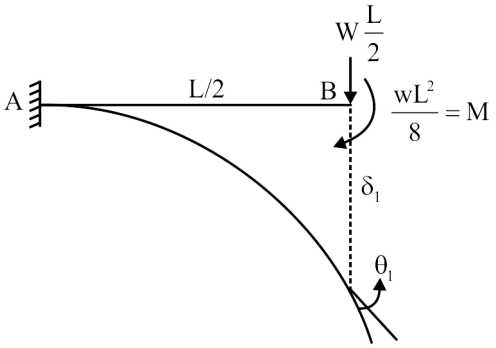
**1. A cantilever of length '$\ell$' carries a uniformly distributed load of 'w' per unit of span over the outer half. The maximum deflection is**

(a) $\frac{w\ell^4}{128EI}$ (b) $\frac{41w\ell^4}{384EI}$

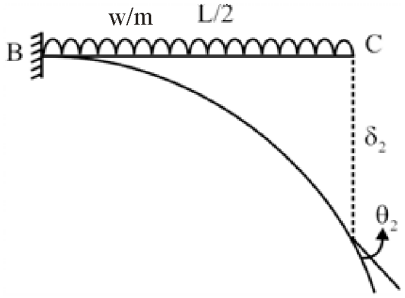
(c) $\frac{11w\ell^4}{272EI}$ (d) $\frac{9w\ell^4}{128EI}$

**Ans. (b) :**




Consider imaginary beam BC




$$\theta_1 = \frac{\left(\frac{wL}{2}\right)\left(\frac{L}{2}\right)^2}{2EI} + \frac{wL^2}{8}\frac{L}{2EI}$$

$$\theta_1 = \frac{wL^3}{8EI}$$

$$\delta_1 = \frac{\left(\frac{wL}{2}\right)\left(\frac{L}{2}\right)^3}{3EI} + \frac{\left(\frac{wL^2}{8}\right)\left(\frac{L}{2}\right)^2}{2EI}$$

$$\delta_1 = \left(\frac{wL^4}{48EI} + \frac{wL^4}{64EI}\right)\downarrow$$

$$\delta_2 = \frac{w\left(\frac{L}{2}\right)^4}{8EI}\downarrow$$

$$\delta_2 = \left(\frac{wL^4}{128EI}\right)\downarrow$$

$$\delta_T = \delta_1 + \delta_2 + \theta_1\frac{L}{2}$$

$$= \frac{wL^4}{48EI} + \frac{wL^4}{64EI} + \frac{wL^4}{128EI} + \frac{wL^4}{16EI}$$

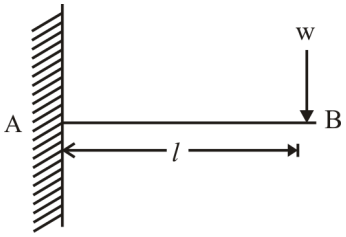
$$= \frac{41wL^4}{384EI}$$

**2. The deflection, at the free end of a cantilever due to a load there, is 1 cm. The deflection at the free end, if this load is distributed such that it varies linearly from zero intensity at the free end to the maximum intensity at the fixed end, is**

(a) 1 cm (b) 0.2 cm

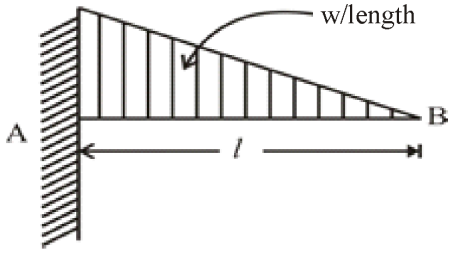
(c) 0.1 cm (d) 0.375 cm

**Ans. (c) :** First condition –




Deflection, $y_B = \frac{wl^3}{3EI} = 1\text{cm}$ .....(i) (given)

Second condition ___




Deflection, $y_B = y_B = \frac{wl^4}{30EI}$

[$\because$ load per unit, W = w$l$]

Then, $y_B = \dfrac{wl \times l^3}{30EI}$

$= \dfrac{W \times l^3}{30EI}$

From equation (i)__

$y_B = \dfrac{1}{10} \times \dfrac{Wl^3}{3EI}$

$= \dfrac{1}{10} \times 1$

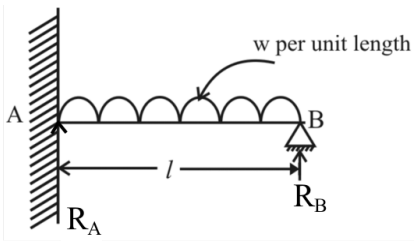
= 0.1 cm

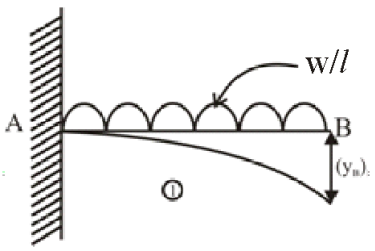
**3. The maximum hogging moment in the case of a cantilever of span '$\ell$' carrying a u.d. load of intensity 'w' per unit length and propped at the free end to the level of fixed end is**

(a) $\dfrac{w\ell^2}{8}$ (b) $\dfrac{w\ell^2}{2}$

(c) $\dfrac{w\ell^2}{4}$ (d) $\dfrac{9w\ell^2}{128}$

**Ans. (a)**

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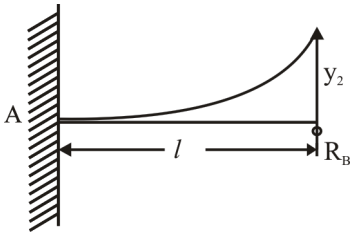



**(2)**

$\dfrac{R_B \ell^3}{3EI} = \dfrac{w\ell^4}{8EI}$

$R_B = \dfrac{3}{8}(wl)$

$\Sigma V = 0$

$R_A - wl + R_B = 0$

$R_A = \dfrac{5}{8} wl(\uparrow)$

So, moment, $M_A = wl\left(\dfrac{l}{2}\right) - R_B \times l = 0$

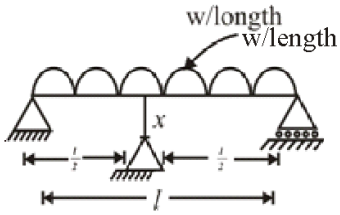
$M_A = \dfrac{wl^2}{8}$

**4. A beam (EI constant) of length '$\ell$' rests on three supports, the central support being 'x' below the straight line joining the ends. The beam carries u.d.l. over the span. The value of 'x' if the central support carries 1/3 of the total load is**

(a) $\dfrac{w\ell^4}{3 \times 48EI}$ (b) $\dfrac{7w\ell^4}{1152EI}$

(c) $\dfrac{5w\ell^4}{3 \times 384EI}$ (d) $\dfrac{11w\ell^4}{272EI}$

**Ans. (b) :**




$\dfrac{5}{384}\dfrac{wl^4}{EI} - \dfrac{1}{3} \times \dfrac{wl^3}{48EI} \times l = x$

$\dfrac{5}{384}\dfrac{wl^4}{EI} - \dfrac{wl^4}{144EI} = x$

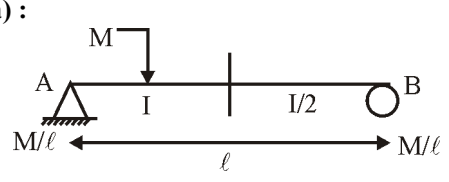
$x = \dfrac{7wl^4}{1152EI}$

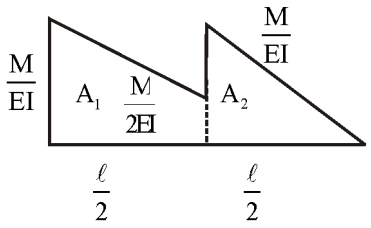
**5. A simply supported beam of span '$\ell$' is subjected to a clockwise moment of 'M' at the left hand support. The moment of inertia of the left half of the beam is I and the right half of the beam is I/2. The slope at the right-hand support is**

(a) $\dfrac{M\ell}{4EI}$ (b) $\dfrac{M\ell}{6EI}$

(c) $\dfrac{M\ell}{3EI}$ (d) $\dfrac{M\ell^2}{4EI}$

**Ans. (a) :**






$$A_1 = \frac{1}{2}\times\frac{\ell}{2}\left(\frac{3}{2}\frac{M}{EI}\right)$$

$$= \frac{3}{8}\frac{M\ell}{EI}$$

$$A_2 = \frac{ML}{4EI}$$

$$x_1 = \left(\frac{2\times\frac{M}{2EI}+\frac{M}{2EI}}{\frac{3M}{2EI}}\right)\times\frac{\ell/2}{3} = \frac{2\ell}{3\times3} = \frac{2\ell}{9}$$

$$x_2 = \frac{\ell}{2}+\frac{\ell/2}{3} = \frac{4\ell}{6} = \frac{2\ell}{3}$$

$$R_B\times\ell = \left(\frac{3M\ell}{8EI}\times\frac{2\ell}{9}\right)+\left(\frac{M\ell}{4EI}\times\frac{2\ell}{3}\right)$$

$$= \frac{M\ell}{12EI}+\frac{M\ell}{6EI} = \frac{M\ell}{4EI}$$

**6. The left-hand support of fixed beam of span 4 m sinks by 2 cm and the right hand supports sinks by 4 cm. The flexural rigidity of the beam is 2 ×10$^2$ kg. cm. The moments developed at the fixed ends are**

(a) 1.5 × 10$^5$ kg cm hogging at left end and sagging at right end

(b) 1.5 × 10$^5$ kg cm sagging at both ends

(c) 1.5 × 10$^5$ kg sagging at left end and hogging at right end

(d) 1.5 × 10$^5$ kg cm hogging at both ends

**Ans. (b) :** $M_{F_{AB}} = \frac{-6EI\delta}{\ell^2}$

$$= -\frac{6\times2\times10^2\times2}{4^2}$$

$$= -1.5\times10^5 \text{ kg cm}$$

$$M_{F_{BA}} = \frac{-6EI\delta}{\ell^2}$$

$$= -\frac{6\times2\times10^2\times4}{4^2}$$

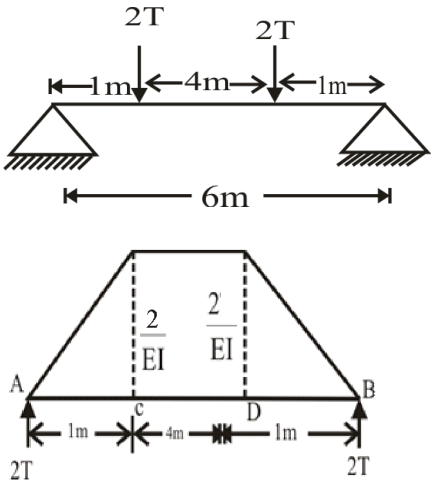
$$= -1.5\times10^5 \text{ kg cm} =$$

**7. A simply supported beam of span 6 m carries two concentrated loads at 2T each at 1 m distance from each of the support. If the flexural rigidity of the beam is constant, the slope at the support is**

(a) $\frac{5}{EI}$ degree (b) $\frac{900}{\pi EI}$ degree

(c) $\frac{\pi}{900EI}$ degree (d) $\frac{EI}{5}$ radius

**Ans. (b) :**




Slope at 'B' due to symmetry–

= [ Area of $\frac{M}{EI}$ diagram between A and B]

$$= \frac{1}{2}\times1\times\frac{2}{EI}+\frac{2}{EI}\times2$$

$$= \frac{1}{EI}+\frac{4}{EI}$$

$$= \frac{5}{EI} \quad \text{(radians)}$$

$$= \frac{5}{EI}\times\frac{180}{\pi}$$

$$= \frac{900}{\pi EI}\text{degree}$$

**8. The left-hand support of a fixed beam, of span 8 m and of uniform cross-section, yields and rotates by 1$^0$ clock-wise. The moment developed at the right-hand supports is (EI is given in kg cm$^2$)**

(a) $\frac{2EI\pi}{1440}$kg.m (b) $\frac{2EI\pi}{144000}$kg.cm

(c) $\frac{2EI}{8}$kg.cm (d) $\frac{2EI\times180}{\pi\times8}$kg.m

**Ans. (b) :** $M = k\theta$

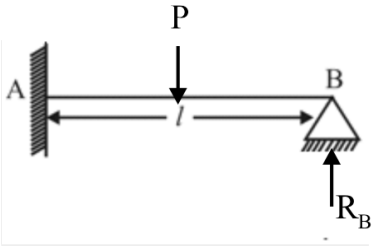
$$= \frac{2EI\pi\,kgcm}{8\times18000} \qquad K = \frac{2EI}{L}$$

**9. In a propped cantilever the number of possible points of contraflexure under any type of loadig is**

(a) two (b) three

(c) one (d) none

**Ans. (c) :**



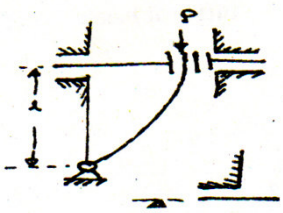

• A propped cantilever beam have one point of contra flexure.

**10. At a point of contraflexure**

(a) the shear force is zero
(b) the deflection is zero
(c) the slope and deflection are zero
(d) the bending moment and shear force are zero
(e) none of above

**Ans. (e) : At a point of contraflexure –**
BM sign changes (+ve) to (-ve) or vice-versa
BM = 0

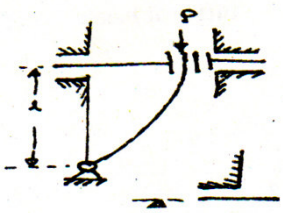
**11. The effective length of the column shown in figure is**



(a) $\frac{\ell}{2}$ (b) $\ell$

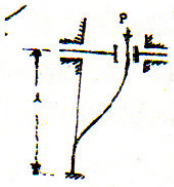
(c) $2\ell$ (d) $\frac{\ell}{\sqrt{2}}$

**Ans. (c) :**



Column free to rotate
Effecting length, $\ell_e = 2\ell$

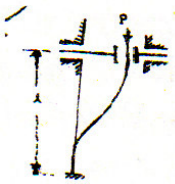
**12. The effective length of the column shown in figure below is**



(a) $\ell$ (b) $0.7\ \ell$

(c) $0.5\ \ell$ (d) $0.8\ \ell$

**Ans. (d):**



For the given column, $\ell_e = 0.8\ell$

**13. The slenderness ratio is the ratio of**

(a) length of column to radium of gyration
(b) effective length of column to the least radius of gyration
(c) effective length of column to least moment of inertia
(d) length of the column to least radius of gyration

**Ans. (b) :** The slenderness ratio is the ratio of effective length of column to least radius of gyration .
Slenderness ratio,

$$\lambda = \frac{le}{k} = \frac{\text{effective length of column}}{\text{Radius of gyration}}$$

**14. The buckling load of a long column given by Euler's formula does not depend on**

(a) elastic property of the material
(b) strength of column material
(c) on the cross-sectional properties
(d) the end conditions

**Ans. (b) :** Euler's formula $(P_e) = \frac{\pi^2 E I_{min}}{L_e^2}$

$$= \frac{\pi^2 . E . k^2 . A}{L_e^2} \quad \left[\because k = \sqrt{\frac{I_{min}}{A}} \text{ or } k^2 . A = I_{min}\right]$$

$P_e \propto E$
$P_e \propto k^2$
$P_e \propto A$
$P_e \propto \frac{1}{l^2}$

So, buckling load from Euler's formula does not depend on strength of column material.

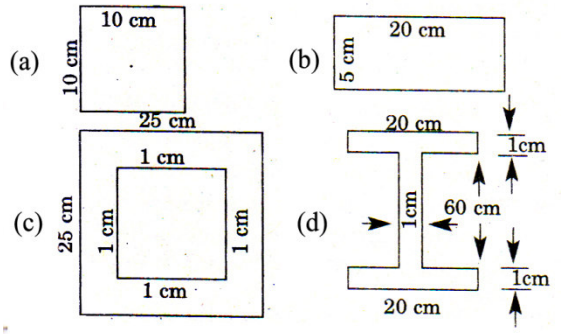
**15. In Rankine's formula for determining the buckling load the constant α does not depend on**

(a) the crushing strength of the material
(b) young's modulus of the material
(c) bulk modulus of the material
(d) $\frac{\sigma_c}{\pi^2 E}$

**Ans. (c) :** Rankin's formula holds good for both long column and short column.
Buckling loud, $(P_e) = \sigma_c . A$

$$\alpha = \frac{\sigma_c}{\pi^2 E}$$

**16. Members with the following cross-section are used as columns. The length and end conditions of all columns are same. The area of all the cross-sections are same. The maximum buckling load is for the column with the cross-section shown in figure.**




**Ans. (d) :** checked by options,

Area = $10 \times 10 = 100\ cm^2$

Buckling load, $P_e = \dfrac{\pi^2 E.k^2.A}{Le^2}$

$$P_e = \frac{\pi^2 E.k^2}{Le^2}.A$$

$P_e \propto 100\ cm^2$

Area, $A = 20 \times 5 = 100\ cm^2$

$P_e = 100\ cm^2$

$P_e = 625\ cm$

$P_e = 62 \times 20 = 1240\ cm$ (max.)

**17. A column is subjected to an eccentric load. The buckling load can be determined by using**

(a) Euler's formula
(b) Rankine-Gorden formula
(c) Straight line formula
(d) Secant formula
(e) None of the above

**Ans. (a) :** A column is subjected to an eccentric load. The buckling load can be determined by using Euler's formula.

**18. Polar moment of inertia is**

(a) The moment of inertia about an axis in the plane of the cross-section
(b) the product of inertia about two mutually perpendicular axis in the plane of the cross-section
(c) about the axis of the member
(d) none of the above

**Ans. (b) :** Polar moment of Inertia may be found by taking the sum of the moments of inertia about two perpendicular axis lying in the plane of the cross-section and passing though this point. The moment of inertia of a plane area with respect to axis perpendicular to the plane of the figure is called a polar moment of inertia with respect to a point where the axis intersects a plane.

**19. Torsional rigidity is**

(a) also known as flexural rigidity
(b) the product of modulus of elasticity and moment of inertia
(c) the torque which develops unit twist per unit length
(d) the product of shear modulus and moment of inertia

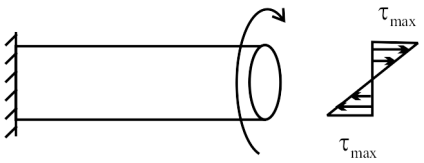
**Ans. (c) :** Torsional rigidity is defined as the torque required to produce a twist of are radian per unit length of the shaft.

Torsional rigidity, $GJ = \dfrac{T}{\theta / l}$

**20. A circular shaft is subjected to torsion. The shear stress in the cross-section.**

(a) Varies parabolically with the maximum stress occurring at the centre
(b) uniform over the cross-section
(c) varies linearly with the radius with the maximum at the circumference and zero at the centre
(d) none of the above

**Ans. (c) :** When a shaft is subjected to a torque or twisting a shearing stress is produced in the shaft the shear stress varies from zero in the axis to a maximum at the outside surface of the shaft.




**21. The power of a shaft can transmit is independent of**

(a) the material used for the shaft
(b) the cross-sectional property
(c) the twist allowed per unit length
(d) the length of the shaft

**Ans. (a) :**

Power transmitted by shaft, $P = \dfrac{2\pi NT}{60}$

We know that, $\dfrac{T}{J} = \dfrac{G\theta}{L} = \dfrac{\tau}{r}$ Or, $T = \dfrac{G\theta J}{L}$

$$P = \frac{2\pi N}{60} \times \frac{G\theta.J}{L}$$

$$P \propto \frac{1}{L}$$

So, power transmitted is independent of the material used for shaft.

**22. A channel section is used for beam with the web of channel horizontal and flanges vertically upwards. The shear stress distribution is**

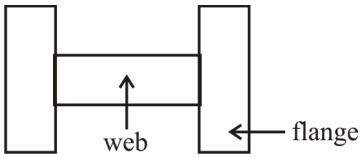
(a) uniform

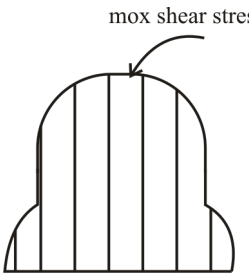
(b) parabolic

(c)

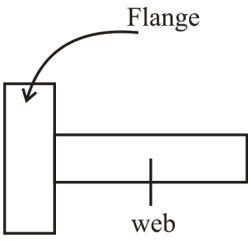
(d) none of these

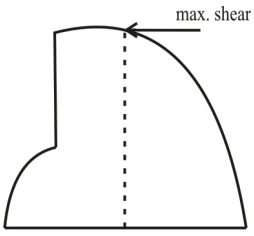
**Ans. (d) :**



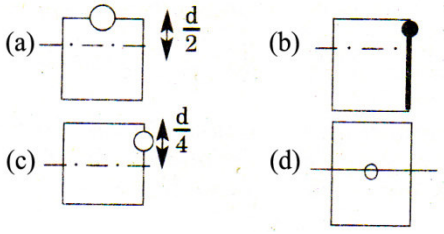

Shear stress distribution



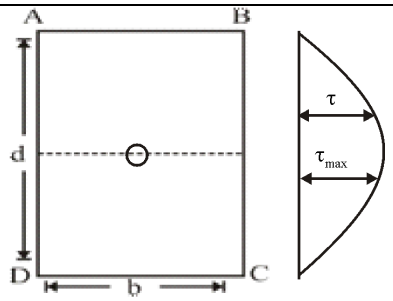

Shear stress distribution

**23. The maximum shear stress in a beam of rectangular corss-section occurs at the point shown in**



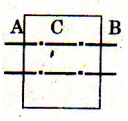

**Ans. (d) :** For rectangular cross - section-

Maximum shear stress, $\tau = \frac{3}{2} \times \frac{V}{bd}$




So, shear stress is zero at top and bottom and shear stress is maximum at a neutral axis.

**24. The variation of shear along the line AB in the cross-section shown in figure (the member is used as a beam)**




(a) linearly varying from zero at A to maximum at B

(b) parabolically varying with maximum at C

(c) triangularly varying with maximum at C

(d) constant along AB

**Ans. (d) :** For a rectangular beam cross-section, the shear stress along the height of cross section can be plotted constant along AB.

**25. The intensity of bending stress in the cross-section at any point distance y from the neutral axis is proportional to (I is moment of inertia)**

(a) $\frac{I}{y}$ (b) I,y

(c) y (d) $\frac{1}{y^2}$

(e) $y^3$

**Ans. (c) :** Bending equation,

$$\frac{\sigma}{y_{max}} = \frac{M}{I} = \frac{E}{R}$$

Bending stress, $\sigma = \frac{M}{I} \times y_{max}$

So, the variation of bending stress in the simple beam is linear. The bending stress varies linearly from zero at the neutral axis to a maximum at the outer surface.

**26. A beam of uniform cross-section is subjected throughout its length to a uniform bending moment. It will bend to**

(a) catenary

(b) a third degree parabola

(c) a circular arc

(d) $4^{th}$ degree parabola

**Ans. (c) :** If bending moment (M) is constant then the radius of curvature of the beam is also constant ( i.e. cross section is also constant through length) so, the shape of elastic curve is a circular arc.

**27. Bernoulli's assumption in theory of simple bending is incorrect.**

(a) the material obeys Hooke's law

(b) the value of young's modulus is same for beam material in tension as well as in compression

(c) the radius of curvature is small compared with the dimension of the cross section

(d) the transverse section of the beam which is place before bending will remain plane even after bending

**Ans. (c) :** Bernoulli's assumption is theory of simple bending-

- Material beam is homogeneous and isotropic.
- The cross-section of the beam remains plane even after bending.
- The material is stressed within elastic limit and obeys Hooke's law.
- The value of modulus of elasticity for the material is same in tension and compression.
- The transverse sections, which are plane and normal to the longitudinal axis before bending, remain plane and normal to the longitudinal axis of the beam offer bending.

**28. The capacity of the beam to resist external transverse loads does not depend on**

(a) the cross-sectional properties of the material

(b) the strength of the beam material

(c) the span of the beam

(d) elastic properties of the material

**Ans. (c) :** According to bending equation-

$$\frac{M}{I}=\frac{E}{R}=\frac{\sigma}{y_{max.}}$$

Or, $$M=\frac{IE}{R}$$

Or, $$M=\frac{\sigma.I}{y_{max.}}$$

Depend upon–

- The cross-sectional properties of the material
- The strength of the beam material
- Elastic properties of the material

**29. The bending stress calculated by using the formula M=fz is absolutely accurate when the bending moment**

(1) variation is linear

(2) variation is parabolic (2nd degree)

(3) is constant

(4) none of the above

(a) 1 (b) 2

(c) 1 and 3 (d) 4

**Ans. (a) :** Bending equation–

$$\frac{M}{I}=\frac{E}{R}=\frac{f}{y}$$

$$f=\frac{M}{z}$$

$$f \propto M$$

Since bending moment variation is linear.

**30. The maximum shearing stress in beams of cross-sections having one axis of symmetry occurs at**

(a) centre of gravity of the section

(b) at about mid-depth of the section

(c) at farthest point from the centre of gravity of the section

(d) at a quarter depth from the top or bottom of the cross-section

**Ans. (b) :** In symmetrical beam section, the maximum shear stress in beams of cross-sections having one axis of symmetry occurs at mid-depth of the section.

**31. The ratio of the strength of 'n' loose beams ( b × d) placed one over the other to the strength of one integral beam (b ×nd) is**

(a) $n^2$ (b) n

(c) $\frac{1}{n}$ (d) $\frac{1}{n^2}$

**Ans. (d) :** Section modulus

$$z=\frac{bd^2}{6}$$

$$z \propto bd^2$$

$$\frac{z_2}{z_1}=\frac{b(nd)^2}{bd^2}$$

$$\frac{z_2}{z_1}=n^2$$

$$\frac{z_1}{z_2}=\frac{1}{n^2}$$

**32. The ratio of flexibility of 'n' loose beams (b × d) placed one over the other to the flexibility of one integral beam (nb ×d) is**

(a) $n^2$ (b) $\frac{1}{n^2}$

(c) 1.0 (d) $n^3$

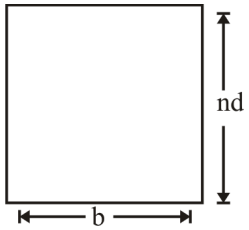
(e) none of those

**Ans. (a) :** As we know,

Flexibility of beam = $\frac{1}{\text{Stiffness}}$

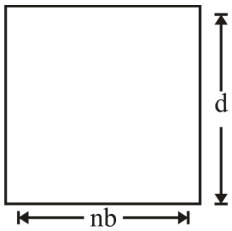
Stiffness of beam = $\frac{M}{\theta}$ or $\left(\frac{BM}{\text{slope}}\right)$

We know that, $$\frac{d^2y}{dx^2}=\frac{M}{EI}$$




Slope, $\theta = \frac{dy}{dx} = \int \frac{M}{EI}$

So, $\frac{\theta}{M} \propto \frac{1}{EI}$




Hence, Flexibility $\propto$ EI

$$\frac{(\text{Flexibility})_1}{(\text{Flexibility})_2} = \frac{(EI)_1}{(EI)_2} = \left(\frac{I_1}{I_2}\right)$$

When, $E_1 = E_2$

$I_1 = \frac{(b)(nd)^3}{12}$ $\quad I_2 = \frac{nbd^3}{12}$

$$\frac{I_1}{I_2} = n^2$$

**33. The ratio of the curvature of the 3 loose beams. (b ×d) placed one over the other to an integral beam (b × 3d) is**

(a) 9 (b) $\frac{1}{9}$

(c) 1.0 (d) 27

(e) $\frac{1}{27}$

**Ans. (d) :** Bending equation–

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y}$$

$$\frac{M}{I} = \frac{E}{R}$$

Radius of curvature = $\frac{1}{R} = K = \frac{M}{E.I.}$

$$K \propto \frac{1}{I}$$

$$\frac{K_2}{K_1} = \frac{1}{b(3d)^3/12} \div \frac{1}{b(d)^3/12} = \frac{1}{27}$$

$$\frac{K_1}{K_2} = 27$$

**34. Two beams are of same length and same weight. One beam is of solid circular section and the other is a hollow section. The internal diameter is half of the external diameter. The ratio of the strength of hollow section to that of solid section is**

(a) 1.0 (b) $\frac{15}{6\sqrt{3}}$

(c) $\frac{6\sqrt{3}}{15}$ (d) 2

**Ans. (b) :** Solid and hollow sections have same weight,

$W_h = W_s$

$y \times V_h = y \times V_s$

$\therefore \quad A_h = A_s$

$$= \frac{\pi}{4}(D^2 - d^2) = \frac{\pi}{4}d_1^2$$

$$D^2 - d^2 = d_1^2$$

Let, $k = \frac{d}{D} = \frac{D}{2 \times D} = \frac{1}{2}$

$$D^2 (1 - K^2) = d^2_1$$

$$D\sqrt{(1-k^2)} = d_1 \qquad \text{.....(i)}$$

$$\frac{z_h}{z_s} = \frac{\frac{\pi}{32}D^3(1-k^4)}{\frac{\pi}{32}d_1^3}$$

$$= \frac{D^3(1-k^4)}{D^3\sqrt{1-k^2}(1-k^2)}$$

$$= \frac{1-k^4}{\sqrt{1-k^2}(1-k^2)}$$

$$\frac{z_h}{z_s} = \frac{1+k^2}{\sqrt{1-k^2}}$$

$$= \frac{1+\left(\frac{1}{2}\right)^2}{\sqrt{1-\left(\frac{1}{2}\right)^2}} = \frac{5}{2\sqrt{3}}$$

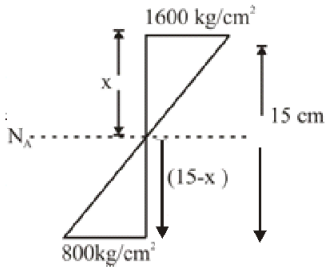
$$\frac{z_h}{z_s} = \frac{15}{6\sqrt{3}}$$

**35. The maximum compressive stress at the top of a beam was 1600 kg/cm$^2$ and the corresponding tensile stress at its bottom was 800 kg/cm$^2$. If the depth of the beam was 15 cm the neutral axis from the top will be**

(a) 5 cm (b) 10 cm

(c) 7.5 cm (d) 12 cm

**Ans. (b) :**




In stress diagram by similar triangle

$$\frac{1600}{x}=\frac{800}{(15-x)}$$

24000 – 1600x = 800x

X = 10 cm

**36. A beam is said to be of uniform strength if**

(a) the bending moment is same throughout the beam

(b) deflection curve is an arc of circle

(c) maximum shear stress is same throughout the beam

(d) the maximum bending stress is same at every section along its longitudinal axis.

**Ans. (d) :** In a beam of uniform strength bending stress developed due to external load at any section of a the beam is same through out beam. A beam of uniform strength can be make by either changing its depth or changing its width along the length of the beam.

**37. If the bending moment at a section of a beam is zero then**

(a) the shear force is zero at that section

(b) the shear force changes sign at the section

(c) the shear force is maximum at that section

(d) nothing is known about the shear force

**Ans. (d) :** In as beam that is point where there is zero bending moment is called the point of contraflexure.
So, at that point of zero bending moment shear force need not be maximum, minimum or zero.

**38. The principal axes of a cross-section are defined as**

(a) the axes about which polar moment of inertia is zero

(b) the axes about which modulus of section are zero

(c) the axes about which the product of inertia is zero

(d) the axes about which radial of gyration are zero

**Ans. (c) :** The principal axes of a cross-section are defined as the axis about which product of inertia is zero.

**39. Wire of diameter'd' is wound round a cylinder. The diameter of the cylinder is 'D'. The bending stress (maximum) induced is the wire is**

(a) $\frac{ED}{d}$ (b) $\frac{Ed}{D}$

(c) $\frac{d}{ED}$ (d) $\frac{D}{Ed}$

**Ans. (b) :** Given,
Wire of diameter, = d
Diameter of cylinder, = D
According to bending equation–

$$\frac{M}{I}=\frac{E}{R}=\frac{\sigma_b}{y_{max.}}$$

$$\therefore \frac{E}{\left(\frac{D}{2}\right)}=\frac{\sigma_b}{\left(\frac{d}{2}\right)}$$

$$\frac{E}{D}=\frac{\sigma_b}{d}$$

$$\sigma_b=\frac{E.d}{D}$$

**40. The radius which a steel strip of 1 cm depth can be bent without the maximum stress exceeding 1400 kg/cm$^2$ is (E = 2 ×10$^6$ kg/cm$^2$) Which letter comes next?**

(a) $\frac{10000}{14}$ cm (b) $\frac{14}{10000}$ cm

(c) $\frac{10000}{7}$ cm (d) $\frac{7}{10000}$ cm

**Ans. (a) :** Given,
depth = d = 1 cm
Max. stress ,$\sigma_{max.} = \sigma_b$ = 1400 kg/cm$^2$
E = 2 × 10$^6$ kg/ cm$^{2=}$
According to bending Equation–

$$\frac{M}{I}=\frac{E}{R}=\frac{\sigma_b}{y_{max.}}$$

$$\therefore \quad \frac{E}{R}=\frac{\sigma_b}{y_{max.}}$$

$$\frac{2\times10^6}{R}=\frac{1400}{\frac{1}{2}}$$

$$\frac{2\times10^6}{R}=2800$$

$$R=\frac{2\times10^6}{2800}$$

$$=\frac{20000}{28}$$

$$R=\frac{10000}{14}\text{ cm}$$

**41. At a point in a strained material the normal stresses on two mutually perpendicular planes are both tensile and they are equal to 10 T/cm$^2$. The stresses on a plane inclined at an angle of 30$^0$ to one of the planes are**

(a) 10 T/cm$^2$ tensile and the shear stress of 10 T/cm$^2$

(b) only shear stress of 10 T/cm$^2$

(c) only normal stress of 10 T/cm$^2$

(d) a normal stress of $5\sqrt{3}$ T/cm$^2$ and a shear stress 5 T/cm$^2$

**Ans. (c) :** $\sigma_x = 10$ T/cm$^2$

$\sigma_y = 10$ T/cm$^2$, $\tau_{xy} = 0$

$\theta = 30^o$

Normal stress,

$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\sigma_n = \frac{10+10}{2} + \left(\frac{10-10}{2}\right) \times \cos 2 \times 30^o + 0 \times \sin 2\theta$$

$\sigma_n = 10 + 0 + 0$

$\sigma_n = 10$ T/cm$^2$

$$\text{Shear stress}, \tau = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$= -\left(\frac{10-10}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$= 0 + 0$

Shear stress, $\tau = 0$

**42. In a strained material the state of stress is a pure shear stress and the stress is 5 T/cm$^2$. The principal stresses are**

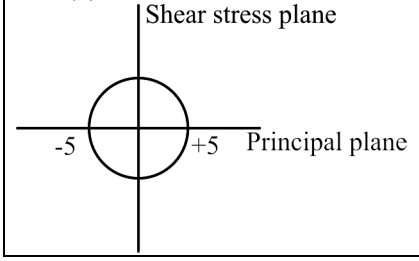
(a) both are equal to of the same nature and acts on plane inclined at 45$^0$ to the planes of shear stress

(b) one is 5 T/cm$^2$ tensile and the other 5 T/cm$^2$ compressive acting on planes inclined at 45$^0$ to the planes of shear stress

(c) one is 5 T/cm$^2$ tensile or compressive and the other is zero on planes inclined at 45$^0$ to the planes of shear stress

(d) both are equal to $5\sqrt{2}$ T/cm$^2$ and of same nature acting on planes inclined at 45$^0$ to the planes of shear stress

**Ans. (b) :**




Pure shear condition

$$\sigma_{P_1}, \sigma_{P_2} = \frac{\sigma_x + \sigma_y}{2} \pm \frac{1}{2}\sqrt{(\sigma_y - \sigma_x)^2 + 4\tau_{xy}^2}$$

$$= \frac{0+0}{2} \pm \frac{1}{2}\sqrt{(0-0)^2 + 4 \times 5^2}$$

$$\pm\frac{1}{2} \times 2 \times 5 = \pm 5 \text{ T/cm}^2$$

$\sigma_{P_1} = +5$ T/cm$^2$, $\sigma_{P_2} = -5$ T/cm$^2$

**43. In a strained material the normal stress on one plane is 10 T/cm$^2$ tensile and the shear stress is 5 T/cm$^2$. On a plane perpendicular to this plane there is only shear stress. The centre for Mohr's circle.**

(a) coincides with the origin

(b) is at a distance representing 10 T/cm$^2$ on the positive side of the axis representing tensile stress

(c) is at a distance representing 5T/cm$^2$ on the positive side of the axis representing tensile stress

(d) at a distance representing $5\sqrt{5}$ T/cm$^2$ on the positive side of the axis representing tensile stress

**Ans. (c) :** $\sigma_x = 10$ T/cm$^2$

$\tau_{xy} = 5$ T/cm$^2$

$\sigma_y = 0$

Principal stress,

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \left(\frac{10t + 0}{2}\right) \pm \sqrt{\left(\frac{10T - 0}{2}\right)^2 + (5T)^2}$$

$$= 5t \pm \sqrt{25t^2 + 25t^2}$$

$$= 5t \pm 5t\sqrt{2}$$

$$\sigma_1 = 5t + 5t\sqrt{2}$$

$= 5t\,[1 + \sqrt{2}\,]$

$= -2.071$ T/cm$^2$

So, centre of mohr's cirde, $= \left(\frac{\sigma_1 + \sigma_2}{2}, 0\right)$

$$= \frac{12.071T - 2.071T}{2}$$

$= 5$ T/cm$^2$

**44. The principal plane is defined as the plane on which**

(a) there is only shear stress

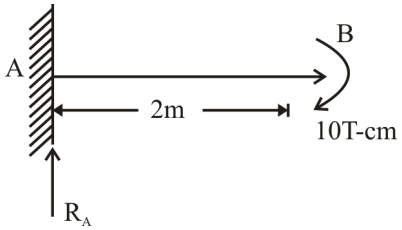
(b) the normal stress is either maximum of minimum

(c) only normal stress either maximum or minimum is acting
(d) both normal and shear stresses are either maximum or minimum

**Ans. (c) :** The principal plane is defined as the plane on which have shear stress is zero the principal plane on which the normal stress is either maximum or minimum is acting.

**45. The shear forces diagram for a cantilever of span 2 m with a couple of 10 T.cm acting at the free end is a rectangle with span of the cantilever a base and the height represents**
(a) 10T (b) 50 kg
(c) zero (d) 5T

**Ans. (c) :**




Shear force = sum of all the vertical farce.
= 0

**46. A step in the bending moment diagram indicates that**
(a) there is concentrated load acting on the beam at that section
(b) there is a change in the intensity of uniformly distributed load on either side of the section
(c) there is a couple acting at that section
(d) no lead accts at that section

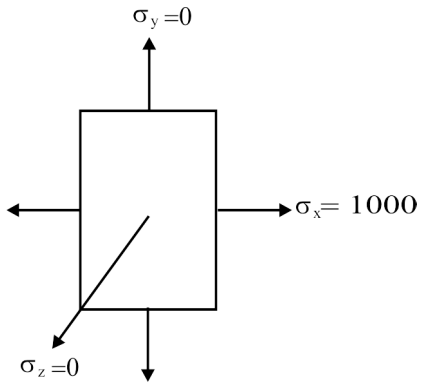
**Ans. (c) :** Bending moment diagram shows the variation of bending moment along the length of the beam.
A step in the bending moment diagram indicts there is a concentrated moment or couple at the section.
A steep in the shear yore diagram indicates there is a concentrated load at the section.

**47. For any material the value of Poisson's ratio is always**
(a) greater than one
(b) between 0.5 and 1.0
(c) between 0 and 0.5
(d) between 0 and 0.05

**Ans. (c) :** For most materials, the value of position's ratio lies in the range 0 to 0.5.

**48. A stress of 1000 kg/cm$^2$ tensile acts on a cube along x direction. If E = 2 × 10$^6$ kg/cm$^2$ and poisson's ratio of 0.25, the stress required to prevent strain a long y direction is**
(a) 4000 kg/cm$^2$ tensile
(b) 250 kg/cm$^2$ tensile
(c) 250 kg/cm$^2$ compressive
(d) zero

**Ans. (b) :** Given,
Longitudinal stress ($\sigma_x$) = 1000 kg/cm$^2$
young modulus E = 2 × 10$^6$ kg/cm$^2$




Total strain in lateral direction

$$\varepsilon_y = \frac{\sigma_y}{E} - \mu\frac{\sigma_x}{E} - \mu\frac{\sigma_z}{E}$$

$\sigma_x = 1000,\ \sigma_y = 0,\ \sigma_z = 0$

$$\varepsilon_y = -\mu\frac{\sigma_x}{E}$$

$$\varepsilon_y = -0.25 \times \frac{1000}{2\times10^6}$$

Negative sign indicates strain in compressive.
To prevent this strain required stress in opposite nature.

$$\sigma = \varepsilon_y \times E$$

$$\sigma = \frac{0.25\times1000}{2\times10^6}\times 2\times10^6$$

$\sigma$ = 250 kg/cm$^2$ tensile

**49. Whatever may be value of Poisson's ratio, the ratio of Bulk modulus to young's modulus will be**
(a) 3
(b) between 33.33 to $\frac{1}{2.94}$
(c) 1 and 3
(d) $\frac{1}{3}$ and $\infty$

**Ans. (d) :** Relation between bulk modulus, young's modulus and poisson ratio.

$$E = 3K(1-2\mu)$$

Ratio of bulk modulus of young modulus

$$\frac{K}{E} = \frac{1}{3(1-2\mu)}$$

Minimum value of poisson ratio ($\mu$) = 0

$$\frac{K}{E} = \frac{1}{3}$$

Maximum value of poisson ratio ($\mu$) = 0.5

$$\frac{K}{E} = \frac{1}{0}$$

$$\frac{K}{E} = \infty$$

**50. A bar of 2 m length is fixed at both ends. If E = $2 \times 10^6$ kg/cm$^2$, the coefficient of expansion is $1.5 \times 10^{-6}/^0$C and the temperature rise is $20^0$C, the stress developed in the material is**

(a) 60 kg/cm$^2$ tensile
(b) 60 kg/cm$^2$ compressive
(c) no stress is developed
(d) 60 kg/cm$^2$ tensile at one face and 60 kg/cm$^2$ compressive on opposite face

**Ans. (b) :** Given,
$L = 2$m
$E = 2 \times 10^6$ kg/ cm$^2$
$\alpha = 1.5 \times 10^{-6}$ /ºC
Rise temperature, $\Delta T$ ºC = 20 ºC
Stress developed in the material,
$\sigma_T = \alpha TE$
$= 1.5 \times 10^{-6} \times 20 \times 2 \times 10^6$
$= 40 \times 1.5 \times 10^{-6} \times 10^6$
$= 60$ kg/cm$^2$ [compressive]

**51. The intensity of pressure is the same in all directions at a point in a fluid**

(a) only when the fluid is frictionless and incompressible
(b) only when the fluid has zero viscosity and is at rest
(c) regardless of the motion of one fluid layer relative to an adjacent layer
(d) for none of the above reasons

**Ans. (b) :** According to pascal's law the pressure or intensity of pressure at a point in a static fluid is equal in all directions.
- This law is valid for the cases of fluid flow where shear stresses do not exist. The cases are fluid at rest.
- No relative motion exists between different fluid layers which means the viscosity of fluid is zero.

**52. With barometer reading 74 cm mercury, 5.05 kgf/cm$^2$ abs is equivalent to**

(a) 13.6 cm of mercury abs
(b) 3.99 kgf/cm$^2$ suction
(c) 5.00 atmosphere
(d) 4.00 kgf/cm$^2$ gauge

**Ans. (d) :** Given,
h = 74 mercury
$P_{abs} = 5.05$ kgf/cm$^2$
$P_{atm} = P\,g\,h$
$= 13.6 \times 10^3 \times 9.81 \times 74 \times 10^{-2}$
$= 98727.84$ N/m$^2$
$= \dfrac{98727.84}{9.81 \times 10000}$ k gf/cm$^2$
$= 1.0064$ k g f/cm$^2$
$\therefore \quad P_{abs} = P_{gauge} + P_{atm}$
$5.05 = P_{gauge} + 1.0064$
$P_{gauge} = 5.050 - 1.0064$
$= 4.04$ k g f/cm$^2$

**53. The magnitude of hydrostatic pressure on one side of a circular surface, 1 m$^2$ area, with centroid 1 m below free water surface**

(a) depends on the orientation of the area
(b) is the product of sp. weight and vertical distance from free surface to the centre of pressure
(c) 1000 kgf
(d) none of the above

**Ans. (c) :** Given,
$\delta = 1000$ kg/m$^3$ Circular surface area, A = 1m$^2$
H = 1m
Hydrostatic pressure, $P = \rho\,g\,h\,A$
$= 1000 \times 9.81 \times 1 \times 1$
$= 1000 \times 9.81$
$= 1000$ k g f [$\therefore$ 1N=9.81 kgf]

**54. The centre of pressure of a vertical circular surface of 1m$^2$ area with centroid 10m below the free water surface, is**

(a) above its centroid
(b) at the centroid
(c) below its centroid
(d) none of these

**Ans. (c) :** Given, Area, A = 1m$^2$
$\bar{h} = 10$m
The centre of pressure lies below the centroid because for any plane surface, the centroid is always positive.

**55. The hydrostatic pressure, in kgf, exerted on one side of an annular area enclosed by concentric circles of radii 2 and 1 m and having its centroid 4 m below free water surface, is**

(a) 12000 $\pi$ (b) 11000 $\pi$
(c) 10000 $\pi$ (d) none of these

**Ans. (a) :** Given, $\bar{h} = 4$m
R = 2m
r = 1m
Hydrostatic pressure, $P = \rho g A \bar{h}$
$= 1000 \times 9.81 \times \pi (R^2 - r^2) \times 4$
$= 1000 \times 9.81 \times \pi \times (2^2 - 1^2) \times 4$
$= 1000 \times 9.81 \times \pi \times 3 \times 4$
$= 12000\ \pi \times 9.81$ N
$= 12000$ kgf

**56. A vertical, gate 3 by 3 m holds water upto its top. The moment in kgf-m, about the top edge of the gate is**

(a) 27000 (b) 20250
(c) 13500 (d) none of there

**Ans. (a) :** Given,
Area = $3 \times 3 = 9$ m$^2$
$\bar{h} = \dfrac{3}{2} = 1.5$ m
$\rho = 1000$ kg/m$^3$

Hydrostatic pressure,

$$F = \rho g A \bar{h}$$
$$= 1000 \times 9.8 \times 9 \times 1.5$$
$$= 13.5 \times 1000 \times 9.8 \text{ N}$$
$$= 13500 \text{ k g f } [\because 1N = 9.81 kgf]$$

So, moment in kgf –m, about the top edge of the gate-

$$\text{Moment, } M = F \times \frac{2}{3}h$$
$$= 13500 \times \frac{2}{3} \times 3 \times 3$$
$$= 27000 \text{ Kgf-m}$$

**57. The horizontal component of the total hydrostatic pressure one a curved surface is equal to the**

(a) weight of the liquid vertical above the curved surface
(b) product of pressure as its centroid and the area
(c) pressure on a vertical projection of the curved surface
(d) scalar sum of all elements components

**Ans. (c) :** The horizontal component of the total hydrostatic pressure on a curved surface. Vertical component is Equal to the weight of liquid supported by the curved surface up to the free liquid surface.

**58. A woodne cube 2 by 2 by 2m, sp. gr. 0.5. floats in water. The volume of the cube submerged under water, in $m^3$ is**

(a) 4 (b) 2
(c) 1 (d) none of these

**Ans. (a) :** Given,
Cube = 2 × 2 × 2 m
Sp. Gravity, s = 0.5
$\rho_{wood} = 1000 \times 0.5$
$= 500 \text{ kg/m}^3$
$\therefore \rho_{wood} \times g \times V_{wood} = \rho_f g \times V_f d$
$500 \times 9.81 \times 8 = 1000 \times 9.81 \times V_f d$

$$V_{fd} = \frac{4000}{1000}$$
$$V_{fd} = 4 \text{ m}^3$$

**59. The line of action of the buoyant force acts through the**

(a) centroid of the volume of the floating body
(b) centroid of displaced volume of fluid
(c) centroid of the horizontal projection of the body
(d) centroid of the vertical projection of the body

**Ans. (b) :** Buoyant force on a submerged or floating body is equal to weight of the liquid displaced by the body and acts vertically upward through the centric of a displaced volume known as the centre of buoyancy.

**60. A body floats is stable equilibrium**

(a) when its metacentric height is zero
(b) only when its centre of gravity below its cenre of buoyancy
(c) when the metacentre is above its centre of gravity
(d) none of these

**Ans. (c) :** A body flouts in stable equilibrium When the meta centre is above its centre of gravity.

**61. When steady flow of a liquid takes place through a convergent section, the liquid is subjected to**

(a) local acceleration
(b) convective acceleration only
(c) both local and convective acceleration
(d) no acceleration

**Ans. (b) :** When steady flow of a liquid takes place through a convergent section, the liquid is subjected convective acceleration i.e. convective acceleration only.

**Types of flow field acceleration–**

| | **Convective acceleration** | **Temporal acceleration** |
|---|---|---|
| Steady uniform flow | 0 | 0 |
| Steady non-uniform | 0 | Exists |
| unsteady uniform | Exists | 0 |
| unsteady non-uniform | Exist | Exists |

**62. Steady uniform flow of water takes place through a straight pipe of uniform diameter. The water is subjected to**

(a) local acceleration only
(b) convective acceleration only
(c) both local and convective accelerations
(d) no acceleration

**Ans. (d) :** Steady uniform flow of water takes place through a straight pipe of uniform diameter. The water is subjected to no acceleration i.e convective and local acceleration is zero

**63. The assumptions about flow required in deriving the Bernoulli's equation are, that it is**

(a) steady, frictionless incompressible, along a streamline
(b) uniform, frictionless, along a streamline, density function of pressure
(c) steady, uniform, incompressible along a streamline
(d) none of these

**Ans. (c) :** Assumptions about flow required in deriving the Bernoulli's equation are steady, uniform, incompressible along a Streamline.

**64. The work that a liquid is capable of doing by virtue of its sustained pressure, in kgf-m/kgf, is equal to**

(a) $\rho$ (b) z
(c) $v^2/2g$ (d) $\rho/\gamma$

**Ans. (d) :** The work that a liquid is capable of doing by virtue of its sustained pressure, in kgf-m/kgf, is equal to $\rho/\gamma$.

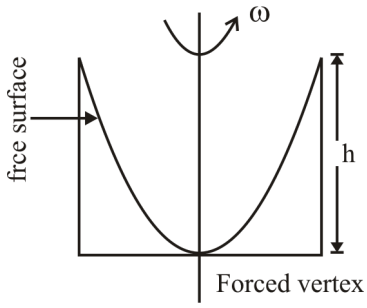
**65. Bernoulli's theorem states that for steady ideal fluid flow**
(a) the velocity is constant along a streamlines
(b) the velocity increases as the pressure increases
(c) the energy is constant along a streamline
(d) none of the above

**Ans. (d) :** Assumption mode in the derivation of Bernalillo's Equation,
- The fluid is ideal i.e viscosity is zero.
- The flow is along the streamline.
- The flow is ideal, viscosity is zero steady.
- The flow is compressible.

Bernalillo's Equation states that in a steady ideal flow of an incompressible fluid, the total energy at any point of the fluid is constant.

**66. A right circular cylinder open at the top is filled with a liquid, sp. gr. 0.80 and rotated about its vertical axis at such a speed that half the liquid spills out. The pressure at the centre of the bottom of the cylinder is**
(a) zero
(b) half its value when the cylinder was full
(c) indeterminable-insufficient date
(d) lesser than a similar case with water as liquid

**Ans. (a) :**




A cylinder rotating about the vertical axis has forced vortex flow. The free surface is parabolic. Haley the volume of cylinder of same height when holy the liquid spills out the centre of the bottom will be exposed and pressure will be zero.

**67. A free vortex**
(a) turns in opposite direction to a forced vortex
(b) has the velocity increasing with the radius
(c) occurs when fluid rotates as a solid
(d) none of the above

**Ans. (d) : Free Vortex–** When no external torque is required to rotate the fluid mass, that type of the flow is called a free vortex. No torque in the free vortex, so free vortex is an irrotational flow. For free vertex, a moment momentum is constant i.e vr = Constant

$$v \propto \frac{1}{r}$$

Example – draining the bath tub.

**68. The discharge of V-notch varies as**
(a) $(\text{head})^{2/3}$ (b) $(\text{head})^{1/2}$
(c) $(\text{head})^{5/2}$ (d) $(\text{head})^{5}$

**Ans. (c) :** V- notch weir is also called the triangular notch or weir.

$$\text{Discharge, } Q = \frac{8}{15} C_d \times \tan\frac{\theta}{2} \cdot \sqrt{2g}\, H^{5/2}$$

$$Q \propto H^{5/2}$$

**69. If the error in reading the head over a rectangular notch is 1%, the resulting percentage error in computed discharge is**
(a) 1 (b) 2.5
(c) 1.5 (d) none of these

**Ans. (c) :** Discharge for rectangular notch-

$$Q = \frac{2}{3} C_d L \sqrt{2g} H^{3/2}$$

Error in reading the head over rectangular notch is 1%

i. e. $\frac{dH}{H} = 1\%$

$$Q = \frac{2}{3} C_d L \sqrt{2g}\ H^{3/2}$$

$Q = K H^{3/2}$

$\therefore$ Differentiating the discharge Equation With respect to 'H'

$$\frac{dQ}{Q} = \frac{3}{2}\frac{dH}{H}$$

$$\frac{dQ}{Q} = \frac{3}{2} \times 1$$

$$\frac{dQ}{Q} = 1.5$$

**70. A rectangular channel, 6 m wide, has a full width weir across it. The number of end contractions are**
(a) 2 (b) 1
(c) zero (d) none of these

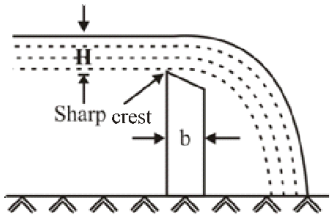
**Ans. (c) :** Given, Rectangular channel wide,
w = 6m
number of end contractions are zero.

**71. Which of the following defines, a sharp crested weir?**
(a) a weir whose top width is zero
(b) a weir whose top width is less than the head over it
(c) a weir in which the water merely touches a line
(d) none of these

**Ans. (c) :** Sharp-crested weir–

The crest of the weir is very such that the water will springs clear of the crest. the weir plate will springs clear of the crest. The weir plate is pevelled at eh crest edges to abstain mercenary thickness. Weir plate should be mode of smooth metal which is free from rust and nicks.

Flow over sharp forested weir is similar as rectangular weir.




**72. A simple pitot tube measure the**

(a) static pressure
(b) velocity at the stagnation point
(c) stagnation pressure head
(d) none of these

**Ans. (c) :** A simple pitot tube can only measures stagnation pressure due to single opening. The basic principle used in this device is that if the speed of flow at a particular pit is reduced to zero which is known as stagnation point. A pitot tube ( pitot probe) measures fluid flow velocity.

**73. For a given discharge through a venturimeter, the reading of the differential manometer, attached to it**

(a) will be the same regardless of the inclination of the meter
(b) increases with increased inclination
(c) will be minimum when the meter is horizontal
(d) none of these

**Ans. (a) :** A Venturimeter is a device used for measuring the rate of flow of a fluid of a liquid flowing through a pipe.

When Venturimeter is inclined then ab θ it shows the same readings. Inclination does not cause any effect on the readings.

Inclination does not cause any effect on the readings. Venturimeter always have smaller convergent portion and larger divergent portion.

**74. A current meter is used to measure**

(a) wind velocities at airports
(b) velocity of liquid flow in open channel
(c) pressure in liquids
(d) none of these

**Ans. (b) :** A current meter is used to measure velocity of liquid flow in open channels.

**75. Reynold's number for pipe flow varies**

(a) as coef. of viscosity
(b) as kinematic viscosity
(c) inversely as the product of viscosity and diameter
(d) inversely as kinematic viscosity

**Ans. (d) :** Reynolds's number, $Re = \frac{\rho V d}{\mu}$

Where, $\rho$ = density of liquid

$v$ = flow velocity

$d$ = dia of pipe

$\mu$ = dynamic viscosity

Kinematic viscosity, $\nu = \frac{\mu}{\rho}$

$$Re = \frac{vd}{\nu}$$

So, Reynolds's number for pipe flow varies inversed as kinematic viscosity.

**76. In laminar flow**

(a) viscosity is not important
(b) fluid particles move in irregular and haphazard paths
(c) coeff viscosity depends on the flow
(d) none of the above is true

**Ans. (d) : Laminar flow–** Laminar flow is defined as that type of flow in which particles move along wall defined paths or stream line and all stream lines are straight and parallel. Thus the particles move in laminas or layers gliding smoothly over the adjacent layer. This type of flow is called steam line flow or viscous flow.

**77. In case of laminar flow through a round tube, the discharge varies**

(a) linearly as viscosity
(b) as the square of the radius
(c) inversely as the pressure drop
(d) inversely as the viscosity

**Ans. (d) :** In laminar flow through a circular pipe.

Discharge, $$Q = \frac{\pi}{8\mu}\left(-\frac{\partial \rho}{\partial x}\right)R^4$$

$$Q \propto \frac{1}{\mu}$$

$$Q \propto R^4$$

**78. Viscous flow takes place through a round tube of 1 $cm^2$ cross-sectional area with a maximum velocity of 2 cm/s. The discharge through the tube, in $cm^3/s$ is**

(a) 2 (b) 1
(c) 4 (d) none of these

**Ans. (a) :** Given,
Area, A = 1 cm
Velocity, V = 2 cm/sec.
Discharge, Q = AV
$= 1 \times 2$
$= 2\ cm^3/sec.$

**79. Turbulent flow generally occurs for cases involving**
(a) very viscous fluids
(b) very slow motions
(c) capillary passage
(d) none of the above

**Ans. (a) :** Turbulent flow occurs when velocities are high and the viscosity is low. Turbulent flow is the flow in which fluid particles flow in a random motion where intermixing is present. In pipe flow, the Reynolds number for far turbulent flow is Re = 4000.

**80. The head loss in turbulent flow in a pipe varies**
(a) directly as the velocity
(b) inversely as the square of the velocity
(c) inversely as the square of the diameter
(d) approximately as the square of the velocity

**Ans. (d) :** Head loss due to friction in turbulent flow is,

$$h = \frac{4fLV^2}{2gD}$$

Or $$h = \frac{fLv^2}{2gD}$$

Where, f = friction factor
$f^l$ = Coefficient of friction
$h \propto v^2$

**81. Water flows into a tank by pipe flow. The loss of head at the exit of the pipe is equal to**
(a) half the velocity head at the exit
(b) velocity head at the exit
(c) zero
(d) none of the above

**Ans. (b) :** Water flows into a tank by pipe flows. The loss of head at exit of the pipe is equal to,

$$h_L = \frac{v^2}{2g}$$

i. e. velocity head at exit.

**82. If two reservoirs whose water levels differ by 2 m are connected by 2 C.I. pipes laid in sense, whose lengths and diameters are 100 m. 40 cm and 200 m and 20 cm rest.**
(a) the loss of head in the system is 2 m of water
(b) the loss of head cannot be determined
(c) there is no flow from one reservoir to the other
(d) none of the above is true

**Ans. (a):** The loss of head in the system is 2 m of water will remains same.

**83. Two reservoirs are connected by a siphon. The flow through the pipe will cease, when**
(a) the pressure at he summit is absolute zero
(b) the summit is at higher elevation than the upper reservoir level
(c) the pressure at the summit falls below the vapour pressure of the fluid
(d) none of the above

**Ans. (b) :** A siphon is long bent used for carrying water from a reservoir at a higher level to another reservoir at a lower level when the two reservoirs are separated by a hill or high level ground. A point 'C' is highest point in siphon is called as summit.

**84. The viscosity of gases**
(a) decreases with temperature
(b) increases with temperature
(c) is independent of temperature
(d) is none of the above

**Ans. (b) :** The viscidity of gases increases with temperature while viscosity of liquid increases with decreases temperature.

**85. Specific head under consistent pressure $C_p$ and under constant volume $C_v$ are related by**
(a) $K = \frac{C_p}{C_v}$ (b) $K = \frac{C_v}{C_p}$
(c) $K = C_pC_v$ (d) $C_p = C_v^k$

**Ans. (a) :** Specific heat under constant pressure $C_p$ and constant volume $C_v$,

Specific heat, $K = \frac{C_p}{C_v}$

**86. The bulk modulus of elasticity of a compressible fluid for isothermal process is equal to**
(a) p (b) kp
(c) k (d) none of these

**Ans. (a) :** The bulk modulus of elasticity of a compressible fluid for isothermal process is Equal to 'p'

$\therefore$ Bulk modulus, $k = \frac{-\Delta p}{dV/V}$ .....(i)

$\therefore$ for Isothermal process, pV = constant
Differentiation both side,

$$pdV + Vdp = 0$$

$$\frac{dV}{V} = -\frac{dp}{p} \quad .....(ii)$$

From (i) and (ii) –

Bulk modulus, $k = \frac{-\Delta p}{(-\Delta V/V)}$

$$k = p$$

**87. Select the expression that does not give the speed of sound wave**

(a) $\sqrt{KRT}$ (b) $\sqrt{\frac{P}{\rho}}$

(c) $\sqrt{\frac{kp}{\rho}}$ (d) $\sqrt{\frac{dp}{d\rho}}$

Where,

T is temperature, ρ is mass density and R is gas constant

**Ans. (c) :** The speed of sound in a gaseous medium

$$v = \sqrt{\frac{KRT}{\mu}}$$

Where,

v = velocity of the sound wave

k = adiabatic index

R = gas constant

μ= molar mass of medium

T = Temperature

In this formula the value of RT can be replaced by PV and the numerator and the denominator are divided by V it becomes $v = \sqrt{\frac{KP}{\rho}}$

But, it is not possible to change the pressure of the medium without changing its temperature so the pressure does not affect the speed of sound.

**88. Mach number is given by**

(a) acoustic speed/stream speed

(b) stream speed/acoustic speed

(c) product of gas constant and temperature

(d) none of these

**Ans. (b) :** Mach number is defined as the ratio of local speed (v) of the gas to speed of sound or acoustic speed.

$$M = \frac{v}{c} = \frac{v}{\sqrt{RT}}$$

**89. A rocket exerts a constant horizontal thrust of 20 kgf on a missile for 5 sec. If the missile having a mass of 1 kg. starts from rest its speed at the end of the period (neglecting the effect of acceleration due to gravity and reduction in the weight of rocket), in m/s is**

(a) 10 (b) 100

(c) 980 (d) none of these

**Ans. (b) :** Given,

Thrust farce, F = 20 kgf

$$\frac{dm}{dt} = \frac{1kg}{5} = 0.2 \text{ kg/sec.}$$

Thrust force on Rocket, $F = V_r \cdot \frac{dm}{dt}$

$$20 = V_r \times 0.2$$

$$V_r = \frac{20}{0.2}$$

$V_r = 100$ m/sec.

**90. A jet of water impinges on a flat fixed plate at an inclination. The force exerted by the jet is maximum, when the inclination of the jet with the plate is**

(a) $90^0$ (b) $45^0$

(c) $180^0$ (d) none of these

**Ans. (a) :** A jet of water impinges on a flat fixed plate at an inclination.

Force exerted by the jet is maximum at ($\theta = 90^o$)

$F_n = \rho a v^2 \sin\theta$

$= \rho a v^2 \sin 90^o$

$F_n = \rho a v^2$

**91. A jet of water impinges tangentially at the inlet of a moving curved vane with an absolute velocity of 10 m/s and leaves the vane at the outlet with an absolute velocity of 2 m/s. The efficiency of the work done is**

(a) indeterminable for want of sufficient data

(b) 96%

(c) 20%

(d) none of these

**Ans. (b) :** Given,

Initial velocity ($v_1$) = 10 m/s

Final velocity ($v_2$) = 2 m/s

Efficiency of the workdone = $(v_1)^2 - (v_2)^2$

$= 10^2 - 2^2 = 96\%$

**92. The specific speed of a turbine is defined as the speed of a geometrically similar similar turbine**

(a) of unit size with unit discharge under unit head

(b) of such size that it develops unit power for unit discharge

(c) of such a size that it develops unit pwoer under unit head

(d) that is done of the above

**Ans. (c) :** Specific speed of a the turbine is the speed at which unit power delivered at unit head .

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

Unit power, P = 1, unit head, H = 1

Ns = N

**93. An impulse turbine**

(a) always operates submerged

(b) makes use of draft tube

(c) is most suited for low head installations

(d) operates by initial complete conversion to kinetic head

**Ans. (d):** If at the inlet of turbine, the energy available is only kinematic energy, the turbine is known as impulse turbine.

- If at the inlet of the turbine, the water possesses kinematic as well as pressure energy the turbine is known as reaction turbine.

**94. Which of the following combinations results in the maximum efficiency of a pelton wheel?**

(a) vane velocity = $\frac{1}{2}\times$ jet velocity and jet deflection $=180^0$

(b) vane velocity = jet velocity and jet deflection $= 90^0$

(c) vane velocity = 2 × jet velocity and jet deflection $= 180^0$

(d) none of the above

**Ans. (a) :** Maximum efficiency of pelten wheel vane velocity = $\frac{1}{2}\times$ Jet velocity and jet deflection = 180º

**95. A reaction turbine discharges 50 $m^3$/s of water under a head of 7.5 m with an over-all efficiency of 80%. The HP developed is**

(a) 5000

(b) 300,000

(c) 4000

(d) none of these

**Ans. (c) :** Given,

Discharge, Q = 50 $m^3$/sec

Head, h = 75 m

Efficiency, $\eta$ = 80%

Power developed,

$P = \eta\rho\, g\, h\, Q$

$= 0.80 \times 1000 \times 9.81 \times 7.5 \times 50$

$= 2943000$ w

$= \frac{2943000}{746}$ HP $\left[\because 1\text{HP} = 746\text{w}\right]$

$= 3945.040$

$\approx 4000$ HP

**96. A centrifugal pump**

(a) continuously extracts energy from the water

(b) lifts water by reciprocating

(c) continuously gives energy to water

(d) does none of the above

**Ans. (d) :** Centrifugal pumps is a hydraulic machine which converts mechanical energy to hydraulic energy by the centrifugal force acting on the fluid.

**97. In a centrifugal pump, water enters the impeller**

(a) axially

(b) radially

(c) tangentially

(d) in none of ways the above

**Ans. (b) :** In a centrifugal pump water enters at the centre of the impeller. The water passes between the vanes and is pushed radically towards the casing and then onwards through the discharge nozzle.

**98. Centrifugal pumps when arranged in series**

(a) increase the discharge only

(b) increase both the discharge and head

(c) increase the head only

(d) increases the head but decreases the discharge

**Ans. (c) :** Centrifuge pumps when managed in series the total head is increased without a change to maximum discharge On other head if centrifugal pumps managed in parallel to any another the discharge is increased without any changes to maximum head.

**99. Which of the following defines the manometeric head of a centrifugal pump?**

(a) total height to which the liquid is lifted

(b) static head minus losses in the pump

(c) difference of energy heads between the outlet of pump casing and the inlet of the pump casing

(d) none of the above

**Ans. (c) :** Manometric head of a centrifugal pump is defined as the difference between total head at the outlet of the pump and total head at the inlet of the pump.

**Manometric head,** $H_m$ = Sanction head ($h_s$) + delivery Head ($h_d$) + Friction head loss in suction pipe ($h_{fs}$) + Friction head loss in delivery pipe ($h_{fd}$) + Velocity head of water in delivery pipe $\left(\frac{v^2}{2g}\right)$.

**100. A centrifugal pump lifts water from sump to an overhead tank through a height of 40m. The head lost due to friction is suction and delivery pipes is 4 m. The work done by the pump efficiency 100%) in kgf-m, kgf of water is**

(a) 44

(b) 40

(c) 36

(d) none of these

**Ans. (a) :** Given,

Efficiency of the pump, $\eta_{pump}$ = 100%

Efficiency = $\frac{\text{water power}}{\text{Shaft power}}$

$\therefore$ If the efficiency is 100%, water power is equal to shaft power,

Therefore, order to lift the water to a height of 40 m and head losses (4m) in suction and delivery pipe So, supply (40+4)=44 m of head shaft power.

Work done by the pump is 44m.